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# **UNITED STATES MILITARY ACADEMY**

**WEST POINT, NEW YORK**



**ACADEMIC PROGRAM**

**CLASS OF 2029**

**Curriculum and Course Descriptions**

**OFFICE OF THE DEAN**

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## Table of Contents

|                                                                                  |     |
|----------------------------------------------------------------------------------|-----|
| INTRODUCTION                                                                     | 4   |
| INTRO: DEAN PREFACE                                                              | 5   |
| INTRO: VICE DEAN FOREWORD                                                        | 6   |
| PART I: THE ACADEMIC PROGRAM                                                     | 7   |
| PART I: THE ACADEMIC PROGRAM - CLASS GRADUATION REQUIREMENTS                     | 31  |
| PART I: THE ACADEMIC PROGRAM - CPS CALCULATION                                   | 42  |
| PART I: THE ACADEMIC PROGRAM - STUDY OF WAR THREAD                               | 47  |
| PART I: THE ACADEMIC PROGRAM - CHARACTER THREAD                                  | 50  |
| PART II: DISCIPLINARY OFFERINGS                                                  | 55  |
| PART III: COURSE DESCRIPTIONS                                                    | 93  |
| PART III: Behavioral Sciences and Leadership (MADN-BSL)                          | 95  |
| PART III: Brigade Tactical Department (MACC-BTD)                                 | 113 |
| PART III: Center for Enhanced Performance (MADN-CEP)                             | 117 |
| PART III: Chemical and Biological Science and Engineering (MADN-CBE)             | 119 |
| PART III: Chemistry & Life Science (MADN-CLS)                                    | 138 |
| PART III: Civil and Environmental Engineering (MADN-CEE)                         | 139 |
| PART III: Civil and Mechanical Engineering (MADN-CME)                            | 156 |
| PART III: Electrical Engineering and Computer Science (MADN-EEC)                 | 157 |
| PART III: English and World Languages (MADN-EWL)                                 | 192 |
| PART III: Geography and Earth Sciences (MADN-GEO)                                | 235 |
| PART III: Geography and Environmental Engineering (MADN-GEN)                     | 248 |
| PART III: History (MADN-HIS)                                                     | 250 |
| PART III: History and War Studies (MADN-HWS)                                     | 251 |
| PART III: Kinesiology (MACC-DPE-KIN)                                             | 280 |
| PART III: Law & Philosophy (MADN-DLP)                                            | 284 |
| PART III: Mathematical Sciences (MADN-MTH)                                       | 299 |
| PART III: Mechanical and Aerospace Engineering (MADN-MAE)                        | 318 |
| PART III: Military Instruction (MACC-DMI)                                        | 335 |
| PART III: Physical Education (MACC-DPE)                                          | 344 |
| PART III: Physics and Nuclear Engineering (MADN-PNE)                             | 354 |
| PART III: Social Sciences (MADN-SOC)                                             | 376 |
| PART III: Systems Engineering (MADN-DSE)                                         | 403 |
| PART III: William E. Simon Center for the Professional Military Ethic (MACC-PME) | 412 |
| PART IV: MAJORS AND MINORS                                                       | 415 |
| PART IV: MAJORS OFFERED                                                          | 416 |
| PART IV: MINORS OFFERED                                                          | 422 |
| PART IV: MAJORS BY DISCIPLINE                                                    | 424 |
| PART IV: FIELD TABLES                                                            | 430 |
| PART IV: Behavioral Sciences and Leadership (MADN-BSL)                           | 431 |
| PART IV: Chemical and Biological Science and Engineering (MADN-CBE)              | 447 |
| PART IV: Chemistry & Life Science (MADN-CLS)                                     | 456 |
| PART IV: Civil and Environmental Engineering (MADN-CEE)                          | 457 |
| PART IV: Civil and Mechanical Engineering (MADN-CME)                             | 466 |
| PART IV: Electrical Engineering and Computer Science (MADN-EEC)                  | 467 |
| PART IV: English and World Languages (MADN-EWL)                                  | 484 |
| PART IV: Geography and Earth Sciences (MADN-GEO)                                 | 685 |
| PART IV: Geography and Environmental Engineering (MADN-GEN)                      | 695 |

## Table of Contents

|                                                          |     |
|----------------------------------------------------------|-----|
| PART IV: History (MADN-HIS)                              | 696 |
| PART IV: History and War Studies (MADN-HWS)              | 697 |
| PART IV: Kinesiology (MACC-DPE-KIN)                      | 719 |
| PART IV: Law & Philosophy (MADN-DLP)                     | 721 |
| PART IV: Mathematical Sciences (MADN-MTH)                | 727 |
| PART IV: Mechanical and Aerospace Engineering (MADN-MAE) | 745 |
| PART IV: Military Instruction (MACC-DMI)                 | 752 |
| PART IV: Physics and Nuclear Engineering (MADN-PNE)      | 753 |
| PART IV: Social Sciences (MADN-SOC)                      | 766 |
| PART IV: Systems Engineering (MADN-DSE)                  | 791 |
| APPENDIX                                                 | 802 |

# INTRODUCTION

## Preface

West Point is committed to developing leaders capable of mastering the modern battlefield through the Academic Program. The challenges of future warfare are many: continuous armed conflict, the proliferation of technology, ethical and legal debates over combat obligations, and the speed of operations—to name a few. These challenges will require Army officers to think critically, innovate rapidly, and make informed decisions. For this reason, West Point's academic program requires cadets to complete both a robust interdisciplinary core curriculum as well as an academic major of their choice. The core curriculum – the broad liberal arts aspect of the cadet's education – covers academic fields ranging from Science, Technology, Engineering, and Math to the Social Sciences and Humanities. The academic major allows a cadet to focus on an academic field of their choice and gain intellectual depth.

This Redbook provides specific descriptions of the Academic Program's components that are designed to develop cadets into capable and forward-thinking leaders. It identifies majors and minors available to each class as well as a listing of all courses offered during each academic year and term. The Redbook also includes descriptions of some of our Academic Individual Advanced Development offerings that enrich what is taught in the classroom. Through these components, the Redbook serves as a reference document to use while deciding on a major and/or minor, refining a course of study, mentoring a cadet, or addressing questions concerning academic requirements and opportunities.

There are many additional resources to help cadets achieve their potential and pursue their intellectual passions. Company Academic Counselors (CACs) and Department Academic Counselors (DACs) are available to work with cadets to develop a personally enriching course of study. Information sessions, open houses, and other opportunities to engage in conversations with faculty and fellow cadets also provide cadets with the opportunity to learn more about majors, minors, and enrichment opportunities.

Faculty can rely on Educating Future Officers to describe how we maintain and enhance our tradition of academic excellence and innovation so that we can achieve our overall mission: “to build, educate, train, and inspire the Corps of Cadets to be commissioned leaders of character committed to the Army values and ready for a lifetime of service to the Army and the Nation.”

I wish you all the best of luck during the upcoming academic year. Thank you for committing to the critical and creative thinking that the Army needs. Live honorably, lead honorably, and demonstrate excellence.



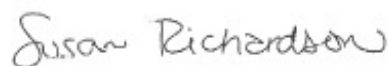
SHANE R. REEVES  
Brigadier General, USA  
Dean of the Academic Board

MADN

01 July 2025

**MEMORANDUM FOR** Cadets, Staff, and Faculty**SUBJECT:** FOREWORD: Academic Program (Redbook), AY 2026

1. The Academic Year (AY) 2026 Redbook is a guide for the Academic Program at the United States Military Academy. Used as a reference, this Redbook enables cadets to design and refine their academic programs and provides faculty information to guide and mentor cadets about and through the academic curriculum. This version contains all courses offered in academic year 2026 and the majors and minors for the Classes of 2025-2029.
2. Parts 1 and 2 of the Redbook present information about the United States Military Academy's educational philosophy, graduation requirements, academic standards, core curriculum, and academic discipline descriptions. Part 3 presents a catalog of all the courses offered at the Military Academy. Parts 4 and 5 provide different approaches to explore majors and minors.
3. Substantive revisions to the majors, minors, and courses as well as any substantive changes to other parts of the Redbook are listed by department in an appendix at the end of the Redbook.
4. Cadets should take ownership of their academic schedule and choose the major and elective courses that fit their intellectual interests and best prepares them to be officers. Cadets in all classes should review and revise their academic schedule, while ensuring they are meeting program requirements and registrar deadlines. Cadets are encouraged to discuss their current program and plans with their Company Academic Counselors (CACs) or Department Academic Counselors (DACs), as appropriate.
5. This volume supersedes all other Redbooks.
6. We welcome suggestions that will improve the Redbook. Please send all comments, suggestions, and edits on the Redbook to [AARS.All@westpoint.edu](mailto:AARS.All@westpoint.edu).



SUSAN RICHARDSON  
Vice Dean for Academic Affairs  
United States Military Academy

# **PART I: THE ACADEMIC PROGRAM**

# PART I: THE ACADEMIC PROGRAM

## USMA EDUCATIONAL PHILOSOPHY

### USMA Mission:

*To build, educate, train, and inspire the Corps of Cadets to be commissioned leaders of character committed to the Army Values and ready for a lifetime of service to the Army and Nation.*

**WPLDS:** The West Point Leader Development System (WPLDS) is a comprehensive and integrated approach designed to harness every facet of the West Point experience while leveraging specific developmental events and activities to reinforce and assess character and honorable living. Upon commissioning, West Point graduates will:

1. Live honorably:
  - Internalize the Cadet Honor Code, West Point Motto, and Army Values.
  - Demonstrate moral courage regardless of personal consequences.
  - Treat others with dignity and respect.
  - Act with humility and professional decorum in all environments through appearance, demeanor, behaviors, and words.
2. Lead honorably:
  - Accomplish the mission consistent with the Army values.
  - Know, adhere to, and enforce standards.
  - Build cohesive and effective teams.
3. Demonstrate excellence:
  - Maximize academic, military, and physical performance.
  - Seek challenges and feedback; reflect and improve.
  - Communicate and interact effectively.
  - Anticipate and solve complex problems.

## ACADEMIC PROGRAM GOALS

In support of WPLDS, the Academic Program has established the following overarching goal and seven Academic Program Goals (APGs) to help guide the academic development of cadets. Each of the seven goals is accompanied by supporting objectives called "What Graduates Can Do" (WGCD) statements. WGCD statements define specific indicators of skills expected of cadets at the time of graduation and are the assessable objectives that we use to determine if our core and major programs are achieving the standards of our program goals as expected. Each are listed below their respective APG in the following list. Our APGs also directly reflect upon, connect with, and contribute to achievement of the institutional outcomes defined within WPLDS.

**The Overarching Academic Goal:** Graduates integrate knowledge and skills from a variety of disciplines to anticipate and respond appropriately to opportunities and challenges in a changing world.

### Academic Program Goals:

**1. Communication:** *Graduates communicate effectively with all audiences.*

WGCD 1.1: Listen actively, read critically, and develop an informed understanding of the communications of others.

WGCD 1.2: Speak and write using Standard American English.

WGCD 1.3: Effectively convey meaningful information to diverse audiences using appropriate forms and media.

WGCD 1.4: Communicate in a foreign language.

WGCD 1.5: Use sound logic and relevant evidence to make convincing arguments.

**2. Critical Thinking and Creativity:** *Graduates think critically and creatively.*



- WGCD 2.1: Identify the essential aspects of a situation and ask relevant questions.  
 WGCD 2.2: Integrate knowledge and skills from a variety of disciplines.  
 WGCD 2.3: Make meaningful connections and distinctions among diverse experiences and concepts.  
 WGCD 2.4: Reason both quantitatively and qualitatively.  
 WGCD 2.5: Think innovatively and accept risk to pursue solutions in the face of ambiguity.  
 WGCD 2.6: Value reflection and creativity; envision possibilities.

**3. Lifelong Learning:** *Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development.*

- WGCD 3.1: Demonstrate the willingness and ability to learn independently.  
 WGCD 3.2: Engage successfully in deliberate self-directed and collaborative learning experiences.  
 WGCD 3.3: Pursue self-awareness and embrace the responsibility for personal intellectual development.  
 WGCD 3.4: Pursue knowledge in areas of personal or professional interest.

**4. Ethical Reasoning:** *Graduates recognize ethical issues and apply ethical perspectives and concepts in decision making.*

- WGCD 4.1: Understand the intellectual foundations of ethical principles.  
 WGCD 4.2: Recognize ethical components of problems and situations.  
 WGCD 4.3: Examine and evaluate different ethical perspectives, principles, and concepts in context.  
 WGCD 4.4: Apply ethical perspectives and concepts in solving complex problems, including those found in military settings.

**5. Science, Technology, Engineering, and Mathematics:** *Graduates apply science, technology, engineering, and mathematics concepts and processes to solve complex problems.*

- WGCD 5.1: Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.  
 WGCD 5.2: Apply the scientific method.  
 WGCD 5.3: Collect and analyze data in support of decision making.  
 WGCD 5.4: Apply an engineering design process to create effective and adaptable solutions.  
 WGCD 5.5: Explain and apply computing and information technology concepts and practices in the context of the Cyber Domain.

**6. Humanities and Social Sciences:** *Graduates apply concepts from the humanities and social sciences to understand and analyze the human condition.*

- WGCD 6.1: Understand, analyze, and know how to influence human behavior.  
 WGCD 6.2: Analyze the history, diversity, complexity, and interaction of cultures.  
 WGCD 6.3: Analyze political, legal, military, and economic influences on social systems.  
 WGCD 6.4: Engage in and reflect on cross cultural experiences.  
 WGCD 6.5: Integrate the methodologies of the humanities and social sciences in decision-making.

**7. Disciplinary Depth:** *Graduates integrate and apply knowledge and methodological approaches gained through in-depth study of an academic discipline.*

- WGCD 7.1: Apply disciplinary tools, methods of inquiry, and theoretical approaches.  
 WGCD 7.2: Identify and explain representative questions and arguments of their chosen disciplines.  
 WGCD 7.3: Recognize limits of a discipline as well as areas in which it contributes to intellectual inquiry and problem solving.  
 WGCD 7.4: Synthesize knowledge and concepts from across their chosen disciplines.  
 WGCD 7.5: Contribute disciplinary knowledge and skills as a part of a collaborative effort engaging challenges that span multiple disciplines.

## ACCREDITATION

The United States Military Academy is accredited by the Middle States Commission on Higher Education, 1007 North Orange Street, 4th Floor, MB #166, Wilmington, DE 19801 (267-284-5011). The Middle States Commission on Higher Education is an institutional accrediting agency recognized by the U.S. Secretary of Education.

- The following USMA majors are accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>:

- Chemical Engineering
- Civil Engineering
- Electrical Engineering
- Engineering Management
- Environmental Engineering
- Mechanical Engineering
- Nuclear Engineering
- Systems Engineering
- The following USMA majors are accredited by the Computing Accreditation Commission of ABET, <http://www.abet.org>:
  - Computer Science
  - Cyber Science
- The USMA major, Geospatial Science, is accredited by the United States Geospatial Intelligence Foundation (USGIF).
- The Chemistry Major with ACS Certification track is approved by the Committee on Professional Training of the American Chemical Society (ACS).

## THE ACADEMIC PROGRAM

The Academic Program is that broad structure within USMA which is responsible for the intellectual development of cadets. It consists of courses, minors, majors, activities, and departments, and is headed by a Brigadier General, the Dean of the Academic Board.

## GRADUATION REQUIREMENTS AND ACADEMIC STANDARDS

Army Regulation 150-1, United States Military Academy Organization, Administration and Operation states that First Class cadets who have successfully completed the requirements of the course of instruction, including academic, military, and physical development programs; have maintained prescribed standards of conduct; and who have demonstrated proper moral-ethical qualities, leadership, and character may receive a diploma signed by the Superintendent, USMA, the Commandant of Cadets, and the Dean of the Academic Board. These cadets will have earned the Bachelor of Science degree and will be designated graduates of USMA. Graduation requirements for all three programs — academic, military, and physical — and institutional (non-program) requirements by class year are available through the following link: Classes of [2024–2026](#), Classes of [2024–2026 Addendum](#), Class of [2027](#), Class of [2028](#), and Class of [2029](#).

## ACADEMIC GRADUATION REQUIREMENTS

To satisfy the academic portion of these graduation requirements, a cadet must:

- Complete successfully or validate each course in the core curriculum, including the common core courses and a core engineering sequence
- Satisfy the requirements of at least one major
- Successfully complete 40 academic courses of at least 3.0 credit hours each. Validated and remedial courses cannot count towards this requirement.
- Achieve a 2.00 Cumulative Quality Point Average (CQPA). The CQPA is calculated based on letter-grade performance for any academic, physical, or military course with credit hours. It generally corresponds to grade point average (GPA) in other colleges and universities.

As part of the West Point experience, a cadet is required to complete requirements and achieve minimum standards in three developmental programs within the USMA West Point Leader Development System (WPLDS). Within the WPLDS, the military program score (MPS), the physical program score (PPS), and the academic program score (APS) combine to form the cadet performance score (CPS). The score is calculated by first normalizing each program score to a mean zero and standard deviation of one and then calculating a weighted average based on a cadet's three normalized scores. A complete explanation of the calculation of the CPS, which ultimately determines a cadet's class rank, can be found at this link ( [CPS Calculation](#) ). The APS is based on performance in courses within the Academic Program and does not include military science and physical education courses. Refer to DPOM 02-01 for APS computation details. Cadets who are deficient in one or more of the three developmental programs for failure to maintain minimum program performance standards may be considered by the Academic Board for separation.

The United States Military Academy's academic curriculum has two significant structural features. The first is a core of

twenty-four academic courses that the Academy considers essential to the broad base of knowledge necessary for all graduates and a three-course core engineering sequence for those cadets who do not choose to major in engineering. This core academic curriculum, when combined with physical education and military science, provides USMA cadets with the necessary knowledge, skills, and attributes necessary for future officers in the U.S. Army. The second structural feature is the opportunity to specialize and explore an area in depth through the selection of an academic major consisting of not less than thirteen required or elective courses.

## WHAT CONSTITUTES A COURSE

Although 'course' is a commonly used word in curriculum descriptions, USMA has the following requirements for an educational experience to be termed a 'course.' Each course is different in many specific ways. In general, however, there are guidelines for any course of instruction which contribute to it being worthy of academic credit. The development of a course along the following lines is what is required for a course to be included for academic credit on a cadet's academic transcript.

**Course outcomes that require new learning experiences.** Learning involves a change in capabilities or dispositions that can be attributed to experience. "Change" usually implies students acquiring a new capability or disposition: what they know (knowledge), how they use what they know (intellectual skills), how they think, what they can do (physical skills), or what they value (attitudes and values). Learning is not normally considered as maintenance of already acquired capabilities. A course of instruction then is the purposeful arrangement of experiences designed to facilitate intended change in students' capabilities or dispositions which we represent by course outcomes.

**A valid comprehensive method of evaluating student mastery of course outcomes.** Student evaluation is a critical component of the learning process and must be present in a course of instruction. We recognize that evaluation methods and the frequency of evaluation will vary as a function of course outcomes; however, the evaluation method should assess students' mastery of course outcomes and should permit valid inferences about student learning.

If a course of instruction meets the preceding two guidelines, then the awarding of credit hours should be based on a calculation of planned time (40 hours of planned time associated with 1.0 credit hour). A 3.0 credit hour course usually requires 40 instructor contact hours with two hours of preparation required for each hour in class: 40 lessons at 3hrs/lesson (1 contact hour in-class and 2 hours of outside preparation) = 120 hours = 3.0 credit hours; 30 lessons at 4hrs/lesson (1.25 contact hours in-class and 2.75 hours of outside preparation) = 120 hours = 3.0 credit hours.

## ACADEMIC COURSE LOADS

The minimum load that all cadets must carry under normal circumstances is five academic courses which are equal to or greater than 15 semester hours of credit. Plebes may take four academic courses in the semester where they have MS100 and PE215. In addition, cadets will meet established requirements for physical education courses and military science core curriculum during a regular academic semester.

During the summer, the Military Academy provides opportunities to pursue Academic Individual Advanced Development (AIAD) and to complete coursework in the Summer Term Academic Plan (STAP). Under some circumstances, STAP may be used to reduce a course load during the academic year to four courses for a term.

## OVERLOADING

Cadets are considered overloading when they are carrying seven academic courses in a term. Overload courses offer the means of adapting an academic program to capitalize upon a cadet's abilities and to satisfy a particular interest. During the Third, Second, and First Class years, cadets may wish to enroll in one overload course for each term. Approval will depend upon demonstrated ability and motivation. Specifically, Dean's List recognition in the preceding term is required. Cadets with a Cumulative Quality Point Average (CQPA) of 2.30 or higher may overload in their First Class year. The designated overload course must be a course that can be dropped without jeopardizing any academic graduation requirements (e.g., core courses, primary major courses).

Cadets are required to designate which of their courses is the designated overload course. No change in this designation will be permitted after the term begins. A cadet who enrolls in an overload course and finds it unsuited to his or her needs may request withdrawal from the overload course through the Academic Affairs and Registrar Services (AARS), Office of the Dean, any time after the first month of the term and up until the beginning of Term End

Examinations (TEE). The Dean may remove a cadet from an overload course for poor performance in that course or in any other course.

## VALIDATION AND ADVANCED COURSE ENROLLMENT

Advanced Placement test scores and previous high school or college academic records are reviewed by the academic departments to determine whether a cadet is eligible for validation of certain courses. Many academic departments use examinations and personal interviews, in addition to screening academic records, to assist in their evaluation. When the process is completed, a cadet may be offered the opportunity to validate a course. Once validated, the decision to take the course is entirely the cadet's to make; cadets should keep in mind, however, that they may have to replace the validated course with another course. The significant aspects of validation are that it is voluntary, and that, while it excuses a cadet from taking a course, it does not reduce graduation course requirements. Validated courses will be reflected without credit hours on the transcript. However, the credit hour content of validated courses is recognized as contributing appropriate disciplinary content towards meeting programmatic rigor for accreditation purposes.

## OVERVIEW OF THE ACADEMIC CURRICULUM

The two charts below further highlight an overview of the academic program. The first chart, figure 1, presents the curriculum in overall blocks of course requirements. The second chart, figure 2, presents the typical sequence of courses taken over eight semesters during a cadet's 47-month experience.

| 37 Core Course Curriculum |                                | 13 Courses for the Major                                                                                                                                                                                                                                                           |                              |
|---------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1                         | Chemistry 1                    | 38                                                                                                                                                                                                                                                                                 | Major course 1               |
| 2                         | Physics 1                      | 39                                                                                                                                                                                                                                                                                 | Major course 2               |
| 3                         | Chem 2 / Physics 2 / Bio       | 40                                                                                                                                                                                                                                                                                 | Major course 3               |
| 4                         | Math (Modeling)                | 41                                                                                                                                                                                                                                                                                 | Major course 4               |
| 5                         | Math (Calculus)                | 42                                                                                                                                                                                                                                                                                 | Major course 5               |
| 6                         | Math (Statistics)              | 43                                                                                                                                                                                                                                                                                 | Major course 6               |
| 7                         | Cyber 1 (IT)                   | 44                                                                                                                                                                                                                                                                                 | Major course 7               |
| 8                         | Cyber 2 / STEM Depth           | 45                                                                                                                                                                                                                                                                                 | Major course 8               |
| 9                         | Physical Geography             | 46                                                                                                                                                                                                                                                                                 | Major course 9               |
| 10                        | Eng Sequence course 1          | 47                                                                                                                                                                                                                                                                                 | Complementary Support Course |
| 11                        | Eng Sequence course 2          | 48                                                                                                                                                                                                                                                                                 | Complementary Support Course |
| 12                        | Eng Sequence course 3          | 49                                                                                                                                                                                                                                                                                 | Complementary Support Course |
| 13                        | Economics                      | 50                                                                                                                                                                                                                                                                                 | Integrative Experience       |
| 14                        | Political Science              | <b>Integrative Curricular Components</b> <ul style="list-style-type: none"> <li>• Character Thread</li> <li>• Human Condition Thread</li> <li>• Study of War Thread</li> <li>• West Point Writing Program</li> <li>• MX400</li> <li>• Three-Course Engineering Sequence</li> </ul> |                              |
| 15                        | International Relations        |                                                                                                                                                                                                                                                                                    |                              |
| 16                        | History 1 (Mil His)            |                                                                                                                                                                                                                                                                                    |                              |
| 17                        | History 2 (U.S. or Regional)   |                                                                                                                                                                                                                                                                                    |                              |
| 18                        | History 3 (Mil Art)            |                                                                                                                                                                                                                                                                                    |                              |
| 19                        | Composition                    |                                                                                                                                                                                                                                                                                    |                              |
| 20                        | Literature                     |                                                                                                                                                                                                                                                                                    |                              |
| 21                        | Philosophy & Ethical Reasoning |                                                                                                                                                                                                                                                                                    |                              |
| 22                        | Foreign Language 1             |                                                                                                                                                                                                                                                                                    |                              |
| 23                        | Foreign Language 2             |                                                                                                                                                                                                                                                                                    |                              |
| 24                        | Psychology                     |                                                                                                                                                                                                                                                                                    |                              |
| 25                        | Constitutional/Military Law    |                                                                                                                                                                                                                                                                                    |                              |
| 26                        | Military Leadership            |                                                                                                                                                                                                                                                                                    |                              |
| 27                        | Military Science 1             |                                                                                                                                                                                                                                                                                    |                              |
| 28                        | Military Science 2             |                                                                                                                                                                                                                                                                                    |                              |
| 29                        | Military Science 3             |                                                                                                                                                                                                                                                                                    |                              |
| 30                        | Officership (MX400)            |                                                                                                                                                                                                                                                                                    |                              |
| 31                        | DPE 1 Boxing                   |                                                                                                                                                                                                                                                                                    |                              |
| 32                        | DPE 2 Military Movement        |                                                                                                                                                                                                                                                                                    |                              |
| 33                        | DPE 3 Personal Fitness         |                                                                                                                                                                                                                                                                                    |                              |
| 34                        | DPE 4 Survival Swimming        |                                                                                                                                                                                                                                                                                    |                              |
| 35                        | DPE 5 Combat Apps              |                                                                                                                                                                                                                                                                                    |                              |
| 36                        | DPE 6 Army Fitness             |                                                                                                                                                                                                                                                                                    |                              |
| 37                        | DPE 7 Lifetime Sport           |                                                                                                                                                                                                                                                                                    |                              |

\*Courses in blue are Math, Science and Engineering (MSE) courses. Courses in red are Humanities and Social Science (HSS) courses. Courses in black are courses affiliated with each academic major. Courses in green are credit-bearing courses from the Military and Physical Programs. Integrative Curricular Components are described below.

Fig 1a: Overview of the Academic Program (with additional MS and PE classes) — Class of 2023-2028\*

**37 Core Course Curriculum**

|    |                           |
|----|---------------------------|
| 1  | Chemistry 1               |
| 2  | Physics 1                 |
| 3  | Chem 2 / Physics 2 / Bio  |
| 4  | Math (Modeling)           |
| 5  | Math (Calculus)           |
| 6  | Math (Statistics)         |
| 7  | Cyber 1                   |
| 8  | Cyber 2 / STEM Depth      |
| 9  | Geography                 |
| 10 | Eng Sequence course 1     |
| 11 | Eng Sequence course 2     |
| 12 | Eng Sequence course 3     |
| 13 | Economics                 |
| 14 | Political Science         |
| 15 | International Relations   |
| 16 | History 1 (U.S. or World) |
| 17 | History 2 (Mil Art I)     |
| 18 | History 3 (Mil Art II)    |
| 19 | Composition               |

|    |                                  |
|----|----------------------------------|
| 20 | Literature                       |
| 21 | Philosophy & Ethical Reasoning   |
| 22 | Foreign Language 1               |
| 23 | Foreign Language 2               |
| 24 | Psychology                       |
| 25 | Constitutional/Military Law      |
| 26 | Military Leadership              |
| 27 | MS 1 Intro to Warfighting        |
| 28 | MS 2 Fund. of Small Unit Ops     |
| 29 | MS 3 Platoon Operations          |
| 30 | Officership (MX400)              |
| 31 | DPE 1 Boxing                     |
| 32 | DPE 2 Personal Fitness           |
| 33 | DPE 3 Military Movement          |
| 34 | DPE 4 Survival Swimming          |
| 35 | DPE 5 Combat Apps                |
| 36 | DPE 6 Army Fitness               |
| 37 | DPE 7 Lifetime Physical Activity |

**13 Courses for the Major**

|    |                              |
|----|------------------------------|
| 38 | Major course 1               |
| 39 | Major course 2               |
| 40 | Major course 3               |
| 41 | Major course 4               |
| 42 | Major course 5               |
| 43 | Major course 6               |
| 44 | Major course 7               |
| 45 | Major course 8               |
| 46 | Major course 9               |
| 47 | Complementary Support Course |
| 48 | Complementary Support Course |
| 49 | Complementary Support Course |
| 50 | Capstone                     |

**Integrative Curricular Components**

- Character Thread
- Study of War Thread
- West Point Writing Program
- MX400
- Three-Course Engineering Sequence

\*Courses in blue are Math, Science and Engineering (MSE) courses. Courses in red are Humanities and Social Science (HSS) courses. Courses in black are courses affiliated with each academic major. Courses in green are credit-bearing courses from the Military and Physical Programs. Integrative Curricular Components are described below.

Fig 1b: Overview of the Academic Program (with additional MS and PE courses) — Class of 2029 and beyond\*



*Class of 2023+ Standard Course Scheduling*

| PLEBE                                                    |                    | YEARLING                                                                                                             |                     | COW                 |                     | FIRSTIE        |                |
|----------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|---------------------|----------------|----------------|
| Fall                                                     | Spring             | Fall                                                                                                                 | Spring              | Fall                | Spring              | Fall           | Spring         |
| MATH MOD                                                 | CALC               | PHILOS<br>PROB/STAT                                                                                                  | PROB/STAT<br>PHILOS | MAJ CRS 3           | MAJ CRS 5           | MAJ CRS 7      | MAJ CRS 9      |
| CHEM 1                                                   | PHYS 1             | SCIENCE 2                                                                                                            | GEOG                | MAJ CRS 4           | MAJ CRS 6           | MAJ CRS 8      | MAJ CRS 10: IE |
| GEOG                                                     | CHEM 1             | PHYS 1                                                                                                               | SCIENCE 2           |                     |                     |                |                |
| EN COMP                                                  | EN LIT             | MAJ CRS 1                                                                                                            | MAJ CRS 2           | 3CES CRS 1          | 3CES CRS 2          | 3CES CRS 3     | MAJ CRS: CSC 2 |
| IT<br>PSYCH                                              | PSYCH<br>IT        | POLI SCI<br>ECON                                                                                                     | ECON<br>POLI SCI    | LDRSH<br>IR         | IR<br>LDRSH         | MAJ CRS: CSC 1 | MAJ CRS: CSC 3 |
| HIST OF ARMY                                             | REG HIST / AM HIST | LANG 1                                                                                                               | LANG 2              | MIL ART<br>IT/CYBER | IT/CYBER<br>MIL ART | MX400<br>LAW   | LAW<br>MX400   |
| : per math placement only; Must satisfy PHYS 1 requisite |                    | Yellow-shaded cells = slots to Schedule in each Major including Complementary Support Courses (CSC), Science2 & 3CES |                     |                     |                     |                |                |
| Credits                                                  | 16.5-17.5          | 17.5                                                                                                                 | 17                  | 16-17               | 15+                 | 15+            | 15+            |
| Credits with MS/PE balanced                              | 17.5-18.5          | 19                                                                                                                   | 18.5                | 18-19               | 16+                 | 16.5+          | 16.5+          |

Fig 2a: Typical Sequence of Courses – 8 Term Academic Program (8TAP) – Class of 2023-2028

*Class of 2029+ Standard Course Scheduling*

| PLEBE                              |                 | YEARLING                                 |           | COW                                      |           | FIRSTIE                    |                |
|------------------------------------|-----------------|------------------------------------------|-----------|------------------------------------------|-----------|----------------------------|----------------|
| Fall                               | Spring          | Fall                                     | Spring    | Fall                                     | Spring    | Fall                       | Spring         |
| AM /WRLD HIST — OPEN               |                 | MAJ CRS 1                                | MAJ CRS 2 | MAJ CRS 3                                | MAJ CRS 5 | MAJ CRS 7                  | MAJ CRS 9      |
| CHEM 1 — PHYS 1 — SCIENCE 2 — GEOG |                 |                                          |           | MAJ CRS 4                                | MAJ CRS 6 | MAJ CRS 8                  | MAJ CRS 10: IE |
| GEOG — CHEM 1 — PHYS 1 — SCIENCE 2 |                 |                                          |           |                                          |           |                            |                |
| EN COMP                            | EN LIT          | PHILOS — PHILOS<br>PROB/STAT — PROB/STAT |           | LDRSH — IR<br>IR — LDRSH                 |           | MAJ CRS: CSC 1             | MAJ CRS: CSC 2 |
| CY — PSYCH<br>PSYCH — CY           |                 | AM POL — ECON<br>ECON — AM POL           |           | 3CES CRS1                                | 3CES CRS2 | 3CES CRS3                  | MAJ CRS: CSC 3 |
| MATH MOD                           | CALC            | LANG 1                                   | LANG 2    | MIL ART1* or 2<br>IT/CYBER — MIL ART1or2 |           | MX400 — LAW<br>LAW — MX400 |                |
| PE116                              | PE215/<br>MS100 | PE117/<br>PE32X — MS200                  |           | PE360/<br>PE450 — MS300                  |           | MIL ART1or2 — PE2XX        |                |

\*Mil Art 1 (HI301) can be taken any semester after plebe year.

|                                                                                    |                                                                                          |                                                                                     |                                                                                         |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Plebe year: one term = 4 acad crses + PE215/MS100, one term = 5 acad crses + PE116 | MIL ART1or2 or a different 6 <sup>th</sup> acad crse could be brought into Yearling Year | Cow Year: one term may include a 6 <sup>th</sup> acad crse depending on MIL ART1or2 | Firstie Year: one term may include a 6 <sup>th</sup> acad crse depending on MIL ART1or2 |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|

|                 |           |      |           |           |           |           |           |           |
|-----------------|-----------|------|-----------|-----------|-----------|-----------|-----------|-----------|
| Credits + MS/PE | 17.0-18.0 | 17.5 | 16.0-18.0 | 16.5-18.5 | 17.0-20.0 | 16.5-19.5 | 15.5-18.0 | 15.5-18.0 |
|-----------------|-----------|------|-----------|-----------|-----------|-----------|-----------|-----------|

Fig 2b: Typical Sequence of Courses – 8 Term Academic Program (8TAP) – Class of 2029 and beyond

## ACADEMIC CORE CURRICULUM

The foundation of the academic program at USMA is 24 core courses and three courses in an engineering sequence for those cadets who do not choose a major in engineering. For most cadets the first year is a common academic experience including, in term 2, the selection of a major, followed by the selection of a three-course engineering sequence in the Fall of Third Class year.

Designed to provide educational breadth, the 24 academic core courses are listed below along with alternate sequences or courses. Cadets may be enrolled in authorized alternative courses to the academic core curriculum based on placement testing and/or validations. Alternative courses are usually more advanced versions of the basic core courses.

Fig 3: The 24 Core Academic Courses (and Alternates) by Discipline

| Area                         | Required Courses |                        | Alternate Courses<br>(based on placement testing/validations) |
|------------------------------|------------------|------------------------|---------------------------------------------------------------|
| Chemistry                    | CH101            | General Chemistry I    | CH151 Advanced General Chemistry I                            |
| Information Technology/Cyber | CY105            | Computing Fundamentals | CY155 Explorations in Computing Fundamentals                  |

|                                                                                                                             |                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Economics</b>                                                                                                            | SS201 Economics: Principles and Problems                                                                                                                                                                                                          |                                                                                                                                                                                                                                            |
| <b>English</b>                                                                                                              | EN101 Composition                                                                                                                                                                                                                                 | EN151 Advanced Composition                                                                                                                                                                                                                 |
| <b>Foreign Language</b>                                                                                                     | Two courses required. Sequence determined by the Department of English and World Languages based on cadet preferences, language ability, and available resources.                                                                                 |                                                                                                                                                                                                                                            |
| <b>History</b>                                                                                                              | HI105 History of the United States<br>OR<br>HI108W Foundations of the Modern World                                                                                                                                                                | HI155 Advanced History of the U.S.<br>OR<br>HI158W Advanced Foundations of the Modern World                                                                                                                                                |
| <b>International Relations</b>                                                                                              | SS307 International Relations                                                                                                                                                                                                                     | SS357 Advanced International Relations                                                                                                                                                                                                     |
| <b>Law</b>                                                                                                                  | LW403 Constitutional and Military Law                                                                                                                                                                                                             |                                                                                                                                                                                                                                            |
| <b>Leadership</b>                                                                                                           | PL100 General Psychology for Leaders<br>AND<br>PL300 Military Leadership                                                                                                                                                                          | PL150 Advanced General Psychology for Leaders<br>AND<br>PL350 Advanced Military Leadership                                                                                                                                                 |
| <b>Literature</b>                                                                                                           | EN102 Literature                                                                                                                                                                                                                                  | EN152 Advanced Literature                                                                                                                                                                                                                  |
| <b>Mathematics</b><br>(Alternate course sequence is based on placement testing and validation)                              | MA103* Mathematical Modeling and Introduction to Calculus<br>*MA100 Precalculus (may be required as an additional course)<br>AND<br>MA104 Calculus I OR<br>MA204 Calculus I & II (validates MA104)<br><br>AND<br>MA206 Probability and Statistics | MA153 Mathematical Modeling and Introduction to Differential Equations<br><br>AND<br>MA205 Calculus II (validates MA104) OR<br>MA255 Advanced Multivariable Calculus (validates MA104)<br>AND<br>MA256 Advanced Probability and Statistics |
| <b>Military History</b>                                                                                                     | HI301 History of the Military Art to 1900<br>AND<br>HI302 History of the Military Art from 1900 to Present                                                                                                                                        | HI351 Advanced History of the Military Art to 1900<br><br>AND<br>HI352 Advanced History of the Military Art 1900 to Present                                                                                                                |
| <b>Officership</b>                                                                                                          | MX400 Officership                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                            |
| <b>Philosophy</b>                                                                                                           | PY201 Philosophy and Ethical Reasoning                                                                                                                                                                                                            | PY251 Advanced Philosophy and Ethical Reasoning                                                                                                                                                                                            |
| <b>Geography</b>                                                                                                            | GE203 Geography                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                            |
| <b>Physics</b>                                                                                                              | PH201 Physics I                                                                                                                                                                                                                                   | PH251 Advanced Physics I                                                                                                                                                                                                                   |
| <b>Political Science</b>                                                                                                    | SS202 American Politics                                                                                                                                                                                                                           | SS252 Advanced American Politics                                                                                                                                                                                                           |
| <b>Science Depth</b><br>(Cadets can choose <u>one</u> course based on academic major and interest)                          | CH102 General Chemistry II<br>CH275 Biology<br>PH202 Physics II<br>PH275 Physics II: Space                                                                                                                                                        | CH152 Advanced General Chemistry II<br>CH375 Advanced Biology<br>PH252 Advanced Physics II                                                                                                                                                 |
| <b>STEM Depth</b><br>(Cadets can choose <u>one</u> course based on academic major and need to satisfy IT/CYBER requirement) | CY305 Cyber Foundations<br>MA204 Calculus I & II<br>MA205 Calculus II<br>CH102 General Chemistry II<br>CH275 Biology<br>PH202 Physics II<br>PH275 Physics II: Space                                                                               | CY355 Cyber Foundations — Computing<br><br>MA255 Advanced Multivariable Calculus<br>CH152 Advanced General Chemistry II<br>CH375 Advanced Biology<br>PH252 Advanced Physics II                                                             |



Fig 3: The 24 Core Academic Courses (and Alternates) by Discipline

Designed to provide educational breadth, the 24 academic core courses are listed above along with alternate sequences or courses. Cadets may be enrolled in authorized alternative courses to the academic core curriculum based on placement testing and/or validations. Alternative courses are usually more advanced versions of the basic core courses.

### **Mathematics, Science, and Engineering (MSE)**

Within the core academic curriculum, there is a strong emphasis on mathematics, science, and engineering. The MSE sequence provides each cadet with a fundamental knowledge of the experimental and analytic techniques of the basic sciences. This sequence begins in the Fourth Class year with two semesters of mathematics, one semester of chemistry, one semester of geography, and one semester of calculus-based physics. Geography or physics may be delayed until the Third Class year depending on faculty resources and cadet preferences. The coursework continues in the Third Class year with one semester of statistics and an additional science depth course that cadets select based on their major and their interests.

### **Humanities and Social Sciences (HSS)**

Additionally, the core curriculum includes a strong pre-professional sequence of courses from the humanities and social sciences to develop an awareness of the people, government, and society that the commissioned officer will serve. This sequence begins in the Fourth Class year with one semester of history, one semester of English composition, one semester of literature, and one semester of psychology. It continues in the Third Class year with one semester each of political science, philosophy, and economics. The Second Class year includes one semester of international relations, two semesters of military history, and one semester of military leadership. The First Class year's contribution to this sequence of professional development is found in a one-semester course in constitutional and military law and one semester of a capstone course titled "Officership."

### **Information Technology/Cyber**

The core curriculum also includes an information technology/cyber sequence (IT/CYBER) designed to ensure that every academy graduate is comfortable with and capable of securely using computers and information technology in an Army that must fight and win in an overarching cyber domain. IT/CYBER skills are first developed through an introductory information technology/cyber course in the Fourth Class year and the integration of computer and cyber applications throughout the core curriculum and particularly in the IT/CYBER requirement in the Second or First Class year. The IT/CYBER requirement may be fulfilled with the core course, CY305, Cyber Foundations or through existing course coverage of cyber topics in particular majors (in this case, the cadet will be required to fulfill a Science, Technology, Engineering and Math [STEM] depth requirement with another STEM course — see listing of core courses above).

### **Foreign Language**

Most cadets will begin their study of a foreign language in Third Class year. If a cadet expresses an interest, however, the sequence may be started in Fourth Class year (contingent on seat availability and discretion of the Registrar). All cadets will take at least two semesters of one of the eight foreign languages offered in residence. Course work will present perspectives from another culture, develop the ability to learn another language, provide an introductory level of proficiency in the language selected, and provide a firm foundation for further language study.

These features mean that the first two academic semesters are a common core experience for the majority of cadets, and most of the second two semesters are common as well. Individual alterations to the typical sequence can be made based on specific needs and capabilities. Cadets are encouraged to work closely with academic counselors when designing their academic programs.

## **Core Engineering Sequences (CES)**

Cadets will seek counsel from their DACs in the beginning of their Third Class year to request a CES that complements their academic major and interests. The Registrar will fill CES enrollments based on cadet preferences and faculty availability. CES assignments typically cannot be changed after the initial assignment unless the cadet finds a cadet 'trade partner' in the same year group who is willing to make a swap.

### **IN - Infrastructure Engineering**

MC300 Fundamentals of Engineering Mechanics and Design  
CE350 Infrastructure Engineering

## CE450 Construction Management

The Infrastructure Engineering Sequence focuses on the design, analysis, and construction of the built environment (i.e., man-made structures and facilities used to accommodate societies' activities). Cadets learn about the importance of the infrastructure sectors, such as water, power, and transportation, and their inter-relationships. The first course, **MC300 Fundamentals of Engineering Mechanics and Design**, provides cadets with foundational engineering mechanics as the basis for all structural elements in the context of learning the engineering design process. In the second course, **CE350 Infrastructure Engineering**, cadets learn to identify, analyze, and assess the built environment, which is the infrastructure foundation for modern society. Expanding upon the structural systems introduced in the first course, cadets engage in learning about water and wastewater, power, transportation, solid waste, communications systems, public administration, and sustainment. In the third course, **CE450 Construction Management**, cadets learn how to plan and execute construction of the built environment. The culminating event that brings together the two previous courses is the designing, planning, and presenting of a construction briefing of a base camp in a combat theater of operations. Integral to all three courses is an active learning environment with emphasis on classroom demonstrations, virtual and physical simulations, and practical exercises.

## CY - Cyber Engineering

CY300 Programming Fundamentals

CY350 Network Engineering and Management

CY450 Cyber Security Engineering

The Cyber Engineering Sequence focuses on fundamental cyber-enabling skills as well as current cybe-related issues, threats, vulnerabilities, and non-technical considerations. As the newest warfighting domain and branch of the Army, cyberspace presents new opportunities and demands for leaders to protect and exploit networks and information resources. Cyber leaders must understand computer systems, networks, and information assurance and be able to analyze, design, and evaluate such systems. The Cyber Engineering Sequence is designed to provide sufficient breadth and depth of knowledge for non-technical majors to establish a nucleus for subsequent learning and adaptation to rapid change. The first course, **CY300 Programming Fundamentals**, provides cadets with broad introduction to computer programming, which will allow them to design, build, and test small to medium programs. The second course, **CY350 Network Engineering and Management**, addresses the analysis, design, building, and testing of modern computer networks. Cadets implement and test wired and wireless networks using standard network models, protocols, and practices. In the final course, **CY450 Cyber Security Engineering**, cadets learn principles of information system security and best practices, and apply techniques and tools to evaluate and improve network security. The capstone integrative experience requires cadets to employ offensive and defensive cyber measures and consider the economic, socio-cultural, and political considerations that impact the technological solutions.

## RO - Robotics Engineering

EE300 Fundamentals of Digital Logic

EE350 Basic Electrical Engineering

EE450 Military Robotic Systems

The Robotics Engineering Sequence focuses on developing a basic understanding of the electrical and electronic technologies used in the military. The fundamentals of electrical power generation and distribution, information processing, computing, communications, and robotics are some of the examples that are developed across the engineering sequence. The first course, **EE300 Fundamentals of Digital Logic**, provides cadets with a basic understanding of digital electronics that are used for information processing, digital communications, and computing. The second course, **EE350 Basic Electrical Engineering**, provides cadets with a basic understanding of the analog electronics that are used for power generation, sensor measurements, electric motors, and radio communications. The final course, **EE450 Military Electronic Systems**, employs a semester-long robotics design project applied to a military setting. Cadets use previous course materials to understand how the robot works, the limitations and difficulties of programming artificial intelligence used in a military setting with a decision-making model to consider the economic, socio-cultural, and political considerations that impact the technological solutions.

## AI - Artificial Intelligence and Data Engineering

CS300 Programming Fundamentals

CS350 Data Engineering

CS450 Engineering AI

The Artificial Intelligence and Data Engineering (AIDE) Core Engineering Sequence (CES) equips cadets with practical skills to tackle the challenges in the modern data-driven world. This sequence focuses on fundamental programming and data engineering skills and the development and application of artificial intelligence models that rely on engineered data. The first course, **CS300 Programming Fundamentals**, provides cadets with a broad introduction to computer programming, allowing them to design, build, and test small to medium programs that use various data sets. The second course, **CS350 Data Engineering**, teaches cadets the skills to retrieve, validate, transform, visualize, and extract insights from data in various formats. Building on the skills developed in CS300 and CS350, **CS450 Engineering AI**, empowers cadets to implement AI models that enhance decision-making and perform rapid prediction. The depth and practical applicability of the skills taught in AIDE start cadets down the path toward mastery of data engineering and AI. This curriculum will prepare cadets to lead and innovate in various industries, providing critical competitive advantages, including strategic superiority for institutions such as the US Military.

## EV - Environmental Engineering

EV300 Environmental Science

EV350 Environmental Engineering Technologies

EV450 Environmental Engineering for Community Development

The Environmental Engineering Sequence focuses on current environmental issues and designing viable, sustainable solutions to them. The sequence emphasizes the science and engineering commonly used to protect public health and the environment. The experience gained allows graduates to address multiple perspectives across relevant Army topics such as installation drinking water and wastewater infrastructure, energy, and environmental stewardship. The first course, **EV300 Environmental Science**, provides Cadets with a broad understanding of what the term “environmental issues” encompasses on a global and local scale. Cadets analyze the influences, especially human, that cause change in the balance of the Earth’s natural and biological cycles. The second course, **EV350 Environmental Engineering Technologies**, explores environmental engineering from a unit process and mass balance approach. Cadets solve fundamental problems in drinking water, wastewater, air pollution, and solid/hazardous waste that would be encountered in a developed country. The final course, **EV450 Environmental Engineering for Community Development**, employs a semester-long design project to engineer solutions for a real-world case study in a developing world community. Cadets use a value-based decision-making model to consider the economic, socio-cultural, and political considerations that impact the technological solutions.

## NE - Nuclear Engineering

NE300 Fundamentals of Nuclear Engineering

NE350 Radiological Engineering Design

NE450 Nuclear Weapons and Weapons Effects

The Nuclear Engineering Sequence studies the practical use of the energy that is released by the nucleus. Applications extend into the fields of power generation, medicine, nuclear weapons, and nuclear weapons effects. The sequence is designed to provide introductory knowledge in the applications of nuclear energy and what Army officers should know about nuclear reactors, radiation health physics, radiation detection, and nuclear weapons and weapons effects. The first course, **NE300 Fundamentals of Nuclear Engineering**, provides cadets with an understanding of the nuclear particles and interactions important to the fission process. Cadets learn about radioactivity, the interactions of radiation with matter, and how to determine the state of criticality of a nuclear system. The second course, **NE350 Radiological Engineering Design**, explores the biological effects of radiation, the fundamentals of radiation protection and detection, the widespread medical and industrial uses of radiation technology, and the transportation, treatment, and disposal of radioactive waste. NE350 employs a semester-long nuclear application design project to engineer solutions for a given realistic situation. Cadets use a decision-making model to consider the economic, social, and political considerations that impact a technological solution. The final course, **NE450 Nuclear Weapons Effects**, introduces cadets to the basic design of nuclear weapons, and the full spectrum of their effects once employed. As future Army officers, cadets gain a necessary understanding of the risks to their soldiers operating in a nuclear environment, and what they need to know to mitigate those risks.

## SE - Systems Engineering

SE300 Introduction to Systems Engineering

SE350 Systems Modeling and Design

SE450 Applied Systems Design and Decision Making

OR

SE301 Foundation of Engineering Design and Systems Management  
EM384 Analytical Methods for Engineering Management  
SE450 Applied Systems Design and Decision Making

The Systems Engineering Sequence exposes Cadets to the fundamentals of the discipline, equips them with some of the more essential tools to facilitate modeling and analysis, and develops an inherent ability to use the Systems Decision Process (SDP) to solve complex problems. The Systems Engineering discipline develops and manages interdisciplinary engineering teams that develop, design, and implement systems. The first course, **SE300/SE301 Introduction to Systems Engineering**, provides the baseline approach to the fundamentals of systems thinking, systems engineering, and systems decision making. Cadets learn the methodological framework and techniques for designing, implementing, managing, and reengineering complex systems or processes and their life cycles, which they are taught using the SDP. The second course, **SE350/EM384 Systems Modeling and Design** focuses on developing a solid foundation of modeling principles, with a particular emphasis on spreadsheet modeling; instilling the ability to collect, clean, analyze system data; and developing the ability to apply both deterministic and stochastic techniques to model and analyze systems in support of the broader decision-making process. Cadets learn to apply these techniques to both military and industry-based systems, as well as systems they may encounter in their own disciplines. The sequence culminates with **SE450 Applied Systems Design and Decision Making**, which builds on the systems design and management skills learned in SE300/SE301, while incorporating material from courses in the USMA core curriculum. Cadets gain systems engineering experience via a semester-long integrative project that uses both the systems engineering design, modeling and analysis skills they have learned thus far, and the skills they have acquired in the core curriculum and their respective majors. Projects are selected to provide Cadets the experience of working for a real client with a real problem that requires an adaptive solution.

## Integrative Curricular Components (ICCs)

ICCs are parts of the curriculum that make thematic connections across selected courses or within the context of a course so that information and skills can be applied to novel and complex issues or challenges. In addition to the ICCs, the Academy offers many other interdisciplinary opportunities that cadets can choose, including the Interdisciplinary Humanities (XH), Science (XS), and Engineering (XE) courses and interdisciplinary minors. Courses can be found in Part 3 of the Redbook and Minors in Part 4.

ICCs include:

- [Character Thread](#)
- [Study of War Thread](#)
- Core engineering sequences
- MX400 (Officership)
- West Point Writing Program (WPWP)

The word “thread” is used to describe a group of courses linked by shared themes.

## Writing

The West Point Writing Program (WPWP) works across the curriculum to provide cadets and faculty with continuity, depth, and coherence in their study of composition, critical thinking, academic argument, writing pedagogy, and professional communication. In practice, it sets, assesses, and supports writing standards that prepare cadets to develop writing proficiency across the curriculum. Accordingly, it describes a pedagogical model and curricular principles that inform courses in the core and in individual majors, designated as First-Year Composition (FYC), Writing-in-the-Core (WiC), Writing-in-the-Major (WiM), and Writing-in-the-Profession (WiP). The overarching goal of the Writing Program is to provide cadets with continuity and coherence in their education so that all graduates are agile, clear, and thoughtful communicators. Particularly through Signature Writing Events, the Program encourages cadets to take ownership of their study and understand that writing effectively is integral to their professional development as officers and their lifelong learning.

## THE MAJOR

The United States Military Academy's curriculum requires cadets to complete at least one disciplinary depth component consisting of an academic major with a minimum of 13 courses. (Part 4 of the Redbook describes each academic major in detail.)

In general, an academic major at West Point consists of at least three components, which are listed below. All courses composing the thirteen-course major must be 3.0 credit hours or higher. Any courses beyond the minimum thirteen required for the major are not subject to this credit hour requirement.

1. The foundation of the major can be found in a study-in-depth component which consists of a minimum of nine courses (ten including the integrative experience; see #3 below) that may be a mixture of required and elective courses.
2. The second component of every major is a collection of three Complementary Support Courses (CSCs). The purpose of these courses is to provide cadets a broader understanding of their academic majors through disciplinary diversity and choice. The courses support the major by either building a foundation of knowledge otherwise absent from the Core Curriculum but required for a complete major or enriching the major by providing alternative viewpoints or modes of thought.
3. The final component of each major is an integrative experience course or courses. The integrative experience is designed to synthesize knowledge and skills nurtured in the core and major programs. Situated within the major, but drawing upon all goals, integrative experiences present cadets with professionally relevant situations that include political, social, economic, and technological issues and challenge cadets to anticipate and respond appropriately to opportunities and challenges in a changing world.

The listing of each academic major in Part 4 of the Redbook also includes entries for the select core requirements. Some academic majors may require certain courses to fulfill these requirements: Science Depth, STEM Depth, IT/Cyber, Core Engineering Sequence, and the West Point Writing Program Writing-in-the-Major course. Courses from the core curriculum may not be counted as part of the thirteen-course requirement for an academic major.

## CHOOSING A MAJOR

Choosing an area of academic specialization is one of the most important academic decisions a cadet will make at the Military Academy. Currently a major requires the commitment of a minimum of 13 courses. Once a cadet starts taking courses to support a major, it is very difficult and, in many cases, impossible to change to another major. Cadets are encouraged to seek guidance in arriving at this decision and to take the necessary time and effort to make that decision a good one.

Once a Fourth Class cadet designates a major, the cadet is passed from the company academic counselor to a counselor in the department that sponsors the chosen discipline. Departmental counselors then help the cadet plan the remaining six terms of his or her academic program, providing guidance on the sequencing of courses best designed to facilitate study of the discipline.

A minimum of 13 courses must be designated to fill the course requirements for the baseline area of academic specialization, arranged in semesters of five academic courses each. Cadets who choose an area requiring more than thirteen courses may need to complete additional academic courses (six or more) in future terms.

## DOUBLE MAJORS

Through validation, advanced placement, or overload, it is possible for a cadet to meet the requirements for more than a single major. Each cadet must officially pursue and gain credit for one major. Beyond that requirement a cadet is free to pursue and gain transcript credit for additional majors provided he or she meets the following guidelines:

- In pursuing two majors, a cadet must meet all curricular requirements, including student learning outcomes for each major, and double-count no more than seven total courses, with no more than four being study-in-depth courses. Integrative experience courses are considered study-in-depth courses.
- Department counselors in both fields must grant approval.
- The graduation transcript will reflect academic majors that have been satisfied.
- Cadets are warned that USMA will not guarantee the completion of additional majors, even though that cadet may have already completed courses towards that goal. Scheduling and resource constraints may prohibit the completion of additional majors.

## ACADEMIC MINORS

Academic minors are designed to encourage and reward intellectual curiosity by allowing cadets to pursue an area of interest beyond their chosen major. Academic minors are not a USMA graduation requirement.

USMA offers a variety of academic minors that focus on: (1) depth-of-study that is not quite as intensive as a major in the discipline, (2) depth-of-study in a field that is not offered as a major, or (3) interdisciplinary knowledge and skills in a field of study that warrants this approach.

The following guidelines apply to academic minors:

- An Academic Minor consists of a minimum of five academic courses of 3.0 credit hours or higher.
- In pursuing a major and one or more minors, a cadet must meet all curricular requirements for the major and each minor. Double-counting practices (counting one course to meet a course requirement in two different curricular components [e.g., core curriculum and a minor, two minors, a major and a minor]) are permitted. However, for any given cadet, at least three courses of at least 3.0 credit hours in any academic minor must be unique to that minor to gain official recognition.
- A department counselor in the academic minor must grant approval for all double counting requests.
- The graduation transcript will reflect all minors that have been satisfied.
- Cadets are warned that USMA will not guarantee that they will be able to complete a minor, even though the cadet may have already completed courses towards that goal. Scheduling and resource constraints may prohibit availability of the courses.

## GRADING PHILOSOPHY AND GRADING POLICIES

As a critical and essential part of the educational philosophy, it is important to articulate explicitly the Dean's academic grading policies and philosophies. The Dean's policies are described in Dean's Policy and Operating Memoranda (DPOMs) and are located in a folder on the Dean's SharePoint page, which is restricted to internal use only by applicable faculty and cadets.

### *Grading Philosophy*

The foundation of USMA faculty's grading philosophy is a commitment to evaluate cadets based on their achievement of announced course outcomes. Satisfactory performance on graded course requirements must therefore reflect satisfactory progress toward meeting course outcomes. USMA faculty establish reasonable academic standards of achievement in advance of cadets taking a course and taking tests. Their goal is not to rank order cadets against each other based on any preconceived concept of an appropriate grade distribution (curving). Instead, they challenge cadets to meet announced standards of performance and assign grades based on their success in doing so. The principal responsibility for academic performance rests with each individual cadet.

### *Grading Policies*

- Instructors are responsible for providing sound instruction, measurement of cadet attainment, and a reasonable amount of additional assistance. Instructors shall strive to motivate and inspire cadets to achieve their full academic potential. Beyond these obligations, the responsibility for academic success or failure rests with each cadet.
- Instructors will provide cadets with a statement of the outcomes for each course. Cadets will be evaluated against these outcomes. Departments will avoid evaluation and grading practices that encourage reliance on curving.
- However compiled numerically, letter grades ranging from A+ to F will be the standard means of communicating academic achievement.
- Instructors will promptly provide cadets an evaluation of each graded course requirement; the evaluation will be a letter grade or a numerical score easily convertible by the cadet to a letter grade.

For more information on these or other grading policies, please refer to DPOM 02-01, Gradekeeping.

## Examinations

- **Written Partial Review (WPR):** This examination is designed to test knowledge of course material covering specified lessons. Each department will determine the material to be covered, the time of the exam, and the weight of the exam. Written Partial Reviews may be scheduled during normal class meetings, during Dean's Hour, or during Dean's exam periods. Cadets having more than one major graded requirement due the same day (more than two requirements for First and Second Class cadets) may request permission to attend an examination on an alternate day. Responsibility for seeking relief rests with the cadet, while the faculty role is one of cooperation in granting permission where reasonably feasible. Relief must be sought at least 72 hours in advance of the scheduled examination.
- **Term End Examinations (TEE):** These examinations, given at the end of the term, test cadets' knowledge of course material presented during that term. At the request of Course Directors, Department Heads may approve of a TEE exemption for a course. The Dean's office will schedule the TEE for each course without a TEE exemption and every cadet will take the TEE in accordance with the established schedule. Cadets who want to change their TEE date must request the change through AARS as individual instructors are not authorized to make this change.

For more information on course examinations, please refer to DPOM 02-03, Classroom and Related Departmental Procedures.

## Grading

In general, the academic departments describe the relative weight of their graded course requirements in terms of marks (points). Graded course requirements include, but are not limited to, daily writs, WPR, research papers, computer exercises, and TEE. Early in each course, each instructor should provide a list of the course requirements and their weights. Because there is no standard scale used by all departments for converting marks to grades, cadets should ascertain from their instructors during the first few lessons of each term how the various departments assign grades.

Cadets can view a report of their grades online at least four times during each term. The first three reports are interim or progress reports. They are provided after the sixth, tenth, and fifteenth weeks of the term. The fourth report reflects final grades, average for the term, and cumulative average. Copies of the report are available in the Academic Affairs and Registrar Services (AARS), Office of the Dean, during the summer.

After TEE's have been graded, department heads assign final course grades using the A+ to F scale. These final course grades are assigned quality points in accordance with the following table:

| Letter Grade | Quality Points | Letter Grade | Quality Points | Letter Grade | Quality Points |
|--------------|----------------|--------------|----------------|--------------|----------------|
| A+           | 4.33           | B            | 3.00           | C-           | 1.67           |
| A            | 4.00           | B-           | 2.67           | D            | 1.00           |
| A-           | 3.67           | C+           | 2.33           | F            | 0.00           |
| B+           | 3.33           | C            | 2.00           |              |                |

Should a cadet resign or be separated during the term before the first TEE, his or her transcript for all courses will reflect a grade of W (Withdrawn) with no credit awarded. Once the TEE cycle begins, cadets will receive a grade in every course in which they are enrolled.

## ACADEMIC STANDARDS

The primary responsibility for attaining satisfactory academic performance rests with the individual cadet. Cadets must strive to achieve their highest level of academic excellence. To meet this responsibility, cadets have an obligation to know their academic status, manage their time, and establish priorities in such a manner as to accomplish this goal. The performance of academic duties is a significant part of the process of preparing for the acceptance of the duties and responsibilities of Army officers. The standard for performance of academic duties is the same as that for the performance of officer duties--excellence and one's personal best.

Cadets must achieve a grade of D or better in all required academic (core and academic major), military science, and physical education courses. Grades of NC (no credit) may be awarded temporarily, but cadets must resolve the

circumstances that resulted in the NC and be awarded a letter grade in order to receive credit for the course.

The Academic Board will consider the cases of all cadets who are recommended by a Department Head as deficient in a course. If the failed course is required by the cadet's academic program, the course must be repeated in order to receive credit. Cadets may elect to repeat a course in which a grade of "D" was earned if their Academic Program Score (APSC) or Cumulative Quality Point Average (CQPA) is less than 2.00. Departments may make a request for an exception to the APSC/CQPA criteria as long as the request is coordinated with AARS and is approved by the department providing the repeated course. When cadets repeat a course, the original course with the grade earned will always remain on the transcript in the semester taken, as this was the historical record of events. A second entry is made for the same course in the term in which it is repeated and the appropriate grade shown. The grade in the course taken earlier must be a "D" or "F" and the latter grade earned will be used to compute the cadet's APSC/CQPA. The APSC/CQPA in subsequent terms is also adjusted. Refer to DPOM 02-01, Gradekeeping.

If a cadet is deficient in a course, the Academic Board may offer a reexamination in that subject as one course of action. Cadets who accept the reexamination will receive additional guidance from the Registrar's office regarding scheduling and expectations. The reexamination shall occur within 60 days of the date of application. Cadets who fail the reexamination may not have another examination and their separation will be finalized. The Academic Board will consider the cases of all cadets who pass the reexamination to determine whether they are otherwise qualified for readmission to the Academy. Passing the reexamination does not remedy the course failure. Cadets readmitted under these procedures will normally repeat the failed course, if it is required. Refer to DPOM 02-02, End of Term Procedures.

### ***Minimum Quality Point Averages***

In addition to passing each required course, cadets must achieve a minimum CQPA of 2.00 in order to graduate. In order to monitor progress in the Academic Program and to signal substandard achievement, the Academic Board has established performance standards based on APS term (APST) and APS cumulative (APSC). The following table presents the minimally acceptable standards (also referred to as Program Evaluative Goals or PEG points) based on APS. Cadets with averages below those stated will be considered deficient in the Academic Program and will be reported to the Academic Board at the end of each term.

### **MINIMALLY ACCEPTABLE APS:**

| <b>CLASS YEAR</b> | <b>TERM</b> | <b>APST</b> | <b>APSC</b> |
|-------------------|-------------|-------------|-------------|
| <b>Fourth</b>     | First Term  | 1.67        | N/A         |
|                   | Second Term | 1.67        | 1.70        |
|                   | STAP        | N/A         | 1.70        |
| <b>Third</b>      | First Term  | 1.67        | 1.80        |
|                   | Second Term | 1.67        | 1.90        |
|                   | STAP        | N/A         | 1.90        |
| <b>Second</b>     | First Term  | 1.67        | 1.95        |
|                   | Second Term | 1.67        | 1.95        |
|                   | STAP        | N/A         | 1.95        |
| <b>First</b>      | First Term  | 1.67        | 2.00        |
|                   | Second Term | 1.67        | 2.00        |
|                   | STAP        | N/A         | 2.00        |

The APST is based on grades in all courses taken during a semester excluding Military Science and Physical Education. The APSC is based on grades in all courses previously taken at the Academy (excluding Military Science and Physical Education) except that grades in repeated courses replace prior grades of D and F.

## **ACADEMIC DEFICIENCIES AND PROBATION**



Cadets who fall below the APSC levels shown in the table for the applicable semester will be reported deficient in the Academic Program to the Academic Board at term end. Cadets deficient in APSC may be considered by the Academic Board for separation for failure to attain minimum standards in the Academic Program.

Cadets deficient in APSC who are retained at the Academy will be placed on academic probation for the following term. Cadets whose APST is below 1.67 will also be placed on academic probation for the following term. Cadets are removed from academic probation at the end of the next term in which both their cumulative and term averages exceed the peg points in the table. Grades earned in the Summer Term Academic Program (STAP) may raise the APSC above the required peg point and remove a cadet from probation. In order to be removed from academic probation for term performance (APST failure), however, cadets must achieve better than a 1.67 in a full academic term (16 weeks).

A cadet placed on academic probation is subject to the following measures during the probationary period:

- Mandatory counseling by an assigned academic counselor within two weeks of the start of the current term. Performance reviews following the tenth- and fifteenth-week grade reports.
- Mandatory review of chain of command duties by the company tactical officer with a view toward reducing time requirements IAW Annex A, USMA Reg 1-1 (Administrative Scheduling Activities) and the Academy Schedule.
- Assessed room tours in lieu of area tours, except in cases of Class 1 offenses.
- Subject to reduced privileges which are reviewed monthly relative to progress and adjustment.
- Limited to participation in one extracurricular activity or Corps Squad sport at a time. This will be reviewed monthly and follow the guidelines in the Academy Schedule.
- Ineligible to participate in events which involve the loss of academic time, either class or evening study period (e.g., away trip sections, spectator at home athletic contests, extracurricular activity events [Director of Cadet Activities (DCA) and religious trips], voluntary lectures or films, Cadet Public Relations Council (CPRC), conferences etc.), except for participation in the one extracurricular activity provided for above. Participation in a mandatory educational trip that is required for a course is permitted.
- Not authorized to use the following facilities during evening study period: day room, post movie, and Eisenhower Hall (except to attend mandatory lectures).
- A cadet's privileges may be withdrawn by the company tactical officer upon the request of an instructor if both agree that this course of action is essential to improve the cadet's grades.

## ACADEMIC INDIVIDUAL ADVANCED DEVELOPMENT (AIAD)

The purpose of an AIAD is to provide a venue for educational experiences that would not be possible within the usual framework of academic, military, and physical programs that comprise the 47-month USMA experience.

An AIAD is an activity offered by the U.S. Military Academy that is:

- Governed by DPOM 06-02: Academic Individual Advanced Development.
- An experiential program of learning.
- The only duty of the cadet during the experience.
- An immersive experience in an environment different from that encountered by cadets during their normal participation in the academic, military, and physical programs of the USMA.
- Offered to cadets through Academic Departments, or other agencies authorized to do so by the Dean.
- AIADs fulfill the cadet IAD graduation requirement.

An AIAD may offer credit recorded on the USMA transcript. AIADs offering academic credit comply with existing administrative standards contained in Dean's Policy and Operating Memorandums (DPOM 02-01, MADN-OPS-ARS, subject: Gradekeeping and DPOM 02-08, MADN-OPS-ARS, subject: Academic Administration), and undergo review by the Curriculum Committee.

For practical purposes, therefore, an AIAD offering academic credit will be treated as a course and Departmental proposals will be made IAW DPOM 05-05, MADN-OPS-ARS, subject: Managing Curricular Change, and guidance provided by the Dean of the Academic Board.

## STUDY ABROAD

West Point's Study Abroad Program (SAP) is, primarily, a language-based program, open to all cadets. Applications for SAP are normally solicited through the Cadet Information System in September or October of each year. Cadets may spend no more than one semester away from the Academy. Host institutions are chosen for their ability to provide language training and education on other subject matters in a secure environment. For locations where English is not

the primary language, cadets are chosen based on their ability to be successful at the university-level in a foreign language taught at West Point. Additional selection criteria include:

- Must be a Second Class cadet during semester of execution;
- LX371 completion or equivalent minimum proficiency in target language (if applicable);
- Minimum cumulative scores of 2.5 in Academic, Physical, and Military Programs;
- Concurrence of TAC Team, BTO, and DAC;
- No Military Development “D’s” or Fails;
- No un-remediated Indoor/Outdoor Obstacle Course Test (IOCT) failure;
- No Regimental or higher disciplinary actions;
- Not enrolled in the Army Weight Control Program;
- Demonstrated maturity to represent USMA in unsupervised conditions;
- 8TAP compatibility.

Final selections are based on a combination of factors that include funding, the number of slots available at partner institutions, 8TAP compatibility, and demonstrated language proficiency. During Study Abroad, cadets must take the equivalent of five USMA courses, three of which must be courses taught in the host institution’s native language. Additionally, course loads must be chosen to ensure there is no impact to on “on-time” graduation.

Cadets interested in participating are encouraged to contact the Department of English and World Languages (DEWL).

## SERVICE ACADEMY EXCHANGE PROGRAM (SAEP)

USMA Cadets wishing to attend other U.S. Service Academics may apply to spend one semester away from USMA on the Service Academy Exchange Program (SAEP). The process starts with an application through the United States Corps of Cadets (USCC) Leader Development Branch (LBD) in January of yearling year. Typically, the LBD will send an email to all eligible cadets with an application form. To be eligible, a yearling must have at least a 3.0 in each pillar (APSC, PPSC, MPSC).

## ACADEMIC ASSISTANCE

Each academic department offers important supplementary programs and assistance to give cadets specific guidance in academic matters. This guidance assists cadets in overcoming academic weaknesses and in exploiting academic strengths.

**Additional Instruction (AI):** Departments are encouraged to arrange times for AI that are mutually agreeable to cadets and the department. It is the responsibility of each cadet to request additional instruction. Specific guidance on AI hours and procedures will be provided by each department. Refer to DPOM 02-03, Classroom and Related Departmental Procedures.

**Academic Counseling:** The Academic Affairs and Registrar Services (AARS), Office of the Dean, coordinates the faculty-based academic counseling programs available to each cadet. Within the Counseling Branch there are counselors available during normal working hours on a walk-in basis. They can discuss elective choices, schedules, course changes, and overloads and can affect any changes in a cadet’s program. One volunteer faculty member serves as a Company Academic Counselor (CAC) for each company and can help cadets on most academic matters or make referrals to the proper authority in the Dean’s Office or academic departments. In addition, the CAC’s have formal academic counseling responsibility for all cadets in each company who have not yet selected a major. Upon selection of a major, cadets are assigned to the Department Academic Counselors (DAC) of the appropriate academic department. Finally, within the cadet chain of command there are a Cadet Company Academic Officer and an Academic Sergeant who can advise on the grading system, company tutors, additional instruction, and other matters.

## ACADEMIC SUPPORT PROGRAMS

**The Moulner Writing Center:** Located on the second floor of Jefferson Hall, the Moulner Writing Center (MWC) sponsors one-on-one consultations, group workshops, and special events for all cadets working on writing and communications projects for any academic course, personal interest, or professional opportunity.

**Center For Enhanced Performance (CEP):** The CEP helps cadets achieve excellence in the Academy’s three pillars — academic, physical, and military — through targeted education and training on specific strategies and intangible mental skills that underlie improved human performance.

- **Peak Performance Program (PPP):** This is offered to all cadets who wish to enhance their academic, athletic, and military performance through psychological and mental skills training. The goal is for the cadet to gain the ability to maximize their potential in any situation especially under pressure and stress. A variety of skills are taught including effective thinking, focus and concentration, visualization and imagery, and team building. Individual sessions are scheduled with the cadet and a performance enhancement trainer. Sessions may also utilize the **Mental Training Lab** to further improve performance.
- **Academic Excellence Program (AEP):** This program is dedicated to fostering student success through personalized support services. AEP offers tutoring, concussion recovery assistance, individualized academic support, scholarship advising, and guidance on major selection, all delivered through one-on-one appointments. Additionally, AEP provides resources and support for students on academic probation to help them get back on track. This program is committed to helping every cadet achieve their academic and military goals.
- **Athletic Academic Success Coordinator (AASC Program):** AASCs provide dedicated support exclusively for Corps Squad Athletes. Utilizing an intrusive counseling style, they focus on enhancing time management, organization, and test preparation. They also help athletes adjust to the rigors of academy life. Through personalized guidance, they ensure that student-athletes can balance their academic and athletic commitments effectively. This program is designed to foster academic success and personal growth for our athletes.
- **Student Success Course (RS101):** This credit-producing course is designed to improve cadet academic, and leadership performance at USMA. Mastery of a variety of strategies leads to this goal. Strategies presented include effective thinking, goal setting, time management, concentration, test taking, and note taking. The strategies mastered are implemented immediately into the cadet's present life at USMA and contribute to life-long learning. The course has five graded assignments and a final pass/no record grade determination.
- **Reading Efficiency (RS102):** This course develops flexible reading strategies. Increasing one's base reading rate while maintaining comprehension is accomplished through the use of computer programs, textbook reading and recreational reading. Various pacing techniques and supervised practice lead to the increase of both reading rate and comprehension.
- **Information Literacy and Critical Thinking (RS103):** This course is designed to improve cadet analytical and intellectual performance at USMA. Mastery of a variety of strategies leads to this goal. Strategies presented include: utilizing critical thinking models, understanding the research process, and evaluation of information sources. The strategies mastered yield short term academic benefits and long-term professional disciplines. The course has four graded assignments and a final pass/no-credit grade determination.
- **Introduction to Performance Psychology (RS104):** This course introduces cadets to the essential principles and skills of performance psychology and provides a framework through which these principles can be applied toward the pursuit of academic, athletic, and military excellence. Cognitive skills for confidence development, attention control methods, relaxation techniques, and imagery are taught and practiced. A detailed personal goal setting plan is completed by the course's end.
- **First Year and Beyond (FYB) Program:** The FYB program is intended to assist select cadets who, based on their performance after 6-week grades in first-term academic courses, are having difficulty adapting to the rigors of the academic program. In addition to the immediate relief and assistance provided during the first term, FYB helps these cadets develop as self-directed learners by continuing to provide educational scaffolding as they move beyond their initial academic term. This scaffolding is deliberately reduced over the cadets' 47-month experience. Please see a CEP counselor to learn more about the FYB program.

## ACADEMIC AWARDS AND RECOGNITION

Excellence in academic pursuits has long been the measure of an individual's self-discipline and self-growth. Recognition of excellence has been the keystone of West Point's educational philosophy for over 200 years. The academic excellence of cadets is recognized throughout their four years at the Military Academy and is expressed in more than 100 awards. Just as such awards reflect an individual commitment to academic excellence in undergraduate study, they also reflect a strong foundation for graduate and post graduate work.

Details on criteria for our awards programs, including Latin Honors, Distinguished Cadets and Superintendent's Award can be found in USMA Regulation 672-2. Dean's List, also explained in USMA Regulation 672-2, is a designation for a cadet with a Term Quality Grade Point Average (TQPA) of 3.0 or greater, considering all academic courses taken during the semester, as well as military science and physical education courses.

**Dean's Company Award:** The Dean's Company Award recognizes academic achievement by a company during the fall and spring terms. Performance in the academic program is determined by the average of the term Academic Program Score (APS) of all cadets in the company. The company with the highest average APS in each regiment receives a gold streamer to be attached to the company guidon. A silver streamer is awarded to any company with a company with an average APS of 3.0 or higher.

## SCHOLARSHIPS AND FELLOWSHIPS

Cadets are encouraged to compete for a number of nationally recognized graduate scholarships and fellowships. The West Point Scholars Program (WPSP) serves four main functions: provide advanced development for cadets, provide intellectual capital to the Army, strengthen partnerships, and bridge the civ-mil divide, and provide strategic recruitment benefit to the Academy. The WPSP Support Cell consists of the Deputy Registrar, who serves as the Primary Executive on behalf of the Dean and under the supervision of the Vice Dean for Operations, the Head Academic Advisor, and the Medical School Advisor. All cadet applications are screened and managed by the executive agent for each scholarship program listed below. Executive agents are academic departments that are selected by the Dean of the Academic Board to house and supervise individual scholarship programs. Each executive agent maintains their own approval process in accordance with the WPSP SOP. The Academic Board will approve the list of applicants. West Point will only sponsor cadets approved through the executive agents. Refer to DPOM 06-03, West Point Scholars Program, for more information.

## GRADUATE SCHOOLING

### ADMINISTRATIVE INFORMATION

**Transcripts:** Permission must be given in writing by the cadet (or former cadet) before transcripts will be released. Transcripts of academic records are available at Academic Affairs and Registrar Services (AARS), Office of the Dean.

**Academic Evaluations:** Upon request, faculty members will provide cadets with academic evaluations using USMA Form 3-230 which is available in the Office of the Dean. These forms permit the Office of the Dean with a cadet's authorization under the provisions of the Privacy Act to provide selected information to designated institutions, agencies, or individuals. In most cases, the Office of the Dean will use these forms and other information in the cadet file as a basis for an official assessment of graduate school potential. Cadets are encouraged to seek academic evaluation by instructors during the Second and First Class years.

**Qualifying Examinations:** Most graduate or professional schools require that prospective students report their score on a nationally recognized qualifying examination before acceptance is granted. Cadets are responsible for arranging to take any of the examinations which may be required for admission to a graduate or professional program.

### GRADUATE SCHOOLING OPPORTUNITIES

**Medical School (West Point Pre-Medical Scholarship Program):** In accordance with Army Regulation (AR) 601-141, the Department of Defense permits up to 3% of cadets from each West Point graduating class to attend medical school under military sponsorship.

The Superintendent is responsible for the selection of cadet applicants for the Uniformed Services University (USU) and Health Professions Scholarship Program (HPSP). The management of this process is accomplished through the West Point Pre-Medical Scholarship Program, coordinated by a health professions advisor in the Office of the Dean in concert with faculty from the Department of Chemical and Biological Science and Engineering (CBSE).

The Association of American Medical Colleges (AAMC) governs the medical school application process for allopathic schools in the United States and Canada. The seventeen core competencies prescribed by the AAMC include proficiency in pre-professional activities, thinking and reasoning, and science competencies. Cadets demonstrate aptitude in these areas through academic success, performance on the Medical College Admission Test® (MCAT), clinical exposure, research, volunteer work and community service. In addition to meeting program requirements, cadets must perform well in the three pillars of academy life: academic, military, and physical. Core competencies and minimum program requirements are listed below.

## AAMC Core Competencies

### Pre-professional Activities      Thinking and Reasoning      Science Competencies

- |                                          |                         |                 |
|------------------------------------------|-------------------------|-----------------|
| -Commitment to learning and growth       | -Critical thinking      | -Human behavior |
| -Cultural awareness                      | -Quantitative reasoning | -Living systems |
| -Cultural humility                       | -Scientific inquiry     |                 |
| -Empathy & compassion                    | -Written communication  |                 |
| -Ethical responsibility to self & others |                         |                 |
| -Interpersonal skills                    |                         |                 |
| -Oral communication                      |                         |                 |
| -Reliability & dependability             |                         |                 |
| -Resilience & adaptability               |                         |                 |
| -Service orientation                     |                         |                 |
| -Teamwork & collaboration                |                         |                 |

## Pre-Medical Program Requirements

- Cumulative Grade Point Average (CQPA)  $\geq 3.2$
- Biology-Chemistry-Physics-Math (BCPM) average  $\geq 3.2$
- MCAT score  $\geq 500$
- MCAT section scores  $\geq 124$
- Significant clinical exposure (target 200 hours of healthcare volunteer work and shadowing)
- Participation in research
- Selfless service demonstrated by volunteer activities and community work

Cadets may enter the program through any major; however, only cadets in CBSE majors are guaranteed seats in the pre-requisite classes. The pre-medical program course guidelines are listed below. Undergraduate course requirements for medical schools vary and cadets are strongly encouraged to explore specific recommendations for their schools of interest. For example, a medical school may require an upper-level science class like genetics that is not included in the USMA guidelines. When pre-medical courses are oversubscribed, the final determination for admission will be made by the CBSE department with input from the health professions advisor in the Office of the Dean. Factors for consideration will include major, commitment to the pre-medical program, participation in the West Point Pre-Medical Society (WPPMS), progress in meeting program requirements, and reaching Academic Program Score Cumulative (APSC) PEG points for each course. The USMA course recommendations and PEG points are as follows:

| <b>USMA Course Requirements</b>         | <b>APSC PEG point</b> |
|-----------------------------------------|-----------------------|
| General chemistry with lab — 1 year     | N/A                   |
| Organic chemistry with lab — 1 year     | 3.0                   |
| Biochemistry — 1 semester               | 3.2                   |
| Biology with lab — 1 semester           | 3.0                   |
| Physiology with lab — 1 semester        | 3.2                   |
| Physics with lab — 1 year               | N/A                   |
| Calculus — 1 semester                   | N/A                   |
| Probability and statistics — 1 semester | N/A                   |
| English — 1 year                        | N/A                   |

Applicants typically apply to many medical schools (15 or more). Omitting courses from the recommended list may limit one's options and compromise opportunities to attend medical school. Cadets who validate required courses (examples: CH101 and CH102) are strongly encouraged to check the guidelines of the medical schools where they plan to apply. Some institutions will not accept the validations, and most schools that accept credit want to see that applicants have pursued a higher-level course for the validated class (example: physical chemistry for cadets who validate inorganic chemistry).

The West Point Pre-Medical Society (WPPMS) provides opportunities for cadets to share interests, attend talks by guest speakers, join in volunteer experiences, and pursue leadership positions. This cadet run club operates under the auspices of a faculty member in the CBSE department in close coordination with the USMA health professions advisor. It is imperative that all pre-med cadets join the WPPMS. Since cadets enter the pre-med scholarship program through a variety of majors, the WPPMS serves as the primary conduit for information flow. All important events are coordinated through the pre-med society, ranging from briefings regarding the medical school program and shadowing opportunities to MCAT preparation and selection by the Medical Program Advisory Committee (MPAC).

Endorsement to attend medical school directly from West Point is made by the MPAC. This multidisciplinary group includes representatives from Keller Army Community Hospital, CBSE, Uniformed Services University, USCC, and the Dean's staff. Formal applications are submitted in the spring of second-class year. The committee meets in April and selects up to three percent of the cadets as primary endorsees. Additional alternates may be selected. Initial selections are provisional pending final second semester grades and MCAT scores. Cadets are required to take the MCAT no later than mid-June and must meet the GPA and MCAT requirements of the program (3.2 GPA, 500 MCAT, 124 section scores) to receive final endorsement. These cadets receive a committee letter written by the health professions advisor for inclusion in their medical school applications.

Once accepted, cadets are fully funded to attend medical school. They may attend the Uniformed Services University (USU) or any civilian medical school in the United States. Students at civilian schools receive funding through the Health Professions Scholarship Program (HPSP).

Please direct any questions to the health professions advisor in the Office of the Dean.

**Other Graduate School Opportunities:** Other graduate school opportunities may be available to cadets in specific disciplines and branches. Please see the appropriate Department Academic Counselors and branch representatives to learn more.

# **PART I: THE ACADEMIC PROGRAM - CLASS GRADUATION REQUIREMENTS**



OFFICE OF THE SUPERINTENDENT  
UNITED STATES MILITARY ACADEMY  
WEST POINT, NEW YORK 10996-1602

MADN-ARS

JAN 12 2021

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Class of 2024 - 2026 Minimum Graduation Requirements

1. **Purpose:** This document articulates the minimum requirements that each Cadet must meet to graduate from the United States Military Academy (USMA).

2. **Discussion:**

The United States Military Academy educates, trains, and inspires cadets across a core set of developmental experiences in the academic, military, physical, and character programs, collectively known as the West Point Leadership Development System (WPLDS). The WPLDS is designed to produce graduates who:

- *Live honorably*
- *Lead honorably*
- *Demonstrate excellence*

To assess if a cadet has demonstrated the minimum virtues and merit expected of a USMA graduate and future US Army officer, USMA's minimum graduation requirements are published below. Note that cadets must always aspire to far exceed these minimum standards, both individually and collectively.

3. **Minimum Requirements:**

a. **Institutional:**

- (1) Achieve a Cumulative Quality Point Average (CQPA) of 2.00.
- (2) Complete a minimum of eight semesters as a full-time cadet, either at the Academy or as a part of an Academic Board-sanctioned semester exchange program.
- (3) Successfully complete the USMA core curriculum in the academic, military, physical, and character programs as defined in the Red Book, Green Book, White Book, and Gold Book, respectively. Also, achieve an Academic Program Score Cumulative (APSC), Military Program Score Cumulative (MPSC), and Physical Program Score Cumulative (PPSC) each of 2.00 or greater.



(4) Successfully complete one of the following broadening experiences: A Military Individual Advanced Development (MIAD), a Physical Individual Advanced Development (PIAD), or an Academic Individual Advanced Development (AIAD).

(5) Meet the Army body composition standards of Army Regulation 600-9 (Army Body Composition Program).

(6) Meet the Army physical readiness requirements.

b. Academic Program: Satisfy the requirements of at least one academic major.

c. Military Program:

(1) Successfully complete Cadet Basic Training (CBT), Cadet Field Training (CFT), and Cadet Leader Development Training (CLDT).

(2) Successfully complete Drill Cadet Leader Training (DCLT), Cadet Troop Leader Training (CTLT), or an Academic Board-approved constructive credit experience.

(3) Successfully complete a West Point summer leader detail during second or first-class summer.

d. Physical Program: Successfully complete the Indoor Obstacle Course Test (IOCT) during the second or first-class year.

e. Character Program:

(1) Abide by the Cadet Honor Code. If a cadet is found to have violated the code and is retained, he/she is required to successfully complete a prescribed remediation program.

(2) Abide by the standards of conduct and respect for others. If a cadet violates these standards and is retained, he/she is required to successfully complete a prescribed remediation program.

4. The Academic Board approved these minimum graduation requirements on 22 October 2020.



DARRYL A. WILLIAMS  
Lieutenant General, US Army  
Superintendent

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OFFICE OF THE SUPERINTENDENT  
UNITED STATES MILITARY ACADEMY  
WEST POINT, NEW YORK 10996-5000

SEP 07 2022

MADN

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Addendum, Class of 2024 - 2026 Minimum Graduation Requirements

1. **Purpose:** This document articulates one addendum to HQ USMA, MADN, memorandum (Class of 2024 - 2026 Minimum Graduation Requirements), originally recommended for approval by the Academic Board on 22 October 2020 and subsequently approved by the Superintendent, LTG Darryl A. Williams.

2. **Addendum - Institutional:** Para 3.a.(6) has been modified to add one requirement regarding the Army Combat Fitness Test (ACFT). The full requirement should read:

(6) Meet the Army physical readiness requirements and pass the ACFT within 120 days before graduation.

3. **Addendum - Academic:** Para 3.b. has been modified to add the following sentence: "Cadets may not have unresolved course failures during their final term." The full requirement should now read:

(1) Satisfy the requirements of at least one academic major.

(2) Cadets may not have unresolved course failures during their final term.

4. The Academic Board recommended approval of these addenda on 14 April 2022 (Academic Program) and 03 August 2022 (Institutional) starting with the Class of 2024.

A handwritten signature in black ink, appearing to read "Steve Gilland", written over the printed name.

STEVEN W. GILLAND  
Lieutenant General, US Army  
Superintendent

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MASP

MAY 31 2023

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Class of 2027 Minimum Graduation Requirements

**1. Purpose:** This document articulates the minimum requirements that each Cadet must meet to graduate from the United States Military Academy (USMA).

**2. Discussion:** The United States Military Academy educates, trains, and inspires cadets across a core set of developmental experiences in the academic, military, physical, and character programs, collectively known as the West Point Leadership Development System (WPLDS). The WPLDS is designed to produce graduates who:

- *Live honorably*
- *Lead honorably*
- *Demonstrate excellence*

To assess if a cadet has demonstrated the minimum virtues and merit expected of a USMA graduate and future US Army officer, USMA's minimum graduation requirements are published below. Note that cadets must always aspire to far exceed these minimum standards, both individually and collectively.

**3. Minimum Requirements:**

a. Institutional:

- (1) Achieve a Cumulative Quality Point Average (CQPA) of 2.00.
- (2) Complete a minimum of eight semesters as a full-time cadet, either at the Academy or as a part of an Academic Board-sanctioned semester exchange program.
- (3) Successfully complete the USMA core curriculum in the academic, military, physical, and character programs as defined in the Red Book, Green Book, White Book, and Gold Book, respectively. Also, achieve an Academic Program Score Cumulative (APSC), Military Program Score Cumulative (MPSC), and Physical Program Score Cumulative (PPSC) each of 2.00 or greater.
- (4) Successfully complete one of the following broadening experiences: a Military Individual Advanced Development (MIAD), a Physical Individual Advanced Development (PIAD), or an Academic Individual Advanced Development (AIAD).

(5) Meet the Army body composition standards of Army Regulation 600-9 (Army Body Composition Program).

(6) Meet the Army physical readiness requirements and pass the Army Combat Fitness Test (ACFT) within 120 days before graduation.

b. Academic Program:

(1) Satisfy the requirements of at least one academic major.

(2) Cadets may not have unresolved course failures during their final term.

c. Military Program:

(1) Successfully complete Cadet Basic Training (CBT), Cadet Field Training (CFT), and Cadet Leader Development Training (CLDT).

(2) Successfully complete Cadet Troop Leader Training (CTLT).

(3) Successfully complete a West Point summer leader detail during second or first-class summer.

d. Physical Program: Successfully complete the Indoor Obstacle Course Test (IOCT) during the second or first-class year.

e. Character Program:

(1) Uphold the Cadet Honor Code. If a cadet is found to have violated the code and is retained, he/she is required to successfully complete a prescribed remediation program.

(2) Uphold the standards of conduct and respect for others. If a cadet is found to have violated these standards and is retained, he/she is required to successfully complete a prescribed remediation program.

4. The Academic Board approved these minimum graduation requirements on 20 April 2023.



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MAY 22 2024

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Class of 2028 Graduation Requirements

1. **Purpose:** This document articulates the minimum requirements that each cadet must meet to graduate from the United States Military Academy (USMA).

2. Discussion: The United States Military Academy educates, trains, and inspires cadets across a core set of developmental experiences in the academic, military, and physical programs and the character domain, collectively known as the West Point Leadership Development System (WPLDS). The WPLDS is designed to produce graduates who: **live honorably; lead honorably; and demonstrate excellence**. To assess if a cadet has demonstrated the minimum virtues and merit expected of a USMA graduate and future US Army officer, USMA's minimum graduation requirements are published below. Note that cadets must always aspire to far exceed these minimum standards, both individually and collectively.

3. Minimum Requirements:

a. Institutional

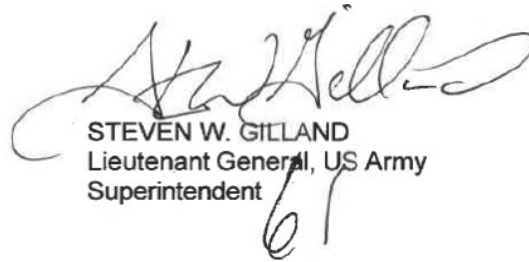
- (1) Achieve a Cumulative Quality Point Average (COPA) of 2.00.
- (2) Complete a minimum of eight semesters as a full-time cadet, either at the Academy or as a part of an Academic Board-sanctioned semester exchange program.
- (3) Successfully complete the USMA core curriculum in the academic, military, and physical programs as defined in the Red Book, Green Book, and White Book, respectively. Also, achieve an: Academic Program Score Cumulative (APSC); Military Program Score Cumulative (MPSC); and Physical Program Score Cumulative (PPSC) each of 2.00 or greater.
- (4) Successfully complete one of the following broadening experiences: a Military Individual Advanced Development (MIAD), a Physical Individual Advanced Development (PIAD), or an Academic Individual Advanced Development (AIAD).
- (5) Meet the Army body composition standards of Army Regulation 600-9 (Army Body Composition Program).
- (6) Meet the Army physical readiness requirements and pass the Army Combat Fitness Test (ACFT) within 120 days before graduation.

b. Academic Program:

- (1) Satisfy the requirements of at least one academic major.
- (2) Cadets may not have unresolved course failures during their final term.

c. Military Program.

- (1) Successfully complete Cadet Basic Training (CBT), Cadet Field Training (CFT), and Cadet Leader Development Training (CLDT).
  - (2) Successfully complete Cadet Troop Leader Training (CTLT).
  - (3) Successfully complete a West Point summer leader detail during second or first Class summer.
- d. Physical Program: Successfully complete the Indoor Obstacle Course Test (IOCT) during the second or first-class year.
- e. Character:
- (1) Uphold the Cadet Honor Code. If a cadet is found to have violated the code and is retained, he/she is required to successfully complete a prescribed remediation program.
  - (2) Uphold the standards of conduct and respect for others. If a cadet is found to have violated these standards and is retained, he/she is required to successfully complete a prescribed remediation program.
4. The Academic Board approved these minimum graduation requirements on 16 May 2024.



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MASP

JUN 05 2025

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Class of 2029 Graduation Requirements

1. **Purpose:** This document articulates the minimum requirements that each cadet must meet to graduate from the United States Military Academy (USMA).

2. Discussion: The United States Military Academy educates, trains, and inspires cadets across a core set of developmental experiences in the academic, military, and physical programs and the character domain, collectively known as the West Point Leadership Development System (WPLDS). The WPLDS is designed to produce graduates who: **live honorably; lead honorably; and demonstrate excellence**. To assess if a cadet has demonstrated the minimum virtues and merit expected of a USMA graduate and future US Army officer, USMA's minimum graduation requirements are published below. Note that cadets must always aspire to far exceed these minimum standards, both individually and collectively.

3. Minimum Requirements:

a. Institutional

- (1) Achieve a Cumulative Quality Point Average (COPA) of 2.00.
- (2) Complete a minimum of eight semesters as a full-time cadet, either at the Academy or as a part of an Academic Board-sanctioned semester exchange program.
- (3) Successfully complete the USMA core curriculum in the academic, military, and physical programs as defined in the Red Book, Green Book, and White Book, respectively. Also, achieve an: Academic Program Score Cumulative (APSC); Military Program Score Cumulative (MPSC); and Physical Program Score Cumulative (PPSC) each of 2.00 or greater.
- (4) Successfully complete one of the following broadening experiences: a Military Individual Advanced Development (MIAD), a Physical Individual Advanced Development (PIAD), or an Academic Individual Advanced Development (AIAD).
- (5) Meet the Army body composition standards of Army Regulation 600-9 (Army Body Composition Program).
- (6) Meet the Army physical readiness requirements and pass the Army Fitness Test (AFT) within 180 days before graduation.

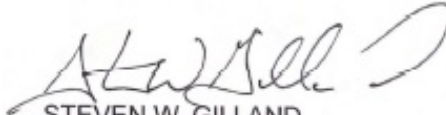
b. Academic Program:

- (1) Satisfy the requirements of at least one academic major.
- (2) Cadets may not have unresolved course failures during their final term.

c. Military Program.



- (1) Successfully complete Cadet Basic Training (CBT), Cadet Field Training (CFT), and Cadet Leader Development Training (CLDT).
  - (2) Successfully complete Cadet Troop Leader Training (CTLT).
  - (3) Successfully complete a West Point summer leader detail during second or first Class summer.
- d. Physical Program: Successfully complete the Indoor Obstacle Course Test (IOCT) during the second or first-class year.
- e. Character:
- (1) Uphold the Cadet Honor Code. If a cadet is found to have violated the code and is retained, he/she is required to successfully complete a prescribed remediation program.
  - (2) Uphold the standards of conduct and respect for others. If a cadet is found to have violated these standards and is retained, he/she is required to successfully complete a prescribed remediation program.
4. The Academic Board approved these minimum graduation requirements on 17 April 2025.



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# **PART I: THE ACADEMIC PROGRAM - CPS CALCULATION**

## **1. Computing the CPS.**

The CPS is computed in a way that provides a measure of a cadet's performance relative to his or her peers. This is meant to adjust for variations in grading systems within and among the three programs.

- a. The APSC, MPSC, and PPSC are standardized so they each have a mean of zero and standard deviation of one.

$$APS^* = \frac{APSC - APSC_{Class\ Mean}}{\sigma_{APSC}}$$

$$MPS^* = \frac{MPSC - MPSC_{Class\ Mean}}{\sigma_{MPSC}}$$

$$PPS^* = \frac{PPSC - PPSC_{Class\ Mean}}{\sigma_{PPSC}}$$

where  $\sigma_{APSC}$ ,  $\sigma_{MPSC}$ ,  $\sigma_{PPSC}$  are the standard deviations of the class's respective scores.

- b. Standardizing the APSC, MPSC, and PPSC ensures that all cadets are evaluated based on their performance relative to their class. It accounts for the deviation of their score from the class average (mean) score. It rewards those who are further above the mean and penalizes those who are further below the mean.

- c. Standardizing does affect determination of Honor Graduates ( $CPS \geq 3.50$ ). This award is meant to be given to cadets based on their performance relative to their peers, not based on an absolute number. Normalizing ensures that only those cadets who truly performed significantly better than their peers become the honor graduates.

- d. The CPS is based on the weighted average of the cumulative program scores. The APSC contributes 55%, the MPSC contributes 30%, and the PPSC contributes 15%.

$$CPS^* = 0.55(APS^*) + 0.30(MPS^*) + 0.15(PPS^*)$$

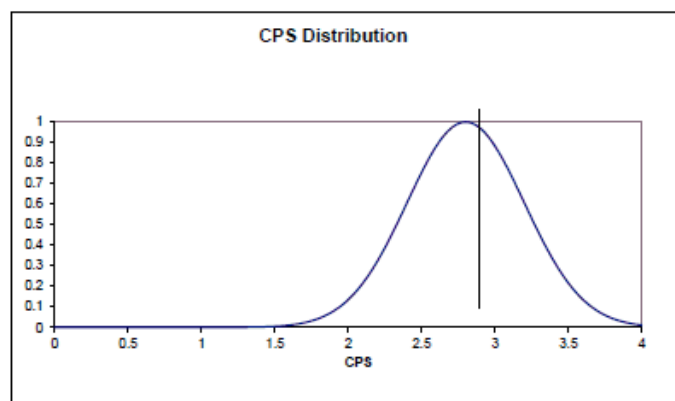
- e. The weighted average of the standardized program scores,  $CPS^*$ , is a normally-distributed random variable with a mean equal to zero. To calculate a representative score on a 4.0 scale,  $CPS^*$  is standardized as follows (Z is the number of standard deviations above or below the class average):

$$Z = \frac{CPS^* - 0.0}{\sigma_{CPS^*}} = \frac{CPS^*}{\sigma_{CPS^*}}$$

f. Based on this result, the CPS is calculated to conform to a distribution with a mean of 2.8 and standard deviation of 0.40:

$$CPS = 0.40Z + 2.80$$

Using this mean and standard deviation ensures that 95% of cadets will have a CPS between 2.0 and 3.6 and 99.7% will be between 1.6 and 4.0. The top 4% of the class will have a  $CPS \geq 3.5$  and be designated as *Honor Graduates*.



## 2. Examples.

To illustrate why the program scores must be standardized, consider the following examples. Suppose the following situation exists:

| Program scores |            |                          |         |         |
|----------------|------------|--------------------------|---------|---------|
|                | Class Mean | Class Standard Deviation | Cadet A | Cadet B |
| APS            | 3.006      | .513                     | 3.006   | 3.326   |
| MPS            | 3.017      | .423                     | 3.017   | 2.878   |
| PPS            | 3.063      | .340                     | 3.063   | 3.454   |

a. **Example 1.** Cadet A is clearly average since his/her program scores are the same as the graduating class' program score averages. Without standardizing, Cadet A's CPS would be

$$CPS = .55(3.006) + .30(3.017) + .15(3.063) = 3.02$$

Cadet A's CPS should be the CPS class mean. In this case, it won't be because each program score has a different standard deviation. Without standardizing, the MPSC and PPSC scores were allowed to carry more weight than intended and allowed Cadet A to be considered above average. Cadet A's CPS after standardizing the program scores is:

$$CPS^* = .55(0.00) + .30(0.00) + .15(0.00) = 0.00$$

$$CPS = 0.40(0.00) + 2.8 = 2.8$$

In this case, Cadet A's CPS is the class average (mean).

- b. **Example 2.** Cadet B's CPS is computed as follows:

$$APS^* = \frac{3.326 - 3.006}{.513} = 0.624$$

$$MPS^* = \frac{2.878 - 3.017}{.423} = -0.329$$

$$PPS^* = \frac{3.454 - 3.063}{.340} = 1.150$$

$$CPS^* = 0.55(0.624) + 0.30(-0.329) + 0.15(1.150) = 0.417$$

Using each cadet's CPS\*, we can compute and verify that the class mean CPS\* is zero and the standard deviation is 0.825. Cadet B's CPS\* score is 0.417, which is clearly above average. Standardize CPS\* and convert it to the 4.0 scale to find the CPS.

$$Z = \frac{0.417}{0.825} \approx 0.506$$

$$CPS = 0.40(0.506) + 2.80 = 3.002$$

Cadet B is above average ( $3.00 > 2.80$ ).

- c. **Example 3** (Are you really an Honor Graduate?). Suppose the following situation exists:

| Program scores |            |                          |         |
|----------------|------------|--------------------------|---------|
|                | Class Mean | Class Standard Deviation | Cadet C |
| APS            | 3.183      | .597                     | 4.000   |
| MPS            | 3.218      | .404                     | 2.895   |
| PPS            | 3.089      | .280                     | 2.873   |

Cadet C appears to be very bright. Does well academically, but performs below average in the physical and military programs. Determine Cadet C's CPS without standardizing the program scores.

$$CPS = .55(4.000) + .30(2.895) + .15(2.873) = 3.50$$

In this case, Cadet C did so well academically that he/she is considered an Honor Graduate. However, this cadet did not truly perform significantly better than his/her peers. The Graduate Outcome Goal requires cadets to excel in all three programs. Now, compute the actual CPS

$$APS^* = \frac{4.000 - 3.183}{.597} = 1.369$$

$$MPS^* = \frac{2.895 - 3.218}{.404} = -0.800$$

$$PPS^* = \frac{2.873 - 3.089}{.340} = -0.635$$

$$CPS^* = 0.55(1.369) + 0.30(-0.800) + 0.15(-0.635) = 0.418$$

$$Z = \frac{0.418}{0.825} \approx 0.507$$

$$CPS = 0.40(0.507) + 2.80 = 3.00$$

Cadet C is above average ( $3.00 > 2.80$ ), but not worthy of being an Honor Graduate.



# **PART I: THE ACADEMIC PROGRAM - STUDY OF WAR THREAD**

## Study of War Thread in the Academic Program United States Military Academy

The Study of War Thread (SoWT) is a multi-disciplinary effort embedded within the core Academic Program that fosters cadets' intellectual understanding of war as a human phenomenon. This knowledge increases the effectiveness of platoon and company operations, enhances officers' understanding of the strategic and moral context of warfare, and provides a foundation for life-long learning for the Academy's graduates throughout their careers as military professionals.

An effective study of war must be a multi-disciplinary effort, as recognized by LTG (Ret.) H. R. McMaster in an external assessment of the West Point curriculum:

*"In general, the study of war and warfare across the curriculum at West Point might best be approached consistent with Sir Michael Howard's 1961 essay on how military professionals should develop their own theory or understanding of war and warfare. First, to study in width. To observe how warfare has developed over a long historical period. Next, to study in depth. To study campaigns and explore them thoroughly, consulting original sources and applying various theories and interdisciplinary approaches ... And lastly, to study in context. Wars must be understood in their social, cultural, economic, human, moral, political, and psychological contexts because as Sir Michael observed "the roots of victory and defeat often have to be sought far from the battlefield." History, political science, social science, philosophy, literature, geography, and psychology all contribute to understanding war in context. And a critical element of the study of war in context is the military's role under the Constitution and an understanding of civil-military relations." (H.R. McMaster, "The Need for a Coherent Interdisciplinary Approach to the Study of War and Warfare at West Point," 2012, p. 2)*

One of the six goals for the Military Program at West Point draws from this same source, that graduates must "demonstrate how to critically think about and understand war and warfare in width, depth, and context." (The Military Program (Greenbook), Academic Year 2020, p. 6) The Academic Program plays an integral role in achieving this goal, developing each cadet's intellectual preparation for the rigors and complexity of combat while helping them internalize their professional identity as commissioned officers. Preparing officers to think broadly and deeply about war as military professionals is essential for lethality, readiness, and future service as strategic thinkers and leaders.

### Structure and Courses

The SoWT is organized around the four fields of expertise described in the Army Profession doctrine: military-technical, geo-cultural/political, moral-ethical, and human-leader development. Cadets progress through a series of core courses building expertise in these four areas and then complete a capstone course on Officership that integrates these fields of expertise into a coherent whole. The capstone course emphasizes the officer's role as a warfighter using historical case studies and current scenarios that highlight the platoon warfighting challenges while always considering the strategic context that provides purpose and a moral setting for their lethal actions. Cadets grapple with concepts such as mission command, multi-domain operations, moral leadership in a complex environment, leadership/lethality at the platoon level, and, when needed, disciplined disobedience to illegal or immoral orders.

The following table lists the core courses that make up the Study of War Thread:

Study of War Thread Core-Course Integration

| Officership Domain | 4th Class Yr                      | 3rd Class Yr        |                            | 2nd Class Yr                     |                                      | 1st Class Yr         |
|--------------------|-----------------------------------|---------------------|----------------------------|----------------------------------|--------------------------------------|----------------------|
|                    | HI101<br>The Army of the Republic | PY201<br>Philosophy | SS202<br>American Politics | SS307<br>International Relations | HI302<br>History of the Military Art | MX400<br>Officership |
| Military-Technical | X                                 |                     |                            |                                  | X                                    | X                    |



|                        |   |   |   |   |   |   |
|------------------------|---|---|---|---|---|---|
| Geo-Cultural/Political | X |   | X | X |   | X |
| Moral-Ethical          | X | X |   | X |   | X |
| Human-Leader           | X |   |   |   | X | X |

### Other Participants

In addition to the courses listed above, the thread will serve as an information sharing mechanism for other stakeholders across the Academy. Such stakeholders include faculty teaching other core courses, the academic departments, faculty responsible for academic majors and minors, the Department of Military Instruction, the Brigade Tactical Department, the Modern War Institute, and other research centers. On an annual basis, the thread can serve as a vehicle for developing common situational awareness of what other organizations are contributing to the study of war and warfare at West Point.

### Study of War Lecture Series

The Study of War Thread will include a lecture series that highlights how course-wide lectures complement cadet education on the various aspects of war and warfare. The series will include events such as the annual Class of '52 Distinguished Lecture and the Combating Terrorism Center's annual Yearling lecture. Other annual events relevant to the study of war that are sponsored by different departments and USCC will be considered for inclusion in the lecture series.

### Thread Goals

The Study of War Thread, as part of the broader West Point experience, provides graduates with an understanding of:

- how the military-technical character of warfare has developed over long historical time periods
- the relationship between the armed forces of the United States and the American political system
- the social, cultural, economic, human, moral-ethical, political, and psychological contexts of war and warfare
- the unique moral-ethical and leadership challenges inherent to war and warfare

### Administration and Assessment

The initial co-chairs are Dr. Scott Silverstone (Professor of International Relations and Director of the Grand Strategy Program in the Department of Social Sciences) and Dr. Jeffrey Peterson (Director, Character Integration Advisory Group, Simon Center for Professional Military Ethic). The committee will meet at least twice a year (about March and October) to facilitate communication, mutual support, and to develop and execute an assessment plan. The co-chairs periodically report to the General Committee to receive feedback on the SoWT and to present assessment results, proposals for changing the thread, and any other issues related to the SoWT.

# **PART I: THE ACADEMIC PROGRAM - CHARACTER THREAD**

## **Character Thread in the Academic Program United States Military Academy**

The character thread is a multi-disciplinary effort embedded within the Academic Program's core curriculum that fosters character development through the content and instructional method of academic core courses. The purpose of the character thread is to support USMA's character-development efforts by identifying and reinforcing character connections across eleven academic core courses and enabling integration with the Cadet Character Education Program (CCEP), prevention curriculum, and other character-development activities conducted across USMA. Faculty members teaching courses in the thread will develop a shared understanding of cadets' strengths and opportunities for improvement related to USMA's character development goals as well as the way thread courses support character development.

The character thread comprises *core thread courses* and *supplemental thread courses*. The thread's core courses have course learning outcomes that directly support USMA's character development goals; supplemental courses contain content that supports USMA's character development goals.

Character thread core courses:

- The Army of the Republic: Leading Citizen Soldiers (HI101)
- General Psychology for Leaders (PL100)
- Philosophy & Ethical Reasoning (PY201)
- Military Leadership (PL300)
- Officership (MX400)

Character thread supplemental courses

- Computing Fundamentals (CY105)
- Literature (EN102)
- Physical Geography (EV203)
- American Politics (SS202)
- History of the Military Art (HI302)
- Constitutional & Military Law (LW403)

The character thread is one element of USMA's common character experiences, which include components of the Academic, Military, and Physical Programs along with prevention activities and Academy events (see appendix a). Common character experiences are developmental opportunities designed to prepare cadets to live honorably, lead honorably, and demonstrate excellence. The character thread promotes development both within a given year (e.g., plebe year) by connecting the thread's courses to simultaneous common character experiences and across its eleven academic core courses by reinforcing character-related content.

Character thread courses support USMA's character development goals by:

1. Introducing cadets to the concept of character writ large and within the specific context of service as an Army officer.
2. Teaching cadets to think critically using different disciplinary approaches that are useful in assessing moral dilemmas.
3. Fostering productive discomfort as cadets learn how to navigate complexity, ambiguity, and opposing perspectives.
4. Preparing cadets to critically reflect on, and civilly discuss, urgent issues facing the nation and the world.
5. Modeling and providing opportunities for cadets to demonstrate, and see the benefits of, empathy and emotional intelligence.
6. Modeling, studying examples of, and providing cadets opportunities to exercise intellectual courage and moral strength.
7. Modeling and encouraging intellectual and ethical humility.
8. Providing opportunities for cadets to explore the professional obligations of U.S. Army officers, the nature of the society in which they serve, and what it means to commit to a lifetime of service to the Army and the nation.
9. Providing cadets opportunities for self-discovery and growth.
10. Providing opportunities for cadets to imagine the probable and possible and to identify the implications of those scenarios.

## **Administration**

Directors of core thread courses and supplemental thread courses will participate in thread meetings (three per year) to identify opportunities to reinforce connections and address institutional focus areas across their courses and within a given cadet year. Leaders of common character experiences not within the Academic Program will also be invited to thread meetings. Course directors, and leaders of these common character experiences will share assessments to create a shared understanding of cadets' strengths and opportunities for improvement related to USMA's character-development goals. Thread leaders will produce a document each year that shares course assessments related to USMA's character-development goals, identifies focus areas for the next academic year, and highlights connections made within the thread and with other common character experiences. This document will be informed by course annual executive summaries that include a section on thread contributions that identify new or existing connections with other thread courses or other common character experiences as well as relevant assessment results and character assessments in other programs.

Course directors of core thread courses and supplemental thread courses will share information about the character thread with faculty members teaching thread courses prior to the start of each semester. At a minimum, course directors should share the purpose of the character thread; their course's contributions to the character thread; and the connection of the course's content to other character thread courses, the Cadet Character Education Program (CCEP), prevention curriculum, and other USMA

character-development activities. Syllabi of thread courses will specify that the course is part of the character thread.

The attached appendix depicts USMA's common character experiences (Appendix A).

## Appendix A: Common Character Experiences

|                                                                                                                                        |                                                                                                    |                          |                       |                      |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------|-----------------------|----------------------|
|  <b>UNITED STATES MILITARY ACADEMY<br/>WEST POINT</b> | <b>Common Character Experiences</b><br><i>"A common path towards living and leading honorably"</i> |                          |                       |                      |
|                                                                                                                                        | <b>Plebe</b>                                                                                       | <b>Yearling</b>          | <b>Cow</b>            | <b>Firstie</b>       |
|                                                                                                                                        | <b>Followership</b>                                                                                | <b>Lead Your Soldier</b> | <b>Lead the Corps</b> | <b>Lead the Army</b> |
| Cadet Summer Training (CST)                                                                                                            |                                                                                                    |                          |                       |                      |
| AY Commandant's Time                                                                                                                   |                                                                                                    |                          |                       |                      |
| Military Science Sequence                                                                                                              |                                                                                                    |                          |                       |                      |
| Physical Program Sequence                                                                                                              |                                                                                                    |                          |                       |                      |
| Academic Program Sequence (Core)                                                                                                       |                                                                                                    |                          |                       |                      |
| Academic Program Sequence (Supplement)                                                                                                 |                                                                                                    |                          |                       |                      |
| Primary Prevention Plan                                                                                                                |                                                                                                    |                          |                       |                      |
| Academy Traditions                                                                                                                     |                                                                                                    |                          |                       |                      |

|                                                                                                                                          |                                                                                                    |                                                      |                                                 |                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------|---------------------------------------------------------------|
|  <b>UNITED STATES MILITARY ACADEMY<br/>WEST POINT</b> | <b>Common Character Experiences</b><br><i>"A common path towards living and leading honorably"</i> |                                                      |                                                 |                                                               |
|                                                                                                                                          | <b>Plebe</b>                                                                                       | <b>Yearling</b>                                      | <b>Cow</b>                                      | <b>Firstie</b>                                                |
|                                                                                                                                          | <b>Followership</b>                                                                                | <b>Lead Your Soldier</b>                             | <b>Lead the Corps</b>                           | <b>Lead the Army</b>                                          |
| <b>CST Character Focus</b>                                                                                                               | CBT Squad30<br>CBT Team Day                                                                        | CFT Squad30<br>CFT Peer Review<br>Leader Dev Academy | CST Cadre and LTP                               | CST Cadre and LTP<br>CLDT Peer Review                         |
| <b>Commandant's Time</b>                                                                                                                 | CCEP<br>ACFT                                                                                       | CCEP<br>Inspiration to Serve<br>ACFT                 | CCEP<br>Leader Challenges (4)<br>ACFT           | CCEP<br>Leader Challenges (4)<br>ACFT                         |
| <b>Military Science</b>                                                                                                                  | MS100                                                                                              | MS200                                                | MS300                                           | MX400                                                         |
| <b>Physical</b>                                                                                                                          | PE116, PE215                                                                                       | PE117, PE32x                                         | PE450, PE360, IOCT                              | PE2xx, IOCT                                                   |
| <b>Academic Core</b>                                                                                                                     | HI101, PL100                                                                                       | PY201                                                | PL300                                           | MX400 (MCC)                                                   |
| <b>Academic Supplement</b>                                                                                                               | EN102, CY105                                                                                       | SS202, EV203                                         | HI302                                           | LW403                                                         |
| <b>Prevention</b>                                                                                                                        | Theme Weeks<br>Deep Dives<br>Bystander Training                                                    | Theme Weeks<br>Deep Dives<br>Bystander Training      | Theme Weeks<br>Deep Dives<br>Bystander Training | Theme Weeks<br>Deep Dives<br>Bystander Training               |
| <b>Academy Events</b>                                                                                                                    | R-Day and A-Day<br>PPW<br>Recognition                                                              | CFT Graduation - CPL<br>YWW                          | Affirmation Ceremony<br>500 <sup>th</sup> Night | Ring Weekend<br>100 <sup>th</sup> Night<br>Grad Wk Ceremonies |

# **PART II: DISCIPLINARY OFFERINGS**

## PART 2: DISCIPLINARY OFFERINGS

This section of the Academic Program (Redbook) presents descriptions of the academic disciplines and specific learning outcomes of all USMA academic majors. A complete listing of the Military Academy's majors and minors appears in Part 4.

### ACADEMIC DISCIPLINE DESCRIPTIONS

#### Aerospace Engineering Major (AEN0 )

**[DA CD: CFA - AERO-SPACE ENGINEERING-SPACE TRAVEL / CIP CD: 14.0201 - Aerospace, Aeronautical and Astronautical/Space Engineering]**

Aerospace Engineering is one of the newest and most exciting of the engineering fields. It deals with the design and analysis of aerospace systems, which could include fixed and rotary-wing aircraft, spacecraft, satellites, uncrewed aircraft systems, and weapons. In a broader sense, aerospace engineering can be applied in the design and analysis of any system that moves through the air to include ground vehicles, and people. Aerospace engineers are in high demand throughout the technology sector with advances in space exploration, hypersonic weapons, advanced air mobility, and uncrewed aircraft systems take on increasingly important roles for the military and society. All cadets experience the same core aerospace engineering program and choose a path in aeronautical or astronautical courses for a depth of study. The goal of the Aerospace Engineering program is to provide high quality instruction in a positive learning environment leading to a degree recognized as being among the best in the nation. The Aerospace Engineering program stresses engineering fundamentals so that graduates are well equipped to understand complex technical problems in a rapidly changing, technology-intensive Army. Once completed, the graduate is well-prepared to excel as an officer and an engineer. The practice-oriented degree is strengthened by the complete integration of design and laboratory experience throughout the curriculum, to include the unique and highly sought after academic flight laboratory program.

To meet this goal, the Program Educational Objectives of the Aerospace Engineering program are:

Within a few years after graduation, aerospace engineering majors are expected to attain:

1. multiple positions of responsibility in which they:
  - a. solve technological problems
  - b. inform and influence others
  - c. build high-performing teams
2. personal and professional growth through formal and informal learning opportunities while being a role model to inspire others
3. experience in providing engineering expertise to the Army and Nation to design solutions and innovate for winning in a complex world

To achieve these Program Educational Objectives, cadets who qualify for graduation with an aerospace engineering major from USMA will demonstrate the following Aerospace Engineering Student Outcomes:

#### Student Learning Outcome 1

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

#### Student Learning Outcome 2

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

#### Student Learning Outcome 3

an ability to communicate effectively with a range of audiences.

#### Student Learning Outcome 4



an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

#### **Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

#### **Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

#### **Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

### **American Politics (PAP3 )**

**[DA CD: ERX - POLITICAL SCIENCE / CIP CD: 45.1002 - American Government and Politics]**

The American Politics program provides students with the ability to explain political outcomes. The program traces the founding and evolution of our American political institutions, emphasizing how philosophical principles influenced the structure and behavior of the nation's governing apparatus. Students receive instruction in political science methods, enabling them to identify and critique methodological approaches to answering political science questions and select an appropriate method of answering their own research questions. Cadets are afforded an opportunity to study contemporary topics in public policy in depth and explain outcomes based on the actions of formal and informal political actors. The American Politics program emphasizes the role of the US military in society, its constitutional roles and responsibilities, and the shifting tenor of civil-military relations across time and levels of policy implementation. Cadets explore the military profession in the context of American political society.

American Politics Student Learning Outcomes and Supporting Objectives:

#### **Student Learning Outcome 1**

Analyze the constitutional origins and historical development of formal institutions (Legislative, Executive, Judicial branches, bureaucracy) and informal actors (media, political parties, interest groups) in American politics.

- a) Identify the fundamental principles of the U.S. Constitution that influence relevant decisions - such as power, conflict, civil rights, and civil liberties - within the American political system.
- b) Examine the political behavior of individuals, actors, and institutions within American politics and the policymaking process.

#### **Student Learning Outcome 2**

Describe the methods by which political scientists research, critically analyze, and explain American politics.

- a) Distinguish research questions appropriate to political science.
- b) Critically evaluate scholarly research based on the merits of research design.
- c) Apply the scientific method using qualitative, quantitative, or experimental methods.
- d) Demonstrate the ability to communicate ideas and provide critical evaluation clearly and effectively.
- e) Demonstrate an appreciation for intellectual pluralism by applying more than one theoretical perspective to a particular phenomenon and assessing the relative value of alternative explanations.

#### **Student Learning Outcome 3**

Characterize U.S. civil-military relations by emphasizing the roles, responsibilities, and culture of the military profession.

- a) Describe the roles and responsibilities of the military, both as a profession and as a bureaucratic entity, as established in the U.S. Constitution and through norms developed over time.
- b) Discuss the major theories of modern U.S. civil-military relations.
- c) Understand the challenges officers face in civil-military relations at the strategic (agenda-setting), operational (formulation and legitimization), and tactical (implementation) levels.

#### **Student Learning Outcome 4**

Describe the U.S. policy-making process.

- a) Compare the strengths and limitations of the different theoretical frameworks - normative and empirical, qualitative and quantitative - in evaluating the domestic policy-making process.
- b) Describe the influences of formal and informal actors on the development and implementation of U.S. foreign and domestic policy.
- c) Explain the theoretical foundations, American principles, and values that influence U.S. foreign and domestic policy.

### **Applied Statistics and Data Science (ASD0 )**

**[DA CD: DHA - STATISTICS / CIP CD: 27.0599 - Statistics, Other]**

Applied statistics uses a variety of computational techniques and methods in order to visualize and explore data, seek

and establish the structure and trends in data, investigate relationships between observed phenomena, and facilitate data interpretation. Data science expands on statistics to encompass the entire life cycle of data, from its specification, gathering, and cleaning through its management and analysis, to its use in making decisions and setting policy. The Data Science and Applied Statistics program at USMA provides Cadets the opportunity to effectively explore structured and unstructured data, define answerable questions, perform statistical analysis, and communicate results both written and orally. The program provides an introduction to the underlying mathematics of Data Science and Applied Statistics while simultaneously offering an exposure to computation and optimization issues inherent in large and disparate data sets.

Graduates who complete an Applied Statistics and Data Science major will be able to:

#### **Student Learning Outcome 1**

Develop and conduct experiments or test hypotheses, analyze and interpret data, and use scientific judgment to draw conclusions.

- Comprehend the basic statistical concepts of data analysis, data collection, modeling and inference.
- Be adept at data visualization using visualization techniques to communicate with others and identify weaknesses in proposed models.
- Employ statistical inference and draw conclusions using formal modeling.

#### **Student Learning Outcome 2**

Identify, formulate, and solve broadly defined technical or scientific problems by applying knowledge of mathematics and science and/or technical topics to areas relevant to Data Science and Applied Statistics.

- Comprehend the underlying structure of common models used in statistical and machine learning as well as the issues of optimization and convergence of algorithms.
- Formulate problems, plan data collection campaigns, and analyze the data to provide insights.
- Demonstrate proficiency in foundational software skills and the associated algorithmic, computational problem-solving strategies.

#### **Student Learning Outcome 3**

Formulate or design a system, process, procedure, or program to meet desired needs.

- Define clear requirements to a problem and use efficient strategies to arrive at an algorithmic solution using a suitable high-level computer language.
- Leverage existing packages and tools to solve computational problems.
- Work with a variety of sources and formats of data.
- Prepare the data for use with a variety of statistical methods and models.
- Ensure the integrity of the data throughout the entire analytical process.

#### **Student Learning Outcome 4**

Communicate effectively with a range of audiences.

- Communicate results both written and orally.

#### **Student Learning Outcome 5**

Understand ethical and professional responsibilities and the impact of technical and/or scientific solutions in global, economic, environmental, and societal contexts.

- Demonstrate understanding of ethical issues in Data Science Research including issues of reproducibility

#### **Student Learning Outcome 6**

Function effectively on teams that establish goals, plan tasks, meet deadlines, and analyze risk and uncertainty.

### **Behavioral Sciences: Applied Psychology (BEA0 )**

**[DA CD: BHS - BEHAVIORAL SCIENCE / CIP CD: 42.2813 - Applied Psychology]**

The Behavioral Sciences Program at West Point is an interdisciplinary, outcome-focused program designed to enhance cadets' ability to analyze and influence human behavior. Through rigorous coursework, cadets develop a comprehensive understanding of behavioral dynamics essential for effective leaders of character. Cadets have the opportunity to specialize by selecting one of three major tracks: Applied Psychology, Organizational Psychology and Leadership, or Military and Society. Each path offers a blend of theoretical knowledge and practical skills, supported by both qualitative and quantitative research training. The program culminates in an applied problem-solving capstone project, where cadets collaborate with real clients to tackle authentic leadership and organizational challenges. This hands-on experience ensures that graduates are not only theoretically informed but also practically adept at addressing complex issues. Aligned with West Point's mission, the Behavioral Sciences Program is committed to producing adaptable, ethically grounded leaders who are prepared to thrive in diverse and challenging environments.

The Behavioral Sciences Program provides cadets with a choice of three major tracks:

- The Applied Psychology major track is focused on applying knowledge of psychology to improve the development,

performance, and well-being of individuals (soldiers).

- The Organizational Psychology and Leadership major track is focused on applying knowledge of psychology to select and develop leaders, and improve the effectiveness of groups, teams, and organizations.
- The Military and Society major track is focused on applying knowledge of military and society to understand how institutions shape civil-military relations, influence military life, leader behaviors, and soldier experiences.

#### Behavioral Sciences Program Student Learning Outcomes and Supporting Objectives:

##### Student Learning Outcome 1

SLO1: Knowledge Application. Graduates will describe, compare, and apply key concepts, theories, and empirical findings in behavioral science to explain human behavior and address behavioral problems in real-world settings.

- SLO1a: Accurately describe and compare major concepts, theories, and research findings in behavioral science.
- SLO1b: Apply behavioral science knowledge to analyze and propose solutions to real-world problems.

##### Student Learning Outcome 2

SLO2: Research and Reasoning. Graduates will locate, evaluate, and use scholarly sources to design, conduct, and interpret basic behavioral research, while following ethical guidelines and demonstrating sound scientific reasoning.

- SLO2a: Design and interpret basic behavioral research using appropriate methods and ethical standards.
- SLO2b: Find and evaluate scholarly sources to support behavioral science arguments and research questions.

##### Student Learning Outcome 3

SLO3: Critical Thinking. Graduates will analyze behavioral problems from multiple perspectives and evaluate potential solutions by clearly connecting them to relevant theories and evidence.

- SLO3a: Analyze complex behavioral problems from multiple theoretical or contextual perspectives.
- SLO3b: Evaluate and justify solutions using relevant behavioral science theories and research.

##### Student Learning Outcome 4

SLO4: Communication. Graduates will communicate behavioral science concepts and findings clearly and professionally in both written and oral formats, adapting their message for different audiences and appropriately using AI tools and technologies to support their communication.

- SLO4a: Deliver clear, professional oral and written communication tailored to audience and purpose.
- SLO4b: Use AI tools and emerging technologies appropriately to enhance communication of behavioral science content.

##### Student Learning Outcome 5

SLO5: Professional Development and Leadership. Graduates will demonstrate how behavioral science knowledge contributes to effective leadership, teamwork, and community engagement, and will identify ways to continue learning and applying psychology in professional settings.

- SLO5a: Explain how behavioral science principles support leadership and team effectiveness.
- SLO5b: Demonstrate commitment to ongoing learning and professional development in behavioral science.

## Biology (BIO0 )

**[DA CD: GPC - MEDICAL SCIENCES GENERAL / CIP CD: 26.0101 - Biology/Biological Sciences, General]**

The Biology Major focuses on the evolution, structure, and processes of living organisms. Advances in molecular biology and biotechnology are providing significant improvements in the quality of our lives even as they alter the fundamental way we view life itself. Evolution, genetic engineering, recombinant DNA research, medical treatment, ecology, and emerging diseases are just a few of today's research areas under the life sciences heading. The Biology Major includes courses that give cadets an in-depth understanding of biology which when combined with math, chemistry and physics can be applied to Army needs. The major focuses on broader understanding of biology and biotechnology. In addition to the core course requirements and three-course engineering sequence (any of the sequences offered) cadets will complete an integrative experience (CH479 Methods and Applications of Biotechnology). This course examines the social, economic, political, ethical, and technological aspects of biotechnology. The Biology Major includes all the courses needed for cadets pursuing the Medical School Option as well as other post-graduate choices.

Applications of the life sciences involve all facets of our lives and are very important to our careers as military officers. A better understanding of biotechnology will allow us to use biosensors to detect weapons of bioterrorism. Biomolecular engineering will make possible the use of cells to manufacture novel biomaterials with specific properties

and functions. Biotechnology and biomedical engineering will improve the medical treatment of battlefield casualties. The most important "system" in the future Army will continue to be the human soldier. Because the soldier is a biological system, biotechnology offers unique potential for enhancing the performance of this most complex, critical, and costly of the Army's systems.

Graduates who complete a Biology Major will be able to:

**Student Learning Outcome 1**

Understand and apply the Scientific Method.

**Student Learning Outcome 2**

Gather, analyze, organize, and present scientific information.

**Student Learning Outcome 3**

Understand the basic principles of life science and chemistry, and their applications to the Army and Society.

**Student Learning Outcome 4**

Understand and apply basic instrumental methods of biological and chemical analysis.

**Student Learning Outcome 5**

Recognize the structure and understand the functions of biomolecules.

**Student Learning Outcome 6**

Know the structures and functions of eukaryotic and prokaryotic cells.

**Student Learning Outcome 7**

Understand the principles of bioenergetics and metabolism.

**Student Learning Outcome 8**

Understand the structure-function relationships at all levels of organization of living organisms.

**Student Learning Outcome 9**

Understand evolutionary biology and its importance.

**Student Learning Outcome 10**

Understand ecology (organization, diversity, and interdependence of living organisms).

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## **Chemical Engineering (CEN2 )**

**[DA CD: CEX - ENGINEERING,CHEMICAL / CIP CD: 14.0701 - Chemical Engineering]**

Chemical engineering is perhaps the broadest and most diverse field in all of engineering. Any commercial process or product that uses or contains molecules probably involved a chemical engineer at some stage of development. This includes all materials used by the military, including such basic items as food, clothing, fuel, water, explosives, metals, polymers, ceramics, semiconductors, medicines, artificial organs, and prostheses, just to name a few. Chemical engineers design these materials at the molecular level, optimize the design for specific applications, and develop efficient methods for production, packaging, and distribution. Chemical engineers are also very concerned with the conversion between matter and energy, particularly since almost all chemical reactions require or produce energy. In terms of contemporary social problems, chemical engineers are at the forefront of efforts to develop new and more efficient energy sources, and we are critical to efforts at environmental remediation, including waste recycling and remediation. Within the military, chemical engineers are uniquely qualified to address problems in fuel and water production and distribution, power generation, as well as detection, decontamination, and protection against chemical and biological agents.

The chemical engineering program is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>, under the commission's General Criteria and Program Criteria for Chemical, Biochemical, Biomolecular, and Similarly Named Engineering Programs.

The mission of the chemical engineering program is to prepare commissioned leaders of character who possess the intellectual capital to leverage new and emerging technologies.

Program Educational Objectives: During a career as commissioned officers in the United States Army and beyond, program graduates:

- (1) Demonstrate effective leadership by leveraging chemical engineering expertise and precise technical communication.
- (2) Contribute to the solution of complex problems in a dynamic environment.
- (3) Apply disciplined technical expertise to succeed in advanced study programs.

On completion of the chemical engineering program, our graduates demonstrate:

**Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

**Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

**Student Learning Outcome 3**

an ability to communicate effectively with a range of audiences.

**Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

**Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

**Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

**Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

**Student Learning Outcome 8**

Understand the chemical engineering curriculum, including chemistry, material and energy balances, safety and environmental factors, thermodynamics of physical and chemical equilibria, heat, mass, and momentum transfer, chemical reaction engineering, continuous and staged separation processes, process dynamics and control, modern experimental and computing techniques, and process design.

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**Chemistry (CHM2 )**

**[DA CD: DDB - CHEMISTRY,ANALYTICAL,GENERAL / CIP CD: 40.0501 - Chemistry, General]**

Chemistry is the branch of sciences that studies the composition, structure, properties, changes and interactions of matter. Every material thing - from the foods we eat, to the medicine we take, to the air we breathe - is a chemical or a mixture of chemicals. Therefore, it is truly the central science and underpins much of the efforts of scientists and engineers to improve life for humankind. Since chemistry is the molecular science, military applications of chemistry rely on the understanding of the structure and changes at the molecular level. These application areas can include the synthesis and development of advanced materials and explosives, solving environmental problems, creating innovative biotechnology solutions, and chemical or biological sensing.

The Chemistry Major includes all the courses recommended by the American Chemical Society and are designed to provide cadets with basic instruction with comparable emphasis on the areas of analytical chemistry, biochemistry, inorganic chemistry, organic chemistry and physical chemistry. Cadets are required to complete a core sequence. In addition, cadets must complete a three-course engineering sequence and may choose from any of the sequences offered. Ten courses are required to complete the major. Thus, the Chemistry Major requires a total of 40 courses to be taken or validated. Cadets choosing this program will complete an integrative experience (CH487 Advanced Chemistry Laboratory) that will examine the social, economic, political, and technological aspects of chemistry. The Chemistry Major with ACS Certification includes one elective course and two research courses (CH489 and CH490). The Chemistry Major with ACS Certification and Honors requires cadets to complete the course requirements for the CHM1A major while attaining a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

Graduates who complete a Chemistry Major will be able to:



**Student Learning Outcome 1**

- Use information resources to gather, organize, and understand scientific material.

**Student Learning Outcome 2**

- Design and execute experiments to address a problem or question.

**Student Learning Outcome 3**

- Analyze and assess scientific data gathered in the laboratory.

**Student Learning Outcome 4**

- Effectively and clearly communicate scientific information in written and oral form to a variety of audiences.

**Student Learning Outcome 5**

- Understand the applications of chemistry in the Army and society.

**Student Learning Outcome 6**

- Recognize the relationship between the properties of a substance, its molecular structure, and its reactivity.

**Student Learning Outcome 7**

- Understand and apply the physical concepts of chemistry.

**Civil Engineering (CVN3 )**

**[DA CD: CCX - CIVIL ENGINEERING / CIP CD: 14.0801 - Civil Engineering, General]**

Civil Engineers design, build, and maintain the infrastructure of modern civilization. Civil Engineers are unique problem solvers, who apply their technical know-how to meet the challenges of disaster relief, deteriorating facilities, traffic congestion, floods, earthquakes, environmental contamination, and community planning. The Civil Engineering program is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>. Civil Engineering majors apply the knowledge of mathematics and science; analyze and solve problems in structural, geotechnical, hydraulic/hydrology, construction engineering; conduct experiments in Civil Engineering technical areas, and analyze and interpret the resulting data; design a Civil Engineering project; use probability and statistics to address uncertainty; include principles of sustainability in design; explain basic concepts in project management, business, public policy, and leadership; analyze issues in professional ethics; and explain the importance of professional licensure. Civil Engineering majors engage in a year-long capstone project requiring them to identify, formulate, and solve open-ended and complex engineering problem in teams. The teams use an iterative design process to define the problem and requirements, analyze alternatives within constraints (e.g., applicable codes and standards), and select the best alternative to solve the problem. The design process takes into consideration public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. Teams communicate their process and solution through written deliverables and presentations to a range of audiences. In the Army, Civil Engineers provide leadership and technical abilities to support combat operations and manage the infrastructure at Army installations worldwide. As part of the United States Army Corps of Engineers, Civil Engineers provide vital public engineering to support our Nation's security, energize the economy, and reduce risks from disasters. Cadets who maintain good standing in the Civil Engineering major take the Fundamentals of Engineering (FE) exam during the spring semester of their First-Class year. Passing the FE exam is the essential first step in becoming a registered professional Civil Engineer.

Graduates who major in Civil Engineering will achieve the following Civil Engineering Program Educational Objectives within a few years after graduation. Civil Engineering majors are expected to attain:

1. MULTIPLE POSITIONS OF RESPONSIBILITY as Army leaders in which they (a) drive outcomes, (b) inform, influence, and inspire others, and (c) build teams.
2. PERSONAL AND PROFESSIONAL GROWTH through formal and informal learning opportunities while serving as professional role models to others.
3. EXPERIENCE IN PROVIDING ENGINEERING EXPERTISE to the Army and Nation to design solutions, solve complex and technical problems, and innovate for winning in a complex environment.

To achieve these objectives, cadets will demonstrate the following Civil Engineering Student Outcomes:  
At the time of graduation, Civil Engineering majors will have attained:

**Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

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**Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

**Student Learning Outcome 3**

an ability to communicate effectively with a range of audiences.

**Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

**Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

**Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

**Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

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**Computer Science (CSC1 )**

**[DA CD: CUE - COMPUTER SCIENCE / CIP CD: 11.0101 - Computer and Information Sciences, General]**

Computer scientists analyze, plan, design, and build computer systems. Within this broad area of computer system design, the computer science program at USMA provides cadets the opportunity to focus on the design of computer software components and the implementation of software systems. The program provides a solid introduction to the general field of computer science, including computer theory, computer programming, algorithm analysis, data structures, software testing and development, computer organization, programming languages, operating systems, and the design of large software systems. According to your interests, you may pursue further study in artificial intelligence, computer graphics, computer networks, cyber security, and other topics. The opportunity to accomplish advanced individual study under the direction of a faculty member is available to those who are interested and qualified. Whether operating a remote sensor network from a fire base in Afghanistan, managing logistics from a Brigade Support Area in Kuwait, pattern matching events for intelligence purposes in Iraq, or simply understanding the computational feasibility of solving a complex problem, the Computer Science major prepares you to succeed as a leader in any branch of the Army and is a superb foundation for advanced civil schooling. The USMA Computer Science major is accredited by the Computing Accreditation Commission of ABET, <http://www.abet.org>.

The Program Educational Objectives (PEO) for Computer Science are that, five to seven years after graduation, graduates who majored in Computer Science will have been successful Army officers who:

- A. Initiated and completed tasks that identify aspects of a complex situation that can be enhanced by using computing technology.
- B. Applied computing knowledge and skills while using an engineering process individually or in diverse teams to develop computing technology applications.
- C. Used effective communication to explain new computing technology to war fighters in support of current and emerging Army war fighting doctrine.
- D. Grown professionally through self-study, continuing education and professional development.

To support these objectives, at the time of graduation, Computer Science graduates will have an ability to:

**Student Learning Outcome 1**

Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.

**Student Learning Outcome 2**

Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the computer science discipline.

**Student Learning Outcome 3**

Communicate effectively in a variety of professional contexts.

**Student Learning Outcome 4**

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Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

**Student Learning Outcome 5**

Function effectively as a member or leader of a team engaged in activities appropriate to the computer science discipline.

**Student Learning Outcome 6**

Apply computer science theory and software development fundamentals to produce computing-based solutions.

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**Cyber Science (CYS1 )**

**[DA CD: BNA - CYBER OPERATIONS / CIP CD: 11.1003 - Computer and Information System Security/Information Assurance]**

The Cyber Science Major is the study of the structure, behavior, and effective use of cyberspace in the presence of intelligent, adaptive, and evolving adversaries, both human and artificial. Cyberspace is made up of hardware, software, information, people, organizations, policies, and laws. Building on a common 12-course foundation, this flexible major allows cadets to choose a cyber in society elective, a cyber operations elective, and 4 cyber electives. The flexibility built into this major gives cadets the ability to craft the cyber experience that best aligns with their interests, whether that be cyber policy, infrastructure defense, cyber operations, or a hybrid. Cadets who successfully complete the NSA prescribed Cyber Operations program of study will earn Cyber Science with Cyber Operations distinction.

Graduates who complete the Cyber Science major will be able to:

**Student Learning Outcome 1**

Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.

**Student Learning Outcome 2**

Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of Cyber Science.

**Student Learning Outcome 3**

Communicate effectively in a variety of professional contexts.

**Student Learning Outcome 4**

Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

**Student Learning Outcome 5**

Function effectively as a member or leader of a team engaged in activities appropriate to Cyber Science.

**Student Learning Outcome 6**

Apply security principles and practices to maintain operations in the presence of risks and threats.

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**Cyber Science: Cyber Operations (CYE0 )**

**[DA CD: BNA - CYBER OPERATIONS / CIP CD: 11.1003 - Computer and Information System Security/Information Assurance]**

The Cyber Science Major is the study of the structure, behavior, and effective use of cyberspace in the presence of intelligent, adaptive, and evolving adversaries, both human and artificial. Cyberspace is made up of hardware, software, information, people, organizations, policies, and laws. Building on a common 10-course foundation, this flexible major allows cadets to choose one of five depth concentrations: Cybersecurity, Network Services, Cyber Operations, Cyber-Physical Systems, and Artificial Intelligence.

This description and outcomes apply to all tracks of the Cyber Science Major, not just Cyber Science: Cyber Operations.

Graduates who complete the Cyber Science major will be able to:

**Student Learning Outcome 1**

- Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.



**Student Learning Outcome 2**

- Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of Cyber Science.

**Student Learning Outcome 3**

- Communicate effectively in a variety of professional contexts.

**Student Learning Outcome 4**

- Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

**Student Learning Outcome 5**

- Function effectively as a member or leader of a team engaged in activities appropriate to Cyber Science.

**Student Learning Outcome 6**

- Apply security principles and practices to maintain operations in the presence of risks and threats.

**Defense and Strategic Studies (DSS2 )**

**[DA CD: BMT - DEFENSE AND STRATEGIC STUDIES / CIP CD: 29.0399 - Military Applied Science, Other]**

The Defense & Strategic Studies (DSS) academic major will continue to reach the Academy's overarching goal through a multidisciplinary approach in its program structure and an interdisciplinary approach within its courses. The program will continue to draw upon military science, history, economics, political science, geography, leadership, information technology and law to understand the nature of war and the military instrument of national power.

**DSS Major Student Learning Outcomes and Supporting Objectives****Student Learning Outcome 1**

Evaluate military strategic decision-making by applying appropriate theoretical, historical, policy and interdisciplinary knowledge.

- a. Cadets can develop a diverse and appropriate base of knowledge to effectively evaluate strategic choices, to include formal strategies and military systems, using a holistic approach.
- b. Cadets can explain the relationship between tactics, operations and strategy with an emphasis on the strategic effect of tactical action.

**Student Learning Outcome 2**

Cadets can frame complex, multifaceted strategic problems relating to the military component of national power and the potential use of force.

- a. Cadets can deconstruct the problem into its essential components and describe how they interrelate by effectively examining the political and military context.
- b. Cadets can identify relevant stakeholders, interests and policy objectives for state and non-state actors.
- c. Cadets can explain the influence that state and non-state actors' organizations, capabilities and systems have on the problem definition.

**Student Learning Outcome 3**

Cadets can apply theoretical frameworks, strategic planning models and critical thinking to construct viable and innovative ways to solve strategic problems.

- a. Cadets can identify and apply relevant theories and models to forming viable and innovative solutions.
- b. Cadets can translate well researched, reliable evidence into useful conclusions.

**Student Learning Outcome 4**

Cadets can Communicate and connect with a diverse range of audiences in order to frame and deliver an insightful strategic analyses.

- a. Cadets can formulate a clear, direct message that communicates the fundamental conclusion(s) up front.
- b. Cadets can persuasively communicate the necessary evidence and reasoning by logically organizing the information or argument.
- c. Cadets can effectively communicate through oral presentation by keeping the audience engaged with creativity, poise and competence.
- d. Cadets can effectively communicate through writing by being precise, concise and clear.

**Economics (ECN2 )**

**[DA CD: EDX - ECONOMICS,GENERAL / CIP CD: 45.0603 - Econometrics and Quantitative Economics]**

The Economics major provides critical thinking skills applied to human behavior and answers the questions: "What is

produced?; How is it produced?; and To whom is it allocated?" Mathematical models are used to develop and test optimal resource allocation mechanisms. These models also provide the necessary foundation to analyze policy regimes within individual markets as well as aggregate economies. These skills are directly applicable to decision makers in tactical command positions who must achieve desirable outcomes with personnel and equipment constraints as well as national strategic decision makers who face similar aggregate constraints. The Economics major also provides deep proficiency in quantitative methods, particularly nonlinear optimization, for more specialized areas such as Operations Research, Resource Management, and Strategist.

## Economics Major Student Learning Outcomes

### Student Learning Outcome 1

Constrained Optimization: Cadets can derive the objective functions and resource constraints facing individuals, firms, and governments, and then solve the relevant optimization problems to describe and predict the behavior of economic agents.

### Student Learning Outcome 2

Market Analysis: Using the outcomes derived from optimization, Cadets can explain how markets and the aggregate economy achieve equilibrium, and then predict how changes in the economic environment impact individual markets and the aggregate economy.

### Student Learning Outcome 3

Market Efficiency: Cadets can apply economic models of individual, firm, and government behavior to evaluate whether equilibrium in a market achieves economic efficiency (the maximization of total surplus).

### Student Learning Outcome 4

Marginal Analysis: Cadets can apply the concept of equalization at the margin to explain and evaluate individual, firm, and government behavior, as well as perform cost-benefit analyses of government policies.

### Student Learning Outcome 5

Incentives: Cadets can explain the central role of incentives in shaping the behavior of economic agents and design incentive structures that promote efficient outcomes.

### Student Learning Outcome 6

Strategic Decision Making: Cadets can model and evaluate economic environments in which the actions of economic agents are interdependent.

### Student Learning Outcome 7

Risk and Uncertainty: Cadets can model and evaluate the impact of uncertain economic outcomes due to either random shocks or the strategic behavior of economic agents, and explain how differential aversion to risk across individuals, firms, and governments affects market outcomes.

### Student Learning Outcome 8

Information: Cadets can model and evaluate the effects of costly, limited, and asymmetric information on economic decision making, and derive policies to mitigate the efficiency losses arising from imperfect and incomplete information.

### Student Learning Outcome 9

Global Economic Linkages: Cadets can model economic interactions of individual consumers, investors, firms and governments in international goods, capital, and currency markets, and evaluate the efficiency costs of government policies that restrain economic exchange among nations.

### Student Learning Outcome 10

Political Economy: Cadets can apply economic models to understand behaviors of agents in political and other "non-economic" settings, and use models drawn from allied social science fields (Political Science, Psychology, Sociology, and Geography) to derive richer explanations of the behavior of individuals, firms and organizations (including governments).

### Student Learning Outcome 11

Government Intervention in Markets: Cadets can use economic models to evaluate the impact(s) of government regulations and policies on market outcomes, as well as design interventions that improve economic efficiency or minimize efficiency losses.

### Student Learning Outcome 12

Macroeconomic Stabilization: Cadets can use economic models to explain and predict fluctuations in key macroeconomic variables, and design fiscal and monetary policies to dampen these fluctuations.

### Student Learning Outcome 13

Economic Growth: Cadets can build models that explain and predict the determinants of economic growth in the long-run, evaluate the impact(s) of government policies on growth, and design policies to promote economic growth.

### Student Learning Outcome 14

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Empirical Methods: Cadets can: apply economic theory to identify testable hypotheses; collect the requisite data and use appropriate empirical methods to test hypotheses; develop empirical models that can identify the causal effect of an economic variable of interest on an outcome of interest; and interpret the findings from statistical and empirical analyses as well as explain their relevance to policy making.

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## **Electrical Engineering (EEN1 )**

**[DA CD: CHX - ENGINEERING,ELECTRICAL / CIP CD: 14.1001 - Electrical, Electronics, and Communications Engineering]**

The impact of the electronics revolution on our daily lives may exceed that of the industrial revolution. The advent of the integrated circuit and the microprocessor have made possible phenomenal advances in such varied fields as medicine, communications, manufacturing, computation, education, energy conversion, and weapons systems. Electrical engineers are at the forefront of this revolution, using the principles of physics, mathematics and the engineering sciences to develop new and innovative applications of electronics. Regardless of branch, officers will surely be involved with electronic systems in military hardware. The courses in the electrical engineering curriculum are directly applicable to the Army you will lead. As a student of electrical engineering you will develop a mastery of the fundamental elements of circuit theory, electromagnetic fields and waves, electronics, digital computer logic and electromechanical energy conversion. You will then study in greater depth subjects selected from the areas of robotics, communications, opto-electronics, alternative energy and cyber engineering. The program emphasizes practical design, hands-on laboratory and computer experience, teamwork, and interdisciplinary projects. The program additionally provides a sound basis for graduate schooling in electrical engineering and related fields as well as fulfilling the disciplinary depth component of the USMA curriculum. The Electrical Engineering Program is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>.

The Electrical Engineering Program objectives are that five to seven years after graduation cadets who major in Electrical Engineering will have been successful Army officers who have:

- Applied their engineering, management, and leadership skills in service of their country.
- Demonstrated intellectual growth through self-study, continuing education, and professional development in the Army.
- Provided technical leadership and disciplinary knowledge as Army officers with a broad understanding of the potential ethical and societal impacts of technology.
- Applied engineering methodology and creativity to Army problems while effectively communicating across mediums and cultures.

Expected Student Outcomes for graduating cadets in the Electrical Engineering major are:

### **Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

### **Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

### **Student Learning Outcome 3**

an ability to communicate effectively with a range of audiences.

### **Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

### **Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

### **Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

### **Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

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## **Engineering Management (ENM1 )**

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**[DA CD: CYX - ENGINEERING, INDUSTRIAL / CIP CD: 15.1501 - Engineering/Industrial Management]**

Engineering Management majors study the engineering relationships among the management tasks of staffing, organizing, planning, financing, and the human element in production, research, engineering, and service organizations. By emphasizing leadership in a technical setting, the program builds on the traditional roles of the basic and applied sciences for engineering and technology management. Engineering managers must understand the interaction of organizational, technical, and behavioral variables in order to build a productive engineering team. Majors get a technical foundation in a specific engineering field of their choice: civil, electrical, nuclear, infrastructure, environmental, software, chemical, or general engineering. The program also provides a solid base of courses in engineering economy, production operations management, supply chain engineering, quantitative business analysis, project management, and computer modeling in order to prepare graduates to lead in a technical environment. The program culminates with a capstone design experience for a real client. The Engineering Management program at West Point is one of the top undergraduate programs in the nation and provides the academic foundations for a wide variety of activities important to Army officers of all branches. The Engineering Management Program is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>.

**Engineering Management Program Educational Objectives:** The Engineering Management Program seeks to prepare future Army officers for productive and rewarding careers in engineering or related professions for service to the nation. Five to seven years after graduation, cadets who majored in Engineering Management will have been successful Army officers who:

1. Successfully lead and participate as a member of multi-disciplinary teams in a diverse cultural environment.
2. Apply critical thinking to their engineering, management, and leadership skills to design solutions to complex problems.
3. Demonstrate intellectual growth and continuous self-improvement through self-study, continuing education, and professional development.
4. Demonstrate effective communicating skills across a variety of mediums and cultures.
5. Act responsibly by upholding strict ethical and moral standards and considering impacts of decisions on social, political, economic, and technological issues.

To achieve these objectives, cadets will demonstrate the following Engineering Management Student Outcomes at the time of graduation:

**Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

**Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

**Student Learning Outcome 3**

an ability to communicate effectively with a range of audiences.

**Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

**Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

**Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

**Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

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**Engineering Psychology (EPS1 )****[DA CD: CYA - HUMAN FACTORS ENGINEERING / CIP CD: 30.2501 - Cognitive Science]**

Engineering Psychology is one of four majors available through the Department of Behavioral Sciences and Leadership. It bridges the disciplines of experimental psychology and engineering by providing cadets with a strong foundational knowledge of the psychology of human performance and how these principles of behavior can be used to engineer and evaluate more effective systems. In the Yearling and Cow years, Engineering Psychology majors enroll in courses designed to teach research and statistical skills, and a series of courses that teach them about the scientific foundation of human behavior. In their Firstie year, they learn to apply this knowledge in the design and evaluation of complex systems.

## Engineering Psychology Major Student Learning Outcomes and Supporting Objectives

### Student Learning Outcome 1

Apply knowledge of human performance to operational readiness and total system design.

- Objective 1: Relate principles of the human nervous and endocrine systems to human performance
- Objective 2: Analyze the characteristics of human perceptual processes in the design of complex systems and work environments
- Objective 3: Apply understanding of human cognitive structure to system design
- Objective 4: Illustrate the limitations of the human body and biomechanics on human performance

### Student Learning Outcome 2

Design, conduct, and analyze research in human performance in person/system interaction.

- Objective 1: Apply descriptive and inferential statistics in analyzing data in research projects
- Objective 2: Synthesize scientific literature in deriving an experimental hypothesis
- Objective 3: Design appropriate research methods to answer a scientific question

### Student Learning Outcome 3

Evaluate research in human performance and person/system interaction.

- Objective 1: Analyze the validity of research methods used in published studies
- Objective 2: Contrast different empirical approaches to a given human performance research question
- Objective 3: Synthesize different research findings into an structured new whole
- Objective 4: Analyze an existing system and design an alternative and more functional system

### Student Learning Outcome 4

Report and defend scientific findings in oral and written format.

- Objective 1: Describe scientific data and theories orally and in writing
- Objective 2: Apply the American Psychological Association's style in preparing written research reports
- Objective 3: Infer extrapolations from research findings to related phenomenon

## English (ENL2 )

**[DA CD: ADX - ENGLISH / CIP CD: 23.0101 - English Language and Literature, General.]**

Cadets who major in English refine skills of analysis and creativity and enrich their imaginative capacities by studying the literary and artistic expressions of primarily Anglo-American and, to a lesser extent, global cultures. Exposed to a range of literary genres and modes of cultural expression, English majors graduate with a broader and deeper understanding of the history of ideas and of the ways in which imaginative literature has influenced human behavior and shaped cultural norms. By exploring the full spectrum of human behaviors and circumstances and by laying the groundwork for the cultivation of emotional intelligence and sympathetic imagination, the study of literature equips graduates to engage the world's intellectual, moral, and emotional complexities with insight and the potential for empathy. English majors refine their listening, speaking, reading, and writing skills in a variety of analytic and creative situations from the textual analysis of poetry and prose to the study of literary history to dramatic performance. The English major with Honors further challenges Cadets to compose a lengthy academic research project in close consultation with a member of the doctoral faculty. With an English major, a Cadet graduates from West Point with the capacity to analyze textual evidence, to account for cultural context and complexity, to communicate precisely, and to imagine creative solutions to difficult problems--attributes that, taken together, constitute excellent preparation for Army service in the twenty-first century and for life more broadly.

### Student Learning Outcome 1

English majors will possess a body of knowledge specific to the study of literature.

Cadets will

- (1) Understand important conventions of literary scholarship, including critical vocabulary, terminology, and research methods.
- (2) Receive a strong foundation in Anglo-American literary history.
- (3) Be exposed to significant works of world literature and to global perspectives.

## Student Learning Outcome 2

English majors will communicate effectively within the discipline, observing audience, vocabulary, conventions, and methodology. Cadets will

- (1) Learn to recognize the difference between rational and emotional responses to texts.
- (2) Speak about literature and culture with depth, nuance, clarity, and precision.
- (3) Practice standard skills of literary scholarship, including close reading, analysis in context, and the application of various critical and theoretical lenses.
- (4) Write effective academic and creative texts that display an appropriate understanding of subject, audience, and purpose.
- (5) Write an extended literary research paper that incorporates secondary sources.

## Student Learning Outcome 3

Student Learning Outcome 3

English majors will apply knowledge of literature within and across disciplinary boundaries

Cadets will

- (1) Gain a knowledge of literature's historical, social, cultural, and political contexts.
- (2) Gain an understanding of the influence of literary models and archetypes on human behavior, especially on the behavior of military professionals; and an awareness of the role of literature within the cultures of other countries and regions.
- (3) Draw meaningful connections between literature and other disciplines in discussion and writing.

## Student Learning Outcome 4

Student Learning Outcome 4

English majors will demonstrate the capability and the desire to pursue continued intellectual development.

Cadets will

- (1) Value the continuing study of literature and other intellectual pursuits.
- (2) Be able to work independently and collaboratively in literary studies or other intellectual fields.

## Student Learning Outcome 5

Student Learning Outcome 5

English majors will cultivate particular habits of mind exercised through literary study.

Cadets will

- (1) Acknowledge and reckon with ambiguity.
- (2) Cultivate patience and the faculty of deep attention.
- (3) Take intellectual and imaginative risks in creative projects.
- (4) Learn to approach problems with resilience, resourcefulness, and invention.

## Environmental Engineering (EVE2 )

**[DA CD: CCO - ENVIRONMENTAL ENGINEERING / CIP CD: 14.1401 - Environmental/Environmental Health Engineering]**

The environmental engineering program is designed for cadets who want to tackle today's most pressing environmental challenges and create innovative, engineered solutions for a sustainable and resilient future. As global concerns such as emerging contaminants, urbanization, and resource scarcity accelerate, environmental engineers play a crucial role in designing sustainable solutions that protect public health and welfare, and prevent future environmental degradation. They apply core knowledge of chemical, biological, and physical processes to remediate environmental contaminants in the air, water, and soil; design water and wastewater treatment systems; manage hazardous and solid waste; improve air quality; assess environmental impacts of projects; and contribute to renewable energy development and circular economy strategies. This discipline is rapidly evolving through its integration with advances in technology and artificial intelligence.

The environmental engineering program emphasizes hands-on learning, critical thinking, and interdisciplinary teamwork via the complete integration of design and laboratory experiences throughout the curriculum. Graduates gain a strong foundation in math, science, and engineering fundamentals so that they are well equipped to understand and combat complex problems in a rapidly changing Army and environmental landscape. Graduates are well-prepared to excel as officers and engineers. The Environmental Engineering Program is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org/>.

The Program Educational Objectives (PEOs) of the Environmental Engineering Program identify what our graduates can accomplish within a few years following graduation. PEOs are reviewed at least biennially by the Environmental Engineering Board of Advisors. PEOs were most recently revised in April 2025 and placed into effect beginning fall term 2025.

Within a few years following graduation, West Point environmental engineering graduates:

1. Are leaders of character able to:
  - a. develop innovative solutions to solve complex cross-disciplinary problems.



- b. minimize environmental impacts.
  - c. inspire teams to execute projects within constraints.
  - d. collaborate with partners and build consensus.
- 2. Possess professional skills enabling them to:
  - a. effectively communicate pertinent information, including environmental topics, to a range of audiences.
  - b. manage resources.
  - c. evaluate, mitigate, manage, and communicate risk.
- 3. Understand the importance of continued self-development to:
  - a. achieve professional licensure and certification.
  - b. pursue continuing education and technical skill development.
  - c. seek formal and informal enrichment experiences including community outreach and environmental stewardship.
  - d. seek mentorship and mentor others.
- 4. Internalize professional values that enable them to have:
  - a. continued inculcation of Army values and understanding of Army priorities.
  - b. the ability to analyze an engineering challenge involving conflicting ethical and professional interests and determine an appropriate course of action.

The Student Learning Outcomes of the Environmental Engineering Program identify what our graduates can accomplish upon graduation.

#### **Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

#### **Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

#### **Student Learning Outcome 3**

an ability to communicate effectively with a range of audiences.

#### **Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

#### **Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

#### **Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

#### **Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

### **Environmental Science (ESC2 )**

**[DA CD: CCQ - ENVIRONMENTAL SCIENCE / CIP CD: 03.0104 - Environmental Science]**

Environmental science is a broad, integrative, science-based discipline which focuses on the interrelationships between people and the environment. Environmental scientists conduct investigations to analyze these interrelationships and to identify, abate, or eliminate human-caused pressures on the environment. The ultimate goal of these investigations is to create a sustainable balance between humans and the natural world that minimizes environmental degradation. This major develops expertise into the processes that sustain our environment by expanding upon the West Point core science education by adding studies in the natural sciences such as biology, ecology, geology, and meteorology, and in the integrative studies of environmental decision making and environmental security. This broad academic background is excellent preparation for challenges faced by a military leader who must balance resource and human requirements. The program seeks to (1) enhance your curiosity about natural processes and your ability to study such processes as a scientist and (2) deepen your knowledge of human influences on the environment and foster evaluation of our individual and collective responsibilities as environmental stewards.

The Environmental Science Student Learning Outcomes are designed to:

**Student Learning Outcome 1**

Enhance curiosity about natural processes and one's ability to study such processes as a scientist.

**Student Learning Outcome 2**

Deepen knowledge of human influences on the environment and foster evaluation of our individual and collective responsibilities as environmental stewards.

**Student Learning Outcome 3**

Develop one's ability to evaluate the connections between the environment and individual, national, and global security.

**Student Learning Outcome 4**

Improve one's facility with the tools of environmental science by developing proficiency in collecting and analyzing lab and field data, deducing patterns, and formulating the next step in an on-going study.

**Student Learning Outcome 5**

Provide a solid foundation in earth, air, water and life sciences and their interconnections.

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**Environmental Science (ESC3 )**

**[DA CD: CCQ - ENVIRONMENTAL SCIENCE / CIP CD: 03.0104 - Environmental Science]**

Environmental science is a broad, integrative, science-based discipline which focuses on the interrelationships between people and the environment. Environmental scientists conduct investigations to analyze these interrelationships and to identify, abate, or eliminate human-caused pressures on the environment. The ultimate goal of these investigations is to create a sustainable balance between humans and the natural world that minimizes environmental degradation. This major develops expertise into the processes that sustain our environment by expanding upon the West Point core science education by adding studies in the natural sciences such as biology, ecology, geology, and meteorology, and in the integrative studies of environmental decision making and environmental security. This broad academic background is excellent preparation for challenges faced by a military leader who must balance resource and human requirements. The program seeks to (1) enhance your curiosity about natural processes and your ability to study such processes as a scientist and (2) deepen your knowledge of human influences on the environment and foster evaluation of our individual and collective responsibilities as environmental stewards.

The Environmental Science Student Learning Outcomes are designed to:

**Student Learning Outcome 1**

Enhance curiosity about natural processes and one's ability to study such processes as a scientist.

**Student Learning Outcome 2**

Deepen knowledge of human influences on the environment and foster evaluation of our individual and collective responsibilities as environmental stewards.

**Student Learning Outcome 3**

Develop one's ability to evaluate the connections between the environment and individual, national, and global security.

**Student Learning Outcome 4**

Improve one's facility with the tools of environmental science by developing proficiency in collecting and analyzing lab and field data, deducing patterns, and formulating the next step in an on-going study.

**Student Learning Outcome 5**

Provide a solid foundation in earth, air, water and life sciences and their interconnections.

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**Foreign Area Studies: Africa (FSI2 )**

**[DA CD: EBX - AREA STUDIES / CIP CD: 05.0101 - African Studies]**

This description and outcomes apply to all Foreign Area Studies Majors, not just Foreign Area Studies: Africa.

A Foreign Area Studies Major is offered to Cadets interested in pursuing an interdisciplinary study of Africa, East Asia, Eurasia, Europe, Latin America, or the Middle East. Cadets choosing one of these area programs will study the



peoples, societies, languages, cultures, geography, history, foreign relations, politics, and economics of a particular region. Cadets will have the opportunity to study in depth the factors that frequently determine national objectives and influence the formulation of governmental policy.

### **Student Learning Outcome 1**

Foreign Area Studies Majors will be proficient in their respective languages.

Cadets will be able to

- (1) Demonstrate foreign language proficiency in listening, speaking, reading, and writing.
- (2) Combine linguistic skills and critical thinking to perform research and conduct analysis.
- (3) Deliver research findings to diverse audiences.

### **Student Learning Outcome 2**

Foreign Area Studies Majors will develop the cultural competence needed to interact effectively with members of the target community.

Cadets will be able to

- (1) Describe and explain crucial figures and events in the history of the target region.
- (2) Discuss significant recent cultural developments.
- (3) Analyze contemporary issues, guided by knowledge of the region's history.
- (4) Account for limitations in existing conceptual frameworks as applied to cultural studies.

### **Student Learning Outcome 3**

Foreign Area Studies Majors will be knowledgeable of the regional dynamics of LX-speaking countries.

Cadets will be able to

- (1) Describe the target region's armed forces.
- (2) Analyze the impact of the physical and human geography on contemporary society.
- (3) Apply concepts of cross-cultural inquiry to evaluate regional change.
- (4) Identify and explain sources of intercultural conflicts.
- (5) Summarize US policies related to the LX-speaking region.

### **Student Learning Outcome 4**

Foreign Area Studies Majors will be critical thinkers.

Cadets will be able to

- (1) Integrate acquired knowledge in area studies and foreign languages with other humanities and social sciences disciplines.
- (2) Transfer language proficiency and intercultural capabilities to the study of and engagement with other language communities.
- (3) Navigate successfully the challenges of living in another culture.

## **Foreign Language: Arabic (FLA2 )**

**[DA CD: AAZ - ARABIC / CIP CD: 16.1101 - Arabic Language and Literature]**

This description and outcomes apply to all foreign language majors and dual language majors, not just Foreign Language: Arabic

The ability to understand and speak a foreign language is a window into the thinking and values of another people. Cadets majoring in foreign languages develop operational proficiency in the foreign language and attain an informed understanding of the culture and regional dynamics of the country or countries where the language is used. These capabilities are invaluable assets to future officers in a globally committed US Army. Majors may focus on the study of a single language: Arabic, Chinese, French, German, Persian, Portuguese, Russian or Spanish; or they may pursue the Dual Foreign Language Major by combining studies of any two of the eight languages. Advanced level study in all languages includes courses in literature, civilization, military readings, and media.

### **Student Learning Outcome 1**

Foreign Language Majors will be proficient in their respective languages.

Cadets will be able to

- (1) Comprehend and analyze language-based cultural products.
- (2) Read critically and analyze texts in the target language.
- (3) Communicate effectively using culturally appropriate forms in the target language.

### **Student Learning Outcome 2**

Foreign Language Majors will develop the cultural competence needed to interact effectively with members of the target community.

Cadets will be able to

- (1) Describe and explain crucial figures and events in the history of the target region.
- (2) Discuss significant recent cultural developments.
- (3) Analyze contemporary issues, guided by knowledge of the region's history.

(4) Account for limitations in existing conceptual frameworks as applied to cultural studies.

### **Student Learning Outcome 3**

Foreign Language Majors will be knowledgeable of the regional dynamics of LX-speaking countries.

Cadets will be able to

- (1) Describe the target region's armed forces.
- (2) State attitudes and beliefs of the LX-speaking population about the culture/policies of the US.
- (3) Summarize US policies related to the LX-speaking region.
- (4) Identify and explain sources of intercultural conflicts.

### **Student Learning Outcome 4**

Foreign Language Majors will be critical thinkers.

Cadets will be able to

- (1) Combine linguistic skills and critical thinking to perform research, conduct analysis, and deliver findings to diverse audiences.
- (2) Integrate acquired knowledge in foreign language studies with other humanities and social sciences disciplines.
- (3) Transfer language proficiency and intercultural capabilities to the study of and engagement with other language communities.
- (4) Recognize the inherent challenges of translation and literary analysis.
- (5) Navigate successfully the challenges of living in another culture.

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## **Geography (GEO1 )**

**[DA CD: DFX - GEOGRAPHY,GENERAL / CIP CD: 45.0701 - Geography]**

Geography is the study of people, places, and the environment and is an ideal major for Cadets interested in the outdoors, global cultures, and the natural world. More specifically, Geographers examine spatial patterns, geographic processes, and natural and human landscapes. Geography is a broad, integrating discipline with methodologies and analytical foundations that span engineering, science, and the humanities. Majoring in Geography requires persistent curiosity and inquiry into human-land-environment interactions including an examination of how natural systems function, how physical landscapes evolve, how human populations adapt, and how humans shape the environment. Cadets explore cultural patterns, population trends, and political, economic, and social systems from a global and regional perspective. Geography majors also develop a greater depth of knowledge of the physical processes that shape the Earth, environmental changes and natural hazards, and anthropogenic pressures on natural resources and the environment. The major integrates the use of geographic skills such as computer cartography, remote sensing, and geographic information systems. Geography is the ideal discipline for an Army officer in a changing world.

Graduates with a major in Geography should be able to:

### **Student Learning Outcome 1**

Describe the development of geography and discuss the discipline's unique place within the social and natural sciences.

### **Student Learning Outcome 2**

Explain how geography connects both physical and human geographic principles in order to understand the interactions between people and the natural environment.

### **Student Learning Outcome 3**

Explain the fundamental concepts (place, space, scale, location, region, landscape) and theories that underlie modern thinking in geography.

### **Student Learning Outcome 4**

Explain the basic physical geography processes that affect human patterns & systems on the earth's surface.

### **Student Learning Outcome 5**

Demonstrate a basic competence in a technical skill of value to geographers, such as foreign language, research methodologies, or geospatial technologies.

### **Student Learning Outcome 6**

Use geospatial information sciences to inform understanding of geographic issues.

### **Student Learning Outcome 7**

Identify global cultural patterns and processes and summarize the regional geography of at least one world realm.

### **Student Learning Outcome 8**

Synthesize and apply knowledge of geography to better understand real world issues, including, but not limited to, topics of concern to the Army.

### **Student Learning Outcome 9**

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Conduct basic geographic research, analyze the findings, and professionally communicate the results orally and in writing.

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## **Geography: Human (GEH1 )**

**[DA CD: DFX - GEOGRAPHY,GENERAL / CIP CD: 45.0701 - Geography]**

Geography: Human (GEH0), Human-Environment (GEE0), Physical (GEP0)

This description and outcomes apply to all three tracks within the Geography major.

Geography is the study of people, places, and the environment and is an ideal major for Cadets interested in the outdoors, global cultures, and the natural world. More specifically, Geographers examine spatial patterns, geographic processes, and natural and human landscapes. Geography is a broad, integrating discipline with methodologies and analytical foundations that span engineering, science, and the humanities. Majoring in geography requires persistent curiosity and inquiry into human-land-environment interactions including an examination of how natural systems function, how physical landscapes evolve, how human populations adapt, and how humans shape the environment. Three tracks allow Cadets to explore geography through the major subfields of the discipline: Human Geography, Physical Geography, or Human-Environment Interaction. The Human Geography track provides a social science perspective that enables Cadets to explore cultural diversity, population trends, and political, economic, and social systems from a global and regional perspective. The Physical Geography track falls within the natural sciences and allows Cadets to develop a greater depth of knowledge on the physical processes that shape the Earth. The Human-Environment Geography track emphasizes the interaction between humans and their environment by delineating the regional challenges inherent with environmental change and natural hazards, identifying anthropogenic pressures on natural resources and the role these resources play in economic wellbeing, and understanding the environments role in regional instability. The major integrates the use of geographic skills such as computer cartography, remote sensing, and geographic information systems. Geography is the ideal discipline for an Army officer in a changing world.

Graduates with a major in Geography should be able to:

### **Student Learning Outcome 1**

Describe the development of geography and discuss the discipline's unique place within the social and natural sciences.

### **Student Learning Outcome 2**

Explain how geography connects both physical and human geographic principles in order to understand the interactions between people and the natural environment.

### **Student Learning Outcome 3**

Explain the fundamental concepts (place, space, scale, location, region, landscape) and theories that underlie modern thinking in geography.

### **Student Learning Outcome 4**

Explain the basic physical geography processes that affect human patterns & systems on the earth's surface.

### **Student Learning Outcome 5**

Demonstrate a basic competence in a technical skill of value to geographers, such as foreign language, research methodologies, or geospatial technologies.

### **Student Learning Outcome 6**

Use geospatial information sciences to inform understanding of geographic issues.

### **Student Learning Outcome 7**

Identify global cultural patterns and processes and summarize the regional geography of at least one world realm.

### **Student Learning Outcome 8**

Synthesize and apply knowledge of geography to better understand real world issues, including, but not limited to, topics of concern to the Army.

### **Student Learning Outcome 9**

Conduct basic geographic research, analyze the findings, and professionally communicate the results orally and in writing.

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## **Geospatial Information Science (GIS2 )**

**[DA CD: BDF - GEOSPATIAL INFORMATION SCIENCE (GIS) / CIP CD: 45.0702 - Geographic Information]**

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## Science and Cartography]

Fundamental to understanding our environment and the geography of the earth is our ability to locate, measure, and quantify geographic phenomena. The discipline of Geospatial Information Science is concerned with the measurement of the earth and all that is on it, natural and man-made. Cadets develop expertise in subjects ranging from traditional methods of land surveying to satellite imaging and positioning systems. The Geospatial Information Science curriculum builds on a firm math science, and geography foundation with specialized courses in land surveying, cartography, photogrammetry, remote sensing, and geographic information systems. Both the civil and military sectors of our society are placing an ever-increasing reliance on the ability to build and query GIS to support a myriad of social/economic and engineering issues. The cadet at the USMA has a rare opportunity to pursue an integrated field of study that is commonly spread over several separate disciplines at other institutions. This major has applicability for the future military officer regardless of branch. Cadets majoring in GIS receive a 3Y (Space Activities) Skill Identifier on their official military record. Additionally, cadets who branch Engineers will qualify for the Geospatial Engineer Officer Identifier. Cadets majoring in GIS also qualify for the United States Geospatial Intelligence Foundations Geospatial Intelligence Certificate.

### GIS Major Student Learning Outcomes and Supporting Objectives

#### Student Learning Outcome 1

Graduates will understand the fundamentals of geospatial data.

- a. Graduates will have an understanding of datums.
- b. Graduates will have an understanding of map projections.
- c. Graduates will have an understanding of map coordinate systems.

#### Student Learning Outcome 2

Graduates will be able to acquire geospatial data.

- a. Graduates will be able to plan a geospatial data collection or acquisition.
- b. Graduates will be familiar with the sources of geospatial data.
- c. Graduates will be able to collect geospatial data using traditional surveying techniques.
- d. Graduates will be able to collect geospatial data using GPS technologies.
- e. Graduates will be able to convert geospatial data to effectively meet project requirements.
- f. Graduates will be able to assess the quality of geospatial data.

#### Student Learning Outcome 3

Graduates will be able to effectively manage geospatial data.

- a. Graduates will be able to design a geospatial database that meets user requirements.
- b. Graduates will be able to build a geospatial database based upon design specifications.
- c. Graduates will be able to critically assess the utility of a geospatial database.
- d. Graduates will be able to modify the design and content of a geospatial database as necessary.

#### Student Learning Outcome 4

Graduates will be able to conduct geospatial analysis.

- a. Graduates will be able to generate digital elevation models.
- b. Graduates will be able to extract thematic data layers from remotely sensed imagery.
- c. Graduates will be able to conduct geospatial queries.
- d. Graduates will be able to take measurements of geospatial features.
- e. Graduates will be able to conduct basic geospatial analytical operations and methods.
- f. Graduates will be able to analyze geospatial surfaces.
- g. Graduates will be able to conduct basic network analysis.

#### Student Learning Outcome 5

Graduates will be able to effectively communicate with geospatial information.

- a. Graduates will understand and be able to use the principles of cartographic design.
- b. Graduates will be able to effectively employ geo-visualization techniques.
- c. Graduates will be able to communicate geospatial information in a non-graphic manner such as briefings and reports.

#### Student Learning Outcome 6

Graduates will understand the profession and responsibilities of the discipline.

- a. Graduates will have an understanding of legal, ethical, and security aspects of geospatial data and information.
- b. Graduates will be familiar with geospatial organizations with an emphasis on the Department of Defense and the U. S. Army.

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## Geospatial Science (GSC0 )

[DA CD: BDF - GEOSPATIAL INFORMATION SCIENCE (GIS) / CIP CD: 45.0702 - Geographic Information Science and Cartography]

Fundamental to understanding our environment and the geography of the earth is our ability to locate, measure, and quantify geographic phenomena. The discipline of Geospatial Science is concerned with the measurement of the earth and all that is on it, natural and man-made. Cadets develop expertise in subjects ranging from traditional methods of land surveying to satellite imaging and positioning systems. The Geospatial Science curriculum builds on a firm math science, and geography foundation with specialized courses in land surveying, cartography, photogrammetry, remote sensing, and geographic information systems. Both the civil and military sectors of our society are placing an ever-increasing reliance on the ability to build and query GS to support a myriad of social/economic and engineering issues. The cadet at the USMA has a rare opportunity to pursue an integrated field of study that is commonly spread over several separate disciplines at other institutions. This major has applicability for the future military officer regardless of branch. Cadets majoring in GS receive a 3Y (Space Activities) Skill Identifier on their official military record. Additionally, cadets who branch Engineers will qualify for the Geospatial Engineer Officer Identifier. Cadets majoring in GS also qualify for the United States Geospatial Intelligence Foundations Geospatial Intelligence Certificate.

## GS Major Student Learning Outcomes and Supporting Objectives

### Student Learning Outcome 1

Graduates will understand the fundamentals of geospatial data.

- a. Graduates will have an understanding of datums.
- b. Graduates will have an understanding of map projections.
- c. Graduates will have an understanding of map coordinate systems.

### Student Learning Outcome 2

Graduates will be able to acquire geospatial data.

- a. Graduates will be able to plan a geospatial data collection or acquisition.
- b. Graduates will be familiar with the sources of geospatial data.
- c. Graduates will be able to collect geospatial data using traditional surveying techniques.
- d. Graduates will be able to collect geospatial data using GPS technologies.
- e. Graduates will be able to convert geospatial data to effectively meet project requirements.
- f. Graduates will be able to assess the quality of geospatial data.

### Student Learning Outcome 3

Graduates will be able to effectively manage geospatial data.

- a. Graduates will be able to design a geospatial database that meets user requirements.
- b. Graduates will be able to build a geospatial database based upon design specifications.
- c. Graduates will be able to critically assess the utility of a geospatial database.
- d. Graduates will be able to modify the design and content of a geospatial database as necessary.

### Student Learning Outcome 4

Graduates will be able to conduct geospatial analysis.

- a. Graduates will be able to generate digital elevation models.
- b. Graduates will be able to extract thematic data layers from remotely sensed imagery.
- c. Graduates will be able to conduct geospatial queries.
- d. Graduates will be able to take measurements of geospatial features.
- e. Graduates will be able to conduct basic geospatial analytical operations and methods.
- f. Graduates will be able to analyze geospatial surfaces.
- g. Graduates will be able to conduct basic network analysis.

### Student Learning Outcome 5

Graduates will be able to effectively communicate with geospatial information.

- a. Graduates will understand and be able to use the principles of cartographic design.
- b. Graduates will be able to effectively employ geo-visualization techniques.
- c. Graduates will be able to communicate geospatial information in a non-graphic manner such as briefings and reports.

### Student Learning Outcome 6

Graduates will understand the profession and responsibilities of the discipline.

- a. Graduates will have an understanding of legal, ethical, and security aspects of geospatial data and information.
- b. Graduates will be familiar with geospatial organizations with an emphasis on the Department of Defense and the U. S. Army.

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## History Major: International (HNT3 )

[DA CD: EGX - HISTORY GENERAL / CIP CD: 54.0199 - History, Other]

This description and the outcomes apply to all history majors, not just History: International.

As Army officers, West Point graduates will perform a broad spectrum of missions vitally important to our nation's

security and interests. They must be intellectually and professionally prepared to face these challenges in an uncertain and dangerous world inhabited by peoples of different languages, religions, and cultures. The Department of History contributes to cadets' intellectual and professional development by imparting historical knowledge, an appreciation of history, and critical thinking and communication skills.

Broad historical knowledge is central to developing informed citizens and soldiers. It helps cadets place their service as future Army officers in the context of U.S., Western, and world history. Additionally, it provides the cultural and historical literacy necessary for officers to serve effectively wherever in the world they may find themselves. This is particularly true in the case of counterinsurgency warfare, where victory depends on achieving legitimacy in the eyes of the indigenous population.

Officers with an appreciation of history recognize that every situation is historically unique. They understand that history is of value not in divining answers about the future but in asking the right questions. History is the means of putting human activities and ideas in context, avoiding false analogies, lending a sense of scope and scale, assessing moral implications, anticipating unintended consequences, and judging the feasibility and suitability of possible courses of action. Consulting history on these issues helps officers arrive at thoughtful, appropriate, and humane solutions to the problems they will face in their careers.

Officers who are critical thinkers challenge accepted wisdom in the search for truth and justice. They are open-minded and able to make independent and informed decisions. They reject simplistic answers that suggest the existence of a black-and-white world; rather, they accept the ambiguity associated with most human endeavors and seek the best solution rather than a single "correct" one. The study of history encourages critical thinking by requiring cadets to:

- formulate critical questions;
- conduct research by gathering and prioritizing information; analyze information within the broad context in which it appears; interpret and synthesize information;
- derive reasoned, evidence-based conclusions;
- assess and adjust their conclusions as conditions change or new information becomes available.

Finally, officers must be able to communicate effectively, both orally and in writing, to influence others. It is of no use to know and appreciate history and to be able to think critically if the officer is incapable of communicating his or her thoughts. The Department of History develops cadets' communication skills through frequent practice in and out of the classroom. Our principal evaluative concern is the content of the message, but we also devote great energy to enabling cadets to communicate with grammatical correctness, stylistic grace, and acceptable format. Cadets may pursue a major in one of three fields: American History, International History, or Military History. Each offers flexibility, permitting cadets to develop a foundation of historical perspective as well as pursue specialized studies in regional areas, languages, or other disciplines.

### **Student Learning Outcome 1**

Communicate effectively in speaking and writing.

1. Demonstrate ability to read and understand information and arguments.
2. Demonstrate ability to understand and respond in verbal debate and discussion.
3. Convey ideas clearly, using conventions of format, structure, voice, tone, and level of formality appropriate to the rhetorical situation and audience in an organized way both orally and in writing.

### **Student Learning Outcome 2**

Understand how individuals, organizations, cultures, and societies behaved and met challenges in the past.

1. Identify and analyze the development of different cultures and civilizations and understand the historical foundations of their ethical development.
2. Analyze the interactions of political, social, economic, ethical, military, and cultural components in the development of civilizations.
3. Analyze and evaluate the role of historical insight in informing the decisions of political and military leaders.

### **Student Learning Outcome 3**

Understand the complexity and ambiguity of change over time. Evaluate complex evidence critically and establish appropriate links between cause and effect.

1. Use historical evidence to assess given situations both preceding and following historical change and describe the limits of that change, identify and analyze ethical components of historical events, identify conflicting points of view, analyze conflicting or ambiguous historical perspectives and evidence, establish causal relationships between facts, and identify points of tension.
2. Apply historical knowledge, critical thinking and research skills to inform their own professional decision-making.

### **Student Learning Outcome 4**

Demonstrate critical thinking and research skills:

1. Engage in independent research.



2. Identify the essential aspects of historical situations and ask relevant questions.
3. Reflect on and evaluate evidence and sources, question assumptions, synthesize and draw informed conclusions on the basis of critical analysis.
4. Employ historical thinking and analysis to explain historical patterns, developments and events.
5. Analyze various approaches to historical research and different historical frameworks.

## **International Affairs: FP & Security Studies (PIF0 )**

**[DA CD: EKB - INTERNATIONAL RELATIONS / CIP CD: 45.0901 - International Relations and Affairs]**

This description and outcomes apply to both the International Affairs: Foreign Policy & Security Studies and International Affairs: Institutions, Governance, and Development major tracks.

Students majoring in International Affairs (IA) study issues of conflict and cooperation in the international system. This involves examination of domestic influences on state behavior, the foreign relations of states, characteristics of the international system, characteristics of states, and the role of non-state actors. Central concerns in IA include power, strategy, war, international cooperation, governance, political institutions and structure, human security, civil society, trade, economic development, ideology, and culture. International Affairs majors think critically about complex political issues, they will learn how to test hypotheses using reliable methods and evidence, evaluate the moral dimension of issues in international relations and comparative politics, appreciate the problem of uncertainty and its significance, and see important events and issues from multiple perspectives. As a social science, IA is organized around alternative theoretical schools of thought that seek to explain international phenomena. The ability to understand, anticipate and shape the complex dynamics of the international environment (between and within states) is of direct relevance to U.S. Army officers, whose careers will be defined by the problems and opportunities that merge within the international system.

### **International Affairs Student Learning Outcomes**

#### **Student Learning Outcome 1**

SLO#1: Think Critically - Graduates can think critically about international affairs and regime types by applying and appreciating the value of intellectual pluralism.

#### **Student Learning Outcome 2**

SLO #2: Read Critically - Graduates can apply the social scientific method while consuming and analyzing international political, economic, and historical literature, including academic and popular writing as well as public discourse.

#### **Student Learning Outcome 3**

SLO #3: Conduct Research - Graduates can identify the social science methodologies appropriate for a given research question and will have the ability to conduct empirical research using these methodologies.

#### **Student Learning Outcome 4**

SLO #4: Gain Disciplinary Knowledge - Graduates can describe the major issues and variety of actors in international affairs and appreciate their incentives as well as the roles that they play in shaping international, regional, and country-specific events and developments.

#### **Student Learning Outcome 5**

SLO #5: Assess and Develop Policy - Graduates can integrate and apply knowledge to describe, explain, analyze, or predict international affairs and domestic political developments as well as identify implications and develop policy options.

#### **Student Learning Outcome 6**

SLO #6: Communicate Effectively - Graduates communicate with clear and effective analysis and arguments about complex international and global issues verbally and in writing.

## **Kinesiology (KIN1 )**

**[DA CD: FCB - OCCUPATIONAL THERAPY, KINESIOLOGY / CIP CD: 31.0505 - Kinesiology and Exercise Science]**

Kinesiology is the scientific study of human movement and is generally divided into the physiological, psychological, and mechanical aspects of movement. The physiological aspects of kinesiology encompass the study of the short-term responses and long-term adaptations of organisms and systems to the challenge of exercise, physical activity, or movement. From a neuromuscular perspective, kinesiology relates to how humans learn and control movements. The psychological aspects of kinesiology relate to the effects of human behavior on physical activity levels and performance. The mechanical aspects of kinesiology encompass the biodynamics of human movements as they relate

to exercise and human performance.

Kinesiology is a broad, interdisciplinary field of study that includes diverse specialties such as exercise physiology, biomechanics, nutrition, exercise psychology, and motor control. Areas of inquiry range in scope from the study of the molecular response of cells to the response and adaptation of the whole body. The kinesiology major encompasses a wide spectrum of performance-related issues involving muscular and cardiovascular physiology, energy balance, exercise adherence, skill acquisition, and fitness testing and prescription.

Upon completion, students will be able to:

**Student Learning Outcome 1**

Identify the anatomical structures that help determine physical competency and human movement

**Student Learning Outcome 2**

Describe the principles of human physical development and adaptation

**Student Learning Outcome 3**

Describe the musculoskeletal principles of work capacity

**Student Learning Outcome 4**

Identify the principles of the mechanics of human movement

**Student Learning Outcome 5**

Describe the principles of exercise psychology and their application to fitness

**Student Learning Outcome 6**

Apply the principles of fitness assessment and exercise prescription

**Student Learning Outcome 7**

Describe the nutritional concepts supporting the energy demands of physical training

## **Law and Legal Studies (LLS3 )**

**[DA CD: PXX - LAW,GENERAL / CIP CD: 22.0000 - Legal Studies, General]**

Law is the study of the means of maintaining social order, balancing individual interests against the interest of society, resolving disputes, and addressing social concerns. The study of law sharpens analytical and problem-solving abilities while developing an appreciation of law as a basic foundation of society. A Law and Legal Studies major will equip cadets with the means to understand conflicting issues, to analyze problems, and then to choose the most appropriate solution. The legal system major is not intended to train lawyers. Rather, it will prepare cadets for success in command or in any other position in which effective analytical, problem solving, and communications skills are essential. From this understanding cadets can expand their breadth of experience and gain insight into current social problems or future challenges. A Legal System major will enhance the ability to think critically, conduct research, and persuasively express oneself orally and in writing. Law provides an excellent preparation for subsequent graduate study in public policy and administration, politics, government, business management, and international relations.

### **Law and Legal Studies Major Student Learning Outcomes and Supporting Objectives**

**Student Learning Outcome 1**

Cadets understand how individuals and organizations use and react to law in the pursuit of social, political, and economic goals

a. Cadets will learn how to read and understand primary sources of American, foreign, and international law, including constitutions, codes, statutes, cases, regulations, and treaties.

b. Cadets will learn to analyze critically legal commentaries, including news reports, editorials, commentaries, documents, and law review articles.

**Student Learning Outcome 2**

Cadets communicate in correct and appropriate legal language when writing and speaking to evince clear and critical thinking.

a. Cadets will develop the oral and written skills to communicate concisely, persuasively, and logically to support or oppose a given decision or course of action.

b. Cadets will learn to apply legal texts and principles when analyzing specific problems and cases in order to reach legally and logically supportable conclusions and decisions.

**Student Learning Outcome 3**



Cadets identify the relationship between legal, ethical, and moral issues, and apply the law properly in decision-making. Cadets will be able to distinguish between the ethical, moral, policy, and legal components of social, political, and economic issues.

#### **Student Learning Outcome 4**

Cadets know how law affects and reflects diverse groups within American society to shape behavior, achievement, and ideas. Cadets will acquire a basic understanding of the substantive and procedural rules of law that underpin the American Legal System.

#### **Student Learning Outcome 5**

Cadets know how law is understood and applied in diverse global societies and cultures. Cadets will understand and internalize the values inherent in the principle of the rule of law as it is applied, or not applied, in a variety of cultures and circumstances.

#### **Student Learning Outcome 6**

Cadets think and act logically and creatively when acting within the framework and constraints of legal requirements. Students will acquire the legal knowledge and tools to function effectively as a military professional at all levels.

### **Mathematical Sciences (MSC1)**

**[DA CD: DHX - MATHEMATICS, GENERAL / CIP CD: 27.0101 - Mathematics, General]**

The mathematical sciences embody those areas of mathematics which have strong interdependence with other disciplines. Their purpose is to clarify concepts and describe scientific phenomena through symbolic language and the rules for its use. Its scope spans the total breadth of knowledge that is capable of being quantified. The full process of the mathematical sciences has expanded from their historical ties with the physical sciences to now include areas such as: the biological, sociological, behavioral, and computer sciences; operations research; and all engineering fields. The Department of Mathematical Sciences offers abundant opportunities for study in a broad range of mathematical subjects. Courses such as differential equations, algebra, mathematical modeling, analysis, numerical computation, statistics, and linear optimization provide a sound mathematical foundation in the science and engineering fields. In addition, follow-on courses such as algebra, analysis, combinatorics, and advanced individual study provide both depth in understanding the foundations of mathematical theory, as well as opportunity for study and research in a selected subject. Whenever possible, the use of technology is emphasized to extend the knowledge required for the consideration of realistic and challenging problems found in today's world. Cadets who major in Mathematical Sciences are required to take a seven-course core mathematics sequence, a statistics elective, an integrative experience course designed to explore real world problems pulled from many disciplines, and two additional mathematics electives for a total of eleven courses in the major. Mathematical Sciences majors are also required to take a three-course engineering sequence of their choice. Cadets will take MA205 - Calculus II as their STEM Depth choice in the core curriculum unless it is validated. Further, cadets will take a three-course Complementary Support sequence that provides breadth to their mathematical understanding (one of the three courses must be IT305 - Theory and Practice of Military IT Systems). Thus, Mathematical Sciences majors will take 41 courses while successfully completing their major and core curriculum requirements. The Mathematical Sciences with Honors major requires two semesters of directed research under senior faculty advisement that replaces the one semester research requirement, and culminates in a written thesis and presentation during the second semester. In order to receive the Mathematical Science with Honors Major, cadets will need to attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major curriculum.

Graduates who complete a Mathematical Sciences major will be able to:

#### **Student Learning Outcome 1**

Demonstrate competence in modeling physical, informational, and social phenomena by

- a. Identifying and articulating assumptions, metrics and constraints
- b. Applying appropriate solutions techniques
- c. Interpreting results within the appropriate context

#### **Student Learning Outcome 2**

Argue and inquire soundly and rigorously; become independent questioners and learners

#### **Student Learning Outcome 3**

Achieve mathematical proficiency in breadth and depth

- a. Understand and apply theorems and algorithms
- b. Understand and apply analytical methods
- c. Understand and apply numerical methods
- d. Understand and apply graphical methods
- e. Understand discrete and continuous structures and processes

#### **Student Learning Outcome 4**

Communicate mathematics, both orally and in writing

**Student Learning Outcome 5**

Use technology to model, visualize, and solve complex problems

**Student Learning Outcome 6**

Develop attitudes - habits of mind

- a. Creative and curious
- b. Experimental disposition
- c. Critical thinking and reasoning
- d. Commitment to life-long learning

**Student Learning Outcome 7**

Understand the role of mathematical sciences (in our world) by analyzing applied problems through disciplinary, multidisciplinary, and interdisciplinary approaches

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**Mechanical Engineering (MEN3 )**

**[DA CD: CKX - MECHANICAL ENGINEERING / CIP CD: 14.1901 - Mechanical Engineering]**

Mechanical Engineering is one of the broadest and most diverse of the engineering fields. It deals with devices and systems for energy conversion, for material transport and for control of motion and forces. A sampling of the topics addressed by the discipline include air, ground, and sea vehicles; power plants; control systems; machinery; machine tools; conventional and nuclear-powered power production facilities; biomedical devices; space vehicles; pollution control; new energy sources; energy conversion; transportation systems; and, military weapons systems. These modern weapons systems are used as vehicles of instruction in many of the courses, making mechanical engineering particularly appropriate for those considering service in most branches of the Army as well as specialties such as engineers, aviation, research and development, project management and logistics. The Department of Civil and Mechanical Engineering offers a major in Mechanical Engineering that is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>. All cadets experience the same core mechanical engineering program and choose three electives for depth of study. The goal of the Mechanical Engineering program is to provide high quality instruction in a positive learning environment leading to a degree recognized as being among the best in the nation. The Mechanical Engineering program stresses engineering fundamentals so that graduates are well equipped to understand complex technical problems in a rapidly changing, technology-intensive Army. Once completed, the graduate is well-prepared to excel as an officer and an engineer. The practice-oriented degree is strengthened by the complete integration of design and laboratory experience throughout the curriculum.

To meet this goal, the Program Educational Objectives of the Mechanical Engineering program are:

Within a few years after graduation, mechanical engineering majors are expected to attain:

1. multiple positions of responsibility in which they:
  - a. drive outcomes
  - b. inform and influence others
  - c. build teams
2. personal and professional growth through formal and informal learning opportunities while being a role model to inspire others
3. experience in providing engineering expertise to the Army and Nation to design solutions and innovate for winning in a complex world

To achieve these Program Educational Objectives, cadets who qualify for graduation with a mechanical engineering major from USMA will demonstrate the following Mechanical Engineering Student Outcomes:

**Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

**Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

**Student Learning Outcome 3**

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an ability to communicate effectively with a range of audiences.

**Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

**Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

**Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

**Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

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**Nuclear Engineering (NEN1 )**

**[DA CD: CLF - ENGINEERING,NUCLEAR / CIP CD: 14.2301 - Nuclear Engineering]**

Nuclear engineering makes practical use of the energy that is released by the atomic nucleus. Applications extend into the fields of electric power, medicine, nuclear weapons, and nuclear weapons effects. At USMA the vehicle for learning the concepts of the field is the commercial nuclear power plant. The approach is interdisciplinary; it draws widely upon mathematics, physics, and mechanics, with special emphasis on applied physics and the thermal-hydraulic aspects of mechanical engineering. The management of engineering is also addressed through decision analysis and economic analysis. The Nuclear Engineering major is designed to provide depth of knowledge in the application of nuclear energy to include power production, radiation health physics, nuclear weapons, and weapons effects. The major is taught through multiple departments and includes interdisciplinary electives from physics, mathematics, mechanical engineering, civil engineering, electrical engineering, and nuclear engineering. The Nuclear Engineering student will gain a broad background for further study in graduate school and Army assignments requiring expertise in mechanical engineering, applied radiation physics, nuclear weapons and weapons effects, or any of a variety of related topics. The Nuclear Engineering program is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>.

The Nuclear Engineering Program Educational Objectives are broad statements that describe what graduates are expected to attain within a few years after graduation. Program educational objectives are based on the needs of the Nuclear Engineering Program's constituencies.

- As Army leaders, graduates solve complex, multi-disciplinary problems for the Army and the Nation.
- Graduates demonstrate the necessary leadership and teamwork skills to work in multi-disciplinary team environments.
- Graduates are prepared to provide appropriate nuclear and radiological engineering expertise to the Army.
- Graduates communicate effectively, orally and in writing.
- Graduates continue to grow intellectually and professionally -- as Army officers and as engineers.

To achieve these Program Educational Objectives, cadets must attain or possess at graduation:

**Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

**Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

**Student Learning Outcome 3**

an ability to communicate effectively with a range of audiences.

**Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

**Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment,

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establish goals, plan tasks, and meet objectives.

#### **Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

#### **Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

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### **Operations Research (ORE2 )**

**[DA CD: CUC - OPERATIONS RESEARCH ANALYST,ENGINEERING / CIP CD: 14.3701 - Operations Research]**

Operations Research (OR) is a scientific approach to decision making with a focus on how best to design and operate systems, usually under conditions requiring the allocation of scarce resources. As a label for a body of methods or as an approach to problem solving, OR is inextricably linked to the direction and management of large systems for people, machines, materials, and money in government, industry, business, and defense. Since its formal inception during WWII, the interdisciplinary field of OR has set itself apart as an applied mathematical science and engineering discipline with a diverse range of applications. Cadets who major in Operations Research are required to take an 11-course core sequence, a two-course research sequence, an additional discipline elective, and a three-course Complementary Support Course sequence that provides breadth to their analytical understanding, for a total of 17 courses in the major. The embedded engineering courses in the major meet the goals of the core engineering sequence. For their STEM depth choice in the core curriculum, cadets majoring in OR will take MA205, Calculus II. Further, CY305: Theory and Practice of Military IT Systems, is required for Operations Research majors as a Complementary Support Course (CSC). In summary, Operations Research majors will take 41 courses while successfully completing their major and core curriculum requirements. In order to receive the Operations Research with Honors major, cadets will need to attain an APSC of at least 3.0 in the core curriculum, an APSC of at least 3.5 in the major curriculum, and take an additional elective course.

Graduates who complete an Operations Research major will be able to:

#### **Student Learning Outcome 1**

1. Demonstrate competence in modeling physical, informational, and social phenomena by:
  - a. Identifying and articulating assumptions, metrics, and constraints
  - b. Applying appropriate solution techniques
  - c. Interpreting results within the appropriate context

#### **Student Learning Outcome 2**

2. Argue and inquire soundly and rigorously; become independent questioners and learners

#### **Student Learning Outcome 3**

3. Achieve proficiency in Operations Research - in breadth and depth:
  - a. Understand and apply probabilistic and statistical models and methods
  - b. Understand and apply simulation methods
  - c. Understand and apply optimization methods

#### **Student Learning Outcome 4**

4. Communicate effectively - orally and in writing

#### **Student Learning Outcome 5**

5. Use technology to model, visualize, and solve complex problems

#### **Student Learning Outcome 6**

6. Develop attitudes - habits of mind
  - a. Creative and curious
  - b. Experimental disposition
  - c. Critical thinking and reasoning
  - d. Commitment to life-long learning

#### **Student Learning Outcome 7**

7. Understand the role of operations research in interdisciplinary problem solving
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### **Philosophy Major (PYL2 )**

**[DA CD: ALX - PHILOSOPHY / CIP CD: 38.0101 - Philosophy]**

The scope of Philosophy is much wider than the areas of Ethics and Just War theory studied in PY201. Besides Ethics, Philosophy investigates the very nature of human knowledge; the relationship between mind and matter; and fundamental principles of the kind that underlie religion, science, and politics. The study of Philosophy also has a significant cultural dimension. It challenges us to understand attitudes, beliefs and arguments from societies more or less remote from our own: the Greek and Roman world; the classical age of Asian thought; 17th-century European thought in its adjustment to modern science; and American contributions to our understanding of philosophical method. Doing Philosophy is a superior example of Critical Thinking at work, thanks to the high value it places in all its courses on systematic investigation, conceptual analysis, and cogent reasoning. West Point's Philosophy major is designed to ensure breadth and to encourage depth. It also provides opportunities to pursue further study in the ethics of war. The intellectual skills and experience that DEP's Philosophy major fosters are beneficial to the Army. The future officer in whom these skills take root will be an asset to any branch.

Cadets who major in Philosophy will be able to: analyze and construct philosophical arguments; show competency in logic and other core areas of philosophy; be familiar with major historical areas, figures, and problems of philosophy; understand philosophical ideas within their historical and cultural contexts; think and write in a rigorous and precise manner; use their critical thinking skills to provide original or innovative approaches to philosophical problems; apply philosophical methods in a variety of contexts at the personal and social levels; apply philosophical methods across disciplinary boundaries to a broad range of issues in the modern world.

### **Student Learning Outcome 1**

Philosophy majors will demonstrate the ability to think and write in a logically rigorous and conceptually precise manner about complex philosophical issues.

### **Student Learning Outcome 2**

Philosophy majors will understand major historical figures and problems in philosophy, underrepresented voices, and philosophical ideas within their historical and cultural contexts.

### **Student Learning Outcome 3**

Philosophy majors will be able to apply philosophical methods within the discipline and across disciplinary boundaries to issues and problems that individuals, communities, and governments face in the modern world.

### **Student Learning Outcome 4**

Philosophy majors will demonstrate competency in logic and other core areas of philosophy.

### **Student Learning Outcome 5**

Philosophy majors will use their critical thinking skills to provide creative approaches to theoretical and practical problems.

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## **Physics (PHY1 )**

**[DA CD: DLX - PHYSICS,GENERAL / CIP CD: 40.0801 - Physics, General]**

Our modern lives have been overwhelmingly affected by the discoveries of physics in the twentieth century, for it is through physics that we have come to understand the fundamentals of nuclear energy, semiconductors, lasers, fiber optics, the interaction of radiation with matter, and even the workings of the universe. It is through this basic understanding that applied scientists and engineers have developed and assembled the myriad technical devices that are so much a part of modern life. The program in physics integrates all these phases of modern technology to develop a fundamental knowledge that can support a variety of technical interests and activities in future years. The major is designed to provide the Cadet a solid foundation in the essential pillars of theoretical physics-classical mechanics, electrodynamics, statistical physics, and quantum mechanics. Additionally, a strong experimental component emphasizes the skills necessary to design and build experimental apparatus and applies these skills to modern physics, lasers and optics. Opportunities are available to perform research at Army and national laboratories during the summer.

### **Physics Major Student Learning Outcomes and Supporting Objectives**

#### **Student Learning Outcome 1**

Cadets can apply the laws of physics to formulate mathematical models of physical systems, solve the resulting equations, and apply the solutions to hypothetical and real-world problems.

a. Cadets can use advanced mathematical methods to solve physics problems.

b. Cadets can identify situations in which relativistic effects are important and apply special relativity to solve mechanics problems.

- c. Cadets can apply Newtonian and Lagrangian mechanics to solve problems in classical physics.
- d. Cadets can solve problems involving electro- and magnetostatics.
- e. Cadets can solve problems involving electrodynamics in vacuum and in homogeneous linear media.
- f. Cadets can identify phenomena and solve problems involving thermodynamics and quantum statistical mechanics.
- g. Cadets can identify situations in which quantum mechanics is necessary and solve problems involving non-relativistic quantum mechanics.
- h. Cadets demonstrate proficiency across the range of skills and knowledge in theoretical physics and mathematics expected of entering graduate students in reputable physics programs.

#### **Student Learning Outcome 2**

Cadets can apply the laws of physics to formulate and test hypotheses in an experimental setting.

- a. Cadets can plan and perform experiments.
- b. Cadets can analyze experimental data.
- c. Cadets demonstrate proficiency across the range of skills and knowledge in experimental physics expected of entering graduate students in reputable physics programs.

#### **Student Learning Outcome 3**

Cadets can complete academic assignments and perform research using accepted ethical and scientific standards.

- a. Cadets provide sufficient citations and notes to clearly distinguish the Cadet's work from the work of others.
- b. Cadets avoid using or obscuring fallacious reasoning in the presentation of solutions to technical problems.
- c. Cadets are aware of acceptable scientific standards and the professional and/or personal consequences of not following them.

#### **Student Learning Outcome 4**

Cadets can communicate logical solutions to scientific and technical problems to superiors, peers, and subordinates.

- a. Cadets can prepare written submissions, posters, laboratory reports, and oral briefings using the style, format, organization, and procedures common to standard scientific presentations.
- b. Cadets can use sound mathematical reasoning, appropriate computational techniques, and statistical methods to explore, represent, and communicate solutions to problems.
- c. Cadets use precise and accepted scientific language in all technical communications.
- d. Cadets read and understand the content of general scientific publications such as *Scientific American* and *Physics Today*. They are able to follow citations to obtain background information on the material in the articles and summarize orally and in writing the strengths and weaknesses of the arguments presented in such publications to non-technical audiences.

#### **Student Learning Outcome 5**

Cadets can employ computing technology to explore analytic solutions, analyze data, create simulations, and present results.

- a. Exploration. Cadets can use computers, software, and programming skills to explore and gain physical insight into the behavior of analytic solutions to physics problems.
- b. Data Analysis. Cadets can use computers to acquire, record, process, and analyze data.
- c. Simulation. Cadets can translate physical models into computer simulations and compare the results to data.
- d. Visualization. Cadets can produce tables, plots, and figures suitable for publication in peer-reviewed journals.

### **Psychology: Applied General Psychology (PSA0 )**

**[DA CD: EPE - PSYCHOLOGY, APPLIED / CIP CD: 42.2813 - Applied Psychology]**

Psychology is one of four majors available through the Department of Behavioral Sciences and Leadership. Psychology is arguably one of the most applicable majors for future Army officers. Graduates gain specialized skills and knowledge related to understanding human behavior, making them uniquely prepared to understand and ultimately lead people. The study of psychology investigates human behavior, cognition, and emotion by analyzing the complex interactions between environmental, social, cultural, and biological influences. Opportunities are available during the summer to apply the lessons learned in the classroom in a variety of settings.

The Psychology major provides cadets a choice of two tracks:

The Applied General Psychology track is focused on applying knowledge of psychology to improve the development, performance, and well-being of individuals(soldiers).

The Organizational Psychology and Leadership track is focused on applying knowledge of psychology to select and develop leaders, and improve the effectiveness of groups, teams, and organizations.

#### **Psychology Major Student Learning Outcomes and Supporting Objectives**

##### **Student Learning Outcome 1**

1. Graduates demonstrate fundamental knowledge and comprehension of the major concepts, theoretical perspectives, historical trends, and empirical findings; and demonstrate the ability to apply their knowledge to address behavioral issues.
  - a. Describe key concepts, principles, and overarching themes in psychology.
  - b. Demonstrate a working knowledge of psychology's content domains.



c. In the Applied General Psychology Track, the emphasis is on applying knowledge of individual and social psychology to address issues relevant to the development, performance, and well-being of people (soldiers).

d. In the Organizational Psychology and Leadership Track, the emphasis is on applying knowledge of human behavior in organizations to: a) select and develop leaders/leadership, and b) enhance organizational effectiveness.

### **Student Learning Outcome 2**

2. Demonstrate scientific reasoning and problem solving skills, including effective research methods.

a. Demonstrate psychology information literacy; and use scientific reasoning to design and conduct basic psychological research, interpret data, and draw appropriate conclusions about psychological phenomena.

b. Engage in innovative and integrative thinking and problem solving.

c. Graduates incorporate sociocultural factors in scientific inquiry.

d. Applied General Psychology Track: the emphasis is on critical thinking in the application of psychological research; and on conducting empirical research studying issues pertinent to the development, performance, and well-being of people.

e. Organizational Psychology and Leadership Track: the emphasis is on critical thinking in the application of research and other writings pertinent to human behavior in organizations; and on conducting empirical research studying issues pertinent to leadership and organizational effectiveness.

### **Student Learning Outcome 3**

3. Graduates demonstrate ethically and socially responsible behaviors for professional and personal settings in a landscape that involves increasing diversity.

a. Graduates apply ethical standards to evaluate psychological science and practice.

b. Graduates build and enhance interpersonal relationships

c. Graduates embrace values consistent with the professional military ethic, which contribute to positive outcomes in work settings and in building a society responsive to multicultural and global concerns.

### **Student Learning Outcome 4**

4. Graduates demonstrate competent written, oral, and interpersonal communication skills.

a. Graduates demonstrate effective writing and presentation skills for different purposes (e.g., they can produce a research study; and present information to a professional audience or decision maker).

b. Graduates interact effectively with others for different purposes (e.g. they demonstrate relevant skills such as counseling, coaching, and/or negotiations).

### **Student Learning Outcome 5**

5. Graduates apply psychology-specific content and skills, effective self-reflection, project-management skills, teamwork skills, and career preparation.

a. Graduates apply psychological content and skills to career goals.

b. Graduates exhibit self-efficacy and self-regulation.

c. Graduates effectively manage individual and team projects and professional requirements.

d. Graduates are inspired to life-long learning in disciplines pertinent to psychology and leadership.

## **Space Science (SSC0 )**

**[DA CD: SSC - SPACE SCIENCE / CIP CD: 29.0305 - Space Systems Operations]**

Our modern lives have increasingly benefitted from the knowledge of the space environment, utilization of satellites and spacecraft, and application of space-based technologies in the 21st century. Space Science is the science of the space environment as well as the science behind the technology of spacecraft, satellites, and space exploration. Some of those technologies and systems that society has benefitted from include: space weather and terrestrial weather forecasting satellites; position, navigation, and timing (PNT) enabled devices such as GPS; satellite communications; missile defense systems; and remote sensing systems used to observe our planet, our heliosphere, and beyond our solar system. Through the understanding of the aforementioned technologies and systems, space scientists develop and assemble a myriad of technical devices that are so much a part of a modern space-enabled profession. The program in Space Science integrates all these facets of space-enabled technology to develop a fundamental knowledge that can support a variety of technical interests and activities in future years. The major is designed to provide the cadet a solid foundation in orbital mechanics, the space environment, attitude determination and control, satellites communications, space structures, astronomy, astrophysics, space physics, remote sensing, and rocket propulsion. Opportunities are available to perform research at DoD and other laboratories that conduct space and missile defense related research during the summer.

### **Space Science Major Student Learning Outcomes**

#### **Student Learning Outcome 1**

1. Cadets can apply mathematical and scientific knowledge to identify, formulate, and solve Space Science problems.

#### **Student Learning Outcome 2**

2. Cadets can conduct experiments, apply the scientific method, as well as analyze and interpret data.

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**Student Learning Outcome 3**

3. Cadets can function on multidisciplinary teams to effectively formulate, state, describe, summarize, and communicate contemporary space issues, National Strategic Space Policy and associated technology; analyze and examine the professional and ethical aspects of the aforementioned.

**Student Learning Outcome 4**

4. Cadets can describe and summarize the technology of a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability.

**Student Learning Outcome 5**

5. Cadets can recognize the need for, and an ability to engage in life-long learning; explain relevant, contemporary issues and concepts needed as a space-enabled professional.

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**Systems and Decision Sciences (SDS0 )**

**[DA CD: BBR - SYSTEMS MANAGEMENT / CIP CD: 30.0601 - Systems Science and Theory]**

The Systems and Decision Sciences (SDS) discipline is concerned with system design, management, and decision analysis of tangible and abstract systems in accordance with performance requirements, budget, and schedules. The program combines elements of traditional engineering, systems engineering, finance, decision analysis, and organizational management courses. Cadets will learn the methods, processes, and tools (MPTs) needed to understand and conduct meaningful decision analysis in support of complex systems. These large systems require an interdisciplinary approach rooted in technical, management, and leadership skills. Industry and domain based concentrations are available, to include Project Management, Logistics Management, Personnel Management, Defense Systems, Financial Systems, Cyber Security, Management Science, International Affairs, and Mathematical Modeling. Systems and Decision Sciences graduates are ready to lead multidisciplinary teams, perform systems thinking, and understand the MPTs needed to model complex, ill-defined, and interdisciplinary problems characterized by global, political, social, military, economic, and technological challenges. A culminating real-world, year-long capstone is required to employ the MPTs associated with complex systems and address issues, such as stakeholder analysis, analytical methods, project and cost management, and the political, social, and environmental, etc., realities of working on large, ill-defined, interdisciplinary problems. This major will produce graduates with technical, high-level business and management skills, and engineering depth to prepare them for future academic and professional opportunities in a society increasingly dominated by technological change.

Each graduate will be able to:

**Student Learning Outcome 1**

Use systems and decision science methods, processes, and tools (MPTs) to solve diverse problems in engineering and non-engineering domains, while considering moral and ethical standards specific to the locale.

**Student Learning Outcome 2**

Employ systems thinking to identify, scope, understand, analyze problems, and specify the needs of multiple stakeholders.

**Student Learning Outcome 3**

Develop innovative system solutions based upon sound decision science techniques and clearly articulate the results to all stakeholders.

**Student Learning Outcome 4**

Perform systems decision-making that considers qualitative and quantitative aspects of the problem.

**Student Learning Outcome 5**

Accurately, clearly, concisely, and persuasively report findings, conclusions, and recommendations to the stakeholders in a multicultural context.

**Student Learning Outcome 6**

Lead interdisciplinary teams to implement effective and efficient solutions by melding traditional and non-traditional engineering domains.

**Student Learning Outcome 7**

Demonstrate the skills and interest for intellectual growth and learning for a career of professional excellence and service to the nation as an officer in the United States Army.

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**Systems and Decision Sciences (SDS1 )**



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**[DA CD: BBR - SYSTEMS MANAGEMENT / CIP CD: 30.0601 - Systems Science and Theory]**

The Systems and Decision Sciences (SDS) discipline is concerned with system design, management, and decision analysis of tangible and abstract systems in accordance with performance requirements, budget, and schedules. The program combines elements of traditional engineering, systems engineering, finance, decision analysis, and organizational management courses. Cadets will learn the methods, processes, and tools (MPTs) needed to understand and conduct meaningful decision analysis in support of complex systems. These large systems require an interdisciplinary approach rooted in technical, management, and leadership skills. Industry and domain based concentrations are available, to include Project Management, Logistics Management, Personnel Management, Defense Systems, Financial Systems, Cyber Security, Management Science, International Affairs, and Mathematical Modeling. Systems and Decision Sciences graduates are ready to lead multidisciplinary teams, perform systems thinking, and understand the MPTs needed to model complex, ill-defined, and interdisciplinary problems characterized by global, political, social, military, economic, and technological challenges. A culminating real-world, year-long capstone is required to employ the MPTs associated with complex systems and address issues, such as stakeholder analysis, analytical methods, project and cost management, and the political, social, and environmental, etc., realities of working on large, ill-defined, interdisciplinary problems. This major will produce graduates with technical, high-level business and management skills, and engineering depth to prepare them for future academic and professional opportunities in a society increasingly dominated by technological change.

Each graduate will be able to:

**Student Learning Outcome 1**

Use systems and decision science methods, processes, and tools (MPTs) to solve diverse problems in engineering and non-engineering domains, while considering moral and ethical standards specific to the locale.

**Student Learning Outcome 2**

Employ systems thinking to identify, scope, understand, analyze problems, and specify the needs of multiple stakeholders.

**Student Learning Outcome 3**

Develop innovative system solutions based upon sound decision science techniques and clearly articulate the results to all stakeholders.

**Student Learning Outcome 4**

Perform systems decision-making that considers qualitative and quantitative aspects of the problem.

**Student Learning Outcome 5**

Accurately, clearly, concisely, and persuasively report findings, conclusions, and recommendations to the stakeholders in a multicultural context.

**Student Learning Outcome 6**

Lead interdisciplinary teams to implement effective and efficient solutions by melding traditional and non-traditional engineering domains.

**Student Learning Outcome 7**

Demonstrate the skills and interest for intellectual growth and learning for a career of professional excellence and service to the nation as an officer in the United States Army.

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**Systems Engineering (SEN1 )****[DA CD: CUX - SYSTEMS ENGINEERING / CIP CD: 14.2701 - Systems Engineering]**

Systems Engineers innovatively solve large, complex problems in a technologically advanced environment by engineering solutions which provide significant value to clients and their organizations. Systems Engineers also lead interdisciplinary teams of engineers and others in the development and implementation of technical solutions to complex issues facing organizations. Systems Engineering majors learn to think systematically, engineer systematically, and approach decisions systematically. The Systems Engineering program is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>.

Thinking systematically involves understanding the entire environment in which the system operates and includes the needs, wants, and desires of all the stakeholders of the system. Engineering systematically involves identifying and understanding the required system functions, developing alternative system solutions, and applying the basic modeling and simulation tools required to analyze the system from an engineering perspective. Approaching decisions systematically involves leading and participating in multi-disciplinary teams to innovate and implement visionary

solutions to these complex problems.

The recent rapid growth and success of systems engineering can be attributed to advances in technology and the transition of society to a highly networked, globally-oriented information age which results in a dramatic increase in the complexity of problems. These problems require systems thinking and a holistic approach to problem solving that is at the heart of the systems engineering discipline. It is the challenge of systems engineers to harness and direct technology toward solving problems most often related to processes and operations. Ultimately, the study and application of systems engineering principles involve innovation and the creative application of analytical models to facilitate sound decision making.

**Systems Engineering Program Educational Objectives:** Within 5 to 7 years of graduation each graduate of the Systems Engineering program will have:

1. Effectively led interdisciplinary teams to solve complex problems.
2. Demonstrated intellectual growth and continued self-improvement through self-study, continuing education, or professional development.
3. Fostered an organizational ethos that promotes the professional, moral, ethical, and respectful treatment of all.
4. Analyzed, designed, implemented, and maintained systems throughout their lifecycles.
5. Clearly communicated engineering solutions and analysis to leaders, both orally and in writing, to enable sound decision-making in the presence of uncertain, biased, or confounding influences.
6. Approached problems holistically while recognizing each system as a whole, with its fit and relationship with the environment being primary concerns.

**Systems Engineering Student Outcomes:** To achieve these objectives, cadets will demonstrate the following Systems Engineering Student Outcomes at the time of graduation:

**Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

**Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

**Student Learning Outcome 3**

an ability to communicate effectively with a range of audiences.

**Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

**Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

**Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

**Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

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## **Systems Engineering (SEN2 )**

**[DA CD: CUX - SYSTEMS ENGINEERING / CIP CD: 14.2701 - Systems Engineering]**

Systems Engineers innovatively solve large, complex problems in a technologically advanced environment by engineering solutions which provide significant value to clients and their organizations. Systems Engineers also lead interdisciplinary teams of engineers and others in the development and implementation of technical solutions to complex issues facing organizations. Systems Engineering majors learn to think systematically, engineer systematically, and approach decisions systematically. The Systems Engineering program is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>.

Thinking systematically involves understanding the entire environment in which the system operates and includes the

needs, wants, and desires of all the stakeholders of the system. Engineering systematically involves identifying and understanding the required system functions, developing alternative system solutions, and applying the basic modeling and simulation tools required to analyze the system from an engineering perspective. Approaching decisions systematically involves leading and participating in multi-disciplinary teams to innovate and implement visionary solutions to these complex problems.

The recent rapid growth and success of systems engineering can be attributed to advances in technology and the transition of society to a highly networked, globally-oriented information age which results in a dramatic increase in the complexity of problems. These problems require systems thinking and a holistic approach to problem solving that is at the heart of the systems engineering discipline. It is the challenge of systems engineers to harness and direct technology toward solving problems most often related to processes and operations. Ultimately, the study and application of systems engineering principles involve innovation and the creative application of analytical models to facilitate sound decision making.

**Systems Engineering Program Educational Objectives:** Within 5 to 7 years of graduation each graduate of the Systems Engineering program will have:

1. Effectively led interdisciplinary teams to solve complex problems.
2. Demonstrated intellectual growth and continued self-improvement through self-study, continuing education, or professional development.
3. Fostered an organizational ethos that promotes the professional, moral, ethical, and respectful treatment of all.
4. Analyzed, designed, implemented, and maintained systems throughout their lifecycles.
5. Clearly communicated engineering solutions and analysis to leaders, both orally and in writing, to enable sound decision-making in the presence of uncertain, biased, or confounding influences.
6. Approached problems holistically while recognizing each system as a whole, with its fit and relationship with the environment being primary concerns.

**Systems Engineering Student Outcomes:** To achieve these objectives, cadets will demonstrate the following Systems Engineering Student Outcomes at the time of graduation:

#### **Student Learning Outcome 1**

an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

#### **Student Learning Outcome 2**

an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

#### **Student Learning Outcome 3**

an ability to communicate effectively with a range of audiences.

#### **Student Learning Outcome 4**

an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

#### **Student Learning Outcome 5**

an ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

#### **Student Learning Outcome 6**

an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

#### **Student Learning Outcome 7**

an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

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### **War Studies Major (WST0 )**

**[DA CD: BDH - WAR STUDIES / CIP CD: 45.0999 - International Relations and National Security Studies]**

War Studies at USMA is an academically integrated interdisciplinary program dedicated to the study of war, strategy, and warfare in three frames: past, present, and future. It rests upon a foundation of a set of War Studies disciplinary courses that synergistically combine history with theory using the Clausewitzian methodology that blends theory and application, complemented by interdisciplinary and single-discipline courses from a range of fields, especially Military

History and Strategic Studies. War Studies at USMA is interdisciplinary in its synthesis and application of multiple disciplines in ways that foster a deeper understanding of war and prepare cadets to plan, direct, and lead in war. Graduates will be well practiced in thinking about war, strategy, and warfare in time and context, finely attuned to the role of contingency in war, and uniquely capable of developing and articulating flexible strategies and plans that reflect both rigorous analysis and wisdom. Ultimately, War Studies majors will be able to integrate knowledge and skills from a variety of disciplines to anticipate and respond appropriately to opportunities and challenges in a changing world.

The War Studies program achieves this vision through the following Student Learning Outcomes:

**Student Learning Outcome 1**

Communicate effectively in speaking and writing.

**Student Learning Outcome 2**

Analyze and explain how theory, strategy, and warfare have evolved over time; and assess their future trajectories.

**Student Learning Outcome 3**

Evaluate war, strategy, and warfare using appropriate interdisciplinary lenses.

**Student Learning Outcome 4**

Analyze the relationship between societies, institutions, war, strategy, and warfare.

**Student Learning Outcome 5**

Demonstrate critical thinking and research skills reflecting a range of methodologies.

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# **PART III: COURSE DESCRIPTIONS**

| Term Legend                                                                                                                                                                           |                   |                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------|
| Academic years start with the Fall term (running August through December) and are named for following calendar year that includes the Spring term (running from January through May). |                   |                                                                                          |
| Term                                                                                                                                                                                  | Short Description | Long Description                                                                         |
| 1                                                                                                                                                                                     | Fall              | Fall Semester                                                                            |
| 2                                                                                                                                                                                     | Spring            | Spring Semester                                                                          |
| 3                                                                                                                                                                                     | STAP 1            | 1st Summer Term                                                                          |
| 4                                                                                                                                                                                     | STAP 2            | 2nd Summer Term                                                                          |
| 5                                                                                                                                                                                     | STAP 3            | 3rd Summer Term                                                                          |
| 7                                                                                                                                                                                     | IAD(A)            | used for courses with varying summer schedules                                           |
| 8                                                                                                                                                                                     | Fall Exch         | used to indicate USMA courses taken while abroad or at a different academy in the Fall   |
| 9                                                                                                                                                                                     | Spring Exch       | used to indicate USMA courses taken while abroad or at a different academy in the Spring |

## Behavioral Sciences and Leadership

### 67 Courses

| MG379                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | LEADING TEAMS              | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                     | <b>Offerings:</b>                            |
| <p>This course is designed to improve cadets' understanding of human behavior in small group/team settings. Course content includes structural characteristics of teams such as size, status, roles and norms in addition to the effects of team tasks and environment. Cadets then use their understanding of these constructs to analyze team phenomena such as cohesion, performance, decision making, problem solving, and conflict resolution. We also devote a number of lessons to current issues such as electronic and virtual groups, high performance work teams, and shared leadership in a team environment. The course is particularly relevant to professional development in that cadets gain a comprehensive understanding of the dynamics of small group and team interaction. This allows them to develop and implement creative leader actions that will maximize unit/team effectiveness. This course is required for cadets pursuing the Management major.</p>                                        |                            | 2026-1 2027-1 2028-1<br>2029-1               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min     |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                       |                                              |
| MG380                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MARKETING                  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-2                     | <b>Offerings:</b>                            |
| <p>This course introduces cadets to the core concepts, tools, and strategies of marketing management, with a focus on applying theory to real-world problems in both military and civilian contexts. Cadets will develop critical thinking and analytical skills through case studies, market-based scenarios, and applied projects that reflect contemporary marketing challenges. Emphasis is placed on understanding the marketing manager's role within organizations and their contribution to strategic decision-making. The course also integrates insights and best practices from industry leaders to prepare cadets for leadership roles in a rapidly changing global economy. This course is required for cadets pursuing the Management major.</p>                                                                                                                                                                                                                                                              |                            | 2026-2 2026-8 2027-2<br>2028-2 2029-2 2029-8 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min     |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                       |                                              |
| MG381                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | INTRODUCTION TO MANAGEMENT | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                     | <b>Offerings:</b>                            |
| <p>This course serves a dual purpose. It is an introduction to the concept of management as well as an introduction to the multidisciplinary nature of the management field of study. It is designed to give cadets a foundation on which to build deeper areas of expertise across the different business disciplines, including marketing, accounting, finance, operations, human resources, and entrepreneurship. This course also focuses on decision making and the managerial activities that organizational leaders use to effectively and efficiently direct the resources of organizations. As a result, the course is structured around the traditional management functions of planning, organizing, leading, and controlling. In addition, cadets will examine the concepts of ethical and global management as they learn to analyze operating environments, assess organizational capabilities, and develop feasible courses of action. This course is required for cadets pursuing the Management major.</p> |                            | 2026-1 2026-8 2027-1<br>2028-1 2029-1 2029-8 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min     |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                       |                                              |
| MG382                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | HUMAN RESOURCE MANAGEMENT  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                     | <b>Offerings:</b>                            |
| <p>This course begins with the premise that people are a firm's most important resource, and that the management of this critical resource ultimately determines the success or failure of the organization. This class exposes cadets to theories and concepts of Human Resource Management (HRM). More broadly, it encompasses topics associated with strategically managing human capital to support the performance objectives of an organization. This course combines explanations of theory with practical lessons that help frame decision making related to HR issues from the perspective of a General Manager (GM). By analyzing HRM practices and organizational outcomes through presented theory and case studies, cadets learn how to apply HRM concepts to positively influence the success of any organization. This course is required for cadets pursuing the Management major.</p>                                                                                                                      |                            | 2026-1 2027-1 2028-1<br>2029-1 2029-8        |

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                                |                                                          |
|--------------|--------------------------------|----------------------------------------------------------|
| <b>MG390</b> | <b>NEGOTIATION FOR LEADERS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

This course immerses cadets in fundamental-level negotiations and bargaining theory and application. The course progresses from dual-party, single-issue, distributive scenarios to multi-party/multi-issue/integrative scenarios. Cadets learn and practice systematic ways to devise an effective strategy prior to entering a negotiation and then actually apply bargaining tools and tactics during the negotiation to accomplish their individual and organizational goals. Cadets learn concepts and frameworks that help them analyze and understand human behavior so that they have a perspective from all parties involved in a negotiation. Examinations are behavioral and written. Emphasis is placed on applying the behavioral principles learned to real-world issues and their impacts on functioning as future Army officers.

2026-1 2026-2 2026-8  
2026-9 2027-1 2027-2  
2027-8 2027-9 2028-1  
2028-2 2028-8 2028-9  
2029-1 2029-2 2029-8  
2029-9

**Lessons:** 20 @ 110 min (1.250 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** Multiple live negotiation stimulations (in-person, virtual, email etc.) throughout semester, culminating with one final two- day simulation exercise.

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>MG395</b> | <b>FUNDAMENTALS OF ACCOUNTING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------------|

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**Scope:** 2026-2

**Offerings:**

The purpose of MG395 is to provide and integrate the analytical tools learned in this and other courses in a management setting. Specifically, this course will provide the fundamentals of understanding, developing, and analyzing financial statements (income statement, statement of retained earnings, balance sheet, and statement of cash flows), using accounting ratio analysis, analyzing inventory, understanding costing systems, and budgeting. By applying the various accounting techniques in a managerial setting, cadets will be better prepared to quantitatively support their managerial decisions. This course is required for cadets pursuing the Management major.

2026-2 2026-8 2027-2  
2027-9 2028-2 2028-9  
2029-2 2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                           |                                                          |
|--------------|---------------------------|----------------------------------------------------------|
| <b>MG410</b> | <b>MANAGERIAL FINANCE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The purpose of MG410 is to provide Management majors with the basic principles of managerial finance, and then to apply these principles in the context of managerial decision-making. Specifically, this course will cover: the fundamentals of the time value of money; the meaning and measurement of risk and return; valuation techniques for stocks and bonds; and standard techniques for financial analysis, to include capital budgeting, discounted cash flow valuation, and weighted average cost of capital. Cadets will leave this course with a solid understanding of how financial managers at the corporate level balance risk and return and thus manage everyday financial decision-making. This course is required for cadets pursuing the Management major.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** SS494

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|              |                              |                                                          |
|--------------|------------------------------|----------------------------------------------------------|
| <b>MG420</b> | <b>OPERATIONS MANAGEMENT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|------------------------------|----------------------------------------------------------|

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**Scope:** 2010-1

**Offerings:**

The purpose of MG420 is to provide cadets with the tools to deal with the quantitative aspects of design and analysis of operations management. Emphasis is on identification, analysis, and solution of production problems using applied quantitative techniques using the case study technique. In addition to case studies, simulations reinforce the problem-solving techniques necessary for today's successful managers. Specific methods and techniques taught and applied are operations strategy, product design and selection, supply chain management, total quality management, forecasting, capacity planning, facility location, facility layout, work system design, inventory management, material requirements planning, and scheduling. This course is required for cadets pursuing the Management major.

No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None



| MG420                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | OPERATIONS MANAGEMENT                                                                                                                                                                                    | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-2                                                                                                                                                                                                   | <b>Offerings:</b>                                             |
| <p>The purpose of MG420 is to provide cadets with the tools to deal with the quantitative aspects of design and analysis of operations management. Emphasis is on identification, analysis, and solution of production problems using applied quantitative techniques through the case study technique. In addition to case studies, simulations reinforce the problem-solving techniques necessary for today's successful managers. Specific methods and techniques taught and applied are operations strategy, product design and selection, supply chain management, total quality management, forecasting, capacity planning, facility location, facility layout, work system design, inventory management, material requirements planning, and scheduling. This course is required for cadets pursuing the Management major.</p>                                                                                                                                                                                                                          |                                                                                                                                                                                                          | 2026-2 2026-8 2027-2<br>2028-2 2029-1 2029-2<br>2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                   |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                                                                                                                                     |                                                               |
| MG421                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | STRATEGIC MANAGEMENT                                                                                                                                                                                     | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-2                                                                                                                                                                                                   | <b>Offerings:</b>                                             |
| <p>This capstone course for Management majors emphasizes the integration of concepts and principles found in all previous management courses as they relate to the strategic management of public, private, and military organizations. This course focuses on all aspects of the strategic management process to include: the identification of opportunities and threats in a competitive environment, the development of organizational core competencies, and the strategic alternatives available to organizations as they seek to achieve their goals in a highly dynamic operating environment filled with complexity, uncertainty, and risk. MG421 uses the case study method that requires comprehensive, in-depth analysis of realistic management situations. This course is required for cadets pursuing the Management major.</p>                                                                                                                                                                                                                 |                                                                                                                                                                                                          | 2026-2 2026-8 2027-2<br>2028-2 2029-2                         |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                   |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | A semester-long integrative group project that includes a written paper and oral presentation. Approval from Management Program Director is required for non-Management majors to enroll in this course. |                                                               |
| MG462                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ENTREPRENEURSHIP                                                                                                                                                                                         | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-2                                                                                                                                                                                                   | <b>Offerings:</b>                                             |
| <p>The Army, like many organizations, exists in a volatile, uncertain, complex, ambiguous, and uncertain (VUCA) global environment, and cadets will be required to lead organizations in this dynamic set of circumstances. An entrepreneurial mindset is a key element of success in such environments. In this course, cadets will develop an understanding of the ways that innovation and entrepreneurship can manifest in conventional and unconventional settings. For the capstone project, cadets will develop a commercially viable business idea and compete with that idea in the Mid-Hudson Regional Business Plan Competition, which will take place in March. If any individual cadet or team place first or second in the Mid-Hudson Regional Business Plan Competition, then they will be propelled into the qualifying rounds for the New York State Business Plan Competition. The qualifying rounds take place in April, and if selected, cadets will compete with their idea at the New York State Business Plan Competition in April.</p> |                                                                                                                                                                                                          | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                   |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                                                                                                                                     |                                                               |
| MG463                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AGILE INNOVATION IN DEFENSE                                                                                                                                                                              | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2024-4                                                                                                                                                                                                   | <b>Offerings:</b>                                             |
| <p>Agile Innovation in Defense (AID) is an experiential educational model that engages teams of Cadets to solve some of the nation's toughest national security and defense problems using lean startup principles. With more than 80 U. S. government organizations and agencies acting as problem sponsors, AID addressed national security and defense problems with the aid of coaches, instructors, and support personnel from National Security Innovation Network (NSIN). Cadets work in teams on curated problems, interfacing directly with problem sponsors with leadership roles in the Department of Defense. Through iterative problem solving, Cadets hone problem-solving skills, learn about high-priority problems in the Department of Defense, and build relationships among subject matter experts and stakeholders.</p>                                                                                                                                                                                                                   |                                                                                                                                                                                                          | No Course Offerings                                           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                   |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | As specified by the professor.                                                                                                                                                                           |                                                               |
| PL100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | GENERAL PSYCHOLOGY FOR LEADERS                                                                                                                                                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2016-1                                    | <b>Offerings:</b>                                                                                             |
| Psychology as a discipline is both a natural and social science that involves the study of the brain, mind, and behavior. General Psychology for Leaders is a course that involves a multidisciplinary study of the human dimension, behavioral sciences, and leadership development. The course examines the cognitive, physical, and social components of the human dimension using the scientific method to promote reflection, development, ethical reasoning, and critical and creative thinking. The learning outcome is that each cadet will be more ethical and effective leader because of a scientific understanding of human behavior. Emphasis is placed on applying the behavioral principals learned to the cadets' current lives and their functioning as future officers. The two course goals are that cadets apply the scientific method to investigate causes or correlates of human behavior, and that they use this scientific understanding of human behavior to explain, predict, and change behavior to become an effective leader of character.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           | 2026-1 2026-2 2026-4<br>2027-1 2027-2 2027-4<br>2027-5 2028-1 2028-2<br>2028-4 2028-5 2029-1<br>2029-2 2029-4 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 39 @ 55 min (2.500 Att/wk)                | <b>Labs:</b> 1 @ 55 min                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                      |                                                                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PL150                                     |                                                                                                               |
| <b>PL150</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>ADV GEN PSYCHOLOGY FOR LEADERS</b>     | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2016-1                                    | <b>Offerings:</b>                                                                                             |
| Psychology as a discipline is both a natural and social science that involves the study of the brain, mind, and behavior. Advanced General Psychology for Leaders is a course that involves an advanced multidisciplinary study of the human dimension, behavioral sciences, and leadership development. The course examines the cognitive, physical, and social components of the human dimension using the scientific method to promote reflection, development, ethical reasoning, and critical and creative thinking. The learning outcome is that each cadet will be a more ethical and effective leader because of a scientific understanding of human behavior. With even more leader development application than PL100, emphasis is placed on applying the behavioral principles learned to the cadets' current lives and their functioning as future officers. The two course goals are that cadets apply the scientific method to investigate causes or correlates of human behavior, and that they use this scientific understanding of human behavior to explain, predict, and change behavior to become an effective leader of character. This course goes beyond PL100 (General Psychology for Leaders) in that students are asked to apply material at a higher level via reading and incorporating significant peer-reviewed research to address real-world Army issues. Students must also communicate their understanding of psychology and how it informs their roles as leaders of character via formal and informal presentations. |                                           | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 40 @ 55 min (2.500 Att/wk)                | <b>Labs:</b> 0 @ 0 min                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                      |                                                                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PL100                                     |                                                                                                               |
| <b>PL250</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>NEUROCOG FNDTNS OF BEHAVIOR</b>        | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                    | <b>Offerings:</b>                                                                                             |
| This course offers an introduction to cognitive neuroscience: the interdisciplinary study of how brain structure and function give rise to behavior and mental processes. We examine the biological bases of behavior through topics such as perception, learning, memory, language, emotion, motivation, sleep, development, psychopathology, and more. Together, we investigate how the brain's structure and activity develop, adapt, and interact to support complex behavior, and how disruptions to these processes impact functioning. The knowledge and skills developed in this course help establish a critical foundation in neurocognition for advanced study and professional practice across many disciplines such as psychology, medicine, education, and human performance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                           | 2026-1 2027-1 2028-1<br>2029-1                                                                                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 40 @ 55 min (2.500 Att/wk)                | <b>Labs:</b> 0 @ 0 min                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                      |                                                                                                               |
| <b>PL289</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>IND STUDY BEH SCI &amp; LDRSHP 1CR</b> | <b>1.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2021-2                                    | <b>Offerings:</b>                                                                                             |
| This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 2 hours of work per week towards completion of the project, for a total of 40 hours of work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 @ 0 min (0.000 Att/wk)                  | <b>Labs:</b> 0 @ 0 min                                                                                        |

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

**PL289A** **IND STUDY BSL - A (1CR)** **1.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2021-2 **Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 2 hours of work per week towards completion of the project, for a total of 40 hours of work.

**Lessons:** 0 @ 0 min (0.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. PL289 must be taken before PL289A. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

**Prerequisite(s):** PL289

**PL289B** **IND STUDY BSL - B (1CR)** **1.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2021-2 **Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 2 hours of work per week towards completion of the project, for a total of 40 hours of work.

**Lessons:** 0 @ 0 min (0.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. PL289A must be taken before PL289B. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

**Prerequisite(s):** PL289A

**PL300** **MILITARY LEADERSHIP** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2026-1 **Offerings:**

PL300 (Military Leadership) is a multidisciplinary study of leadership within military organizations. Cadets gain a theoretical and pragmatic understanding of the behavioral and organizational sciences, which they apply to develop their own leadership skills, cultivate leadership abilities in others, and enhance the effectiveness of military organizations. The course surveys a carefully selected set of the most relevant and useful ideas, concepts, theories and knowledge from these sciences. A representative list of topics explored includes: leadership development, character and its development, power and influence, team dynamics, organizational change, and transformational leadership. In addition to applying knowledge of these topics, cadets develop a leadership portfolio which helps refine and inform their personal approach to leading. The portfolio includes several assignments that require personal reflection, critical thinking, and mentor engagement. PL300 is designed to maximize cadets' practical application of theory in their roles as leaders in the Corps of Cadets and as future Army officers.

**Lessons:** 29 @ 75 min (2.500 Att/wk) **Labs:** 1 @ 75 min

**Special Requirements:** None

**Prerequisite(s):** PL100  
-Or-  
PL150

**Disqualifier(s):** PL350

**PL350** **ADVANCED MILITARY LEADERSHIP** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2026-1 **Offerings:**

PL350 (Advanced Military Leadership) is a multidisciplinary study of leadership within military organizations. Cadets gain a theoretical and pragmatic understanding of the behavioral and organizational sciences, which they apply to develop their own leadership skills, cultivate leadership abilities in others, and enhance the effectiveness of military organizations. The course surveys a carefully selected set of the most relevant and useful ideas, concepts, theories and knowledge from these sciences. A representative list of topics explored includes: leadership development, character and its development, power and influence, team dynamics, organizational change, and transformational leadership. In addition to applying knowledge of these topics, cadets develop a leadership portfolio which helps refine and inform their personal approach to leading. The portfolio includes several assignments that require personal reflection, critical thinking, and mentor engagement. PL350 is designed to maximize cadets' practical application of theory in their roles as leaders in the Corps of Cadets and as future Army officers. PL350 offers some advanced topics and experiential opportunities beyond that of PL300. Enrollment is limited to cadets recognized as having exceptional leadership potential within the Corps.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 29 @ 75 min (2.000 Att/wk)      **Labs:** 1 @ 75 min

**Special Requirements:** Enrollment based on a formal endorsement from Academy staff and faculty (e.g., TACs, AWPAD, and instructors). Cadets wishing to self-nominate for PL350 can also seek approval from the PL300/PL350 Course Director.

**Prerequisite(s):** PL100  
-Or-  
PL150

**Disqualifier(s):** PL300

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| <b>PL360</b> | <b>PSYCH ELITE PERFORMANCE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course focuses on the psychological components, theories, and skills that support sustained elite performance. The goal of the course is to foster cadets' understanding, development, implementation, and coaching of the psychological skills that influence and generate success in high-level performance situations across domains (academic, physical, athletic, military, and leadership). Topics include performance mindset and awareness, psychological skills training fundamentals, confidence and self-talk, motivation and goal setting, attention control, energy and stress management, mindfulness, imagery, and special topics related to performance psychology. Throughout the course, particular emphasis is placed on how these topics relate to and can be applied in cadets' roles as current and future leaders. Cadets will both understand theoretical bases underlying these topics, and apply them, through applied mental training exercises, leading and taking part in squad discussions on the course topics applied to leadership specifically, and through a semester-long peer mentorship case study with a fellow cadet.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** PL100  
-Or-  
PL150  
-Or-  
PL100B

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| <b>PL361</b> | <b>RESEARCH METHODS I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course introduces cadets to the foundational principles and practices of scientific research in the behavioral sciences. Cadets will explore the philosophy and terminology of the scientific method, gaining a solid understanding of essential research concepts. The course covers a range of research designs, including both experimental and non-experimental methodologies, to illustrate how different approaches shape the study of behavior. Cadets will also learn basic data analysis techniques, including both descriptive and inferential statistics. Throughout the course, cadets will apply their knowledge by working in groups to design and conduct research projects using observational methodologies on topics of their choice.

2026-1 2026-8 2027-1  
2028-1 2029-1 2029-8  
2029-9

**Lessons:** 28 @ 75 min (2.000 Att/wk)      **Labs:** 2 @ 75 min

**Special Requirements:** Semester-long group research project.

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| <b>PL363</b> | <b>QUALITATIVE RESEARCH METHODS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

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| Qualitative Research Methods facilitates discussion, awareness, and understanding of social research methodology. In this course we seek to understand the basics of social research with a specific focus on qualitative social research methods. This objective is achieved through discussions, applications, and demonstrating the ability to solve social problems via viable social research. This includes the development, design, and application of several analysis papers, culminating in a written research proposal and an oral presentation. Methods covered include ethnography, interviewing, content analysis, (un)obtrusive measures, visual analysis, and ethics.                                                                                                                                                                                                                                                                                                                                                                            |                                                                       | 2026-2 2027-1 2028-1 2029-1                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Some data gathering fieldwork required.                               |                                                    |
| <b>PL370</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>US SOCIETY &amp; THE SERVICEMEMBER</b>                             | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                | <b>Offerings:</b>                                  |
| This course introduces the scientific study of society and the interactions among people within American society. Providing a broad introduction to American social ideals and civic responsibilities, the course then explores civic and lived life. Through sociological analysis of social institutions and their impact on individuals, cadets will identify patterns across situations and determine the social significance of our modern institutions, including government, family, schools, and the military. This course includes writing, discussions, exams, presentations, and hands-on projects. **This is a pilot course that must be reviewed by the Curriculum Committee NLT AY28-1 to continue.**                                                                                                                                                                                                                                                                                                                                              |                                                                       | 2026-2 2027-2 2028-2                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                  |                                                    |
| <b>PL371</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>INTRODUCTORY SOCIOLOGY</b>                                         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2004-1                                                                | <b>Offerings:</b>                                  |
| Sociology is the scientific study of society and the interactions among humans. The goal of Introductory Sociology is to provide a survey of the field of sociology and educate and inspire cadets to examine contemporary situations that involve social interaction and use sociological concepts, theories, and research to explain what is taking place, identify social threads and patterns across the situations, and determine the personal as well as the social significance of their analysis. Sociology demands that the student transcend the taken-for-granted, subjective world view and develop a sociological imagination by revealing the linkages and relationships among social facts and connect public issues to self awareness. PL371 is a survey course with the identification of common threads across social situations, and determining the self and social significance of facts. The teaching and learning strategy involves reading, writing, discussions, presentations, and other active-learning, hands and heads-on projects. |                                                                       | 2029-1 2029-2                                      |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Three papers synthesizing course readings; lead one class discussion. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PL100<br>-Or-<br>PL150<br>-Or-<br>PL100B                              |                                                    |
| <b>PL372</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>MILITARY FAMILIES</b>                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                | <b>Offerings:</b>                                  |
| This course offers a multidisciplinary examination of contemporary American families, with a special focus on the unique experiences of military families. Grounded in a sociological perspective, the course integrates insights from psychology, history, anthropology, biology, and economics to deepen understanding of the family as a social institution and its interactions with other institutions such as work, education, and the military. Cadets will analyze the historical and current forces shaping families, with particular attention to the challenges and dynamics faced by military families. Through the evaluation of contemporary issues, policies, and scholarly research, cadets will apply scientific inquiry and critical thinking to assess and anticipate the evolving roles and needs of American military families.                                                                                                                                                                                                             |                                                                       | 2026-2 2027-2 2028-2 2029-2                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                  |                                                    |
| <b>PL373</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>LIFE-SPAN DEVELOPMENTAL PSYCH</b>                                  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                | <b>Offerings:</b>                                  |

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| <p>This course examines individual development and the forces that shape human growth from birth to late life, including death, and bereavement. Cadets will explore development through multiple theoretical perspectives, emphasizing physical, cognitive, and socioemotional changes across the lifespan. Topics include moral development, self-concept, identity formation, and the influence of family and society. Special attention is given to the application of developmental principles in military contexts, preparing cadets to understand and lead individuals from various backgrounds, foster resilience, and support the well-being of those under their command throughout their military careers.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p>                                                                                                                                                                                                                                                              |                            | 2026-1 2027-1 2028-1<br>2029-1                                                                         |
| <b>PL376</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>PSYCHOPATHOLOGY</b>     | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                     | <b>Offerings:</b>                                                                                      |
| <p>This course introduces cadets to the study of psychological disorders, with a focus on understanding the defining characteristics of a range of mental health conditions. Cadets will explore how these disorders develop and are maintained, as well as evidence-based approaches to their treatment. The course examines psychological disorders from multiple theoretical perspectives, including genetic, biological, cognitive, behavioral, sociocultural, and psychoanalytic viewpoints. Through this comprehensive approach, cadets will gain the knowledge and skills necessary to recognize, understand, and address mental health challenges within military and broader contexts.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> A case study paper that provides an in-depth analysis of a selected psychological disorder.</p>                                                                                                                                                                                             |                            | 2026-2 2027-2 2027-8<br>2028-2 2028-8 2029-1<br>2029-2 2029-8                                          |
| <b>PL379</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>GROUP DYNAMICS</b>      | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2010-1                     | <b>Offerings:</b>                                                                                      |
| <p>This course is designed to improve cadets' understanding of human behavior in small group/team settings. Course content includes structural characteristics of teams such as size, status, roles and norms in addition to the effects of task and environment. Cadets then use their understanding of these constructs to analyze team phenomena such as cohesion, performance, decision making, problem solving and conflict resolution. We also devote a number of lessons to current issues such as electronic and virtual groups, high performance work teams and shared leadership in a team environment. The course is particularly relevant to professional development in that cadets gain a comprehensive understanding of the dynamics of small group and team interaction. This allows them to develop and implement creative leader actions that will maximize unit/team effectiveness.</p> <p><b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p> <p><b>Prerequisite(s):</b> PL100<br/>-Or-<br/>PL150<br/>-Or-<br/>PL100B</p> |                            | No Course Offerings                                                                                    |
| <b>PL383</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>SOCIAL PSYCHOLOGY</b>   | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                     | <b>Offerings:</b>                                                                                      |
| <p>This course surveys the field of contemporary social psychology. Cadets examine the impact of social structure and group membership on social behavior, while focusing on processes such as attribution, cognition, and learning that underlie social behavior. The course is intended to enable cadets to more effectively analyze and explain human behavior in a given situation. Specific topics include the self, attitudes and attitude change, conformity, obedience, compliance, deviance, helping behavior, aggression, attraction and romance, groups and intergroup relations, and collective behavior. The classroom experience is heavily discussion-oriented in order to maximize the applications of social psychological concepts, theories, and perspectives to daily life and military leadership.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p>                                                                                                                                                            |                            | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-9<br>2028-1 2028-2 2028-9<br>2029-1 2029-2 2029-8<br>2029-9 |
| <b>PL384</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>SOCIOLOGICAL THEORY</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-2                     | <b>Offerings:</b>                                                                                      |

Sociological theory is a set of interrelated ideas that allow for the systemization of knowledge of the social world, the explanation of that world, and predictions about the future of the world. In some ways, all of us are amateur theorists, interpreting the meanings of the events and encounters that shape the world and ourselves. In PL384 cadets will learn in-depth how theories can help make sense of our times and to choose courses of action to realize our collective and individual dreams. In this course, theory is brought down-to-earth, to show how a sociological imagination (in other words, a theoretical consciousness that embraces self-awareness) is valuable to self and society. Questions like "Are families disintegrating?" "Why are some people discriminated against?" "What accounts for the crime rate?" "Are religion and economics compatible?" "Why is the sexual division of labor so persistent?" "Are wars inevitable?" can be addressed. Theories are thus tentative answers to the questions that preoccupy us as members of families, professions, communities, nations and, increasingly, as global citizens.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** PL371

2029-2

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| <b>PL386</b> | <b>EXPERIMENTAL PSYCHOLOGY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course provides cadets with detailed practical knowledge and skills in the experimental analysis of cognition, behavior and human performance. Particular emphasis is placed on design of laboratory and field experiments, laboratory automation and instrumented data acquisition, computer data analysis, and on the distinction between observational versus experimental research. The course includes practical exercises in several content areas of experimental and engineering psychology.

2026-1 2027-1 2027-8  
2028-1 2028-8 2029-1

**Lessons:** 26 @ 75 min (2.000 Att/wk)      **Labs:** 4 @ 75 min

**Special Requirements:** None

**Corequisite(s):** MA376

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| <b>PL387</b> | <b>FOUNDATIONS OF COUNSELING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course provides cadets with an introduction to counseling fundamentals emphasizing the practical application of theories, principles, and techniques essential for military leadership. Through hands-on experiences, including video and audio recording exercises, cadets will develop critical counseling skills necessary for guiding, supporting, and motivating individuals and teams. The curriculum addresses personal, performance, career, and disciplinary counseling within the unique context of military service, preparing cadets to lead effectively in both academic and operational environments. Topics include the counseling process, basic and advanced communication skills, interpersonal relationship dynamics, goal setting, intervention strategies, crisis intervention, multicultural awareness, and assertiveness. Assessments include behavioral demonstrations of counseling competencies and written examinations, ensuring cadets are equipped to foster resilience, cohesion, and high performance as future Army leaders.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2 2029-8  
2029-9

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>PL389</b> | <b>IND STUDY BEH SCI &amp; LDRSHP 2CR</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2021-2

**Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 4 hours of work per week towards completion of the project, for a total of 80 hours of work.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

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| <b>PL389A</b> | <b>IND STUDY BSL A (2CR)</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2021-2                                                                                                                                                                                                                                                        | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 4 hours of work per week towards completion of the project, for a total of 80 hours of work.                                                                                                                                                                          |                                                                                                                                                                                                                                                               | 2029-1                                                        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. PL389 must be taken before PL389A. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.  |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PL389                                                                                                                                                                                                                                                         |                                                               |
| <b>PL389B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>IND STUDY BSL B (2CR)</b>                                                                                                                                                                                                                                  | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2021-2                                                                                                                                                                                                                                                        | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 4 hours of work per week towards completion of the project, for a total of 80 hours of work.                                                                                                                                                                          |                                                                                                                                                                                                                                                               | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. PL389A must be taken before PL389B. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PL389A                                                                                                                                                                                                                                                        |                                                               |
| <b>PL390</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>BIOLOGICAL PSYCHOLOGY</b>                                                                                                                                                                                                                                  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                                                                                                                                                        | <b>Offerings:</b>                                             |
| This course introduces the cadet to the physiological and anatomical structures and processes that underlie human behavior with an emphasis on human performance. The course examines the structure of the nervous and endocrine systems, the mechanisms of seeing and hearing, movement, stress and arousal, learning, memory, biological causes of abnormal behavior, sleep, language, and the effects on performance of damage to neural structures. Cadets are introduced to the scientific examination of real-world bio-psychological problems in laboratory assignments in examining tissue, brains and eyes to provide three-dimensional realism to classroom instruction. This course provides the basis for competence in later engineering psychology electives. Biological Psychology is the "hardware" introduction to engineering psychology. |                                                                                                                                                                                                                                                               | 2026-1 2026-8 2027-1<br>2027-8 2028-1 2028-8<br>2029-1 2029-8 |
| <b>Lessons:</b> 38 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 2 @ 55 min                                                                                                                                                                                                                                       |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                                                                                                                                                                          |                                                               |
| <b>PL391</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>SENSATION/PERCEPTN/PSYCPHYS</b>                                                                                                                                                                                                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                                                                                                                                                        | <b>Offerings:</b>                                             |
| This course covers the acquisition and analysis of information by the human nervous system from examination of the physical properties of light and sound, the functioning of the visual, auditory systems and the kinesthetic processes, and the theoretical background of contemporary perceptual research. The following general topics are covered: psychophysical methods, including measurement, scaling and signal detection theory; physiology of the visual, auditory and kinesthetic systems; recognition of color and brightness, pitch and loudness, patterns, features, and the role of visual channels; visual detection and tracking; the role of kinesthesia in military applications; and research methodology in perception. Laboratory assignments provide an opportunity to apply concepts learned to real-world challenges.            |                                                                                                                                                                                                                                                               | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 27 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 3 @ 75 min                                                                                                                                                                                                                                       |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                                                                                                                                                                          |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PL386 PL390<br>-Or-<br>MA376 PL250                                                                                                                                                                                                                            |                                                               |



| PL392                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | COGNITIVE PSYCHOLOGY           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                         | <b>Offerings:</b>                                                              |
| This course addresses the processes of human information gathering, learning and memory using an information processing model. The course deals with cognitive theory and application, including stage models of processing and memory, machine models and artificial intelligence, and research methodology in these areas. Emphasis is placed on practical military applications in such areas as pattern recognition and detection, text processing, visual search and associated problems. Laboratory experiences stress development of experimental paradigms in this area of investigation and the use of test instrumentation, and computer software models to investigate cognitive processing.                                                                       |                                | 2026-2 2027-1 2027-2<br>2027-8 2028-1 2028-2<br>2028-8 2029-2 2029-8<br>2029-9 |
| <b>Lessons:</b> 38 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 2 @ 55 min        |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                           |                                                                                |
| PL393                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CRIMINOLOGY-CRIM JUST SYSTM    | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                         | <b>Offerings:</b>                                                              |
| Criminology is the scientific study of the making of laws, the breaking of laws, and the reaction to the breaking of laws. When a crime appears to have been committed and authorities have been notified, the criminal justice system is set in motion. The criminal justice system is the societal response to crime and includes three major activities: law enforcement, the judicial process, and corrections. The course provides an overview of (a) the theories offered to explain crime and delinquent behavior (b) the criminal justice system's responses to those behaviors, and (c) the relationships between the varied explanations of criminal behavior and society's response to those behaviors. The focus of the course is primarily on the United States. |                                | 2026-1 2027-1 2028-1<br>2029-1                                                 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min         |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                           |                                                                                |
| PL394                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ANTHROPOMETRICS & BIOMECHANICS | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                         | <b>Offerings:</b>                                                              |
| Virtually every activity in which humans engage involves interacting with our environment, often in a physical manner. This course will teach cadets the fundamentals of anthropometrics and biomechanics so they can plan for and design for the physiological capabilities and limitations of humans. Anthropometrics is the study of human measurement. Biomechanics is the study of forces on our muscular and skeletal system. Using these disciplines, cadets will conduct bodily and workspace analyses to improve the environments, equipment, and systems with which humans interact. The course will emphasize work performed in military settings.                                                                                                                 |                                | 2026-2 2027-2 2028-2<br>2029-2                                                 |
| <b>Lessons:</b> 36 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 4 @ 55 min        |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                           |                                                                                |
| PL398                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | LEADERSHIP THEORY & PRACTICE   | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                         | <b>Offerings:</b>                                                              |
| This course is devoted to exploring concepts and theories of leadership and leader development to enhance cadet understanding, implementation, and assessment of leadership practices at the individual, team, and organizational level. The course examines the evolution of leadership theory, emphasizes scientific research and empirical support for existing leadership theories, and produces critical and creative thinkers. Additionally, cadets will study emerging leadership perspectives and engage in case studies to enhance their understanding and application of effective leadership styles in the volatile, uncertain, chaotic, and ambiguous (VUCA) world.                                                                                               |                                | 2026-2 2027-2 2028-2<br>2029-1 2029-2 2029-8                                   |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min         |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                           |                                                                                |
| PL399                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | BEH SCI & LEADERSHIP PRACTICUM | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2006-4                         | <b>Offerings:</b>                                                              |

The Department of Behavioral Sciences and Leadership's Academic Individual Advanced Development (AIAD) program is designed to give cadets practical experience in their field of study and to reflect on their experiences by completing specified academic requirements. Recent AIADs have involved internships with the American Psychological Association; studies of psychological support to NATO operations in France and stress in military operations in Norway; as well as other topics in CONUS, China, Germany, and Australia. Scope, depth and material covered will meet the requirement of a 3-credit hour course in the department. Grades are determined based on preparatory briefings and essays, a journal of daily activities or Weblog with instructors, the quality of the work performed during the internship, student evaluation of the experience and a final paper, briefing, or exam that incorporates their experience with a topic from their field of study, due upon return.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

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|               |                                           |                                                          |
|---------------|-------------------------------------------|----------------------------------------------------------|
| <b>PL399A</b> | <b>BEH SCI &amp; LEADERSHIP PRACTICUM</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|-------------------------------------------|----------------------------------------------------------|

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**Scope:**

2007-4

**Offerings:**

The Department of Behavioral Sciences and Leadership's Academic Individual Advanced Development (AIAD) program is designed to give cadets practical experience in their field of study and to reflect on their experiences by completing specified academic requirements. Recent AIADs have involved internships with the American Psychological Association; studies of psychological support to NATO operations in France and stress in military operations in Norway; as well as other topics at West Point, elsewhere in the continental United States, or overseas. Scope, depth and material covered will meet the requirement of a two-credit hour course in the department. Grades are determined based on preparatory briefings and essays, a journal of daily activities or Weblog with instructors, the quality of the work performed during the internship, student evaluation of the experience and a final paper, briefing, or exam that incorporates their experience with a topic from their field of study, due upon return.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

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|---------------|-------------------------------------------|----------------------------------------------------------|
| <b>PL399B</b> | <b>BEH SCI &amp; LEADERSHIP PRACTICUM</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:**

2018-7

**Offerings:**

Scope, depth and material covered will meet the requirement of a 2-credit hour course in the department. Grades are determined based on preparatory briefings and essays, a journal of daily activities or Weblog with instructors, the quality of the work performed during the internship, student evaluation of the experience and a final paper, briefing, or exam that incorporates their experience with a topic from their field of study, due at the end of the term.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

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|--------------|----------------------------------|----------------------------------------------------------|
| <b>PL462</b> | <b>ADVANCED RESEARCH METHODS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:**

2026-1

**Offerings:**

This advanced research methods course provides cadets with in-depth education in the principles and practices of scientific research in the behavioral sciences. Cadets will further develop their research skills and mastery of the scientific method, with a focus on designing, conducting, and interpreting complex research projects. Emphasis is placed on advanced research methodologies and data analysis techniques essential for evaluating evidence and solving real-world problems relevant to military leadership and operations. Through hands-on research experience, peer collaboration, and critical analysis of current literature, cadets develop the analytical and methodological expertise necessary to lead evidence-based initiatives that enhance unit effectiveness, inform military training, and advance our understanding of human performance in military contexts.

**Lessons:** 22 @ 75 min (2.000 Att/wk)

**Labs:** 8 @ 75 min

**Special Requirements:**

Semester-long group research project.

**Prerequisite(s):**

PL361

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>PL470</b> | <b>TOPICS-BEHAVIOR SCI/LDRSHIP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:**

2019-2

**Offerings:**

This course explores an advanced topic in Behavioral Sciences and Leadership. Specific subject matter will vary with the expertise of the senior faculty member conducting the course.

2026-8 2026-9 2027-1  
2027-2 2027-8 2027-9  
2028-1 2028-2 2028-8  
2028-9 2029-1 2029-2  
2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** As specified by the professor.

**Prerequisite(s):** PL100  
-Or-  
PL150  
-Or-  
PL100B

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|---------------|------------------------------------|----------------------------------------------------------|
| <b>PL470A</b> | <b>TOPICS-BEHAVIOR SCI/LDRSHIP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2024-1 **Offerings:**

This course explores an advanced topic in Behavioral Sciences and Leadership. Specific subject matter will vary with the expertise of the senior faculty member conducting the course.

2026-2 2029-1 2029-8  
2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** As specified by the professor. Cadets must successfully complete PL470 before enrolling in PL470A.

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|--------------|-----------------------------|----------------------------------------------------------|
| <b>PL471</b> | <b>LEADERSHIP IN COMBAT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1 **Offerings:**

PL471 provides an interdisciplinary examination of leadership in combat at the tactical level. The course establishes a theoretical foundation for understanding the human dimension of combat and investigates the factors that influence effective leadership of soldiers in high-stress, dynamic environments. Cadets will engage with a diverse collection of readings, films, first-hand accounts from combat veterans, and guided self-study to critically analyze leadership challenges and strategies in combat situations. Emphasis is placed on developing insights and skills essential for leading soldiers under the unique pressures of military operations.

2026-1 2026-8 2027-1  
2028-1 2029-2 2029-8

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** PL300  
-Or-  
PL350

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|              |                                      |                                                          |
|--------------|--------------------------------------|----------------------------------------------------------|
| <b>PL472</b> | <b>CROSS-CULTURAL ORG'L BEHAVIOR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2010-2 **Offerings:**

This course examines the individual, group and organizational level influences on human behavior in the international arena. Cadets will gain an understanding of these influences and use the insights gained to formulate leader actions to effectively motivate and manage in a global environment. The course emphasizes the practical application of management theories and research findings in the international situations that cadets encounter in their personal lives and in the field Army with an increasing emphasis on the global environment. Course content includes foundations of individual behavior, diversity, motivation, decision making, rewards, feedback and power and influence in an international setting. We will also examine organizational influences on ethical behavior in the global arena with an emphasis on creating ethical climates in the organizations we belong to.

No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** PL300

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|---------------|--------------------------------------|----------------------------------------------------------|
| <b>PL474X</b> | <b>TECHNOLOGY, DATA, AND SOCIETY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1 **Offerings:**

This course introduces cadets to concepts of studying the impact of connected technologies and data collection on society. It enhances lethality and warfighting capabilities by improving critical thinking and decision-making abilities, preparing cadets for the challenges they will face on the modern battlefield as technology continues to advance. The course supports the pillars of Academic, Physical, and Military development to prepare cadets for future leadership challenges. This approach is accomplished through synthesizing arguments in short, two-page papers, conducting peer reviews of other students' papers, and applying course readings to real-world examples through a variety of case study formats. A term-end exam is completed by the course's end. \*\*This is a pilot course and must receive CC review NLT AY27-2 to continue.\*\*

2026-2 2027-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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**PL475 HUMAN-COMPUTER INTERACTION**
**3.0 Credit Hours**  
**(BS=0.0,ET=1.0,MA=0.0)**

**Scope:** 2026-1

**Offerings:**

Computer use in the world today is at an all-time high. Consequently, the need for user-friendly computers is crucial. Somewhat ironically, human capacity for memory has often been explained using the computer metaphor, while the computer designer often attempts to instill human-like qualities into their computer designs. This course focuses on the interface between the human and computer. Initial focus is placed on understanding the theoretical foundations of human processes. The course then examines how these processes interact with computer usage. Students will learn design principles that enhance compatibility with computer systems.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 36 @ 55 min (2.500 Att/wk)      **Labs:** 4 @ 55 min

**Special Requirements:** None

**Prerequisite(s):** PL392

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**PL476 EDUCATIONAL PSYCHOLOGY**
**3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)**

**Scope:** 1979-2

**Offerings:**

In this course, cadets will develop their instructional skills and formulate a conceptual basis for their instructional practices as army officers. The course is oriented toward the study of psychological theories of learning and application of these theories to the design, delivery, and evaluation of adult education and training. The course is subdivided into major areas of study. Learning theory focuses on the study of the learning process with balanced treatment given to behavioristic and cognitive perspectives. Instructional design emphasizes a systems approach to planning and decision making in learning situations.

No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Course practicum in which cadets design, deliver, and evaluate a learning experience. The practicum involves submission of a written design plan and evaluation (15-20 pages) and the delivery of instruction (55 minutes).

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**PL479 LEADING CHANGING ORGANIZATIONS**
**3.0 Credit Hours**  
**(BS=0.0,ET=3.0,MA=0.0)**

**Scope:** 2026-1

**Offerings:**

This course explores the challenges and strategies of leading organizational change in dynamic environments shaped by technological innovation, evolving societal values, and complex global issues. Cadets will gain a foundational understanding of organizational functioning and empirical evaluation, then apply evidence-based approaches to design and assess improvement initiatives. Drawing from industrial/organizational psychology, behavioral science, business, management, and military doctrine, the course uses case studies, empirical analysis, and collaborative projects to examine real-world successes and failures. Cadets will develop practical leadership strategies for implementing and evaluating change, preparing them to drive transformation and adaptability as future Army leaders.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2  
2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Cadets prepare an individual research report examining an actual large-scale change within the context of a modern organization.

**Corequisite(s):** PL300  
-Or-  
PL350

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**PL482 ARMED FORCES AND SOCIETY**
**3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)**

**Scope:** 2026-1

**Offerings:**

This course examines the intersection of armed forces and society, focusing on two key domains: the relationship between military organizations and the broader societal context, and the study of the military as a distinct social institution with unique demands on its members. Grounded in a military sociological perspective, cadets will analyze the military's role within society and culture in the United States and internationally. The course emphasizes critical reading, writing, and discussion of contemporary military and societal issues, and challenges cadets to apply their understanding to address real-world problems. Cadets will integrate course concepts into a focused research project that explores an aspect of the human dimension of the armed forces and society.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Research paper that applies course concepts to analysis of the current military.

**Prerequisite(s):** PL300  
-Or-  
PL350

2026-1 2026-8 2027-1  
2028-1 2029-1 2029-2  
2029-8

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|--------------|----------------------------------|----------------------------------------------------------|
| <b>PL485</b> | <b>HUMAN FACTORS ENGINEERING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.5,ET=1.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course surveys the theories and methods of human factors engineering (ergonomics). Human factors engineering is concerned with the application of technology and the design of equipment for human use. This course emphasizes the cognitive dimension of human factors engineering. The focus is on understanding the capabilities and limitations of humans as they interact with equipment and facilities. This course lays the foundations for the systematic application of information about humans to the design of equipment and workspace environments.

**Lessons:** 24 @ 75 min (2.000 Att/wk)      **Labs:** 6 @ 75 min

**Special Requirements:** Term project--research report and oral presentation. Students will perform an in-depth analysis of an existing human-machine system. Results will be reported in written and oral formats.

2026-1 2027-1 2028-1  
2029-1

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|---------------|---------------------------------------|----------------------------------------------------------|
| <b>PL488B</b> | <b>CAPSTONE SEM IN BEHAVIORAL SCI</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This capstone seminar engages cadets in advanced study of behavioral science topics through small group discussions and problem-based learning projects. Focusing on applied psychology, organizational psychology, leadership, and the intersection of military and society, the course addresses issues and challenges relevant to the Army and West Point. Cadets collaborate on real-world projects that require critical analysis and practical application of behavioral science principles, preparing them for leadership roles as Army officers. Course topics are dynamic and revised annually to reflect emerging issues and priorities within the military context.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Limited to Behavioral Sciences Majors in their Firstie year.

2026-2 2027-2 2028-2  
2029-1 2029-2

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|---------------|----------------------------------|----------------------------------------------------------|
| <b>PL488C</b> | <b>COLLOQUIUM-BSL-LEADERSHIP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|----------------------------------|----------------------------------------------------------|

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**Scope:** 1984-1

**Offerings:**

The colloquium focuses on advanced study of behavioral science topics and issues using small group discussions of important books and articles of both traditional and contemporary topics in psychology, sociology, organizational leadership, and engineering psychology. It is a reading and discussion course. Subcourse topics are not fixed and are subject to annual revision.

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Two to four critical essays and oral presentations based on readings.

No Course Offerings

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|---------------|---------------------------------|----------------------------------------------------------|
| <b>PL488D</b> | <b>COLLOQUIUM-BSL-SOCIOLOGY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|---------------------------------|----------------------------------------------------------|

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**Scope:** 2020-1

**Offerings:**

The colloquium focuses on advanced study of behavioral science topics and issues using small group discussions of important books and articles of both traditional and contemporary topics in psychology, sociology, organizational leadership, and engineering psychology. It is a reading and discussion course. Subcourse topics are not fixed and are subject to annual revision.

**Lessons:** 20 @ 110 min (1.250 Att/wk)      **Labs:** 0 @ 0 min

2029-2

**Special Requirements:** Two to four critical essays and oral presentations based on readings.

**Prerequisite(s):** PL371

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| <b>PL488E</b> | <b>COLLOQUIUM-BSL-ENGIN PSYCH</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The colloquium focuses on advanced study of behavioral science topics and issues using small group discussions of important books and articles of both traditional and contemporary topics in psychology, sociology, organizational leadership, and engineering psychology. It is a reading and discussion course. Subcourse topics are not fixed and are subject to annual revision.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 28 @ 55 min (2.500 Att/wk)      **Labs:** 12 @ 55 min

**Special Requirements:** None

**Prerequisite(s):** PL485

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|--------------|-------------------------------------------|----------------------------------------------------------|
| <b>PL489</b> | <b>IND STUDY BEH SCI &amp; LDRSHP 3CR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-------------------------------------------|----------------------------------------------------------|

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**Scope:** 2021-2

**Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 6 hours of work per week towards completion of the project, for a total of 120 hours of work.

2026-1 2026-2 2027-1  
2027-2 2027-8 2028-1  
2028-2 2028-8 2029-1  
2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

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|---------------|------------------------------|----------------------------------------------------------|
| <b>PL489A</b> | <b>IND STUDY BSL A (3CR)</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|------------------------------|----------------------------------------------------------|

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**Scope:** 2021-2

**Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 6 hours of work per week towards completion of the project, for a total of 120 hours of work.

2027-2 2028-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. PL489 must be taken before PL489A. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

**Prerequisite(s):** PL489

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|               |                              |                                                          |
|---------------|------------------------------|----------------------------------------------------------|
| <b>PL489B</b> | <b>IND STUDY BSL B (3CR)</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|------------------------------|----------------------------------------------------------|

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**Scope:** 2021-2

**Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Behavioral Sciences and Leadership, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform an average of 6 hours of work per week towards completion of the project, for a total of 120 hours of work.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                               |                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Enrollment by permission of one of the Program Directors in the Department of Behavioral Sciences and Leadership. PL489A must be taken before PL489B. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PL489A                                                                                                                                                                                                                                                        |                                                    |
| <b>PL490</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ENGINEERING PSYCHOLOGY</b>                                                                                                                                                                                                                                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                                                                                                                                                                                                                                                        | <b>Offerings:</b>                                  |
| This course integrates the material previously covered in the Human Factors curriculum, especially PL485, Human Factors Engineering. It uses the theoretical bases and practical applications of Human Factors Engineering in the treatment of design problems. Emphasis in this course is on the design of systems to fit human capabilities. Course project is a design project of a contemporary applied problem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                               | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 16 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 14 @ 75 min                                                                                                                                                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Term project--team design projects applying selected engineering psychology concepts to a contemporary problem.                                                                                                                                               |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PL485                                                                                                                                                                                                                                                         |                                                    |
| <b>PL497</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>THESIS I IN THE DEPT. OF BSL</b>                                                                                                                                                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2021-1                                                                                                                                                                                                                                                        | <b>Offerings:</b>                                  |
| The course provides cadets with an opportunity to enhance their skills in one or a combination of the disciplines hosted in the Department of Behavioral Sciences & Leadership. Under the supervision of a faculty advisor, each cadet defines a topic, conducts a review of the literature, and develops a research plan. Regular meetings with faculty advisors are scheduled to discuss issues in methodology, review progress in the review of the literature, and critique draft papers. At the end of the semester, cadets present a fully developed research proposal for their thesis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                               | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                        |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Literature review and research proposal; oral defense of proposal.                                                                                                                                                                                            |                                                    |
| <b>PL498</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>THESIS II IN THE DEPT. OF BSL</b>                                                                                                                                                                                                                          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2021-1                                                                                                                                                                                                                                                        | <b>Offerings:</b>                                  |
| This course continues the work on the thesis completed in PL497. Under the supervision of a faculty advisor, each cadet executes the proposed thesis plan, submits a written thesis, and presents their project during Projects Day and/or at a professional conference. Regular meetings with faculty advisors are scheduled to discuss issues in methodology, complete data analyses, and critique draft papers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                               | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                        |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Written research report with an oral defense.                                                                                                                                                                                                                 |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PL100 PL361<br>-Or-<br>PL150 PL361<br>-Or-<br>PL100B PL361                                                                                                                                                                                                    |                                                    |
| <b>PL499</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>LEADERS IN ACTION</b>                                                                                                                                                                                                                                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2005-2                                                                                                                                                                                                                                                        | <b>Offerings:</b>                                  |
| This course is designed to enhance cadets' leadership performance through the application of essential leadership skills in challenging, on-going, real-world projects, and scenario-driven leadership laboratory exercises. The course uses a series of "concept study > actions > reflections" (CAR) cycles to focus students on the enhancing (and hindering) factors that typically surface when an individual has responsibility for executing a project and must "do" leadership. Cadets move through a CAR cycle in three related stages. First, cadets consider specific concepts, theories and models of leadership covered in prior courses. Then, using a pool of projects resourced by the faculty expressly for this course, cadets wrestle with real-world leadership projects (such as leading an organizational unit through an unexpected change), keeping these issues and insights in mind. Finally, both during and after the project, cadets engage in self-reflection exercises (e.g., journals) and meet with faculty mentors, to help process and make sense of their leadership experience on both a personal and conceptual level. |                                                                                                                                                                                                                                                               | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                        |                                                    |

**Special Requirements:** Course-long research projects and written and oral reports.

**Prerequisite(s):** PL398

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## Brigade Tactical Department

### 14 Courses

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                   |                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------|
| <b>MD100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CADET BASIC TRAINING</b>       | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2014-0                            | <b>Offerings:</b>                                             |
| New Cadets negotiate a demanding training regimen in soldiering and cadetship. The New Cadet makes the transition into the Corps of Cadets and learns the standards of behavior and values consistent with the mission of the United States Military Academy. The New Cadet is evaluated by the cadet chain of command and summer Tactical Officer during each of two details and receives a grade for MD100, which is a numerically weighted average of the submitted grades.                                                                                                                                              |                                   | 2027-0 2028-0 2029-0                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min            |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                              |                                                               |
| <b>MD101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>4TH CLASS MILITARY PERF I</b>  | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2014-1                            | <b>Offerings:</b>                                             |
| Fourth Class cadets learn to be a follower and to support the chain of command and to develop understanding of and commitment to the duty concept. Fourth Class cadets serve as members of a squad in their assigned companies. The Fourth Class cadet's primary task is to strive for individual excellence in academic, military, and physical activities while developing teamwork among their classmates. The Fourth Class cadet is evaluated by the cadet chain of command and company Tactical Officer and receives a grade for MD101, which is a numerically weighted average of the submitted grades.               |                                   | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min            |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                              |                                                               |
| <b>MD102</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>4TH CLASS MILITARY PERF II</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2014-2                            | <b>Offerings:</b>                                             |
| Fourth Class cadets learn to be a follower and to support the chain of command and to develop understanding of and commitment to the duty concept. Fourth Class cadets serve as members of a squad in their assigned companies. The Fourth Class cadet's primary task is to strive for increased individual excellence in academic, military, and physical activities while optimizing the teamwork among their classmates. The Fourth Class cadet is evaluated by the cadet chain of command and company Tactical Officer and receives a grade for MD102, which is a numerically weighted average of the submitted grades. |                                   | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min            |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                              |                                                               |
| <b>MD200</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CADET FIELD TRAINING</b>       | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2014-0                            | <b>Offerings:</b>                                             |
| Third Class cadets participate in a demanding training program which further develops proficiency in Pre-Commissioning Tasks and exposes the Third Class cadet to tactical tasks that require teamwork. Various training situations provide opportunities for Third Class cadets to serve as leaders where they are observed by the cadet chain of command and officers. The Third Class cadet is evaluated by the cadet chain of command and summer Tactical Officer during each of two details and receives a grade for MD200, which is a numerically weighted average of the submitted grades.                           |                                   | 2027-0 2028-0 2029-0                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min            |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                              |                                                               |
| <b>MD201</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>3RD CLASS MILITARY PERF I</b>  | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2008-1                            | <b>Offerings:</b>                                             |

Third Class cadets serve as team leaders. They learn to lead others by exercising responsibility for the professional development of one or two subordinates, while serving as a member of a larger military unit. Drawing on recent experiences as Fourth Class cadets, Third Class cadets offer expertise and counsel which facilitates the transition process and development of cadet privates. The Third Class cadet is evaluated by the cadet chain of command and company Tactical Officers and receives a grade for MD201, which is a numerically weighted average of the submitted grades.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

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|              |                                   |                                                    |
|--------------|-----------------------------------|----------------------------------------------------|
| <b>MD202</b> | <b>3RD CLASS MILITARY PERF II</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------|

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**Scope:** 2014-1

**Offerings:**

Third Class cadets serve as team leaders. They learn to lead others by exercising responsibility for the professional development of one or two subordinates, while serving as a member of a larger military unit. Drawing on recent experiences as Fourth Class cadets, Third Class cadets offer expertise and counsel which further facilitates the transition process and development of cadet privates. The Third Class cadet is evaluated by the cadet chain of command and company Tactical Officers and receives a grade for MD202, which is a numerically weighted average of the submitted grades.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

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|              |                                       |                                                    |
|--------------|---------------------------------------|----------------------------------------------------|
| <b>MD300</b> | <b>WEST POINT DETAIL CHAIN OF CMD</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------------------|----------------------------------------------------|

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**Scope:** 2014-0

**Offerings:**

First and Second Class cadets will serve as members of the cadet Chain of Command for a West Point Leader Detail. First Class cadets, as Senior NCOs or officers, learn the basic duties and responsibilities of a commissioned officer by leading in a cadet summer training assignment. They set the example for all subordinates by establishing and attaining sound goals, maintaining standards of behavior, and demonstrating values within the organization. They develop subordinates in such a fashion as to foster teamwork, cohesion, and the desire to excel in all areas of endeavor. Second Class cadets serve as cadet noncommissioned officers. This affords them the opportunity to learn and experience the roles and functions of non-commissioned officers in military units while training and leading their subordinates in both individual and collective training. First and Second Class cadets are evaluated by the cadet chain of command and company Tactical Officer and receive a grade for MD300, which is a numerically weighted average of the submitted grades.

2027-0 2028-0 2029-0

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

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|              |                                  |                                                    |
|--------------|----------------------------------|----------------------------------------------------|
| <b>MD301</b> | <b>2ND CLASS MILITARY PERF I</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|----------------------------------|----------------------------------------------------|

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**Scope:** 2008-1

**Offerings:**

Second Class cadets serve in cadet NCO leadership and staff positions in the Corps of Cadets. Second Class cadets lead through subordinate leaders; that is, they lead small military units in which they exercise responsibility for other members through their influence upon subordinate leaders. The Second Class cadet is evaluated by the cadet chain of command and company Tactical Officer and receives a grade for MD301, which is a numerically weighted average of the submitted grades.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

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|              |                                   |                                                    |
|--------------|-----------------------------------|----------------------------------------------------|
| <b>MD302</b> | <b>2ND CLASS MILITARY PERF II</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2008-2

**Offerings:**

Second Class cadets serve in cadet NCO leadership and staff positions in the Corps of Cadets. Second Class cadets lead through subordinate leaders; they lead small military units in which they exercise responsibility for other members through their influence upon subordinate leaders. The Second Class cadet is evaluated by the cadet chain of command and company Tactical Officer and receives a grade for MD302, which is a numerically weighted average of the submitted grades.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------|
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                                      |
| <b>MD400</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>CDT TRP LEADERSHIP TNG (CTLT)</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2014-0                               | <b>Offerings:</b>                                                    |
| First and second class cadets learn the basic duties and responsibilities of a commissioned and noncommissioned officer by leading in an active Army unit during CTLT. CTLT provides first and second class cadets with a realistic leadership environment to observe, while performing duties normally given newly assigned second lieutenants in the active Army. In CTLT, the cadet is evaluated by the officer chain of command in his/her unit and receives a MD400 grade of Pass/Fail.                                                                                                                           |                                      | 2027-0 2028-0 2029-0                                                 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min               |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                                      |
| <b>MD401</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>1ST CLASS MILITARY PERF I</b>     | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2014-1                               | <b>Offerings:</b>                                                    |
| First Class cadets serve in First Sergeant, Command Sergeant Major or officer leadership positions from platoon through brigade level and learn to lead through their personal influence upon both a chain of command and staff. They learn that success as a leader is based on the performance of the unit and demonstrate their capacity to exercise personal self-discipline in the absence of close supervision. The First Class cadet is evaluated by the cadet chain of command and company Tactical Officer and receives a grade for MD401, which is a numerically weighted average of the submitted grades.   |                                      | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min               |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                                      |
| <b>MD402</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>1ST CLASS MILITARY PERF II</b>    | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2014-2                               | <b>Offerings:</b>                                                    |
| First Class cadets serve in First Sergeant, Command Sergeant Major or officer leadership positions from platoon through brigade level and learn to lead through their personal influence upon both a chain of command and staff. They learn that success as a leader is based on the performance of the unit. They demonstrate their capacity to exercise personal self-discipline in the absence of close supervision. The First Class cadet is evaluated by the cadet chain of command and company Tactical Officer and receives a grade for MD402, which is a numerically weighted average of the submitted grades. |                                      | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-0 2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min               |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                                      |
| <b>MD403</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>CBT/CFT CADRE</b>                 | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2014-1                               | <b>Offerings:</b>                                                    |
| MD403 is for First Class cadets who fail MD402 and are therefore placed in a conditioned status at the end of their First Class year. First Class cadets enrolled in MD403 will serve two consecutive details in either Cadet Basic Training or Cadet Field Training. The requirements outlined for MD300 apply to MD403; evaluation is conducted in the same fashion. The grades from each detail are each equally weighted and combined to derive an overall MD403 grade.                                                                                                                                            |                                      | 2026-1 2027-0 2027-1<br>2028-0 2028-1 2029-0<br>2029-1               |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min               |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                                      |
| <b>MD404</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>1ST CL MILITARY PERF-DEC GR</b>   | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2008-1                               | <b>Offerings:</b>                                                    |
| MD404 is offered to cadets who remain at the Academy beyond eight semesters and have participated in all previous military development courses. Cadet officers in MD404 will receive a grade based on performance and requirements specified in MD402. Evaluation is conducted in the same fashion as in MD402.                                                                                                                                                                                                                                                                                                        |                                      | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min               |                                                                      |

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| Special Requirements: | None |
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## Center for Enhanced Performance

### 5 Courses

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                          |                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------|
| <b>RS100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>STUDENT SUCCESS COURSE, PREP</b>      | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2020-1                                   | <b>Offerings:</b>                                             |
| <p>This course is designed to improve cadet academic performance and mental agility. Mastery of a variety of these strategies leads to development of a more self-regulated leader of character. Strategies presented include: help seeking (AI, Library, tutors), organization and time management, class preparation (text study system), test preparation, overcoming nerves, learned optimism and importance of attitude. The strategies mastered are implemented immediately into the cadets' present life at USMA and contribute to life-long learning. The course has no graded assignments. A final pass/fail grade determination is recorded on the cadet transcript.</p> |                                          | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10 @ 55 min (2.500 Att/wk)               | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                     |                                                               |
| <b>RS101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>STUDENT SUCCESS COURSE</b>            | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2020-1                                   | <b>Offerings:</b>                                             |
| <p>This course is designed to improve cadet academic, physical, and leadership performance at USMA. Mastery of a variety of strategies leads to this goal. Strategies presented include: effective thinking, goal setting, time management, textbook study system, concentration, test taking, visualization, memory, note taking, stress management, and others. The strategies mastered are implemented immediately into the cadets' present life at USMA and contribute to life-long learning. The course has graded assignments, a complete student success plan and a course portfolio. A final pass/no-credit grade determination is recorded on the cadet transcript.</p>   |                                          | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 20 @ 55 min (2.500 Att/wk)               | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                     |                                                               |
| <b>RS102</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>READING EFFICIENCY</b>                | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2020-1                                   | <b>Offerings:</b>                                             |
| <p>This course develops flexible reading strategies. Increasing one's base reading rate while maintaining comprehension is accomplished through the use of computer programs, textbook reading, and recreational reading. Various pacing techniques and supervised practice lead to the increase of both reading rate and comprehension, but most of all to the development of a reader who has a variety of strategies to use depending upon the type of reading required.</p>                                                                                                                                                                                                    |                                          | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10 @ 55 min (2.500 Att/wk)               | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                     |                                                               |
| <b>RS103</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INFO LITERACY &amp; CRIT THINKING</b> | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2020-1                                   | <b>Offerings:</b>                                             |
| <p>This course is designed to improve cadet information literacy and critical thinking performance at USMA. Mastery of a variety of strategies leads to this goal. Strategies presented include: effective thinking, goal setting, time management, understanding of the research process, academic/library research skills, evaluation of information sources, critical reading and reasoning. The strategies mastered are implemented immediately into the cadets' present life at USMA and contribute to life-long learning. The course has no graded assignments. A final pass/no-credit grade determination is recorded on the cadet transcript.</p>                          |                                          | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2        |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 20 @ 75 min (2.000 Att/wk)               | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                     |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | EN101<br>-Or-<br>EN151                   |                                                               |
| <b>RS104</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INTRO PERFORMANCE PSYCHOLOGY</b>      | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |

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**Scope:** 2020-2**Offerings:**

This course introduces cadets to the essential principles and skills of performance psychology and provides a framework through which these principles can be applied toward the pursuit of academic, athletic, and military excellence. The continuous cycle of conscious thought processes leading to unconscious mood states affecting physiological change and thus performance execution is examined in the context of each student's own experience of successful performance, and then specific skills which optimally leverage the mind-body connection are taught and practiced. Cognitive skills for confidence development, attention control methods, relaxation techniques, and imagery are taught and practiced. A detailed personal goal setting plan is completed by the course's end.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 10 @ 55 min (0.625 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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# Chemical and Biological Science and Engineering

## 64 Courses

| CH101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | GENERAL CHEMISTRY I       | 4.0 Credit Hours<br>(BS=4.0,ET=0.0,MA=0.0)                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-5                    | <b>Offerings:</b>                                                                                      |
| <p>This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and associated energies. Fundamental concepts, principles, theories, and laws of chemistry are emphasized. Stoichiometry, states of matter, solutions, foundational thermodynamics, acid-base and redox reactions are addressed. The course also provides the student with an introduction to materials chemistry, environmental chemistry, and military chemistry. An extensive laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to reinforce fundamental concepts introduced in the classroom.</p>                                                                           |                           | 2026-1 2026-2 2026-5<br>2027-1 2027-2 2027-5<br>2027-9 2028-1 2028-2<br>2028-5 2028-9 2029-1<br>2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 8 @ 120 min  |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                      |                                                                                                        |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CH151                     |                                                                                                        |
| CH101X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | GENERAL CHEMISTRY I       | 3.5 Credit Hours<br>(BS=3.5,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                    | <b>Offerings:</b>                                                                                      |
| <p>USED FOR SCHEDULING PURPOSES ONLY. This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and the associated energies involved. Fundamental concepts, principles, theories, and laws of chemistry are stressed. Stoichiometry, states of matter, solutions, kinetics, thermodynamics, acid-base and redox equilibria, electro-, organic, and nuclear chemistry are stressed. The course also provides the student with a strong foundation in materials chemistry, the chemistry of life, environmental chemistry, and military chemistry. A laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to illustrate fundamental concepts.</p> |                           | No Course Offerings                                                                                    |
| <b>Lessons:</b> 31 @ 80 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 9 @ 120 min  |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                      |                                                                                                        |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CH101 CH101Y CH151        |                                                                                                        |
| CH101Y                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | GENERAL CHEMISTRY I       | 3.5 Credit Hours<br>(BS=3.5,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                    | <b>Offerings:</b>                                                                                      |
| <p>USED FOR SCHEDULING PURPOSES ONLY. This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and the associated energies involved. Fundamental concepts, principles, theories, and laws of chemistry are stressed. Stoichiometry, states of matter, solutions, kinetics, thermodynamics, acid-base and redox equilibria, electro-, organic, and nuclear chemistry are stressed. The course also provides the student with a strong foundation in materials chemistry, the chemistry of life, environmental chemistry, and military chemistry. A laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to illustrate fundamental concepts.</p> |                           | No Course Offerings                                                                                    |
| <b>Lessons:</b> 31 @ 80 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 9 @ 120 min  |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                      |                                                                                                        |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CH101 CH101X CH151        |                                                                                                        |
| CH102                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | GENERAL CHEMISTRY II      | 4.0 Credit Hours<br>(BS=4.0,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                    | <b>Offerings:</b>                                                                                      |
| <p>This course extends the foundational disciplinary content and practices from General Chemistry I into chemical equilibrium acid/base chemistry, electrochemistry, thermodynamics (entropy and free energy) and kinetics. Basic principles governing organic chemistry is also addressed. The laboratory is integrated within the course. The initial labs develop skills which are they applied to an authentic research problem.</p>                                                                                                                                                                                                                                                                                                                                                                                                               |                           | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                                          |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 10 @ 120 min |                                                                                                        |

**Special Requirements:** None

**Prerequisite(s):** CH101  
-Or-  
CH151

**Disqualifier(s):** CH152

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|               |                             |                                                          |
|---------------|-----------------------------|----------------------------------------------------------|
| <b>CH102X</b> | <b>GENERAL CHEMISTRY II</b> | <b>3.5 Credit Hours</b><br><b>(BS=3.5,ET=0.0,MA=0.0)</b> |
|---------------|-----------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

USED FOR SCHEDULING PURPOSES ONLY. This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and the associated energies involved. Fundamental concepts, principles, theories, and laws of chemistry are stressed. Stoichiometry, states of matter, solutions, kinetics, thermodynamics, acid-base and redox equilibria, electro-, organic, and nuclear chemistry are stressed. The course also provides the student with a strong foundation in materials chemistry, the chemistry of life, environmental chemistry, and military chemistry. A laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to illustrate fundamental concepts.

No Course Offerings

**Lessons:** 31 @ 80 min (2.500 Att/wk)    **Labs:** 9 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** CH151  
-Or-  
CH101

**Disqualifier(s):** CH102  
-Or-  
CH102Y  
-Or-  
CH152

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|               |                             |                                                          |
|---------------|-----------------------------|----------------------------------------------------------|
| <b>CH102Y</b> | <b>GENERAL CHEMISTRY II</b> | <b>3.5 Credit Hours</b><br><b>(BS=3.5,ET=0.0,MA=0.0)</b> |
|---------------|-----------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

USED FOR SCHEDULING PURPOSES ONLY. This course provides a solid background in chemistry principles and applications. It includes a study of the nature of matter, its atomic and molecular structure, and the associated energies involved. Fundamental concepts, principles, theories, and laws of chemistry are stressed. Stoichiometry, states of matter, solutions, kinetics, thermodynamics, acid-base and redox equilibria, electro-, organic, and nuclear chemistry are stressed. The course also provides the student with a strong foundation in materials chemistry, the chemistry of life, environmental chemistry, and military chemistry. A laboratory program is integrated within the course and is designed to develop an appreciation of classical and modern investigative techniques and to illustrate fundamental concepts.

No Course Offerings

**Lessons:** 31 @ 80 min (2.500 Att/wk)    **Labs:** 9 @ 120 min

**Special Requirements:** NONE

**Prerequisite(s):** CH151  
-Or-  
CH101

**Disqualifier(s):** CH102  
-Or-  
  
-Or-  
CH152  
-Or-  
CH102X

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|              |                                |                                                          |
|--------------|--------------------------------|----------------------------------------------------------|
| <b>CH151</b> | <b>ADV GENERAL CHEMISTRY I</b> | <b>4.0 Credit Hours</b><br><b>(BS=4.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

An advanced coverage of the concepts and principles covered in CH101 including a more in-depth laboratory program.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 8 @ 120 min

**Special Requirements:** None

**Disqualifier(s):** CH101



| CH152                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ADV GENERAL CHEMISTRY II                                                                                                                                                                           | 4.0 Credit Hours<br>(BS=4.0,ET=0.0,MA=0.0)                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                                                     |
| An advanced coverage of the concepts and principles covered in CH101-102 including a more in-depth laboratory program with emphasis on instrumental analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                    | No Course Offerings                                                                   |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 10 @ 120 min                                                                                                                                                                          |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                                                                                                                                                                               |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CH151<br>-Or-<br>CH101                                                                                                                                                                             |                                                                                       |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CH102                                                                                                                                                                                              |                                                                                       |
| CH275                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | BIOLOGY                                                                                                                                                                                            | 4.0 Credit Hours<br>(BS=4.0,ET=0.0,MA=0.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                                                     |
| This course provides a broad understanding of biological principles, applications and the relevance of biological science to the military and society. This course consists of an examination of the unity and diversity of life. The course utilizes a reductionist approach to biological study by beginning with an introduction to life at the cellular level and proceeding through Mendelian Genetics, central dogma, DNA technologies, and Darwinian evolution. The course culminates in the application of basic biological principles to human structure and function. Emphasis is placed on course material that is relevant to current environmental issues and disease particularly as these areas apply to military operations. A laboratory program is integrated within the course and is designed to enhance understanding of classical and modern investigative techniques and to illustrate fundamental concepts. |                                                                                                                                                                                                    | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-8<br>2028-1 2028-2 2028-8<br>2029-1 2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 8 @ 120 min                                                                                                                                                                           |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                                                                                                                                                                               |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CH101<br>-Or-<br>CH151                                                                                                                                                                             |                                                                                       |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CH375                                                                                                                                                                                              |                                                                                       |
| CH289                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | INTRODUCTION TO RESEARCH I                                                                                                                                                                         | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                                                     |
| Introduces the methods of research in chemistry, chemical engineering, or life science that includes use of the research literature and instruction in intermediate experimental and theoretical procedures and techniques specific to the cadet's program of study. Under the direct supervision of faculty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                    | 2026-1 2026-2 2026-7<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2                  |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                             |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 40 hours per semester towards completion of the project. |                                                                                       |
| CH289A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | INTRODUCTION TO RESEARCH I(A)                                                                                                                                                                      | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                                                     |
| Introduces the methods of research in chemistry, chemical engineering, or life science that includes use of the research literature and instruction in intermediate experimental and theoretical procedures and techniques specific to the cadet's program of study. Under the direct supervision of faculty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                    | No Course Offerings                                                                   |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                             |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 40 hours per semester towards completion of the project. |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CH289                                                                                                                                                                                              |                                                                                       |
| CH290                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | INTRODUCTION TO RESEARCH II                                                                                                                                                                        | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                    |                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| Continues the development of research methods in chemistry, chemical engineering, or life science that includes use of the research literature and instruction in intermediate experimental and theoretical procedures and techniques specific to the cadet's program of study. Under the direct supervision of faculty.                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                    | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 40 hours per semester towards completion of the project. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CH289                                                                                                                                                                                              |                                                               |
| <b>CH290A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INTRODUCTION TO RESEARCH II(A)</b>                                                                                                                                                              | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| Continues the development of research methods in chemistry, chemical engineering, or life science that includes use of the research literature and instruction in intermediate experimental and theoretical procedures and techniques specific to the cadet's program of study. Under the direct supervision of faculty.                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                    | 2026-1 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 40 hours per semester towards completion of the project. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CH290                                                                                                                                                                                              |                                                               |
| <b>CH290B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INTRODUCTION TO RESEARCH II(B)</b>                                                                                                                                                              | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| Continues the development of research methods in chemistry, chemical engineering, or life science that includes use of the research literature and instruction in intermediate experimental and theoretical procedures and techniques specific to the cadet's program of study. Under the direct supervision of faculty.                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                    | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 40 hours per semester towards completion of the project. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CH290 CH290A                                                                                                                                                                                       |                                                               |
| <b>CH291</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>HEALTHCARE PROFESSIONS SEM I</b>                                                                                                                                                                | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| Healthcare is a complex and rewarding field with numerous career opportunities for military service members. This course is intended for cadets planning to attend medical school and/or seriously interested in a healthcare career. The course will examine important aspects of medicine that transcend the basic sciences including medical history, economics, ethics, and challenges to healthcare in the United States. An introduction to military medicine will be provided with information on deployments, career opportunities, and changes in military healthcare. Coverage of these important topics will provide exposure to key issues and help cadets prepare for futures as medical professionals. |                                                                                                                                                                                                    | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 10 @ 75 min (0.750 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Completion of the fourth-class year.                                                                                                                                                               |                                                               |
| <b>CH292</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>HEALTHCARE PROFESSIONS SEM II</b>                                                                                                                                                               | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| Healthcare is a complex and rewarding field with numerous career opportunities for military service members. This follow-on course is intended for 2nd class cadets planning to attend medical school. Course content will focus on the medical application process and topics of interest to future physicians. Specific subjects will include writing a personal statement, interviewing, selecting a medical school, completing the AMCAS application, and the basics of clinical practice. The seminar will facilitate the development of knowledge and skills key to the medical school application process while providing an opportunity to monitor MCAT preparation.                                         |                                                                                                                                                                                                    | 2026-2 2027-2 2028-2<br>2029-2                                |

**Lessons:** 10 @ 75 min (0.750 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Completion of the third-class year.

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|--------------|-------------------------------|----------------------------------------------------|
| <b>CH300</b> | <b>BIOMEDICAL ENGINEERING</b> | <b>3.0 Credit Hours<br/>(BS=2.0,ET=1.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course is designed to develop an understanding of the field of biomedical engineering. The course covers application of engineering principles to the study of medical physiology. Topics include biomaterials and biomaterial processing, interaction of biomaterials with tissues, tissue engineering, transport modeling, and medical imaging. Mathematical modeling is an integral part of the course.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):**

CH102 MA205 PH206  
-Or-  
CH102 MA255 PH206  
-Or-  
CH102 MA205 PH256  
-Or-  
CH102 MA255 PH256  
-Or-  
CH102 MA205 PH202X  
-Or-  
CH102 MA255 PH202X  
-Or-  
CH102 MA205 PH202  
-Or-  
CH102 MA255 PH202  
-Or-  
CH102 MA205 PH252  
-Or-  
CH102 MA255 PH252  
-Or-  
CH152 MA205 PH202  
-Or-  
CH152 MA255 PH202  
-Or-  
CH152 MA205 PH252  
-Or-  
CH152 MA255 PH252  
-Or-  
CH102 MA204X PH202  
-Or-  
CH102 MA205 PH275  
-Or-  
CH102 MA255 PH275  
-Or-  
CH152 MA205 PH275  
-Or-  
CH152 MA255 PH275  
-Or-  
CH102 MA204 PH202  
-Or-  
CH102 MA204 PH252  
-Or-  
CH152 MA204 PH202  
-Or-  
CH152 MA204 PH252

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| <b>CH350</b> | <b>BIOPROCESS ENGINEERING</b> | <b>3.0 Credit Hours<br/>(BS=1.0,ET=2.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course is designed to develop an understanding of the field of biochemical engineering. The course covers the application of engineering principles to the study of bioprocesses and biochemical reactor design. Topics include enzymes and enzyme kinetics, cell growth and cell growth kinetics, suspension and immobilization of cultures, bioreactor design, scale-up, and control, and recovery and purification technology. Mathematical modeling is an integral part of the course.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** CH102 MA205 PH206  
 -Or-  
 CH102 MA255 PH206  
 -Or-  
 CH102 MA205 PH256  
 -Or-  
 CH102 MA255 PH256  
 -Or-  
 CH102 MA205 PH202X  
 -Or-  
 CH102 MA255 PH202X  
 -Or-  
 CH102 MA205 PH202  
 -Or-  
 CH102 MA255 PH202  
 -Or-  
 CH102 MA205 PH252  
 -Or-  
 CH102 MA255 PH252  
 -Or-  
 CH152 MA205 PH202  
 -Or-  
 CH152 MA255 PH202  
 -Or-  
 CH152 MA205 PH252  
 -Or-  
 CH152 MA255 PH252  
 -Or-  
 CH102 MA204X PH202  
 -Or-  
 CH102 MA205 PH275  
 -Or-  
 CH102 MA255 PH275  
 -Or-  
 CH152 MA205 PH275  
 -Or-  
 CH152 MA255 PH275  
 -Or-  
 CH102 MA204 PH202  
 -Or-  
 CH102 MA204 PH252  
 -Or-  
 CH152 MA204 PH202  
 -Or-  
 CH152 MA204 PH252

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| <b>CH362</b> | <b>MASS &amp; ENERGY BALANCES</b> | <b>3.5 Credit Hours</b><br>(BS=0.0,ET=3.5,MA=0.0) |
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**Scope:** 2026-1

**Offerings:**

Introduction to mass and energy balances in single phase and multiphase, nonreactive and reactive systems. Course topics include an introduction to engineering calculations and process variables, use of computers in solving chemical engineering problems, fundamentals of material balances in single-phase and multi-phase systems, energy balances on nonreactive and reactive processes, applications of combined material and energy balances, introduction to chemical engineering unit operations, and a general introduction to the field of chemical engineering.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** CH102  
 -Or-  
 CH152

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|--------------|-----------------------------|---------------------------------------------------|
| <b>CH363</b> | <b>SEPARATION PROCESSES</b> | <b>3.5 Credit Hours</b><br>(BS=0.0,ET=3.5,MA=0.0) |
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**Scope:** 2026-1

**Offerings:**

This course covers methods for the physical separation of chemicals. Topics include dew point and bubble point calculations, adiabatic flash, distillation, chromatography, liquid-liquid and gas-liquid absorption. Students are taught the significance of staging of unit operations. Heavy emphasis is placed on theory of operation, numerical methods of solution, and simulation.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** CH362

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|--------------|--------------------------------------|----------------------------------------------------------|
| <b>CH364</b> | <b>CHEMICAL REACTION ENGINEERING</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course studies the effects of chemical reaction kinetics on systems of engineering significance. It introduces selection and operation of commercial chemical reactors, emphasizing chemical kinetics and transport phenomena. It studies currently practiced engineering techniques associated with each of these reactors. Topics covered in this course include ideal reactors including batch, CSTR and PFR, isothermal and nonisothermal. Other topics may include catalytic reactors, bioreactors, reactors, transient and steady state design, pressure drop in reactors, recycle, stability, and numerical methods.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** CH362

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>CH365</b> | <b>CHEMICAL ENG THERMODYNAMICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course covers the body of thermodynamic knowledge necessary for understanding modern chemical process simulation. Students learn the theory behind the thermodynamic methods used in the software. The course includes calculus- and numerical-based thermodynamics approaches for determining the properties of substances, solutions, and multiphase mixtures. Topics include equations of state, pure component properties, transport properties, properties of mixtures, fugacity, excess properties, activity coefficients, and phase equilibria. The problems in the course emphasize engineering applications. Topics covered in class are related to real systems through the use of chemical process simulators.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** CH363 CH364 MA366 ME301  
-Or-  
CH363 CH364 MA364 ME301  
-Or-  
CH363 CH364 MA365 ME301  
-Or-  
CH363 CH364 MA364 MC311

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>CH367</b> | <b>INTRO / AUTOMATIC PROC CONTROL</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course covers the principles necessary to understand the automatic control of chemical processes. Students learn the current mathematical models and mechanical details of various control elements, including sensors, transmitters, actuators, and controllers. Application of mathematical models will be covered with dynamic modeling techniques as well as real-time training using process simulators. The course will also cover tuning of controllers as well as safe response to process upsets. A capstone project will involve dynamic modelling of an integrated process control system.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** EE301 MA366  
-Or-  
EE301 MA364  
-Or-  
EE301 MA365

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| CH371                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | INTRO TO ANALYTICAL CHEM      | 3.5 Credit Hours<br>(BS=3.5,ET=0.0,MA=0.0)   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                        | <b>Offerings:</b>                            |
| <p>The course teaches the fundamental concepts of analytical chemistry. Topics include acid-base equilibria, redox potentials, compleximetric titrimetry, separations, electrochemistry, and absorption spectroscopy. The course provides an overview of modern analytical techniques being used in various fields. The course emphasizes the development of rigorous laboratory techniques and introduces the cadet to computer based data acquisition. Cadet laboratory work is evaluated in terms of the student's ability to accurately determine the identity and quantity of an unknown sample.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                               |                                              |
| <b>Lessons:</b> 29 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 17 @ 120 min     | 2026-2 2027-2 2028-2<br>2029-2               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                          |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CH102<br>-Or-<br>CH152        |                                              |
| CH375                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ADVANCED BIOLOGY              | 4.0 Credit Hours<br>(BS=4.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                        | <b>Offerings:</b>                            |
| <p>This course focuses on the evolution, diversity, structure, and processes of living organisms. The curriculum starts with a description of the structure and function of cells and their metabolism. These topics are used to explain DNA replication, transcription, translation and gene expression. In turn these concepts serve as a foundation of knowledge examining natural selection, population genetics, biodiversity, and ecology. Emphasis is placed on related course material to current environmental issues and disease, particularly as these areas apply to military operations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                               |                                              |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 8 @ 120 min      | 2026-1 2027-1 2028-1<br>2029-1               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                          |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CH101<br>-Or-<br>CH151        |                                              |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CH102<br>-Or-<br>CH152        |                                              |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CH275                         |                                              |
| CH376                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ADV. TECH IN LIGHT MICROSCOPY | 3.0 Credit Hours<br>(BS=3.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                        | <b>Offerings:</b>                            |
| <p>This course provides a deep understanding of the wide variety of advanced techniques of light microscopy used universally to investigate living and fixed systems at the cellular and molecular level. This course consists of an examination of optical-based microscopy techniques based on light's behavior as a ray, as a wave, and as a particle. Microscopy techniques below the diffraction limit of light and ethics in image analysis are also studied. This course uses both theoretical and hands-on methods to develop a deeper understanding of the fundamental properties underlying the use of microscopy as a tool for the study of the amazingly diverse world too small for humans to see with the unaided eye. The course culminates in the use of laser scanning techniques to probe the microscopic world below the Abbe diffraction limit of light which was once believed to be impossible with light microscopy. Emphasis is placed on advanced microscopy concepts, equipment, problems and traditional microscopy challenges particularly as these areas apply to challenges facing military and civilian researchers in the world today.</p> |                               |                                              |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min        | 2026-1 2027-1 2027-2<br>2028-1 2028-2 2029-2 |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                          |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CH275<br>-Or-<br>CH375        |                                              |
| CH383                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ORGANIC CHEMISTRY I           | 3.5 Credit Hours<br>(BS=3.5,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                        | <b>Offerings:</b>                            |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------|
| Organic chemistry I is an introduction to the relationship between chemical structure and the physical and chemical properties of molecules. A qualitative description of structure and bonding is presented. The relationships between free energy changes and equilibria, and between activation energy and rate of reaction are developed. Stereochemistry and isomerism are explored. The concept of the mechanism of reaction is presented and the relationships between mechanism, the least energy path, stable intermediates and transition states are exemplified by the reactions of the alkanes, alkenes, alkyl halides, and alcohols. The use of instrumental methods of structural analysis is also introduced. |                                     | 2026-1 2026-8 2027-1<br>2028-1 2029-1             |
| <b>Lessons:</b> 34 @ 80 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 15 @ 120 min           |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CH102<br>-Or-<br>CH152              |                                                   |
| <b>CH384</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ORGANIC CHEMISTRY II</b>         | <b>3.5 Credit Hours</b><br>(BS=3.5,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                              | <b>Offerings:</b>                                 |
| The reactions of the important functional groups are explored: conjugated alkenes; aldehydes; ketones; carboxylic acids; and amines. The concept of aromaticity is explored and its mechanistic implications are developed. Selected topics in carbohydrate and lipid chemistry are also studied. Functional group interconversions and synthetic strategy are presented. The laboratory capstone synthesis introduces cadets to multi-step synthetic sequences.                                                                                                                                                                                                                                                             |                                     | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 34 @ 80 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 15 @ 120 min           |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CH383                               |                                                   |
| <b>CH385</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>INTRODUCTION TO CELL BIOLOGY</b> | <b>3.5 Credit Hours</b><br>(BS=3.5,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                              | <b>Offerings:</b>                                 |
| The course will cover the structure and function of prokaryotic and eukaryotic cells. The course will present a detailed discussion on the molecular biology of DNA replication, transcription, translation, the control of gene expression, cell-to-cell signaling, and the cytoskeleton. Emphasis will be placed on research methods and techniques that have lead to our understanding of how the cell works.                                                                                                                                                                                                                                                                                                             |                                     | 2026-1 2027-1 2027-8<br>2028-1 2028-8 2029-1      |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 7 @ 120 min            |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CH375                               |                                                   |
| <b>CH387</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>HUMAN PHYSIOLOGY</b>             | <b>3.5 Credit Hours</b><br>(BS=3.5,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                              | <b>Offerings:</b>                                 |
| This course consists of an in-depth study of human physiology and the interrelationships between major organs and systems of the body. This course will concentrate on homeostatic reflex mechanisms of the human body. Major topics covered include endocrinology, neural physiology, muscles, cardiovascular physiology, respiratory physiology, renal physiology, digestion, immunology, and reproductive physiology. The laboratory program reinforces the foundational principles of thermoregulation, muscle, cardiac, respiratory, neural, sensory and renal physiology and introduces cadets to basic laboratory measurements and diagnostics for each of those subjects.                                            |                                     | 2026-2 2026-8 2027-2<br>2028-2 2029-2             |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 8 @ 120 min            |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CH375<br>-Or-<br>CH275              |                                                   |
| <b>CH388</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>GENETICS</b>                     | <b>3.0 Credit Hours</b><br>(BS=3.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                              | <b>Offerings:</b>                                 |

Genetic analysis has transformed the ability to investigate organisms from single genes to complete genomes. This course examines the structure and function relationships of inheritance and genetic variation while introducing fundamental methodologies for the modern analysis of genes and genomes. Students will learn the fundamental principles of molecular genetics, heredity and genetic diseases, high throughput sequencing, and quantitative genetics. It is expected that students will enter with a basic understanding of the characteristics of life, and the nature of DNA & RNA organization, replication, and transcription. Students will develop problem-solving skills to extract and utilize genetic information for use in scientific research and medicine while considering the bioethics of this rapidly evolving field.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** All cadets will complete a Mendelian Project.

**Corequisite(s):** CH375

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|              |                                |                                                          |
|--------------|--------------------------------|----------------------------------------------------------|
| <b>CH389</b> | <b>ADVANCED LAB PROJECTS I</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------|----------------------------------------------------------|

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**Scope:** 2025-7

**Offerings:**

The development of usable protocols, procedures, or laboratory experiments to advance current research projects directed by a member of the faculty. Individual cadets must gain the consent of the faculty member and present project title and scope of proposed effort for Program Director approval.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2  
2029-7

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform 80 hours per semester towards completion of the project.

**Prerequisite(s):** CH101  
-Or-  
CH151

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>CH390</b> | <b>ADVANCED LAB PROJECTS II</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

The development of usable protocols, procedures, or laboratory experiments to advance current research projects directed by a member of the faculty. Project can be either a continuation of CH389 or a new project limited to the scope of 2.0 credit hours. Individual cadets must gain the consent of the faculty member and present project title and scope of proposed effort for Program Director approval.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform 80 hours per semester towards completion of the project.

**Prerequisite(s):** CH389

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|              |                             |                                                          |
|--------------|-----------------------------|----------------------------------------------------------|
| <b>CH391</b> | <b>ADV LAB PROJECTS III</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The development of usable protocols, procedures, or laboratory experiments to advance current research projects directed by a member of the faculty. Project can be either a continuation of CH390 or a new project limited to the scope of 2.0 credit hours. Individual cadets must gain the consent of the faculty member and present project title and scope of proposed effort for Program Director approval.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform 80 hours per semester towards completion of the project.

**Prerequisite(s):** CH390

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|              |                                 |                                                          |
|--------------|---------------------------------|----------------------------------------------------------|
| <b>CH392</b> | <b>ADVANCED LAB PROJECTS IV</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**



The development of usable protocols, procedures, or laboratory experiments to advance current research projects directed by a member of the faculty. Project can be either a continuation of CH391 or a new project limited to the scope of 2.0 credit hours. Individual cadets must gain the consent of the faculty member and present project title and scope of proposed effort for Program Director approval.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform 80 hours per semester towards completion of the project.

**Prerequisite(s):**

CH391

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|---------------|-----------------------------------|----------------------------------------------------------|
| <b>CH392A</b> | <b>ADVANCED LAB PROJECTS IV A</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:**

2026-1

**Offerings:**

The development of usable protocols, procedures, or laboratory experiments to advance current research projects directed by a member of the faculty. Project can be either a continuation of CH391, CH392 or a new project limited to the scope of 2.0 credit hours. Individual cadets must gain the consent of the faculty member and present project title and scope of proposed effort for Program Director approval.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform 80 hours per semester towards completion of the project. Cadets should complete CH392 before enrolling in CH392A.

**Prerequisite(s):**

CH391 CH392

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|---------------|-----------------------------------|----------------------------------------------------------|
| <b>CH392B</b> | <b>ADVANCED LAB PROJECTS IV B</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|-----------------------------------|----------------------------------------------------------|

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**Scope:**

2026-1

**Offerings:**

The development of usable protocols, procedures, or laboratory experiments to advance current research projects directed by a member of the faculty. Project can be either a continuation of CH391, CH392, CH392A or a new project limited to the scope of 2.0 credit hours. Individual cadets must gain the consent of the faculty member and present project title and scope of proposed effort for Program Director approval.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform 80 hours per semester towards completion of the project. Cadets should complete CH392A before enrolling in CH392B.

**Prerequisite(s):**

CH391 CH392 CH392A

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|               |                                   |                                                          |
|---------------|-----------------------------------|----------------------------------------------------------|
| <b>CH392C</b> | <b>ADVANCED LAB PROJECTS IV C</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|-----------------------------------|----------------------------------------------------------|

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**Scope:**

2026-1

**Offerings:**

The development of usable protocols, procedures, or laboratory experiments to advance current research projects directed by a member of the faculty. Project can be either a continuation of CH391, CH392, CH392A, CH392B or a new project limited to the scope of 2.0 credit hours. Individual cadets must gain the consent of the faculty member and present project title and scope of proposed effort for Program Director approval.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform 80 hours per semester towards completion of the project. Cadets should complete CH392B before enrolling in CH392C.

**Prerequisite(s):**

CH391 CH392 CH392A CH392B

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>CH399</b> | <b>TOPICS IN CHEM/LS/CHMENG</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------------|----------------------------------------------------------|

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**Scope:**

2026-1

**Offerings:**

This course provides in-depth study of a special topic in chemistry, chemical engineering and life science not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the Visiting Professor, Rotating PhD, or a senior faculty member. This course may also be offered as an AIAD course at USMA.

2026-8 2027-8 2028-8  
2029-8

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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|               |                                 |                                                    |
|---------------|---------------------------------|----------------------------------------------------|
| <b>CH399A</b> | <b>TOPICS IN CHEM/LS/CHMENG</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course provides in-depth study of a special topic in chemistry, chemical engineering and life science not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the Visiting Professor, Rotating PhD, or a senior faculty member. This course may also be offered as an AIAD course at USMA.

No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 120 @ 0 min

**Special Requirements:** None

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|              |                                       |                                                    |
|--------------|---------------------------------------|----------------------------------------------------|
| <b>CH400</b> | <b>CHEM ENG PROFESSIONAL PRACTICE</b> | <b>1.5 Credit Hours<br/>(BS=0.0,ET=1.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The course will meet once per week and will cover topics such as ethics, continuing education, and global and social issues within chemical engineering. Special emphasis will be placed on preparation for the Fundamentals of Engineering Exam using practice problems and graded practice exams. The course also covers professional plant engineering using plant simulators and mock exercises to teach proper troubleshooting and response techniques.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 20 @ 55 min (1.250 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** CH365 CH459 CH485

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|              |                                |                                                    |
|--------------|--------------------------------|----------------------------------------------------|
| <b>CH402</b> | <b>CHEM ENG PROCESS DESIGN</b> | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course provides a capstone experience that brings together material from previous courses to examine contemporary problems in chemical engineering process design. The course provides instruction in the conceptual design of processes to achieve design goals, as well as the economic optimization of the process. The course emphasizes the use of computer simulations, theory of unit operations, process control, safety, environmental and economic factors. The effect of changes in design on the process economics will be investigated. Written design reports for the capstone design project are required.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (3.000 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** The completion of significant out-of-class design problems requiring the equivalent of 2.5 credit hour of student effort. Compensatory time is provided to complete the design requirement.

**Prerequisite(s):** CH365 CH459 CH485

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|--------------|-------------------------------------------|----------------------------------------------------|
| <b>CH450</b> | <b>BIOENGINEERING MDLING &amp; ANLSIS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course provides a broad understanding of bioengineering disciplines to include biomechanics, biomaterials, tissue engineering, biocatalysis, biochemical engineering, and biosensors. Fundamental concepts of molecular kinetics, thermodynamics, and mass transport are applied in problem sets in each bioengineering sub-discipline and capstone design project providing students the opportunity for modeling, analysis, and design from the biomolecular to physiological length scale and across multiple time scales. Modeling software such as MATLAB and Mathematica is extensively used.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

|                         |                    |
|-------------------------|--------------------|
| <b>Prerequisite(s):</b> | CH102 MA205 PH206  |
|                         | -Or-               |
|                         | CH102 MA255 PH206  |
|                         | -Or-               |
|                         | CH102 MA205 PH256  |
|                         | -Or-               |
|                         | CH102 MA255 PH256  |
|                         | -Or-               |
|                         | CH102 MA205 PH202X |
|                         | -Or-               |
|                         | CH102 MA255 PH202X |
|                         | -Or-               |
|                         | CH102 MA205 PH202  |
|                         | -Or-               |
|                         | CH102 MA255 PH202  |
|                         | -Or-               |
|                         | CH102 MA205 PH252  |
|                         | -Or-               |
|                         | CH102 MA255 PH252  |
|                         | -Or-               |
|                         | CH152 MA205 PH202  |
|                         | -Or-               |
|                         | CH152 MA205 PH252  |
|                         | -Or-               |
|                         | CH152 MA255 PH202  |
|                         | -Or-               |
|                         | CH152 MA255 PH252  |
|                         | -Or-               |
|                         | CH102 MA204X PH202 |
|                         | -Or-               |
|                         | CH102 MA205 PH275  |
|                         | -Or-               |
|                         | CH102 MA255 PH275  |
|                         | -Or-               |
|                         | CH152 MA205 PH275  |
|                         | -Or-               |
|                         | CH152 MA255 PH275  |
|                         | -Or-               |
|                         | CH102 MA204 PH202  |
|                         | -Or-               |
|                         | CH102 MA204 PH251  |
|                         | -Or-               |
|                         | CH152 MA204 PH202  |
|                         | -Or-               |
|                         | CH152 MA204 PH252  |

|              |                     |                                                    |
|--------------|---------------------|----------------------------------------------------|
| <b>CH457</b> | <b>MICROBIOLOGY</b> | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b> |
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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2026-1                   | <b>Offerings:</b>              |
| <p>This course introduces the diversity of microorganisms in all three domains of life. The course covers prokaryotic cell structure and function, growth, genetics, and metabolism. The course will survey five major groups of microorganisms: eubacteria, archaea, protozoa, fungi and viruses including ecology, their role in human disease and their applications in medicine, industry and warfare. Cadets have the opportunity to explore both a viral and a bacterial disease in-depth and present their findings in a briefing and a paper. The 18-hour laboratory program focuses on practical applications of concepts covered in class, with a particular emphasis on the eubacteria. The lab program culminates with a hands-on laboratory examination.</p> |                          | 2026-1 2027-1 2028-1<br>2029-1 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 7 @ 120 min |                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                     |                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CH375                    |                                |

|              |                             |                                                    |
|--------------|-----------------------------|----------------------------------------------------|
| <b>CH459</b> | <b>CHEM ENGR LABORATORY</b> | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                        | 2026-1                    | <b>Offerings:</b>              |
| <p>This course provides laboratory experience in selected chemical engineering unit operations, such as gas absorption, evaporation, distillation, liquid-liquid extraction, cooling tower, heat exchanger, and chemical reactors. Process control and process safety are emphasized in laboratory and classroom instruction. Written and oral reports required.</p> |                           | 2026-1 2027-1 2028-1<br>2029-1 |
| <b>Lessons:</b> 7 @ 120 min (3.000 Att/wk)                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 40 @ 120 min |                                |

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|------------------------------|-------------------|
| <b>Special Requirements:</b> | None              |
| <b>Prerequisite(s):</b>      | CH362 CH363 CH364 |

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|--------------|----------------------|----------------------------------------------------|
| <b>CH460</b> | <b>HUMAN ANATOMY</b> | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1 | <b>Offerings:</b>              |
| This course is designed to provide cadets with a detailed study of the anatomical structure of the human body. Body structure will be studied by organ systems and will involve a balance between gross anatomical study and histology. Form-function relationships will be emphasized. The laboratory study will involve working with human skeletal collections and virtual dissection of cadavers and preserved specimens. The 14-hour laboratory program focuses on structural identification (naming) of human and mammalian anatomy and various imaging modalities (e.g., radiographs, CT scans), and computer programs. Cadets that successfully complete this course will have a good understanding of human body structure, construction, and function. |        | 2026-2 2027-2 2028-2<br>2029-2 |

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 7 @ 120 min

|                              |                                                                                                          |
|------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Special Requirements:</b> | None                                                                                                     |
| <b>Prerequisite(s):</b>      | CH102 CH375 CH387<br>-Or-<br>CH102 CH385 CH387<br>-Or-<br>CH152 CH375 CH387<br>-Or-<br>CH152 CH385 CH387 |

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|--------------|--------------------------|----------------------------------------------------|
| <b>CH471</b> | <b>POLYMER CHEMISTRY</b> | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1 | <b>Offerings:</b>              |
| This course is an introduction to modern polymer chemistry. It provides an introduction to macromolecules and their properties. It covers polymerization methods, copolymerization, the morphology of polymers, and the testing and characterization of polymer products. The course also introduces polymer additives, natural and biomedical polymers and modern polymer applications, emphasizing the military uses of polymer products. |        | 2026-2 2027-2 2028-2<br>2029-2 |

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 7 @ 120 min

|                              |                        |
|------------------------------|------------------------|
| <b>Special Requirements:</b> | None                   |
| <b>Prerequisite(s):</b>      | CH102<br>-Or-<br>CH152 |

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|--------------|----------------------------|----------------------------------------------------|
| <b>CH472</b> | <b>INORGANIC CHEMISTRY</b> | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1 | <b>Offerings:</b>              |
| This course features an in-depth study of main group and transition elements and their compounds, with emphasis on chemical bonding and both atomic and molecular structures. The fundamentals of quantum chemistry to include the valence bond and molecular orbital theories as applied to inorganic chemistry are studied. An introduction to symmetry/group theory, coordination chemistry/crystal field theory, chemistry in aqueous and nonaqueous solutions, and organometallic compounds are also included in the course. Chemical principles and spectroscopic techniques will also be emphasized. Journal articles from the chemical literature are used to supplement the text with topics of current interest. |        | 2026-1 2027-1 2028-1<br>2029-1 |

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 7 @ 120 min

|                              |                     |
|------------------------------|---------------------|
| <b>Special Requirements:</b> | None                |
| <b>Prerequisite(s):</b>      | -Or-<br>CH384 CH481 |

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|--------------|---------------------|----------------------------------------------------|
| <b>CH473</b> | <b>BIOCHEMISTRY</b> | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b> |
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|---------------|--------|-------------------|
| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

This course is an introduction to biochemical systems and concentrates on studying them from the molecular approach. Three themes are emphasized: 1) Structure - Function relationships, 2) Metabolism, and 3) Regulation of the systems and processes studied. The fundamental goals of the course are to provide students the basic knowledge of biochemistry and to give them a framework for analyzing problems and questions in life science studies. Additional emphasis is placed on familiarizing students with the experimental techniques used in biochemistry and their application to current issues of interest.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** None

**Corequisite(s):** CH384

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>CH474</b> | <b>INSTRU METHODS OF ANALYSIS</b> | <b>3.5 Credit Hours</b><br><b>(BS=3.5,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

A laboratory course designed to develop proficiency in the selection and use of modern instrumental methods of chemical analysis. Topics include atomic spectroscopy, molecular absorption and fluorescence spectroscopy, infrared and Raman spectroscopy, nuclear magnetic resonance and mass spectrometry, and chromatography. The laboratory program includes a Capstone experimental procedure and methodology design component. Cadet laboratory work is evaluated in terms of the student's ability to determine the proper instrumental methodology to analyze a chemical sample.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 29 @ 55 min (2.500 Att/wk)      **Labs:** 15 @ 120 min

**Special Requirements:** One project report on a selected research topic.

**Prerequisite(s):**  
CH371 PH204  
-Or-  
CH371 PH254  
-Or-  
CH371 PH202  
-Or-  
CH371 PH252  
-Or-  
CH371 PH206  
-Or-  
CH371 PH256  
-Or-  
CH371 PH202X  
-Or-  
CH371 PH275

**Corequisite(s):** CH384

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|--------------|--------------------------------------|----------------------------------------------------------|
| <b>CH479</b> | <b>METHODS &amp; APPS OF BIOTECH</b> | <b>3.5 Credit Hours</b><br><b>(BS=3.5,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

This course is intended to reinforce topics learned in other life science courses by studying laboratory and practical applications of biotechnology. Laboratories will concentrate on biotechnology methods including purification, separation, and identification of DNA, RNA and protein. Other biotechnology techniques that will be studied include recombinant DNA techniques, PCR, and DNA sequencing. Classroom lessons will include discussions of assigned readings on the modern applications of biotechnology.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 23 @ 55 min (2.500 Att/wk)      **Labs:** 24 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** CH388 CH457

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|              |                             |                                                          |
|--------------|-----------------------------|----------------------------------------------------------|
| <b>CH481</b> | <b>PHYSICAL CHEMISTRY I</b> | <b>3.5 Credit Hours</b><br><b>(BS=3.5,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

The major areas of study in this course are chemical thermodynamics with a special focus on chemical equilibrium, and chemical kinetics, introduction to intermolecular interactions. Some of the specific topics covered include properties of real gases, the kinetic theory of gases, the laws of thermodynamics as related to chemical systems, diffusion as a description of mass transport, rates of chemical reactions, and molecular reaction dynamics. The laboratory program illustrates the fundamental topics covered through precision measurements, utilizing modern instrumental and computational methods.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** None

**Corequisite(s):** CH383

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|              |                              |                                                          |
|--------------|------------------------------|----------------------------------------------------------|
| <b>CH482</b> | <b>PHYSICAL CHEMISTRY II</b> | <b>3.5 Credit Hours</b><br><b>(BS=3.5,ET=0.0,MA=0.0)</b> |
|--------------|------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

This course builds on the concepts covered in CH481 through a study of the quantum mechanics of atoms and molecules, their interaction with radiation, and statistical thermodynamics. Some of the specific topics covered include the electronic structure of atoms and molecules, molecular geometry, molecular symmetry, several types of spectroscopy used for identification and monitoring of the local molecular environment, and the details of molecular motion. Various levels of theory are used to obtain increasingly more accurate descriptions of atomic and molecular systems with user-friendly software tools. Statistical thermodynamics enables understanding about the connection between the microscopic details in quantum mechanics and the macroscopic observations made in the laboratory. The laboratory program illustrates the fundamental topics through use of modern instrumental and computational methods.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):**  
-Or-  
CH481

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|              |                               |                                                          |
|--------------|-------------------------------|----------------------------------------------------------|
| <b>CH485</b> | <b>HEAT AND MASS TRANSFER</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
|--------------|-------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

This course includes the study of the mechanisms of energy and mass transport, with special emphasis on applications in engineering systems. Coverage includes Fourier's Law of Heat Conduction, and Fick's Law of Diffusion, the development of shell energy and species balances, and the use of these equations to solve for temperature and concentration profiles in chemical engineering systems. An important emphasis in the course is the use of transport equations to understand species diffusion, convection, and chemical reaction in equipment design.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):**  
MA366 ME362  
-Or-  
MA364 ME362  
-Or-  
MA365 ME362  
-Or-  
MA364 MC312

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|              |                                      |                                                          |
|--------------|--------------------------------------|----------------------------------------------------------|
| <b>CH487</b> | <b>ADVANCED CHEMISTRY LABORATORY</b> | <b>3.0 Credit Hours</b><br><b>(BS=3.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

In this laboratory course cadets will further develop their knowledge and understanding of organic and inorganic syntheses, quantitative and qualitative instrumental analysis, and applications of physical chemistry principles to molecular structure and kinetics. They will carry out experiments based on current needs and applications of the Army. Cadets and faculty will also discuss current research and present their work.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 5 @ 55 min (2.500 Att/wk)    **Labs:** 35 @ 120 min

**Special Requirements:** Library research and written reports are required. All cadets will complete an integrative experience group project that will investigate the social, political and economical implications of chemistry.

**Prerequisite(s):** CH474

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|              |                              |                                                          |
|--------------|------------------------------|----------------------------------------------------------|
| <b>CH489</b> | <b>INDIVIDUAL RESEARCH I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|------------------------------|----------------------------------------------------------|

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**Scope:** 2025-7

**Offerings:**

This undergraduate research course is designed to significantly advance the cadet's knowledge and comprehension of science and/or engineering by answering a real world scientific question. Course work includes defining a problem, understanding related issues, designing an experimental approach, analyzing data, and drawing conclusions. By applying the scientific method to attempt to solve an actual problem, cadets will expand their critical thinking and intellectual capability. Cadets are supervised by a faculty advisor with expertise in the chosen research area. Cadets conduct research individually but may be part of a larger group working on a project with a broad scope. The minimum requirement for moving onto CH490 is a defined problem and hypothesis, a background in related research, and an experimental design. The Head of the Department will approve cadet projects. Lessons and labs will be established through consultation between cadet and advisor. Requirements include both written and oral progress reports.

2026-1 2026-2 2026-7  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 120 hours per semester towards completion of the project.

**Prerequisite(s):**

CH102  
-Or-  
CH152

**CH489A**

**IND RESEARCH I**

**3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:**

2026-1

**Offerings:**

2026-1

Same scope as CH489, but a continuation of research.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 120 hours per semester towards completion of the project.

**Corequisite(s):**

CH489

**CH490**

**INDIVIDUAL RESEARCH II**

**3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:**

2025-7

**Offerings:**

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

With the same outcome goal as CH489 of significantly advancing the cadet's knowledge and comprehension of science and engineering into answering a real world scientific question, this course typically involves experimentation, data analysis, data evaluation, and publishing results. Cadets are supervised by a faculty advisor with expertise in the chosen research area. Cadets conduct research individually but may be part of a larger group working on a project with a broad scope. This course could conclude in a poster presentation, publication and/or a conference presentation as the undergraduate researcher contributes to the larger scientific community. Requirements include both a written final report and an oral presentation.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 120 hours per semester towards completion of the project.

**Prerequisite(s):**

CH489

**CH490A**

**INDIVIDUAL RESEARCH II - A**

**3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:**

2026-1

**Offerings:**

No Course Offerings

With the same outcome goal as CH489 and CH490 of significantly advancing the cadet's knowledge and comprehension of science and engineering into answering a real world scientific question, this course typically involves experimentation, data analysis, data evaluation, and publishing results. Cadets are supervised by a faculty advisor with expertise in the chosen research area. Cadets conduct research individually but may be part of a larger group working on a project with a broad scope. This course could conclude in a poster presentation, publication and/or a conference presentation as the undergraduate researcher contributes to the larger scientific community. Requirements include both a written final report and an oral presentation.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 120 hours per semester towards completion of the project. Cadets must complete CH490 before taking CH490A.

**Prerequisite(s):**

CH490

| CH491                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ADVANCED INDIVIDUAL STUDY I                                                                                                                                                                                                                                | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-7                                                                                                                                                                                                                                                     | <b>Offerings:</b>                                             |
| <p>This undergraduate research course is structured similarly to both CH489 and CH490. In this course, a cadet may continue on an established research problem or begin a new line of investigation. The minimum requirement for moving onto CH491 is a defined problem and hypothesis, a background in related research, and an experimental design. Requirements include both written and oral progress reports. Written recommendation from Department Head must be presented to AARS and approved by the Dean of the Academic Board as this course constitutes a third semester of independent study.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                            | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 120 hours per semester towards completion of the project.                                                        |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CH490                                                                                                                                                                                                                                                      |                                                               |
| CH492                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ADVANCED INDIVIDUAL STUDY II                                                                                                                                                                                                                               | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-7                                                                                                                                                                                                                                                     | <b>Offerings:</b>                                             |
| <p>The Advanced Individual Study provides cadets the unique opportunity to complete a carefully defined question or problem researched over the course of the previous 1.5 to 2 years. Based in research, this problem may be critical, experimental, applied, or creative in nature, and represents an effort to make an original contribution to the field. The Research Thesis is a culmination of a research effort that goes beyond normal requirements of the major and represents the cadet's best work in their discipline. Each thesis must demonstrate clear critical thinking, a mastery of disciplinary material, clarity in communication of complex ideas, and professionalism in production. Cadets must complete a written thesis and present an oral thesis defense to the faculty and staff. Additionally cadets having completed a year of research on one topic would be able to explore another researcher topic altogether. Written recommendation from Department Head must be presented to AARS and approved by the Dean of the Academic Board.</p> |                                                                                                                                                                                                                                                            | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 120 hours per semester towards completion of the project.                                                        |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CH491                                                                                                                                                                                                                                                      |                                                               |
| CH492A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ADVANCED INDIVIDUAL STUDY II-A                                                                                                                                                                                                                             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                                                                                                                                                     | <b>Offerings:</b>                                             |
| <p>The Advanced Individual Study provides cadets the unique opportunity to complete a carefully defined question or problem researched over the course of the previous 1.5 to 2 years. Based in research, this problem may be critical, experimental, applied, or creative in nature, and represents an effort to make an original contribution to the field. The Research Thesis is a culmination of a research effort that goes beyond normal requirements of the major and represents the cadet's best work in their discipline. Each thesis must demonstrate clear critical thinking, a mastery of disciplinary material, clarity in communication of complex ideas, and professionalism in production. Cadets must complete a written thesis and present an oral thesis defense to the faculty and staff. Additionally cadets having completed a year of research on one topic would be able to explore another researcher topic altogether. Written recommendation from Department Head must be presented to AARS and approved by the Dean of the Academic Board.</p> |                                                                                                                                                                                                                                                            | 2027-1 2028-1 2029-1                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LESSONS and LABS: Established by consultation between the cadet and his/her faculty advisor. Cadets are expected to perform an average of 120 hours per semester towards completion of the project. Cadet must complete CH492 prior to taking this course. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CH492                                                                                                                                                                                                                                                      |                                                               |
| CH498                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ADV STUDY IN PRE-MED SCIENCE                                                                                                                                                                                                                               | 0.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                                                                                                                                                     | <b>Offerings:</b>                                             |



The cadet pursues advanced study of topics in preparation for the Medical College Admissions Test (MCAT) and the medical school application process on an individual or small group basis, independent of a formal classroom setting. MCAT preparation includes the following areas: Biological and Biochemical Foundations of Living Systems, Chemical and Physical Foundations of Biological Systems, Psychological, Social, and Biological Foundations of Behavior, and Critical Analysis and Reasoning Skills. Primary and secondary medical school applications are completed during this period. Like graduate level study and preparation for professional examinations, the scope of work is tailored to the interests and abilities of the cadet in consultation with a faculty advisor.

2026-3 2027-3 2028-3  
2029-3

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Selection by Medical Program Advisory Committee (MPAC) to attend medical school directly from West Point.

**CH499**

**TOPICS IN CHEM/LS/CHMENG W/LAB**

**3.5 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)**

**Scope:**

2026-1

**Offerings:**

This course provides in-depth study of a special topic in chemistry, chemical engineering and life science not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the Visiting Professor, Rotating PhD, or a senior faculty member. This course may also be offered as an AIAD course at USMA. This course will contain significant lab content to justify 3.5 credit hours.

2029-3

**Lessons:** 40 @ 55 min (2.500 Att/wk)

**Labs:** 7 @ 120 min

**Special Requirements:**

None

**Prerequisite(s):**

CH102  
-Or-  
CH152

## Chemistry & Life Science

0 Courses

No Records Found

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## Civil and Environmental Engineering

### 57 Courses

| CE189                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | INDEPENDENT STUDY IN CE (1CR)                                                                                                                                                                                                                    | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                                                                                                                                           | <b>Offerings:</b>                                      |
| The cadet pursues study of a research or design topic in civil engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but include defining the problem, studying the fundamentals involved, organizing an approach, performing research, achieving a solution, submitting a written report, and giving a formal briefing. |                                                                                                                                                                                                                                                  | 2026-1 2026-2 2027-2<br>2028-2 2029-2                  |
| <b>Lessons:</b> 13 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                           |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Enrollment by permission of Civil Engineering Program Director. Other requirements as determined by the Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. |                                                        |
| CE189A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | INDEPENDENT STUDY IN CE (1CR)                                                                                                                                                                                                                    | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                                                                                                                                           | <b>Offerings:</b>                                      |
| The cadet pursues study of a research or design topic in civil engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but include defining the problem, studying the fundamentals involved, organizing an approach, performing research, achieving a solution, submitting a written report, and giving a formal briefing. |                                                                                                                                                                                                                                                  | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-2 |
| <b>Lessons:</b> 13 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                           |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Enrollment by permission of Civil Engineering Program Director. Appropriate ET credit will be determined by the Civil Engineering Program Director on a case by case basis. Other requirements as determined by the Faculty Advisor.             |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CE189                                                                                                                                                                                                                                            |                                                        |
| CE189B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | INDEPENDENT STUDY IN CE (1CR)                                                                                                                                                                                                                    | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                                                                                                                                           | <b>Offerings:</b>                                      |
| The cadet pursues study of a research or design topic in civil engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but include defining the problem, studying the fundamentals involved, organizing an approach, performing research, achieving a solution, submitting a written report, and giving a formal briefing. |                                                                                                                                                                                                                                                  | 2026-1 2026-2                                          |
| <b>Lessons:</b> 13 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                           |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Enrollment by permission of Civil Engineering Program Director. Appropriate ET credit will be determined by the Civil Engineering Program Director on a case by case basis. Other requirements as determined by the Faculty Advisor.             |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CE189A                                                                                                                                                                                                                                           |                                                        |
| CE201                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | INTRO TO CIVIL ENGINEERING                                                                                                                                                                                                                       | 1.0 Credit Hours<br>(BS=0.0,ET=1.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                                                                                                                                           | <b>Offerings:</b>                                      |
| This course introduces cadets to the civil engineering profession. The course consists of classroom instruction and activities that addresses who civil engineers are, what they build, how they design, and how the West Point Civil Engineering program prepares cadets to enter the profession.                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                  | 2026-1 2027-1 2028-1<br>2029-1                         |
| <b>Lessons:</b> 6 @ 75 min (0.625 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 4 @ 75 min                                                                                                                                                                                                                          |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                                                                                                                                                                             |                                                        |

| <b>CE289</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INDEPENDENT STUDY IN CE (2CR)</b>                                                                                                                                                                                                             | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| The cadet pursues study of a research or design topic in civil engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but include defining the problem, studying the fundamentals involved, organizing an approach, performing research, achieving a solution, submitting a written report, and giving a formal briefing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                  | 2026-1 2026-2 2027-1<br>2028-1 2029-1                         |
| <b>Lessons:</b> 27 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                           |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of Civil Engineering Program Director. Other requirements as determined by the Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. |                                                               |
| <b>CE289A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INDEPENDENT STUDY IN CE (2CR)</b>                                                                                                                                                                                                             | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| The cadet pursues study of a research or design topic in civil engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but include defining the problem, studying the fundamentals involved, organizing an approach, performing research, achieving a solution, submitting a written report, and giving a formal briefing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                  | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 27 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                           |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of Civil Engineering Program Director. Appropriate ET credit will be determined by the Civil Engineering Program Director on a case by case basis. Other requirements as determined by the Faculty Advisor.             |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CE289                                                                                                                                                                                                                                            |                                                               |
| <b>CE289B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INDEPENDENT STUDY IN CE (2CR)</b>                                                                                                                                                                                                             | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| The cadet pursues study of a research or design topic in civil engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but include defining the problem, studying the fundamentals involved, organizing an approach, performing research, achieving a solution, submitting a written report, and giving a formal briefing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                  | No Course Offerings                                           |
| <b>Lessons:</b> 27 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                           |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of Civil Engineering Program Director. Appropriate ET credit will be determined by the Civil Engineering Program Director on a case by case basis. Other requirements as determined by the Faculty Advisor.             |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CE289B                                                                                                                                                                                                                                           |                                                               |
| <b>CE350</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INFRASTRUCTURE ENGINEERING</b>                                                                                                                                                                                                                | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| This course identifies, analyzes, and assesses built infrastructure which is the foundation for modern society. The complex and interconnected nature of infrastructures is investigated and demands on critical components are calculated. Students explore the non-technical factors necessary for the functioning of infrastructure including supplies, trained personnel, and cross-sector dependencies. The course provides a basis for understanding the complexity and cost of maintaining, rebuilding and developing infrastructure. Major blocks of instruction include water and wastewater, power, transportation, solid waste, communications systems, and sustainability. Several in-class scenarios are provided to synthesize the connectivity between the major items of infrastructure. Finally, as infrastructure is one of the six variables in the joint operating environment, the knowledge gained is employed to analyze infrastructure in the context of combat operations. |                                                                                                                                                                                                                                                  | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 35 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 5 @ 55 min                                                                                                                                                                                                                          |                                                               |

**Special Requirements:** None

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|              |                               |                                                          |
|--------------|-------------------------------|----------------------------------------------------------|
| <b>CE364</b> | <b>MECHANICS OF MATERIALS</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course studies the behavior of a variety of materials under normal, shear, torsional, bending and combined loads. The concepts of stress, strain, failure theory and failure mechanisms are explored. The loading, geometry, functional environment and material properties of machine or structural parts are used to relate the forces applied to a body to the resulting internal forces and deformations so that performance can be evaluated. Practical applications involving the design and adequacy of mechanical and structural elements under various loading and environmental conditions are emphasized. The course includes several laboratory experiences that require students to develop and conduct experiments, analyze and interpret data, and use engineering judgement to draw conclusions.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 25 @ 75 min (1.560 Att/wk)    **Labs:** 10 @ 75 min

**Special Requirements:** None

**Prerequisite(s):** MC300

**Corequisite(s):** MA205  
-Or-  
MA255  
-Or-  
MA204

**Disqualifier(s):** MC364

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>CE371</b> | <b>SOIL MECHANICS/FNDTN ENGNRG</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Soil Mechanics is the study of soil properties which govern the use of soil as a construction or foundation material. The course is devoted to describing soils, analyzing soil stresses, determining consolidation settlement, designing earth embankments, determining earth pressures, and designing foundations based upon applicable engineering principles and recognition of the fundamental concepts of soil behavior. The course includes several laboratory experiences that require students to develop and conduct experiments to examine soil properties, analyze and interpret data, use engineering judgement to draw conclusions, and extract necessary parameters for design.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 24 @ 55 min (1.500 Att/wk)    **Labs:** 16 @ 110 min

**Special Requirements:** None

**Prerequisite(s):** MA206  
-Or-  
MA206X  
-Or-  
MA256

**Corequisite(s):** CE364  
-Or-  
MC364

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>CE380</b> | <b>HYDROLOGY/HYDRAULIC DESIGN</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course studies both hydrology, which is the study of occurrence, movement and distribution of rainfall, and hydraulic design, which is the application of fluid mechanics and other science and engineering disciplines in the design of structures and development of water resources. Hydrologic principles are applied to model and analyze the distribution and movement of rainfall in a watershed. Hydraulic principles are applied to analyze and design flow through systems of reservoirs, channels and culverts. The course makes extensive use of computer simulation models used in engineering practice. The course includes several laboratory experiences that require students to develop and conduct experiments, analyze and interpret data, and use engineering judgement to draw conclusions.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** Three design problems; term project; compensatory time provided. One day-long field trip.

**Prerequisite(s):**

-Or-  
 -Or-  
 MA206 MC300  
 -Or-  
 MA206 MC302  
 -Or-  
 MA206X MC300  
 -Or-  
 MA256 MC300  
 -Or-  
 ME362

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|--------------|--------------------------------------|----------------------------------------------------|
| <b>CE389</b> | <b>INDEPENDENT STUDY IN CE (3CR)</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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|---------------|--------|-------------------|
| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

The cadet pursues study of a research or design topic in civil engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but include defining the problem, studying the fundamentals involved, organizing an approach, performing research, achieving a solution, submitting a written report, and giving a formal briefing.

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of Civil Engineering Program Director. Other requirements as determined by the Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

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|---------------|--------------------------------------|----------------------------------------------------|
| <b>CE389A</b> | <b>INDEPENDENT STUDY IN CE (3CR)</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

The cadet pursues study of a research or design topic in civil engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but include defining the problem, studying the fundamentals involved, organizing an approach, performing research, achieving a solution, submitting a written report, and giving a formal briefing.

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of Civil Engineering Program Director. Other requirements as determined by the Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**Prerequisite(s):** CE389

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|--------------|---------------------------------|----------------------------------------------------|
| <b>CE399</b> | <b>CIVIL ENG PRAC-FIELD ENG</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
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|---------------|--------|-------------------|
| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

This course provides cadets with an opportunity to learn and practice the field aspects of civil engineering. Topics include plane surveying, introduction to construction materials, wood frame building construction, heavy equipment operations, concrete placement and finishing, roadway construction, steel fabrication, reinforced concrete construction, bridge construction, power production, and environmental systems. Cadets perform actual construction projects as part of course requirements. LESSONS and LABS: 12 lessons of varying length, scheduled across three weeks of full-day instruction during the summer.

**Lessons:** 12 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** TDY travel to the course location at the U.S. Air Force Academy.

**Prerequisite(s):** CE302  
 -Or-  
 CE300  
 -Or-  
 MC302  
 -Or-  
 MC300

| CE401                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CIV ENG PROF PRAC AND APP                                                                        | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                           | <b>Offerings:</b>                                                              |
| This course prepares cadets to enter the civil engineering profession. The course consists of classroom discussions and activities that address career planning and analyzing issues in engineering ethics and sustainability. The course introduces processes to identify, formulate, and solve complex engineering problems. The processes develop the recognition of ethical and professional responsibilities in engineering judgments considering the impact of solutions in global, economic, environmental, and societal contexts.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                  | 2026-2 2027-2 2028-2<br>2029-2                                                 |
| <b>Lessons:</b> 30 @ 75 min (1.880 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                           |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                                             |                                                                                |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CE494                                                                                            |                                                                                |
| CE403                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | STRUCTURAL ANALYSIS                                                                              | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                           | <b>Offerings:</b>                                                              |
| This course addresses the analysis of basic structural forms such as beams, trusses, and frames, which are found in bridges and buildings. Classical deflection techniques such as direct integration and virtual work; and indeterminate analysis techniques such as the force method and displacement method (specifically direct stiffness) are used to determine forces and deflections in elastic structures. Structural analysis computer programs are introduced and directly applied in the solution of graded analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                  | 2026-1 2027-1 2028-1<br>2029-1                                                 |
| <b>Lessons:</b> 35 @ 55 min (2.188 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 5 @ 55 min                                                                          |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | One half-day field trip. Compensatory time provided.                                             |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CE364<br>-Or-<br>MC364                                                                           |                                                                                |
| CE404                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DSN STEEL STRUCTURES                                                                             | 3.5 Credit Hours<br>(BS=0.0,ET=3.5,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                           | <b>Offerings:</b>                                                              |
| This course teaches the engineering thought process through the design of steel structures. The course synthesizes the fundamentals of statics, mechanics of materials, and structural analysis and applies them to the design of structural members, with emphasis on satisfying real-world needs. Topics include an introduction to the design of structural systems, design of steel tension and compression members, design of beams and beam-columns, and an introduction to connection design. Students read, interpret and create structural plans using an intelligent 3D digital modeling and design software. Students analyze structural systems and elements using finite element modeling software. All design is performed in accordance with codes and specifications used in current engineering practice. A comprehensive design problem requires development of a design methodology, consideration of alternative solutions, and design of an optimal steel structure to meet stated functional requirements. |                                                                                                  | 2026-2 2027-2 2028-2<br>2029-2                                                 |
| <b>Lessons:</b> 32 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 3 @ 75 min                                                                          |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                                             |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CE403<br>-Or-<br>CE453                                                                           |                                                                                |
| CE450                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CONSTRUCTION MANAGEMENT                                                                          | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                           | <b>Offerings:</b>                                                              |
| This course focuses on the implementation portion of the design process. The management of construction is covered to include scope of work, rough order-of-magnitude estimating, scheduling, planning, progress reporting, resource constraining, and quality control. The roles of the contractor, owner, and designer are explained. An iterative decision-making design process is developed and applied in multiple contexts that requires developing requirements, performing analysis and synthesis, generating multiple solutions, evaluating solutions against requirements, considering risks, and making trade-offs, for the purpose of obtaining a high-quality solution under given circumstances.                                                                                                                                                                                                                                                                                                                  |                                                                                                  | 2026-1 2026-2 2027-1<br>2027-2 2027-8 2028-1<br>2028-2 2028-8 2029-1<br>2029-2 |
| <b>Lessons:</b> 24 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 6 @ 75 min                                                                          |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | One design project requiring a formal oral and written presentation; compensatory time provided. |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CE350 MC300                                                                                      |                                                                                |

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>CE472</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ADV SOIL MECHNCS/FNDTN ENGRNG</b>                                                                                                                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                                                         | <b>Offerings:</b>                                              |
| Students will extend what they learned in Soil Mechanics and Foundation Engineering and design advanced foundations in this course. Topics covered are: slope stability, field testing, field instrumentation, designing braced excavations, designing piles and drilled shafts, designing flexible walls, designing earth retaining structures, and designing earth structures using geosynthetics.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                | 2026-1 2027-1 2028-1 2029-1                                    |
| <b>Lessons:</b> 27 @ 75 min (1.690 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 3 @ 75 min                                                                                                                                                                                                        |                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                                                                                                                                                                                                           |                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CE371                                                                                                                                                                                                                          |                                                                |
| <b>CE483</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>DSN CONC STRUCTURES</b>                                                                                                                                                                                                     | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b>             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                                                         | <b>Offerings:</b>                                              |
| The course introduces the materials and mechanical properties of concrete, and the design of reinforced concrete structures. Mix design and strength testing labs develop the concept of proportioning constituents for quality concrete and provide a background in techniques of material testing, quality control, and sound construction practices. The study of reinforced concrete and masonry includes analysis and design of simple structures, resulting in an appreciation for the strength and serviceability of these structures. Current codes and standards are used to guide the practical design of beams, slabs, columns, and footings. The course includes several laboratory experiences that require students to develop and conduct experiments, analyze and interpret data, and use engineering judgement to draw conclusions. |                                                                                                                                                                                                                                | 2026-1 2027-1 2028-1 2029-1                                    |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 8 @ 120 min                                                                                                                                                                                                       |                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                                                                                                                                                                                                           |                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CE403                                                                                                                                                                                                                          |                                                                |
| <b>CE489</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ADV IND STUDY CIVIL ENGRNG</b>                                                                                                                                                                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                                                         | <b>Offerings:</b>                                              |
| The cadet, on an individual or small group basis, pursues advanced study of a research or design topic in civil engineering. The scope of the course is tailored to the needs of the project and desires of the cadet, in consultation with the Faculty Advisor. The cadet is required to define and analyze the problem, study the fundamentals involved, organize an approach, determine a procedure, perform research and/or achieve a solution, submit a written report, and give a formal briefing.                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                | 2026-1 2026-2 2026-9 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 3 @ 120 min                                                                                                                                                                                                       |                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | As determined by faculty advisor. Many CE489 projects will have a significant laboratory requirement. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.  |                                                                |
| <b>CE489A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>ADV IND STUDY CIVIL ENGRNG</b>                                                                                                                                                                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                                                                                                                                                                         | <b>Offerings:</b>                                              |
| The cadet, on an individual or small group basis, pursues advanced study of a research or design topic in civil engineering. The scope of the course is tailored to the needs of the project and desires of the cadet, in consultation with the Faculty Advisor. The cadet is required to define and analyze the problem, study the fundamentals involved, organize an approach, determine a procedure, perform research and/or achieve a solution, submit a written report, and give a formal briefing.                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                | 2026-1 2026-2 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                         |                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | As determined by faculty advisor. Many CE489A projects will have a significant laboratory requirement. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. |                                                                |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | CE489                                                                                                                                                                                                                          |                                                                |
| <b>CE489B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>ADV IND STUDY CIVIL ENGRNG</b>                                                                                                                                                                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>             |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                              |                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                       | <b>Offerings:</b>                                  |
| The cadet, on an individual or small group basis, pursues advanced study of a research or design topic in civil engineering. The scope of the course is tailored to the needs of the project and desires of the cadet, in consultation with the Faculty Advisor. The cadet is required to define and analyze the problem, study the fundamentals involved, organize an approach, determine a procedure, perform research and/or achieve a solution, submit a written report, and give a formal briefing. Cadets will typically enroll in CE489B if they have already completed CE489 and CE489A.                                                                                                                                                                                                                                                                 |                                                                                                                              | No Course Offerings                                |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                       |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | As determined by the faculty advisor. Many CE489B projects will have a significant laboratory requirement.                   |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CE489A                                                                                                                       |                                                    |
| <b>CE490</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>TOPICS IN CIVIL ENGINEERING</b>                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                       | <b>Offerings:</b>                                  |
| This course provides in-depth study of a special topic in engineering mechanics or in structural, geotechnical, environmental, water resources, construction, or transportation engineering not offered elsewhere in the USMA curriculum. The course is intended to broaden the cadet's exposure to the civil engineering discipline. Course content will be based on the special expertise of the visiting professor or a senior civil engineering faculty member.                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                              | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 38 @ 55 min (2.375 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 2 @ 55 min                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | TBD. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. |                                                    |
| <b>CE490A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>TOPICS IN CIVIL ENGINEERING</b>                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                       | <b>Offerings:</b>                                  |
| This course provides in-depth study of a special topic in engineering mechanics or in structural, geotechnical, environmental, water resources, construction, or transportation engineering not offered elsewhere in the USMA curriculum. The course is intended to broaden the cadet's exposure to the civil engineering discipline. Course content will be based on the special expertise of the visiting professor or a senior civil engineering faculty member.                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                              | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                                                       |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | TBD                                                                                                                          |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CE490                                                                                                                        |                                                    |
| <b>CE491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ADV STRUCTURAL ANALYSIS</b>                                                                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                       | <b>Offerings:</b>                                  |
| This course builds upon the material covered in CE403/453 to develop a better understanding of structural behavior. Two-dimensional analysis of trusses and frames is reviewed and extended into 3D. Matrix analysis methods, including an introduction to continuum finite elements are developed as the basis for modern computer-based structural analysis. An introduction to nonlinear analysis is presented. Coursework involves extensive use of the computer as an analytical tool. Students use state-of-the-art structural engineering analysis and design software.                                                                                                                                                                                                                                                                                   |                                                                                                                              | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 37 @ 55 min (2.310 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 3 @ 55 min                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Graded homework is assigned to reinforce concepts covered in class.                                                          |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CE403                                                                                                                        |                                                    |
| <b>CE493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>CIV ENG CAPSTONE DESIGN I</b>                                                                                             | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                       | <b>Offerings:</b>                                  |
| This course challenges students to identify, formulate, and solve a complex engineering problem in teams. The design problems will be open-ended in nature and involve the application of math, science, and engineering principles acquired in previous coursework. In teams, students will use an iterative design process to define the problem and requirements, analyze alternatives within constraints (e.g., applicable codes and standards), and select the best alternative to solve the problem. The design process will take into consideration public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. Teams will communicate their process and solution through written deliverables and presentations to a range of audiences. Work on this problem will continue in the next course: CE494. |                                                                                                                              | 2026-1 2027-1 2028-1<br>2029-1                     |

|                                            |                           |
|--------------------------------------------|---------------------------|
| <b>Lessons:</b> 10 @ 75 min (0.630 Att/wk) | <b>Labs:</b> 20 @ 160 min |
| <b>Special Requirements:</b>               | None                      |
| <b>Prerequisite(s):</b>                    | CE371 CE380 CE403         |
| <b>Corequisite(s):</b>                     | CE404 CE483               |

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|--------------|-----------------------------------|----------------------------------------------------|
| <b>CE494</b> | <b>CIV ENG CAPSTONE DESIGN II</b> | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-1 | <b>Offerings:</b>           |
| This course continues the engineering design process initiated in CE493. In teams, students will model, test, implement, and communicate their solution to a complex engineering problem. The iterative design process will consider applicable codes and standards along with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. Teams will communicate their process and solution through written deliverables and presentations to a range of audiences. |        | 2026-2 2027-2 2028-2 2029-2 |

|                                            |                           |
|--------------------------------------------|---------------------------|
| <b>Lessons:</b> 10 @ 75 min (2.000 Att/wk) | <b>Labs:</b> 20 @ 160 min |
| <b>Special Requirements:</b>               | None                      |
| <b>Prerequisite(s):</b>                    | CE493                     |

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|--------------|-----------------------------------|----------------------------------------------------|
| <b>CE495</b> | <b>TRANSPORTATION ENGINEERING</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1 | <b>Offerings:</b>           |
| This course provides cadets with a solid introduction to the principles of transportation engineering with a focus on highway engineering and traffic analysis. The material learned will provide the basic skill set that will allow students to solve transportation problems that are likely to appear in professional practice (civilian and military), on the Fundamentals of Engineering exam (FE), and on the Principles and Practice of Engineering exam (PE). |        | 2026-1 2027-1 2028-1 2029-1 |

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|--------------------------------------------|-------------------------------|
| <b>Lessons:</b> 29 @ 75 min (1.810 Att/wk) | <b>Labs:</b> 1 @ 75 min       |
| <b>Special Requirements:</b>               | One in-class design exercise. |
| <b>Prerequisite(s):</b>                    | CE371 CE380                   |

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|---------------|------------------------------------|----------------------------------------------------|
| <b>CV300X</b> | <b>INTRO TO SUSTAINABLE DESIGN</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-2 | <b>Offerings:</b> |
| Public infrastructure designed by civil and environmental engineers underpins modern society. However, day-to-day, its benefits often are taken for granted and its hidden costs become more evident over time. In this introductory course to the Environmental Engineering and Infrastructure Engineering Core Engineering Sequences, students will discuss the maintenance of public trust through the inclusion of resilience and sustainability concepts in infrastructure design and design implementation. Students will study methods to qualify and quantify the social, economic, and environmental benefits and costs of infrastructure emplacement and operation. Students will be introduced to the engineering design process in which they will seek to minimize examined costs and maximize value to partners using a methodical decision-making process. The course will culminate with a term project addressing an Army Civil and/or Environmental Engineering need. |        | 2026-2            |

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|--------------------------------------------|--------------------------------------------|
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk) | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>               | Course project with in-class presentation. |

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|--------------|--------------------------------|----------------------------------------------------|
| <b>CV311</b> | <b>THERMAL-FLUID SYSTEMS I</b> | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1 | <b>Offerings:</b> |
| Thermal-Fluid Systems is an integrated study of fundamental topics in thermodynamics and fluid mechanics. The course introduces conservation principles for mass, energy, and linear momentum as well as the 2nd Law of Thermodynamics. Principles are applied to incompressible flow in pipes, open channels and turbomachinery and power generation systems, focusing on engineering applications. Laboratory exercises are integrated into classroom work. This course includes completion of a comprehensive, out-of-class design problem. This design problem provides the opportunity for students to apply engineering science and the engineering design process to a hands-on project. Lesson length varies between 75 minutes and 55 minutes (and corresponding preparation time varies as well). **This course is a pilot in AY26-2 and AY27-2. It must undergo review by the Curriculum Committee NLT AY28-1 to continue. ET = 3.5CR pending ABET-PEV review.** |        | 2026-2 2027-2     |

**Lessons:** 25 @ 55 min (2.500 Att/wk)    **Labs:** 15 @ 75 min

**Special Requirements:** None

**Disqualifier(s):** ME301  
-Or-  
ME362  
-Or-  
MC311

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|--------------|--------------------------------------|----------------------------------------------------------|
| <b>EV201</b> | <b>INTRO TO ENV ENG &amp; DESIGN</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-2

**Offerings:**

2026-1

This course introduces Cadets to the environmental engineering profession to gain an appreciation for the interdisciplinary nature of the field. Fundamentals regarding engineering design are presented and a variety of tools are used to analyze and visualize solutions to environmental challenges. The course provides a platform for the incorporation of required skills students use throughout the rest of their engineering course of study. Cadets solve a series of progressively more complex engineering problems by applying the design process while addressing public health, safety, and welfare. The course introduces a variety of techniques to communicate effectively with a range of audiences as Cadets function effectively on teams and apply new knowledge and learning strategies.

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Open to EVE1 majors. Any exceptions must be pre-approved by the Environmental Engineering Curriculum Coordinator.

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|---------------|---------------------------------------|----------------------------------------------------------|
| <b>EV289A</b> | <b>INTRO IND STUDY &amp; RESEARCH</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

The course is an individually supervised research and study program designed to familiarize cadets with introductory scientific procedures, techniques, and topics. The cadet and advisor prepare a research and study proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Activities vary by project, but the primary purpose is to acquaint students with essential skills required for independent research. If required for a specific degree, the proposal will include a justification for engineering science or design credit.

**Lessons:** 0 @ 0 min (0.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the appropriate CEE program director. Other requirements as determined by the faculty advisor and EV289A course director. Discipline-specific final product is required.

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|---------------|------------------------------------------|----------------------------------------------------------|
| <b>EV289B</b> | <b>INTRO IND STUDY &amp; RESEARCH II</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

2026-1 2027-1 2027-2  
2028-1 2028-2 2029-1  
2029-2

The course is an individually supervised research and study program designed to familiarize cadets with introductory scientific procedures, techniques, and topics. The cadet and advisor prepare a research and study proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Activities vary by project, but the primary purpose is to acquaint students with essential skills required for independent research. If required for a specific degree, the proposal will include a justification for engineering science or design credit.

**Lessons:** 0 @ 0 min (0.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the appropriate CEE program director. Other requirements as determined by the faculty advisor and EV289B course director. Discipline-specific final product is required. Cadets will take EV289A before enrolling in EV289B.

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|---------------|-------------------------------------------|----------------------------------------------------------|
| <b>EV289C</b> | <b>INTRO IND STUDY &amp; RESEARCH III</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The course is an individually supervised research and study program designed to familiarize cadets with introductory scientific procedures, techniques, and topics. The cadet and advisor prepare a research and study proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Activities vary by project, but the primary purpose is to acquaint students with essential skills required for independent research. If required for a specific degree, the proposal will include a justification for engineering science or design credit.

2027-2 2028-2 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

Enrollment by permission of the appropriate CEE program director. Other requirements as determined by the faculty advisor and EV289B course director. Discipline-specific final product is required. Cadets will take EV289A/B before enrolling in EV289C.

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|--------------|------------------------------|----------------------------------------------------------|
| <b>EV300</b> | <b>ENVIRONMENTAL SCIENCE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:**

2026-1

**Offerings:**

As the introductory course to the Environmental Engineering Sequence, EV300 provides a broad understanding of current global and local environmental issues. It specifically focuses on natural ecosystems processes, the effects of pollution on human health, assessing the level of risk associated with pollution, and the environmental effects of energy use, air pollution, global climate change, acid rain, and smog. Discussions of anthropogenic influences on the environment also consider social, economic, technological and political impacts. Cadets learn to evaluate environmental issues through current events and interactive debates. A course project requires cadets to apply the scientific method to evaluate a current environmental problem and provides an opportunity to integrate multiple course topics with an in-depth study of an issue of interest.

2026-1 2026-8 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2**Lessons:** 40 @ 55 min (2.500 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

Design and conduct an environmental study.

**Corequisite(s):**EV203  
-Or-  
EV203X  
-Or-  
GE203**Disqualifier(s):**-Or-  
EV301

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|              |                                     |                                                          |
|--------------|-------------------------------------|----------------------------------------------------------|
| <b>EV301</b> | <b>ENVIRONMENTAL SUSTAINABILITY</b> | <b>3.0 Credit Hours</b><br><b>(BS=1.0,ET=2.0,MA=0.0)</b> |
|--------------|-------------------------------------|----------------------------------------------------------|

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**Scope:**

2026-1

**Offerings:**

EV301 provides the cadet with an overview of current global and local environmental issues through the lens of sustainable practices. It specifically focuses on natural ecosystem processes, the effects of pollution on human health, assessing the level of human health risk associated with pollution, and the strain on natural resources that stem from population growth. Discussions of anthropogenic influences on sustainability include social, economic, technological, and political factors. Cadets evaluate environmental issues through review of scientific literature, current events, and interactive debates. A course project requires cadets to conduct an in-depth life cycle assessment (LCA) of an activity or product. The project results in a foundation of engineering science that facilitates follow-on engineering design.

2026-2 2027-2 2028-2  
2029-2**Lessons:** 40 @ 55 min (2.500 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

Complete a course project involving lifecycle analysis.

**Corequisite(s):**EV203  
-Or-  
EV203X  
-Or-  
GE203**Disqualifier(s):**

EV300

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|              |                                        |                                                          |
|--------------|----------------------------------------|----------------------------------------------------------|
| <b>EV350</b> | <b>ENVIRONMENTAL ENGR TECHNOLOGIES</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|----------------------------------------|----------------------------------------------------------|

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**Scope:**

2026-1

**Offerings:**

This course builds on environmental issues introduced in EV300 and further explores environmental engineering from a unit process and materials balance approach. Analyzing water (transport, quality, drinking water treatment, and wastewater treatment) and air (transport, quality, and pollutant minimization), the cadet is exposed to the breadth of the environmental discipline. A laboratory experience is integral to the course. In the laboratory, physical, chemical, and biological quality are discussed and measured. An introductory environmental engineering design project on river water quality and wastewater treatment plant design is developed within the semester.

**Lessons:** 34 @ 55 min (2.500 Att/wk)      **Labs:** 3 @ 110 min

**Special Requirements:** One design project.

**Prerequisite(s):** CH101 EV300 MA104  
-Or-  
CH151 EV300 MA104  
-Or-  
CH101 EV301 MA104  
-Or-  
CH151 EV301 MA104

**Disqualifier(s):** EV385

2026-2 2027-2 2027-8  
2028-2 2028-8 2029-2  
2029-9

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|---------------|---------------------------------------|----------------------------------------------------------|
| <b>EV389A</b> | <b>INTER IND STUDY &amp; RESEARCH</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|---------------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

The course is an individually supervised research and study program designed to familiarize cadets with advanced scientific procedures, techniques, and topics. The cadet and advisor prepare a research and study proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Activities vary by project, but the primary purpose is to acquaint students with essential skills required for independent research. If required for a specific degree, the proposal will include a justification for engineering science or design credit.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the appropriate GENE program director. Other requirements as determined by the faculty advisor and EV389A course director. Discipline-specific final product is required.

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|---------------|------------------------------------------|----------------------------------------------------------|
| <b>EV389C</b> | <b>INTER IND STUDY &amp; RESEARCH II</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|------------------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

The course is an individually supervised research and study program designed to familiarize cadets with advanced scientific procedures, techniques, and topics. The cadet and advisor prepare a research and study proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Activities vary by project, but the primary purpose is to acquaint students with essential skills required for independent research. If required for a specific degree, the proposal will include a justification for engineering science or design credit.

2026-1 2026-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the appropriate CEE program director. Other requirements as determined by the faculty advisor and EV389C course director. Discipline-specific final product is required. Cadets will take EV389A before enrolling in EV389C.

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>EV394</b> | <b>HYDROGEOLOGY/HYDRAULIC SYSTEMS</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course covers the principles governing the movement of subterranean water (groundwater), the interaction of this water with the porous medium, and the transport of chemical constituents (contaminants) in the subsurface. Lesson blocks explore traditional background elements of hydraulic engineering to include flow systems for the conveyance of groundwater and drainage systems for groundwater. Computer models are used to evaluate groundwater problems and conduct sensitivity analyses.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 23 @ 95 min (2.000 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** One course project.

|                         |                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite(s):</b> | EV203 MA206<br>-Or-<br>EV203 MA366<br>-Or-<br>EV203X MA206<br>-Or-<br>EV203X MA366<br>-Or-<br>EV203 MA205 MA364<br>-Or-<br>EV203X MA205 MA364<br>-Or-<br>EV203 MA255 MA365<br>-Or-<br>EV203X MA255 MA365<br>-Or-<br>EV203 MA204 MA365<br>-Or-<br>GE203 MA204 MA365<br>-Or-<br>GE203 MA205 MA365<br>-Or-<br>GE203 MA255 MA365<br>-Or-<br>GE203 MA205 MA364 |
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|--------------|-------------------------------------|----------------------------------------------------|
| <b>EV396</b> | <b>ENVIRONMENTAL BIOLOGICAL SYS</b> | <b>3.5 Credit Hours<br/>(BS=1.0,ET=2.5,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1 | <b>Offerings:</b>              |
| This course will examine biology from a practical environmental engineering and environmental science perspective. The foci of the course are applied public health, microbiology and microbial energetics. Specific topics include the biological health issues associated with drinking water, microbial aspects of industrial and domestic waste treatment and protection or restoration of natural water bodies from environmental contaminants. Students are also introduced to medical geography and the spatial biological health issues associated with a deployment. Laboratory exercises are used to introduce the student to water quality analyses and practices commonly used in the fields of environmental engineering and the environmental sciences. |        | 2026-2 2027-2 2028-2<br>2029-2 |

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 12 @ 55 min

**Special Requirements:** None

|                         |                                                                                  |
|-------------------------|----------------------------------------------------------------------------------|
| <b>Prerequisite(s):</b> | CH102 EV300<br>-Or-<br>CH152 EV300<br>-Or-<br>CH102 EV301<br>-Or-<br>CH152 EV301 |
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| <b>EV397</b> | <b>AIR POLLUTION ENGINEERING</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1 | <b>Offerings:</b>              |
| This course employs a design approach to air pollution control. It begins by defining air pollution problems, to include pollutant types, sources, legislation, and effects on both local and global scales. The course then examines the design of various means of controlling particulate and gaseous air pollution from both mobile and stationary sources. Finally, students study the link between meteorology and air pollution, as well as pollutant dispersion modeling in the atmosphere. A course project involves collection and analysis of indoor air quality data at selected locations on the installation. |        | 2026-2 2027-2 2028-2<br>2029-2 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Field Trip(s).

|                         |                                          |
|-------------------------|------------------------------------------|
| <b>Prerequisite(s):</b> | EV203<br>-Or-<br>EV203X<br>-Or-<br>GE203 |
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| <b>EV400</b> | <b>ENVIRONMENTAL ENGINEERING SEM</b> | <b>1.0 Credit Hours<br/>(BS=0.0,ET=1.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                    | <b>Offerings:</b>                                             |
| This seminar will meet once each week and will include all first class cadets majoring in environmental engineering. The seminar topics will address a variety of fundamental engineering science, design, and professional practice topics including engineering ethics, economics, and licensing. Periodically, guest lecturers from the military, industrial, and academic communities will provide their perspective on these topics.                                                                                                                                                                                                                                                                                                                                                                                      |                                                                           | No Course Offerings                                           |
| <b>Lessons:</b> 13 @ 55 min (1.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                    |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                      |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | EV490                                                                     |                                                               |
| <b>EV401</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>PHYS &amp; CHEM TREATMENT</b>                                          | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                    | <b>Offerings:</b>                                             |
| This course takes a process approach to environmental engineering using engineering science and design of drinking water treatment systems as the primary foci. Building upon concepts gained in environmental chemistry, cadets study physical and chemical processes used in environmental engineering. Discussion includes the theories behind these processes and the design procedures involved in their application. Cadets develop comprehensive concept design of drinking water treatment processes. While the focus of the course is drinking water treatment, the processes developed are also applicable to wastewater treatment, groundwater remediation, air pollution control, and the treatment of solid and hazardous wastes.                                                                                 |                                                                           | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 23 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 7 @ 120 min                                                  |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | One term project, one field trip.                                         |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MC311<br>-Or-<br>CV311                                                    |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | XS391                                                                     |                                                               |
| <b>EV402</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>BIOCHEMICAL TREATMENT</b>                                              | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                    | <b>Offerings:</b>                                             |
| This course applies principles from environmental biotechnology, microbial ecology, and environmental microbiology to educate cadets how to treat wastewaters and sludges. The course explores the design of longstanding wastewater treatment technologies and approaches, while introducing cadets to emerging applications of biochemical wastewater treatment and resource recovery. Modeling approaches for designing wastewater treatment works using biochemical reactors are also introduced. A comprehensive, multi-step laboratory project serves as the culminating course project.                                                                                                                                                                                                                                 |                                                                           | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 7 @ 120 min                                                  |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Engineering design project with a written report.                         |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | -Or-<br>-Or-<br>EV396 ME311<br>-Or-<br>EV396 MC311<br>-Or-<br>CV311 EV396 |                                                               |
| <b>EV450</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>ENV ENG FOR COMMUNITY DEVELOP</b>                                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                    | <b>Offerings:</b>                                             |
| This course is the capstone experience for a three-course environmental engineering sequence. With a focus on water and sanitation challenges in the developing world, students assess various technologies and their ability to meet community needs. The course highlights the engineering design process to develop appropriate solutions and introduces decision modeling with consideration of social, political, and economic factors. A semester-long term project leverages real world case studies to provide cadet teams an opportunity to apply knowledge and creatively design sustainable solutions to ill-defined problems. Students must make logical assumptions throughout the project, present and evaluate solution designs, and prepare a formal written report defending their selected course of action. |                                                                           | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                    |                                                               |

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| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Must be a First Class cadet. An engineering design project is completed in multi-disciplinary design groups.                                                                                                                |                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | EV350<br>-Or-<br>EV401                                                                                                                                                                                                      |                                                                      |
| <b>EV481</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>WATER RESOURCES PLAN &amp; DESIGN</b>                                                                                                                                                                                    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2026-1                                                                                                                                                                                                                      | <b>Offerings:</b>                                                    |
| In this course, students explore the effective use of water as a manageable natural resource. The course begins with instruction on the tools required by water resource managers to make sound decisions in their field. The course assesses current needs for water and the structural (engineered) and non-structural approaches available to meet these needs. Elements of engineering design and the design process are introduced. The bulk of the course is concerned with assessment of the impacts of various water resources development activities on the economic, socio-cultural and ecological sectors of the environment. Methods for conducting tradeoff analyses among the engineered and environmental aspects of projects are developed and applied in a term project. The course makes use of case studies of current water resource projects. Visiting speakers discuss topics relevant to water resource management.                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                             | 2026-2 2027-2 2028-2<br>2029-2                                       |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                      |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Written and oral research reports on a contemporary water resources project.                                                                                                                                                |                                                                      |
| <b>EV488</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>SOLID &amp; HAZ WASTE TREAT &amp; REMD</b>                                                                                                                                                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2026-1                                                                                                                                                                                                                      | <b>Offerings:</b>                                                    |
| This course examines the treatment, storage and disposal of solid and hazardous wastes. Both regulatory requirements and evolving technology associated with solving modern solid waste disposal problems are discussed. Design of remediation processes for cleanup of contaminated waste sites are presented. Students will complete a term project related to subsurface remediation that will involve design and experimentation to assess the efficiency of a hazardous waste treatment and remediation process. **The 3.0CR version of this course is a pilot and must undergo review by the Curriculum Committee NLT AY26-1 to continue.**                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                             | 2026-2 2027-2 2028-2<br>2029-2                                       |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                      |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Subsurface remediation term project                                                                                                                                                                                         |                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | EV402                                                                                                                                                                                                                       |                                                                      |
| <b>EV489A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ADVANCED INDIVIDUAL STUDY I</b>                                                                                                                                                                                          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2026-1                                                                                                                                                                                                                      | <b>Offerings:</b>                                                    |
| The course is an individually supervised research and study program designed to provide cadets the opportunity to pursue advanced scientific procedures, techniques, and/or topics within their discipline. The cadet and faculty advisor prepare a research proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Lessons and labs are established by consultation between the cadet and faculty advisor. The program for each cadet will culminate in one of two outcomes: 1) discipline-appropriate final written report (e.g., senior thesis) with presentation in a public forum (e.g., Projects Day); or 2) enrollment in EV489B for continued research and independent study at a 3.0-credit level. If required for a specific degree, the proposal will include a justification for engineering science or design credit. Program Director will determine the applicable MA, BS, and ET credits for each offering, as required. |                                                                                                                                                                                                                             | 2026-1 2026-2 2026-7<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                      |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Enrollment by permission of the appropriate CEE program director.<br>Discipline-appropriate final written report with presentation in a public forum required prior to completion of a cadet's culminating research course. |                                                                      |
| <b>EV489B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ADVANCED INDIVIDUAL STUDY II</b>                                                                                                                                                                                         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2026-1                                                                                                                                                                                                                      | <b>Offerings:</b>                                                    |



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| <p>The course is an individually supervised research and study program designed to provide cadets the opportunity to pursue advanced scientific procedures, techniques, and/or topics within their discipline. The cadet and faculty advisor prepare a research proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Lessons and labs are established by consultation between the cadet and faculty advisor. The program for each cadet will culminate in one of two outcomes: 1) discipline-appropriate final written report (e.g., senior thesis) with presentation in a public forum (e.g., Projects Day); or 2) enrollment in EV489C for continued research and independent study at a 3.0-credit level. If required for a specific degree, the proposal will include a justification for engineering science or design credit. Program Director will determine the applicable MA, BS, and ET credits for each offering, as required.</p> <p><b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> Enrollment by permission of the appropriate CEE program director.<br/>Discipline-appropriate final written report with presentation in a public forum required prior to completion of a cadet's culminating research course.</p> <p><b>Prerequisite(s):</b> EV480<br/>-Or-<br/>EV489A</p> |                                       | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>EV489C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ADV IND STUDY AND RESEARCH III</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                | <b>Offerings:</b>                                             |
| <p>The course is an individually supervised research and study program designed to provide cadets the opportunity to pursue advanced scientific procedures, techniques, and/or topics within their discipline. The cadet and faculty advisor prepare a research proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Lessons and labs are established by consultation between the cadet and faculty advisor. The program for each cadet will culminate in one of two outcomes: 1) discipline-appropriate final written report (e.g., senior thesis) with presentation in a public forum (e.g., Projects Day); or 2) enrollment in EV489D for continued research and independent study at a 3.0-credit level. If required for a specific degree, the proposal will include a justification for engineering science or design credit. Program Director will determine the applicable MA, BS, and ET credits for each offering, as required.</p> <p><b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> Enrollment by permission of the appropriate GENE program director.<br/>Discipline-appropriate final written report with presentation in a public forum required prior to completion of a cadet's culminating research course.</p> <p><b>Prerequisite(s):</b> EV489A EV489B</p>            |                                       | 2026-1 2027-2 2028-2<br>2029-2                                |
| <b>EV489D</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ADV IND STUDY AND RESEARCH IV</b>  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                | <b>Offerings:</b>                                             |
| <p>The course is an individually supervised research and study program designed to provide cadets the opportunity to pursue advanced scientific procedures, techniques, and/or topics within their discipline. The cadet and faculty advisor prepare a research proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Lessons and labs are established by consultation between the cadet and faculty advisor. The program for each cadet will culminate in a discipline-appropriate final written report (e.g., senior thesis) with presentation in a public forum (e.g., Projects Day). If required for a specific degree, the proposal will include a justification for engineering science or design credit. Program Director will determine the applicable MA, BS, and ET credits for each offering, as required.</p> <p><b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> Enrollment by permission of the appropriate GENE program director.<br/>Discipline-appropriate final written report with presentation in a public forum required prior to completion of a cadet's culminating research course.</p> <p><b>Prerequisite(s):</b> EV489A EV489B EV489C</p>                                                                                                                          |                                       | 2026-2                                                        |
| <b>EV490</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>ENVIRON ENG DESIGN</b>             | <b>3.5 Credit Hours</b><br>(BS=0.0,ET=3.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                | <b>Offerings:</b>                                             |

This is the first in a sequence of two courses that comprise the environmental engineering integrative experience. In this senior engineering design course, teams of cadets apply the engineering design process to develop alternative solutions to complex, open-ended environmental engineering problems. In addition, cadets are introduced to techniques, skills, and modern engineering tools required for the engineering design process. The course culminates with a decision brief and report in which teams present a conceptual (35%) design for their recommended solution alternative, as well as their analysis of the engineering, social, economic, and environmental criteria that led them to recommend this course of action. The approved design alternative will be completed the following semester in EV491 Advanced Environmental Engineering Design. \*The 3.5CR version of this course is a pilot for AY24-1, AY25-1, and AY26-1 and will be updated on approval by the General Committee in AY26.\*

**Lessons:** 30 @ 160 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** Standing as a first class cadet.

**Prerequisite(s):** EV481

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| <b>EV491</b> | <b>ADV ENVIRON ENG DESIGN</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This is the second in a sequence of two courses that comprise the environmental engineering integrative experience. In this senior engineering design course, teams of cadets refine and finalize the conceptual designs they produced at the end of EV490 to solve a complex, open-ended environmental engineering problem. Ensuring that the complete design meets specified engineering, social, environmental, and economic criteria, they present the results both orally and in a written report. This course also introduces cadets to the engineering profession, to include engineering ethics and an emphasis on continued study to earn professional certifications. \*The 3.5CR version of this course is a pilot for AY24-2 and will be updated on approval by the General Committee in AY25.\*

**Lessons:** 30 @ 160 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** One design problem. Standing as a first class cadet is required for enrollment.

**Prerequisite(s):** EV490

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| <b>MC300</b> | <b>FUND OF ENGR MECH AND DESIGN</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2025-4

**Offerings:**

The engineering design process and the method of design are introduced. Principles of equilibrium are used to analyze forces on statically determinate rigid bodies and structures to include trusses and frames. The behavior of deformable bodies under axial and flexural loading is examined. The concepts of stress, strain, and material properties are introduced and are used to relate external forces applied to a body to the resulting internal forces and deformations so that performance can be evaluated. Practical applications involving the design and adequacy of mechanical and structural elements under various loading conditions are emphasized.

**Lessons:** 20 @ 75 min (1.250 Att/wk) **Labs:** 10 @ 75 min

**Special Requirements:** None

**Corequisite(s):** PH205  
-Or-  
PH255  
-Or-  
PH201  
-Or-  
PH201X  
-Or-  
PH251

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|              |                                    |                                                          |
|--------------|------------------------------------|----------------------------------------------------------|
| <b>XS391</b> | <b>PRIN &amp; APPL OF ENV CHEM</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.5,ET=2.5,MA=0.0)</b> |
|--------------|------------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

This course examines chemical interactions of pollutants in air, soil, and water systems. The focus of the course is problem solving with the following topic coverage: approximately 80% applied aquatic chemistry, 15% environmental organic chemistry, and 5% applied analytical chemistry. Specific topics include the chemistry applied in drinking water production and the chemical aspects of industrial and hazardous waste treatment. The fate of heavy metals and organic contaminants in soil and aqueous systems is also discussed.

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** One in-class lab.

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**Prerequisite(s):**

CH102 MA104

-Or-

CH152 MA104

-Or-

CH102 MA204

-Or-

CH152 MA204

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## Civil and Mechanical Engineering

### 1 Courses

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| ME490                                                                                                                                                                                                                                                                                      | TOPICS IN MECHANICAL ENGNRG                                                                                                  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                              | 1990-2                                                                                                                       | <b>Offerings:</b>                          |
| This course provides in-depth study of a special topic in engineering mechanics or mechanical engineering not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the Visiting Professor or a senior mechanical engineering faculty member. |                                                                                                                              | No Course Offerings                        |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                                                                                                       |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                               | TBD. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. |                                            |

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# Electrical Engineering and Computer Science

## 116 Courses

| AI105                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | COMPUTING FUNDAMENTALS                                                      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                      | <b>Offerings:</b>                                                                                                              |
| <p>This course provides an introduction to the fundamentals of computing, cyberspace, and artificial intelligence (AI). The course presents basic program design and construction techniques, with consideration given to principles of software engineering. Problem solving using computing devices and AI tools is a central theme throughout the course as students employ various design methodologies. Students utilize an integrated development environment and contemporary application software. Emphasis is placed on critical thinking, creativity, and learning how to learn. Students are introduced to legal, ethical, professional, and security issues and the challenges, opportunities, and attributes of the cyber domain, including AI.</p>                                                     |                                                                             | 2026-1 2026-2 2026-4<br>2026-5 2027-1 2027-2<br>2027-4 2027-9 2028-1<br>2028-2 2028-4 2028-9<br>2029-1 2029-2 2029-4<br>2029-5 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                      |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                                                        |                                                                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CY105<br>-Or-<br>CY155<br>-Or-<br>CY105X<br>-Or-<br>AI105X<br>-Or-<br>AI155 |                                                                                                                                |
| AI105X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | COMPUTING FUNDAMENTALS                                                      | 3.0 Credit Hours<br>(BS=0.0,ET=2.0,MA=0.0)                                                                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                      | <b>Offerings:</b>                                                                                                              |
| <p>This course provides an introduction to the fundamentals of computing and Cyberspace. The course presents basic program design and construction techniques, with consideration given to principles of software engineering. Problem solving using computing devices as tools is a central theme throughout the course as students employ various design methodologies. Students utilize an integrated development environment and contemporary application software. Emphasis is placed on critical thinking, creativity, and learning how to learn. Students are introduced to legal, ethical, professional, and security issues and the challenges, opportunities, and attributes of the cyber domain. *This is a pilot course and must undergo review by the Curriculum Committee NLT AY26-2 to continue.*</p> |                                                                             | 2026-1                                                                                                                         |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                      |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                                                        |                                                                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CY105<br>-Or-<br>CY105X<br>-Or-<br>AI105<br>-Or-<br>AI105X                  |                                                                                                                                |
| AI155                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | EXPLORATIONS IN COMPUTING FUND                                              | 3.0 Credit Hours<br>(BS=0.0,ET=0.5,MA=0.0)                                                                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                                                      | <b>Offerings:</b>                                                                                                              |
| <p>This course provides a more advanced opportunity to explore the fundamentals of computing and Cyberspace. The course presents basic program design and construction techniques, with consideration given to principles of software engineering. Problem solving using computing devices as tools is a central theme throughout the course as students employ various design methodologies. Students utilize an integrated development environment and contemporary application software. Emphasis is placed on critical thinking, creativity, and learning how to learn. Students are introduced to legal, ethical, professional, and security issues and the challenges, opportunities, and attributes of the cyber domain.</p>                                                                                  |                                                                             | No Course Offerings                                                                                                            |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                      |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                                                        |                                                                                                                                |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CY105<br>-Or-<br>CY155<br>-Or-<br>AI105<br>-Or-<br>AI105X<br>-Or-<br>CY105X |                                                        |
| <b>AI300</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>PROGRAMMING FUNDAMENTALS</b>                                             | <b>3.0 Credit Hours<br/>(BS=0.0,ET=2.5,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                                                      | <b>Offerings:</b>                                      |
| Cadets learn fundamental computing concepts that will allow them to design, build and test small to medium programs using a high-level programming language. Key concepts include applying appropriate aspects of a structured problem-solving process, applying a standardized design notation such as the Unified Modeling Language (UML) to communicate their design, and iteratively testing their program. **This is a placeholder course for the new Artificial Intelligence and Data Engineering Core Engineering Sequence. Course Scope will be updated NLT AY26-1.**                                                                                                                                                                            |                                                                             | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                      |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                        |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AI105 AI105X<br>-Or-<br>AI155                                               |                                                        |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CY300                                                                       |                                                        |
| <b>AI350</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>DATA ENGINEERING</b>                                                     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-2                                                                      | <b>Offerings:</b>                                      |
| Cadets engineer a data-processing pipeline in a high-level programming language, which includes the acquisition of data from various sources and the cleaning, transformation, and storage of that data via flat files and other various storage systems. Cadets explore the data and visualize it to gain understanding and insights to ultimately make predictions, solve problems, and aid in decision-making. **This is a provisionally approved course for the Artificial Intelligence and Data Engineering Core Engineering Sequence. AI350 must undergo review by the Curriculum Committee NLT AY26-1 to continue.**                                                                                                                              |                                                                             | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                      |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                        |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AI300                                                                       |                                                        |
| <b>AI450</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>ENGINEERING AI</b>                                                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-2                                                                      | <b>Offerings:</b>                                      |
| Cadets explore the potential of data-driven artificial intelligence (AI) algorithms to solve complex engineering problems. Cadets build models that rival and even surpass human-level performance on a range of tasks. Cadets analyze problems, design and compare models suited to the objective, and implement their approach in software. The course culminates in an open-ended project of the cadets' choosing. In this project, cadets build a model that achieves superior results on a task that would otherwise require human cognition to perform. **This is a provisionally approved course for the Artificial Intelligence and Data Engineering core engineering sequence. It must undergo review by the Curriculum Committee NLT AY26-2.** |                                                                             | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                      |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | A course project is required. Compensatory time is given.                   |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AI350<br>-Or-<br>CS486<br>-Or-<br>MA477                                     |                                                        |
| <b>CS380</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>COMPUTER SYSTEMS &amp; ORGANIZATN</b>                                    | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>     |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2023-1                                                    | <b>Offerings:</b>                                                              |
| This course provides an introduction to computer systems, computer organization and related concepts. Emphasis is placed on understanding the implications of computer hardware and architecture on the performance and security of written code. Students learn basic C programming and assembly language. Topics covered include basic computer organization, architecture, reverse engineering and parallel computing. In addition to theory, students gain practical real-world experience using tools for profiling and debugging. By the end of this programming-intensive course, students will understand how the fundamental principles of computer systems impact their ability to write efficient code.                                                                                                                                                                                                                                                                                                  |                                                           | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                  |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 9 @ 120 min                                  |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                                      |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | -Or-<br>CS384 CY300<br>-Or-<br>CY300 CY384                |                                                                                |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | EE360                                                     |                                                                                |
| <b>CS384</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>DATA STRUCTURES</b>                                    | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=3.0,MA=0.0)                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2019-2                                                    | <b>Offerings:</b>                                                              |
| This course is designed to build on the cadet's basic programming knowledge. Major emphasis is placed on object-based design, programming methodology, algorithms and algorithm analysis, data structures, and abstract data types as tools for the analysis, design, and implementation of software modules to meet specified requirements. Cadets will learn and employ several well-known algorithms and data structures. Techniques of searching, sorting, recursion, and hashing will be examined. Data structures such as sets, heaps, linked lists, stacks, queues, and trees will be covered. A block-structured programming language reflecting comprehensive support for good software engineering principles will be the foundation of application-oriented exercises. Cadets will design software solutions by employing problem decomposition and selecting the appropriate algorithms and abstract data types.                                                                                        |                                                           | 2026-1 2026-2 2027-1<br>2027-2 2027-8 2028-1<br>2028-2 2028-8 2029-1<br>2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                    |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                                      |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CS360<br>-Or-<br>CS360A<br>-Or-<br>CS301<br>-Or-<br>CY300 |                                                                                |
| <b>CS385</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>DESIGN &amp; ANALYS-ALGORITHMS</b>                     | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2007-2                                                    | <b>Offerings:</b>                                                              |
| This course studies analysis of algorithms and the relevance of analysis to the design of efficient computer algorithms. Algorithmic approaches covered include greedy, divide and conquer, and dynamic programming. Topics include sorting, searching, graph algorithms, and disjoint set structure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                           | 2026-2 2027-2 2027-9<br>2028-2 2028-9 2029-2<br>2029-8 2029-9                  |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                    |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                                      |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CS384 MA372                                               |                                                                                |
| <b>CS386X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>APPLIED NEURAL NETWORKS</b>                            | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=3.0,MA=0.0)                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024-2                                                    | <b>Offerings:</b>                                                              |
| This course builds solely on basic programming skills to introduce the field of deep learning, neural network architectures, and learning algorithms in a way accessible to cadets pursuing a variety of majors. Cadets will develop an understanding of emerging trends and research in the field. Major emphasis is placed on applying neural networks to problems in a variety of domains by training and tuning models for tasks such as pattern recognition, time series prediction, data mining, and optimization. Hands-on exercises, programming assignments, and case studies will provide cadets with valuable experience using state-of-the-art software libraries and pre-trained models (PTMs). The course culminates in an open-ended project of the cadets' choosing but intended to be an application of course content to cadets' personal research projects. *This is a pilot course and must be reviewed by the Curriculum Committee NLT AY26-1 to continue. ET credit pending ABET-PEV review." |                                                           | No Course Offerings                                                            |

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** CY300

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| <b>CS387</b> | <b>IND STUDY IN COMPUTER SCI 1CR</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2022-2

**Offerings:**

The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor.

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| <b>CS387A</b> | <b>IND STUDY IN COMP SCI 1CR - A</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2022-2

**Offerings:**

The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):** CS387

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| <b>CS387B</b> | <b>IND STUDY IN COMP SCI 1CR - B</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|--------------------------------------|----------------------------------------------------------|

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**Scope:** 2022-2

**Offerings:**

The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):** CS387 CS387A

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| <b>CS388</b> | <b>IND STUDY IN COMPUTER SCI 2CR</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2022-2

**Offerings:**

The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2026-2 2027-1 2027-2  
2028-1 2028-2 2029-1  
2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min



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| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                    |
| <b>CS388A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>IND STUDY IN COMP SCI 2CR - A</b>                                                                                        | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2022-2                                                                                                                      | <b>Offerings:</b>                                  |
| The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                             | 2027-2 2028-2                                      |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CS388                                                                                                                       |                                                    |
| <b>CS388B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>IND STUDY IN COMP SCI 2CR - B</b>                                                                                        | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2022-2                                                                                                                      | <b>Offerings:</b>                                  |
| The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                             | 2027-2 2028-2                                      |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CS388 CS388A                                                                                                                |                                                    |
| <b>CS388C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>IND STUDY IN COMP SCI 2CR - C</b>                                                                                        | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-1                                                                                                                      | <b>Offerings:</b>                                  |
| The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                             | 2029-1                                             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CS388 CS388A CS388B                                                                                                         |                                                    |
| <b>CS388D</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>IND STUDY IN COMP SCI 2CR - D</b>                                                                                        | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-2                                                                                                                      | <b>Offerings:</b>                                  |
| The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                             | 2029-2                                             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                    |

**Prerequisite(s):** CS388 CS388A CS388B CS388C

|              |                                      |                                                    |
|--------------|--------------------------------------|----------------------------------------------------|
| <b>CS389</b> | <b>IND STUDY IN COMPUTER SCIENCE</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------|

**Scope:** 2026-1

**Offerings:**

The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor.

|               |                                      |                                                    |
|---------------|--------------------------------------|----------------------------------------------------|
| <b>CS389A</b> | <b>IND STUDY IN COMPUTER SCIENCE</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|--------------------------------------|----------------------------------------------------|

**Scope:** 2026-5

**Offerings:**

independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2026-1 2026-8 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):** CS389

|               |                                      |                                                    |
|---------------|--------------------------------------|----------------------------------------------------|
| <b>CS389B</b> | <b>IND STUDY IN COMP SCI 3CR - B</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|--------------------------------------|----------------------------------------------------|

**Scope:** 2022-2

**Offerings:**

The student pursues study of a topic in Computer Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Computer Science Program Director. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):** CS389 CS389A

|              |                         |                                                    |
|--------------|-------------------------|----------------------------------------------------|
| <b>CS393</b> | <b>DATABASE SYSTEMS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|-------------------------|----------------------------------------------------|

**Scope:** 2022-2

**Offerings:**

This course addresses the analysis, design implementation, and management of relational databases. Structured query language (SQL) is covered in depth along with standard problem domain and data modeling techniques. Implementation techniques and considerations are discussed and practiced extensively. Key concepts include analysis and design using a standardized notation, data model to logical schema conversion techniques, normalization, data security, client-server architectures and web-based access to database systems. Students are exposed to user and privilege management, database backup and recovery, and security vulnerabilities such as SQL injection.

2026-1 2027-1 2027-8  
2028-1 2028-8 2029-1  
2029-2 2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):**

-Or-

-Or-

-Or-

CY300 CY305

-Or-

CY300 CY355

-Or-

CY300 CY305X

**Disqualifier(s):**

IT393

**CS394****DISTRIB APPLICATION ENGRNG****3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)****Scope:**

2020-1

**Offerings:**

2029-1

Building on the foundations of algorithm implementation, data structures, data representation, and object oriented programming this course focuses on the principles of designing, implementing, and testing a modern distributed application. Cadets study the construction and interaction of user interface, network, web server, database, and other components to produce an integrated working secure system. Cadets will learn new tools and skills working as a team to analyze, design, and implement a system that solves a given problem. This is one of the courses that a Computer Science major can choose from a list of elective courses and the focus is on data structure concepts and object oriented programming.

**Lessons:** 30 @ 75 min (2.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

None

**Corequisite(s):**

CS403

**Disqualifier(s):**

IT394

**CS400****PRO CONSIDERATIONS IN COMPUTIN****3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)****Scope:**

2020-2

**Offerings:**2026-2 2027-2 2028-2  
2029-2

This course addresses professional considerations for computing professionals, primarily focusing on non-technical considerations and the development of communication skills. Coursework includes significant emphasis on written work that is based on relevant reading assignments, class discussions, individual research, distinguished guest speakers, and personal reflection. Content will address current, emerging, and relevant topics in the computing profession. Students will develop the ability to recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles. They will also learn to identify and discuss local and global impacts of computing solutions on individuals, organizations, and society. Students will demonstrate the ability to communicate effectively in writing in a variety of professional contexts, including an iterative writing experience.

**Lessons:** 40 @ 55 min (2.500 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

None

**Corequisite(s):**

XE401

**CS403****SOFTWARE TESTING & DEVELOPMENT****3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)****Scope:**

2019-2

**Offerings:**2026-2 2026-8 2027-2  
2027-8 2028-2 2028-8  
2029-1 2029-2 2029-8

This course builds on the fundamental programming skills from prerequisite courses to explore advanced concepts used in modern object oriented software design to create software that is robust, reusable, and extensible in varying problem domains. Cadets gain confidence in their abilities to model, implement, and test solutions to demanding programming problems.

**Lessons:** 30 @ 75 min (2.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

Design projects; compensatory time provided.

**Prerequisite(s):** CS384 IT305  
 -Or-  
 CS384 IT355  
 -Or-  
 CS384 CY305  
 -Or-  
 CS384 CY355  
 -Or-  
 CS384 CY305X

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|              |                          |                                                          |
|--------------|--------------------------|----------------------------------------------------------|
| <b>CS473</b> | <b>COMPUTER GRAPHICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------|----------------------------------------------------------|

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**Scope:** 2019-2

**Offerings:**

This course concerns computer programs that draw two- and three-dimensional objects on computer output devices and receive input from users through graphical input devices. Cadets implement interactive programs through a commonly available graphical application programmers' interface (API). They learn about graphical hardware devices and the elegant algorithms that underlie the API, including elementary computational geometry, continuous time physical simulation, homogeneous transformations, parametric forms, clipping, shading, color, and surface rendering. These concepts are all illustrated with examples of military data visualization including two-dimensional maps and three-dimensional battle simulation and terrain visualization.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** CS384 MA205 PH203  
 -Or-  
 CS384 MA255 PH203  
 -Or-  
 CS384 MA255 PH253  
 -Or-  
 CS384 MA205 PH253  
 -Or-  
 CS384 MA205 PH201  
 -Or-  
 CS384 MA255 PH201  
 -Or-  
 CS384 MA255 PH251  
 -Or-  
 CS384 MA205 PH251  
 -Or-  
 CS384 MA104 PH205  
 -Or-  
 CS384 MA104 PH255  
 -Or-  
 CS384 MA104 PH201X  
 -Or-  
 CS384 MA205 PH201X  
 -Or-  
 CS384 MA255 PH201X  
 -Or-  
 CS384 MA104 PH201  
 -Or-  
 CS384 MA104 PH251  
 -Or-  
 CS384 MA204X PH201  
 -Or-  
 CS384 MA204 PH201  
 -Or-  
 CS384 MA204 PH251

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|              |                                      |                                                          |
|--------------|--------------------------------------|----------------------------------------------------------|
| <b>CS474</b> | <b>INTRO TO THEORETICAL COMP SCI</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------------|

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**Scope:** 2023-2

**Offerings:**

This course introduces computer science theory through the study of abstract machines, grammars, languages, decidability, and NP-completeness. Students evaluate fundamental limits of these machines and grammars and classify languages according to the Chomsky hierarchy; apply various techniques to prove facts about these machines, grammars, and languages; recognize the difference between problems that are and are not solvable; and determine when a problem is NP-complete. Throughout, the course links fundamental computer science theory to modern-day practical computing devices and computational problems.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                |                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------|
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                           |                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | EE360 MA372                                    |                                                                       |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | CS385                                          |                                                                       |
| <b>CS478</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>PROGRAMMING LANGUAGES</b>                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2022-2                                         | <b>Offerings:</b>                                                     |
| Concepts of high-level programming language design are explored in detail. Cadets will examine the fundamental issues of programming language design and use this knowledge as a framework for comparison of different high-level languages. Cadets will study concepts from some or all of the imperative, functional, object-oriented, concurrent, and logic programming language paradigms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                | 2026-1 2027-1 2028-1 2029-1                                           |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                         |                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                           |                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CS384                                          |                                                                       |
| <b>CS481</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>OPERATING SYSTEMS</b>                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2020-1                                         | <b>Offerings:</b>                                                     |
| The operating system controls the computer itself and provides a secure and useful interface for users and application programs. The operating system controls all the computer resources: processors, main storage, secondary storage, I/O devices, and files. It determines which programs will be in memory at any given time and the order in which programs will run. The operating system should resolve conflicts between processes, attempt to optimize the performance of the computer, allow the computer to communicate with other computers, and maintain a record of actions performed as it goes about its system tasks. This course investigates the basic design issues encountered in order to produce an operating system that can address the above problems in an efficient manner. These concepts are reinforced by a series of programming projects that include both design and implementation. |                                                | 2026-1 2027-1 2028-1 2029-1                                           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                         |                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Programming projects; compensatory time given. |                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CS380 CS403                                    |                                                                       |
| <b>CS483</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>DIGITAL FORENSICS</b>                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2019-2                                         | <b>Offerings:</b>                                                     |
| Digital Forensics will explore the evidence left behind when malicious activity occurs on an information system. The material in this course will build on your knowledge of Operating Systems, file formats, file system structure, computer architecture, and networking. The course begins with an overview of these areas, then examines how to find and extract digital evidence. During the course, you will be challenged with three projects (subjects to be chosen by you) and in class challenges that will allow you to demonstrate your understanding of the material.                                                                                                                                                                                                                                                                                                                                     |                                                | 2026-2 2027-2 2028-2 2029-2                                           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                         |                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                           |                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CS380 CS481                                    |                                                                       |
| <b>CS484</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ADVANCED COMPUTER NETWORKS</b>              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2024-2                                         | <b>Offerings:</b>                                                     |
| This course builds on an understanding of basic networking topics and provides cadets with an advanced understanding of computer networks. This course explores more deeply the problems facing modern computer networks. It covers advanced networking techniques that are enabling rapid innovation to solve challenges in routing, congestion control, traffic engineering, mobility, and reliability. Specific topics covered will vary but may include software-defined networking and programmable network devices.                                                                                                                                                                                                                                                                                                                                                                                              |                                                | 2026-1 2026-2 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2 2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                         |                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                           |                                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                             |                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CY350<br>-Or-<br>CY350X                                                                                                                                                                                                     |                                                        |
| <b>CS485</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>SPEC TOPICS IN COMPUTER SCI</b>                                                                                                                                                                                          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2019-2                                                                                                                                                                                                                      | <b>Offerings:</b>                                      |
| This course provides in-depth study of a special topic in computer science not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the visiting professor or a senior computer science faculty member.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                             | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                      |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | To be determined by the program director.                                                                                                                                                                                   |                                                        |
| <b>CS485A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>SPEC TOPICS IN COMPUTER SCI A</b>                                                                                                                                                                                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2023-1                                                                                                                                                                                                                      | <b>Offerings:</b>                                      |
| This course provides in-depth study of a special topic in computer science not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the visiting professor or a senior computer science faculty member.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                             | 2026-1 2027-1 2028-1                                   |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                      |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | To be determined by the program director. Cadets should complete CS485 before taking CS485A.                                                                                                                                |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CS485                                                                                                                                                                                                                       |                                                        |
| <b>CS486</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ARTIFICIAL INTELLIGENCE</b>                                                                                                                                                                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2022-2                                                                                                                                                                                                                      | <b>Offerings:</b>                                      |
| The course provides an introduction to the field of Artificial Intelligence (AI). Cadets will develop an appreciation for the domain of AI and an understanding of the current interest and research in the field. The historical ideas and techniques of AI and the resulting set of concepts will be covered. Classic programs will be covered as well as underlying theory. Topics include a history of computer problem solving, heuristic search techniques, knowledge representation, knowledge engineering, predicate calculus, and expert and/or rule based systems. Advanced topics that may be covered include intelligent agents, genetic algorithms, neural networks, fuzzy logic, robotics, vision, natural language processing, learning, and the programming languages of AI. The course will emphasize the practical application of artificial intelligence to industry and business as well as DoD. |                                                                                                                                                                                                                             | 2026-1 2027-1 2028-1<br>2029-1                         |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                      |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Term project/paper; compensatory time given.                                                                                                                                                                                |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CS384 EE360 MA206<br>-Or-<br>CS384 EE300 MA206<br>-Or-<br>CS384 EE360 MA256<br>-Or-<br>CS384 EE300 MA256<br>-Or-<br>AI350 MA206<br>-Or-<br>CS384 MA206<br>-Or-<br>AI350 MA256<br>-Or-<br>CS384 MA256<br>-Or-<br>MA477 MA486 |                                                        |
| <b>CS489</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ADV IND STUDY COMPUTER SCI</b>                                                                                                                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1990-1                                                                                                                                                                                                                      | <b>Offerings:</b>                                      |

The detailed syllabus of this elective will be tailored to the specific project and to qualifications of the cadet. The research or study program will be proposed by the cadet or selected from those proposed by the department. The cadet will formalize a proposal, design a viable research plan, and conduct research under the guidance and supervision of a faculty advisor. The Head of the Department will approve cadet projects. Lessons and labs established by consultation between cadet and advisor.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

At least 3.0 average in CS courses normally required. Grades based largely on research paper/presentation to faculty. Participation in Eastern Collegiate Science Conference/publication of research are options.

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------|
| <b>CS489A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>ADV IND STUDY COMPUTER SCI</b>  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1995-1                             | <b>Offerings:</b>                                             |
| Same as CS489.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                    | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Same as CS489.                     |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CS489                              |                                                               |
| <b>CS489B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>ADV IND STUDY COMPUTER SCI</b>  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1995-1                             | <b>Offerings:</b>                                             |
| Temp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                    | 2026-2                                                        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                               |                                                               |
| <b>CS490</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>COMPUTR SCI SUMMER RESEARCH</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1990-4                             | <b>Offerings:</b>                                             |
| This course is designed to familiarize the cadet with advanced techniques for independent research in computer science. The course will normally require research, development, and implementation of a novel idea or concept. An oral presentation and a written project report will be completed under the supervision of a USMA faculty member who serves as project advisor. The course requires three full weeks of study, completed in conjunction with the Academic Individual Advanced Development program. Scope, depth, and material covered will meet the requirements of a three-credit course in computer science. |                                    | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Oral and written reports.          |                                                               |
| <b>CS490A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>COMPUTR SCI SUMMER RESEARCH</b> | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1990-4                             | <b>Offerings:</b>                                             |
| This course is designed to familiarize the cadet with advanced techniques for independent research in computer science. The course will normally require research, development, and implementation of a novel idea or concept. An oral presentation and a written project report will be completed under the supervision of a USMA faculty member who serves as project advisor. The course requires three weeks of study, completed in conjunction with the Academic Individual Advanced Development program. Scope, depth, and material covered will be equivalent to two credits of course work in computer science.         |                                    | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                               |                                                               |
| <b>CS490B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>COMPUTR SCI SUMMER RESEARCH</b> | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                         |                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1990-4                                  | <b>Offerings:</b>                                  |
| <p>This course is designed to familiarize the cadet with advanced techniques for independent research in computer science. No Course Offerings</p> <p>The course will normally require research, development, and implementation of a novel idea or concept. An oral presentation and a written project report will be completed under the supervision of a USMA faculty member who serves as project advisor. The course requires three weeks of study, completed in conjunction with the Academic Individual Advanced Development program. Scope, depth, and material covered will be equivalent to one credit of course work in computer science.</p>                                                                                                                                                             |                                         |                                                    |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0 @ 0 min (0.000 Att/wk)                | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Oral and written reports.               |                                                    |
| <b>CY105</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>COMPUTING FUNDAMENTALS</b>           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=2.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                  | <b>Offerings:</b>                                  |
| <p>This course provides an introduction to the fundamentals of computing and Cyberspace. The course presents basic program design and construction techniques, with consideration given to principles of software engineering. Problem solving using computing devices as tools is a central theme throughout the course as students employ various design methodologies. Students utilize an integrated development environment and contemporary application software. Emphasis is placed on critical thinking, creativity, and learning how to learn. Students are introduced to legal, ethical, professional, and security issues and the challenges, opportunities, and attributes of the cyber domain.</p>                                                                                                      |                                         |                                                    |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 40 @ 55 min (2.500 Att/wk)              | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                    |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | IT105<br>-Or-<br>IT155<br>-Or-<br>CY155 |                                                    |
| <b>CY105X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>COMPUTING FUNDAMENTALS</b>           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=2.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1                                  | <b>Offerings:</b>                                  |
| <p>This course provides an introduction to the fundamentals of computing and Cyberspace. The course presents basic program design and construction techniques, with consideration given to principles of software engineering. Problem solving using computing devices as tools is a central theme throughout the course as students employ various design methodologies. Students utilize an integrated development environment and contemporary application software. Emphasis is placed on critical thinking, creativity, and learning how to learn. Students are introduced to legal, ethical, professional, and security issues and the challenges, opportunities, and attributes of the cyber domain. *This is a pilot course and must undergo review by the Curriculum Committee NLT AY26-2 to continue.*</p> |                                         |                                                    |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 40 @ 55 min (2.500 Att/wk)              | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                    |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CY105<br>-Or-<br>CY155                  |                                                    |
| <b>CY155</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>EXPLORATIONS IN COMPUTING FUND</b>   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.5,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2021-1                                  | <b>Offerings:</b>                                  |
| <p>This course provides a more advanced opportunity to explore the fundamentals of computing and Cyberspace. The course presents basic program design and construction techniques, with consideration given to principles of software engineering. Problem solving using computing devices as tools is a central theme throughout the course as students employ various design methodologies. Students utilize an integrated development environment and contemporary application software. Emphasis is placed on critical thinking, creativity, and learning how to learn. Students are introduced to legal, ethical, professional, and security issues and the challenges, opportunities, and attributes of the cyber domain.</p>                                                                                  |                                         |                                                    |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 30 @ 75 min (0.000 Att/wk)              | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                    |                                                    |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                       |                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | IT105<br>-Or-<br>IT155<br>-Or-<br>CY105                                                                               |                                                                                                                                                               |
| <b>CY300</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>PROGRAMMING FUNDAMENTALS</b>                                                                                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=2.5,MA=0.0)</b>                                                                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-2                                                                                                                | <b>Offerings:</b>                                                                                                                                             |
| Cadets learn fundamental computing concepts that will allow them to design, build and test small to medium programs using a high-level programming language. Key concepts include applying appropriate aspects of a structured problem solving process, applying a standardized design notation such as the Unified Modeling Language (UML) to communicate their design, and iteratively testing their program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                       | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-8<br>2028-1 2028-2 2028-8<br>2029-1 2029-2 2029-8<br>2029-9                                                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                                |                                                                                                                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                                  |                                                                                                                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | IT105<br>-Or-<br>IT155<br>-Or-<br>CY105<br>-Or-<br>CY155<br>-Or-<br>AI105<br>-Or-<br>AI105X<br>-Or-<br>AI155          |                                                                                                                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CS301<br>-Or-<br>IT300<br>-Or-<br>CS300                                                                               |                                                                                                                                                               |
| <b>CY305</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CYBER FOUNDATIONS</b>                                                                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=1.5,MA=0.0)</b>                                                                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                | <b>Offerings:</b>                                                                                                                                             |
| This course examines the critical role of emerging technologies in modern warfare, with particular emphasis on cyber and computing capabilities. Through case studies, students investigate how technological innovations are integrated onto the battlefield by analyzing historical precedents, evaluating contemporary practices, and exploring future trends. Each case study is complemented by experiential learning activities that reinforce and illustrate central themes including the transient nature of technological superiority, the increasing accessibility of technological advantage through human-machine teaming, and the importance of early exploration, risk-aware adoption, and innovation driven from both the top down and bottom up. Core topics, such as cyberspace operations, electronic warfare, information operations, robotics, and artificial intelligence, are covered with a focus on their operational advantages, risks, and strategic implications. The course also explores emerging technologies more broadly, presents frameworks for assessing their impacts, and introduces the Army's modernization and acquisition processes. By the end of the course, students develop analytical skills, strategic awareness, and the confidence to critically assess and apply technology in support of military operations. *The AY26-1 offering is a pilot that must be reviewed by the Curriculum Committee NLT AY26-2 to continue.* |                                                                                                                       | 2026-1 2026-2 2026-3<br>2026-4 2026-5 2026-9<br>2027-1 2027-2 2027-4<br>2027-5 2028-1 2028-2<br>2028-4 2029-1 2029-2<br>2029-3 2029-4 2029-5<br>2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                                |                                                                                                                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                                  |                                                                                                                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | IT105<br>-Or-<br>IT155<br>-Or-<br>IT105X<br>-Or-<br>CY105<br>-Or-<br>CY155<br>-Or-<br>AI105<br>-Or-<br>AI105X<br>-Or- |                                                                                                                                                               |

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| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | AI155<br>IT305<br>-Or-<br>CY355                                                                                                       |                                                                                       |
| <b>CY305X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>CYBER FOUNDATIONS</b>                                                                                                              | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=1.5,MA=0.0)</b>                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                                                                                                                                | <b>Offerings:</b>                                                                     |
| <p>This course provides a comprehensive overview of the critical role of innovative technologies in modern warfare, with a particular focus on cyber and computing capabilities. Students will explore pathways for integrating new technologies onto the battlefield, examining historical precedents and current practices. Key topics include cyber operations, electronic warfare, information operations, robotics, and artificial intelligence -- with an emphasis on assessing their advantages, risks, and implications for future conflicts. Additionally, the course covers emerging technologies, technology evaluation, and the Army's modernization and acquisition processes. By the end of this course, students will possess analytical skills, strategic insights, and a critical mindset that enables them to evaluate and leverage cutting-edge technologies in military operations. **This is a pilot course and must go through the CC for approval NLT AY25-2 to continue.**</p> |                                                                                                                                       | No Course Offerings                                                                   |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 40 @ 55 min (2.500 Att/wk)                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                                  |                                                                                       |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CY305                                                                                                                                 |                                                                                       |
| <b>CY350</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>COMPUTER NETWORKS</b>                                                                                                              | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b>                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2024-1                                                                                                                                | <b>Offerings:</b>                                                                     |
| <p>This course provides cadets with an introduction to computer networks by breaking the subject into comprehensible parts and building a survey of the state of the art. The goal of the course is to provide each cadet with basic concepts necessary to understand the design and operation of computer networks. Taking a layered approach, it examines the Internet with an emphasis on the TCP/IP protocol suite. Additionally, basic principles including multiplexing, switching, flow control, and error control are covered. Internetworking and its application to both local and wide area networks are also investigated. The course offers an understanding of the current status and future directions of technology and how technology relates to standards.</p>                                                                                                                                                                                                                       |                                                                                                                                       | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2<br>2029-8 2029-9 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 30 @ 75 min (2.000 Att/wk)                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                                  |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CY300                                                                                                                                 |                                                                                       |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | -Or-<br>CY350X                                                                                                                        |                                                                                       |
| <b>CY355</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>CYBER FOUNDATIONS - COMPUTING</b>                                                                                                  | <b>3.0 Credit Hours</b><br><b>(BS=0.5,ET=1.5,MA=0.0)</b>                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2020-1                                                                                                                                | <b>Offerings:</b>                                                                     |
| <p>Provides a more in-depth study of computing for cadets who have demonstrated ability beyond the level of CY305. The course covers material presented in CY305 at an accelerated pace to provide cadets additional opportunities for application and hands-on experience with cyber principles and concepts such as encryption and machine learning, with less emphasis on networking.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                       | 2026-1 2027-1 2028-1<br>2029-1                                                        |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 30 @ 75 min (2.000 Att/wk)                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CY355 is designed primarily for CS and IT majors and Cyber minors. Enrollment by other cadets with permission of the Department Head. |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | IT105<br>-Or-<br>IT155<br>-Or-<br>CY105<br>-Or-<br>CY155<br>-Or-<br>AI105<br>-Or-<br>AI105X<br>-Or-                                   |                                                                                       |

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| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | AI155<br>CY305<br>-Or-<br>IT305      |                                                          |
| <b>CY383</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>SECURE INTERFACE DESIGN</b>       | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2022-2                               | <b>Offerings:</b>                                        |
| <p>This course provides a practical introduction to user interface development and usability engineering of interactive applications. The disciplines of Human-Computer Interaction (HCI), Cybersecurity, and Software Engineering guide these endeavors. Major emphasis is on the principles and techniques for human-centered, secure design and implementation of graphical user interfaces (GUIs) within a software development lifecycle. Cadets will extend their knowledge of programming in a high-level language by learning how to use an interface builder to create a fully functional GUI. Cadets will learn and practice human-centered problem analysis techniques and testing methodologies to ensure that their interfaces are usable and secure. A hypothetico-deductive approach to design is emphasized throughout their development efforts. Fundamentals taught in this course will prepare cadets for more advanced software development, development of physical devices, or a deeper theoretical look at HCI topics.</p> |                                      | 2026-2 2027-2 2028-2<br>2029-2                           |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 30 @ 75 min (2.000 Att/wk)           | <b>Labs:</b> 0 @ 0 min                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                 |                                                          |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CY300                                |                                                          |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | IT383                                |                                                          |
| <b>CY384</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>NETWORK SYSTEMS PROGRAMMING</b>   | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2021-2                               | <b>Offerings:</b>                                        |
| <p>This course applies fundamental programming skills to automate interactions with a computer, a local operating system, or the Internet and so use and manage resources and services. Examples of the resources and services that the programming in this course will address include file systems, web servers, mail servers, database servers, image and audio files, compressed and encrypted files and files used in common office environments (documents, presentations, spreadsheets).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                      | No Course Offerings                                      |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 30 @ 75 min (2.000 Att/wk)           | <b>Labs:</b> 0 @ 0 min                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                 |                                                          |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CY300                                |                                                          |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | IT384                                |                                                          |
| <b>CY385</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>CYBER ALGORITHMIC FOUNDATIONS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2022-2                               | <b>Offerings:</b>                                        |
| <p>This course grounds cadets in algorithms and key topics in the theory of computation, with a focus on how key theoretical techniques help the cyber professional discern what is and is not feasible in cyberspace. Topics include analysis of algorithms, how to use algorithmic complexities to choose between algorithms, algorithmic approaches, and an introduction to formal languages, automata, computational theory, decidability, and the Chomsky hierarchy.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                      | 2026-2 2027-2 2028-2<br>2029-2                           |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 30 @ 75 min (2.000 Att/wk)           | <b>Labs:</b> 0 @ 0 min                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                 |                                                          |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | EE360 MA372                          |                                                          |
| <b>CY387</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>IND STUDY IN CYBER SCI 1CR</b>    | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2022-2                               | <b>Offerings:</b>                                        |
| <p>The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.</p>                                                                                                                                                                                                                                                                                                                                           |                                      | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2   |

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| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                        |
| <b>CY387A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>IND STUDY IN CYBER SCI 1CR - A</b>                                                                                    | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022-2                                                                                                                   | <b>Offerings:</b>                                      |
| The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                          | 2029-2                                                 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CY387                                                                                                                    |                                                        |
| <b>CY387B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>IND STUDY IN CYBER SCI 1CR - B</b>                                                                                    | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022-2                                                                                                                   | <b>Offerings:</b>                                      |
| The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                          | No Course Offerings                                    |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CY387 CY387A                                                                                                             |                                                        |
| <b>CY388</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>IND STUDY IN CYBER SCI 2CR</b>                                                                                        | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022-2                                                                                                                   | <b>Offerings:</b>                                      |
| The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                          | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                        |
| <b>CY388A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>IND STUDY IN CYBER SCI 2CR - A</b>                                                                                    | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022-2                                                                                                                   | <b>Offerings:</b>                                      |
| The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                          | 2027-2 2028-2 2029-2                                   |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                        |

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| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CY388                                                                                                                    |                                                                      |
| <b>CY388B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>IND STUDY IN CYBER SCI 2CR - B</b>                                                                                    | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022-2                                                                                                                   | <b>Offerings:</b>                                                    |
| The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                          | No Course Offerings                                                  |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CY388 CY388A                                                                                                             |                                                                      |
| <b>CY389</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>IND STUDY IN CYBER SCI 3CR</b>                                                                                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022-2                                                                                                                   | <b>Offerings:</b>                                                    |
| The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                          | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                                      |
| <b>CY389A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>IND STUDY IN CYBER SCI 3CR - A</b>                                                                                    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022-2                                                                                                                   | <b>Offerings:</b>                                                    |
| The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                          | 2026-1 2026-8 2027-2<br>2028-2 2029-1 2029-2                         |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                                      |
| <b>CY389B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>IND STUDY IN CYBER SCI 3CR - B</b>                                                                                    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2022-2                                                                                                                   | <b>Offerings:</b>                                                    |
| The student pursues study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                          | No Course Offerings                                                  |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                                   |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor. |                                                                      |
| <b>CY392</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>DEFENSIVE CYBER OPERATIONS</b>                                                                                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                   |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                    | <b>Offerings:</b>                                                                                      |
| This course gives cadets the ability to apply adversarial thinking in order to anticipate and defend resources. The course maintains a breadth of cyber topics, including secure programing best practices, system administration, and network scanning. Cadets will learn how to secure computer systems, patch vulnerabilities, and detect and remove malware from computer systems. **This is a provisionally approved course for the updated Cyber Science major, CYS1. It will need full approval by the Curriculum Committee in AY26-2 to continue. It may be used as a pilot in AY26-1.**                                                                                                                                                                                                                                                                |                                           | 2026-1 2027-1 2028-1                                                                                   |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                      |                                                                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CY350<br>-Or-<br>CY350X                   |                                                                                                        |
| <b>CY394</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CLOUD COMPUTING</b>                    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2022-2                                    | <b>Offerings:</b>                                                                                      |
| This course provides a solid background in cloud computing principles and applications. It includes understanding cloud computing fundamentals, cloud architecture considerations, cloud technology service selection, and cloud computing security controls. The course does not focus on vendor specific cloud solutions like Amazon Web Services or Microsoft Azure, but instead focuses on the tools and management of cloud computing systems that could be deployed on these vendor specific platforms. Cadets will examine various cloud services and recommend solutions to problems utilizing cloud services. They will also learn how to take existing distributed applications and configure and deploy them to cloud environments. The culmination of the course is a group project that solves a problem utilizing cloud services and technologies |                                           | 2026-2 2027-2 2028-2<br>2029-2                                                                         |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                      |                                                                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CS393 CY300<br>-Or-<br>CY300 CY355        |                                                                                                        |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | IT394                                     |                                                                                                        |
| <b>CY450</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CYBER SECURITY ENGINEERING</b>         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2019-2                                    | <b>Offerings:</b>                                                                                      |
| The focus for this course is to design, build and test secure networked computer systems. Topics covered include operating system and network security, secure network architecture, and offensive and defensive information operations. Practical exercises that give students hands-on experience with current network security tools and techniques complement a series of laboratory exercises that have small groups of cadets secure their own small network. In a culminating exercise, cadets design, build and test defensive measures to protect a production network from intrusions.                                                                                                                                                                                                                                                                |                                           | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-9<br>2028-1 2028-2 2028-9<br>2029-1 2029-2 2029-8<br>2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                      |                                                                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CS484<br>-Or-<br>CY350<br>-Or-<br>CY350X  |                                                                                                        |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | CS482                                     |                                                                                                        |
| <b>CY460</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CYBER POLICY, STRATEGY, &amp; OPNS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=1.5,MA=0.0)</b>                                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2024-1                                    | <b>Offerings:</b>                                                                                      |
| This course addresses the entire spectrum of information warfare from the political, legal, and ethical aspects to the technology and techniques of cyber attack. The Political Science and Computer Science faculty jointly teach this course. The course covers how digitization has changed the world and the national security environment of the United States. Students also learn how attack and defense are conducted in cyberspace through classroom discussion and hands-on exercises in the IWAR Laboratory. The course culminates with a group project in which cadets are given a real scenario and possible U.S. objectives and then develop and brief an information operation plan.                                                                                                                                                             |                                           | 2026-1 2026-8 2027-1<br>2028-1 2029-1 2029-8                                                           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |

**Special Requirements:** None

**Prerequisite(s):**

-Or-

-Or-  
CY105  
-Or-  
CY155  
-Or-  
IT105  
-Or-  
AI105  
-Or-  
AI105X  
-Or-  
AI155

**Corequisite(s):** SS307  
-Or-  
SS357

**Disqualifier(s):** IT460

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|              |                                |                                                          |
|--------------|--------------------------------|----------------------------------------------------------|
| <b>CY465</b> | <b>ORGANIZATIONAL SECURITY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------|----------------------------------------------------------|

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**Scope:** 2023-2

**Offerings:**

This course focuses on techniques and considerations related to protecting organizations from cybersecurity threats and managing risk to support successful accomplishment of the organization's mission. Topics include cyber governance and policy, personnel security, security operations, systems administration, cybersecurity planning, procurement, and business continuity.

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One oral group presentation, one 8-10 page research paper, and two shorter preliminary papers; compensatory time provided.

**Prerequisite(s):** CY450 PL300 SS307  
-Or-  
CY450 PL350 SS307  
-Or-  
CY450 PL300 SS357  
-Or-  
CY450 PL350 SS357

**Corequisite(s):** LW403

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>CY482</b> | <b>OFFENSIVE CYBER OPERATIONS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------------|

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**Scope:** 2026-2

**Offerings:**

This course focuses on conducting cyber attacks using tools, custom software, and scripts. In it, cadets will identify vulnerabilities in target systems, write malware to exploit these vulnerabilities, gain unauthorized access, and achieve actions on the objective all while avoiding detection. \*\*This is a provisionally approved course for the updated Cyber Science major, CYS1, and will need full review by the Curriculum Committee NLT AY27-1 to continue. It may be used as a pilot course in AY26-2.\*\*

**Lessons:** 0 @ 0 min (0.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** CS481

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|              |                                       |                                                          |
|--------------|---------------------------------------|----------------------------------------------------------|
| <b>CY485</b> | <b>SPECIAL TOPIC IN CYBER SCIENCE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------------------|----------------------------------------------------------|

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**Scope:** 2023-1

**Offerings:**

This course provides in-depth study of a special topic in cyber science not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the visiting professor or a senior faculty member.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** To be determined by the program director

**CY489** **ADV IND STUDY CYBER SCI 3CR** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2022-2

**Offerings:**

The student pursues study of an advanced topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, and conducts research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor.

**CY489A** **ADV IND STUDY CYBER SCI 3CR-A** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2022-2

**Offerings:**

The student pursues an advanced study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, and conducts research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2027-2 2028-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):** CY489

**CY489B** **ADV IND STUDY CYBER SCI 3CR-B** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2024-1

**Offerings:**

The student pursues an advanced study of a topic in Cyber Science on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, and conducts research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Cyber Science Program Director. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):** CY489 CY489A

**EE300** **FUNDAMENTALS OF DIGITAL LOGIC** **3.0 Credit Hours**  
(BS=0.0,ET=3.0,MA=0.0)

**Scope:** 2026-1

**Offerings:**

This is a course for non-electrical engineering majors that covers the analysis, design, simulation, and construction of digital logic circuits and systems. The course provides the necessary tools to design digital hardware circuits such as clocks, security devices, and computer hardware. The course begins with the study of number systems and digitization, Boolean algebra, and the design of combinational logic circuits. The first half of the course focuses on combinational logic designs. The second half of the course introduces sequential logic circuits like memory systems, counters, and shift registers. Laboratory work reinforces the course material by requiring cadets to design and implement basic digital circuits. Throughout the course, cadets are challenged to understand and leverage increasing levels of abstraction when designing complex digital systems.

2026-1 2026-2 2027-1  
2028-1 2029-1

**Lessons:** 34 @ 55 min (2.500 Att/wk)

**Labs:** 6 @ 120 min

**Special Requirements:** None



**Prerequisite(s):** IT105  
 -Or-  
 IT155  
 -Or-  
 CY105  
 -Or-  
 CY155  
 -Or-  
 AI105  
 -Or-  
 AI105X  
 -Or-  
 AI155

**Disqualifier(s):** EE360

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>EE301</b> | <b>FUNDAMENTALS OF ELEC ENGIN</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

This introductory course in electrical engineering for the non-electrical engineering major provides a foundation in basic circuit theory and analysis, power in circuits and electric power systems, analog and digital electronics, and cyber-physical security. Lectures, laboratory work, practical applications, and classroom demonstrations emphasize and illustrate the fundamental theories and concepts presented in the course. Engineering design is reflected in laboratory work and a design project.

2026-1 2026-2 2026-3  
 2026-4 2026-9 2027-1  
 2027-2 2027-3 2027-8  
 2027-9 2028-1 2028-2  
 2028-3 2028-8 2028-9  
 2029-1 2029-2 2029-3  
 2029-4 2029-5 2029-8  
 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** MA205  
 -Or-  
 MA255  
 -Or-  
 MA366  
 -Or-  
 MA204X  
 -Or-  
 MA204

**Corequisite(s):** PH202  
 -Or-  
 PH252  
 -Or-  
 PH206  
 -Or-  
 PH256  
 -Or-  
 PH275

**Disqualifier(s):** EE302 EE350

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|              |                               |                                                          |
|--------------|-------------------------------|----------------------------------------------------------|
| <b>EE302</b> | <b>INTRO ELECTRICAL ENGIN</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
|--------------|-------------------------------|----------------------------------------------------------|

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**Scope:** 2009-1

**Offerings:**

This first course in electrical engineering provides a solid introduction to electric circuit theory. Fundamental principles and network theorems are developed using DC resistive circuits. The complete responses of RC, RL, and RLC circuits are obtained using classical and Laplace-transform techniques to solve the related differential equations. Electrical system transfer functions, time-domain and frequency-domain relationships, stability, frequency response, steady-state AC analysis, and power are also studied. Laboratory work, practical applications, and classroom demonstrations emphasize and illustrate the fundamentals presented in the course.

2026-2 2027-1 2027-2  
 2028-2 2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** None

|                         |              |
|-------------------------|--------------|
| <b>Corequisite(s):</b>  | MA205 PH202  |
|                         | -Or-         |
|                         | MA205 PH252  |
|                         | -Or-         |
|                         | MA255 PH202  |
|                         | -Or-         |
|                         | MA255 PH252  |
|                         | -Or-         |
|                         | MA205 PH204  |
|                         | -Or-         |
|                         | MA255 PH204  |
|                         | -Or-         |
|                         | MA205 PH206  |
|                         | -Or-         |
|                         | MA205 PH256  |
|                         | -Or-         |
|                         | MA255 PH206  |
|                         | -Or-         |
|                         | MA255 PH256  |
|                         | -Or-         |
|                         | MA205 PH202X |
|                         | -Or-         |
|                         | MA255 PH202X |
|                         | -Or-         |
|                         | MA204X PH202 |
|                         | -Or-         |
|                         | MA204X PH252 |
|                         | -Or-         |
|                         | MA204 PH202  |
|                         | -Or-         |
|                         | MA204 PH252  |
| <b>Disqualifier(s):</b> | EE350        |
|                         | -Or-         |
|                         | EE301        |

|              |                                     |                                                          |
|--------------|-------------------------------------|----------------------------------------------------------|
| <b>EE350</b> | <b>BASIC ELECTRICAL ENGINEERING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|-------------------------------------|----------------------------------------------------------|

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1 | <b>Offerings:</b>                            |
| This is a course for non-electrical engineering majors that provides a foundation in basic circuit theory and analysis, power in circuits, analog electronics, and microcontrollers. Lectures, laboratory work, classroom demonstrations and discussions showing practical applications illustrate the fundamental theories and concepts presented in the course. Students apply their foundational knowledge toward progressively complex integrations of circuits and sensors to model practical robotic systems. Engineering science is reflected in laboratory work. |        | 2026-1 2026-2 2027-1<br>2027-2 2028-2 2029-2 |

**Lessons:** 33 @ 55 min (2.500 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** MA104 PH205  
-Or-  
MA104 PH255  
-Or-  
MA104 PH201X  
-Or-  
MA104 PH201  
-Or-  
MA104 PH251  
-Or-  
MA104 PH275

**Disqualifier(s):** EE302  
-Or-  
EE301

|              |                                      |                                                          |
|--------------|--------------------------------------|----------------------------------------------------------|
| <b>EE360</b> | <b>DIGITAL LOGIC W/ EMBEDDED SYS</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------------|

|               |        |                   |
|---------------|--------|-------------------|
| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

This course introduces the principles and practices of digital system design, from foundational logic circuits to basic processor architecture. Students begin by exploring binary and hexadecimal number systems, Boolean algebra, and their application in designing combinational and sequential circuits. Design techniques such as Karnaugh maps and finite state machines are applied to implement circuits using medium-scale integration (MSI) devices and field-programmable gate arrays (FPGAs). Students also learn the basics of the Very High-Speed Integrated Circuit Hardware Description Language (VHDL) for simulating digital systems and programming hardware. In the final block, the course transitions to microprocessor fundamentals, covering processor components, instruction execution, and machine-level programming, providing a bridge between digital logic design and computer architecture.

2026-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)**Labs:** 7 @ 120 min**Special Requirements:**

None

**Prerequisite(s):**

CS105  
-Or-  
CS155  
-Or-  
IT105  
-Or-  
IT155  
-Or-  
CY105  
-Or-  
CY155  
-Or-  
AI105  
-Or-  
AI105X  
-Or-  
AI155

**Disqualifier(s):**

EE300

**EE360****FUNDAMENTALS OF DIGITAL LOGIC**
**3.5 Credit Hours**  
**(BS=0.0,ET=3.5,MA=0.0)**
**Scope:**

2026-2

**Offerings:**

This course introduces the principles and practices of digital system design, from foundational logic circuits to basic processor architecture. Students begin by exploring binary and hexadecimal number systems, Boolean algebra, and their application in designing combinational and sequential circuits. Design techniques such as Karnaugh maps and finite state machines are applied to implement circuits using medium-scale integration (MSI) devices and field-programmable gate arrays (FPGAs). Students also learn the basics of the Very High-Speed Integrated Circuit Hardware Description Language (VHDL) for simulating digital systems and programming hardware. In the final block, the course transitions to microprocessor fundamentals, covering processor components, instruction execution, and machine-level programming, providing a bridge between digital logic design and computer architecture.

2026-2 2026-8 2026-9  
2027-1 2027-2 2027-8  
2027-9 2028-1 2028-2  
2028-8 2028-9 2029-1  
2029-2 2029-8 2029-9

**Lessons:** 40 @ 55 min (2.500 Att/wk)**Labs:** 7 @ 120 min**Special Requirements:**

None

**Prerequisite(s):**

CS105  
-Or-  
CS155  
-Or-  
IT105  
-Or-  
IT155  
-Or-  
CY105  
-Or-  
CY155  
-Or-  
AI105  
-Or-  
AI105X  
-Or-  
AI155

**Disqualifier(s):**

EE300

**EE362****INTRODUCTION TO ELECTRONICS**
**3.5 Credit Hours**  
**(BS=0.0,ET=3.5,MA=0.0)**
**Scope:**

2014-1

**Offerings:**

This course continues cadet education in electrical engineering through the study of basic electronic devices and circuits. It begins with an introduction to semiconductor physics. It then covers the operation of the pn-junction diode and the metal-oxide semiconductor field-effect transistor (MOSFET) in DC, large-signal, and small-signal regimes. The course emphasizes single-stage amplifier design. The course concludes with an introduction to bipolar junction transistors (BJT) and the design, analysis, simulation, building, and testing of a two-stage audio amplifier. Six laboratory exercises and computer-aided design and analysis using modern circuit simulation software supplement the lectures with practical circuit analysis, design, construction and testing.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** All cadets design, build and test a multistage audio amplifier (0.5 design credits).

**Prerequisite(s):** EE302  
-Or-  
EE301

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|              |                                      |                                                          |
|--------------|--------------------------------------|----------------------------------------------------------|
| <b>EE375</b> | <b>COMPUTER ARCHITECTURE W/MICRO</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------------|

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**Scope:** 2020-1 **Offerings:**

This course provides an introduction to computer organization and design. It builds on digital logic theory and devices to develop more complex systems. Graded assignments emphasize understanding and applying the basics of computer system organization, design, and operation. Students analyze contemporary computer organization by examining the operation of a program at the register level. Students learn, simulate, and program a modern processor. Assembly language programming provides system function control that bridges the gap between hardware and software. C programming demonstrates a high-level language greatly utilized in computing. The course introduces topics including RISC architectures, arithmetic processing, input/output, memory design, and parallel computing.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Projects using Assembly Language and C.

**Prerequisite(s):** EE360

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|              |                                 |                                                          |
|--------------|---------------------------------|----------------------------------------------------------|
| <b>EE377</b> | <b>ELECTRICAL POWER ENGNRNG</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|---------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1 **Offerings:**

This course provides a study of the fundamentals in three areas of electric power engineering: electromechanical energy conversion, electric power systems, and power electronics. Steady-state behavior in single-phase and balanced three-phase power circuits is emphasized. Transformers, AC & DC machines, transmission lines, power systems, power electronic devices, and electrical safety are studied. Laboratory exercises demonstrate the electrical, mechanical, and physical characteristics of several of the systems studied.

**Lessons:** 36 @ 55 min (2.500 Att/wk)      **Labs:** 4 @ 120 min

**Special Requirements:** Computer-aided analysis of a small power system is included.

**Prerequisite(s):** EE302  
-Or-  
EE301

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|              |                                     |                                                          |
|--------------|-------------------------------------|----------------------------------------------------------|
| <b>EE380</b> | <b>EMBEDDED SYSTEMS DEVELOPMENT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|-------------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1 **Offerings:**

This course teaches students how to employ microcontrollers in the designs of embedded systems through applied exercises. Microcontrollers are programmed at the register level using the C language. Students conduct a detailed study of common microcontroller peripherals with an emphasis on their application to real-time control design. Cadets are exposed to addressing, serial and parallel input and output, timing, interrupts, A-to-D and D-to-A conversion, universal asynchronous receiver transmitter protocol, inter-integrated circuit protocol, serial peripheral interface protocol, and low power design. Throughout the course, students practice top-down and state machine design for moderately complex digital systems.

**Lessons:** 24 @ 75 min (1.600 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** EE360

**Disqualifier(s):** EE487

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| EE380                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EMBEDDED SYSTEMS DEVELOPMENT                                                                                                     | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-2                                                                                                                           | <b>Offerings:</b>                                             |
| <p>This course teaches students how to employ microcontrollers and single-board computers in the designs of embedded systems through applied exercises. Microcontrollers are programmed at the register level using the C language. Students conduct a detailed study of common microcontroller peripherals with an emphasis on their application to real-time control design. Cadets are exposed to addressing, serial and parallel input and output, timing, interrupts, A-to-D and D-to-A conversion. Subsequently, students study single-board computers and apply them in more complex designs. The basics of operating systems are taught using Linux and Robot Operating Systems. Emphasis is placed on linking peripherals to the processor and using multiprocessing. Throughout the course, students practice top-down and state machine design for moderately complex digital systems.</p>                                                                                                                                                                                                                                                                                                                          |                                                                                                                                  | No Course Offerings                                           |
| <b>Lessons:</b> 24 @ 75 min (1.600 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                  | <b>Labs:</b> 7 @ 120 min                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                                                             |                                                               |
| EE381                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | SIGNALS AND SYSTEMS                                                                                                              | 3.5 Credit Hours<br>(BS=0.0,ET=3.5,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                           | <b>Offerings:</b>                                             |
| <p>This course provides a general study of linear system theory and signal representation techniques as preparation for continued study in communications, control, and electronic systems. Topics include the resolution of continuous time signals and discrete time sequences into their images as frequency functions using Fourier series and transforms. The study includes singularity functions, convolution, convergence properties, and transform properties. The Laplace transform and its inverse provide a method for determining the system function for systems described by differential equations, while the z-transform and its inverse provide a method of analysis for difference equations. The course includes a brief study of communication system principles to include sampling and a study of analog and digital (both finite and infinite impulse response) filter design. Laboratory exercises emphasize programming with software tools to generate, analyze, and process signals. Students gain experience developing scripts and algorithms to implement signal processing techniques, with opportunities to apply AI-assisted methods and receive instructor guidance in problem solving.</p> |                                                                                                                                  | 2026-1 2026-8 2027-1<br>2028-1 2029-1 2029-8                  |
| <b>Lessons:</b> 40 @ 55 min (3.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                  | <b>Labs:</b> 6 @ 120 min                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                                                             |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | EE301<br>-Or-<br>EE302                                                                                                           |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MA206 MA364<br>-Or-<br>MA206 MA365<br>-Or-<br>MA206X MA364<br>-Or-<br>MA206X MA365<br>-Or-<br>MA256 MA364<br>-Or-<br>MA256 MA365 |                                                               |
| EE383                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ELECTROMAGN FIELDS & WAVES                                                                                                       | 3.5 Credit Hours<br>(BS=0.0,ET=3.5,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                           | <b>Offerings:</b>                                             |
| <p>This course is an introduction to electromagnetic fields, which are the foundation of electrical engineering. The course begins with transmission line analysis using circuit models and reviews the mathematical tools (vector algebra and calculus) that are used to describe electromagnetic phenomena. Maxwell's equations are solved to describe time-harmonic fields under various boundary conditions and at interfaces between dissimilar media. Additional topics include the applications of electromagnetic field theory to transmission lines, antennas and waveguides, and the role of electromagnetics in science, technology and society. Laboratory periods provide opportunities for instructor-assisted problem solving, technology demonstrations, and an experimental lab with an associated lab report. Finally, students complete a computer aided design project on transmission lines.</p>                                                                                                                                                                                                                                                                                                          |                                                                                                                                  | 2026-2 2026-8 2027-2<br>2028-2 2029-1 2029-2<br>2029-8 2029-9 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                  | <b>Labs:</b> 4 @ 120 min                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                                                             |                                                               |

|                         |                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite(s):</b> | PH204<br>-Or-<br>PH254<br>-Or-<br>PH202<br>-Or-<br>PH252<br>-Or-<br>PH206<br>-Or-<br>PH256<br>-Or-<br>PH202X<br>-Or-<br>PH275 |
| <b>Corequisite(s):</b>  | MA364<br>-Or-<br>MA365                                                                                                        |

|              |                                   |                                                    |
|--------------|-----------------------------------|----------------------------------------------------|
| <b>EE387</b> | <b>IND STUDY IN ELECT ENG 1CR</b> | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------|

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2022-2                                                                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                                                                                                                                                    | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor. |                                                               |

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|---------------|---------------------------------------|----------------------------------------------------|
| <b>EE387A</b> | <b>IND STUDY IN ELECT ENG 1CR - A</b> | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2022-2                                                                                                                                                                                                                                             | <b>Offerings:</b>                            |
| The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                                                                                                                                                                                                                                                    | 2026-1 2026-2 2027-2<br>2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                             |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor. |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | EE387                                                                                                                                                                                                                                              |                                              |

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| <b>EE387B</b> | <b>IND STUDY IN ELECT ENG 1CR - B</b> | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2022-2                 | <b>Offerings:</b>   |
| The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |                        | No Course Offerings |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min |                     |

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| <b>Special Requirements:</b> | Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor. |
| <b>Prerequisite(s):</b>      | EE387 EE387A                                                                                                                                                                                                                                       |

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|--------------|-----------------------------------|----------------------------------------------------|
| <b>EE388</b> | <b>IND STUDY IN ELECT ENG 2CR</b> | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2022-2 | <b>Offerings:</b>                                             |
| The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |        | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

|                              |                                                                                                                                                                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Special Requirements:</b> | Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor. |
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|---------------|---------------------------------------|----------------------------------------------------|
| <b>EE388A</b> | <b>IND STUDY IN ELECT ENG 2CR - A</b> | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2022-2 | <b>Offerings:</b>                                      |
| The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |        | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2 |

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

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| <b>Special Requirements:</b> | Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor. |
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**Prerequisite(s):** EE388

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|---------------|---------------------------------------|----------------------------------------------------|
| <b>EE388B</b> | <b>IND STUDY IN ELECT ENG 2CR - B</b> | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|---------------------------------------|----------------------------------------------------|

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2022-2 | <b>Offerings:</b>   |
| The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head. |        | No Course Offerings |

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

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| <b>Special Requirements:</b> | Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor. |
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**Prerequisite(s):** EE388 EE388A

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|--------------|-----------------------------------|----------------------------------------------------|
| <b>EE389</b> | <b>IND STUDY IN ELECT ENG 3CR</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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|---------------|--------|-------------------|
| <b>Scope:</b> | 2022-2 | <b>Offerings:</b> |
|---------------|--------|-------------------|

The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor.

**EE389A**

**IND STUDY IN ELECT ENG 3CR - A**

**3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)**

**Scope:**

2022-2

**Offerings:**

The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2026-1 2027-2 2028-2  
2029-1

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):**

EE389

**EE389B**

**IND STUDY IN ELECT ENG 3CR - B**

**3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)**

**Scope:**

2022-2

**Offerings:**

The student pursues study of a topic in Electrical Engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the student based on resources and in consultation with a faculty advisor. The cadet will formalize a proposal, design a viable research plan, or conduct research under the guidance and supervision of a faculty member. In consultation with a faculty advisor, the student will write a proposal that outlines the scope of the project, includes graded requirements, and establishes lesson and lab meetings, as appropriate. Proposals must be approved by the Department Head.

2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Enrollment by permission of the Electrical Engineering Program Director. Appropriate ET credit will be determined by the Electrical Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):**

EE389 EE389A

**EE400**

**EE PROFESSIONAL CONSIDERATIONS**

**3.0 Credit Hours**  
**(BS=0.0,ET=3.0,MA=0.0)**

**Scope:**

2019-2

**Offerings:**

This course addresses the concerns of professional electrical engineers such as engineering ethics, economics, licensing, manufacturability, sustainability, reliability, safety, and design methodologies. It includes Fundamentals of Engineering Exam preparation and supports the USMA writing program as a Writing in the Major course. The course includes all first class cadets majoring in electrical engineering. Guest lecturers from military, industrial, and academic communities will present some of the material.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

**Corequisite(s):**

XE401

**EE450**

**MILITARY ROBOTIC SYSTEMS**

**3.0 Credit Hours**  
**(BS=0.0,ET=3.0,MA=0.0)**



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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                         | <b>Offerings:</b>                                             |
| This is the capstone course of a three course engineering sequence designed to introduce students to the fundamentals of robotics engineering. Students apply these fundamentals, along with foundations of digital logic and circuit design acquired in the sequence, to interface various sensors and actuators with a simple microprocessor using experiments that demonstrate some basic applications of a simple robot. Finally, students apply the engineering design process to guide their design of a robot from a moderately complex state machine to a fully integrated robotic system capable of performing some practice military robotic applications.                                                                             |                                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 6 @ 120 min                                                                                                       |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A design project is required. Compensatory time given.                                                                         |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EE300 EE350<br>-Or-<br>EE300 EE302<br>-Or-<br>EE350 EE360<br>-Or-<br>EE302 EE360<br>-Or-<br>EE300 EE301<br>-Or-<br>EE301 EE360 |                                                               |
| <b>EE462</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ELECTRONIC DESIGN</b>                                                                                                       | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                         | <b>Offerings:</b>                                             |
| This course focuses on the design, simulation, building, and testing of a wide variety of application-oriented circuits based upon the bipolar junction transistor (BJT) and operational amplifier (OPAMP). Applications of the BJT include current sources, active loads, differential amplifiers, and power amplifiers. OPAMP applications include mathematical operations and active filters. Themes common to both the BJT and OPAMP are frequency response and feedback. In addition, printed circuit board design is introduced through a skill-builder power supply kit. The classroom material is supplemented with seven labs, computer-aided simulations using modern circuit simulation software, and a comprehensive design project. |                                                                                                                                | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 7 @ 120 min                                                                                                       |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A major design project requires cadets to design, build, and test an electrical system.                                        |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EE360 EE362                                                                                                                    |                                                               |
| <b>EE477</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>DIGITAL COMMUNICATIONS SYSTEMS</b>                                                                                          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                         | <b>Offerings:</b>                                             |
| This course examines modern digital communications systems. The study of digital communications systems includes waveform sampling, time multiplexing, line coding, digital modulation, and clock recovery techniques. Time and frequency domain analysis are the basis for study of bandwidth considerations, filtering, and channel and communication system modeling. Mathematics topics include Fourier analysis, probability, information theory, correlation, random variables, and entropy. Digital communication resilience is explored through channel capacity, line coding, and error correcting codes. Four labs are utilized to put topics into practice.                                                                           |                                                                                                                                | 2026-2 2026-8 2027-2<br>2028-2 2029-2                         |
| <b>Lessons:</b> 37 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 3 @ 120 min                                                                                                       |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Course project.                                                                                                                |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EE362 EE381 MA206<br>-Or-<br>EE362 EE381 MA256                                                                                 |                                                               |
| <b>EE480</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>OPTICAL FIBER COMMUNICATIONS</b>                                                                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                                                                                                         | <b>Offerings:</b>                                             |
| This course unlocks the secrets of fiber optics, the technology powering the modern internet and telecommunications. Students will master the fundamentals of lightwave transmission through optical fibers, including modal fields, attenuation, and dispersion. Additionally, students will delve into key communication components like transmitters, modulators, and receivers. Students will gain hands-on experience with software tools and measurement equipment to characterize fiber and device properties. The course culminates with students designing, building, and characterizing a fiber optic communications link.                                                                                                             |                                                                                                                                | 2026-1 2027-1 2028-1<br>2029-1                                |

**Lessons:** 32 @ 55 min (2.500 Att/wk)    **Labs:** 8 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** EE383  
-Or-  
XE383

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| <b>EE482</b> | <b>WIRELESS COMM SYS ENGINEERING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2014-2

**Offerings:**

This course provides an introduction to wireless systems engineering with applications to voice and data networks. Description of well known systems such as cell phones, pagers, and wireless LAN's is presented along with the design considerations for deployment of wireless networks. Wireless radio channel modeling along with common impairments such as multipath fading are introduced and modulation techniques well suited to the wireless applications are presented. Receivers for the various modulation schemes are analyzed in terms of performance and the trade-offs offered by source and channel coding are presented. Multiple access techniques used in wireless applications are introduced and the design of networks described. The course concludes with an analysis and description of deployed systems along with their standards and services provided.

2026-2 2028-2

**Lessons:** 38 @ 55 min (2.500 Att/wk)    **Labs:** 2 @ 110 min

**Special Requirements:** Course Project.

**Prerequisite(s):** EE381

**Corequisite(s):** EE383

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|--------------|------------------------------|----------------------------------------------------------|
| <b>EE483</b> | <b>PHOTONICS ENGINEERING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course introduces the fundamental principles of photonics to equip students with the skills to design, build, and test their own free-space optical communication system. Students gain hands-on experience with key photonic devices like transmitters, modulators, lenses, waveguides, gratings, and detectors. The course weaves in key ideas from electromagnetics and quantum mechanics throughout the course.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 33 @ 55 min (2.500 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** None

**Corequisite(s):** EE362 EE383  
-Or-  
EE362 PH382

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>EE484</b> | <b>FUNDAMENTALS OF ROBOTICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-2

**Offerings:**

This course introduces students to the fundamentals of robotics and exposes them to the best practices followed by the robotics community (academic and research). The lesson content is broadly organized into two categories - the theory underpinning robots and exploration through practical implementation. Some of the concepts taught are sensor fusion, robot manipulation, state estimation using Kalman filters, and the use of an open-source middleware. Students will become familiar with the command line tools, gain practice on software programming, and learn to interface with simple electronics through single board computers. \*The ET credits are proposed and will be validated through Academy processes.

2026-2 2027-2 2028-2

**Lessons:** 23 @ 75 min (2.000 Att/wk)    **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** EE360 MA364  
-Or-  
EE360 MA365  
-Or-  
EE360 MA372  
-Or-  
EE301 MA364  
-Or-  
EE301 MA365

| EE484X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | FUNDAMENTALS OF ROBOTICS                                                                                | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-2                                                                                                  | <b>Offerings:</b>                          |
| <p>This course introduces students to the fundamentals of robotics and exposes them to the best practices followed by the robotics community (academic and research). The lesson content is broadly organized into two categories - the theory underpinning robots and exploration through practical implementation. Some of the concepts taught are sensor fusion, robot manipulation, state estimation using Kalman filters, and the use of an open-source middleware. Students will become familiar with the command line tools, gain practice on software programming, and learn to interface with simple electronics through single board computers. **This is a pilot course and will need CC review NLT AY27. ET credits will be assigned after ABET PEV review.**</p>                                                                                                                               |                                                                                                         | 2029-2                                     |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 30 @ 75 min (2.000 Att/wk)                                                                              | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                    |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EE360 MA364<br>-Or-<br>EE360 MA365<br>-Or-<br>EE360 MA372<br>-Or-<br>EE301 MA364<br>-Or-<br>EE301 MA365 |                                            |
| EE485                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SPEC TOPICS IN EE                                                                                       | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2011-1                                                                                                  | <b>Offerings:</b>                          |
| <p>This course provides an in-depth study of special topics in electrical engineering not offered elsewhere in the USMA curriculum. Course content will be based on expertise of a senior electrical engineering faculty member or a Visiting Professor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                         | 2027-2 2028-2 2029-2<br>2029-8 2029-9      |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 36 @ 55 min (2.500 Att/wk)                                                                              | <b>Labs:</b> 4 @ 120 min                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | To be determined by the senior faculty member or visiting professor.                                    |                                            |
| EE486                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SOLID STATE ELECTRONICS                                                                                 | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-2                                                                                                  | <b>Offerings:</b>                          |
| <p>The course covers device physics, operating principles and applications of diodes, bipolar junction transistors, and field effect transistors (FET). It begins with basic properties of crystalline solids, energy diagrams, and thermal physics. P-N junction diodes are the first semiconductor device explored with further study into metal oxide semiconductor (MOS) capacitor and MOSFET based devices. The course also covers the layout of complementary metal oxide semiconductor (CMOS) gates on an integrated circuit chip. Throughout the course, a number of general electronic devices are introduced including digital memories, charge coupled devices, solar cells, photodiodes, light emitting diodes and lasers. The laboratories are focused on P-N junction device parameter extraction, CMOS digital circuit characterizations, and the application of optoelectronic devices.</p> |                                                                                                         | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 36 @ 55 min (2.500 Att/wk)                                                                              | <b>Labs:</b> 4 @ 55 min                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                    |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EE301<br>-Or-<br>EE302                                                                                  |                                            |
| EE487                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | EMBEDDED SYSTEMS DEVELOPMENT                                                                            | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-2                                                                                                  | <b>Offerings:</b>                          |
| <p>This course teaches students how to employ microcontrollers and single-board computers in the designs of embedded systems through applied exercises. Microcontrollers are programmed at the register level using the C language. Students conduct a detailed study of common microcontroller peripherals with an emphasis on their application to real-time control design. Cadets are exposed to addressing, serial and parallel input and output, timing, interrupts, A-to-D and D-to-A conversion. Subsequently, students study single-board computers and apply them in more complex designs. The basics of operating systems are taught using Linux and Robot Operating Systems. Emphasis is placed on linking peripherals to the processor and using multiprocessing. Throughout the course, students practice top-down and state machine design for moderately complex digital systems.</p>       |                                                                                                         | 2029-2                                     |

**Lessons:** 24 @ 75 min (1.600 Att/wk)      **Labs:** 7 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** CS380  
-Or-  
EE375  
-Or-  
EE475

**Disqualifier(s):** EE380

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|              |                                    |                                                          |
|--------------|------------------------------------|----------------------------------------------------------|
| <b>EE489</b> | <b>ADV IND STUDY IN ELECT ENGR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|------------------------------------|----------------------------------------------------------|

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**Scope:** 1974-1

**Offerings:**

Course requirements will be tailored to the needs and qualifications of the individual cadet. The course will normally involve a project requiring research, experimentation, and the submission of a report under the guidance of a departmental advisor. Alternatively, study may take the form of a tutorial course covering material not available in the regular elective course offerings.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** To be determined. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**Prerequisite(s):** EE363  
-Or-  
EE363A  
-Or-  
EE362

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|---------------|------------------------------------|----------------------------------------------------------|
| <b>EE489A</b> | <b>ADV IND STUDY IN ELECT ENGR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 1974-1

**Offerings:**

Course requirements will be tailored to the needs and qualifications of the individual cadet. The course will normally involve a project requiring research, experimentation, and the submission of a report under the guidance of a departmental advisor. Alternatively, study may take the form of a tutorial course covering material not available in the regular elective course offerings.

2026-2 2027-1 2027-2  
2028-1 2028-2 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**Prerequisite(s):** EE489

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>EE490</b> | <b>ELEC ENGRNG SUMMER RESEARCH</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 1990-4

**Offerings:**

This course is designed to familiarize the cadet with advanced techniques for independent research in electrical engineering. The course will normally require research, development, and experimental implementation of a novel idea or concept. An oral presentation and a written project report will be completed under the supervision of a usma faculty member who serves as project advisor. The course requires three full weeks of study, completed in conjunction with the academic individual advanced development program. Scope, depth, and material covered will meet the requirements of a three-credit course in electrical engineering.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Oral and written reports. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**Prerequisite(s):** EE363A

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|---------------|------------------------------------|----------------------------------------------------------|
| <b>EE490A</b> | <b>ELEC ENGRNG SUMMER RESEARCH</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 1990-4

**Offerings:**

This course is designed to familiarize the cadet with advanced techniques for independent research in computer science. No Course Offerings  
The course will normally require research, development, and implementation of a novel idea or concept. An oral presentation and a written project report will be completed under the supervision of a usma faculty member who serves as project advisor. The course requires three weeks of study, completed in conjunction with the academic individual advanced development program. Scope, depth, and material covered will be equivalent to two credits of course work in electrical engineering.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Oral and written reports. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**EE490B**

**ELEC ENGRNG SUMMER RESEARCH**

**1.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:**

1990-4

**Offerings:**

This course is designed to familiarize the cadet with advanced techniques for independent research in electrical engineering. The course will normally require research, development, and experimental implementation of a novel idea or concept. An oral presentation and a written project report will be completed under the supervision of a usma faculty member who serves as project advisor. The course requires three weeks of study, completed in conjunction with the academic individual advanced development program. Scope, depth, and material covered will be equivalent to one credit of course work in electrical engineering.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Oral and written reports. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**Prerequisite(s):**

EE363A

**XE383**

**ELECTROMAGNETIC WAVES**

**3.0 Credit Hours**  
(BS=3.0,ET=0.0,MA=0.0)

**Scope:**

2018-2

**Offerings:**

This course is an introduction to electromagnetic waves, which are the foundation of electrical engineering and applied physics. The course begins with transmission line analysis using circuit models and reviews the mathematical tools (vector algebra and calculus) that are used to describe electromagnetic phenomena. Maxwell's equations are solved to describe time-harmonic fields under various boundary conditions and at interfaces between dissimilar media. Additional topics include the applications of electromagnetic wave theory to transmission lines, antennas and waveguides, as well as the role of electromagnetics in science, technology and society. Laboratory exercises are conducted to experimentally characterize transmission lines and antennas, and to provide instructor-assisted problem solving sessions. Additionally, Cadets complete a computer project on finding the numerical solutions to Maxwell's equations.

No Course Offerings

**Lessons:** 32 @ 55 min (2.500 Att/wk)

**Labs:** 4 @ 120 min

**Special Requirements:**

None

**Prerequisite(s):**

PH382

**Corequisite(s):**

MA364  
-Or-  
MA365

**Disqualifier(s):**

EE383

**XE401**

**INTEGRATIVE SYSTEM DESIGN I**

**3.5 Credit Hours**  
(BS=0.0,ET=3.5,MA=0.0)

**Scope:**

2020-1

**Offerings:**

This course is the first part of a two-semester team-based capstone design experience in electrical engineering, computer science and information technology. It provides an integrative experience, presenting each cadet team with a professionally relevant, open-ended situation including professional, ethical, social, security, legal, economic, and political dimensions, where an engineering approach has strong potential to produce benefits. Under the guidance of a faculty advisor for each project team, cadets develop client-focused products, applying the principles of design and implementation to effect an optimal outcome for the circumstances presented to the team by creating a product or service that meets requirements and constraints negotiated with the client.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 150 min (2.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

A senior design project is required in this course.

**Prerequisite(s):** CS403  
-Or-  
EE362  
-Or-  
IT394  
-Or-  
CY394

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|              |                                     |                                                          |
|--------------|-------------------------------------|----------------------------------------------------------|
| <b>XE402</b> | <b>INTEGRATIVE SYSTEM DESIGN II</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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| <b>Scope:</b> | 2019-2 | <b>Offerings:</b> |
|---------------|--------|-------------------|

This course is team-based capstone design experience in electrical engineering, computer science and information technology. It provides an integrative experience, presenting each cadet team with a professionally relevant, open-ended situation including professional, ethical, social, security, legal, economic, and political dimensions, where an engineering approach has strong potential to produce benefits. Under the guidance of a faculty advisor for each project team, cadets develop client-focused products, applying the principles of design and implementation to effect an optimal outcome for the circumstances presented to the team by creating a product or service that meets requirements and constraints negotiated with the client.

**Lessons:** 30 @ 150 min (2.000 Att/wk)    **Labs:** 0 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** XE401

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>XE442</b> | <b>ALTERNATIVE ENERGY ENGINEERING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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|---------------|--------|-------------------|
| <b>Scope:</b> | 2015-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

This course provides a study of the fundamentals of alternative energy generation, storage, integration and efficient use. Solar power (both solar thermal and photovoltaic), wind power, hydro power, fuel cells and other sources of energy are covered. Focus is placed on energy conversion, modeling alternative energy sources, and integration of these sources into the power grid. The technical, economic, and global challenges associated with these alternative energies is covered in depth.

**Lessons:** 36 @ 55 min (2.500 Att/wk)    **Labs:** 4 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** EE301  
-Or-  
EE302

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|--------------|-------------------------------------|----------------------------------------------------------|
| <b>XE472</b> | <b>DYNAMIC MODELING AND CONTROL</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

This course covers dynamic modeling and control of linear systems. The course provides an overview of classical control theory as the foundation for control applications in electrical, mechanical, and aeronautical systems. Topics here include system modeling using Laplace transform, frequency domain, and state variable methods. Mathematical models are developed for electrical, mechanical, aeronautical, chemical and other physical control systems. Control systems analysis and design techniques are studied within the context of how each system is physically controlled in practice. Laboratory exercises include feedback design and system identification. Computer design exercises include dynamic modeling and control of various engineering systems.

**Lessons:** 36 @ 55 min (2.500 Att/wk)    **Labs:** 4 @ 120 min

**Special Requirements:** Computer interactive exercises.

**Prerequisite(s):** EE301  
-Or-  
EE302  
-Or-  
EE350 MA205  
-Or-  
EE350 MA255  
-Or-  
EE350 MA204X  
-Or-  
EE350 MA204

| XE492                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DISRUPTIVE INNOVATIONS                                               | 3.0 Credit Hours<br>(BS=0.5,ET=2.5,MA=0.0) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2013-1                                                               | <b>Offerings:</b>                          |
| <p>The course begins by developing the background understanding of what disruptive technology is and a historical context about successes and failures of social, cultural, and religious acceptance of technological innovation. To develop this framework, students read several texts underlying the innovator's dilemma, how scientific revolutions are structured, and cultural distinctions found between the sciences and humanities. For each class meeting, students read current scientific and technical literature and come prepared to discuss current events related to technological innovation. Each student researches potential disruptive technologies and prepares a compelling argument of why the specific technologies are disruptive so they can defend their choice and rationale. Cadets also interact with national level innovators throughout academia, industry, and government.</p>                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                      | 2026-1 2027-1 2028-1                       |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                               |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                 |                                            |
| XE497                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CRITICAL SCIENTIFIC REASONING                                        | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2013-1                                                               | <b>Offerings:</b>                          |
| <p>The purpose of XE497, Critical Scientific Reasoning, is to improve the students' ability to analyze complex problems in a variety of applied physical science applications using mathematical, scientific, and engineering principles and clearly articulate their analysis and results verbally and in writing. The process of pursuing this goal will make cadets better officers, scholars, and citizens. Several methods will be applied to assist in the pursuit of these goals. Fundamental scientific laws, principles, and theorems and their application to scientific and engineering problem solving will be reviewed. Breadth across a variety of scientific and engineering disciplines will be achieved by studying and discussing current research activities from a variety of fields as well as examining the limitations to scientific advancement in each field. The course will draw from several disciplines including Biology, Chemistry, Civil Engineering, Computing Sciences, Electrical Engineering, Mathematical Science, Mechanical Engineering and Physics. In order to take advantage of the diverse skills of the USMA faculty and selected experts from outside USMA, some classes will be led by guest instructors, each of whom will recommend readings in support of his or her topic.</p> |                                                                      | No Course Offerings                        |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                               |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Department Head approval to enroll. Open only to First Class cadets. |                                            |

## English and World Languages

### 180 Courses

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------|
| <b>EN100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>FOUNDATIONAL WRITING</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-1                      | <b>Offerings:</b>                                                    |
| This foundational course will provide cadets with the basic critical reading and writing skills necessary to enter EN101. EN100 aims to develop an awareness of substance, organization, style, and correctness through close, critical reading and by focusing on sentence- and paragraph- level composition. Through daily writing, exercises, and one-on-one conferences, cadets will develop the clear, logical, and grammatically correct writing that will enable them to succeed in EN101.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                             | No Course Offerings                                                  |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min      |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                        |                                                                      |
| <b>EN101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>COMPOSITION</b>          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-1                      | <b>Offerings:</b>                                                    |
| This course aims to develop clear, logical, and grammatically correct expression in written discourse. Augmented by one-on-one writing conferences, daily writing and revision reinforce instruction in the writing process. Organization, substance, style, and correctness in written communication are major concerns of the course, but the course improves other modes of discourse through a process-oriented approach to composition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                             | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min      |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                        |                                                                      |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | EN151                       |                                                                      |
| <b>EN102</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>LITERATURE</b>           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024-5                      | <b>Offerings:</b>                                                    |
| As an introduction to the study of literature, EN102 provides Cadets the opportunity to refine the skills of oral and written communication practiced in EN101 and to develop certain foundational capacities essential to undergraduate learning and to professional development: critical thinking, close reading, empathy, intellectual agility, emotional intelligence, and creativity. By studying the most challenging and complex employment of language in its most difficult forms, Cadets confront ambiguity and hone their interpretive skills through the judicious interpretation of evidence. Rigorous immersion in literature enables Cadets to increase their command of the English language and to deepen their appreciation for the power and beauty of literary expression. By engaging with major literary genres Cadets cultivate awareness of the central importance of literature to their own culture and to the cultures of various regions and nations across the globe. In their encounter with diverse perspectives Cadets acquire insight into the human condition and begin to grapple with the ethical issues that are the focus of study in PY201. |                             | 2026-2 2026-4 2026-5<br>2027-2 2027-5 2028-2<br>2028-5 2029-2 2029-5 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min      |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                        |                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | EN101<br>-Or-<br>EN151      |                                                                      |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | EN152                       |                                                                      |
| <b>EN151</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>ADVANCED COMPOSITION</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-1                      | <b>Offerings:</b>                                                    |



This advanced course, into which the Department of English and Philosophy places select cadets, further develops the skills of accomplished writers. This course demands a rigorous and disciplined approach to writing through intensive study and research. It also demands attention to the nuances and persuasiveness of language itself. The course challenges cadets to push their writing beyond basic, first-year academic prose through an emphasis on advanced figures of speech and thought, through a deliberate attention to style, and through an in-depth approach to reading and research. This course expects cadets to think critically and to discover and articulate new ideas and original arguments through persuasive, polished, and mature writing. Finally, this course instills habits of mind that will that will serve cadets throughout their academic and military careers. Because this is an advanced version of EN101, cadets will be placed in EN151 individually, based on the results of a placement exam given in the summer prior to enrollment and by approval of the Head of the Department of English and Philosophy.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Cadets who have credit for EN101 are disqualified from taking EN151.

**Disqualifier(s):** EN101

2026-1 2027-1 2028-1  
2029-1

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|--------------|----------------------------|----------------------------------------------------------|
| <b>EN152</b> | <b>ADVANCED LITERATURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

EN152 is an advanced version of EN102. The course revolves around the innovative interpretation of literary texts, though it simultaneously acquaints Cadets with strategies for grounding these interpretations in concrete textual evidence and logical inferences. EN152 promotes close, deliberate reading of literary works, presents a comprehensive overview of the terms and tenets of literary studies, and emphasizes how writing, as a process, allows for the further development of one's critical thinking and analytical skills. Finally, EN152 explores the power of literature to promote ethical awareness, cultivate empathy, and emphasize a shared sense of humanity.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** EN102

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| <b>EN300</b> | <b>LITERARY METHODOLOGIES</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course provides cadets the methodological tools required to analyze and evaluate primary and secondary sources. Through the study of representative primary sources ranging from the ancient to the postmodern, cadets will learn the critical vocabularies and theoretical contexts necessary for the meaningful study of literature. Attention to the nature and history of literary genres, the historical development of literary criticism, and a variety of theoretical approaches to literature will provide cadets with the foundational knowledge required of an English major.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** EN102

**Disqualifier(s):** EN333

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>EN311</b> | <b>ANCIENT TO EARLY MODERN LIT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course will focus primarily on the development of the Western literary canon from antiquity to the Early Modern period. Cadets may read from the works of Greek and Roman authors such as Homer, Sophocles, Thucydides, Cicero, Virgil, Ovid, from Medieval authors such as St. Thomas Aquinas, Sir Thomas Malory, Geoffrey Chaucer, and from Early Modern authors such as John Donne, George Herbert, Ben Jonson, and Miguel de Cervantes. Cadets will explore a variety of genres including, but not limited to, epic poetry, histories, drama, and lyric poetry. This course may take its focus from a central theme or idea and trace that theme or idea in the course texts.

2026-1 2028-1 2030-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** EN102  
-Or-  
EN152

| EN321                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AMERICAN LITERATURE I            | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-1                           | <b>Offerings:</b>                          |
| <p>The course will focus on the development of American literature from early contact to the Civil War. Students will read from works by such authors as the Puritans, Jefferson, Lincoln, the Transcendentalists, Dickinson, Whitman, and Melville, as well as literature outside of the New England canon: for example, works by Native Americans, French and Spanish colonizers, and African captives. All works will be considered in the context of cultural and intellectual history. We will consider a broad range of genres and modes of writing, including (but not limited to) colonial theory, ethnography, autobiography, fiction, essays, and poetry. A central concern of the course will be the question of what constitutes American literature.</p> |                                  | 2026-1 2028-1 2030-1                       |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min           |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | A few essays of moderate length. |                                            |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PY201<br>-Or-<br>PY251           |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EN343                            |                                            |
| EN322                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AMERICAN LITERATURE II           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-1                           | <b>Offerings:</b>                          |
| <p>This course will examine both traditional and nontraditional writings from the Civil War to the present. We will examine post-Civil War literature and the myriad, often contradictory desires--economic, aesthetic, sexual, spiritual, and intellectual--to which it gives expression. The course will provide a framework within which students may examine the literature in an historical context. As does American Literature I, the course stresses the diversity of experience and poetics that characterizes American literature. In addition, students will trace the evolution of important literary movements and philosophical influences, as well as the metamorphosis of certain genres over time.</p>                                               |                                  | 2026-2 2026-8 2028-2<br>2029-8 2030-2      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min           |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | A few essays of moderate length. |                                            |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PY201<br>-Or-<br>PY251           |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EN348                            |                                            |
| EN331                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | BRITISH LITERATURE I             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-1                           | <b>Offerings:</b>                          |
| <p>This course is an introduction to the study of British literature, ranging from the Anglo-Saxon period through the eighteenth century. Cadets will encounter representative masterworks from the Old English, Medieval, Renaissance, and Neoclassical periods, exploring in the process the development of literary forms, the culture of the British Isles, and the English language itself. Possible areas of emphasis include narrative and lyric poetry from all these periods, drama from the Middle Ages and Renaissance, the periodical essay from the Neoclassical period, and the emergence of the novel as a distinct form of literature in the eighteenth century.</p>                                                                                  |                                  | 2027-1 2029-1                              |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min           |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | None                             |                                            |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PY201<br>-Or-<br>PY251           |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EN341                            |                                            |
| EN332                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | BRITISH LITERATURE II            | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-2                           | <b>Offerings:</b>                          |

This course continues the survey initiated in British Literature I by considering major authors and works of the nineteenth and twentieth centuries. Through representative but necessarily selective readings, cadets will trace the development of British literature from the Romantic Period into the Victorian Age and then to the present day. Possible areas of emphasis include poetry of the English Romantics; Victorian poetry and prose, to include the novel; and poetry, short fiction, and drama from the twentieth century. Study will emphasize the relation of the works considered to the cultural history of Great Britain and the British Empire and will attend as well to the wider influence of the British tradition.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** PY201  
-Or-  
PY251

**Disqualifier(s):** EN346

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| <b>EN340</b> | <b>CONTEMPORARY LITERATURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course examines literature of the later 20th and the 21st centuries. In addition, it may explore the implications of contemporary information technologies for traditional literary forms, the role of globalization in literary production and reception, or the relation of literature to pressing current issues such as persistent violent conflict, immigration, and climate change. Possible genres and modes of writing include (but are not limited to) short fiction, the novel, poetry, graphic narrative, and electronic literature. A central concern of the course will be the question of what constitutes literature in the contemporary period, a time marked by developing and often differing trends.

2027-1 2027-8 2028-8  
2029-1 2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** EN102  
-Or-  
EN152

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| <b>EN351</b> | <b>WORLD LITERATURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course enhances cadets' cultural awareness and refines their disciplinary knowledge and interpretive skills by introducing them to major literary texts from around the globe. As an advanced exercise in comparative study and synthesis, World Literature builds on core courses such as EN302 and foreign language offerings. The prose and poetry of a variety of periods and a range of countries provide contexts for and contrasts to the Anglo-American tradition. In a given semester typical texts could include epics and tragedies of Ancient Greece and Rome, Russian novels, works of medieval Islamic literature, haiku of Japan, Continental European novels of the nineteenth century, or postmodern fiction of South America. This course familiarizes students not only with important literary forms and genres but also with cultural and historical contexts for many of the most pressing issues in our volatile world.

2026-8 2026-9 2027-1  
2027-8 2027-9 2029-1  
2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** PY201  
-Or-  
PY251

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| <b>EN353</b> | <b>WAR LITERATURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course ranges widely across cultures and historical periods in studying how human creative imagination has dealt with war. The works in this course are especially illuminating to professional soldiers.

2026-8 2027-2 2029-2  
2029-8

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** PY201  
-Or-  
PY251

**Disqualifier(s):** EN374

| EN354                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SPECIAL TOPICS COLLOQUIUM                                      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-2                                                         | <b>Offerings:</b>                                              |
| This course explores a special topic in Literature. Specific subject matter will vary with the expertise of the senior faculty member conducting the course.                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                | 2026-2 2027-2 2028-2 2029-2                                    |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                         |                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                                           |                                                                |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PY201<br>-Or-<br>PY251                                         |                                                                |
| EN354A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | SPECIAL TOPICS COLLOQUIUM - A                                  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-2                                                         | <b>Offerings:</b>                                              |
| This course explores a special topic in Literature. Specific subject matter will vary with the expertise of the senior faculty member conducting the course.                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                | 2026-2 2026-8 2027-2 2027-8 2028-2 2028-8 2029-2 2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                         |                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | EN354 must be taken before a cadet can take EN354A.            |                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | EN354                                                          |                                                                |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PY201<br>-Or-<br>PY251                                         |                                                                |
| EN354B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | SPECIAL TOPICS COLLOQUIUM-B                                    | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-2                                                         | <b>Offerings:</b>                                              |
| This course explores a special topic in Literature. Specific subject matter will vary with the expertise of the senior faculty member conducting the course.                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                | 2026-2 2027-2 2028-2 2029-2                                    |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                         |                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | EN354 and EN354A must be taken before a cadet can take EN354B. |                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | EN354 EN354A                                                   |                                                                |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PY201<br>-Or-<br>PY251                                         |                                                                |
| EN355                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CRITICISM COLLOQUIUM                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-1                                                         | <b>Offerings:</b>                                              |
| This course introduces cadets to the theory of interpretation and the practice of literary criticism. Through the study of critics ranging from the ancient to the postmodern, Cadets investigate mimetic, pragmatic, expressive, and objective schools. They also cultivate their own philosophies of interpretation and apply them to primary texts. Readings may focus on aesthetic, cultural, and ethical dimensions of literature, on the role of the critic, and on the proliferation of competing theories during the latter half of the twentieth century. |                                                                | 2027-8 2028-8                                                  |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                         |                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | A few essays of moderate length.                               |                                                                |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PY201<br>-Or-<br>PY251                                         |                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | EN344                                                          |                                                                |
| EN361                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | POETRY                                                         | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                     |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                           | <b>Offerings:</b>                                  |
| Embracing a wide variety of authors, works, periods, traditions, and forms, this course considers the literary genre through which human beings have expressed their most intensely imaginative visions of themselves and the world, and connections between the two. Some consideration of poetics and prosody will complement the cadets' reading of verse that ranges from Japanese haiku through the Shakespearean sonnet to the free-verse creations of modern and contemporary poets.                                      |                                  | 2026-1 2028-1 2030-1                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A few essays of moderate length. |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PY201<br>-Or-<br>PY251           |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EN391                            |                                                    |
| <b>EN362</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>FILM AND FILM THEORY</b>      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-2                           | <b>Offerings:</b>                                  |
| This course examines film as the major new art form of the twentieth century. Screenings of important films and readings in film theory introduce cadets to the origins, evolution, and cultural influence of cinema. Cadets explore connections between film and the other arts as well as the relationship between art and technology. Topics may include the Hollywood studio system, the transition to sound, world cinema, auteur theory, screenwriting, censorship, and propaganda.                                        |                                  | 2026-2 2028-2 2029-1<br>2029-8 2029-9              |
| <b>Lessons:</b> 34 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 6 @ 120 min         |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A few essays of moderate length. |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PY201<br>-Or-<br>PY251           |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EN342                            |                                                    |
| <b>EN363</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>THE NOVEL</b>                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-2                           | <b>Offerings:</b>                                  |
| In this course the word novel designates any extended fictional narrative, almost always in prose. Cadets will explore the novel of kind or time or both, and, in addition to becoming better readers, will work toward understanding the culturally complex world around them.                                                                                                                                                                                                                                                  |                                  | 2026-2 2028-2 2030-2                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A few essays of moderate length  |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PY201<br>-Or-<br>PY251           |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EN385                            |                                                    |
| <b>EN364</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>DRAMA</b>                     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                           | <b>Offerings:</b>                                  |
| This course surveys significant plays from a variety of periods and traditions to give cadets an appreciation of a genre that exists as both written literature and creative interpretation. Works to be studied range from the classical tragedies of ancient Greece through the great products of the English renaissance to modern efforts by British and American playwrights. Although the primary focus rests upon the Anglo-American tradition, the course will not neglect dramatists from other countries and cultures. |                                  | 2027-1 2029-1                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A few essays of moderate length. |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PY201<br>-Or-<br>PY251           |                                                    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                       |                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EN367                                                                                                                 |                                                    |
| <b>EN370</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>SHAKESPEARE</b>                                                                                                    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-2                                                                                                                | <b>Offerings:</b>                                  |
| This course surveys representative Shakespearean plays, including great tragedies, histories, and comedies. Study stresses the nature of Shakespeare's genius and the relation of his works to the cultures of all ages.                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                       | 2027-2 2029-2 2029-8                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A few essays of moderate length.                                                                                      |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PY201<br>-Or-<br>PY251                                                                                                |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EN394                                                                                                                 |                                                    |
| <b>EN371</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>SINGLE-AUTHOR COLLOQUIUM</b>                                                                                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-2                                                                                                                | <b>Offerings:</b>                                  |
| This elective course provides in-depth study of a single author's life and work. The featured author will vary depending on instructor expertise and preference as well as department need. The course allows cadets to gain an insight into the life and work of important authors deeper than the necessarily limited exposure within a survey or genre course affords. The course invites cadets to explore the relationships between biography and creative output, to understand individual works within the context of a writer's larger oeuvre, and to see individual authors as both agents and products of their world. |                                                                                                                       | 2026-2 2028-2 2030-2                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                                                                  |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | EN102<br>-Or-<br>EN152                                                                                                |                                                    |
| <b>EN371A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>SINGLE AUTHOR COLLOQUIUM</b>                                                                                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                                                                                                                | <b>Offerings:</b>                                  |
| This elective course provides in-depth study of a single author's life and work. The featured author will vary depending on instructor expertise and preference as well as department need. The course allows cadets to gain an insight into the life and work of important authors deeper than the necessarily limited exposure within a survey or genre course affords. The course invites cadets to explore the relationships between biography and creative output, to understand individual works within the context of a writer's larger oeuvre, and to see individual authors as both agents and products of their world. |                                                                                                                       | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Students may take EN371A only after completing EN371, and with permission of the English Program Director.            |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | EN371                                                                                                                 |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | EN102<br>-Or-<br>EN152                                                                                                |                                                    |
| <b>EN371B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>SINGLE AUTHOR COLLOQUIUM-B</b>                                                                                     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                                                                                                                | <b>Offerings:</b>                                  |
| This elective course provides in-depth study of a single author's life and work. The featured author will vary depending on instructor expertise and preference as well as department need. The course allows cadets to gain an insight into the life and work of important authors deeper than the necessarily limited exposure within a survey or genre course affords. The course invites cadets to explore the relationships between biography and creative output, to understand individual works within the context of a writer's larger oeuvre, and to see individual authors as both agents and products of their world. |                                                                                                                       | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Students may take EN371B only after completing EN371 and EN371A, and with permission of the English Program Director. |                                                    |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | EN371 EN371A                                         |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | EN102<br>-Or-<br>EN152                               |                                                    |
| <b>EN400</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>SEMINAR IN ADV LITERARY STUDY</b>                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2025-1                                               | <b>Offerings:</b>                                  |
| This course provides cadets with the opportunity for advanced study in the discipline. Through a focus on a particular topic in literature, cadets will build on the foundation established in EN300. They will deepen their mastery of critical methods and theoretical models and grow as scholars by closely examining an author, period, theme, issue, or debate from multiple perspectives. Through intensive study of primary and secondary texts, this course broadens the knowledge base by bridging disciplinary approaches and setting the stage for cadets' continued educational development. |                                                      | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 16 @ 140 min (1.000 Att/wk)                          | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | A major project and a few essays of moderate length. |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | EN300<br>-Or-<br>EN333<br>-Or-<br>EP333              |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | EN433                                                |                                                    |
| <b>EN401</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>SENIOR THESIS I</b>                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2025-1                                               | <b>Offerings:</b>                                  |
| This course permits cadets with the requisite energy and talent to initiate a yearlong project requiring research in depth that culminates in a substantial thesis of high scholarly quality.                                                                                                                                                                                                                                                                                                                                                                                                             |                                                      | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 30 @ 75 min (2.000 Att/wk)                           | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                                 |                                                    |
| <b>EN402</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>SENIOR THESIS II</b>                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2025-2                                               | <b>Offerings:</b>                                  |
| This course permits cadets to complete a yearlong project requiring research in depth that culminates in a substantial thesis of high scholarly quality.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                      | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 30 @ 75 min (2.000 Att/wk)                           | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Oral defense of thesis.                              |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | EN401                                                |                                                    |
| <b>EN433</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>SEMINAR IN ADV LITERARY STUDY</b>                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2025-1                                               | <b>Offerings:</b>                                  |
| This course provides cadets with the opportunity for advanced study in the discipline. Through a focus on a particular topic in literature, cadets will build on the foundation established in EN333. They will deepen their mastery of critical methods and theoretical models and grow as scholars by closely examining an author, period, theme, issue, or debate from multiple perspectives. Through intensive study of primary and secondary texts, this course broadens the knowledge base by bridging disciplinary approaches and setting the stage for cadets' continued educational development. |                                                      | No Course Offerings                                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 40 @ 55 min (2.500 Att/wk)                           | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | A major project and a few essays of moderate length. |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | EN333<br>-Or-<br>EP333                               |                                                    |

| EN490                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | INDEPENDENT STUDY: LITERATURE                                         | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-1                                                                | <b>Offerings:</b>                          |
| This optional elective offers the cadet an opportunity for in-depth study of an advanced topic in Literature under the mentorship of a senior faculty advisor. The scope and topic of the course are developed in consultation with the faculty advisor and appropriately build upon academic work already completed in the regular Literature electives. Since such a course is beyond normal teaching duties, an agreement to serve as a faculty advisor will be at the discretion of the faculty member. Enrollment is subject to Department approval.                                                                                                                                                                                                                                                                                                                      |                                                                       | 2026-9 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                  |                                            |
| EN490A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | INDEP STUDY: LITERATURE (A)                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-1                                                                | <b>Offerings:</b>                          |
| This optional elective offers the cadet an opportunity for in-depth study of an advanced topic in Literature under the mentorship of a senior faculty advisor. The scope and topic of the course are developed in consultation with the faculty advisor and appropriately build upon academic work already completed in the regular Literature electives. Since such a course is beyond normal teaching duties, an agreement to serve as a faculty advisor will be at the discretion of the faculty member. Enrollment is subject to Department approval.                                                                                                                                                                                                                                                                                                                      |                                                                       | No Course Offerings                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cadets should successfully complete EN490 before enrolling in EN490A. |                                            |
| LA203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ARABIC I (STANDARD)                                                   | 4.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-1                                                                | <b>Offerings:</b>                          |
| In the standard course sequence, Cadets acquire a basic proficiency in speaking, listening, reading and writing skills in Arabic. Learning activities focus on situations Cadets are likely to encounter in Arabic society. Cadets are taught how to express simple ideas and basic needs, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. In addition to speaking, listening and reading skills, Cadets learn how to write sentences, paragraphs and/or short compositions on familiar topics. Through readings and discussions, Cadets are introduced to the cultures and history of the Arabic-speaking world. Cadets acquire a command of basic Arabic vocabulary and gain a general understanding of how the language works, and they become able to apply that knowledge when learning other foreign languages. |                                                                       | 2026-1 2027-1 2028-1<br>2029-1             |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                  |                                            |
| LA204                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ARABIC II (STANDARD)                                                  | 4.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-2                                                                | <b>Offerings:</b>                          |
| Continuation of LA203.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                       | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                  |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LA203                                                                 |                                            |
| LA371                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | INTENSIVE INTERMEDIATE ARABIC                                         | 4.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-1                                                                | <b>Offerings:</b>                          |



In the intensive intermediate course, cadets develop proficiency in those skills necessary for communicating effectively in Arabic and for pursuing upper-level courses. Cadets develop speaking skills that enable them to engage in conversations on a variety of topics with other class members and with native speakers. Cadets reinforce and expand their language skills by reading, viewing, discussing, and writing about contemporary life, current events, and other cultural and historical topics as presented in selected materials of the Arabic-speaking world. In addition, cadets gain an overview of the profession of arms in Arabic-speaking regions by reading, discussing, and writing about pertinent materials that focus on the mission and history of the military in those countries. Cadets also review the basic rules of Arabic grammar and continue to acquire a corpus of Arabic vocabulary. They will be able to use computer-assisted learning resources to strengthen and maintain their language proficiency. This course serves as a bridge to advanced elective Arabic courses.

**Lessons:** 80 @ 55 min (5.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LA204

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|--------------|-------------------------------------------|----------------------------------------------------------|
| <b>LA372</b> | <b>ARABIC FOR ORAL &amp; WRITTEN COMM</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-------------------------------------------|----------------------------------------------------------|

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**Scope:** 2025-2

**Offerings:**

This course addresses students' specific needs in the development of listening, speaking, reading, and writing skills in Arabic. Cadets expand their active vocabulary, gain greater command of complex grammatical structures, and develop appropriate styles of written and oral communication. Special emphasis is placed on conversational and expository speaking. Cadets increase their oral proficiency through dialogues, role play, group discussions, formal presentations, and simulations of everyday language tasks likely to be encountered in the target region. Guided writing activities develop Cadets' competence in the application of critical language structures. Listening and reading comprehension are strengthened via engagement with film, music, short stories, news reports, and other popular media. Course topics typically focus on regional culture, the military, and society.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LA371

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>LA470</b> | <b>SPECIAL TOPIC IN ARABIC</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LA475

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|--------------|--------------------------|----------------------------------------------------------|
| <b>LA472</b> | <b>COLLOQUIAL ARABIC</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course introduces the dialect of a particular Arab country. Oral proficiency gained in this course is complementary to previously learned modern standard Arabic. The course may be taken twice for credit if two different dialects are offered. Consult department counselor.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):**  
-Or-  
LA475

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|---------------|------------------------|----------------------------------------------------------|
| <b>LA472A</b> | <b>ADVANCED ARABIC</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course introduces the dialect of a particular Arab country. Oral proficiency gained in this course is complementary to previously learned modern standard Arabic. The course may be taken twice for credit if two different dialects are offered. Consult department counselor. This course covers material not included in LA472 taken abroad

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Must have already taken LA472 abroad

**Prerequisite(s):** LA475

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>LA475</b> | <b>ARABIC RDG/WRTG THRU MEDIA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

In this course cadets enhance their reading and writing skills through study and discussion of contemporary Arabic media (e.g. the Internet, television, film, radio, newspapers and magazines), as well as short literary selections. Reading strategies and textual analysis are addressed. Writing tasks develop organization, substance, and style. Graded work typically includes oral and written summaries of authentic texts and short compositions or reaction papers. The course is conducted in Arabic.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LA372

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>LA476</b> | <b>MILITARY SPKG/RDG - ARABIC</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Cadets gain an understanding of the profession of arms in the Arabic-speaking world through lectures and selected reading materials (e.g. journal articles, Internet media, training manuals, biographies, and historical documents). Course content may encompass the mission and role, training, operations, tactics, and organization of the armed forces. Oral proficiency is enhanced through in-class discussion as well as role-plays and simulations focusing on scenarios likely to be encountered while an officer is deployed in an Arabic-speaking region. Media complement instruction. Graded work may include briefings, role-plays, and simulation. The course is conducted in Arabic.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LA475

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|--------------|----------------------------|----------------------------------------------------------|
| <b>LA483</b> | <b>ARAB CIVILIZATION I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course and the following one, LA484, constitute an integrated study of the culture, history, and geography of the Arabic-speaking world. Readings, lectures, discussions, and audio-visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the courses focus on the values and attitudes, the customs and traditions, and the social structures of Arabic people. At the same time, cadets continue to develop greater proficiency in Arabic. Graded work may include giving oral presentations, writing short essays or preparing a term paper. A majority of the work is done in Arabic.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):**  
-Or-  
LA475

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|--------------|-----------------------------|----------------------------------------------------------|
| <b>LA484</b> | <b>ARAB CIVILIZATION II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Continuation of LA483.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------|
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                         |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LA483                        |                                                    |
| <b>LA485</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>ARABIC LITERATURE I</b>   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-1                       | <b>Offerings:</b>                                  |
| In this course cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of the target society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater proficiency in the target language. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in the target language.                                                                                                                                                                                                                                                                                        |                              | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min       |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                         |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | LA475                        |                                                    |
| <b>LA486</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>ARABIC LITERATURE II</b>  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-1                       | <b>Offerings:</b>                                  |
| In this course cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of the target society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater proficiency in the target language. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in the target language.                                                                                                                                                                                                                                                                                        |                              | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min       |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                         |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LA485                        |                                                    |
| <b>LA492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>ARABIC LITERATURE III</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-2                       | <b>Offerings:</b>                                  |
| In this course cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of the target society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater proficiency in the target language. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in the target language.                                                                                                                                                                                                                                                                                        |                              | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min       |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                         |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LA486                        |                                                    |
| <b>LC203</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>CHINESE I (STANDARD)</b>  | <b>4.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-1                       | <b>Offerings:</b>                                  |
| In the standard course sequence, Cadets acquire a basic proficiency in speaking, listening, reading and writing skills in Chinese. Learning activities focus on situations Cadets are likely to encounter in Chinese society. Cadets are taught how to express simple ideas and basic needs, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. In addition to speaking, listening and reading skills, Cadets learn how to write sentences, paragraphs and/or short compositions on familiar topics. Through readings and discussions, Cadets are introduced to the cultures and history of the Chinese-speaking world. Cadets acquire a command of basic Chinese vocabulary and gain a general understanding of how the language works, and they become able to apply that knowledge when learning other foreign languages. |                              | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min       |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                         |                                                    |

| LC204                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CHINESE II (STANDARD)          | 4.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-2                         | <b>Offerings:</b>                            |
| Continuation of LC203.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                | 2026-2 2027-2 2028-2<br>2029-2               |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min         |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                           |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LC203                          |                                              |
| LC371                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | INTENSIVE INTERMEDIATE CHINESE | 4.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                         | <b>Offerings:</b>                            |
| In the intensive intermediate course, cadets develop proficiency in those skills necessary for communicating effectively in Chinese and for pursuing upper-level courses. Cadets develop speaking skills that enable them to engage in conversations on a variety of topics with other class members and with native speakers. Cadets reinforce and expand their language skills by reading, viewing, discussing, and writing about contemporary life, current events, and other cultural and historical topics as presented in selected materials of the Chinese-speaking world. In addition, cadets gain an overview of the profession of arms in Chinese-speaking regions by reading, discussing, and writing about pertinent materials that focus on the mission and history of the military in those countries. Cadets also review the basic rules of Chinese grammar and continue to acquire a corpus of Chinese vocabulary. They will be able to use computer-assisted learning resources to strengthen and maintain their language proficiency. This course serves as a bridge to advanced elective Chinese courses. |                                | 2026-1 2027-1 2027-8<br>2028-1 2028-8 2029-1 |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min         |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                           |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LC204                          |                                              |
| LC372                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | CHINESE FOR ORAL & WRIT COMM   | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-2                         | <b>Offerings:</b>                            |
| This course addresses students' specific needs in the development of listening, speaking, reading, and writing skills in Chinese. Cadets expand their active vocabulary, gain greater command of complex grammatical structures, and develop appropriate styles of written and oral communication. Special emphasis is placed on conversational and expository speaking. Cadets increase their oral proficiency through dialogues, role play, group discussions, formal presentations, and simulations of everyday language tasks likely to be encountered in the target region. Guided writing activities develop Cadets' competence in the application of critical language structures. Listening and reading comprehension are strengthened via engagement with film, music, short stories, news reports, and other popular media. Course topics typically focus on regional culture, the military, and society.                                                                                                                                                                                                          |                                | 2026-2 2027-2 2028-2<br>2029-2               |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min         |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                           |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LC371                          |                                              |
| LC470                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | SPECIAL TOPIC IN CHINESE       | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                         | <b>Offerings:</b>                            |
| This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                | No Course Offerings                          |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min         |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                           |                                              |
| LC470A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SPECIAL TOPICS IN CHINESE      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                         | <b>Offerings:</b>                            |

This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities. No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>LC475</b> | <b>CHINESE RDG/WRTG THRU MEDIA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

In this course cadets enhance their reading and writing skills through study and discussion of contemporary Chinese media (e.g. the Internet, television, film, radio, newspapers and magazines), as well as short literary selections. Reading strategies and textual analysis are addressed. Writing tasks develop organization, substance, and style. Graded work typically includes oral and written summaries of authentic texts and short compositions or reaction papers. The course is conducted in Chinese.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>LC476</b> | <b>MILITARY SPKG/RDG - CHINESE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Cadets gain an understanding of the profession of arms in the Chinese-speaking world through lectures and selected reading materials (e.g. journal articles, Internet media, training manuals, biographies, and historical documents). Course content may encompass the mission and role, training, operations, tactics, and organization of the armed forces. Oral proficiency is enhanced through in-class discussion as well as role-plays and simulations focusing on scenarios likely to be encountered while an officer is deployed in a Chinese-speaking region. Media complement instruction. Graded work may include briefings, role-plays, and simulation. The course is conducted in Chinese.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>LC483</b> | <b>CHINESE CIVILIZATION I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course and the following one, LC484, constitute an integrated study of the culture, history, and geography of the Chinese-speaking world. Readings, lectures, discussions, and audio-visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the courses focus on the values and attitudes, the customs and traditions, and the social structures of Chinese-speaking people. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays or preparing a term paper. A majority of the work is done in Chinese.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LC372

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| <b>LC484</b> | <b>CHINESE CIVILIZATION II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Continuation of LC483.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LC475

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| <b>LC485</b> | <b>CHINESE LITERATURE I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-1                             | <b>Offerings:</b>                                 |
| In this course cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of the target society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater proficiency in the target language. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in the target language. |                                    | 2026-1 2027-1 2028-1 2029-1                       |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min             |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                               |                                                   |
| <b>LC486</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CHINESE LITERATURE II</b>       | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-2                             | <b>Offerings:</b>                                 |
| In this course cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of the target society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater proficiency in the target language. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in the target language. |                                    | 2026-2 2027-2 2028-2 2029-2                       |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min             |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                               |                                                   |
| <b>LC492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CHINESE LITERATURE III</b>      | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-2                             | <b>Offerings:</b>                                 |
| In this course cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of the target society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater proficiency in the target language. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in the target language. |                                    | No Course Offerings                               |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min             |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                               |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | LC486                              |                                                   |
| <b>LE101</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ACDMC RDG/WRTG INTL CDTs I</b>  | <b>3.5 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-1                             | <b>Offerings:</b>                                 |
| This course seeks to enhance the language proficiency of non-native English speakers within the cognitively rigorous demands of a military-academic environment. While essentially a writing course, significant rhetorical, oratorical, and analytical skills are developed through extensive reading and systematic analysis of culturally relevant texts. At the same time, research and documentation skills are stressed to develop positive control over linguistic and professional conventions expected of Cadets in subsequent or concurrent courses at the Academy.               |                                    | 2026-1 2027-1 2028-1 2029-1                       |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 15 @ 60 min           |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                               |                                                   |
| <b>LE102</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ACDMC RDG/WRTG INTL CDTs II</b> | <b>3.5 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-2                             | <b>Offerings:</b>                                 |
| Continuation of LE101.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                    | 2026-2 2027-2 2028-2 2029-2                       |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 15 @ 60 min           |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                               |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | LE101                              |                                                   |

| LF203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FRENCH I (STANDARD)            | 4.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                         | <b>Offerings:</b>                          |
| In the standard course sequence, Cadets acquire a basic proficiency in speaking, listening, reading, and writing skills in French. Learning activities focus on situations Cadets are likely to encounter in French society. Cadets are taught how to express simple ideas and basic needs, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. In addition to speaking, listening and reading skills, Cadets learn how to write short sentences on familiar topics. Through readings and discussions, Cadets are introduced to the culture and history of the French-speaking world. Cadets acquire a command of basic French vocabulary and gain a general understanding of how the language works, and become able to apply that knowledge when learning other foreign languages.                                                                                                                                                                                                                                                                              |                                | 2026-1 2027-1 2028-1<br>2029-1             |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min         |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                           |                                            |
| LF204                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FRENCH II (STANDARD)           | 4.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-2                         | <b>Offerings:</b>                          |
| Continuation of LF203.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min         |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                           |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LF203                          |                                            |
| LF371                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | INTENSIVE INTERMEDIATE FRENCH  | 4.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                         | <b>Offerings:</b>                          |
| In the intensive intermediate course, cadets develop proficiency in those skills necessary for communicating effectively in French and for pursuing upper-level courses. Cadets develop speaking skills that enable them to engage in conversations on a variety of topics with other class members and with native speakers. Cadets reinforce and expand their language skills by reading, viewing, discussing, and writing about contemporary life, current events, and other cultural and historical topics as presented in selected materials of the French-speaking world. In addition, cadets gain an overview of the profession of arms in French-speaking regions by reading, discussing, and writing about pertinent materials that focus on the mission and history of the military in those countries. Cadets also review the basic rules of French grammar and continue to acquire a corpus of French vocabulary. They will be able to use computer-assisted learning resources to strengthen and maintain their language proficiency. This course serves as a bridge to advanced elective French courses. |                                | 2026-1 2027-1 2028-1<br>2029-1             |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min         |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                           |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LF204                          |                                            |
| LF372                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FRENCH FOR ORAL & WRITTEN COMM | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-2                         | <b>Offerings:</b>                          |
| This course addresses students' specific needs in the development of listening, speaking, reading, and writing skills in French. Cadets expand their active vocabulary, gain greater command of complex grammatical structures, and develop appropriate styles of written and oral communication. Special emphasis is placed on conversational and expository speaking. Cadets increase their oral proficiency through dialogues, role play, group discussions, formal presentations, and simulations of everyday language tasks likely to be encountered in the target region. Guided writing activities develop Cadets' competence in the application of critical language structures. Listening and reading comprehension are strengthened via engagement with film, music, short stories, news reports, and other popular media. Course topics typically focus on regional culture, the military, and society.                                                                                                                                                                                                     |                                | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min         |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                           |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LF371                          |                                            |

| LF470                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SPECIAL TOPIC IN FRENCH    | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-1                     | <b>Offerings:</b>                          |
| This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                            |                                            |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min     |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                       |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | LF372                      |                                            |
| LF475                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FRENCH RDG/WRTG THRU MEDIA | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-1                     | <b>Offerings:</b>                          |
| In this course cadets enhance their reading and writing skills through study and discussion of contemporary French media (e.g. the Internet, television, film, radio, newspapers and magazines), as well as short literary selections. Reading strategies and textual analysis are addressed. Writing tasks develop organization, substance, and style. Graded work typically includes oral and written summaries of authentic texts and short compositions or reaction papers. The course is conducted in French.                                                                                                                                                                                                                          |                            |                                            |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min     | 2026-1 2027-1 2028-1<br>2029-1             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                       |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | LF372                      |                                            |
| LF476                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MILITARY SPKG/RDG - FRENCH | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-2                     | <b>Offerings:</b>                          |
| Cadets gain an understanding of the profession of arms in the French-speaking world through lectures and selected reading materials (e.g. journal articles, Internet media, training manuals, biographies, and historical documents). Course content may encompass the mission and role, training, operations, tactics, and organization of the armed forces. Oral proficiency is enhanced through in-class discussion as well as role-plays and simulations focusing on scenarios likely to be encountered while an officer is deployed in a French-speaking region. Media complement instruction. Graded work may include briefings, role-plays, and simulation. The course is conducted in French.                                       |                            |                                            |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min     | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                       |                                            |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LF475                      |                                            |
| LF483                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FRENCH CIVILIZATION I      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-1                     | <b>Offerings:</b>                          |
| This course constitutes an integrated study of the culture, history, and geography of France from its beginnings to the end of World War II. Readings, lectures, discussions, and audio-visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the people of France. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in French. |                            |                                            |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min     | 2027-1 2028-1 2029-1                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                       |                                            |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -Or-<br>LF475              |                                            |
| LF484                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FRENCH CIVILIZATION II     | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2025-2                     | <b>Offerings:</b>                          |



This course constitutes an integrated study of the culture, history, and geography of France from the late 19th Century to the present. Readings, lectures, discussions, and audio-visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of France. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. This course is conducted in French.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** LF475

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|--------------|-------------------------------|----------------------------------------------------------|
| <b>LF485</b> | <b>SURVEY OF FRENCH LIT I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course is a survey of French literature tracing its development from the Middle Ages through the 18th century. Cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of French society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater proficiency in French. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in French.

2026-1 2027-1 2028-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):**  
-Or-  
LF475

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>LF486</b> | <b>SURVEY OF FRENCH LIT II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course is a survey of French literature of the 19th - 21st centuries. Cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of French society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. This course is conducted in French.

2026-2 2027-2 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** LF475

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|--------------|----------------------------------|----------------------------------------------------------|
| <b>LF492</b> | <b>MASTERWORKS OF FRENCH LIT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Cadets develop competence in the knowledge and comprehension of representative French literary works and their relationship to the cultural context of French society. Selected examples of various literary genres that focus on events pertaining to the two World Wars, conflicts in the former French colonies and other experiences are read, discussed, and analyzed. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in French.

2027-2 2028-2 2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** LF475

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|--------------|----------------------------|----------------------------------------------------------|
| <b>LG203</b> | <b>GERMAN I (STANDARD)</b> | <b>4.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

In the standard course sequence, Cadets acquire a basic proficiency in speaking, listening, reading and writing skills in German. Learning activities focus on situations Cadets are likely to encounter in German society. Cadets are taught how to express simple ideas and basic needs, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. In addition to speaking, listening and reading skills, Cadets learn how to write sentences, paragraphs and/or short compositions on familiar topics. Through readings and discussions, Cadets are introduced to the cultures and history of the German-speaking world. Cadets acquire a command of basic German vocabulary and gain a general understanding of how the language works, and they become able to apply that knowledge when learning other foreign languages.

**Lessons:** 80 @ 55 min (5.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|--------------|-----------------------------|----------------------------------------------------------|
| <b>LG204</b> | <b>GERMAN II (STANDARD)</b> | <b>4.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

Continuation of LG203.

**Offerings:**

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 80 @ 55 min (5.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG203

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|--------------|--------------------------------------|----------------------------------------------------------|
| <b>LG371</b> | <b>INTENSIVE INTERMEDIATE GERMAN</b> | <b>4.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

In the intensive intermediate course, cadets develop proficiency in those skills necessary for communicating effectively in German and for pursuing upper-level courses. Cadets develop speaking skills that enable them to engage in conversations on a variety of topics with other class members and with native speakers. Cadets reinforce and expand their language skills by reading, viewing, discussing, and writing about contemporary life, current events, and other cultural and historical topics as presented in selected materials of the German-speaking world. In addition, cadets gain an overview of the profession of arms in German-speaking regions by reading, discussing, and writing about pertinent materials that focus on the mission and history of the military in those countries. Cadets also review the basic rules of German grammar and continue to acquire a corpus of German vocabulary. They will be able to use computer-assisted learning resources to strengthen and maintain their language proficiency. This course serves as a bridge to advanced elective German courses.

**Lessons:** 80 @ 55 min (5.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG204

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|--------------|-------------------------------------------|----------------------------------------------------------|
| <b>LG372</b> | <b>GERMAN FOR ORAL &amp; WRITTEN COMM</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-------------------------------------------|----------------------------------------------------------|

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**Scope:** 2025-2

This course addresses students' specific needs in the development of listening, speaking, reading, and writing skills in German. Cadets expand their active vocabulary, gain greater command of complex grammatical structures, and develop appropriate styles of written and oral communication. Special emphasis is placed on conversational and expository speaking. Cadets increase their oral proficiency through dialogues, role play, group discussions, formal presentations, and simulations of everyday language tasks likely to be encountered in the target region. Guided writing activities develop Cadets' competence in the application of critical language structures. Listening and reading comprehension are strengthened via engagement with film, music, short stories, news reports, and other popular media.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG371

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>LG470</b> | <b>SPECIAL TOPIC IN GERMAN</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities. No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>LG475</b> | <b>GERMAN RDG/WRTG THRU MEDIA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

In this course cadets enhance their reading and writing skills through study and discussion of contemporary German media (e.g. the Internet, television, film, radio, newspapers and magazines), as well as short literary selections. Reading strategies and textual analysis are addressed. Writing tasks develop organization, substance, and style. Graded work typically includes oral and written summaries of authentic texts and short compositions or reaction papers. The course is conducted in German.

2026-1 2026-8 2027-1  
2028-1 2029-1 2029-8

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG372

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>LG476</b> | <b>MILITARY SPKG/RDG - GERMAN</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Cadets gain an understanding of the profession of arms in the German-speaking world through lectures and selected reading materials (e.g. journal articles, Internet media, training manuals, biographies, and historical documents). Course content may encompass the mission and role, training, operations, tactics, and organization of the armed forces. Oral proficiency is enhanced through in-class discussion as well as role-plays and simulations focusing on scenarios likely to be encountered while an officer is deployed in a German-speaking region. Media complement instruction. Graded work may include briefings, role-plays, and simulation. The course is conducted in German.

2026-2 2026-8 2027-2  
2028-2 2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG475

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|--------------|------------------------------|----------------------------------------------------------|
| <b>LG483</b> | <b>GERMAN CIVILIZATION I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course constitutes an integrated study of the culture, history, and geography of Germany, Austria, and Switzerland from their beginnings to the end of World War II. Readings, lectures, discussions, and audio and visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the people of Germany, Austria, and Switzerland. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in German.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG475

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|--------------|-------------------------------|----------------------------------------------------------|
| <b>LG484</b> | <b>GERMAN CIVILIZATION II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course constitutes an integrated study of the culture, history, and geography of Germany, Austria, and Switzerland since the end of World War II. Readings, lectures, discussions, and audio-visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the people of Germany, Austria, and Switzerland. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in German.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG475

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|              |                               |                                                          |
|--------------|-------------------------------|----------------------------------------------------------|
| <b>LG485</b> | <b>SURVEY OF GERMAN LIT I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course is a survey of German literature tracing its development from the 19th century through post-World War II. Cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of German society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in German.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG475

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>LG486</b> | <b>SURVEY OF GERMAN LIT II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course is a survey of German literature from the Enlightenment to the early 19th century. Cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of German society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in German.

2026-2 2027-2 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG475

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|--------------|----------------------------------------|----------------------------------------------------------|
| <b>LG492</b> | <b>20TH &amp; 21ST CENTURY GERMANY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Cadets develop competence in the knowledge and comprehension of representative German literary works and their relationship to the cultural context of German society. Selected examples of various literary genres that focus on the experiences of the two World Wars, a divided nation, and reunification are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in German.

2027-2 2028-2 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LG475

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>LN287</b> | <b>INTRO IND STUDY - FGN LANGS</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

The cadet pursues research topics in a foreign language of his or her choice. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor.

2026-1 2026-2 2027-1  
2028-1 2029-1

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the Language Section Program Director in consultation with the faculty advisor involved in the research project.

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>LN287A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INTRO IND STUDY - FGN LANGS</b>                                                                                                                                                                               | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| The cadet pursues research topics in a foreign language of his or her choice. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor.                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                  | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                           |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Enrollment by permission of the Language Section Program Director in consultation with the faculty advisor involved in the research project. LN287A will only be offered if a cadet has already completed LN287. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LN287                                                                                                                                                                                                            |                                                               |
| <b>LN380</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>NATURE OF MODERN LANGUAGES</b>                                                                                                                                                                                | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| Cadets learn that human language is a rule-based and universal system. They examine languages like those taught at USMA from the perspective of linguists, teachers and Army officers. Topics include the origin of and the basis for language, the nature of grammar, language sounds, the phenomenon of meaning, and how language attains communication. Knowledge gained is frequently interdisciplinary and relevant to courses offered at USMA in psychology, communication, English and foreign or second languages. Graded work may include giving oral presentations and completing a term project or paper. |                                                                                                                                                                                                                  | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                           |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Individual oral or written reports.                                                                                                                                                                              |                                                               |
| <b>LN387</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>IND STUDY - FGN LANGS</b>                                                                                                                                                                                     | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| The cadet pursues research topics in a foreign language of his or her choice. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor.                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                  | 2026-1 2026-2 2027-1<br>2028-1                                |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                           |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Enrollment by permission of the Language Section Program Director in consultation with the faculty advisor involved in the research project. LN387A will only be offered if a cadet has already completed LN387. |                                                               |
| <b>LN387A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>IND STUDY - FGN LANGS</b>                                                                                                                                                                                     | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| The cadet pursues research topics in a foreign language of his or her choice. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor.                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                  | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                           |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Enrollment by permission of the Language Section Program Director in consultation with the faculty advisor involved in the research project. LN387A will only be offered if a cadet has already completed LN387. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LN387                                                                                                                                                                                                            |                                                               |
| <b>LN400</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>LANGUAGE IN CULTURAL CONTEXT</b>                                                                                                                                                                              | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                                                                                                                                                                                                           | <b>Offerings:</b>                                             |
| Cadets travel to selected sites where cultural and linguistic immersion is an opportunity. Cadets engage in structured activities and instruction in the target language. They visit sites of cultural and historical significance, and pursue a program of learning as approved by the Department of Foreign Languages that is similar to other IAD course experiences except for the number of credit hours awarded.                                                                                                                                                                                               |                                                                                                                                                                                                                  | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                           |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                                                                                                                                                                                             |                                                               |

| LN440A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ARABIC IN CULTURAL CONTEXT  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2024-7                      | <b>Offerings:</b>                          |
| Cadets travel to and reside in a linguistic and cultural community for three weeks. There they use their knowledge of Arabic, its varieties and connected cultures to accomplish learning tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; and directed language learning activities. A department instructor accompanies the participating cadets who are obliged to use Arabic during this extended stay.   |                             | 2026-7 2027-7 2028-7<br>2029-7             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min      |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                        |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LA204                       |                                            |
| LN440C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CHINESE IN CULTURAL CONTEXT | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2024-7                      | <b>Offerings:</b>                          |
| Cadets travel to and reside in a linguistic and cultural community for three weeks. There they use their knowledge of Chinese, its varieties and connected cultures to accomplish learning tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; and directed language learning activities. A department instructor accompanies the participating cadets who are obliged to use Chinese during this extended stay. |                             | 2026-7 2027-7 2028-7<br>2029-7             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min      |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                        |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LC204                       |                                            |
| LN440F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | FRENCH IN CULTURAL CONTEXT  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2024-7                      | <b>Offerings:</b>                          |
| Cadets travel to and reside in a linguistic and cultural community for three weeks. There they use their knowledge of French, its varieties and connected cultures to accomplish learning tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; and directed language learning activities. A department instructor accompanies the participating cadets who are obliged to use French during this extended stay.   |                             | 2026-7 2027-7 2028-7<br>2029-7             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min      |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                        |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LF204                       |                                            |
| LN440G                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | GERMAN IN CULTURAL CONTEXT  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025-7                      | <b>Offerings:</b>                          |
| Cadets travel to and reside in a linguistic and cultural community for three weeks. There they use their knowledge of German, its varieties and connected cultures to accomplish learning tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; and directed language learning activities. A department instructor accompanies the participating cadets who are obliged to use German during this extended stay.   |                             | No Course Offerings                        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min      |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                        |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LG204                       |                                            |
| LN440H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | HEBREW IN CULTURAL CONTEXT  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-7                                | <b>Offerings:</b>                                  |
| Cadets achieve communication skills in Hebrew in a real-world context through assigned language-learning tasks, instructor's guidance and immersion in the culture of the target language. Each cadet will develop and improve language proficiency while increasing cross-cultural competence in the Hebrew-speaking world.                                                                                                                                                                                                                                                                                                                        |                                       | No Course Offerings                                |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                  |                                                    |
| <b>LN440P</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>PORTUGUESE IN CULTURAL CONTEXT</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024-7                                | <b>Offerings:</b>                                  |
| Cadets travel to and reside in a linguistic and cultural community for three weeks. There they use their knowledge of Portuguese, its varieties and connected cultures to accomplish learning tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; and directed language learning activities. A department instructor accompanies the participating cadets who are obliged to use Portuguese during this extended stay.      |                                       | 2026-7 2027-7 2028-7<br>2029-7                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                  |                                                    |
| <b>LN440R</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>RUSSIAN IN CULTURAL CONTEXT</b>    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024-7                                | <b>Offerings:</b>                                  |
| Cadets travel to and reside in a linguistic and cultural community for three weeks. There they use their knowledge of Russian, its varieties and connected cultures to accomplish learning tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; and directed language learning activities. A department instructor usually accompanies participating cadets who complete all work in the language during this extended stay. |                                       | 2026-7 2027-7 2028-7<br>2029-7                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | LR204                                 |                                                    |
| <b>LN440S</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>SPANISH IN CULTURAL CONTEXT</b>    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024-7                                | <b>Offerings:</b>                                  |
| Cadets travel to and reside in a linguistic and cultural community for three weeks. There they use their knowledge of Spanish, its varieties and connected cultures to accomplish learning tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; and directed language learning activities. A Department instructor accompanies the participating cadets, who are obliged to use Spanish during this extended stay.           |                                       | 2026-7 2027-7 2028-7<br>2029-7                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                  |                                                    |
| <b>LN440Z</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>PERSIAN IN CULTURAL CONTEXT</b>    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024-7                                | <b>Offerings:</b>                                  |
| Cadets travel to and reside in a linguistic and cultural community where they use their knowledge of Persian, its varieties and connected cultures to accomplish research tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; as well as research in the language. A Department instructor may accompany participating cadets, who complete all work in the language during this extended stay.                             |                                       | 2026-7 2027-7 2028-7<br>2029-7                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                  |                                                    |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LZ204                                    |                                                    |
| <b>LN441G</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>STUDY GERMAN LANG &amp; CULTURE</b>   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-7                                   | <b>Offerings:</b>                                  |
| Cadets experience an intensive program of study and cultural activities tailored to their skill level at a language institute in a site in a German-speaking country. Classes meet three to four hours per day in small groups of six to eight students. Classes address speaking, listening, reading and writing, and emphasize improvement of speaking and listening proficiency. Students may reside with German host families and conduct local cultural excursions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                          | 2029-7                                             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                     |                                                    |
| <b>LN450</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ADVANCED LANGUAGE IN CONTEXT</b>      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2024-7                                   | <b>Offerings:</b>                                  |
| Cadets travel to and reside in a linguistic and cultural community where they use their knowledge of a second language, its varieties and connected cultures to accomplish research tasks, solve problems and live daily routines. A structured program of immersion may include visits to sites of military, political, historical, or social significance; official orientations and lectures; meetings with local or national civilian and military leaders; as well as research in the language. A department instructor accompanies participating cadets who complete all work in the language during this extended stay.                                                                                                                                                                                                                                                                                                                                       |                                          | 2026-7 2027-7 2028-7<br>2029-7                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                     |                                                    |
| <b>LN451</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ADV LANG &amp; CULTURE IN CONTEXT</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                                   | <b>Offerings:</b>                                  |
| Cadets travel to and, over an extended period, reside in a linguistic and cultural community where they develop further their foreign language proficiency, cultural competence and regional capability. A structured program of experiential learning includes visits to sites of cultural, geographic, political, historical, or social significance. Participation in military training exercises, involvement in service learning, and attendance at cultural events may be part of the immersion experience. Cadets write reflective essays, keep personal/public journals, complete task-based writing assignments, deliver briefings, and produce research papers. A department instructor may conduct a site visit while cadets are abroad.                                                                                                                                                                                                                  |                                          | 2026-1 2026-2 2029-1<br>2029-2                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                     |                                                    |
| <b>LN482H</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>SPOKEN HEBREW</b>                     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-2                                   | <b>Offerings:</b>                                  |
| This course aims to develop entry-level oral proficiency in Hebrew (approx. 800 words), the ability to read printed Hebrew for all vocabulary covered, and the ability to write simple sentences in Hebrew. Most of the course work will be oral.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                          | 2027-2 2028-2                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                     |                                                    |
| <b>LN482J</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INTRODUCTORY JAPANESE</b>             | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                                   | <b>Offerings:</b>                                  |
| In the standard course, cadets acquire a basic proficiency in speaking, listening, reading, and writing skills in Japanese. Learning activities focus on situations cadets are likely to encounter in the target society. Cadets are taught how to express simple ideas and basic needs and minimum courtesy in areas of immediate need or familiar topics, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. Although instruction places greater emphasis on reading, listening, and speaking skills, Cadets also learn how to write simple statements on familiar topics by hand and by typing. Through readings and discussions, Cadets are introduced to the culture, history, and geography of the Japanese-speaking world. Cadets acquire a command of basic Japanese vocabulary and gain a general understanding of how the language works and how to apply that knowledge when learning other foreign |                                          | 2026-2 2029-2                                      |



languages. Cadets also learn military terminology important to the Japanese Self Defense Force and the roles and responsibilities of the JSDF. The course will only be taught when resources are available.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Cadets must complete two semesters of foreign language study at USMA prior to taking this elective. This course requires special permission from the DEWL Department Head.

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>LN487</b> | <b>ADV IND STUDY-FOREIGN LANGS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|------------------------------------|----------------------------------------------------------|

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**Scope:** 2025-1

**Offerings:**

LN487 and LN488 are essentially honors or tutorial courses available only to exceptionally motivated and qualified cadets who have exhausted all other language-specific courses and who wish to pursue a special field of interest in language, linguistics or a language-related field. The minimum completion requirement is a term paper, based on individual research of a length and on a topic upon which instructor and cadet have agreed.

2026-1 2026-8 2027-1  
2027-8 2028-1 2028-8  
2029-1 2029-8 2029-9

**Lessons:** 17 @ 55 min (1.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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|               |                                    |                                                          |
|---------------|------------------------------------|----------------------------------------------------------|
| <b>LN487A</b> | <b>ADV IND STUDY-FOREIGN LANGS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

LN487A and LN488A are essentially honors or tutorial courses available only to exceptionally motivated and qualified cadets who have exhausted all other language-specific courses and who wish to pursue a special field of interest in language, linguistics or a language-related field. The minimum completion requirement is a term paper, based on individual research of a length and on a topic upon which instructor and cadet have agreed.

2028-1 2029-1

**Lessons:** 17 @ 55 min (0.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** SPECIAL REQUIREMENTS: A cadet must take LN487 prior to taking LN487A. Department Head approval required.

**Prerequisite(s):** LN487

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>LN488</b> | <b>ADV IND STUDY-FOREIGN LANGS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

LN487 and LN488 are essentially honors or tutorial courses available only to exceptionally motivated and qualified cadets who have exhausted all other language-specific courses and who wish to pursue a special field of interest in language, linguistics or a language-related field. The minimum completion requirement is a term paper, based on individual research of a length and on a topic upon which instructor and cadet have agreed.

2026-2 2026-9 2027-1  
2027-2 2027-8 2027-9  
2028-1 2028-2 2028-8  
2028-9 2029-2 2029-8  
2029-9

**Lessons:** 17 @ 55 min (1.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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|               |                                    |                                                          |
|---------------|------------------------------------|----------------------------------------------------------|
| <b>LN488A</b> | <b>ADV IND STUDY-FOREIGN LANGS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

LN487A and LN488A are essentially honors or tutorial courses available only to exceptionally motivated and qualified cadets who have exhausted all other language-specific courses and who wish to pursue a special field of interest in language, linguistics or a language-related field. The minimum completion requirement is a term paper, based on individual research of a length and on a topic upon which instructor and cadet have agreed.

2027-2 2028-2 2029-2  
2029-8 2029-9

**Lessons:** 17 @ 55 min (1.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A cadet must take LN488 prior to taking LN488A. Department Head approval required.

**Prerequisite(s):** LN488

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|               |                                          |                                                          |
|---------------|------------------------------------------|----------------------------------------------------------|
| <b>LN490A</b> | <b>ARABIC LANGUAGE &amp; CULTURE CAP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|------------------------------------------|----------------------------------------------------------|

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-2                                    | <b>Offerings:</b>                                 |
| In this capstone course concentrators integrate their knowledge of Arabic language and culture of the Arabic-speaking world with other aspects of the curriculum. They attend lectures, participate in seminar discussions and complete a project of international import. Cadets develop a regionally focused topic, complete research and present findings for possible application at the joint command level. They make use of their acquired language skills while completing a course that is interdisciplinary in nature and meets academic program goals.    |                                           | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                    |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                      |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LA475                                     |                                                   |
| <b>LN490C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CHINESE LANGUAGE &amp; CULTURE CAP</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-2                                    | <b>Offerings:</b>                                 |
| In this capstone course concentrators integrate their knowledge of Chinese language and culture of the Chinese-speaking world with other aspects of the curriculum. They attend lectures, participate in seminar discussions, and complete a project of international import. Cadets develop a regionally focused topic, complete research and present findings for possible application at the joint command level. They make use of their acquired language skills while completing a course that is interdisciplinary in nature and meets academic program goals. |                                           | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                    |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                      |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LC475                                     |                                                   |
| <b>LN490F</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>FRENCH LANGUAGE &amp; CULTURE CAP</b>  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-2                                    | <b>Offerings:</b>                                 |
| In this capstone course concentrators integrate their knowledge of French language and culture of the French-speaking world with other aspects of the curriculum. They attend lectures, participate in seminar discussions, and complete a project of international import. Cadets develop a regionally focused topic, complete research and present findings for possible application at the joint command level. They make use of their acquired language skills while completing a course that is interdisciplinary in nature and meets academic program goals.   |                                           | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                    |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                      |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LF475                                     |                                                   |
| <b>LN490G</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>GERMAN LANGUAGE &amp; CULTURE CAP</b>  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-2                                    | <b>Offerings:</b>                                 |
| In this capstone course concentrators integrate their knowledge of German language and culture of the German-speaking world with other aspects of the curriculum. They attend lectures, participate in seminar discussions, and complete a project of international import. Cadets develop a regionally focused topic, complete research and present findings for possible application at the joint command level. They make use of their acquired language skills while completing a course that is interdisciplinary in nature and meets academic program goals.   |                                           | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                    |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                      |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | LG475                                     |                                                   |
| <b>LN490P</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>PORTUGUESE LANG &amp; CULTURE CAP</b>  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-2                                    | <b>Offerings:</b>                                 |

In this capstone course concentrators integrate their knowledge of Portuguese language and culture of the Portuguese-speaking world with other aspects of the curriculum. They attend lectures, participate in seminar discussions and complete a project of international import. Cadets develop a regionally focused topic, complete research and present findings for possible application at the joint command level. They make use of their acquired language skills while completing a course that is interdisciplinary in nature and meets academic program goals.

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LP475

2026-2 2027-2 2028-2  
2029-2

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|---------------|-------------------------------------------|----------------------------------------------------------|
| <b>LN490R</b> | <b>RUSSIAN LANGUAGE &amp; CULTURE CAP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

In this capstone course concentrators integrate their knowledge of Russian language and culture of the Russian-speaking world with other aspects of the curriculum. They attend lectures, participate in seminar discussions and complete a project of international import. Cadets develop a regionally focused topic, complete research and present findings for possible application at the joint command level. They make use of their acquired language skills while completing a course that is interdisciplinary in nature and meets academic program goals.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LR475

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|---------------|-------------------------------------------|----------------------------------------------------------|
| <b>LN490S</b> | <b>SPANISH LANGUAGE &amp; CULTURE CAP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

In this capstone course concentrators integrate their knowledge of Spanish language and culture of the Spanish-speaking world with other aspects of the curriculum. They attend lectures, participate in seminar discussions and complete a project of international import. Cadets develop a regionally focused topic, complete research and present findings for possible application at the joint command level. They make use of their acquired language skills while completing a course that is interdisciplinary in nature and meets academic program goals.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS475

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|---------------|-------------------------------------------|----------------------------------------------------------|
| <b>LN490Z</b> | <b>PERSIAN LANGUAGE &amp; CULTURE CAP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

In this capstone course concentrators integrate their knowledge of Persian language and culture of the Persian-speaking world with other aspects of the curriculum. They attend lectures, participate in seminar discussions and complete a project of international import. Cadets develop a regionally focused topic, complete research and present findings for possible application at the joint command level. They make use of their acquired language skills while completing a course that is interdisciplinary in nature and meets academic program goals.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LZ475

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|--------------|------------------------------------------|----------------------------------------------------------|
| <b>LN491</b> | <b>SEM ABROAD: ADV LANG &amp; CULT I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

Cadets attend a military academy or an undergraduate institution abroad and enroll in courses that enhance their language proficiency and cultural literacy. Courses may focus on language acquisition, literature, military science, history or the social sciences. If appropriate, cadets participate in military development activities. They attend lectures and seminars and complete all course requirements. The course grade becomes part of their Academic Summary.

2026-8 2026-9 2027-1  
2027-2 2027-8 2027-9  
2028-1 2028-2 2028-8  
2028-9 2029-1 2029-8  
2029-9

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                      |                                                                                                        |
| <b>LN492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SEM ABROAD: ADV LANG &amp; CULT II</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-1                                    | <b>Offerings:</b>                                                                                      |
| Cadets attend a military academy or an undergraduate institution abroad and enroll in courses that enhance their language proficiency and cultural literacy. Courses may focus on language acquisition, literature, military science, history or the social sciences. If appropriate, cadets participate in military development activities. They attend lectures and seminars and complete all course requirements. The course grade becomes part of their Academic Summary. |                                           | 2026-8 2026-9 2027-1<br>2027-2 2027-8 2027-9<br>2028-1 2028-2 2028-8<br>2028-9 2029-1 2029-8<br>2029-9 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                      |                                                                                                        |
| <b>LN493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SEM ABROAD: ADV LANG&amp;CULT III</b>  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-1                                    | <b>Offerings:</b>                                                                                      |
| Cadets attend a military academy or an undergraduate institution abroad and enroll in courses that enhance their language proficiency and cultural literacy. Courses may focus on language acquisition, literature, military science, history or the social sciences. If appropriate, cadets participate in military development activities. They attend lectures and seminars and complete all course requirements. The course grade becomes part of their Academic Summary. |                                           | 2026-8 2026-9 2027-1<br>2027-2 2027-8 2027-9<br>2028-1 2028-2 2028-8<br>2028-9 2029-1 2029-8<br>2029-9 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                      |                                                                                                        |
| <b>LN494</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SEM ABROAD: ADV LANG &amp; CULT IV</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-1                                    | <b>Offerings:</b>                                                                                      |
| Cadets attend a military academy or an undergraduate institution abroad and enroll in courses that enhance their language proficiency and cultural literacy. Courses may focus on language acquisition, literature, military science, history or the social sciences. If appropriate, cadets participate in military development activities. They attend lectures and seminars and complete all course requirements. The course grade becomes part of their Academic Summary. |                                           | 2026-8 2026-9 2027-1<br>2027-2 2027-8 2027-9<br>2028-1 2028-2 2028-8<br>2028-9 2029-1 2029-8<br>2029-9 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                      |                                                                                                        |
| <b>LN495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SEM ABROAD: ADV LANG &amp; CULT V</b>  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-1                                    | <b>Offerings:</b>                                                                                      |
| Cadets attend a military academy or an undergraduate institution abroad and enroll in courses that enhance their language proficiency and cultural literacy. Courses may focus on language acquisition, literature, military science, history or the social sciences. If appropriate, cadets participate in military development activities. They attend lectures and seminars and complete all course requirements. The course grade becomes part of their Academic Summary. |                                           | 2026-8 2026-9 2027-1<br>2027-2 2027-8 2027-9<br>2028-1 2028-2 2028-8<br>2028-9 2029-1 2029-8<br>2029-9 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                      |                                                                                                        |
| <b>LP203</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>PORTUGUESE I (STANDARD)</b>            | <b>4.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                                      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-1                                    | <b>Offerings:</b>                                                                                      |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------|
| <p>In the standard course sequence, Cadets acquire a basic proficiency in speaking, listening, reading and writing skills in Portuguese. Learning activities focus on situations Cadets are likely to encounter in Portuguese society. Cadets are taught how to express simple ideas and basic needs, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. In addition to speaking, listening and reading skills, Cadets learn how to write sentences, paragraphs and/or short compositions on familiar topics. Through readings and discussions, Cadets are introduced to the cultures and history of the Portuguese-speaking world. Cadets acquire a command of basic Portuguese vocabulary and gain a general understanding of how the language works, and they become able to apply that knowledge when learning other foreign languages.</p> <p><b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p>                                                                                                                                                                                                                                                                                                                                 |                                       | 2026-1 2027-1 2028-1<br>2029-1                    |
| <b>LP204</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>PORTUGUESE II (STANDARD)</b>       | <b>4.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-2                                | <b>Offerings:</b>                                 |
| Continuation of LP203.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                  |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LP203                                 |                                                   |
| <b>LP371</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>INTENSIVE INTERMED. PORTUGUESE</b> | <b>4.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                                | <b>Offerings:</b>                                 |
| <p>In the intensive intermediate course, cadets develop proficiency in those skills necessary for communicating effectively in Portuguese and for pursuing upper-level courses. Cadets develop speaking skills that enable them to engage in conversations on a variety of topics with other class members and with native speakers. Cadets reinforce and expand their language skills by reading, viewing, discussing, and writing about contemporary life, current events, and other cultural and historical topics as presented in selected materials of the Portuguese-speaking world. In addition, cadets gain an overview of the profession of arms in Portuguese-speaking regions by reading, discussing, and writing about pertinent materials that focus on the mission and history of the military in those countries. Cadets also review the basic rules of Portuguese grammar and continue to acquire a corpus of Portuguese vocabulary. They will be able to use computer-assisted learning resources to strengthen and maintain their language proficiency. This course serves as a bridge to advanced elective Portuguese courses.</p> <p><b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p> <p><b>Prerequisite(s):</b> LP204</p> <p><b>Disqualifier(s):</b> LP361 LP362</p> |                                       | 2026-1 2027-1 2028-1<br>2029-1                    |
| <b>LP372</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>PORTUGUESE FOR ORAL/WRIT COMM</b>  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-2                                | <b>Offerings:</b>                                 |
| <p>This course addresses students' specific needs in the development of listening, speaking, reading, and writing skills in Portuguese. Cadets expand their active vocabulary, gain greater command of complex grammatical structures, and develop appropriate styles of written and oral communication. Special emphasis is placed on conversational and expository speaking. Cadets increase their oral proficiency through dialogues, role play, group discussions, formal presentations, and simulations of everyday language tasks likely to be encountered in the target region. Guided writing activities develop Cadets' competence in the application of critical language structures. Listening and reading comprehension are strengthened via engagement with film, music, short stories, news reports, and other popular media. Course topics typically focus on regional culture, the military, and society.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p> <p><b>Prerequisite(s):</b> LP371</p>                                                                                                                                                                                                                                                                    |                                       | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>LP470</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>SPECIAL TOPIC IN PORTUGUESE</b>    | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-1                                | <b>Offerings:</b>                                  |
| This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                       | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                  |                                                    |
| <b>LP475</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>PORTUGUESE RDG/WRTG THRU MEDIA</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-1                                | <b>Offerings:</b>                                  |
| In this course cadets enhance their reading and writing skills through study and discussion of contemporary Portuguese media (e.g. the Internet, television, film, radio, newspapers and magazines), as well as short literary selections. Reading strategies and textual analysis are addressed. Writing tasks develop organization, substance, and style. Graded work typically includes oral and written summaries of authentic texts and short compositions or reaction papers. The course is conducted in Portuguese.                                                                                                                                                                                                     |                                       | 2026-1 2027-1 2028-1 2029-1                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LP372                                 |                                                    |
| <b>LP476</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>MILITARY SPKG/RDG - PORTUGUESE</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-2                                | <b>Offerings:</b>                                  |
| Cadets gain an understanding of the profession of arms in the Portuguese-speaking world through lectures and selected reading materials (e.g. journal articles, Internet media, training manuals, biographies, and historical documents). Course content may encompass the mission and role, training, operations, tactics, and organization of the armed forces. Oral proficiency is enhanced through in-class discussion as well as role-plays and simulations focusing on scenarios likely to be encountered while an officer is deployed in a Portuguese-speaking region. Media complement instruction. Graded work may include briefings, role-plays, and simulation. The course is conducted in Portuguese.              |                                       | 2026-2 2027-2 2028-2 2029-2                        |
| <b>Lessons:</b> 40 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LP372<br>-Or-<br>LP475                |                                                    |
| <b>LP483</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>PORTUGUESE CIVILIZATION I</b>      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-1                                | <b>Offerings:</b>                                  |
| This course constitutes an integrated study of the culture, history, and geography of Portugal and the Lusophone countries. Readings, lectures, discussions, and audio-visual materials encompass the representative artistic and intellectual accomplishments, political institutions, economy, and popular culture of those regions. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the Portuguese-speaking people. At the same time, cadets continue to develop greater proficiency in Portuguese. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Portuguese. |                                       | 2027-1 2028-1                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LP372                                 |                                                    |
| <b>LP484</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>PORTUGUESE CIVILIZATION II</b>     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-2                                | <b>Offerings:</b>                                  |
| This course constitutes an integrated study of the culture, history, and geography of Brazil. Readings, lectures, discussions, and audio-visual materials encompass the representative artistic and intellectual accomplishments, political institutions, economy, and popular culture of Brazil. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the Brazilian people. At the same time, cadets continue to develop greater proficiency in Portuguese. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Portuguese.                                                |                                       | 2026-2 2027-2 2028-2 2029-2                        |

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| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                  |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LP372                                 |                                                   |
| <b>LP485</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>SURVEY OF PORTUGUESE LIT I</b>     | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-1                                | <b>Offerings:</b>                                 |
| In this course cadets gain basic competence in the knowledge and comprehension of representative Brazilian and Portuguese short stories and of their relationship to the cultural contexts of Brazilian and Portuguese society. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Portuguese.                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 2027-1 2028-1                                     |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                  |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LP372                                 |                                                   |
| <b>LP492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>LIT OF PORT-SPKG WORLD</b>         | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-2                                | <b>Offerings:</b>                                 |
| Cadets develop competence in the knowledge and comprehension of representative Portuguese and Brazilian literary works and their relationship to the cultural contexts of Portuguese and Brazilian society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Portuguese.                                                                                                                                                                                                                                                                                         |                                       | 2026-2 2027-2 2028-2                              |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                  |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LP372                                 |                                                   |
| <b>LR203</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>RUSSIAN I (STANDARD)</b>           | <b>4.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-1                                | <b>Offerings:</b>                                 |
| In the standard course sequence, Cadets acquire a basic proficiency in speaking, listening, reading and writing skills in Russian. Learning activities focus on situations Cadets are likely to encounter in Russian society. Cadets are taught how to express simple ideas and basic needs, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. In addition to speaking, listening and reading skills, Cadets learn how to write sentences, paragraphs and/or short compositions on familiar topics. Through readings and discussions, Cadets are introduced to the cultures and history of the Russian-speaking world. Cadets acquire a command of basic Russian vocabulary and gain a general understanding of how the language works, and they become able to apply that knowledge when learning other foreign languages. |                                       | 2026-1 2027-1 2028-1<br>2029-1                    |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                  |                                                   |
| <b>LR204</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>RUSSIAN II (STANDARD)</b>          | <b>4.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-2                                | <b>Offerings:</b>                                 |
| Continuation of LR203.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                       | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                  |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LR203                                 |                                                   |
| <b>LR371</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INTENSIVE INTERMEDIATE RUSSIAN</b> | <b>4.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                                  | <b>Offerings:</b>                                 |
| In the intensive intermediate course, cadets develop proficiency in those skills necessary for communicating effectively in Russian and for pursuing upper-level courses. Cadets develop speaking skills that enable them to engage in conversations on a variety of topics with other class members and with native speakers. Cadets reinforce and expand their language skills by reading, viewing, discussing, and writing about contemporary life, current events, and other cultural and historical topics as presented in selected materials of the Russian-speaking world. In addition, cadets gain an overview of the profession of arms in Russian-speaking regions by reading, discussing, and writing about pertinent materials that focus on the mission and history of the military in those countries. Cadets also review the basic rules of Russian grammar and continue to acquire a corpus of Russian vocabulary. They will be able to use computer-assisted learning resources to strengthen and maintain their language proficiency. This course serves as a bridge to advanced elective Russian courses. |                                         | 2026-1 2027-1 2027-8<br>2028-1 2028-8 2029-1      |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                  |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                    |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LR204                                   |                                                   |
| <b>LR372</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>RUSSIAN FOR ORAL &amp; WRIT COMM</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-2                                  | <b>Offerings:</b>                                 |
| This course addresses students' specific needs in the development of listening, speaking, reading, and writing skills in Russian. Cadets expand their active vocabulary, gain greater command of complex grammatical structures, and develop appropriate styles of written and oral communication. Special emphasis is placed on conversational and expository speaking. Cadets increase their oral proficiency through dialogues, role play, group discussions, formal presentations, and simulations of everyday language tasks likely to be encountered in the target region. Guided writing activities develop Cadets' competence in the application of critical language structures. Listening and reading comprehension are strengthened via engagement with film, music, short stories, news reports, and other popular media. Course topics typically focus on regional culture, the military, and society.                                                                                                                                                                                                          |                                         | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                  |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                    |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LR371                                   |                                                   |
| <b>LR470</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>SPECIAL TOPIC IN RUSSIAN</b>         | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                                  | <b>Offerings:</b>                                 |
| This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                         | No Course Offerings                               |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                  |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                    |                                                   |
| <b>LR475</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>RUSSIAN RDG/WRITG THRU MEDIA</b>     | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                                  | <b>Offerings:</b>                                 |
| In this course cadets enhance their reading and writing skills through study and discussion of contemporary Russian media (e.g. the Internet, television, film, radio, newspapers and magazines), as well as short literary selections. Reading strategies and textual analysis are addressed. Writing tasks develop organization, substance, and style. Graded work typically includes oral and written summaries of authentic texts and short compositions or reaction papers. The course is conducted in Russian.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                         | 2026-1 2027-1 2027-8<br>2028-1 2028-8 2029-1      |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                  |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                    |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LR372                                   |                                                   |
| <b>LR476</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>MILITARY SPKG/RDG - RUSSIAN</b>      | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-2                                  | <b>Offerings:</b>                                 |



Cadets gain an understanding of the profession of arms in the Russian-speaking world through lectures and selected reading materials (e.g. journal articles, Internet media, training manuals, biographies, and historical documents). Course content may encompass the mission and role, training, operations, tactics, and organization of the armed forces. Oral proficiency is enhanced through in-class discussion as well as role-plays and simulations focusing on scenarios likely to be encountered while an officer is deployed in a Russian-speaking region. Media complement instruction. Graded work may include briefings, role-plays, and simulation. The course is conducted in Russian.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** LR475

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| <b>LR483</b> | <b>RUSSIAN CIV I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course constitutes an integrated study of the culture, history, and geography of Russia and the Soviet Union from its beginnings to the end of World War II. Readings, lectures, discussions, and audio-visual materials encompass this civilization's representative artistic and intellectual accomplishments, its political institutions, economy, and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures in Russia. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Russian.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** LR475

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| <b>LR484</b> | <b>RUSSIAN CIV II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course constitutes an intensive study of the culture, history, and geography of Russia and the Soviet Union since the end of World War II. Readings, lectures, discussions, and audio-visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of Russia. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Russian.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** LR476

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| <b>LR485</b> | <b>SURVEY OF RUSSIAN LITERATURE I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course is a survey of Russian literature, tracing its development from the early 19th century to the beginning of WWI. Cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of Russian society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Russian.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):**  
-Or-  
LR475

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|--------------|----------------------------------|----------------------------------------------------------|
| <b>LR486</b> | <b>SURVEY OF RUSSIAN LIT. II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course is a survey of Russian and Soviet literature from the time of the Russian Revolution through the post-World War II "Thaw" period. Cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of that society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Russian.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** LR476

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| <b>LR492</b> | <b>RUSSIAN LIFE IN FICTION</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-2 | <b>Offerings:</b> |
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Cadets develop competence in the knowledge and comprehension of representative Russian literary works and their relationship to the cultural context of Russian society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency in the Russian language. Video and film presentations supplement readings. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Russian.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** LR476

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| <b>LS203</b> | <b>SPANISH I (STANDARD)</b> | <b>4.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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In the standard course sequence, cadets acquire a basic proficiency in speaking, listening, reading and writing skills in Spanish. Learning activities focus on situations Cadets are likely to encounter in Spanish society. Cadets are taught how to express simple ideas and basic needs, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. In addition to speaking, listening and reading skills, Cadets learn how to write sentences, paragraphs and/or short compositions on familiar topics. Through readings and discussions, Cadets are introduced to the cultures and history of the Spanish-speaking world. Cadets acquire a command of basic Spanish vocabulary and gain a general understanding of how the language works, and they become able to apply that knowledge when learning other foreign languages.

**Lessons:** 80 @ 55 min (5.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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|--------------|------------------------------|----------------------------------------------------------|
| <b>LS204</b> | <b>SPANISH II (STANDARD)</b> | <b>4.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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Continuation of LS203.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 80 @ 55 min (5.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS203

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>LS371</b> | <b>INTENSIVE INTERMEDIATE SPANISH</b> | <b>4.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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In the intensive intermediate course, cadets develop proficiency in those skills necessary for communicating effectively in Spanish and for pursuing upper-level courses. Cadets develop speaking skills that enable them to engage in conversations on a variety of topics with other class members and with native speakers. Cadets reinforce and expand their language skills by reading, viewing, discussing, and writing about contemporary life, current events, and other cultural and historical topics as presented in selected materials of the Spanish-speaking world. In addition, cadets gain an overview of the profession of arms in Spanish-speaking regions by reading, discussing, and writing about pertinent materials that focus on the mission and history of the military in those countries. Cadets also review the basic rules of Spanish grammar and continue to acquire a corpus of Spanish vocabulary. They will be able to use computer-assisted learning resources to strengthen and maintain their language proficiency. This course serves as a bridge to advanced elective Spanish courses.

**Lessons:** 80 @ 55 min (5.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS204

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| <b>LS372</b> | <b>SPANISH FOR ORAL &amp; WRIT COMM</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course addresses students' specific needs in the development of listening, speaking, reading, and writing skills in Spanish. Cadets expand their active vocabulary, gain greater command of complex grammatical structures, and develop appropriate styles of written and oral communication. Special emphasis is placed on conversational and expository speaking. Cadets increase their oral proficiency through dialogues, role play, group discussions, formal presentations, and simulations of everyday language tasks likely to be encountered in the target region. Guided writing activities develop Cadets' competence in the application of critical language structures. Listening and reading comprehension are strengthened via engagement with film, music, short stories, news reports, and other popular media. Course topics typically focus on regional culture, the military, and society.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS371

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>LS470</b> | <b>SPECIAL TOPIC IN SPANISH</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS372

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>LS475</b> | <b>SPANISH RDG/WRTG THRU MEDIA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

In this course cadets enhance their reading and writing skills through study and discussion of contemporary Spanish media (e.g. the Internet, television, film, radio, newspapers and magazines), as well as short literary selections. Reading strategies and textual analysis are addressed. Writing tasks develop organization, substance, and style. Graded work typically includes oral and written summaries of authentic texts and short compositions or reaction papers. The course is conducted in Spanish.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS372

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>LS476</b> | <b>MILITARY SPKG/RDG - SPANISH</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Cadets gain an understanding of the profession of arms in the Spanish-speaking world through lectures and selected reading materials (e.g. journal articles, Internet media, training manuals, biographies, and historical documents). Course content may encompass the mission and role, training, operations, tactics, and organization of the armed forces. Oral proficiency is enhanced through in-class discussion as well as role-plays and simulations focusing on scenarios likely to be encountered while an officer is deployed in a Spanish-speaking region. Media complement instruction. Graded work may include briefings, role-plays, and simulation. The course is conducted in Spanish.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS475

2026-2 2027-2 2028-2  
2029-2

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>LS483</b> | <b>SPANISH CIV AND CULTURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course constitutes an integrated study of the culture, history, and geography of Spain. Readings, lectures, discussions, and audio-visual materials encompass Spain's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the Spanish people. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. The work is done in Spanish.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS475

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|--------------|--------------------------------------|----------------------------------------------------------|
| <b>LS484</b> | <b>SPANISH AMERICAN CIV AND CULT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course constitutes an integrated study of the culture, history, and geography of the countries of Spanish America. Readings, lectures, discussions, and audio-visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economies, and popular cultures. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the people in Spanish America. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. The work is done in Spanish.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS475

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>LS485</b> | <b>SPANISH-AMERICAN LITERATURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

In this course cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the Spanish-American cultural context. Selected examples of various literary genres are read and discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. The work is done in Spanish.

2026-1 2027-1 2028-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):**  
-Or-  
LS475

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>LS486</b> | <b>THE LITERATURE OF SPAIN</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

In this course cadets gain basic competence in the knowledge and comprehension of representative Spanish literary works, from the middle ages to the present, and their relationship to the cultural context of Spanish society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater proficiency in Spanish. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. The work is done in Spanish.

2027-2 2028-2 2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):**  
-Or-  
LS475

|              |                                       |                                                          |
|--------------|---------------------------------------|----------------------------------------------------------|
| <b>LS492</b> | <b>20TH/21ST CENTURY HISPANIC LIT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

In this course cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the Hispanic context. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. The work is done in Spanish.

2026-2 2028-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LS475

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| <b>LX300</b> | <b>3RD SEMESTER FOREIGN LANG</b> | <b>0.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

Cadets may enroll in a third semester of foreign language in any course for which the cadet is qualified.

2029-1

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:**

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| <b>LX400</b> | <b>4TH SEMESTER FOREIGN LANG</b> | <b>0.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

Cadets may enroll in a fourth semester of foreign language in any course for which qualified.

2029-1

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:**

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>LX476</b> | <b>MILITARY SPKG/RDG IN WORLD LAN</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Cadets may enroll in the Military Speaking and Reading course in any of the eight offered World Languages.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>LZ203</b> | <b>PERSIAN I (STANDARD)</b> | <b>4.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------|
| <p>In the standard course sequence, Cadets acquire a basic proficiency in speaking, listening, reading and writing skills in Persian. Learning activities focus on situations Cadets are likely to encounter in Persian society. Cadets are taught how to express simple ideas and basic needs, comprehend the language in everyday contexts, and read simplified texts and brief, authentic selections. In addition to speaking, listening and reading skills, Cadets learn how to write sentences, paragraphs and/or short compositions on familiar topics. Through readings and discussions, Cadets are introduced to the cultures and history of the Persian-speaking world. Cadets acquire a command of basic Persian vocabulary and gain a general understanding of how the language works, and they become able to apply that knowledge when learning other foreign languages.</p> <p><b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b>                      None</p>                                                                                                                                                                                                                                                                                                               |                                         | 2026-1 2027-1 2028-1<br>2029-1                    |
| <b>LZ204</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>PERSIAN II (STANDARD)</b>            | <b>4.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-2                                  | <b>Offerings:</b>                                 |
| Continuation of LZ203.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                         | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                  |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                    |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | LZ203                                   |                                                   |
| <b>LZ371</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INTENSIVE INTERMEDIATE PERSIAN</b>   | <b>4.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-1                                  | <b>Offerings:</b>                                 |
| <p>In the intensive intermediate course, cadets develop proficiency in those skills necessary for communicating effectively in Persian and for pursuing upper-level courses. Cadets develop speaking skills that enable them to engage in conversations on a variety of topics with other class members and with native speakers. Cadets reinforce and expand their language skills by reading, viewing, discussing, and writing about contemporary life, current events, and other cultural and historical topics as presented in selected materials of the Persian-speaking world. In addition, cadets gain an overview of the profession of arms in Persian-speaking regions by reading, discussing, and writing about pertinent materials that focus on the mission and history of the military in those countries. Cadets also review the basic rules of Persian grammar and continue to acquire a corpus of Persian vocabulary. They will be able to use computer-assisted learning resources to strengthen and maintain their language proficiency. This course serves as a bridge to advanced elective Persian courses.</p> <p><b>Lessons:</b> 80 @ 55 min (5.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b>                      None</p> <p><b>Prerequisite(s):</b>                                LZ204</p> |                                         | 2026-1 2027-1 2028-1<br>2029-1                    |
| <b>LZ372</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>PERSIAN FOR ORAL &amp; WRIT COMM</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-2                                  | <b>Offerings:</b>                                 |
| <p>This course addresses students' specific needs in the development of listening, speaking, reading, and writing skills in Persian. Cadets expand their active vocabulary, gain greater command of complex grammatical structures, and develop appropriate styles of written and oral communication. Special emphasis is placed on conversational and expository speaking. Cadets increase their oral proficiency through dialogues, role play, group discussions, formal presentations, and simulations of everyday language tasks likely to be encountered in the target region. Guided writing activities develop Cadets' competence in the application of critical language structures. Listening and reading comprehension are strengthened via engagement with film, music, short stories, news reports, and other popular media. Course topics typically focus on regional culture, the military, and society.</p> <p><b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b>                      None</p> <p><b>Prerequisite(s):</b>                                LZ371</p>                                                                                                                                                                                                          |                                         | 2026-2 2027-2 2028-2<br>2029-2                    |
| <b>LZ470</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>SPECIAL TOPIC IN PERSIAN</b>         | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-1                                  | <b>Offerings:</b>                                 |

This course is taught by a member of senior faculty and provides cadets an opportunity to further develop their language proficiency, regional expertise, and cultural capabilities. 2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

|              |                                    |                                                   |
|--------------|------------------------------------|---------------------------------------------------|
| <b>LZ475</b> | <b>PERSIAN RDG/WRTG THRU MEDIA</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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Cadets strengthen their reading and writing proficiency through study and discussion derived from contemporary Persian media (e.g., Internet, film, newsprint and magazines) and short literary selections. Reading strategies and textual analysis are introduced and practiced. Writing tasks address organization, substance and grammatical accuracy. Graded work typically includes oral and paragraph-length written summaries of Persian texts and short compositions or reaction papers.

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LZ371  
-Or-  
LN487

|              |                                    |                                                   |
|--------------|------------------------------------|---------------------------------------------------|
| <b>LZ476</b> | <b>MILITARY SPKG/RDG - PERSIAN</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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|---------------|--------|-------------------|
| <b>Scope:</b> | 2025-2 | <b>Offerings:</b> |
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Cadets gain an understanding of the profession of arms in the Persian-speaking world through lectures and selected reading materials (e.g. journal articles, Internet media, training manuals, biographies, and historical documents). Course content may encompass the mission and role, training, operations, tactics, and organization of the armed forces. Oral proficiency is enhanced through in-class discussion as well as role-plays and simulations focusing on scenarios likely to be encountered while an officer is deployed in a Persian-speaking region. Media complement instruction. Graded work may include briefings, role-plays, and simulation. The course is conducted in Persian.

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LZ372

|              |                               |                                                   |
|--------------|-------------------------------|---------------------------------------------------|
| <b>LZ483</b> | <b>PERSIAN CIVILIZATION I</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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This course constitutes an integrated study of the culture, history, and geography of the old Persian Empire from their beginnings in the first millennium BCE to the formation of the modern nation-states of Afghanistan, Iran, and Tajikistan in the 16th century. Readings, lectures, discussions, and audio and visual materials encompass this civilization's representative artistic and intellectual accomplishments, its present-day political institutions, economy, and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the people of Afghanistan, Iran, and Tajikistan. At the same time, cadets continue to develop greater language proficiency. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Persian.

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LZ372

|              |                                |                                                   |
|--------------|--------------------------------|---------------------------------------------------|
| <b>LZ484</b> | <b>PERSIAN CIVILIZATION II</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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|---------------|--------|-------------------|
| <b>Scope:</b> | 2025-2 | <b>Offerings:</b> |
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This course constitutes an integrated study of the culture, history, geography and political systems of the Persian-speaking world. Readings, lectures, discussions and audio-visual materials encompass the civilization's representative artistic and intellectual accomplishments, its present day political institutions, economy and popular culture. In addition, the course focuses on the values and attitudes, the customs and traditions, and the social structures of the people in the Persian-speaking world. At the same time, Cadets continue to develop greater proficiency in the target language. Graded work will include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Persian.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LZ371  
-Or-  
LN487

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|              |                             |                                                          |
|--------------|-----------------------------|----------------------------------------------------------|
| <b>LZ485</b> | <b>PERSIAN LITERATURE I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
|               |        | 2027-1 2028-1     |

This course is a survey of Persian literature that has impacted Iranian politics, cultural discourse, and the collective Persian memory from the Middle Ages through the 19th century. Cadets gain a basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of Iranian society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings, where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Persian.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LZ372

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| <b>LZ486</b> | <b>PERSIAN LITERATURE II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-2 | <b>Offerings:</b> |
|               |        | 2026-2            |

This course is a survey of Persian literature that has impacted Iranian politics, cultural discourse, and the collective Persian memory from the onset of the 20th century to the present day. Cadets gain basic competence in the knowledge and comprehension of representative literary works and their relationship to the cultural context of that society. Selected examples of various literary genres are read, discussed, and analyzed. At the same time, cadets continue to develop greater language proficiency. Video and film presentations supplement readings where possible. Graded work may include giving oral presentations, writing short essays, or preparing a term paper. A majority of the work is done in Persian.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LZ372

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>WR200</b> | <b>DEVELOPMENTAL AND PROF WRITING</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b>   |
|               |        | No Course Offerings |

This remedial course teaches cadets basic critical reading and writing skills necessary to communicate successfully across academic disciplines and as officers within the Army profession. In addition to teaching and reinforcing grammatical correctness, argument organization, and written clarity, WR200 encourages cadets to take ownership of their development as writers and communicators. Through intensive instruction and practical exercise, cadets will hone tools essential to their continued development as writers of clear, logical, correct, and compelling writing.

**Lessons:** 16 @ 55 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                                     |                                                          |
|--------------|-------------------------------------|----------------------------------------------------------|
| <b>WR303</b> | <b>WRITING PROCESS AND PEDAGOGY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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WR303 is the gateway seminar for the WPWP's Writing Fellows Program. The course consists of three main components: seminar-based examinations of significant scholarship in college composition and its pedagogy, individual mentorships with faculty across the disciplines, and teaching practica in the form of peer-to-peer consultations in the Mounger Writing Center. Cadets develop jointly as scholars and teachers by researching the craft of writing and communication in academic as well as professional environments; by contributing originally to high-level scholarly conversations; and above all, by collaborating to develop further resources for thinking, writing, and communicating that benefit all West Pointers.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by invitation only. As part of the course, cadets conduct peer consultations in the Mounger Writing Center.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

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| <b>WR313</b> | <b>ADVANCED WRITING PEDAGOGY</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

WR313 is the advanced seminar for the WPWP's Writing Fellows Program. Similar to the Program's gateway seminar (WR303), the course focuses on seminar-based examinations of significant scholarship in college composition and its pedagogy as well as teaching practica, mainly in the form of peer-to-peer consultations in the Mounger Writing Center. Cadets revise and build upon major assignments in WR303 while also completing independent projects that seek to make permanent contributions to writing and communications education at the Academy.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 26 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Cadets conduct peer-to-peer consultations in the Mounger Writing Center as well as various workshops and briefings pertaining to the larger West Point Writing Program. Cadets must have successfully completed WR303.

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| <b>ZH303</b> | <b>BRITISH LITERATURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.

No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|--------------|----------------------------|----------------------------------------------------------|
| <b>ZH313</b> | <b>AMERICAN LITERATURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.

No Course Offerings

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                             |                                                          |
|--------------|-----------------------------|----------------------------------------------------------|
| <b>ZH323</b> | <b>TOPICS IN LITERATURE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.

No Course Offerings

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                           |                                                          |
|--------------|---------------------------|----------------------------------------------------------|
| <b>ZH333</b> | <b>LITERARY CRITICISM</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

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This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.

No Course Offerings

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:**                      None

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**ZH373****WESTERN ART**

**3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)**

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**Scope:**                                              2025-1

**Offerings:**

This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.

No Course Offerings

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:**                      None

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**ZH383****EASTERN ART**

**3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)**

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**Scope:**                                              2025-1

**Offerings:**

This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.

No Course Offerings

**Lessons:** 40 @ 55 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:**                      None

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# Geography and Earth Sciences

## 41 Courses

| GE203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GEOGRAPHY                                                                                                                                                                                            | 3.0 Credit Hours<br>(BS=2.5,ET=0.0,MA=0.0)                                                   |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2026-1                                                                                                                                                                                               | <b>Offerings:</b>                                                                            |
| <p>This core course provides cadets with a fundamental understanding of scientific principles and processes of earth science, meteorology, climatology, geomorphology and environmental systems, as well as an introduction to cultural geography. Further, the course introduces cadets to technical skills - (terrain analysis, image interpretation and spectral analysis, remote sensing, global positioning system, geographic information systems cartography) - to delineate the geographic distribution of landforms, weather, climate, and culture systems; and evaluate their potential impact on military operations. Lessons are reinforced by use of in- and out-of-class practical exercises, terrain walks and computer exercises to demonstrate the interrelationship between physical and human processes, and their impact on the environment. Historical and contemporary vignettes are employed to demonstrate how weather, climate, terrain, soils, vegetation and culture are important, cogent and frequently decisive in military operations.</p> |                                                                                                                                                                                                      | 2026-1 2026-2 2026-5<br>2027-1 2027-2 2027-5<br>2028-1 2028-2 2028-5<br>2029-1 2029-2 2029-5 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                                                                                                                                                                                 |                                                                                              |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | EV203                                                                                                                                                                                                |                                                                                              |
| GE205                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | FOUNDATIONS IN GEOGRAPHY                                                                                                                                                                             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2026-1                                                                                                                                                                                               | <b>Offerings:</b>                                                                            |
| <p>This course presents the basic concepts, theories and methods of inquiry in the discipline of geography as foundation for advanced study in Human/Regional Geography; Environmental Geography; or Geospatial Information Science. The course includes models and concepts from the many sub-disciplinary (systematic) areas of geography to include cultural, historical, economic, urban, political and military geography. The application of concepts to real-world issues is emphasized. Research skills and techniques used by professional geographers are presented. Cadets use these approaches to spatially analyze and map the distribution of human and environmental phenomena. Several short papers will be assigned.</p>                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                      | 2026-1 2027-1 2028-1<br>2029-1                                                               |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Requires Department Head approval for all cadets not selecting a MAJ in the Department of Geography & Earth Sciences.                                                                                |                                                                                              |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | EV303                                                                                                                                                                                                |                                                                                              |
| GE299A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | INTRO IND STUDY & RESEARCH                                                                                                                                                                           | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2026-1                                                                                                                                                                                               | <b>Offerings:</b>                                                                            |
| <p>The course is an individually supervised research and study program designed to familiarize cadets with introductory scientific procedures, techniques, and topics. The cadet and advisor prepare a research and study proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Activities vary by project, but the primary purpose is to acquaint students with essential skills required for independent research. If required for a specific degree, the proposal will include a justification for engineering science or design credit.</p>                                                                                                                                 |                                                                                                                                                                                                      | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Enrollment by permission of the appropriate GEO program director. Other requirements as determined by the faculty advisor and GE299A course director. Discipline-specific final product is required. |                                                                                              |
| GE300                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GLOBAL ENVIRONMENTAL SYSTEMS                                                                                                                                                                         | 3.0 Credit Hours<br>(BS=3.0,ET=0.0,MA=0.0)                                                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2026-1                                                                                                                                                                                               | <b>Offerings:</b>                                                                            |

GE300 course provides a broad, global perspective on environmental challenges and their impact on humanity and the species we share the planet with. It will examine critical global issues such as resource management, pollution, conservation, corporate social responsibility, community resilience, and innovative solutions for addressing these concerns. Through targeted discussions, debates, case study reviews, scientific literature, short papers, and media, cadets will explore how science, economics, policy, and social and ethical considerations intersect with global development goals. Applying responsible environmental and business practices helps restore ecosystem health, promotes resource stewardship, and enhances societal well-being. This course will begin to develop a comprehensive understanding of the environmental, economic, and social challenges we face with global resources in the coming decades.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

2026-1 2027-1 2028-1  
2029-1

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>GE305</b> | <b>GEOGRAPHIC RESEARCH METHODS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course introduces academic geographic inquiry and the methods, techniques, and ethical considerations needed to effectively design, plan, and conduct geographic research. The course starts with why research is important and the ethics of doing research, which includes an introduction to the institutional review board process. Cadets will then learn how to conduct and write a literature review and develop research questions. Finally, quantitative, qualitative and spatial methods will be introduced. This course is designed to be an applied introduction to geographical research techniques.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** EV203 EV365  
-Or-  
GE203 GE340  
-Or-  
EV203 GE340

**Disqualifier(s):** EV367

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>GE310</b> | <b>GEOG INFORMATION SYSTEMS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course introduces basic theories and technologies of Geographic Information Systems (GIS), including the representation, acquisition, manipulation, analysis, and visualization of geographically related information. Used by environmentalists, engineers, land-use planners, architects, managers of large land holdings, and the military, these highly-intricate "decision support" systems assist managers in answering important "what if" questions. Hands-on experiences of GIS software are emphasized through lab and practical exercise applications.

2026-1 2026-2 2026-9  
2027-1 2027-2 2027-8  
2027-9 2028-1 2028-2  
2028-8 2028-9 2029-1  
2029-2

**Lessons:** 25 @ 75 min (0.000 Att/wk)    **Labs:** 5 @ 75 min

**Special Requirements:** Short oral reports, one database design; compensatory time provided.

**Disqualifier(s):** EV398

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| <b>GE321</b> | <b>CLIMATOLOGY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Climatology investigates the earth's atmospheric phenomena, giving special attention to the dynamic physical processes which produce weather and result in distinctive climates across the planet. A primary focus of the course is to examine how the climate system can impact humans, including an examination of human health, agriculture, and military operations. A similar emphasis is placed on ways in which humans can alter the climate through urbanization, pollution, and increasing greenhouse gas concentrations. Climate change policy and mitigation are also explored through scientific readings, and as differing viewpoints are presented, lively discussion and debate are encouraged. Numerous case studies are offered throughout the course, allowing students to apply climate data and information to problem solving in real-world situations.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

|                         |                                         |
|-------------------------|-----------------------------------------|
| <b>Prerequisite(s):</b> | EV203<br>-Or-<br>GE305<br>-Or-<br>GE203 |
| <b>Disqualifier(s):</b> | EV389B                                  |

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|--------------|--------------------|----------------------------------------------------|
| <b>GE324</b> | <b>METEOROLOGY</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                 | <b>Offerings:</b>                     |
| <p>This course introduces meteorological processes, systems, and patterns with emphasis on spatial distributions. The course begins with a comprehensive look at the structure of the atmosphere to include the energy budget, heat transfer mechanisms, as well as an examination of daily and seasonal patterns of temperature. A thorough look at atmospheric moisture and stability precedes a study of cloud and precipitation processes followed by a study of the atmosphere in motion, namely air pressure, governing forces, winds, small and local-scale wind systems and the general circulation of the planet. Specific phenomena are then examined, including mid-latitude cyclones, thunderstorms/lightning, tornadoes, severe thunderstorms, hurricanes, air pollution, and a brief look at climate and climate change. The end of the course focuses on the art and science of weather forecasting and its applicability to military operations. In-class labs.</p> |                        | 2026-2 2026-8 2027-2<br>2028-2 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min |                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                   |                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | EV203<br>-Or-<br>GE203 |                                       |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | EV387                  |                                       |

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|--------------|------------------------|----------------------------------------------------|
| <b>GE331</b> | <b>AQUATIC SCIENCE</b> | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                              | <b>Offerings:</b>                            |
| <p>This course provides disciplinary depth in the science of oceans, estuaries, lakes, rivers, and water ecosystems through the study of physical, chemical and biological principles related to marine and freshwater biomes. Communities of marine and freshwater organisms at various ecological zones are explored, as they are affected by light, nutrients, water chemistry, and other physical and chemical properties. The impacts of humans on these water ecosystems are also evaluated. The course provides the student with a strong foundation in the science of the hydrosphere while introducing students to environmental science lab and field research methods.</p> |                                                                                                                                     | 2026-1 2027-1 2028-1<br>2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | In-class labs and field trips; term project examining aspects of one of the world's aquatic ecosystems. Compensatory time provided. |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EV301 GE203<br>-Or-<br>EV203 EV301<br>-Or-<br>GE203 GE300<br>-Or-<br>EV203 GE300                                                    |                                              |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EV310                                                                                                                               |                                              |

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|--------------|----------------------|----------------------------------------------------|
| <b>GE332</b> | <b>GEOMORPHOLOGY</b> | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-1                                                                               | <b>Offerings:</b>    |
| <p>This course studies the processes that create landforms on the surface of the earth and their regional and global distributions. The course focuses on processes and their inter-relationships with geologic structure, soils and climate. Processes emphasized include glaciers, streams, downslope motion caused by gravity, groundwater, coastlines, and eolian landscapes. Each student prepares a final report synthesizing these processes and how they relate to real-world applications.</p> |                                                                                      | 2026-2 2027-2 2028-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 0 @ 0 min                                                               |                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Two field trips; one written report and one oral report; compensatory time provided. |                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | EV203<br>-Or-<br>GE203                                                               |                      |

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| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EV388B                                                                                                                              |                                                                                                                                |
| <b>GE333</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>PHYSICAL GEOLOGY</b>                                                                                                             | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b>                                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                              | <b>Offerings:</b>                                                                                                              |
| <p>This course primarily emphasizes learning to identify minerals and rocks and then applying this knowledge to analyze the significant geologic processes that act on and within the Earth. These processes include plate tectonics, rock mechanics, structural geology, geologic mapping, ground and surface water, deep time, and elements of mining and petroleum engineering. Field trips are conducted to illustrate how local geology influenced development and construction in the Hudson Valley. The course is capstoned by an open-ended geologic mapping project which requires the creative application of geology to propose a practical solution to a stated need. Cadets use a geologic exploration simulation to convert given resources into a product containing elements that professional geologists would create.</p> |                                                                                                                                     | 2026-1 2026-8 2027-1<br>2029-2                                                                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 27 @ 75 min (2.500 Att/wk)                                                                                                          | <b>Labs:</b> 13 @ 75 min                                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | One all-day field trip; one term project.                                                                                           |                                                                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EV203<br>-Or-<br>GE203                                                                                                              |                                                                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EV388A                                                                                                                              |                                                                                                                                |
| <b>GE333A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>GEOLOGY FIELD COURSE</b>                                                                                                         | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b>                                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                              | <b>Offerings:</b>                                                                                                              |
| <p>The geology field course is a three-week long summer Individual Advanced Development Program. It is taught at a variety of field locations as funding and travel opportunities allow, but always directly driven by the location's link to course concepts. Geologic concepts are initially presented in a classroom setting and supplemented with laboratory exercises. The majority of the course, however, is conducted at geologic sites in the field where concepts are illustrated and expanded. The course provides the cadet with knowledge of and appreciation for the science of geology as well as practical experience in geological mapping. Field trips to active mines and active geologic research locations relate classroom learning to the real world.</p>                                                            |                                                                                                                                     | 2026-7 2027-7 2028-7<br>2029-7                                                                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0 @ 0 min (0.000 Att/wk)                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | TDY travel to the course location; excursions to remote field locations; two graded field mapping exercises; graded field notebook. |                                                                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EV203<br>-Or-<br>GE203                                                                                                              |                                                                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EV399A                                                                                                                              |                                                                                                                                |
| <b>GE340</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>GEOGRAPHY OF GLOBAL CULTURES</b>                                                                                                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                              | <b>Offerings:</b>                                                                                                              |
| <p>This course provides the geographic foundation for study in interdisciplinary and management academic areas. Contemporary regions of the world political map serve as the framework within which geographic concepts and analytical techniques are applied. Each cadet will develop an awareness of the diversity and distribution of people on the earth, human organization and exploitation of territory, and interactions among culture groups. Particular emphasis is placed on social institutions, their impact on economic development, and the subsequent identification and analysis of developed, emerging, and underdeveloped states.</p>                                                                                                                                                                                    |                                                                                                                                     | 2026-1 2026-2 2026-8<br>2026-9 2027-1 2027-2<br>2027-8 2027-9 2028-1<br>2028-2 2028-8 2028-9<br>2029-1 2029-2 2029-8<br>2029-9 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 30 @ 75 min (2.000 Att/wk)                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                                                |                                                                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GE203<br>-Or-<br>EV203                                                                                                              |                                                                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | EV365                                                                                                                               |                                                                                                                                |
| <b>GE341</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>GEOGRAPHY OF NORTH AMERICA</b>                                                                                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2026-1                                                                                                                              | <b>Offerings:</b>                                                                                                              |

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| <p>This course provides a regional geography of North America, with balanced coverage of the human and physical geography of the United States and Canada. Lectures are appropriately supplemented with movies, slides, and maps to facilitate understanding of important themes that are prevalent in various subregions. Emphasis is placed on cultural patterns and contemporary environmental issues.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> One oral report.</p> <p><b>Disqualifier(s):</b> EV384</p>                                                                                                                                                                                                                                   |                                     | 2027-1 2028-1 2029-1                                                                         |
| <b>GE342</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>GEOGRAPHY OF ASIA</b>            | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-1                              | <b>Offerings:</b>                                                                            |
| <p>The course studies the physical and cultural environment of Asia with emphasis on those geographic elements related to the region's progress, developing nations, and emerging world and regional powers. Topics covered include a consideration of the physical and resource base, environmental and cultural factors, spatial organization of agricultural and industrial economies, population patterns and problems, and examination of the realm's several major subregions.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> 1 field trip; one written report and one oral presentation.</p> <p><b>Disqualifier(s):</b> EV372</p>                                                                                                             |                                     | 2026-2 2026-8 2027-2<br>2028-2 2029-2                                                        |
| <b>GE343</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>GEOGRAPHY OF RUSSIA</b>          | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-1                              | <b>Offerings:</b>                                                                            |
| <p>This course examines the political, economic, and cultural geography of Russia and its adjacent neighbors; the Baltic states, east central European region, transcaucasus, and central Asia. Topics covered include: the commonwealth of independent states; ecocide in the former Soviet Union; disposition of the former soviet military; and ethnic rivalries. The objective of the course is to provide the student with an understanding of the recent past of the traditional soviet system in order to understand, as well as geographically evaluate, Russia's and the other former republics' situation today.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> 1 field trip; one research paper.</p> <p><b>Disqualifier(s):</b> EV371</p> |                                     | 2026-1 2026-8 2026-9<br>2027-1 2027-8 2027-9<br>2028-1 2028-8 2028-9<br>2029-1 2029-8 2029-9 |
| <b>GE344</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>GEOGRAPHY OF AFRICA</b>          | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-1                              | <b>Offerings:</b>                                                                            |
| <p>This course examines the cultural and natural diversity of African landscapes, with an emphasis on development, population issues, disease, and the origin, dispersal, spatial organization, and interaction of important cultural groups. Africa's physical landscapes will also be introduced as the palette upon which Africa's complex human mosaic has developed. Students will explore, from a geographic perspective, why Africa has seemingly been plagued with problems of economic development, health, and political instability.</p> <p><b>Lessons:</b> 15 @ 140 min (1.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> One written research report with brief oral presentation. One field trip is possible.</p> <p><b>Disqualifier(s):</b> EV375</p>                       |                                     | 2026-2 2027-2 2027-9<br>2028-2 2028-9 2029-2<br>2029-9                                       |
| <b>GE345</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>GEOGRAPHY OF THE MIDDLE EAST</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-1                              | <b>Offerings:</b>                                                                            |
| <p>This course examines the cultural and natural diversity of Southwest Asian landscapes. The realm's cultures and ethnicities are studied in a geographic context, with an emphasis on the origin, dispersal, spatial organization, and interaction of important cultural groups. Among issues examined are the distribution and strategic significance of critical mineral and energy resources, population and resource disparities, cultural conflict, and economic development. Students will learn how geographic issues impact the prospects for peace and stability in the region.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p>                                                                                                                                            |                                     | 2026-1 2026-8 2026-9<br>2027-1 2027-8 2027-9<br>2028-1 2028-8 2028-9<br>2029-1 2029-8 2029-9 |

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| <b>Special Requirements:</b> | One written research report with brief oral presentation. One field trip is possible. |
| <b>Disqualifier(s):</b>      | EV376                                                                                 |

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| <b>GE346</b> | <b>GEOGRAPHY OF EUROPE</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1 | <b>Offerings:</b>                                                                            |
| The course examines European cultural landscapes, focusing on the environmental and cultural diversity exhibited among the states of modern Europe. Nationalism and the territorial imperative, long recognized as major forces in Europe, are studied from a geographic perspective to include patterns and processes of both regional continuity and change. Emphasis is given to the rapidly developing urbanization and mutual interdependence among countries of Western Europe. West and East European agricultural/industrial resource bases and developmental strategies are compared and contrasted. Specific topics are tailored to current issues and include regional conflict, economic development and trade, and problems of energy and the environment. This course concludes with a study of contemporary European extraregional spatial relationships with other major world culture regions. |        | 2026-2 2026-8 2026-9<br>2027-2 2027-8 2027-9<br>2028-2 2028-8 2028-9<br>2029-2 2029-8 2029-9 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One field trip; one research paper.

**Disqualifier(s):** EV386

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| <b>GE347</b> | <b>GEOGRAPHY OF LATIN AMERICA</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1 | <b>Offerings:</b>                                                                            |
| This course studies the physical and cultural landscape of Latin America, giving special treatment to the diversity and cultural identity of the region. Topics covered include an historical geography of the region, including pre-columbian civilizations, Iberian, African, and European influences; the geography of transportation networks, agriculture, urbanization, and population. National boundaries, major landforms and climatic conditions are discussed to describe their effect on civilization. This course also investigates the historical relationship between the United States and Latin America and covers recent U.S. military interventions in the region. |        | 2026-1 2026-8 2026-9<br>2027-1 2027-8 2027-9<br>2028-1 2028-8 2028-9<br>2029-1 2029-8 2029-9 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One oral report; one research paper.

**Disqualifier(s):** EV373

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| <b>GE351</b> | <b>LAND USE PLAN &amp; MGT</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.5,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1 | <b>Offerings:</b>              |
| An introduction to land use planning and management with focus on the land-law interfaces between the physical, cultural, and legal realms. The course surveys the policies and legislative basis for land use controls at the local, federal and regional levels to include national parks and forests, agricultural lands, rangelands, and military training areas. Natural resource management issues and strategies are explored. The importance of geographic concepts is emphasized in the conduct of applied case studies addressing land use conflicts and environmental strategies. |        | 2026-1 2027-1 2028-1<br>2029-1 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One field trip; one oral presentation; compensatory time provided.

**Prerequisite(s):** EV203  
-Or-  
GE203

**Disqualifier(s):** EV391A

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| <b>GE352</b> | <b>URBAN GEOGRAPHY</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2026-1 | <b>Offerings:</b>                     |
| This course examines the location, function, structure, growth and interaction of urban areas. Spatial techniques are used to explore the internal attributes of cities, as well as their connectivity to other places. While the primary focus is on urbanization in the United States, primate cities abroad are often used for comparative purposes. Emphasis is placed on contemporary urban problems, particularly environmental issues and social disparities. |        | 2026-2 2026-8 2027-2<br>2028-2 2029-2 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min



**Special Requirements:** One oral report.

**Disqualifier(s):** EV390B

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>GE354</b> | <b>NATURAL HAZARDS AND RISK</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course focuses on natural phenomena that pose hazards to people. The cause, nature, and occurrence frequency of natural hazards such as flooding, earthquakes, hurricanes, and volcanic activity will be examined. Emphasis will also be placed on how people perceive and respond to these hazards. Land use policies and practices in these hazard areas will also receive attention. Students participate in map based laboratory exercises and have the opportunity to write a short paper advising a government official how to mitigate local geohazards.

2026-2 2026-8 2026-9  
2027-2 2027-8 2027-9  
2028-2 2028-8 2028-9  
2029-2 2029-8 2029-9

**Lessons:** 37 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One research paper; compensatory time provided.

**Prerequisite(s):** EV203  
-Or-  
GE203

**Disqualifier(s):** EV391B

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|--------------|-----------------------|----------------------------------------------------------|
| <b>GE373</b> | <b>REMOTE SENSING</b> | <b>3.0 Credit Hours</b><br><b>(BS=1.0,ET=2.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Remote Sensing is collecting information about something without being in physical contact with it. In this course, students will learn remote sensing principles that are applicable across engineering and humanities disciplines. The focus of the course is the electro-magnetic spectrum and how it is leveraged to collect information about a surface. Through lectures and practical exercises students will become familiar with different remote sensing applications, technologies, and the physical basis of satellite and aerial remote sensing. Pertinent to this study is a basic understanding of orbital mechanics, data analysis, and sensor design. The course will help students apply remotely sensed data to solve current problems.

2026-1 2026-8 2027-1  
2028-1 2029-1

**Lessons:** 21 @ 75 min (2.000 Att/wk)    **Labs:** 9 @ 75 min

**Special Requirements:** None

**Prerequisite(s):** EV203 PH201  
-Or-  
EV203 PH251  
-Or-  
GE203 PH201  
-Or-  
GE203 PH251

**Disqualifier(s):** EV377

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|--------------|--------------------|----------------------------------------------------------|
| <b>GE378</b> | <b>CARTOGRAPHY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Cartography covers theoretical concepts and practical aspects for visualizing and analyzing digital spatial data. The course covers the principles of cartographic communication and enables the student to apply map design principles along with computer mapping techniques to solve contemporary problems in geography, economics, international relations, and applied sciences. A final course design project requires the student to utilize various online resources to obtain open-source geodata and demonstrate their ability to synthesize sound mapping principles.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 21 @ 75 min (2.000 Att/wk)    **Labs:** 9 @ 75 min

**Special Requirements:** Lessons dedicated to Course design project work and presentation.

**Disqualifier(s):** EV378

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|--------------|-----------------------|----------------------------------------------------------|
| <b>GE379</b> | <b>PHOTOGRAMMETRY</b> | <b>3.0 Credit Hours</b><br><b>(BS=3.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Photogrammetry relates the principles of photography, precise image measurement, and computation for the development of accurate spatial information. Photogrammetry is heavily utilized in the public, private, and defense sectors for the creation of high-resolution maps, three-dimensional (3D) models, and for detailed geometric analyses (e.g., change detection associated with imaged objects or the surface of the Earth). This course covers the fundamentals of optics, photography, image measurement, coordinate transformations, calculating geometry from vertical and tilted aerial imagery, stereoscopic visualization, photogrammetric control and flight mission planning, and the development of geospatial products such as digital surface models (DSM), 3D point clouds, and orthorectified basemaps. Laboratory exercises and a final project provide practical hands-on experience with topics ranging from photogrammetric analysis of historical film imagery to the creation of digital 3D point clouds and surfaces using unmanned aircraft systems (UAS) and modern digital cameras.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 24 @ 75 min (2.000 Att/wk)      **Labs:** 6 @ 75 min

**Special Requirements:** None

**Prerequisite(s):** CY105 EV203  
-Or-  
CY155 EV203  
-Or-  
CY105 GE203  
-Or-  
CY155 GE203  
-Or-  
AI105 GE203  
-Or-  
AI105X GE203  
-Or-  
AI155 GE203

**Disqualifier(s):** EV379

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| <b>GE380</b> | <b>SURVEYING</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.5,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

A framework for understanding and applying practical surveying methods is developed. Consideration of error theory and the concepts of precision and accuracy yields understanding of the probabilistic nature of measurements. The principles of differential leveling, electronic distance measurement, angular measurement, global navigation satellite system (GNSS) positioning, and terrestrial laser scanning (a.k.a. Lidar) are studied and applied using state-of-the-art surveying equipment and software tools. Plane surveys are principally explored, although the fundamentals of geodetic surveys are also presented. Traverse, triangulation, trilateration, level networks and the proper adjustment of related measurements are examined. Control survey, land survey, topographic survey, horizontal and vertical curve design, and Geographic Information System (GIS) software applications are included. Extensive use of laboratory periods permits the application of surveying fundamentals, methods and planning skills to actual field situations.

2026-1 2026-9 2027-1  
2027-8 2027-9 2028-1  
2028-8 2028-9 2029-1  
2029-8 2029-9

**Lessons:** 21 @ 55 min (2.500 Att/wk)      **Labs:** 19 @ 120 min

**Special Requirements:** None

**Disqualifier(s):** EV380

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|---------------|-----------------------------------------|----------------------------------------------------------|
| <b>GE399A</b> | <b>INTER IND STUDY &amp; RESEARCH I</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The course is an individually supervised research and study program designed to familiarize cadets with advanced scientific procedures, techniques, and topics. The cadet and advisor prepare a research and study proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Activities vary by project, but the primary purpose is to acquaint students with essential skills required for independent research. If required for a specific degree, the proposal will include a justification for engineering science or design credit.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of the appropriate GEO program director. Other requirements as determined by the faculty advisor and GE399A course director. Discipline-specific final product is required.

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| <b>GE399B</b> | <b>INTER IND STUDY &amp; RESEARCH II</b> | <b>2.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                               | <b>Offerings:</b>                                             |
| <p>The course is an individually supervised research and study program designed to familiarize cadets with advanced scientific procedures, techniques, and topics. The cadet and advisor prepare a research and study proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Activities vary by project, but the primary purpose is to acquaint students with essential skills required for independent research. If required for a specific degree, the proposal will include a justification for engineering science or design credit.</p> |                                                                                                                                                                                                      | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Enrollment by permission of the appropriate GEO program director. Other requirements as determined by the faculty advisor and GE399B course director. Discipline-specific final product is required. |                                                               |
| <b>GE431</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>ECOLOGY</b>                                                                                                                                                                                       | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                               | <b>Offerings:</b>                                             |
| <p>This course examines ecosystems through the study of ecological principles related to an organism's relationship to its environment, community, and ecosystem. Species, population, community, and ecosystem level interactions and dynamics are emphasized. The fundamental influences of energy flow and material cycling are examined, as well as the unique role of wetlands within ecosystems. The course includes several field trips, which lead to a culminating term project designed to integrate previously acquired environmental science technical skills and ecological principles.</p>                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                      | 2026-1 2027-1 2028-1<br>2029-2                                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | In-class labs and out-of-class field trips; term paper examining aspects of one of the world's ecosystems.                                                                                           |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | CH275 EV310 EV350<br>-Or-<br>CH375 EV310 EV350<br>-Or-<br>CH275 EV350 GE331<br>-Or-<br>CH375 EV350 GE331                                                                                             |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EV471                                                                                                                                                                                                |                                                               |
| <b>GE462</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>ENVIRONMENT AND DEVELOPMENT</b>                                                                                                                                                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                               | <b>Offerings:</b>                                             |
| <p>This course looks at how the environment and development are closely connected. It explores the ways environmental conditions shape development choices and how development efforts, in turn, affect the environment. While physical geographers tend to focus on natural systems and processes, and human geographers study how people use space, this course brings those approaches together. It emphasizes how communities respond to environmental limits and how their development decisions can change landscapes and ecosystems. To make sense of these relationships, students will study natural processes like land formation and climate, as well as the social, economic, and political forces that drive development.</p>                                                                                                                                                                                            |                                                                                                                                                                                                      | 2026-2 2028-2 2029-2                                          |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | None                                                                                                                                                                                                 |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EV203 EV365<br>-Or-<br>GE203 GE340<br>-Or-<br>EV203 GE340                                                                                                                                            |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EV486                                                                                                                                                                                                |                                                               |
| <b>GE477</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>ADVANCED REMOTE SENSING</b>                                                                                                                                                                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                               | <b>Offerings:</b>                                             |

This course examines advanced remote sensing theory and digital image processing techniques suitable for the processing of remotely sensed data. Emphasis is on the processing and analysis of state-of-the-art high spatial and spectral resolution data gathered by airborne and satellite sensors. Topics covered include geometric and radiometric image rectification, registration and resampling techniques, image enhancements, data merging, image segmentation, and automated feature extraction. A wide range of practical exercises and in-class laboratory assignments provides hands-on experience with a variety of remotely sensed imagery ranging from multi-spectral to hyper-spectral data. The course culminates with a capstone term project that allows cadets to apply digital image processing skills to a scientific problem.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** In-class labs; term project. Compensatory time provided.

**Prerequisite(s):** EV203 EV377  
-Or-  
GE203 GE373  
-Or-  
EV203 GE373

**Disqualifier(s):** EV477

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>GE478</b> | <b>MILITARY GEOSPATIAL OPERATIONS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course is designed to teach the most current state of geospatial operations in the military. It is built to provide the student an improved understanding of the cornerstone to the digital force - the "common operational picture" or COP. This course is divided into five major blocks of instruction: (1) a linked discussion of geospatial operations' development, organizations and data systems; (2) the geographic information system (GIS) as a military tool - system input, management, data analysis and production outputs; (3) Army geospatial operations in the garrison environment; (4) Army geospatial operations in combat environments; and (5) geospatial operations for joint/coalition forces. The course includes several relevant practical exercises and laboratories, a field trip, guest lectures and one panel discussion. Due to the currency of the material discussed a secret security clearance is required for all participants.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** EV203  
-Or-  
GE203

**Disqualifier(s):** EV478

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|--------------|----------------------------------------|----------------------------------------------------------|
| <b>GE490</b> | <b>SPEC TOPICS-GEOG &amp; ENVRNMNT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course explores an advanced topic in Human and Regional Geography, Environmental Geography, Environmental Science, or Geospatial Science. Specific subject matter will vary with the expertise of the visiting professor or senior faculty member conducting the course.

2026-2

**Lessons:** 15 @ 140 min (1.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** As specified by the professor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**Prerequisite(s):** EV203  
-Or-  
GE203

**Disqualifier(s):** EV485

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|--------------|---------------------------|----------------------------------------------------------|
| <b>GE491</b> | <b>MILITARY GEOGRAPHY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

History is replete with examples of the impact of terrain, weather and climate on military operations at all scales. National strategies are influenced heavily by geographic realities of relative location, spatial interaction, population dynamics and resource distribution. This course emphasizes the development of a geographic method for systematic analysis of the battlefield that is appropriate for platoon leader and corps commander alike. Students evaluate the elements of national power and examine their geostrategic influences, past and present. The role of the environment in shaping today's Army and its missions is discussed. Jungle, cold region, alpine, riverine, desert, temperate and urban operational environments are examined for their effect on military planning and execution. Finally, cadets review case studies of the impact of these diverse environments on military operations at the tactical level.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** One oral presentation and one written research project.

**Prerequisite(s):** EV203  
-Or-  
GE203

**Disqualifier(s):** EV482

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|--------------|-------------------------------|----------------------------------------------------------|
| <b>GE494</b> | <b>ENVIRONMENTAL SECURITY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This interdisciplinary seminar uses Environmental Security in a case study approach to analyze how environmental issues affect U.S. National Security. Cadets explore environmental security topics such as water, natural resource shortages, energy use and dependency, and global climate change using an interdisciplinary approach from social, political, economic, and scientific-technological perspectives. The course culminates in a student team analysis of a developing country relating environmental security issues to U.S. National Security interests. The final project includes a formal brief and written paper.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Standing as a first class cadet required for enrollment.

**Disqualifier(s):** EV487

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>GE495</b> | <b>HONORS SEMINAR IN GEOGRAPHY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course will examine major research initiatives in the discipline and delineate their data requirements. The primary objective of this course is to identify and outline the senior thesis, which is the culminating event for the Honors Program. Hence, cadets participating in this course will explore research methods and data sources used by geographers, conduct a critical analysis of seminal literature in the field, define a research problem, identify and evaluate data sources, and assemble a research proposal. The final product of this course will be a written research proposal that will define the senior thesis (written during GE499A). The cadet will make a formal presentation of this proposal to senior geography faculty. The course is conducted in a seminar and one-and-one format. Lessons and labs are established by consultation between the cadet and faculty advisor.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Senior Thesis or as determined by the faculty advisor.

**Prerequisite(s):** EV203  
-Or-  
GE203

**Disqualifier(s):** EV480

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>GE496</b> | <b>COLLOQUIUM IN GEOGRAPHY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The colloquium is a directed readings course using small group discussions of important literature, methodological traditions, and contemporary research trends in the field of geography. Dependent on instructor preference and individual student interest, in-depth readings will be pursued in one or more of the following areas of geographic study: cultural, political, regional or military geography. Compensatory time is given to permit extra readings.

2026-2 2026-8 2027-2  
2027-8 2027-9 2028-2  
2028-8 2028-9 2029-2  
2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A research proposal and its oral presentation.

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| <b>Prerequisite(s):</b> | EV203 EV365<br>-Or-<br>GE203 GE340<br>-Or-<br>EV203 GE340 |
| <b>Disqualifier(s):</b> | EV483                                                     |

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|--------------|---------------------------------------|----------------------------------------------------|
| <b>GE498</b> | <b>ADV GEOGRAPHIC INFORMATION SYS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                    | <b>Offerings:</b>                     |
| <p>This course examines the analytical methods used in Geographic Systems (GS) and provides cadets with a clear understanding of the theoretical/conceptual aspects of algorithms found in GS software. Lectures focus on the underlying mathematical basis for widely used spatial analytical techniques. Among the topics covered are neighborhood operations, map transformation, spatial interpolation, terrain analysis, network analysis, spatial overlay, fuzzy sets, neural networks, and expert systems. In-class practical exercises and laboratory assignments compliment the lectures by providing hands-on experience with a variety of advanced analytical techniques. The course culminates with a capstone term project that allows cadets to identify a scientific problem, formulate a hypothesis, use GS to solve the problem, and then present the results of their analysis.</p> |                                           | 2026-1 2026-8 2027-1<br>2028-1 2029-1 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 30 @ 75 min (2.000 Att/wk)                | <b>Labs:</b> 0 @ 0 min                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Term project. Compensatory time provided. |                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EV398<br>-Or-<br>GE310                    |                                       |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EV498                                     |                                       |

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|---------------|------------------------------------|----------------------------------------------------|
| <b>GE499A</b> | <b>ADVANCED INDIVIDUAL STUDY I</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|------------------------------------|----------------------------------------------------|

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                                                                                                                                                                                                              | <b>Offerings:</b>                                             |
| <p>The course is an individually supervised research and study program designed to provide cadets the opportunity to pursue advanced scientific procedures, techniques, and/or topics within their discipline. The cadet and faculty advisor prepare a research proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Lessons and labs are established by consultation between the cadet and faculty advisor. The program for each cadet will culminate in one of two outcomes: 1) discipline-appropriate final written report (e.g., senior thesis) with presentation in a public forum (e.g., Projects Day); or 2) enrollment in GE499B for continued research and independent study at a 3.0-credit level. If required for a specific degree, the proposal will include a justification for engineering science or design credit. Program Director will determine the applicable MA, BS, and ET credits for each offering, as required.</p> |                                                                                                                                                                                                                                     | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Enrollment by permission of the appropriate GEO program director.<br/>Discipline-appropriate final written report with presentation in a public forum required prior to completion of a cadet's culminating research course.</p> |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | EV489A                                                                                                                                                                                                                              |                                                               |

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|---------------|-------------------------------------|----------------------------------------------------|
| <b>GE499B</b> | <b>ADVANCED INDIVIDUAL STUDY II</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                   | <b>Offerings:</b>                                             |
| <p>The course is an individually supervised research and study program designed to provide cadets the opportunity to pursue advanced scientific procedures, techniques, and/or topics within their discipline. The cadet and faculty advisor prepare a research proposal setting forth the objectives, scope, and anticipated accomplishments of his/her efforts for the semester. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Once approved, the proposal serves as a basis for the cadet's research and study program. Progress in research reports and observations by the faculty advisor form the basis for grades. Lessons and labs are established by consultation between the cadet and faculty advisor. The program for each cadet will culminate in one of two outcomes: 1) discipline-appropriate final written report (e.g., senior thesis) with presentation in a public forum (e.g., Projects Day); or 2) enrollment in EV489C for continued research and independent study at a 3.0-credit level. If required for a specific degree, the proposal will include a justification for engineering science or design credit. Program Director will determine the applicable MA, BS, and ET credits for each offering, as required.</p> |                          | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 @ 0 min (0.000 Att/wk) | <b>Labs:</b> 0 @ 0 min                                        |

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| <b>Special Requirements:</b> | Enrollment by permission of the appropriate GEO program director.<br>Discipline-appropriate final written report with presentation in a public forum<br>required prior to completion of a cadet's culminating research course. |
| <b>Prerequisite(s):</b>      | EV480<br>-Or-<br>EV489A<br>-Or-<br>GE499A                                                                                                                                                                                      |
| <b>Disqualifier(s):</b>      | EV489B                                                                                                                                                                                                                         |

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## Geography and Environmental Engineering

### 6 Courses

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| <b>EV365</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>GEOGRAPHY OF GLOBAL CULTURES</b>                                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2019-2                                                                                | <b>Offerings:</b>                                  |
| <p>This course provides the geographic foundation for study in interdisciplinary and management academic areas. Contemporary regions of the world political map serve as the framework within which geographic concepts and analytical techniques are applied. Each cadet will develop an awareness of the diversity and distribution of people on the earth, human organization and exploitation of territory, and interactions among culture groups. Particular emphasis is placed on social institutions, their impact on economic development, and the subsequent identification and analysis of developed, emerging, and underdeveloped states.</p>                                     |                                                                                       | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | EV203<br>-Or-<br>EV203X                                                               |                                                    |
| <b>EV373</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>GEOGRAPHY OF LATIN AMERICA</b>                                                     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2023-1                                                                                | <b>Offerings:</b>                                  |
| <p>This course studies the physical and cultural landscape of Latin America, giving special treatment to the diversity and cultural identity of the region. Topics covered include an historical geography of the region, including pre-columbian civilizations, Iberian, African, and European influences; the geography of transportation networks, agriculture, urbanization, and population. National boundaries, major landforms and climatic conditions are discussed to describe their effect on civilization. This course also investigates the historical relationship between the United States and Latin America and covers recent U.S. military interventions in the region.</p> |                                                                                       | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | One oral report; one research paper.                                                  |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | EV373A                                                                                |                                                    |
| <b>EV375</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>GEOGRAPHY OF AFRICA</b>                                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2024-2                                                                                | <b>Offerings:</b>                                  |
| <p>This course examines the cultural and natural diversity of African landscapes, with an emphasis on development, population issues, disease, and the origin, dispersal, spatial organization, and interaction of important cultural groups. Africa's physical landscapes will also be introduced as the palette upon which Africa's complex human mosaic has developed. Students will explore, from a geographic perspective, why Africa has seemingly been plagued with problems of economic development, health, and political instability.</p>                                                                                                                                          |                                                                                       | No Course Offerings                                |
| <b>Lessons:</b> 15 @ 140 min (1.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | One written research report with brief oral presentation. One field trip is possible. |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | EV374                                                                                 |                                                    |
| <b>EV376</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>GEOGRAPHY OF THE MIDDLE EAST</b>                                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2019-2                                                                                | <b>Offerings:</b>                                  |
| <p>This course examines the cultural and natural diversity of Southwest Asian landscapes. The realm's cultures and ethnicities are studied in a geographic context, with an emphasis on the origin, dispersal, spatial organization, and interaction of important cultural groups. Among issues examined are the distribution and strategic significance of critical mineral and energy resources, population and resource disparities, cultural conflict, and economic development. Students will learn how geographic issues impact the prospects for peace and stability in the region.</p>                                                                                               |                                                                                       | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                |                                                    |



|                              |                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------|
| <b>Special Requirements:</b> | One written research report with brief oral presentation. One field trip is possible. |
| <b>Disqualifier(s):</b>      | EV374                                                                                 |

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|--------------|----------------------------|----------------------------------------------------|
| <b>EV386</b> | <b>GEOGRAPHY OF EUROPE</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|----------------------------|----------------------------------------------------|

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2019-2 | <b>Offerings:</b>   |
| The course examines European cultural landscapes, focusing on the environmental and cultural diversity exhibited among the states of modern Europe. Nationalism and the territorial imperative, long recognized as major forces in Europe, are studied from a geographic perspective to include patterns and processes of both regional continuity and change. Emphasis is given to the rapidly developing urbanization and mutual interdependence among countries of Western Europe. West and East European agricultural/industrial resource bases and developmental strategies are compared and contrasted. Specific topics are tailored to current issues and include regional conflict, economic development and trade, and problems of energy and the environment. This course concludes with a study of contemporary European extraregional spatial relationships with other major world culture regions. |        | No Course Offerings |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One field trip; one research paper.

**Disqualifier(s):** EV386F

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|--------------|----------------------------------------|----------------------------------------------------|
| <b>EV485</b> | <b>SPEC TOPICS-GEOG &amp; ENVRNMNT</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|----------------------------------------|----------------------------------------------------|

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                        | 2024-1 | <b>Offerings:</b>   |
| This course explores an advanced topic in Human and Regional Geography, Environmental Geography, Environmental Science, Environmental Engineering, or Geospatial Information Science. Specific subject matter will vary with the expertise of the visiting professor or senior faculty member conducting the course. |        | No Course Offerings |

**Lessons:** 15 @ 140 min (1.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** As specified by the professor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**Prerequisite(s):** EV203  
-Or-  
EV203X

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## History

### 4 Courses

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                   |                                                    |
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| <b>HI367</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>IMPERIAL AND SOVIET RUSSIA</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2020-2                            | <b>Offerings:</b>                                  |
| <p>This course examines the political, social, and cultural history of Russia as it emerged from the Mongol era up to the present day. It explores the development of the Tsarist political and social systems, the emergence of literary, artistic, and revolutionary movements, and the development of Russia's position in European politics from the time of Peter I through WWI. It also covers the rise of the Soviet Union, the leadership's attempts to implement communist ideology and responses to that attempt, Russia's relationship with various national and ethnic groups, and the emergence of the Soviet Union as a superpower. The course concludes with the collapse of the Soviet Union and the emergence of new states in the 1990s.</p> |                                   | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min            |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | A 1500-word research paper.       |                                                    |
| <b>ZH315</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>MODERN REGIONAL HISTORY</b>    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2010-1                            | <b>Offerings:</b>                                  |
| <p>For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce a historical research paper to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering modern historical developments of the area/region where the cadet is studying. This course covers broad historical processes and developments of the region over a long period of time. The course effectively encompasses a recognized historical era, for example "Modern", "Early Modern", or "Ancient".</p>                                                                                                                   |                                   | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min            |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                              |                                                    |
| <b>ZH325</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>TOPICS IN REGIONAL HISTORY</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2010-1                            | <b>Offerings:</b>                                  |
| <p>For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce written historical submissions to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering special topics in the area/region where the cadet is studying. Topics include (but are not limited to) specific eras of history; cultural &amp; ethnic aspects of history; art &amp; literary history; and other scientific &amp; technical history topics.</p>                                                                                                                                                                |                                   | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min            |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                              |                                                    |
| <b>ZH365</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>POLITICS AND DIPLOMACY</b>     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2010-1                            | <b>Offerings:</b>                                  |
| <p>For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce a historical research paper to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering special political and diplomatic history topics in the area/region where the cadet is studying. Topics include internal political development and/or international relations and diplomatic history.</p>                                                                                                                                                                                                                          |                                   | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min            |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                              |                                                    |

# History and War Studies

## 107 Courses

| HI105                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | HISTORY OF THE UNITED STATES                                                                               | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                                                                                                     | <b>Offerings:</b>                                             |
| History of the United States addresses the social, political, economic, foreign relations, and sectional developments of the nation from its colonial roots through the 21st century. The course consists of two blocks of instruction, each followed by a major examination. This course will help cadets know themselves and begin to develop strategic competence. The course also develops methods of historical research and analysis, critical thinking, lucid writing, and effective participation in classroom discussion.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                            | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                                                                       |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | HI155                                                                                                      |                                                               |
| HI108L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | RGNL STUD WORLD HIST - L. AMER                                                                             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                                                                                                     | <b>Offerings:</b>                                             |
| HI108X, Regional Studies in World History, is a detailed study of the development and critical events in the history of one of six regions, each with its own letter designation. As the stems are all a part of one course, cadets may not take multiple stems as electives. The focus on one region enables cadets to develop a deeper understanding of a different culture and unfamiliar ideas and concepts. The course also develops methods of historical research and analysis, critical thinking, lucid writing, and effective participation in classroom discussion. HI108L combines with HI105 or HI155 to form the plebe history sequence of the Core Academic Program. The Latin America stem (identifier L) challenges cadets to study how social, political, economic, technological, and military factors have influenced international and multicultural relations and how these relations have influenced the development of specific societies, polities, cultures, economies, technologies, military systems, and gender roles throughout Latin America.                   |                                                                                                            | No Course Offerings                                           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Critical Analysis of historical literature and a research paper of 1500 words, compensatory time provided. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | HI101<br>-Or-<br>HI151                                                                                     |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | HI101<br>-Or-<br>HI151                                                                                     |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | HI108A<br>-Or-<br>HI108E<br>-Or-<br>HI108M<br>-Or-<br>HI108R<br>-Or-<br>HI108U                             |                                                               |
| HI108R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | RGNL STUDY WORLD HIST - RUSSIA                                                                             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                                                                                                     | <b>Offerings:</b>                                             |
| HI108X, Regional Studies in World History, is a detailed study of the development and critical events in the history of one of six regions, each with its own letter designation. As the stems are all a part of one course, cadets may not take multiple stems as electives. The focus on one region enables cadets to develop a deeper understanding of a different culture and unfamiliar ideas and concepts. The course also develops methods of historical research and analysis, critical thinking, lucid writing, and effective participation in classroom discussion. HI108R combines with HI105 or HI155 to form the plebe history sequence of the Core Academic Program. The Russia stem (identifier R) challenges cadets to study how social, political, economic, technological, and military factors have influenced international and multicultural relations and how these relations have influenced the development of specific societies, polities, cultures, economies, technologies, military systems, and gender roles throughout Russia, Central Asia, and the Caucasus. |                                                                                                            | 2026-1                                                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                     |                                                               |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                            |                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Critical Analysis of historical literature and a research paper of 1500 words, compensatory time provided. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | HI101<br>-Or-<br>HI151                                                                                     |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | HI101<br>-Or-<br>HI151                                                                                     |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | HI108A<br>-Or-<br>HI108E<br>-Or-<br>HI108L<br>-Or-<br>HI108M<br>-Or-<br>HI108U                             |                                                               |
| <b>HI108W</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>FOUNDATIONS OF MODERN WORLD</b>                                                                         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026-1                                                                                                     | <b>Offerings:</b>                                             |
| HI108W will focus on trade, technology, politics, society, and the state. These themes will help cadets understand how states and societies innovate, why states go to war, how they marshal the resources to do so, and how these factors have changed over time. In doing so, this course will help cadets know themselves, know others, and begin to develop strategic competence. The course also develops methods of historical research and analysis, critical thinking, lucid writing, and effective participation in classroom discussion. |                                                                                                            | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                                                                                       |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | HI105<br>-Or-<br>HI155<br>-Or-<br>HI158W                                                                   |                                                               |
| <b>HI155</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>ADVANCED HISTORY OF THE U.S.</b>                                                                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026-1                                                                                                     | <b>Offerings:</b>                                             |
| This course encompasses the same chronological period and thematic coverage as HI105, but it assumes some familiarity with American history and consequently places special emphasis on historical analysis and criticism. Moreover, students acquire a broader understanding of American history and the historian's methods.                                                                                                                                                                                                                     |                                                                                                            | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                                                                                       |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | HI105                                                                                                      |                                                               |
| <b>HI158W</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>ADV FOUNDATIONS MODERN WORLD</b>                                                                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026-1                                                                                                     | <b>Offerings:</b>                                             |
| HI158W will focus on trade, technology, politics, society, and the state. These themes will help cadets understand how states and societies innovate, why states go to war, how they marshal the resources to do so, and how these factors have changed over time. In doing so, this course will help cadets know themselves, know others, and begin to develop strategic competence. The course also develops methods of historical research and analysis, critical thinking, lucid writing, and effective participation in classroom discussion. |                                                                                                            | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                                     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                                                                                       |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | HI108W<br>-Or-<br>HI105<br>-Or-<br>HI155                                                                   |                                                               |

| HI301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | HIST OF MILITARY ART TO 1900                                                                                                                                                                                                                                                                        | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                                                                                                                                                                                              | <b>Offerings:</b>                          |
| This one-term elective course traces the evolution of the art of war to 1900. Emphasis is placed on the changing character of warfare as nations adjusted to social, political, ideological, economic, and technological developments. The historical method is used as a tool for understanding war as a complex, ambiguous, and unpredictable human phenomenon. Analysis focuses on causation, the interrelationship of events as warfare evolved over the ages, operational and logistical aspects of military history, and the role of society in warfare. |                                                                                                                                                                                                                                                                                                     | 2026-1                                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                                                              |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | A research essay and an analytical review essay; compensatory time provided for both essays.                                                                                                                                                                                                        |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | HI105<br>-Or-<br>HI155<br>-Or-<br>HI108<br>-Or-<br>HI108A<br>-Or-<br>HI108E<br>-Or-<br>HI108L<br>-Or-<br>HI108M<br>-Or-<br>HI108R<br>-Or-<br>HI108U<br>-Or-<br>HI158A<br>-Or-<br>HI158E<br>-Or-<br>HI158L<br>-Or-<br>HI158M<br>-Or-<br>HI158R<br>-Or-<br>HI158U<br>-Or-<br>HI108W<br>-Or-<br>HI158W |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | HI351<br>-Or-<br>HI301H<br>-Or-<br>HI301X                                                                                                                                                                                                                                                           |                                            |

| HI301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | HIST OF MILITARY ART TO 1900                 | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2027-1                                       | <b>Offerings:</b>                                                              |
| This core course traces the evolution of the art of war to 1900. Emphasis is placed on the changing character of warfare as nations adjusted to social, political, ideological, economic, and technological developments. The historical method is used as a tool for understanding war as a complex, ambiguous, and unpredictable human phenomenon. Analysis focuses on causation, the interrelationship of events as warfare evolved over the ages, operational and logistical aspects of military history, and the role of society in warfare. |                                              | 2027-1 2027-2 2027-8<br>2027-9 2028-1 2028-2<br>2028-8 2028-9 2029-1<br>2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                       |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | A research essay is required in this course. |                                                                                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                     |                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | HI105<br>-Or-<br>HI155<br>-Or-<br>HI108<br>-Or-<br>HI108A<br>-Or-<br>HI108E<br>-Or-<br>HI108L<br>-Or-<br>HI108M<br>-Or-<br>HI108R<br>-Or-<br>HI108U<br>-Or-<br>HI158A<br>-Or-<br>HI158E<br>-Or-<br>HI158L<br>-Or-<br>HI158M<br>-Or-<br>HI158R<br>-Or-<br>HI158U<br>-Or-<br>HI108W<br>-Or-<br>HI158W |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | HI351<br>-Or-<br>HI301H                                                                                                                                                                                                                                                                             |                                                               |
| <b>HI301H</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>HIST OF MILITARY ART TO 1900</b>                                                                                                                                                                                                                                                                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2026-1                                                                                                                                                                                                                                                                                              | <b>Offerings:</b>                                             |
| Temporary course for History Majors enrolled in HI301                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                     | No Course Offerings                                           |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                                                              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | HI301: Two research papers, one of at least 300 words and one of 1500 words; HI302: A 1500-word research paper tied to a WWII colloquium; compensatory time provided.                                                                                                                               |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | HI301<br>-Or-<br>HI301X<br>-Or-<br>HI351                                                                                                                                                                                                                                                            |                                                               |
| <b>HI301X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>HIST OF MILITARY ART TO 1900</b>                                                                                                                                                                                                                                                                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2026-1                                                                                                                                                                                                                                                                                              | <b>Offerings:</b>                                             |
| This core course traces the evolution of the art of war to 1900. Emphasis is placed on the changing character of warfare as nations adjusted to social, political, ideological, economic, and technological developments. The historical method is used as a tool for understanding war as a complex, ambiguous, and unpredictable human phenomenon. Analysis focuses on causation, the interrelationship of events as warfare evolved over the ages, operational and logistical aspects of military history, and the role of society in warfare. |                                                                                                                                                                                                                                                                                                     | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                                                              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | A research essay is required in this course.                                                                                                                                                                                                                                                        |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | HI351<br>-Or-<br>HI301                                                                                                                                                                                                                                                                              |                                                               |

| HI302                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | HIST OF MIL ART 1900- PRESENT                                                                                                                                                                                                                                                                       | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                                                                                    |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2025-4                                                                                                                                                                                                                                                                                              | <b>Offerings:</b>                                                                                                                                             |
| This one-term, upperclass core course traces the evolution of the art of war from 1900 to the wars of the present day. Emphasis is placed on the changing character of warfare as nations adjusted to social, political, ideological, economic, and technological developments. The historical method is used as a tool for understanding war as a complex, ambiguous, and unpredictable human phenomenon. Analysis focuses on causation, the interrelationship of events as warfare evolved over the ages, operational and logistical aspects of military history, and the role of society in warfare. |                                                                                                                                                                                                                                                                                                     | 2026-1 2026-2 2026-4<br>2026-8 2026-9 2027-1<br>2027-2 2027-4 2027-8<br>2027-9 2028-1 2028-2<br>2028-4 2028-8 2028-9<br>2029-1 2029-2 2029-4<br>2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                                                              |                                                                                                                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | A research essay and an analytical review essay, compensatory time provided for both essays.                                                                                                                                                                                                        |                                                                                                                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | HI105<br>-Or-<br>HI155<br>-Or-<br>HI108<br>-Or-<br>HI108A<br>-Or-<br>HI108E<br>-Or-<br>HI108L<br>-Or-<br>HI108M<br>-Or-<br>HI108R<br>-Or-<br>HI108U<br>-Or-<br>HI158A<br>-Or-<br>HI158E<br>-Or-<br>HI158L<br>-Or-<br>HI158M<br>-Or-<br>HI158R<br>-Or-<br>HI158U<br>-Or-<br>HI108W<br>-Or-<br>HI158W |                                                                                                                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | HI352<br>-Or-<br>HI302X<br>-Or-<br>HI302H                                                                                                                                                                                                                                                           |                                                                                                                                                               |

| HI302H                                                 | HIST OF MIL ART 1900- PRESENT                                                                                                                                         | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Scope:</b>                                          | 2026-1                                                                                                                                                                | <b>Offerings:</b>                          |
| Temporary course for History Majors enrolled in HI302. |                                                                                                                                                                       | No Course Offerings                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                |                                            |
| <b>Special Requirements:</b>                           | HI301: Two Research papers, one of at least 300 words and one of 1500 words; HI302: A 1500-word research paper tied to a WWII colloquium; compensatory time provided. |                                            |

**Prerequisite(s):** HI105 HI301  
 -Or-  
 HI105 HI301H  
 -Or-  
 HI155 HI301  
 -Or-  
 HI155 HI301H  
 -Or-  
 HI108 HI301

**Disqualifier(s):** HI302  
 -Or-  
 HI352  
 -Or-  
 HI302X

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| <b>HI302X</b> | <b>HIST OF MIL ART 1900- PRESENT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This two term, upperclass core course traces the evolution of the art of war from the hundred years war through the Napoleonic era to the American civil war and the wars of the twentieth century. Emphasis is placed on the changing nature of warfare as nations adjust to social, political, economic and technological developments. Analysis focuses on causation, the interrelationship of events as warfare evolved over the ages, operational and logistical aspects of military history, and the role of society in warfare.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** HI302  
 -Or-  
 HI302H

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| <b>HI337</b> | <b>CHINA: EMPIRE, REPUBLIC, &amp; MAO</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This course traces the history of China from its emergence as the dominant culture in East Asia in the 10th century to the present. It examines how the legacies of the Tang empire in China played out in the multi-state system that endured from 960 to the Mongol conquest, and then were reclaimed in the form of the Late Imperial state under the Ming and "foreign" Manchu rule. It considers the search for a "new China" under the Republic, which was established following the collapse of the Late Imperial state, and how that search fared during a time of competing warlords and the rise of nationalism. It shows why Mao came to represent a new utopian vision and how that vision tragically failed. Finally, the course explores how the search for "new China" and historical legitimacy continues today both on the mainland and in the continuation of the Republic in Taiwan.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

**Prerequisite(s):** HI105  
 -Or-  
 HI155  
 -Or-  
 HI108  
 -Or-  
 HI108A  
 -Or-  
 HI108E  
 -Or-  
 HI108L  
 -Or-  
 HI108M  
 -Or-  
 HI108R  
 -Or-  
 HI108U  
 -Or-  
 HI158A  
 -Or-  
 HI158E  
 -Or-  
 HI158L  
 -Or-



HI158M  
-Or-  
HI158R  
-Or-  
HI158U  
-Or-  
HI108W  
-Or-  
HI158W

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| <b>HI338</b> | <b>WARFARE IN AGE OF REVOLUTIONS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This course examines the theory and practice of warfare in Europe during the Age of Revolutions, roughly considered to be 1750 to 1814. Political revolutions such as the American and French Revolutions, along with other revolutions such as the Agricultural and the Industrial, and the intellectual ferment spawned by the Age of Enlightenment, all resulted in significant changes in the conduct of warfare. This course will examine those events, with particular focus on their relevance to the art of warfare. Themes include changes in military organization, doctrine, technology, and the accompanying social, political, and economic factors that influenced the armies of the day. The course will also cover the wars and campaigns that took place during this timeframe, including the American and French Revolutions and the wars of Napoleon.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI339</b> | <b>THE MODERN MIDDLE EAST</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This course enables cadets to explore the social, political, economic, and military interactions in the formation of the Modern Middle East. The first block examines the decline of the Gunpowder Empires and the subsequent penetration of European colonialism into the Islamic world (India, North Africa, Egypt, and the Levant), with emphasis on the factors that led to military decline of the Turkic world and the relative economic and military advantages of the European powers. During this block, students will discuss the Middle East's modernizing and reform efforts that European colonialism helped to catalyze, to include democratization, constitutions, capitalism, and industrialization. The second block covers the events that follow the World Wars and subsequent decolonization of the Middle East against the backdrop of the Cold War. Cadets will closely examine the Arab-Israeli conflict, the rise of Arab Nationalism and the tension between military revolutionary dictatorship and attempts at constitutional monarchy and republics. The final phase will begin with the Iranian revolution of 1979 and the Soviet invasion of Afghanistan. It will consider the rise of political Islam as a revolutionary ideology and the post-Cold War challenges leading to current wars and insurrections.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI340</b> | <b>COLONIAL AMERICA</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This course examines the international, political, social, cultural, and economic origins and development of colonial North America prior to the War for Independence, with attention to French and Spanish as well as British colonies. It explores the development of American identities and the significance of colonization and intercultural encounters for all the peoples, Native and European, of North America.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI341</b> | <b>THE AGE OF EXPLORATION</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This course concentrates on the on the 'age of exploration' and its impact on the Early Modern World, 1453-1715. It provides students interested in the history of Early Modern Europe, the Atlantic world, the history of Africa and colonial Latin America a general understanding of the ideologies and institutions that enabled Europe to colonize parts of Africa and the Americas during this important period in world history. Specific topics include: medieval precedents of early modern imperialism; theories of monarchy and empire; ideologies of conquest and colonization; the continuity of Native cultures and beliefs; the relevance of race and slavery in understanding European influence in Africa and the Americas; and the creation of an Atlantic economy.

2026-2 2026-8 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI343</b> | <b>MODERN GERMANY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course is a survey of the German lands from the dawn of the modern era through contemporary times. The course will combine social, political, economic, and cultural history in examining crucial themes and developments related to the German-speaking regions. Cadets will consider German nation and state formation; social, demographic, and economic transformation; imperialism, war and ideological change; the transformation of male and female roles; and trends in high and popular culture. The course will include a significant segment on twentieth-century Germany and the role the German state played in determining the course of world history, whether as the Nazi state that unleashed the Holocaust or as the West German Cold War bulwark. German history has much to teach us, and has led to enormous debates about the nature of the modern era.

2026-2 2026-8 2026-9  
2027-8 2027-9 2028-2  
2028-8 2028-9 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI344</b> | <b>MODERN DIPLOMACY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The course focuses on the major diplomatic developments in Europe from 1814 through the end of the Cold War in 1991. It traces the emergence of the European state system after the Treaty of Westphalia and the impact of the revolution in France on European diplomatic relations. It examines the diplomatic system established at the Congress of Vienna through the crises and conflicts of the mid-19th century. The course also examines the various factors that led to the First World War, the developments of the interwar period, the origins and conduct of the Second World War, and the origins of the Cold War. The final lessons will explore Europe's role in the Cold War, the rise of international organizations, trans-national diplomacy, the end of the Cold War, and recent modifications to Europe's role in world affairs.

2026-8 2027-1 2027-9  
2028-9 2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI345</b> | <b>MODERN AFRICA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course takes a thematic approach to African history, describing the forces which led to the partitioning of the continent, the practices of European colonialism/imperialism, the emergence of independent African states, and political, economic, and social developments in contemporary Africa. The goal of the course is to focus on critical events, relationships, and themes on the continent that continue to effect current events.

2026-1 2028-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI346</b> | <b>INDIA, PAKISTAN, &amp; BANGLADESH</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course explores the complex history of South Asia upon which the modern states of India, Pakistan, and Bangladesh are founded. The course examines cultural roots, religious developments, and traditions of power and authority in the multiplicity of states before the introduction of Islam. It goes on to consider the impact of later Muslim control much of South Asia. It covers British rule of South Asia and focuses particularly on the violent partition at independence in 1947 into India and Pakistan. It will examine the legacies of Gandhi, Jinnah as the founder of Pakistan, and Winston Churchill as a figure in British India policy. Lastly the course studies the development of states since independence, including the transition of East Pakistan into Bangladesh, with particular attention to ongoing struggles over religious and secular identities.

2026-2 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI347</b> | <b>ASIAN WARFARE AND POLITICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course explores the interaction between warfare and political systems in East Asia. It begins with the transition from military monarchy to bureaucratic empire in the Warring States Period. It then maps the rise of nomadic confederations in the Inner Asian steppe and their strategic interaction with the Han state. It traces how the collapse of the Han state led to military turmoil in East Asia, the rise of hybrid states, a new cosmopolitan empire, and then a multi-state system. It considers how in Japan, the importation of the bureaucratic state led first to centralization and then to the rise of the samurai and a feudal structure. Next, the course examines the development of a new form of nomadic confederation under the Mongols, and how Mongol warfare led to a more centralized state in China, and turmoil and a federalist system in Japan. In the modern period, the course considers how the challenge of Western military force led to political turmoil and the rise of the Communists in China, but in Japan led to the building of the Imperial Army, noted for its competence and for its atrocities. The course concludes with reflection on how the experience of war in East Asia continues to affect the region's politics and political structures.

2027-1 2027-8 2028-8

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A1500-word research paper.

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| <b>HI348</b> | <b>MODERN LATIN AMERICA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course surveys the cultural, economic, political, and social evolution of Latin America from the era of independence to the present. The course begins with a brief examination of Pre-Colombian and colonial events and structures. Students will study the economic development of modern Latin America and its influence on social, political, and military change. Case studies of national histories, such as Mexico, Cuba, Brazil, Argentina, and other countries help to illuminate the broad themes that underlie modern Latin American history. The course will examine Latin American relations with the United States and other nations of the world.

2026-8 2026-9 2027-1  
2027-8 2027-9 2028-8  
2028-9 2029-1 2029-8  
2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI351</b> | <b>ADV HIST OF MILITARY ART -1900</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

HI351-352 parallels HI301-302. However, in addition to accelerated study of HI301-302 material, the cadet will study selected periods in greater depth and breadth. This course offers the cadet a more profound understanding of men and women as warriors and of the evolution of the art of war than would otherwise be available.

No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One 500-word critical analysis and one 1500-word research paper; compensatory time provided.

**Prerequisite(s):**

- HI105
- Or-
- HI155
- Or-
- HI108
- Or-
- HI108A
- Or-
- HI108E
- Or-
- HI108L
- Or-
- HI108M
- Or-
- HI108R
- Or-
- HI108U
- Or-
- HI158A
- Or-

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|                                                                                                                                                                                                                                                                                                                                       | HI158E<br>-Or-<br>HI158L<br>-Or-<br>HI158M<br>-Or-<br>HI158R<br>-Or-<br>HI158U<br>-Or-<br>HI108W<br>-Or-<br>HI158W                                                                                                                                                                                  |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                               | HI301<br>-Or-<br>HI301X<br>-Or-<br>HI301H                                                                                                                                                                                                                                                           |                                                    |
| <b>HI352</b>                                                                                                                                                                                                                                                                                                                          | <b>ADV HIST OF MIL ART 1900- PRES</b>                                                                                                                                                                                                                                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                                                                                                                              | <b>Offerings:</b>                                  |
| HI351-352 parallels HI301-302. However, in addition to accelerated study of HI301-302 material, the cadet will study selected periods in greater depth and breadth. This course offers the cadet a more profound understanding of men and women as warriors and of the evolution of the art of war than would otherwise be available. |                                                                                                                                                                                                                                                                                                     | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                       | 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                             |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                          | One 500-word critical analysis and one 1500-word research paper; compensatory time provided.                                                                                                                                                                                                        |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                               | HI105<br>-Or-<br>HI155<br>-Or-<br>HI108<br>-Or-<br>HI108A<br>-Or-<br>HI108E<br>-Or-<br>HI108L<br>-Or-<br>HI108M<br>-Or-<br>HI108R<br>-Or-<br>HI108U<br>-Or-<br>HI158A<br>-Or-<br>HI158E<br>-Or-<br>HI158L<br>-Or-<br>HI158M<br>-Or-<br>HI158R<br>-Or-<br>HI158U<br>-Or-<br>HI108W<br>-Or-<br>HI158W |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                               | HI302<br>-Or-<br>HI302H<br>-Or-<br>HI302X                                                                                                                                                                                                                                                           |                                                    |
| <b>HI355</b>                                                                                                                                                                                                                                                                                                                          | <b>WARFARE, INDUSTRY, &amp; EMPIRE</b>                                                                                                                                                                                                                                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                   | <b>Offerings:</b>                                 |
| This course examines the history of warfare around the globe from the end of the Napoleonic Wars through the First World War and its aftermath. It connects the study of military strategy, tactics, and campaigns with the political, economic, social, and cultural contexts and dynamics that shaped military developments. It explores the impact of changing technology on the conduct of war, the development of nationalism and imperialism, "cabinet wars" between European nation-states, wars of national unification, and wars of empire. This course explores several themes particularly useful to any modern soldier, including the nature and intensity of national and imperial wars and the effect of changing technology on society and the conduct of war.                                                                                                                                                                                                   |                                          | 2027-2 2029-2                                     |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 30 @ 75 min (2.000 Att/wk)               | <b>Labs:</b> 0 @ 0 min                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A 1500-word research paper.              |                                                   |
| <b>HI356</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>WAR AT SEA, SKY, SPACE</b>            | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2027-2                                   | <b>Offerings:</b>                                 |
| This course examines the history of the Sea, Air, and Space Domains in the 20th Century through exploring how US military and political leaders conceived of their utility and how each influenced the conduct and outcome of armed conflict. The course proceeds chronologically beginning with the introduction of Mahanian thought in the US Navy in the 1890s, adding in the development of military aircraft in the 1910s, and first uses of space during the Second World War. It traces the evolving role of each domain through the World Wars, Korea, Vietnam, late Cold War and immediate post-Cold War conflicts. Cadets will study the interactions between the three domains and how they shaped the contemporary operating environment. Cadets will be able to understand and appreciate the context of the modern joint force and how each domain is inherently interconnected and needed to find tactical, operational, and strategic success in war and peace. |                                          | 2027-2 2028-2 2029-2                              |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 30 @ 75 min (2.000 Att/wk)               | <b>Labs:</b> 0 @ 0 min                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                     |                                                   |
| <b>HI357</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>DECOLONIZATION &amp; THE COLD WAR</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                   | <b>Offerings:</b>                                 |
| During the Cold War, American policies of containment and collective security collided with attempts at communist expansion, wars of national liberation that resulted in the end of European empires throughout much of Africa and Asia (a process often called "decolonization"), as well as postcolonial conflicts shaped by the Cold War superpower confrontation. Cadets will examine the strategic conditions and political considerations influencing the use of force and diplomacy during this period. They will gain an appreciation for the various actors' decisions and experiences, how the Cold War and decolonization influenced societies across the globe, and the legacies of these conflicts.                                                                                                                                                                                                                                                               |                                          | 2026-1                                            |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 30 @ 75 min (2.000 Att/wk)               | <b>Labs:</b> 0 @ 0 min                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A 1500-word research paper.              |                                                   |
| <b>HI357</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>ARMED CONFLICT IN THE COLD WAR</b>    | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2027-1                                   | <b>Offerings:</b>                                 |
| This course examines the history of armed conflict during the Cold War and the influence of ideology and technology on war in the second half of the 20th century. The course proceeds chronologically beginning with an introduction to the ideologies and alliances of the US and USSR, looking at the rise of non-aligned states and competing ideologies in the developing world, and first battles of the Cold War. It traces how the goals and objectives of the superpowers collided with local interests resulting in unexpected results and frustration. Cadets will study how technology became both a constraining and enabling factor in war during the period. They will be able to understand and appreciate the context of how warfare evolved during this period and how the legacies of Cold War conflicts shape the contemporary operating environment.                                                                                                       |                                          | 2027-1 2028-1 2029-1                              |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 30 @ 75 min (2.000 Att/wk)               | <b>Labs:</b> 0 @ 0 min                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                     |                                                   |
| <b>HI359</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>ERA OF THE SECOND WORLD WAR</b>       | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                   | <b>Offerings:</b>                                 |

This course examines the Interwar Years, 1919-1939, and the Second World War from a global perspective while using a thematic approach to compare the different experiences of each of the major belligerents. Whether covering the Versailles Treaty, the rise of Adolf Hitler, the US Army during the Great Depression, home fronts, or the Holocaust, the cadets in the course will examine the social, political, cultural, and economic factors that contributed to how belligerents waged war, and, in turn, how war affected each of these factors across the globe. The course covers how and why the belligerents planned and executed particular strategies and operations in the European, Pacific, and China-Burma-India theaters to achieve their coalition and national goals. Finally, this course examines the interrelationship of sea, air, and land forces, and the complexities of providing logistical support to joint and combined operations on an unprecedented scale.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI361</b> | <b>MEDIEVAL EUROPE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1 | <b>Offerings:</b> |
| <p>The millennium between the "fall" of the Roman Empire and the Voyages of Discovery--the Middle Ages--has often been characterized as brutish and inferior. Yet, this tough, fascinating society offered immeasurable potential for growth and adaptation. The personages and events of the European medieval world spawned many of the ideas and institutions of modernity. Topics for study will include the barbarian invasions, Byzantine Empire, Carolingian Europe, feudalism, medieval technology, Christian Church, medieval warfare, Crusades, rise of universities, crises of the 14th century, growth of monarchical power, and economic and social change.</p> |        | 2027-2 2029-2     |

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI362</b> | <b>POLITICS/SOC-EARLY MOD EURO</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1 | <b>Offerings:</b>   |
| <p>After tracing the legacy of the middle ages, this course concentrates on the development of the modern nation state in Europe, 1453-1648. Specific topics include the renaissance, humanism, the reformation, the age of religious wars, and the contrast between the growth of absolutist and constitutional governments. The latter portion of the course concentrates on the political, social, economic, diplomatic, and military trends that shaped modern Europe.</p> |        | No Course Offerings |

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>HI364</b> | <b>MODERN WESTERN EUROPE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1 | <b>Offerings:</b>                                                              |
| <p>This course is an introduction to European history from 1789 to the present. The course considers how and why Europe -- a small, relatively poor, and politically fragmented place -- became the engine of globalization and an important civilization in its own right. Our approach is broadly cultural, using politics, economics, society, religion, and other arenas to understand the events and people of Modern Western Europe. Chief topics: French Revolution, liberalism and the industrial revolution, socialism and the rise of labor, modern colonialism, world wars, communism and capitalism, decolonization, Cold War, and the European Union.</p> |        | 2026-2 2026-8 2026-9<br>2027-8 2027-9 2028-2<br>2028-8 2028-9 2029-8<br>2029-9 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI367</b> | <b>IMPERIAL AND SOVIET RUSSIA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1 | <b>Offerings:</b>                                                              |
| <p>This course examines the political, social, and cultural history of Russia as it emerged from the Mongol era up to the present day. It explores the development of the Tsarist political and social systems, the emergence of literary, artistic, and revolutionary movements, and the development of Russia's position in European politics from the time of Peter I through WWI. It also covers the rise of the Soviet Union, the leadership's attempts to implement communist ideology and responses to that attempt, Russia's relationship with various national and ethnic groups, and the emergence of the Soviet Union as a superpower. The course concludes with the collapse of the Soviet Union and the emergence of new states in the 1990s.</p> |        | 2026-8 2026-9 2027-1<br>2027-8 2027-9 2028-8<br>2028-9 2029-1 2029-8<br>2029-9 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

**HI368** **MOD CENTRAL & E. EUR,1896-1989** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2026-1

**Offerings:**

Between 1896 and 1989, Central and Eastern Europe experienced two world wars, at least three major revolutions, and radical industrial and environmental dislocations. The region witnessed everything from the birth of its modern culture to the creation of new post-World War I nation-states, to the Holocaust, to massive forced population shifts, to the creation of the communist Eastern Bloc, to the popular overthrow of Communism in 1989. Radical regimes on the right and left brought incredible change, quashed hopes, and produced both progress and suffering of unprecedented proportion. This course will examine life in late-19th and 20th century Habsburg Europe and its successor states of Poland, Hungary, Czechoslovakia, and Yugoslavia. It will do so comparatively, highlighting themes of nation-creation, everyday life, social transition, war, revolution, and ethnic cleansing.

2026-1 2028-1

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

**HI369** **AMERICAN FRONTIERS** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2026-1

**Offerings:**

HI 369 enables cadets to explore the social, political, economic, and military interactions between many diverse cultures in North America during the period of European and U.S. expansion since 1500. The course does this by examining the history of Native America and the "American" West, which included much of colonial British North America, and much of the American South through the 1830s, along with Spanish, French, and other European frontiers in North America. The course integrates Native American, Latino, and economic history in the study of migration, cultural contact, and "international" relations on the frontiers of North America. The course also explores change and diversity in cultural perspectives by examining myths of the West from a range of ethnic and other viewpoints. The course is an elective in the American History stem of the history program, but can be taken for credit in the International stem as well.

2027-2 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**HI370** **ANCIENT & MEDIEVAL WARFARE** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2026-1

**Offerings:**

This course focuses on warfare from the dawn of recorded history through the fourteenth century. Thus, it will provide cadets with opportunities to study the campaigns of Alexander, the military methods of the Romans, the military aspects of feudalism, the Scottish war of independence, and other topics which are not covered in the core military courses. Although the course includes in-depth analyses of certain battles and campaigns, it places more emphasis on "war and society" issues such as the relationship between military participation and social standing in human societies, the connections between armies and governments, and the impact of economic, technological and social change on military structures. Also, HI370 will shift some emphasis away from the operational level of war to the analysis of the strategic and tactical levels of war, and away from use of secondary sources to use of primary materials.

2026-2 2026-8 2027-8  
2027-9 2028-2 2028-8  
2028-9

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** Two critical analyses of at least 750 words each; compensatory time provided.

**HI372** **US FGN RELATIONS SINCE 1898** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2026-1

**Offerings:**

This course examines American foreign relations from the nation's entry into the world arena as a major power in 1898 through both World Wars, and the Cold War, to its station in today's multipolar world. It is a study of the forces, events, personalities, and principles that have shaped America's role in the world and provided the framework for the development of current foreign policy.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper; compensatory time provided.

**HI374** **HISTORY OF AFRICA** **3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                    | <b>Offerings:</b>                                 |
| This course begins with a survey of pre-colonial Africa, including the evolution of early human cultures, the rise and fall of African civilization and states, the spread of Islam, and the contact between Africans and Europeans. It will then focus on the region south of the Sahara, describing the forces which led to the partitioning of the continent, the practices of European colonialism/imperialism, the emergence of independent African states, and political, economic, and social developments in contemporary Africa.                                                                                                                                                                                                                                                                                                                                                                       |                                           | No Course Offerings                               |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A 1500-word research paper.               |                                                   |
| <b>HI375</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>HISTORY OF CHINA</b>                   | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                    | <b>Offerings:</b>                                 |
| This course begins with the emergence of distinct Chinese civilization and the rich legacies of early and dynastic China. It will then trace the history of the imperial period from the creation of the state, through the rise of different dynasties, barbarian invasion, and Mongol occupation, until the height of imperial power under the rule of the foreign Manchus. It will move to consider why China had such difficulty with its Western contacts in the 19th century and why China has experienced such turmoil in the 20th century. The course concludes by examining Mao's rise to power and the tragic events that followed. The emergence of Taiwan as a separate regional power is also discussed.                                                                                                                                                                                           |                                           | No Course Offerings                               |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A 1500-word research paper.               |                                                   |
| <b>HI376</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>EARLY MODERN WARFARE</b>               | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                    | <b>Offerings:</b>                                 |
| This course examines the history of warfare in Europe from the Renaissance through the campaigns of Frederick the Great. It combines the study of military campaigns with that of the political, economic, social, and cultural factors shaping military developments. It explores the so-called "Military Revolution" of the sixteenth and seventeenth centuries with particular emphasis on the relationships between military developments and state building, the rise of absolutism in France and the Wars of Louis XIV, and the rise of Prussia and the Wars of Frederick the Great. Study of the so-called "age of limited war" sets the stage for future study of the American Revolution and the Wars of the French Revolutions and Napoleon. This course contains several themes particularly useful to the modern soldier. Among them are the nature, intensity, and complexity of wars of religion. |                                           | 2027-2 2029-2                                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A 1500-word research paper.               |                                                   |
| <b>HI377</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>EMPIRES &amp; REVOLUTIONS</b>          | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2027-2                                    | <b>Offerings:</b>                                 |
| This course examines the military, economic, political, and cultural expansion of empires throughout the world beginning in the fifteenth century. It then explores the doctrines, ideologies, and practices of imperialism and colonial rule before moving on to consider how colonized peoples resisted colonial rule and were eventually able to mount anticolonial revolutions that created independent nation-states. The course ends with the collapse of European empires during the period of the Cold War.                                                                                                                                                                                                                                                                                                                                                                                             |                                           | 2027-2 2029-2                                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                    |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                      |                                                   |
| <b>HI378</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>INDUSTRY, IDEOLOGY &amp; THE STATE</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2028-1                                    | <b>Offerings:</b>                                 |
| Taking a global history approach, this course examines how industrial technologies, economic developments, and new political ideologies gave rise to the modern nation-state from 1780 to the present. It then explores on a global scale how these developments challenged societies and cultures through increasingly integrated trade networks and imperial projects, and then how they contributed to the World Wars of the 20th century. Lastly, the course analyzes how these developments led to the rise of a truly global economic and political system that has wrought far-reaching social and cultural changes across the world. The course will consider developments in Asia, Africa, Latin America, Europe, and the United States.                                                                                                                                                               |                                           | 2028-1                                            |



**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>HI379</b> | <b>HISTORY OF LATIN AMERICA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

No Course Offerings

This course surveys the political, economic, and social evolution of Latin America by tracing its history from Pre-Colombian times through the 20th century. Beginning with an exploration of the three races that blend to create a unique Latin American culture, the course surveys the wide scope of Latin American history that has created Latin America, as we know it today. Specific attention is paid to the period of Spanish and Portuguese conquest and colonization, Latin America's fight for independence, and the subsequent struggle to build sovereign, modernized nations in the 19th century. This course ends by looking at the social revolutions and guerrilla insurgencies of the 20th century, and Latin America's attempt to overcome its historical problems of political instability, economic underdevelopment, and social stratification.

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A 1500-word research paper.

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| <b>HI381</b> | <b>HISTORY OF IRREGULAR WARFARE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

2026-1 2026-8 2027-1  
2028-1 2029-1

The course examines unconventional warfare from a historical perspective, particularly conflicts involving opponents with a significant disparity in their conventional military capabilities. Through several case studies, the course explores why belligerents succeed or fail in unconventional warfare and how ideology, technology, and social, political, and economic factors help determine the outcome of wars between regular and irregular forces. Covering a broad period of history, selected case studies include wars of conquest or colonization, revolutionary wars, and peacekeeping or constabulary operations.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A research paper of at least 1500 words; compensatory time provided.

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| <b>HI383</b> | <b>MIDDLE EASTERN WARFARE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

2026-2 2028-2

This course introduces cadets to a variety of issues affecting the nature and conduct of war in the geographic and cultural region bounded by Turco-Arabic languages. A special emphasis is to understand the utility of warfare in a specific social context, that of evolving Turco-Arabic methods of governance, military institutions, and military operations from the time of Muhammad to the Treaty of Karlowitz. Significant military issues in this geographically, demographically, ethnically, and culturally diverse region are explored to gain historical perspective into the various social, political, and religious legitimacies groups of people have exploited in the quest to dominate Islamicate civilizations from 600-1700 CE.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A research paper of at least 1500 words; compensatory time provided.

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| <b>HI385</b> | <b>WAR &amp; ITS THEORISTS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

No Course Offerings

This course engages with the development of military theory as an intersection of military and intellectual history. As a military history course, it addresses the history of the practice of warfare. As an intellectual history course, it explores the relationship between ideas and human behavior. The fusion of these two approaches engages with influential theorists and practitioners (including Clausewitz, Corbett, Liddell Hart, Brodie, and others) to examine how the intellectual currents of their respective eras intersected with their ideas on the utility of armed force, and how subsequent eras reinterpreted theories and ideas to align with their own views of war.

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Two 800-1000-word papers; compensatory time provided.

**Corequisite(s):** HI301  
-Or-  
HI351

| HI387                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | TOPICS IN AMERICAN HISTORY      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                          | <b>Offerings:</b>                            |
| <p>This is a special topics course that offers the opportunity to examine an aspect of American history that does not align with regularly-scheduled American history elective courses. Topics will vary from one iteration to the next. Consult the Department of History Academic Counselors for more information.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                 |                                              |
| No Course Offerings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |                                              |
| HI388                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | TOPICS IN INTERNATIONAL HISTORY | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                          | <b>Offerings:</b>                            |
| <p>This is a special topics course that offers the opportunity to examine an aspect of international history that does not align with regularly-scheduled international history elective courses. Topics will vary from one iteration to the next. Consult the Department of History Academic Counselors for more information.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                 |                                              |
| No Course Offerings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |                                              |
| HI389                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | TOPICS IN MILITARY HISTORY      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-7                          | <b>Offerings:</b>                            |
| <p>This is a special topics course that offers the opportunity to examine an aspect of military history that does not align with regularly-scheduled military history elective courses. Topics will vary from one iteration to the next. Consult the Department of History Academic Counselors for more information.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                 |                                              |
| No Course Offerings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 |                                              |
| HI391                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | WORLD RELIGIONS                 | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                          | <b>Offerings:</b>                            |
| <p>This course analyses the emergence, development and present cultural expression of the major religions of the world, emphasizing their 19th and 20th century experience. It also examines the development of religion in the ancient world and in pre-literate and non-technical societies. Cadets study the world's religions as molded by and as molders of the social, political and economic forces unique to particular cultures. Special attention is paid to the role of each religion in the formulation and adaptation of public and foreign policy.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> A 1500-word research paper.</p>                                                                                                                                                                                                                                                                                                                                                    |                                 | 2026-1 2026-8 2026-9<br>2027-2 2028-1 2029-1 |
| HI392                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AMERICAN HISTORICAL MEMORY      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                          | <b>Offerings:</b>                            |
| <p>The stories we tell about our past matter, often on a grand scale. Narratives of the past produced by historians, artists, politicians, institutions, and communities reveal much about individual and collective identities within the United States. Such narratives often help shape local, regional, and national politics, institutions, identities, and values for both good and ill. This course examines American historical memory and the myriad ways in which American remembrance and commemoration of the past has interacted with, reflected, and affected contemporary conditions, communities, institutions, and values more than they have accurately narrated, portrayed, and commemorated past events. Ultimately this course will examine in a broad sense the role of history in shaping national identity and nationalism within the United States, and the place of history within American society and culture.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)    <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p> |                                 | 2027-2 2029-2                                |
| HI393                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AMERICA IN DEPRESSION AND WAR   | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-2                                                                           | <b>Offerings:</b>                                  |
| Between 1919 and 1960 the United States demobilized after one war, experienced dramatic but uneven economic growth, suffered the Great Depression, fought a second world war, and emerged as the premier world power in the Cold War that followed. Amid massive economic and international change, the nation underwent significant social and political change, with the growth of activist government, conflict between business and labor, the emergence of a widely based middle class, and the contested growth of civil rights movements. This course explores how and why the United States changed between the end of the First World War and the election of John F. Kennedy, connecting domestic and international contexts and trajectories.                                                                                                 |                                                                                  | 2026-2 2028-2                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                             |                                                    |
| <b>HI394</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>REVOLUTIONARY AMERICA</b>                                                     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                                                           | <b>Offerings:</b>                                  |
| This course examines the social, political, and economic origins and consequences of the American Revolution through the adoption of the Constitution. It explores the development of an American identity and the meaning of the Revolution for all Americans, to include women, African Americans, and the poor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                  | 2026-2 2028-2                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | A 1500-word research paper or historiographic essay; compensatory time provided. |                                                    |
| <b>HI395</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>HIST OF CIVIL WAR AMERICA</b>                                                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                                                           | <b>Offerings:</b>                                  |
| This course focuses on the causes and consequences of the American Civil War. Cadets will analyze the road to war, the war itself, and Reconstruction to place the entire period in its broader historical context. The course covers the ante-bellum South and North, focusing on the peculiar effect of slavery on society. Cadets will examine the home fronts to see the populace's reaction to war as both the Union and the Confederacy engage in conflict. In approaching Reconstruction, students will focus on the political, economic, and racial policies that were implemented to rebuild the nation.                                                                                                                                                                                                                                        |                                                                                  | 2026-2 2028-2                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | A 1500-word research paper; compensatory time provided.                          |                                                    |
| <b>HI396</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>THEODORE ROOSEVELT'S AMERICA</b>                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                                                           | <b>Offerings:</b>                                  |
| Between 1865 and 1919, the United States emerged from the Civil War and became a rising world power. Mass industrialization during the Gilded Age made it a powerhouse in the global economy but also raised myriad social and political issues that eventually transformed the United States during the Progressive Era. Consolidating control over its continental territory in the last quarter of the 19th century, the United States then expanded and intervened overseas in 1898 and beyond, eventually becoming embroiled in the cataclysm of the First World War in 1917-1918. Theodore Roosevelt's America: the Gilded Age, Progressive Era, and World War I analyzes these and related issues, with particular attention to connections between changes at home and abroad, and between changes in political, social, and economic dimensions |                                                                                  | 2026-1 2028-1 2029-8                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                             |                                                    |
| <b>HI397</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>THE US FROM COLD WAR TO TODAY</b>                                             | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2027-1                                                                           | <b>Offerings:</b>                                  |
| In 1960 the United States was deeply engaged in the Cold War with communism and the Soviet Union. Since then, the nation has experienced massive social, political, economic, cultural, and international change, with the growth of activist government and backlash against it, conflict between business and labor, economic stagnation and the return of a boom-and-bust economy, the contested growth of civil rights and other reform movements, the end of the Cold War and the return of ethnonationalism and great power competition. This course explores how and why the United States changed between the election of John F. Kennedy and the present, focusing on 1960-2001 and connecting domestic and international contexts and trajectories.                                                                                            |                                                                                  | 2027-1 2029-1                                      |

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>HI398</b> | <b>CIVIL RIGHTS IN AMER HIST</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

2027-2 2028-2 2029-2

This course examines the history of civil rights and the Civil Rights Movement in America from 1865 to the present, focusing on the earliest efforts to achieve civil rights for African Americans and the constant evolution of those efforts. This course focuses on the political, economic, and social issues that shaped understandings about civil rights, citizenship, race, class and gender, enriching the understanding of the fight for Civil Rights in American History beyond the Civil Rights Movement of the 1950s and 1960s. While this course will focus primarily on African American civil rights and civil rights movements in American history, various other civil rights movements may be included for comparative analysis. Accordingly, this course will historicize issues related to race, ethnicity, and gender in ways that will better prepare future officers to lead in the modern Army.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>HI399</b> | <b>HISTORY STAFF RIDE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-7

**Offerings:**

2026-7 2027-7 2028-7  
2029-7

History Staff Ride analyzes various campaigns and battles focusing on enhancing cadet understanding of the relationship between the strategic, operational and tactical levels of war and gaining a heightened appreciation for the importance of leadership on the battlefield. Cadets begin with classroom preparation and intense study prior to the staff ride, and through travel, experience the battlefields first-hand. They gain a historical understanding of the campaigns, and by walking the terrain, develop the ability to analyze complex battlefield problems and conditions. The ability to plan, prepare and execute a staff ride, with a focus on cadet led presentations and discussions, will also be evaluated. In addition, cadets participate in a number of cultural activities during travel.

**Lessons:** 0 @ 0 min (0.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>HI399A</b> | <b>HISTORY STAFF RIDE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-7

**Offerings:**

2026-7 2027-7 2028-7  
2029-7

History Staff Ride analyzes various campaigns and battles focusing on enhancing cadet understanding of the relationship between the strategic, operational and tactical levels of war and gaining a heightened appreciation for the importance of leadership on the battlefield. Cadets begin with classroom preparation and intense study prior to the staff ride, and through travel, experience the battlefields first-hand. They gain a historical understanding of the campaigns, and by walking the terrain, develop the ability to analyze complex battlefield problems and conditions. The ability to plan, prepare and execute a staff ride, with a focus on cadet led presentations and discussions, will also be evaluated. In addition, cadets participate in a number of cultural activities during travel.

**Lessons:** 0 @ 0 min (0.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** HI399

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| <b>HI400</b> | <b>UNITY AND CONFLICT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

2027-2

This course is designed to teach cadets to analyze how humans have historically conceptualized themselves and others, and how they mobilized mutual associations and groups to claim political legitimacy and power. In particular, the course will provide historical comprehension of what these concepts are, theories of how, why, and when they emerged, and how they interconnect or break down. HI400 will examine modern developments such as the Enlightenment, the scientific revolution, the growth of the state, European-stimulated social movements and mass politics, and European imperialism to study why these changes inspired varied but overlapping methods for unifying peoples, creating boundaries, and defining differences. The course is global in content and multidisciplinary in method. \*\*This is a pilot course and must be reviewed by the Curriculum Committee NLT AY29-1 to continue.\*\*

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

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| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                                                       |                                                    |
| <b>HI460</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>SENIOR FACULTY COURSE</b>                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                                                     | <b>Offerings:</b>                                  |
| This course is taught by a senior member in the Department of History in a field of that historian's expertise. The course offers students the opportunity to study under the guidance of a historian in topics not normally offered by the Department of History. This course will include an exploration of the way in which history has been written; including examining the changing interpretations, traditions, methods, and frameworks of historians.                   |                                                                            | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A 1500-word research paper.                                                |                                                    |
| <b>HI460A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>SENIOR FACULTY COURSE</b>                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                                                     | <b>Offerings:</b>                                  |
| This course is taught by a senior member in the Department of History in a field of that historian's expertise. The course offers students the opportunity to study under the guidance of a historian in topics not normally offered by the Department of History.                                                                                                                                                                                                              |                                                                            | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A 1500-word research paper.                                                |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | HI460                                                                      |                                                    |
| <b>HI462</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>THE HISTORY OF INNOVATION</b>                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                                                     | <b>Offerings:</b>                                  |
| Innovations in technology, science, thought and ideology have radically changed the course of history across the world. This course examines why these innovations occur and then how they are practically applied in a military, social, political, economic, and cultural context. This course will include an exploration of the way in which history has been written; including examining the changing interpretations, traditions, methods, and frameworks of historians. |                                                                            | 2026-2                                             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A 1500-word research paper.                                                |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | IT305<br>-Or-<br>IT355<br>-Or-<br>CY305<br>-Or-<br>CY355<br>-Or-<br>CY305X |                                                    |
| <b>HI464</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>VISITING PROFESSOR ELECTIVE</b>                                         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                                                     | <b>Offerings:</b>                                  |
| This course is taught by the visiting Ewing Chair in Military History on a topic of that historian's expertise. The course offers students the opportunity to study under the guidance of a distinguished historian in topics not normally offered by the Department of History. The course may be taught by a distinguished visiting professor on the occasion that the Ewing Chair is unable to do so.                                                                        |                                                                            | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                                                       |                                                    |
| <b>HI475</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>HISTORY OF DAILY LIFE</b>                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                                                                     | <b>Offerings:</b>                                  |

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| <p>This course explores the everyday experiences of people throughout history, focusing on how daily life has evolved across different cultures, societies, and time periods. From ancient civilizations to modern times, students will examine the routines, customs, and technologies that shaped the daily existence of ordinary people. Topics will include food, clothing, family structures, work, leisure, and social norms, as well as how major historical events, such as wars, revolutions, and industrialization, influenced the everyday lives of individuals. By engaging with primary sources, material culture, and scholarly interpretations, students will gain a deeper understanding of how people navigated their worlds, from ancient households to contemporary urban centers. This course emphasizes the significance of the "mundane" as a window into broader social, political, and cultural developments, highlighting how daily life reflects and contributes to the fabric of history. **This is a pilot course for AY26-1 and AY28-1 and must undergo review by the Curriculum Committee NLT AY29-2 to continue.**</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p> |                                       | 2028-1                                            |
| <b>HI487</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>EARLY AMERICAN REPUB 1787-1826</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2027-1                                | <b>Offerings:</b>                                 |
| <p>Although the Constitution outlined the form of federal government in the United States, it left unanswered many questions concerning how that government should function. This course examines how, between 1787 and 1826, evolving political thought, economic development, changing social conditions, and sectionalism influenced successive generations' debates about the role of government in American life.</p> <p><b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> None</p> <p><b>Disqualifier(s):</b> HI390</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 2027-1 2029-1                                     |
| <b>HI494</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ADV IND STUDY IN HISTORY (1CR)</b> | <b>1.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                                | <b>Offerings:</b>                                 |
| <p>This course is an individually supervised research and study program. The cadet may engage in independent research or an internship and research either at West Point or another location, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a substantial work of scholarship. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member.</p> <p><b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> Enrollment by permission of the Head of the Department. Other requirements as determined by the Faculty Advisor.</p>                                                                                                                                                                                                                                                                                                                                                                 |                                       | No Course Offerings                               |
| <b>HI495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ADV IND STUDY IN HISTORY (2CR)</b> | <b>2.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                                | <b>Offerings:</b>                                 |
| <p>This course is an individually supervised research and study program. The cadet may engage in independent research or an internship and research either at West Point or another location, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project, but the primary purpose is for the cadet to engage in independent study, usually in preparation for a more substantial project in the future. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member.</p> <p><b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)      <b>Labs:</b> 0 @ 0 min</p> <p><b>Special Requirements:</b> Enrollment by permission of the Head of the Department. Other requirements as determined by the Faculty Advisor.</p>                                                                                                                                                                                                                                                                                                                                          |                                       | No Course Offerings                               |
| <b>HI496</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ADV IND STUDY IN HISTORY (3CR)</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                                | <b>Offerings:</b>                                 |
| <p>This course is an individually supervised research and study program. The cadet may engage in independent research or an internship and research either at West Point or another location, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a substantial work of scholarship. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                       | 2027-1 2028-1 2029-1                              |

**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

Enrollment by permission of the Head of the Department. Other requirements as determined by the Faculty Advisor.

**HI496A****ADV IND STUDY IN HISTORY (3CR)****3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)**Scope:**

2026-1

**Offerings:**

This course is an individually supervised research and study program. The cadet may engage in independent research or an internship and research either at West Point or another location, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a substantial work of scholarship. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member.

**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

Enrollment by permission of the Head of the Department. Other requirements as determined by the Faculty Advisor.

**HI498****COLLOQUIUM IN HISTORY****3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)**Scope:**

2026-1

**Offerings:**

The colloquium employs seminar discussions of important books and scholarly articles to enhance understanding of major historical issues. Subcourses are designed to provide in-depth study of various topics in American, European, military, and international and strategic history. Cadets select a subcourse topic as the basis for their reading program after consultation with their faculty advisor or departmental counselor. Subcourse topics may vary each year in accordance with student interest and faculty expertise. The colloquium satisfies the 400-level course requirement for the history fields of study. Cadets who major in history should complete a colloquium that will support their subsequent enrollment in HI499, Senior Thesis in History.

2026-1 2027-1 2028-1  
2029-1**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

An historiographical essay of 1500 words; compensatory time provided.

**HI498A****COLLOQUIUM IN HISTORY****3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)**Scope:**

2026-1

**Offerings:**

The colloquium employs seminar discussions of important books and scholarly articles to enhance understanding of major historical issues. Subcourses are designed to provide in-depth study of various topics in American, European, military, and international and strategic history. Cadets select a subcourse topic as the basis for their reading program after consultation with their faculty advisor or departmental counselor. Subcourse topics may vary each year in accordance with student interest and faculty expertise. The colloquium satisfies the 400-level course requirement for the history fields of study. Cadets who major in history should complete a colloquium that will support their subsequent enrollment in HI499, Senior Thesis in History.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

An historiographical essay of 1500 words; compensatory time provided.

**Prerequisite(s):**

HI498

**HI499****SENIOR THESIS****3.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)**Scope:**

2026-1

**Offerings:**

The course provides cadets selecting the major in history with an opportunity to enhance their skills in historical research and analysis. For this reason the course serves as excellent preparation for graduate study in history and related disciplines. Based upon their background and research interests cadets are organized into small thesis-writing seminars. Under the supervision of a seminar advisor, each cadet defines a topic, develops a research plan, accomplishes research, and drafts a thesis. The seminar meets occasionally to discuss issues in historiography and methodology, review progress in research, and critique draft papers. At the end of the semester cadets present their findings and defend their theses before a committee of faculty and fellow students.

2026-2 2027-2 2028-2  
2029-2**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

A research paper of 3500 words; compensatory time provided. Presentation and defense of thesis before a committee of faculty.

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | HI498                                                                                                                         |                                                               |
| <b>HI499A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>SENIOR THESIS - A</b>                                                                                                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                                                                                                                        | <b>Offerings:</b>                                             |
| The course provides cadets selecting the major in history with an opportunity to enhance their skills in historical research and analysis. For this reason the course serves as excellent preparation for graduate study in history and related disciplines. Based upon their background and research interests cadets are organized into small thesis-writing seminars. Under the supervision of a seminar advisor, each cadet defines a topic, develops a research plan, accomplishes research, and drafts a thesis. The seminar meets occasionally to discuss issues in historiography and methodology, review progress in research, and critique draft papers. At the end of the semester cadets present their findings and defend their theses before a committee of faculty and fellow students.                                 |                                                                                                                               | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | A research paper of 3500 words; compensatory time provided. Presentation and defense of thesis before a committee of faculty. |                                                               |
| <b>HX400</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INTEGRATIVE EXPERIENCE IN HIST</b>                                                                                         | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                                                                                                                        | <b>Offerings:</b>                                             |
| This course is an administrative placeholder for history major integrative experience credit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                               | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                          |                                                               |
| <b>WS301</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INTRODUCTION TO WAR STUDIES</b>                                                                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                                                                                                                        | <b>Offerings:</b>                                             |
| The greatest study of war, Carl von Clausewitz's On War, outlined a methodology for learning about war by study and reflection, integrating history and theory. WS301 introduces cadets to this methodology, providing the intellectual foundation they need for further coursework in the War Studies major. Cadets enrolled in this course will analyze both classic and recent works in Strategic Studies, along with historical case studies. Applying the theoretical content to examples of practice enables cadets to better understand both the theory and the history, and to test the value of generalizations, principles, or conceptual linkages proposed by theory.                                                                                                                                                       |                                                                                                                               | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                          |                                                               |
| <b>WS330</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INTRODUCTION TO GRAND STRATEGY</b>                                                                                         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                                                                                                                        | <b>Offerings:</b>                                             |
| This seminar examines the fundamentals of grand strategy in theory and in practice. It addresses the historical moment in which the modern concept of grand strategy was born and explores the major debates that have animated scholars and practitioners since. We focus on the core building blocks of grand strategy: the elements of national power that are available to be coordinated and applied in a coherent fashion across policy contexts. We analyze historical case studies to explore how the modern concept of grand strategy can be retrospectively applied to explain state behavior across time and diverse contexts. The seminar ultimately aims to prepare future Army officers to apply concepts from grand strategy to analyze and address complex problem sets in the contemporary international environment. |                                                                                                                               | 2027-1 2028-1 2029-1                                          |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                          |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | SS457                                                                                                                         |                                                               |
| <b>WS335</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>TOPICS IN GRAND STRATEGY</b>                                                                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                                                                                                                        | <b>Offerings:</b>                                             |



WS335 Topics in Grand Strategy is the second of a two-part sequence designed to build on the course material covered in WS330. The course will cover applied regional and thematic topics in the study of grand strategy in a contemporary and/or historical context to illustrate key theoretical concepts. Grades are determined based on presentations, essays and other writing exercises, class participation, and a final exam, essay, and/or briefing.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** SS458

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|--------------|----------------------------|----------------------------------------------------------|
| <b>WS345</b> | <b>MILITARY INNOVATION</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This interdisciplinary course examines the subject of military innovation from a theoretical, strategic, historical, and policy-oriented perspective. WS345 addresses several key questions: Why do militaries innovate? How does this process of innovation occur? Why do attempts at military innovation succeed or fail? To answer these questions, this course introduces the innovation concept and ties innovation to the levels of war. It provides the historical narrative to military innovation, while emphasizing the contemporary operating environment by exploring the possibility of a recent Revolution in Military Affairs through emerging technologies and the international security environment.

2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** DS345

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>WS350</b> | <b>STRATEGIC COMMUNICATION</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Leaders must possess the ability to communicate complex ideas clearly and concisely, both verbally and in writing. WS350 is grounded in the application of sound communication and narrative techniques relevant to the tactical, operational, and strategic levels of war. The course introduces the concept of information as an element of national power used to achieve U.S. strategic objectives. It then turns to how narratives are used to develop and execute strategy. Cadets gain foundational knowledge of how information is used to develop understanding, communicate, and act decisively. Students study interactions between the military and the media to understand influence campaigns and their effects during competition and conflict. The course culminates in crafting communication plans to tackle vital strategic challenges. Through a combination of theoretical understanding and practical application, cadets learn to develop strategic communication plans, engage with media effectively, and leverage information as a dynamic element of combat power.

2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** DS350

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| <b>WS358</b> | <b>STRATEGY AND GENERALSHIP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course focuses on the evolution of generalship in the western military experience and the performance of general officers who have sought to translate success at different levels of war into achieving policy & strategic goals. The course examines how generals have crafted and executed military strategy from the early modern era to the present. It explores various factors that shaped strategic thinking and influenced the way generals commanded their forces in these different historical eras and analyzes the application of those ideas and systems in selected case studies.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** HI358

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| <b>WS360</b> | <b>SPECIAL OPS IN THEORY &amp; PRACTI</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course explores the nature and dynamics of low-intensity conflict in contrast to conventional and high-intensity warfare. Specifically, cadets will examine the strategy and tactics of insurgency, counterinsurgency, international terrorism, and peace operations. Another component of this course examines Special Operations Forces (SOF) by exploring the unique methods of special operators and their relationship with irregular warfare. Cadets will examine how U.S. SOF are organized, how special operations in general succeed, and why SOF are well-suited for low-intensity conflict. Several relevant guest speakers are integrated into the instruction throughout the course.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** DS360

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| <b>WS365</b> | <b>REBELLN, INSURGNCY, &amp; CIV WAR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This course explores the social, political, economic, and military dynamics of irregular warfare, focusing on the tactical to strategic intricacies of conflict among state and non-state actors. Cadets gain insights into the complexities of irregular warfare in competition and conflict through a comprehensive examination of theoretical frameworks and practical case studies.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** DS460

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| <b>WS370</b> | <b>US STRATEGY AND POLICY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This course introduces cadets to foundational concepts within US national security policy, strategic theory, and military strategy. Cadets analyze the decisions of US policymakers through practical case studies and contemporary events by applying major theoretical concepts in defense strategy and policy through individual critical thinking, group collaboration, and oral and written communication. Cadets will be able to understand, analyze, and effectively communicate the relation of tactical action to national policy within the context of the US defense establishment. \*This course will be conducted as a pilot course in AY26-2 and AY27-2. If there are substantive changes to the course, it will need review by the Curriculum Committee NLT AY28-1 to continue in AY28-2.\*

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** DS370

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| <b>WS385</b> | <b>WAR AND ITS THEORISTS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This course engages with the development of military theory as an intersection of military and intellectual history. As a military history course, it addresses the history of the practice of warfare. As an intellectual history course, it explores the relationship between ideas and human behavior. The fusion of these two approaches engages with influential theorists and practitioners (including Clausewitz, Corbett, Liddell Hart, Brodie, and others) to examine how the intellectual currents of their respective eras intersected with their ideas on the utility of armed force, and how subsequent eras reinterpreted theories and ideas to align with their own views of war.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** HI385

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|--------------|--------------------------------------|----------------------------------------------------------|
| <b>WS389</b> | <b>WAR IN THE CONTEMPORARY WORLD</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2027-1 | <b>Offerings:</b> |
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Why do wars begin? How are they fought? Why and how do they end? Since the first and only use of atomic weapons in 1945, scholars and practitioners alike have argued for decades that the character of war is changing and that the pace of change is steadily increasing. That change can be disorienting, but there are many aspects about war's nature that are enduring and should be recognized as points of continuity even as war today continues to evolve. This course examines armed conflict in recent decades, analyzing trends in technological, sociological, political, and cultural change to better understand the modern character of war: its causes, conduct, and consequences. Cadets taking this course will be better prepared to understand change and continuity in modern warfare as they enter the profession of arms.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|--------------|-------------------|----------------------------------------------------------|
| <b>WS398</b> | <b>FUTURE WAR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

While the violent, interactive, and fundamentally political nature of war remains unchanged, the character of war has evolved over time and the pace of change appears to be accelerating. Among various other responsibilities related to studying and directing war, military professionals prepare the institution for future challenges by analyzing current and emerging trends to discern how the character of war is changing and will continue to change. This course introduces cadets to the study of future war through rigorous analysis of various theories of future war, current and emerging trends in the international environment and the defense and security communities, and emerging technologies. \*\*This course will be taught as a pilot in AY26-1 and AY27-1. It must be reviewed by the Curriculum Committee NLT AY27-2 to continue.\*\*

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** DS498

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|--------------|----------------------------------|----------------------------------------------------------|
| <b>WS399</b> | <b>INTERNSHIP IN WAR STUDIES</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The Academic Individual Advanced Development (AIAD) program is designed to give cadets practical experience in their field of study and to reflect on their experiences by completing specified requirements. Current AIADs that cadets may potentially execute for credit include domestic and international staff rides and internships to areas of strategic relevance, though AIAD availability will vary by year. Instructors, students, and host agencies will develop a specific academic plan prior to executing the AIAD that will require either department or curriculum review depending on the nature of the trip. Students will normally complete specially assigned readings, a daily reflective journal based on their practical experiences and write a term paper that benefits the host organization or expands on the AIAD's purpose. The assigned instructor will provide feedback on student submissions and assign term grades accordingly.

2027-7 2028-7 2029-7

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** DS399

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>WS455</b> | <b>COMPARATIVE DEFENSE POLICY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course's objective is to analyze the defense policies of various countries and the outcomes of those defense policies, to include national security objectives, national military objectives, military doctrine, force structure, and military capabilities. Countries studied will include actual and potential coalition partners and potential adversaries. Cadets will examine the political, economic, and social influences on each military establishment. Cultural influences on the development and implementation of the defense policies for countries studied will be examined, including the effects each country's culture has on the missions, structure, roles, and capabilities of the military. Cadets will develop their own framework of analysis to critically analyze the defense policies and cultures of other countries, and will be able to clearly articulate that analysis through written and oral means. Guest speakers include liaison officers and Foreign Area Officers to provide insight into the specific military establishments of those countries studied.

2026-8 2028-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** DS455

| WS475                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | FORECAST AND GAME IN DECISION | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                        | <b>Offerings:</b>                          |
| This course introduces cadets to the interdisciplinary nature of decision-making by addressing how leaders make decisions in situations of incomplete, conflicting, and incorrect information using forecasting, modeling, and gaming. The curriculum enables cadets to synthesize long-term developments and potential future scenarios through forecasting and trend analysis. It requires cadets to apply lessons in a series of complex, competitive games. Cadets will leave the class with a basic understanding of the complexities involved in strategic decision-making and methods to frame, understand, and address that complexity.                                                                                                                                                                                                                                                                                                  |                               | 2028-2 2029-2                              |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min        |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                          |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DS475                         |                                            |
| WS485                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | JOINT AND MULTINATIONAL OPS   | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                        | <b>Offerings:</b>                          |
| This course provides a historical, theoretical, and practical understanding of the roles of sea, air, space, and cyber power, alliances, and coalitions in military strategy and national security policy. After establishing a strong theoretical and historical foundation in alliance warfare and conflict and competition within each of the domains, students apply their knowledge to assess the strategies of multiple state and non-state actors, and to make practical recommendations on the application of military power in a joint and multinational context during competition, crisis, and conflict.                                                                                                                                                                                                                                                                                                                              |                               | 2028-1 2029-1                              |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min        |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                          |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DS485                         |                                            |
| WS489                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | IND STUDY IN WAR STUDIES      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                        | <b>Offerings:</b>                          |
| The course provides an environment that is conducive to independent effort in a subject area of special interest to the cadet. Original research or specialized study can be accomplished in any of the many fields within War Studies. The course is conducted in three phases. First, the cadet and the individual advisor from the War Studies faculty will reach agreement on a subject area for research. Research methods will be studied under the direction of the faculty member. Research may involve field trips and personal interviews with experts in the area of study. In the second phase, the cadet will engage in independent research and prepare a draft analytical paper or report detailing the findings. During this period, frequent consultation with the faculty advisor occurs regarding the progress in the project. In the third phase, the cadet will present and define the findings before a faculty committee. |                               | 2028-1 2029-1                              |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min        |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                          |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DS489                         |                                            |
| WS490                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SPECIAL TOPICS IN WAR STUDIES | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                        | <b>Offerings:</b>                          |
| The Special Topics Course provides cadets an opportunity for reading and analysis in depth within a topic area of special interest and timely relevance to War Studies. The course director will determine the approach dependent on the topic and enrollment. Courses will normally develop the cadet's understanding of the topic through study of theory, history, doctrine, and historical and contemporary case studies. A generous portion of the course will normally address modern complex problems related to the topic area and assignments will emphasize analytical writing. Topics will vary by semester.                                                                                                                                                                                                                                                                                                                          |                               | 2028-1 2029-1                              |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min        |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                          |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | DS490                         |                                            |

| <b>WS495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>METHODS &amp; CAPSTONE WAR STUDIES</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                                    | <b>Offerings:</b>                                  |
| <p>This is a research methods course designed to support the interdisciplinary nature of the research-based courses in the War Studies Program, WS496 Thesis in War Studies and WS497 Client-Based Capstone in War Studies. In this course, students will learn different research methodologies for the strategic studies field and develop effective writing skills in a seminar format. Partway through the semester students will choose an interdisciplinary research topic related to the War Studies field. The topic, approved by the Course Director, will either be as part of a student-selected thesis for WS496 or an assigned client-based project for WS497. Thesis students must also select faculty members from across the institution to serve as their thesis advisors as part of WS495. Student assignments include practical exercises, a research proposal, a literature review and a draft research methodology.</p>                                                                                                                                                                                                                                                                                                                               |                                           | 2028-1 2029-1                                      |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                                      |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | DS495                                     |                                                    |
| <b>WS496</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>THESIS IN WAR STUDIES</b>              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                                    | <b>Offerings:</b>                                  |
| <p>Thesis in War Studies is an integrative course in strategy designed to provide War Studies majors with practical experience in addressing real, complex and ambiguous strategic issues. Students will write and defend an interdisciplinary thesis that in some way relates to the military instrument of power or national security. Thesis cadets will continue the work they began with their faculty advisors in the fall by revising their literature review and research methodology, then complete data collection, data analysis, thesis writing and ultimately conduct an oral defense of their thesis before an interdisciplinary faculty board.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           | 2028-2 2029-2                                      |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                                      |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | DS496                                     |                                                    |
| <b>WS497</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>CLIENT-BASED CAPSTONE IN WS</b>        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                                    | <b>Offerings:</b>                                  |
| <p>Client-Based Capstone in War Studies is an integrative course in strategy designed to provide War Studies majors with practical experience in addressing real, complex and ambiguous strategic issues. Students will work in small groups to solve such a problem for an external client organization focusing on the use of the military instrument of power or national security strategy. Student groups will determine stakeholder needs, define the client's problem, and conduct appropriate research to develop a viable solution or set of recommendations. Groups will meet regularly with their client, complete a final written report and provide a formal presentation with an oral defense to the faculty. Alternatively, select students may work with an interdisciplinary team from another academic department's capstone course to contribute analysis from the strategic studies perspective to the group's project. Capstone cadets will continue the work they began with their faculty advisors in the fall by revising their literature review and research methodology, then complete data collection, data analysis, final report writing and ultimately conduct an oral defense of their work before an interdisciplinary faculty board.</p> |                                           | 2028-2 2029-2                                      |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                                      |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | DS497                                     |                                                    |
| <b>XH405</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>THE HOLOCAUST AND ITS LEGACY</b>       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                                    | <b>Offerings:</b>                                  |
| <p>This is a multi-disciplinary course administered through the Center for Holocaust and Genocide Studies. The Holocaust and its aftermath make enormous demands on us not just emotionally, but intellectually, requiring that we consider it from historical, military, psychological, philosophical, political, scientific, representational and legal perspectives. XH405 is a multi-disciplinary response to these challenges. The course will explore the causes, course, and consequences of the Holocaust, examining the processes that led to the Nazi genocide against the Jews, with a particular focus on the role of the military. It will examine the Holocaust from a variety of perspectives (perpetrators, victims, bystanders, resisters and rescuers), and consider the moral and ethical choices made by members of each group. The course will require an in-depth understanding of German and European history, and it will impart an appreciation for the cultures and mentalities of the interwar and wartime era. The course will utilize primary sources, films, documentaries,</p>                                                                                                                                                              |                                           | 2026-2 2027-2 2028-2 2029-2                        |

testimonies, and propaganda. It will conclude with consideration of the political and legal responses to the Holocaust in the later 1940s, and the later incorporation of the Holocaust into the global public consciousness.

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A research paper of 1500 words. Compensatory time provided.

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>XH415</b> | <b>GENOCIDE AND MASS ATROCITY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

2026-1 2027-1 2028-1  
2029-2

This is a multi-disciplinary course administered through the Center for Holocaust and Genocide Studies. It is a multi-disciplinary attempt to understand the dynamics which produce mass atrocity. Cadets will learn of the causes, course, and consequences of selected genocides, examining the processes that led to genocide, with a particular focus on the roles played by militaries. The course opens with the intellectual theories about the phenomena of ethnic cleansing and genocide. It will then move to the analysis of how ethnic cleansing and genocide are tied to conflict and militaries. Drawing from a variety of scholarly disciplines and methodologies, cadets will investigate the moral, legal, historical, and diplomatic problems these terms pose. For the remainder of the course, cadets will examine more directly the case studies of specific incidences of ethnic cleansing and genocide. These case studies will vary year-to-year, but they will include at least two examples of ethnic cleansing or genocide on the American frontier, in German Southwest Africa, the Ottoman Empire, the Soviet Union, Rwanda, Cambodia, Yugoslavia, China and in Europe. Cadets will study primary sources, films, documentaries, testimonies, and propaganda. All iterations of the course will consider the development of Western mass politics and the emergence of racial nationalism and new imperialism in late 19th century Europe.

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A research paper of 1500 words. Compensatory time provided.

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|--------------|----------------------------------------|----------------------------------------------------------|
| <b>XH491</b> | <b>INTERDISCIPLINARY CAPSTONE-HIST</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

No Course Offerings

This course is for history majors participating in special interdisciplinary capstone projects. The projects may involve one or more academic departments outside of the History Department. Interdisciplinary Capstone Projects must be approved by the History Department Head.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Limited to History Majors in their Firstie Year.

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>ZH315</b> | <b>MODERN REGIONAL HISTORY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

2026-8 2026-9 2027-8  
2027-9 2028-8 2028-9  
2029-8 2029-9

For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce a historical research paper to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering modern historical developments of the area/region where the cadet is studying. This course covers broad historical processes and developments of the region over a long period of time. The course effectively encompasses a recognized historical era, for example "Modern", "Early Modern", or "Ancient".

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>ZH325</b> | <b>TOPICS IN REGIONAL HISTORY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

2026-8 2026-9 2027-8  
2027-9 2028-8 2028-9  
2029-8 2029-9

For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce written historical submissions to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering special topics in the area/region where the cadet is studying. Topics include (but are not limited to) specific eras of history; cultural & ethnic aspects of history; art & literary history; and other scientific & technical history topics.

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

| ZH335                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MILITARY HISTORY           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                     | <b>Offerings:</b>                                             |
| For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce a historical research paper to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering modern military history developments of the area/region where the cadet is studying.                                                                                                                                                                                                                           |                            | 2026-8                                                        |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                       |                                                               |
| ZH345                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | TOPICS IN MILITARY HISTORY | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                     | <b>Offerings:</b>                                             |
| For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce a historical research paper to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering special military history topics in the area/region where the cadet is studying. Topics include (but are not limited to) specific wars & campaigns; types of war (insurgency, air, naval, ect.); as well as other courses that incorporate military history with aspects of social science or military science. |                            | 2026-8 2027-8 2027-9<br>2028-8 2028-9 2029-8<br>2029-9        |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                       |                                                               |
| ZH355                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | FOREIGN PERSPECTIVES       | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                     | <b>Offerings:</b>                                             |
| For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce a historical research paper to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering any aspect of United States History or Western Civilization from the perspective of the nation/region where the cadet is studying.                                                                                                                                                                             |                            | 2026-8 2026-9 2027-8<br>2027-9 2028-8 2028-9<br>2029-8 2029-9 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                       |                                                               |
| ZH365                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | POLITICS AND DIPLOMACY     | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                     | <b>Offerings:</b>                                             |
| For Cadets attending foreign military academies or other academic institutions. Cadets will attend classroom instruction and produce a historical research paper to be presented upon return to USMA. Instruction may be in English or foreign language. This class serves as the equivalent to a foreign course covering special political and diplomatic history topics in the area/region where the cadet is studying. Topics include internal political development and/or international relations and diplomatic history.                                                                                                           |                            | 2026-8 2026-9 2027-8<br>2027-9 2028-8 2028-9<br>2029-8 2029-9 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min     |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                       |                                                               |

## Kinesiology

### 13 Courses

| KN355                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | FUNCTIONAL ANATOMY                                                                                                            | 3.5 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2013-1                                                                                                                        | <b>Offerings:</b>                                             |
| A knowledge of basic and applied anatomy is essential to the study of human beings engaged in motor performance. An individual who understands the anatomical bases that underlie human movement and who can systematically analyze movement and determine interventions is more likely to improve technique and reduce the risk of injury. Therefore, this course is designed to introduce the structures of human anatomy and explain how these structures are involved in human movement. In support of class room instruction cadets will be introduced to basic laboratory techniques and collection, analysis and interpretation of data demonstrating anatomical and mechanical function of muscles, joints, and limbs. On successful completion of the course, cadets should be able to identify and understand the anatomical structures essential for human movement and apply their anatomical knowledge to human movement problems in athletic, educational, clinical, and/or work settings. |                                                                                                                               | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 7 @ 110 min                                                                                                      |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                                                                          |                                                               |
| KN360                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | BIOMECHANICS OF HUMAN MOVEMENT                                                                                                | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2011-2                                                                                                                        | <b>Offerings:</b>                                             |
| A knowledge of basic and applied biomechanics is essential to the study of human beings engaged in motor performance. An individual who understands the mechanical bases that underlie human movement and who can systematically analyze movement and determine interventions is more likely to improve technique and reduce the risk of injury. Specifically, this course will provide cadets with: 1) a basic knowledge of the biomechanical foundations of human movement; 2) the knowledge and skills necessary to complete a systematic analysis and evaluation of human motor performance; and, 3) the ability to determine and provide interventions that are likely to improve movement.                                                                                                                                                                                                                                                                                                         |                                                                                                                               | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                                                                          |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PH202<br>-Or-<br>PH252<br>-Or-<br>PH205<br>-Or-<br>PH255<br>-Or-<br>PH201X<br>-Or-<br>PH201<br>-Or-<br>PH251<br>-Or-<br>PH275 |                                                               |
| KN365                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NUTRITION                                                                                                                     | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020-1                                                                                                                        | <b>Offerings:</b>                                             |
| Performance Nutrition is designed to teach the basic concepts and functions of nutrition as well as their application to human performance. This includes emphasis in food chemistry, digestion, absorption, and utilization of nutrients, nutrient timing, and nutritive supplementation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                               | 2026-2 2027-1 2027-2<br>2027-8 2028-1 2028-2<br>2028-8 2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                                                                          |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PL100<br>-Or-<br>PL150<br>-Or-<br>PL100B                                                                                      |                                                               |



| <b>KN455</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>PSYCHOLOGY OF EXERCISE</b>             | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2021-2                                    | <b>Offerings:</b>                                  |
| This course comprehensively examines theory and research related to exercise psychology, and introduces sport psychology as an associated discipline. The course is designed to provide a broad overview of exercise psychology and increase understanding of how psychological factors influence adherence and performance in exercise and sport. Additionally, the course addresses associated topics including addictive and unhealthy behaviors, burnout and overtraining, aggression in sport, and character development through sport.                                                                                                                                             |                                           | 2026-1 2027-1 2028-1 2029-1                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                      |                                                    |
| <b>KN460</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>EXERCISE PHYSIOLOGY</b>                | <b>3.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2011-1                                    | <b>Offerings:</b>                                  |
| This course is designed to apply the principles of chemistry, physics, anatomy, and physiology towards an understanding of the impact of work on the human body. Study will include both mechanical and physiological response to acute bouts of work and functional adaptation to repeated bouts of work. In support of class room instruction students will be introduced to basic laboratory techniques and collection, analysis and interpretation of physiological data. In addition, material will be presented that will emphasize the influence of age, disease-states, or the environment on the physiological response as a clue to physiological mechanisms and significance. |                                           | 2026-1 2027-1 2028-1 2029-1 2029-2 2029-8 2029-9   |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 8 @ 110 min                  |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                      |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CH387 KN355                               |                                                    |
| <b>KN470</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>FITNESS ASSESSMENT AND RX</b>          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2013-1                                    | <b>Offerings:</b>                                  |
| This course is designed to increase theoretical and practical knowledge and understanding of the administrative, medical, and biological aspects of developing physical competency through physical activity and exercise. Students will apply the scientific theories behind exercise assessment and prescription towards developing functional independence across the spectrum of activities for daily living, recreation, sports performance, and prevention and rehabilitation of disease covering various populations across the life span.                                                                                                                                        |                                           | 2026-1 2027-1 2028-1 2029-1                        |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                      |                                                    |
| <b>KN475</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>MUSCULAR FUNCTION &amp; ADAPTATION</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2024-2                                    | <b>Offerings:</b>                                  |
| This course is designed to expose students to the physiology of muscular function and adaptation associated with a fundamentally sound resistance and conditioning program. The course will emphasize the description, assessment, and analysis of sport movement. It will facilitate student understanding of the design of progressive resistance training programs to enhance performance variables, bridging theory to practice. Basic anatomy will be reviewed, and muscle function and adaptation will be the focus, to include how muscle responds to training, detraining, and overtraining.                                                                                     |                                           | 2026-2 2027-2 2028-2 2029-2 2029-8 2029-9          |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                      |                                                    |
| <b>KN480</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>T/P OF ADVANCED PERFORMANCE</b>        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2010-2                                    | <b>Offerings:</b>                                  |
| This integrative experience course was designed to provide cadets with advanced content knowledge in human adaptation to exercise and to serve as the USMA Integrative Experience. KN480 will address the overarching academic program goal: "to anticipate and respond effectively to the uncertainties of a changing technological, social, political, and economic world."                                                                                                                                                                                                                                                                                                            |                                           | 2026-2 2027-2 2028-2 2029-2                        |
| <b>Lessons:</b> 30 @ 75 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                    |                                                    |

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| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                  |                                                         |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | KN455 KN460                           |                                                         |
| <b>KN485</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>TOPICS: EXERCISE/SPORT SCIENCE</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2008-2                                | <b>Offerings:</b>                                       |
| This course provides in-depth study of a special topic in exercise and sport sciences not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the Visiting Professor, Rotating PhD, or a senior DPE faculty member.                                                                                                                                                                                                                                        |                                       | 2026-2 2027-2 2028-2 2029-2                             |
| <b>Lessons:</b> 30 @ 75 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min                |                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                       |                                                         |
| <b>KN491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>INDIV RESEARCH IN KINESIOLOGY</b>  | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2017-1                                | <b>Offerings:</b>                                       |
| This elective course provides an opportunity for a cadet to conduct an in-depth research project, study program, or special project in exercise and sport science. The cadet will formalize a proposal, develop a viable research plan, and conduct the project under the guidance and supervision of a faculty advisor. The Director - Human Performance Laboratory will approve all individual research projects. The course will require a commitment of approximately 40 hours.                       |                                       | 2026-1 2026-2 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                |                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                  |                                                         |
| <b>KN492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>INDIV RESEARCH IN KINESIOLOGY</b>  | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2017-1                                | <b>Offerings:</b>                                       |
| This elective course provides an opportunity for a cadet to conduct an in-depth research project, study program, or special project in exercise and sport science. The cadet will formalize a proposal, develop a viable research plan, and conduct the project under the guidance and supervision of a faculty advisor. The Director - Human Performance Laboratory will approve all individual research projects. The course will require a commitment of approximately 80 hours.                       |                                       | 2026-1 2026-2 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                |                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                  |                                                         |
| <b>KN493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>INDIV RESEARCH IN KINESIOLOGY</b>  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2017-1                                | <b>Offerings:</b>                                       |
| This elective course provides an opportunity for a cadet to conduct an in-depth research project, study program, or special project in exercise and sport science. The cadet will formalize a proposal, develop a viable research plan, and conduct the project under the guidance and supervision of a faculty advisor. The Director - Human Performance Laboratory will approve all individual research projects. The course will require a commitment of approximately 120 hours.                      |                                       | 2026-1 2026-2 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                |                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                  |                                                         |
| <b>KN495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>HONORS THESIS</b>                  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2010-2                                | <b>Offerings:</b>                                       |
| This course provides Kinesiology majors with an opportunity to enhance their skills in clinical research and analysis. Under the supervision of a thesis advisor, cadets will implement the research proposal developed in KN493. Cadets will meet regularly as a group with their seminar advisors to discuss issues in design, methodology, and data analysis. At the end of the semester cadets will present their findings and defend their theses before a committee of faculty and fellow students. |                                       | 2026-2 2027-2 2028-2 2029-2                             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                |                                                         |

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|------------------------------|-----------------------------------------------------------|
| <b>Special Requirements:</b> | None                                                      |
| <b>Prerequisite(s):</b>      | KN491 PL361<br>-Or-<br>KN492 PL361<br>-Or-<br>KN493 PL361 |

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## Law & Philosophy

### 61 Courses

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| <b>LW199</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>CIVIL RIGHTS STAFF RIDE</b>                                                                                                                                                                                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-7                                                                                                                                                                                                               | <b>Offerings:</b>                                                                                                              |
| This course, which also qualifies as an Academic Individual Advanced Development (AIAD) opportunity, applies the law and history in an immersive, interdisciplinary study of civil rights in the United States and the resilience of foundational values over time. It consists of a week in the classroom and a two-week staff ride incorporating relevant historical locations into student learning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                      | 2026-7 2027-7 2028-7<br>2029-7                                                                                                 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                               |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | One written partial review, four researched written/oral presentations during the staff ride and a final thought paper. Permission of instructor required for enrollment.                                            |                                                                                                                                |
| <b>LW310</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INTRO TO LEGAL METHOD</b>                                                                                                                                                                                         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2024-5                                                                                                                                                                                                               | <b>Offerings:</b>                                                                                                              |
| This course provides an introduction to the study of jurisprudence and, thereby, an intellectual foundation for legal studies. Jurisprudence explores the theory and philosophy of law, its relations to morality, and its limits. The intent of the course is to provide cadets a platform on which to examine the nature of law, legal reasoning, and legal institutions. Topics covered include positivism and natural law theory, theories of criminal justice, concepts of liberty, responsibility, and human rights. Cadets also will learn the fundamentals of legal research and writing.                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                      | 2026-1 2027-1 2028-1<br>2029-1                                                                                                 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                               |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                                                                                                                 |                                                                                                                                |
| <b>LW399</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INDIV ADV DEVELOPMENT IN LAW</b>                                                                                                                                                                                  | <b>1.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                                                                                                                                                                                                               | <b>Offerings:</b>                                                                                                              |
| The Academic Individual Advanced Development (AIAD) program is designed to introduce cadets to the practice of law in the military. The course consists of a three-week internship in one of a variety of legal offices. Possible internships include clerkships with the Army Court of Criminal Appeals, the U.S. Court of Military Appeals, the U.S. Supreme Court, the Department of Defense, and the Department of the Army agencies. Cadets may also intern in Staff Judge Advocate or Trial Defense Offices at military installations worldwide.                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                      | No Course Offerings                                                                                                            |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                               |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Grades are determined based on a journal of daily activities, the quality of the work actually performed during the internship, and a briefing which is presented to the department faculty upon the cadet's return. |                                                                                                                                |
| <b>LW403</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>CONSTITUTIONAL/MILITARY LAW</b>                                                                                                                                                                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2024-5                                                                                                                                                                                                               | <b>Offerings:</b>                                                                                                              |
| This course studies the United States Constitution and Military Law. Cadets will acquire information and skills in order to recognize and resolve constitutional and legal problems. The course provides analytical models for dealing with problems regarding societal and military order. Finally, the course seeks to enable the cadet to make an intelligent commitment to the values and preferences embodied in the Constitution and our system of military and civilian law. Examples from military law are used to model fundamental principles examined in the course. Significant court decisions are explored to support the course goals. Specific substantive areas include: separation of powers, judicial review, war powers, the law of armed conflict, equal protection, privacy, individual rights, searches and inspections, military justice processes, and military criminal law. |                                                                                                                                                                                                                      | 2026-1 2026-2 2026-5<br>2026-8 2027-1 2027-2<br>2027-5 2027-8 2028-1<br>2028-2 2028-5 2028-8<br>2029-1 2029-2 2029-5<br>2029-8 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                               |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                                                                                                                 |                                                                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | SS202<br>-Or-<br>SS252                                                                                                                                                                                               |                                                                                                                                |

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| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LW403L                             |                                                                                              |
| <b>LW403L</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>CONSTITUTIONAL/MILITARY LAW</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                             | <b>Offerings:</b>                                                                            |
| <p>This course studies the United States Constitution and Military Law. Cadets will acquire information and skills in order to recognize and resolve constitutional and legal problems. The course provides analytical models for dealing with problems regarding societal and military order. Finally, the course seeks to enable the cadet to make an intelligent commitment to the values and preferences embodied in the Constitution and our system of military and civilian law. Examples from military law are used to model fundamental principles examined in the course. Significant court decisions are explored to support the course goals. Specific substantive areas include: separation of powers, judicial review, war powers, the law of armed conflict, equal protection, privacy, individual rights, searches and inspections, military justice processes, and military criminal law. This version of LW403 is designed specifically for Law majors who have had LW310 as a pre-req.</p> |                                    | 2026-1 2027-1 2028-1<br>2029-1                                                               |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 40 @ 55 min (2.500 Att/wk)         | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                               |                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LW310 SS202<br>-Or-<br>LW310 SS252 |                                                                                              |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LW403                              |                                                                                              |
| <b>LW410</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>COMPARATIVE LEGAL SYSTEMS</b>   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                             | <b>Offerings:</b>                                                                            |
| <p>This course uses a comparative approach to study the three major legal systems of the world: the English common law system, the civil law system (and its branches) of continental Europe, and the Islamic legal system. These three systems are the foundation for the laws and legal systems of most of the world today, including Latin America, Africa, the Middle East and east Asia. Similarities and differences between these systems and the American legal system are explored. Social, political, and economic factors which distinguish these systems and more recently have begun to integrate them are covered. Emphasis is placed on the sources of law, the procedures for resolving legal disputes, and basic principles of civil and criminal justice.</p>                                                                                                                                                                                                                              |                                    | 2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 30 @ 75 min (2.000 Att/wk)         | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                               |                                                                                              |
| <b>LW461</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>THE 14TH AMENDMENT</b>          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-2                             | <b>Offerings:</b>                                                                            |
| <p>Equal protection is an important aspect of the rule of law, and underlies the American approach to justice. This course uses case law to explore the struggle for civil rights in the United States by those discriminated against based on race, religion, national origin, sex, and sexual orientation. The course will focus on legal decisions that empowered change and the movement toward equality, as well as an understanding of the importance of diversity.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                    | 2026-2 2027-2 2028-2<br>2029-2                                                               |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 40 @ 55 min (2.500 Att/wk)         | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                               |                                                                                              |
| <b>LW462</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>CYBER LAW</b>                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                             | <b>Offerings:</b>                                                                            |
| <p>Cyberspace has become an emerging area for legal regulation and military operations. This domain poses challenges and legal issues that impact criminal law, constitutional law, and international law. This course approaches cyber law from multiple perspectives including both international and domestic regulations. Specifically, the course will focus on international and customary law aspects of cyber operations along with U.S. law pertaining to cyberspace.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                    | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-8<br>2028-1 2028-2 2028-8<br>2029-1 2029-2 2029-8 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 40 @ 55 min (2.500 Att/wk)         | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                               |                                                                                              |
| <b>LW472</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>CRIMINAL LAW</b>                | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                                                                                                                                                 | <b>Offerings:</b>                                                                            |
| <p>This course will examine the legal, social, religious, cultural, and political motivations that justice systems use to characterize certain actions as "criminal." The course will revolve around the traditional reasons for criminal law, namely blameworthiness and punishment, and also examine how institutions use criminal law to serve their narrow interests. This course will introduce theories surrounding criminal law and illustrate how cadets may apply law immediately in their roles as officers. The course will examine federal and state criminal codes and also the Uniform Code of Military Justice. From a legal perspective based on the U.S. Constitution and other criminal codes, some of the topics covered include the death penalty, insanity, corporate crime, conspiracy, murder, necessity, and self-defense.</p>                                                                                 |                                                                                                                                                        | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 30 @ 75 min (2.000 Att/wk)                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                                                   |                                                                                              |
| <b>LW473</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ENVIRONMENTAL LAW</b>                                                                                                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                                                                                                                                                 | <b>Offerings:</b>                                                                            |
| <p>Environmental law has become an integral part of the legal system in the United States today. This course provides an introduction to environmental issues and the framework of the major federal environmental statutes (the National Environmental Policy Act, Clean Water Act, Clean Air Act, Endangered Species Act, etc.), and how the law works in practice. The course also covers environmental issues in the military and the growing subject of International Environmental Law. This course provides a solid understanding of the legislative, administrative and judicial system of environmental law today.</p>                                                                                                                                                                                                                                                                                                        |                                                                                                                                                        | 2026-1 2027-1 2028-1<br>2029-1                                                               |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 30 @ 75 min (2.000 Att/wk)                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Four written partial reviews and a TEE. Two homework assignments.                                                                                      |                                                                                              |
| <b>LW474</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>LAW OF ARMED CONFLICT</b>                                                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                                                                                                                                                 | <b>Offerings:</b>                                                                            |
| <p>This course is designed to develop in each student an understanding of basic law of armed conflict (LOAC), with an emphasis on issues that might arise on the battlefield at a tactical level. The ethical and historical background of LOAC will be examined, including Geneva Conventions and protocols, and how LOAC is enforced on international and national levels, to include prosecution under the Uniform Code of Military Justice. Illustrative examples will include the Nuremberg Tribunal, My Lai, and the Gulf War. The emphasis is on the LOAC responsibilities of the junior officer.</p>                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                        | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-8<br>2028-1 2028-2 2028-8<br>2029-1 2029-2 2029-8 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 40 @ 55 min (2.500 Att/wk)                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Two research papers of moderate length (approximately five pages each) may be required. Topics will be determined in consultation with the instructor. |                                                                                              |
| <b>LW475</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ADV CONSTITUTIONAL LAW SEM</b>                                                                                                                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                                                                                                                                                 | <b>Offerings:</b>                                                                            |
| <p>This seminar course covers a broad range of traditional and contemporary constitutional law topics. In addition to studying U.S. Supreme Court cases in particular areas of constitutional law, cadets are given an opportunity to study the historical foundations of the U.S. Constitution and underlying theories and principles of constitutionalism. The seminar format demands active participation in classroom debate, role playing, and critical thinking about complex issues of law and policy. As part of the seminar curriculum, each cadet will assume the role of a Supreme Court Justice. In this role, the cadet will study a real case pending before the Supreme Court and will write an abbreviated opinion reflecting the cadet's decision based on principled reasoning. The seminar typically travels to the Supreme Court to hear argument in the studied case as part of the opinion writing exercise.</p> |                                                                                                                                                        | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2                                       |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 30 @ 75 min (2.000 Att/wk)                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Preparation of a Supreme Court "opinion" (10, double spaced, typewritten pages).                                                                       |                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LW403<br>-Or-<br>LW403L                                                                                                                                |                                                                                              |
| <b>LW476</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ADVANCED LAW OF ARMED CONFLICT</b>                                                                                                                  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-2                                                                                                                                                 | <b>Offerings:</b>                                                                            |

The modern battlefield has had a significant impact on the Law of Armed Conflict in a variety of ways. This course builds on the lessons from LW474 and explores complex issues in modern conflict like Human Rights, Drone Warfare, Civilians on the Battlefield, and War Crimes. The lessons in this course will challenge the cadets to think critically and creatively about the application of the law on the modern battlefield.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** LW474

2026-2 2027-2 2028-2  
2029-2

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| <b>LW477</b> | <b>MILITARY JUSTICE: FOUNDATIONS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course introduces future officers to the characteristics that mark American military justice as distinctive, useful, protective, and in some ways inherently problematic. In critically examining military law's practice and procedure, cadets will view military justice as an evolving system within a framework that combines Constitutional structure and requirements, civil-military relations among uniformed professionals, the civilian Commander-in-Chief, and Congress.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>LW481</b> | <b>INTERNATIONAL LAW</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

The field of international law is one of the most dynamic areas of the law, and its principles are often applied in addressing the complex security problems facing our nation. This course will familiarize cadets with the body of rules and expectations which govern the rights and obligations of states and international organizations, during both times of peace and conflict. In particular, this course will emphasize the aspects of international law that are relevant to the operational and tactical problems that officers will confront as they deploy overseas. This course will integrate some themes from other courses in the Department of Law and reinforce some fundamental principles officers will confront as they seek to understand the relationship between law, diplomacy, and military operations.

2026-1 2026-2 2026-8  
2026-9 2027-1 2027-2  
2027-8 2027-9 2028-1  
2028-2 2028-8 2028-9  
2029-1 2029-2 2029-8  
2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>LW482</b> | <b>NATIONAL SECURITY LAW</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This seminar examines the legal framework for national security decisions. Cadets will analyze the delicate balance of liberty and security that must exist to preserve a democratic society. Particular areas include: constitutional separation of powers and shared responsibility for national security; the legality and scope of war and other uses of armed force short of war; access to and protection of sensitive information; intelligence collection and clandestine activities; the role of the media, responses to terrorism and international organized crime; and the formulation of national security policy and law.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2  
2029-8

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>LW488</b> | <b>BUSINESS LAW</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course introduces cadets to the basics of business and commercial law. Contractual principles under the common law and Uniform Commercial Code are emphasized. Current legal issues in the following areas are explored: consumer protection; real, personal, and intellectual property law; antitrust law; and employment discrimination. Included is a survey of the basic principles of government contracting law. Additionally, cadets engage in business negotiations exercises. This course employs both case study and problem-solving methods of instruction.

2026-1 2027-1 2028-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

| LW489                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SEMINAR & INDEP RESEARCH                                                                                                                                  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                             |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025-2                                                                                                                                                    | <b>Offerings:</b>                                                                                      |
| The course guides the student to conduct intensive study in a legal subject area for which the cadet has a special interest. Each student enrolled in the course will work with a designated member of the Law faculty to develop and pursue a specific research topic. The course will be conducted in three phases, followed by a formal presentation. In the first phase, the student will work with the faculty advisor to develop a research topic, and plan a research strategy. Students in the second phase will conduct research under the guidance of the faculty member. In the third phase the student will document the research he or she has completed in a paper and refine the paper through a draft/review process with the faculty member. The final phase requires an oral presentation and defense of the research. |                                                                                                                                                           | 2026-1 2027-2 2028-2                                                                                   |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                                                                                                                                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | One paper approximately 30 pages or more in length, oral presentation and defense of findings and conclusions.                                            |                                                                                                        |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LW310 LW403                                                                                                                                               |                                                                                                        |
| LW490                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SPECIAL TOPICS IN THE LAW                                                                                                                                 | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025-1                                                                                                                                                    | <b>Offerings:</b>                                                                                      |
| An in-depth seminar course concentrating on a single area of the law. The course is conducted by the Department's Visiting Professor or a Law faculty member when the visiting professor is unavailable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                           | 2026-2 2026-8 2027-2<br>2027-8 2027-9 2028-1<br>2028-2 2028-8 2028-9<br>2029-1 2029-2 2029-8<br>2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | One or more essays, as determined by the course instructor.                                                                                               |                                                                                                        |
| LW490A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SPECIAL TOPICS IN THE LAW - A                                                                                                                             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025-1                                                                                                                                                    | <b>Offerings:</b>                                                                                      |
| An in-depth seminar course concentrating on a single area of the law. The course is conducted by the Department's Visiting Professor or a Law faculty member when the visiting professor is unavailable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                           | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2                                                           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | One or more essays, as determined by the course instructor.                                                                                               |                                                                                                        |
| LW490B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SPECIAL TOPICS IN THE LAW - B                                                                                                                             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                                                                                                                                    | <b>Offerings:</b>                                                                                      |
| An in-depth seminar course concentrating on a single area of the law. The course is conducted by the Department's Visiting Professor or a Law faculty member when the visiting professor is unavailable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                           | 2026-1 2027-1                                                                                          |
| <b>Lessons:</b> 30 @ 0 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min                                                                                                                                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                                                                                                      |                                                                                                        |
| LW495                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | JURISPRUDENCE AND LEGAL THEORY                                                                                                                            | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025-2                                                                                                                                                    | <b>Offerings:</b>                                                                                      |
| This is the capstone course for both the American Legal Studies and International and Comparative Legal Studies Majors. The course is an advanced seminar in legal philosophy as applied to contemporary domestic and international legal issues. It analyzes these issues using the perspectives of jurisprudence (the ideas and reasoning of jurists) and legal theory (using insight from disciplines such as science, economics, and political theory to address legal problems). It explores theoretical and practical approaches to identifying, developing, and preserving the rule of law. The course integrates legal coursework throughout the Academy curriculum and the Cadet's respective legal studies major.                                                                                                              |                                                                                                                                                           | 2026-2 2027-2 2028-2<br>2029-2                                                                         |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                    |                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Three written partial reviews and a final paper which analyzes a contemporary legal problem using the analytical tools of jurisprudence and legal theory. |                                                                                                        |



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| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LW310 LW403<br>-Or-<br>LW403L                                                                                                                                                                                        |                                                                                                                                |
| <b>LW498</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>THESIS I: PROPOSAL &amp; RESEARCH</b>                                                                                                                                                                             | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                       |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                                                                                                                                                                                                               | <b>Offerings:</b>                                                                                                              |
| <p>The purpose of the Senior Thesis is to provide cadets with the opportunity to create a project that is academically, professionally, and personally meaningful to them and that reflects their thinking and abilities as developed at West Point and in the Department of Law. Through the scholarly project that results from this course, cadets will be expected to show how they and their work have progressed and that their work is of professional quality. Cadets will choose a faculty advisor with whom they will work over two semesters. In collaboration with the faculty advisor, cadets will explore their chosen areas of law with a goal of producing a project, usually a thirty page paper that is of professional quality. This paper will be completed during LW499. Cadets will meet individually with their advisors on a regular basis to discuss the law, progress on the thesis, and developmental issues.</p> |                                                                                                                                                                                                                      | 2026-1 2027-1 2028-1<br>2029-1                                                                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cadets will not be required to attend classes, but may be required to individually attend a small number of conferences with their advisors and will be expected to submit written progress reports to the advisors. |                                                                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LW310 LW403<br>-Or-<br>LW403L                                                                                                                                                                                        |                                                                                                                                |
| <b>LW499</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>THESIS II: PAPER &amp; DEFENSE</b>                                                                                                                                                                                | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                       |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                                                                                                                                                                                                               | <b>Offerings:</b>                                                                                                              |
| <p>This course continues the work on the thesis commenced in LW498. At the end of the course, cadets will submit their theses to the Department of Law and orally defend their theses before a faculty committee.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                      | 2026-2 2027-2 2028-2<br>2029-2                                                                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cadets will not be required to attend classes, but may be required to individually attend a small number of conferences with their advisors and will be expected to submit written progress reports to the advisors. |                                                                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | LW498                                                                                                                                                                                                                |                                                                                                                                |
| <b>PY189X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>IND STUDY IN PHILOSOPHY (1CR)</b>                                                                                                                                                                                 | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                       |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                                                                                                                                                                                                               | <b>Offerings:</b>                                                                                                              |
| <p>The cadet pursues research topics in Philosophy of his or her choice. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. This cadet-created plan should equate to 40 hours' worth of research culminating in an appropriate term-end project worth 1CR, such as a poster presentation on Project's Day.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                      | 2029-1                                                                                                                         |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                                                                                                                                                 |                                                                                                                                |
| <b>PY201</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>PHILOSOPHY &amp; ETHICAL REASONING</b>                                                                                                                                                                            | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b>                                                                       |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2024-5                                                                                                                                                                                                               | <b>Offerings:</b>                                                                                                              |
| <p>This course helps third-class Cadets develop their capacities to think clearly and critically and to write in a logically rigorous and conceptually precise manner. The course acquaints Cadets with various viewpoints on major philosophic issues, a significant part of it is devoted to developing Cadets' ethical reasoning ability. The study of ethical theory provides the foundation for this ability as well as the foundation for their Academy's character development efforts. The course also includes a substantial block on the ethical application of military power through the study of the Just War Tradition.</p>                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                      | 2026-1 2026-2 2026-5<br>2026-8 2027-1 2027-2<br>2027-5 2027-8 2028-1<br>2028-2 2028-5 2028-8<br>2029-1 2029-2 2029-5<br>2029-8 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                                                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | A few essays of moderate length.                                                                                                                                                                                     |                                                                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | EN102<br>-Or-<br>EN152                                                                                                                                                                                               |                                                                                                                                |

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| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PY251                                 |                                                               |
| <b>PY251</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>ADVANCED PHILOSOPHY AND ETHICS</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2025-1                                | <b>Offerings:</b>                                             |
| This course is an advanced introduction to philosophy and ethics. It helps third-class Cadets develop their capacities to think clearly and critically and to write in a logically rigorous and conceptually precise manner. The course acquaints Cadets with various viewpoints on major philosophic issues, and a significant part of the course is devoted to developing Cadets' ethical reasoning ability. The study of ethical theory provides the foundation for this ability as well as the foundation for the Academy's character development efforts. The course also includes a substantial block on the ethical application of military power through the study of the Just War Tradition.                                           |                                       | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                  |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | EN102<br>-Or-<br>EN152                |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PY201                                 |                                                               |
| <b>PY300</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>PHILOSOPHICAL METHODS</b>          | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2025-1                                | <b>Offerings:</b>                                             |
| This course provides cadets the methodological tools required to analyze and understand the important moments and topics in philosophy, developing the philosophical language necessary for success within the philosophy curriculum. Through the study of philosophy within the western tradition, cadets will learn about major developments in logic, metaphysics, epistemology, and ethics, among other areas. This background provides the foundational knowledge required of a philosophy major.                                                                                                                                                                                                                                          |                                       | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A few essays of moderate length.      |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PY201<br>-Or-<br>PY251                |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PY333                                 |                                                               |
| <b>PY305</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>LOGICAL REASONING</b>              | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2025-2                                | <b>Offerings:</b>                                             |
| This course blends two areas of study that are often kept separate in university courses on logic: informal logic and formal (or symbolic) logic. Informal logic's emphasis is on natural language arguments relatively simple in structure, on rules of valid inference as codified in what is called traditional logic, and on the identification of mistakes in reasoning that make arguments logically weak though possibly persuasive (fallacies). Formal logic builds a symbolic representation of sentences and arguments, describes rigorous tests for determining whether symbolized arguments are valid, and provides the means to assess arguments of far greater complexity than the rules of traditional logic are able to manage. |                                       | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | A few essays of moderate length.      |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PY201<br>-Or-<br>PY251                |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PY359                                 |                                                               |
| <b>PY310</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>REALITY AND KNOWLEDGE</b>          | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2025-1                                | <b>Offerings:</b>                                             |

This course will address the perennial questions concerning the nature of reality (metaphysics) and what we can know about it (epistemology). How do we acquire knowledge of the physical world, the nonphysical world? Are there noncorporeal entities (souls, deities, angels)? If so, what can we claim to know about them? How are belief and knowledge related? A systematic and comprehensive approach to these problems and others will entail reading works by Plato, Aristotle, Descartes, Locke, Leibniz, Hume, and Kant, as well as more recent metaphysicians and epistemologists.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** PY201  
-Or-  
PY251

**Disqualifier(s):** PY383

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|--------------|---------------|---------------------------------------------------|
| <b>PY320</b> | <b>ETHICS</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-2 | <b>Offerings:</b> |
| <p>This course offers a systematic examination and comparison of standard Ethical doctrines as well as an analysis of some of the fundamental concepts and assumptions belonging to the nature of ethical thinking itself (Meta-ethics). The ethical doctrines to be studied include those associated with renowned philosophers such as Aristotle (virtue theory), Kant (deontology), and Mill (utilitarianism). The focus will be not only on original texts which advance the doctrines but also on criticisms and defenses of them by contemporary philosophers. The texts of Meta-ethics to be studied belong to the analytic tradition of Philosophy and concern the meaning and status of normative language in general. PY320 provides a worthwhile background Cadets may apply in any course in applied Ethics, such as PY325 Military Ethics and PY326 Cyber Ethics. It will also prove useful to Cadets in other academic majors, particularly in Political Theory, Law, and History.</p> |        | 2027-2 2029-2     |

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** PY201  
-Or-  
PY251

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|--------------|------------------------|---------------------------------------------------|
| <b>PY325</b> | <b>MILITARY ETHICS</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1 | <b>Offerings:</b>                                                                                      |
| <p>The fundamental values and principles of the warrior ethos can be traced back to ancient Greece and Rome. These values provide the moral boundaries of the military profession and distinguish members of this profession from other individuals and groups who employ violence to achieve their ends. Cadets in this course will examine the moral principles that define the profession of arms, both in terms of when the use of force is permissible (or even obligatory) to achieve political objectives, and what, if any, limits ought to govern how that force is used.</p> |        | 2026-1 2026-8 2026-9<br>2027-1 2027-8 2027-9<br>2028-1 2028-8 2028-9<br>2029-1 2029-2 2029-8<br>2029-9 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** PY201  
-Or-  
PY251

**Disqualifier(s):** PY365

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|--------------|-----------------------------|---------------------------------------------------|
| <b>PY326</b> | <b>ETHICS OF TECHNOLOGY</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025-2 | <b>Offerings:</b>                            |
| <p>This course explores the ethical, social, and legal issues arising from emerging technologies, with a focus on the internet, artificial intelligence, and robotics. Students will examine how these technologies shape society, influence personal and professional life, and challenge existing ethical frameworks. Topics include data privacy, cybersecurity, algorithmic bias, the ethics of AI decision-making, the impact of automation on jobs, and the potential consequences of autonomous systems in warfare and everyday life. The course will also address legal regulations and policy debates, encouraging students to critically evaluate the balance between innovation and the responsibility to safeguard human rights and societal well-being.</p> |        | 2026-1 2027-1 2027-8<br>2028-1 2028-8 2029-1 |

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PY201<br>-Or-<br>PY251                                         |                                                   |
| <b>PY329</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>TOPICS IN ETHICS</b>                                        | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-2                                                         | <b>Offerings:</b>                                 |
| This course provides cadets an opportunity for reading and analysis in depth of some of the seminal philosophical works in ethics. Taught in seminar format, the course challenges first-class and second-class cadets to take responsibility for discussion and analysis and for drawing connections between ideas as they occur throughout history and across cultures. The cadets will gain a deeper understanding of the human condition and of the complex world of values. |                                                                | 2026-2 2028-2                                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                         |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                           |                                                   |
| <b>PY329A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>TOPICS IN ETHICS-A</b>                                      | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-2                                                         | <b>Offerings:</b>                                 |
| This course provides cadets an opportunity for reading and analysis in depth of some of the seminal philosophical works in ethics. Taught in seminar format, the course challenges first-class and second-class cadets to take responsibility for discussion and analysis and for drawing connections between ideas as they occur throughout history and across cultures. The cadets will gain a deeper understanding of the human condition and of the complex world of values. |                                                                | No Course Offerings                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                         |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PY329 must be taken before a cadet can take PY329A.            |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PY329                                                          |                                                   |
| <b>PY329B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>TOPICS IN ETHICS-B</b>                                      | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                                                         | <b>Offerings:</b>                                 |
| This course provides cadets an opportunity for reading and analysis in depth of some of the seminal philosophical works in ethics. Taught in seminar format, the course challenges first-class and second-class cadets to take responsibility for discussion and analysis and for drawing connections between ideas as they occur throughout history and across cultures. The cadets will gain a deeper understanding of the human condition and of the complex world of values. |                                                                | No Course Offerings                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                         |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PY329 and PY329A must be taken before a cadet can take PY329B. |                                                   |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PY329 PY329A                                                   |                                                   |
| <b>PY330</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>POLITICAL PHILOSOPHY</b>                                    | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                                                         | <b>Offerings:</b>                                 |
| Examining the major theories and problems in the history of political philosophy from Plato to Rawls and emphasizing contemporary theory, this course includes such topics as liberty, equality, political authority, the obligation to obey the State, civil disobedience, anarchism, liberalism, conservatism, democracy, meritocracy, and global politics.                                                                                                                    |                                                                | 2027-1 2029-1                                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                         |                                                   |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | A few essays of moderate length.                               |                                                   |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PY201<br>-Or-<br>PY251                                         |                                                   |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PY363                                                          |                                                   |
| <b>PY345</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>PHILOSOPHY OF RELIGION</b>                                  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                                                         | <b>Offerings:</b>                                 |

This course examines the nature of religion and its truth claims from the perspective of philosophical analysis. It examines such perennial questions as: is there a God? What are the arguments for and against the existence of a Supreme Being? How can a good God permit Evil? Is there life after death? Is it rational to believe in God or does faith stand above or against reason? What is the relationship of religion to ethics? Is the Good good because God commands it, or does God command the Good because it is good?

2026-1 2028-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** PY201  
-Or-  
PY251

**Disqualifier(s):** PY381

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| <b>PY350</b> | <b>PHILOSOPHY OF SCIENCE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

Mathematics and the sciences (especially the natural sciences) have often been portrayed in the modern era as paradigmatic sources of knowledge. Nevertheless, one can still pose a number of lively and much-debated questions: what makes something a science? Is there a single "scientific method" or ideal way of discovering, confirming, or disconfirming scientific truths? Are there limitations to the knowledge the sciences can provide? Indeed, do the sciences provide knowledge? Does science make any presuppositions about the nature of the world or about what exists (ontology)? What is the nature of mathematics? Does it apply to a world of ideal objects, to rules for using symbols, or to the physical world? What kinds of things are numbers? Readings will include works by Peirce, Frege, the Vienna Circle, and Kuhn, as well as contemporary readings in the philosophy of science and mathematics and in the philosophies of physics, biology, the social sciences, and logic.

2026-2 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** PY201  
-Or-  
PY251

**Disqualifier(s):** PY386

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|--------------|---------------------------|----------------------------------------------------------|
| <b>PY355</b> | <b>PHILOSOPHY OF MIND</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course will jointly address major topics in the traditional philosophy of mind and questions created by recent developments in artificial intelligence: what is mind? What is the relationship of a mind to the physical world, including the brain? What is consciousness and self-consciousness? What are the definitions of mental states and processes, such as perception, desire, belief, emotion, reasoning, and action, and their relationship? Can computers be constructed to think or behave like human beings, or to have consciousness? Readings will come from classical sources, such as Descartes, as well as contemporary literature in philosophy, cognitive science, and artificial intelligence.

2027-2 2027-9 2028-9  
2029-2 2029-8 2029-9

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

**Corequisite(s):** PY201  
-Or-  
PY251

**Disqualifier(s):** PY366

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| <b>PY360</b> | <b>ANCIENT PHILOSOPHY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

The heritage from ancient Greece and Rome provides the foundation for the Western concept of the universe and the place of people in it. This course examines the origins of philosophy; the essentially secular view of man and the world established during the classical period; and major figures whose views continue to shape Western thought.

2026-1 2026-9 2028-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** A few essays of moderate length.

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| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PY201<br>-Or-<br>PY251                    |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PY388                                     |                                                    |
| <b>PY369</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>ASIAN PHILOSOPHICAL TRADITIONS</b>     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-2                                    | <b>Offerings:</b><br>2026-2 2028-2                 |
| This course focuses on philosophies originating in Asia, possibly including, but not necessarily limited to, those of India, China, Japan, and Korea. These philosophical traditions offer a valuable opportunity to broaden a student's intellectual perspective because they are largely free from the influence of the Hellenic philosophers that are so foundational to the Anglo-European canon. The course emphasizes engagement with primary source texts and attempts to locate those texts within the social, political, and historical contexts from which they emerged. The particular content of the course may vary in terms of the traditions explored and historical timeframe but will always do so in a way that respects the richness of the many distinct philosophies that have emerged from the Asian continent. |                                           |                                                    |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | A few essays of moderate length.          |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PY201<br>-Or-<br>PY251                    |                                                    |
| <b>PY370</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>17TH &amp; 18TH CENTURY PHILOSOPHY</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-1                                    | <b>Offerings:</b><br>2027-1 2029-1                 |
| This course examines a selection of texts written by central figures in the formative centuries of modern European philosophy. Their ideas have had continuing influence on philosophers down to our present day, as well as profound influence on the development of political thought and on the scientific understanding of human beings. Two schools of thought will be covered: Rationalism and Empiricism. Associated with the first school are the continental philosophers Descartes (widely accepted as the founder of Modern Philosophy), Spinoza and Leibniz. The school of Empiricism includes the British philosophers Hobbes, Locke, Berkeley and Hume.                                                                                                                                                                 |                                           |                                                    |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | None                                      |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -Or-<br>EP375                             |                                                    |
| <b>PY375</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>KANT &amp; 19TH CENTURY PHILOSOPHY</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-2                                    | <b>Offerings:</b><br>2027-2 2029-2                 |
| This course gives primary attention to the systematic philosophy of the German thinker, Immanuel Kant, whose influence on Nineteenth Century thinking was widespread and who is commonly recognized as one of the pillars of Modern Philosophy. The course will also devote attention to other important areas of philosophical thinking in the Nineteenth Century, whether within the Kantian tradition or lying outside it in other movements, such as Utilitarianism or Pragmatism, which had a continuing and significant influence on later philosophical thinking.                                                                                                                                                                                                                                                              |                                           |                                                    |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | None                                      |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | PY376                                     |                                                    |
| <b>PY380</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>20TH CENTURY PHILOSOPHY</b>            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-2                                    | <b>Offerings:</b><br>2027-2 2029-2                 |
| This course will introduce cadets to a representative sample major of figures and topics which have set the stage for understanding contemporary Philosophy in the so-called Analytic Tradition. Major figures include Frege, Russell, Wittgenstein, Moore, the philosophers of the Vienna Circle, and American philosophers such as Quine, Putnam, Davidson and Kripke. Topics include the ideal of a logically perfect language, meaning and reference, the nature of truth, the distinction between analytic and synthetic statements, the common sense analysis of metaphysical concepts, and the rule-centered social nature of language. As appropriate, leading figures and ideas drawn from Continental Philosophy will be introduced.                                                                                        |                                           |                                                    |

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** PY201  
-Or-  
PY251

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| <b>PY390</b> | <b>INTER-DEPARTMENT SEMINAR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-2 | <b>Offerings:</b> |
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PY390 brings together a faculty member from DEP and one other department with Cadets majoring in those two departments with the aim of conducting a joint investigation of an important topic or cluster of topics, or the work of a single author, of recognized significance and shared interest. It will be taught every other year, on each occasion combining Philosophy with another discipline. Examples of second disciplines include History, Political Theory, Psychology, and Law. Examples of topics include: justice, philosophies of history, the evolution of human rights theory, the relationship between morality and law, cognition and mental phenomena, and evolving conceptions of citizenship. Examples of single author investigations include: Locke, Rousseau, Hume. The course will count as credit towards the major in both of the paired departments.

2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** PY201  
-Or-  
PY251

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| <b>PY390A</b> | <b>INTER-DEPARTMENT SEMINAR-A</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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PY390 brings together a faculty member from DEP and one other department with Cadets majoring in those two departments with the aim of conducting a joint investigation of an important topic or cluster of topics, or the work of a single author, of recognized significance and shared interest. It will be taught every other year, on each occasion combining Philosophy with another discipline. Examples of second disciplines include History, Political Theory, Psychology, and Law. Examples of topics include: justice, philosophies of history, the evolution of human rights theory, the relationship between morality and law, cognition and mental phenomena, and evolving conceptions of citizenship. Examples of single author investigations include: Locke, Rousseau, Hume. The course will count as credit towards the major in both of the paired departments.

No Course Offerings

**Lessons:** 30 @ 75 min (0.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** PY390 must be taken before a cadet can take PY390A.

**Prerequisite(s):** PL390

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| <b>PY390B</b> | <b>INTER-DEPARTMENT SEMINAR-B</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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PY390 brings together a faculty member from DEP and one other department with Cadets majoring in those two departments with the aim of conducting a joint investigation of an important topic or cluster of topics, or the work of a single author, of recognized significance and shared interest. It will be taught every other year, on each occasion combining Philosophy with another discipline. Examples of second disciplines include History, Political Theory, Psychology, and Law. Examples of topics include: justice, philosophies of history, the evolution of human rights theory, the relationship between morality and law, cognition and mental phenomena, and evolving conceptions of citizenship. Examples of single author investigations include: Locke, Rousseau, Hume. The course will count as credit towards the major in both of the paired departments.

No Course Offerings

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** PY390 and PY390A must be taken before a cadet can take PY390B.

**Prerequisite(s):** PY390 PY390A

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| <b>PY395</b> | <b>SPECIAL TOPICS IN PHILOSOPHY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2025-1 | <b>Offerings:</b> |
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This course explores an advanced topic in Philosophy. Specific subject matter will vary with the expertise of the senior faculty member conducting the course. 2027-1 2029-1 2029-8

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>PY400</b> | <b>SENIOR SEMINAR IN PHILOSOPHY</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2 **Offerings:**

This course provides cadets with the opportunity for advanced study in the discipline. Through the advanced study of a topic in philosophy, cadets will build on the foundation established in PY300 and throughout their academic career at West Point. They will deepen their mastery of philosophical concepts and methods and grow as scholars by applying those concepts and methods to a number of different disciplinary perspectives. Through intensive study of primary and secondary texts, this course broadens the knowledge base by bridging disciplinary approaches and setting the stage for cadets' continued educational development. 2026-2 2027-2 2028-2 2029-2

**Lessons:** 16 @ 140 min (1.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** A major project and a few essays of moderate length.

**Corequisite(s):** PY300  
-Or-  
PY333

**Disqualifier(s):** PY433

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| <b>PY433</b> | <b>PHILOSOPHY SENIOR SEMINAR</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1 **Offerings:**

This course provides cadets with the opportunity for advanced study in the discipline. Through the advanced study of a topic in philosophy, cadets will build on the foundation established in PY333 and throughout their academic career at West Point. They will deepen their mastery of philosophical concepts and methods and grow as scholars by applying those concepts and methods to a number of different disciplinary perspectives. Through intensive study of primary and secondary texts, this course broadens the knowledge base by bridging disciplinary approaches and setting the stage for cadets' continued educational development. No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** A major project and a few essays of moderate length.

**Corequisite(s):** PY333  
-Or-  
EP333

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| <b>PY490</b> | <b>THESIS RESEARCH</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1 **Offerings:**

This course permits cadets with the requisite energy and talent to initiate a yearlong project requiring research in depth that culminates in a substantial thesis of high scholarly quality. 2026-1 2027-1 2028-1 2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>PY491</b> | <b>SENIOR THESIS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2 **Offerings:**

This course permits cadets to complete a yearlong project requiring research in depth that culminates in a substantial thesis of high scholarly quality. 2026-2 2027-2 2028-2 2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** Oral defense of thesis.

**Prerequisite(s):** PY490



| <b>PY493X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>PHILOSOPHY CAPSTONE A</b>         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                               | <b>Offerings:</b>                                  |
| <p>This course permits cadets to participate in a capstone project that requires application of philosophical research and thought to a real-world problem. It is intended to develop the participants' philosophical research, writing and logical argumentation skills and have them gain collaborative communication skills as they work with industry partners. It is normally paired with PY494X, making it a year-long effort, but it can also stand alone depending on the scope of the project. If done together with PY494X, the two credits together will count towards the honor track of the Philosophy Major in lieu of PY490/91. *This is a pilot course that must be reviewed by the Curriculum Committee NLT AY26-2 to continue.**</p> |                                      | 2026-1 2027-1                                      |
| <b>Lessons:</b> 30 @ 75 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min               |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PY201<br>-Or-<br>PY251               |                                                    |
| <b>PY494X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>PHILOSOPHY CAPSTONE B</b>         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-2                               | <b>Offerings:</b>                                  |
| <p>This course serves as the spring course of the capstone project, normally paired with PY493X. It permits cadets to continue their participation in a year's long capstone project applying philosophical research to a real-world problem. In this second course, the participants will continue their research and collaboration and present and/or publish their work in academic and professional settings. It can also serve as a standalone course during the spring term, depending on the scope of the project. If paired with PY493X, these two credits will count toward the honor track of the Philosophy Major in lieu of PY490/91.</p>                                                                                                  |                                      | 2026-2 2027-2                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min               |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PY201<br>-Or-<br>PY251               |                                                    |
| <b>PY495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INDEPENDENT STUDY: PHILOSOPHY</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-1                               | <b>Offerings:</b>                                  |
| <p>This optional elective offers the cadet an opportunity for in-depth study of an advanced topic in Philosophy under the mentorship of a senior faculty advisor. The scope and topic of the course are developed in consultation with the faculty advisor and appropriately build upon academic work already completed in the regular Philosophy electives. Since such a course is beyond normal teaching duties, an agreement to serve as a faculty advisor will be at the discretion of the faculty member. Enrollment is subject to Department approval.</p>                                                                                                                                                                                       |                                      | 2029-1 2029-2                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min               |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                    |
| <b>PY495A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>INDEPENDENT STUDY: PHILOSOPHY</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-2                               | <b>Offerings:</b>                                  |
| <p>This optional elective offers the cadet an opportunity for in-depth study of an advanced topic in Philosophy under the mentorship of a senior faculty advisor. The scope and topic of the course are developed in consultation with the faculty advisor and appropriately build upon academic work already completed in the regular Philosophy electives. Since such a course is beyond normal teaching duties, an agreement to serve as a faculty advisor will be at the discretion of the faculty member. Enrollment is subject to Department approval.</p>                                                                                                                                                                                       |                                      | 2029-2                                             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min               |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                 |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PY495                                |                                                    |
| <b>XH400</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INTERCOLLEGIATE SEMINAR</b>       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                        | <b>Offerings:</b>                                 |
| The focus of the course is a topic such importance that it deserves a multi-disciplinary examination. Previous iterations of the course carried out with Bard College, examined the nature and cultural traditions of a Just War, and the complex nature of human intolerance. The course has included faculty and Cadets from the departments of English and Philosophy, History, Law, and Social Sciences. Intended for cows and firsties, it counts for credit towards the major in their respective departments. The course is conducted as a seminar course meeting in two-hour blocks during which the Cadets have chief responsibility for the discussion. A parallel course at the other institution engages undergraduates and faculty representing a comparably wide range of disciplines. |                               | No Course Offerings                               |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 20 @ 110 min (1.000 Att/wk)   | <b>Labs:</b> 0 @ 0 min                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                          |                                                   |
| <b>ZH343</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>PHILOSOPHICAL PROBLEMS</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                        | <b>Offerings:</b>                                 |
| This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                               | No Course Offerings                               |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 40 @ 55 min (0.000 Att/wk)    | <b>Labs:</b> 0 @ 0 min                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                          |                                                   |
| <b>ZH353</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>HISTORY OF PHILOSOPHY</b>  | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                        | <b>Offerings:</b>                                 |
| This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                               | No Course Offerings                               |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 40 @ 55 min (0.000 Att/wk)    | <b>Labs:</b> 0 @ 0 min                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                          |                                                   |
| <b>ZH363</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>ETHICS</b>                 | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025-1                        | <b>Offerings:</b>                                 |
| This course, taken abroad by the cadet as part of a foreign study program approved by West Point, falls within the disciplinary area covered by the Department of English and has been determined by the Department as suitable to earn West Point academic credit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                               | No Course Offerings                               |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 40 @ 55 min (0.000 Att/wk)    | <b>Labs:</b> 0 @ 0 min                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                          |                                                   |

## Mathematical Sciences

### 61 Courses

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                           |                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>MA100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>PRECALCULUS MATHEMATICS</b>                            | <b>3.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.5)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2016-1                                                    | <b>Offerings:</b>                                                              |
| This course prepares cadets with background deficiencies in algebra and trigonometry for the core mathematics program. The course develops fundamental skills in algebra, trigonometry, and functions, through an introduction to mathematical modeling and problem solving. Since this course does not count toward graduation requirements; cadets enrolled in MA100 will forfeit an elective opportunity.                                                                                                                                                                                                                                                                                                                                                                                                            |                                                           | 2026-2 2027-2 2028-2<br>2029-2                                                 |
| <b>Lessons:</b> 41 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 17 @ 55 min                                  |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                                                      |                                                                                |
| <b>MA103</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>MATH MODELING/INTRO CALCULUS</b>                       | <b>4.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=4.5)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-1                                                    | <b>Offerings:</b>                                                              |
| This course presents fundamental concepts in data-driven decision-making, using mathematical models to solve real-world, interdisciplinary problems. Students gain experience transforming ill-defined and complex problems into appropriate mathematical models by identifying reasonable and necessary assumptions. The course emphasizes proper interpretation of findings and effective communication of technical concepts to general audiences. The course exercises a wide array of mathematical concepts while nurturing creativity and critical thinking. Specific topics may include data analytics, classification, optimization, matrices, vectors, and continuous/discrete functions. Students use technology to visualize problems, explore potential solutions, and perform time-consuming calculations. |                                                           | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1                         |
| <b>Lessons:</b> 56 @ 55 min (4.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 8 @ 55 min                                   |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                                                      |                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | -Or-<br>MA153                                             |                                                                                |
| <b>MA104</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>CALCULUS I</b>                                         | <b>4.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=4.5)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026-1                                                    | <b>Offerings:</b>                                                              |
| MA104 introduces the core concepts of differential and integral calculus, including limits, derivatives, integrals, and their applications. Students develop the ability to analyze and model change in one variable, solve real-world optimization and accumulation problems, and interpret the results of mathematical models. The course emphasizes conceptual understanding, problem-solving, and communication of results, with applications that may be drawn from areas such as the natural sciences, social sciences, engineering, and real-world decision-making contexts.                                                                                                                                                                                                                                     |                                                           | 2026-1 2026-2 2027-1<br>2027-2 2027-9 2028-1<br>2028-2 2028-9 2029-1<br>2029-2 |
| <b>Lessons:</b> 56 @ 55 min (4.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 8 @ 55 min                                   |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                                                      |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | MA103<br>-Or-<br>MA101<br>-Or-<br>MA103X<br>-Or-<br>MA153 |                                                                                |
| <b>MA153</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>MATH MODELING/INTRO DIF EQ</b>                         | <b>4.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=4.5)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2016-1                                                    | <b>Offerings:</b>                                                              |

This is the first course of a two-semester advanced mathematics sequence for selected cadets who have validated single variable calculus and demonstrated strength in the mathematical sciences. It is designed to provide a foundation for the continued study of mathematics, sciences, and engineering. This course emphasizes the interaction between mathematics and the physical sciences through modeling with differential equations. Topics may include a study of first order differential equations, first order difference equations, second order linear equations, systems of first order linear and non-linear equations, numerical methods, and non-linear equations and stability. An understanding of course material is enhanced through the use of a computer algebra system.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 56 @ 55 min (4.000 Att/wk)    **Labs:** 8 @ 55 min

**Special Requirements:** None

**Disqualifier(s):** MA103  
-Or-  
MA101

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|              |                          |                                                          |
|--------------|--------------------------|----------------------------------------------------------|
| <b>MA204</b> | <b>CALCULUS I AND II</b> | <b>4.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=4.5)</b> |
|--------------|--------------------------|----------------------------------------------------------|

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**Scope:** 2025-2

**Offerings:**

An alternative version of the second semester of the mathematics core curriculum. This course, intended for students who have had a full year of high school calculus, provides a short review of single-variable calculus before progressing into multi-variable differential and integral calculus. It will also cover vectors and geometry of Euclidean space, vector functions, and vector calculus. Throughout the course mathematical models motivate the study of topics such as optimization, accumulation, change in several variables, and other topics from the natural, social, and decision sciences. Students who successfully complete the course will receive validation credit for MA104 and equivalency credit for MA205.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 56 @ 55 min (4.000 Att/wk)    **Labs:** 8 @ 55 min

**Special Requirements:** None

**Prerequisite(s):** MA103  
-Or-  
MA153

**Disqualifier(s):** MA104  
-Or-  
MA205  
-Or-  
MA204X  
-Or-  
MA255

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|               |                          |                                                          |
|---------------|--------------------------|----------------------------------------------------------|
| <b>MA204X</b> | <b>CALCULUS I AND II</b> | <b>4.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=4.5)</b> |
|---------------|--------------------------|----------------------------------------------------------|

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**Scope:** 2024-2

**Offerings:**

An alternative version of the second semester of the mathematics core curriculum. This course, intended for students who have had a full year of high school calculus, provides a short review of single-variable calculus before progressing into multi-variable differential and integral calculus. It will also cover vectors and geometry of Euclidean space, vector functions, and vector calculus. Throughout the course mathematical models motivate the study of topics such as optimization, accumulation, change in several variables, and other topics from the natural, social, and decision sciences. Students who successfully complete the course will receive validation credit for MA104 and equivalency credit for MA205. \*\*This is a pilot course and must be approved by the Curriculum Committee NLT AY25-1 to continue.\*\*

No Course Offerings

**Lessons:** 56 @ 55 min (4.000 Att/wk)    **Labs:** 8 @ 55 min

**Special Requirements:** None

**Prerequisite(s):** MA103  
-Or-  
MA153

**Disqualifier(s):** -Or-  
MA205  
-Or-  
MA255  
-Or-  
MA104

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|              |                    |                                                          |
|--------------|--------------------|----------------------------------------------------------|
| <b>MA205</b> | <b>CALCULUS II</b> | <b>4.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=4.0)</b> |
|--------------|--------------------|----------------------------------------------------------|

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                 |                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                                          | <b>Offerings:</b>                                                                                                                                                                            |
| This course provides a foundation for the continued study of mathematics and for the subsequent study of the physical sciences, social sciences, and engineering. MA205 covers topics in multivariable differential and integral calculus, vectors and geometry of Euclidean space, vector functions, and vector calculus. Throughout the course mathematical models motivate the study of topics such as optimization, accumulation, change in several variables, and other topics from the natural, social, and decision sciences. An understanding of course material is enhanced through the use of computer algebra systems.                                                |                                                 | 2026-1 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2                                                                                                                                       |
| <b>Lessons:</b> 52 @ 55 min (3.250 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 4 @ 55 min                         |                                                                                                                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                            |                                                                                                                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MA104                                           |                                                                                                                                                                                              |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MA255                                           |                                                                                                                                                                                              |
| <b>MA206</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>PROBABILITY &amp; STATISTICS</b>             | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=3.0)                                                                                                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                          | <b>Offerings:</b>                                                                                                                                                                            |
| This is the final course in the mathematics core curriculum. The course develops each cadet's ability to structure their reasoning under conditions of uncertainty and presents fundamental probability and statistical concepts that support the USMA core curriculum. Coverage includes data analysis, probabilistic models, independence, simulation, random variables and their distributions, hypothesis testing, confidence intervals, and linear regression. The course also introduces engineering applications of probability and statistics techniques. Applied problems motivate concepts, and technology enhances understanding, problem solving, and communication. |                                                 | 2026-1 2026-2 2026-3<br>2026-5 2026-8 2026-9<br>2027-1 2027-2 2027-3<br>2027-5 2027-8 2027-9<br>2028-1 2028-2 2028-3<br>2028-5 2028-8 2028-9<br>2029-1 2029-2 2029-3<br>2029-5 2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                          |                                                                                                                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                            |                                                                                                                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | -Or-<br>MA255<br>-Or-<br>MA104<br>-Or-<br>MA204 |                                                                                                                                                                                              |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MA256                                           |                                                                                                                                                                                              |
| <b>MA255</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ADV MULTIVARIABLE CALCULUS</b>               | <b>4.5 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=4.5)                                                                                                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2016-2                                          | <b>Offerings:</b>                                                                                                                                                                            |
| This is the second course of a two-semester advanced mathematics sequence for selected cadets who have validated single variable calculus and demonstrated strength in the mathematical sciences. It is designed to provide a foundation for the continued study of mathematics, sciences, and engineering. This course consists of an advanced coverage of topics in multivariable calculus. Topics may include a study of infinite sequences and series, vectors and the geometry of space, vector functions, partial derivatives, multiple integrals, and vector calculus. An understanding of course material is enhanced through the use of a computer algebra system.      |                                                 | 2026-2 2027-2 2028-2<br>2029-2                                                                                                                                                               |
| <b>Lessons:</b> 56 @ 55 min (4.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 8 @ 55 min                         |                                                                                                                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                            |                                                                                                                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MA153                                           |                                                                                                                                                                                              |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MA205                                           |                                                                                                                                                                                              |
| <b>MA256</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ADV PROBABILITY AND STATISTICS</b>           | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=3.0)                                                                                                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2026-1                                          | <b>Offerings:</b>                                                                                                                                                                            |

This is the advanced version of the final course in the mathematics core curriculum. The course develops each cadet's ability to structure their reasoning under conditions of uncertainty and presents fundamental probability and statistical concepts that support the USMA core curriculum. Coverage includes data analysis, probabilistic models, independence, simulation, random variables and their distributions, hypothesis testing, confidence intervals, and linear regression. The course also introduces engineering applications of probability and statistics techniques. Applied problems motivate concepts, and technology enhances understanding, problem solving, and communication.

2027-1 2027-2 2028-1  
2028-2 2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MA104  
-Or-  
MA255  
-Or-  
MA204

**Disqualifier(s):** MA206

## MA289 INTRO IND STUDY IN MATH (1CR)

**1.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2024-2

**Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Mathematical Sciences, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform a total of 40 hours of work for the semester. A final presentation and final project deliverable are required.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Mathematical Sciences. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

## MA289A INTRO IND STUDY IN MATH (1CR)

**1.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2024-2

**Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Mathematical Sciences, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform a total of 40 hours of work for the semester. A final presentation and final project deliverable are required.

2026-1 2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Mathematical Sciences. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

**Prerequisite(s):** MA289

## MA289B INTRO IND STUDY IN MATH (1CR)

**1.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2024-2

**Offerings:**

This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Mathematical Sciences, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform a total of 40 hours of work for the semester. A final presentation and final project deliverable are required.

2026-1 2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of one of the Program Directors in the Department of Mathematical Sciences. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required.

**Prerequisite(s):** MA289 MA289A

| MA289C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | INTRO IND STUDY IN MATH (1CR)                                                                                                                                                                                | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-2                                                                                                                                                                                                       | <b>Offerings:</b>                                                    |
| This course is an individually supervised research and study program. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty member. Activities vary by project, but the primary purpose is for the cadet to engage in independent study and produce a work of scholarship. With the approval of one of the Program Directors in the Department of Mathematical Sciences, the cadet chooses a project of interest and is supervised by a faculty member. Cadets are expected to perform a total of 40 hours of work for the semester. A final presentation and final project deliverable are required.                             |                                                                                                                                                                                                              | 2026-1 2029-2                                                        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                       |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Enrollment by permission of one of the Program Directors in the Department of Mathematical Sciences. Other requirements as determined by the Faculty Advisor. Discipline-specific final project is required. |                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MA289 MA289A MA289B                                                                                                                                                                                          |                                                                      |
| MA363                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ORDINARY DIFFERENTIAL EQUATION                                                                                                                                                                               | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0)                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2023-2                                                                                                                                                                                                       | <b>Offerings:</b>                                                    |
| This course introduces the mathematical theory of ordinary differential equations and its application to problems in science and engineering. The course emphasizes the conceptual and theoretical understanding of differential equations necessary for advanced study in mathematical sciences. The course covers first and second order differential equations, systems of differential equations, and infinite series. Students will utilize a range of fundamental solution techniques - analytical, qualitative, and numerical. Special focus will be placed on the theory of existence and uniqueness of solutions.                                                                            |                                                                                                                                                                                                              | 2026-1 2027-2 2028-2                                                 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                       |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Several special problems.                                                                                                                                                                                    |                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204                                                                                                                                                                      |                                                                      |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -Or-<br>MA364<br>-Or-<br>MA365                                                                                                                                                                               |                                                                      |
| MA364                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ENGINEERING MATHEMATICS                                                                                                                                                                                      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0)                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2023-2                                                                                                                                                                                                       | <b>Offerings:</b>                                                    |
| This course provides additional mathematical techniques and deepens the understanding of concepts in mathematics to support continued study in science and engineering. Emphasis is placed upon using mathematics to gain insight into natural and man-made phenomena that give rise to problems in differential equations and infinite series representations of functions. Analytic and numerical solutions to differential equations and systems of differential equations are found using a variety of techniques. Linear algebra topics include solutions to homogeneous and non-homogeneous systems of equations. An introduction to classical partial differential equations is also included. |                                                                                                                                                                                                              | 2026-2 2026-8 2027-2<br>2027-9 2028-2 2028-9<br>2029-2 2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                       |                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Several special problems.                                                                                                                                                                                    |                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MA205<br>-Or-<br>MA204X<br>-Or-<br>MA204                                                                                                                                                                     |                                                                      |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MA363<br>-Or-<br>MA366<br>-Or-<br>MA365<br>-Or-<br>MA255                                                                                                                                                     |                                                                      |

| MA365                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ADV MATH FOR ENGRS/SCIENTISTS                             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0)                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2019-2                                                    | <b>Offerings:</b>                                                                                                    |
| This is a course designed for the advanced mathematics student that has completed courses in differential equations and vector calculus (those that have completed MA153 and MA255) and will study ME, EE, NE, Physics or Space Science. MA365 begins where the advanced mathematics program ends. The advanced engineering course offering includes topics in linear algebra, complex variables, Fourier series, partial differential equations, and computational mathematics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                           | 2026-1 2027-1 2028-1<br>2029-1 2029-8 2029-9                                                                         |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | None                                                      |                                                                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MA255                                                     |                                                                                                                      |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MA364<br>-Or-<br>MA366                                    |                                                                                                                      |
| MA367                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | MATH FOR THE SOCIAL SCIENCES                              | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0)                                                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2020-2                                                    | <b>Offerings:</b>                                                                                                    |
| MA367 is a required course for those majoring in economics and is available to any who would like to develop their understanding of the mathematical tools applied to common methods of inquiry in several of the social sciences including, but not limited to, psychology and political science. The course develops student ability to use and develop mathematical models to quantify the relationship between variables, actors, and outcomes. The course helps formalize and quantify the properties of these relationships. The course continues to develop math skills introduced in the core math program, and it introduces more advanced math topics that serve as fundamental skills required for modeling in upper level microeconomics, macroeconomics, and econometrics courses. Specific topics include multivariable optimization, including linear and nonlinear programming, differential equations, stochastic modeling, and an introduction to dynamic programming. This course is designed to demonstrate the relevance of mathematics to the modern practice of analysis in the social sciences and to help prepare future Army leaders to use sound logic and relevant evidence to make convincing arguments. |                                                           | 2026-2 2027-2 2028-2<br>2029-2                                                                                       |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | None                                                      |                                                                                                                      |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204 |                                                                                                                      |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MA371 MA381<br>-Or-<br>MA371 MA481<br>-Or-<br>MA381 MA481 |                                                                                                                      |
| MA371                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | LINEAR ALGEBRA                                            | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0)                                                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2013-1                                                    | <b>Offerings:</b>                                                                                                    |
| This course emphasizes both the computational and theoretical aspects of linear algebra one encounters in many subjects ranging from economics to engineering. The course covers solutions of linear systems of equations and the algebra of matrices. The foundational aspects of vector spaces and linear transformations to include linear dependence and independence, subspaces, bases and dimension, inner products, and orthonormalization are developed. This is rounded out with a detailed investigation of eigenvalues and eigenvectors as they relate to diagonalization, quadratic equations, and systems of differential equations. The Invertible Matrix Theorem is explored as the conceptual/theoretical thread of the course. A computer algebra system is used to explore concepts and compute solutions to problems. Applications of the course material are included in the form of special problems to illustrate its wide scope.                                                                                                                                                                                                                                                                               |                                                           | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-8<br>2027-9 2028-1 2028-2<br>2028-8 2028-9 2029-1<br>2029-2 2029-8 2029-9 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Several special problems.                                 |                                                                                                                      |



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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MA255<br>-Or-<br>MA104<br>-Or-<br>MA204X<br>-Or-<br>MA204 |                                                                                                               |
| <b>MA372</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>INTRODUCTION TO DISCRETE MATH</b>                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2013-1                                                    | <b>Offerings:</b>                                                                                             |
| The purpose of this course is to introduce topics in Discrete Mathematics, providing a foundation for further study and application. The topics covered are useful to both the applied mathematician and the computer scientist. They include propositional logic, elements of set theory, combinatorics, relations, functions, partitions, methods of proof, induction and recursion, digraphs, trees, finite state machines, and algebraic systems. Specific applications to computer science are presented.                                                                                                                                              |                                                           | 2026-2 2026-8 2027-1<br>2027-2 2027-8 2028-1<br>2028-2 2028-8 2029-2<br>2029-8                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                      |                                                                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MA206<br>-Or-<br>MA206X<br>-Or-<br>MA256                  |                                                                                                               |
| <b>MA376</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>APPLIED STATISTICS</b>                                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2020-1                                                    | <b>Offerings:</b>                                                                                             |
| This course builds on the foundations presented in the core probability and statistics course to provide a broad introduction to some of the most common models and techniques in applied statistics. The mathematical basis for each of the models and techniques is presented with particular emphasis on the development of the required test statistics and their distributions. Topics covered include hypothesis testing, analysis of variance, categorical data analysis, regression analysis, and nonparametric methods.                                                                                                                            |                                                           | 2026-1 2026-2 2027-1<br>2027-2 2027-8 2027-9<br>2028-1 2028-2 2028-8<br>2028-9 2029-1 2029-2<br>2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | One (or more) special problem(s).                         |                                                                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MA206<br>-Or-<br>MA206X<br>-Or-<br>MA256                  |                                                                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SE375                                                     |                                                                                                               |
| <b>MA381</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>NONLINEAR OPTIMIZATION</b>                             | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2013-1                                                    | <b>Offerings:</b>                                                                                             |
| This course provides an undergraduate presentation of nonlinear topics in mathematical programming that builds on multivariable Calculus II. The emphasis of this course is on developing a conceptual understanding of the fundamental topics introduced. These topics include general convexity, convex functions, derivative-based multivariable search techniques, minima and maxima of convex functions, gradients, hessian matrices, Lagrange Multipliers, Fritz-John and Kuhn-Tucker optimality conditions, and constrained and unconstrained optimization. Computer software is used to explore and expose various key ideas throughout the course. |                                                           | 2026-1 2026-8 2027-1<br>2028-1 2029-1 2029-8<br>2029-9                                                        |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | One (or more) special problem(s).                         |                                                                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204 |                                                                                                               |

| MA383                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | FOUNDATIONS OF MATH                                                         | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0)                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2013-1                                                                      | <b>Offerings:</b>                                                                     |
| This course introduces the student to the methods and language of upper division mathematics. It presents formal set theory, and introduces the student to the methods of formulating and writing mathematical proofs. Finally, it provides the student a rigorous introduction to the theory of relations, functions, and infinite sets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                             | 2026-1 2026-8 2027-1<br>2027-8 2028-1 2028-8<br>2029-1 2029-8 2029-9                  |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                      |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                        |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MA205<br>-Or-<br>MA205X<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204 |                                                                                       |
| MA385                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CHAOS AND FRACTALS                                                          | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2013-2                                                                      | <b>Offerings:</b>                                                                     |
| This course introduces topics in fractal geometry and chaotic dynamical systems, providing a foundation for applications and further study. The topics from fractal geometry include the military applications of image analysis and data storage. The chaotic dynamical systems studied in the course are one-, two-, and three-dimensional, nonlinear, discrete and continuous dynamical systems. Topics include the logistics equation, the Henon attractor, the Lorenz equations, bifurcation theory, Julia sets, and the Mandelbrot set. These topics have applications in many fields of science, and examples from biology, meteorology, engineering, and the social sciences are studied. The course integrates concepts introduced in the core mathematics courses.                                             |                                                                             | 2027-1 2029-1                                                                         |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                      |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | One (or more) special problem(s).                                           |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204                   |                                                                                       |
| MA386                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | INTRO TO NUMERICAL ANALYSIS                                                 | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0)                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020-1                                                                      | <b>Offerings:</b>                                                                     |
| This course develops an understanding of the methods for solving mathematical problems using a digital computer. Algorithms leading to solution of mathematical problems will be examined for consistency, stability, and convergence. After a brief review of calculus theory, a study of error analysis and computer arithmetic will provide the framework for the study of the following topics: solutions of equations of one variable, solutions of linear and nonlinear systems of equations, the use of polynomials to approximate discrete data, curve fitting, numerical integration and differentiation, and the approximation of continuous functions. Special problems will incorporate computer graphics and the use of mathematical software libraries to produce numerical solutions of applied problems. |                                                                             | 2026-1 2026-8 2027-1<br>2027-8 2027-9 2028-1<br>2028-8 2028-9 2029-1<br>2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                      |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Several special problems.                                                   |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | -Or-<br>-Or-<br>-Or-<br>-Or-<br>IT105 MA205<br>-Or-<br>IT105 MA255<br>-Or-  |                                                                                       |

IT155 MA205  
 -Or-  
 IT155 MA255  
 -Or-  
 IT105 MA366  
 -Or-  
 IT155 MA366  
 -Or-  
 CY105 MA205  
 -Or-  
 CY105 MA255  
 -Or-  
 CY105 MA366  
 -Or-  
 CY155 MA205  
 -Or-  
 CY155 MA255  
 -Or-  
 CY155 MA366  
 -Or-  
 CY105 MA204X  
 -Or-  
 CY155 MA204X  
 -Or-  
 CY105 MA204  
 -Or-  
 CY155 MA204  
 -Or-  
 AI105 MA205  
 -Or-  
 AI105 MA255  
 -Or-  
 AI105 MA204  
 -Or-  
 AI155 MA204  
 -Or-  
 AI155 MA205  
 -Or-  
 AI155 MA255  
 -Or-  
 AI105X MA204  
 -Or-  
 AI105 MA205  
 -Or-  
 AI105X MA255

|              |                                |                                                    |
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| <b>MA387</b> | <b>MATHEMATICAL ANALYSIS I</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
|--------------|--------------------------------|----------------------------------------------------|

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                             | 2013-2                 | <b>Offerings:</b>              |
| A one semester course providing a rigorous introduction to the calculus of a single variable. The course is designed to introduce the student to the foundations of the calculus necessary for advanced undergraduate and graduate studies in applied mathematics and engineering. Course coverage includes a treatment of the structure of the real number system, sequences, continuous functions, and differentiation. |                        | 2026-2 2027-2 2028-2<br>2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min |                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                              | None                   |                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                   | MA383                  |                                |

|              |                     |                                                    |
|--------------|---------------------|----------------------------------------------------|
| <b>MA388</b> | <b>SABERMETRICS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
|--------------|---------------------|----------------------------------------------------|

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2022-2                 | <b>Offerings:</b>              |
| This course builds on the statistical foundation of the core mathematics sequence by exploring the application of statistical concepts to sports analytics. Students develop skills and apply statistical techniques appropriate for baseball and other sports including: regression, forecasting, and stochastic processes. Guest lectures and a course trip section to discuss Sabermetrics at the baseball Hall of Fame in Cooperstown, NY are part of this course (when available). Software packages (Mathematica, Excel) are used as decision support tools to investigate application problems and augment understanding of course material. |                        | 2026-2 2027-2 2028-2<br>2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min |                                |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                                                                                                                                                                                                          |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | MA206<br>-Or-<br>MA256                                                                                                                                                                                                        |                                                               |
| <b>MA389</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>INDIV STUDY IN MATHEMATICS</b>                                                                                                                                                                                             | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2017-1                                                                                                                                                                                                                        | <b>Offerings:</b>                                             |
| This course is intended for individually supervised research and study, in order to familiarize cadets with techniques used in advanced scientific study. The primary purpose is to prepare students for independent research in mathematics with the essential skills required. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                                                                                                                                                                                                               | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | Cadets must complete either an individual written research report or present an oral report to members of the department faculty at the end of the semester.                                                                  |                                                               |
| <b>MA389A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>INDIV STUDY IN MATHEMATICS-A</b>                                                                                                                                                                                           | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2024-1                                                                                                                                                                                                                        | <b>Offerings:</b>                                             |
| This course is intended for individually supervised research and study, in order to familiarize cadets with techniques used in advanced scientific study. The primary purpose is to prepare students for independent research in mathematics with the essential skills required. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                                                                                                                                                                                                               | 2026-1 2029-1 2029-2                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | Cadets must complete either an individual written research report or present an oral report to members of the department faculty at the end of the semester. Cadets must successfully complete MA389 before taking MA389A.    |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | MA389                                                                                                                                                                                                                         |                                                               |
| <b>MA389B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>INDIV STUDY IN MATHEMATICS-B</b>                                                                                                                                                                                           | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2024-1                                                                                                                                                                                                                        | <b>Offerings:</b>                                             |
| This course is intended for individually supervised research and study, in order to familiarize cadets with techniques used in advanced scientific study. The primary purpose is to prepare students for independent research in mathematics with the essential skills required. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                                                                                                                                                                                                               | 2026-1 2029-1 2029-2                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | Cadets must complete either an individual written research report or present an oral report to members of the department faculty at the end of the semester. Cadets must take MA389 and MA389A before taking MA389B.          |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | MA389 MA389A                                                                                                                                                                                                                  |                                                               |
| <b>MA389C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>INDIV STUDY IN MATHEMATICS-C</b>                                                                                                                                                                                           | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2025-1                                                                                                                                                                                                                        | <b>Offerings:</b>                                             |
| This course is intended for individually supervised research and study, in order to familiarize cadets with techniques used in advanced scientific study. The primary purpose is to prepare students for independent research in mathematics with the essential skills required. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                                                                                                                                                                                                               | 2026-1 2029-1 2029-2                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | Cadets must complete either an individual written research report or present an oral report to members of the department faculty at the end of the semester. Cadets must take MA389, MA389A, and MA389B before taking MA389C. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | MA389 MA389A MA389B                                                                                                                                                                                                           |                                                               |

| MA389D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | INDIV STUDY IN MATHEMATICS-D                                                                                                                                                                                                          | 2.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025-2                                                                                                                                                                                                                                | <b>Offerings:</b>                          |
| This course is intended for individually supervised research and study, in order to familiarize cadets with techniques used in advanced scientific study. The primary purpose is to prepare students for independent research in mathematics with the essential skills required. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                       | 2029-2                                     |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Cadets must complete either an individual written research report or present an oral report to members of the department faculty at the end of the semester. Cadets must take MA389, MA389A, MA389B, and MA389C before taking MA389D. |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MA389 MA389A MA389B MA389C                                                                                                                                                                                                            |                                            |
| MA391                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MATHEMATICAL MODELING                                                                                                                                                                                                                 | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025-1                                                                                                                                                                                                                                | <b>Offerings:</b>                          |
| This course is designed to give cadets the opportunity to develop skills in model construction and model analysis while addressing interesting scenarios with practical applications from a wide variety of fields. The course addresses the complex process of translating real-world events into mathematical language, solving the resulting mathematical model (iterating as necessary), and interpreting the results in terms of real-world issues. Topics may include model development from data, optimization, dynamic models, and deterministic and stochastic model development. Interdisciplinary projects based on actual modeling scenarios are used to integrate the various topics into a coherent theme. |                                                                                                                                                                                                                                       | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Several special projects.                                                                                                                                                                                                             |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204                                                                                                                                                                             |                                            |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MA206<br>-Or-<br>MA206X<br>-Or-<br>MA256                                                                                                                                                                                              |                                            |
| MA394                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | FUNDAMENTALS/NETWORK SCIENCE                                                                                                                                                                                                          | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2023-1                                                                                                                                                                                                                                | <b>Offerings:</b>                          |
| MA394 exposes cadets to the basic concepts of networks and gives them an opportunity to apply techniques learned in the course to real-world problems. Students will develop skills and problem-solving strategies for modeling complex networks associated with physical, informational, and social phenomena. Software packages are used as decision support tools to investigate application problems and augment understanding of the course material.                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                       | 2027-1 2028-1 2029-1<br>2029-8 2029-9      |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                                                                                                                                                                                                                  |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MA371                                                                                                                                                                                                                                 |                                            |
| MA396                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NUM METH SOLUTIONS DIFF EQNS                                                                                                                                                                                                          | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=3.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2013-2                                                                                                                                                                                                                                | <b>Offerings:</b>                          |
| The focus of this course is to find numerical solutions of differential equations that result when modeling physical phenomena. The numerical solution of both initial value problems and boundary-value problems that arise with ordinary differential equations are covered. Techniques for solving partial differential equations are introduced. Software packages (Mathematica, Maple, Matlab, etc.) have proved to be very useful tools for many numerical techniques and are used to augment an understanding of course material.                                                                                                                                                                                 |                                                                                                                                                                                                                                       | 2026-2                                     |

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One (or more) special problem(s).

**Prerequisite(s):**

-Or-

-Or-

-Or-

-Or-

IT105 MA205

-Or-

IT105 MA255

-Or-

IT155 MA205

-Or-

IT155 MA255

-Or-

CY105 MA205

-Or-

CY105 MA255

-Or-

CY155 MA205

-Or-

CY155 MA255

-Or-

CY105 MA204X

-Or-

CY155 MA204X

-Or-

CY105 MA204

-Or-

CY155 MA204

-Or-

AI105 MA204

-Or-

AI105 MA205

-Or-

AI105 MA255

-Or-

AI105X MA204

-Or-

AI105X MA204

-Or-

AI105X MA255

-Or-

AI155 MA204

-Or-

AI155 MA205

-Or-

AI155 MA255

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| <b>MA461</b> | <b>GRAPH THEORY AND NETWORKS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=3.0)</b> |
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**Scope:** 2013-1

**Offerings:**

This course introduces the student to the techniques, algorithms, and structures used in graph theory and network flows in order to solve real world discrete optimization problems. Basic definitions relating to graphs and digraphs, together with a large number of examples and applications are provided. Cadets learn to implement new graph theory techniques in their area of study. Emphasis is on modeling, algorithms, and optimization.

2026-1 2028-1 2029-1  
2029-8 2029-9

**Lessons:** 40 @ 55 min (0.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Two special problems.

**Prerequisite(s):**

MA206

-Or-

MA206X

-Or-

MA256

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|--------------|----------------------|----------------------------------------------------------|
| <b>MA462</b> | <b>COMBINATORICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=3.0)</b> |
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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-2                                                                 | <b>Offerings:</b>                                                              |
| This course introduces the basic techniques and modes of combinatorial problem-solving important to the field of computer science and mathematical sciences such as operations research. Applications of combinatorics are also related to fields such as genetics, organic chemistry, electrical engineering and political science. Combinatorial enumeration and logical structure are stressed. Applications and examples provide the structure of progression through topics which include counting methods, generating functions, recurrence relations, and enumeration techniques.                                                                                                                               |                                                                        | 2026-8 2027-1 2029-8                                                           |
| <b>Lessons:</b> 40 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                 |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Completion of the mathematics core curriculum required for enrollment. |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA206<br>-Or-<br>MA256<br>-Or-<br>MA206X                               |                                                                                |
| <b>MA464</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>APPLIED ALGEBRA W/ CRYPTOLOGY</b>                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-2                                                                 | <b>Offerings:</b>                                                              |
| We study the underlying algebra of computer science structures as well as sets, set functions, Boolean algebra, finite state machines, groups, and modular arithmetic. We introduce and study mathematical aspects of cryptology with an emphasis on cryptanalysis of encryption ciphers. We study early paper-and-pencil systems through current computer algorithms for encryption. We employ algebraic principles in both design and analysis of encryption systems, be it matrix, linear feedback shift register sequence, or linear congruential random number generator sequence efforts. Further, we investigate the mathematics of breaking machine ciphers and of designing modern public-key crypto systems. |                                                                        | 2026-2 2027-2 2028-2<br>2029-2                                                 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                 |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                   |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA206<br>-Or-<br>MA256<br>-Or-<br>MA206X                               |                                                                                |
| <b>MA466</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ABSTRACT ALGEBRA</b>                                                | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-2                                                                 | <b>Offerings:</b>                                                              |
| This is an introductory course in modern algebra for cadets who plan to do graduate work in mathematics or theoretical work in the physical sciences or engineering. The emphasis of the course is on group theory, considering such topics as cyclic and abelian groups, normal sub-groups and factor groups, series of groups, and solvable groups. Selected applications are interspersed with the material on group theory. The course concludes with an introduction to rings and fields. One special problem is provided to allow the student to do independent research in an area of the student's interest.                                                                                                   |                                                                        | 2027-1 2028-1 2029-2                                                           |
| <b>Lessons:</b> 40 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                 |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | One special problem.                                                   |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA206<br>-Or-<br>MA206X<br>-Or-<br>MA256                               |                                                                                |
| <b>MA476</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>MATHEMATICAL STATISTICS</b>                                         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-2                                                                 | <b>Offerings:</b>                                                              |
| This course builds on the foundation presented in the core probability and statistics course to provide a mathematical presentation of the important topics in mathematical statistics. The course begins with a review of probability concepts from the core course, adding additional topics such as transformations of random variables and moment generating functions. To provide the mathematical basis for much of statistical practice, certain limit theorems and sampling distributions are proven. The central focus of the course is distribution theory, to include the theory of estimation and the theory of hypothesis testing.                                                                        |                                                                        | 2026-2 2027-2 2027-8<br>2027-9 2028-2 2028-8<br>2028-9 2029-2 2029-8<br>2029-9 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                 |                                                                                |

**Special Requirements:** One (or more) special problem(s).

**Prerequisite(s):** MA205 MA206  
 -Or-  
 MA206 MA255  
 -Or-  
 MA205 MA256  
 -Or-  
 MA255 MA256  
 -Or-  
 MA205 MA206X  
 -Or-  
 MA206X MA255  
 -Or-  
 MA204X MA206  
 -Or-  
 MA204X MA256  
 -Or-  
 MA204 MA206  
 -Or-  
 MA204 MA256

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| <b>MA477</b> | <b>THEORY &amp; APPL OF DATA SCIENCE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=3.0)</b> |
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| <b>Scope:</b> | 2024-1 | <b>Offerings:</b> |
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This course builds on the foundations presented in the core probability and statistics course and the applied statistics course to develop a broad base of Advanced Data Science to some of the most common techniques in the field. The mathematical basis for each method is presented with focus on both the statistical theory and application. Topics covered may include classification and regression trees, regularization methods, splines and localized regression, and model validation.

**Lessons:** 30 @ 75 min (2.000 Att/wk)     **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MA371 MA376  
 -Or-  
 CY300 MA376  
 -Or-  
 MA376 MA486

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| <b>MA478</b> | <b>GENERALIZED LINEAR MODELS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=3.0)</b> |
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| <b>Scope:</b> | 2022-2 | <b>Offerings:</b> |
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This course provides an introduction to statistical modeling beyond that gained in MA376. Students will learn statistical models for analyzing quantitative and qualitative data. Methods will generally be taught in the generalized linear model framework and may include binomial and multinomial regression models, models for count data and models for survival data. Students will also be exposed to techniques for handling problems that arise when analyzing real data: for example missing data, unstructured data and influential observations.

**Lessons:** 30 @ 75 min (2.000 Att/wk)     **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MA376

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| <b>MA481</b> | <b>LINEAR OPTIMIZATION</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=3.0)</b> |
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| <b>Scope:</b> | 2020-1 | <b>Offerings:</b> |
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This course emphasizes the applications of optimal solutions to linear algebraic systems using the simplex method of linear programming. This includes an in-depth development of the simplex method, the theory of duality, an analysis of the dual problem, convex hull concepts, integer programming, sensitivity analysis and the revised simplex procedure. Additional computational techniques that are applicable to specific mathematical models such as the transportation problem, assignment problem and network problems are also studied. Problems illustrating applications are emphasized throughout the course. Use of existing computer software to solve problems is also emphasized.

**Lessons:** 40 @ 55 min (2.500 Att/wk)     **Labs:** 0 @ 0 min

**Special Requirements:** Several special problems.



**Prerequisite(s):** MA371

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| <b>MA484</b> | <b>PARTIAL DIFF EQUATIONS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=3.0)</b> |
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**Scope:** 2020-1

**Offerings:**

The course is devoted to the solution of the classical partial differential equations of mathematical physics and most engineering fields. For example, these equations describe such diverse phenomena as the flow of heat in a metal plate, the gravitational field of the solar system, the vibration of a structural beam, and the energy levels of the hydrogen atom. The subject matter has application in many fields and should be of interest to mathematics, science, and engineering concentrators. Specific topics covered are the heat, wave, and potential equations, Fourier series, series solutions to ordinary differential equations, special functions, and boundary value problems.

2027-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One special problem.

**Prerequisite(s):** MA205  
-Or-  
MA205X  
-Or-  
MA255  
-Or-  
MA204X  
-Or-  
MA204

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| <b>MA485</b> | <b>APPLIED COMPLEX VARIABLES</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=3.0)</b> |
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**Scope:** 2013-2

**Offerings:**

This course presents a logical development of complex variable theory sufficient for the development and solution of a number of interesting and practical problems. Residue theory is developed and applied to problems in integration and in the solution of partial differential equations via transform techniques. Conformal mapping theory is used to solve partial differential equations for which the solution is a harmonic function satisfying prescribed boundary conditions. These classical Dirichlet-Neumann problems model phenomena arising in the study of electrostatic potential, equilibrium thermodynamics, incompressible fluids, elasticity, and other areas of continuum mechanics.

2028-1 2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One special problem.

**Prerequisite(s):** MA205  
-Or-  
MA255  
-Or-  
MA204X  
-Or-  
MA204

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>MA486</b> | <b>MATHEMATICAL COMPUTATION</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=3.0)</b> |
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**Scope:** 2019-1

**Offerings:**

This course provides an undergraduate presentation of computational problem solving that builds on math and science classes from USMA. The emphasis of this course is on developing a conceptual understanding of the fundamental topics in algorithm development as well as developing an understanding of a high level programming language. MA486 exposes students to mathematical applications best understood through computational analysis. Contemporary topics of application may include linear algebra, numerical analysis, statistics, data analytics and visualization, simulation, optimization, and machine learning. Students will be exposed to topics in computer science including data structures, object oriented programming, debugging, complexity, as well as testing and verification. We will use a high level computer programming language such as R or Python to explore and expose various topics.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

|                                                                                                                                                                                                                                                                                                                                    |                                                           |                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                            | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204 |                                                                                                                      |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                             | MA206<br>-Or-<br>MA256                                    |                                                                                                                      |
| <b>MA487</b>                                                                                                                                                                                                                                                                                                                       | <b>MATHEMATICAL ANALYSIS II</b>                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                                                                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                      | 2013-1                                                    | <b>Offerings:</b>                                                                                                    |
| Continuation of MA387. Course coverage includes Riemann and Stieltjes integration, infinite series, sequences and series of functions, uniform convergence, and power series.                                                                                                                                                      |                                                           | 2029-1                                                                                                               |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                       | None                                                      |                                                                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                            | MA387                                                     |                                                                                                                      |
| <b>MA488</b>                                                                                                                                                                                                                                                                                                                       | <b>SPECIAL TOPICS IN MATHEMATICS</b>                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                                                                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                      | 2019-2                                                    | <b>Offerings:</b>                                                                                                    |
| This course provides an in-depth study of a special topic in mathematics not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the visiting professor or a senior mathematical science faculty member.                                                                            |                                                           | 2026-2 2026-9 2027-8<br>2027-9 2028-8 2028-9<br>2029-1 2029-8 2029-9                                                 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                       | To be determined by the program director.                 |                                                                                                                      |
| <b>MA488A</b>                                                                                                                                                                                                                                                                                                                      | <b>SPECIAL TOPICS IN MATHEMATICS</b>                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                                                                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                      | 2020-1                                                    | <b>Offerings:</b>                                                                                                    |
| This course provides an in-depth study of a special topic in mathematics not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the visiting professor or a senior mathematical science faculty member.                                                                            |                                                           | No Course Offerings                                                                                                  |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                       | To be determined by the program director.                 |                                                                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                            | MA488                                                     |                                                                                                                      |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                             | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204 |                                                                                                                      |
| <b>MA489</b>                                                                                                                                                                                                                                                                                                                       | <b>ADV INDIV STUDY IN MATH</b>                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b>                                                                   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                      | 2013-1                                                    | <b>Offerings:</b>                                                                                                    |
| This is essentially a tutorial course or an individual project, offered only to a limited number of highly qualified cadets who have completed available mathematics elective courses and have expressed a wish to pursue advanced study in a field of mathematics. The course work will be tailored to suit the individual needs. |                                                           | 2026-1 2026-2 2026-9<br>2027-1 2027-2 2027-8<br>2027-9 2028-1 2028-2<br>2028-8 2028-9 2029-1<br>2029-2 2029-8 2029-9 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 0 @ 0 min                                    |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                       | None                                                      |                                                                                                                      |

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------|
| <b>MA489A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ADV INDIV STUDY IN MATH-A</b>                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2024-1                                                                     | <b>Offerings:</b>                                  |
| This is essentially a tutorial course or an individual project, offered only to a limited number of highly qualified cadets who have completed available mathematics electives courses and have expressed a wish to pursue advanced study in a field of mathematics. The course work will be tailored to suit the individual needs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                            | 2026-1 2029-1 2029-2                               |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cadets should successfully complete MA489 before taking MA489A.            |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MA489                                                                      |                                                    |
| <b>MA489B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ADV INDIV STUDY IN MATH-B</b>                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2024-1                                                                     | <b>Offerings:</b>                                  |
| This is essentially a tutorial course or an individual project, offered only to a limited number of highly qualified cadets who have completed available mathematics elective courses and have expressed a wish to pursue advanced study in a field of mathematics. The course work will be tailored to suit the individual needs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                            | 2029-2                                             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cadets should successfully complete MA489 and MA489A before taking MA489B. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | MA489 MA489B                                                               |                                                    |
| <b>MA490</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ETHICS IN MATH, DATA SCI &amp; ENG</b>                                  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2024-2                                                                     | <b>Offerings:</b>                                  |
| This course is intended to serve as an integrative experience for cadets of all majors and FOSs. Cadets having completed the core math program will be given the opportunity to explore ethical issues including legitimate data use, algorithmic fairness, and further develop skills in model construction and analysis while addressing problems and scenarios with practical applications from science, social sciences, engineering, computer science and/or mathematics. Interdisciplinary projects based on actual modeling scenarios are used to explore ethical issues and integrate the various topics into a coherent theme.                                                                                                                                                                                                                                                                      |                                                                            | 2026-2 2027-2 2028-2 2029-2                        |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Several special projects.                                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | MA206<br>-Or-<br>MA256<br>-Or-<br>MA206X                                   |                                                    |
| <b>MA491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>RESEARCH SEMNR-APPLD MATH</b>                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2013-1                                                                     | <b>Offerings:</b>                                  |
| The student integrates the mathematical concepts and techniques learned in previous courses with the principles developed throughout the whole USMA Curriculum to solve a current problem of interest to the individual, to the Academy, or to agencies in the Department of the Army. Cadets may select problems from a list of suitable projects provided by the Department of Mathematical Sciences. Cadets choose a faculty advisor who has an interest and background in the problem. Cadets may work individually or in small teams, depending on the nature of the research. Regular workshop sessions will be held. Cadets will be given an opportunity to present their research at the Service Academies Student Mathematics Conference and/or other undergraduate conferences. Research reports will be reviewed, edited, and compiled into the USMA Transactions on Cadet Mathematical Research. |                                                                            | 2026-1 2026-2 2027-2 2028-2 2029-2                 |
| <b>Lessons:</b> 17 @ 55 min (1.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Weekly meetings @ 55 min; one research paper (80 hours).                   |                                                    |
| <b>MA493A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>OPNL CALC AND TRANSFORMS</b>                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2013-1                                                    | <b>Offerings:</b>                                  |
| This course is the logical extension and synthesis of MA484 and MA485. It employs the integral calculus of complex functions and the theory of residues to investigate solutions to a number of partial differential equations arising from electrostatics, thermostatics, elasticity, gravitation, and other fields of continuum mechanics. The Poisson-Integral Formula is applied to the solution of boundary-value problems. Fourier and Laplace transforms are studied in detail and are used to develop general techniques for the solution of many ordinary, partial, and integral equations which result from the above applications. |                                                           | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Several special projects.                                 |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204                   |                                                    |
| <b>MA493B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>REAL VARIABLE THEORY</b>                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2013-1                                                    | <b>Offerings:</b>                                  |
| Continuation of MA487. Topics include sequences and series of functions, equicontinuity power series, Fourier series, the exponential and logarithmic function, and the Gamma function. The last portion of the course will be devoted to individual research projects.                                                                                                                                                                                                                                                                                                                                                                       |                                                           | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | One special project.                                      |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204 |                                                    |
| <b>MA493C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>TOPICS IN NUMERICAL ANALYSIS</b>                       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2013-1                                                    | <b>Offerings:</b>                                  |
| A continuation of MA396. Topics include boundary-value problems for ordinary and/or partial differential equations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                           | 2028-1                                             |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | One term-end research project.                            |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204 |                                                    |
| <b>MA493D</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>INTRODUCTION TO TOPOLOGY</b>                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2013-1                                                    | <b>Offerings:</b>                                  |
| The course begins with cardinality and the modern definition of a function. Then the basic properties of topological spaces--compactness, connectedness, and continuity--will be emphasized. Special attention will be given to metric topologies on Euclidean spaces. Complete metric spaces and function spaces will be introduced.                                                                                                                                                                                                                                                                                                         |                                                           | 2027-1                                             |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                      |                                                    |

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| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204        |                                                    |
| <b>MA493E</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>TOPICS IN ANALYSIS</b>                                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-1                                                           | <b>Offerings:</b>                                  |
| This course provides cadets the opportunity to pursue in detail subjects of special interest.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                  | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                             |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204        |                                                    |
| <b>MA498</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>SR THESIS I: RSCRCH &amp; PROPOSAL</b>                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-1                                                           | <b>Offerings:</b>                                  |
| The purpose of the Senior Thesis is to provide cadets with an unique opportunity to create a scholarly product that is academically, professionally, and personally meaningful to them and that reflects their thinking and abilities as developed at West Point and in the Department of Mathematical Sciences. Cadets will choose a faculty advisor with whom they will collaborate over two semesters. Cadets will meet on a regular basis with their advisor to discuss mathematics, progress on their research and thesis, and developmental issues. The objectives of the research are: (1) to synthesize and cohere the cadet's studies; (2) to apply methodological skills of research design, conceptual reasoning, analysis, and research gained to a selected area of substantive interest; (3) to extend the cadet's in-depth study of the selected area of interest beyond the level obtained in the Mathematical Sciences Major; (4) to design and conduct focused research beyond the constrained opportunities in elective courses; and (5) to develop cadet skills in conceptual reasoning, critical analysis, and effective writing. |                                                                  | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 17 @ 55 min (1.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Weekly meeting @ 55 min; one research proposal and presentation. |                                                    |
| <b>MA499</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>SR THESIS II: PAPER &amp; DEFENSE</b>                         | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=3.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-2                                                           | <b>Offerings:</b>                                  |
| This course continues the work on the thesis commenced in MA 498. At the end of the course, cadets will submit a written thesis to the Department of Mathematical Sciences. In addition, cadets will defend that thesis before a faculty committee. Cadets will be given an opportunity to present their research at the Service Academies Student Mathematics Conference and/or other undergraduate conferences. Theses will be reviewed, edited, and compiled into the USMA Transactions on Cadet Mathematical Research.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                  | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 17 @ 55 min (1.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                           |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Weekly meetings @ 55 min; one research paper and defense.        |                                                    |

# Mechanical and Aerospace Engineering

## 54 Courses

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| <b>AE189</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>IND STUDY IN AEROSPACE ENG 1CR</b>                                                                                                                                                                                                        | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                                                                                                                                                                                                                                       | <b>Offerings:</b>                                  |
| The cadet pursues advanced study of a topic in aerospace engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results. |                                                                                                                                                                                                                                              | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 7 @ 55 min (0.440 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 7 @ 55 min                                                                                                                                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Enrollment by permission of Aerospace Engineering Program Director. Appropriate ET credit will be determined by the Aerospace Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor. |                                                    |
| <b>AE201</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>INTRO TO AEROSPACE ENGINEERING</b>                                                                                                                                                                                                        | <b>3.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                                                                                                                                                                                                                                       | <b>Offerings:</b>                                  |
| This course provides an overview of the field of Aerospace engineering and an introduction to engineering analysis through modeling, simulation, and experimentation. AE201 provides an introduction to structured programming for engineering applications and an overview of solid modeling using Computer Aided Design. *Pending ABET-PEV review of ET=3.5CR. The ET credit is proposed and will be validated through Academy processes.                                                                                                                                |                                                                                                                                                                                                                                              | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 5 @ 120 min                                                                                                                                                                                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                                                                                                                                                                                                                                         |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MA205<br>-Or-<br>MA255<br>-Or-<br>MA204X<br>-Or-<br>MA204                                                                                                                                                                                    |                                                    |
| <b>AE287</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>INTRODUCTION TO AERONAUTICS</b>                                                                                                                                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                                                                                                                                                                                                                                       | <b>Offerings:</b>                                  |
| This introductory course provides an understanding of how fixed-wing aircraft work and the forces and airflow involved in flight. Equations of motion are derived for a rigid aircraft in steady state level flight, maneuvering flight, and during takeoff and landing. Theoretical concepts are demonstrated in laboratory sessions that include actual flights in the Department of Civil and Mechanical Engineering's fixed-wing aircraft, and the evolution of flight during a museum field trip.                                                                     |                                                                                                                                                                                                                                              | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 27 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 3 @ 120 min                                                                                                                                                                                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | One lab-related trip section, compensatory time given. Two lab exercises in a fixed-wing aircraft.                                                                                                                                           |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | ME387                                                                                                                                                                                                                                        |                                                    |
| <b>AE289</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>IND STUDY IN AEROSPACE ENG 2CR</b>                                                                                                                                                                                                        | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026-1                                                                                                                                                                                                                                       | <b>Offerings:</b>                                  |
| The cadet pursues advanced study of a topic in aerospace engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results. |                                                                                                                                                                                                                                              | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 14 @ 55 min (0.880 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 14 @ 55 min                                                                                                                                                                                                                     |                                                    |

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| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Enrollment by permission of Aerospace Engineering Program Director. Appropriate ET credit will be determined by the Aerospace Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor. |                                                    |
| <b>AE354</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>PROPULSION</b>                                                                                                                                                                                                                            | <b>3.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2027-2                                                                                                                                                                                                                                       | <b>Offerings:</b>                                  |
| This course applies the fundamentals of thermodynamics to the study of air-breathing (turbojets, turbofans, and turboprops/turboshafts) and non air-breathing (rocket) air vehicle propulsion systems. Propulsive forces and performance parameters for these propulsion systems will be explored. Propeller characteristics and performance of electric and hybrid-electric propulsion systems are introduced. ***This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY27-1. Pending ABET-PEV review of ET=3.5CR.***                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                              | 2027-2 2028-2 2029-2                               |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 4 @ 120 min                                                                                                                                                                                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                                                                                                                                                                                                                         |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ME301                                                                                                                                                                                                                                        |                                                    |
| <b>AE364</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>AEROSPACE STRUCTURES</b>                                                                                                                                                                                                                  | <b>3.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2027-1                                                                                                                                                                                                                                       | <b>Offerings:</b>                                  |
| The course covers the introduction of mechanics of materials and introduction to linear elasticity, including stress and strain, generalized Hooke's law, thin-walled pressure vessels, and principal stresses. The course also covers the form and function of aerospace structural components, materials allowable, and factors and margins of safety. Flight and ground loads and the bending of beams having asymmetric cross-sections are covered as they relate to aerospace structures. Shear flow analyses of stress in idealized semi monocoque cross-sections and the elastic buckling of columns and thin-wall structures are introduced. ***This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY26-2. Pending ABET-PEV review of ET=3.5CR.*** |                                                                                                                                                                                                                                              | 2027-1 2028-1 2029-1                               |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 6 @ 120 min                                                                                                                                                                                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                                                                                                                                                                                                                         |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MC300                                                                                                                                                                                                                                        |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MC364                                                                                                                                                                                                                                        |                                                    |
| <b>AE388</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>VTOL AERONAUTICS</b>                                                                                                                                                                                                                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2027-1                                                                                                                                                                                                                                       | <b>Offerings:</b>                                  |
| The aerodynamics of vertical flight is analyzed for hover, translating, and partial power flight. Theory and experimental results are used to predict aircraft performance. The course analyzes the dynamic response of a rotor system and the performance aspects of the vehicle as a whole. This is followed by a design workshop, during which cadets complete the initial sizing of a vertical aircraft to meet specific mission requirements. The course includes one flight lab in a helicopter, a laboratory examining rotor power and thrust utilizing a whirl stand apparatus, and one field trip to a commercial helicopter company. ***This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY26-2. Pending ABET-PEV review of ET=3.0CR.***       |                                                                                                                                                                                                                                              | 2027-2 2028-2 2029-2                               |
| <b>Lessons:</b> 28 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 2 @ 120 min                                                                                                                                                                                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ME202 pre-requisite may be waived with validation exercise on structured programming and AE Program Director approval.                                                                                                                       |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ME202                                                                                                                                                                                                                                        |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ME388                                                                                                                                                                                                                                        |                                                    |
| <b>AE389</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>IND STUDY IN AEROSPACE ENG</b>                                                                                                                                                                                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026-1                                                                                                                                                                                                                                       | <b>Offerings:</b>                                  |

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| <p>The cadet pursues advanced study of a topic in aerospace engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results.</p> <p><b>Lessons:</b> 20 @ 55 min (1.250 Att/wk)      <b>Labs:</b> 20 @ 55 min</p> <p><b>Special Requirements:</b> Enrollment by permission of Aerospace Engineering Program Director. Appropriate ET credit will be determined by the Aerospace Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor.</p>                                                                                                                                                                          |                                | 2026-1 2027-1 2028-1 2029-1                       |
| <b>AE400</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>AEROSPACE SEMINAR</b>       | <b>1.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2028-1                         | <b>Offerings:</b>                                 |
| <p>The cadet pursues advanced study of a topic in aerospace engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results. ***This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY27-2. Pending ABET-PEV review of ET=1.0CR.***</p> <p><b>Lessons:</b> 20 @ 55 min (1.250 Att/wk)      <b>Labs:</b> 20 @ 55 min</p> <p><b>Special Requirements:</b> None</p>                                                                                                                                                                                                                    |                                | 2028-1 2029-1                                     |
| <b>AE473</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>STABILITY AND CONTROL</b>   | <b>3.5 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2028-1                         | <b>Offerings:</b>                                 |
| <p>This course introduces aircraft static and dynamic stability and control. The course develops and provides the tools required to analyze and design the stability and control attributes of an aircraft in preparation for aircraft design. ***This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY27-2. Pending ABET-PEV review of ET=3.5CR.***</p> <p><b>Lessons:</b> 40 @ 55 min (0.000 Att/wk)      <b>Labs:</b> 4 @ 120 min</p> <p><b>Special Requirements:</b> One flight laboratory exercise in a fixed-wing aircraft.</p> <p><b>Prerequisite(s):</b> XE472</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                | 2028-2 2029-2                                     |
| <b>AE481</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>AERODYNAMICS</b>            | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2027-1                         | <b>Offerings:</b>                                 |
| <p>A course on the fundamentals of incompressible inviscid fluid mechanics and 2-D aerodynamics. Topics include fluid statics, flow kinematics, integral and differential forms of the governing equations, potential flow theory, and thin-airfoil theory. Applications emphasize aeronautics-relevant topics. This course provides knowledge of the fundamentals of fluid mechanics and aerodynamics. Fluid mechanics as applied to the theory of flight prepare students to understand external flows over aircraft including wing design, drag build-up, and viscous phenomena. The course concludes with an introduction to modern computational fluid dynamic tools. ***This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY26-2. Pending ABET-PEV review of ET=3.0CR.***</p> <p><b>Lessons:</b> 28 @ 75 min (2.000 Att/wk)      <b>Labs:</b> 2 @ 120 min</p> <p><b>Special Requirements:</b> Two laboratory exercises involving a wind tunnel.</p> <p><b>Corequisite(s):</b> ME362</p> <p><b>Disqualifier(s):</b> ME481</p> |                                | 2027-2 2028-2 2029-2                              |
| <b>AE483</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>AEROSPACE SYSTEM DESIGN</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2027-2                         | <b>Offerings:</b>                                 |



This course covers the fundamentals of aircraft and spacecraft design. Topics include understanding the mission and payload requirements, sizing methods, understanding design space, and tradeoffs. Students will be introduced to typical design and analysis tools used in the industry. \*\*\*This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY27-1. Pending ABET-PEV review of ET=3.0CR.\*\*\*

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>AE489</b> | <b>ADV IND STUDY IN AEROSPACE ENG</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

The cadet pursues advanced study of a topic in aerospace engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 20 @ 55 min (1.250 Att/wk)    **Labs:** 20 @ 55 min

**Special Requirements:** Enrollment by permission of Aerospace Engineering Program Director. Appropriate ET credit will be determined by the Aerospace Engineering Program Director on a case-by-case basis. Other requirements as determined by the Faculty Advisor.

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| <b>AE498</b> | <b>AEROSPACE ENGINEERING DESIGN I</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2028-1

**Offerings:**

This course serves as the first half of the aerospace engineering capstone design experience. Cadets apply the aerospace system design process and their knowledge of math, science, and engineering mechanics to design solutions to real-world engineering problems. They must address public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. Students begin capstone assignments early in the course and continue their projects with AE499. \*\*\*This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY27-2. Pending ABET-PEV review of ET=3.5CR.\*\*\*

2028-1 2029-1

**Lessons:** 20 @ 150 min (2.000 Att/wk)    **Labs:** 10 @ 150 min

**Special Requirements:** None

**Prerequisite(s):** AE483

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|--------------|----------------------------------------|----------------------------------------------------------|
| <b>AE499</b> | <b>AEROSPACE ENGINEERING DESIGN II</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2028-2

**Offerings:**

This course provides experience in the integration of math, science, and engineering principles into a comprehensive engineering design project. Open-ended, client-based design problems emphasize a multidisciplinary approach to total system design providing multiple paths to a number of feasible and acceptable solutions which meet the stated performance requirements. Design teams are required to develop product specifications, generate alternatives, make practical engineering approximations, perform appropriate analysis to support the technical feasibility of the design, and make decisions leading to an optimal system design. System integration, human factors engineering, computer-aided design, maintainability, and fabrication techniques are addressed. This course provides an integrative experience in support of the overarching academic program goal, and is often interdisciplinary in nature. \*\*\*This is a provisionally approved course for the Aerospace Engineering Major (AEN0) and must undergo review by the Curriculum Committee NLT AY28-1. Pending ABET-PEV review of ET=3.5CR.\*\*\*

2028-2 2029-2

**Lessons:** 10 @ 150 min (1.000 Att/wk)    **Labs:** 20 @ 150 min

**Special Requirements:** None

**Prerequisite(s):** AE498

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| <b>MC311</b> | <b>THERMAL-FLUID SYSTEMS I</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Thermal-Fluid Systems I is an integrated study of fundamental topics in thermodynamics and fluid mechanics. The course introduces conservation principles for mass, energy, and linear momentum as well as the 2nd Law of Thermodynamics. Principles are applied to incompressible flow in pipes and turbomachinery, power generation systems, refrigeration cycles, and total air-conditioning focusing on the control volume approach. Laboratory exercises are integrated into classroom work. This course includes completion of a comprehensive, out-of-class design problem. This design problem provides the opportunity for students to apply engineering science and the engineering design process to a hands-on project. Lesson length varies between 75 minutes and 55 minutes (and corresponding preparation time varies as well).

**Lessons:** 38 @ 75 min (2.500 Att/wk)      **Labs:** 2 @ 75 min

**Special Requirements:** None

**Prerequisite(s):**

-Or-  
 -Or-  
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 -Or-  
 -Or-  
 -Or-  
 -Or-  
 -Or-  
 CH101 MA205 PH205  
 -Or-  
 CH101 MA255 PH205  
 -Or-  
 CH101 MA205 PH255  
 -Or-  
 CH101 MA255 PH255  
 -Or-  
 CH151 MA205 PH205  
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 CH151 MA255 PH205  
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 CH151 MA205 PH255  
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 CH151 MA255 PH255  
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 CH101 MA366 PH205  
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 CH101 MA366 PH255  
 -Or-  
 CH151 MA366 PH205  
 -Or-  
 CH151 MA366 PH255  
 -Or-  
 CH101 MA205 PH201  
 -Or-  
 CH101 MA205 PH201X  
 -Or-  
 CH101 MA255 PH201X  
 -Or-  
 CH101 MA366 PH201X  
 -Or-  
 CH151 MA205 PH201X  
 -Or-  
 CH151 MA255 PH201X  
 -Or-  
 CH151 MA366 PH201X  
 -Or-  
 CH101 MA205 PH201  
 -Or-  
 CH101 MA205 PH251  
 -Or-  
 CH101 MA255 PH201  
 -Or-  
 CH101 MA255 PH251  
 -Or-  
 CH101 MA366 PH201

-Or-  
CH101 MA366 PH251  
-Or-  
CH151 MA205 PH201  
-Or-  
CH151 MA255 PH201  
-Or-  
CH151 MA205 PH251  
-Or-  
CH151 MA255 PH251  
-Or-  
CH151 MA366 PH201  
-Or-  
CH151 MA363 PH251  
-Or-  
CH101 MA204 PH201  
-Or-  
CH151 MA204 PH201  
-Or-  
CH101 MA204 PH251  
-Or-  
CH151 MA204 PH251

**Disqualifier(s):**

ME301 ME362

| ME189                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | INTRO IND STUDY IN ME                                                                                                                                                                                                                          | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| The cadet pursues study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results. |                                                                                                                                                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 7 @ 55 min (0.440 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 7 @ 55 min                                                                                                                                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Enrollment by permission of Mechanical Engineering Program Director. Appropriate ET credit will be determined by the Mechanical Engineering Program Director on a case by case basis. Other requirements as determined by the Faculty Advisor. |                                                               |
| ME189A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | INTRO IND STUDY IN ME (A)                                                                                                                                                                                                                      | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| The cadet pursues study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results. |                                                                                                                                                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 7 @ 55 min (0.440 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 7 @ 55 min                                                                                                                                                                                                                        |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Enrollment by permission of Mechanical Engineering Program Director. Appropriate ET credit will be determined by the Mechanical Engineering Program Director on a case by case basis. Other requirements as determined by the Faculty Advisor. |                                                               |
| ME189B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | INTRO IND STUDY IN ME (B)                                                                                                                                                                                                                      | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| The cadet pursues study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results. |                                                                                                                                                                                                                                                | 2027-1 2028-1 2029-1                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                         |                                                               |

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| <b>Special Requirements:</b> | Enrollment by permission of Mechanical Engineering Program Director. Appropriate ET credit will be determined by the Mechanical Engineering Program Director on a case by case basis. Other requirements as determined by the Faculty Advisor. |
| <b>Prerequisite(s):</b>      | ME189A                                                                                                                                                                                                                                         |

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| <b>ME201</b> | <b>INTRO TO MECH ENGINEERING</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2026-1                   | <b>Offerings:</b>           |
| This course introduces product design as an iterative decision making process. It also introduces project planning and team dynamics, technical communications, and using computing tools including structured programming and 2-D/3-D visualization and analysis to aid the design process. Cadets solve a series of progressively more challenging problems by applying the design process while addressing social, political, economic and technical issues. |                          | 2026-1 2027-1 2028-1 2029-1 |
| <b>Lessons:</b> 16 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 24 @ 55 min |                             |

**Special Requirements:** None

**Corequisite(s):** PH205  
-Or-  
PH255  
-Or-  
PH201X  
-Or-  
PH201  
-Or-  
PH251

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|--------------|-------------------------------------|----------------------------------------------------|
| <b>ME202</b> | <b>INTRO COMPUTATIONAL ANALYSIS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                           | 2026-1                   | <b>Offerings:</b>           |
| ME202 students explore computational methods of engineering practice. Students are introduced to: structured programming, instrumentation and data collection in experimental methods, numerical methods of engineering analysis, and simulation-based analysis. Considerable emphasis is placed on data analysis for understanding technical concepts. |                          | 2026-2 2027-2 2028-2 2029-2 |
| <b>Lessons:</b> 16 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 24 @ 55 min |                             |

**Special Requirements:** None

**Corequisite(s):** MA205  
-Or-  
MA255  
-Or-  
MA204X  
-Or-  
MA204

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|--------------|-------------------------------|----------------------------------------------------|
| <b>ME289</b> | <b>INDIVIDUAL STUDY IN ME</b> | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                   | <b>Offerings:</b>                                       |
| The cadet pursues study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results. |                          | 2026-1 2026-2 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2 |
| <b>Lessons:</b> 14 @ 55 min (0.880 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 14 @ 55 min |                                                         |

**Special Requirements:** Enrollment by permission of Mechanical Engineering Program Director. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. Other requirements as determined by the Faculty Advisor.

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|---------------|-----------------------------------|----------------------------------------------------|
| <b>ME289A</b> | <b>INDIVIDUAL STUDY IN ME (A)</b> | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

The cadet pursues study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 14 @ 55 min (0.880 Att/wk)

**Labs:** 14 @ 55 min

**Special Requirements:**

Enrollment by permission of Mechanical Engineering Program Director. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. Other requirements as determined by the Faculty Advisor.

**ME301**

**THERMODYNAMICS**

**3.5 Credit Hours**  
(BS=0.0,ET=3.5,MA=0.0)

**Scope:**

2026-1

**Offerings:**

Thermodynamics concerns the study of energy. In this course, the student will gain a basic engineering knowledge of energy applications and limitations. This course provides the groundwork for subsequent studies in engineering sciences and an appreciation of numerous problems associated with energy. Emphasis is placed on practical application to power generation, thermal and air pollution, refrigeration, air conditioning, automotive and aircraft engines, and combustion. Laboratory exercises are integrated into classroom work. This course includes completion of a comprehensive, out-of-class design problem. This design problem provides the opportunity to apply the principles of thermodynamics which are taught in the classroom to a realistic problem encountered by practicing engineers.

2026-1 2026-2 2026-8  
2027-1 2027-2 2027-8  
2028-1 2028-2 2028-8  
2029-1 2029-2

**Lessons:** 40 @ 75 min (3.000 Att/wk)

**Labs:** 0 @ 120 min

**Special Requirements:**

None

**Prerequisite(s):**

CH101 MA205  
-Or-  
CH101 MA255  
-Or-  
CH151 MA205  
-Or-  
CH151 MA255  
-Or-  
CH101 MA204X  
-Or-  
CH151 MA204X  
-Or-  
CH101 MA204  
-Or-  
CH151 MA204

**Disqualifier(s):**

MC311 MC312

**ME306**

**DYNAMICS**

**3.0 Credit Hours**  
(BS=0.0,ET=3.0,MA=0.0)

**Scope:**

2026-1

**Offerings:**

Dynamics examines the motion of particles, systems of particles, and rigid bodies under the influence of forces. It focuses on the use of Newton's Second Law, in three major, progressive blocks of instruction: from scalar, then vector, treatments of rectilinear and curvilinear motion of single particles; through vector motion of systems of particles; to general three-dimensional motion of rigid bodies. The course also provides brief introductions to energy methods: work-energy and impulse-momentum.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

**Prerequisite(s):**

PH201  
-Or-  
PH251

**Corequisite(s):**

CE300  
-Or-  
MC300

**Disqualifier(s):**

MC306

**ME362**

**FLUID MECHANICS**

**3.5 Credit Hours**  
(BS=0.0,ET=3.5,MA=0.0)

**Scope:**

2026-1

**Offerings:**

Fluid Mechanics is the study of the behavior of liquids and gases under all conditions of rest and motion. The basic conservation laws are developed and applied to problems encountered in stream and pipe flow, modeling, low speed aerodynamics, forces on submerged surfaces, turbomachinery, and flow measurement. The course emphasizes physical concepts as well as mathematics and is augmented by laboratory work and a wide variety of training aids. This course provides the foundation for further study in aerodynamics, energy systems engineering, automotive engineering, civil engineering, and environmental engineering.

2026-1 2026-2 2026-8  
2027-1 2027-2 2027-8  
2027-9 2028-1 2028-2  
2028-8 2028-9 2029-1  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)    **Labs:** 4 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** MA205 PH205  
-Or-  
MA255 PH205  
-Or-  
MA205 PH255  
-Or-  
MA255 PH255  
-Or-  
MA205 PH201X  
-Or-  
MA255 PH201X  
-Or-  
MA205 PH201  
-Or-  
MA205 PH251  
-Or-  
MA255 PH201  
-Or-  
MA255 PH251  
-Or-  
MA204X PH201  
-Or-  
MA204X PH251  
-Or-  
MA204 PH201  
-Or-  
MA204 PH251

**Disqualifier(s):** MC311 MC312

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|--------------|------------------------------|----------------------------------------------------|
| <b>ME380</b> | <b>ENGINEERING MATERIALS</b> | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Course explores the relationship between the microscopic structure and macroscopic properties of materials used in engineering applications. The origin of mechanical and physical properties is studied. Emphasized is an understanding of the fundamental aspects of atomic and microstructural concepts for proper materials selection and enhancement of engineering properties. Materials under study are metals, ceramics, polymers, composites, nano-sized/structured materials, biomaterials, smart materials, and semi- and super-conductors. Laboratory exercises are incorporated throughout the course to provide practical experience in making decisions concerning material composition and processing in order to optimize engineering properties. Experiences from the field are detailed to demonstrate application of concepts.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 5 @ 75 min

**Special Requirements:** None

**Prerequisite(s):** MC300

**Disqualifier(s):** MC380

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|--------------|------------------------------------|----------------------------------------------------|
| <b>ME387</b> | <b>INTRODUCTION TO AERONAUTICS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This introductory course provides an understanding of how fixed-wing aircraft work and the forces and airflow involved in flight. Equations of motion are derived for a rigid aircraft in steady state level flight, maneuvering flight, and during takeoff and landing. Theoretical concepts are demonstrated in laboratory sessions that include actual flights in the Department of Civil and Mechanical Engineering's fixed-wing aircraft, and the evolution of flight during a museum field trip.

2027-1

**Lessons:** 28 @ 75 min (2.000 Att/wk)    **Labs:** 2 @ 120 min

**Special Requirements:** One lab-related trip section, compensatory time given

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|--------------|-------------------------------|----------------------------------------------------------|
| <b>ME388</b> | <b>HELICOPTER AERONAUTICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
|---------------|--------|-------------------|

The aerodynamics of helicopter flight is analyzed for hover, translating, and partial power flight. Theory and experimental results are used to predict aircraft performance. The course analyzes the dynamic response of the rotor system and the performance aspects of the vehicle as a whole. This is followed by a design workshop, during which cadets complete the initial sizing of a helicopter to meet specific mission requirements. The course includes one flight lab in a helicopter, a laboratory examining rotor power and thrust utilizing a whirl stand apparatus, and one field trip to a commercial helicopter company.

**Lessons:** 28 @ 75 min (2.000 Att/wk)      **Labs:** 2 @ 120 min

**Special Requirements:** ME202 pre-requisite may be waived with validation exercise on structured programming and ME Program Director approval

**Prerequisite(s):** ME370  
-Or-  
ME202

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| <b>ME389</b> | <b>INTRO TO ADV STUDY IN MECH ENG</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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The cadet pursues advanced study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results.

**Lessons:** 20 @ 55 min (1.250 Att/wk)      **Labs:** 20 @ 55 min

**Special Requirements:** Enrollment by permission of Mechanical Engineering Program Director. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. Other requirements as determined by Faculty Advisor.

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|---------------|---------------------------------------|----------------------------------------------------------|
| <b>ME389A</b> | <b>INTRO TO ADV STUDY IN MECH ENG</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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The cadet pursues advanced study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results.

**Lessons:** 20 @ 55 min (0.000 Att/wk)      **Labs:** 20 @ 55 min

**Special Requirements:** Enrollment by permission of Mechanical Engineering Program Director. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. Other requirements as determined by Faculty Advisor.

**Corequisite(s):** ME389

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>ME400</b> | <b>MECHANICAL ENGINEERING SEMINAR</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=1.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2026-1 | <b>Offerings:</b> |
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This seminar consists of a series of guest speakers. It will include all cadets majoring in mechanical engineering. Guest Speaker topics will address the concerns of professional mechanical engineers such as engineering ethics, continuing education, engineering economy, social and safety considerations, and professional registration. Guest lecturers will be primarily mechanical engineering practitioners, providing the students an opportunity to interact with professionals in their major field of interest.

**Lessons:** 10 @ 75 min (0.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

| ME403                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MANUFACTURING/MACHINE COMP DSN             | 3.5 Credit Hours<br>(BS=0.0,ET=3.5,MA=0.0)                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026-1                                     | <b>Offerings:</b>                                             |
| <p>This course introduces manufacturing machines and machine component design. Students develop their understanding of the engineering science behind static and fatigue failure theories and use it to design or choose mechanical components such as fasteners, springs, bearings, gears, and shafts. Instructors and laboratory technicians train cadets on safe use of machines such as mills, lathes, grinders, belt sanders, drill presses, bandsaws, and welding equipment. The course culminates in a team design and manufacturing project that uses the technical engineering and hands-on skills learned throughout the semester.</p>                                   |                                            | 2026-1 2026-2 2027-1<br>2028-1 2029-1                         |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 35 @ 55 min (2.500 Att/wk)                 | <b>Labs:</b> 8 @ 120 min                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                       |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -Or-<br>MC300 MC364<br>-Or-<br>CE364 MC300 |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ME201                                      |                                                               |
| ME404                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MECHANICAL ENGINEERING DESIGN              | 3.5 Credit Hours<br>(BS=0.0,ET=3.5,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026-1                                     | <b>Offerings:</b>                                             |
| <p>This course serves as the first half of the mechanical engineering capstone design experience. Cadets apply the Mechanical Engineering Design Process and their knowledge of math, science, and engineering mechanics to design solutions to real-world engineering problems. They must address public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. Students begin capstone assignments early in the course and continue their projects with ME496.</p>                                                                                                                                                               |                                            | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 20 @ 150 min (2.000 Att/wk)                | <b>Labs:</b> 10 @ 150 min                                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                       |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ME403                                      |                                                               |
| ME472                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ENERGY CONVERSION SYSTEMS                  | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026-1                                     | <b>Offerings:</b>                                             |
| <p>Fundamental concepts are extended to the engineering analysis of coal, oil and natural gas fossil fuel systems to assess the dominant sources of energy and technologies in the electric power, transportation, industrial, and residential and commercial energy sectors. Renewable and alternative energy resources including solar, wind, biomass, hydro, geothermal, nuclear and ocean energy are assessed, along with analysis of conventional and emerging technologies to harness them. National and global energy issues are discussed with technical, economical, environmental, societal and geopolitical considerations and in the context of Army energy needs.</p> |                                            | 2027-2 2028-2 2029-2                                          |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 30 @ 75 min (2.000 Att/wk)                 | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None.                                      |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MC311<br>-Or-<br>ME301                     |                                                               |
| ME480                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | HEAT TRANSFER                              | 3.5 Credit Hours<br>(BS=0.0,ET=3.5,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026-1                                     | <b>Offerings:</b>                                             |
| <p>The three modes of heat transfer, conduction, convection, and radiation, are studied in detail and applications are made to various engineering systems. The principles of conduction and convection are used to study the mechanisms of heat transfer during boiling, condensation and the design of heat exchangers.</p>                                                                                                                                                                                                                                                                                                                                                      |                                            | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 40 @ 55 min (2.500 Att/wk)                 | <b>Labs:</b> 8 @ 120 min                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                                       |                                                               |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                   |                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA364 MC311<br>-Or-<br>MA365 MC311<br>-Or-<br>MA364 ME301<br>-Or-<br>MA365 ME301<br>-Or-<br>MA364X MC311<br>-Or-<br>MA364X ME301                                                                                                                  |                                                               |
| <b>ME481</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>AIRCRAFT AERODYNAMICS AND DSGN</b>                                                                                                                                                                                                             | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                                                                                                                                                                                                                                            | <b>Offerings:</b>                                             |
| The course approaches the design of an aircraft from the principles of aerodynamics, stability, and control. A flight laboratory in the departments fixed-wing airplanes provide an opportunity to obtain data and analyze the stability and control of an actual aircraft. Lift, drag, and aerodynamic moments are studied for airfoils (2-D) and finite wings (3-D) in the subsonic flow regime. Theoretical concepts are demonstrated in laboratory sessions that include low-speed wind tunnel testing.                                                                                                            |                                                                                                                                                                                                                                                   | 2026-2 2027-2 2027-9<br>2028-2 2028-9 2029-2<br>2029-9        |
| <b>Lessons:</b> 28 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 2 @ 120 min                                                                                                                                                                                                                          |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                                                                                                                                              |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ME387<br>-Or-<br>AE287                                                                                                                                                                                                                            |                                                               |
| <b>ME486</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>VIBRATION ENGINEERING</b>                                                                                                                                                                                                                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                                                                                                                                                                                                                                            | <b>Offerings:</b>                                             |
| In this course students develop a foundation in the analysis and design of free and forced single and multi-degree of freedom systems. Applications include modeling, damping, resonance, force transmissibility, vibration absorbers, matrix formulation and modal analysis. Emphasis is placed on vibrations examples from several engineering fields. Laboratory experiences provide students with the opportunity to apply principles taught in the classroom to physical systems, allowing the comparison of theoretical and actual vibrating systems. In-class demonstrations supplement the theory development. |                                                                                                                                                                                                                                                   | 2026-2 2027-1 2027-2<br>2028-1 2028-2 2029-1<br>2029-2        |
| <b>Lessons:</b> 36 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 4 @ 120 min                                                                                                                                                                                                                          |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                                                                                                                                                                              |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA364 ME306<br>-Or-<br>MA364 MC306<br>-Or-<br>MA365 ME306<br>-Or-<br>MA365 MC306                                                                                                                                                                  |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MC486                                                                                                                                                                                                                                             |                                                               |
| <b>ME489</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ADV STUDY IN MECH ENGRNG</b>                                                                                                                                                                                                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1                                                                                                                                                                                                                                            | <b>Offerings:</b>                                             |
| The cadet pursues advanced study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results.                                            |                                                                                                                                                                                                                                                   | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 20 @ 55 min (1.250 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 20 @ 55 min                                                                                                                                                                                                                          |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Enrollment by permission of Mechanical Engineering Program Director. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. Other requirements as determined by Faculty Advisor. |                                                               |

| ME489A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ADV STUDY IN MECH ENGRNG                                                                                                                                                                                                                              | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                                                                                | <b>Offerings:</b>                                             |
| The cadet pursues advanced study of a topic in mechanical engineering on an individual or small group basis, independent of a formal classroom setting. Similar to graduate level research, the scope of the selected project is tailored to the interests of the cadet based on resources and in consultation with a faculty advisor. To develop research skills, the cadet is integral in all phases of project completion by defining objectives, studying fundamentals and background material, outlining the approach, conducting analysis, and communicating results.                                           |                                                                                                                                                                                                                                                       | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 20 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 20 @ 55 min                                                                                                                                                                                                                              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Enrollment by permission of Mechanical Engineering Program Director. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ME489                                                                                                                                                                                                                                                 |                                                               |
| ME489B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | INDEPENDENT STUDY, ADVANCED                                                                                                                                                                                                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                                                                                | <b>Offerings:</b>                                             |
| This course will cover advanced topics in M. E.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                       | No Course Offerings                                           |
| <b>Lessons:</b> 40 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Must have ME489 and ME489A                                                                                                                                                                                                                            |                                                               |
| ME490                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | TOPICS IN MECHANICAL ENGRNG                                                                                                                                                                                                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                                                                                | <b>Offerings:</b>                                             |
| This course provides in-depth study of a special topic in engineering mechanics or mechanical engineering not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the Visiting Professor or a senior mechanical engineering faculty member.                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                       | 2026-8                                                        |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | TBD. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.                                                                                                                          |                                                               |
| ME490A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | TOPICS IN MECHANICAL ENGRNG                                                                                                                                                                                                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                                                                                | <b>Offerings:</b>                                             |
| This course provides in-depth study of a special topic in engineering mechanics or mechanical engineering not offered elsewhere in the USMA curriculum. Course content will be based on the special expertise of the Visiting Professor or a senior mechanical engineering faculty member.                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                       | No Course Offerings                                           |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | TBD                                                                                                                                                                                                                                                   |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ME490                                                                                                                                                                                                                                                 |                                                               |
| ME491                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | AUTOMOTIVE POWERPLANTS                                                                                                                                                                                                                                | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026-1                                                                                                                                                                                                                                                | <b>Offerings:</b>                                             |
| Students engage in the analysis, testing, and evaluation of internal combustion engines and their subsystems with a goal toward understanding the principles affecting performance and efficiency. Spark ignition and compression ignition engine systems are studied in detail, with laboratory events designed to connect theory and practice. Material explores modern technologies in use and under development which function to increase powerplant efficiency and decrease emissions. A series of design problems is interspersed throughout the course along with a semester long engineering design project. |                                                                                                                                                                                                                                                       | 2026-2 2026-8 2027-2<br>2028-2 2029-2                         |
| <b>Lessons:</b> 26 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 5 @ 75 min                                                                                                                                                                                                                               |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ME202 pre-requisite may be waived with validation exercise on structured programming and ME Program Director approval                                                                                                                                 |                                                               |

**Prerequisite(s):**

-Or-  
MC311 ME202  
-Or-  
ME202 ME301

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|              |                                           |                                                          |
|--------------|-------------------------------------------|----------------------------------------------------------|
| <b>ME492</b> | <b>AUTO PWR TRAINS &amp; VEH DYNAMICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|-------------------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

Students engage in the analysis, testing, and evaluation of automotive powertrains and their subsystems with a goal toward understanding the principles affecting vehicle dynamics and platform performance. Wheeled systems are studied in detail, with laboratory events designed to connect theory and practice. A series of design problems is interspersed throughout the course along with a semester long engineering design project.

2026-1 2026-8 2027-1  
2028-1 2029-1

**Lessons:** 26 @ 75 min (2.000 Att/wk)    **Labs:** 4 @ 75 min

**Special Requirements:** Knowledge of structured programming and solid modeling is encouraged

**Prerequisite(s):**

PH201  
-Or-  
PH205  
-Or-  
PH251  
-Or-  
PH255

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|              |                            |                                                          |
|--------------|----------------------------|----------------------------------------------------------|
| <b>ME493</b> | <b>WEAPONS ENGINEERING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

Weapons Engineering is an introductory course in armament and ordnance theory with an emphasis on analysis, design, and testing of ground combat weapon systems. The course covers the underlying principles affecting the generation and application of combat power at scales ranging from small arms (pistols, rifles) to large cannons (tank guns, howitzers). Theory is verified with laboratory exercises and experimental data. Concept analysis and component design problems are interspersed throughout the course, and a semester long Engineering Design Project (EDP) unifies the main course elements with a contemporarily relevant problem.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 6 @ 75 min

**Special Requirements:** None

**Corequisite(s):**

MC311  
-Or-  
ME362

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|              |                                 |                                                          |
|--------------|---------------------------------|----------------------------------------------------------|
| <b>ME496</b> | <b>MECHANICAL SYSTEM DESIGN</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course provides experience in the integration of math, science, and engineering principles into a comprehensive engineering design project. Open-ended, client-based design problems emphasize a multidisciplinary approach to total system design providing multiple paths to a number of feasible and acceptable solutions which meet the stated performance requirements. Design teams are required to develop product specifications, generate alternatives, make practical engineering approximations, perform appropriate analysis to support the technical feasibility of the design, and make decisions leading to an optimal system design. System integration, human factors engineering, computer-aided design, maintainability, and fabrication techniques are addressed. This course provides an integrative experience in support of the overarching academic program goal, and is often interdisciplinary in nature.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 10 @ 150 min (2.000 Att/wk)    **Labs:** 20 @ 150 min

**Special Requirements:** None

**Prerequisite(s):**

ME404  
-Or-  
XE485

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|              |                            |                                                          |
|--------------|----------------------------|----------------------------------------------------------|
| <b>XE200</b> | <b>SCIENTIFIC ARGUMENT</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course teaches cadets how to define and communicate new ideas to solve scientific problems in the context of challenging, modern technological problems facing the Army, Department of Defense, and society. Through participation in a speaker series, students engage with leaders from the DoD, academia, and industry to define their passions in STEM and select a cutting-edge problem for further study. Lessons on technical communication are interwoven to develop technical communication skills, focusing on writing detailed literature reviews, developing compelling technical proposals, and giving engaging presentations. Throughout the semester, students construct a research project proposal and presentation to argue an innovative solution to their chosen problem.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 10 @ 75 min (1.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Cadets must be a STEM major or minor in their Third Class year or above to enroll in this course.

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|--------------|-------------------------------------|----------------------------------------------------------|
| <b>XE310</b> | <b>INTRO TO BIOMECH ENGINEERING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course introduces a broad spectrum of biomechanical concepts, including the musculoskeletal system, biological materials, locomotion, and computational biomechanics, through the application of engineering. Contemporary topics will be discussed and analyzed, and the concepts will be applied within the scope of a project in experimental biomechanics, which will employ standard, cutting edge biomechanical software and instrumentation techniques.

2026-2 2028-2 2029-2

**Lessons:** 25 @ 75 min (2.000 Att/wk)    **Labs:** 5 @ 75 min

**Special Requirements:** None

**Prerequisite(s):**

- MA205 PH205
- Or-
- MA255 PH205
- Or-
- MA205 PH255
- Or-
- MA255 PH255
- Or-
- MA205 PH201X
- Or-
- MA255 PH201X
- Or-
- MA205 PH201
- Or-
- MA205 PH251
- Or-
- MA255 PH201
- Or-
- MA255 PH251
- Or-
- MA204X PH201
- Or-
- MA204X PH251
- Or-
- MA204 PH201
- Or-
- MA204 PH251

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|--------------|----------------------------------------|----------------------------------------------------------|
| <b>XE365</b> | <b>ADV EXP METHODS &amp; DATA PROC</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|----------------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

This course examines a range of advanced interdisciplinary topics in engineering. The focus is on development of the ability to effectively design and analyze experiments. Topics include processing of audio, thermal, and other data specific to the interests of the instructor team for each offering. Advanced topics such as compressible flow and turbulence will be explored in this course, but additional topics from civil engineering, computing services, electrical engineering, mathematical science, mechanical engineering, and physics may be explored. A heavy emphasis is placed on the ability to effectively assess potential uncertainties in experiments and to communicate technical results to an expert audience.

2027-1 2028-1 2029-1

**Lessons:** 28 @ 55 min (1.500 Att/wk)    **Labs:** 12 @ 55 min

**Special Requirements:** Admission to the course may be competitive and require permission from the Course Director.

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>XE465</b> | <b>TOPICS: ADVANCED TECHNOLOGY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                                                                                                                                                                                                                                                                      | <b>Offerings:</b>                                  |
| This course is taught by the Class of 1950 Chair of Advanced Technology, a visiting scholar with a distinguished record of academic and professional achievement in the field of engineering, science and technology. The seminars focus on topical issues that either reflect the Chair's area of expertise or are conducted by an expert in the field. Students will apply mathematics, science, and engineering fundamentals to evaluate equipment, processes, and concepts being used in the Army. The course has a final design briefing that is an integrative experience. Admission into course is with permission of Department Head. |                                                                                                                                                                                                                                                                             | No Course Offerings                                |
| <b>Lessons:</b> 20 @ 110 min (1.250 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                                      |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FCS Decision Brief to distinguished guests; Industry field trip.                                                                                                                                                                                                            |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MA205 PH202<br>-Or-<br>MA205 PH252<br>-Or-<br>MA255 PH202<br>-Or-<br>MA255 PH252<br>-Or-<br>MA204X PH202<br>-Or-<br>MA204X PH252<br>-Or-<br>MA204X PH275<br>-Or-<br>MA205 PH275<br>-Or-<br>MA255 PH275<br>-Or-<br>MA204 PH202<br>-Or-<br>MA204 PH252<br>-Or-<br>MA204 PH275 |                                                    |
| <b>XE475</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>MECHATRONICS</b>                                                                                                                                                                                                                                                         | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                                                                                                                                                                                                                                                                      | <b>Offerings:</b>                                  |
| XE 475 is a comprehensive introductory course in the field of mechatronics. Mechatronics is the crossroads in engineering where mechanical engineering, electrical engineering, computer science, and controls engineering meet to create new and exciting real-world systems. Knowledge of mechanical and electrical components, controls theory, and design are integrated to solve actual physical design applications.                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                             | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 5 @ 120 min                                                                                                                                                                                                                                                    |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                                                                                                                                                                                                                                        |                                                    |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | XE472                                                                                                                                                                                                                                                                       |                                                    |
| <b>XE485</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ENGINEERING CAPSTONE DESIGN I</b>                                                                                                                                                                                                                                        | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                                                                                                                                                                                                                                                                      | <b>Offerings:</b>                                  |
| This course challenges students to solve a complex engineering problem in teams. The design problems will be open-ended in nature and involve the application of math, science, and engineering principles acquired in previous coursework. The students will be assigned to a team and project, and use an iterative design process to define the problem, analyze alternatives, and select the best alternative to solve the problem. Teams will communicate their recommendation for a design solution and determine the applicable standards associated with their problem. Work on this problem will continue in the next course: XE495. |                                                                                                                                                                                                                                                                             | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 10 @ 150 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 20 @ 150 min                                                                                                                                                                                                                                                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Enrollment in XE485 is subject to Course Director approval.                                                                                                                                                                                                                 |                                                    |
| <b>XE495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ENGINEERING CAPSTONE DESIGN II</b>                                                                                                                                                                                                                                       | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2026-1                                                                                                                                                                                                                                                                      | <b>Offerings:</b>                                  |

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This course continues the process initiated in XE485. Students will implement, test, model, and communicate their solution to solve a complex engineering problem in teams. In addition, the teams will document their work and reflect on how the project met the design requirements as outlined in ABET. The teams will also communicate the social, technical, economic, and political impact of their designs

**Lessons:** 2 @ 150 min (2.000 Att/wk)      **Labs:** 28 @ 150 min

**Special Requirements:**                      Enrollment in XE495 is subject to Course Director approval.

2026-2

2027-2

2028-2

2029-2

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## Military Instruction

### 30 Courses

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| <b>DS320</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>INTRO TO STRATEGIC STUDIES</b>                                                                    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2023-2                                                                                               | <b>Offerings:</b>                                             |
| DS320 is the introductory course of the Defense & Strategic Studies major and therefore covers the foundational concepts within the field. This course addresses the enduring and contemporary issues of strategy by studying key components of international relations, geopolitics, military theory, and grand strategy. Cadets will apply course concepts through traditional coursework such as papers, presentations, and exams. Lastly, the course aims to develop foundational skills in research, writing, and critical thinking to enable success as a DSS graduate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                      | 2026-1 2027-1 2029-1                                          |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 30 @ 75 min (2.000 Att/wk)                                                                           | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                                                                 |                                                               |
| <b>DS345</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>MILITARY INNOVATION</b>                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2020-1                                                                                               | <b>Offerings:</b>                                             |
| This interdisciplinary course examines the subject of military innovation from a theoretical, strategic, historical, and policy oriented perspective. DS 345 addresses several key questions: Why do militaries innovate? How does this process of innovation occur? Why do attempts at military innovation succeed or fail? To answer these questions, this course introduces the innovation concept and ties innovation to the levels of war. It provides the historical narrative to military innovation, while emphasizing the contemporary operating environment by exploring the possibility of a recent Revolution in Military Affairs through emerging technologies and the international security environment.                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                      | 2026-1 2027-1                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 30 @ 75 min (2.000 Att/wk)                                                                           | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | An analytical paper and class presentation on a cadet-selected recent or future operational concept. |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MS345                                                                                                |                                                               |
| <b>DS350</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>STRAT &amp; PERSUASIVE COMMUNICATN</b>                                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025-1                                                                                               | <b>Offerings:</b>                                             |
| Leaders must possess the ability to communicate complex ideas clearly and concisely, both verbally and in writing. DS350 is grounded in the application of sound communication and narrative techniques relevant to the tactical, operational, and strategic levels of war. The course introduces the concept of information as an element of national power used to achieve U.S. strategic objectives. It then turns to how narratives are used to develop and execute strategy. Cadets gain foundational knowledge of how information is used to develop understanding, communicate, and act decisively. Students study interactions between the military and the media to understand influence campaigns and their effects during competition and conflict. The course culminates in crafting communication plans to tackle vital strategic challenges. Through a combination of theoretical understanding and practical application, cadets learn to develop strategic communication plans, engage with media effectively, and leverage information as a dynamic element of combat power. |                                                                                                      | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2029-1<br>2029-2 2029-8 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 30 @ 75 min (2.000 Att/wk)                                                                           | <b>Labs:</b> 0 @ 0 min                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                                                                 |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MS350                                                                                                |                                                               |
| <b>DS360</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SPECIAL OPS: THEORY &amp; PRACTICE</b>                                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2023-2                                                                                               | <b>Offerings:</b>                                             |
| This course explores the nature and dynamics of low-intensity conflict in contrast to conventional and high-intensity warfare. Specifically, cadets will examine the strategy and tactics of insurgency, counterinsurgency, international terrorism, and peace operations. Another component of this course examines Special Operations Forces (SOF) by exploring the unique methods of special operators and their relationship with irregular warfare. Cadets will examine how U.S. SOF are organized, how special operations in general succeed, and why SOF are well-suited for low-intensity conflict. Several relevant guest speakers are integrated into the instruction throughout the course.                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                      | 2026-1 2027-1                                                 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 30 @ 75 min (2.000 Att/wk)                                                                           | <b>Labs:</b> 0 @ 0 min                                        |

**Special Requirements:** Research paper and oral presentations.

**Disqualifier(s):** MS360

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|--------------|-------------------------------|----------------------------------------------------------|
| <b>DS370</b> | <b>US STRATEGY AND POLICY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

This course introduces cadets to foundational concepts within US national security policy, strategic theory, and military strategy. Cadets analyze the decisions of US policymakers through practical case studies and contemporary events by applying major theoretical concepts in defense strategy and policy through individual critical thinking, group collaboration, and oral and written communication. Cadets will be able to understand, analyze, and effectively communicate the relation of tactical action to national policy within the context of the US defense establishment.

2026-2 2027-2 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** DS320

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|--------------|-------------------------------------|----------------------------------------------------------|
| <b>DS399</b> | <b>STRATEGIC STUDIES INTERNSHIP</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-------------------------------------|----------------------------------------------------------|

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**Scope:** 2023-2

**Offerings:**

The Academic Individual Advanced Development (AIAD) program is designed to give cadets practical experience in their field of study and to reflect on their experiences by completing specified requirements. Current AIADs that cadets may potentially execute for credit include domestic and international staff rides and internships to areas of strategic relevance, though AIAD availability will vary by year. Instructors, students, and host agencies will develop a specific academic plan prior to executing the AIAD that will require either department or curriculum review depending on the nature of the trip. Students will normally complete specially assigned readings, a daily reflective journal based on their practical experiences, and write a term paper that benefits the host organization or expands on the AIAD's purpose. The assigned instructor will provide feedback on student submissions and assign term grades accordingly.

2026-7

**Lessons:** 0 @ 0 min (0.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** Program Director approval

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>DS455</b> | <b>COMPARATIVE DEFENSE POLICY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2023-2

**Offerings:**

This course's objective is to analyze the defense policies of various countries and the outcomes of those defense policies, to include national security objectives, national military objectives, military doctrine, force structure, and military capabilities. Countries studied will include actual and potential coalition partners and potential adversaries. Cadets will examine the political, economic, and social influences on each military establishment. Cultural influences on the development and implementation of the defense policies for countries studied will be examined, including the effects each country's culture has on the missions, structure, roles, and capabilities of the military. Cadets will develop their own framework of analysis to critically analyze the defense policies and cultures of other countries, and will be able to clearly articulate that analysis through written and oral means. Guest speakers include liaison officers and Foreign Area Officers to provide insight into the specific military establishments of those countries studied.

2026-1 2026-2 2026-8  
2026-9 2027-1 2027-2  
2027-8 2027-9 2029-1  
2029-2 2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** A comparative study with a 2500-word paper and a class presentation; compensatory time provided.

**Prerequisite(s):** SS307  
-Or-  
SS357

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| <b>DS460</b> | <b>IRREGULAR WARFARE THEORY &amp; PRA</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This course explores the social, political, economic, and military dynamics of irregular warfare, focusing on the tactical to strategic intricacies of conflict among state and non-state actors. Cadets gain insights into the complexities of irregular warfare in competition and conflict through a comprehensive examination of theoretical frameworks and practical case studies.

2026-2 2027-2 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min



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| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                  |                                                    |
| <b>DS475</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>FORECAST &amp; GAM IN DECISION-MAK</b>             | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025-1                                                | <b>Offerings:</b>                                  |
| This course introduces cadets to the interdisciplinary nature of decision-making by addressing how leaders make decisions in situations of incomplete, conflicting, and incorrect information using forecasting, modeling, and gaming. The curriculum enables cadets to synthesize long-term developments and potential future scenarios through forecasting and trend analysis. It requires cadets to apply lessons in a series of complex, competitive games. Cadets will leave the class with a basic understanding of the complexities involved in strategic decision-making and methods to frame, understand, and address that complexity.                                                                                                                                                                                                                                                                                                                                                                                                  |                                                       | 2026-2 2027-2 2029-2                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                  |                                                    |
| <b>DS485</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>DOMAINS OF WAR</b>                                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2023-2                                                | <b>Offerings:</b>                                  |
| This course provides a historical, theoretical, and practical understanding of the roles of sea, air, space, and cyber power in military strategy and national security policy. After establishing a strong theoretical and historical foundation in each of the domains, students apply acquired knowledge to assess the strategies of multiple state and non-state actors, and to make practical recommendations on the application of military power across all domains to affect the outcomes of competition, crisis, and conflict.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                       | 2026-1 2027-1 2029-1                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                  |                                                    |
| <b>DS489</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>ADV IND STUD-DEF/STRAT STUDIES</b>                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2013-1                                                | <b>Offerings:</b>                                  |
| The course provides an environment that is conducive to independent effort in a subject area of special interest to the cadet. Original research or specialized study can be accomplished in any of the many fields within Defense and Strategic Studies. The course is conducted in three phases. First, the cadet and the individual advisor from the Defense and Strategic Studies faculty will reach agreement on a subject area for research. Research methods will be studied under the direction of the faculty member. Research may involve field trips and personal interviews with experts in the area of study. In the second phase, the cadet will engage in independent research and prepare a draft analytical paper or report detailing the findings. During this period, frequent consultation with the faculty advisor occurs regarding the progress in the project. In the third phase, the cadet will present and define the findings before a faculty committee.                                                             |                                                       | 2027-1 2029-1 2029-2                               |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | One paper or report of variable length; oral defense. |                                                    |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MS489                                                 |                                                    |
| <b>DS489A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>ADV IND STUD-DEF/STRAT STUDY A</b>                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2023-1                                                | <b>Offerings:</b>                                  |
| The course provides an environment that is conducive to independent effort in a subject area of special interest to the cadet. Original research or specialized study can be accomplished in any of the many fields within Defense and Strategic Studies. The course is conducted in three phases. First, the cadet and the individual advisor from the Defense and Strategic Studies faculty will reach agreement on a subject area for research. Research methods will be studied under the direction of the faculty member. Research may involve field trips and personal interviews with experts in the area of study. In the second phase, the cadet will engage in independent research and prepare a draft analytical paper or report detailing the findings. During this period, frequent consultation with the faculty advisor occurs regarding the progress in the project. In the third phase, the cadet will present and define the findings before a faculty committee. DS489A can only be taken after a cadet has completed DS489. |                                                       | 2027-1                                             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | One paper or report of variable length; oral defense. |                                                    |

| <b>DS490</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>SPECIAL TOPICS: STRAT STUDIES</b>                                                     | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2020-1                                                                                   | <b>Offerings:</b>                                  |
| The Special Topics Course provides cadets an opportunity for reading and analysis in depth within a topic area of special interest and timely relevance to Defense and Strategic Studies. The course director will determine the approach dependent on the topic and enrollment. Courses will normally develop the cadet's understanding of the topic through study of theory, history, doctrine, and historical and contemporary case studies. A generous portion of the course will normally address modern complex problems related to the topic area and assignments will emphasize analytical writing. Topics will vary by semester. A past Special Topics Course was "Sea and Air Power."                                                                                                                                                                                                                                                                |                                                                                          | 2027-1 2029-1                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                     |                                                    |
| <b>DS495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>RESEARCH METHODS STRAT STUDIES</b>                                                    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2019-2                                                                                   | <b>Offerings:</b>                                  |
| This is a research methods course designed to support the interdisciplinary nature of the research-based courses in the Defense & Strategic Studies Program, DS496 Strategic Studies Thesis and DS498 Strategic Studies Capstone. In this course, students will learn different research methodologies for the strategic studies field and develop effective writing skills in a seminar format. Partway through the semester students will choose an interdisciplinary research topic related to the Defense & Strategic Studies field. The topic, approved by the Course Director, will either be as part of a student-selected thesis for DS496 or an assigned client-based project for DS497. Thesis students must also select faculty members from across the institution to serve as their thesis advisors as part of DS495. Student assignments include practical exercises, a research proposal, a literature review and a draft research methodology. |                                                                                          | 2026-1 2027-1                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                     |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | DS320                                                                                    |                                                    |
| <b>DS496</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>STRATEGIC STUDIES THESIS</b>                                                          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2022-2                                                                                   | <b>Offerings:</b>                                  |
| Strategic Studies Thesis is an integrative course in strategy designed to provide Defense & Strategic Studies majors with practical experience in addressing real, complex and ambiguous strategic issues. Students will write and defend an interdisciplinary thesis that in some way relates to the military instrument of power or national security. Thesis cadets will continue the work they began with their faculty advisors in the fall by revising their literature review and research methodology, then complete data collection, data analysis, thesis writing and ultimately conduct an oral defense of their thesis before an interdisciplinary faculty board.                                                                                                                                                                                                                                                                                  |                                                                                          | 2026-2 2027-2 2029-2                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Program Director approval                                                                |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | DS489 DS495                                                                              |                                                    |
| <b>DS496A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>STRATEGIC STUDIES THESIS - A</b>                                                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2026-1                                                                                   | <b>Offerings:</b>                                  |
| Strategic Studies Thesis is an integrative course in strategy designed to provide Defense & Strategic Studies majors with practical experience in addressing real, complex and ambiguous strategic issues. Students will write and defend an interdisciplinary thesis that in some way relates to the military instrument of power or national security. Thesis cadets will continue the work they began with their faculty advisors in the fall by revising their literature review and research methodology, then complete data collection, data analysis, thesis writing and ultimately conduct an oral defense of their thesis before an interdisciplinary faculty board.                                                                                                                                                                                                                                                                                  |                                                                                          | 2026-1 2027-1                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                   |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | This course is intended to be the first course in a two-part thesis sequence with DS496. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | DS489 DS495                                                                              |                                                    |
| <b>DS497</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>STRATEGIC STUDIES CAPSTONE</b>                                                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                            |                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2022-2                                                                                     | <b>Offerings:</b>                                  |
| Strategic Studies Capstone is an integrative course in strategy designed to provide Defense & Strategic Studies majors with practical experience in addressing real, complex and ambiguous strategic issues. Students will work in small groups to solve such a problem for an external client organization focusing on the use of the military instrument of power or national security strategy. Student groups will determine stakeholder needs, define the client's problem, and conduct appropriate research to develop a viable solution or set of recommendations. Groups will meet regularly with their client, complete a final written report and provide a formal presentation with an oral defense to the faculty. Alternatively, select students may work with an interdisciplinary team from another academic department's capstone course to contribute analysis from the strategic studies perspective to the group's project. Capstone cadets will continue the work they began with their faculty advisors in the fall by revising their literature review and research methodology, then complete data collection, data analysis, final report writing and ultimately conduct an oral defense of their work before an interdisciplinary faculty board. |                                                                                            | 2026-2 2027-2 2029-2                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min                                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Program Director Approval                                                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | DS489 DS495                                                                                |                                                    |
| <b>DS497A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>STRATEGIC STUDIES CAPSTONE - A</b>                                                      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2026-1                                                                                     | <b>Offerings:</b>                                  |
| Strategic Studies Capstone is an integrative course in strategy designed to provide Defense & Strategic Studies majors with practical experience in addressing real, complex and ambiguous strategic issues. Students will work in small groups to solve such a problem for an external client organization focusing on the use of the military instrument of power or national security strategy. Student groups will determine stakeholder needs, define the client's problem, and conduct appropriate research to develop a viable solution or set of recommendations. Groups will meet regularly with their client, complete a final written report and provide a formal presentation with an oral defense to the faculty. Alternatively, select students may work with an interdisciplinary team from another academic department's capstone course to contribute analysis from the strategic studies perspective to the group's project. Capstone cadets will continue the work they began with their faculty advisors in the fall by revising their literature review and research methodology, then complete data collection, data analysis, final report writing and ultimately conduct an oral defense of their work before an interdisciplinary faculty board. |                                                                                            | 2026-1 2027-1                                      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min                                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | This course is intended to be the first course in a two-part capstone sequence with DS497. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | DS489 DS495                                                                                |                                                    |
| <b>DS498</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>LEADERSHIP IN FUTURE WAR</b>                                                            | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2022-1                                                                                     | <b>Offerings:</b>                                  |
| Clausewitz states that the violent, interactive, and fundamentally political nature of war is unchanging. The character of war, however, changes with technology, society, and the international political system. This course integrates previous instruction from DS320, DS370, and academy core courses on leadership and strategy while exploring how emerging technologies interact with the contemporary international environment to influence the character of war. Cadets analyze how technologies like artificial intelligence, autonomous weapons, social media, cyber, and drones might influence the future of war and create the need for adaptive leaders. In line with strategic guidance, this course emphasizes intellectual leadership and professionalism in the art and science of warfighting, deepens Cadets' understanding of history, and embraces the role of emerging technologies to better prepare leaders to compete with adversaries. DS498 seeks to deepen this understanding through case study analysis, written and oral communication, and collaborative projects.                                                                                                                                                                    |                                                                                            | 2026-2 2027-2 2029-2                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min                                                                     |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                                                                       |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | DS370 PL300<br>-Or-<br>DS370 PL350                                                         |                                                    |
| <b>ML100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>INTRO TO WARFIGHTING LAB</b>                                                            | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2020-3                                                                                     | <b>Offerings:</b>                                  |

ML100 Introduction to Warfighting Laboratory is a forty-seven-day period of instruction for Fourth Class cadets at the United States Military Academy. ML100 is offered during the summer term as part of Cadet Basic Training (CBT). CBT is a demanding progression of training requirements that forms the foundation for all future instruction at USMA. It instructs, trains, inspires, and transitions civilians and prior service military personnel into Cadets and future officers. CBT also inculcates attributes expected of future officers, including foundational military competencies and the values of character and commitment.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

|              |                                |                                                    |
|--------------|--------------------------------|----------------------------------------------------|
| <b>ML200</b> | <b>INTRO TO SMALL UNIT OPS</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------|----------------------------------------------------|

**Scope:**

2024-0

**Offerings:**

ML200 Introduction to Small Unit Operations Laboratory is a six-to-eight week program of instruction for Third Class cadets at the United States Military Academy. ML200 is offered during the summer term as part of Cadet Field Training (CFT I and CFT II). ML200 expands on the knowledge gained in MSL100 and MS100 by providing hands on application of concepts and lessons in a tactical field environment. Cadets perform select BOLC-A tasks in order to further develop the military and tactical knowledge necessary for future field training and lays the foundation for subsequent Military Science courses. The purpose of ML200 is to emphasize the foundational military competencies; prepare Third Class cadets to assume duties as NCOs in the Corps of Cadets; instill the warrior ethos in each cadet; and inspire each cadet to professional excellence through physically and mentally demanding training.

2029-0

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

**Prerequisite(s):**

MS100  
-Or-  
MS100X

|              |                 |                                                    |
|--------------|-----------------|----------------------------------------------------|
| <b>ML300</b> | <b>CLDT LAB</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------|----------------------------------------------------|

**Scope:**

2021-1

**Offerings:**

Cadets are enrolled in ML300 during their second or first class summer. This course trains, mentors, and assesses basic leadership skills focusing on Troop Leading Procedures (TLPs), effective communication, and tactical decision making in order to develop competent and confident small unit leaders capable of operating in an uncertain and rapidly changing environment. In a series of tactical scenarios that reflect the Decisive Action Training Environment, cadets experience a minimum of two assessed leadership positions which provides the cadet with a common experience to further enhance his/her leadership ability to solve difficult tactical problems.

2027-0 2028-0 2029-0

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

**Prerequisite(s):**

MS100 MS200 MS300  
-Or-  
MS100 MS200 MS350X  
-Or-  
MS100 MS250X MS350X  
-Or-  
MS100 MS250X MS300  
-Or-  
MS100 MS250 MS300  
-Or-  
MS100 MS200 MS350  
-Or-  
MS100 MS250 MS350  
-Or-  
MS100X MS250 MS300  
-Or-  
MS100X MS200 MS350  
-Or-  
MS100X MS250 MS350

|               |                                       |                                                    |
|---------------|---------------------------------------|----------------------------------------------------|
| <b>ML300A</b> | <b>CDT LEAD DEV TRAIN (SANDHURST)</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|---------------------------------------|----------------------------------------------------|

**Scope:**

2021-0

**Offerings:**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                       |                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------|
| ML300A CLDT Sandhurst is a 28-day period of instruction for rising Second- and First-Class cadets held at the Royal Military Academy at Sandhurst in the United Kingdom. ML300A is offered during each summer term to Cadets who demonstrate excellence across all three pillars. It is a demanding progression of training requirements that serves as a culminating event for all military training at USMA. CLDT ML300A evaluates Cadet leadership potential through execution and assessment of core warrior skills and a tactical field problem in order to improve their competence and fitness for future service as leaders within the Corps of Cadets and as commissioned officers in the United States Army. Cadets who successfully complete ML300B will receive credit for their ML300 requirement in lieu of participating in CLDT at USMA. |                                       | 2027-0 2028-0 2029-0                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ML100 ML200 MS100 MS200               |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ML300                                 |                                                               |
| <b>ML401X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>ADV MILITARY LEADERSHIP LAB</b>    | <b>0.5 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                | <b>Offerings:</b>                                             |
| This course provides an enhanced understanding of the roles and responsibilities of junior officers to expand beyond the basic warfighting mission taught in the military science curriculum. ML401X introduces cadets to the requirements of junior officers beyond tactical mission planning and challenges cadets to produce products required of them during their first assignments. Throughout the course, cadets demonstrate their knowledge through a series of lessons that introduce cadet to the systems and processes the Army uses in daily operations and how to understand and develop each of the products within the systems. Cadets who successfully complete ML401X are postured to excel in their first Army assignments.                                                                                                            |                                       | 2026-1 2026-2                                                 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 10 @ 120 min             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |
| <b>MS100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>INTRODUCTION TO WARFIGHTING</b>    | <b>1.5 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2019-1                                | <b>Offerings:</b>                                             |
| This course provides cadets with the foundation of military and tactical knowledge necessary for future field training and development in subsequent military science courses. Cadets will gain a solid foundation built on basic Army concepts such as Shoot, Move, and Communicate. Cadets will also learn fundamental Army unit organizations, capabilities and missions, and develop an understanding of the roles of NCOs and Officers. Cadets who have successfully completed MS100, will understand their role as Soldiers and will be well prepared as they transition from follower to leader during the next chapter of their military education, Cadet Field Training.                                                                                                                                                                        |                                       | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |
| <b>MS100X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>INTRODUCTION TO WARFIGHTING</b>    | <b>1.5 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2026-1                                | <b>Offerings:</b>                                             |
| This course provides cadets with the foundation of military and tactical knowledge necessary for future field training and development in subsequent military science courses. Cadets will gain a solid foundation built on basic Army concepts such as Shoot, Move, and Communicate. Cadets will also learn fundamental Army unit organizations, capabilities and missions, and develop an understanding of the roles of NCOs and Officers. Cadets who have successfully completed MS100, will understand their role as Soldiers and will be well prepared as they transition from follower to leader during the next chapter of their military education, Cadet Field Training. *This is a pilot version of MS100 and must go through the Curriculum Committee for review NLT AY26-2 to continue.*                                                     |                                       | 2026-1 2026-2                                                 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MS100                                 |                                                               |
| <b>MS200</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>FUNDAMENTALS OF SMALL UNIT OPS</b> | <b>1.5 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2024-0                                | <b>Offerings:</b>                                             |

This course introduces cadets to the small unit leader's role in the Army by developing the foundational tactical knowledge, critical thinking and problem-solving skills necessary for adaptive leaders in current and future operational environments. Fundamentals of Small Unit Operations builds upon the knowledge, competencies and experience cadets gain in MS100 and Cadet Field Training. It explores Army leadership, troop leading procedures, and small-unit tactics to develop and hone decision-making skills. Throughout the course, cadets demonstrate their knowledge through a series of tactical decision exercises, mission briefing assignments, written assessments, and simulations lab activities. Cadets who successfully complete MS200 are able to apply problem solving techniques to unstructured tactical problems and scenarios, use and incorporate battle drills, demonstrate foundational military competencies, and execute principles of patrolling and planning in a multi-domain operational environment.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MS100  
-Or-  
MS100X

**Disqualifier(s):** MS250  
-Or-  
MS250X

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>MS250</b> | <b>ADV FUND OF SMALL UNIT OPS</b> | <b>1.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

This course provides an enhanced exploration of MS200 course material by introducing cadets to the small unit leader's role in the Army by developing the foundational tactical knowledge, critical thinking, and problem-solving skills necessary for adaptive leaders in current and future operational environments. Advanced Fundamentals of Small Unit Operations builds upon the knowledge, competencies and experience cadets gain in MS100 and Cadet Field Training while further challenging cadets on its respective application to modern Army leadership, troop leading procedures, small-unit tactics, and decision-making skills within the context of complex and real-world problem sets. Throughout the course, cadets demonstrate their knowledge through a series of tactical decision exercises, development of operational planning products and orders, mission briefings, written assessments, and simulations lab activities. Cadets who successfully complete MS250 are postured to excel in follow-on advanced military instruction and can apply problem solving techniques to unstructured tactical problems and scenarios, use and incorporate battle drills, demonstrate foundational military competencies, and execute principles of patrolling and planning in a multi-domain operational environment.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Disqualifier(s):** MS200  
-Or-  
MS250X

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|              |                           |                                                          |
|--------------|---------------------------|----------------------------------------------------------|
| <b>MS300</b> | <b>PLATOON OPERATIONS</b> | <b>1.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------|----------------------------------------------------------|

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**Scope:** 2022-0

**Offerings:**

This course builds upon foundational tactical planning and decision-making skills taught during MS200. MS300 further develops the cadet's knowledge of doctrinal tactical principles and general professional knowledge, using Troop Leading Procedures (TLPs) as a framework for planning and preparation. Cadets are challenged to apply tactical knowledge, competencies, and decision-making to solve complex situations that require critical thinking and creative problem-solving skills. Instruction in the fundamentals of small unit operations emphasizes both offensive and defensive tactics. Additionally, cadets are expected to demonstrate an increased understanding of the TLPs and mental agility through execution of tactical decision-making exercises. In addition to tactics, cadets continue their general instruction in the various Army systems, procedures and functions that are important aspects of officership. Finally, cadets examine the small unit leader's role in ensuring that the moral and ethical decision making process is integrated into all operations.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MS100 MS200  
-Or-  
MS100 MS200  
-Or-  
MS100 MS250X  
-Or-  
MS100 MS250  
-Or-  
MS100X MS200  
-Or-  
MS100X MS250

**Disqualifier(s):** MS350  
-Or-  
MS350X

|              |                                    |                                                    |
|--------------|------------------------------------|----------------------------------------------------|
| <b>MS350</b> | <b>ADVANCED PLATOON OPERATIONS</b> | <b>1.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|------------------------------------|----------------------------------------------------|

**Scope:** 2026-1

**Offerings:**

This course provides an enhanced exploration of MS300 course material by building upon foundational tactical planning and decision-making skills taught during MS200/MS250. MS350 further develops the cadet's knowledge of doctrinal tactical principles and general professional knowledge, using Troop Leading Procedures (TLPs) as a framework for planning and preparation. Cadets are challenged to apply tactical knowledge, competencies, and decision-making to solve complex situations that require critical thinking and creating problem-solving skills. Instruction in the fundamentals of small unit operations emphasizes both offensive and defensive tactics. Additionally, cadets are expected to demonstrate an increased understanding of the TLPs and mental agility through execution of tactical decision-making exercises. In addition to tactics, cadets continue their general instruction in the various Army systems, procedures, and functions that are important aspects of officership. Finally, cadets examine the small unit leader's role in ensuring that the moral and ethical decision-making process is integrated into all operations.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MS250  
-Or-  
MS250X  
-Or-  
MS200

**Disqualifier(s):** MS300  
-Or-  
MS350X

## Physical Education

### 47 Courses

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                     |                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------|
| <b>PE100</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>PHYS ED FOUNDATIONS-MEN</b>      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                 |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1998-2                              | <b>Offerings:</b><br>No Course Offerings                                           |
| <b>Lessons:</b> 68 @ 45 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 9 @ 45 min             |                                                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                |                                                                                    |
| <b>PE105</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>FUNDAMENTAL COMBATIVE SKILLS</b> | <b>0.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                 |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-0                              | <b>Offerings:</b><br>2029-0                                                        |
| In this 3.5 hour time block Cadets are taught rudimentary grappling techniques. In this block of instruction the New Cadets are introduced to the following positions: Guard, Top Mount, and Side Control. The New Cadets learn how to hold and escape from these positions and also learn submission holds to apply to their partner when they are in the dominant position.                                                                                                                                                                                                                                                                                                          |                                     |                                                                                    |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min              |                                                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                |                                                                                    |
| <b>PE107</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>FOUNDATIONS OF MOVEMENT</b>      | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                 |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2008-1                              | <b>Offerings:</b><br>No Course Offerings                                           |
| This introductory movement and fitness course is designed to improve a cadet's upper body strength, hip flexor strength, and core body stabilization and to prepare him or her for success in PE 117 (Military Movement). The class focuses on developing the specific strength and skills needed to pass PE 117 (Military Movement) and the Indoor Obstacle Course Test.                                                                                                                                                                                                                                                                                                              |                                     |                                                                                    |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 20 @ 55 min            |                                                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                |                                                                                    |
| <b>PE108</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>FOUNDATIONS OF FITNESS</b>       | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                 |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2007-3                              | <b>Offerings:</b><br>No Course Offerings                                           |
| The purpose of this course is to physically develop cadets utilizing progressive and sequential resistance exercises and cardiorespiratory conditioning exercises that will enable them to pass the Army Physical Fitness Test and Indoor Obstacle Course Test. This course will provide cadets, who have a deficient Physical Performance Score Cumulative, an opportunity to supplement/raise their PPSC. Cadets will develop a sense of self-responsibility for their personal fitness and a lifetime commitment to maintain their physical fitness. This course will only be offered during Summer Term Academic Program.                                                          |                                     |                                                                                    |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 20 @ 90 min            |                                                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                |                                                                                    |
| <b>PE109</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>FUNDAMENTALS OF AQUATICS</b>     | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                 |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2008-1                              | <b>Offerings:</b><br>2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| This course has two phases: phase one is a remedial swimming exploration curriculum designed to prepare cadets classified as non-swimmers for the survival swimming curriculum. The remedial phase is designed to help cadets acquire in-water experiences, and gradually refine the basic motor skills needed to be comfortable, safe, and effective in and around the aquatic environment. Phase two emphasizes the military applications of swimming and survival skills to include the elements of breath control, buoyancy positions, stroke assessment, and swimming endurance. Successful completion fulfills the survival swimming graduation requirement for selected cadets. |                                     |                                                                                    |



**Lessons:** 40 @ 50 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>PE115</b> | <b>FUNDAMENTALS OF COMBATIVES</b> | <b>0.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------------|

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**Scope:** 2013-1

**Offerings:**

This course exposes Cadets to variety of basic standing and ground skills. Cadets learn how to engage in the free movement range, clinch range, and grappling range. In the free movement range Cadets learn how to strike with their hands and defend themselves. In the Clinch range cadets learn how to close with their opponent, achieve the Clinch, then control their opponent using knees and movement. In the Grappling range Cadets learn how to move into and out of positions and apply submissions and chokes. Body mechanics, aggressiveness, and affective reactions are stressed.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 19 @ 50 min

**Special Requirements:** None

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|              |               |                                                          |
|--------------|---------------|----------------------------------------------------------|
| <b>PE116</b> | <b>BOXING</b> | <b>0.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------|----------------------------------------------------------|

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**Scope:** 2006-1

**Offerings:**

A course in which the offensive and defensive skills of amateur boxing are taught. Course content includes stances, movement, basic punches (i.e. jab, cross, hook, and upper cut), defenses, strategies, and tactics. Instruction on refereeing, judging, and serving as a corner second are presented. Boxers are evaluated, assessed and provided feedback on their ability to box. The course exposes participants to the coping strategies necessary to deal with a physical threat.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 19 @ 50 min

**Special Requirements:** None

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|              |                          |                                                          |
|--------------|--------------------------|----------------------------------------------------------|
| <b>PE117</b> | <b>MILITARY MOVEMENT</b> | <b>0.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------|----------------------------------------------------------|

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**Scope:** 2025-1

**Offerings:**

Military Movement is a 19-lesson course designed to expose Cadets to a variety of basic movement skills aimed at developing basic muscular and motor patterns. The course is critical in the promotion of strength, power, flexibility, balance, agility, coordination, social interaction, fear management, and teamwork and serves as a basis for many other athletic and military activities that Cadets will encounter during their time at the USMA as well as in their Army career.

2026-1 2026-2 2026-3  
2027-1 2027-2 2027-3  
2028-1 2028-2 2028-3  
2029-1 2029-2 2029-3

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 19 @ 50 min

**Special Requirements:** None

**Disqualifier(s):** PE117X

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|               |                                |                                                          |
|---------------|--------------------------------|----------------------------------------------------------|
| <b>PE117R</b> | <b>MILITARY MOVEMENT - REM</b> | <b>0.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|--------------------------------|----------------------------------------------------------|

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**Scope:** 2025-1

**Offerings:**

Military Movement is a 19-lesson course designed to expose Cadets to a variety of basic movement skills aimed at developing basic muscular and motor patterns. The course is critical in the promotion of strength, power, flexibility, balance, agility, coordination, social interaction, fear management, and teamwork and serves as a basis for many other athletic and military activities that Cadets will encounter during their time at the USMA as well as in their Army career.

No Course Offerings

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 19 @ 50 min

**Special Requirements:** None

**Prerequisite(s):** PE117

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|               |                          |                                                          |
|---------------|--------------------------|----------------------------------------------------------|
| <b>PE117X</b> | <b>MILITARY MOVEMENT</b> | <b>0.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|--------------------------|----------------------------------------------------------|

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**Scope:** 2026-1

**Offerings:**

Military Movement is a 19-lesson course designed to expose Cadets to a variety of basic movement skills aimed at developing basic muscular and motor patterns. The course is critical in the promotion of strength, power, flexibility, balance, agility, coordination, social interaction, fear management, and teamwork and serves as a basis for many other athletic and military activities that Cadets will encounter during their time at the USMA as well as in their Army career. \*\*This is a pilot version of PE117 and must be reviewed by the Curriculum Committee NLT AY26-2 for continued enrollment in AY27+.\*\*\*

2026-1 2026-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

None

**Disqualifier(s):**

PE117

**PE205****ADVANCED COMBATIVE SKILLS****0.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)**Scope:**

2013-0

**Offerings:**

This four hour time block is designed to enhance Cadets' grappling ability. Cadets review fundamental grappling skills from the previous summer and are taught new grappling skills in positioning and submissions to further develop their ability to perform well in future Combatives classes.

2029-0

**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

None

**PE215****FUNDAMENTALS/PERSONAL FITNESS****1.5 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)**Scope:**

2025-1

**Offerings:**

The course emphasizes basic human physiology concepts and the mechanisms by which the body adapts to and benefits from physical training. Cadets will learn how to assess, develop, and maintain their individual physical, nutritional, and sleep readiness as defined by the Army's Holistic Health and Fitness (H2F) system. Activities will focus on improving strength, endurance, mobility, power and speed while utilizing body composition to enhance overall physical performance. Cadets will understand how to properly fuel for, prepare, and recover from physical activity utilizing appropriate nutritional behaviors and sleep strategies. Human performance optimization will be discussed in the context of motivation, commitment and perseverance. Cadets will leave the course with exposure to H2F concepts, and a comprehensive profile of their individual readiness.

2026-1 2026-2 2026-5  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2**Lessons:** 20 @ 55 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

None

**Disqualifier(s):**

PE150

**PE220****AEROBIC FITNESS****0.5 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)**Scope:**

2013-1

**Offerings:**

The course provides cadets with information and experiences to create an aerobic optimal performance plan. Cadets are exposed to numerous aerobic fitness activities and participate in events focused on military applications. The principles of exercise physiology serve as the foundation for the course as students design and participate in various aerobic conditioning assessment activities.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 18 @ 50 min**Special Requirements:**

None

**PE222****BADMINTON/PICKLEBALL****0.5 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)**Scope:**

2011-2

**Offerings:**

This is a 19 lesson course featuring nine lessons of Pickleball and ten lessons of Badminton. Pickleball is a sport played by two, three, or four people. Pickleball uses a wooden paddle and whiffleball and is very similar to tennis. The course focus is on the rules of play and basic skill development of service and service return, forehand and backhand drives, volley and half-volley, drop shot, lob, and overhead smash. Additional instruction in basic offensive/defensive strategy and tactics is provided. Badminton is a sport played by two, three, or four people on the same size court as pickleball. Badminton uses a lightweight strung racquet and shuttlecock (birdie). The course focus is also on the rules of play and basic skill development of service and service return, forehand, backhand, drop shot, lob, and overhead smash. Offensive/defensive strategy and tactics are discussed. Grading is determined by a final exam in each.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2  
2029-8 2029-9

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                       |                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 18 @ 50 min              |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                  |                                                                                              |
| <b>PE223</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>BASKETBALL</b>                     | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2025-1                                | <b>Offerings:</b>                                                                            |
| This course is designed to provide Cadets with the ability and knowledge necessary to play and coach the game of basketball. It is intended to assist students in developing basic skills for playing as well as teaching the fundamental basketball skills while nurturing an appreciation for basketball as a lifetime physical activity. In addition, leadership skills, team development, and the value of teamwork are stressed through live competition. Cadets are evaluated on their demonstration and knowledge of fundamental skills, performance on the James Naismith Basketball Obstacle Course, individual and team performance, and a written term end exam.                       |                                       | 2026-1 2026-2 2027-1<br>2027-2 2027-9 2028-1<br>2028-2 2028-9 2029-1<br>2029-2 2029-8 2029-9 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 18 @ 50 min              |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                  |                                                                                              |
| <b>PE224</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>ADVANCED CLOSE QUARTERS COMBAT</b> | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2011-1                                | <b>Offerings:</b>                                                                            |
| This course is designed to educate and expose Cadets in the realm of edge weapons and impact weapons. Cadets learn realistic offensive tactics, defensive tactics, weapon movement patterns, and footwork in order to engage an enemy who has an edge weapon or impact weapon or to use an edge weapon or impact weapon on the enemy in a hostile situation. The cadets are evaluated on their functional ability to perform the skill learned in the course and also on their ability to show tactful aggression and fear management.                                                                                                                                                            |                                       | No Course Offerings                                                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 18 @ 50 min              |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                  |                                                                                              |
| <b>PE226</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>COMBAT GRAPPLING</b>               | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2013-1                                | <b>Offerings:</b>                                                                            |
| This course is designed to provide the cadets with realistic grappling applications and ground fighting skills to enhance their knowledge and warrior ethos to prepare them as future soldiers for unarmed combat and CQC situations. This course focuses on five different grappling positions: Mount, Guard, Side Control, Rear Mount, and scarf hold. Cadets learn how to maintain the positions, escape from the positions, submit from the positions, and strike from the positions. Cadets are evaluated in their function ability to perform skill-sets learned in class, and are also evaluated in their ability to manage their fear, keep their composure, and dominate their opponent. |                                       | 2026-1 2026-2 2026-8<br>2027-2 2028-2 2029-2                                                 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 19 @ 50 min              |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                  |                                                                                              |
| <b>PE228</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>MODERN ARMY COMBATIVES L1 CERT</b> | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2010-1                                | <b>Offerings:</b>                                                                            |
| PE 228 is designed to certify cadets to teach U.S. Army Level I Combative Skills. Cadets develop a foundation of basic combative grappling skills and an aggressive mind set needed to engage and defeat an enemy in Close Quarters Combat (CQC). The course focuses on 15 basic grappling skills. Cadets are evaluated on their demonstration and knowledge of the required skills. Successful completion of the course qualifies the individual to conduct Skill Level I Combatives training for soldiers.                                                                                                                                                                                      |                                       | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                                |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 18 @ 50 min              |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                  |                                                                                              |
| <b>PE230</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>CYCLING</b>                        | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2013-1                                | <b>Offerings:</b>                                                                            |

Cycling as a lifetime sport, is designed to take the beginner through a progressive program of bicycle training and instruction to include: proper mounting, balance, turning, ascending, and descending individually and in a group. The course labs are hands-on and focused on learning through practical application and drills on the bike. All riders are also introduced to basic bike maintenance and required to demonstrate baseline skills in preventive maintenance checks and services (PMCS). Classroom instruction is focused on the introduction of cycling principles and as a feedback forum for the riding labs.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 18 @ 50 min

**Special Requirements:**

None

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

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**PE232 EMERGENCY WATER SAFETY**

**0.5 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2010-1

**Offerings:**

The purpose of this course is to introduce cadets who are already proficient swimmers, to first responder training in methodologies generic to Water Rescue, CPR, and Emergency First Aid. Cadets are exposed to a variety of distress and drowning scenarios, and will be able to demonstrate strategies and site specific response techniques essential to safely performing a water related assist and/or rescue. Course focus is on military application in both still water and theatre specific environments, with a special emphasis on emergency management protocol. Written evaluation, five skill-specific performance components, and successful completion of two comprehensive exit scenarios are required for successful course completion.

2026-1 2026-2 2029-1  
2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 18 @ 50 min

**Special Requirements:**

None

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**PE234 LIFEGUARDING**

**0.5 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2010-1

**Offerings:**

This course focuses on a holistic approach to the duties and responsibilities of a trained professional lifeguard and exposes cadets to key elements and strategies related to accident prevention, surveillance methodology, and performance. Additional content and activities focus on emergency response, search and rescue, and duty specific incident/accident management. Cadets who successfully complete certification requirements may obtain professional accreditation/licensure in Lifeguarding, CPR/PR, Oxygen Administration, and Automated External Defibrillation (AED). Additional accreditation/licensure may also be available in both Open Water and Water park Lifeguarding. Written evaluation, four skill-specific performance evaluations, and successful completion of three comprehensive exit scenarios are required for successful course completion.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 18 @ 50 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

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**PE236 GROUP EXERCISE LEADERSHIP**

**0.5 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2013-1

**Offerings:**

Using music as the controlling factor, this course is designed to give participants an opportunity to experience different modalities of exercise such as high/low impact, step, kickboxing, circuit training, spinning, yoga/pilates and water exercise in an Exercise to Music group fitness setting. Participants will be assessed on knowledge of applicable fitness principles, exercise safety, lesson construction and a team-teaching experience of one's choice.

2026-1 2029-1

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 18 @ 50 min

**Special Requirements:**

None

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**PE238 GOLF**

**0.5 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:** 2010-1

**Offerings:**

This course is designed to provide the beginner and novice golfer with the skills, knowledge, and techniques needed to play golf. The basic techniques taught are the full swing, pitching, chipping, and putting. Course grading is based upon a series of skill tests, a written examination, and a golf swing analysis.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2  
2029-8

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 18 @ 50 min

**Special Requirements:**

None

| PE242                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ICE SKATING              | 0.5 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2013-1                   | <b>Offerings:</b>                                                              |
| This course is designed to provide cadets who have little or no previous skating experience with the basic skills necessary to safely participate as a recreational skater. The forward and backward stroke, snow plow, "T" stop, and hockey stop, as well as forward and backward crossovers are presented. Additional skills taught are turns, spins and jumps. Grading is based upon the cadet's ability to demonstrate the skills taught during the course. A compulsory skating routine is also used for evaluating student proficiency. Additionally, a short creative routine of optional figures chosen by the cadet is evaluated. |                          | No Course Offerings                                                            |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 18 @ 50 min |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                     |                                                                                |
| PE244                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | JUDO                     | 0.5 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2010-1                   | <b>Offerings:</b>                                                              |
| The purpose of this course is to introduce judo as a competitive sport and the application of judo skills for self defense and combatives training. The course content will include falling skills and basic throwing, pinning, and submission skills. Judo customs, courtesies, terminology, and competitive rules will be introduced. Students will gain an entry level knowledge and understanding of the basic skills, safety concerns, and rules needed to participate in competitive Judo. Students will be graded on a demonstration of basic skills and knowledge of competitive rules and terminology.                            |                          | 2026-8 2029-1 2029-2<br>2029-8 2029-9                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 18 @ 50 min |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                     |                                                                                |
| PE245                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | OLYMPIC WEIGHTLIFTING    | 0.5 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2022-1                   | <b>Offerings:</b>                                                              |
| This introductory course provides cadets with the knowledge and hands-on experience for conducting Olympic Weightlifting. Cadets develop the critical skills needed for coaching and demonstrating safe and proper Olympic Weightlifting technique that will benefit both them and their soldiers. This course is designed to instill a lifetime desire for continued participation in training and competing in Olympic Weightlifting.                                                                                                                                                                                                    |                          | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 2029-8           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 19 @ 50 min |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                     |                                                                                |
| PE246                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | RACQUETBALL              | 0.5 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2011-1                   | <b>Offerings:</b>                                                              |
| This course introduces the basic skill and strategy fundamentals of racquetball. Cadets learn to identify and demonstrate the basic fundamentals of: personal playing safety; rules of play; forehand and backhand stroke techniques; kill, passing, and defensive shots; serve, serve return techniques and strategies. Singles play, doubles and "cut throat" are examined. Grading is determined by performance on two skills tests (rally & ceiling shot), and a written final exam.                                                                                                                                                   |                          | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                  |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 18 @ 50 min |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                     |                                                                                |
| PE248                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | INDOOR ROCK CLIMBING     | 0.5 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2010-1                   | <b>Offerings:</b>                                                              |
| This course develops fundamental rock climbing skills, techniques and safety awareness. This course introduces basic rock climbing systems, rappelling, belaying, knots, top roping, and assorted climbing skills. Course grading is based on climbing skills, rappelling skills, knowledge of basic rock climbing systems, and the application of judgment and safety practices in various situations.                                                                                                                                                                                                                                    |                          | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2<br>2029-8 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 18 @ 50 min |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                     |                                                                                |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------|
| <b>PE250</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>SCUBA</b>                | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2011-1                      | <b>Offerings:</b>                                             |
| This course is designed to provide cadets with the basic skills and knowledge needed to safely participate in SCUBA diving and pursue certification as a National Association of Underwater Instructors (NAUI) Basic SCUBA Diver. Successful completion of this course leads to Confined Water Certification and the ability to enroll in Open Water training. The requirements of this course include the successful demonstration of skin and SCUBA diving skills, the ability to practice and adhere to safe diving activities, and the completion of a comprehensive, written final examination. Cadets who possess Scuba certification or are members of the Cadet Sky Diving Club are ineligible for this course. |                             | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 6 @ 50 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 12 @ 50 min    |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                        |                                                               |
| <b>PE252</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>SKIING-ALPINE</b>        | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2013-2                      | <b>Offerings:</b>                                             |
| This course is designed to teach beginning skiers to ski in balance and control in all terrain and snow conditions. Knowledge of skiing equipment, proper body position, stopping, gliding, edging, sliding, turning, and carving is taught. The course grade is based upon skiing performance assessments administered on the slope.                                                                                                                                                                                                                                                                                                                                                                                   |                             | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 18 @ 50 min    |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                        |                                                               |
| <b>PE254</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>SKIING-CROSS COUNTRY</b> | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2010-1                      | <b>Offerings:</b>                                             |
| This course introduces cadets to the basic skills and techniques of cross-country skiing. It emphasizes skill development and the benefits of skiing as a lifetime fitness activity. Cadets are required to successfully demonstrate the diagonal stride, skating, turning, uphill techniques, and downhill techniques. Course grading is determined by instructor, peer and self-assessment of skiing ability and a written examination.                                                                                                                                                                                                                                                                               |                             | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 18 @ 50 min    |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                        |                                                               |
| <b>PE256</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>SNOWBOARDING</b>         | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2010-1                      | <b>Offerings:</b>                                             |
| This course is designed to provide cadets with the basic skills and knowledge needed to safely participate in snowboarding. The course focuses on teaching beginning snowboarders to ride in balance and control in various terrain and snow conditions. Knowledge of boarding equipment, as well as skills in proper stance and balance, stopping, gliding, edging, turning, carving and basic freestyle maneuvers will be covered.                                                                                                                                                                                                                                                                                    |                             | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 18 @ 50 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 0 @ 0 min      |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                        |                                                               |
| <b>PE258</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>SOCCER</b>               | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2025-1                      | <b>Offerings:</b>                                             |
| This course is designed to provide Cadets with the skills and knowledge necessary for playing soccer. A variety of individual skills and techniques are taught, as well as individual/team offensive and defensive strategies. The value of small sided games are used as building blocks that lead to full 11 aside matches. Grading for the course is based upon three skills test (juggling, passing, shooting) and a written examination.                                                                                                                                                                                                                                                                           |                             | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 18 @ 50 min    |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                        |                                                               |
| <b>PE260</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>SPORTS PHYSIOLOGY</b>    | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  |                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2010-1                           | <b>Offerings:</b>                                                                     |
| The objectives of this course are to introduce cadets to applied concepts of Sports Physiology, conduct personal fitness assessments in DPE's Center for Physical Development Excellence facility, and perhaps complete an independent study examining a sports physiology issue. The cadet becomes familiar with the varied aspects of Sports Physiology and is able to demonstrate baseline and advanced knowledge of 'core principles.' Critical thinking and analysis is used in all endeavors. The personal assessments conducted in the lab and the independent study approach provides cadets with insightful physiological information that can enhance their personal fitness performance. |                                  | 2026-1 2029-1                                                                         |
| <b>Lessons:</b> 18 @ 50 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 0 @ 0 min           |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                             |                                                                                       |
| <b>PE262</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>STRENGTH DEVELOPMENT</b>      | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-1                           | <b>Offerings:</b>                                                                     |
| This course provides Cadets with the knowledge and hands-on experience for conducting a variety of resistance training exercises. Cadets develop the critical skills needed for teaching safe and proper resistance training techniques that will benefit both themselves and their Soldiers. Cadets are assessed in the following areas: demonstrate proficiency performing multi-joint strength exercises, understand foundational programming criteria, and effectively provide peer coaching.                                                                                                                                                                                                   |                                  | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2<br>2029-8        |
| <b>Lessons:</b> 6 @ 50 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Labs:</b> 12 @ 50 min         |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                             |                                                                                       |
| <b>PE264</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>TENNIS</b>                    | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2013-1                           | <b>Offerings:</b>                                                                     |
| This course is designed to develop basic tennis skills including the forehand and backhand groundstrokes, serve, and volley. Students also learn the basic rules and etiquette of tennis as well as simple singles and doubles strategies. Grading is based on skills tests, an in-class tournament, instructor stroke evaluations, and a written final examination.                                                                                                                                                                                                                                                                                                                                |                                  | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2<br>2029-8        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 18 @ 50 min         |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                             |                                                                                       |
| <b>PE266</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>VOLLEYBALL</b>                | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2010-1                           | <b>Offerings:</b>                                                                     |
| This course is designed to teach cadets the fundamentals of volleyball. The individual skills taught are the underhand pass, set, serve, block, and spike. Additionally, the course covers defensive and offensive formations, the transition game, officiating techniques and United States Volleyball Association (USVA) rules. Grading is based upon skill testing on the underhand pass, set and serve, a written test, and a round robin tournament.                                                                                                                                                                                                                                           |                                  | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 2029-8                  |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 18 @ 50 min         |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                             |                                                                                       |
| <b>PE267</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>YOGA</b>                      | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024-1                           | <b>Offerings:</b>                                                                     |
| This introductory course provides cadets with the knowledge and experience for participating in Yoga. Cadets develop the skills needed for executing, teaching, and demonstrating safe and proper Yoga poses that will benefit both them and their soldiers. This course is designed to instill a lifetime desire for continued participation in Yoga and Mindfulness.                                                                                                                                                                                                                                                                                                                              |                                  | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2<br>2029-8 2029-9 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 19 @ 50 min         |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                             |                                                                                       |
| <b>PE268</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>CURRENT LIFETIME ACTIVITY</b> | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                       |                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2010-1                                | <b>Offerings:</b>                                             |
| The lifetime sports skills series are credit courses intended to provide cadets an opportunity to learn additional sports skills which can be incorporated into lifetime fitness goals and activities. They will further the development of neuromuscular and kinesthetic abilities through the acquisition of and participation in sport skills and can be part of the cadet personal fitness programs. Specific lifetime sport activities are selected on the basis of current trends, resources, instructor expertise, and cadet needs.                                                                                               |                                       | 2026-2 2027-2 2028-2<br>2029-1 2029-2                         |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 18 @ 50 min              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |
| <b>PE320</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>SURVIVAL SWIMMING - ELEMENTARY</b> | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2010-1                                | <b>Offerings:</b>                                             |
| The Survival Swimming-Elementary course is designed to develop aquatic proficiency for cadets who swam 150 yards in 4 minutes or more on their initial entry swim classification test. The Program of Instruction (POI) is divided into two areas: basic swimming and combat/survival swimming. Emphasis in all levels is on the military applications of swimming and survival skills to include the elements of breath control, buoyancy positions, stroke assessment, and swimming endurance. Grading is primarily based on criterion-referenced scales in basic and survival swimming skills.                                        |                                       | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 19 @ 50 min              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |
| <b>PE321</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>SURVIVAL SWIMMING - LOW</b>        | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2010-1                                | <b>Offerings:</b>                                             |
| The Survival Swimming-Low Intermediate course is designed to develop aquatic proficiency for cadets who swam 150 yards between 3 minutes 16 seconds and 3 minutes 59 seconds on their initial entry swim classification test. The Program of Instruction (POI) is divided into two areas: basic swimming and combat/survival swimming. Emphasis in all levels is on the military applications of swimming and survival skills to include the elements of breath control, buoyancy positions, stroke assessment, and swimming endurance. Grading is primarily based on criterion-referenced scales in basic and survival swimming skills. |                                       | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 19 @ 50 min              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |
| <b>PE322</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>SURVIVAL SWIMMING - HIGH</b>       | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2010-1                                | <b>Offerings:</b>                                             |
| The Survival Swimming-High Intermediate course is designed to develop aquatic proficiency for cadets who swam 150 yards between 2 minutes 30 seconds and 3 minute 15 seconds on their initial entry swim classification test. The Program of Instruction (POI) is divided into two areas: basic swimming and combat/survival swimming. Emphasis in all levels is on the military applications of swimming and survival skills to include the elements of breath control, buoyancy positions, stroke assessment, and swimming endurance. Grading is primarily based on criterion-referenced scales in basic and survival swimming skills. |                                       | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 19 @ 50 min              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |
| <b>PE323</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>SURVIVAL SWIMMING - ADVANCED</b>   | <b>0.5 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2010-1                                | <b>Offerings:</b>                                             |
| The Survival Swimming-Advanced course is designed to develop aquatic proficiency for cadets who swam 150 yards in less than 2 minutes 30 seconds on their initial entry swim test. The Program of Instruction (POI) is divided into two areas: basic swimming and combat/survival swimming. Emphasis in all levels is on the military applications of swimming and survival skills to include the elements of breath control, buoyancy positions, stroke assessment, and swimming endurance. Grading is primarily based on criterion-referenced scales in basic and survival swimming skills.                                            |                                       | 2026-1 2026-2 2026-3<br>2027-3 2028-3 2029-1<br>2029-2        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 19 @ 50 min              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | None                                  |                                                               |



| PE360                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | COMBAT APPLICATIONS            | 0.5 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2013-1                         | <b>Offerings:</b>                                                              |
| This course provides cadets with a comprehensive set of basic combatives skills suited for a combat scenario. Cadets will learn to respond appropriately to aggression by utilizing proper body mechanics, skills, aggressiveness, and fear management. Two combat ranges of hand-to-hand fighting are taught: 1) Grappling range - cadets learn to fight and win on the ground and, 2) Clinch range - cadets learn to close the distance and control the fight between themselves and an attacker. Cadets will be evaluated on their ability to perform selected combative skills and their capacity to exhibit the warrior ethos and fear management.                                                                                                                                                                                        |                                | 2026-1 2026-2 2027-1<br>2027-2 2027-3 2028-1<br>2028-2 2028-3 2029-1<br>2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 20 @ 55 min       |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                           |                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | PE460                          |                                                                                |
| PE450                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ARMY FITNESS DEVELOPMENT       | 1.5 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2025-1                         | <b>Offerings:</b>                                                              |
| This course prepares future officers for their role as fitness leaders by equipping them with content and experiences that cultivate their ability to plan, implement, and assess unit physical training under varied conditions. Course material emphasizes basic concepts of human physiology and the mechanisms by which the body adapts to and benefits from physical readiness training. Cadets will apply their knowledge of the United States Army's Holistic Health and Fitness (H2F) system in pursuit of optimizing unit (group) performance. Cadets must assess the physical requirements of units and Soldiers, program platoon level physical readiness training, and evaluate the efficacy of that training. Course graduates have the knowledge to enhance the health and human performance of the future units they will lead. |                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                  |
| <b>Lessons:</b> 20 @ 55 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min         |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                           |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | PE150<br>-Or-<br>PE215         |                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | PE350                          |                                                                                |
| PE471                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ADV SP DEV/PHY IND ADV DEV     | 2.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2010-7                         | <b>Offerings:</b>                                                              |
| Advanced Sport Development is an intense physical program designed for cadets with an interest in total fitness and a comprehensive scuba experience. This program consists of four subcourses: Aerobic Fitness (mountain biking, hiking, kayaking, etc.), Sports Physiology, Muscular Fitness, and SCUBA.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                | 2026-7 2027-7 2028-7<br>2029-7                                                 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min         |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                           |                                                                                |
| PE472                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | OUTER LIMITS - MOUNTAIN LEADER | 2.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2005-4                         | <b>Offerings:</b>                                                              |
| The Outer Limits – Mountain Leader course is designed to provide cadets with the basic skills and knowledge needed to safely participate in basic to advanced rock climbing. Successful completion of this course allows cadets to participate in many levels of basic to advanced levels of lead rock climbing and prepare them for future experiences in a variety of climbing adventures to include ice climbing and mountaineering co-related adventures.                                                                                                                                                                                                                                                                                                                                                                                  |                                | 2026-7 2027-7 2028-7<br>2029-7                                                 |
| <b>Lessons:</b> 15 @ 300 min (5.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min         |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                           |                                                                                |

## Physics and Nuclear Engineering

### 74 Courses

| NE189                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | INTRO IND STUDY NUCLEAR ENGR                                                                                                                                                                                                                       | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2019-1                                                                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| <p>This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in nuclear engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in nuclear engineering. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.</p>                                                    |                                                                                                                                                                                                                                                    | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Enrollment by permission of Nuclear Engineering Program Director. Other requirements as determined by the Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. |                                                               |
| NE189A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | INTRO IND STUDY NUCLEAR ENGR                                                                                                                                                                                                                       | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2019-1                                                                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| <p>This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in nuclear engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in nuclear engineering. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. NE189A and NE189B are follow-on variants of NE189.</p> |                                                                                                                                                                                                                                                    | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Enrollment by permission of Nuclear Engineering Program Director. Other requirements as determined by the Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NE189                                                                                                                                                                                                                                              |                                                               |
| NE189B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | INTRO IND STUDY NUCLEAR ENGR                                                                                                                                                                                                                       | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2019-1                                                                                                                                                                                                                                             | <b>Offerings:</b>                                             |
| <p>This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in nuclear engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in nuclear engineering. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. NE189A and NE189B are follow-on variants of NE189.</p> |                                                                                                                                                                                                                                                    | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Enrollment by permission of Nuclear Engineering Program Director. Other requirements as determined by the Faculty Advisor.                                                                                                                         |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NE189A                                                                                                                                                                                                                                             |                                                               |
| NE289                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | INTERM IND STUDY NUCLEAR ENGR                                                                                                                                                                                                                      | 2.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2019-1                                                                                                                                                                                                                                             | <b>Offerings:</b>                                             |

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in nuclear engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in nuclear engineering. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Enrollment by permission of Nuclear Engineering Program Director. Other requirements as determined by the Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**NE289A**

**INTERM IND STUDY NUCLEAR ENGR**

**2.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:**

2019-1

**Offerings:**

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in nuclear engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in nuclear engineering. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. NE289A and NE289B are follow-on variants of NE289.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Enrollment by permission of Nuclear Engineering Program Director. Other requirements as determined by the Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

**Prerequisite(s):**

NE289

**NE289B**

**INTERM IND STUDY NUCLEAR ENGR**

**2.0 Credit Hours**  
(BS=0.0,ET=0.0,MA=0.0)

**Scope:**

2019-1

**Offerings:**

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in nuclear engineering on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in nuclear engineering. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. NE289A and NE289B are follow-on variants of NE289.

2026-1

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Enrollment by permission of Nuclear Engineering Program Director. Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):**

NE289A

**NE300**

**FUNDAMENTALS OF NUCLEAR ENGR**

**3.0 Credit Hours**  
(BS=0.0,ET=3.0,MA=0.0)

**Scope:**

2022-1

**Offerings:**

This course provides the student with an understanding of the fundamental physical principles involved in radioactive decay, radiation interaction with matter, nuclear fission, and nuclear fuel material properties. The course covers neutron interactions with matter, fission, neutron lifecycle, neutron moderation, and reactor criticality. This course is essential for the nuclear engineer and is an excellent choice for the applied scientist.

2026-1 2026-2 2026-8  
2026-9 2027-1 2027-2  
2027-8 2027-9 2028-1  
2028-2 2028-8 2028-9  
2029-1 2029-2 2029-8  
2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

None

**Corequisite(s):** PH202  
 -Or-  
 PH252  
 -Or-  
 PH205  
 -Or-  
 PH255  
 -Or-  
 PH201X  
 -Or-  
 PH201  
 -Or-  
 PH251  
 -Or-  
 PH275

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>NE350</b> | <b>RADIOLOGICAL ENGR DESIGN</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2019-2 | <b>Offerings:</b> |
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This course focuses on nuclear engineering systems including radiation protection, shielding, and the uses of radioactive sources in industrial processes. Specific topics emphasize the operation of radiation detectors, shielding principles, health effects of radiation, radiological dispersion devices, and nuclear incidents. A design project applies the concepts presented in this course to the solution of practical problems.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** One design project.

**Prerequisite(s):** NE300

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| <b>NE355</b> | <b>NUCLEAR REACTOR ENGINEERING</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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| <b>Scope:</b> | 2019-2 | <b>Offerings:</b> |
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This course focuses on nuclear reactor systems, the release of nuclear energy in the reactor core, and its removal as heat for producing electric power. Specific topics emphasize reactor kinetics, heterogeneous reactors, control rods and shim, reactor poisons, heat transfer, and alternative energy systems. The fundamentals of transport theory and the solution to the transport equation using Monte Carlo N-Particle (MCNPX) transport code are introduced. The laboratory component includes a student-designed lab.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 8 @ 120 min

**Special Requirements:** One paper and a student-designed lab project.

**Prerequisite(s):** NE300

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| <b>NE361</b> | <b>COMPUTATIONAL DESIGN IN NE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2019-2 | <b>Offerings:</b> |
|---------------|--------|-------------------|

This is a required course for nuclear engineering and nuclear engineering science majors. This course consists of an introduction to radioactive decay, radiation interactions and transport, and detailed instruction in the use and application of advanced nuclear engineering computational tools culminating in the design of a nuclear engineering system. This course is designed to provide an introduction to the science and theory behind modeling nuclear phenomena along with practical exposure to industry- standard computational tools used to design reactors, radiation shields, detectors and other nuclear systems. Specific topics include: radiation interactions and the Boltzmann transport equation, theory of deterministic and Monte Carlo methods for radiation transport, radioactive decay, problem solving using deterministic and Monte Carlo computer simulations, design of nuclear engineering systems, and analysis and validation of computational results.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NE300 PH206<br>-Or-<br>NE300 PH256<br>-Or-<br>NE300 PH202X<br>-Or-<br>NE300 PH202<br>-Or-<br>NE300 PH252<br>-Or-<br>NE300 PH275 |                                                               |
| <b>NE389</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>INDIVIDUAL STUDY IN NE</b>                                                                                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2019-1                                                                                                                          | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced nuclear or radiological engineering procedures and techniques. The primary purpose is to acquaint students with the essential skills required for independent research in nuclear or radiological engineering. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.                                                                                      |                                                                                                                                 | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                                                                                                          |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.         |                                                               |
| <b>NE389A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>INDIVIDUAL STUDY IN NE</b>                                                                                                   | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2019-1                                                                                                                          | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced nuclear or radiological engineering procedures and techniques. The primary purpose is to acquaint students with the essential skills required for independent research in nuclear or radiological engineering. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.                                                                                      |                                                                                                                                 | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Labs:</b> 0 @ 0 min                                                                                                          |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.         |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | NE389                                                                                                                           |                                                               |
| <b>NE400</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>NUCLEAR ENGINEERING SEMINAR</b>                                                                                              | <b>1.0 Credit Hours<br/>(BS=0.0,ET=1.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2008-2                                                                                                                          | <b>Offerings:</b>                                             |
| This seminar will meet once each week and will include all first class cadets majoring in nuclear engineering. The seminar topics will address the concerns of professional nuclear engineers such as engineering ethics, economics, and licensing procedures. Guest lecturers will discuss topics of current interest in the field of nuclear engineering to include DoD initiatives in the FA52 (Nuclear Combating Weapons of Mass Destruction). Much of the seminar material will be presented by guest lecturers from the military, industrial, and academic communities. |                                                                                                                                 | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 16 @ 55 min (1.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                          |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                                                                                            |                                                               |
| <b>NE450</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>NUCLEAR WEAPONS EFFECTS</b>                                                                                                  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2019-2                                                                                                                          | <b>Offerings:</b>                                             |
| This course focuses on the operation of nuclear and fusion weapons, and the effects of a nuclear weapon detonation. Specific topics emphasize blast effects, thermal radiation, initial radiation and fallout, electromagnetic pulse, biological effects of radiation, and the policy issues associated with weapons of mass destruction. Extension problems with design components apply the concepts presented in NE450 to the solution of practical problems.                                                                                                              |                                                                                                                                 | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                          |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Extension problems with design components.                                                                                      |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | NE300                                                                                                                           |                                                               |

| NE452                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | INSTRUMENTATION AND SHIELDING  | 3.5 Credit Hours<br>(BS=0.0,ET=3.5,MA=0.0)                    |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2020-1                         | <b>Offerings:</b>                                             |
| <p>This course focuses on nuclear instrumentation and radiation detectors, and on biological and material radiation protection through shielding. Specific topics include a study of radiation, and radiation detection devices to include: ionization chambers, proportional counters, Geiger-Mueller counters, scintillation detectors, semiconductor diode detectors, germanium and sodium iodide gamma-ray detectors, and neutron detectors. Radiation shielding, as a force protection measure, includes the design, analysis, and confirmation of radiation shields using point kernel and removal diffusion methods. Emphasis is placed on practical application of the radiation detection instruments and the associated acquisition software.</p>                                                                                                                                                                                                                                        |                                | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 8 @ 120 min       |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                           |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -Or-<br>NE355<br>-Or-<br>NE300 |                                                               |
| NE461                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ADV COMPUTATIONAL DESIGN IN NE | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2024-1                         | <b>Offerings:</b>                                             |
| <p>This is a mandatory course for Nuclear Engineering majors. The course relies on skills and techniques introduced in NE361 (Computational Design in Nuclear Engineering) and NE355 (Nuclear Reactor Engineering) along with advanced computer codes to design solutions to problems involving advanced reactor concepts, nuclear fuel cycles, nuclear shielding, unstructured mesh Monte Carlo problems, and radiographic imaging. The course includes reinforcement of a variety of computational solutions to the Boltzmann transport equation to include discrete ordinates and Monte Carlo simulations of subcritical and critical assemblies for thermal and fast reactor designs. The course includes a significant design experience where cadets will optimize a complex shielding problem using an unstructured mesh applied to a solid model CAD design. Analysis includes social, political, economic and environmental factors surrounding advanced reactors and shield designs.</p> |                                | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                           |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NE361<br>-Or-<br>NE355         |                                                               |
| NE474                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | RADIOLOGICAL SAFETY            | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2020-1                         | <b>Offerings:</b>                                             |
| <p>This course focuses on application of radiation interactions with matter, biological effects of ionizing radiation, and radiological dose assessment. Specific topics emphasize radiation transformations, kinetics and particle interactions, early and late biological effects of radiation, internal and external exposure and dose calculations, radiation safety regulations, and application of health physics principles to reduce hazards in nuclear engineering.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                           |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NE300                          |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NE374                          |                                                               |
| NE489                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ADV IND STDY NUCLEAR ENGNRG    | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2013-1                         | <b>Offerings:</b>                                             |
| <p>This course is an individually supervised research and study program to familiarize students with advanced nuclear or radiological engineering procedures and techniques. The primary purpose is to acquaint students with the essential features of independent research in nuclear or radiological engineering. With the approval of the Head of the Department, the cadet chooses a research project currently in progress in the Department and is supervised by a faculty member conducting the research. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.</p>                                                                                                                                                                                                                                                                                                                                                      |                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |

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| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Cadets must complete a written research report and present an oral report to members of the department faculty at the end of the semester. Cadets enrolled in NE489 are expected to present their research at a national or regional undergraduate conference. |                                                               |
| <b>NE489A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>ADV IND STUDY NUCLEAR ENGNRG</b>                                                                                                                                                                                                                            | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2013-2                                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| Same as NE489.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Same as NE489. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.                                                                                                                         |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | NE489                                                                                                                                                                                                                                                          |                                                               |
| <b>NE495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ADV NUC SYSTEM DESIGN PROJ I</b>                                                                                                                                                                                                                            | <b>3.5 Credit Hours</b><br>(BS=0.0,ET=3.5,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2020-1                                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| This is the first course in a two-semester capstone design experience. The course provides experience in the integration of math, science, and engineering principles into a comprehensive nuclear system design project. The design project emphasizes a multidisciplinary approach to total system design providing multiple paths to a number of feasible and acceptable solutions which meet the stated performance requirements. Design teams are required to develop product specifications, generate alternatives, make practical engineering approximations, and perform appropriate analysis to support the technical feasibility of the design, make decisions leading to an optimal system design, and brief their interim results during in-process reviews (IPRs). Topics such as engineering economics and the Code of Federal Regulations are introduced. Computational codes such as MCNP and other nuclear industry codes specific to the project will be introduced. |                                                                                                                                                                                                                                                                | 2026-1 2027-1 2028-1<br>2029-1                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Comprehensive team design project; compensatory time provided. Permission of the Head of the Department of Physics and Nuclear Engineering required.                                                                                                           |                                                               |
| <b>NE496</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ADV NUC SYSTEM DESIGN PROJ II</b>                                                                                                                                                                                                                           | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=3.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2019-2                                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| This is the second course in a two-semester capstone design experience. The course provides experience in the integration of math, science, and engineering principles into a comprehensive nuclear system design project. The design project emphasizes a multidisciplinary approach to total system design providing multiple paths to a number of feasible and acceptable solutions which meet the stated performance requirements. Design teams are required to develop product specifications, generate alternatives, make practical engineering approximations, and perform appropriate analysis to support the technical feasibility of the design, make decisions leading to an optimal system design, and brief their interim results during in-process reviews (IPRs). During this course, the design project is completed and presented to the project sponsor.                                                                                                             |                                                                                                                                                                                                                                                                | 2026-2 2027-2 2028-2<br>2029-2                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Comprehensive team design project; compensatory time provided.                                                                                                                                                                                                 |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | NE495                                                                                                                                                                                                                                                          |                                                               |
| <b>PH189</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INTRO IND STUDY PHYSICS</b>                                                                                                                                                                                                                                 | <b>1.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2019-1                                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in physics on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |

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| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Enrollment by permission of Physics Program Director. Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>PH189A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>INTRO IND STUDY PHYSICS</b>                                                                                 | <b>1.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-1                                                                                                         | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in physics on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. PH189A and PH189B are follow-on variants of PH189. |                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Enrollment by permission of Physics Program Director. Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PH189                                                                                                          |                                                               |
| <b>PH189B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>INTRO IND STUDY PHYSICS</b>                                                                                 | <b>1.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-1                                                                                                         | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in physics on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. PH189A and PH189B are follow-on variants of PH189. |                                                                                                                | 2026-1 2029-1 2029-2                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Enrollment by permission of Physics Program Director. Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PH189A                                                                                                         |                                                               |
| <b>PH201</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>PHYSICS I</b>                                                                                               | <b>4.0 Credit Hours</b><br>(BS=4.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2023-1                                                                                                         | <b>Offerings:</b>                                             |
| This required, calculus-based core physics course consists of a comprehensive study of classical mechanics. The course is designed to promote scientific literacy and to develop the use of the scientific method to solve complex problems. Topics include vectors, kinematics, Newton's laws, work, energy, torque, momentum, and rotational motion. An integrated laboratory program illustrates basic scientific techniques and serves to stimulate intellectual curiosity through discovery laboratories. The core physics program is designed to demonstrate the relevance of physics to military technology and to help prepare future Army leaders to anticipate and adapt to technological change.                                                                                                 |                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 10 @ 120 min                                                                                      |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                           |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | MA104                                                                                                          |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PH251<br>-Or-<br>PH205<br>-Or-<br>PH201X                                                                       |                                                               |
| <b>PH202</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>PHYSICS II</b>                                                                                              | <b>4.0 Credit Hours</b><br>(BS=4.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2023-1                                                                                                         | <b>Offerings:</b>                                             |



This calculus-based core physics course consists of a comprehensive study of electricity and magnetism. Topics include electrostatics, Gauss's Law, magnetic fields, Ampere's Law, Faraday's Law, circuits (direct current and alternating current), electromagnetic waves, geometric optics, physical optics, and elements of modern physics. An integrated laboratory program illustrates basic scientific techniques and serves to stimulate intellectual curiosity through discovery laboratories. The core physics program is designed to demonstrate the relevance of physics to military technology and to help prepare future Army leaders to anticipate and adapt to technological change.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 10 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** PH201  
-Or-  
PH251  
-Or-  
PH205  
-Or-  
PH255  
-Or-  
PH201X

**Disqualifier(s):** PH252  
-Or-  
PH202X  
-Or-  
PH275

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2  
2029-4 2029-5 2029-8  
2029-9

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|--------------|---------------------------|----------------------------------------------------------|
| <b>PH251</b> | <b>ADVANCED PHYSICS I</b> | <b>4.0 Credit Hours</b><br><b>(BS=4.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------|----------------------------------------------------------|

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**Scope:** 2023-1 **Offerings:**

This required, calculus-based advanced core physics course consists of a comprehensive study of classical mechanics. The course is designed to promote scientific literacy and to develop the use of the scientific method to solve complex problems. Topics include vectors, kinematics, Newton's laws, work, energy, torque, momentum, and rotational motion. An integrated laboratory program illustrates basic scientific techniques and serves to stimulate intellectual curiosity through discovery laboratories. The core physics program is designed to demonstrate the relevance of physics to military technology and to help prepare future Army leaders to anticipate and adapt to technological change.

**Lessons:** 40 @ 55 min (3.000 Att/wk)      **Labs:** 10 @ 120 min

**Special Requirements:** None

**Corequisite(s):** MA104

**Disqualifier(s):** PH201  
-Or-  
PH205  
-Or-  
PH255

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

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|--------------|----------------------------|----------------------------------------------------------|
| <b>PH252</b> | <b>ADVANCED PHYSICS II</b> | <b>4.0 Credit Hours</b><br><b>(BS=4.0,ET=0.0,MA=0.0)</b> |
|--------------|----------------------------|----------------------------------------------------------|

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**Scope:** 2023-1 **Offerings:**

This calculus-based advanced core physics course consists of a comprehensive study of electricity and magnetism. Topics include electrostatics, Gauss's Law, magnetic fields, Ampere's Law, Faraday's Law, circuits (direct current and alternating current), electromagnetic waves, geometric optics, physical optics, and elements of modern physics. An integrated laboratory program illustrates basic scientific techniques and serves to stimulate intellectual curiosity through discovery laboratories. The core physics program is designed to demonstrate the relevance of physics to military technology and to help prepare future Army leaders to anticipate and adapt to technological change.

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 10 @ 120 min

**Special Requirements:** None

**Prerequisite(s):** PH251  
-Or-  
PH201  
-Or-  
PH205  
-Or-  
PH255  
-Or-  
PH201X

2026-1 2027-1 2027-2  
2028-1 2028-2 2029-1  
2029-2

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PH202<br>-Or-<br>PH256<br>-Or-<br>PH275                                                                        |                                                               |
| <b>PH275</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>PHYSICS II: SPACE</b>                                                                                       | <b>4.0 Credit Hours<br/>(BS=4.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2026-1                                                                                                         | <b>Offerings:</b>                                             |
| Calculus-based core physics course serving as an introduction to space physics. The course includes an introduction to orbital mechanics, electricity and magnetism, and optics. The course is designed to promote scientific literacy of the physics principles governing the space domain to translate space-based effects into operational impacts. All topics will build an understanding of the capabilities, limitations, and vulnerabilities of space-based assets available to the modern warfighter. Additionally, an integrated laboratory program illustrates basic scientific techniques and serves to stimulate intellectual curiosity through discovery laboratories related to these military space assets. This core physics course is designed to demonstrate the relevance of physics to military space-based technology and to help prepare future Army leaders to anticipate and adapt to technological change. |                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 10 @ 75 min                                                                                       |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | None                                                                                                           |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MA104 PH201<br>-Or-<br>MA104 PH251                                                                             |                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PH202<br>-Or-<br>PH202X<br>-Or-<br>PH252<br>-Or-<br>SP275X                                                     |                                                               |
| <b>PH289</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INTERM IND STUDY PHYSICS</b>                                                                                | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2019-1                                                                                                         | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in physics on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.                                                                                                                                                                            |                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of Physics Program Director. Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>PH289A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INTERM IND STUDY PHYSICS</b>                                                                                | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2019-1                                                                                                         | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in physics on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. PH289A and PH289B are follow-on variants of PH289.                                                                                                                         |                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Enrollment by permission of Physics Program Director. Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PH289                                                                                                          |                                                               |
| <b>PH289B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INTERM IND STUDY PHYSICS</b>                                                                                | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-1                                                                                                         | <b>Offerings:</b>                                                                            |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in physics on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. PH289A and PH289B are follow-on variants of PH289. |                                                                                                                | No Course Offerings                                                                          |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                         |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Enrollment by permission of Physics Program Director. Other requirements as determined by the Faculty Advisor. |                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PH289A                                                                                                         |                                                                                              |
| <b>PH361</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>EXPERIMENTAL PHYSICS</b>                                                                                    | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2005-1                                                                                                         | <b>Offerings:</b>                                                                            |
| This course provides instruction and experimental experiences designed to exercise the student's knowledge of classical and modern physics and to extend his or her familiarity with equipment and techniques used in a physical science laboratory. Cadets, working in groups, execute and report on experimental projects. The program of instruction includes familiarization with electronics and instrumentation, data analysis, and laboratory procedures and practices. Knowledge and skills acquired in this course are essential for subsequent laboratory work in solid state physics, nuclear physics, and optics.                                                                                                                                                                               |                                                                                                                | No Course Offerings                                                                          |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 8 @ 120 min                                                                                       |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                           |                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -Or-<br>-Or-<br>PH202<br>-Or-<br>PH252<br>-Or-<br>PH206<br>-Or-<br>PH256<br>-Or-<br>PH202X<br>-Or-<br>PH275    |                                                                                              |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PH365                                                                                                          |                                                                                              |
| <b>PH365</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>MODERN PHYSICS</b>                                                                                          | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b>                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-2                                                                                                         | <b>Offerings:</b>                                                                            |
| This course introduces special relativity and the fundamental concepts of quantum physics with application to atomic physics and nuclear physics in order to prepare cadets for advanced study of science and engineering, especially quantum mechanics, statistical physics, nuclear physics, solid state physics, laser physics, medical radiation physics, and nuclear engineering. This course will also be of interest to any cadet who wishes to gain a deeper appreciation of the natural world or of the technology of the 21st Century.                                                                                                                                                                                                                                                            |                                                                                                                | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-8<br>2028-1 2028-2 2028-8<br>2029-1 2029-2 2029-8 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                         |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                           |                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -Or-<br>-Or-<br>PH204<br>-Or-<br>PH254<br>-Or-<br>PH202<br>-Or-<br>PH252                                       |                                                                                              |

-Or-  
PH206  
-Or-  
PH256  
-Or-  
PH202X  
-Or-  
PH275  
-Or-  
SP275X

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------|
| <b>PH366</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>APPLIED QUANTUM PHYSICS</b>          | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2013-2                                  | <b>Offerings:</b>                                  |
| This course uses the experimental and laboratory skills developed in PH361 to explore the applications of the 20th Century developments studied in PH365. The topics covered will vary but may include molecular structure, the properties of solids including metals and semiconductors, nuclear physics, and elementary particle physics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                         | No Course Offerings                                |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 8 @ 120 min                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                    |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PH361 PH484                             |                                                    |
| <b>PH381</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>INTRMED CLASSICAL MECHANICS</b>      | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2020-1                                  | <b>Offerings:</b>                                  |
| This course continues the development of physical principles introduced in the core physics curriculum. Direct application of Newton's laws is used to analyze phenomena such as projectile motion with air resistance, charged particle motion, and motion in a central force field. Harmonic, driven, and damped oscillations are studied in depth, as are systems of coupled oscillators. The formalism of Lagrangian mechanics is studied in depth. The mathematical tools of classical mechanics are introduced, to include vector fields, line integrals, the calculus of variations, linear algebra, and eigenvalue equations. Cadets will be required to develop and demonstrate the ability to use a computer algebra system to solve advanced problems and plot the solutions.                                                         |                                         | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                  |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                    |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PH363<br>-Or-<br>MA364<br>-Or-<br>MA365 |                                                    |
| <b>PH382</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>INTERMEDIATE ELECTRODYNAMICS</b>     | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2020-1                                  | <b>Offerings:</b>                                  |
| This course continues the study of classical electrodynamics introduced in the introductory physics sequence by developing the differential forms of the Maxwell equations and applying them to boundary value problems in two and three dimensions. In addition, scalar and vector potentials are introduced, multipole field expansions are developed for complex sources, electromagnetic fields in dielectric and magnetic media are studied, the propagation of electromagnetic waves in conducting and nonconducting media is considered and electromagnetic radiation is introduced. The course concludes with the study of the connection between special relativity and electrodynamics. This course provides an essential foundation for courses in optics, lasers, quantum mechanics, statistical mechanics, and solid state physics. |                                         | 2026-1 2026-8 2027-1<br>2028-1 2029-1              |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                  |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                    |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | PH363<br>-Or-<br>MA364<br>-Or-<br>MA365 |                                                    |
| <b>PH384</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>APPLIED OPTICS</b>                   | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | 2019-2                               | <b>Offerings:</b>                                             |
| This course provides intermediate development in the concepts of geometric, wave, and quantum optics. Primary coverage includes common optical devices, light transmission through optical media, diffraction, interference and polarization. Applied topics typically include: Fourier optics, holography, imaging, atmospheric effects, optical and infrared detectors, nonlinear optics, and tactical employment considerations. |                                      | 2026-2 2027-2 2028-2<br>2029-2 2029-8 2029-9                  |
| <b>Lessons:</b> 25 @ 75 min (1.700 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                          | <b>Labs:</b> 5 @ 75 min              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                        | None                                 |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                             | MA364<br>-Or-<br>MA365               |                                                               |
| <b>PH389</b>                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>INDIVIDUAL STUDY IN PHYSICS</b>   | <b>3.0 Credit Hours<br/>(BS=1.5,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | 2019-1                               | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                      | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min               |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                        | none                                 |                                                               |
| <b>PH389A</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INDIVIDUAL STUDY IN PHYSICS-A</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | 2019-1                               | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                      | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min               |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                        | none                                 |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                             | PH389                                |                                                               |
| <b>PH389B</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INDIVIDUAL STUDY IN PHYSICS-B</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | 2024-2                               | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                      | 2026-1                                                        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min               |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                        | None                                 |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                             | PH389 PH389A                         |                                                               |
| <b>PH389C</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INDIVIDUAL STUDY IN PHYSICS-C</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | 2025-1                               | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                      | 2029-1                                                        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 0 @ 0 min               |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                        | None                                 |                                                               |

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| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PH389 PH389A PH389B                                                                   |                                                    |
| <b>PH389D</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>INDIVIDUAL STUDY IN PHYSICS-D</b>                                                  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2025-2                                                                                | <b>Offerings:</b>                                  |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The primary purpose is to acquaint students with the essential skills required for independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.                                                                                                                                                                                                                                                                    |                                                                                       | 2029-2                                             |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PH389 PH389A PH389B PH389C                                                            |                                                    |
| <b>PH456</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>SCIENCE AND POLICY</b>                                                             | <b>3.0 Credit Hours<br/>(BS=2.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2020-1                                                                                | <b>Offerings:</b>                                  |
| This course challenges cadets to draw upon their core academic experience to analyze complex policy issues. The relationship and interaction between social, political, economic, and technological dimensions of these issues are explored. Emphasis is given to gaining an understanding of both the power and limitations of science and scientific thinking when confronting problems in the policy arena.                                                                                                                                                                                                                                                                                         |                                                                                       | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Enrollment in this course requires approval of the Head of the Department of Physics. |                                                    |
| <b>PH481</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>STATISTICAL PHYSICS</b>                                                            | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2020-1                                                                                | <b>Offerings:</b>                                  |
| This course applies basic concepts of probability and statistics to systems consisting of a large number of particles to determine measurable macroscopic quantities such as temperature, pressure, energy, and heat capacity. Emphasis is placed on the calculation of the canonical and grand canonical partition functions for various model physical systems. Particular attention is focused on three ideal gas systems: a gas consisting of massive Maxwell-Boltzmann particles, a gas consisting of massless bosons (i.e., photons), and a gas consisting of fermions.                                                                                                                          |                                                                                       | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MA206 PH484<br>-Or-<br>MA206X PH484<br>-Or-<br>MA256 PH484                            |                                                    |
| <b>PH482</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ADVANCED CLASSICAL MECHAN</b>                                                      | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2019-2                                                                                | <b>Offerings:</b>                                  |
| This course continues the development of concepts introduced in PH381. Hamiltonian mechanics is explored using the calculus of variations to provide a foundation for connecting classical mechanics, quantum mechanics, and statistical mechanics. The two-body central force problem, the mechanics of rotating systems, and scattering theory are studied in depth. The mathematical techniques associated with cylindrical, spherical, and curvilinear coordinates are introduced, as are the basic principles of nonlinear dynamics and chaos. Cadets will be required to develop and demonstrate the ability to use a computer algebra system to solve advanced problems and plot the solutions. |                                                                                       | 2027-2 2028-2 2029-2                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Labs:</b> 0 @ 0 min                                                                |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | None                                                                                  |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PH381                                                                                 |                                                    |
| <b>PH484</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>INTERMEDIATE QUANTUM MECHANICS</b>                                                 | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2020-1                            | <b>Offerings:</b>                                       |
| This is the first in a two-course sequence in Quantum Mechanics. This course begins with a basic introduction to the fundamental postulates of quantum theory. These postulates are then used to develop Heisenberg's uncertainty principle and Schroedinger's equation. Solutions to Schroedinger's equation are sought, first for relatively simple systems such as square wells and harmonic oscillators, and then for the hydrogen atom. The properties of the hydrogen atom are studied in detail. The course also covers approximation methods used for physical systems with small perturbing forces acting on them. |                                   | 2026-1 2027-1 2028-1 2029-1                             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min            |                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                              |                                                         |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PH365                             |                                                         |
| <b>PH485</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>LASER PHYSICS</b>              | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2020-1                            | <b>Offerings:</b>                                       |
| This course provides a combined theoretical and experimental investigation into the realm of coherent optical radiation generation, amplification, propagation, and application. Cadets apply the basic principles of electromagnetism, optics, and modern physics to analyze specific laser systems. Emphasis is placed on evaluation of laser systems, and experiments are performed to demonstrate properties of specific systems. Selected topics in modern optical engineering are explored to illustrate basic applications of various systems. Laser theory is investigated using semiclassical methods.             |                                   | 2026-1 2027-1 2028-1 2029-1                             |
| <b>Lessons:</b> 25 @ 75 min (1.700 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 5 @ 75 min           |                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                              |                                                         |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PH384                             |                                                         |
| <b>PH486</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>EXPERIMENTAL PHYSICS</b>       | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-2                            | <b>Offerings:</b>                                       |
| This course provides instruction and experimental experiences designed to exercise the cadet's knowledge of classical and modern physics and to extend their familiarity with equipment and techniques used in a physical science laboratory. Cadets, working in groups or by themselves, integrate a variety of physics theories and experimental techniques. The program of instruction includes applications of electronics and instrumentation, data analysis, and laboratory procedures and practices.                                                                                                                 |                                   | 2026-2 2027-2 2028-2 2029-2                             |
| <b>Lessons:</b> 24 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 16 @ 120 min         |                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                              |                                                         |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PH484 PH485                       |                                                         |
| <b>PH487</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ADVANCED QUANTUM MECHANICS</b> | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2019-2                            | <b>Offerings:</b>                                       |
| This is the second in a two-course sequence on Quantum Mechanics. This course covers the application of the fundamentals learned in the first course of the sequence. The topics include, but are not limited to perturbation theory (time independent and time dependent), the variational principle, the WKB approximation, the adiabatic approximation, and quantum scattering.                                                                                                                                                                                                                                          |                                   | 2026-2 2027-2 2028-2 2029-2                             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min            |                                                         |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                              |                                                         |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PH484                             |                                                         |
| <b>PH489</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ADV INDIV STUDY IN PHYSICS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>      |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2013-1                            | <b>Offerings:</b>                                       |
| This course is an individually supervised research and study program to familiarize students with advanced scientific procedures and techniques. The primary purpose is to acquaint students with the essential features of independent research in physics. With the approval of the Head of the Department, the cadet chooses a research project currently in progress in the Department, and is supervised by a faculty member conducting the research.                                                                                                                                                                  |                                   | 2026-1 2026-2 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2 |

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| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Cadets must complete a written research report and present an oral report to members of the department faculty at the end of the semester. Cadets enrolled in PH489 are expected to present their research at a national or regional undergraduate conference. |                                                               |
| <b>PH489A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ADV INDIV STUDY IN PHYSICS</b>                                                                                                                                                                                                                              | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2013-2                                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| This course is a second course in an individually supervised research and study program to familiarize students with advanced scientific procedures and techniques. The primary purpose is to foster the student's continued development of the essential features of independent research in physics. With the approval of the Head of the Department, the student continues with a research project currently in progress in the Department, and is supervised by a faculty member conducting the research.                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                                                                                                                                                                                           |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PH489                                                                                                                                                                                                                                                          |                                                               |
| <b>PH489B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ADV INDIV STUDY IN PHYSICS</b>                                                                                                                                                                                                                              | <b>3.0 Credit Hours</b><br>(BS=1.0,ET=2.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1990-2                                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| Similar to PH489 and PH489A but offered to students who have already completed those versions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                | No Course Offerings                                           |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                                                                                                                                                                                           |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PH489A                                                                                                                                                                                                                                                         |                                                               |
| <b>PH495</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>SPECIAL TOPICS IN PHYSICS</b>                                                                                                                                                                                                                               | <b>3.0 Credit Hours</b><br>(BS=3.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2011-2                                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| This course is taught by the Class of 1967 Endowed Chair or another faculty member who is not occupying an authorized USMA position, including any visiting scholar with a distinguished record of academic and professional achievement in the field of engineering, science, and technology. The Special Topics in Physics course focuses on topical issues that reflect the technical expertise of the Chair or visiting scholar. Students will apply math, science, and engineering fundamentals they have learned to these studies.                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                | 2026-2 2027-2 2028-2                                          |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                                                                                                                                                                                           |                                                               |
| <b>SP189</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>INTRO IND STUDY SPACE SCIENCE</b>                                                                                                                                                                                                                           | <b>1.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2018-7                                                                                                                                                                                                                                                         | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                                                                                                                                                                                                                                                | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                         |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Enrollment by permission of Space and Missile Defense Program Director. Other requirements as determined by the Faculty Advisor.                                                                                                                               |                                                               |
| <b>SP189A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>INTRO IND STUDY SPACE SCIENCE</b>                                                                                                                                                                                                                           | <b>1.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)             |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                     |                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2019-1                                                                                                                              | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP189A and B are follow-on variants of SP189.                                                                                                                                                            |                                                                                                                                     | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Enrollment by permission of Space and Missile Defense Program Director.<br>Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SP189                                                                                                                               |                                                               |
| <b>SP189B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>INTRO IND STUDY SPACE SCIENCE</b>                                                                                                | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2019-1                                                                                                                              | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP189A and B are follow-on variants of SP189.                                                                                                                                                            |                                                                                                                                     | 2026-1 2026-2 2029-1<br>2029-2                                |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Enrollment by permission of Space and Missile Defense Program Director.<br>Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SP189A                                                                                                                              |                                                               |
| <b>SP189C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>INTRO IND STUDY SPACE SCIENCE</b>                                                                                                | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2025-2                                                                                                                              | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP189A/B/C are follow-on variants of SP189.                                                                                                                                                              |                                                                                                                                     | 2029-2                                                        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Enrollment by permission of Space and Missile Defense Program Director.<br>Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SP189B                                                                                                                              |                                                               |
| <b>SP275X</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>SPACE PHYSICS</b>                                                                                                                | <b>4.0 Credit Hours<br/>(BS=4.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2023-2                                                                                                                              | <b>Offerings:</b>                                             |
| Calculus-based core physics course serving as an introduction to space physics. The course includes an introduction to orbital mechanics, electricity and magnetism, and space weather. The course is designed to promote scientific literacy and to develop the use of the scientific method to solve problems. All topics will feed into understanding the capabilities, limitations, and vulnerabilities of the warfighter's space-based assets. Additionally, an integrated laboratory program illustrates basic scientific techniques and serves to stimulate intellectual curiosity through discovery laboratories that are entirely related to the warfighter's space-based assets. This core physics course is designed to demonstrate the relevance of physics to military space-based technology and to help prepare future Army leaders to anticipate and adapt to technological change. **This is a pilot course and must be reviewed by the Curriculum Committee NLT 24-2 to continue.** |                                                                                                                                     | No Course Offerings                                           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Labs:</b> 8 @ 75 min                                                                                                             |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | None                                                                                                                                |                                                               |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | PH201<br>-Or-<br>PH251                                                                                                              |                                                                                       |
| <b>SP289</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>INTERM IND STUDY SPACE SCI</b>                                                                                                   | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2018-7                                                                                                                              | <b>Offerings:</b>                                                                     |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.                                                    |                                                                                                                                     | 2026-1 2026-2 2027-1<br>2027-2 2027-7 2028-1<br>2028-2 2028-7 2029-1<br>2029-2 2029-7 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Enrollment by permission of Space and Missile Defense Program Director.<br>Other requirements as determined by the faculty advisor. |                                                                                       |
| <b>SP289A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>INTERM IND STUDY SPACE SCIENCE</b>                                                                                               | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2019-1                                                                                                                              | <b>Offerings:</b>                                                                     |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP289A and SP289B are follow-on variants of SP289. |                                                                                                                                     | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2                         |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Enrollment by permission of Space and Missile Defense Program Director.<br>Other requirements as determined by the Faculty Advisor. |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | SP289                                                                                                                               |                                                                                       |
| <b>SP289B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>INTERM IND STUDY SPACE SCIENCE</b>                                                                                               | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2019-1                                                                                                                              | <b>Offerings:</b>                                                                     |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP289A and SP289B are follow-on variants of SP289. |                                                                                                                                     | 2026-1 2029-1 2029-2                                                                  |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                                                                       |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Enrollment by permission of Space and Missile Defense Program Director.<br>Other requirements as determined by the Faculty Advisor. |                                                                                       |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | SP289A                                                                                                                              |                                                                                       |
| <b>SP289C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>INTERM IND STUDY SPACE SCIENCE</b>                                                                                               | <b>2.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>                                    |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2025-2                                                                                                                              | <b>Offerings:</b>                                                                     |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP289A/B/C are follow-on variants of SP289.        |                                                                                                                                     | 2029-2                                                                                |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                                                                       |

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| <b>Special Requirements:</b> | Enrollment by permission of Space and Missile Defense Program Director.<br>Other requirements as determined by the Faculty Advisor. |
| <b>Prerequisite(s):</b>      | SP289B                                                                                                                              |

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| <b>SP389</b> | <b>IND RESEARCH SPACE SCIENCE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-1

**Offerings:**

2026-1 2026-2 2026-7  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research.

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of Space and Missile Defense Program Director.  
Other requirements as determined by the Faculty Advisor.

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| <b>SP389A</b> | <b>IND RESEARCH SPACE SCIENCE - A</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-1

**Offerings:**

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP389A and SP389B are follow-on variants of SP389.

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of Space and Missile Defense Program Director.  
Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):** SP389

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| <b>SP389B</b> | <b>IND RESEARCH SPACE SCIENCE - B</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|---------------|---------------------------------------|----------------------------------------------------------|

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**Scope:** 2019-1

**Offerings:**

2026-1 2026-2 2027-2  
2028-2 2029-1 2029-2

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP389A and SP389B are follow-on variants of SP389.

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Enrollment by permission of Space and Missile Defense Program Director.  
Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):** SP389A

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| <b>SP389C</b> | <b>IND RESEARCH SPACE SCIENCE - C</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-1

**Offerings:**

2029-1 2029-2

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP389A, SP389B, and SP389C are follow-on variants of SP389.

**Lessons:** 0 @ 0 min (0.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**Enrollment by permission of Space and Missile Defense Program Director.  
Other requirements as determined by the Faculty Advisor.**Prerequisite(s):**

SP389B

**SP471****ASTRONAUTICS****3.0 Credit Hours**  
**(BS=3.0,ET=0.0,MA=0.0)****Scope:**

2024-1

**Offerings:**

This course is an introduction to the history, principles, and challenges of space and astronautics by way of examination of the following topics: historical satellite and space-related events; orbits; orbital mechanics; science of spacecraft subsystems including, but not limited to, satellite communications, remote-sensing payloads, power subsystems, thermal subsystems, and attitude determination and control subsystems; launch vehicles; and rocket and propulsion systems.

2026-1 2026-8 2027-1  
2028-1 2029-1**Lessons:** 30 @ 75 min (2.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

None

**Prerequisite(s):**

-Or-

-Or-

-Or-

PH202

-Or-

PH252

-Or-

SP275X

-Or-

PH275

**SP472****SPACE PHYSICS****3.0 Credit Hours**  
**(BS=3.0,ET=0.0,MA=0.0)****Scope:**

2024-1

**Offerings:**

Space Physics is concerned with understanding the environment between the sun and the Earth's upper atmosphere as well as the environment within the entire heliosphere. This course provides understanding of the structure and dynamics of the Sun, including solar events such as sun spots, prominences, coronal mass ejections and solar flares. That understanding is then used as a basis to understand the solar wind and its influence on the heliosphere to include influences on our terrestrial magnetosphere, ionosphere and thermosphere. In addition, magnetospheric storms and auroral precipitation are among phenomena studied in the context of their effects on spacecraft, satellites, and also terrestrial systems and life in general.

2026-2 2027-2 2028-2  
2029-2**Lessons:** 30 @ 75 min (2.000 Att/wk)**Labs:** 0 @ 0 min**Special Requirements:**

None

**Prerequisite(s):**

-Or-

-Or-

-Or-

-Or-

-Or-

-Or-

MA205 PH202

-Or-

MA255 PH202

-Or-

MA205 PH252

-Or-

MA255 PH252

-Or-

MA205 SP275X

-Or-

MA255 SP275X

-Or-

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MA204X PH202<br>-Or-<br>MA204X PH252<br>-Or-<br>MA205 PH275<br>-Or-<br>MA255 PH275<br>-Or-<br>MA204X PH275<br>-Or-<br>MA204 PH202<br>-Or-<br>MA204 PH252<br>-Or-<br>MA204 PH275<br><br><b>Disqualifier(s):</b> PH472                                                                                                                                                                |                                                    |
| <b>SP473</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>OBSERVATIONAL ASTRONOMY</b>                                                                                                                                                                                                                                                                                                                                                      | <b>3.5 Credit Hours<br/>(BS=3.5,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2024-1                                                                                                                                                                                                                                                                                                                                                                              | <b>Offerings:</b>                                  |
| This course is an introduction to the history, principles, and application of astronomical observation focusing on visual, photometric, and spectroscopic techniques. Cadets will be introduced to the methods and tools employed in modern astronomical observatories. Emphasis is placed on data analysis and statistics, geometrical and physical optics of telescopes, time systems, catalogs, and images obtained by telescopes equipped with light detectors. The course will include real-time observation of celestial objects, starting with objects in our solar system and expanding to the Milky Way. |                                                                                                                                                                                                                                                                                                                                                                                     | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Labs:</b> 8 @ 75 min                                                                                                                                                                                                                                                                                                                                                             |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Labs are held at night to properly observe celestial objects without solar interference.                                                                                                                                                                                                                                                                                            |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -Or-<br><br>-Or-<br><br>-Or-<br><br>-Or-<br><br>-Or-<br>MA205 SP275X<br>-Or-<br>MA205 PH202<br>-Or-<br>MA205 PH252<br>-Or-<br>MA255 PH202<br>-Or-<br>MA255 PH252<br>-Or-<br>MA255 SP275X<br>-Or-<br>MA204X PH202<br>-Or-<br>MA204X PH252<br>-Or-<br>MA204X PH275<br>-Or-<br>MA205 PH275<br>-Or-<br>MA255 PH275<br>-Or-<br>MA204 PH202<br>-Or-<br>MA204 PH252<br>-Or-<br>MA204 PH275 |                                                    |
| <b>SP474</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>ASTROPHYSICS</b>                                                                                                                                                                                                                                                                                                                                                                 | <b>3.0 Credit Hours<br/>(BS=3.0,ET=0.0,MA=0.0)</b> |

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2024-1                 | <b>Offerings:</b>              |
| This course is an introduction to Astrophysics and is a study of stellar structure and evolution, galactic structure, and cosmology. Phenomena of interest include quasars, black holes, supernovae and the cosmic microwave background radiation. Special emphasis will be placed on those galactic phenomena, such as Galactic Cosmic Rays (GCRs) and Anomalous Cosmic Rays (ACRs), which have an impact on our heliosphere. Supporting topics include an overview of orbital mechanics and relevant physics, observation techniques, and nuclear physics. |                        | 2026-2 2027-2 2028-2<br>2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min |                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                   |                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA205 PH202            |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA205 PH252            |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA255 PH202            |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA255 PH252            |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA205 SP275X           |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA255 SP275X           |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA204X PH202           |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA204X PH252           |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA204X PH275           |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA205 PH275            |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA255 PH275            |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA204 PH202            |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA204 PH252            |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | -Or-                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MA204 PH275            |                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | PH472                  |                                |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                     |                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>SP489</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>ADV IND RESEARCH SPACE SCIENCE</b>                                                                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2019-1                                                                                                                              | <b>Offerings:</b>                                             |
| This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. |                                                                                                                                     | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Labs:</b> 0 @ 0 min                                                                                                              |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Enrollment by permission of Space and Missile Defense Program Director.<br>Other requirements as determined by the Faculty Advisor. |                                                               |
| <b>SP489A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ADV IND RESEARCH SPACE SCIENCE</b>                                                                                               | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b>            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2019-1                                                                                                                              | <b>Offerings:</b>                                             |

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP489A and SP489B are follow-on variants to SP489.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Enrollment by permission of Space and Missile Defense Program Director.  
Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):**

SP489

**SP489B**

**ADV IND RESEARCH SPACE SCIENCE**

**3.0 Credit Hours**  
**(BS=0.0,ET=0.0,MA=0.0)**

**Scope:**

2019-1

**Offerings:**

2026-2 2029-2

This course is an individually supervised research and study program to familiarize cadets with advanced scientific procedures and techniques. The cadet pursues study of a research or design topic in space and missile defense on an individual or small group basis, independent of a formal classroom setting. The scope of the course is tailored to the needs of the project and interests of the cadet in consultation with a faculty advisor. Activities vary by project but the primary purpose is to acquaint students with the essential skills required for independent research in space and missile defense. With the approval of the Head of the Department, the cadet chooses a research project of interest and is supervised by a faculty member conducting the research. SP489A and SP489B are follow-on variants to SP489.

**Lessons:** 0 @ 0 min (0.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Enrollment by permission of Space and Missile Defense Program Director.  
Other requirements as determined by the Faculty Advisor.

**Prerequisite(s):**

SP489A

## Social Sciences

### 92 Courses

| SS201                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ECONOMICS-PRINCIPLES/PROBLEMS    | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2019-2                           | <b>Offerings:</b>                                                                                                              |
| <p>This course presents the basic principles of economic analysis and their application to contemporary economic problems and supports the further study of economics and related disciplines in the social sciences. The course is organized into two primary branches: microeconomics, the study of the behavior of individuals, households, and firms in making decisions, in a market economy and the application of this theory to contemporary issues in both domestic and global markets; and macroeconomics, the study of the performance, structure, and behavior of the whole economy including issues of national income, output, consumption, unemployment, inflation, as well as fiscal and monetary policy. In addition, the course includes instruction on personal finance, focusing specifically on major purchase decisions (home and auto), insurance, basic investing, and investing for retirement. This instruction is intended not only to improve individual financial decision-making but also to equip Cadets with the ability to address the financial issues of their future Soldiers. Cadets develop analytical tools in order to assess the economic implications of policy decisions by military and government officials and to improve their own decision-making process.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  | 2026-1 2026-2 2026-5<br>2026-8 2027-1 2027-2<br>2027-5 2027-8 2028-1<br>2028-2 2028-5 2028-8<br>2029-1 2029-2 2029-5<br>2029-8 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min           |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                             |                                                                                                                                |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | MA104                            |                                                                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SS251                            |                                                                                                                                |
| SS202                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AMERICAN POLITICS                | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2019-2                           | <b>Offerings:</b>                                                                                                              |
| <p>This course explores the American political system - its philosophical underpinnings, the structure and behavior of formal government institutions, and the influence of informal political actors within the political construct. The course introduces the discipline of political science by exploring a broad range of literature: classics of American politics, leading political theory, and contemporary reading. Cadets will apply their knowledge of political ideas, institutions, and behavior to public policy making and demonstrate critical analysis of contemporary debates in American politics. Finally, this course provides cadets with an understanding of the professional norms and responsibilities associated with their role as military officers within the American system of government.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                  | 2026-1 2026-2 2026-5<br>2026-8 2027-1 2027-2<br>2027-9 2028-1 2028-2<br>2028-9 2029-1 2029-2<br>2029-5 2029-8                  |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min           |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Analytical writing requirements. |                                                                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SS252                            |                                                                                                                                |
| SS252                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ADVANCED AMERICAN POLITICS       | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                                                                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2021-1                           | <b>Offerings:</b>                                                                                                              |
| <p>This course explores the American political system, its philosophical underpinnings, the structure and behavior of formal government institutions, and the influence of informal political actors within the political construct. The course introduces the discipline of political science by exploring a broad range of literature: classics of American politics, leading political theory, and contemporary reading. Cadets will apply their knowledge of political ideas, institutions, and behavior to public policy-making and demonstrate critical analysis of contemporary debates in American politics. Cadets enrolled in SS252 are expected to conduct advanced undergraduate research that includes identifying additional readings related to policy debates for incorporation into policy-related classes, as well as a miniature literature review as part of their culminating research paper. These analytical research requirements exceed those levied on SS202 cadets. Further, SS252 cadets are exposed to additional seminal works in political science throughout the course, including classical political philosophy, primary historical documents from early American politicians and political thinkers, and contemporary works on civil-military relations. Finally, this course provides cadets with an understanding of the professional norms and responsibilities associated with their role as military officers within the American system of government. Aided through additional readings, this emphasis on civil-military relations focuses on contemporary and historical debates over the appropriate role of military involvement in policy development and the political process.</p> |                                  | 2026-1 2027-1 2028-1<br>2029-1                                                                                                 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min           |                                                                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Analytical writing requirements. |                                                                                                                                |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                   |                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------------------------------------------|
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SS202                             |                                                    |
| <b>SS289A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>COLLOQUIUM (AMER POLITICS)</b> | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2025-1                            | <b>Offerings:</b>                                  |
| <p>The colloquium provides cadets an opportunity for reading and analysis in depth in a topic area of special interest and timely relevance to their concentration. The course employs the seminar approach in which the instructor meets with small groups to discuss assigned readings, and cadets present their own analyses to the group. Course directors develop topics and determine the semesters in which they will be offered. Department Academic Counselors then forward course offerings and descriptions to Social Science majors and those majoring in areas related to the colloquium topic. Topics will vary by year to advantage the expertise of visiting professors or senior fellows. **This is a pilot course and must receive approval by the CC NLT AY26 to continue.**</p>                                                                                                                                |                                   | 2026-1 2029-1 2029-2                               |
| <b>Lessons:</b> 15 @ 75 min (1.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min            |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                              |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SS202<br>-Or-<br>SS252            |                                                    |
| <b>SS289B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>INTRO TO IA SENIOR THESIS</b>  | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2023-2                            | <b>Offerings:</b>                                  |
| <p>This course prepares students to participate in the International Affairs capstone Senior Thesis in the following term. The course will be conducted through a combination of class (group) meetings and 1-on-1 individual meetings with Course Director, both types to be scheduled at the latter's discretion. It familiarizes cadets with the research procedures, techniques, and topics for future study undertaken in support of independent research, as determined by the faculty advisor (course director). In general, cadets will explore a topic in detail to create a literature review and understand the current methodology used to study that topic. Grades will be determined through individual assignments, including class participation, response papers, and research building-block assignments. The outcome of the course is a research design that forms the basis for the cadet's senior thesis.</p> |                                   | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 10 @ 55 min (0.750 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min            |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                              |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SS307<br>-Or-<br>SS357            |                                                    |
| <b>SS291B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>INTRO TO IA SENIOR PROJECT</b> | <b>1.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2023-1                            | <b>Offerings:</b>                                  |
| <p>This introductory course prepares students who will participate in the International Affairs capstone Senior Project in the following term. It familiarizes cadets with the research procedures, techniques, and topics for future study, undertaken in support of an outside client. The scope of the course depends on the needs of the project and the needs of the cadets, as determined by the faculty advisor (course director). In general, cadets will select topics, form teams, meet with the client, and begin the research process. Grades will be determined through individual and group assignments, including class participation, response papers, and research building-block assignments. The outcome of the course is a research design that forms the basis for the cadet team's research paper for the client</p>                                                                                         |                                   | 2026-1 2027-1 2028-1<br>2029-1                     |
| <b>Lessons:</b> 10 @ 75 min (0.750 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min            |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | None                              |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SS307<br>-Or-<br>SS357            |                                                    |
| <b>SS307</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INTERNATIONAL RELATIONS</b>    | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2020-2                            | <b>Offerings:</b>                                  |

The objectives of this course are to provide cadets with an introduction to the fundamental concepts of international relations and the analytical tools necessary to evaluate two fundamental questions: "Why do states do what they do?" and "What causes conflict and cooperation in the international system?" Emphasizing intellectual pluralism, SS307 focuses on the value of applying different theoretical perspectives to historic cases and contemporary events in international affairs. Cadets will employ international relations theory to examine key issues such as the role of power and the use of force in the international system, the international political economy, the role of identity and culture in state behavior, foreign policy decision-making, and the role of morality and ethics in international affairs.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** At discretion of current course director.

**Prerequisite(s):** SS201 SS202  
-Or-  
SS201 SS252  
-Or-  
SS202 SS251  
-Or-  
SS251 SS252

**Disqualifier(s):** SS357

2026-1 2026-2 2026-3  
2026-8 2027-1 2027-2  
2027-3 2027-8 2027-9  
2028-1 2028-2 2028-3  
2028-8 2028-9 2029-1  
2029-2 2029-3 2029-8

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|              |                                    |                                                          |
|--------------|------------------------------------|----------------------------------------------------------|
| <b>SS357</b> | <b>ADV INTERNATIONAL RELATIONS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|------------------------------------|----------------------------------------------------------|

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**Scope:** 2020-2

**Offerings:**

The objectives of this course are to provide cadets with an introduction to the fundamental concepts of international relations and the analytical tools necessary to evaluate two fundamental questions: "Why do states do what they do?" and "What causes conflict and cooperation in the international system?" Emphasizing intellectual pluralism, SS357 focuses on the value of applying different theoretical perspectives to historic cases and contemporary events in international affairs. Cadets will employ international relations theory to examine key issues such as the role of power and the use of force in the international system, the international political economy, the role of identity and culture in state behavior, foreign policy decision-making, and the role of morality and ethics in international affairs. While the course content of SS357 is similar to that of SS307 (International Relations), SS357 is designed as an introductory course primarily for cadets majoring in International Affairs.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** At the discretion of the current course director.

**Disqualifier(s):** SS307

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|              |                                 |                                                          |
|--------------|---------------------------------|----------------------------------------------------------|
| <b>SS360</b> | <b>POL SCI RESEARCH METHODS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------------|----------------------------------------------------------|

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**Scope:** 2024-1

**Offerings:**

This course is an introduction to research methods for political science majors. It serves two main purposes. First, this course introduces research design and the myriad methodologies employed by scholars as they engage in debates within the political science community and beyond. This course will cover many aspects of research design, but will focus primarily on the ability to ask good questions and to craft research plans to best answer those questions. Second, the course is an introduction to some of the major debates within political science. The course equips students with the tools to understand and conduct research in political science in upper-level electives.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** One research design project.

**Corequisite(s):** SS202  
-Or-  
SS252

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|--------------|--------------------|----------------------------------------------------------|
| <b>SS364</b> | <b>GAME THEORY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2005-2

**Offerings:**

Game theory is designed to provide students with the tools necessary to think through the various courses of action available as they face a uncertain situations, determine market reaction to each alternative, identify the costs and benefits of each course of action and select the course of action that minimizes cost while maximizing benefits. The purpose of this course is to introduce cadets to the application of strategic thinking to tactical scenarios. This course consists of two components that are taught concurrently. The first component is the introduction of basic game theory and the second component is the application of those theories to tactical and strategic choice scenarios.

2026-2 2026-8 2027-1  
2027-2 2027-8 2028-1  
2028-2 2028-8 2029-2  
2029-8

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** A research paper is required.

**Prerequisite(s):** MA206  
-Or-  
MA256  
-Or-  
MA206X

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| <b>SS366</b> | <b>COMPARATIVE POLITICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2020-2

**Offerings:**

SS366 examines critical questions in political science: Why do regimes succeed, fail, and change? What causes democracy to emerge? Why are some countries torn by conflict while others are peaceful? Cadets will analyze states, regimes, society, identity, and political action in order to address these vital issues related to the internal workings of a country. As well as being central to the discipline of political science, these questions also play an important role in world politics and the formulation of U.S. foreign policy. Through examining countries around the world, SS366 uses real-world examples and empirics to discern potentially generalizable relationships. Studying Comparative Politics will provide cadets the tools to critically analyze these important questions.

2026-1 2026-2 2026-8  
2027-1 2027-2 2027-8  
2028-1 2028-2 2028-8  
2029-1 2029-2 2029-8

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** At the discretion of the current course director.

**Prerequisite(s):** SS202  
-Or-  
SS252

**Corequisite(s):** SS307  
-Or-  
SS357

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| <b>SS368</b> | <b>ECONOMETRICS I</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2024-1

**Offerings:**

This course is designed to teach students how to quantify, test, and employ economic theories as they are used in real world applications. The course covers the use of economic theory and data in the construction, estimation, and interpretation of econometric models. Special emphasis is placed on estimation of parameters of economic models and statistical inference using estimated models to determine the validity of economic theories. The primary mathematical tool employed in the course is multiple regression analysis. A number of applications demonstrate the use of the techniques studied.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Computer lab exercises conducted during regular class periods.

**Prerequisite(s):** MA206  
-Or-  
MA256

**Corequisite(s):** MA367

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| <b>SS370</b> | <b>MASS MEDIA &amp; AMER POLITICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-2

**Offerings:**

This seminar introduces cadets to what is perhaps the single most influential private institution in the American political system, oftentimes referred to as the fourth branch or fourth estate of American government. This course examines the major concepts, theoretical frameworks, and substantive issues surrounding the study of the media as a conduit between the people and the government and as a political actor in its own right. In particular, the roles, motivations, and effects of the constitutionally protected media on American political institutions and policymaking are extensively probed. The latter part of the course is dedicated to the study of the relationship between the military and the media, and prominent guest speakers are featured throughout the course to add context and practical application to the theories learned in class. The class includes a trip section to New York City to meet with journalists and news executives from national media outlets to round out the educational experience.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Research paper.

**Prerequisite(s):** SS202  
-Or-  
SS252

| SS372                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | POLITICS OF CHINA                                                                                        | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2020-2                                                                                                   | <b>Offerings:</b>                          |
| SS372 is a seminar-style course designed to provide a framework for understanding and analyzing China's history, economics, domestic politics, and foreign policy. To do so, cadets study the history and current structure of the Chinese regime in order to explore the sources of power and legitimacy in the People's Republic of China. Domestic challenges and opportunities receive attention as the class critically analyzes policy choices. In the foreign policy realm, the class examines sources of continuity and change, seeking to explain current actions and predict future responses.                                                                                                                                                                                                     |                                                                                                          | 2026-1 2026-8 2027-1<br>2028-1 2029-1      |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                   |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | At the discretion of the current course director                                                         |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SS307<br>-Or-<br>SS357                                                                                   |                                            |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SS366                                                                                                    |                                            |
| SS373                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | THE AMERICAN PRESIDENCY                                                                                  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2020-1                                                                                                   | <b>Offerings:</b>                          |
| This seminar examines the concept of executive power and authority with particular emphasis on the institution of the presidency in the American political system. The course will analyze the constitutional origins and evolution of the presidency. We will place particular emphasis on the formal rules and informal norms that developed since the Founding and frame presidential behavior. We will analyze the various factors that influence the perpetual transformation of the institutional organization and operation of the modern executive branch. The course will examine the dynamic relationships the executive branch maintains with other branches of government, the media, the public, and other key stakeholders and how these relationships shape the development of public policy. |                                                                                                          | 2026-1 2027-1 2028-1<br>2029-1             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                   |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Case study of presidential leadership, with graded bibliography and outline; compensatory time provided. |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SS202<br>-Or-<br>SS252                                                                                   |                                            |
| SS374                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | POLITICS OF NORTHEAST ASIA                                                                               | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2020-2                                                                                                   | <b>Offerings:</b>                          |
| SS374 is a seminar course designed to challenge cadets to think about why East Asian actors ? states, non-state actors, organizations, and individuals ? do what they do, both within and beyond the region. Students will apply the social scientific method and various social science theories to explain how and why East Asian actors make certain policy choices. Students will explore how pursuits of national interests create competition, cooperation, and conflict across political, economic, and military spectrums. Cadets will integrate and apply knowledge to describe, explain, analyze, or predict international affairs and domestic political developments as well as identify implications and develop feasible policy options.                                                       |                                                                                                          | No Course Offerings                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                   |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | At the discretion of the current course director.                                                        |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SS307<br>-Or-<br>SS357                                                                                   |                                            |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SS366                                                                                                    |                                            |
| SS375                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | POL OF THE POST-SOVIET STATES                                                                            | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2020-2                                                                                                   | <b>Offerings:</b>                          |

SS375 explores the fascinating - and often confounding - politics of Russia and the other states that once comprised the Soviet Union. Recognizing the role that historical legacies play in shaping the region's contemporary politics, the course begins with a review of the political, economic, and social history of the Soviet empire and its abrupt collapse in 1991. The course then shifts to an exploration of contemporary post-Soviet politics, utilizing political science theories and methods to explore a wide variety of political phenomena in the region including democratization, authoritarianism, mass mobilization, economic transition and reform, governance and corruption, nationalist conflict, and more. Finally, the course examines the international relations of the post-Soviet states, with particular emphasis on the ends, ways, and means that Russia uses to advance its strategic interests in the region and beyond.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** At the discretion of the current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

**Corequisite(s):** SS366

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| <b>SS376</b> | <b>SEMINAR IN AMERICAN GOVERNMENT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2026-1

**Offerings:**

This course is organized around questions addressed by the major subfields of American Politics. It builds on the knowledge from research methods and political thought to introduce cadets to some of the major puzzles addressed by political scientists in American Politics. Cadets will read influential primary sources and empirical studies that shape our understanding of the questions posed in this course. In doing so, cadets will become familiar with the wide variety of theoretical approaches, research designs, and methodologies that scholars of American Politics use to advance our knowledge of the field.

2026-1 2027-1 2028-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** SS202  
-Or-  
SS252

**Corequisite(s):** SS360

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| <b>SS377</b> | <b>POLITICS OF EUROPE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2020-2

**Offerings:**

In SS377, cadets explore the domestic and international politics of the European states, using a wide array of political science theories and methods to explain and predict state behavior in the region. They will examine the political and economic institutions of European states as well as explain how those institutions developed over time. They will also examine the development and functioning of European international institutions with special emphasis on the institutions of the European Union. Finally, cadets will use theory to predict how European institutions might evolve in the future under the stress of various contemporary challenges. This course builds on the theoretical foundations laid by SS366 and SS307/SS357.

2026-1 2028-1 2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** At the discretion of the current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

**Corequisite(s):** SS366

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| <b>SS378</b> | <b>GREAT POWER COMPETITION</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2020-2

**Offerings:**

One of the defining features of international relations at any point in history has been the distribution of power among the major states and the character of the political, economic, and military relationships among them. This course draws from the major schools of international relations theory to explore the underlying causes of war and peace among great powers, from the clash between Athens and Sparta in the 5th Century B.C., to great power politics of the 19th Century, the World Wars of the early 20th Century, and the Cold War between the United States and the USSR. The course concludes by examining how we might apply the theoretical and historical lessons of earlier periods to the challenge of great power politics among the United States, China, and Russia in the 21st Century.

2026-2 2026-8 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** At discretion of current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

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| <b>SS379</b> | <b>UNITED STATES CONGRESS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2019-2 | <b>Offerings:</b> |
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This course deals with the origins, institutional evolution, and behavior of the United States Congress. It focuses on the role of Congress in representation and policymaking within the American political system. The course addresses the development of the United States Congress, congressional campaigns and elections, the tension inherent in Members of Congress acting both as representatives for their constituents and policymakers for the nation, the structure and behavior of committees, the responsibilities of legislative leaders, and the interaction between Congress and the Executive and Judicial branches. Case studies, practical exercises, and guest lecturers are used to highlight these topics.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Analytical papers and briefings; compensatory time provided.

**Prerequisite(s):** SS202  
-Or-  
SS252

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| <b>SS380</b> | <b>LABOR ECONOMICS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2021-1 | <b>Offerings:</b> |
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This course studies the nature and determinants of pay and employment. The course emphasizes the role of institutions which are significant in determining the pattern and speed of adjustment in the labor market. The traditional tools of micro and macroeconomics and econometrics are employed. Military manpower is examined as an application of the theories developed during the course.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Research paper and oral presentations.

**Prerequisite(s):** SS368 SS382

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| <b>SS382</b> | <b>INTERMEDIATE MICROECONOMICS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2021-1 | <b>Offerings:</b> |
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This course is a theory course in which cadets develop a thorough understanding of microeconomic modeling and models; it is a prerequisite for most downstream economics courses. The course develops a methodology that economists use to study the interaction among individual economic agents (such as consumers, firms and the government) and the allocation of scarce resources among these agents. The goal is for cadets to understand optimization, markets, and to some extent policy-making, using an integrated, theoretical model. Ultimately the consequence of a change in the market environment, in public policy or in the global economy can be assessed vis-à-vis its impact on individual economic agents.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MA205 SS201  
-Or-  
MA205 SS251  
-Or-  
MA255 SS201  
-Or-  
MA255 SS251  
-Or-  
MA204X SS201  
-Or-  
MA204 SS201

**Corequisite(s):** MA367

| SS383                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | POLITICS OF THE MIDDLE EAST                                                                                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2020-2                                                                                                                                | <b>Offerings:</b>                            |
| Many countries in the Middle East suffer from significant challenges including weak states and institutions, authoritarian persistence, and economic instability. Scholars have explained these issues by assessing the role played by political culture, economic structures, colonialism and foreign interventions and states have tried to find solutions through Arab Nationalism, Pan-Arabism, Islamism and other avenues. This course examines these challenges, offers explanations for these problems, and explores potential solutions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                       | 2026-2 2027-2 2028-2<br>2029-2 2029-8 2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                                                                                                                |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | At the discretion of the current course director.                                                                                     |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SS307<br>-Or-<br>SS357                                                                                                                |                                              |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SS366                                                                                                                                 |                                              |
| SS384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | POLITICS OF LATIN AMERICA                                                                                                             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2020-2                                                                                                                                | <b>Offerings:</b>                            |
| SS384 examines the politics of Latin American countries by applying the theories, methodologies, and conceptual tools provided by the disciplines of comparative politics and international relations. In the course students will study phenomena such as democratic regimes, authoritarian regimes, political transitions, political participation, political institutions, socio-economic development, and foreign policy within Latin America and its member countries. At the conclusion of this course, students can apply comparative political science approaches to understand Latin American politics, have a deeper understanding of Latin American history and policies, and develop a greater familiarity with the tools of social science.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                       | 2027-1 2027-9 2028-1<br>2028-9               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                                                                                                                |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | At the discretion of the current course director.                                                                                     |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SS307<br>-Or-<br>SS357                                                                                                                |                                              |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SS366                                                                                                                                 |                                              |
| SS385                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | HISTORY OF ECONOMICS                                                                                                                  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2021-1                                                                                                                                | <b>Offerings:</b>                            |
| This course conducts a general study of the history of economics by examining specific economic thinkers and the economic development of the United States and its institutions placed within a historical context. Institutions influence our nation's economic system, culture, political process, conflict resolution mechanisms, financial systems, and labor/management boards to name a few. Due to changes in these institutions and our nation's laws over time as a product of historical events and periods, the economy differs in terms of efficiency, growth potential, and equity in allocating resources. The first course block examines, compares, and contrasts a variety of history's prominent economic thinkers and their philosophies from Adam Smith to modern economists. It explores how we think about economics in terms of historical context, theory, and the formation of various schools of economic thought. The second course block conducts an overview of the United States economy and its development through the historical context and lens of significant events and distinctive periods of American history starting with the colonies and ending with contemporary challenges facing the country. This block explores how the US government has taken deliberate steps to adjust the capitalist economic system. Included are examinations of efforts to promote economic growth, the causes and impacts of significant economic events, and steps taken to stabilize and reform the economies of nations following major military conflicts. A small part of the block is devoted to war financing. By examining these significant changes or turning points in our nation's history, cadets can better apply economic ideas to the events or periods to help understand the current relevance of the material. |                                                                                                                                       | 2026-2 2027-2 2028-2<br>2029-2               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                                                                                                                |                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Historical economic policy critique (1000 words); current event presentation; small group lesson presentation; small group exercises. |                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SS201<br>-Or-<br>SS251                                                                                                                |                                              |
| SS386                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | POLITICAL THOUGHT                                                                                                                     | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                    |                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2017-2                                                                                             | <b>Offerings:</b>                                                                            |
| This course introduces students to the fundamental questions of political life, as expressed in classic works of political philosophy. The course presents the history of political thought as a series of debates over the meaning of justice, man's place in nature, and the human good. Students explore three main approaches to these questions - ancient, medieval, and modern - and they compare and contrast Western ways of wrestling with fundamental questions to others that have arisen outside the West. The course culminates in a consideration of the place of the American regime in the history of political thought.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                    | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-9<br>2028-1 2028-2 2028-9<br>2029-1 2029-2 2029-8 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                             |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Colloquium and seminar essays, participation in the Discussion Leader Model, and Commonplace Book. |                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SS202<br>-Or-<br>SS252                                                                             |                                                                                              |
| <b>SS386</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>POLITICAL THOUGHT AND IDEAS</b>                                                                 | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b>                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2029-2                                                                                             | <b>Offerings:</b>                                                                            |
| This course examines the fundamental questions of Western political philosophy. In order to better understand why these problems are of vital relevance to contemporary civilization in the late modern West, students consider six themes: the nature of politics and how theorists, citizens, and statesmen have understood political things; the nature of freedom and the conditions necessary for its establishment, maintenance, preservation, and improvement; republicanism in antiquity and modernity; liberal democracy and constitutional order; the relationship between religion and politics; and, the fundamental presuppositions of traditional, modern, and contemporary social science. This course allows students to achieve critical understanding of the ancient and modern foundations of Western political thought and how these ideas have contributed to American republicanism, liberal democracy, and representative government; to clarify a range of modern political problems at home and abroad that challenge civilization; to acquire a competence reading, writing about, and discussing classic works of political philosophy, fostering life-long learning on masterpieces of human reflection; to develop cross-disciplinary capacity to study politics by (a) integrating basic chronological knowledge of what has happened in Western intellectual life, (b) understanding how to study ideas as vital components of traditional liberal education and officer development, (c) writing a Seminar Essay that synthesizes course learning, (d) making use of language study, grammar, etymology, and philology, and (e) keeping a Commonplace Book; to apply political thought to contemporary circumstances; and, to contrast Western principles with a major non-Western tradition of political thought. |                                                                                                    | No Course Offerings                                                                          |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                             |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Colloquium and seminar essays, participation in the Discussion Leader Model, and Commonplace Book. |                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SS202<br>-Or-<br>SS252                                                                             |                                                                                              |
| <b>SS387</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>PUBLIC ECONOMICS</b>                                                                            | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b>                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2022-1                                                                                             | <b>Offerings:</b>                                                                            |
| This course is an applied microeconomics course focusing on theoretical models to answer real world policy issues in the public sector. In particular, this course examines issues in public expenditures, social insurance, social welfare, redistribution, taxation, and public choice. Using economic models, this course helps students sharpen their analytical skills to solve complex governance challenges. Cadets should gain a better understanding of unique challenges that governments face when providing services while trying to balance economic efficiency with social equity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                    | 2026-1 2026-8 2027-1<br>2027-8 2028-1 2028-8<br>2029-1 2029-8 2029-9                         |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Labs:</b> 0 @ 0 min                                                                             |                                                                                              |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Analysis paper-- public expenditure program.                                                       |                                                                                              |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -Or-<br><br>-Or-<br>SS201 SS368 SS382<br>-Or-<br>SS251 SS368 SS382                                 |                                                                                              |
| <b>SS388</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>INTERMEDIATE MACROECONOMICS</b>                                                                 | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b>                                     |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2021-1                                                                                             | <b>Offerings:</b>                                                                            |



This course is dedicated to the study of aggregate economic activity. The course examines the determinants of long run growth, and then addresses short run economic fluctuations. The course uses various models, including the Solow Growth Model, the IS-IM model, and the Aggregate Demand - Aggregate Supply model. The microeconomic foundations for macroeconomics are discussed, and current macroeconomic policy issues are debated. These issues are discussed within the context of both open and closed economies.

2026-1 2026-2 2026-8  
2027-1 2027-2 2028-1  
2028-2 2029-1 2029-2  
2029-8

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** One policy analysis paper (1000 words); compensatory time provided.

**Prerequisite(s):** SS201  
-Or-  
SS251

**Corequisite(s):** MA367

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| <b>SS390</b> | <b>BEHAVIORAL ECONOMICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------|----------------------------------------------------------|

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**Scope:** 2022-1

**Offerings:**

This course will cover how insights from psychology and behavioral economics relate to the foundational economic model of rational choice. This course will cover four main topic areas. First, the course will examine how individual preferences tend to deviate from a rational choice model. Second, the course will explore how cognitive limitations alter individual choice. Third, the course will look at how social preferences and social influence impact decisions. Fourth and finally, this course will investigate how public policy interacts with individual behavioral tendencies.

No Course Offerings

**Lessons:** 40 @ 55 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Corequisite(s):** SS368  
-Or-  
SS382

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| <b>SS392</b> | <b>RIGHTS, LIBERTIES &amp; US CITIZEN</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2025-2

**Offerings:**

This seminar focuses on political actors, institutions and processes as they relate to the concepts of civil rights and civil liberties. Emphasis is placed on how political participation and engagement influence civil rights and civil liberties in the American political system. The course considers how institutions and informal actors shape the domestic policymaking process pertaining to civil rights and civil liberties, and also includes discussions about how these issues apply to the roles and responsibilities of Army officers.

2026-2 2027-2 2028-2

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** SS202 SS360  
-Or-  
SS252 SS360

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| <b>SS394</b> | <b>FUND. FINANCIAL DATA ANALYSIS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------------|

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**Scope:** 2025-1

**Offerings:**

Fundamentals of Financial Data Analysis teaches cadets the principles of financial data, the tools of understanding financial information, and the application of econometric methods to analyze financial markets, estimate relationships, and make informed finance-related decisions. Cadets will use companies' and market financial data to measure performance and assess financial position. Additionally, cadets will be exposed to the fundamentals of valuing equity, debt, and diverse investment instruments. To accomplish this, students will explore various econometric techniques commonly used in financial research, such as linear regression, time series analysis, and forecasting. The course places emphasis on understanding the theoretical underpinnings of these methods and their practical implementation in asset pricing, evaluating stock performance and volatility, and risk management.

2026-1 2026-2 2026-8  
2027-1 2027-2 2027-8  
2028-1 2028-2 2029-1  
2029-2 2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** Financial Statement Analysis. Computer lab exercises conducted during regular class periods.

**Prerequisite(s):** SS201  
-Or-  
SS251

| SS395                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | INTERNATIONAL SECURITY                                                   | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020-1                                                                   | <b>Offerings:</b>                          |
| <p>This course uses the foundation provided by SS307/SS357 to explore one of the critical forces that drives world politics - international security. The purpose of this course is to provide students a solid foundation in the field of international security studies - one of the major sub-fields of the international relations discipline - including a grasp of key concepts and case studies relevant to the field today and throughout history. Students will examine fundamental questions in this field, including "What is security?" "What causes war, and what prevents it?" and "How do actors in the international system use force as a tool to achieve goals and influence others?" As a "Writing in the Major" course for the Security Studies and Foreign Policy track of the International Affairs major, students should expect heavy emphasis on writing assignments as a key tool for developing, communicating, and validating their understanding of course material.</p>                                                                                                    |                                                                          | 2026-1 2027-1 2028-1 2029-1                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 30 @ 75 min (2.000 Att/wk)                                               | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | At discretion of current course director.                                |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SS357<br>-Or-<br>SS307                                                   |                                            |
| SS399                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SOCSCI INTERNSHIP/PRACTCAL EXP                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2005-4                                                                   | <b>Offerings:</b>                          |
| <p>The Academic Individual Advanced Development (AIAD) program is designed to give cadets practical experience in their field of study and to reflect on their experiences by completing specified academic requirements. Recent internships involve participation in the American, European and Russian (AMEURU) program hosted by the University of Maastricht, the Tri-Service Global Spectrum trip to Vietnam, study at the German Marshall Center, the American Institute on Political and Economic Systems (AIPES) in the Czech Republic, the International Institute for Political and Economic Studies (IIPES) in Greece, and the International Studies Program (ISP) in Eastern Europe. Scope, depth and material covered will meet the requirement of a 3-credit hour course in Social Sciences. Grades are determined based on preparatory briefings and essays, a journal of daily activities, the quality of the work performed during the internship, and a final paper, briefing, or exam that incorporates their experience with a topic from their field of study, due upon return.</p> |                                                                          | 2026-7 2027-7 2028-7 2029-7                |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 @ 0 min (0.000 Att/wk)                                                 | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Admission to the course requires an interview with the AIAD coordinator. |                                            |
| SS457                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | INTRODUCTION TO GRAND STRATEGY                                           | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2022-2                                                                   | <b>Offerings:</b>                          |
| <p>This seminar examines the fundamentals of grand strategy in theory and in practice. It addresses the historical moment in which the modern concept of grand strategy was born and explores the major debates that have animated scholars and practitioners since. We focus on the core building blocks of grand strategy: the elements of national power that are available to be coordinated and applied in a coherent fashion across policy contexts. We analyze historical case studies to explore how the modern concept of grand strategy can be retrospectively applied to explain state behavior across time and diverse contexts. The seminar ultimately aims to prepare future Army officers to apply concepts from grand strategy to analyze and address complex problem sets in the contemporary international environment.</p>                                                                                                                                                                                                                                                            |                                                                          | 2029-2                                     |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 30 @ 75 min (2.000 Att/wk)                                               | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | At the discretion of the course director.                                |                                            |
| SS458                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | TOPICS IN GRAND STRATEGY                                                 | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020-1                                                                   | <b>Offerings:</b>                          |
| <p>SS458 Applied Topics in Grand Strategy is the second of a two-part sequence designed to build on the course material covered in SS457. The course will cover applied regional and thematic topics in the study of grand strategy in a contemporary and/or historical context to illustrate key theoretical concepts. Scope, depth and material covered will meet the requirement of a 3-hour credit course in the Grand Strategy Program. Grades are determined based on presentations, essays and other writing exercises, class participation, and a final exam, essay, and/or briefing.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                          | 2026-1 2029-1                              |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 30 @ 75 min (2.000 Att/wk)                                               | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | At the discretion of the current course director.                        |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SS457                                                                    |                                            |

| SS461                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | LEADERSHIP ECONOMICS          | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)              |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2020-2                        | <b>Offerings:</b>                                       |
| <p>Good leadership has always been defined by what worked yesterday, yet no formal discipline exists to teach you principles for dealing with unanticipated challenges and uncertainty. As you'll learn, leadership is the act of choosing to shape the world, shifting it from its current path. However, there is no single leadership formula precisely because the innovations of leaders are new creations. Instead, we will construct a framework of human action that rests on decades of research in behavioral economics and neuroscience, formalized with the mathematical tools of microeconomics. With this framework, we assess the role and appropriate application of autonomy, character and honor, humility, empathy, resilience, and more. We will also read and consider how the latest self-help and business movements fit within this framework, including "psychological capital" (PsyCap), "Immunity to Change," "Grit," and others. The key takeaway of the course will be a formalization of what it means to be people of action.</p>                                                                                                                                                             |                               | 2027-1 2027-2 2028-2 2029-1                             |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 30 @ 75 min (2.000 Att/wk)    | <b>Labs:</b> 0 @ 0 min                                  |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                          |                                                         |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SS368 SS382                   |                                                         |
| SS462                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ECONOMIC GROWTH & DEVELOPMENT | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2020-2                        | <b>Offerings:</b>                                       |
| <p>This course provides an overview of major topics in the study of economic development. It combines the study of economic theories on the determinants of growth and development with a discussion of related and current empirical work. The goal is not to provide exhaustive coverage of all relevant topics, but instead to expose cadets to current issues and debates in the field of development economics. By the end of the course, cadets should be familiar with indicators of growth and development, know seminal theories and be able to assess them in light of empirical evidence, be able to read, understand, and critique current empirical research in economic development, and be able to think innovatively about policy design by drawing on and integrating theoretical and empirical research. Ultimately, cadets will be empowered as future decision-makers to better design, implement, and evaluate policy interventions.</p>                                                                                                                                                                                                                                                                |                               | 2026-1 2028-1 2029-1                                    |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 30 @ 75 min (2.000 Att/wk)    | <b>Labs:</b> 0 @ 0 min                                  |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                          |                                                         |
| SS463                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | INVESTMENT THEORY & APPL      | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2021-1                        | <b>Offerings:</b>                                       |
| <p>Investment Theory and Application is designed to give cadets an appreciation of the theoretical underpinnings for investing as well as the way investors think about and approach the markets today. This course pulls from concepts in statistics, microeconomics, econometrics, accounting, and finance. Broadly, the course is organized into three interconnected blocks. Before studying the movement of prices and the measure of risk and return in the markets, investors need to understand how to value (price) a security (stock, bond, or option). After understanding these basic valuation techniques, investors can begin appreciating the relationship between risk and return in the markets and how that builds to some of the foundational theories and models in finance: modern portfolio theory, the efficiency market hypothesis, and the capital asset pricing model. Finally, after building the theoretical story of efficient markets, we will explore the counterpoint. The course concludes with a rigorous final project, where students wade through imperfect information to propose and defend recommendations for the current active portfolio within West Point's Investment Club.</p> |                               | 2026-2 2027-2 2028-2 2029-2                             |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 30 @ 75 min (2.000 Att/wk)    | <b>Labs:</b> 0 @ 0 min                                  |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                          |                                                         |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SS368 SS382 SS494             |                                                         |
| SS464                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | HOMELAND SECURITY             | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2024-1                        | <b>Offerings:</b>                                       |
| <p>The purpose of SS464 is to address the complex challenges of homeland security through an interdisciplinary approach. The goal of this course is to provide future leaders with a thorough understanding of the homeland security policy area. This course explores how the evolving nature of the terrorist threat, particularly catastrophic terrorism, poses unprecedented and complex challenges to how America provides for its security. The course examines how homeland security policy intersects with other domestic and foreign policy issues, how our federalist system of government affects homeland security, and how moral, ethical, and civil liberties concerns complicate the development of effective homeland security policies. By analyzing the threat and developing an understanding of the unique policy problems and tools of homeland security, the course enables students to critically assess national efforts in such areas as border</p>                                                                                                                                                                                                                                                 |                               | 2026-1 2026-2 2027-1 2027-2 2028-1 2028-2 2029-1 2029-2 |

security, domestic counterterrorism policy, critical infrastructure protection, and emergency preparedness and response. Students will learn about the major policy and institutional reforms underway in the homeland security policy area, examine whether these changes are improving or will improve U.S. security policy, and develop their own views on the direction of national homeland security policy. The course will enable students to think critically about how the United States' overseas efforts to combat terrorism, preempt emerging threats, and counter the proliferation of weapons of mass destruction relate to domestic homeland security efforts. By the end of the course, students will gain a solid intellectual foundation to think critically and creatively about America's efforts to prevent terrorist attacks within the United States, reduce our vulnerability to terrorist attack, and minimize the damage and recover from attacks that may occur.

**Lessons:** 20 @ 110 min (1.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Admission to the course will be capped with priority given to Terrorism Studies Minor students.

**Prerequisite(s):** SS307  
-Or-  
SS357

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|--------------|----------------------------------|---------------------------------------------------|
| <b>SS465</b> | <b>TERRORISM: NEW CHALLENGES</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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**Scope:** 2019-2

**Offerings:**

The purpose of SS465, is to address the challenges of terrorism in the current and future global security environment through an interdisciplinary approach. Specifically, this course examines the unique challenge terrorism poses to liberal democratic states, policy makers and to military professionals. By analyzing the different perspectives of terrorism, given a variety of political and strategic contexts, students better understand terrorist motivations, strategies, means and ends. Finally, the course explores how a liberal democratic state can best fight terrorism in this new threat environment.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2 2029-8  
2029-9

**Lessons:** 20 @ 110 min (1.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Admission to the course is subject to the approval of the Comparative Politics Academy Professor.

**Prerequisite(s):** SS307  
-Or-  
SS357

**Disqualifier(s):** SS474

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|--------------|----------------------------|---------------------------------------------------|
| <b>SS466</b> | <b>COMBATING TERRORISM</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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**Scope:** 2019-2

**Offerings:**

The Combating Terrorism course represents a unique opportunity for students to conduct in-depth and integrated study on the most pressing past, present, and future counterterrorism challenges to the United States and its interests. The objectives of this course are: (1) to synthesize and apply the cadet's studies across their core academic major and elective course work to the thematic issue of counterterrorism; (2) to apply methodological skills of research design, conceptual reasoning, analysis, and research gained to counterterrorism; (3) to extend the cadet's in-depth study of the selected area of interest beyond the level obtained in course work with regards to counterterrorism; (4) to design and conduct focused research; and (5) to develop cadet skills in conceptual reasoning, critical analysis, and effective writing.

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

**Lessons:** 20 @ 110 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** Ten 2-3 page analyses of current events; one 20-page research paper; significant class participation.

**Prerequisite(s):** SS465  
-Or-  
SS474

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|--------------|--------------------------------|---------------------------------------------------|
| <b>SS468</b> | <b>POLITICAL PARTICIPATION</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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**Scope:** 2020-1

**Offerings:**

This course provides a broad understanding of the dynamics of political participation. The goals of this course are two-fold. First, it comprehensively examines both individual and group participation, as well as the many ways in which participation manifests itself in the democratic process, namely in the form of electoral (voting, campaigning) and non-electoral behavior ("civicness", group action, etc). As such, this course will include topics in public opinion, the electoral process, and voting behavior. Second, the approach is both empirical - and theoretical. This course examines results of electoral behavior (primarily U.S. national and state elections), complemented with competing theoretical approaches which serve to explain and better understand this behavior.

2027-1 2028-1 2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One 2000-word paper. Compensatory time provided.

**Prerequisite(s):** SS202 SS360  
-Or-  
SS202 SS386  
-Or-  
SS252 SS360  
-Or-  
SS252 SS386

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|--------------|------------------------|----------------------------------------------------------|
| <b>SS469</b> | <b>ECONOMETRICS II</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2022-1

**Offerings:**

This course is designed to teach students advanced concepts in estimation and statistical inference. Building upon the material covered in SS368, students will learn how to test for failure of the data to meet the assumptions of the basic regression model and how to allow for these departures from the standard assumptions during estimation. Among the topics covered will be Generalized Least Squares, Time Series, Instrumental Variables, and Simultaneous Equations estimation. Application of the techniques to the estimation of economic models using actual economic data is an integral part of the instruction. The course makes substantial use of statistical software packages.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** End-of-semester research paper and presentation required.

**Prerequisite(s):** SS368

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|--------------|--------------------------|----------------------------------------------------------|
| <b>SS470</b> | <b>MONEY AND BANKING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2020-1

**Offerings:**

SS470 is a senior level economics course whose primary purpose is to provide depth in the student's background and understanding of macroeconomics and international economics. The focus of the course is on the financial sector of the economy, which provides the means to transfer savings from firms, households, and governments to investors who want to purchase new capital goods. The course begins by discussing the various types of financial institutions and examines the importance of financial intermediation. The course then identifies how to measure the risks faced by financial institutions and how to manage these same risks.

2026-1 2026-2 2027-1  
2028-1 2029-1 2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** The final eight lessons of the course comprise an extended computer banking simulation, allowing cadets to apply the theories from previous lessons.

**Prerequisite(s):** SS388

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|--------------|-------------------------------------------|----------------------------------------------------------|
| <b>SS472</b> | <b>SOLDIER &amp; STATE: AM CIV-MIL RE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2021-1

**Offerings:**

This seminar explores the unique role of the soldier within our democratic republic. We begin by situating the profession of arms within the Executive branch as an agent to its direct principal, the Legislative branch. We proceed by examining the similarities and differences between the military and other agents of the administrative state. We explore the military's role in providing professional expertise in the policy process and examine current trends that threaten to undermine this advisory position. Using a historical framework, we will examine the evolution of civil-military relations in times of war, peace, and perpetual conflict. We will place particular emphasis on the theories and norms of civil-military relations in a post-9/11 world including navigating the tensions inherent to the dual role of the soldier as war fighter and state builder..

2026-2 2027-1 2027-8  
2028-1 2028-2 2028-8

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One major research paper; compensatory time provided.

**Prerequisite(s):** SS202  
-Or-  
SS252

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>SS473</b> | <b>AMERICAN FOREIGN POLICY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2022-1

**Offerings:**

SS473 examines the historical evolution, development, implementation, and consequences of American foreign policy. It analyzes the actors that craft and influence policy, concentrating both on formal government institutions--such as the President, Congress, and the foreign policy bureaucracy--and informal institutions such as public opinion, political parties, interest groups, and the media. The course examines key events in American Foreign Policy history through the lens of constraints and 'policy choice.' What choices did U.S. foreign policy makers confront? What policies did they choose to implement and why? What were the consequences of those decisions? Students will ultimately apply these lessons as they evaluate historic foreign policy decisions, debate contemporary dilemmas, and contemplate the future of American foreign policy.

**Lessons:** 30 @ 75 min (2.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

Cadets are required to submit a term paper that analyzes a historical event in American foreign policy history.

**Prerequisite(s):**

SS202 SS307  
-Or-  
SS202 SS357  
-Or-  
SS252 SS307  
-Or-  
SS252 SS357

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|--------------|------------------------------------|----------------------------------------------------------|
| <b>SS475</b> | <b>COMP POLITICAL INSTITUTIONS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:**

2020-2

**Offerings:**

2026-1 2027-1 2028-1  
2029-1

This courses uses the comparative method, in conjunction with political science theory, to explore and explain the wide variation in political institutions around the world. The course devotes significant attention to understanding the most basic institutional characteristic of a state: its regime type. The first portion of the course wrestles with key definitions: what are the defining characteristics of democracies? What are the defining characteristics of dictatorships and other forms of authoritarian rule? How do we understand regimes that do not fall neatly into either category? The second portion of the course explores the causes of regime transitions: when, where, how, and why does democracy emerge? When is democracy likely to collapse? When do dictatorships fail, and what type of regime results? Finally, the course explores the wide array of outcomes that are influenced by a country's political institutions: economic development, inequality, intrastate conflict, and others. As a "Writing in the Major" course for the Institutions, Governance, and Development track of the International Affairs Major, students in SS475 will develop their substantive knowledge of political institutionalism alongside their critical reading and writing skills through a series of developmental writing assignments.

**Lessons:** 30 @ 75 min (2.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

At the discretion of the current course director.

**Prerequisite(s):**

SS202 SS307 SS366  
-Or-  
SS252 SS357 SS366  
-Or-  
SS202 SS357 SS366

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|--------------|-------------------|----------------------------------------------------------|
| <b>SS476</b> | <b>CIVIL WARS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:**

2024-1

**Offerings:**

2027-1 2028-1 2029-1

SS476 covers the entire life-cycle of civil conflict from onset, to dynamics, to termination. Through the use of a rich theoretical and empirical literature on conflict and a range of political science methodologies, it examines questions such as: what causes civil wars? Why do certain patterns of violence occur in civil wars? What makes civil wars so hard to end? What is the impact of third parties on termination of civil wars? The course emphasizes the theory and practice of civil war settlement and termination with particular attention to the role of third parties.

**Lessons:** 30 @ 75 min (2.000 Att/wk)

**Labs:** 0 @ 0 min

**Special Requirements:**

At discretion of current course director.

**Prerequisite(s):**

SS307  
-Or-  
SS357

**Corequisite(s):**

SS366

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| <b>SS477</b> | <b>ECONOMICS OF NATIONAL SECURITY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:**

2019-2

**Offerings:**

This is a capstone course for the economics major that is designed to integrate microeconomics, macroeconomics, and econometrics. Coursework and classroom discussion require students to apply theoretical concepts and models from the major's toolbox courses, electives, and the core curriculum to policy issues inherent to the provision of national security. The course prepares students to address the broader economic challenges of national security and provides them with a baseline understanding of the economic underpinnings and ramifications of defense policy. Students use relevant databases, econometrics, and the skills they have learned as economics majors to prepare and present term research papers that analyze major defense and public policy decisions.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Term Research Paper; compensatory time provided.

**Prerequisite(s):** SS368 SS382 SS388  
-Or-  
SS380  
-Or-  
SS387  
-Or-  
SS484

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|--------------|---------------------------------------|---------------------------------------------------|
| <b>SS478</b> | <b>DIST PROF OF SECURITY STUD SEM</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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**Scope:** 2019-2

**Offerings:**

This course is taught by a visiting scholar with a distinguished record of academic achievement and professional service in the national security arena. The seminar focuses on topical issues that reflect the professor's area of expertise. Dr. Kori Schake, a former Director of Defense Strategy and Requirements for the National Security Council and presently a distinguished fellow at the Hoover Institution, serves as our current visiting scholar. In the seminar, students take part in detailed discussions, deliver presentations, conduct research, and prepare analytical papers. Students also benefit from guest speakers who share their experiences and expertise with the seminar. Previous Distinguished Professors include General (Retired) Barry R. McCaffrey, former Director of the White House Office of National Drug Control Policy; General (Retired) John Galvin, former commander of SACEUR and noted author; Professor Richard Shultz, Director of International Security Studies at the Fletcher School of Law and Diplomacy; Professor Linda Brady, Director of the School of International Affairs at Georgia Institute of Technology; and Admiral (Retired) Stansfield Turner, former President of the Naval War College and Director of the Central Intelligence Agency.

No Course Offerings

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Research paper and oral presentations.

**Prerequisite(s):** SS307  
-Or-  
SS357

|              |                                   |                                                   |
|--------------|-----------------------------------|---------------------------------------------------|
| <b>SS479</b> | <b>INTERNATIONAL ORGANIZATION</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
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**Scope:** 2020-2

**Offerings:**

While occasionally individual states in the international system have sought to find the solution to problems that they face alone - be they security-related, financial, social or environmental - the nature of the 21st century's geopolitical issues almost demand a collective, transnational response. But what entity or entities serve to facilitate this transnational cooperation? Enter international organizations, or IOs. This course is centered on introducing students to the landscape of global IOs that exist in the world today. To that end, this course investigates the emergence, purposes, and contemporary statuses of a variety of the major IOs that mark the international relations landscape today. These will include global IOs (like the United Nations, the World Bank, the International Monetary Fund, the World Trade Organization, the International Criminal Court), regional IOs (like the European Union [EU], African Union [AU], and the Association of Southeast Asian Nations [ASEAN]); and a variety of non-governmental organizations (NGOs) (like Doctors Without Borders, Amnesty International, and the International Committee of the Red Cross). The course also assesses the impact and relevance of international organizations as an issue of concern to Army officers and national security professionals.

No Course Offerings

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** At the discretion of the current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

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|--------------|------------------------------------|---------------------------------------------------|
| <b>SS480</b> | <b>PUBLIC POLICYMAKING PROCESS</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0) |
|--------------|------------------------------------|---------------------------------------------------|

**Scope:** 2019-2

**Offerings:**

SS480 is the capstone course for American Politics majors in the Social Sciences Department. This course examines the major concepts, theoretical frameworks, and substantive dilemmas of the public policymaking process. The aim of this course is to arm students with a myriad of tools to understand, evaluate, and contextualize specific political problems in the public policy arena. Students will be expected to integrate the concepts of not only basic, toolbox, and elective courses, but knowledge acquired from other courses from other disciplines. Public policy spans the disciplines of politics, economics, sociology, philosophy, and psychology, as policymakers wrestle with developing and implementing value-laden decisions in a world of scarcity and uncertainty. As such, the student of public policy must use a variety of social science tools - and increasingly, physical science tools, to dissect policy problems, develop viable and feasible alternatives, and fashion methods of adoption and implementation. Consequently, this course is designed to build upon the student's conceptual and analytical base in the quest to establish and refine a systematic approach to public policy analysis, formulation, adoption, and implementation.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Analytical writing requirements; compensatory time provided.

**Prerequisite(s):** SS202 SS360 SS386  
-Or-  
SS252 SS360 SS386

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>SS481</b> | <b>POLITICS OF DEFENSE POLICY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-2

**Offerings:**

This seminar is a survey of the politics that shape America's policy decisions over war and peace. We study the domestic influences of foreign policy and the international political dynamics that shape why and how America intervenes. It is an examination of American national security policy and institutions using theoretical, historical, and practical perspectives. Drawing from various literatures, we examine and evaluate the choices our nation makes in defense policy decisions. We address questions concerning military innovation and adaptation, change and transition in the armed services, defense resources, and capacities of actors in the defense policy arena. Using the lens of "grand strategy," we examine how defense policy decisions are influenced by a broad and complex array of political and economic factors and how these decisions shape future domestic and foreign policy environments.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 40 @ 55 min (1.250 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Analytical writing requirements; compensatory time provided.

**Prerequisite(s):** SS202 SS360 SS386  
-Or-  
SS252 SS360 SS386

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|--------------|----------------------------------|----------------------------------------------------------|
| <b>SS483</b> | <b>NATIONAL SECURITY SEMINAR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2020-2

**Offerings:**

This course helps cadets to improve their ability to analyze the national security choices of the United States. An underlying premise is that it is useful to think strategically about national security. To think strategically means to keep in mind the need to identify and to reconcile ends, ways, and means when seeking to advance the national interests of the United States. Sound in theory, this approach is hard to implement in practice. This course introduces cadets to actors, processes, and issues that influence US national security decisions and gives cadets a greater appreciation for the challenges associated with making and implementing national security policy.

2026-2 2027-2 2028-2  
2029-2 2029-8

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** At discretion of current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

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|--------------|--------------------------------|----------------------------------------------------------|
| <b>SS484</b> | <b>INTERNATIONAL ECONOMICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-2

**Offerings:**

This course integrates economic principles taken in SS382 and SS388. International Economics promotes understanding of the economic causes and effects of international trade, examines the justifications for and effectiveness of a variety of trade policies, explains and critiques the international flow of money, and explores the impact of these topics upon individual firms in the marketplace. The course's methodology rests on theoretical concepts and models such as profit maximization, market equilibrium, preference maximization, and macroeconomic equilibrium. The course is divided into four blocks. The first three blocks investigate the theory of international trade in goods and comparative advantage, the practice of international trade and international political economy, and the workings of international monetary markets. The final block compels cadets to apply their estimates of the international macroeconomic environment to choices made by national governments.

2026-2 2027-2 2028-2  
2029-2 2029-8 2029-9



**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** One in-class case study and one analytical paper (1500 words); compensatory time provided.

**Prerequisite(s):** SS382 SS388

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>SS485</b> | <b>POLITICS SUB-SAHARAN AFR</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2020-2 | <b>Offerings:</b> |
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This is a discussion-based seminar course designed to survey the origins and dimensions of contemporary issues within the post-colonial, sub-Saharan African state. The concepts of democracy, institutionalization, political economy, war, and peace remain a complex landscape for many African states. Focusing on theoretical work and case studies, cadets will examine sub-Saharan Africa's historical experiences, its economic heritage, and the international context in which it is embedded. At the same time, cadets will explore how Africans have shaped their own political and economic situations. Using social science methods, cadets will gain an understanding of how context shapes political behavior, in general, and how historical and political forces have influenced African politics, in particular.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** At the discretion of the current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

**Corequisite(s):** SS366

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|--------------|-----------------------|----------------------------------------------------------|
| <b>SS486</b> | <b>STATE BUILDING</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2020-2 | <b>Offerings:</b> |
|---------------|--------|-------------------|

SS486 examines states and state building in the modern world. Whether one focuses on international relations, comparative politics, or American politics, the state remains a critical component to our understanding of political life. Cadets in this course engage a robust theoretical literature that forms the foundation for an understanding of contemporary and future state building efforts. Throughout the course, we explore major theories of state formation such as warfare, economic development, and institutions. Next, the course adopts the view of the state and investigates a critical challenge confronting civilian and military leaders - why do some states become strong and others weak? Subsequently, we adopt the view of the civil society and examine state building dynamics from the vantage of those whom the state seeks to rule. Finally, the course examines how the state building project functions around the world through a variety of regional case studies. Cadets completing this course join a small but critical set of military officers prepared to lead and advise the next state building project.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** At discretion of current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

**Corequisite(s):** SS366

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| <b>SS487</b> | <b>INT'L POLITICAL ECONOMY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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| <b>Scope:</b> | 2020-2 | <b>Offerings:</b> |
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SS487 serves as an introduction to the complex and fascinating fundamental relationships between politics, economics, and society at the international and domestic levels. How do political institutions, actors, and policies shape international economic phenomena? And how do these economic phenomena influence political forces in turn? Though international political economy falls within the broader subfield of international relations, we will soon discover that domestic politics and economics cannot be separated from international political economy. Thus, students will explore the mechanisms that link politics and economics across a variety of issues that are central to the field of IPE, including the politics and economics of trade, economic development, international monetary affairs, and emerging challenges in the global economy.

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** At the discretion of the current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

| SS489A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SENIOR RESEARCH AMER POLITICS                         | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2020-2                                                | <b>Offerings:</b>                          |
| <p>The course provides an environment that is conducive to independent effort in a subject area of special interest to the cadet. Original research or specialized study will be accomplished in the field of American Politics. The course is conducted in three phases. First, the cadet and the individual tutor from the Social Sciences faculty will reach agreement on a subject area for research. Research methods will be studied under the direction of the faculty member. Research may involve field trips and personal interviews with experts in the area of study. In the second phase, the cadet will engage in independent research and prepare a draft analytical paper or report detailing the findings. During this period, frequent consultation with the faculty advisor occurs regarding the progress in the project. In the third phase, the cadet will present and defend the findings before a faculty committee. This course is the first in a two-part thesis program in the fall and encompasses individual projects in the spring.</p> |                                                       | 2026-1 2027-1 2028-1<br>2029-1             |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 30 @ 75 min (2.000 Att/wk)                            | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | One paper or report of variable length; oral defense. |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SS360                                                 |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SS489                                                 |                                            |
| SS489B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | SENIOR RESEARCH INTERN AFFAIRS                        | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2020-2                                                | <b>Offerings:</b>                          |
| <p>The course provides an environment that is conducive to independent effort in a subject area of special interest to the cadet. Original research or specialized study will be accomplished in the field of International Affairs. The course is conducted in three phases. First, the cadet and the individual tutor from the Social Sciences faculty will reach agreement on a subject area for research. Research methods will be studied under the direction of the faculty member. Research may involve field trips and personal interviews with experts in the area of study. In the second phase, the cadet will engage in independent research and prepare a draft analytical paper or report detailing the findings. During this period, frequent consultation with the faculty advisor occurs regarding the progress in the project. In the third phase, the cadet will present and defend the findings before a faculty committee. This course is the first in a two-part thesis program.</p>                                                           |                                                       | 2026-1 2026-8 2027-1<br>2028-1 2029-1      |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 30 @ 75 min (2.000 Att/wk)                            | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | One paper or report of variable length; oral defense. |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SS360                                                 |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SS489                                                 |                                            |
| SS489C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | RESEARCH METHODS IN ECONOMICS                         | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2021-1                                                | <b>Offerings:</b>                          |
| <p>This course is the first phase of the Senior Thesis in Economics sequence and offers cadets an opportunity to develop their research skills, through a combination of lectures and tutorial-style guided independent study, with the goal of preparing for the Senior Thesis in Economics. Cadets will develop economic research projects from the idea-phase through the literature review, data acquisition/analysis, and writing phases in an iterative process of producing intermediate drafts of project components and receiving instructor feedback. This will culminate in an initial draft of their senior thesis project. At the same time, lectures and assignments will cover various topics related to economic research, including strategies for generating research ideas, writing in economics, programming in STATA, data management strategies, and common empirical approaches for causal inference.</p>                                                                                                                                     |                                                       | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 30 @ 75 min (2.000 Att/wk)                            | <b>Labs:</b> 0 @ 0 min                     |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | One paper or report of variable length; oral defense. |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SS368                                                 |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SS489                                                 |                                            |
| SS490A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | COLLOQUIUM (AMER POLITICS)                            | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2019-2                                                | <b>Offerings:</b>                          |

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| The colloquium provides cadets an opportunity for reading and analysis in depth in a topic area of special interest and timely relevance to their concentration. The course employs the seminar approach in which the instructor meets with small groups to discuss assigned readings, and cadets present their own analyses to the group. Course directors develop topics and determine the semesters in which they will be offered. Department Academic Counselors then forward course offerings and descriptions to Social Science majors and those majoring in areas related to the colloquium topic.                                                                                                                                                                                                                                                                |                                                                                                                                                                | 2026-1 2027-2 2029-1                               |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                         |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | These will vary by topic. Typically three analytical papers of 1000-2000 words based on selected readings; class attendance adjusted to provide research time. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SS202<br>-Or-<br>SS252                                                                                                                                         |                                                    |
| <b>SS490B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>COLLOQUIUM (INTERNTL AFFAIRS)</b>                                                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020-2                                                                                                                                                         | <b>Offerings:</b>                                  |
| The colloquium provides cadets an opportunity for reading and analysis in depth in a topic area of special interest and timely relevance to their concentration. The course employs the seminar approach in which the instructor meets with small groups to discuss assigned readings, and cadets present their own analyses to the group. Course directors develop topics and determine the semesters in which they will be offered. Department Academic Counselors then forward course offerings and descriptions to Social Science majors and those majoring in areas related to the colloquium topic.                                                                                                                                                                                                                                                                |                                                                                                                                                                | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                         |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | These will vary by topic. Typically three analytical papers of 1000-2000 words based on selected readings; class attendance adjusted to provide research time. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SS307 SS366<br>-Or-<br>SS357 SS366                                                                                                                             |                                                    |
| <b>SS490C</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>COLLOQUIUM (ECONOMICS)</b>                                                                                                                                  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020-2                                                                                                                                                         | <b>Offerings:</b>                                  |
| The colloquium provides cadets an opportunity for reading and analysis in depth in a topic area of special interest and timely relevance to their concentration. The course employs the seminar approach in which the instructor meets with small groups to discuss assigned readings, and cadets present their own analyses to the group. Course directors develop topics and determine the semesters in which they will be offered. Department Academic Counselors then forward course offerings and descriptions to Social Science majors and those majoring in areas related to the colloquium topic.                                                                                                                                                                                                                                                                |                                                                                                                                                                | 2026-2                                             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                         |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | These will vary by topic. Typically three analytical papers of 1000-2000 words based on selected readings; class attendance adjusted to provide research time. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SS307<br>-Or-<br>SS357                                                                                                                                         |                                                    |
| <b>SS490E</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>SOCIAL SCIENCES COLLOQUIUM</b>                                                                                                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020-2                                                                                                                                                         | <b>Offerings:</b>                                  |
| The colloquium provides cadets an opportunity for reading and analysis in depth in a topic area of special interest and timely relevance to their concentration. The course employs the seminar approach in which the instructor meets with small groups to discuss assigned readings, and cadets present their own analyses to the group. Course directors develop topics and determine the semesters in which they will be offered. Department Academic Counselors then forward course offerings and descriptions to Social Science majors and those majoring in areas related to the colloquium topic. Topics will vary by year but recent SS490 colloquiums include: Nationalism and Ethnic Conflict; Politics and Film; the Politics of Intelligence; Politics and Government of South and Southeast Asia; Philosophy, Religion, and Terror; and Winning the Peace. |                                                                                                                                                                | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                                         |                                                    |

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| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | These will vary by topic. Typically three analytical papers of 1000-2000 words based on selected readings; class attendance adjusted to provide research time. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SS307 SS366<br>-Or-<br>SS357 SS366                                                                                                                             |                                                    |
| <b>SS491</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SENIOR PROJECT</b>                                                                                                                                          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2020-2                                                                                                                                                         | <b>Offerings:</b>                                  |
| SS491 is one of three capstone courses that cadets may choose to take in their senior year to serve as the integrative experience within the International Affairs Major. In SS491, cadets will be placed into small groups under the direction of a faculty member to complete an in-depth research project based on cadet research interests. Research projects may be oriented toward the needs and interests of an external "client" organization in the government, civilian academia, or the private sector. In addition to producing a written research report, each research group will present their projects to the West Point community and, if applicable, to the external client organization or other audiences.                                                                                |                                                                                                                                                                | 2026-2 2027-2 2028-2<br>2029-2                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                                                         |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | At the discretion of the current course director and International Affairs Program Director.                                                                   |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SS307<br>-Or-<br>SS357                                                                                                                                         |                                                    |
| <b>SS492</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>DIST PROF DEF ECON SEMINAR</b>                                                                                                                              | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2019-2                                                                                                                                                         | <b>Offerings:</b>                                  |
| This course is taught by the Bernard Rogers Distinguished Professor of Defense Economics, a scholar with a distinguished record of academic achievement and professional service in the arena of Defense Economics. This course is focused on topical issues that allow students to benefit from the specific expertise of the Rogers Chair. Students typically take part in seminar discussions, conduct research, and prepare analytical papers. Potential topics are Army procurement policy, contract design, the growth of military technology, the Department of Defense budget process, and corporate finance in the defense sector.                                                                                                                                                                   |                                                                                                                                                                | No Course Offerings                                |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                                                         |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                                                                                                                                           |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SS368 SS382 SS388                                                                                                                                              |                                                    |
| <b>SS493</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>SENIOR STUDIES - AMER POLITICS</b>                                                                                                                          | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2019-2                                                                                                                                                         | <b>Offerings:</b>                                  |
| This course provides cadets an opportunity for reading and analysis in depth in a topic area of special interest and timely relevance to their concentration. The course employs the seminar approach in which a senior faculty member meets with small groups to discuss assigned readings, and cadets present their own analyses to the group. Course directors develop topics and determine the semesters in which they will be offered. Department Academic Counselors then forward course offerings and descriptions to Social Science majors and those majoring in areas related to the senior studies topic. Topics will vary by year but recent senior studies include: Politics of Race, Gender, Sexuality and Politics, Studies in Grand Strategy, State and Local Politics, and Judicial Politics. |                                                                                                                                                                | 2026-2 2027-2 2028-2<br>2029-1                     |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                                                                                                                                         |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | One research paper (minimum length of twenty typed, double-spaced pages).                                                                                      |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SS360 SS386                                                                                                                                                    |                                                    |
| <b>SS493A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>SENIOR STUDIES - SOC SCIENCES</b>                                                                                                                           | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2020-2                                                                                                                                                         | <b>Offerings:</b>                                  |

This course provides cadets an opportunity for reading and analysis in depth in a topic area of special interest and timely relevance to their concentration. The course employs the seminar approach in which a senior faculty member meets with small groups to discuss assigned readings, and cadets present their own analyses to the group. Course directors develop topics and determine the semesters in which they will be offered. Department Academic Counselors then forward course offerings and descriptions to Social Science majors and those majoring in areas related to the senior studies topic. Topics will vary by year but recent senior studies include: Politics of Race, Gender, Sexuality and Politics, Studies in Grand Strategy, State and Local Politics, and Judicial Politics.

**Lessons:** 30 @ 75 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** One research paper (minimum length of twenty typed, double-spaced pages).

**Prerequisite(s):** SS360 SS386

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| <b>SS494</b> | <b>PRINCIPLES OF FINANCE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-2

**Offerings:**

This course applies economic principles to the financial decisions that businesses make every day, and to the capital markets in which households and firms interact. The course covers topics including project analysis using net present value techniques, risk and return of assets and projects, efficient capital markets, corporate capital structure and dividend policy, and valuation of assets. Cadets will learn methods to analyze individual projects as well as business enterprises as a whole. As the U. S. Military continues to privatize many functions, knowledge of techniques used by corporations is becoming essential for our future Army leaders. This course is the second of a two-course financial economics sequence for which SS394 ? Financial Statement Analysis is a prerequisite, except for cadets with permission of the instructor.

2026-1 2026-8 2027-1  
2027-8 2028-1 2028-8  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Financial analysis.

**Prerequisite(s):** SS201 SS394  
-Or-  
SS251 SS394

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| <b>SS495</b> | <b>SENIOR FACULTY COLLOQUIUM</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2020-2

**Offerings:**

SS495 is one of three capstone courses that cadets may choose to take in their senior year to serve as the integrative experience within the International Affairs Major. In SS495, a senior faculty member in the International Affairs Program will develop an in-depth exploration of a selected topic to present to cadets in a seminar format. As cadets study the course material and pursue an individual research project relating to the seminar theme, they will draw on and integrate the skills and knowledge learned in previous core, major, and CSC courses.

2026-2 2027-2 2028-2  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** At the discretion of the current course director.

**Prerequisite(s):** SS307  
-Or-  
SS357

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| <b>SS497</b> | <b>ISSUES IN MICROECONOMIC THEORY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-2

**Offerings:**

This course provides cadets an opportunity for reading and analysis in depth in a topic area of special interest and timely relevance to their concentration. The course employs the seminar approach in which a senior faculty member meets with small groups to discuss assigned readings, and cadets present their own analyses to the group. Course directors develop topics and determine the semesters in which they will be offered. Department Academic Counselors then forward course offerings and descriptions to Social Science majors and those majoring in areas related to the senior studies topic. Topics will vary by year but recent senior studies include: Homeland Security, Advanced Terrorism, and Environmental Economics.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** These will vary by topic. Typically three analytical papers or projects of 3000-4500 words based on selected readings; class attendance adjusted to provide research time.

**Prerequisite(s):** SS360  
-Or-  
SS368

| SS498                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SENIOR THESIS                                                                                  | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2020-2                                                                                         | <b>Offerings:</b>                          |
| <p>This course is taken in the spring term of the senior year and comprises the second and final phase of the Senior Thesis in Economics, International Affairs, or American Politics. This course is reserved for cadets participating in the thesis program for their selected major. International Affairs Majors must take SS489 in the Fall term to enroll in SS498. Economics Majors must take an extra elective (normally SS469 Econometrics II) during the Fall term to enroll in SS498. American Politics Majors must take an extra elective (normally SS480 Public Policy Making Process) during the Fall term to enroll in SS498.</p>                                                                                                                                            |                                                                                                | No Course Offerings                        |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                         |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Cadets must receive approval from their major's Program Director in order to enroll in SS498.  |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SS360 SS489                                                                                    |                                            |
| SS498A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SENIOR THESIS IN AMER POLITICS                                                                 | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2020-2                                                                                         | <b>Offerings:</b>                          |
| <p>This course is taken in the spring term of the senior year and comprises the second and final phase of the Senior Thesis in American Politics. Cadets enrolled in SS498A normally will complete SS489A Senior Research Seminar in American Politics in the fall semester of their senior year, where they will complete a prospectus, literature review, annotated bibliography, outlines, and initial draft of their senior thesis. In SS498A, students will continue work on an independent study basis with their thesis advisor and committee, conducting further research and updating drafts to produce a final written thesis product generally ranging from 30-50 pages in length. Students defend their thesis before a committee in the last two weeks of classes.</p>         |                                                                                                | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                         |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Cadets must receive approval from their major's Program Director in order to enroll in SS498A. |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SS489A<br>-Or-<br>SS489                                                                        |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SS498                                                                                          |                                            |
| SS498B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SENIOR THESIS IN INTER AFFAIRS                                                                 | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2020-2                                                                                         | <b>Offerings:</b>                          |
| <p>This course is taken in the spring term of the senior year and comprises the second and final phase of the Senior Thesis in International Affairs. Cadets enrolled in SS498B normally will complete SS489B Senior Research Seminar in International Affairs in the fall semester of their senior year, where they will complete a prospectus, literature review, annotated bibliography, outlines, and initial draft of their senior thesis. In SS498B, students will continue work on an independent study basis with their thesis advisor and committee, conducting further research and updating drafts to produce a final written thesis product generally ranging from 30-50 pages in length. Students defend their thesis before a committee in the last two weeks of classes.</p> |                                                                                                | 2026-2 2027-2 2028-2<br>2029-2             |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                         |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Cadets must receive approval from their major's Program Director in order to enroll in SS498B. |                                            |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SS489B<br>-Or-<br>SS489                                                                        |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SS498                                                                                          |                                            |
| SS498C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | SENIOR THESIS IN ECONOMICS                                                                     | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2021-1                                                                                         | <b>Offerings:</b>                          |

This course is taken in the spring term of the senior year and comprises the second and final phase of the Senior Thesis in Economics sequence. Cadets enrolled in SS498C normally will have completed SS489C Research Methods in Economics, where they will have developed a promising initial draft of their senior thesis. In SS498, students will continue work on an independent study basis with their thesis advisor and committee, conducting further research and updating drafts to produce a final written thesis product generally ranging from 25-50 pages in length. Students defend their thesis before a committee in the last two weeks of classes.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Cadets must receive approval from their major's Program Director in order to enroll in SS498C.

**Prerequisite(s):** SS489C  
-Or-  
SS489

**Disqualifier(s):** SS498

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|              |                            |                                                          |
|--------------|----------------------------|----------------------------------------------------------|
| <b>XH200</b> | <b>AMBITION AND HUBRIS</b> | <b>1.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|----------------------------|----------------------------------------------------------|

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**Scope:** 2024-2 **Offerings:**

This course considers the role of ambition in a good life, how the best-laid plans of ambitious individuals sometimes go wrong, and how leaders in all fields of human endeavor can balance ambition and humility to set goals that are both lofty and achievable, while responding properly to setbacks and failure. Students engage these and related questions through close study of a single text in a small seminar setting. The text for each year's course will be selected by the professor offering the seminar; it will in each case be a fundamental text from the professor's field that has had a decisive impact on his or her development. In the culminating assignment, students write an essay describing how, in light of the lessons they have learned from the work they have studied, they intend to improve their own character and the world around them.

**Lessons:** 15 @ 75 min (1.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Cadets must apply and be selected to enroll in this course

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|              |                |                                                          |
|--------------|----------------|----------------------------------------------------------|
| <b>XH300</b> | <b>PURPOSE</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2022-1 **Offerings:**

The purpose of XH300 (formerly XH497) is to improve cadets' ability to evaluate complex issues involving ethical judgements and choice among scarce resources, reach reasoned positions on these issues, and effectively argue their positions verbally and in writing. The process of pursuing this goal will make cadets better officers, scholars, and citizens. The course will employ several methods to assist in this pursuit. First, it will achieve breadth by focusing on current issues from a variety of fields, examining the "hard choices" that confront society, government, military leaders, and individual citizens. Among the disciplines from which the course will draw are Philosophy, Law, Political Science, Economics, Physics, Biology, and English. Each cadet will also be assigned an individual mentor from among the faculty of the Departments of Social Sciences, History, Law, or English. Requirements include a briefing on a current issue in the cadet's major field, a book review, and a personal statement summarizing academic and other goals.

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Admission to the course requires an interview and the approval of the department head.

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>XH397</b> | <b>GRAND STRATEGY FIELD STUDY</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2022-7 **Offerings:**

The Grand Strategy Field Study AIAD experience provides the opportunity to apply theories of grand strategy to real-world historical and contemporary events. Through travel to a place of grand strategic importance - whether the ruins of the Athenian Empire, the decisive battlefields of World War II, or such new arenas of Great Power competition as the Indo-Pacific - cadets will gain a deep first-hand understanding of the geopolitical, cultural, economic, and technological factors shaping grand strategy throughout history and in the present day. Scope, depth, and material covered will satisfy the requirement of a 3-credit hour course in the Grand Strategy program.

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                            |                                                          |
|--------------|----------------------------|----------------------------------------------------------|
| <b>XH407</b> | <b>VISION AND RHETORIC</b> | <b>1.5 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|----------------------------|----------------------------------------------------------|

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**Scope:** 2023-1 **Offerings:**

The purpose of XH407 Vision and Rhetoric is to build upon the foundation of critical thought established in XH300 Purpose. Cadets apply the concepts developed in XH300 to contemporary issues facing the United States Army, Department of Defense, Nation, and wider world. Cadets continue to refine their understanding of their place in the society and develop skills to convey this sense of self to others, both verbally and in writing. Cadets will continue to be advised by both Department of Social Sciences faculty and academic advisors in their home departments.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 15 @ 75 min (1.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Admission to the course requires an interview and the approval of the department head.

**Prerequisite(s):** XH497  
-Or-  
XH300

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|--------------|------------------------------|----------------------------------------------------------|
| <b>XH476</b> | <b>TUTORIAL IN AM FNDTNS</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2024-1

**Offerings:**

This tutorial employs one-on-one or small group discussions to examine selected texts from the period of the American founding utilizing diverse disciplinary perspectives. In consultation with a faculty supervisor, cadets select an area of focus (e.g., Tom Paine's Common Sense, the Declaration of Independence, the Constitution, etc.) as the basis for a tailored reading program. Areas of focus may vary year to year in accordance with student interest and faculty expertise. The course meets for approximately 750 minutes (10 meetings x 75 minutes) and concludes with an oral examination by a committee of faculty members representing different disciplines.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 10 @ 75 min (1.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Cadets must be enrolled in the American Foundations minor.

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|--------------|-----------------------------------|----------------------------------------------------------|
| <b>ZH337</b> | <b>REGIONAL POLITICAL SYSTEMS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------------|

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**Scope:** 2020-1

**Offerings:**

For cadets attending foreign military academies and academic institutions. Instruction may be in English or a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course covering the politics, societies, and structures of states in different regions of the world. The course also covers the study of the relationship between the state and society in these regions. Regions include--but are not limited to--the Middle East, East Asia, Southwest Asia, Central Asia, North Africa, South Africa, Latin America, South America, and Europe.

2026-8 2026-9 2027-8  
2027-9 2028-8 2028-9  
2029-1 2029-2 2029-8  
2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                                          |                                                          |
|--------------|------------------------------------------|----------------------------------------------------------|
| <b>ZH347</b> | <b>INT'L ORGNZTNS &amp; INSTITUTIONS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|------------------------------------------|----------------------------------------------------------|

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**Scope:** 2020-1

**Offerings:**

For cadets attending foreign military academies and academic institutions. Instruction may be in English or a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course about international regimes, international institutions, and / or international organizations and the structure, role, and relevance of these actors in the international system. In addition, course content may include material about the relationship between international organizations and institutions and states. International organizations and institutions studied may include--but are not limited to--the United Nations, NATO, the European Union, International Economic Organizations, the International Criminal Court, and the Kyoto Protocol / other Climate Change Institutions.

2026-8 2026-9 2027-8  
2027-9 2028-8 2028-9  
2029-2 2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|--------------|---------------------------------|----------------------------------------------------------|
| <b>ZH367</b> | <b>TOPICS IN MICROECONOMICS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-2

**Offerings:**



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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------|
| For cadets attending foreign military academies and academic institutions. Instruction may be in English or in a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course about topics covered in the study of microeconomics. Topics include, but are not limited to, history of economic thought, manpower and labor economics, public and social policy issues, energy and natural resource issues, gender, law, and applied microeconomic issues.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                       | 2026-8 2026-9 2027-8<br>2027-9 2028-8 2028-9<br>2029-2 2029-8 2029-9           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                  |                                                                                |
| <b>ZH377</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>TOPICS IN MACROECONOMICS</b>       | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2019-2                                | <b>Offerings:</b>                                                              |
| For cadets attending foreign military academies and academic institutions. Instruction may be in English or in a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course about topics covered in the study of macroeconomics. Topics include, but are not limited to, international trade, foreign exchange, the international monetary system, global capital markets, and globalization.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                       | 2026-8 2026-9 2027-8<br>2027-9 2028-8 2028-9<br>2029-2 2029-8 2029-9           |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                  |                                                                                |
| <b>ZH407</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>TOPICS/AMERICAN FOREIGN POLICY</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2020-1                                | <b>Offerings:</b>                                                              |
| For cadets attending foreign military academies and academic institutions. Instruction may be in English or a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course covering the development, implementation, and consequences of American foreign policy. It analyzes the actors who make American foreign policy, concentrating both on government sources such as the President, Congress, and the foreign policy bureaucracy, as well as external sources such as public opinion, interest groups, and the media. Topics include - but are not limited to - U.S. relations with China, Russia, and the European Union, energy politics, the Arab-Israeli crisis, weapons of mass destruction and rogue states, terrorism, democracy promotion, and the global response to US foreign policy. In exploring each of these current challenges and dilemmas, this course attempts to understand the policies and strategies the U.S. utilizes to secure its interests and achieve its objectives. |                                       | No Course Offerings                                                            |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                  |                                                                                |
| <b>ZH427</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>TOPICS IN COMPARATIVE POLITICS</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2020-1                                | <b>Offerings:</b>                                                              |
| For cadets attending foreign military academies and academic institutions. Instruction may be in English or a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course covering the history and development of state social structures, political cultures, and systems and structures of government. Topics include, but are not limited to, democratization, regional anthropology, and conflict resolution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 2026-8 2026-9 2027-8<br>2027-9 2028-8 2028-9<br>2029-1 2029-2 2029-8<br>2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | None                                  |                                                                                |
| <b>ZH447</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>TOPICS: INTERNATIONAL POLITICS</b> | <b>3.0 Credit Hours</b><br>(BS=0.0,ET=0.0,MA=0.0)                              |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2020-1                                | <b>Offerings:</b>                                                              |
| For cadets attending foreign military academies and academic institutions. Instruction may be in English or in a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course about topics covered in the study of international relations. Topics include--but are not limited to--international security studies, international political economy, economic development, and the history of the development of modern international relations and the international system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                       | 2026-8 2026-9 2027-8<br>2027-9 2028-8 2028-9<br>2029-1 2029-2 2029-8<br>2029-9 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Labs:</b> 0 @ 0 min                |                                                                                |

**Special Requirements:** None

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|--------------|---------------------------------------|----------------------------------------------------|
| <b>ZH467</b> | <b>TOPICS-INTERNATIONAL ECONOMICS</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
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**Scope:** 2019-2

**Offerings:**

For cadets attending foreign military academies and academic institutions. Instruction may be in English or a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course about international economic systems, international institutions, and / or international organizations and the structure, role, and relevance of these actors in the global economic system. In addition, course content may include material about the relationship between international organizations and institutions and states. International organizations and institutions studied may include - but are not limited to - the United Nations, World Bank, International Monetary Fund, the European Union, World Trade Organization, the Bretton Woods system and International Financial Organizations. Topics include - but are not limited to - international political economy, economic development, regional economics, and the history of the modern international economic system.

2026-8 2026-9 2027-8  
2027-9 2028-8 2028-9  
2029-2 2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                                      |                                                    |
|--------------|--------------------------------------|----------------------------------------------------|
| <b>ZH477</b> | <b>TOPICS-INT'L BUSINESS/FINANCE</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------|

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**Scope:** 2019-2

**Offerings:**

For cadets attending foreign military academies and academic institutions. Instruction may be in English or a foreign language. Cadets will attend classes and produce papers and other academic work as required by the course instructor and the institution's academic requirements. This class serves as the equivalent to a foreign course about international and foreign financial systems, international accounting and foreign business practices. In addition, course content may include material about the relationship between businesses, institutions and states in foreign countries. Topics include--but are not limited to--corporate finance, financial statements and accounting, currency issues, central banking, and commercial and retail banking.

2026-8 2026-9 2027-8  
2027-9 2028-8 2028-9  
2029-2 2029-8 2029-9

**Lessons:** 30 @ 75 min (2.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** None

## Systems Engineering

### 26 Courses

| EM381                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ENGINEERING ECONOMY                                                                                                                          | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2017-1                                                                                                                                       | <b>Offerings:</b>                                                                                                    |
| <p>This course prepares cadets to consider the economic dimension in the evaluation of engineering alternatives; a consideration vital to the Systems Decision Process, engineering management, systems acquisition and many other application areas. While emphasis is on the analytical consideration of money and its impact on the areas above, the course also incorporates professional ethics in the engineering economic analysis process. The course is taught in four lesson blocks. The Time Value of Money (TVM) block -includes the quantitative methods for economic analysis of engineering alternatives by introducing cost concepts, interest concepts, the cash flow diagram and developing interest formulas. The Analysis Methods block develops techniques for project evaluation and comparison and ways to account for risk and uncertainty. The After Tax Cash Flow block incorporates the real-world effect of taxes, depreciation and inflation into the analysis methods. The Capital Budgeting block completes a comprehensive introduction to engineering economy by introducing the concept of economic service life and project financing. A one lesson introduction to personal finance is included to demonstrate how many of the concepts used in the business world can also be applied for personal planning. Course concepts are applied using Excel in both graded and ungraded labs. Cadets will spend several lessons in a computer lab environment.</p> |                                                                                                                                              | 2026-1 2026-2 2026-3<br>2027-1 2027-2 2028-1<br>2028-2 2029-1 2029-2                                                 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                                                                       |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                                                                                         |                                                                                                                      |
| EM384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ANYL METH FOR ENGR MANAGEMENT                                                                                                                | 3.0 Credit Hours<br>(BS=0.0,ET=3.0,MA=0.0)                                                                           |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2024-1                                                                                                                                       | <b>Offerings:</b>                                                                                                    |
| <p>EM384 focuses on the application of deterministic and probabilistic models used by analysts to make engineering and management decisions. Cadets learn to apply various modeling techniques to represent and solve real-world organizational problems in the military and industry. Topics include: linear and integer programming, network modeling, decision making under uncertainty, and simulation modeling. Cadets apply concepts and tools using Microsoft Excel in a variety of computing environments. The techniques taught in this course have been applied to an increasingly wide variety of complex problems in business, government, military, health care, and education. Ethical responsibilities in describing the results of analyses to decision makers are integrated throughout the course. Cadets make innovative use of spreadsheets to develop and analyze models. Cadets are tested on the application of course concepts from the five blocks of instruction during four graded labs, four problem sets, and a comprehensive term-end exam.</p>                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                              | 2026-1 2026-2 2026-8<br>2027-1 2027-2 2027-3<br>2027-8 2028-1 2028-2<br>2028-3 2028-8 2029-1<br>2029-2 2029-4 2029-8 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Labs:</b> 0 @ 0 min                                                                                                                       |                                                                                                                      |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                                                                                         |                                                                                                                      |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | -Or-<br>-Or-<br>-Or-<br>-Or-<br>CY105<br>-Or-<br>CY155<br>-Or-<br>IT105<br>-Or-<br>IT155<br>-Or-<br>AI105<br>-Or-<br>AI105X<br>-Or-<br>AI155 |                                                                                                                      |

**Corequisite(s):** MA206  
-Or-  
MA256  
-Or-  
MA206X  
-Or-  
MA206Y

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|--------------|---------------------------|----------------------------------------------------------|
| <b>EM411</b> | <b>PROJECT MANAGEMENT</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
|--------------|---------------------------|----------------------------------------------------------|

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**Scope:** 2019-2

**Offerings:**

This course develops skills required to lead an organization to the achievement of their objectives through the proper application of the management of planning, implementing and controlling the organization activities, personnel and resources. The course focuses on the Implementation phase of the Systems Decision Process (SDP). Topics include project selection, roles and responsibilities of the project manager, planning the project, budgeting the project, scheduling the project, allocating resources to the project, monitoring and controlling the project, evaluating and terminating the project, risk assessment and management, organizational structure and human resources. Case studies illustrate problems and how to solve them. Course assignments are designed to help students learn and apply project management techniques taught in the course. The class design project will provide students with the opportunity to integrate project management software, Microsoft Project, into the preparation of an Engineering Management Project Plan.

2026-1 2026-2 2026-8  
2027-1 2027-2 2027-8  
2027-9 2028-1 2028-2  
2028-8 2028-9 2029-1  
2029-2 2029-8 2029-9

**Lessons:** 35 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

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|              |                                   |                                                          |
|--------------|-----------------------------------|----------------------------------------------------------|
| <b>EM420</b> | <b>PRODUCTION OPERATIONS MGMT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|-----------------------------------|----------------------------------------------------------|

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**Scope:** 2020-1

**Offerings:**

This course deals with the quantitative aspects of design and analysis of production operations management. Emphasis is on identification, analysis, and solution of production problems using applied quantitative techniques. Practical exercises reinforce the problem-solving techniques necessary for today's successful military and civilian engineering managers and systems engineers. Specific methods and techniques taught and applied are operations strategy, product design and selection, total quality management, capacity planning, facility location, facility layout, work system design, lean systems and scheduling. This course is required for those pursuing the Engineering Management major and an elective for the Systems Engineering, Systems and Decision Sciences, Management and other engineering majors.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MA206  
-Or-  
MA256  
-Or-  
MA206X

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|--------------|---------------------------|----------------------------------------------------------|
| <b>EM481</b> | <b>SYSTEMS SIMULATION</b> | <b>3.5 Credit Hours</b><br><b>(BS=0.0,ET=3.5,MA=0.0)</b> |
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**Scope:** 2020-2

**Offerings:**

Cadets learn and explore discrete event simulation techniques and tools used to analyze and improve complex systems. Applications include operations, transportation, manufacturing and logistics systems. Topics include functional modeling with functional flow diagrams, simulation theory, the modeling process, input data analysis, generation of random numbers, verification and validation of simulation models, experimental design, output analysis, and application using simulation software. The course concepts provide cadets the tools to evaluate military and civilian systems. Emphasis is placed on using simulation in the Systems Decision Process (SDP). Cadets demonstrate proficiency and develop communication skills through design projects and briefings. Cadets spend several lessons in a computer lab environment.

2026-1 2026-2 2026-8  
2027-1 2027-2 2027-8  
2028-1 2028-2 2028-8  
2029-1 2029-2 2029-8

**Lessons:** 35 @ 75 min (2.500 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** In-progress reviews and course project; compensatory time provided.

**Prerequisite(s):** MA206  
-Or-  
MA256  
-Or-  
MA206X

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SE481                                   |                                                                                                                               |
| <b>EM482</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>SUPPLY CHAIN ENG &amp; INFO MGMT</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2020-2                                  | <b>Offerings:</b>                                                                                                             |
| <p>This course teaches cadets the strategic importance of supply chain design, planning, operation, business processes, and information management systems. Cadets will become familiar with engineering a supply chain network---from conducting inventory management to establishing proper sourcing and transportation strategies to understanding capacity and facility locations to constructing the proper information technology framework needed to be successful. Cadets will develop the ability to evaluate how information flows can be a substitute for the stock of physical resources. Additionally, cadets will understand why such information systems succeed or fail through the explanation of concepts, insights, practical tools and the information technology that supports decision making. This course will focus on understanding the key drivers of a supply chain such as inventory, facilities, transportation, sourcing, pricing, and information. Cadets will also learn to assess the impact of strategic alliances and globalization on supply chain strategies and best practices, to include smart pricing, customer value, and new product and supply chain design.</p>                                                                                                                                            |                                         | 2026-2 2027-2 2028-2 2029-2                                                                                                   |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 30 @ 75 min (2.000 Att/wk)              | <b>Labs:</b> 0 @ 0 min                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                                    |                                                                                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SM482                                   |                                                                                                                               |
| <b>SE301</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>FNDTN ENGIN DSGN &amp; SYS MGMT</b>  | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2025-1                                  | <b>Offerings:</b>                                                                                                             |
| <p>SE301 serves as the "roadmap" course for all cadets taking the Engineering Management, Systems Engineering, or Systems &amp; Decision Sciences majors as well as all cadets enrolled in the Core Engineering Sequence. This course presents the methodological framework and techniques for designing, implementing, managing and reengineering complex systems or processes. Cadets learn engineering design and engineering management processes and gain an appreciation for future environments and system life-cycles. Cadets analyze case studies and complete practice problems to illustrate mastery of course topics. Cadets also use spreadsheet software for modeling and analyzing design alternatives. SE301 introduces a Systems Decision Process while incorporating material from courses in the USMA core curriculum and also previews the modeling and decision making tools that cadets will learn in follow-on Department of Systems Engineering courses. The course is designed to allow Cadets the opportunity to learn engineering design and engineering management processes on an individual level so that each Cadet will have the experience necessary to succeed in future Systems Engineering courses. Cadets will spend a number of lessons in a computer lab environment.</p>                                        |                                         | 2026-1 2026-2 2026-5 2026-8 2027-1 2027-2 2027-5 2027-7 2027-8 2028-1 2028-2 2028-5 2028-7 2028-8 2029-1 2029-2 2029-5 2029-8 |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 30 @ 75 min (2.000 Att/wk)              | <b>Labs:</b> 0 @ 0 min                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                                    |                                                                                                                               |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SE300                                   |                                                                                                                               |
| <b>SE302</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>FUNDAMENTALS OF SYSTEMS ENG</b>      | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                                                                            |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2020-2                                  | <b>Offerings:</b>                                                                                                             |
| <p>SE302 prepares students to effectively model, analyze, and understand complex, interdisciplinary, and ill-defined problems as systems so they can design and implement effective solutions. The course covers principles and methods for requirements management, functional and nonfunctional analysis, risk management, testing, cost estimation, and technical system architecture from industry and DoD including IDEF0 modeling, the Systems Modeling Language (SysML), and the Department of Defense Architectural Framework (DoDAF). The Engineering Vee model is used as a framework throughout the course. The course also includes a review of Model-Based Systems Engineering (MBSE) methodologies. The techniques taught in this course have been applied to an increasingly wide variety of complex, ill-defined problems in business, government, military, health care, and national capacity development. Ethical responsibilities in describing the results of analyses to decision makers are integrated throughout the course. Cadets develop communication skills through written reports and presentations. A course project will challenge cadets to apply their modeling and analysis skills to a real world complex, ill-defined problem in political, military, economic, social, cultural, and informational contexts.</p> |                                         | 2026-1 2027-1 2028-1 2029-1                                                                                                   |
| <b>Lessons:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 40 @ 55 min (2.500 Att/wk)              | <b>Labs:</b> 0 @ 0 min                                                                                                        |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | None                                    |                                                                                                                               |
| <b>SE370</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>COMPUTER AIDED SYSTEMS ENG</b>       | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                                                                            |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2020-2                                                                                                                          | <b>Offerings:</b>                                                              |
| Cadets learn how to use information and technology in support of systems decision-making. They learn how to manipulate data in spreadsheets as well as through a computer language (R) to support decisions. The course introduces cadets to several types of exploratory data analysis to include numeric, text, relational, and geospatial. Cadets learn how to manage analysis in both the traditional and cloud environments, as well as the tradeoffs associated with working in each. Communication skills are developed through both written and oral projects and development of interactive graphical presentations. Cadets will spend most lessons in a computer lab environment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                 | 2026-2 2026-8 2027-2<br>2027-9 2028-2 2028-9<br>2029-1 2029-2 2029-8<br>2029-9 |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min                                                                                                          |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Two design projects.                                                                                                            |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | IT105<br>-Or-<br>IT155<br>-Or-<br>IT105X<br>-Or-<br>CY105<br>-Or-<br>CY155<br>-Or-<br>AI105<br>-Or-<br>AI105X<br>-Or-<br>AI105X |                                                                                |
| <b>SE375</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>STATISTICS FOR ENGINEERS</b>                                                                                                 | <b>3.0 Credit Hours<br/>(BS=0.0,ET=2.0,MA=1.0)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2014-1                                                                                                                          | <b>Offerings:</b>                                                              |
| This course is an integral part of the Systems Engineering major that emphasizes both the statistical analyses of data and a statistical methodology important to systems analysis and design. The over-arching course goal is to develop cadets into critical consumers and providers of statistical information as it relates to the techniques, activities, and modeling applications that typify systems engineering concerns.. The course builds on the core probability and statistics course and introduces statistics applications fundamental to the design and analysis of simulations and engineering systems. Specific topics include point and interval estimation, parametric and non-parametric tests of hypotheses, analysis of variance, linear regression, and survey design of experiments, specifically analysis of power and determination of sample size. The course emphasizes the importance of knowing and understanding the assumptions associated with the use of inferential statistics as well as the usefulness of statistical software packages. The basic principles learned in this course will facilitate data analysis in support of Army acquisition and system redesign decision-making. Ethical implications in the analysis and presentation of experimental results, as well as interactions with decision makers, are addressed. |                                                                                                                                 | 2026-1 2027-1 2027-8<br>2028-1 2028-8 2029-1<br>2029-2                         |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min                                                                                                          |                                                                                |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                                                                                                            |                                                                                |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MA206<br>-Or-<br>MA256<br>-Or-<br>MA206X                                                                                        |                                                                                |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MA376                                                                                                                           |                                                                                |
| <b>SE385</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>DECISION ANALYSIS</b>                                                                                                        | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b>                             |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2020-2                                                                                                                          | <b>Offerings:</b>                                                              |
| The course presents basic techniques of decision-making concentrating on both theoretical and modeling aspects. This course develops innovative systems engineers who can integrate the art and science of decision making for single and multiple objective environments to support the Decision Making phase of the Systems Decision Process (SDP). The focus of the course is modeling problem structure, uncertainty, risk and preference in the context of decision-making. Topics include mathematical foundations and axioms of decision analysis, influence diagrams, decision trees, simulation, sensitivity analysis, subjective probability assessments, value of information, classification of risk attitude given uncertainty. Cadets will also use value focused thinking to support decisions in multiple objective and resource allocation environments. A series of several computer laboratory exercises provides a key bridge between the mathematical theory and the application of skills to open-ended decision problems. Communication skills are developed with both written reports and oral presentations.                                                                                                                                                                                                                                     |                                                                                                                                 | 2026-2 2026-8 2027-2<br>2027-8 2028-2 2028-8<br>2029-1 2029-2 2029-8           |
| <b>Lessons:</b> 40 @ 55 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Labs:</b> 0 @ 0 min                                                                                                          |                                                                                |

**Special Requirements:** Course design project; compensatory time provided.

**Prerequisite(s):** MA206  
-Or-  
MA256  
-Or-  
MA206X

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|--------------|-----------------------------|----------------------------------------------------------|
| <b>SE387</b> | <b>DETERMINISTIC MODELS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=2.5,MA=0.5)</b> |
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**Scope:** 2020-2

**Offerings:**

This course is the first of a two-course sequence that emphasizes modeling and analysis of real-world systems. This course focuses on techniques without consideration of uncertainty or probabilistic effects. The course introduces modeling concepts, scale and measurement principles, mathematical analysis methods, and computational approaches used by systems engineers, operations researchers, and management professionals to support systems design and the Systems Decision Process (SDP). Emphasis is placed on creative application of these topics including system representations, problem formulation, solution methods, analysis of results, and interpretation. Topics include axiomatic design, modularity, value function creation, linear programming, sensitivity analysis, networks, transportation models, and human-systems integration assessment. Cadets will spend several lessons in a computer lab environment.

2026-1 2027-1 2027-8  
2028-1 2028-8 2029-1  
2029-2

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** IT105  
-Or-  
IT155  
-Or-  
CS105  
-Or-  
CS155  
-Or-  
IT105X  
-Or-  
CY105  
-Or-  
CY155  
-Or-  
AI105  
-Or-  
AI105X  
-Or-  
AI155

**Disqualifier(s):** EM384

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|--------------|--------------------------|----------------------------------------------------------|
| <b>SE388</b> | <b>STOCHASTIC MODELS</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=2.5,MA=0.5)</b> |
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**Scope:** 2020-2

**Offerings:**

This course is the second of a two-course sequence that emphasizes modeling and analysis of real-world systems. Building on the systems modeling approaches introduced in SE387, this course introduces uncertainty into design, modeling, parameter estimations, and data as they effect many of the classical stochastic models used by systems engineers, operations researchers and management professionals to capture and describe quantitative effects of uncertainty on systems design and analysis, and on decision-making as part of the Systems Decision Process (SDP). Topics include stochastic value modeling, law of averages, reliability, realization analysis, Bayesian updating, conditional probability models, and simulation. This course prepares cadets for the quantitative reasoning and analysis techniques required in follow-on courses, including SE481, EM484, SE485 and SE402/403. Cadets will spend several lessons in a computer lab environment.

2026-2 2026-8 2027-2  
2028-2 2029-1 2029-2  
2029-8

**Lessons:** 30 @ 75 min (2.000 Att/wk)    **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** MA206  
-Or-  
MA256  
-Or-  
MA206X

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|--------------|---------------------------------------|----------------------------------------------------------|
| <b>SE400</b> | <b>PROFESSIONAL ENGINEERING SEMIN</b> | <b>1.0 Credit Hours</b><br><b>(BS=0.0,ET=1.0,MA=0.0)</b> |
|--------------|---------------------------------------|----------------------------------------------------------|

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| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2020-2                                                                                                                                                                                                               | <b>Offerings:</b>                                  |
| This seminar course for SE and EM majors meets to discuss important aspects of the engineering profession to include engineering ethics, licensing procedures, and professional development plans. The seminar also includes presentations by guest lecturers from the military, DoD industrial base, and academic communities. Further, this seminar helps prepare Cadets for the Fundamentals of Engineering Exam.                                                                                                                                                                                                                                         |                                                                                                                                                                                                                      | 2026-2 2027-2 2028-2 2029-2                        |
| <b>Lessons:</b> 6 @ 120 min (0.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                               |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                                                                                                                                                 |                                                    |
| <b>SE402</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>SYSTEMS DESIGN &amp; MANAGEMENT I</b>                                                                                                                                                                             | <b>4.0 Credit Hours<br/>(BS=0.0,ET=4.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-1                                                                                                                                                                                                               | <b>Offerings:</b>                                  |
| Systems Design and Management I is the first course in a two-semester capstone experience for Systems Engineering, Engineering Management, Systems and Decision Sciences, and Operations Research majors. SE402 integrates the principles, concepts and models explored in previous core and engineering topic courses. The course applies the principles of systems design, engineering management, and/or reengineering to a real-world system. Cadets work under the supervision of a faculty member to address a problem presented by a real-world client, providing them an integrative experience for their education in engineering design.           |                                                                                                                                                                                                                      | 2029-1                                             |
| <b>Lessons:</b> 30 @ 160 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                               |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                                                                                                                                                 |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -Or-<br>EM384<br>-Or-<br>SE388<br>-Or-<br>SE301                                                                                                                                                                      |                                                    |
| <b>SE402</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>SYSTEMS DESIGN &amp; MANAGEMENT I</b>                                                                                                                                                                             | <b>4.0 Credit Hours<br/>(BS=0.0,ET=4.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2026-1                                                                                                                                                                                                               | <b>Offerings:</b>                                  |
| Systems Design and Management I is the first course in a two-semester capstone experience for Systems Engineering, Engineering Management, Systems and Decision Sciences, and Operations Research majors. SE402 integrates the principles, concepts and models explored in previous core and engineering topic courses. The course applies the principles of systems design, engineering management, and/or reengineering to a real-world system. Cadets work under the supervision of a faculty member to address a problem presented by a real-world client, providing them an integrative experience for their education in engineering design.           |                                                                                                                                                                                                                      | 2026-1 2027-1 2027-8 2028-1 2028-8                 |
| <b>Lessons:</b> 40 @ 160 min (2.500 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                               |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | This class will meet with a 1:2 ratio of out-of-class to in-class work. This is to facilitate dedicated capstone project work. Instructors will provide course drops if out-of-class expectations exceed this ratio. |                                                    |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -Or-<br>EM384<br>-Or-<br>SE388<br>-Or-<br>SE301                                                                                                                                                                      |                                                    |
| <b>SE403</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>SYSTEMS DESIGN &amp; MANAGEMENT II</b>                                                                                                                                                                            | <b>4.0 Credit Hours<br/>(BS=0.0,ET=4.0,MA=0.0)</b> |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025-2                                                                                                                                                                                                               | <b>Offerings:</b>                                  |
| Systems Design and Management II is the second course in a two-semester capstone experience for Systems Engineering, Engineering Management, Systems and Decision Sciences, and Operations Research majors. SE403 integrates the principles, concepts and models explored in previous core and engineering courses. The course applies the principles of systems design, engineering management, and/or reengineering to a real-world system of direct concern to a real-world client. Cadets work under the supervision of a faculty member to continue work on the same project begun in SE402, culminating the integrative experience in their education. |                                                                                                                                                                                                                      | 2029-2                                             |
| <b>Lessons:</b> 30 @ 160 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                               |                                                    |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                                                                                                                                                 |                                                    |



**Prerequisite(s):**

-Or-  
SE402

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|              |                                           |                                                    |
|--------------|-------------------------------------------|----------------------------------------------------|
| <b>SE403</b> | <b>SYSTEMS DESIGN &amp; MANAGEMENT II</b> | <b>4.0 Credit Hours<br/>(BS=0.0,ET=4.0,MA=0.0)</b> |
|--------------|-------------------------------------------|----------------------------------------------------|

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**Scope:** 2026-2

**Offerings:**

2026-2 2027-2 2028-2

Systems Design and Management II is the second course in a two-semester capstone experience for Systems Engineering, Engineering Management, Systems and Decision Sciences, and Operations Research majors. SE403 integrates the principles, concepts and models explored in previous core and engineering courses. The course applies the principles of systems design, engineering management, and/or reengineering to a real-world system of direct concern to a real-world client. Cadets work under the supervision of a faculty member to continue work on the same project begun in SE402, culminating the integrative experience in their education.

**Lessons:** 40 @ 160 min (2.500 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** This class will meet with a 1:2 ratio of out-of-class to in-class work. This is to facilitate dedicated capstone project work. Instructors will provide course drops if out-of-class expectations exceed this ratio.

**Prerequisite(s):**

-Or-  
SE402

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|               |                            |                                                    |
|---------------|----------------------------|----------------------------------------------------|
| <b>SE440X</b> | <b>SYSTEM ARCHITECTURE</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
|---------------|----------------------------|----------------------------------------------------|

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**Scope:** 2025-2

**Offerings:**

2026-2

SE440 focuses on preparing students to effectively model, analyze, and understand complex systems in an effort to design and implement effective solutions. The course covers principles and methods for Model-Based Systems Engineering (MBSE) and Digital Engineering. The course will apply tools for technical system architecting from industry and DoD including the Systems Modeling Language (SysML) modeling and the Department of Defense Architectural Framework (DoDAF). Cadets apply concepts and tools using advanced modeling software which includes Enterprise Architect. Ethical responsibilities in describing the results of analyses to decision makers are integrated throughout the course. Cadets develop communication skills through two written reports and make innovative use of modeling packages to develop and analyze systems. A course project will challenge cadets to apply their modeling and analysis skills to a real-world complex system. \*\*\*Pending ABET-PEV review of ET=3.0CR\*\*\*

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** SE302

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|              |                                       |                                                    |
|--------------|---------------------------------------|----------------------------------------------------|
| <b>SE450</b> | <b>APPLIED SYS DSGN/DECISN MAKING</b> | <b>3.0 Credit Hours<br/>(BS=0.0,ET=3.0,MA=0.0)</b> |
|--------------|---------------------------------------|----------------------------------------------------|

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**Scope:** 2025-2

**Offerings:**

2026-1 2026-2 2027-1  
2027-2 2028-1 2028-2  
2029-1 2029-2

This course is the third course of the three-course systems engineering sequence. The course serves as the culminating systems engineering experience for non-engineering cadets and integrates the principles, concepts, and models explored in previous courses. Cadets apply the Systems Decision Process to devise problem solutions that are effective and adaptable. Cadets work in groups to complete a culminating engineering design experience involving the solution of an incompletely defined problem with no single correct answer. Cadets must consider the economic, political, social and ethical constraints of the system and use creativity to generate potential design alternatives. Cadet groups will use models to analyze the alternative solutions and make a recommendation based on economic analysis and system performance. The course requires assessment of the recommended solution and a written plan for implementation.

**Lessons:** 30 @ 75 min (2.000 Att/wk) **Labs:** 0 @ 0 min

**Special Requirements:** None

**Prerequisite(s):** SE350  
-Or-  
EM384 SE301

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|              |                        |                                                    |
|--------------|------------------------|----------------------------------------------------|
| <b>SE485</b> | <b>COMBAT MODELING</b> | <b>3.5 Credit Hours<br/>(BS=0.0,ET=3.5,MA=0.0)</b> |
|--------------|------------------------|----------------------------------------------------|

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**Scope:** 2020-2

**Offerings:**

This course explores the theoretical and practical issues in combat modeling and simulation - the study of combat systems, tactics, and the battlefield environment in conflicts between opposing forces. The course focuses on models and algorithms used in state-of-the-art combat simulations, and techniques for analyzing their effects. Major topics of investigation include the model development process, combat attrition models, combat methodologies, analysis of human performance data, verification and validation, and measures of effectiveness. Cadets learn to manipulate 3D visual and system characteristic databases to build and test virtual prototypes of new combat system designs. Application of design of experiments and statistical analysis methods assist cadets in assessing the effectiveness of weapons systems, doctrine, and tactics on the future battlefield. The cadet can apply the concepts learned in this course to evaluate potential new Army combat systems, force structures, or doctrinal changes. The techniques taught in this course are a significant part of the Systems Decision Process (SDP) as they encourage creative and independent thought that applies mathematical, physical, and computer sciences to solve future technological problems. Ethical implications in the development and use of combat models also are discussed.

2026-1 2027-1 2028-1  
2029-1

**Lessons:** 35 @ 75 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** Course projects will require significant (about 18 hours) of out of class time.

**Prerequisite(s):** EM384  
-Or-  
MA376  
-Or-  
SE375

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|              |                                       |                                                          |
|--------------|---------------------------------------|----------------------------------------------------------|
| <b>SE489</b> | <b>AD IND STY IN SYS ENG/ENG MGMT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------------------|----------------------------------------------------------|

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**Scope:** 2008-1

**Offerings:**

This is a tutorial course in which an individual cadet or a group of cadets study in depth an advanced topic in systems engineering or engineering management under the direct mentorship of a faculty advisor. The scope of the course is tailored to the desires of the cadet(s) in consultation with a faculty advisor. Cadets will coordinate with a faculty mentor who has an interest and background in the research area and who will assist in scoping and developing course content. Communication skills are developed and assessed through both written reports and oral presentations.

2026-1 2026-2 2026-7  
2026-9 2027-1 2027-2  
2028-1 2028-2 2029-1  
2029-2

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** As determined by faculty advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

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|              |                                      |                                                          |
|--------------|--------------------------------------|----------------------------------------------------------|
| <b>SE490</b> | <b>AD TOPICS IN SYS ENG/ENG MGMT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|--------------------------------------|----------------------------------------------------------|

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**Scope:** 2008-2

**Offerings:**

This course provides in-depth study of a special topic or topics in systems engineering or engineering management not offered elsewhere in the USMA curriculum. This course is intended to broaden a cadet's or group of cadets' exposure to the systems engineering or engineering management discipline. The Department of Systems Engineering visiting professor or senior faculty member assigned to the course is responsible for developing the course topic or topics and advertising the course to prospective cadets.

2026-2 2027-1 2027-2  
2028-1 2028-2 2029-1  
2029-2

**Lessons:** 40 @ 55 min (2.500 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** To be announced. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

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|              |                                       |                                                          |
|--------------|---------------------------------------|----------------------------------------------------------|
| <b>SE491</b> | <b>RSRCH PROJ IN SYS ENG/ENG MGMT</b> | <b>3.0 Credit Hours</b><br><b>(BS=0.0,ET=0.0,MA=0.0)</b> |
|--------------|---------------------------------------|----------------------------------------------------------|

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**Scope:** 2008-1

**Offerings:**

The cadet, or cadet team, integrates the concepts and techniques learned in previous Systems Engineering or Engineering Management courses to solve a current problem of interest to the Academy, the Department of the Army, or other agencies in the Department of Defense. Subject to approval from the course and program directors, cadets may select project topics which are follow-on research from their summer AIAD experience, a topic of interest to them, or one that is compatible with on-going research within the Department of Systems Engineering and/or the Operations Research Center of Excellence. Cadets will coordinate with a faculty mentor who has an interest and background in the research area and who will assist in scoping the project and directing the research effort. Cadets may work individually or in small teams, depending on the nature of the research. The course will culminate with a student presentation and a written report.

2026-1 2026-2 2027-1  
2027-2 2027-8 2028-1  
2028-2 2028-8 2029-1  
2029-2 2029-8

**Lessons:** 0 @ 0 min (0.000 Att/wk)      **Labs:** 0 @ 0 min

**Special Requirements:** As determined by Faculty Advisor. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required.

| SE492                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | IND ADV DEVELOPMENT COURSE                                                                                                               | 1.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2001-4                                                                                                                                   | <b>Offerings:</b>                          |
| This course offers the opportunity for cadets to receive academic credit for study and/or work completed during the Academic Individual Advanced Development (AIAD) program. The content of the course and the nature of academic credit will be determined by the Head of Department in consultation with the cadet and the summer host agency. Communication skills are developed with both written reports and oral presentations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                          | No Course Offerings                        |
| <b>Lessons:</b> 0 @ 0 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Labs:</b> 0 @ 0 min                                                                                                                   |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | To be announced. The program director will determine the applicable amount MA, BS and ET credit for each course offering, when required. |                                            |
| SM484                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SYSTEM DYNAMICS SIMULATION                                                                                                               | 3.5 Credit Hours<br>(BS=0.0,ET=3.5,MA=0.0) |
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2020-1                                                                                                                                   | <b>Offerings:</b>                          |
| Simulation modeling can be used to study the effects of changes to existing systems or processes, or evaluate the performance of new systems prior to their implementation. The techniques taught in this course are a significant part of the Systems Decision Process (SDP) as they introduce the concept of dynamic systems thinking and analysis. By their nature, large scale systems are dynamic. These systems involve complex cause and effect relationships that form feedback loops between the variables of interest. These systems produce outcomes that are not always intuitive. The cadets use the properties of dynamic systems and analytical techniques to design continuous models of complex systems or processes, implement these models, and perform an analysis of the results. Topics include applications of System Dynamics, client/modeler relationships, problem articulation, functional modeling through causal loop diagrams and stock and flow diagrams, modeling and simulation in a PC-based continuous event simulation package, policy design, policy testing, and policy implementation. These concepts and principles are applied to military and civilian applications such as physical systems, human decision processes, population, and economic/business processes. Cadets develop communication skills by presenting their design results in both written reports and oral presentations. The course also addresses ethical implications in the development and application of dynamic models as well as interactions with decision makers. Cadets will spend several lessons in a computer lab environment. |                                                                                                                                          | 2026-1 2027-1 2028-1<br>2029-1             |
| <b>Lessons:</b> 35 @ 75 min (0.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Labs:</b> 0 @ 0 min                                                                                                                   |                                            |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Course design project.                                                                                                                   |                                            |
| <b>Disqualifier(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | EM484                                                                                                                                    |                                            |

# William E. Simon Center for the Professional Military Ethic

## 1 Courses

| MX400                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | OFFICERSHIP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3.0 Credit Hours<br>(BS=0.0,ET=0.0,MA=0.0)                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>Scope:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2023-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Offerings:</b>                                             |
| <p>MX400 is the Superintendent's capstone course. With its emphasis on each officer's duty to provide moral leadership, MX400 challenges cadets to become commissioned leaders of character who demonstrate virtue, honor, patriotism, and subordination to civilian authority. In MX400, cadets look both to the past and the future, reflecting on their own character-development experiences as part of the West Point Leader Development System (WPLDS), while also studying the enduring and emerging ethical challenges of the profession they are about to enter. MX400 empowers cadets to understand and embrace their ongoing development as leaders of character and their emerging identities as Army officers. Upon completion of MX400, each cadet should internalize their professional identity and feel confident in his or her ability to apply critical thinking in complex situations and present interdisciplinary solutions to complex military problems. MX400 supports the West Point Writing Program, as one of the two Writing-in-the-Profession courses, with the Pershing Reflective Essay.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2026-1 2026-2 2027-1<br>2027-2 2028-1 2028-2<br>2029-1 2029-2 |
| <b>Lessons:</b> 30 @ 75 min (2.000 Att/wk)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Labs:</b> 0 @ 0 min                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                               |
| <b>Special Requirements:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MX400 is limited to First Class Cadets who have completed PY201, SS201, SS202, and MS300 and have completed or are concurrently enrolled in PL300/PL350 and HI302.                                                                                                                                                                                                                                                                                                                                                                                                       |                                                               |
| <b>Prerequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MS300 PY201 SS201 SS202<br>-Or-<br>MS300 PY201 SS201 SS252<br>-Or-<br>MS300 PY251 SS201 SS202<br>-Or-<br>MS300 PY251 SS201 SS252<br>-Or-<br>MS300 PY201 SS202 SS251<br>-Or-<br>MS300 PY201 SS251 SS252<br>-Or-<br>MS300 PY251 SS251 SS252<br>-Or-<br>MS300 PY251 SS202 SS251<br>-Or-<br>MS350X PY201 SS201 SS202<br>-Or-<br>MS350X PY201 SS201 SS252<br>-Or-<br>MS350X PY251 SS201 SS202<br>-Or-<br>MS350X PY251 SS201 SS252<br>-Or-<br>MS350 PY201 SS201 SS202<br>-Or-<br>MS350 PY201 SS201 SS252<br>-Or-<br>MS350 PY251 SS201 SS202<br>-Or-<br>MS350 PY251 SS201 SS252 |                                                               |
| <b>Corequisite(s):</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | HI302 PL300<br>-Or-<br>HI302H PL300<br>-Or-<br>HI302X PL300<br>-Or-<br>HI302 PL350<br>-Or-<br>HI302H PL350<br>-Or-<br>HI302X PL350<br>-Or-<br>HI352 PL300<br>-Or-<br>HI352 PL350                                                                                                                                                                                                                                                                                                                                                                                         |                                                               |





# **PART IV: MAJORS AND MINORS**

## 2029 MAJOR Offerings

Majors available to the Class of 2029 are listed below along with the department that has primary responsibility for them.

By Department:

| Dept         | Code                 | Description                                          | Transcript Description                         |
|--------------|----------------------|------------------------------------------------------|------------------------------------------------|
| MACC-DPE-KIN | <a href="#">KIN1</a> | Kinesiology Major                                    | Kinesiology                                    |
| MADN-BSL     | <a href="#">BEA0</a> | Behavioral Sciences: Applied Psychology Major        | Behavioral Sciences: Applied Psychology        |
| MADN-BSL     | <a href="#">BEM0</a> | Behavioral Sciences: Military & Society Major        | Behavioral Sciences: Military & Society        |
| MADN-BSL     | <a href="#">BEO0</a> | Behavioral Sciences: Organizational Psychology Major | Behavioral Sciences: Organizational Psychology |
| MADN-BSL     | <a href="#">EPS1</a> | Engineering Psychology Major                         | Engineering Psychology                         |
| MADN-BSL     | <a href="#">MGN0</a> | Management Major                                     | Management                                     |
| MADN-CBE     | <a href="#">BIO0</a> | Biology Major                                        | Biology                                        |
| MADN-CBE     | <a href="#">CEN2</a> | Chemical Engineering Major                           | Chemical Engineering                           |
| MADN-CBE     | <a href="#">CES2</a> | Chemical Engineering Studies Major                   | Chemical Engineering Studies                   |
| MADN-CBE     | <a href="#">CHM2</a> | Chemistry Major                                      | Chemistry                                      |
| MADN-CEE     | <a href="#">CVN3</a> | Civil Engineering Major                              | Civil Engineering                              |
| MADN-CEE     | <a href="#">CNG2</a> | Civil Engineering Studies                            | Civil Engineering Studies                      |
| MADN-CEE     | <a href="#">EVE2</a> | Environmental Engineering Major                      | Environmental Engineering                      |
| MADN-CEE     | <a href="#">EES2</a> | Environmental Engineering Studies Major              | Environmental Engineering Studies              |
| MADN-DLP     | <a href="#">LLS3</a> | Law and Legal Studies Major                          | Law and Legal Studies                          |
| MADN-DLP     | <a href="#">PYL2</a> | Philosophy Major                                     | Philosophy                                     |
| MADN-DSE     | <a href="#">ENM1</a> | Engineering Management Major                         | Engineering Management                         |
| MADN-DSE     | <a href="#">SDS1</a> | Systems and Decision Sciences Major                  | Systems and Decision Sciences                  |
| MADN-DSE     | <a href="#">SEN2</a> | Systems Engineering Major                            | Systems Engineering                            |
| MADN-EEC     | <a href="#">CSC1</a> | Computer Science Major                               | Computer Science                               |
| MADN-EEC     | <a href="#">CYS1</a> | Cyber Science Major                                  | Cyber Science                                  |
| MADN-EEC     | <a href="#">CYO0</a> | Cyber Science: Cyber Operations Major                | Cyber Science: Cyber Operations                |
| MADN-EEC     | <a href="#">EIT2</a> | Electronic & Information Technology Systems Major    | Elec & Info Tech Sys                           |
| MADN-EEC     | <a href="#">EEN1</a> | Electrical Engineering Major                         | Electrical Engineering                         |
| MADN-EWL     | <a href="#">ENL2</a> | English Major                                        | English                                        |
| MADN-EWL     | <a href="#">FSI2</a> | Foreign Area Studies Major: Africa                   | Foreign Area Studies: Africa                   |
| MADN-EWL     | <a href="#">FSA2</a> | Foreign Area Studies Major: East Asia                | Foreign Area Studies: East Asia                |
| MADN-EWL     | <a href="#">FSU2</a> | Foreign Area Studies Major: Eurasia                  | Foreign Area Studies: Eurasia                  |
| MADN-EWL     | <a href="#">FSE2</a> | Foreign Area Studies Major: Europe                   | Foreign Area Studies: Europe                   |
| MADN-EWL     | <a href="#">FSL2</a> | Foreign Area Studies Major: Latin America            | Foreign Area Studies: Latin America            |
| MADN-EWL     | <a href="#">FSM2</a> | Foreign Area Studies Major: Middle East              | Foreign Area Studies: Middle East              |
| MADN-EWL     | <a href="#">FZA1</a> | Foreign Language Major: Persian & Arabic             | Foreign Language: Persian & Arabic             |
| MADN-EWL     | <a href="#">FLA2</a> | Foreign Language Major: Arabic                       | Foreign Language: Arabic                       |
| MADN-EWL     | <a href="#">FAC2</a> | Foreign Language Major: Arabic & Chinese             | Foreign Language: Arabic & Chinese             |
| MADN-EWL     | <a href="#">FAF2</a> | Foreign Language Major: Arabic & French              | Foreign Language: Arabic & French              |
| MADN-EWL     | <a href="#">FAG1</a> | Foreign Language Major: Arabic & German              | Foreign Language: Arabic & German              |
| MADN-EWL     | <a href="#">FAZ2</a> | Foreign Language Major: Arabic & Persian             | Foreign Language: Arabic & Persian             |
| MADN-EWL     | <a href="#">FAP2</a> | Foreign Language Major: Arabic & Portuguese          | Foreign Language: Arabic & Portuguese          |
| MADN-EWL     | <a href="#">FAR2</a> | Foreign Language Major: Arabic & Russian             | Foreign Language: Arabic & Russian             |
| MADN-EWL     | <a href="#">FAS2</a> | Foreign Language Major: Arabic & Spanish             | Foreign Language: Arabic & Spanish             |



|          |                      |                                                |                                          |
|----------|----------------------|------------------------------------------------|------------------------------------------|
| MADN-EWL | <a href="#">FLC2</a> | Foreign Language Major: Chinese                | Foreign Language: Chinese                |
| MADN-EWL | <a href="#">FCA1</a> | Foreign Language Major: Chinese & Arabic       | Foreign Language: Chinese & Arabic       |
| MADN-EWL | <a href="#">FCF2</a> | Foreign Language Major: Chinese & French       | Foreign Language: Chinese & French       |
| MADN-EWL | <a href="#">FCG2</a> | Foreign Language Major: Chinese & German       | Foreign Language: Chinese & German       |
| MADN-EWL | <a href="#">FCZ2</a> | Foreign Language Major: Chinese & Persian      | Foreign Language: Chinese & Persian      |
| MADN-EWL | <a href="#">FCP2</a> | Foreign Language Major: Chinese & Portuguese   | Foreign Language: Chinese & Portuguese   |
| MADN-EWL | <a href="#">FCR2</a> | Foreign Language Major: Chinese & Russian      | Foreign Language: Chinese & Russian      |
| MADN-EWL | <a href="#">FCS2</a> | Foreign Language Major: Chinese & Spanish      | Foreign Language: Chinese & Spanish      |
| MADN-EWL | <a href="#">FLF2</a> | Foreign Language Major: French                 | Foreign Language: French                 |
| MADN-EWL | <a href="#">FFA1</a> | Foreign Language Major: French & Arabic        | Foreign Language: French & Arabic        |
| MADN-EWL | <a href="#">FFC1</a> | Foreign Language Major: French & Chinese       | Foreign Language: French & Chinese       |
| MADN-EWL | <a href="#">FFG2</a> | Foreign Language Major: French & German        | Foreign Language: French & German        |
| MADN-EWL | <a href="#">FFZ2</a> | Foreign Language Major: French & Persian       | Foreign Language: French & Persian       |
| MADN-EWL | <a href="#">FFP2</a> | Foreign Language Major: French & Portuguese    | Foreign Language: French & Portuguese    |
| MADN-EWL | <a href="#">FFR2</a> | Foreign Language Major: French & Russian       | Foreign Language: French & Russian       |
| MADN-EWL | <a href="#">FFS2</a> | Foreign Language Major: French & Spanish       | Foreign Language: French & Spanish       |
| MADN-EWL | <a href="#">FLG2</a> | Foreign Language Major: German                 | Foreign Language: German                 |
| MADN-EWL | <a href="#">FGA2</a> | Foreign Language Major: German & Arabic        | Foreign Language: German & Arabic        |
| MADN-EWL | <a href="#">FGC1</a> | Foreign Language Major: German & Chinese       | Foreign Language: German & Chinese       |
| MADN-EWL | <a href="#">FGF1</a> | Foreign Language Major: German & French        | Foreign Language: German & French        |
| MADN-EWL | <a href="#">FGZ2</a> | Foreign Language Major: German & Persian       | Foreign Language: German & Persian       |
| MADN-EWL | <a href="#">FGP2</a> | Foreign Language Major: German & Portuguese    | Foreign Language: German & Portuguese    |
| MADN-EWL | <a href="#">FGR2</a> | Foreign Language Major: German & Russian       | Foreign Language: German & Russian       |
| MADN-EWL | <a href="#">FGS2</a> | Foreign Language Major: German & Spanish       | Foreign Language: German & Spanish       |
| MADN-EWL | <a href="#">FLZ1</a> | Foreign Language Major: Persian                | Foreign Language: Persian                |
| MADN-EWL | <a href="#">FZC1</a> | Foreign Language Major: Persian and Chinese    | Foreign Language: Persian and Chinese    |
| MADN-EWL | <a href="#">FZF1</a> | Foreign Language Major: Persian and French     | Foreign Language: Persian and French     |
| MADN-EWL | <a href="#">FZG1</a> | Foreign Language Major: Persian and German     | Foreign Language: Persian and German     |
| MADN-EWL | <a href="#">FZP1</a> | Foreign Language Major: Persian and Portuguese | Foreign Language: Persian and Portuguese |
| MADN-EWL | <a href="#">FZR1</a> | Foreign Language Major: Persian and Russian    | Foreign Language: Persian and Russian    |
| MADN-EWL | <a href="#">FZS1</a> | Foreign Language Major: Persian and Spanish    | Foreign Language: Persian and Spanish    |
| MADN-EWL | <a href="#">FLP2</a> | Foreign Language Major: Portuguese             | Foreign Language: Portuguese             |
| MADN-EWL | <a href="#">FPA1</a> | Foreign Language Major: Portuguese & Arabic    | Foreign Language: Portuguese & Arabic    |
| MADN-EWL | <a href="#">FPC1</a> | Foreign Language Major: Portuguese & Chinese   | Foreign Language: Portuguese & Chinese   |
| MADN-EWL | <a href="#">FPF1</a> | Foreign Language Major: Portuguese & French    | Foreign Language: Portuguese & French    |
| MADN-EWL | <a href="#">FPG1</a> | Foreign Language Major: Portuguese & German    | Foreign Language: Portuguese & German    |
| MADN-EWL | <a href="#">FPZ2</a> | Foreign Language Major: Portuguese & Persian   | Foreign Language: Portuguese & Persian   |
| MADN-EWL | <a href="#">FPR2</a> | Foreign Language Major: Portuguese & Russian   | Foreign Language: Portuguese & Russian   |
| MADN-EWL | <a href="#">FPS2</a> | Foreign Language Major: Portuguese & Spanish   | Foreign Language: Portuguese & Spanish   |
| MADN-EWL | <a href="#">FLR2</a> | Foreign Language Major: Russian                | Foreign Language: Russian                |
| MADN-EWL | <a href="#">FRA1</a> | Foreign Language Major: Russian & Arabic       | Foreign Language: Russian & Arabic       |
| MADN-EWL | <a href="#">FRC1</a> | Foreign Language Major: Russian & Chinese      | Foreign Language: Russian & Chinese      |
| MADN-EWL | <a href="#">FRF1</a> | Foreign Language Major: Russian & French       | Foreign Language: Russian & French       |
| MADN-EWL | <a href="#">FRG1</a> | Foreign Language Major: Russian & German       | Foreign Language: Russian & German       |
| MADN-EWL | <a href="#">FRZ2</a> | Foreign Language Major: Russian & Persian      | Foreign Language: Russian & Persian      |
| MADN-EWL | <a href="#">FRP1</a> | Foreign Language Major: Russian & Portuguese   | Foreign Language: Russian & Portuguese   |
| MADN-EWL | <a href="#">FRS2</a> | Foreign Language Major: Russian & Spanish      | Foreign Language: Russian & Spanish      |
| MADN-EWL | <a href="#">FLS2</a> | Foreign Language Major: Spanish                | Foreign Language: Spanish                |

|          |                      |                                                                        |                                        |
|----------|----------------------|------------------------------------------------------------------------|----------------------------------------|
| MADN-EWL | <a href="#">FSB1</a> | Foreign Language Major: Spanish & Arabic                               | Foreign Language: Spanish & Arabic     |
| MADN-EWL | <a href="#">FSC1</a> | Foreign Language Major: Spanish & Chinese                              | Foreign Language: Spanish & Chinese    |
| MADN-EWL | <a href="#">FSF1</a> | Foreign Language Major: Spanish & French                               | Foreign Language: Spanish & French     |
| MADN-EWL | <a href="#">FSG1</a> | Foreign Language Major: Spanish & German                               | Foreign Language: Spanish & German     |
| MADN-EWL | <a href="#">FSZ2</a> | Foreign Language Major: Spanish & Persian                              | Foreign Language: Spanish & Persian    |
| MADN-EWL | <a href="#">FSP1</a> | Foreign Language Major: Spanish & Portuguese                           | Foreign Language: Spanish & Portuguese |
| MADN-EWL | <a href="#">FSR1</a> | Foreign Language Major: Spanish & Russian                              | Foreign Language: Spanish & Russian    |
| MADN-GEO | <a href="#">ESC3</a> | Environmental Science Major                                            | Environmental Science                  |
| MADN-GEO | <a href="#">GEO1</a> | Geography Major                                                        | Geography                              |
| MADN-GEO | <a href="#">GSC0</a> | Geospatial Science Major                                               | Geospatial Science                     |
| MADN-HWS | <a href="#">HNT3</a> | History Major: International                                           | History: International                 |
| MADN-HWS | <a href="#">HMH3</a> | History Major: Military                                                | History: Military                      |
| MADN-HWS | <a href="#">HUS3</a> | History Major: United States                                           | History: United States                 |
| MADN-HWS | <a href="#">WST0</a> | War Studies Major                                                      | War Studies                            |
| MADN-MAE | <a href="#">AEN0</a> | Aerospace Engineering Major                                            | Aerospace Engineering                  |
| MADN-MAE | <a href="#">MEN3</a> | Mechanical Engineering Major                                           | Mechanical Engineering                 |
| MADN-MAE | <a href="#">MES3</a> | Mechanical Engineering Studies Major                                   | Mechanical Engineering Studies         |
| MADN-MTH | <a href="#">ASD0</a> | Applied Statistics and Data Science Major                              | Applied Statistics and Data Science    |
| MADN-MTH | <a href="#">MSC1</a> | Mathematical Sciences Major                                            | Mathematical Sciences                  |
| MADN-MTH | <a href="#">MST1</a> | Mathematical Studies Major                                             | Mathematical Studies                   |
| MADN-MTH | <a href="#">ORE2</a> | Operations Research Major                                              | Operations Research                    |
| MADN-MTH | <a href="#">ORS1</a> | Operations Research Studies Major                                      | Operations Research Studies            |
| MADN-PNE | <a href="#">NEN1</a> | Nuclear Engineering Major                                              | Nuclear Engineering                    |
| MADN-PNE | <a href="#">NES1</a> | Nuclear Engineering Science Major                                      | Nuclear Engineering Science            |
| MADN-PNE | <a href="#">PHY1</a> | Physics Major                                                          | Physics                                |
| MADN-PNE | <a href="#">PHS0</a> | Physics Studies Major                                                  | Physics Studies                        |
| MADN-PNE | <a href="#">SSC0</a> | Space Science Major                                                    | Space Science                          |
| MADN-SOC | <a href="#">PAP3</a> | American Politics Major                                                | American Politics                      |
| MADN-SOC | <a href="#">ECN2</a> | Economics Major                                                        | Economics                              |
| MADN-SOC | <a href="#">PIF0</a> | International Affairs Major: Foreign Policy and Security Studies       | International Affairs                  |
| MADN-SOC | <a href="#">PII0</a> | International Affairs Major: Institutions, Governance, and Development | International Affairs                  |

By Major:

| Code                 | Description                                          | Transcript Description                         | Dept     |
|----------------------|------------------------------------------------------|------------------------------------------------|----------|
| <a href="#">AEN0</a> | Aerospace Engineering Major                          | Aerospace Engineering                          | MADN-MAE |
| <a href="#">ASD0</a> | Applied Statistics and Data Science Major            | Applied Statistics and Data Science            | MADN-MTH |
| <a href="#">BEA0</a> | Behavioral Sciences: Applied Psychology Major        | Behavioral Sciences: Applied Psychology        | MADN-BSL |
| <a href="#">BEM0</a> | Behavioral Sciences: Military & Society Major        | Behavioral Sciences: Military & Society        | MADN-BSL |
| <a href="#">BEO0</a> | Behavioral Sciences: Organizational Psychology Major | Behavioral Sciences: Organizational Psychology | MADN-BSL |
| <a href="#">BIO0</a> | Biology Major                                        | Biology                                        | MADN-CBE |
| <a href="#">CEN2</a> | Chemical Engineering Major                           | Chemical Engineering                           | MADN-CBE |
| <a href="#">CES2</a> | Chemical Engineering Studies Major                   | Chemical Engineering Studies                   | MADN-CBE |
| <a href="#">CHM2</a> | Chemistry Major                                      | Chemistry                                      | MADN-CBE |
| <a href="#">CNG2</a> | Civil Engineering Studies                            | Civil Engineering Studies                      | MADN-CEE |
| <a href="#">CSC1</a> | Computer Science Major                               | Computer Science                               | MADN-EEC |

|      |                                                   |                                        |          |
|------|---------------------------------------------------|----------------------------------------|----------|
| CVN3 | Civil Engineering Major                           | Civil Engineering                      | MADN-CEE |
| CY00 | Cyber Science: Cyber Operations Major             | Cyber Science: Cyber Operations        | MADN-EEC |
| CYS1 | Cyber Science Major                               | Cyber Science                          | MADN-EEC |
| ECN2 | Economics Major                                   | Economics                              | MADN-SOC |
| EEN1 | Electrical Engineering Major                      | Electrical Engineering                 | MADN-EEC |
| EES2 | Environmental Engineering Studies Major           | Environmental Engineering Studies      | MADN-CEE |
| EIT2 | Electronic & Information Technology Systems Major | Elec & Info Tech Sys                   | MADN-EEC |
| ENL2 | English Major                                     | English                                | MADN-EWL |
| ENM1 | Engineering Management Major                      | Engineering Management                 | MADN-DSE |
| EPS1 | Engineering Psychology Major                      | Engineering Psychology                 | MADN-BSL |
| ESC3 | Environmental Science Major                       | Environmental Science                  | MADN-GEO |
| EVE2 | Environmental Engineering Major                   | Environmental Engineering              | MADN-CEE |
| FAC2 | Foreign Language Major: Arabic & Chinese          | Foreign Language: Arabic & Chinese     | MADN-EWL |
| FAF2 | Foreign Language Major: Arabic & French           | Foreign Language: Arabic & French      | MADN-EWL |
| FAG1 | Foreign Language Major: Arabic & German           | Foreign Language: Arabic & German      | MADN-EWL |
| FAP2 | Foreign Language Major: Arabic & Portuguese       | Foreign Language: Arabic & Portuguese  | MADN-EWL |
| FAR2 | Foreign Language Major: Arabic & Russian          | Foreign Language: Arabic & Russian     | MADN-EWL |
| FAS2 | Foreign Language Major: Arabic & Spanish          | Foreign Language: Arabic & Spanish     | MADN-EWL |
| FAZ2 | Foreign Language Major: Arabic & Persian          | Foreign Language: Arabic & Persian     | MADN-EWL |
| FCA1 | Foreign Language Major: Chinese & Arabic          | Foreign Language: Chinese & Arabic     | MADN-EWL |
| FCF2 | Foreign Language Major: Chinese & French          | Foreign Language: Chinese & French     | MADN-EWL |
| FCG2 | Foreign Language Major: Chinese & German          | Foreign Language: Chinese & German     | MADN-EWL |
| FCP2 | Foreign Language Major: Chinese & Portuguese      | Foreign Language: Chinese & Portuguese | MADN-EWL |
| FCR2 | Foreign Language Major: Chinese & Russian         | Foreign Language: Chinese & Russian    | MADN-EWL |
| FCS2 | Foreign Language Major: Chinese & Spanish         | Foreign Language: Chinese & Spanish    | MADN-EWL |
| FCZ2 | Foreign Language Major: Chinese & Persian         | Foreign Language: Chinese & Persian    | MADN-EWL |
| FFA1 | Foreign Language Major: French & Arabic           | Foreign Language: French & Arabic      | MADN-EWL |
| FFC1 | Foreign Language Major: French & Chinese          | Foreign Language: French & Chinese     | MADN-EWL |
| FFG2 | Foreign Language Major: French & German           | Foreign Language: French & German      | MADN-EWL |
| FFP2 | Foreign Language Major: French & Portuguese       | Foreign Language: French & Portuguese  | MADN-EWL |
| FFR2 | Foreign Language Major: French & Russian          | Foreign Language: French & Russian     | MADN-EWL |
| FFS2 | Foreign Language Major: French & Spanish          | Foreign Language: French & Spanish     | MADN-EWL |
| FFZ2 | Foreign Language Major: French & Persian          | Foreign Language: French & Persian     | MADN-EWL |
| FGA2 | Foreign Language Major: German & Arabic           | Foreign Language: German & Arabic      | MADN-EWL |
| FGC1 | Foreign Language Major: German & Chinese          | Foreign Language: German & Chinese     | MADN-EWL |
| FGF1 | Foreign Language Major: German & French           | Foreign Language: German & French      | MADN-EWL |
| FGP2 | Foreign Language Major: German & Portuguese       | Foreign Language: German & Portuguese  | MADN-EWL |
| FGR2 | Foreign Language Major: German & Russian          | Foreign Language: German & Russian     | MADN-EWL |
| FGS2 | Foreign Language Major: German & Spanish          | Foreign Language: German & Spanish     | MADN-EWL |
| FGZ2 | Foreign Language Major: German & Persian          | Foreign Language: German & Persian     | MADN-EWL |
| FLA2 | Foreign Language Major: Arabic                    | Foreign Language: Arabic               | MADN-EWL |
| FLC2 | Foreign Language Major: Chinese                   | Foreign Language: Chinese              | MADN-EWL |
| FLF2 | Foreign Language Major: French                    | Foreign Language: French               | MADN-EWL |
| FLG2 | Foreign Language Major: German                    | Foreign Language: German               | MADN-EWL |
| FLP2 | Foreign Language Major: Portuguese                | Foreign Language: Portuguese           | MADN-EWL |
| FLR2 | Foreign Language Major: Russian                   | Foreign Language: Russian              | MADN-EWL |
| FLS2 | Foreign Language Major: Spanish                   | Foreign Language: Spanish              | MADN-EWL |
| FLZ1 | Foreign Language Major: Persian                   | Foreign Language: Persian              | MADN-EWL |

|      |                                                |                                          |              |
|------|------------------------------------------------|------------------------------------------|--------------|
| FPA1 | Foreign Language Major: Portuguese & Arabic    | Foreign Language: Portuguese & Arabic    | MADN-EWL     |
| FPC1 | Foreign Language Major: Portuguese & Chinese   | Foreign Language: Portuguese & Chinese   | MADN-EWL     |
| FPF1 | Foreign Language Major: Portuguese & French    | Foreign Language: Portuguese & French    | MADN-EWL     |
| FPG1 | Foreign Language Major: Portuguese & German    | Foreign Language: Portuguese & German    | MADN-EWL     |
| FPR2 | Foreign Language Major: Portuguese & Russian   | Foreign Language: Portuguese & Russian   | MADN-EWL     |
| FPS2 | Foreign Language Major: Portuguese & Spanish   | Foreign Language: Portuguese & Spanish   | MADN-EWL     |
| FPZ2 | Foreign Language Major: Portuguese & Persian   | Foreign Language: Portuguese & Persian   | MADN-EWL     |
| FRA1 | Foreign Language Major: Russian & Arabic       | Foreign Language: Russian & Arabic       | MADN-EWL     |
| FRC1 | Foreign Language Major: Russian & Chinese      | Foreign Language: Russian & Chinese      | MADN-EWL     |
| FRF1 | Foreign Language Major: Russian & French       | Foreign Language: Russian & French       | MADN-EWL     |
| FRG1 | Foreign Language Major: Russian & German       | Foreign Language: Russian & German       | MADN-EWL     |
| FRP1 | Foreign Language Major: Russian & Portuguese   | Foreign Language: Russian & Portuguese   | MADN-EWL     |
| FRS2 | Foreign Language Major: Russian & Spanish      | Foreign Language: Russian & Spanish      | MADN-EWL     |
| FRZ2 | Foreign Language Major: Russian & Persian      | Foreign Language: Russian & Persian      | MADN-EWL     |
| FSA2 | Foreign Area Studies Major: East Asia          | Foreign Area Studies: East Asia          | MADN-EWL     |
| FSB1 | Foreign Language Major: Spanish & Arabic       | Foreign Language: Spanish & Arabic       | MADN-EWL     |
| FSC1 | Foreign Language Major: Spanish & Chinese      | Foreign Language: Spanish & Chinese      | MADN-EWL     |
| FSE2 | Foreign Area Studies Major: Europe             | Foreign Area Studies: Europe             | MADN-EWL     |
| FSF1 | Foreign Language Major: Spanish & French       | Foreign Language: Spanish & French       | MADN-EWL     |
| FSG1 | Foreign Language Major: Spanish & German       | Foreign Language: Spanish & German       | MADN-EWL     |
| FSI2 | Foreign Area Studies Major: Africa             | Foreign Area Studies: Africa             | MADN-EWL     |
| FSL2 | Foreign Area Studies Major: Latin America      | Foreign Area Studies: Latin America      | MADN-EWL     |
| FSM2 | Foreign Area Studies Major: Middle East        | Foreign Area Studies: Middle East        | MADN-EWL     |
| FSP1 | Foreign Language Major: Spanish & Portuguese   | Foreign Language: Spanish & Portuguese   | MADN-EWL     |
| FSR1 | Foreign Language Major: Spanish & Russian      | Foreign Language: Spanish & Russian      | MADN-EWL     |
| FSU2 | Foreign Area Studies Major: Eurasia            | Foreign Area Studies: Eurasia            | MADN-EWL     |
| FSZ2 | Foreign Language Major: Spanish & Persian      | Foreign Language: Spanish & Persian      | MADN-EWL     |
| FZA1 | Foreign Language Major: Persian & Arabic       | Foreign Language: Persian & Arabic       | MADN-EWL     |
| FZC1 | Foreign Language Major: Persian and Chinese    | Foreign Language: Persian and Chinese    | MADN-EWL     |
| FZF1 | Foreign Language Major: Persian and French     | Foreign Language: Persian and French     | MADN-EWL     |
| FZG1 | Foreign Language Major: Persian and German     | Foreign Language: Persian and German     | MADN-EWL     |
| FZP1 | Foreign Language Major: Persian and Portuguese | Foreign Language: Persian and Portuguese | MADN-EWL     |
| FZR1 | Foreign Language Major: Persian and Russian    | Foreign Language: Persian and Russian    | MADN-EWL     |
| FZS1 | Foreign Language Major: Persian and Spanish    | Foreign Language: Persian and Spanish    | MADN-EWL     |
| GEO1 | Geography Major                                | Geography                                | MADN-GEO     |
| GSC0 | Geospatial Science Major                       | Geospatial Science                       | MADN-GEO     |
| HMH3 | History Major: Military                        | History: Military                        | MADN-HWS     |
| HNT3 | History Major: International                   | History: International                   | MADN-HWS     |
| HUS3 | History Major: United States                   | History: United States                   | MADN-HWS     |
| KIN1 | Kinesiology Major                              | Kinesiology                              | MACC-DPE-KIN |
| LLS3 | Law and Legal Studies Major                    | Law and Legal Studies                    | MADN-DLP     |
| MEN3 | Mechanical Engineering Major                   | Mechanical Engineering                   | MADN-MAE     |
| MES3 | Mechanical Engineering Studies Major           | Mechanical Engineering Studies           | MADN-MAE     |
| MGN0 | Management Major                               | Management                               | MADN-BSL     |
| MSC1 | Mathematical Sciences Major                    | Mathematical Sciences                    | MADN-MTH     |
| MST1 | Mathematical Studies Major                     | Mathematical Studies                     | MADN-MTH     |
| NEN1 | Nuclear Engineering Major                      | Nuclear Engineering                      | MADN-PNE     |
| NES1 | Nuclear Engineering Science Major              | Nuclear Engineering Science              | MADN-PNE     |

|                      |                                                                        |                               |          |
|----------------------|------------------------------------------------------------------------|-------------------------------|----------|
| <a href="#">ORE2</a> | Operations Research Major                                              | Operations Research           | MADN-MTH |
| <a href="#">ORS1</a> | Operations Research Studies Major                                      | Operations Research Studies   | MADN-MTH |
| <a href="#">PAP3</a> | American Politics Major                                                | American Politics             | MADN-SOC |
| <a href="#">PHS0</a> | Physics Studies Major                                                  | Physics Studies               | MADN-PNE |
| <a href="#">PHY1</a> | Physics Major                                                          | Physics                       | MADN-PNE |
| <a href="#">PIF0</a> | International Affairs Major: Foreign Policy and Security Studies       | International Affairs         | MADN-SOC |
| <a href="#">PII0</a> | International Affairs Major: Institutions, Governance, and Development | International Affairs         | MADN-SOC |
| <a href="#">PYL2</a> | Philosophy Major                                                       | Philosophy                    | MADN-DLP |
| <a href="#">SDS1</a> | Systems and Decision Sciences Major                                    | Systems and Decision Sciences | MADN-DSE |
| <a href="#">SEN2</a> | Systems Engineering Major                                              | Systems Engineering           | MADN-DSE |
| <a href="#">SSC0</a> | Space Science Major                                                    | Space Science                 | MADN-PNE |
| <a href="#">WST0</a> | War Studies Major                                                      | War Studies                   | MADN-HWS |

## 2029 MINOR Offerings

Minors available to the Class of 2029 are listed below along with the department that has primary responsibility for them.

By Department:

| Dept     | Code                  | Description                            | Transcript Description                 |
|----------|-----------------------|----------------------------------------|----------------------------------------|
| MADN-CBE | <a href="#">BEN1N</a> | Bioengineering Minor                   | Bioengineering Minor                   |
| MADN-DSE | <a href="#">EMN0N</a> | Engineering Management Minor           | Engineering Management Minor           |
| MADN-EEC | <a href="#">CYB1N</a> | Cyber Security Minor                   | Cyber Security Minor                   |
| MADN-EEC | <a href="#">ROB0N</a> | Robotics Minor                         | Robotics Minor                         |
| MADN-EWL | <a href="#">RSI1N</a> | Regional Studies Minor - Africa        | Regional Studies Minor - Africa        |
| MADN-EWL | <a href="#">RSA1N</a> | Regional Studies Minor - East Asia     | Regional Studies Minor - East Asia     |
| MADN-EWL | <a href="#">RSU1N</a> | Regional Studies Minor - Eurasia       | Regional Studies Minor - Eurasia       |
| MADN-EWL | <a href="#">RSE1N</a> | Regional Studies Minor - Europe        | Regional Studies Minor - Europe        |
| MADN-EWL | <a href="#">RSL1N</a> | Regional Studies Minor - Latin America | Regional Studies Minor - Latin America |
| MADN-EWL | <a href="#">RSM1N</a> | Regional Studies Minor - Middle East   | Regional Studies Minor - Middle East   |
| MADN-HWS | <a href="#">GHM1N</a> | General History Minor                  | General History Minor                  |
| MADN-MTH | <a href="#">AST1N</a> | Applied Statistics Minor               | Applied Statistics Minor               |
| MADN-MTH | <a href="#">MSM0N</a> | Mathematics Minor                      | Mathematics Minor                      |
| MADN-PNE | <a href="#">NSM0N</a> | Nuclear Science Minor                  | Nuclear Science Minor                  |
| MADN-PNE | <a href="#">PHT0N</a> | Photonics Minor                        | Photonics Minor                        |
| MADN-PNE | <a href="#">SPA0N</a> | Space Science Minor                    | Space Science Minor                    |
| MADN-SOC | <a href="#">AMF0N</a> | American Foundations Minor             | American Foundations Minor             |
| MADN-SOC | <a href="#">TST1N</a> | Terrorism Studies Minor                | Terrorism Studies Minor                |

By Minor:

| Code                  | Description                            | Transcript Description                 | Dept     |
|-----------------------|----------------------------------------|----------------------------------------|----------|
| <a href="#">AMF0N</a> | American Foundations Minor             | American Foundations Minor             | MADN-SOC |
| <a href="#">AST1N</a> | Applied Statistics Minor               | Applied Statistics Minor               | MADN-MTH |
| <a href="#">BEN1N</a> | Bioengineering Minor                   | Bioengineering Minor                   | MADN-CBE |
| <a href="#">CYB1N</a> | Cyber Security Minor                   | Cyber Security Minor                   | MADN-EEC |
| <a href="#">EMN0N</a> | Engineering Management Minor           | Engineering Management Minor           | MADN-DSE |
| <a href="#">GHM1N</a> | General History Minor                  | General History Minor                  | MADN-HWS |
| <a href="#">MSM0N</a> | Mathematics Minor                      | Mathematics Minor                      | MADN-MTH |
| <a href="#">NSM0N</a> | Nuclear Science Minor                  | Nuclear Science Minor                  | MADN-PNE |
| <a href="#">PHT0N</a> | Photonics Minor                        | Photonics Minor                        | MADN-PNE |
| <a href="#">ROB0N</a> | Robotics Minor                         | Robotics Minor                         | MADN-EEC |
| <a href="#">RSA1N</a> | Regional Studies Minor - East Asia     | Regional Studies Minor - East Asia     | MADN-EWL |
| <a href="#">RSE1N</a> | Regional Studies Minor - Europe        | Regional Studies Minor - Europe        | MADN-EWL |
| <a href="#">RSI1N</a> | Regional Studies Minor - Africa        | Regional Studies Minor - Africa        | MADN-EWL |
| <a href="#">RSL1N</a> | Regional Studies Minor - Latin America | Regional Studies Minor - Latin America | MADN-EWL |
| <a href="#">RSM1N</a> | Regional Studies Minor - Middle East   | Regional Studies Minor - Middle East   | MADN-EWL |
| <a href="#">RSU1N</a> | Regional Studies Minor - Eurasia       | Regional Studies Minor - Eurasia       | MADN-EWL |
| <a href="#">SPA0N</a> | Space Science Minor                    | Space Science Minor                    | MADN-PNE |

|       |                         |                         |          |
|-------|-------------------------|-------------------------|----------|
| TST1N | Terrorism Studies Minor | Terrorism Studies Minor | MADN-SOC |
|-------|-------------------------|-------------------------|----------|

## 2029 MAJORS By Discipline

### Humanities and Social Sciences (HSS)

Majors available to the Class of 2029 are listed below along with the department that has primary responsibility for them. The majors are listed as a Humanities and Social Sciences (HSS) or Mathematics, Science, and Engineering (MSE) discipline.

HSS By Department:

| Dept     | Description                                          | Transcript Description                         | Code |
|----------|------------------------------------------------------|------------------------------------------------|------|
| MADN-BSL | Behavioral Sciences: Applied Psychology Major        | Behavioral Sciences: Applied Psychology        | BEA0 |
| MADN-BSL | Behavioral Sciences: Military & Society Major        | Behavioral Sciences: Military & Society        | BEM0 |
| MADN-BSL | Behavioral Sciences: Organizational Psychology Major | Behavioral Sciences: Organizational Psychology | BE00 |
| MADN-BSL | Engineering Psychology Major                         | Engineering Psychology                         | EPS1 |
| MADN-BSL | Management Major                                     | Management                                     | MGN0 |
| MADN-DLP | Law and Legal Studies Major                          | Law and Legal Studies                          | LLS3 |
| MADN-DLP | Philosophy Major                                     | Philosophy                                     | PYL2 |
| MADN-EWL | English Major                                        | English                                        | ENL2 |
| MADN-EWL | Foreign Area Studies Major: Africa                   | Foreign Area Studies: Africa                   | FSI2 |
| MADN-EWL | Foreign Area Studies Major: East Asia                | Foreign Area Studies: East Asia                | FSA2 |
| MADN-EWL | Foreign Area Studies Major: Eurasia                  | Foreign Area Studies: Eurasia                  | FSU2 |
| MADN-EWL | Foreign Area Studies Major: Europe                   | Foreign Area Studies: Europe                   | FSE2 |
| MADN-EWL | Foreign Area Studies Major: Latin America            | Foreign Area Studies: Latin America            | FSL2 |
| MADN-EWL | Foreign Area Studies Major: Middle East              | Foreign Area Studies: Middle East              | FSM2 |
| MADN-EWL | Foreign Language Major: Persian & Arabic             | Foreign Language: Persian & Arabic             | FZA1 |
| MADN-EWL | Foreign Language Major: Arabic                       | Foreign Language: Arabic                       | FLA2 |
| MADN-EWL | Foreign Language Major: Arabic & Chinese             | Foreign Language: Arabic & Chinese             | FAC2 |
| MADN-EWL | Foreign Language Major: Arabic & French              | Foreign Language: Arabic & French              | FAF2 |
| MADN-EWL | Foreign Language Major: Arabic & German              | Foreign Language: Arabic & German              | FAG1 |
| MADN-EWL | Foreign Language Major: Arabic & Persian             | Foreign Language: Arabic & Persian             | FAZ2 |
| MADN-EWL | Foreign Language Major: Arabic & Portuguese          | Foreign Language: Arabic & Portuguese          | FAP2 |
| MADN-EWL | Foreign Language Major: Arabic & Russian             | Foreign Language: Arabic & Russian             | FAR2 |
| MADN-EWL | Foreign Language Major: Arabic & Spanish             | Foreign Language: Arabic & Spanish             | FAS2 |
| MADN-EWL | Foreign Language Major: Chinese                      | Foreign Language: Chinese                      | FLC2 |
| MADN-EWL | Foreign Language Major: Chinese & Arabic             | Foreign Language: Chinese & Arabic             | FCA1 |
| MADN-EWL | Foreign Language Major: Chinese & French             | Foreign Language: Chinese & French             | FCF2 |
| MADN-EWL | Foreign Language Major: Chinese & German             | Foreign Language: Chinese & German             | FCG2 |
| MADN-EWL | Foreign Language Major: Chinese & Persian            | Foreign Language: Chinese & Persian            | FCZ2 |
| MADN-EWL | Foreign Language Major: Chinese & Portuguese         | Foreign Language: Chinese & Portuguese         | FCP2 |
| MADN-EWL | Foreign Language Major: Chinese & Russian            | Foreign Language: Chinese & Russian            | FCR2 |
| MADN-EWL | Foreign Language Major: Chinese & Spanish            | Foreign Language: Chinese & Spanish            | FCS2 |
| MADN-EWL | Foreign Language Major: French                       | Foreign Language: French                       | FLF2 |
| MADN-EWL | Foreign Language Major: French & Arabic              | Foreign Language: French & Arabic              | FFA1 |
| MADN-EWL | Foreign Language Major: French & Chinese             | Foreign Language: French & Chinese             | FFC1 |
| MADN-EWL | Foreign Language Major: French & German              | Foreign Language: French & German              | FFG2 |
| MADN-EWL | Foreign Language Major: French & Persian             | Foreign Language: French & Persian             | FFZ2 |
| MADN-EWL | Foreign Language Major: French & Portuguese          | Foreign Language: French & Portuguese          | FFP2 |
| MADN-EWL | Foreign Language Major: French & Russian             | Foreign Language: French & Russian             | FFR2 |



|          |                                                                  |                                          |                      |
|----------|------------------------------------------------------------------|------------------------------------------|----------------------|
| MADN-EWL | Foreign Language Major: French & Spanish                         | Foreign Language: French & Spanish       | <a href="#">FFS2</a> |
| MADN-EWL | Foreign Language Major: German                                   | Foreign Language: German                 | <a href="#">FLG2</a> |
| MADN-EWL | Foreign Language Major: German & Arabic                          | Foreign Language: German & Arabic        | <a href="#">FGA2</a> |
| MADN-EWL | Foreign Language Major: German & Chinese                         | Foreign Language: German & Chinese       | <a href="#">FGC1</a> |
| MADN-EWL | Foreign Language Major: German & French                          | Foreign Language: German & French        | <a href="#">FGF1</a> |
| MADN-EWL | Foreign Language Major: German & Persian                         | Foreign Language: German & Persian       | <a href="#">FGZ2</a> |
| MADN-EWL | Foreign Language Major: German & Portuguese                      | Foreign Language: German & Portuguese    | <a href="#">FGP2</a> |
| MADN-EWL | Foreign Language Major: German & Russian                         | Foreign Language: German & Russian       | <a href="#">FGR2</a> |
| MADN-EWL | Foreign Language Major: German & Spanish                         | Foreign Language: German & Spanish       | <a href="#">FGS2</a> |
| MADN-EWL | Foreign Language Major: Persian                                  | Foreign Language: Persian                | <a href="#">FLZ1</a> |
| MADN-EWL | Foreign Language Major: Persian and Chinese                      | Foreign Language: Persian and Chinese    | <a href="#">FZC1</a> |
| MADN-EWL | Foreign Language Major: Persian and French                       | Foreign Language: Persian and French     | <a href="#">FZF1</a> |
| MADN-EWL | Foreign Language Major: Persian and German                       | Foreign Language: Persian and German     | <a href="#">FZG1</a> |
| MADN-EWL | Foreign Language Major: Persian and Portuguese                   | Foreign Language: Persian and Portuguese | <a href="#">FZP1</a> |
| MADN-EWL | Foreign Language Major: Persian and Russian                      | Foreign Language: Persian and Russian    | <a href="#">FZR1</a> |
| MADN-EWL | Foreign Language Major: Persian and Spanish                      | Foreign Language: Persian and Spanish    | <a href="#">FZS1</a> |
| MADN-EWL | Foreign Language Major: Portuguese                               | Foreign Language: Portuguese             | <a href="#">FLP2</a> |
| MADN-EWL | Foreign Language Major: Portuguese & Arabic                      | Foreign Language: Portuguese & Arabic    | <a href="#">FPA1</a> |
| MADN-EWL | Foreign Language Major: Portuguese & Chinese                     | Foreign Language: Portuguese & Chinese   | <a href="#">FPC1</a> |
| MADN-EWL | Foreign Language Major: Portuguese & French                      | Foreign Language: Portuguese & French    | <a href="#">PPF1</a> |
| MADN-EWL | Foreign Language Major: Portuguese & German                      | Foreign Language: Portuguese & German    | <a href="#">FPG1</a> |
| MADN-EWL | Foreign Language Major: Portuguese & Persian                     | Foreign Language: Portuguese & Persian   | <a href="#">FPZ2</a> |
| MADN-EWL | Foreign Language Major: Portuguese & Russian                     | Foreign Language: Portuguese & Russian   | <a href="#">FPR2</a> |
| MADN-EWL | Foreign Language Major: Portuguese & Spanish                     | Foreign Language: Portuguese & Spanish   | <a href="#">FPS2</a> |
| MADN-EWL | Foreign Language Major: Russian                                  | Foreign Language: Russian                | <a href="#">FLR2</a> |
| MADN-EWL | Foreign Language Major: Russian & Arabic                         | Foreign Language: Russian & Arabic       | <a href="#">FRA1</a> |
| MADN-EWL | Foreign Language Major: Russian & Chinese                        | Foreign Language: Russian & Chinese      | <a href="#">FRC1</a> |
| MADN-EWL | Foreign Language Major: Russian & French                         | Foreign Language: Russian & French       | <a href="#">FRF1</a> |
| MADN-EWL | Foreign Language Major: Russian & German                         | Foreign Language: Russian & German       | <a href="#">FRG1</a> |
| MADN-EWL | Foreign Language Major: Russian & Persian                        | Foreign Language: Russian & Persian      | <a href="#">FRZ2</a> |
| MADN-EWL | Foreign Language Major: Russian & Portuguese                     | Foreign Language: Russian & Portuguese   | <a href="#">FRP1</a> |
| MADN-EWL | Foreign Language Major: Russian & Spanish                        | Foreign Language: Russian & Spanish      | <a href="#">FRS2</a> |
| MADN-EWL | Foreign Language Major: Spanish                                  | Foreign Language: Spanish                | <a href="#">FLS2</a> |
| MADN-EWL | Foreign Language Major: Spanish & Arabic                         | Foreign Language: Spanish & Arabic       | <a href="#">FSB1</a> |
| MADN-EWL | Foreign Language Major: Spanish & Chinese                        | Foreign Language: Spanish & Chinese      | <a href="#">FSC1</a> |
| MADN-EWL | Foreign Language Major: Spanish & French                         | Foreign Language: Spanish & French       | <a href="#">FSF1</a> |
| MADN-EWL | Foreign Language Major: Spanish & German                         | Foreign Language: Spanish & German       | <a href="#">FSG1</a> |
| MADN-EWL | Foreign Language Major: Spanish & Persian                        | Foreign Language: Spanish & Persian      | <a href="#">FSZ2</a> |
| MADN-EWL | Foreign Language Major: Spanish & Portuguese                     | Foreign Language: Spanish & Portuguese   | <a href="#">FSP1</a> |
| MADN-EWL | Foreign Language Major: Spanish & Russian                        | Foreign Language: Spanish & Russian      | <a href="#">FSR1</a> |
| MADN-GEO | Geography Major                                                  | Geography                                | <a href="#">GEO1</a> |
| MADN-HWS | History Major: International                                     | History: International                   | <a href="#">HNT3</a> |
| MADN-HWS | History Major: Military                                          | History: Military                        | <a href="#">HMH3</a> |
| MADN-HWS | History Major: United States                                     | History: United States                   | <a href="#">HUS3</a> |
| MADN-HWS | War Studies Major                                                | War Studies                              | <a href="#">WST0</a> |
| MADN-SOC | American Politics Major                                          | American Politics                        | <a href="#">PAP3</a> |
| MADN-SOC | International Affairs Major: Foreign Policy and Security Studies | International Affairs                    | <a href="#">PIF0</a> |

|          |                                                                        |                       |      |
|----------|------------------------------------------------------------------------|-----------------------|------|
| MADN-SOC | International Affairs Major: Institutions, Governance, and Development | International Affairs | PIIO |
|----------|------------------------------------------------------------------------|-----------------------|------|

## HSS By Major:

| Code                 | Description                                          | Transcript Description                         | Dept     |
|----------------------|------------------------------------------------------|------------------------------------------------|----------|
| <a href="#">BEA0</a> | Behavioral Sciences: Applied Psychology Major        | Behavioral Sciences: Applied Psychology        | MADN-BSL |
| <a href="#">BEM0</a> | Behavioral Sciences: Military & Society Major        | Behavioral Sciences: Military & Society        | MADN-BSL |
| <a href="#">BEO0</a> | Behavioral Sciences: Organizational Psychology Major | Behavioral Sciences: Organizational Psychology | MADN-BSL |
| <a href="#">ENL2</a> | English Major                                        | English                                        | MADN-EWL |
| <a href="#">EPS1</a> | Engineering Psychology Major                         | Engineering Psychology                         | MADN-BSL |
| <a href="#">FAC2</a> | Foreign Language Major: Arabic & Chinese             | Foreign Language: Arabic & Chinese             | MADN-EWL |
| <a href="#">FAF2</a> | Foreign Language Major: Arabic & French              | Foreign Language: Arabic & French              | MADN-EWL |
| <a href="#">FAG1</a> | Foreign Language Major: Arabic & German              | Foreign Language: Arabic & German              | MADN-EWL |
| <a href="#">FAP2</a> | Foreign Language Major: Arabic & Portuguese          | Foreign Language: Arabic & Portuguese          | MADN-EWL |
| <a href="#">FAR2</a> | Foreign Language Major: Arabic & Russian             | Foreign Language: Arabic & Russian             | MADN-EWL |
| <a href="#">FAS2</a> | Foreign Language Major: Arabic & Spanish             | Foreign Language: Arabic & Spanish             | MADN-EWL |
| <a href="#">FAZ2</a> | Foreign Language Major: Arabic & Persian             | Foreign Language: Arabic & Persian             | MADN-EWL |
| <a href="#">FCA1</a> | Foreign Language Major: Chinese & Arabic             | Foreign Language: Chinese & Arabic             | MADN-EWL |
| <a href="#">FCF2</a> | Foreign Language Major: Chinese & French             | Foreign Language: Chinese & French             | MADN-EWL |
| <a href="#">FCG2</a> | Foreign Language Major: Chinese & German             | Foreign Language: Chinese & German             | MADN-EWL |
| <a href="#">FCP2</a> | Foreign Language Major: Chinese & Portuguese         | Foreign Language: Chinese & Portuguese         | MADN-EWL |
| <a href="#">FCR2</a> | Foreign Language Major: Chinese & Russian            | Foreign Language: Chinese & Russian            | MADN-EWL |
| <a href="#">FCS2</a> | Foreign Language Major: Chinese & Spanish            | Foreign Language: Chinese & Spanish            | MADN-EWL |
| <a href="#">FCZ2</a> | Foreign Language Major: Chinese & Persian            | Foreign Language: Chinese & Persian            | MADN-EWL |
| <a href="#">FFA1</a> | Foreign Language Major: French & Arabic              | Foreign Language: French & Arabic              | MADN-EWL |
| <a href="#">FFC1</a> | Foreign Language Major: French & Chinese             | Foreign Language: French & Chinese             | MADN-EWL |
| <a href="#">FFG2</a> | Foreign Language Major: French & German              | Foreign Language: French & German              | MADN-EWL |
| <a href="#">FFP2</a> | Foreign Language Major: French & Portuguese          | Foreign Language: French & Portuguese          | MADN-EWL |
| <a href="#">FFR2</a> | Foreign Language Major: French & Russian             | Foreign Language: French & Russian             | MADN-EWL |
| <a href="#">FFS2</a> | Foreign Language Major: French & Spanish             | Foreign Language: French & Spanish             | MADN-EWL |
| <a href="#">FFZ2</a> | Foreign Language Major: French & Persian             | Foreign Language: French & Persian             | MADN-EWL |
| <a href="#">FGA2</a> | Foreign Language Major: German & Arabic              | Foreign Language: German & Arabic              | MADN-EWL |
| <a href="#">FGC1</a> | Foreign Language Major: German & Chinese             | Foreign Language: German & Chinese             | MADN-EWL |
| <a href="#">FGF1</a> | Foreign Language Major: German & French              | Foreign Language: German & French              | MADN-EWL |
| <a href="#">FGP2</a> | Foreign Language Major: German & Portuguese          | Foreign Language: German & Portuguese          | MADN-EWL |
| <a href="#">FGR2</a> | Foreign Language Major: German & Russian             | Foreign Language: German & Russian             | MADN-EWL |
| <a href="#">FGS2</a> | Foreign Language Major: German & Spanish             | Foreign Language: German & Spanish             | MADN-EWL |
| <a href="#">FGZ2</a> | Foreign Language Major: German & Persian             | Foreign Language: German & Persian             | MADN-EWL |
| <a href="#">FLA2</a> | Foreign Language Major: Arabic                       | Foreign Language: Arabic                       | MADN-EWL |
| <a href="#">FLC2</a> | Foreign Language Major: Chinese                      | Foreign Language: Chinese                      | MADN-EWL |
| <a href="#">FLF2</a> | Foreign Language Major: French                       | Foreign Language: French                       | MADN-EWL |
| <a href="#">FLG2</a> | Foreign Language Major: German                       | Foreign Language: German                       | MADN-EWL |
| <a href="#">FLP2</a> | Foreign Language Major: Portuguese                   | Foreign Language: Portuguese                   | MADN-EWL |
| <a href="#">FLR2</a> | Foreign Language Major: Russian                      | Foreign Language: Russian                      | MADN-EWL |
| <a href="#">FLS2</a> | Foreign Language Major: Spanish                      | Foreign Language: Spanish                      | MADN-EWL |
| <a href="#">FLZ1</a> | Foreign Language Major: Persian                      | Foreign Language: Persian                      | MADN-EWL |
| <a href="#">FPA1</a> | Foreign Language Major: Portuguese & Arabic          | Foreign Language: Portuguese & Arabic          | MADN-EWL |

|      |                                                                        |                                          |          |
|------|------------------------------------------------------------------------|------------------------------------------|----------|
| FPC1 | Foreign Language Major: Portuguese & Chinese                           | Foreign Language: Portuguese & Chinese   | MADN-EWL |
| FPF1 | Foreign Language Major: Portuguese & French                            | Foreign Language: Portuguese & French    | MADN-EWL |
| FPG1 | Foreign Language Major: Portuguese & German                            | Foreign Language: Portuguese & German    | MADN-EWL |
| FPR2 | Foreign Language Major: Portuguese & Russian                           | Foreign Language: Portuguese & Russian   | MADN-EWL |
| FPS2 | Foreign Language Major: Portuguese & Spanish                           | Foreign Language: Portuguese & Spanish   | MADN-EWL |
| FPZ2 | Foreign Language Major: Portuguese & Persian                           | Foreign Language: Portuguese & Persian   | MADN-EWL |
| FRA1 | Foreign Language Major: Russian & Arabic                               | Foreign Language: Russian & Arabic       | MADN-EWL |
| FRC1 | Foreign Language Major: Russian & Chinese                              | Foreign Language: Russian & Chinese      | MADN-EWL |
| FRF1 | Foreign Language Major: Russian & French                               | Foreign Language: Russian & French       | MADN-EWL |
| FRG1 | Foreign Language Major: Russian & German                               | Foreign Language: Russian & German       | MADN-EWL |
| FRP1 | Foreign Language Major: Russian & Portuguese                           | Foreign Language: Russian & Portuguese   | MADN-EWL |
| FRS2 | Foreign Language Major: Russian & Spanish                              | Foreign Language: Russian & Spanish      | MADN-EWL |
| FRZ2 | Foreign Language Major: Russian & Persian                              | Foreign Language: Russian & Persian      | MADN-EWL |
| FSA2 | Foreign Area Studies Major: East Asia                                  | Foreign Area Studies: East Asia          | MADN-EWL |
| FSB1 | Foreign Language Major: Spanish & Arabic                               | Foreign Language: Spanish & Arabic       | MADN-EWL |
| FSC1 | Foreign Language Major: Spanish & Chinese                              | Foreign Language: Spanish & Chinese      | MADN-EWL |
| FSE2 | Foreign Area Studies Major: Europe                                     | Foreign Area Studies: Europe             | MADN-EWL |
| FSF1 | Foreign Language Major: Spanish & French                               | Foreign Language: Spanish & French       | MADN-EWL |
| FSG1 | Foreign Language Major: Spanish & German                               | Foreign Language: Spanish & German       | MADN-EWL |
| FSI2 | Foreign Area Studies Major: Africa                                     | Foreign Area Studies: Africa             | MADN-EWL |
| FSL2 | Foreign Area Studies Major: Latin America                              | Foreign Area Studies: Latin America      | MADN-EWL |
| FSM2 | Foreign Area Studies Major: Middle East                                | Foreign Area Studies: Middle East        | MADN-EWL |
| FSP1 | Foreign Language Major: Spanish & Portuguese                           | Foreign Language: Spanish & Portuguese   | MADN-EWL |
| FSR1 | Foreign Language Major: Spanish & Russian                              | Foreign Language: Spanish & Russian      | MADN-EWL |
| FSU2 | Foreign Area Studies Major: Eurasia                                    | Foreign Area Studies: Eurasia            | MADN-EWL |
| FSZ2 | Foreign Language Major: Spanish & Persian                              | Foreign Language: Spanish & Persian      | MADN-EWL |
| FZA1 | Foreign Language Major: Persian & Arabic                               | Foreign Language: Persian & Arabic       | MADN-EWL |
| FZC1 | Foreign Language Major: Persian and Chinese                            | Foreign Language: Persian and Chinese    | MADN-EWL |
| FZF1 | Foreign Language Major: Persian and French                             | Foreign Language: Persian and French     | MADN-EWL |
| FZG1 | Foreign Language Major: Persian and German                             | Foreign Language: Persian and German     | MADN-EWL |
| FZP1 | Foreign Language Major: Persian and Portuguese                         | Foreign Language: Persian and Portuguese | MADN-EWL |
| FZR1 | Foreign Language Major: Persian and Russian                            | Foreign Language: Persian and Russian    | MADN-EWL |
| FZS1 | Foreign Language Major: Persian and Spanish                            | Foreign Language: Persian and Spanish    | MADN-EWL |
| GEO1 | Geography Major                                                        | Geography                                | MADN-GEO |
| HMH3 | History Major: Military                                                | History: Military                        | MADN-HWS |
| HNT3 | History Major: International                                           | History: International                   | MADN-HWS |
| HUS3 | History Major: United States                                           | History: United States                   | MADN-HWS |
| LLS3 | Law and Legal Studies Major                                            | Law and Legal Studies                    | MADN-DLP |
| MGN0 | Management Major                                                       | Management                               | MADN-BSL |
| PAP3 | American Politics Major                                                | American Politics                        | MADN-SOC |
| PIF0 | International Affairs Major: Foreign Policy and Security Studies       | International Affairs                    | MADN-SOC |
| PII0 | International Affairs Major: Institutions, Governance, and Development | International Affairs                    | MADN-SOC |
| PYL2 | Philosophy Major                                                       | Philosophy                               | MADN-DLP |

|      |                   |             |          |
|------|-------------------|-------------|----------|
| WST0 | War Studies Major | War Studies | MADN-HWS |
|------|-------------------|-------------|----------|

## 2029 MAJORS By Discipline

### Mathematics, Science, and Engineering (MSE)

MSE By Department:

| Dept         | Description                                       | Transcript Description              | Code |
|--------------|---------------------------------------------------|-------------------------------------|------|
| MACC-DPE-KIN | Kinesiology Major                                 | Kinesiology                         | KIN1 |
| MADN-CBE     | Biology Major                                     | Biology                             | BIO0 |
| MADN-CBE     | Chemical Engineering Major                        | Chemical Engineering                | CEN2 |
| MADN-CBE     | Chemical Engineering Studies Major                | Chemical Engineering Studies        | CES2 |
| MADN-CBE     | Chemistry Major                                   | Chemistry                           | CHM2 |
| MADN-CEE     | Civil Engineering Major                           | Civil Engineering                   | CVN3 |
| MADN-CEE     | Civil Engineering Studies                         | Civil Engineering Studies           | CNG2 |
| MADN-CEE     | Environmental Engineering Major                   | Environmental Engineering           | EVE2 |
| MADN-CEE     | Environmental Engineering Studies Major           | Environmental Engineering Studies   | EES2 |
| MADN-DSE     | Engineering Management Major                      | Engineering Management              | ENM1 |
| MADN-DSE     | Systems and Decision Sciences Major               | Systems and Decision Sciences       | SDS1 |
| MADN-DSE     | Systems Engineering Major                         | Systems Engineering                 | SEN2 |
| MADN-EEC     | Computer Science Major                            | Computer Science                    | CSC1 |
| MADN-EEC     | Cyber Science Major                               | Cyber Science                       | CYS1 |
| MADN-EEC     | Cyber Science: Cyber Operations Major             | Cyber Science: Cyber Operations     | CY00 |
| MADN-EEC     | Electronic & Information Technology Systems Major | Elec & Info Tech Sys                | EIT2 |
| MADN-EEC     | Electrical Engineering Major                      | Electrical Engineering              | EEN1 |
| MADN-GEO     | Environmental Science Major                       | Environmental Science               | ESC3 |
| MADN-GEO     | Geospatial Science Major                          | Geospatial Science                  | GSC0 |
| MADN-MAE     | Aerospace Engineering Major                       | Aerospace Engineering               | AEN0 |
| MADN-MAE     | Mechanical Engineering Major                      | Mechanical Engineering              | MEN3 |
| MADN-MAE     | Mechanical Engineering Studies Major              | Mechanical Engineering Studies      | MES3 |
| MADN-MTH     | Applied Statistics and Data Science Major         | Applied Statistics and Data Science | ASD0 |
| MADN-MTH     | Mathematical Sciences Major                       | Mathematical Sciences               | MSC1 |
| MADN-MTH     | Mathematical Studies Major                        | Mathematical Studies                | MST1 |
| MADN-MTH     | Operations Research Major                         | Operations Research                 | ORE2 |
| MADN-MTH     | Operations Research Studies Major                 | Operations Research Studies         | ORS1 |
| MADN-PNE     | Nuclear Engineering Major                         | Nuclear Engineering                 | NEN1 |
| MADN-PNE     | Nuclear Engineering Science Major                 | Nuclear Engineering Science         | NES1 |
| MADN-PNE     | Physics Major                                     | Physics                             | PHY1 |
| MADN-PNE     | Physics Studies Major                             | Physics Studies                     | PHS0 |
| MADN-PNE     | Space Science Major                               | Space Science                       | SSC0 |
| MADN-SOC     | Economics Major                                   | Economics                           | ECN2 |

MSE By Major:

| Code                 | Description                                       | Transcript Description              | Dept         |
|----------------------|---------------------------------------------------|-------------------------------------|--------------|
| <a href="#">AEN0</a> | Aerospace Engineering Major                       | Aerospace Engineering               | MADN-MAE     |
| <a href="#">ASD0</a> | Applied Statistics and Data Science Major         | Applied Statistics and Data Science | MADN-MTH     |
| <a href="#">BIO0</a> | Biology Major                                     | Biology                             | MADN-CBE     |
| <a href="#">CEN2</a> | Chemical Engineering Major                        | Chemical Engineering                | MADN-CBE     |
| <a href="#">CES2</a> | Chemical Engineering Studies Major                | Chemical Engineering Studies        | MADN-CBE     |
| <a href="#">CHM2</a> | Chemistry Major                                   | Chemistry                           | MADN-CBE     |
| <a href="#">CNG2</a> | Civil Engineering Studies                         | Civil Engineering Studies           | MADN-CEE     |
| <a href="#">CSC1</a> | Computer Science Major                            | Computer Science                    | MADN-EEC     |
| <a href="#">CVN3</a> | Civil Engineering Major                           | Civil Engineering                   | MADN-CEE     |
| <a href="#">CYO0</a> | Cyber Science: Cyber Operations Major             | Cyber Science: Cyber Operations     | MADN-EEC     |
| <a href="#">CYS1</a> | Cyber Science Major                               | Cyber Science                       | MADN-EEC     |
| <a href="#">ECN2</a> | Economics Major                                   | Economics                           | MADN-SOC     |
| <a href="#">EEN1</a> | Electrical Engineering Major                      | Electrical Engineering              | MADN-EEC     |
| <a href="#">EES2</a> | Environmental Engineering Studies Major           | Environmental Engineering Studies   | MADN-CEE     |
| <a href="#">EIT2</a> | Electronic & Information Technology Systems Major | Elec & Info Tech Sys                | MADN-EEC     |
| <a href="#">ENM1</a> | Engineering Management Major                      | Engineering Management              | MADN-DSE     |
| <a href="#">ESC3</a> | Environmental Science Major                       | Environmental Science               | MADN-GEO     |
| <a href="#">EVE2</a> | Environmental Engineering Major                   | Environmental Engineering           | MADN-CEE     |
| <a href="#">GSC0</a> | Geospatial Science Major                          | Geospatial Science                  | MADN-GEO     |
| <a href="#">KIN1</a> | Kinesiology Major                                 | Kinesiology                         | MACC-DPE-KIN |
| <a href="#">MEN3</a> | Mechanical Engineering Major                      | Mechanical Engineering              | MADN-MAE     |
| <a href="#">MES3</a> | Mechanical Engineering Studies Major              | Mechanical Engineering Studies      | MADN-MAE     |
| <a href="#">MSC1</a> | Mathematical Sciences Major                       | Mathematical Sciences               | MADN-MTH     |
| <a href="#">MST1</a> | Mathematical Studies Major                        | Mathematical Studies                | MADN-MTH     |
| <a href="#">NEN1</a> | Nuclear Engineering Major                         | Nuclear Engineering                 | MADN-PNE     |
| <a href="#">NES1</a> | Nuclear Engineering Science Major                 | Nuclear Engineering Science         | MADN-PNE     |
| <a href="#">ORE2</a> | Operations Research Major                         | Operations Research                 | MADN-MTH     |
| <a href="#">ORS1</a> | Operations Research Studies Major                 | Operations Research Studies         | MADN-MTH     |
| <a href="#">PHS0</a> | Physics Studies Major                             | Physics Studies                     | MADN-PNE     |
| <a href="#">PHY1</a> | Physics Major                                     | Physics                             | MADN-PNE     |
| <a href="#">SDS1</a> | Systems and Decision Sciences Major               | Systems and Decision Sciences       | MADN-DSE     |
| <a href="#">SEN2</a> | Systems Engineering Major                         | Systems Engineering                 | MADN-DSE     |
| <a href="#">SSC0</a> | Space Science Major                               | Space Science                       | MADN-PNE     |

# PART IV: FIELD TABLES

## Behavioral Sciences and Leadership

### 2029 Behavioral Sciences: Applied Psychology Major Curriculum

| Code | Short Description                       | Description                                   | Transcript Description                  | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------------------|-----------------------------------------------|-----------------------------------------|--------------|--------------|
| BEA0 | Behavioral Sciences: Applied Psychology | Behavioral Sciences: Applied Psychology Major | Behavioral Sciences: Applied Psychology | 6            | 7            |

### 2029 Behavioral Sciences: Applied Psychology Major Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                                                                                                                                           | Description                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | Choose 2 of 2                  |
| PL383                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SOCIAL PSYCHOLOGY              |
| PL387                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FOUNDATIONS OF COUNSELING      |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| <b>Two-Course Research Methods Sequence</b>                                                                                                                                                                                                                                                                                                                                                                                                            | Choose 2 of 3                  |
| PL361                                                                                                                                                                                                                                                                                                                                                                                                                                                  | RESEARCH METHODS I             |
| PL363                                                                                                                                                                                                                                                                                                                                                                                                                                                  | QUALITATIVE RESEARCH METHODS   |
| PL462                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ADVANCED RESEARCH METHODS      |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| <b>Applied General Psychology Track</b>                                                                                                                                                                                                                                                                                                                                                                                                                | Choose 3 of 3                  |
| PL250                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NEUROCOG FNDTNS OF BEHAVIOR    |
| PL373                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LIFE CYCLE & HUMAN DEVEL       |
| PL376                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PSYCHOPATHOLOGY                |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| <b>Depth of Discipline Electives</b>                                                                                                                                                                                                                                                                                                                                                                                                                   | Choose 2 of 12                 |
| MG379                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LEADING TEAMS                  |
| PL360                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PSYCH ELITE PERFORMANCE        |
| PL370                                                                                                                                                                                                                                                                                                                                                                                                                                                  | US SOCIETY & THE SERVICEMEMBER |
| PL372                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MILITARY FAMILIES              |
| PL390                                                                                                                                                                                                                                                                                                                                                                                                                                                  | BIOLOGICAL PSYCHOLOGY          |
| PL391                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SENSATION/PERCEPTN/PSYCPHYS    |
| PL392                                                                                                                                                                                                                                                                                                                                                                                                                                                  | COGNITIVE PSYCHOLOGY           |
| PL393                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CRIMINOLOGY-CRIM JUST SYSTM    |
| PL398                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LEADERSHIP THEORY & PRACTICE   |
| PL470                                                                                                                                                                                                                                                                                                                                                                                                                                                  | TOPICS-BEHAVIOR SCI/LDRSHIP    |
| PL471                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LEADERSHIP IN COMBAT           |
| PL479                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LEADING ORGNZS THRU CHANGE     |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| <b>Complementary Support Courses</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
| Select three complementary support courses in consultation with your DAC, who will provide you with suggested topic areas to consider. Language courses are listed as "LX" and "LA" but cadets can choose any advanced language course for which they qualify. You should limit complementary support courses from within the Department of Behavioral Sciences and Leadership, so as to ensure you meet the academic intent of complementary support. |                                |
| <b>Behavioral Sciences CSCs</b>                                                                                                                                                                                                                                                                                                                                                                                                                        | Choose 3 of 76                 |
| Cadets cannot double count CH375 or PH202/PH275 to satisfy both the CSC requirement and the Science Depth requirement.                                                                                                                                                                                                                                                                                                                                 |                                |

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|       |                                |
|-------|--------------------------------|
| CH375 | ADVANCED BIOLOGY               |
| CH383 | ORGANIC CHEMISTRY I            |
| CH384 | ORGANIC CHEMISTRY II           |
| CH385 | INTRODUCTION TO CELL BIOLOGY   |
| CH387 | HUMAN PHYSIOLOGY               |
| CH473 | BIOCHEMISTRY                   |
| CY383 | SECURE INTERFACE DESIGN        |
| CY460 | CYBER POLICY, STRATEGY, & OPNS |
| EM411 | PROJECT MANAGEMENT             |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN364 | DRAMA                          |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE352 | URBAN GEOGRAPHY                |
| GE491 | MILITARY GEOGRAPHY             |
| GE494 | ENVIRONMENTAL SECURITY         |
| HI398 | CIVIL RIGHTS IN AMER HIST      |
| KN355 | FUNCTIONAL ANATOMY             |
| KN360 | BIOMECHANICS OF HUMAN MOVEMENT |
| KN455 | PSYCHOLOGY OF EXERCISE         |
| LA371 | INTENSIVE INTERMEDIATE ARABIC  |
| LA372 | ARABIC FOR ORAL & WRITTEN COMM |
| LA475 | ARABIC RDG/WRTG THRU MEDIA     |
| LA476 | MILITARY SPKG/RDG - ARABIC     |
| LA483 | ARAB CIVILIZATION I            |
| LA484 | ARAB CIVILIZATION II           |
| LA485 | ARABIC LITERATURE I            |
| LA486 | ARABIC LITERATURE II           |
| LN380 | NATURE OF MODERN LANGUAGES     |
| LW310 | INTRO TO LEGAL METHOD          |
| LW461 | CIVIL RIGHTS AND THE LAW       |
| LW472 | CRIMINAL LAW                   |
| LW474 | LAW OF ARMED CONFLICT          |
| LW481 | INTERNATIONAL LAW              |
| LW482 | NATIONAL SECURITY LAW          |
| LW488 | BUSINESS LAW                   |
| MA367 | MATH FOR THE SOCIAL SCIENCES   |
| MA391 | MATHEMATICAL MODELING          |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE   |
| MA476 | MATHEMATICAL STATISTICS        |
| MA490 | ETHICS IN MATH, DATA SCI & ENG |
| MG380 | MARKETING                      |
| MG381 | INTRODUCTION TO MANAGEMENT     |
| MG382 | HUMAN RESOURCE MANAGEMENT      |
| MG390 | NEGOTIATION FOR LEADERS        |
| PH202 | PHYSICS II                     |
| PH275 | PHYSICS II: SPACE              |
| PL394 | ANTHROPOMETRICS & BIOMECHANICS |
| PL475 | HUMAN-COMPUTER INTERACTION     |
| PL482 | ARMED FORCES AND SOCIETY       |
| PY305 | LOGICAL REASONING              |
| PY310 | REALITY AND KNOWLEDGE          |
| PY320 | ETHICS                         |
| PY325 | MILITARY ETHICS                |
| PY326 | ETHICS OF TECHNOLOGY           |
| PY345 | PHILOSOPHY OF RELIGION         |
| PY350 | PHILOSOPHY OF SCIENCE          |



|       |                                |
|-------|--------------------------------|
| PY355 | PHILOSOPHY OF MIND             |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS370 | MASS MEDIA & AMER POLITICS     |
| SS392 | RIGHTS, LIBERTIES & US CITIZEN |
| SS395 | INTERNATIONAL SECURITY         |
| SS464 | HOMELAND SECURITY              |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS466 | COMBATING TERRORISM            |
| SS476 | CIVIL WARS                     |
| WS330 | INTRODUCTION TO GRAND STRATEGY |
| WS350 | STRATEGIC COMMUNICATION        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS370 | US STRATEGY AND POLICY         |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| WS475 | FORECAST AND GAME IN DECISION  |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |
| XH415 | GENOCIDE AND ETHNIC CLEANSING  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

PL488B

CAPSTONE SEM IN BEHAVIORAL SCI

**AND****Science Depth**

Although CH275 will be recommended, cadets can select among the following science depth courses.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| CH375 | ADVANCED BIOLOGY     |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                                |
|-------|--------------------------------|
| PL462 | EXPERIMENTAL APP IN PSYCHOLOGY |
|-------|--------------------------------|

### 2029 Behavioral Sciences: Applied Psychology Major w/ Honors Curriculum

| Code  | Short Description                            | Description                                             | Transcript Description                           | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|---------------------------------------------------------|--------------------------------------------------|--------------|--------------|
| BEA0H | Behavioral Sciences: Applied Psych w/ Honors | Behavioral Sciences: Applied Psychology Major w/ Honors | Behavioral Sciences: Applied Psychology w/Honors | 2            | 0            |

### 2029 Behavioral Sciences: Applied Psychology Major w/ Honors Tracks

| Subject Area                                            | Description               |
|---------------------------------------------------------|---------------------------|
| <b>Complete one of the following two-course tracks:</b> |                           |
| <b>Psychological Research</b>                           | Choose 2 of 2             |
| PL497                                                   | SEMINAR IN BEHAVIORAL SCI |
| PL498                                                   | ADV STUDY-BEHAVIOR SCI    |
| <b>OR</b>                                               |                           |
| <b>Biological Psychology</b>                            | Choose 2 of 2             |
| CH383                                                   | ORGANIC CHEMISTRY I       |
| CH384                                                   | ORGANIC CHEMISTRY II      |
| <b>AND</b>                                              |                           |

Complete the requirements for the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

### 2029 Behavioral Sciences: Applied Psychology Major w/ Honors Remarks Pending full review of the Curriculum Committee in AY25-2. Elective courses for the major may change.

### 2029 Behavioral Sciences: Military & Society Major Curriculum

| Code | Short Description                       | Description                                   | Transcript Description                  | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------------------|-----------------------------------------------|-----------------------------------------|--------------|--------------|
| BEM0 | Behavioral Sciences: Military & Society | Behavioral Sciences: Military & Society Major | Behavioral Sciences: Military & Society | 6            | 7            |

### 2029 Behavioral Sciences: Military & Society Major Tracks

| Subject Area                                | Description               |
|---------------------------------------------|---------------------------|
| <b>Required Courses</b>                     | Choose 2 of 2             |
| PL383                                       | SOCIAL PSYCHOLOGY         |
| PL387                                       | FOUNDATIONS OF COUNSELING |
| <b>AND</b>                                  |                           |
| <b>Two-Course Research Methods Sequence</b> | Choose 2 of 3             |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| PL361                                                                                                                                                                                                                                                                                                                                                                                                                                                  | RESEARCH METHODS I             |
| PL363                                                                                                                                                                                                                                                                                                                                                                                                                                                  | QUALITATIVE RESEARCH METHODS   |
| PL462                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ADVANCED RESEARCH METHODS      |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| <b>Military &amp; Society Track</b>                                                                                                                                                                                                                                                                                                                                                                                                                    | Choose 3 of 3                  |
| PL370                                                                                                                                                                                                                                                                                                                                                                                                                                                  | US SOCIETY & THE SERVICEMEMBER |
| PL372                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MILITARY FAMILIES              |
| PL482                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ARMED FORCES AND SOCIETY       |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| <b>Depth of Discipline Electives</b>                                                                                                                                                                                                                                                                                                                                                                                                                   | Choose 2 of 13                 |
| MG379                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LEADING TEAMS                  |
| PL250                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NEUROCOG FNDTNS OF BEHAVIOR    |
| PL360                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PSYCH ELITE PERFORMANCE        |
| PL373                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LIFE CYCLE & HUMAN DEVEL       |
| PL376                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PSYCHOPATHOLOGY                |
| PL390                                                                                                                                                                                                                                                                                                                                                                                                                                                  | BIOLOGICAL PSYCHOLOGY          |
| PL391                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SENSATION/PERCEPTN/PSYCPHYS    |
| PL392                                                                                                                                                                                                                                                                                                                                                                                                                                                  | COGNITIVE PSYCHOLOGY           |
| PL393                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CRIMINOLOGY-CRIM JUST SYSTM    |
| PL398                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LEADERSHIP THEORY & PRACTICE   |
| PL470                                                                                                                                                                                                                                                                                                                                                                                                                                                  | TOPICS-BEHAVIOR SCI/LDRSHIP    |
| PL471                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LEADERSHIP IN COMBAT           |
| PL479                                                                                                                                                                                                                                                                                                                                                                                                                                                  | LEADING ORGNZS THRU CHANGE     |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| <b>Complementary Support Courses</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
| Select three complementary support courses in consultation with your DAC, who will provide you with suggested topic areas to consider. Language courses are listed as "LX" and "LA" but cadets can choose any advanced language course for which they qualify. You should limit complementary support courses from within the Department of Behavioral Sciences and Leadership, so as to ensure you meet the academic intent of complementary support. |                                |
| <b>Behavioral Sciences CSCs</b>                                                                                                                                                                                                                                                                                                                                                                                                                        | Choose 3 of 77                 |
| Cadets cannot double count CH375 or PH202/PH275 to satisfy both the CSC requirement and the Science Depth requirement.                                                                                                                                                                                                                                                                                                                                 |                                |
| CH375                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ADVANCED BIOLOGY               |
| CH383                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORGANIC CHEMISTRY I            |
| CH384                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ORGANIC CHEMISTRY II           |
| CH385                                                                                                                                                                                                                                                                                                                                                                                                                                                  | INTRODUCTION TO CELL BIOLOGY   |
| CH387                                                                                                                                                                                                                                                                                                                                                                                                                                                  | HUMAN PHYSIOLOGY               |
| CH473                                                                                                                                                                                                                                                                                                                                                                                                                                                  | BIOCHEMISTRY                   |
| CY383                                                                                                                                                                                                                                                                                                                                                                                                                                                  | SECURE INTERFACE DESIGN        |
| EM411                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PROJECT MANAGEMENT             |
| EN361                                                                                                                                                                                                                                                                                                                                                                                                                                                  | POETRY                         |
| EN362                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FILM AND FILM THEORY           |
| EN364                                                                                                                                                                                                                                                                                                                                                                                                                                                  | DRAMA                          |
| GE340                                                                                                                                                                                                                                                                                                                                                                                                                                                  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE352                                                                                                                                                                                                                                                                                                                                                                                                                                                  | URBAN GEOGRAPHY                |
| GE491                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MILITARY GEOGRAPHY             |
| GE494                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ENVIRONMENTAL SECURITY         |
| HI398                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CIVIL RIGHTS IN AMER HIST      |
| KN355                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FUNCTIONAL ANATOMY             |
| KN360                                                                                                                                                                                                                                                                                                                                                                                                                                                  | BIOMECHANICS OF HUMAN MOVEMENT |
| KN455                                                                                                                                                                                                                                                                                                                                                                                                                                                  | PSYCHOLOGY OF EXERCISE         |
| LA371                                                                                                                                                                                                                                                                                                                                                                                                                                                  | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ARABIC FOR ORAL & WRITTEN COMM |
| LA475                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ARABIC RDG/WRTG THRU MEDIA     |

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|       |                                |
|-------|--------------------------------|
| LA476 | MILITARY SPKG/RDG - ARABIC     |
| LA483 | ARAB CIVILIZATION I            |
| LA484 | ARAB CIVILIZATION II           |
| LA485 | ARABIC LITERATURE I            |
| LA486 | ARABIC LITERATURE II           |
| LN380 | NATURE OF MODERN LANGUAGES     |
| LW310 | INTRO TO LEGAL METHOD          |
| LW461 | CIVIL RIGHTS AND THE LAW       |
| LW472 | CRIMINAL LAW                   |
| LW474 | LAW OF ARMED CONFLICT          |
| LW481 | INTERNATIONAL LAW              |
| LW482 | NATIONAL SECURITY LAW          |
| LW488 | BUSINESS LAW                   |
| MA367 | MATH FOR THE SOCIAL SCIENCES   |
| MA391 | MATHEMATICAL MODELING          |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE   |
| MA476 | MATHEMATICAL STATISTICS        |
| MA490 | ETHICS IN MATH, DATA SCI & ENG |
| MG380 | MARKETING                      |
| MG381 | INTRODUCTION TO MANAGEMENT     |
| MG382 | HUMAN RESOURCE MANAGEMENT      |
| MG390 | NEGOTIATION FOR LEADERS        |
| PH202 | PHYSICS II                     |
| PH275 | PHYSICS II: SPACE              |
| PL394 | ANTHROPOMETRICS & BIOMECHANICS |
| PL475 | HUMAN-COMPUTER INTERACTION     |
| PY305 | LOGICAL REASONING              |
| PY310 | REALITY AND KNOWLEDGE          |
| PY320 | ETHICS                         |
| PY325 | MILITARY ETHICS                |
| PY326 | ETHICS OF TECHNOLOGY           |
| PY345 | PHILOSOPHY OF RELIGION         |
| PY350 | PHILOSOPHY OF SCIENCE          |
| PY355 | PHILOSOPHY OF MIND             |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS370 | MASS MEDIA & AMER POLITICS     |
| SS392 | RIGHTS, LIBERTIES & US CITIZEN |
| SS395 | INTERNATIONAL SECURITY         |
| SS464 | HOMELAND SECURITY              |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS466 | COMBATING TERRORISM            |
| SS476 | CIVIL WARS                     |
| WS330 | INTRODUCTION TO GRAND STRATEGY |
| WS350 | STRATEGIC COMMUNICATION        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS370 | US STRATEGY AND POLICY         |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| WS475 | FORECAST AND GAME IN DECISION  |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |
| XH415 | GENOCIDE AND ETHNIC CLEANSING  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

PL488B

CAPSTONE SEM IN BEHAVIORAL SCI

**AND**

**Science Depth**

Although CH275 will be recommended, cadets can select among the following science depth courses.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| CH375 | ADVANCED BIOLOGY     |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                                |
|-------|--------------------------------|
| PL462 | EXPERIMENTAL APP IN PSYCHOLOGY |
|-------|--------------------------------|

**2029 Behavioral Sciences: Military & Society Major Remarks Pending full review of the Curriculum Committee in AY25-2. Elective courses for the major may change.**

**2029 Behavioral Sciences: Military & Society Major w/ Honors Curriculum**

| Code  | Short Description                                | Description                                             | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------------------------|---------------------------------------------------------|----------------------------------------------|--------------|--------------|
| BEM0H | Behavioral Sciences: Military & Society w/Honors | Behavioral Sciences: Military & Society Major w/ Honors | Behavioral Sci: Military & Society w/ Honors | 2            | 0            |

**2029 Behavioral Sciences: Military & Society Major w/ Honors Tracks**

| Subject Area                                            | Description               |
|---------------------------------------------------------|---------------------------|
| <b>Complete one of the following two-course tracks:</b> |                           |
| <b>Psychological Research</b>                           | Choose 2 of 2             |
| PL497                                                   | SEMINAR IN BEHAVIORAL SCI |
| PL498                                                   | ADV STUDY-BEHAVIOR SCI    |
| <b>OR</b>                                               |                           |
| <b>Biological Psychology</b>                            | Choose 2 of 2             |
| CH383                                                   | ORGANIC CHEMISTRY I       |
| CH384                                                   | ORGANIC CHEMISTRY II      |
| <b>AND</b>                                              |                           |

Complete the requirements for the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Behavioral Sciences: Military & Society Major w/ Honors Remarks Pending full review of the Curriculum Committee in AY25-2. Elective courses for the major may change.

### 2029 Behavioral Sciences: Organizational Psychology Major Curriculum

| Code | Short Description                              | Description                                          | Transcript Description                         | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------------------|------------------------------------------------------|------------------------------------------------|--------------|--------------|
| BEO0 | Behavioral Sciences: Organizational Psychology | Behavioral Sciences: Organizational Psychology Major | Behavioral Sciences: Organizational Psychology | 6            | 7            |

### 2029 Behavioral Sciences: Organizational Psychology Major Tracks

| Subject Area                                            | Description                    |
|---------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                 | Choose 2 of 2                  |
| PL383                                                   | SOCIAL PSYCHOLOGY              |
| PL387                                                   | FOUNDATIONS OF COUNSELING      |
| <b>AND</b>                                              |                                |
| <b>Two-Course Research Methods Sequence</b>             | Choose 2 of 3                  |
| PL361                                                   | RESEARCH METHODS I             |
| PL363                                                   | QUALITATIVE RESEARCH METHODS   |
| PL462                                                   | ADVANCED RESEARCH METHODS      |
| <b>AND</b>                                              |                                |
| <b>Organizational Psychology &amp; Leadership Track</b> | Choose 3 of 3                  |
| MG379                                                   | LEADING TEAMS                  |
| PL398                                                   | LEADERSHIP THEORY & PRACTICE   |
| PL479                                                   | LEADING CHANGING ORGANIZATIONS |
| <b>AND</b>                                              |                                |
| <b>Depth of Discipline Electives</b>                    | Choose 2 of 12                 |
| PL250                                                   | NEUROCOG FNDTNS OF BEHAVIOR    |
| PL360                                                   | PSYCH ELITE PERFORMANCE        |
| PL370                                                   | US SOCIETY & THE SERVICEMEMBER |
| PL372                                                   | MILITARY FAMILIES              |
| PL373                                                   | LIFE CYCLE & HUMAN DEVEL       |
| PL376                                                   | PSYCHOPATHOLOGY                |
| PL390                                                   | BIOLOGICAL PSYCHOLOGY          |
| PL391                                                   | SENSATION/PERCEPTN/PSYCPHYS    |
| PL392                                                   | COGNITIVE PSYCHOLOGY           |
| PL393                                                   | CRIMINOLOGY-CRIM JUST SYSTM    |
| PL470                                                   | TOPICS-BEHAVIOR SCI/LDRSHIP    |
| PL471                                                   | LEADERSHIP IN COMBAT           |
| <b>AND</b>                                              |                                |
| <b>Complementary Support Courses</b>                    |                                |

Select three complementary support courses in consultation with your DAC, who will provide you with suggested topic areas to consider. Language courses are listed as "LX" and "LA" but cadets can choose any advanced language course for which they qualify. You should limit complementary support courses from within the Department of Behavioral Sciences and Leadership, so as to ensure you meet the academic intent of complementary support.

**Behavioral Sciences CSCs**

Choose 3 of 76

Cadets cannot double count CH375 or PH202/PH275 to satisfy both the CSC requirement and the Science Depth requirement.

|       |                                |
|-------|--------------------------------|
| CH375 | ADVANCED BIOLOGY               |
| CH383 | ORGANIC CHEMISTRY I            |
| CH384 | ORGANIC CHEMISTRY II           |
| CH385 | INTRODUCTION TO CELL BIOLOGY   |
| CH387 | HUMAN PHYSIOLOGY               |
| CH473 | BIOCHEMISTRY                   |
| CY383 | SECURE INTERFACE DESIGN        |
| CY460 | CYBER POLICY, STRATEGY, & OPNS |
| EM411 | PROJECT MANAGEMENT             |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN364 | DRAMA                          |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE352 | URBAN GEOGRAPHY                |
| GE491 | MILITARY GEOGRAPHY             |
| GE494 | ENVIRONMENTAL SECURITY         |
| HI398 | CIVIL RIGHTS IN AMER HIST      |
| KN355 | FUNCTIONAL ANATOMY             |
| KN360 | BIOMECHANICS OF HUMAN MOVEMENT |
| KN455 | PSYCHOLOGY OF EXERCISE         |
| LA371 | INTENSIVE INTERMEDIATE ARABIC  |
| LA372 | ARABIC FOR ORAL & WRITTEN COMM |
| LA475 | ARABIC RDG/WRTG THRU MEDIA     |
| LA476 | MILITARY SPKG/RDG - ARABIC     |
| LA483 | ARAB CIVILIZATION I            |
| LA484 | ARAB CIVILIZATION II           |
| LA485 | ARABIC LITERATURE I            |
| LA486 | ARABIC LITERATURE II           |
| LN380 | NATURE OF MODERN LANGUAGES     |
| LW310 | INTRO TO LEGAL METHOD          |
| LW461 | CIVIL RIGHTS AND THE LAW       |
| LW472 | CRIMINAL LAW                   |
| LW474 | LAW OF ARMED CONFLICT          |
| LW481 | INTERNATIONAL LAW              |
| LW482 | NATIONAL SECURITY LAW          |
| LW488 | BUSINESS LAW                   |
| MA367 | MATH FOR THE SOCIAL SCIENCES   |
| MA391 | MATHEMATICAL MODELING          |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE   |
| MA476 | MATHEMATICAL STATISTICS        |
| MA490 | ETHICS IN MATH, DATA SCI & ENG |
| MG380 | MARKETING                      |
| MG381 | INTRODUCTION TO MANAGEMENT     |
| MG382 | HUMAN RESOURCE MANAGEMENT      |
| MG390 | NEGOTIATION FOR LEADERS        |
| PH202 | PHYSICS II                     |
| PH275 | PHYSICS II: SPACE              |
| PL394 | ANTHROPOMETRICS & BIOMECHANICS |
| PL475 | HUMAN-COMPUTER INTERACTION     |

|       |                                |
|-------|--------------------------------|
| PL482 | ARMED FORCES AND SOCIETY       |
| PY305 | LOGICAL REASONING              |
| PY310 | REALITY AND KNOWLEDGE          |
| PY320 | ETHICS                         |
| PY325 | MILITARY ETHICS                |
| PY326 | ETHICS OF TECHNOLOGY           |
| PY345 | PHILOSOPHY OF RELIGION         |
| PY350 | PHILOSOPHY OF SCIENCE          |
| PY355 | PHILOSOPHY OF MIND             |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS370 | MASS MEDIA & AMER POLITICS     |
| SS392 | RIGHTS, LIBERTIES & US CITIZEN |
| SS395 | INTERNATIONAL SECURITY         |
| SS464 | HOMELAND SECURITY              |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS466 | COMBATING TERRORISM            |
| SS476 | CIVIL WARS                     |
| WS330 | INTRODUCTION TO GRAND STRATEGY |
| WS350 | STRATEGIC COMMUNICATION        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS370 | US STRATEGY AND POLICY         |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| WS475 | FORECAST AND GAME IN DECISION  |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |
| XH415 | GENOCIDE AND ETHNIC CLEANSING  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

PL488B

CAPSTONE SEM IN BEHAVIORAL SCI

**AND****Science Depth**

Although CH275 will be recommended, cadets can select among the following science depth courses.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| CH375 | ADVANCED BIOLOGY     |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**



Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

PL462

EXPERIMENTAL APP IN PSYCHOLOGY

## 2029 Behavioral Sciences: Organizational Psychology Major Remarks Pending full review of the Curriculum Committee in AY25-2. Elective courses for the major may change.

### 2029 Behavioral Sciences: Organizational Psychology Major w/ Honors Curriculum

| Code  | Short Description                            | Description                                                    | Transcript Description                         | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------------------|------------------------------------------------|--------------|--------------|
| BEO0H | Behavioral Sci: Organization Psych w/ Honors | Behavioral Sciences: Organizational Psychology Major w/ Honors | Behavioral Sci: Organizational Psych w/ Honors | 2            | 0            |

### 2029 Behavioral Sciences: Organizational Psychology Major w/ Honors Tracks

| Subject Area                                            | Description               |
|---------------------------------------------------------|---------------------------|
| <b>Complete one of the following two-course tracks:</b> |                           |
| <b>Psychological Research</b>                           | Choose 2 of 2             |
| PL497                                                   | SEMINAR IN BEHAVIORAL SCI |
| PL498                                                   | ADV STUDY-BEHAVIOR SCI    |
| <b>OR</b>                                               |                           |
| <b>Biological Psychology</b>                            | Choose 2 of 2             |
| CH383                                                   | ORGANIC CHEMISTRY I       |
| CH384                                                   | ORGANIC CHEMISTRY II      |
| <b>AND</b>                                              |                           |

Complete the requirements for the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Behavioral Sciences: Organizational Psychology Major w/ Honors Remarks Pending full review of the Curriculum Committee in AY25-2. Elective courses for the major may change.

### 2029 Engineering Psychology Major Curriculum

| Code | Short Description      | Description                  | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------|------------------------------|------------------------|--------------|--------------|
| EPS1 | Engineering Psychology | Engineering Psychology Major | Engineering Psychology | 10           | 3            |

### 2029 Engineering Psychology Major Tracks

| Subject Area            | Description   |
|-------------------------|---------------|
| <b>Required Courses</b> | Choose 8 of 8 |

|        |                                |
|--------|--------------------------------|
| MA376  | APPLIED STATISTICS             |
| PL386  | EXPERIMENTAL PSYCHOLOGY        |
| PL390  | BIOLOGICAL PSYCHOLOGY          |
| PL391  | SENSATION/PERCEPTN/PSYCPHYS    |
| PL392  | COGNITIVE PSYCHOLOGY           |
| PL394  | ANTHROPOMETRICS & BIOMECHANICS |
| PL475  | HUMAN-COMPUTER INTERACTION     |
| PL488E | COLLOQUIUM-BSL-ENGIN PSYCH     |

**AND****Complementary Support Courses**

Choose 3 of 48

For a study in depth choose three complementary support courses within any of the focus areas (DAC will show you a list). For a broadening study, choose any three complementary support courses from across the focus areas. You should limit complementary support courses from within the Department of Behavioral Sciences and Leadership, so as to ensure you meet the academic intent of complementary support. That is to experience the field of engineering psychology from the perspective of professionals outside your field with whom you will work with as part of a multidisciplinary design team. Language courses are listed as "LX" and "LA" but cadets can choose any advanced language course for which they qualify.

|       |                                |
|-------|--------------------------------|
| CH383 | ORGANIC CHEMISTRY I            |
| CH384 | ORGANIC CHEMISTRY II           |
| CH387 | HUMAN PHYSIOLOGY               |
| CH388 | GENETICS                       |
| CH460 | HUMAN ANATOMY                  |
| CY383 | SECURE INTERFACE DESIGN        |
| CY460 | CYBER POLICY, STRATEGY, & OPNS |
| EM381 | ENGINEERING ECONOMY            |
| EM411 | PROJECT MANAGEMENT             |
| EM481 | SYSTEMS SIMULATION             |
| EN362 | FILM AND FILM THEORY           |
| EV397 | AIR POLLUTION ENGINEERING      |
| GE373 | REMOTE SENSING                 |
| LA475 | ARABIC RDG/WRTG THRU MEDIA     |
| LA476 | MILITARY SPKG/RDG - ARABIC     |
| LW462 | CYBER LAW                      |
| LW473 | ENVIRONMENTAL LAW              |
| LW482 | NATIONAL SECURITY LAW          |
| LW488 | BUSINESS LAW                   |
| LX300 | 3RD SEMESTER FOREIGN LANG      |
| LX400 | 4TH SEMESTER FOREIGN LANG      |
| MA367 | MATH FOR THE SOCIAL SCIENCES   |
| MA391 | MATHEMATICAL MODELING          |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE   |
| MA476 | MATHEMATICAL STATISTICS        |
| ME202 | COMPUTER AIDED DESIGN          |
| ME490 | TOPICS IN MECHANICAL ENGNRG    |
| MG379 | LEADING TEAMS                  |
| MG382 | HUMAN RESOURCE MANAGEMENT      |
| MG420 | OPERATIONS MANAGEMENT          |
| NE474 | RADIOLOGICAL SAFETY            |
| PL360 | PSYCH ELITE PERFORMANCE        |
| PL363 | QUALITATIVE RESEARCH METHODS   |
| PL471 | LEADERSHIP IN COMBAT           |
| PL479 | LEADING ORGNZS THRU CHANGE     |
| PY320 | ETHICS                         |
| PY326 | ETHICS OF TECHNOLOGY           |
| PY350 | PHILOSOPHY OF SCIENCE          |
| PY355 | PHILOSOPHY OF MIND             |

|       |                                |
|-------|--------------------------------|
| SE301 | FNDTN ENGIN DSGN & SYS MGMT    |
| SE387 | DETERMINISTIC MODELS           |
| SE485 | COMBAT MODELING                |
| SM484 | SYSTEM DYNAMICS SIMULATION     |
| SS464 | HOMELAND SECURITY              |
| WS345 | MILITARY INNOVATION            |
| WS350 | STRATEGIC COMMUNICATION        |
| WS365 | REBELLN, INSURGENCY, & CIV WAR |

**AND****Integrative Experience for the Major**

Choose 2 of 2

|       |                           |
|-------|---------------------------|
| PL485 | HUMAN FACTORS ENGINEERING |
| PL490 | ENGINEERING PSYCHOLOGY    |

**AND****Science Depth**

Cadets in this major may take any approved choice to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                        |
|-------|------------------------|
| PL490 | ENGINEERING PSYCHOLOGY |
|-------|------------------------|

## 2029 Engineering Psychology Major w/ Honors Curriculum

| Code  | Short Description                | Description                            | Transcript Description           | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------|----------------------------------------|----------------------------------|--------------|--------------|
| EPS1H | Engineering Psychology w/ Honors | Engineering Psychology Major w/ Honors | Engineering Psychology w/ Honors | 2            | 0            |

### 2029 Engineering Psychology Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                                       | Description               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Complete any one of the following two-course tracks.                                                                                                                                                                                                                                                                                               |                           |
| <b>Engineering Psychology Research</b>                                                                                                                                                                                                                                                                                                             | Choose 2 of 2             |
| PL497                                                                                                                                                                                                                                                                                                                                              | SEMINAR IN BEHAVIORAL SCI |
| PL498                                                                                                                                                                                                                                                                                                                                              | ADV STUDY-BEHAVIOR SCI    |
| <b>OR</b>                                                                                                                                                                                                                                                                                                                                          |                           |
| Choose 2 courses from a focus area in the major for a study in depth of an aspect of Engineering Psychology. The 2 courses are recommended to come from a focus area. Notwithstanding, the 2 courses can be decided in consultation with an academic counselor to meet specific intellectual interests within the field of Engineering Psychology. |                           |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                         |                           |
| Complete the requirements of the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.                                                                                                                                                                                                                   |                           |

### 2029 Management Major Curriculum

| Code | Short Description | Description      | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------|------------------|------------------------|--------------|--------------|
| MGNO | Management        | Management Major | Management             | 8            | 5            |

### 2029 Management Major Tracks

| Subject Area                         | Description                 |
|--------------------------------------|-----------------------------|
| <b>Required Courses</b>              | Choose 7 of 7               |
| MG379                                | LEADING TEAMS               |
| MG380                                | MARKETING                   |
| MG381                                | INTRODUCTION TO MANAGEMENT  |
| MG382                                | HUMAN RESOURCE MANAGEMENT   |
| MG395                                | FUNDAMENTALS OF ACCOUNTING  |
| MG410                                | MANAGERIAL FINANCE          |
| MG420                                | OPERATIONS MANAGEMENT       |
| <b>AND</b>                           |                             |
| <b>Additional Electives</b>          | Choose 2 of 6               |
| MA204                                | CALCULUS I AND II           |
| MA205                                | CALCULUS II                 |
| MG390                                | NEGOTIATION FOR LEADERS     |
| MG462                                | ENTREPRENEURSHIP            |
| PL470                                | TOPICS-BEHAVIOR SCI/LDRSHIP |
| PL479                                | LEADING ORGNZS THRU CHANGE  |
| <b>AND</b>                           |                             |
| <b>Complementary Support Courses</b> | Choose 3 of 52              |

|       |                               |
|-------|-------------------------------|
| EM381 | ENGINEERING ECONOMY           |
| EM411 | PROJECT MANAGEMENT            |
| EM420 | PRODUCTION OPERATIONS MGMT    |
| EM481 | SYSTEMS SIMULATION            |
| EM482 | SUPPLY CHAIN ENG & INFO MGMT  |
| GE305 | GEOGRAPHIC RESEARCH METHODS   |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE341 | GEOGRAPHY OF NORTH AMERICA    |
| GE342 | GEOGRAPHY OF ASIA             |
| GE343 | GEOGRAPHY OF RUSSIA           |
| GE344 | GEOGRAPHY OF AFRICA           |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST  |
| GE346 | GEOGRAPHY OF EUROPE           |
| GE347 | GEOGRAPHY OF LATIN AMERICA    |
| GE491 | MILITARY GEOGRAPHY            |
| HI392 | AMERICAN HISTORICAL MEMORY    |
| LW472 | CRIMINAL LAW                  |
| LW481 | INTERNATIONAL LAW             |
| LW488 | BUSINESS LAW                  |
| MA367 | MATH FOR THE SOCIAL SCIENCES  |
| MA376 | APPLIED STATISTICS            |
| MA388 | SABERMETRICS                  |
| PL361 | RESEARCH METHODS I            |
| PL372 | MILITARY FAMILIES             |
| PL373 | LIFE CYCLE & HUMAN DEVEL      |
| PL383 | SOCIAL PSYCHOLOGY             |
| PL387 | FOUNDATIONS OF COUNSELING     |
| PL392 | COGNITIVE PSYCHOLOGY          |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM   |
| PL398 | LEADERSHIP THEORY & PRACTICE  |
| PL482 | ARMED FORCES AND SOCIETY      |
| PY320 | ETHICS                        |
| PY325 | MILITARY ETHICS               |
| PY329 | TOPICS IN ETHICS              |
| SE370 | COMPUTER AIDED SYSTEMS ENG    |
| SE375 | STATISTICS FOR ENGINEERS      |
| SE385 | DECISION ANALYSIS             |
| SS364 | GAME THEORY                   |
| SS368 | ECONOMETRICS I                |
| SS380 | LABOR ECONOMICS               |
| SS382 | INTERMEDIATE MICROECONOMICS   |
| SS385 | HISTORY OF ECONOMICS          |
| SS387 | PUBLIC ECONOMICS              |
| SS388 | INTERMEDIATE MACROECONOMICS   |
| SS390 | BEHAVIORAL ECONOMICS          |
| SS394 | FUND. FINANCIAL DATA ANALYSIS |
| SS461 | LEADERSHIP ECONOMICS          |
| SS462 | ECONOMIC GROWTH & DEVELOPMENT |
| SS463 | INVESTMENT THEORY & APPL      |
| SS470 | MONEY AND BANKING             |
| SS494 | PRINCIPLES OF FINANCE         |
| WS370 | US STRATEGY AND POLICY        |
| AND   |                               |

---

**Integrative Experience for the Major** Choose 1 of 1

|       |                      |
|-------|----------------------|
| MG421 | STRATEGIC MANAGEMENT |
|-------|----------------------|

---

**AND**

**Science Depth**

Cadets in this major may take any approved choice to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |               |
|-------|---------------|
| MG379 | LEADING TEAMS |
|-------|---------------|

**2029 Management Major w/ Honors Curriculum**

| Code  | Short Description    | Description                | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------|----------------------------|------------------------|--------------|--------------|
| MGN0H | Management w/ Honors | Management Major w/ Honors | Management w/ Honors   | 2            | 0            |

**2029 Management Major w/ Honors Tracks**

| Subject Area            | Description               |
|-------------------------|---------------------------|
| <b>Required Courses</b> | Choose 2 of 2             |
| PL497                   | SEMINAR IN BEHAVIORAL SCI |
| PL498                   | ADV STUDY-BEHAVIOR SCI    |
| <b>AND</b>              |                           |

Complete the requirements for the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## Chemical and Biological Science and Engineering

### 2029 Bioengineering Minor Curriculum

| Code  | Short Description    | Description          | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------|----------------------|------------------------|--------------|--------------|
| BEN1N | Bioengineering Minor | Bioengineering Minor | Bioengineering Minor   | 0            | 5            |

### 2029 Bioengineering Minor Tracks

| Subject Area                                                                                                                                                    | Description                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                         | Choose 3 of 4                  |
| CH300                                                                                                                                                           | BIOMEDICAL ENGINEERING         |
| CH350                                                                                                                                                           | BIOPROCESS ENGINEERING         |
| CH450                                                                                                                                                           | BIOENGINEERING MDLING & ANLSIS |
| XE310                                                                                                                                                           | INTRO TO BIOMECH ENGINEERING   |
| <b>AND</b>                                                                                                                                                      |                                |
| <b>Bioengineering Minor Electives</b>                                                                                                                           | Choose 2 of 21                 |
| Prerequisites must be satisfied. Additional electives are available on approval of the program director and must meet a minimum of 3.0 credit hours per course. |                                |
| CH362                                                                                                                                                           | MASS & ENERGY BALANCES         |
| CH364                                                                                                                                                           | CHEMICAL REACTION ENGINEERING  |
| CH387                                                                                                                                                           | HUMAN PHYSIOLOGY               |
| CH471                                                                                                                                                           | POLYMER CHEMISTRY              |
| CH473                                                                                                                                                           | BIOCHEMISTRY                   |
| CH479                                                                                                                                                           | METHODS & APPS OF BIOTECH      |
| CH489                                                                                                                                                           | INDIVIDUAL RESEARCH I          |
| EE381                                                                                                                                                           | SIGNALS AND SYSTEMS            |
| EE489                                                                                                                                                           | ADV IND STUDY IN ELECT ENGR    |
| EV396                                                                                                                                                           | ENVIRONMENTAL BIOLOGICAL SYS   |
| EV402                                                                                                                                                           | BIOCHEMICAL TREATMENT          |
| EV489A                                                                                                                                                          | ADVANCED INDIVIDUAL STUDY I    |
| KN360                                                                                                                                                           | BIOMECHANICS OF HUMAN MOVEMENT |
| KN493                                                                                                                                                           | INDIV RESEARCH IN KINESIOLOGY  |
| MA391                                                                                                                                                           | MATHEMATICAL MODELING          |
| MA477                                                                                                                                                           | THEORY & APPL OF DATA SCIENCE  |
| MA486                                                                                                                                                           | MATHEMATICAL COMPUTATION       |
| MA489                                                                                                                                                           | ADV INDIV STUDY IN MATH        |
| ME362                                                                                                                                                           | FLUID MECHANICS                |
| ME489                                                                                                                                                           | ADV STUDY IN MECH ENGRNG       |
| XE475                                                                                                                                                           | MECHATRONICS                   |

### 2029 Biology Major Curriculum

| Code | Short Description | Description | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------|-------------|------------------------|--------------|--------------|
|------|-------------------|-------------|------------------------|--------------|--------------|

BIOO Biology

Biology Major

Biology

11

2

## 2029 Biology Major Tracks

| Subject Area                                                                             | Description                    |
|------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                  | Choose 7 of 7                  |
| CH383                                                                                    | ORGANIC CHEMISTRY I            |
| CH384                                                                                    | ORGANIC CHEMISTRY II           |
| CH385                                                                                    | INTRODUCTION TO CELL BIOLOGY   |
| CH387                                                                                    | HUMAN PHYSIOLOGY               |
| CH388                                                                                    | GENETICS                       |
| CH457                                                                                    | MICROBIOLOGY                   |
| CH473                                                                                    | BIOCHEMISTRY                   |
| <b>AND</b>                                                                               |                                |
| <b>Electives</b>                                                                         | Choose 2 of 14                 |
| CH362                                                                                    | MASS & ENERGY BALANCES         |
| CH371                                                                                    | INTRO TO ANALYTICAL CHEM       |
| CH376                                                                                    | ADV. TECH IN LIGHT MICROSCOPY  |
| CH399                                                                                    | TOPICS IN CHEM/LS/CHMENG       |
| CH450                                                                                    | BIOENGINEERING MDLING & ANLSIS |
| CH460                                                                                    | HUMAN ANATOMY                  |
| CH471                                                                                    | POLYMER CHEMISTRY              |
| CH474                                                                                    | INSTRU METHODS OF ANALYSIS     |
| CH481                                                                                    | PHYSICAL CHEMISTRY I           |
| CH499                                                                                    | TOPICS IN CHEM/LS/CHMENG W/LAB |
| GE341                                                                                    | GEOGRAPHY OF NORTH AMERICA     |
| KN355                                                                                    | FUNCTIONAL ANATOMY             |
| NE474                                                                                    | RADIOLOGICAL SAFETY            |
| PH365                                                                                    | MODERN PHYSICS                 |
| PL390                                                                                    | BIOLOGICAL PSYCHOLOGY          |
| <b>AND</b>                                                                               |                                |
| <b>Complementary Support Courses</b>                                                     | Choose 3 of 6                  |
| The CSCs will provide foundational math and science support for cadets in this major.    |                                |
| CH102                                                                                    | GENERAL CHEMISTRY II           |
| MA204                                                                                    | CALCULUS I AND II              |
| MA205                                                                                    | CALCULUS II                    |
| MA255                                                                                    | ADV MULTIVARIABLE CALCULUS     |
| PH202                                                                                    | PHYSICS II                     |
| PH252                                                                                    | ADVANCED PHYSICS II            |
| <b>AND</b>                                                                               |                                |
| <b>Integrative Experience for the Major</b>                                              | Choose 1 of 1                  |
| CH479                                                                                    | METHODS & APPS OF BIOTECH      |
| <b>AND</b>                                                                               |                                |
| <b>Science Depth</b>                                                                     |                                |
| Cadets in this major must take CH375 to satisfy the Science Depth requirement.           |                                |
| CH375                                                                                    | INTRODUCTION TO BIOLOGY        |
| <b>STEM Depth</b>                                                                        |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.          |                                |
| CY305                                                                                    | CYBER FOUNDATIONS              |
| CY355                                                                                    | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                               |                                |
| <b>Curriculum Requirements</b>                                                           |                                |
| This section describes how cadets in this major satisfy various curriculum requirements. |                                |
| <b>IT/CYBER Requirement</b>                                                              |                                |



Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may take any approved core engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

CH479

METHODS &amp; APPS OF BIOTECH

## 2029 Biology Major w/ Honors Curriculum

| Code  | Short Description | Description             | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------|-------------------------|------------------------|--------------|--------------|
| BIO0H | Biology w/ Honors | Biology Major w/ Honors | Biology w/ Honors      | 2            | 0            |

### 2029 Biology Major w/ Honors Tracks

| Subject Area            | Description            |
|-------------------------|------------------------|
| <b>Required Courses</b> | Choose 2 of 2          |
| CH489                   | INDIVIDUAL RESEARCH I  |
| CH490                   | INDIVIDUAL RESEARCH II |
| <b>AND</b>              |                        |

Complete the requirements of the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Chemical Engineering Major Curriculum

| Code | Short Description    | Description                | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------|----------------------------|------------------------|--------------|--------------|
| CEN2 | Chemical Engineering | Chemical Engineering Major | Chemical Engineering   | 16           | 3            |

### 2029 Chemical Engineering Major Tracks

| Subject Area            | Description                    |
|-------------------------|--------------------------------|
| <b>Required Courses</b> | Choose 12 of 12                |
| CH362                   | MASS & ENERGY BALANCES         |
| CH363                   | SEPARATION PROCESSES           |
| CH364                   | CHEMICAL REACTION ENGINEERING  |
| CH365                   | CHEMICAL ENG THERMODYNAMICS    |
| CH367                   | INTRO / AUTOMATIC PROC CONTROL |

|                                                                                                                                                                                    |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| CH383                                                                                                                                                                              | ORGANIC CHEMISTRY I            |
| CH400                                                                                                                                                                              | CHEM ENG PROFESSIONAL PRACTICE |
| CH459                                                                                                                                                                              | CHEM ENGR LABORATORY           |
| CH485                                                                                                                                                                              | HEAT AND MASS TRANSFER         |
| MC300                                                                                                                                                                              | FUND OF ENGR MECH AND DESIGN   |
| ME301                                                                                                                                                                              | THERMODYNAMICS                 |
| ME362                                                                                                                                                                              | FLUID MECHANICS                |
| <b>AND</b>                                                                                                                                                                         |                                |
| <b>Electives</b>                                                                                                                                                                   | Choose 3 of 20                 |
| Prerequisites must be satisfied. Additional electives are available on approval of the program director and must meet minimum engineering content of 3.0 credit hours per course.  |                                |
| CE364                                                                                                                                                                              | MECHANICS OF MATERIALS         |
| CH300                                                                                                                                                                              | BIOMEDICAL ENGINEERING         |
| CH350                                                                                                                                                                              | BIOPROCESS ENGINEERING         |
| CH450                                                                                                                                                                              | BIOENGINEERING MDLING & ANLSIS |
| EE360                                                                                                                                                                              | FUNDAMENTALS OF DIGITAL LOGIC  |
| EE377                                                                                                                                                                              | ELECTRICAL POWER ENGRNG        |
| EM411                                                                                                                                                                              | PROJECT MANAGEMENT             |
| EM420                                                                                                                                                                              | PRODUCTION OPERATIONS MGMT     |
| EM481                                                                                                                                                                              | SYSTEMS SIMULATION             |
| ME306                                                                                                                                                                              | DYNAMICS                       |
| ME380                                                                                                                                                                              | ENGINEERING MATERIALS          |
| ME472                                                                                                                                                                              | ENERGY CONVERSION SYSTEMS      |
| ME480                                                                                                                                                                              | HEAT TRANSFER                  |
| ME491                                                                                                                                                                              | AUTOMOTIVE POWERPLANTS         |
| NE300                                                                                                                                                                              | FUNDAMENTALS OF NUCLEAR ENGR   |
| NE350                                                                                                                                                                              | RADIOLOGICAL ENGR DESIGN       |
| NE450                                                                                                                                                                              | NUCLEAR WEAPONS EFFECTS        |
| SE301                                                                                                                                                                              | FNDTN ENGIN DSGN & SYS MGMT    |
| SM484                                                                                                                                                                              | SYSTEM DYNAMICS SIMULATION     |
| XE475                                                                                                                                                                              | MECHATRONICS                   |
| <b>AND</b>                                                                                                                                                                         |                                |
| <b>Complementary Support Courses</b>                                                                                                                                               | Choose 3 of 5                  |
| The following courses provide foundational math and applied science for cadets in the chemical engineering major. Cadets who complete MA255 will enroll in MA365 in lieu of MA364. |                                |
| EE301                                                                                                                                                                              | FUNDAMENTALS OF ELEC ENGIN     |
| MA364                                                                                                                                                                              | ENGINEERING MATHEMATICS        |
| MA365                                                                                                                                                                              | ADV MATH FOR ENGRS/SCIENTISTS  |
| PH202                                                                                                                                                                              | PHYSICS II                     |
| PH252                                                                                                                                                                              | ADVANCED PHYSICS II            |
| <b>AND</b>                                                                                                                                                                         |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                        | Choose 1 of 1                  |
| CH402                                                                                                                                                                              | CHEM ENG PROCESS DESIGN        |
| <b>AND</b>                                                                                                                                                                         |                                |
| <b>Science Depth</b>                                                                                                                                                               |                                |
| Cadets in this major must take CH102 to satisfy the Science Depth requirement.                                                                                                     |                                |
| CH102                                                                                                                                                                              | GENERAL CHEMISTRY II           |
| <b>STEM Depth</b>                                                                                                                                                                  |                                |
| Cadets in this major must take MA204, MA205 or MA255 to satisfy the STEM Depth requirement                                                                                         |                                |
| MA204                                                                                                                                                                              | CALCULUS I AND II              |
| MA205                                                                                                                                                                              | CALCULUS II                    |
| MA255                                                                                                                                                                              | ADV MULTIVARIABLE CALCULUS     |
| <b>AND</b>                                                                                                                                                                         |                                |

**Curriculum Requirements**

This section describes how cadets in this major will satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the core curriculum, EE301 and CH367.

CH367

INTRO / AUTOMATIC PROC CONTROL

EE301

FUNDAMENTALS OF ELEC ENGIN

**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

CH459

CHEM ENGR LABORATORY

**2029 Chemical Engineering Major w/ Honors Curriculum**

| Code  | Short Description              | Description                          | Transcript Description         | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------|--------------------------------------|--------------------------------|--------------|--------------|
| CEN2H | Chemical Engineering w/ Honors | Chemical Engineering Major w/ Honors | Chemical Engineering w/ Honors | 1            | 0            |

**2029 Chemical Engineering Major w/ Honors Tracks**

| Subject Area           | Description           |
|------------------------|-----------------------|
| <b>Required Course</b> | Choose 1 of 1         |
| CH489                  | INDIVIDUAL RESEARCH I |
| <b>AND</b>             |                       |

Complete the requirements of the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

**2029 Chemical Engineering Studies Major Curriculum**

| Code | Short Description            | Description                        | Transcript Description       | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------|------------------------------------|------------------------------|--------------|--------------|
| CES2 | Chemical Engineering Studies | Chemical Engineering Studies Major | Chemical Engineering Studies | 13           | 3            |

**2029 Chemical Engineering Studies Major Tracks**

| Subject Area            | Description            |
|-------------------------|------------------------|
| <b>Required Courses</b> | Choose 9 of 9          |
| CH362                   | MASS & ENERGY BALANCES |

|       |                                |
|-------|--------------------------------|
| CH363 | SEPARATION PROCESSES           |
| CH364 | CHEMICAL REACTION ENGINEERING  |
| CH365 | CHEMICAL ENG THERMODYNAMICS    |
| CH367 | INTRO / AUTOMATIC PROC CONTROL |
| CH383 | ORGANIC CHEMISTRY I            |
| MC300 | FUND OF ENGR MECH AND DESIGN   |
| ME301 | THERMODYNAMICS                 |
| ME362 | FLUID MECHANICS                |

**AND****Electives**

Choose 3 of 22

Prerequisites must be satisfied. Additional electives are available on approval of the program director and must meet minimum engineering content of 3.0 credit hours per course.

|       |                                |
|-------|--------------------------------|
| CE364 | MECHANICS OF MATERIALS         |
| CH300 | BIOMEDICAL ENGINEERING         |
| CH350 | BIOPROCESS ENGINEERING         |
| CH450 | BIOENGINEERING MDLING & ANLSIS |
| CH459 | CHEM ENGR LABORATORY           |
| CH485 | HEAT AND MASS TRANSFER         |
| EE360 | FUNDAMENTALS OF DIGITAL LOGIC  |
| EE377 | ELECTRICAL POWER ENGRNG        |
| EM411 | PROJECT MANAGEMENT             |
| EM420 | PRODUCTION OPERATIONS MGMT     |
| EM481 | SYSTEMS SIMULATION             |
| ME306 | DYNAMICS                       |
| ME380 | ENGINEERING MATERIALS          |
| ME472 | ENERGY CONVERSION SYSTEMS      |
| ME480 | HEAT TRANSFER                  |
| ME491 | AUTOMOTIVE POWERPLANTS         |
| NE300 | FUNDAMENTALS OF NUCLEAR ENGR   |
| NE350 | RADIOLOGICAL ENGR DESIGN       |
| NE450 | NUCLEAR WEAPONS EFFECTS        |
| SE301 | FNDTN ENGIN DSGN & SYS MGMT    |
| SM484 | SYSTEM DYNAMICS SIMULATION     |
| XE475 | MECHATRONICS                   |

**AND****Complementary Support Courses**

Choose 3 of 5

The following courses provide foundational math and applied science for cadets in the chemical engineering major. Cadets who complete MA255 will enroll in MA365 in lieu of MA364.

|       |                               |
|-------|-------------------------------|
| EE301 | FUNDAMENTALS OF ELEC ENGIN    |
| MA364 | ENGINEERING MATHEMATICS       |
| MA365 | ADV MATH FOR ENGRS/SCIENTISTS |
| PH202 | PHYSICS II                    |
| PH252 | ADVANCED PHYSICS II           |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|       |                         |
|-------|-------------------------|
| CH402 | CHEM ENG PROCESS DESIGN |
|-------|-------------------------|

**AND****Science Depth**

Cadets in this major must take CH102 to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
|-------|----------------------|

**STEM Depth**

Cadets in this major must take MA204, MA205 or MA255 to satisfy the STEM Depth requirement

|       |                   |
|-------|-------------------|
| MA204 | CALCULUS I AND II |
|-------|-------------------|

|            |                            |
|------------|----------------------------|
| MA205      | CALCULUS II                |
| MA255      | ADV MULTIVARIABLE CALCULUS |
| <b>AND</b> |                            |

**Curriculum Requirements**

This section describes how cadets in this major will satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the core curriculum, EE301 and CH367.

|       |                                |
|-------|--------------------------------|
| CH367 | INTRO / AUTOMATIC PROC CONTROL |
| EE301 | FUNDAMENTALS OF ELEC ENGIN     |

**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                             |
|-------|-----------------------------|
| CH365 | CHEMICAL ENG THERMODYNAMICS |
|-------|-----------------------------|

## 2029 Chemical Engineering Studies Major Remarks The is the alternate major for the Chemical Engineering Major.

### 2029 Chemistry Major Curriculum

| Code | Short Description | Description     | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------|-----------------|------------------------|--------------|--------------|
| CHM2 | Chemistry         | Chemistry Major | Chemistry              | 12           | 1            |

### 2029 Chemistry Major Tracks

| Subject Area                                                                                                                     | Description                |
|----------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Required Courses</b>                                                                                                          | Choose 9 of 9              |
| CH371                                                                                                                            | INTRO TO ANALYTICAL CHEM   |
| CH383                                                                                                                            | ORGANIC CHEMISTRY I        |
| CH384                                                                                                                            | ORGANIC CHEMISTRY II       |
| CH471                                                                                                                            | POLYMER CHEMISTRY          |
| CH472                                                                                                                            | INORGANIC CHEMISTRY        |
| CH473                                                                                                                            | BIOCHEMISTRY               |
| CH474                                                                                                                            | INSTRU METHODS OF ANALYSIS |
| CH481                                                                                                                            | PHYSICAL CHEMISTRY I       |
| CH482                                                                                                                            | PHYSICAL CHEMISTRY II      |
| <b>AND</b>                                                                                                                       |                            |
| <b>Complementary Support Courses</b>                                                                                             |                            |
| Cadets must take MA204/MA205/MA255 and PH202 and an additional elective to satisfy the Complementary Support Course requirement. |                            |
| <b>Complementary Support Courses - Required</b>                                                                                  | Choose 2 of 4              |
| MA204                                                                                                                            | CALCULUS I AND II          |
| MA205                                                                                                                            | CALCULUS II                |
| MA255                                                                                                                            | ADV MULTIVARIABLE CALCULUS |
| PH202                                                                                                                            | PHYSICS II                 |

**AND****Complementary Support Course - Elective**

Choose 1 of 15

Electives from other programs and departments may be chosen upon approval of the Chemistry Program Director. A Cadet cannot double count a course from this list to satisfy both the CSC requirement and the requirements for Science Depth or STEM Depth or the Core Engineering Sequence.

|       |                                |
|-------|--------------------------------|
| CH362 | MASS & ENERGY BALANCES         |
| CH375 | ADVANCED BIOLOGY               |
| CH385 | INTRODUCTION TO CELL BIOLOGY   |
| CH450 | BIOENGINEERING MDLING & ANLSIS |
| CS393 | DATABASE SYSTEMS               |
| EE486 | SOLID STATE ELECTRONICS        |
| EV350 | ENVIRONMNTL ENGR TECHNOLOGIES  |
| MA364 | ENGINEERING MATHEMATICS        |
| MA365 | ADV MATH FOR ENGRS/SCIENTISTS  |
| MA376 | APPLIED STATISTICS             |
| MC300 | FUND OF ENGR MECH AND DESIGN   |
| ME380 | ENGINEERING MATERIALS          |
| PH365 | MODERN PHYSICS                 |
| PL390 | BIOLOGICAL PSYCHOLOGY          |
| XS391 | PRIN & APPL OF ENV CHEM        |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|       |                               |
|-------|-------------------------------|
| CH487 | ADVANCED CHEMISTRY LABORATORY |
|-------|-------------------------------|

**AND****Science Depth**

Cadets in this major must take CH102 as the Science Depth course.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
|-------|----------------------|

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements****IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                               |
|-------|-------------------------------|
| CH487 | ADVANCED CHEMISTRY LABORATORY |
|-------|-------------------------------|

## 2029 Chemistry Major w/ ACS Certification Curriculum

| Code  | Short Description              | Description                          | Transcript Description         | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------|--------------------------------------|--------------------------------|--------------|--------------|
| CHM2A | Chemistry w/ ACS Certification | Chemistry Major w/ ACS Certification | Chemistry w/ ACS Certification | 0            | 2            |

### 2029 Chemistry Major w/ ACS Certification Tracks

| Subject Area                | Description                                                                        |
|-----------------------------|------------------------------------------------------------------------------------|
| <b>Research Requirement</b> | Choose 2 of 10                                                                     |
|                             | Cadets must take at least 4.5 credit hours of research from the following courses. |
| CH289                       | INTRODUCTION TO RESEARCH I                                                         |
| CH290                       | INTRODUCTION TO RESEARCH II                                                        |
| CH389                       | ADVANCED LAB PROJECTS I                                                            |
| CH390                       | ADVANCED LAB PROJECTS II                                                           |
| CH391                       | ADV LAB PROJECTS III                                                               |
| CH392                       | ADVANCED LAB PROJECTS IV                                                           |
| CH489                       | INDIVIDUAL RESEARCH I                                                              |
| CH490                       | INDIVIDUAL RESEARCH II                                                             |
| CH491                       | ADVANCED INDIVIDUAL STUDY I                                                        |
| CH492                       | ADVANCED INDIVIDUAL STUDY II                                                       |

## 2029 Chemistry Major w/ ACS Certification (Honors) Curriculum

| Code  | Short Description         | Description                                   | Transcript Description    | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------|-----------------------------------------------|---------------------------|--------------|--------------|
| CHM2H | Chemistry w/ ACS (Honors) | Chemistry Major w/ ACS Certification (Honors) | Chemistry w/ ACS (Honors) | 2            | 0            |

### 2029 Chemistry Major w/ ACS Certification (Honors) Tracks

| Subject Area            | Description            |
|-------------------------|------------------------|
| <b>Required Courses</b> | Choose 2 of 2          |
| CH489                   | INDIVIDUAL RESEARCH I  |
| CH490                   | INDIVIDUAL RESEARCH II |
| <b>AND</b>              |                        |

Complete the requirements of the Chemistry major as shown above, take above two research courses, and attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## Chemistry & Life Science

No Major found



## Civil and Environmental Engineering

### 2029 Civil Engineering Major Curriculum

| Code | Short Description | Description             | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------|-------------------------|------------------------|-----------------|-----------------|
| CVN3 | Civil Engineering | Civil Engineering Major | Civil Engineering      | 15              | 3               |

### 2029 Civil Engineering Major Tracks

| Subject Area                       | Description                    |
|------------------------------------|--------------------------------|
| <b>Required Courses</b>            | Choose 11 of 11                |
| CE201                              | INTRO TO CIVIL ENGINEERING     |
| CE350                              | INFRASTRUCTURE ENGINEERING     |
| CE364                              | MECHANICS OF MATERIALS         |
| CE371                              | SOIL MECHANICS/FNDTN ENGNRG    |
| CE380                              | HYDROLOGY/HYDRAULIC DESIGN     |
| CE401                              | CIV ENG PROF PRAC AND APP      |
| CE403                              | STRUCTURAL ANALYSIS            |
| CE404                              | DSN STEEL AND WOOD STRUCTURES  |
| CE450                              | CONSTRUCTION MANAGEMENT        |
| CE483                              | DSN CONC AND MASON STRUCTURES  |
| MC300                              | FUND OF ENGR MECH AND DESIGN   |
| <b>AND</b>                         |                                |
| <b>Civil Engineering Electives</b> | Choose 2 of 24                 |
| CE389                              | INDEPENDENT STUDY IN CE (3CR)  |
| CE389A                             | INDEPENDENT STUDY IN CE (3CR)  |
| CE399                              | CIVIL ENG PRAC-FIELD ENG       |
| CE472                              | ADV SOIL MECHNCS/FNDTN ENGNRG  |
| CE489                              | ADV IND STUDY CIVIL ENGRING    |
| CE489A                             | ADV IND STUDY CIVIL ENGRING    |
| CE490                              | TOPICS IN CIVIL ENGINEERING    |
| CE491                              | ADV STRUCTURAL ANALYSIS        |
| CE495                              | TRANSPORTATION ENGINEERING     |
| EE301                              | FUNDAMENTALS OF ELEC ENGIN     |
| EV350                              | ENVIRONMNTL ENGR TECHNOLOGIES  |
| EV394                              | HYDROGEOLOGY/HYDRAULIC SYSTEMS |
| EV401                              | PHYS & CHEM TREATMENT          |
| EV481                              | WATER RESOURCES PLAN & DESIGN  |
| GE310                              | GEOG INFORMATION SYSTEMS       |
| GE380                              | SURVEYING                      |
| ME202                              | COMPUTER AIDED DESIGN          |
| ME301                              | THERMODYNAMICS                 |
| ME306                              | DYNAMICS                       |
| ME380                              | ENGINEERING MATERIALS          |
| ME472                              | ENERGY CONVERSION SYSTEMS      |
| ME486                              | VIBRATION ENGINEERING          |
| ME491                              | AUTOMOTIVE POWERPLANTS         |
| XE490                              | SUSTAINABILITY ENGINEERING     |
| <b>AND</b>                         |                                |

**Complementary Support Course - Required - Fluid Mechanics**

Choose 1 of 1

The CSCs for this major include three required courses which provide foundational math, science and engineering support for cadets in this major.

ME362

FLUID MECHANICS

**AND****Complementary Support Course - Required - Math**

Choose 2 of 5

MA204

CALCULUS I AND II

MA205

CALCULUS II

MA255

ADV MULTIVARIABLE CALCULUS

MA364

ENGINEERING MATHEMATICS

MA365

ADV MATH FOR ENGRS/SCIENTISTS

**AND****Integrative Experience for the Major**

Choose 2 of 2

CE493

CIV ENG CAPSTONE DESIGN I

CE494

CIV ENG CAPSTONE DESIGN II

**AND****Science Depth**

Cadets in this major must take PH206 or PH275 to satisfy the Science Depth requirement.

PH202

PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take CY305 to satisfy the STEM Depth requirement.

CY305

CYBER FOUNDATIONS

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305.

CY305

CYBER FOUNDATIONS

**Core Engineering Sequence**

Cadets in this major will have the civil engineering sequence embedded in their major.

CE350

INFRASTRUCTURE ENGINEERING

CE450

CONSTRUCTION MANAGEMENT

MC300

FUND OF ENGR MECH AND DESIGN

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

CE401

CIV ENG PROF PRAC AND APP

**2029 Civil Engineering Studies Curriculum**

| Code | Short Description         | Description               | Transcript Description    | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------|---------------------------|---------------------------|--------------|--------------|
| CNG2 | Civil Engineering Studies | Civil Engineering Studies | Civil Engineering Studies | 5            | 9            |

## 2029 Civil Engineering Studies Tracks

| Subject Area                                                                                                                                                                          | Description                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                               | Choose 2 of 2                  |
| CE201                                                                                                                                                                                 | INTRO TO CIVIL ENGINEERING     |
| CE401                                                                                                                                                                                 | CIV ENG PROF PRAC AND APP      |
| <b>AND</b>                                                                                                                                                                            |                                |
| <b>Required Course Electives</b>                                                                                                                                                      | Choose 4 of 6                  |
| CE364                                                                                                                                                                                 | MECHANICS OF MATERIALS         |
| CE371                                                                                                                                                                                 | SOIL MECHANICS/FNDTN ENGNRG    |
| CE380                                                                                                                                                                                 | HYDROLOGY/HYDRAULIC DESIGN     |
| CE403                                                                                                                                                                                 | STRUCTURAL ANALYSIS            |
| CE404                                                                                                                                                                                 | DSN STEEL STRUCTURES           |
| CE483                                                                                                                                                                                 | DSN CONC STRUCTURES            |
| <b>AND</b>                                                                                                                                                                            |                                |
| <b>Civil Engineering Electives</b>                                                                                                                                                    | Choose 2 of 24                 |
| CE389                                                                                                                                                                                 | INDEPENDENT STUDY IN CE (3CR)  |
| CE389A                                                                                                                                                                                | INDEPENDENT STUDY IN CE (3CR)  |
| CE399                                                                                                                                                                                 | CIVIL ENG PRAC-FIELD ENG       |
| CE472                                                                                                                                                                                 | ADV SOIL MECHNCS/FNDTN ENGRNG  |
| CE489                                                                                                                                                                                 | ADV IND STUDY CIVIL ENGRING    |
| CE489A                                                                                                                                                                                | ADV IND STUDY CIVIL ENGRING    |
| CE490                                                                                                                                                                                 | TOPICS IN CIVIL ENGINEERING    |
| CE491                                                                                                                                                                                 | ADV STRUCTURAL ANALYSIS        |
| CE495                                                                                                                                                                                 | TRANSPORTATION ENGINEERING     |
| EE301                                                                                                                                                                                 | FUNDAMENTALS OF ELEC ENGIN     |
| EV350                                                                                                                                                                                 | ENVIRONMNTL ENGR TECHNOLOGIES  |
| EV394                                                                                                                                                                                 | HYDROGEOLOGY/HYDRAULIC SYSTEMS |
| EV401                                                                                                                                                                                 | PHYS & CHEM TREATMENT          |
| EV481                                                                                                                                                                                 | WATER RESOURCES PLAN & DESIGN  |
| GE310                                                                                                                                                                                 | GEOG INFORMATION SYSTEMS       |
| GE380                                                                                                                                                                                 | SURVEYING                      |
| ME202                                                                                                                                                                                 | COMPUTER AIDED DESIGN          |
| ME301                                                                                                                                                                                 | THERMODYNAMICS                 |
| ME306                                                                                                                                                                                 | DYNAMICS                       |
| ME380                                                                                                                                                                                 | ENGINEERING MATERIALS          |
| ME472                                                                                                                                                                                 | ENERGY CONVERSION SYSTEMS      |
| ME486                                                                                                                                                                                 | VIBRATION ENGINEERING          |
| ME491                                                                                                                                                                                 | AUTOMOTIVE POWERPLANTS         |
| XE490                                                                                                                                                                                 | SUSTAINABILITY ENGINEERING     |
| <b>AND</b>                                                                                                                                                                            |                                |
| <b>Complementary Support Course - Required - Fluid Mechanics</b>                                                                                                                      | Choose 1 of 1                  |
| The CSCs for this major are divided between two required and one elective courses. The CSCs will provide foundational math, science and engineering support for cadets in this major. |                                |
| ME362                                                                                                                                                                                 | FLUID MECHANICS                |
| <b>AND</b>                                                                                                                                                                            |                                |
| <b>Complementary Support Course - Required - Math</b>                                                                                                                                 | Choose 1 of 2                  |
| For cadets in the advanced math program, MA255 will satisfy the math requirement while MA104 validation will count towards the core.                                                  |                                |
| MA255                                                                                                                                                                                 | ADV MULTIVARIABLE CALCULUS     |
| MA366                                                                                                                                                                                 | APPLIED ENGINEERING MATH       |

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Complementary Support Course - Elective</b>                                                                                                                               | Choose 1 of 6                 |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II          |
| CH275                                                                                                                                                                        | BIOLOGY                       |
| MA365                                                                                                                                                                        | ADV MATH FOR ENGRS/SCIENTISTS |
| MA371                                                                                                                                                                        | LINEAR ALGEBRA                |
| MA376                                                                                                                                                                        | APPLIED STATISTICS            |
| MA386                                                                                                                                                                        | INTRO TO NUMERICAL ANALYSIS   |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 2 of 2                 |
| CE493                                                                                                                                                                        | CIV ENG CAPSTONE DESIGN I     |
| CE494                                                                                                                                                                        | CIV ENG CAPSTONE DESIGN II    |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Science Depth</b>                                                                                                                                                         |                               |
| Cadets in this major must take PH206 or PH275 to satisfy the Science Depth requirement.                                                                                      |                               |
| PH202                                                                                                                                                                        | PHYSICS II                    |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take CY305 to satisfy the STEM Depth requirement.                                                                                                  |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Curriculum Requirements</b>                                                                                                                                               |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305.                                 |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major are required to take the Infrastructure engineering sequence.                                                                                           |                               |
| CE350                                                                                                                                                                        | INFRASTRUCTURE ENGINEERING    |
| CE450                                                                                                                                                                        | CONSTRUCTION MANAGEMENT       |
| MC300                                                                                                                                                                        | FUND OF ENGR MECH AND DESIGN  |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| CE401                                                                                                                                                                        | CIV ENG PROF PRAC AND APP     |

## 2029 Civil Engineering Studies Remarks Alternate Major for Civil Engineering.

### 2029 Civil Engineering w/ Honors Curriculum

| Code  | Short Description           | Description                 | Transcript Description      | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------|-----------------------------|-----------------------------|--------------|--------------|
| CVN3H | Civil Engineering w/ Honors | Civil Engineering w/ Honors | Civil Engineering w/ Honors | 0            | 1            |

### 2029 Civil Engineering w/ Honors Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

**Coursework Requirements**

Complete 42 total academic courses, which requires one additional course than the requirements for the major as shown above. The additional course can be selected from the Math and Basic Science Electives, the Civil Engineering Electives, or as approved by the Civil Engineering Program Director.

**AND****Requirements for Graduation**

To graduate with Individual Honors a cadet must submit an individual paper or report which can be any of the following:

- (1) A project report for an individual CE489 Advanced Study Project.
- (2) An individual paper written for a regional or national student paper competition.
- (3) An individual paper, suitable for publication or presentation at a professional conference, drawn from one of the following sources:

..... A CE489 Advanced Study Project.

..... A Senior Design Project.

..... An engineering-related Academic Individual Advanced Development (AIAD).

..... An experience relevant to the cadet's program of study and approved by the associated Program Director.

Cadets desiring to aspire to Individual Honors will coordinate with their Department Academic Counselor to develop a plan no later than the end of second class year for completing the individual paper or report. The Department Academic Counselor will certify the completion of the significant individual paper component of the Academic Honors Program.

**AND****Grade Requirements**

Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

**AND****Examination Requirement**

Pass the Fundamentals of Engineering Examination in the final semester of study.

## 2029 Environmental Engineering Major Curriculum

| Code | Short Description         | Description                     | Transcript Description    | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|---------------------------|---------------------------------|---------------------------|-----------------|-----------------|
| EVE2 | Environmental Engineering | Environmental Engineering Major | Environmental Engineering | 16              | 2               |

### 2029 Environmental Engineering Major Tracks

| Subject Area            | Description               |
|-------------------------|---------------------------|
| <b>Required Courses</b> | Choose 11 of 11           |
| CV311                   | THERMAL-FLUID SYSTEMS I   |
| EV201                   | INTRO TO ENV ENG & DESIGN |

|       |                                |
|-------|--------------------------------|
| EV301 | ENVIRONMENTAL SUSTAINABILITY   |
| EV394 | HYDROGEOLOGY/HYDRAULIC SYSTEMS |
| EV396 | ENVIRONMENTAL BIOLOGICAL SYS   |
| EV397 | AIR POLLUTION ENGINEERING      |
| EV401 | PHYS & CHEM TREATMENT          |
| EV402 | BIOCHEMICAL TREATMENT          |
| EV481 | WATER RESOURCES PLAN & DESIGN  |
| EV488 | SOLID & HAZ WASTE TREAT & REMD |
| XS391 | PRIN & APPL OF ENV CHEM        |

**AND****Environmental Engineering Field Electives** Choose 2 of 26

Electives selected must include a combined minimum of 6 Engineering Topics (ET) credits. Cadets who will graduate with honors must select EV489A as one of the two courses.

|        |                                |
|--------|--------------------------------|
| CE350  | INFRASTRUCTURE ENGINEERING     |
| CE364  | MECHANICS OF MATERIALS         |
| CE371  | SOIL MECHANICS/FNDTN ENGNRG    |
| CE380  | HYDROLOGY/HYDRAULIC DESIGN     |
| CE450  | CONSTRUCTION MANAGEMENT        |
| CE472  | ADV SOIL MECHNCS/FNDTN ENGRNG  |
| CH362  | MASS & ENERGY BALANCES         |
| CH363  | SEPARATION PROCESSES           |
| CH364  | CHEMICAL REACTION ENGINEERING  |
| EE301  | FUNDAMENTALS OF ELEC ENGIN     |
| EE377  | ELECTRICAL POWER ENGRNG        |
| EM381  | ENGINEERING ECONOMY            |
| EM411  | PROJECT MANAGEMENT             |
| EV489A | ADVANCED INDIVIDUAL STUDY I    |
| EV489B | ADVANCED INDIVIDUAL STUDY II   |
| GE310  | GEOG INFORMATION SYSTEMS       |
| GE380  | SURVEYING                      |
| GE498  | ADV GEOGRAPHIC INFORMATION SYS |
| MC300  | FUND OF ENGR MECH AND DESIGN   |
| ME202  | COMPUTER AIDED DESIGN          |
| ME380  | ENGINEERING MATERIALS          |
| ME472  | ENERGY CONVERSION SYSTEMS      |
| ME480  | HEAT TRANSFER                  |
| SE301  | FNDTN ENGIN DSGN & SYS MGMT    |
| SE385  | DECISION ANALYSIS              |
| XE442  | ALTERNATIVE ENERGY ENGINEERING |

**AND****Complementary Support Courses**

The following courses provide foundational math, engineering and applied science for cadets in the environmental engineering major.

**Complementary Support Course - Math** Choose 2 of 5

Cadets who take MA104 will take MA205 and MA364. Cadets who validate MA104 (counting towards the Math requirement) will take MA204 or MA255 and MA365 for the Complementary Support Course - Math requirement for this major.

|       |                               |
|-------|-------------------------------|
| MA204 | CALCULUS I AND II             |
| MA205 | CALCULUS II                   |
| MA255 | ADV MULTIVARIABLE CALCULUS    |
| MA364 | ENGINEERING MATHEMATICS       |
| MA365 | ADV MATH FOR ENGRS/SCIENTISTS |

**AND****Complementary Support Courses - Physics** Choose 1 of 3

Cadets must take either PH202, PH252, or PH275.

|       |                     |
|-------|---------------------|
| PH202 | PHYSICS II          |
| PH252 | ADVANCED PHYSICS II |
| PH275 | PHYSICS II: SPACE   |

**AND**

**Integrative Experience for the Major** Choose 2 of 2

|       |                        |
|-------|------------------------|
| EV490 | ENVIRON ENG DESIGN     |
| EV491 | ADV ENVIRON ENG DESIGN |

**AND**

#### Science Depth

Cadets in this major must take CH102 to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
|-------|----------------------|

#### STEM Depth

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND**

#### Curriculum Requirements

This section describes how cadets in this major will satisfy various curriculum requirements.

#### IT/CYBER Requirement

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the core curriculum and CY305 or CY355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

#### Core Engineering Sequence

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

#### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                              |
|-------|------------------------------|
| EV301 | ENVIRONMENTAL SUSTAINABILITY |
|-------|------------------------------|

## 2029 Environmental Engineering Major w/ Honors Curriculum

| Code  | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| EVE2H | Environmental Engineering w/ Honors | Environmental Engineering Major w/ Honors | Environmental Engineering w/ Honors | 0            | 0            |

## 2029 Environmental Engineering Major w/ Honors Tracks

| Subject Area              | Description                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Grade Requirements</b> | Complete the requirements of the major as shown above [including taking EV489A as one of two required Environmental Engineering Field Electives] and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major. |

## 2029 Environmental Engineering Studies Major Curriculum

| Code | Short Description                 | Description                                | Transcript Description            | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-----------------------------------|--------------------------------------------|-----------------------------------|-----------------|-----------------|
| EES2 | Environmental Engineering Studies | Environmental Engineering Studies<br>Major | Environmental Engineering Studies | 7               | 9               |

### 2029 Environmental Engineering Studies Major Tracks

| Subject Area                                      | Description                    |
|---------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                           | Choose 7 of 7                  |
| EV201                                             | INTRO TO ENV ENG & DESIGN      |
| EV301                                             | ENVIRONMENTAL SUSTAINABILITY   |
| EV350                                             | ENVIRONMNTL ENGR TECHNOLOGIES  |
| EV396                                             | ENVIRONMENTAL BIOLOGICAL SYS   |
| EV397                                             | AIR POLLUTION ENGINEERING      |
| EV401                                             | PHYS & CHEM TREATMENT          |
| XS391                                             | PRIN & APPL OF ENV CHEM        |
| <b>AND</b>                                        |                                |
| <b>Environmental Engineering Field Electives</b>  | Choose 5 of 16                 |
| CE350                                             | INFRASTRUCTURE ENGINEERING     |
| CE380                                             | HYDROLOGY/HYDRAULIC DESIGN     |
| CE450                                             | CONSTRUCTION MANAGEMENT        |
| CV311                                             | THERMAL-FLUID SYSTEMS I        |
| EE301                                             | FUNDAMENTALS OF ELEC ENGIN     |
| EM381                                             | ENGINEERING ECONOMY            |
| EV394                                             | HYDROGEOLOGY/HYDRAULIC SYSTEMS |
| EV402                                             | BIOCHEMICAL TREATMENT          |
| EV450                                             | ENV ENG FOR COMMUNITY DEVELOP  |
| EV481                                             | WATER RESOURCES PLAN & DESIGN  |
| EV488                                             | SOLID & HAZ WASTE TREAT & REMD |
| EV489A                                            | ADVANCED INDIVIDUAL STUDY I    |
| EV489B                                            | ADVANCED INDIVIDUAL STUDY II   |
| GE310                                             | GEOG INFORMATION SYSTEMS       |
| MC300                                             | FUND OF ENGR MECH AND DESIGN   |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Courses</b>              | Choose 3 of 12                 |
| Cadets must select three courses from this table. |                                |
| CE380                                             | HYDROLOGY/HYDRAULIC DESIGN     |
| CV311                                             | THERMAL-FLUID SYSTEMS I        |
| EE301                                             | FUNDAMENTALS OF ELEC ENGIN     |
| EM381                                             | ENGINEERING ECONOMY            |
| MA204                                             | CALCULUS I AND II              |
| MA205                                             | CALCULUS II                    |
| MA255                                             | ADV MULTIVARIABLE CALCULUS     |
| MA364                                             | ENGINEERING MATHEMATICS        |
| MA365                                             | ADV MATH FOR ENGRS/SCIENTISTS  |
| PH202                                             | PHYSICS II                     |



|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II           |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 2                 |
| EV450                                                                                                                                                                        | ENV ENG FOR COMMUNITY DEVELOP |
| EV481                                                                                                                                                                        | WATER RESOURCES PLAN & DESIGN |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Science Depth</b>                                                                                                                                                         |                               |
| Cadets in this major must take CH102 to satisfy the Science Depth requirement.                                                                                               |                               |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II          |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Curriculum Requirements</b>                                                                                                                                               |                               |
| This section describes how cadets in this major will satisfy various curriculum requirements.                                                                                |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the core curriculum and CY305 or CY355.                                                |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major will satisfy the core engineering requirement as part of their major courses.                                                                           |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| EV301                                                                                                                                                                        | ENVIRONMENTAL SUSTAINABILITY  |

## 2029 Environmental Engineering Studies Major Remarks Alternate major for Environmental Engineering.

Civil and Mechanical Engineering

No Major found

## Electrical Engineering and Computer Science

### 2029 Computer Science Major Curriculum

| Code | Short Description | Description            | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------|------------------------|------------------------|-----------------|-----------------|
| CSC1 | Computer Science  | Computer Science Major | Computer Science       | 13              | 4               |

### 2029 Computer Science Major Tracks

| Subject Area                                         | Description                    |
|------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                              | Choose 10 of 10                |
| CS380                                                | COMPUTER SYSTEMS & ORGANIZATN  |
| CS384                                                | DATA STRUCTURES                |
| CS385                                                | DESIGN & ANALYS-ALGORITHMS     |
| CS400                                                | CS PROFESSIONAL CONSIDERATIONS |
| CS403                                                | SOFTWARE TESTING & DEVELOPMENT |
| CS474                                                | INTRO TO THEORETICAL COMP SCI  |
| CS478                                                | PROGRAMMING LANGUAGES          |
| CS481                                                | OPERATING SYSTEMS              |
| CY300                                                | PROGRAMMING FUNDAMENTALS       |
| CY350                                                | COMPUTER NETWORKS              |
| <b>AND</b>                                           |                                |
| <b>Computer Science Major Electives</b>              | Choose 2 of 20                 |
| AI350                                                | DATA ENGINEERING               |
| AI450                                                | ENGINEERING AI                 |
| CS389                                                | IND STUDY IN COMPUTER SCIENCE  |
| CS389A                                               | IND STUDY IN COMPUTER SCIENCE  |
| CS393                                                | DATABASE SYSTEMS               |
| CS473                                                | COMPUTER GRAPHICS              |
| CS483                                                | DIGITAL FORENSICS              |
| CS484                                                | ADVANCED COMPUTER NETWORKS     |
| CS485                                                | SPEC TOPICS IN COMPUTER SCI    |
| CS486                                                | ARTIFICIAL INTELLIGENCE        |
| CY392                                                | DEFENSIVE CYBER OPERATIONS     |
| CY394                                                | CLOUD COMPUTING                |
| CY482                                                | OFFENSIVE CYBER OPERATIONS     |
| EE380                                                | EMBEDDED SYSTEMS DEVELOPMENT   |
| EE484                                                | FUNDAMENTALS OF ROBOTICS       |
| MA386                                                | INTRO TO NUMERICAL ANALYSIS    |
| MA394                                                | FUNDAMENTALS/NETWORK SCIENCE   |
| MA477                                                | THEORY & APPL OF DATA SCIENCE  |
| PL475                                                | HUMAN-COMPUTER INTERACTION     |
| XE492                                                | DISRUPTIVE INNOVATIONS         |
| <b>AND</b>                                           |                                |
| <b>Complementary Support Courses (CS Foundation)</b> | Choose 2 of 2                  |
| EE360                                                | FUNDAMENTALS OF DIGITAL LOGIC  |
| MA372                                                | INTRODUCTION TO DISCRETE MATH  |
| <b>AND</b>                                           |                                |

**Complementary Support Course  
(Breadth Elective)**

Choose 1 of 22

|       |                                |
|-------|--------------------------------|
| CY460 | CYBER POLICY, STRATEGY, & OPNS |
| CY465 | ORGANIZATIONAL SECURITY        |
| EE301 | FUNDAMENTALS OF ELEC ENGIN     |
| EM411 | PROJECT MANAGEMENT             |
| HI462 | THE HISTORY OF INNOVATION      |
| LW462 | CYBER LAW                      |
| MA205 | CALCULUS II                    |
| MA371 | LINEAR ALGEBRA                 |
| MA376 | APPLIED STATISTICS             |
| MA386 | INTRO TO NUMERICAL ANALYSIS    |
| MA391 | MATHEMATICAL MODELING          |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE   |
| MA464 | APPLIED ALGEBRA W/ CRYPTOLOGY  |
| MG379 | LEADING TEAMS                  |
| MG462 | ENTREPRENEURSHIP               |
| PL479 | LEADING CHANGING ORGANIZATIONS |
| PY305 | LOGICAL REASONING              |
| PY326 | ETHICS OF TECHNOLOGY           |
| PY350 | PHILOSOPHY OF SCIENCE          |
| PY355 | PHILOSOPHY OF MIND             |
| SE385 | DECISION ANALYSIS              |
| SS364 | GAME THEORY                    |

**AND****Integrative Experience for the Major**

Choose 2 of 2

|       |                              |
|-------|------------------------------|
| XE401 | INTEGRATIVE SYSTEM DESIGN I  |
| XE402 | INTEGRATIVE SYSTEM DESIGN II |

**AND****Science Depth**

Cadets in this major may take any of the approved options to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major will satisfy the STEM Depth requirement by taking CY355.

|       |                               |
|-------|-------------------------------|
| CY355 | CYBER FOUNDATIONS - COMPUTING |
|-------|-------------------------------|

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY355.

|       |                               |
|-------|-------------------------------|
| CY355 | CYBER FOUNDATIONS - COMPUTING |
|-------|-------------------------------|

**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                                |
|-------|--------------------------------|
| CS400 | CS PROFESSIONAL CONSIDERATIONS |
|-------|--------------------------------|

## 2029 Computer Science Major w/ Honors Curriculum

| Code  | Short Description          | Description                      | Transcript Description     | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------|----------------------------------|----------------------------|--------------|--------------|
| CSC1H | Computer Science w/ Honors | Computer Science Major w/ Honors | Computer Science w/ Honors | 0            | 1            |

## 2029 Computer Science Major w/ Honors Tracks

| Subject Area                             | Description                   |
|------------------------------------------|-------------------------------|
| <b>Computer Science Honors Electives</b> | Choose 1 of 18                |
| AI350                                    | DATA ENGINEERING              |
| AI450                                    | ENGINEERING AI                |
| CS473                                    | COMPUTER GRAPHICS             |
| CS483                                    | DIGITAL FORENSICS             |
| CS484                                    | ADVANCED COMPUTER NETWORKS    |
| CS485                                    | SPEC TOPICS IN COMPUTER SCI   |
| CS486                                    | ARTIFICIAL INTELLIGENCE       |
| CS489                                    | ADV IND STUDY COMPUTER SCI    |
| CS489A                                   | ADV IND STUDY COMPUTER SCI    |
| CY392                                    | DEFENSIVE CYBER OPERATIONS    |
| CY394                                    | CLOUD COMPUTING               |
| CY482                                    | OFFENSIVE CYBER OPERATIONS    |
| EE380                                    | EMBEDDED SYSTEMS DEVELOPMENT  |
| EE484                                    | FUNDAMENTALS OF ROBOTICS      |
| MA394                                    | FUNDAMENTALS/NETWORK SCIENCE  |
| MA477                                    | THEORY & APPL OF DATA SCIENCE |
| PL475                                    | HUMAN-COMPUTER INTERACTION    |
| XE492                                    | DISRUPTIVE INNOVATIONS        |

### AND

#### Research Requirement

Consists of both a written document and an oral presentation of a depth and quality suitable for submission to a professional conference.

The research will normally be accomplished as an extension of a project begun in the CS Honors Elective. The research must reflect individual effort, although it may build on an existing group project (especially the context of XE401/402).

Neither the project/research work nor the resulting paper and presentation need be completed during the same semester they are begun, but must be complete by the end of the TEE period of semester 8.

The project/research must be conducted under the supervision/mentorship of a member of the faculty, normally the instructor of the corresponding course. The final written document and oral presentation must be approved by both the research mentor and the Computer Science Program Director.

### AND

#### Grade Requirements

Complete the requirements of the major as shown above, attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Cyber Science Major Curriculum

| Code | Short Description | Description         | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------|---------------------|------------------------|-----------------|-----------------|
| CYS1 | Cyber Science     | Cyber Science Major | Cyber Science          | 11              | 6               |

## 2029 Cyber Science Major Tracks

| Subject Area                                              | Description                    |
|-----------------------------------------------------------|--------------------------------|
| <b>Cyber Foundations Required Courses</b>                 | Choose 7 of 7                  |
| CS380                                                     | COMPUTER SYSTEMS & ORGANIZATN  |
| CS384                                                     | DATA STRUCTURES                |
| CS400                                                     | PRO CONSIDERATIONS IN COMPUTIN |
| CY300                                                     | PROGRAMMING FUNDAMENTALS       |
| CY350                                                     | COMPUTER NETWORKS              |
| CY460                                                     | CYBER POLICY, STRATEGY, & OPNS |
| CY465                                                     | ORGANIZATIONAL SECURITY        |
| <b>AND</b>                                                |                                |
| <b>Cyber Operations Elective</b>                          | Choose 1 of 2                  |
| Cyber operations provides technical depth to the program. |                                |
| CY392                                                     | DEFENSIVE CYBER OPERATIONS     |
| CY482                                                     | OFFENSIVE CYBER OPERATIONS     |
| <b>AND</b>                                                |                                |
| <b>Cyber Electives</b>                                    | Choose 4 of 21                 |
| CS386                                                     | NEURAL NETWORKS                |
| CS393                                                     | DATABASE SYSTEMS               |
| CS403                                                     | SOFTWARE TESTING & DEVELOPMENT |
| CS481                                                     | OPERATING SYSTEMS              |
| CS483                                                     | DIGITAL FORENSICS              |
| CS484                                                     | ADVANCED COMPUTER NETWORKS     |
| CS486                                                     | ARTIFICIAL INTELLIGENCE        |
| CY385                                                     | CYBER ALGORITHMIC FOUNDATIONS  |
| CY392                                                     | DEFENSIVE CYBER OPERATIONS     |
| CY394                                                     | CLOUD COMPUTING                |
| CY482                                                     | OFFENSIVE CYBER OPERATIONS     |
| CY485                                                     | SPECIAL TOPIC IN CYBER SCIENCE |
| CY489                                                     | ADV IND STUDY CYBER SCI 3CR    |
| EE380                                                     | EMBEDDED SYSTEMS DEVELOPMENT   |
| LW462                                                     | CYBER LAW                      |
| MA376                                                     | APPLIED STATISTICS             |
| MA394                                                     | FUNDAMENTALS/NETWORK SCIENCE   |
| MA464                                                     | APPLIED ALGEBRA W/ CRYPTOLOGY  |
| MA477                                                     | THEORY & APPL OF DATA SCIENCE  |
| PY326                                                     | ETHICS OF TECHNOLOGY           |

|                                                                                                                                                           |                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| WS485                                                                                                                                                     | JOINT AND MULTINATIONAL OPS    |
| <b>AND</b>                                                                                                                                                |                                |
| <b>Complementary Support Courses (Cyber Foundation)</b>                                                                                                   | Choose 2 of 2                  |
| EE360                                                                                                                                                     | FUNDAMENTALS OF DIGITAL LOGIC  |
| MA372                                                                                                                                                     | INTRODUCTION TO DISCRETE MATH  |
| <b>AND</b>                                                                                                                                                |                                |
| <b>Complementary Support Course (Cyber in Society Elective)</b>                                                                                           | Choose 1 of 3                  |
| The interactions of the cyber domain in society and the military are a critical part of understanding the impacts of cyber.                               |                                |
| LW462                                                                                                                                                     | CYBER LAW                      |
| PY326                                                                                                                                                     | ETHICS OF TECHNOLOGY           |
| WS485                                                                                                                                                     | JOINT AND MULTINATIONAL OPS    |
| <b>AND</b>                                                                                                                                                |                                |
| <b>Integrative Experience for the Major</b>                                                                                                               | Choose 2 of 2                  |
| XE401                                                                                                                                                     | INTEGRATIVE SYSTEM DESIGN I    |
| XE402                                                                                                                                                     | INTEGRATIVE SYSTEM DESIGN II   |
| <b>AND</b>                                                                                                                                                |                                |
| <b>Science Depth</b>                                                                                                                                      |                                |
| Cadets may take any Academy-approved science depth choice.                                                                                                |                                |
| CH102                                                                                                                                                     | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                     | BIOLOGY                        |
| PH202                                                                                                                                                     | PHYSICS II                     |
| PH252                                                                                                                                                     | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                     | PHYSICS II: SPACE              |
| <b>AND</b>                                                                                                                                                |                                |
| <b>Curriculum Requirements</b>                                                                                                                            |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                  |                                |
| <b>STEM Depth</b>                                                                                                                                         |                                |
| Cadets in this major will satisfy the STEM Depth requirement by taking CY355.                                                                             |                                |
| CY355                                                                                                                                                     | CYBER FOUNDATIONS - COMPUTING  |
| <b>IT/Cyber Requirement</b>                                                                                                                               |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY355.              |                                |
| CY355                                                                                                                                                     | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                          |                                |
| Cadets in this major will satisfy the core engineering requirement with the embedded CYBER Sequence courses.                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                               |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and CS400. |                                |
| CS400                                                                                                                                                     | PRO CONSIDERATIONS IN COMPUTIN |

## 2029 Cyber Science Major w/ Honors Curriculum

| Code  | Short Description       | Description                   | Transcript Description  | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------|-------------------------------|-------------------------|--------------|--------------|
| CYS1H | Cyber Science w/ Honors | Cyber Science Major w/ Honors | Cyber Science w/ Honors | 1            | 0            |

### 2029 Cyber Science Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Cyber Honors Electives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Choose 1 of 11                 |
| CS403                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SOFTWARE TESTING & DEVELOPMENT |
| CS483                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | DIGITAL FORENSICS              |
| CS484                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ADVANCED COMPUTER NETWORKS     |
| CS485                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SPEC TOPICS IN COMPUTER SCI    |
| CS486                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ARTIFICIAL INTELLIGENCE        |
| CS489                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ADV IND STUDY COMPUTER SCI     |
| CY482                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | OFFENSIVE CYBER OPERATIONS     |
| CY485                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SPECIAL TOPIC IN CYBER SCIENCE |
| CY489                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ADV IND STUDY CYBER SCI 3CR    |
| XE492                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | DISRUPTIVE INNOVATIONS         |
| XE497                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CRITICAL SCIENTIFIC REASONING  |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                |
| <b>Research Requirement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |
| The research will normally be accomplished as an extension of a research effort or project begun in the honors elective and must reflect individual effort, although it may build on an existing group project. The research must be conducted under the supervision of a member of the faculty and must result in a written paper and oral presentation of a depth and quality suitable for submission to a professional conference. Neither the research nor the resulting paper and presentation need be completed during the semester they are begun, but must be completed by the last day of classes of semester 8. The final written document and oral presentation must be approved by both the research mentor and Cyber Science Program Director. |                                |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                |
| <b>Grade Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
| Attain an APSC of at least 3.0 in courses taken for the core curriculum and an APSC of at least 3.5 in courses taken for the major.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                |

### 2029 Cyber Science: Cyber Operations Major Curriculum

| Code | Short Description               | Description                           | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------|---------------------------------------|---------------------------------|--------------|--------------|
| CY00 | Cyber Science: Cyber Operations | Cyber Science: Cyber Operations Major | Cyber Science: Cyber Operations | 17           | 1            |

### 2029 Cyber Science: Cyber Operations Major Tracks

| Subject Area                              | Description                    |
|-------------------------------------------|--------------------------------|
| <b>Cyber Foundations Required Courses</b> | Choose 7 of 7                  |
| CS380                                     | COMPUTER SYSTEMS & ORGANIZATN  |
| CS384                                     | DATA STRUCTURES                |
| CS400                                     | PRO CONSIDERATIONS IN COMPUTIN |
| CY300                                     | PROGRAMMING FUNDAMENTALS       |
| CY350                                     | COMPUTER NETWORKS              |



|                                                                                                                                              |                                |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| CY460                                                                                                                                        | CYBER POLICY, STRATEGY, & OPNS |
| CY465                                                                                                                                        | ORGANIZATIONAL SECURITY        |
| <b>AND</b>                                                                                                                                   |                                |
| <b>Cyber Operations Required Courses</b>                                                                                                     | Choose 6 of 6                  |
| Cyber operations provides technical depth to the program.                                                                                    |                                |
| CS403                                                                                                                                        | SOFTWARE TESTING & DEVELOPMENT |
| CS481                                                                                                                                        | OPERATING SYSTEMS              |
| CS483                                                                                                                                        | DIGITAL FORENSICS              |
| CY385                                                                                                                                        | CYBER ALGORITHMIC FOUNDATIONS  |
| CY392                                                                                                                                        | DEFENSIVE CYBER OPERATIONS     |
| CY482                                                                                                                                        | OFFENSIVE CYBER OPERATIONS     |
| <b>AND</b>                                                                                                                                   |                                |
| <b>Complementary Support Courses (Cyber Foundation)</b>                                                                                      | Choose 2 of 2                  |
| EE360                                                                                                                                        | FUNDAMENTALS OF DIGITAL LOGIC  |
| MA372                                                                                                                                        | INTRODUCTION TO DISCRETE MATH  |
| <b>AND</b>                                                                                                                                   |                                |
| <b>Complementary Support Course (Cyber in Society Elective)</b>                                                                              | Choose 1 of 3                  |
| The interactions of the cyber domain in society and the military are a critical part of understanding the impacts of cyber.                  |                                |
| LW462                                                                                                                                        | CYBER LAW                      |
| PY326                                                                                                                                        | ETHICS OF TECHNOLOGY           |
| WS485                                                                                                                                        | JOINT AND MULTINATIONAL OPS    |
| <b>AND</b>                                                                                                                                   |                                |
| <b>Integrative Experience for the Major</b>                                                                                                  | Choose 2 of 2                  |
| XE401                                                                                                                                        | INTEGRATIVE SYSTEM DESIGN I    |
| XE402                                                                                                                                        | INTEGRATIVE SYSTEM DESIGN II   |
| <b>AND</b>                                                                                                                                   |                                |
| <b>Science Depth</b>                                                                                                                         |                                |
| Cadets may take any Academy-approved science depth choice.                                                                                   |                                |
| CH102                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                        | PHYSICS II: SPACE              |
| <b>AND</b>                                                                                                                                   |                                |
| <b>Curriculum Requirements</b>                                                                                                               |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                     |                                |
| <b>STEM Depth</b>                                                                                                                            |                                |
| Cadets in this major will satisfy the STEM Depth requirement by taking CY355.                                                                |                                |
| CY355                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>IT/Cyber Requirement</b>                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY355. |                                |
| CY355                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                             |                                |
| Cadets in this major will satisfy the core engineering requirement with the embedded CYBER Sequence courses.                                 |                                |
| <b>Writing-in-the-Major</b>                                                                                                                  |                                |

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and CS400.

CS400

PRO CONSIDERATIONS IN  
COMPUTIN

## 2029 Cyber Science: Cyber Operations Major w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|-----------------|-----------------|
| CY00H | Cyber Science: Cyber Operations<br>w/ Honors | Cyber Science: Cyber Operations<br>Major w/ Honors | Cyber Science: Cyber Operations<br>w/ Honors | 0               | 0               |

## 2029 Cyber Science: Cyber Operations Major w/ Honors Tracks

| Subject Area                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Research Requirement</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | The research will normally be accomplished as an extension of a research effort or project begun in an advanced computing course and must reflect individual effort, although it may build on an existing group project. The research must be conducted under the supervision of a member of the faculty and must result in a written paper and oral presentation of a depth and quality suitable for submission to a professional conference. Neither the research nor the resulting paper and presentation need be completed during the same semester they are begun, but must be completed by the end of the TEE period of semester 8. The final written document and oral presentation must be approved by both the research mentor and the Cyber Science Program Director. |
| <b>AND</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Grade Requirements</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                             | Attain an APSC of at least 3.0 in courses taken for the core curriculum and an APSC of at least 3.5 in courses taken for the major.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## 2029 Cyber Security Minor Curriculum

| Code  | Short Description    | Description          | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|----------------------|----------------------|------------------------|-----------------|-----------------|
| CYB1N | Cyber Security Minor | Cyber Security Minor | Cyber Security Minor   | 0               | 5               |

## 2029 Cyber Security Minor Tracks

| Subject Area                          | Description                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>General Information</b>            |                                                                                             |
|                                       | Cadets majoring in Cyber Science will not be permitted to obtain a minor in Cyber Security. |
| <b>Cyber Foundations Course Track</b> |                                                                                             |

A total of three courses in the overall program of study must be unique to the minor. You must take both courses in the Technical Track. Wherever possible, these courses may be double-counted, such as for cadets pursuing the Cyber Three-Course Engineering Sequence, or EECS, Math, and Systems majors. You must also take at least one course from the Non-Technical Track. Other courses from the Non-Technical Track may be taken as Cyber Application Electives.

**Technical Track**

Choose 2 of 2

CY350

COMPUTER NETWORKS

CY450

CYBER SECURITY ENGINEERING

**AND****Non-Technical Track**

Choose 1 of 3

CY460

CYBER POLICY, STRATEGY, &amp; OPNS

LW462

CYBER LAW

PY326

ETHICS OF TECHNOLOGY

**AND****Cyber Application Electives**

Choose 2 of 36

You must select two courses as electives, focusing on areas where cyber applies to specific domains or learning about thought processes/tools from other disciplines that can be relevant to cyber security. Cyber requires a balance of both technical and non-technical knowledge and skills, as well as application. Cadets may achieve this in any way practical.

CH457

MICROBIOLOGY

CS393

DATABASE SYSTEMS

CS484

ADVANCED COMPUTER NETWORKS

CY300

PROGRAMMING FUNDAMENTALS

CY392

NETWORK SERVICES MANAGEMENT

CY460

CYBER POLICY, STRATEGY, &amp; OPNS

EE360

FUNDAMENTALS OF DIGITAL LOGIC

EE377

ELECTRICAL POWER ENGRNG

EM420

PRODUCTION OPERATIONS MGMT

EN374

THE ARTS OF WAR

GE310

GEOG INFORMATION SYSTEMS

GE373

REMOTE SENSING

GE478

MILITARY GEOSPATIAL OPERATIONS

LA475

ARABIC RDG/WRTG THRU MEDIA

LC475

CHINESE RDG/WRTG THRU MEDIA

LF475

FRENCH RDG/WRTG THRU MEDIA

LG475

GERMAN RDG/WRTG THRU MEDIA

LP475

PORTUGUESE RDG/WRTG THRU MEDIA

LR475

RUSSIAN RDG/WRTG THRU MEDIA

LS475

SPANISH RDG/WRTG THRU MEDIA

LW462

CYBER LAW

LW482

NATIONAL SECURITY LAW

LZ475

PERSIAN RDG/WRTG THRU MEDIA

MA371

LINEAR ALGEBRA

MA394

FUNDAMENTALS/NETWORK SCIENCE

MA464

APPLIED ALGEBRA W/ CRYPTOLOGY

NE300

FUNDAMENTALS OF NUCLEAR ENGR

PY326

ETHICS OF TECHNOLOGY

SS464

HOMELAND SECURITY

SS465

TERRORISM: NEW CHALLENGES

SS483

NATIONAL SECURITY SEMINAR

SS486

STATE BUILDING

WS358

STRATEGY AND GENERALSHIP

WS370

US STRATEGY AND POLICY

XE472

DYNAMIC MODELING AND CONTROL

XE492

DISRUPTIVE INNOVATIONS

## 2029 Electrical Engineering Major Curriculum

| Code | Short Description      | Description                  | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|------------------------|------------------------------|------------------------|-----------------|-----------------|
| EEN1 | Electrical Engineering | Electrical Engineering Major | Electrical Engineering | 13              | 4               |

## 2029 Electrical Engineering Major Tracks

| Subject Area            | Description                    |
|-------------------------|--------------------------------|
| <b>Required Courses</b> | Choose 9 of 9                  |
| EE302                   | INTRO ELECTRICAL ENGIN         |
| EE360                   | FUNDAMENTALS OF DIGITAL LOGIC  |
| EE362                   | INTRODUCTION TO ELECTRONICS    |
| EE377                   | ELECTRICAL POWER ENGNRNG       |
| EE381                   | SIGNALS AND SYSTEMS            |
| EE383                   | ELECTROMAGN FIELDS & WAVES     |
| EE400                   | EE PROFESSIONAL CONSIDERATIONS |
| EE462                   | ELECTRONIC DESIGN              |
| XE401                   | INTEGRATIVE SYSTEM DESIGN I    |
| <b>AND</b>              |                                |
| <b>EECS Electives</b>   | Choose 1 of 18                 |
| CS393                   | DATABASE SYSTEMS               |
| CY300                   | PROGRAMMING FUNDAMENTALS       |
| EE389                   | IND STUDY IN ELECT ENG 3CR     |
| EE389A                  | IND STUDY IN ELECT ENG 3CR - A |
| EE477                   | DIGITAL COMMUNICATIONS SYSTEMS |
| EE480                   | OPTICAL FIBER COMMUNICATIONS   |
| EE482                   | WIRELESS COMM SYS ENGINEERING  |
| EE483                   | PHOTONICS ENGINEERING          |
| EE484                   | FUNDAMENTALS OF ROBOTICS       |
| EE485                   | SPEC TOPICS IN EE              |
| EE486                   | SOLID STATE ELECTRONICS        |
| EE489                   | ADV IND STUDY IN ELECT ENGR    |
| EE489A                  | ADV IND STUDY IN ELECT ENGR    |
| EE490                   | ELEC ENGRNG SUMMER RESEARCH    |
| XE442                   | ALTERNATIVE ENERGY ENGINEERING |
| XE472                   | DYNAMIC MODELING AND CONTROL   |
| XE475                   | MECHATRONICS                   |
| XE492                   | DISRUPTIVE INNOVATIONS         |
| <b>AND</b>              |                                |
| <b>Depth Option</b>     |                                |

Choose one of the following five depth options.

**Depth Option 1 - Robotics**

Choose 4 of 4

EE477

DIGITAL COMMUNICATIONS  
SYSTEMS

EE484

FUNDAMENTALS OF ROBOTICS

XE472

DYNAMIC MODELING AND  
CONTROL

XE475

MECHATRONICS

**OR**

**Depth Option 2 - Electronic Warfare and Communications**

Choose 3 of 3

EE477

DIGITAL COMMUNICATIONS  
SYSTEMS

EE480

OPTICAL FIBER  
COMMUNICATIONS

EE482

WIRELESS COMM SYS  
ENGINEERING

**OR**

**Depth Option 3 - Power and Energy**

Choose 3 of 3

EE486

SOLID STATE ELECTRONICS

XE442

ALTERNATIVE ENERGY  
ENGINEERING

XE472

DYNAMIC MODELING AND  
CONTROL

**OR**

**Depth Option 4 - OptoElectronics**

Choose 3 of 3

EE480

OPTICAL FIBER  
COMMUNICATIONS

EE483

PHOTONICS ENGINEERING

EE486

SOLID STATE ELECTRONICS

**OR**

**Depth Option 5 - Cyber Engineering**

Choose 4 of 4

If this Depth Option is chosen, EE477 must be selected as the EECS Elective.

CY300

PROGRAMMING FUNDAMENTALS

CY350

COMPUTER NETWORKS

CY450

CYBER SECURITY ENGINEERING

EE477

DIGITAL COMMUNICATIONS  
SYSTEMS

**AND**

**Complementary Support Courses**

Choose 3 of 4

For the Math CSC course, most cadets will take MA364, but some may qualify for MA365.

EE380

EMBEDDED SYSTEMS  
DEVELOPMENT

MA364

ENGINEERING MATHEMATICS

MA365

ADV MATH FOR  
ENGRS/SCIENTISTS

ME301

THERMODYNAMICS

**AND**

**Integrative Experience for the Major**

Choose 1 of 1

XE402

INTEGRATIVE SYSTEM DESIGN II

**AND**

**Science Depth**

Cadets in this major must take PH206 to satisfy the Science Depth requirement.

PH202

PHYSICS II

**STEM Depth**

Cadets in this major must take MA204 or MA205 to satisfy the STEM Depth requirement.

|       |                   |
|-------|-------------------|
| MA204 | CALCULUS I AND II |
| MA205 | CALCULUS II       |

**AND**

### Curriculum Requirements

This section describes how cadets in this major will satisfy various curriculum requirements.

#### IT/CYBER Requirement

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the core curriculum, EE380 and several other courses in the EE major.

|       |                                |
|-------|--------------------------------|
| EE360 | FUNDAMENTALS OF DIGITAL LOGIC  |
| EE362 | INTRODUCTION TO ELECTRONICS    |
| EE377 | ELECTRICAL POWER ENGRNG        |
| EE380 | EMBEDDED SYSTEMS DEVELOPMENT   |
| EE381 | SIGNALS AND SYSTEMS            |
| EE383 | ELECTROMAGN FIELDS & WAVES     |
| EE400 | EE PROFESSIONAL CONSIDERATIONS |
| EE462 | ELECTRONIC DESIGN              |
| XE401 | INTEGRATIVE SYSTEM DESIGN I    |
| XE402 | INTEGRATIVE SYSTEM DESIGN II   |

#### Core Engineering Sequence

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

#### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                                |
|-------|--------------------------------|
| EE400 | EE PROFESSIONAL CONSIDERATIONS |
|-------|--------------------------------|

## 2029 Electrical Engineering Major w/ Honors Curriculum

| Code  | Short Description                | Description                            | Transcript Description           | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------|----------------------------------------|----------------------------------|--------------|--------------|
| EEN1H | Electrical Engineering w/ Honors | Electrical Engineering Major w/ Honors | Electrical Engineering w/ Honors | 0            | 1            |

### 2029 Electrical Engineering Major w/ Honors Tracks

| Subject Area                                                             | Description                    |
|--------------------------------------------------------------------------|--------------------------------|
| <b>Required Course</b>                                                   | Choose 1 of 17                 |
| Choose one additional elective from the 17 EECS electives for the major. |                                |
| CS393                                                                    | DATABASE SYSTEMS               |
| CY300                                                                    | PROGRAMMING FUNDAMENTALS       |
| EE389                                                                    | IND STUDY IN ELECT ENG 3CR     |
| EE389A                                                                   | IND STUDY IN ELECT ENG 3CR - A |
| EE477                                                                    | DIGITAL COMMUNICATIONS SYSTEMS |
| EE480                                                                    | OPTICAL FIBER COMMUNICATIONS   |

|        |                                |
|--------|--------------------------------|
| EE482  | WIRELESS COMM SYS ENGINEERING  |
| EE483  | PHOTONICS ENGINEERING          |
| EE485  | SPEC TOPICS IN EE              |
| EE486  | SOLID STATE ELECTRONICS        |
| EE489  | ADV IND STUDY IN ELECT ENGR    |
| EE489A | ADV IND STUDY IN ELECT ENGR    |
| EE490  | ELEC ENGRNG SUMMER RESEARCH    |
| XE442  | ALTERNATIVE ENERGY ENGINEERING |
| XE472  | DYNAMIC MODELING AND CONTROL   |
| XE475  | MECHATRONICS                   |
| XE492  | DISRUPTIVE INNOVATIONS         |

**AND**

Participate in either an undergraduate research experience or report on their engineering design experience.

Research-focused programs will typically include enrollment in the Advanced Individual Study in Electrical Engineering, EE489, EE489A, EE490, or XE492 the grade for which will be based on a research paper suitable for submission to a conference.

The engineering design experience can result from participation in the Integrative System Design I and II series. The requirement for the engineering paper will be in addition to the XE401-XE402 coursework.

**AND****Grade Requirements**

Cadets must complete the requirements of the major as shown above, attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Electronic & Information Technology Systems Major Curriculum

| Code | Short Description    | Description                                       | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------|---------------------------------------------------|------------------------|--------------|--------------|
| EIT2 | Elec & Info Tech Sys | Electronic & Information Technology Systems Major | Elec & Info Tech Sys   | 3            | 13           |

### 2029 Electronic & Information Technology Systems Major Tracks

| Subject Area                                                                                                                                                                                                                                                | Description                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                                                                                                     | Choose 1 of 2                  |
| CS400                                                                                                                                                                                                                                                       | PRO CONSIDERATIONS IN COMPUTIN |
| EE400                                                                                                                                                                                                                                                       | EE PROFESSIONAL CONSIDERATIONS |
| <b>AND</b>                                                                                                                                                                                                                                                  |                                |
| <b>EITS Foundations</b>                                                                                                                                                                                                                                     |                                |
| Complete one foundation option.                                                                                                                                                                                                                             |                                |
| <b>Foundation Option 1 - Computing</b>                                                                                                                                                                                                                      | Choose 4 of 8                  |
| Choose CY300 as the Programming Foundation course; either CS484 or CY350 as the Networking Foundation course; CS482, CS483 or CY450 as the Network Security Foundation course; and either EE300 or EE360 as the Complementary Support Course for Computing. |                                |
| CS482                                                                                                                                                                                                                                                       | CYBER SECURITY ENGINEERING     |
| CS483                                                                                                                                                                                                                                                       | DIGITAL FORENSICS              |

|       |                               |
|-------|-------------------------------|
| CS484 | ADVANCED COMPUTER NETWORKS    |
| CY300 | PROGRAMMING FUNDAMENTALS      |
| CY450 | CYBER SECURITY ENGINEERING    |
| EE300 | FUNDAMENTALS OF DIGITAL LOGIC |
| EE360 | FUNDAMENTALS OF DIGITAL LOGIC |

**OR****Foundation Option 2 - Electrical Engineering**

Choose 4 of 14

Choose EE300 or EE360 as the Digital Logic Foundation course; EE301, EE302 or EE350 as the Electrical Engineering Foundation course; EE362, EE377, EE381, EE383, EE450, or EE462 as the Electrical Engineering Foundation elective; and either MA204/MA205 or MA255 as the Complementary Support Course for Electrical Engineering.

|       |                               |
|-------|-------------------------------|
| EE300 | FUNDAMENTALS OF DIGITAL LOGIC |
| EE301 | FUNDAMENTALS OF ELEC ENGIN    |
| EE302 | INTRO ELECTRICAL ENGIN        |
| EE350 | BASIC ELECTRICAL ENGINEERING  |
| EE360 | FUNDAMENTALS OF DIGITAL LOGIC |
| EE362 | INTRODUCTION TO ELECTRONICS   |
| EE377 | ELECTRICAL POWER ENGRNG       |
| EE381 | SIGNALS AND SYSTEMS           |
| EE383 | ELECTROMAGN FIELDS & WAVES    |
| EE450 | MILITARY ROBOTIC SYSTEMS      |
| EE462 | ELECTRONIC DESIGN             |
| MA204 | CALCULUS I AND II             |
| MA205 | CALCULUS II                   |
| MA255 | ADV MULTIVARIABLE CALCULUS    |

**AND****EITS Electives**

Choose 7 of 54

A table of electives, consisting of the following courses, will be provided by your DAC. Not including previously specified courses, complete 7 additional EITS elective courses from this list. You must work with your DAC to select courses that provide coherent threads through the major

|        |                                |
|--------|--------------------------------|
| CS380  | COMPUTER ORGANIZATION          |
| CS384  | DATA STRUCTURES                |
| CS385  | DESIGN & ANALYS-ALGORITHMS     |
| CS393  | DATABASE SYSTEMS               |
| CS394  | DISTRIB APPLICATION ENGRNG     |
| CS403  | SOFTWARE TESTING & DEVELOPMENT |
| CS473  | COMPUTER GRAPHICS              |
| CS474  | FUNDAMENTALS-COMPUTER THEORY   |
| CS478  | PROGRAMMING LANGUAGES          |
| CS481  | OPERATING SYSTEMS              |
| CS482  | CYBER SECURITY ENGINEERING     |
| CS483  | DIGITAL FORENSICS              |
| CS484  | ADVANCED COMPUTER NETWORKS     |
| CS486  | ARTIFICIAL INTELLIGENCE        |
| CS489  | ADV IND STUDY COMPUTER SCI     |
| CS489A | ADV IND STUDY COMPUTER SCI     |
| CS490  | COMPUTR SCI SUMMER RESEARCH    |
| CY300  | PROGRAMMING FUNDAMENTALS       |
| CY350  | COMPUTER NETWORKS              |
| CY383  | SECURE INTERFACE DESIGN        |
| CY384  | NETWORK SYSTEMS PROGRAMMING    |
| CY385  | CYBER ALGORITHMIC FOUNDATIONS  |
| CY392  | DEFENSIVE CYBER OPERATIONS     |
| CY394  | CLOUD COMPUTING                |
| CY450  | CYBER SECURITY ENGINEERING     |



|        |                                |
|--------|--------------------------------|
| CY460  | CYBER POLICY, STRATEGY, & OPNS |
| CY465  | ORGANIZATIONAL SECURITY        |
| CY482  | OFFENSIVE CYBER OPERATIONS     |
| EE300  | FUNDAMENTALS OF DIGITAL LOGIC  |
| EE301  | FUNDAMENTALS OF ELEC ENGIN     |
| EE302  | INTRO ELECTRICAL ENGIN         |
| EE350  | BASIC ELECTRICAL ENGINEERING   |
| EE360  | FUNDAMENTALS OF DIGITAL LOGIC  |
| EE362  | INTRODUCTION TO ELECTRONICS    |
| EE377  | ELECTRICAL POWER ENGRNG        |
| EE380  | EMBEDDED SYSTEMS DEVELOPMENT   |
| EE381  | SIGNALS AND SYSTEMS            |
| EE383  | ELECTROMAGN FIELDS & WAVES     |
| EE450  | MILITARY ROBOTIC SYSTEMS       |
| EE462  | ELECTRONIC DESIGN              |
| EE477  | DIGITAL COMMUNICATIONS SYSTEMS |
| EE480  | OPTICAL FIBER COMMUNICATIONS   |
| EE482  | WIRELESS COMM SYS ENGINEERING  |
| EE483  | PHOTONICS ENGINEERING          |
| EE485  | SPEC TOPICS IN EE              |
| EE486  | SOLID STATE ELECTRONICS        |
| EE489  | ADV IND STUDY IN ELECT ENGR    |
| EE489A | ADV IND STUDY IN ELECT ENGR    |
| MA386  | INTRO TO NUMERICAL ANALYSIS    |
| MA464  | APPLIED ALGEBRA W/ CRYPTOLOGY  |
| XE442  | ALTERNATIVE ENERGY ENGINEERING |
| XE472  | DYNAMIC MODELING AND CONTROL   |
| XE475  | MECHATRONICS                   |
| XE492  | DISRUPTIVE INNOVATIONS         |

**AND****Complementary Support Courses** Choose 2 of 24

|       |                               |
|-------|-------------------------------|
| CY350 | COMPUTER NETWORKS             |
| LW462 | CYBER LAW                     |
| MA204 | CALCULUS I AND II             |
| MA205 | CALCULUS II                   |
| MA255 | ADV MULTIVARIABLE CALCULUS    |
| MA364 | ENGINEERING MATHEMATICS       |
| MA365 | ADV MATH FOR ENGRS/SCIENTISTS |
| MA371 | LINEAR ALGEBRA                |
| MA372 | INTRODUCTION TO DISCRETE MATH |
| MA376 | APPLIED STATISTICS            |
| MA383 | FOUNDATIONS OF MATH           |
| MA385 | CHAOS AND FRACTALS            |
| MA386 | INTRO TO NUMERICAL ANALYSIS   |
| MA388 | SABERMETRICS                  |
| MA391 | MATHEMATICAL MODELING         |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE  |
| MA461 | GRAPH THEORY AND NETWORKS     |
| MA462 | COMBINATORICS                 |
| MA464 | APPLIED ALGEBRA W/ CRYPTOLOGY |
| MA466 | ABSTRACT ALGEBRA              |
| MA476 | MATHEMATICAL STATISTICS       |
| MA477 | THEORY & APPL OF DATA SCIENCE |
| PY326 | ETHICS OF TECHNOLOGY          |
| WS485 | JOINT AND MULTINATIONAL OPS   |
| XH497 | CRITICAL THOUGHT              |

**AND****Integrative Experience for the Major**

Choose 2 of 2

XE401

INTEGRATIVE SYSTEM DESIGN I

XE402

INTEGRATIVE SYSTEM DESIGN II

**AND****Science Depth**

Cadets in this major may take any approved choice to satisfy the Science Depth core curriculum requirement.

CH102

GENERAL CHEMISTRY II

CH152

ADV GENERAL CHEMISTRY II

CH275

BIOLOGY

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take CY305 or CY355 to satisfy the STEM Depth core curriculum requirement.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305 or CY355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in the major will satisfy the Core Engineering Sequence requirement as part of their major courses.

**Writing in the Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

CS400

CS PROFESSIONAL CONSIDERATIONS

EE400

EE PROFESSIONAL CONSIDERATIONS

## 2029 Electronic & Information Technology Systems Major Remarks Alternative Major for Computer Science, all five Cyber Science concentrations, and Electrical Engineering.

### 2029 Robotics Minor Curriculum

| Code  | Short Description | Description    | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------|----------------|------------------------|--------------|--------------|
| ROB0N | Robotics Minor    | Robotics Minor | Robotics Minor         | 2            | 3            |

### 2029 Robotics Minor Tracks

| Subject Area         | Description   |
|----------------------|---------------|
| Robotics Core Course | Choose 2 of 2 |

|                                                                                                                                    |                                |
|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| XE472                                                                                                                              | DYNAMIC MODELING AND CONTROL   |
| XE475                                                                                                                              | MECHATRONICS                   |
| <b>AND</b>                                                                                                                         |                                |
| <b>Electives</b>                                                                                                                   | Choose 3 of 28                 |
| Advanced Individual Study projects in CS489, EE489, ME389 and ME489 must be robot-related and approved by the EE Program Director. |                                |
| CS380                                                                                                                              | COMPUTER SYSTEMS & ORGANIZATN  |
| CS384                                                                                                                              | DATA STRUCTURES                |
| CS403                                                                                                                              | SOFTWARE TESTING & DEVELOPMENT |
| CS484                                                                                                                              | ADVANCED COMPUTER NETWORKS     |
| CS486                                                                                                                              | ARTIFICIAL INTELLIGENCE        |
| CS489                                                                                                                              | ADV IND STUDY COMPUTER SCI     |
| CY300                                                                                                                              | PROGRAMMING FUNDAMENTALS       |
| CY450                                                                                                                              | CYBER SECURITY ENGINEERING     |
| EE301                                                                                                                              | FUNDAMENTALS OF ELEC ENGIN     |
| EE302                                                                                                                              | INTRO ELECTRICAL ENGIN         |
| EE350                                                                                                                              | BASIC ELECTRICAL ENGINEERING   |
| EE360                                                                                                                              | FUNDAMENTALS OF DIGITAL LOGIC  |
| EE380                                                                                                                              | EMBEDDED SYSTEMS DEVELOPMENT   |
| EE450                                                                                                                              | MILITARY ELECTRONIC SYSTEMS    |
| EE477                                                                                                                              | DIGITAL COMMUNICATIONS SYSTEMS |
| EE482                                                                                                                              | WIRELESS COMM SYS ENGINEERING  |
| EE489                                                                                                                              | ADV IND STUDY IN ELECT ENGR    |
| LW462                                                                                                                              | CYBER LAW                      |
| MA371                                                                                                                              | LINEAR ALGEBRA                 |
| MC300                                                                                                                              | FUND OF ENGR MECH AND DESIGN   |
| ME202                                                                                                                              | COMPUTER AIDED DESIGN          |
| ME306                                                                                                                              | DYNAMICS                       |
| ME389                                                                                                                              | INTRO TO ADV STUDY IN MECH ENG |
| ME489                                                                                                                              | ADV STUDY IN MECH ENGRNG       |
| PY326                                                                                                                              | ETHICS OF TECHNOLOGY           |
| PY350                                                                                                                              | PHILOSOPHY OF SCIENCE          |
| PY355                                                                                                                              | PHILOSOPHY OF MIND             |
| XE492                                                                                                                              | DISRUPTIVE INNOVATIONS         |

## English and World Languages

### 2029 English Major Curriculum

| Code | Short Description | Description   | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------|---------------|------------------------|-----------------|-----------------|
| ENL2 | English           | English Major | English                | 2               | 11              |

### 2029 English Major Tracks

| Subject Area                                                                                                                                                                                                    | Description                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Required Course</b>                                                                                                                                                                                          | Choose 1 of 1               |
| EN300                                                                                                                                                                                                           | LITERARY METHODOLOGIES      |
| <b>AND</b>                                                                                                                                                                                                      |                             |
| <b>English Electives</b>                                                                                                                                                                                        |                             |
| Choose 8 of 18 English Electives while meeting the distribution requirements below.                                                                                                                             |                             |
| <b>English Electives - Distribution Requirements</b>                                                                                                                                                            |                             |
| Take at least one course in each of the following three categories.                                                                                                                                             |                             |
| <b>Distribution Requirement - Early Period Survey</b>                                                                                                                                                           | Choose 1 of 3               |
| EN311                                                                                                                                                                                                           | ANCIENT TO EARLY MODERN LIT |
| EN321                                                                                                                                                                                                           | AMERICAN LITERATURE I       |
| EN331                                                                                                                                                                                                           | BRITISH LITERATURE I        |
| <b>AND</b>                                                                                                                                                                                                      |                             |
| <b>Distribution Requirement - Late Period Survey</b>                                                                                                                                                            | Choose 1 of 3               |
| EN322                                                                                                                                                                                                           | AMERICAN LITERATURE II      |
| EN332                                                                                                                                                                                                           | BRITISH LITERATURE II       |
| EN340                                                                                                                                                                                                           | CONTEMPORARY LITERATURE     |
| <b>AND</b>                                                                                                                                                                                                      |                             |
| <b>Distribution Requirement - Genre Course</b>                                                                                                                                                                  | Choose 1 of 4               |
| EN361                                                                                                                                                                                                           | POETRY                      |
| EN362                                                                                                                                                                                                           | FILM AND FILM THEORY        |
| EN363                                                                                                                                                                                                           | THE NOVEL                   |
| EN364                                                                                                                                                                                                           | DRAMA                       |
| <b>AND</b>                                                                                                                                                                                                      |                             |
| <b>English Electives - Choice</b>                                                                                                                                                                               | Choose 5 of 17              |
| Fulfill the remaining English Elective requirement by choosing 5 additional courses from this list. One course from this list may not satisfy both the "Choice" requirement and the "Distribution" requirement. |                             |
| EN311                                                                                                                                                                                                           | ANCIENT TO EARLY MODERN LIT |
| EN321                                                                                                                                                                                                           | AMERICAN LITERATURE I       |
| EN322                                                                                                                                                                                                           | AMERICAN LITERATURE II      |
| EN331                                                                                                                                                                                                           | BRITISH LITERATURE I        |
| EN332                                                                                                                                                                                                           | BRITISH LITERATURE II       |
| EN340                                                                                                                                                                                                           | CONTEMPORARY LITERATURE     |
| EN351                                                                                                                                                                                                           | WORLD LITERATURE            |
| EN353                                                                                                                                                                                                           | WAR LITERATURE              |
| EN354                                                                                                                                                                                                           | SPECIAL TOPICS COLLOQUIUM   |
| EN355                                                                                                                                                                                                           | CRITICISM COLLOQUIUM        |

|       |                               |
|-------|-------------------------------|
| EN361 | POETRY                        |
| EN362 | FILM AND FILM THEORY          |
| EN363 | THE NOVEL                     |
| EN364 | DRAMA                         |
| EN370 | SHAKESPEARE                   |
| EN371 | SINGLE-AUTHOR COLLOQUIUM      |
| EN490 | INDEPENDENT STUDY: LITERATURE |

**AND****Complementary Support Courses**

In consultation with their DAC, Cadets will choose one track and three CSC courses from within that track. Working with their DAC, Cadets studying abroad can potentially get CSC credit for any coursework deemed appropriate by the English Program Director.

**Literature and History**

Choose 3 of 8

|       |                                 |
|-------|---------------------------------|
| HI340 | COLONIAL AMERICA                |
| HI348 | MODERN LATIN AMERICA            |
| HI364 | MODERN WESTERN EUROPE           |
| HI368 | MOD CENTRAL & E. EUR, 1896-1989 |
| HI392 | AMERICAN HISTORICAL MEMORY      |
| HI395 | HIST OF CIVIL WAR AMERICA       |
| HI398 | CIVIL RIGHTS IN AMER HIST       |
| HI487 | EARLY AMERICAN REPUB 1787-1826  |

**OR****Literature and the Natural World**

Choose 3 of 11

If CH275 or CH375 is counted as one of the required science courses, it cannot count as a CSC.

|       |                              |
|-------|------------------------------|
| CE380 | HYDROLOGY/HYDRAULIC DESIGN   |
| CH275 | BIOLOGY                      |
| CH375 | ADVANCED BIOLOGY             |
| CH457 | MICROBIOLOGY                 |
| GE324 | METEOROLOGY                  |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE431 | ECOLOGY                      |
| NE474 | RADIOLOGICAL SAFETY          |
| PH456 | SCIENCE AND POLICY           |
| PL250 | NEUROCOG FNDTNS OF BEHAVIOR  |
| PL390 | BIOLOGICAL PSYCHOLOGY        |

**OR****Literature and Philosophy**

Choose 3 of 15

|       |                                |
|-------|--------------------------------|
| PY305 | LOGICAL REASONING              |
| PY310 | REALITY AND KNOWLEDGE          |
| PY320 | ETHICS                         |
| PY326 | ETHICS OF TECHNOLOGY           |
| PY330 | POLITICAL PHILOSOPHY           |
| PY345 | PHILOSOPHY OF RELIGION         |
| PY350 | PHILOSOPHY OF SCIENCE          |
| PY355 | PHILOSOPHY OF MIND             |
| PY360 | ANCIENT PHILOSOPHY             |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| PY370 | 17TH & 18TH CENTURY PHILOSOPHY |
| PY375 | KANT & 19TH CENTURY PHILOSOPHY |
| PY380 | 20TH CENTURY PHILOSOPHY        |
| PY390 | INTER-DEPARTMENT SEMINAR       |
| SS386 | POLITICAL THOUGHT              |

**OR****Literature and Foreign Language**

Choose 3 of 6

Students adopting this track take a third semester of foreign language plus two subsequent courses in their language.

|       |                            |
|-------|----------------------------|
| LA485 | ARABIC LITERATURE I        |
| LC485 | CHINESE LITERATURE I       |
| LF485 | SURVEY OF FRENCH LIT I     |
| LN380 | NATURE OF MODERN LANGUAGES |
| LX300 | 3RD SEMESTER FOREIGN LANG  |
| LX400 | 4TH SEMESTER FOREIGN LANG  |

**OR**

**Literature, Culture and Society**

Choose 3 of 18

|       |                                |
|-------|--------------------------------|
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| HI344 | MODERN DIPLOMACY               |
| HI391 | WORLD RELIGIONS                |
| HI392 | AMERICAN HISTORICAL MEMORY     |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW461 | CIVIL RIGHTS AND THE LAW       |
| LW462 | CYBER LAW                      |
| LW472 | CRIMINAL LAW                   |
| PL383 | EXPERIMENTAL SOCIAL PSYCHOLOGY |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM    |
| PL482 | ARMED FORCES AND SOCIETY       |
| SS372 | POLITICS AND GOV OF CHINA      |
| SS392 | RIGHTS, LIBERTIES & US CITIZEN |
| SS473 | AMERICAN FOREIGN POLICY        |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS476 | CIVIL WARS                     |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |
| XH415 | GENOCIDE AND ETHNIC CLEANSING  |

**AND**

**Integrative Experience for the Major**

Choose 1 of 1

|       |                               |
|-------|-------------------------------|
| EN400 | SEMINAR IN ADV LITERARY STUDY |
|-------|-------------------------------|

**AND**

**Science Depth**

Cadets in this major may take any approved choice to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND**

**Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

EN300

LITERARY METHODOLOGIES

### 2029 English Major w/ Honors Curriculum

| Code  | Short Description | Description             | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------|-------------------------|------------------------|--------------|--------------|
| ENL2H | English w/ Honors | English Major w/ Honors | English w/ Honors      | 2            | 0            |

### 2029 English Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                | Description      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Required Courses</b>                                                                                                                                                                     | Choose 2 of 2    |
| Cadets must achieve at least a B in EN401 in order to proceed into EN402. The thesis adviser will normally recommend the cadet for Honors consideration if the EN402 grade is A- or better. |                  |
| EN401                                                                                                                                                                                       | SENIOR THESIS I  |
| EN402                                                                                                                                                                                       | SENIOR THESIS II |
| <b>AND</b>                                                                                                                                                                                  |                  |
| Complete the requirements of the major as shown above; attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.                                                            |                  |
| <b>AND</b>                                                                                                                                                                                  |                  |
| <b>Approval of the Department Head</b>                                                                                                                                                      |                  |
| On the basis of recommendations by the cadet's DAC and senior-thesis adviser.                                                                                                               |                  |

### 2029 Foreign Area Studies Major: Africa Curriculum

| Code | Short Description            | Description                        | Transcript Description       | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------|------------------------------------|------------------------------|--------------|--------------|
| FSI2 | Foreign Area Studies: Africa | Foreign Area Studies Major: Africa | Foreign Area Studies: Africa | 5            | 8            |

### 2029 Foreign Area Studies Major: Africa Tracks

| Subject Area            | Description                    |
|-------------------------|--------------------------------|
| <b>Required Courses</b> | Choose 5 of 5                  |
| GE340                   | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                   | GEOGRAPHY OF AFRICA            |
| HI345                   | MODERN AFRICA                  |
| SS366                   | COMPARATIVE POLITICS           |
| SS485                   | GOV & POLITICS SUB-SAHARAN AFR |
| <b>AND</b>              |                                |
| <b>Language Track</b>   |                                |

You must choose one of the following two language tracks.

**French Electives**

Choose 4 of 19

|        |                                |
|--------|--------------------------------|
| LF371  | INTENSIVE INTERMEDIATE FRENCH  |
| LF372  | FRENCH FOR ORAL & WRITTEN COMM |
| LF470  | SPECIAL TOPIC IN FRENCH        |
| LF475  | FRENCH RDG/WRTG THRU MEDIA     |
| LF476  | MILITARY SPKG/RDG - FRENCH     |
| LF483  | FRENCH CIVILIZATION I          |
| LF484  | FRENCH CIVILIZATION II         |
| LF485  | SURVEY OF FRENCH LIT I         |
| LF486  | SURVEY OF FRENCH LIT II        |
| LF492  | MASTERWORKS OF FRENCH LIT      |
| LN440F | FRENCH IN CULTURAL CONTEXT     |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**OR**

**Portuguese Electives**

Choose 4 of 18

|        |                                |
|--------|--------------------------------|
| LN440P | PORTUGUESE IN CULTURAL CONTEXT |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |
| LP371  | INTENSIVE INTERMED. PORTUGUESE |
| LP372  | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP470  | SPECIAL TOPIC IN PORTUGUESE    |
| LP475  | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476  | MILITARY SPKG/RDG - PORTUGUESE |
| LP483  | PORTUGUESE CIVILIZATION I      |
| LP484  | PORTUGUESE CIVILIZATION II     |
| LP485  | SURVEY OF PORTUGUESE LIT I     |
| LP492  | LIT OF PORT-SPKG WORLD         |

**AND**

**Complementary Support Courses**

Choose 3 of 22

Choose 3 CSCs, each from a different discipline. If GE491 or SS486 is selected as a CSC, cadets must choose a different course from the Integrated Experience list (i.e. GE491 or SS486 cannot double-count as a CSC and the IE for the major).

|       |                             |
|-------|-----------------------------|
| EN300 | LITERARY METHODOLOGIES      |
| EN311 | ANCIENT TO EARLY MODERN LIT |
| EN340 | CONTEMPORARY LITERATURE     |
| EN351 | WORLD LITERATURE            |
| EN353 | WAR LITERATURE              |
| EN361 | POETRY                      |
| EN362 | FILM AND FILM THEORY        |
| EN363 | THE NOVEL                   |
| EN364 | DRAMA                       |



|       |                             |
|-------|-----------------------------|
| GE491 | MILITARY GEOGRAPHY          |
| HI344 | MODERN DIPLOMACY            |
| HI391 | WORLD RELIGIONS             |
| LW410 | COMPARATIVE LEGAL SYSTEMS   |
| LW473 | ENVIRONMENTAL LAW           |
| LW481 | INTERNATIONAL LAW           |
| SS465 | TERRORISM: NEW CHALLENGES   |
| SS475 | COMP POLITICAL INSTITUTIONS |
| SS484 | INTERNATIONAL ECONOMICS     |
| SS486 | STATE BUILDING              |
| SS487 | INT'L POLITICAL ECONOMY     |
| WS358 | STRATEGY AND GENERALSHIP    |
| WS455 | COMPARATIVE DEFENSE POLICY  |

**AND****Integrative Experience for the Major** Choose 1 of 4

Cadets selecting an LN490 course must align this course with the appropriate language track chosen for the major.

|        |                               |
|--------|-------------------------------|
| GE491  | MILITARY GEOGRAPHY            |
| LN490F | FRENCH LANGUAGE & CULTURE CAP |
| LN490P | PORTUGUESE LANG & CULTURE CAP |
| SS486  | STATE BUILDING                |

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|        |                               |
|--------|-------------------------------|
| LN490F | FRENCH LANGUAGE & CULTURE CAP |
| LN490P | PORTUGUESE LANG & CULTURE CAP |

| Code  | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FSI2H | Foreign Area Studies: Africa w/ Honors | Foreign Area Studies Major: Africa w/ Honors | Foreign Area Studies: Africa w/ Honors | 1            | 1            |

### 2029 Foreign Area Studies Major: Africa w/ Honors Tracks

| Subject Area               | Description                                                                              |
|----------------------------|------------------------------------------------------------------------------------------|
| <b>Additional Elective</b> |                                                                                          |
|                            | Take an additional course in French or Portuguese.                                       |
| <b>AND</b>                 |                                                                                          |
| <b>Honors Thesis</b>       | Choose 1 of 1                                                                            |
|                            | Take LN488 and complete an honors thesis under the direction of a senior faculty member. |
| LN488                      | ADV IND STUDY-FOREIGN LANGS                                                              |
| <b>AND</b>                 |                                                                                          |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

### 2029 Foreign Area Studies Major: East Asia Curriculum

| Code | Short Description               | Description                           | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------|---------------------------------------|---------------------------------|--------------|--------------|
| FSA2 | Foreign Area Studies: East Asia | Foreign Area Studies Major: East Asia | Foreign Area Studies: East Asia | 5            | 8            |

### 2029 Foreign Area Studies Major: East Asia Tracks

| Subject Area             | Description                    |
|--------------------------|--------------------------------|
| <b>Required Courses</b>  | Choose 5 of 5                  |
| GE340                    | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342                    | GEOGRAPHY OF ASIA              |
| HI337                    | CHINA: EMPIRE, REPUBLIC, & MAO |
| SS366                    | COMPARATIVE POLITICS           |
| SS372                    | POLITICS AND GOV OF CHINA      |
| <b>AND</b>               |                                |
| <b>Chinese Electives</b> | Choose 4 of 19                 |
| LC371                    | INTENSIVE INTERMEDIATE CHINESE |
| LC372                    | CHINESE FOR ORAL & WRIT COMM   |
| LC470                    | SPECIAL TOPIC IN CHINESE       |
| LC475                    | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                    | MILITARY SPKG/RDG - CHINESE    |
| LC483                    | CHINESE CIVILIZATION I         |
| LC484                    | CHINESE CIVILIZATION II        |
| LC485                    | CHINESE LITERATURE I           |
| LC486                    | CHINESE LITERATURE II          |
| LC492                    | CHINESE LITERATURE III         |

|        |                                |
|--------|--------------------------------|
| LN440C | CHINESE IN CULTURAL CONTEXT    |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**AND****Complementary Support Courses**

Choose 3 of 23

Choose 3 of 17, each from a different discipline. If GE491 or SS486 is selected as a CSC, cadets must choose a different course from the Integrated Experience list (i.e. GE491 or SS486 cannot double-count as a CSC and the IE for the major).

|       |                             |
|-------|-----------------------------|
| EN300 | LITERARY METHODOLOGIES      |
| EN340 | CONTEMPORARY LITERATURE     |
| EN351 | WORLD LITERATURE            |
| EN353 | WAR LITERATURE              |
| EN361 | POETRY                      |
| EN362 | FILM AND FILM THEORY        |
| EN363 | THE NOVEL                   |
| EN364 | DRAMA                       |
| GE491 | MILITARY GEOGRAPHY          |
| HI344 | MODERN DIPLOMACY            |
| HI347 | ASIAN WARFARE AND POLITICS  |
| HI391 | WORLD RELIGIONS             |
| LW410 | COMPARATIVE LEGAL SYSTEMS   |
| LW473 | ENVIRONMENTAL LAW           |
| LW481 | INTERNATIONAL LAW           |
| SS374 | POL & GOV OF KOREAS & JAPAN |
| SS465 | TERRORISM: NEW CHALLENGES   |
| SS475 | COMP POLITICAL INSTITUTIONS |
| SS484 | INTERNATIONAL ECONOMICS     |
| SS486 | STATE BUILDING              |
| SS487 | INT'L POLITICAL ECONOMY     |
| WS358 | STRATEGY AND GENERALSHIP    |
| WS455 | COMPARATIVE DEFENSE POLICY  |

**AND****Integrative Experience for the Major**

Choose 1 of 3

|        |                                |
|--------|--------------------------------|
| GE491  | MILITARY GEOGRAPHY             |
| LN490C | CHINESE LANGUAGE & CULTURE CAP |
| SS486  | STATE BUILDING                 |

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND**

**Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN490C

CHINESE LANGUAGE &amp; CULTURE CAP

**2029 Foreign Area Studies Major: East Asia w/ Honors Curriculum**

| Code  | Short Description                         | Description                                     | Transcript Description                    | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------------------|-------------------------------------------------|-------------------------------------------|--------------|--------------|
| FSA2H | Foreign Area Studies: East Asia w/ Honors | Foreign Area Studies Major: East Asia w/ Honors | Foreign Area Studies: East Asia w/ Honors | 1            | 1            |

**2029 Foreign Area Studies Major: East Asia w/ Honors Tracks**

| Subject Area                                                                             | Description                 |
|------------------------------------------------------------------------------------------|-----------------------------|
| <b>Additional Elective</b>                                                               |                             |
| Take an additional course in Chinese.                                                    |                             |
| <b>AND</b>                                                                               |                             |
| <b>Honors Thesis</b>                                                                     | Choose 1 of 1               |
| Take LN488 and complete an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                                    | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                                               |                             |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

**2029 Foreign Area Studies Major: Eurasia Curriculum**

| Code | Short Description             | Description                         | Transcript Description        | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------|-------------------------------------|-------------------------------|--------------|--------------|
| FSU2 | Foreign Area Studies: Eurasia | Foreign Area Studies Major: Eurasia | Foreign Area Studies: Eurasia | 5            | 8            |

**2029 Foreign Area Studies Major: Eurasia Tracks**

| Subject Area                                                                                                                                                                                                                                    | Description                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                                                                                         | Choose 5 of 5                  |
| GE340                                                                                                                                                                                                                                           | GEOGRAPHY OF GLOBAL CULTURES   |
| GE343                                                                                                                                                                                                                                           | GEOGRAPHY OF RUSSIA            |
| HI367                                                                                                                                                                                                                                           | IMPERIAL AND SOVIET RUSSIA     |
| SS366                                                                                                                                                                                                                                           | COMPARATIVE POLITICS           |
| SS375                                                                                                                                                                                                                                           | GOV & POL RUSSIA & NEIGHBORS   |
| <b>AND</b>                                                                                                                                                                                                                                      |                                |
| <b>Language Track</b>                                                                                                                                                                                                                           |                                |
| You must choose one of the following two language tracks.                                                                                                                                                                                       |                                |
| <b>Russian Electives</b>                                                                                                                                                                                                                        | Choose 4 of 19                 |
| LN440R                                                                                                                                                                                                                                          | RUSSIAN IN CULTURAL CONTEXT    |
| LN451                                                                                                                                                                                                                                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                                                                                                                                                                                                                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                                                                                                                                                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG & CULT V  |
| LR371                                                                                                                                                                                                                                           | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                                                                                                                                                                                                                                           | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470                                                                                                                                                                                                                                           | SPECIAL TOPIC IN RUSSIAN       |
| LR475                                                                                                                                                                                                                                           | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                                                                                                                                                                                                                                           | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                                                                                                                                                                                                                                           | RUSSIAN CIV I                  |
| LR484                                                                                                                                                                                                                                           | RUSSIAN CIV II                 |
| LR485                                                                                                                                                                                                                                           | SURVEY OF RUSSIAN LITERATURE I |
| LR486                                                                                                                                                                                                                                           | SURVEY OF RUSSIAN LIT. II      |
| LR492                                                                                                                                                                                                                                           | RUSSIAN LIFE IN FICTION        |
| <b>OR</b>                                                                                                                                                                                                                                       |                                |
| <b>Persian Electives</b>                                                                                                                                                                                                                        | Choose 4 of 18                 |
| LN440Z                                                                                                                                                                                                                                          | PERSIAN IN CULTURAL CONTEXT    |
| LN451                                                                                                                                                                                                                                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                                                                                                                                                                                                                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                                                                                                                                                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                                                                                                                                                                                                                           | SEM ABROAD: ADV LANG & CULT V  |
| LZ371                                                                                                                                                                                                                                           | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                                                                                                                                                                                                                                           | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                                                                                                                                                                                                                                           | SPECIAL TOPIC IN PERSIAN       |
| LZ475                                                                                                                                                                                                                                           | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                                                                                                                                                                                                                                           | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                                                                                                                                                                                                                                           | PERSIAN CIVILIZATION I         |
| LZ484                                                                                                                                                                                                                                           | PERSIAN CIVILIZATION II        |
| LZ485                                                                                                                                                                                                                                           | PERSIAN LITERATURE I           |
| LZ486                                                                                                                                                                                                                                           | PERSIAN LITERATURE II          |
| <b>AND</b>                                                                                                                                                                                                                                      |                                |
| <b>Complementary Support Courses</b>                                                                                                                                                                                                            | Choose 3 of 24                 |
| Choose 3 CSCs, each from a different discipline. If GE491 or SS486 is selected as a CSC, cadets must choose a different course from the Integrated Experience list (i.e. GE491 or SS486 cannot double-count as a CSC and the IE for the major). |                                |

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN311 | ANCIENT TO EARLY MODERN LIT  |
| EN340 | CONTEMPORARY LITERATURE      |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN362 | FILM AND FILM THEORY         |
| EN363 | THE NOVEL                    |
| EN364 | DRAMA                        |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI362 | POLITICS/SOC-EARLY MOD EURO  |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major** Choose 1 of 4

Cadets selecting an LN490 course must align this course with the appropriate language track chosen for the major.

|        |                                |
|--------|--------------------------------|
| GE491  | MILITARY GEOGRAPHY             |
| LN490R | RUSSIAN LANGUAGE & CULTURE CAP |
| LN490Z | PERSIAN LANGUAGE & CULTURE CAP |
| SS486  | STATE BUILDING                 |

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN490R

RUSSIAN LANGUAGE &amp; CULTURE CAP

LN490Z

PERSIAN LANGUAGE &amp; CULTURE CAP

## 2029 Foreign Area Studies Major: Eurasia w/ Honors Curriculum

| Code  | Short Description                       | Description                                   | Transcript Description                  | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------|-----------------------------------------------|-----------------------------------------|--------------|--------------|
| FSU2H | Foreign Area Studies: Eurasia w/ Honors | Foreign Area Studies Major: Eurasia w/ Honors | Foreign Area Studies: Eurasia w/ Honors | 1            | 1            |

## 2029 Foreign Area Studies Major: Eurasia w/ Honors Tracks

| Subject Area                                                                                                                         | Description                 |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Additional Elective</b>                                                                                                           |                             |
| Take an additional course in Russian.                                                                                                |                             |
| <b>AND</b>                                                                                                                           |                             |
| <b>Honors Thesis</b>                                                                                                                 | Choose 1 of 1               |
| Take LN488 and complete an honors thesis under the direction of a senior faculty member.                                             |                             |
| LN488                                                                                                                                | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                                                                                           |                             |
| Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major. |                             |

## 2029 Foreign Area Studies Major: Europe Curriculum

| Code | Short Description            | Description                        | Transcript Description       | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------|------------------------------------|------------------------------|--------------|--------------|
| FSE2 | Foreign Area Studies: Europe | Foreign Area Studies Major: Europe | Foreign Area Studies: Europe | 4            | 9            |

## 2029 Foreign Area Studies Major: Europe Tracks

| Subject Area            | Description                  |
|-------------------------|------------------------------|
| <b>Required Courses</b> | Choose 4 of 4                |
| GE340                   | GEOGRAPHY OF GLOBAL CULTURES |
| GE346                   | GEOGRAPHY OF EUROPE          |
| SS366                   | COMPARATIVE POLITICS         |
| SS377                   | POLITICS & GOV OF EUROPE     |
| <b>AND</b>              |                              |
| <b>Language Track</b>   |                              |

You must choose one of the following four language tracks.

### **French Electives**

Choose 5 of 20

Choose 4 of 20 French and 1 of 1 History.

|        |                                |
|--------|--------------------------------|
| HI364  | MODERN WESTERN EUROPE          |
| LF371  | INTENSIVE INTERMEDIATE FRENCH  |
| LF372  | FRENCH FOR ORAL & WRITTEN COMM |
| LF470  | SPECIAL TOPIC IN FRENCH        |
| LF475  | FRENCH RDG/WRTG THRU MEDIA     |
| LF476  | MILITARY SPKG/RDG - FRENCH     |
| LF483  | FRENCH CIVILIZATION I          |
| LF484  | FRENCH CIVILIZATION II         |
| LF485  | SURVEY OF FRENCH LIT I         |
| LF486  | SURVEY OF FRENCH LIT II        |
| LF492  | MASTERWORKS OF FRENCH LIT      |
| LN440F | FRENCH IN CULTURAL CONTEXT     |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

### **OR**

### **German Electives**

Choose 5 of 20

Choose 4 of 20 German and 1 of 1 History.

|        |                                |
|--------|--------------------------------|
| HI343  | MODERN GERMANY                 |
| LG371  | INTENSIVE INTERMEDIATE GERMAN  |
| LG372  | GERMAN FOR ORAL & WRITTEN COMM |
| LG470  | SPECIAL TOPIC IN GERMAN        |
| LG475  | GERMAN RDG/WRTG THRU MEDIA     |
| LG476  | MILITARY SPKG/RDG - GERMAN     |
| LG483  | GERMAN CIVILIZATION I          |
| LG484  | GERMAN CIVILIZATION II         |
| LG485  | SURVEY OF GERMAN LIT I         |
| LG486  | SURVEY OF GERMAN LIT II        |
| LG492  | 20TH & 21ST CENTURY GERMANY    |
| LN440G | GERMAN IN CULTURAL CONTEXT     |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

### **OR**

### **Portuguese Electives**

Choose 5 of 20

Choose 4 of 20 Portuguese and 1 of 2 History.

|        |                                |
|--------|--------------------------------|
| HI344  | MODERN DIPLOMACY               |
| HI364  | MODERN WESTERN EUROPE          |
| LN440P | PORTUGUESE IN CULTURAL CONTEXT |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |



|       |                                 |
|-------|---------------------------------|
| LN493 | SEM ABROAD: ADV LANG&CULT III   |
| LN494 | SEM ABROAD: ADV LANG & CULT IV  |
| LN495 | SEM ABROAD: ADV LANG & CULT V   |
| LP371 | INTENSIVE INTERMED. PORTUGUESE  |
| LP372 | PORTUGUESE FOR ORAL/WRIT COMM   |
| LP470 | SPECIAL TOPIC IN PORTUGUESE     |
| LP475 | PORTUGUESE RDG/WRITG THRU MEDIA |
| LP476 | MILITARY SPKG/RDG - PORTUGUESE  |
| LP483 | PORTUGUESE CIVILIZATION I       |
| LP484 | PORTUGUESE CIVILIZATION II      |
| LP485 | SURVEY OF PORTUGUESE LIT I      |
| LP492 | LIT OF PORT-SPKG WORLD          |

**OR****Spanish Electives**

Choose 5 of 21

Choose 4 of 21 Spanish and 1 of 2 History.

|        |                                |
|--------|--------------------------------|
| HI344  | MODERN DIPLOMACY               |
| HI364  | MODERN WESTERN EUROPE          |
| LN440S | SPANISH IN CULTURAL CONTEXT    |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |
| LS371  | INTENSIVE INTERMEDIATE SPANISH |
| LS372  | SPANISH FOR ORAL & WRIT COMM   |
| LS470  | SPECIAL TOPIC IN SPANISH       |
| LS475  | SPANISH RDG/WRITG THRU MEDIA   |
| LS476  | MILITARY SPKG/RDG - SPANISH    |
| LS483  | SPANISH CIV AND CULTURE        |
| LS484  | SPANISH AMERICAN CIV AND CULT  |
| LS485  | SPANISH-AMERICAN LITERATURE    |
| LS486  | THE LITERATURE OF SPAIN        |
| LS492  | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Complementary Support Courses**

Choose 3 of 28

Choose 3 CSCs, each from a different discipline. DACs will counsel cadets on appropriate CSC choices based on language track and courses only counting towards one requirement (track versus CSC). If GE491 or SS486 is selected as a CSC, cadets must choose a different course from the Integrated Experience list (i.e. GE491 or SS486 cannot double-count as a CSC and the IE for the major).

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN311 | ANCIENT TO EARLY MODERN LIT    |
| EN340 | CONTEMPORARY LITERATURE        |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN363 | THE NOVEL                      |
| EN364 | DRAMA                          |
| GE491 | MILITARY GEOGRAPHY             |
| HI338 | WARFARE IN AGE OF REVOLUTIONS  |
| HI344 | MODERN DIPLOMACY               |
| HI364 | MODERN WESTERN EUROPE          |
| HI368 | MOD CENTRAL & E. EUR,1896-1989 |

|       |                              |
|-------|------------------------------|
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS385 | WAR AND ITS THEORISTS        |
| WS455 | COMPARATIVE DEFENSE POLICY   |
| XH405 | THE HOLOCAUST AND ITS LEGACY |

**AND****Integrative Experience for the Major** Choose 1 of 6

Cadets selecting an LN490 course must align this course with the appropriate language track chosen for the major.

|        |                                |
|--------|--------------------------------|
| GE491  | MILITARY GEOGRAPHY             |
| LN490F | FRENCH LANGUAGE & CULTURE CAP  |
| LN490G | GERMAN LANGUAGE & CULTURE CAP  |
| LN490P | PORTUGUESE LANG & CULTURE CAP  |
| LN490S | SPANISH LANGUAGE & CULTURE CAP |
| SS486  | STATE BUILDING                 |

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|        |                                |
|--------|--------------------------------|
| LN490F | FRENCH LANGUAGE & CULTURE CAP  |
| LN490G | GERMAN LANGUAGE & CULTURE CAP  |
| LN490P | PORTUGUESE LANG & CULTURE CAP  |
| LN490S | SPANISH LANGUAGE & CULTURE CAP |

## 2029 Foreign Area Studies Major: Europe w/ Honors Curriculum

| Code  | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FSE2H | Foreign Area Studies: Europe w/ Honors | Foreign Area Studies Major: Europe w/ Honors | Foreign Area Studies: Europe w/ Honors | 1            | 1            |

## 2029 Foreign Area Studies Major: Europe w/ Honors Tracks

| Subject Area                                                                             | Description                 |
|------------------------------------------------------------------------------------------|-----------------------------|
| <b>Additional Elective</b>                                                               |                             |
| Take an additional course in French, German, Portuguese or Spanish.                      |                             |
| <b>AND</b>                                                                               |                             |
| <b>Honors Thesis</b>                                                                     | Choose 1 of 1               |
| Take LN488 and complete an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                                    | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                                               |                             |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Foreign Area Studies Major: Latin America Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FSL2 | Foreign Area Studies: Latin America | Foreign Area Studies Major: Latin America | Foreign Area Studies: Latin America | 5            | 8            |

## 2029 Foreign Area Studies Major: Latin America Tracks

| Subject Area                                              | Description                    |
|-----------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                   | Choose 5 of 5                  |
| GE340                                                     | GEOGRAPHY OF GLOBAL CULTURES   |
| GE347                                                     | GEOGRAPHY OF LATIN AMERICA     |
| HI348                                                     | MODERN LATIN AMERICA           |
| SS366                                                     | COMPARATIVE POLITICS           |
| SS384                                                     | POLITICS & GOVT-LATIN AMER     |
| <b>AND</b>                                                |                                |
| <b>Language Track</b>                                     |                                |
| You must choose one of the following two language tracks. |                                |
| <b>Portuguese Electives</b>                               | Choose 4 of 18                 |
| LN440P                                                    | PORTUGUESE IN CULTURAL CONTEXT |

|       |                                |
|-------|--------------------------------|
| LN451 | ADV LANG & CULTURE IN CONTEXT  |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LN491 | SEM ABROAD: ADV LANG & CULT I  |
| LN492 | SEM ABROAD: ADV LANG & CULT II |
| LN493 | SEM ABROAD: ADV LANG&CULT III  |
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LP371 | INTENSIVE INTERMED. PORTUGUESE |
| LP372 | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP470 | SPECIAL TOPIC IN PORTUGUESE    |
| LP475 | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476 | MILITARY SPKG/RDG - PORTUGUESE |
| LP483 | PORTUGUESE CIVILIZATION I      |
| LP484 | PORTUGUESE CIVILIZATION II     |
| LP485 | SURVEY OF PORTUGUESE LIT I     |
| LP492 | LIT OF PORT-SPKG WORLD         |

**OR****Spanish Electives**

Choose 4 of 19

|        |                                |
|--------|--------------------------------|
| LN440S | SPANISH IN CULTURAL CONTEXT    |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |
| LS371  | INTENSIVE INTERMEDIATE SPANISH |
| LS372  | SPANISH FOR ORAL & WRIT COMM   |
| LS470  | SPECIAL TOPIC IN SPANISH       |
| LS475  | SPANISH RDG/WRTG THRU MEDIA    |
| LS476  | MILITARY SPKG/RDG - SPANISH    |
| LS483  | SPANISH CIV AND CULTURE        |
| LS484  | SPANISH AMERICAN CIV AND CULT  |
| LS485  | SPANISH-AMERICAN LITERATURE    |
| LS486  | THE LITERATURE OF SPAIN        |
| LS492  | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Complementary Support Courses**

Choose 3 of 23

Choose 3 CSCs, each from a different discipline. If GE491 or SS486 is selected as a CSC, cadets must choose a different course from the Integrated Experience list (i.e. GE491 or SS486 cannot double-count as a CSC and the IE for the major).

|       |                             |
|-------|-----------------------------|
| EN300 | LITERARY METHODOLOGIES      |
| EN311 | ANCIENT TO EARLY MODERN LIT |
| EN340 | CONTEMPORARY LITERATURE     |
| EN351 | WORLD LITERATURE            |
| EN353 | WAR LITERATURE              |
| EN361 | POETRY                      |
| EN362 | FILM AND FILM THEORY        |
| EN363 | THE NOVEL                   |
| EN364 | DRAMA                       |
| GE491 | MILITARY GEOGRAPHY          |
| HI344 | MODERN DIPLOMACY            |
| HI391 | WORLD RELIGIONS             |
| LW410 | COMPARATIVE LEGAL SYSTEMS   |

|       |                             |
|-------|-----------------------------|
| LW473 | ENVIRONMENTAL LAW           |
| LW481 | INTERNATIONAL LAW           |
| SS465 | TERRORISM: NEW CHALLENGES   |
| SS475 | COMP POLITICAL INSTITUTIONS |
| SS484 | INTERNATIONAL ECONOMICS     |
| SS486 | STATE BUILDING              |
| SS487 | INT'L POLITICAL ECONOMY     |
| WS358 | STRATEGY AND GENERALSHIP    |
| WS385 | WAR AND ITS THEORISTS       |
| WS455 | COMPARATIVE DEFENSE POLICY  |

**AND****Integrative Experience for the Major** Choose 1 of 4

Cadets selecting an LN490 course must align this course with the appropriate language track chosen for the major.

|        |                                |
|--------|--------------------------------|
| GE491  | MILITARY GEOGRAPHY             |
| LN490P | PORTUGUESE LANG & CULTURE CAP  |
| LN490S | SPANISH LANGUAGE & CULTURE CAP |
| SS486  | STATE BUILDING                 |

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|        |                                |
|--------|--------------------------------|
| LN490P | PORTUGUESE LANG & CULTURE CAP  |
| LN490S | SPANISH LANGUAGE & CULTURE CAP |

## 2029 Foreign Area Studies Major: Latin America w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FSL2H | Foreign Area Studies: Latin America w/ Honors | Foreign Area Studies Major: Latin America w/ Honors | Foreign Area Studies: Latin America w/ Honors | 1            | 1            |

### 2029 Foreign Area Studies Major: Latin America w/ Honors Tracks

| Subject Area                                                                             | Description                 |
|------------------------------------------------------------------------------------------|-----------------------------|
| <b>Additional Elective</b>                                                               |                             |
| Take an additional course in Portuguese or Spanish.                                      |                             |
| <b>AND</b>                                                                               |                             |
| <b>Honors Thesis</b>                                                                     | Choose 1 of 1               |
| Take LN488 and complete an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                                    | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                                               |                             |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

### 2029 Foreign Area Studies Major: Middle East Curriculum

| Code | Short Description                 | Description                             | Transcript Description            | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------------|-----------------------------------------|-----------------------------------|--------------|--------------|
| FSM2 | Foreign Area Studies: Middle East | Foreign Area Studies Major: Middle East | Foreign Area Studies: Middle East | 5            | 8            |

### 2029 Foreign Area Studies Major: Middle East Tracks

| Subject Area                                              | Description                    |
|-----------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                   | Choose 5 of 5                  |
| GE340                                                     | GEOGRAPHY OF GLOBAL CULTURES   |
| GE345                                                     | GEOGRAPHY OF THE MIDDLE EAST   |
| HI339                                                     | THE MODERN MIDDLE EAST         |
| SS366                                                     | COMPARATIVE POLITICS           |
| SS383                                                     | POLITICS & GOVT-MIDDLE EAST    |
| <b>AND</b>                                                |                                |
| <b>Language Track</b>                                     |                                |
| You must choose one of the following two language tracks. |                                |
| <b>Arabic Electives</b>                                   | Choose 4 of 20                 |
| LA371                                                     | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                                     | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                                                     | SPECIAL TOPIC IN ARABIC        |
| LA472                                                     | COLLOQUIAL ARABIC              |
| LA475                                                     | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                                                     | MILITARY SPKG/RDG - ARABIC     |
| LA483                                                     | ARAB CIVILIZATION I            |
| LA484                                                     | ARAB CIVILIZATION II           |
| LA485                                                     | ARABIC LITERATURE I            |
| LA486                                                     | ARABIC LITERATURE II           |

|        |                                |
|--------|--------------------------------|
| LA492  | ARABIC LITERATURE III          |
| LN440A | ARABIC IN CULTURAL CONTEXT     |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**OR****Persian Electives**

Choose 4 of 18

|        |                                |
|--------|--------------------------------|
| LN440Z | PERSIAN IN CULTURAL CONTEXT    |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |
| LZ371  | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372  | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470  | SPECIAL TOPIC IN PERSIAN       |
| LZ475  | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476  | MILITARY SPKG/RDG - PERSIAN    |
| LZ483  | PERSIAN CIVILIZATION I         |
| LZ484  | PERSIAN CIVILIZATION II        |
| LZ485  | PERSIAN LITERATURE I           |
| LZ486  | PERSIAN LITERATURE II          |

**AND****Complementary Support Courses**

Choose 3 of 38

Choose 3 CSCs, each from a different discipline. If GE491 or SS486 is selected as a CSC, cadets must choose a different course from the Integrated Experience list (i.e. GE491 or SS486 cannot double-count as a CSC and the IE for the major).

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN311 | ANCIENT TO EARLY MODERN LIT    |
| EN340 | CONTEMPORARY LITERATURE        |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN363 | THE NOVEL                      |
| EN364 | DRAMA                          |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE491 | MILITARY GEOGRAPHY             |
| HI344 | MODERN DIPLOMACY               |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI383 | MIDDLE EASTERN WARFARE         |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS477 | ECONOMICS OF NATIONAL SECURITY |

|       |                                |
|-------|--------------------------------|
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 4

Cadets selecting an LN490 course must align this course with the appropriate language track chosen for the major.

|        |                                |
|--------|--------------------------------|
| GE491  | MILITARY GEOGRAPHY             |
| LN490A | ARABIC LANGUAGE & CULTURE CAP  |
| LN490Z | PERSIAN LANGUAGE & CULTURE CAP |
| SS486  | STATE BUILDING                 |

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|        |                                |
|--------|--------------------------------|
| LN490A | ARABIC LANGUAGE & CULTURE CAP  |
| LN490Z | PERSIAN LANGUAGE & CULTURE CAP |



## 2029 Foreign Area Studies Major: Middle East w/ Honors Curriculum

| Code  | Short Description                           | Description                                       | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|---------------------------------------------------|---------------------------------------------|--------------|--------------|
| FSM2H | Foreign Area Studies: Middle East w/ Honors | Foreign Area Studies Major: Middle East w/ Honors | Foreign Area Studies: Middle East w/ Honors | 1            | 1            |

### 2029 Foreign Area Studies Major: Middle East w/ Honors Tracks

| Subject Area                                                                             | Description                 |
|------------------------------------------------------------------------------------------|-----------------------------|
| <b>Additional Elective</b>                                                               |                             |
| Take an additional course in Arabic or Persian.                                          |                             |
| <b>AND</b>                                                                               |                             |
| <b>Honors Thesis</b>                                                                     | Choose 1 of 1               |
| Take LN488 and complete an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                                    | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                                               |                             |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Foreign Language Major: Persian & Arabic Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FZA1 | Foreign Language: Persian & Arabic | Foreign Language Major: Persian & Arabic | Foreign Language: Persian & Arabic | 2            | 13           |

### 2029 Foreign Language Major: Persian & Arabic Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 18                 |
| LN440Z                 | PERSIAN IN CULTURAL CONTEXT    |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |

|       |                                |
|-------|--------------------------------|
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LZ371 | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372 | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470 | SPECIAL TOPIC IN PERSIAN       |
| LZ475 | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476 | MILITARY SPKG/RDG - PERSIAN    |
| LZ483 | PERSIAN CIVILIZATION I         |
| LZ484 | PERSIAN CIVILIZATION II        |
| LZ485 | PERSIAN LITERATURE I           |
| LZ486 | PERSIAN LITERATURE II          |

**AND****Secondary Language Track**

Choose 4 of 14

|       |                                |
|-------|--------------------------------|
| LA203 | ARABIC I (STANDARD)            |
| LA204 | ARABIC II (STANDARD)           |
| LA371 | INTENSIVE INTERMEDIATE ARABIC  |
| LA372 | ARABIC FOR ORAL & WRITTEN COMM |
| LA472 | COLLOQUIAL ARABIC              |
| LA475 | ARABIC RDG/WRTG THRU MEDIA     |
| LA476 | MILITARY SPKG/RDG - ARABIC     |
| LA483 | ARAB CIVILIZATION I            |
| LA484 | ARAB CIVILIZATION II           |
| LA485 | ARABIC LITERATURE I            |
| LA486 | ARABIC LITERATURE II           |
| LA492 | ARABIC LITERATURE III          |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 36

Choose 3 of 36, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN351 | WORLD LITERATURE               |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491 | MILITARY GEOGRAPHY             |
| HI339 | THE MODERN MIDDLE EAST         |
| HI344 | MODERN DIPLOMACY               |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI383 | MIDDLE EASTERN WARFARE         |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS383 | POLITICS OF THE MIDDLE EAST    |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS477 | ECONOMICS OF NATIONAL SECURITY |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |

|       |                                |
|-------|--------------------------------|
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490Z | PERSIAN LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in the major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Persian & Arabic with Honors Curriculum

| Code  | Short Description                            | Description                                          | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FZA1H | Foreign Language: Persian & Arabic w/ Honors | Foreign Language Major: Persian & Arabic with Honors | Foreign Language: Persian & Arabic w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Persian & Arabic with Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Arabic Curriculum

| Code | Short Description        | Description                    | Transcript Description   | Req Crse Cnt | Opt Crse Cnt |
|------|--------------------------|--------------------------------|--------------------------|--------------|--------------|
| FLA2 | Foreign Language: Arabic | Foreign Language Major: Arabic | Foreign Language: Arabic | 2            | 11           |

### 2029 Foreign Language Major: Arabic Tracks

| Subject Area                                 | Description                    |
|----------------------------------------------|--------------------------------|
| <b>Required Course</b>                       | Choose 1 of 1                  |
| LN380                                        | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                   |                                |
| <b>Electives</b>                             | Choose 8 of 20                 |
| LA371                                        | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                        | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                                        | SPECIAL TOPIC IN ARABIC        |
| LA472                                        | COLLOQUIAL ARABIC              |
| LA475                                        | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                                        | MILITARY SPKG/RDG - ARABIC     |
| LA483                                        | ARAB CIVILIZATION I            |
| LA484                                        | ARAB CIVILIZATION II           |
| LA485                                        | ARABIC LITERATURE I            |
| LA486                                        | ARABIC LITERATURE II           |
| LA492                                        | ARABIC LITERATURE III          |
| LN440A                                       | ARABIC IN CULTURAL CONTEXT     |
| LN451                                        | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                        | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                        | SEM ABROAD: ADV LANG & CULT II |
| LN493                                        | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                        | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                        | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                   |                                |
| <b>Complementary Support Courses</b>         | Choose 3 of 37                 |
| Choose one each from a different discipline. |                                |
| EN300                                        | LITERARY METHODOLOGIES         |

|        |                                |
|--------|--------------------------------|
| EN340  | CONTEMPORARY LITERATURE        |
| EN351  | WORLD LITERATURE               |
| EN353  | WAR LITERATURE                 |
| EN361  | POETRY                         |
| EN362  | FILM AND FILM THEORY           |
| EN363  | THE NOVEL                      |
| EN364  | DRAMA                          |
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344  | GEOGRAPHY OF AFRICA            |
| GE345  | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491  | MILITARY GEOGRAPHY             |
| HI339  | THE MODERN MIDDLE EAST         |
| HI344  | MODERN DIPLOMACY               |
| HI381  | HISTORY OF IRREGULAR WARFARE   |
| HI383  | MIDDLE EASTERN WARFARE         |
| HI391  | WORLD RELIGIONS                |
| LN482H | SPOKEN HEBREW                  |
| LW410  | COMPARATIVE LEGAL SYSTEMS      |
| LW473  | ENVIRONMENTAL LAW              |
| LW481  | INTERNATIONAL LAW              |
| SS366  | COMPARATIVE POLITICS           |
| SS383  | POLITICS & GOVT-MIDDLE EAST    |
| SS465  | TERRORISM: NEW CHALLENGES      |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS477  | ECONOMICS OF NATIONAL SECURITY |
| SS479  | INTERNATIONAL ORGANIZATION     |
| SS484  | INTERNATIONAL ECONOMICS        |
| SS486  | STATE BUILDING                 |
| SS487  | INT'L POLITICAL ECONOMY        |
| WS358  | STRATEGY AND GENERALSHIP       |
| WS360  | SPECIAL OPS IN THEORY & PRACTI |
| WS365  | REBELLN, INSURGNCY, & CIV WAR  |
| WS455  | COMPARATIVE DEFENSE POLICY     |
| ZH337  | REGIONAL POLITICAL SYSTEMS     |
| ZH347  | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367  | TOPICS IN MICROECONOMICS       |
| ZH377  | TOPICS IN MACROECONOMICS       |
| ZH407  | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427  | TOPICS IN COMPARATIVE POLITICS |
| ZH447  | TOPICS: INTERNATIONAL POLITICS |
| ZH467  | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477  | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490A | ARABIC LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Arabic & Chinese Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FAC2 | Foreign Language: Arabic & Chinese | Foreign Language Major: Arabic & Chinese | Foreign Language: Arabic & Chinese | 2            | 13           |

### 2029 Foreign Language Major: Arabic & Chinese Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 20                 |
| LA371                  | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                  | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                  | SPECIAL TOPIC IN ARABIC        |
| LA472                  | COLLOQUIAL ARABIC              |
| LA475                  | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                  | MILITARY SPKG/RDG - ARABIC     |
| LA483                  | ARAB CIVILIZATION I            |
| LA484                  | ARAB CIVILIZATION II           |
| LA485                  | ARABIC LITERATURE I            |
| LA486                  | ARABIC LITERATURE II           |
| LA492                  | ARABIC LITERATURE III          |
| LN440A                 | ARABIC IN CULTURAL CONTEXT     |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |

|                                                  |                                |
|--------------------------------------------------|--------------------------------|
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 13                 |
| LC203                                            | CHINESE I (STANDARD)           |
| LC204                                            | CHINESE II (STANDARD)          |
| LC371                                            | INTENSIVE INTERMEDIATE CHINESE |
| LC372                                            | CHINESE FOR ORAL & WRIT COMM   |
| LC475                                            | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                                            | MILITARY SPKG/RDG - CHINESE    |
| LC483                                            | CHINESE CIVILIZATION I         |
| LC484                                            | CHINESE CIVILIZATION II        |
| LC485                                            | CHINESE LITERATURE I           |
| LC486                                            | CHINESE LITERATURE II          |
| LC492                                            | CHINESE LITERATURE III         |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 43                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN340                                            | CONTEMPORARY LITERATURE        |
| EN351                                            | WORLD LITERATURE               |
| EN353                                            | WAR LITERATURE                 |
| EN361                                            | POETRY                         |
| EN362                                            | FILM AND FILM THEORY           |
| EN363                                            | THE NOVEL                      |
| EN364                                            | DRAMA                          |
| GE340                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                            | GEOGRAPHY OF AFRICA            |
| GE345                                            | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491                                            | MILITARY GEOGRAPHY             |
| HI339                                            | THE MODERN MIDDLE EAST         |
| HI344                                            | MODERN DIPLOMACY               |
| HI381                                            | HISTORY OF IRREGULAR WARFARE   |
| HI383                                            | MIDDLE EASTERN WARFARE         |
| HI391                                            | WORLD RELIGIONS                |
| LN482H                                           | SPOKEN HEBREW                  |
| LW410                                            | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                            | ENVIRONMENTAL LAW              |
| LW481                                            | INTERNATIONAL LAW              |
| SS366                                            | COMPARATIVE POLITICS           |
| SS383                                            | POLITICS & GOVT-MIDDLE EAST    |
| SS465                                            | TERRORISM: NEW CHALLENGES      |
| SS475                                            | COMP POLITICAL INSTITUTIONS    |
| SS477                                            | ECONOMICS OF NATIONAL SECURITY |
| SS479                                            | INTERNATIONAL ORGANIZATION     |
| SS484                                            | INTERNATIONAL ECONOMICS        |
| SS486                                            | STATE BUILDING                 |
| SS487                                            | INT'L POLITICAL ECONOMY        |
| WS358                                            | STRATEGY AND GENERALSHIP       |
| WS360                                            | SPECIAL OPS IN THEORY & PRACTI |
| WS365                                            | REBELLN, INSURGNCY, & CIV WAR  |
| WS455                                            | COMPARATIVE DEFENSE POLICY     |
| ZH337                                            | REGIONAL POLITICAL SYSTEMS     |

|       |                                |
|-------|--------------------------------|
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490A | ARABIC LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Arabic & Chinese w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FAC2H | Foreign Language: Arabic & Chinese w/ Honors | Foreign Language Major: Arabic & Chinese w/ Honors | Foreign Language: Arabic & Chinese w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Arabic & Chinese w/ Honors Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|



**Honors Thesis Course**

Choose 1 of 1

Write an honors thesis under the direction of a senior faculty member.

LN488

ADV IND STUDY-FOREIGN LANGS

**AND**

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

**2029 Foreign Language Major: Arabic & French Curriculum**

| Code | Short Description                 | Description                             | Transcript Description            | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------------|-----------------------------------------|-----------------------------------|--------------|--------------|
| FAF2 | Foreign Language: Arabic & French | Foreign Language Major: Arabic & French | Foreign Language: Arabic & French | 2            | 13           |

**2029 Foreign Language Major: Arabic & French Tracks**

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 20                 |
| LA371                           | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                           | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                           | SPECIAL TOPIC IN ARABIC        |
| LA472                           | COLLOQUIAL ARABIC              |
| LA475                           | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                           | MILITARY SPKG/RDG - ARABIC     |
| LA483                           | ARAB CIVILIZATION I            |
| LA484                           | ARAB CIVILIZATION II           |
| LA485                           | ARABIC LITERATURE I            |
| LA486                           | ARABIC LITERATURE II           |
| LA492                           | ARABIC LITERATURE III          |
| LN440A                          | ARABIC IN CULTURAL CONTEXT     |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LF203                           | FRENCH I (STANDARD)            |
| LF204                           | FRENCH II (STANDARD)           |
| LF371                           | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                           | FRENCH FOR ORAL & WRITTEN COMM |
| LF475                           | FRENCH RDG/WRTG THRU MEDIA     |

|       |                             |
|-------|-----------------------------|
| LF476 | MILITARY SPKG/RDG - FRENCH  |
| LF483 | FRENCH CIVILIZATION I       |
| LF484 | FRENCH CIVILIZATION II      |
| LF485 | SURVEY OF FRENCH LIT I      |
| LF486 | SURVEY OF FRENCH LIT II     |
| LF492 | MASTERWORKS OF FRENCH LIT   |
| LN487 | ADV IND STUDY-FOREIGN LANGS |
| LN488 | ADV IND STUDY-FOREIGN LANGS |

**AND****Complementary Support Courses**

Choose 3 of 44

Choose 3 CSCs, each from a different discipline.

|        |                                |
|--------|--------------------------------|
| EN300  | LITERARY METHODOLOGIES         |
| EN340  | CONTEMPORARY LITERATURE        |
| EN351  | WORLD LITERATURE               |
| EN353  | WAR LITERATURE                 |
| EN361  | POETRY                         |
| EN362  | FILM AND FILM THEORY           |
| EN363  | THE NOVEL                      |
| EN364  | DRAMA                          |
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344  | GEOGRAPHY OF AFRICA            |
| GE345  | GEOGRAPHY OF THE MIDDLE EAST   |
| HI339  | THE MODERN MIDDLE EAST         |
| HI344  | MODERN DIPLOMACY               |
| HI381  | HISTORY OF IRREGULAR WARFARE   |
| HI383  | MIDDLE EASTERN WARFARE         |
| HI391  | WORLD RELIGIONS                |
| LN482H | SPOKEN HEBREW                  |
| LW410  | COMPARATIVE LEGAL SYSTEMS      |
| LW473  | ENVIRONMENTAL LAW              |
| LW481  | INTERNATIONAL LAW              |
| SS366  | COMPARATIVE POLITICS           |
| SS383  | POLITICS & GOVT-MIDDLE EAST    |
| SS465  | TERRORISM: NEW CHALLENGES      |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS477  | ECONOMICS OF NATIONAL SECURITY |
| SS479  | INTERNATIONAL ORGANIZATION     |
| SS484  | INTERNATIONAL ECONOMICS        |
| SS486  | STATE BUILDING                 |
| SS487  | INT'L POLITICAL ECONOMY        |
| WS358  | STRATEGY AND GENERALSHIP       |
| WS360  | SPECIAL OPS IN THEORY & PRACTI |
| WS365  | REBELLN, INSURGNCY, & CIV WAR  |
| WS455  | COMPARATIVE DEFENSE POLICY     |
| ZH337  | REGIONAL POLITICAL SYSTEMS     |
| ZH347  | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367  | TOPICS IN MICROECONOMICS       |
| ZH377  | TOPICS IN MACROECONOMICS       |
| ZH407  | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427  | TOPICS IN COMPARATIVE POLITICS |
| ZH447  | TOPICS: INTERNATIONAL POLITICS |
| ZH467  | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477  | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 1

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| LN490A                                                                                                                                                                       | ARABIC LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Science Depth</b>                                                                                                                                                         |                               |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                               |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II          |
| CH275                                                                                                                                                                        | BIOLOGY                       |
| PH202                                                                                                                                                                        | PHYSICS II                    |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II           |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Curriculum Requirement</b>                                                                                                                                                |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: Arabic & French w/ Honors Curriculum

| Code  | Short Description                           | Description                                       | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|---------------------------------------------------|---------------------------------------------|--------------|--------------|
| FAF2H | Foreign Language: Arabic & French w/ Honors | Foreign Language Major: Arabic & French w/ Honors | Foreign Language: Arabic & French w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Arabic & French w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Arabic & German Curriculum

| Code | Short Description                 | Description                             | Transcript Description            | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------------|-----------------------------------------|-----------------------------------|--------------|--------------|
| FAG1 | Foreign Language: Arabic & German | Foreign Language Major: Arabic & German | Foreign Language: Arabic & German | 2            | 13           |

## 2029 Foreign Language Major: Arabic & German Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 20                 |
| LA371                           | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                           | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                           | SPECIAL TOPIC IN ARABIC        |
| LA472                           | COLLOQUIAL ARABIC              |
| LA475                           | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                           | MILITARY SPKG/RDG - ARABIC     |
| LA483                           | ARAB CIVILIZATION I            |
| LA484                           | ARAB CIVILIZATION II           |
| LA485                           | ARABIC LITERATURE I            |
| LA486                           | ARABIC LITERATURE II           |
| LA492                           | ARABIC LITERATURE III          |
| LN440A                          | ARABIC IN CULTURAL CONTEXT     |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LG203                           | GERMAN I (STANDARD)            |
| LG204                           | GERMAN II (STANDARD)           |
| LG371                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG475                           | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                           | MILITARY SPKG/RDG - GERMAN     |
| LG483                           | GERMAN CIVILIZATION I          |
| LG484                           | GERMAN CIVILIZATION II         |
| LG485                           | SURVEY OF GERMAN LIT I         |
| LG486                           | SURVEY OF GERMAN LIT II        |
| LG492                           | 20TH & 21ST CENTURY GERMANY    |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |

|                                                  |                                  |
|--------------------------------------------------|----------------------------------|
| LN488                                            | ADV IND STUDY-FOREIGN LANGS      |
| <b>AND</b>                                       |                                  |
| <b>Complementary Support Courses</b>             | Choose 3 of 46                   |
| Choose 3 CSCs, each from a different discipline. |                                  |
| EN300                                            | LITERARY METHODOLOGIES           |
| EN340                                            | CONTEMPORARY LITERATURE          |
| EN351                                            | WORLD LITERATURE                 |
| EN353                                            | WAR LITERATURE                   |
| EN361                                            | POETRY                           |
| EN362                                            | FILM AND FILM THEORY             |
| EN363                                            | THE NOVEL                        |
| EN364                                            | DRAMA                            |
| GE340                                            | GEOGRAPHY OF GLOBAL CULTURES     |
| GE344                                            | GEOGRAPHY OF AFRICA              |
| GE345                                            | GEOGRAPHY OF THE MIDDLE EAST     |
| GE491                                            | MILITARY GEOGRAPHY               |
| HI339                                            | THE MODERN MIDDLE EAST           |
| HI344                                            | MODERN DIPLOMACY                 |
| HI368                                            | MOD CENTRAL & E. EUR,1896-1989   |
| HI381                                            | HISTORY OF IRREGULAR WARFARE     |
| HI383                                            | MIDDLE EASTERN WARFARE           |
| HI391                                            | WORLD RELIGIONS                  |
| LN482H                                           | SPOKEN HEBREW                    |
| LW410                                            | COMPARATIVE LEGAL SYSTEMS        |
| LW473                                            | ENVIRONMENTAL LAW                |
| LW481                                            | INTERNATIONAL LAW                |
| SS366                                            | COMPARATIVE POLITICS             |
| SS383                                            | POLITICS & GOVT-MIDDLE EAST      |
| SS465                                            | TERRORISM: NEW CHALLENGES        |
| SS475                                            | COMP POLITICAL INSTITUTIONS      |
| SS477                                            | ECONOMICS OF NATIONAL SECURITY   |
| SS479                                            | INTERNATIONAL ORGANIZATION       |
| SS484                                            | INTERNATIONAL ECONOMICS          |
| SS486                                            | STATE BUILDING                   |
| SS487                                            | INT'L POLITICAL ECONOMY          |
| WS358                                            | STRATEGY AND GENERALSHIP         |
| WS360                                            | SPECIAL OPS IN THEORY & PRACTICE |
| WS365                                            | REBELLN, INSURGNCY, & CIV WAR    |
| WS385                                            | WAR AND ITS THEORISTS            |
| WS455                                            | COMPARATIVE DEFENSE POLICY       |
| XH405                                            | THE HOLOCAUST AND ITS LEGACY     |
| ZH337                                            | REGIONAL POLITICAL SYSTEMS       |
| ZH347                                            | INT'L ORGNZTNS & INSTITUTIONS    |
| ZH367                                            | TOPICS IN MICROECONOMICS         |
| ZH377                                            | TOPICS IN MACROECONOMICS         |
| ZH407                                            | TOPICS/AMERICAN FOREIGN POLICY   |
| ZH427                                            | TOPICS IN COMPARATIVE POLITICS   |
| ZH447                                            | TOPICS: INTERNATIONAL POLITICS   |
| ZH467                                            | TOPICS-INTERNATIONAL ECONOMICS   |
| ZH477                                            | TOPICS-INT'L BUSINESS/FINANCE    |
| <b>AND</b>                                       |                                  |
| <b>Integrative Experience for the Major</b>      | Choose 1 of 1                    |
| LN490A                                           | ARABIC LANGUAGE & CULTURE CAP    |
| <b>AND</b>                                       |                                  |
| <b>Science Depth</b>                             |                                  |

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

#### **STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

#### **AND**

#### **Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

#### **IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

#### **Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

#### **Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## **2029 Foreign Language Major: Arabic & German w/ Honors Curriculum**

| <b>Code</b> | <b>Short Description</b>                    | <b>Description</b>                                | <b>Transcript Description</b>               | <b>Req Crse Cnt</b> | <b>Opt Crse Cnt</b> |
|-------------|---------------------------------------------|---------------------------------------------------|---------------------------------------------|---------------------|---------------------|
| FAG1H       | Foreign Language: Arabic & German w/ Honors | Foreign Language Major: Arabic & German w/ Honors | Foreign Language: Arabic & German w/ Honors | 1                   | 0                   |

### **2029 Foreign Language Major: Arabic & German w/ Honors Tracks**

| <b>Subject Area</b>         | <b>Description</b>                                                     |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## **2029 Foreign Language Major: Arabic & Persian Curriculum**

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|-----------------|-----------------|
| FAZ2 | Foreign Language: Arabic & Persian | Foreign Language Major: Arabic & Persian | Foreign Language: Arabic & Persian | 2               | 13              |

### 2029 Foreign Language Major: Arabic & Persian Tracks

| Subject Area                                     | Description                    |
|--------------------------------------------------|--------------------------------|
| <b>Required Course</b>                           | Choose 1 of 1                  |
| LN380                                            | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                       |                                |
| <b>Electives</b>                                 | Choose 6 of 20                 |
| LA371                                            | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                            | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                                            | SPECIAL TOPIC IN ARABIC        |
| LA472                                            | COLLOQUIAL ARABIC              |
| LA475                                            | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                                            | MILITARY SPKG/RDG - ARABIC     |
| LA483                                            | ARAB CIVILIZATION I            |
| LA484                                            | ARAB CIVILIZATION II           |
| LA485                                            | ARABIC LITERATURE I            |
| LA486                                            | ARABIC LITERATURE II           |
| LA492                                            | ARABIC LITERATURE III          |
| LN440A                                           | ARABIC IN CULTURAL CONTEXT     |
| LN451                                            | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                            | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                            | SEM ABROAD: ADV LANG & CULT II |
| LN493                                            | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 13                 |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LZ203                                            | PERSIAN I (STANDARD)           |
| LZ204                                            | PERSIAN II (STANDARD)          |
| LZ371                                            | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                                            | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                                            | SPECIAL TOPIC IN PERSIAN       |
| LZ475                                            | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                                            | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                                            | PERSIAN CIVILIZATION I         |
| LZ484                                            | PERSIAN CIVILIZATION II        |
| LZ485                                            | PERSIAN LITERATURE I           |
| LZ486                                            | PERSIAN LITERATURE II          |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 43                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN340                                            | CONTEMPORARY LITERATURE        |
| EN351                                            | WORLD LITERATURE               |

|            |                                |
|------------|--------------------------------|
| EN353      | WAR LITERATURE                 |
| EN361      | POETRY                         |
| EN362      | FILM AND FILM THEORY           |
| EN363      | THE NOVEL                      |
| EN364      | DRAMA                          |
| GE340      | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344      | GEOGRAPHY OF AFRICA            |
| GE345      | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491      | MILITARY GEOGRAPHY             |
| HI339      | THE MODERN MIDDLE EAST         |
| HI344      | MODERN DIPLOMACY               |
| HI381      | HISTORY OF IRREGULAR WARFARE   |
| HI383      | MIDDLE EASTERN WARFARE         |
| HI391      | WORLD RELIGIONS                |
| LN482H     | SPOKEN HEBREW                  |
| LW410      | COMPARATIVE LEGAL SYSTEMS      |
| LW473      | ENVIRONMENTAL LAW              |
| LW481      | INTERNATIONAL LAW              |
| SS366      | COMPARATIVE POLITICS           |
| SS383      | POLITICS & GOVT-MIDDLE EAST    |
| SS465      | TERRORISM: NEW CHALLENGES      |
| SS475      | COMP POLITICAL INSTITUTIONS    |
| SS477      | ECONOMICS OF NATIONAL SECURITY |
| SS479      | INTERNATIONAL ORGANIZATION     |
| SS484      | INTERNATIONAL ECONOMICS        |
| SS486      | STATE BUILDING                 |
| SS487      | INT'L POLITICAL ECONOMY        |
| WS358      | STRATEGY AND GENERALSHIP       |
| WS360      | SPECIAL OPS IN THEORY & PRACTI |
| WS365      | REBELLN, INSURGNCY, & CIV WAR  |
| WS455      | COMPARATIVE DEFENSE POLICY     |
| ZH337      | REGIONAL POLITICAL SYSTEMS     |
| ZH347      | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367      | TOPICS IN MICROECONOMICS       |
| ZH377      | TOPICS IN MACROECONOMICS       |
| ZH407      | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427      | TOPICS IN COMPARATIVE POLITICS |
| ZH447      | TOPICS: INTERNATIONAL POLITICS |
| ZH467      | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477      | TOPICS-INT'L BUSINESS/FINANCE  |
| <b>AND</b> |                                |

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**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490A | ARABIC LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

---

**AND**
**Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |



**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

**2029 Foreign Language Major: Arabic & Persian w/ Honors Curriculum**

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FAZ2H | Foreign Language: Arabic & Persian w/ Honors | Foreign Language Major: Arabic & Persian w/ Honors | Foreign Language: Arabic & Persian w/ Honors | 1            | 0            |

**2029 Foreign Language Major: Arabic & Persian w/ Honors Tracks**

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

**2029 Foreign Language Major: Arabic & Portuguese Curriculum**

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------|--------------|
| FAP2 | Foreign Language: Arabic & Portuguese | Foreign Language Major: Arabic & Portuguese | Foreign Language: Arabic & Portuguese | 2            | 13           |

**2029 Foreign Language Major: Arabic & Portuguese Tracks**

| Subject Area           | Description   |
|------------------------|---------------|
| <b>Required Course</b> | Choose 1 of 1 |

|                                                  |                                |
|--------------------------------------------------|--------------------------------|
| LN380                                            | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                       |                                |
| <b>Electives</b>                                 | Choose 6 of 20                 |
| LA371                                            | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                            | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                                            | SPECIAL TOPIC IN ARABIC        |
| LA472                                            | COLLOQUIAL ARABIC              |
| LA475                                            | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                                            | MILITARY SPKG/RDG - ARABIC     |
| LA483                                            | ARAB CIVILIZATION I            |
| LA484                                            | ARAB CIVILIZATION II           |
| LA485                                            | ARABIC LITERATURE I            |
| LA486                                            | ARABIC LITERATURE II           |
| LA492                                            | ARABIC LITERATURE III          |
| LN440A                                           | ARABIC IN CULTURAL CONTEXT     |
| LN451                                            | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                            | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                            | SEM ABROAD: ADV LANG & CULT II |
| LN493                                            | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 12                 |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LP203                                            | PORTUGUESE I (STANDARD)        |
| LP204                                            | PORTUGUESE II (STANDARD)       |
| LP371                                            | INTENSIVE INTERMED. PORTUGUESE |
| LP372                                            | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475                                            | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476                                            | MILITARY SPKG/RDG - PORTUGUESE |
| LP483                                            | PORTUGUESE CIVILIZATION I      |
| LP484                                            | PORTUGUESE CIVILIZATION II     |
| LP485                                            | SURVEY OF PORTUGUESE LIT I     |
| LP492                                            | LIT OF PORT-SPKG WORLD         |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 43                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN340                                            | CONTEMPORARY LITERATURE        |
| EN351                                            | WORLD LITERATURE               |
| EN353                                            | WAR LITERATURE                 |
| EN361                                            | POETRY                         |
| EN362                                            | FILM AND FILM THEORY           |
| EN363                                            | THE NOVEL                      |
| EN364                                            | DRAMA                          |
| GE340                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                            | GEOGRAPHY OF AFRICA            |
| GE345                                            | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491                                            | MILITARY GEOGRAPHY             |
| HI339                                            | THE MODERN MIDDLE EAST         |
| HI344                                            | MODERN DIPLOMACY               |
| HI381                                            | HISTORY OF IRREGULAR WARFARE   |

|        |                                  |
|--------|----------------------------------|
| HI383  | MIDDLE EASTERN WARFARE           |
| HI391  | WORLD RELIGIONS                  |
| LN482H | SPOKEN HEBREW                    |
| LW410  | COMPARATIVE LEGAL SYSTEMS        |
| LW473  | ENVIRONMENTAL LAW                |
| LW481  | INTERNATIONAL LAW                |
| SS366  | COMPARATIVE POLITICS             |
| SS383  | POLITICS & GOVT-MIDDLE EAST      |
| SS465  | TERRORISM: NEW CHALLENGES        |
| SS475  | COMP POLITICAL INSTITUTIONS      |
| SS477  | ECONOMICS OF NATIONAL SECURITY   |
| SS479  | INTERNATIONAL ORGANIZATION       |
| SS484  | INTERNATIONAL ECONOMICS          |
| SS486  | STATE BUILDING                   |
| SS487  | INT'L POLITICAL ECONOMY          |
| WS358  | STRATEGY AND GENERALSHIP         |
| WS360  | SPECIAL OPS IN THEORY & PRACTICE |
| WS365  | REBELLN, INSURGNCY, & CIV WAR    |
| WS455  | COMPARATIVE DEFENSE POLICY       |
| ZH337  | REGIONAL POLITICAL SYSTEMS       |
| ZH347  | INT'L ORGNZTNS & INSTITUTIONS    |
| ZH367  | TOPICS IN MICROECONOMICS         |
| ZH377  | TOPICS IN MACROECONOMICS         |
| ZH407  | TOPICS/AMERICAN FOREIGN POLICY   |
| ZH427  | TOPICS IN COMPARATIVE POLITICS   |
| ZH447  | TOPICS: INTERNATIONAL POLITICS   |
| ZH467  | TOPICS-INTERNATIONAL ECONOMICS   |
| ZH477  | TOPICS-INT'L BUSINESS/FINANCE    |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490A

ARABIC LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Arabic & Portuguese w/ Honors Curriculum

| Code  | Short Description                           | Description                                           | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|-------------------------------------------------------|---------------------------------------------|--------------|--------------|
| FAP2H | Foreign Lang: Arabic & Portuguese w/ Honors | Foreign Language Major: Arabic & Portuguese w/ Honors | Foreign Lang: Arabic & Portuguese w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Arabic & Portuguese w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Arabic & Russian Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FAR2 | Foreign Language: Arabic & Russian | Foreign Language Major: Arabic & Russian | Foreign Language: Arabic & Russian | 2            | 13           |

## 2029 Foreign Language Major: Arabic & Russian Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 20                 |
| LA371                  | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                  | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                  | SPECIAL TOPIC IN ARABIC        |
| LA472                  | COLLOQUIAL ARABIC              |
| LA475                  | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                  | MILITARY SPKG/RDG - ARABIC     |
| LA483                  | ARAB CIVILIZATION I            |

|        |                                |
|--------|--------------------------------|
| LA484  | ARAB CIVILIZATION II           |
| LA485  | ARABIC LITERATURE I            |
| LA486  | ARABIC LITERATURE II           |
| LA492  | ARABIC LITERATURE III          |
| LN440A | ARABIC IN CULTURAL CONTEXT     |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LR203 | RUSSIAN I (STANDARD)           |
| LR204 | RUSSIAN II (STANDARD)          |
| LR371 | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372 | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475 | RUSSIAN RDG/WRITG THRU MEDIA   |
| LR476 | MILITARY SPKG/RDG - RUSSIAN    |
| LR483 | RUSSIAN CIV I                  |
| LR484 | RUSSIAN CIV II                 |
| LR485 | SURVEY OF RUSSIAN LITERATURE I |
| LR486 | SURVEY OF RUSSIAN LIT. II      |
| LR492 | RUSSIAN LIFE IN FICTION        |

**AND****Complementary Support Courses**

Choose 3 of 43

Choose 3 of 38, each from a different discipline.

|        |                              |
|--------|------------------------------|
| EN300  | LITERARY METHODOLOGIES       |
| EN340  | CONTEMPORARY LITERATURE      |
| EN351  | WORLD LITERATURE             |
| EN353  | WAR LITERATURE               |
| EN361  | POETRY                       |
| EN362  | FILM AND FILM THEORY         |
| EN363  | THE NOVEL                    |
| EN364  | DRAMA                        |
| GE340  | GEOGRAPHY OF GLOBAL CULTURES |
| GE344  | GEOGRAPHY OF AFRICA          |
| GE345  | GEOGRAPHY OF THE MIDDLE EAST |
| GE491  | MILITARY GEOGRAPHY           |
| HI339  | THE MODERN MIDDLE EAST       |
| HI344  | MODERN DIPLOMACY             |
| HI381  | HISTORY OF IRREGULAR WARFARE |
| HI383  | MIDDLE EASTERN WARFARE       |
| HI391  | WORLD RELIGIONS              |
| LN482H | SPOKEN HEBREW                |
| LW410  | COMPARATIVE LEGAL SYSTEMS    |
| LW473  | ENVIRONMENTAL LAW            |
| LW481  | INTERNATIONAL LAW            |
| SS366  | COMPARATIVE POLITICS         |
| SS383  | POLITICS & GOVT-MIDDLE EAST  |
| SS465  | TERRORISM: NEW CHALLENGES    |

|       |                                |
|-------|--------------------------------|
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS477 | ECONOMICS OF NATIONAL SECURITY |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490A | ARABIC LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Arabic & Russian w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FAR2H | Foreign Language: Arabic & Russian w/ Honors | Foreign Language Major: Arabic & Russian w/ Honors | Foreign Language: Arabic & Russian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Arabic & Russian w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Arabic & Spanish Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FAS2 | Foreign Language: Arabic & Spanish | Foreign Language Major: Arabic & Spanish | Foreign Language: Arabic & Spanish | 2            | 13           |

### 2029 Foreign Language Major: Arabic & Spanish Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 20                 |
| LA371                  | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                  | ARABIC FOR ORAL & WRITTEN COMM |
| LA470                  | SPECIAL TOPIC IN ARABIC        |
| LA472                  | COLLOQUIAL ARABIC              |
| LA475                  | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                  | MILITARY SPKG/RDG - ARABIC     |
| LA483                  | ARAB CIVILIZATION I            |
| LA484                  | ARAB CIVILIZATION II           |
| LA485                  | ARABIC LITERATURE I            |
| LA486                  | ARABIC LITERATURE II           |
| LA492                  | ARABIC LITERATURE III          |
| LN440A                 | ARABIC IN CULTURAL CONTEXT     |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |

|                                                  |                                |
|--------------------------------------------------|--------------------------------|
| LN493                                            | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 13                 |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LS203                                            | SPANISH I (STANDARD)           |
| LS204                                            | SPANISH II (STANDARD)          |
| LS371                                            | INTENSIVE INTERMEDIATE SPANISH |
| LS372                                            | SPANISH FOR ORAL & WRIT COMM   |
| LS475                                            | SPANISH RDG/WRTG THRU MEDIA    |
| LS476                                            | MILITARY SPKG/RDG - SPANISH    |
| LS483                                            | SPANISH CIV AND CULTURE        |
| LS484                                            | SPANISH AMERICAN CIV AND CULT  |
| LS485                                            | SPANISH-AMERICAN LITERATURE    |
| LS486                                            | THE LITERATURE OF SPAIN        |
| LS492                                            | 20TH/21ST CENTURY HISPANIC LIT |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 43                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN340                                            | CONTEMPORARY LITERATURE        |
| EN351                                            | WORLD LITERATURE               |
| EN353                                            | WAR LITERATURE                 |
| EN361                                            | POETRY                         |
| EN362                                            | FILM AND FILM THEORY           |
| EN363                                            | THE NOVEL                      |
| EN364                                            | DRAMA                          |
| GE340                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                            | GEOGRAPHY OF AFRICA            |
| GE345                                            | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491                                            | MILITARY GEOGRAPHY             |
| HI339                                            | THE MODERN MIDDLE EAST         |
| HI344                                            | MODERN DIPLOMACY               |
| HI381                                            | HISTORY OF IRREGULAR WARFARE   |
| HI383                                            | MIDDLE EASTERN WARFARE         |
| HI391                                            | WORLD RELIGIONS                |
| LN482H                                           | SPOKEN HEBREW                  |
| LW410                                            | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                            | ENVIRONMENTAL LAW              |
| LW481                                            | INTERNATIONAL LAW              |
| SS366                                            | COMPARATIVE POLITICS           |
| SS383                                            | POLITICS & GOVT-MIDDLE EAST    |
| SS465                                            | TERRORISM: NEW CHALLENGES      |
| SS475                                            | COMP POLITICAL INSTITUTIONS    |
| SS477                                            | ECONOMICS OF NATIONAL SECURITY |
| SS479                                            | INTERNATIONAL ORGANIZATION     |
| SS484                                            | INTERNATIONAL ECONOMICS        |
| SS486                                            | STATE BUILDING                 |
| SS487                                            | INT'L POLITICAL ECONOMY        |
| WS358                                            | STRATEGY AND GENERALSHIP       |
| WS360                                            | SPECIAL OPS IN THEORY & PRACTI |
| WS365                                            | REBELLN, INSURGNCY, & CIV WAR  |
| WS455                                            | COMPARATIVE DEFENSE POLICY     |



|       |                                |
|-------|--------------------------------|
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490A | ARABIC LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Arabic & Spanish w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FAS2H | Foreign Language: Arabic & Spanish w/ Honors | Foreign Language Major: Arabic & Spanish w/ Honors | Foreign Language: Arabic & Spanish w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Arabic & Spanish w/ Honors Tracks

| Subject Area                                                                                                                                               | Description                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                                                                                                                | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member.                                                                                     |                             |
| LN488                                                                                                                                                      | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                                                                                                                 |                             |
| Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major. |                             |

## 2029 Foreign Language Major: Arabic w/ Honors Curriculum

| Code  | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|-------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FLA2H | Foreign Language: Arabic w/ Honors | Foreign Language Major: Arabic w/ Honors | Foreign Language: Arabic w/ Honors | 1            | 1            |

## 2029 Foreign Language Major: Arabic w/ Honors Tracks

| Subject Area                                                                   | Description                    |
|--------------------------------------------------------------------------------|--------------------------------|
| <b>Additional Elective</b>                                                     | Choose 1 of 15                 |
| Take an additional advanced-level elective, not already taken, from this list. |                                |
| LA470                                                                          | SPECIAL TOPIC IN ARABIC        |
| LA476                                                                          | MILITARY SPKG/RDG - ARABIC     |
| LA483                                                                          | ARAB CIVILIZATION I            |
| LA484                                                                          | ARAB CIVILIZATION II           |
| LA485                                                                          | ARABIC LITERATURE I            |
| LA486                                                                          | ARABIC LITERATURE II           |
| LA492                                                                          | ARABIC LITERATURE III          |
| LN440A                                                                         | ARABIC IN CULTURAL CONTEXT     |
| LN451                                                                          | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                                                          | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                                                          | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                                                          | SEM ABROAD: ADV LANG & CULT II |
| LN493                                                                          | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                                                          | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                                                          | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                                                     |                                |
| <b>Honors Thesis</b>                                                           | Choose 1 of 1                  |
| Write an honors thesis under the direction of a senior faculty member.         |                                |
| LN488                                                                          | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                     |                                |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Foreign Language Major: Chinese Curriculum

| Code | Short Description         | Description                     | Transcript Description    | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|---------------------------|---------------------------------|---------------------------|-----------------|-----------------|
| FLC2 | Foreign Language: Chinese | Foreign Language Major: Chinese | Foreign Language: Chinese | 2               | 11              |

### 2029 Foreign Language Major: Chinese Tracks

| Subject Area                                 | Description                    |
|----------------------------------------------|--------------------------------|
| <b>Required Course</b>                       | Choose 1 of 1                  |
| LN380                                        | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                   |                                |
| <b>Electives</b>                             | Choose 8 of 19                 |
| LC371                                        | INTENSIVE INTERMEDIATE CHINESE |
| LC372                                        | CHINESE FOR ORAL & WRIT COMM   |
| LC470                                        | SPECIAL TOPIC IN CHINESE       |
| LC475                                        | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                                        | MILITARY SPKG/RDG - CHINESE    |
| LC483                                        | CHINESE CIVILIZATION I         |
| LC484                                        | CHINESE CIVILIZATION II        |
| LC485                                        | CHINESE LITERATURE I           |
| LC486                                        | CHINESE LITERATURE II          |
| LC492                                        | CHINESE LITERATURE III         |
| LN440C                                       | CHINESE IN CULTURAL CONTEXT    |
| LN451                                        | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                        | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                        | SEM ABROAD: ADV LANG & CULT II |
| LN493                                        | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                        | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                        | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                   |                                |
| <b>Complementary Support Courses</b>         | Choose 3 of 29                 |
| Choose one each from a different discipline. |                                |
| EN300                                        | LITERARY METHODOLOGIES         |
| EN340                                        | CONTEMPORARY LITERATURE        |
| EN351                                        | WORLD LITERATURE               |
| EN353                                        | WAR LITERATURE                 |
| EN361                                        | POETRY                         |
| EN362                                        | FILM AND FILM THEORY           |
| EN363                                        | THE NOVEL                      |
| EN364                                        | DRAMA                          |
| GE340                                        | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342                                        | GEOGRAPHY OF ASIA              |
| GE491                                        | MILITARY GEOGRAPHY             |
| HI337                                        | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI344                                        | MODERN DIPLOMACY               |
| HI347                                        | ASIAN WARFARE AND POLITICS     |
| HI391                                        | WORLD RELIGIONS                |
| LW410                                        | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                        | ENVIRONMENTAL LAW              |
| LW481                                        | INTERNATIONAL LAW              |
| PY369                                        | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366                                        | COMPARATIVE POLITICS           |
| SS372                                        | POLITICS AND GOV OF CHINA      |

|       |                             |
|-------|-----------------------------|
| SS374 | POL & GOV OF KOREAS & JAPAN |
| SS465 | TERRORISM: NEW CHALLENGES   |
| SS475 | COMP POLITICAL INSTITUTIONS |
| SS484 | INTERNATIONAL ECONOMICS     |
| SS486 | STATE BUILDING              |
| SS487 | INT'L POLITICAL ECONOMY     |
| WS358 | STRATEGY AND GENERALSHIP    |
| WS455 | COMPARATIVE DEFENSE POLICY  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490C

CHINESE LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Chinese & Arabic Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FCA1 | Foreign Language: Chinese & Arabic | Foreign Language Major: Chinese & Arabic | Foreign Language: Chinese & Arabic | 2            | 13           |

### 2029 Foreign Language Major: Chinese & Arabic Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

|                                                  |                                |
|--------------------------------------------------|--------------------------------|
| <b>Required Course</b>                           | Choose 1 of 1                  |
| LN380                                            | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                       |                                |
| <b>Electives</b>                                 | Choose 6 of 19                 |
| LC371                                            | INTENSIVE INTERMEDIATE CHINESE |
| LC372                                            | CHINESE FOR ORAL & WRIT COMM   |
| LC470                                            | SPECIAL TOPIC IN CHINESE       |
| LC475                                            | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                                            | MILITARY SPKG/RDG - CHINESE    |
| LC483                                            | CHINESE CIVILIZATION I         |
| LC484                                            | CHINESE CIVILIZATION II        |
| LC485                                            | CHINESE LITERATURE I           |
| LC486                                            | CHINESE LITERATURE II          |
| LC492                                            | CHINESE LITERATURE III         |
| LN440C                                           | CHINESE IN CULTURAL CONTEXT    |
| LN451                                            | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                            | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                            | SEM ABROAD: ADV LANG & CULT II |
| LN493                                            | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 14                 |
| LA203                                            | ARABIC I (STANDARD)            |
| LA204                                            | ARABIC II (STANDARD)           |
| LA371                                            | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                            | ARABIC FOR ORAL & WRITTEN COMM |
| LA472                                            | COLLOQUIAL ARABIC              |
| LA475                                            | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                                            | MILITARY SPKG/RDG - ARABIC     |
| LA483                                            | ARAB CIVILIZATION I            |
| LA484                                            | ARAB CIVILIZATION II           |
| LA485                                            | ARABIC LITERATURE I            |
| LA486                                            | ARABIC LITERATURE II           |
| LA492                                            | ARABIC LITERATURE III          |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 46                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN340                                            | CONTEMPORARY LITERATURE        |
| EN351                                            | WORLD LITERATURE               |
| EN353                                            | WAR LITERATURE                 |
| EN361                                            | POETRY                         |
| EN362                                            | FILM AND FILM THEORY           |
| EN364                                            | DRAMA                          |
| GE340                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342                                            | GEOGRAPHY OF ASIA              |
| GE344                                            | GEOGRAPHY OF AFRICA            |
| GE491                                            | MILITARY GEOGRAPHY             |
| HI337                                            | CHINA: EMPIRE, REPUBLIC, & MAO |

|            |                                |
|------------|--------------------------------|
| HI339      | THE MODERN MIDDLE EAST         |
| HI344      | MODERN DIPLOMACY               |
| HI347      | ASIAN WARFARE AND POLITICS     |
| HI381      | HISTORY OF IRREGULAR WARFARE   |
| HI383      | MIDDLE EASTERN WARFARE         |
| HI391      | WORLD RELIGIONS                |
| LW410      | COMPARATIVE LEGAL SYSTEMS      |
| LW473      | ENVIRONMENTAL LAW              |
| LW481      | INTERNATIONAL LAW              |
| PY369      | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366      | COMPARATIVE POLITICS           |
| SS372      | POLITICS AND GOV OF CHINA      |
| SS374      | POL & GOV OF KOREAS & JAPAN    |
| SS465      | TERRORISM: NEW CHALLENGES      |
| SS475      | COMP POLITICAL INSTITUTIONS    |
| SS477      | ECONOMICS OF NATIONAL SECURITY |
| SS479      | INTERNATIONAL ORGANIZATION     |
| SS484      | INTERNATIONAL ECONOMICS        |
| SS486      | STATE BUILDING                 |
| SS487      | INT'L POLITICAL ECONOMY        |
| WS358      | STRATEGY AND GENERALSHIP       |
| WS360      | SPECIAL OPS IN THEORY & PRACTI |
| WS365      | REBELLN, INSURGNCY, & CIV WAR  |
| WS455      | COMPARATIVE DEFENSE POLICY     |
| ZH337      | REGIONAL POLITICAL SYSTEMS     |
| ZH347      | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367      | TOPICS IN MICROECONOMICS       |
| ZH377      | TOPICS IN MACROECONOMICS       |
| ZH407      | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427      | TOPICS IN COMPARATIVE POLITICS |
| ZH447      | TOPICS: INTERNATIONAL POLITICS |
| ZH467      | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477      | TOPICS-INT'L BUSINESS/FINANCE  |
| <b>AND</b> |                                |

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**Integrative Experience for the Major** Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490C | CHINESE LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                   |
|-------|-------------------|
| CY305 | CYBER FOUNDATIONS |
|-------|-------------------|

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: Chinese & Arabic w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FCA1H | Foreign Language: Chinese & Arabic w/ Honors | Foreign Language Major: Chinese & Arabic w/ Honors | Foreign Language: Chinese & Arabic w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Chinese & Arabic w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Chinese & French Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FCF2 | Foreign Language: Chinese & French | Foreign Language Major: Chinese & French | Foreign Language: Chinese & French | 2            | 13           |

## 2029 Foreign Language Major: Chinese & French Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LC371                  | INTENSIVE INTERMEDIATE CHINESE |
| LC372                  | CHINESE FOR ORAL & WRIT COMM   |
| LC470                  | SPECIAL TOPIC IN CHINESE       |

|        |                                |
|--------|--------------------------------|
| LC475  | CHINESE RDG/WRTG THRU MEDIA    |
| LC476  | MILITARY SPKG/RDG - CHINESE    |
| LC483  | CHINESE CIVILIZATION I         |
| LC484  | CHINESE CIVILIZATION II        |
| LC485  | CHINESE LITERATURE I           |
| LC486  | CHINESE LITERATURE II          |
| LC492  | CHINESE LITERATURE III         |
| LN440C | CHINESE IN CULTURAL CONTEXT    |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LF203 | FRENCH I (STANDARD)            |
| LF204 | FRENCH II (STANDARD)           |
| LF371 | INTENSIVE INTERMEDIATE FRENCH  |
| LF372 | FRENCH FOR ORAL & WRITTEN COMM |
| LF475 | FRENCH RDG/WRTG THRU MEDIA     |
| LF476 | MILITARY SPKG/RDG - FRENCH     |
| LF483 | FRENCH CIVILIZATION I          |
| LF484 | FRENCH CIVILIZATION II         |
| LF485 | SURVEY OF FRENCH LIT I         |
| LF486 | SURVEY OF FRENCH LIT II        |
| LF492 | MASTERWORKS OF FRENCH LIT      |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 29

Choose 3 CSCs, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN340 | CONTEMPORARY LITERATURE        |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN363 | THE NOVEL                      |
| EN364 | DRAMA                          |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342 | GEOGRAPHY OF ASIA              |
| GE491 | MILITARY GEOGRAPHY             |
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI344 | MODERN DIPLOMACY               |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366 | COMPARATIVE POLITICS           |
| SS372 | POLITICS AND GOV OF CHINA      |



|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| SS374                                                                                                                                                                        | POL & GOV OF KOREAS & JAPAN    |
| SS465                                                                                                                                                                        | TERRORISM: NEW CHALLENGES      |
| SS475                                                                                                                                                                        | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                                                                                        | INTERNATIONAL ECONOMICS        |
| SS486                                                                                                                                                                        | STATE BUILDING                 |
| SS487                                                                                                                                                                        | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                                                                                        | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                                                                                        | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                  |
| LN490C                                                                                                                                                                       | CHINESE LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Curriculum Requirement</b>                                                                                                                                                |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES     |

## 2029 Foreign Language Major: Chinese & French w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FCF2H | Foreign Language: Chinese & French w/ Honors | Foreign Language Major: Chinese & French w/ Honors | Foreign Language: Chinese & French w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Chinese & French w/ Honors Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

**Honors Thesis Course** Choose 1 of 1  
 Write an honors thesis under the direction of a senior faculty member.  
 LN488 ADV IND STUDY-FOREIGN LANGS  
**AND**

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Chinese & German Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FCG2 | Foreign Language: Chinese & German | Foreign Language Major: Chinese & German | Foreign Language: Chinese & German | 2            | 13           |

### 2029 Foreign Language Major: Chinese & German Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LC371                           | INTENSIVE INTERMEDIATE CHINESE |
| LC372                           | CHINESE FOR ORAL & WRIT COMM   |
| LC470                           | SPECIAL TOPIC IN CHINESE       |
| LC475                           | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                           | MILITARY SPKG/RDG - CHINESE    |
| LC483                           | CHINESE CIVILIZATION I         |
| LC484                           | CHINESE CIVILIZATION II        |
| LC485                           | CHINESE LITERATURE I           |
| LC486                           | CHINESE LITERATURE II          |
| LC492                           | CHINESE LITERATURE III         |
| LN440C                          | CHINESE IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LG203                           | GERMAN I (STANDARD)            |
| LG204                           | GERMAN II (STANDARD)           |
| LG371                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG475                           | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                           | MILITARY SPKG/RDG - GERMAN     |

|                                                                                                         |                                |
|---------------------------------------------------------------------------------------------------------|--------------------------------|
| LG483                                                                                                   | GERMAN CIVILIZATION I          |
| LG484                                                                                                   | GERMAN CIVILIZATION II         |
| LG485                                                                                                   | SURVEY OF GERMAN LIT I         |
| LG486                                                                                                   | SURVEY OF GERMAN LIT II        |
| LG492                                                                                                   | 20TH & 21ST CENTURY GERMANY    |
| LN487                                                                                                   | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                   | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                                              |                                |
| <hr/>                                                                                                   |                                |
| <b>Complementary Support Courses</b>                                                                    | Choose 3 of 32                 |
| Choose 3 CSCs, each from a different discipline.                                                        |                                |
| EN300                                                                                                   | LITERARY METHODOLOGIES         |
| EN340                                                                                                   | CONTEMPORARY LITERATURE        |
| EN351                                                                                                   | WORLD LITERATURE               |
| EN353                                                                                                   | WAR LITERATURE                 |
| EN361                                                                                                   | POETRY                         |
| EN362                                                                                                   | FILM AND FILM THEORY           |
| EN363                                                                                                   | THE NOVEL                      |
| EN364                                                                                                   | DRAMA                          |
| GE340                                                                                                   | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342                                                                                                   | GEOGRAPHY OF ASIA              |
| GE491                                                                                                   | MILITARY GEOGRAPHY             |
| HI337                                                                                                   | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI344                                                                                                   | MODERN DIPLOMACY               |
| HI347                                                                                                   | ASIAN WARFARE AND POLITICS     |
| HI368                                                                                                   | MOD CENTRAL & E. EUR,1896-1989 |
| HI391                                                                                                   | WORLD RELIGIONS                |
| LW410                                                                                                   | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                                                                                   | ENVIRONMENTAL LAW              |
| LW481                                                                                                   | INTERNATIONAL LAW              |
| PY369                                                                                                   | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366                                                                                                   | COMPARATIVE POLITICS           |
| SS372                                                                                                   | POLITICS AND GOV OF CHINA      |
| SS374                                                                                                   | POL & GOV OF KOREAS & JAPAN    |
| SS465                                                                                                   | TERRORISM: NEW CHALLENGES      |
| SS475                                                                                                   | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                   | INTERNATIONAL ECONOMICS        |
| SS486                                                                                                   | STATE BUILDING                 |
| SS487                                                                                                   | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                   | STRATEGY AND GENERALSHIP       |
| WS385                                                                                                   | WAR AND ITS THEORISTS          |
| WS455                                                                                                   | COMPARATIVE DEFENSE POLICY     |
| XH405                                                                                                   | THE HOLOCAUST AND ITS LEGACY   |
| <b>AND</b>                                                                                              |                                |
| <hr/>                                                                                                   |                                |
| <b>Integrative Experience for the Major</b>                                                             | Choose 1 of 1                  |
| LN490C                                                                                                  | CHINESE LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                              |                                |
| <hr/>                                                                                                   |                                |
| <b>Science Depth</b>                                                                                    |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement. |                                |
| CH102                                                                                                   | GENERAL CHEMISTRY II           |
| CH275                                                                                                   | BIOLOGY                        |
| PH202                                                                                                   | PHYSICS II                     |
| PH252                                                                                                   | ADVANCED PHYSICS II            |
| PH275                                                                                                   | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                       |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                         |                                |

CY305 CYBER FOUNDATIONS  
CY355 CYBER FOUNDATIONS - COMPUTING

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305 CYBER FOUNDATIONS  
CY355 CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380 NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Chinese & German w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FCG2H | Foreign Language: Chinese & German w/ Honors | Foreign Language Major: Chinese & German w/ Honors | Foreign Language: Chinese & German w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Chinese & German w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Chinese & Persian Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FCZ2 | Foreign Language: Chinese & Persian | Foreign Language Major: Chinese & Persian | Foreign Language: Chinese & Persian | 2            | 13           |

## 2029 Foreign Language Major: Chinese & Persian Tracks

| Subject Area                                     | Description                    |
|--------------------------------------------------|--------------------------------|
| <b>Required Course</b>                           | Choose 1 of 1                  |
| LN380                                            | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                       |                                |
| <b>Electives</b>                                 | Choose 6 of 19                 |
| LC371                                            | INTENSIVE INTERMEDIATE CHINESE |
| LC372                                            | CHINESE FOR ORAL & WRIT COMM   |
| LC470                                            | SPECIAL TOPIC IN CHINESE       |
| LC475                                            | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                                            | MILITARY SPKG/RDG - CHINESE    |
| LC483                                            | CHINESE CIVILIZATION I         |
| LC484                                            | CHINESE CIVILIZATION II        |
| LC485                                            | CHINESE LITERATURE I           |
| LC486                                            | CHINESE LITERATURE II          |
| LC492                                            | CHINESE LITERATURE III         |
| LN440C                                           | CHINESE IN CULTURAL CONTEXT    |
| LN451                                            | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                            | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                            | SEM ABROAD: ADV LANG & CULT II |
| LN493                                            | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 13                 |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LZ203                                            | PERSIAN I (STANDARD)           |
| LZ204                                            | PERSIAN II (STANDARD)          |
| LZ371                                            | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                                            | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                                            | SPECIAL TOPIC IN PERSIAN       |
| LZ475                                            | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                                            | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                                            | PERSIAN CIVILIZATION I         |
| LZ484                                            | PERSIAN CIVILIZATION II        |
| LZ485                                            | PERSIAN LITERATURE I           |
| LZ486                                            | PERSIAN LITERATURE II          |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 29                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN340                                            | CONTEMPORARY LITERATURE        |
| EN351                                            | WORLD LITERATURE               |
| EN353                                            | WAR LITERATURE                 |
| EN361                                            | POETRY                         |
| EN362                                            | FILM AND FILM THEORY           |
| EN363                                            | THE NOVEL                      |
| EN364                                            | DRAMA                          |
| GE340                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342                                            | GEOGRAPHY OF ASIA              |
| GE491                                            | MILITARY GEOGRAPHY             |
| HI337                                            | CHINA: EMPIRE, REPUBLIC, & MAO |

|       |                                |
|-------|--------------------------------|
| HI344 | MODERN DIPLOMACY               |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366 | COMPARATIVE POLITICS           |
| SS372 | POLITICS AND GOV OF CHINA      |
| SS374 | POL & GOV OF KOREAS & JAPAN    |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490C | CHINESE LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: Chinese & Persian w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FCZ2H | Foreign Language: Chinese & Persian w/ Honors | Foreign Language Major: Chinese & Persian w/ Honors | Foreign Language: Chinese & Persian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Chinese & Persian w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Chinese & Portuguese Curriculum

| Code | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FCP2 | Foreign Language: Chinese & Portuguese | Foreign Language Major: Chinese & Portuguese | Foreign Language: Chinese & Portuguese | 2            | 13           |

### 2029 Foreign Language Major: Chinese & Portuguese Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LC371                  | INTENSIVE INTERMEDIATE CHINESE |
| LC372                  | CHINESE FOR ORAL & WRIT COMM   |
| LC470                  | SPECIAL TOPIC IN CHINESE       |
| LC475                  | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                  | MILITARY SPKG/RDG - CHINESE    |
| LC483                  | CHINESE CIVILIZATION I         |
| LC484                  | CHINESE CIVILIZATION II        |
| LC485                  | CHINESE LITERATURE I           |
| LC486                  | CHINESE LITERATURE II          |
| LC492                  | CHINESE LITERATURE III         |
| LN440C                 | CHINESE IN CULTURAL CONTEXT    |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |
| LN494                  | SEM ABROAD: ADV LANG & CULT IV |
| LN495                  | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>             |                                |

|                                 |                                |
|---------------------------------|--------------------------------|
| <b>Secondary Language Track</b> | Choose 4 of 12                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LP203                           | PORTUGUESE I (STANDARD)        |
| LP204                           | PORTUGUESE II (STANDARD)       |
| LP371                           | INTENSIVE INTERMED. PORTUGUESE |
| LP372                           | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475                           | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476                           | MILITARY SPKG/RDG - PORTUGUESE |
| LP483                           | PORTUGUESE CIVILIZATION I      |
| LP484                           | PORTUGUESE CIVILIZATION II     |
| LP485                           | SURVEY OF PORTUGUESE LIT I     |
| LP492                           | LIT OF PORT-SPKG WORLD         |

**AND****Complementary Support Courses** Choose 3 of 28

Choose 3 CSCs, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN340 | CONTEMPORARY LITERATURE        |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN363 | THE NOVEL                      |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342 | GEOGRAPHY OF ASIA              |
| GE491 | MILITARY GEOGRAPHY             |
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI344 | MODERN DIPLOMACY               |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366 | COMPARATIVE POLITICS           |
| SS372 | POLITICS AND GOV OF CHINA      |
| SS374 | POL & GOV OF KOREAS & JAPAN    |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

**AND****Integrative Experience for the Major** Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490C | CHINESE LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |



|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Curriculum Requirement</b>                                                                                                                                                |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: Chinese & Portuguese w/ Honors Curriculum

| Code  | Short Description                            | Description                                            | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FCP2H | Foreign Lang: Chinese & Portuguese w/ Honors | Foreign Language Major: Chinese & Portuguese w/ Honors | Foreign Lang: Chinese & Portuguese w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Chinese & Portuguese w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Chinese & Russian Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FCR2 | Foreign Language: Chinese & Russian | Foreign Language Major: Chinese & Russian | Foreign Language: Chinese & Russian | 2            | 13           |

## 2029 Foreign Language Major: Chinese &amp; Russian Tracks

| Subject Area                                     | Description                    |
|--------------------------------------------------|--------------------------------|
| <b>Required Course</b>                           | Choose 1 of 1                  |
| LN380                                            | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                       |                                |
| <b>Electives</b>                                 | Choose 6 of 19                 |
| LC371                                            | INTENSIVE INTERMEDIATE CHINESE |
| LC372                                            | CHINESE FOR ORAL & WRIT COMM   |
| LC470                                            | SPECIAL TOPIC IN CHINESE       |
| LC475                                            | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                                            | MILITARY SPKG/RDG - CHINESE    |
| LC483                                            | CHINESE CIVILIZATION I         |
| LC484                                            | CHINESE CIVILIZATION II        |
| LC485                                            | CHINESE LITERATURE I           |
| LC486                                            | CHINESE LITERATURE II          |
| LC492                                            | CHINESE LITERATURE III         |
| LN440C                                           | CHINESE IN CULTURAL CONTEXT    |
| LN451                                            | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                            | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                            | SEM ABROAD: ADV LANG & CULT II |
| LN493                                            | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 13                 |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LR203                                            | RUSSIAN I (STANDARD)           |
| LR204                                            | RUSSIAN II (STANDARD)          |
| LR371                                            | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                                            | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475                                            | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                                            | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                                            | RUSSIAN CIV I                  |
| LR484                                            | RUSSIAN CIV II                 |
| LR485                                            | SURVEY OF RUSSIAN LITERATURE I |
| LR486                                            | SURVEY OF RUSSIAN LIT. II      |
| LR492                                            | RUSSIAN LIFE IN FICTION        |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 28                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN340                                            | CONTEMPORARY LITERATURE        |
| EN351                                            | WORLD LITERATURE               |
| EN353                                            | WAR LITERATURE                 |
| EN361                                            | POETRY                         |
| EN362                                            | FILM AND FILM THEORY           |
| EN363                                            | THE NOVEL                      |
| GE340                                            | GEOGRAPHY OF GLOBAL CULTURES   |

|       |                                |
|-------|--------------------------------|
| GE342 | GEOGRAPHY OF ASIA              |
| GE491 | MILITARY GEOGRAPHY             |
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI344 | MODERN DIPLOMACY               |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366 | COMPARATIVE POLITICS           |
| SS372 | POLITICS AND GOV OF CHINA      |
| SS374 | POL & GOV OF KOREAS & JAPAN    |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490C

CHINESE LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

CH102

GENERAL CHEMISTRY II

CH275

BIOLOGY

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Chinese & Russian w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FCR2H | Foreign Language: Chinese & Russian w/ Honors | Foreign Language Major: Chinese & Russian w/ Honors | Foreign Language: Chinese & Russian w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Chinese & Russian w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Chinese & Spanish Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FCS2 | Foreign Language: Chinese & Spanish | Foreign Language Major: Chinese & Spanish | Foreign Language: Chinese & Spanish | 2            | 13           |

## 2029 Foreign Language Major: Chinese & Spanish Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LC371                  | INTENSIVE INTERMEDIATE CHINESE |
| LC372                  | CHINESE FOR ORAL & WRIT COMM   |
| LC470                  | SPECIAL TOPIC IN CHINESE       |
| LC475                  | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                  | MILITARY SPKG/RDG - CHINESE    |
| LC483                  | CHINESE CIVILIZATION I         |
| LC484                  | CHINESE CIVILIZATION II        |
| LC485                  | CHINESE LITERATURE I           |
| LC486                  | CHINESE LITERATURE II          |
| LC492                  | CHINESE LITERATURE III         |
| LN440C                 | CHINESE IN CULTURAL CONTEXT    |

|       |                                |
|-------|--------------------------------|
| LN451 | ADV LANG & CULTURE IN CONTEXT  |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LN491 | SEM ABROAD: ADV LANG & CULT I  |
| LN492 | SEM ABROAD: ADV LANG & CULT II |
| LN493 | SEM ABROAD: ADV LANG&CULT III  |
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LS203 | SPANISH I (STANDARD)           |
| LS204 | SPANISH II (STANDARD)          |
| LS371 | INTENSIVE INTERMEDIATE SPANISH |
| LS372 | SPANISH FOR ORAL & WRIT COMM   |
| LS475 | SPANISH RDG/WRTG THRU MEDIA    |
| LS476 | MILITARY SPKG/RDG - SPANISH    |
| LS483 | SPANISH CIV AND CULTURE        |
| LS484 | SPANISH AMERICAN CIV AND CULT  |
| LS485 | SPANISH-AMERICAN LITERATURE    |
| LS486 | THE LITERATURE OF SPAIN        |
| LS492 | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Complementary Support Courses**

Choose 3 of 29

Choose 3 CSCs, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN340 | CONTEMPORARY LITERATURE        |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN363 | THE NOVEL                      |
| EN364 | DRAMA                          |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342 | GEOGRAPHY OF ASIA              |
| GE491 | MILITARY GEOGRAPHY             |
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI344 | MODERN DIPLOMACY               |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366 | COMPARATIVE POLITICS           |
| SS372 | POLITICS AND GOV OF CHINA      |
| SS374 | POL & GOV OF KOREAS & JAPAN    |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  |                                |
| Choose 1 of 1                                                                                                                                                                |                                |
| LN490C                                                                                                                                                                       | CHINESE LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Curriculum Requirement</b>                                                                                                                                                |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES     |

## 2029 Foreign Language Major: Chinese & Spanish w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FCS2H | Foreign Language: Chinese & Spanish w/ Honors | Foreign Language Major: Chinese & Spanish w/ Honors | Foreign Language: Chinese & Spanish w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Chinese & Spanish w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Chinese w/ Honors Curriculum

| Code  | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FLC2H | Foreign Language: Chinese w/ Honors | Foreign Language Major: Chinese w/ Honors | Foreign Language: Chinese w/ Honors | 1            | 1            |

## 2029 Foreign Language Major: Chinese w/ Honors Tracks

| Subject Area               | Description                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------|
| <b>Additional Elective</b> | Choose 1 of 16<br>Take an additional advanced-level elective, not already taken, from this list. |
| LC470                      | SPECIAL TOPIC IN CHINESE                                                                         |
| LC475                      | CHINESE RDG/WRTG THRU MEDIA                                                                      |
| LC476                      | MILITARY SPKG/RDG - CHINESE                                                                      |
| LC483                      | CHINESE CIVILIZATION I                                                                           |
| LC484                      | CHINESE CIVILIZATION II                                                                          |
| LC485                      | CHINESE LITERATURE I                                                                             |
| LC486                      | CHINESE LITERATURE II                                                                            |
| LC492                      | CHINESE LITERATURE III                                                                           |
| LN440C                     | CHINESE IN CULTURAL CONTEXT                                                                      |
| LN451                      | ADV LANG & CULTURE IN CONTEXT                                                                    |
| LN487                      | ADV IND STUDY-FOREIGN LANGS                                                                      |
| LN491                      | SEM ABROAD: ADV LANG & CULT I                                                                    |
| LN492                      | SEM ABROAD: ADV LANG & CULT II                                                                   |
| LN493                      | SEM ABROAD: ADV LANG&CULT III                                                                    |
| LN494                      | SEM ABROAD: ADV LANG & CULT IV                                                                   |
| LN495                      | SEM ABROAD: ADV LANG & CULT V                                                                    |
| <b>AND</b>                 |                                                                                                  |
| <b>Honors Thesis</b>       | Choose 1 of 1<br>Write an honors thesis under the direction of a senior faculty member.          |
| LN488                      | ADV IND STUDY-FOREIGN LANGS                                                                      |
| <b>AND</b>                 |                                                                                                  |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Foreign Language Major: French Curriculum

| Code | Short Description        | Description                    | Transcript Description   | Req Crse Cnt | Opt Crse Cnt |
|------|--------------------------|--------------------------------|--------------------------|--------------|--------------|
| FLF2 | Foreign Language: French | Foreign Language Major: French | Foreign Language: French | 2            | 11           |

## 2029 Foreign Language Major: French Tracks

| Subject Area                                 | Description                    |
|----------------------------------------------|--------------------------------|
| <b>Required Course</b>                       | Choose 1 of 1                  |
| LN380                                        | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                   |                                |
| <b>Electives</b>                             | Choose 8 of 19                 |
| LF371                                        | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                                        | FRENCH FOR ORAL & WRITTEN COMM |
| LF470                                        | SPECIAL TOPIC IN FRENCH        |
| LF475                                        | FRENCH RDG/WRTG THRU MEDIA     |
| LF476                                        | MILITARY SPKG/RDG - FRENCH     |
| LF483                                        | FRENCH CIVILIZATION I          |
| LF484                                        | FRENCH CIVILIZATION II         |
| LF485                                        | SURVEY OF FRENCH LIT I         |
| LF486                                        | SURVEY OF FRENCH LIT II        |
| LF492                                        | MASTERWORKS OF FRENCH LIT      |
| LN440F                                       | FRENCH IN CULTURAL CONTEXT     |
| LN451                                        | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                        | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                        | SEM ABROAD: ADV LANG & CULT II |
| LN493                                        | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                        | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                        | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                   |                                |
| <b>Complementary Support Courses</b>         | Choose 3 of 32                 |
| Choose one each from a different discipline. |                                |
| EN300                                        | LITERARY METHODOLOGIES         |
| EN311                                        | ANCIENT TO EARLY MODERN LIT    |
| EN340                                        | CONTEMPORARY LITERATURE        |
| EN351                                        | WORLD LITERATURE               |
| EN353                                        | WAR LITERATURE                 |
| EN361                                        | POETRY                         |
| EN362                                        | FILM AND FILM THEORY           |
| EN363                                        | THE NOVEL                      |
| EN364                                        | DRAMA                          |
| GE340                                        | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                        | GEOGRAPHY OF AFRICA            |
| GE346                                        | GEOGRAPHY OF EUROPE            |
| GE491                                        | MILITARY GEOGRAPHY             |
| HI338                                        | WARFARE IN AGE OF REVOLUTIONS  |
| HI344                                        | MODERN DIPLOMACY               |
| HI345                                        | MODERN AFRICA                  |
| HI361                                        | MEDIEVAL EUROPE                |
| HI364                                        | MODERN WESTERN EUROPE          |
| HI391                                        | WORLD RELIGIONS                |
| LW410                                        | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                        | ENVIRONMENTAL LAW              |
| LW481                                        | INTERNATIONAL LAW              |
| SS366                                        | COMPARATIVE POLITICS           |
| SS377                                        | POLITICS & GOV OF EUROPE       |
| SS465                                        | TERRORISM: NEW CHALLENGES      |



|       |                             |
|-------|-----------------------------|
| SS475 | COMP POLITICAL INSTITUTIONS |
| SS484 | INTERNATIONAL ECONOMICS     |
| SS486 | STATE BUILDING              |
| SS487 | INT'L POLITICAL ECONOMY     |
| WS358 | STRATEGY AND GENERALSHIP    |
| WS455 | COMPARATIVE DEFENSE POLICY  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490F

FRENCH LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: French & Arabic Curriculum

| Code | Short Description                 | Description                             | Transcript Description            | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-----------------------------------|-----------------------------------------|-----------------------------------|-----------------|-----------------|
| FFA1 | Foreign Language: French & Arabic | Foreign Language Major: French & Arabic | Foreign Language: French & Arabic | 2               | 13              |

### 2029 Foreign Language Major: French & Arabic Tracks

| Subject Area           | Description                |
|------------------------|----------------------------|
| <b>Required Course</b> | Choose 1 of 1              |
| LN380                  | NATURE OF MODERN LANGUAGES |

**AND****Electives**

Choose 6 of 19

|        |                                |
|--------|--------------------------------|
| LF371  | INTENSIVE INTERMEDIATE FRENCH  |
| LF372  | FRENCH FOR ORAL & WRITTEN COMM |
| LF470  | SPECIAL TOPIC IN FRENCH        |
| LF475  | FRENCH RDG/WRTG THRU MEDIA     |
| LF476  | MILITARY SPKG/RDG - FRENCH     |
| LF483  | FRENCH CIVILIZATION I          |
| LF484  | FRENCH CIVILIZATION II         |
| LF485  | SURVEY OF FRENCH LIT I         |
| LF486  | SURVEY OF FRENCH LIT II        |
| LF492  | MASTERWORKS OF FRENCH LIT      |
| LN440F | FRENCH IN CULTURAL CONTEXT     |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**AND****Secondary Language Track**

Choose 4 of 14

|       |                                |
|-------|--------------------------------|
| LA203 | ARABIC I (STANDARD)            |
| LA204 | ARABIC II (STANDARD)           |
| LA371 | INTENSIVE INTERMEDIATE ARABIC  |
| LA372 | ARABIC FOR ORAL & WRITTEN COMM |
| LA472 | COLLOQUIAL ARABIC              |
| LA475 | ARABIC RDG/WRTG THRU MEDIA     |
| LA476 | MILITARY SPKG/RDG - ARABIC     |
| LA483 | ARAB CIVILIZATION I            |
| LA484 | ARAB CIVILIZATION II           |
| LA485 | ARABIC LITERATURE I            |
| LA486 | ARABIC LITERATURE II           |
| LA492 | ARABIC LITERATURE III          |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 47

Choose 3 CSCs, each from a different discipline.

|       |                               |
|-------|-------------------------------|
| EN300 | LITERARY METHODOLOGIES        |
| EN311 | ANCIENT TO EARLY MODERN LIT   |
| EN340 | CONTEMPORARY LITERATURE       |
| EN351 | WORLD LITERATURE              |
| EN353 | WAR LITERATURE                |
| EN361 | POETRY                        |
| EN362 | FILM AND FILM THEORY          |
| EN363 | THE NOVEL                     |
| EN364 | DRAMA                         |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE344 | GEOGRAPHY OF AFRICA           |
| GE346 | GEOGRAPHY OF EUROPE           |
| GE491 | MILITARY GEOGRAPHY            |
| HI338 | WARFARE IN AGE OF REVOLUTIONS |
| HI339 | THE MODERN MIDDLE EAST        |

|       |                                |
|-------|--------------------------------|
| HI344 | MODERN DIPLOMACY               |
| HI345 | MODERN AFRICA                  |
| HI361 | MEDIEVAL EUROPE                |
| HI364 | MODERN WESTERN EUROPE          |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI383 | MIDDLE EASTERN WARFARE         |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS477 | ECONOMICS OF NATIONAL SECURITY |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490F | FRENCH LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

**2029 Foreign Language Major: French & Arabic w/ Honors Curriculum**

| Code  | Short Description                           | Description                                       | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|---------------------------------------------------|---------------------------------------------|--------------|--------------|
| FFA1H | Foreign Language: French & Arabic w/ Honors | Foreign Language Major: French & Arabic w/ Honors | Foreign Language: French & Arabic w/ Honors | 1            | 0            |

**2029 Foreign Language Major: French & Arabic w/ Honors Tracks**

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

**2029 Foreign Language Major: French & Chinese Curriculum**

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FFC1 | Foreign Language: French & Chinese | Foreign Language Major: French & Chinese | Foreign Language: French & Chinese | 2            | 13           |

**2029 Foreign Language Major: French & Chinese Tracks**

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LF371                  | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                  | FRENCH FOR ORAL & WRITTEN COMM |
| LF470                  | SPECIAL TOPIC IN FRENCH        |
| LF475                  | FRENCH RDG/WRTG THRU MEDIA     |

|        |                                |
|--------|--------------------------------|
| LF476  | MILITARY SPKG/RDG - FRENCH     |
| LF483  | FRENCH CIVILIZATION I          |
| LF484  | FRENCH CIVILIZATION II         |
| LF485  | SURVEY OF FRENCH LIT I         |
| LF486  | SURVEY OF FRENCH LIT II        |
| LF492  | MASTERWORKS OF FRENCH LIT      |
| LN440F | FRENCH IN CULTURAL CONTEXT     |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LC203 | CHINESE I (STANDARD)           |
| LC204 | CHINESE II (STANDARD)          |
| LC371 | INTENSIVE INTERMEDIATE CHINESE |
| LC372 | CHINESE FOR ORAL & WRIT COMM   |
| LC475 | CHINESE RDG/WRTG THRU MEDIA    |
| LC476 | MILITARY SPKG/RDG - CHINESE    |
| LC483 | CHINESE CIVILIZATION I         |
| LC484 | CHINESE CIVILIZATION II        |
| LC485 | CHINESE LITERATURE I           |
| LC486 | CHINESE LITERATURE II          |
| LC492 | CHINESE LITERATURE III         |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 31

Choose 3 CSCs, each from a different discipline.

|       |                               |
|-------|-------------------------------|
| EN300 | LITERARY METHODOLOGIES        |
| EN311 | ANCIENT TO EARLY MODERN LIT   |
| EN340 | CONTEMPORARY LITERATURE       |
| EN351 | WORLD LITERATURE              |
| EN353 | WAR LITERATURE                |
| EN361 | POETRY                        |
| EN362 | FILM AND FILM THEORY          |
| EN363 | THE NOVEL                     |
| EN364 | DRAMA                         |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE344 | GEOGRAPHY OF AFRICA           |
| GE346 | GEOGRAPHY OF EUROPE           |
| GE491 | MILITARY GEOGRAPHY            |
| HI338 | WARFARE IN AGE OF REVOLUTIONS |
| HI344 | MODERN DIPLOMACY              |
| HI345 | MODERN AFRICA                 |
| HI361 | MEDIEVAL EUROPE               |
| HI364 | MODERN WESTERN EUROPE         |
| HI391 | WORLD RELIGIONS               |
| LW410 | COMPARATIVE LEGAL SYSTEMS     |
| LW473 | ENVIRONMENTAL LAW             |
| LW481 | INTERNATIONAL LAW             |

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| SS366                                                                                                                                                                        | COMPARATIVE POLITICS          |
| SS377                                                                                                                                                                        | POLITICS & GOV OF EUROPE      |
| SS465                                                                                                                                                                        | TERRORISM: NEW CHALLENGES     |
| SS475                                                                                                                                                                        | COMP POLITICAL INSTITUTIONS   |
| SS484                                                                                                                                                                        | INTERNATIONAL ECONOMICS       |
| SS486                                                                                                                                                                        | STATE BUILDING                |
| SS487                                                                                                                                                                        | INT'L POLITICAL ECONOMY       |
| WS358                                                                                                                                                                        | STRATEGY AND GENERALSHIP      |
| WS455                                                                                                                                                                        | COMPARATIVE DEFENSE POLICY    |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                 |
| LN490F                                                                                                                                                                       | FRENCH LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Science Depth</b>                                                                                                                                                         |                               |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                               |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II          |
| CH275                                                                                                                                                                        | BIOLOGY                       |
| PH202                                                                                                                                                                        | PHYSICS II                    |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II           |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Curriculum Requirement</b>                                                                                                                                                |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: French & Chinese w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FFC1H | Foreign Language: French & Chinese w/ Honors | Foreign Language Major: French & Chinese w/ Honors | Foreign Language: French & Chinese w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: French & Chinese w/ Honors Tracks

| Subject Area                                                                                                                                               | Description                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                                                                                                                | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member.                                                                                     |                             |
| LN488                                                                                                                                                      | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                                                                                                                 |                             |
| Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major. |                             |

## 2029 Foreign Language Major: French & German Curriculum

| Code | Short Description                 | Description                             | Transcript Description            | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------------|-----------------------------------------|-----------------------------------|--------------|--------------|
| FFG2 | Foreign Language: French & German | Foreign Language Major: French & German | Foreign Language: French & German | 2            | 13           |

## 2029 Foreign Language Major: French & German Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LF371                           | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                           | FRENCH FOR ORAL & WRITTEN COMM |
| LF470                           | SPECIAL TOPIC IN FRENCH        |
| LF475                           | FRENCH RDG/WRTG THRU MEDIA     |
| LF476                           | MILITARY SPKG/RDG - FRENCH     |
| LF483                           | FRENCH CIVILIZATION I          |
| LF484                           | FRENCH CIVILIZATION II         |
| LF485                           | SURVEY OF FRENCH LIT I         |
| LF486                           | SURVEY OF FRENCH LIT II        |
| LF492                           | MASTERWORKS OF FRENCH LIT      |
| LN440F                          | FRENCH IN CULTURAL CONTEXT     |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LG203                           | GERMAN I (STANDARD)            |
| LG204                           | GERMAN II (STANDARD)           |
| LG371                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG475                           | GERMAN RDG/WRTG THRU MEDIA     |

|       |                             |
|-------|-----------------------------|
| LG476 | MILITARY SPKG/RDG - GERMAN  |
| LG483 | GERMAN CIVILIZATION I       |
| LG484 | GERMAN CIVILIZATION II      |
| LG485 | SURVEY OF GERMAN LIT I      |
| LG486 | SURVEY OF GERMAN LIT II     |
| LG492 | 20TH & 21ST CENTURY GERMANY |
| LN487 | ADV IND STUDY-FOREIGN LANGS |
| LN488 | ADV IND STUDY-FOREIGN LANGS |

**AND****Complementary Support Courses**

Choose 3 of 34

Choose 3 CSCs, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN311 | ANCIENT TO EARLY MODERN LIT    |
| EN340 | CONTEMPORARY LITERATURE        |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN363 | THE NOVEL                      |
| EN364 | DRAMA                          |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE346 | GEOGRAPHY OF EUROPE            |
| GE491 | MILITARY GEOGRAPHY             |
| HI338 | WARFARE IN AGE OF REVOLUTIONS  |
| HI344 | MODERN DIPLOMACY               |
| HI345 | MODERN AFRICA                  |
| HI361 | MEDIEVAL EUROPE                |
| HI364 | MODERN WESTERN EUROPE          |
| HI368 | MOD CENTRAL & E. EUR,1896-1989 |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS385 | WAR AND ITS THEORISTS          |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490F

FRENCH LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |



|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Curriculum Requirement</b>                                                                                                                                                |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: French & German w/ Honors Curriculum

| Code  | Short Description                           | Description                                       | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|---------------------------------------------------|---------------------------------------------|--------------|--------------|
| FFG2H | Foreign Language: French & German w/ Honors | Foreign Language Major: French & German w/ Honors | Foreign Language: French & German w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: French & German w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: French & Persian Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FFZ2 | Foreign Language: French & Persian | Foreign Language Major: French & Persian | Foreign Language: French & Persian | 2            | 13           |

## 2029 Foreign Language Major: French &amp; Persian Tracks

| Subject Area                                     | Description                    |
|--------------------------------------------------|--------------------------------|
| <b>Required Course</b>                           | Choose 1 of 1                  |
| LN380                                            | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                       |                                |
| <b>Electives</b>                                 | Choose 6 of 19                 |
| LF371                                            | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                                            | FRENCH FOR ORAL & WRITTEN COMM |
| LF470                                            | SPECIAL TOPIC IN FRENCH        |
| LF475                                            | FRENCH RDG/WRTG THRU MEDIA     |
| LF476                                            | MILITARY SPKG/RDG - FRENCH     |
| LF483                                            | FRENCH CIVILIZATION I          |
| LF484                                            | FRENCH CIVILIZATION II         |
| LF485                                            | SURVEY OF FRENCH LIT I         |
| LF486                                            | SURVEY OF FRENCH LIT II        |
| LF492                                            | MASTERWORKS OF FRENCH LIT      |
| LN440F                                           | FRENCH IN CULTURAL CONTEXT     |
| LN451                                            | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                            | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                            | SEM ABROAD: ADV LANG & CULT II |
| LN493                                            | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 13                 |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LZ203                                            | PERSIAN I (STANDARD)           |
| LZ204                                            | PERSIAN II (STANDARD)          |
| LZ371                                            | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                                            | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                                            | SPECIAL TOPIC IN PERSIAN       |
| LZ475                                            | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                                            | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                                            | PERSIAN CIVILIZATION I         |
| LZ484                                            | PERSIAN CIVILIZATION II        |
| LZ485                                            | PERSIAN LITERATURE I           |
| LZ486                                            | PERSIAN LITERATURE II          |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 36                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN311                                            | ANCIENT TO EARLY MODERN LIT    |
| EN340                                            | CONTEMPORARY LITERATURE        |
| EN351                                            | WORLD LITERATURE               |
| EN353                                            | WAR LITERATURE                 |
| EN361                                            | POETRY                         |
| EN362                                            | FILM AND FILM THEORY           |
| EN363                                            | THE NOVEL                      |

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EN364                                                                                                                                                                        | DRAMA                         |
| GE340                                                                                                                                                                        | GEOGRAPHY OF GLOBAL CULTURES  |
| GE344                                                                                                                                                                        | GEOGRAPHY OF AFRICA           |
| GE346                                                                                                                                                                        | GEOGRAPHY OF EUROPE           |
| GE491                                                                                                                                                                        | MILITARY GEOGRAPHY            |
| HI338                                                                                                                                                                        | WARFARE IN AGE OF REVOLUTIONS |
| HI344                                                                                                                                                                        | MODERN DIPLOMACY              |
| HI345                                                                                                                                                                        | MODERN AFRICA                 |
| HI361                                                                                                                                                                        | MEDIEVAL EUROPE               |
| HI364                                                                                                                                                                        | MODERN WESTERN EUROPE         |
| HI391                                                                                                                                                                        | WORLD RELIGIONS               |
| LW410                                                                                                                                                                        | COMPARATIVE LEGAL SYSTEMS     |
| LW473                                                                                                                                                                        | ENVIRONMENTAL LAW             |
| LW481                                                                                                                                                                        | INTERNATIONAL LAW             |
| SS366                                                                                                                                                                        | COMPARATIVE POLITICS          |
| SS377                                                                                                                                                                        | POLITICS & GOV OF EUROPE      |
| SS465                                                                                                                                                                        | TERRORISM: NEW CHALLENGES     |
| SS475                                                                                                                                                                        | COMP POLITICAL INSTITUTIONS   |
| SS484                                                                                                                                                                        | INTERNATIONAL ECONOMICS       |
| SS486                                                                                                                                                                        | STATE BUILDING                |
| SS487                                                                                                                                                                        | INT'L POLITICAL ECONOMY       |
| WS358                                                                                                                                                                        | STRATEGY AND GENERALSHIP      |
| WS455                                                                                                                                                                        | COMPARATIVE DEFENSE POLICY    |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                 |
| LN490F                                                                                                                                                                       | FRENCH LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Science Depth</b>                                                                                                                                                         |                               |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                               |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II          |
| CH275                                                                                                                                                                        | BIOLOGY                       |
| PH202                                                                                                                                                                        | PHYSICS II                    |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II           |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Curriculum Requirement</b>                                                                                                                                                |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: French & Persian w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FFZ2H | Foreign Language: French & Persian w/ Honors | Foreign Language Major: French & Persian w/ Honors | Foreign Language: French & Persian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: French & Persian w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: French & Portuguese Curriculum

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------|--------------|
| FFP2 | Foreign Language: French & Portuguese | Foreign Language Major: French & Portuguese | Foreign Language: French & Portuguese | 2            | 13           |

### 2029 Foreign Language Major: French & Portuguese Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LF371                  | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                  | FRENCH FOR ORAL & WRITTEN COMM |
| LF470                  | SPECIAL TOPIC IN FRENCH        |
| LF475                  | FRENCH RDG/WRTG THRU MEDIA     |
| LF476                  | MILITARY SPKG/RDG - FRENCH     |
| LF483                  | FRENCH CIVILIZATION I          |
| LF484                  | FRENCH CIVILIZATION II         |
| LF485                  | SURVEY OF FRENCH LIT I         |
| LF486                  | SURVEY OF FRENCH LIT II        |
| LF492                  | MASTERWORKS OF FRENCH LIT      |
| LN440F                 | FRENCH IN CULTURAL CONTEXT     |

|       |                                |
|-------|--------------------------------|
| LN451 | ADV LANG & CULTURE IN CONTEXT  |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LN491 | SEM ABROAD: ADV LANG & CULT I  |
| LN492 | SEM ABROAD: ADV LANG & CULT II |
| LN493 | SEM ABROAD: ADV LANG&CULT III  |
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |

**AND****Secondary Language Track**

Choose 4 of 12

|       |                                |
|-------|--------------------------------|
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LP203 | PORTUGUESE I (STANDARD)        |
| LP204 | PORTUGUESE II (STANDARD)       |
| LP371 | INTENSIVE INTERMED. PORTUGUESE |
| LP372 | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475 | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476 | MILITARY SPKG/RDG - PORTUGUESE |
| LP483 | PORTUGUESE CIVILIZATION I      |
| LP484 | PORTUGUESE CIVILIZATION II     |
| LP485 | SURVEY OF PORTUGUESE LIT I     |
| LP492 | LIT OF PORT-SPKG WORLD         |

**AND****Complementary Support Courses**

Choose 3 of 31

Choose 3 CSCs, each from a different discipline.

|       |                               |
|-------|-------------------------------|
| EN300 | LITERARY METHODOLOGIES        |
| EN311 | ANCIENT TO EARLY MODERN LIT   |
| EN340 | CONTEMPORARY LITERATURE       |
| EN351 | WORLD LITERATURE              |
| EN353 | WAR LITERATURE                |
| EN361 | POETRY                        |
| EN362 | FILM AND FILM THEORY          |
| EN363 | THE NOVEL                     |
| EN364 | DRAMA                         |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE344 | GEOGRAPHY OF AFRICA           |
| GE346 | GEOGRAPHY OF EUROPE           |
| GE491 | MILITARY GEOGRAPHY            |
| HI338 | WARFARE IN AGE OF REVOLUTIONS |
| HI344 | MODERN DIPLOMACY              |
| HI345 | MODERN AFRICA                 |
| HI361 | MEDIEVAL EUROPE               |
| HI364 | MODERN WESTERN EUROPE         |
| HI391 | WORLD RELIGIONS               |
| LW410 | COMPARATIVE LEGAL SYSTEMS     |
| LW473 | ENVIRONMENTAL LAW             |
| LW481 | INTERNATIONAL LAW             |
| SS366 | COMPARATIVE POLITICS          |
| SS377 | POLITICS & GOV OF EUROPE      |
| SS465 | TERRORISM: NEW CHALLENGES     |
| SS475 | COMP POLITICAL INSTITUTIONS   |
| SS484 | INTERNATIONAL ECONOMICS       |
| SS486 | STATE BUILDING                |
| SS487 | INT'L POLITICAL ECONOMY       |
| WS358 | STRATEGY AND GENERALSHIP      |

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| WS455                                                                                                                                                                        | COMPARATIVE DEFENSE POLICY    |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Integrative Experience for the Major</b>                                                                                                                                  |                               |
|                                                                                                                                                                              | Choose 1 of 1                 |
| LN490F                                                                                                                                                                       | FRENCH LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Science Depth</b>                                                                                                                                                         |                               |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                               |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II          |
| CH275                                                                                                                                                                        | BIOLOGY                       |
| PH202                                                                                                                                                                        | PHYSICS II                    |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II           |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Curriculum Requirement</b>                                                                                                                                                |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: French & Portuguese w/ Honors Curriculum

| Code  | Short Description                           | Description                                           | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|-------------------------------------------------------|---------------------------------------------|--------------|--------------|
| FFP2H | Foreign Lang: French & Portuguese w/ Honors | Foreign Language Major: French & Portuguese w/ Honors | Foreign Lang: French & Portuguese w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: French & Portuguese w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: French & Russian Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FFR2 | Foreign Language: French & Russian | Foreign Language Major: French & Russian | Foreign Language: French & Russian | 2            | 13           |

## 2029 Foreign Language Major: French & Russian Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LF371                           | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                           | FRENCH FOR ORAL & WRITTEN COMM |
| LF470                           | SPECIAL TOPIC IN FRENCH        |
| LF475                           | FRENCH RDG/WRTG THRU MEDIA     |
| LF476                           | MILITARY SPKG/RDG - FRENCH     |
| LF483                           | FRENCH CIVILIZATION I          |
| LF484                           | FRENCH CIVILIZATION II         |
| LF485                           | SURVEY OF FRENCH LIT I         |
| LF486                           | SURVEY OF FRENCH LIT II        |
| LF492                           | MASTERWORKS OF FRENCH LIT      |
| LN440F                          | FRENCH IN CULTURAL CONTEXT     |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LR203                           | RUSSIAN I (STANDARD)           |
| LR204                           | RUSSIAN II (STANDARD)          |
| LR371                           | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                           | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475                           | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                           | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                           | RUSSIAN CIV I                  |
| LR484                           | RUSSIAN CIV II                 |
| LR485                           | SURVEY OF RUSSIAN LITERATURE I |
| LR486                           | SURVEY OF RUSSIAN LIT. II      |
| LR492                           | RUSSIAN LIFE IN FICTION        |

**AND****Complementary Support Courses**

Choose 3 of 31

Choose 3 CSCs, each from a different discipline.

|       |                               |
|-------|-------------------------------|
| EN300 | LITERARY METHODOLOGIES        |
| EN311 | ANCIENT TO EARLY MODERN LIT   |
| EN340 | CONTEMPORARY LITERATURE       |
| EN351 | WORLD LITERATURE              |
| EN353 | WAR LITERATURE                |
| EN361 | POETRY                        |
| EN362 | FILM AND FILM THEORY          |
| EN363 | THE NOVEL                     |
| EN364 | DRAMA                         |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE344 | GEOGRAPHY OF AFRICA           |
| GE346 | GEOGRAPHY OF EUROPE           |
| GE491 | MILITARY GEOGRAPHY            |
| HI338 | WARFARE IN AGE OF REVOLUTIONS |
| HI344 | MODERN DIPLOMACY              |
| HI345 | MODERN AFRICA                 |
| HI361 | MEDIEVAL EUROPE               |
| HI364 | MODERN WESTERN EUROPE         |
| HI391 | WORLD RELIGIONS               |
| LW410 | COMPARATIVE LEGAL SYSTEMS     |
| LW473 | ENVIRONMENTAL LAW             |
| LW481 | INTERNATIONAL LAW             |
| SS366 | COMPARATIVE POLITICS          |
| SS377 | POLITICS & GOV OF EUROPE      |
| SS465 | TERRORISM: NEW CHALLENGES     |
| SS475 | COMP POLITICAL INSTITUTIONS   |
| SS484 | INTERNATIONAL ECONOMICS       |
| SS486 | STATE BUILDING                |
| SS487 | INT'L POLITICAL ECONOMY       |
| WS358 | STRATEGY AND GENERALSHIP      |
| WS455 | COMPARATIVE DEFENSE POLICY    |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490F | FRENCH LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**



Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

### Core Engineering Sequence

Cadets in this major may choose any approved three course engineering sequence.

### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: French & Russian w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FFR2H | Foreign Language: French & Russian w/ Honors | Foreign Language Major: French & Russian w/ Honors | Foreign Language: French & Russian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: French & Russian w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: French & Spanish Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FFS2 | Foreign Language: French & Spanish | Foreign Language Major: French & Spanish | Foreign Language: French & Spanish | 2            | 13           |

### 2029 Foreign Language Major: French & Spanish Tracks

| Subject Area           | Description                   |
|------------------------|-------------------------------|
| <b>Required Course</b> | Choose 1 of 1                 |
| LN380                  | NATURE OF MODERN LANGUAGES    |
| <b>AND</b>             |                               |
| <b>Electives</b>       | Choose 6 of 19                |
| LF371                  | INTENSIVE INTERMEDIATE FRENCH |

|        |                                |
|--------|--------------------------------|
| LF372  | FRENCH FOR ORAL & WRITTEN COMM |
| LF470  | SPECIAL TOPIC IN FRENCH        |
| LF475  | FRENCH RDG/WRTG THRU MEDIA     |
| LF476  | MILITARY SPKG/RDG - FRENCH     |
| LF483  | FRENCH CIVILIZATION I          |
| LF484  | FRENCH CIVILIZATION II         |
| LF485  | SURVEY OF FRENCH LIT I         |
| LF486  | SURVEY OF FRENCH LIT II        |
| LF492  | MASTERWORKS OF FRENCH LIT      |
| LN440F | FRENCH IN CULTURAL CONTEXT     |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LS203 | SPANISH I (STANDARD)           |
| LS204 | SPANISH II (STANDARD)          |
| LS371 | INTENSIVE INTERMEDIATE SPANISH |
| LS372 | SPANISH FOR ORAL & WRIT COMM   |
| LS475 | SPANISH RDG/WRTG THRU MEDIA    |
| LS476 | MILITARY SPKG/RDG - SPANISH    |
| LS483 | SPANISH CIV AND CULTURE        |
| LS484 | SPANISH AMERICAN CIV AND CULT  |
| LS485 | SPANISH-AMERICAN LITERATURE    |
| LS486 | THE LITERATURE OF SPAIN        |
| LS492 | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Complementary Support Courses**

Choose 3 of 31

Choose 3 CSCs, each from a different discipline.

|       |                               |
|-------|-------------------------------|
| EN300 | LITERARY METHODOLOGIES        |
| EN311 | ANCIENT TO EARLY MODERN LIT   |
| EN340 | CONTEMPORARY LITERATURE       |
| EN351 | WORLD LITERATURE              |
| EN353 | WAR LITERATURE                |
| EN361 | POETRY                        |
| EN362 | FILM AND FILM THEORY          |
| EN363 | THE NOVEL                     |
| EN364 | DRAMA                         |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE344 | GEOGRAPHY OF AFRICA           |
| GE346 | GEOGRAPHY OF EUROPE           |
| GE491 | MILITARY GEOGRAPHY            |
| HI338 | WARFARE IN AGE OF REVOLUTIONS |
| HI344 | MODERN DIPLOMACY              |
| HI345 | MODERN AFRICA                 |
| HI361 | MEDIEVAL EUROPE               |
| HI364 | MODERN WESTERN EUROPE         |
| HI391 | WORLD RELIGIONS               |

|       |                             |
|-------|-----------------------------|
| LW410 | COMPARATIVE LEGAL SYSTEMS   |
| LW473 | ENVIRONMENTAL LAW           |
| LW481 | INTERNATIONAL LAW           |
| SS366 | COMPARATIVE POLITICS        |
| SS377 | POLITICS & GOV OF EUROPE    |
| SS465 | TERRORISM: NEW CHALLENGES   |
| SS475 | COMP POLITICAL INSTITUTIONS |
| SS484 | INTERNATIONAL ECONOMICS     |
| SS486 | STATE BUILDING              |
| SS487 | INT'L POLITICAL ECONOMY     |
| WS358 | STRATEGY AND GENERALSHIP    |
| WS455 | COMPARATIVE DEFENSE POLICY  |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490F | FRENCH LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: French & Spanish w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FFS2H | Foreign Language: French & Spanish w/ Honors | Foreign Language Major: French & Spanish w/ Honors | Foreign Language: French & Spanish w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: French & Spanish w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: French w/ Honors Curriculum

| Code  | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|-------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FLF2H | Foreign Language: French w/ Honors | Foreign Language Major: French w/ Honors | Foreign Language: French w/ Honors | 1            | 1            |

### 2029 Foreign Language Major: French w/ Honors Tracks

| Subject Area                                                                   | Description                    |
|--------------------------------------------------------------------------------|--------------------------------|
| <b>Additional Elective</b>                                                     | Choose 1 of 16                 |
| Take an additional advanced-level elective, not already taken, from this list. |                                |
| LF470                                                                          | SPECIAL TOPIC IN FRENCH        |
| LF475                                                                          | FRENCH RDG/WRTG THRU MEDIA     |
| LF476                                                                          | MILITARY SPKG/RDG - FRENCH     |
| LF483                                                                          | FRENCH CIVILIZATION I          |
| LF484                                                                          | FRENCH CIVILIZATION II         |
| LF485                                                                          | SURVEY OF FRENCH LIT I         |
| LF486                                                                          | SURVEY OF FRENCH LIT II        |
| LF492                                                                          | MASTERWORKS OF FRENCH LIT      |
| LN440F                                                                         | FRENCH IN CULTURAL CONTEXT     |
| LN451                                                                          | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                                                          | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                                                          | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                                                          | SEM ABROAD: ADV LANG & CULT II |
| LN493                                                                          | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                                                          | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                                                          | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                                                     |                                |
| <b>Honors Thesis</b>                                                           | Choose 1 of 1                  |
| Write an honors thesis under the direction of a senior faculty member.         |                                |
| LN488                                                                          | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                     |                                |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Foreign Language Major: German Curriculum

| Code | Short Description        | Description                    | Transcript Description   | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|--------------------------|--------------------------------|--------------------------|-----------------|-----------------|
| FLG2 | Foreign Language: German | Foreign Language Major: German | Foreign Language: German | 2               | 11              |

## 2029 Foreign Language Major: German Tracks

| Subject Area                                 | Description                    |
|----------------------------------------------|--------------------------------|
| <b>Required Course</b>                       | Choose 1 of 1                  |
| LN380                                        | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                   |                                |
| <b>Electives</b>                             | Choose 8 of 19                 |
| LG371                                        | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                                        | GERMAN FOR ORAL & WRITTEN COMM |
| LG470                                        | SPECIAL TOPIC IN GERMAN        |
| LG475                                        | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                                        | MILITARY SPKG/RDG - GERMAN     |
| LG483                                        | GERMAN CIVILIZATION I          |
| LG484                                        | GERMAN CIVILIZATION II         |
| LG485                                        | SURVEY OF GERMAN LIT I         |
| LG486                                        | SURVEY OF GERMAN LIT II        |
| LG492                                        | 20TH & 21ST CENTURY GERMANY    |
| LN440G                                       | GERMAN IN CULTURAL CONTEXT     |
| LN451                                        | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                        | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                        | SEM ABROAD: ADV LANG & CULT II |
| LN493                                        | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                        | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                        | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                   |                                |
| <b>Complementary Support Courses</b>         | Choose 3 of 32                 |
| Choose one each from a different discipline. |                                |
| EN300                                        | LITERARY METHODOLOGIES         |
| EN311                                        | ANCIENT TO EARLY MODERN LIT    |
| EN340                                        | CONTEMPORARY LITERATURE        |
| EN351                                        | WORLD LITERATURE               |
| EN353                                        | WAR LITERATURE                 |
| EN361                                        | POETRY                         |
| EN362                                        | FILM AND FILM THEORY           |
| EN363                                        | THE NOVEL                      |
| EN364                                        | DRAMA                          |
| GE340                                        | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                        | GEOGRAPHY OF EUROPE            |
| GE491                                        | MILITARY GEOGRAPHY             |
| HI343                                        | MODERN GERMANY                 |

|        |                                |
|--------|--------------------------------|
| HI344  | MODERN DIPLOMACY               |
| HI364  | MODERN WESTERN EUROPE          |
| HI368  | MOD CENTRAL & E. EUR,1896-1989 |
| HI391  | WORLD RELIGIONS                |
| LN482H | SPOKEN HEBREW                  |
| LW410  | COMPARATIVE LEGAL SYSTEMS      |
| LW473  | ENVIRONMENTAL LAW              |
| LW481  | INTERNATIONAL LAW              |
| SS366  | COMPARATIVE POLITICS           |
| SS377  | POLITICS & GOV OF EUROPE       |
| SS465  | TERRORISM: NEW CHALLENGES      |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS484  | INTERNATIONAL ECONOMICS        |
| SS486  | STATE BUILDING                 |
| SS487  | INT'L POLITICAL ECONOMY        |
| WS358  | STRATEGY AND GENERALSHIP       |
| WS385  | WAR AND ITS THEORISTS          |
| WS455  | COMPARATIVE DEFENSE POLICY     |
| XH405  | THE HOLOCAUST AND ITS LEGACY   |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490G | GERMAN LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: German & Arabic Curriculum

| Code | Short Description                 | Description                             | Transcript Description            | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------------|-----------------------------------------|-----------------------------------|--------------|--------------|
| FGA2 | Foreign Language: German & Arabic | Foreign Language Major: German & Arabic | Foreign Language: German & Arabic | 2            | 13           |

### 2029 Foreign Language Major: German & Arabic Tracks

| Subject Area                                      | Description                    |
|---------------------------------------------------|--------------------------------|
| <b>Required Course</b>                            | Choose 1 of 1                  |
| LN380                                             | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                        |                                |
| <b>Electives</b>                                  | Choose 6 of 19                 |
| LG371                                             | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                                             | GERMAN FOR ORAL & WRITTEN COMM |
| LG470                                             | SPECIAL TOPIC IN GERMAN        |
| LG475                                             | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                                             | MILITARY SPKG/RDG - GERMAN     |
| LG483                                             | GERMAN CIVILIZATION I          |
| LG484                                             | GERMAN CIVILIZATION II         |
| LG485                                             | SURVEY OF GERMAN LIT I         |
| LG486                                             | SURVEY OF GERMAN LIT II        |
| LG492                                             | 20TH & 21ST CENTURY GERMANY    |
| LN440G                                            | GERMAN IN CULTURAL CONTEXT     |
| LN451                                             | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                             | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                             | SEM ABROAD: ADV LANG & CULT II |
| LN493                                             | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                             | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                             | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                        |                                |
| <b>Secondary Language Track</b>                   | Choose 4 of 14                 |
| LA203                                             | ARABIC I (STANDARD)            |
| LA204                                             | ARABIC II (STANDARD)           |
| LA371                                             | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                             | ARABIC FOR ORAL & WRITTEN COMM |
| LA472                                             | COLLOQUIAL ARABIC              |
| LA475                                             | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                                             | MILITARY SPKG/RDG - ARABIC     |
| LA483                                             | ARAB CIVILIZATION I            |
| LA484                                             | ARAB CIVILIZATION II           |
| LA485                                             | ARABIC LITERATURE I            |
| LA486                                             | ARABIC LITERATURE II           |
| LA492                                             | ARABIC LITERATURE III          |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Courses</b>              | Choose 3 of 48                 |
| Choose 3 of 48, each from a different discipline. |                                |
| EN300                                             | LITERARY METHODOLOGIES         |
| EN351                                             | WORLD LITERATURE               |

|        |                                |
|--------|--------------------------------|
| EN362  | FILM AND FILM THEORY           |
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344  | GEOGRAPHY OF AFRICA            |
| GE346  | GEOGRAPHY OF EUROPE            |
| GE491  | MILITARY GEOGRAPHY             |
| HI339  | THE MODERN MIDDLE EAST         |
| HI343  | MODERN GERMANY                 |
| HI344  | MODERN DIPLOMACY               |
| HI364  | MODERN WESTERN EUROPE          |
| HI368  | MOD CENTRAL & E. EUR,1896-1989 |
| HI381  | HISTORY OF IRREGULAR WARFARE   |
| HI383  | MIDDLE EASTERN WARFARE         |
| HI391  | WORLD RELIGIONS                |
| LN482H | SPOKEN HEBREW                  |
| LW410  | COMPARATIVE LEGAL SYSTEMS      |
| LW473  | ENVIRONMENTAL LAW              |
| LW481  | INTERNATIONAL LAW              |
| SS366  | COMPARATIVE POLITICS           |
| SS377  | POLITICS & GOV OF EUROPE       |
| SS465  | TERRORISM: NEW CHALLENGES      |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS477  | ECONOMICS OF NATIONAL SECURITY |
| SS479  | INTERNATIONAL ORGANIZATION     |
| SS484  | INTERNATIONAL ECONOMICS        |
| SS486  | STATE BUILDING                 |
| SS487  | INT'L POLITICAL ECONOMY        |
| WS358  | STRATEGY AND GENERALSHIP       |
| WS360  | SPECIAL OPS IN THEORY & PRACTI |
| WS365  | REBELLN, INSURGNCY, & CIV WAR  |
| WS385  | WAR AND ITS THEORISTS          |
| WS455  | COMPARATIVE DEFENSE POLICY     |
| XH405  | THE HOLOCAUST AND ITS LEGACY   |
| ZH337  | REGIONAL POLITICAL SYSTEMS     |
| ZH347  | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367  | TOPICS IN MICROECONOMICS       |
| ZH377  | TOPICS IN MACROECONOMICS       |
| ZH407  | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427  | TOPICS IN COMPARATIVE POLITICS |
| ZH447  | TOPICS: INTERNATIONAL POLITICS |
| ZH467  | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477  | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490G | GERMAN LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                   |
|-------|-------------------|
| CY305 | CYBER FOUNDATIONS |
|-------|-------------------|



CY355 CYBER FOUNDATIONS - COMPUTING

**AND**

**Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305 CYBER FOUNDATIONS

CY355 CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380 NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: German & Arabic w/ Honors Curriculum

| Code  | Short Description                           | Description                                       | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|---------------------------------------------------|---------------------------------------------|--------------|--------------|
| FGA2H | Foreign Language: German & Arabic w/ Honors | Foreign Language Major: German & Arabic w/ Honors | Foreign Language: German & Arabic w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: German & Arabic w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: German & Chinese Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FGC1 | Foreign Language: German & Chinese | Foreign Language Major: German & Chinese | Foreign Language: German & Chinese | 2            | 13           |

## 2029 Foreign Language Major: German & Chinese Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| <b>Required Course</b>                            | Choose 1 of 1                  |
| LN380                                             | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                        |                                |
| <b>Electives</b>                                  | Choose 6 of 19                 |
| LG371                                             | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                                             | GERMAN FOR ORAL & WRITTEN COMM |
| LG470                                             | SPECIAL TOPIC IN GERMAN        |
| LG475                                             | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                                             | MILITARY SPKG/RDG - GERMAN     |
| LG483                                             | GERMAN CIVILIZATION I          |
| LG484                                             | GERMAN CIVILIZATION II         |
| LG485                                             | SURVEY OF GERMAN LIT I         |
| LG486                                             | SURVEY OF GERMAN LIT II        |
| LG492                                             | 20TH & 21ST CENTURY GERMANY    |
| LN440G                                            | GERMAN IN CULTURAL CONTEXT     |
| LN451                                             | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                             | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                             | SEM ABROAD: ADV LANG & CULT II |
| LN493                                             | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                             | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                             | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                        |                                |
| <b>Secondary Language Track</b>                   | Choose 4 of 13                 |
| LC203                                             | CHINESE I (STANDARD)           |
| LC204                                             | CHINESE II (STANDARD)          |
| LC371                                             | INTENSIVE INTERMEDIATE CHINESE |
| LC372                                             | CHINESE FOR ORAL & WRIT COMM   |
| LC475                                             | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                                             | MILITARY SPKG/RDG - CHINESE    |
| LC483                                             | CHINESE CIVILIZATION I         |
| LC484                                             | CHINESE CIVILIZATION II        |
| LC485                                             | CHINESE LITERATURE I           |
| LC486                                             | CHINESE LITERATURE II          |
| LC492                                             | CHINESE LITERATURE III         |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Courses</b>              | Choose 3 of 26                 |
| Choose 3 of 26, each from a different discipline. |                                |
| EN300                                             | LITERARY METHODOLOGIES         |
| EN351                                             | WORLD LITERATURE               |
| EN362                                             | FILM AND FILM THEORY           |
| GE340                                             | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                             | GEOGRAPHY OF EUROPE            |
| GE491                                             | MILITARY GEOGRAPHY             |
| HI343                                             | MODERN GERMANY                 |
| HI344                                             | MODERN DIPLOMACY               |
| HI364                                             | MODERN WESTERN EUROPE          |
| HI368                                             | MOD CENTRAL & E. EUR,1896-1989 |
| HI391                                             | WORLD RELIGIONS                |
| LN482H                                            | SPOKEN HEBREW                  |
| LW410                                             | COMPARATIVE LEGAL SYSTEMS      |

|       |                              |
|-------|------------------------------|
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS377 | POLITICS & GOV OF EUROPE     |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS385 | WAR AND ITS THEORISTS        |
| WS455 | COMPARATIVE DEFENSE POLICY   |
| XH405 | THE HOLOCAUST AND ITS LEGACY |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490G | GERMAN LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: German & Chinese w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FGC1H | Foreign Language: German & Chinese w/ Honors | Foreign Language Major: German & Chinese w/ Honors | Foreign Language: German & Chinese w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: German & Chinese w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: German & French Curriculum

| Code | Short Description                 | Description                             | Transcript Description            | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------------|-----------------------------------------|-----------------------------------|--------------|--------------|
| FGF1 | Foreign Language: German & French | Foreign Language Major: German & French | Foreign Language: German & French | 2            | 13           |

### 2029 Foreign Language Major: German & French Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LG371                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG470                           | SPECIAL TOPIC IN GERMAN        |
| LG475                           | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                           | MILITARY SPKG/RDG - GERMAN     |
| LG483                           | GERMAN CIVILIZATION I          |
| LG484                           | GERMAN CIVILIZATION II         |
| LG485                           | SURVEY OF GERMAN LIT I         |
| LG486                           | SURVEY OF GERMAN LIT II        |
| LG492                           | 20TH & 21ST CENTURY GERMANY    |
| LN440G                          | GERMAN IN CULTURAL CONTEXT     |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LF203                           | FRENCH I (STANDARD)            |

|       |                                |
|-------|--------------------------------|
| LF204 | FRENCH II (STANDARD)           |
| LF371 | INTENSIVE INTERMEDIATE FRENCH  |
| LF372 | FRENCH FOR ORAL & WRITTEN COMM |
| LF475 | FRENCH RDG/WRTG THRU MEDIA     |
| LF476 | MILITARY SPKG/RDG - FRENCH     |
| LF483 | FRENCH CIVILIZATION I          |
| LF484 | FRENCH CIVILIZATION II         |
| LF485 | SURVEY OF FRENCH LIT I         |
| LF486 | SURVEY OF FRENCH LIT II        |
| LF492 | MASTERWORKS OF FRENCH LIT      |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 26

Choose 3 of 26, each from a different discipline.

|        |                                |
|--------|--------------------------------|
| EN300  | LITERARY METHODOLOGIES         |
| EN351  | WORLD LITERATURE               |
| EN362  | FILM AND FILM THEORY           |
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346  | GEOGRAPHY OF EUROPE            |
| GE491  | MILITARY GEOGRAPHY             |
| HI343  | MODERN GERMANY                 |
| HI344  | MODERN DIPLOMACY               |
| HI364  | MODERN WESTERN EUROPE          |
| HI368  | MOD CENTRAL & E. EUR,1896-1989 |
| HI391  | WORLD RELIGIONS                |
| LN482H | SPOKEN HEBREW                  |
| LW410  | COMPARATIVE LEGAL SYSTEMS      |
| LW473  | ENVIRONMENTAL LAW              |
| LW481  | INTERNATIONAL LAW              |
| SS366  | COMPARATIVE POLITICS           |
| SS377  | POLITICS & GOV OF EUROPE       |
| SS465  | TERRORISM: NEW CHALLENGES      |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS484  | INTERNATIONAL ECONOMICS        |
| SS486  | STATE BUILDING                 |
| SS487  | INT'L POLITICAL ECONOMY        |
| WS358  | STRATEGY AND GENERALSHIP       |
| WS385  | WAR AND ITS THEORISTS          |
| WS455  | COMPARATIVE DEFENSE POLICY     |
| XH405  | THE HOLOCAUST AND ITS LEGACY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490G

GERMAN LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                   |
|-------|-------------------|
| CY305 | CYBER FOUNDATIONS |
|-------|-------------------|

CY355 CYBER FOUNDATIONS - COMPUTING

**AND**

**Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305 CYBER FOUNDATIONS

CY355 CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: German & French w/ Honors Curriculum

| Code  | Short Description                           | Description                                       | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|---------------------------------------------------|---------------------------------------------|--------------|--------------|
| FGF1H | Foreign Language: German & French w/ Honors | Foreign Language Major: German & French w/ Honors | Foreign Language: German & French w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: German & French w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: German & Persian Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FGZ2 | Foreign Language: German & Persian | Foreign Language Major: German & Persian | Foreign Language: German & Persian | 2            | 13           |

### 2029 Foreign Language Major: German & Persian Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| <b>Required Course</b>                            | Choose 1 of 1                  |
| LN380                                             | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                        |                                |
| <b>Electives</b>                                  | Choose 6 of 19                 |
| LG371                                             | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                                             | GERMAN FOR ORAL & WRITTEN COMM |
| LG470                                             | SPECIAL TOPIC IN GERMAN        |
| LG475                                             | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                                             | MILITARY SPKG/RDG - GERMAN     |
| LG483                                             | GERMAN CIVILIZATION I          |
| LG484                                             | GERMAN CIVILIZATION II         |
| LG485                                             | SURVEY OF GERMAN LIT I         |
| LG486                                             | SURVEY OF GERMAN LIT II        |
| LG492                                             | 20TH & 21ST CENTURY GERMANY    |
| LN440G                                            | GERMAN IN CULTURAL CONTEXT     |
| LN451                                             | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                             | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                             | SEM ABROAD: ADV LANG & CULT II |
| LN493                                             | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                             | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                             | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                        |                                |
| <b>Secondary Language Track</b>                   | Choose 4 of 13                 |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LZ203                                             | PERSIAN I (STANDARD)           |
| LZ204                                             | PERSIAN II (STANDARD)          |
| LZ371                                             | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                                             | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                                             | SPECIAL TOPIC IN PERSIAN       |
| LZ475                                             | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                                             | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                                             | PERSIAN CIVILIZATION I         |
| LZ484                                             | PERSIAN CIVILIZATION II        |
| LZ485                                             | PERSIAN LITERATURE I           |
| LZ486                                             | PERSIAN LITERATURE II          |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Courses</b>              | Choose 3 of 26                 |
| Choose 3 of 26, each from a different discipline. |                                |
| EN300                                             | LITERARY METHODOLOGIES         |
| EN351                                             | WORLD LITERATURE               |
| EN362                                             | FILM AND FILM THEORY           |
| GE340                                             | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                             | GEOGRAPHY OF EUROPE            |
| GE491                                             | MILITARY GEOGRAPHY             |
| HI343                                             | MODERN GERMANY                 |
| HI344                                             | MODERN DIPLOMACY               |
| HI364                                             | MODERN WESTERN EUROPE          |
| HI368                                             | MOD CENTRAL & E. EUR,1896-1989 |
| HI391                                             | WORLD RELIGIONS                |
| LN482H                                            | SPOKEN HEBREW                  |
| LW410                                             | COMPARATIVE LEGAL SYSTEMS      |

|       |                              |
|-------|------------------------------|
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS377 | POLITICS & GOV OF EUROPE     |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS385 | WAR AND ITS THEORISTS        |
| WS455 | COMPARATIVE DEFENSE POLICY   |
| XH405 | THE HOLOCAUST AND ITS LEGACY |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490G | GERMAN LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: German & Persian w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FGZ2H | Foreign Language: German & Persian w/ Honors | Foreign Language Major: German & Persian w/ Honors | Foreign Language: German & Persian w/ Honors | 1            | 0            |



### 2029 Foreign Language Major: German & Persian w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: German & Portuguese Curriculum

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------|--------------|
| FGP2 | Foreign Language: German & Portuguese | Foreign Language Major: German & Portuguese | Foreign Language: German & Portuguese | 2            | 13           |

### 2029 Foreign Language Major: German & Portuguese Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LG371                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG470                           | SPECIAL TOPIC IN GERMAN        |
| LG475                           | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                           | MILITARY SPKG/RDG - GERMAN     |
| LG483                           | GERMAN CIVILIZATION I          |
| LG484                           | GERMAN CIVILIZATION II         |
| LG485                           | SURVEY OF GERMAN LIT I         |
| LG486                           | SURVEY OF GERMAN LIT II        |
| LG492                           | 20TH & 21ST CENTURY GERMANY    |
| LN440G                          | GERMAN IN CULTURAL CONTEXT     |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 12                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |

|       |                                |
|-------|--------------------------------|
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LP203 | PORTUGUESE I (STANDARD)        |
| LP204 | PORTUGUESE II (STANDARD)       |
| LP371 | INTENSIVE INTERMED. PORTUGUESE |
| LP372 | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475 | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476 | MILITARY SPKG/RDG - PORTUGUESE |
| LP483 | PORTUGUESE CIVILIZATION I      |
| LP484 | PORTUGUESE CIVILIZATION II     |
| LP485 | SURVEY OF PORTUGUESE LIT I     |
| LP492 | LIT OF PORT-SPKG WORLD         |

**AND****Complementary Support Courses**

Choose 3 of 26

Choose 3 of 26, each from a different discipline.

|        |                                |
|--------|--------------------------------|
| EN300  | LITERARY METHODOLOGIES         |
| EN351  | WORLD LITERATURE               |
| EN362  | FILM AND FILM THEORY           |
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346  | GEOGRAPHY OF EUROPE            |
| GE491  | MILITARY GEOGRAPHY             |
| HI343  | MODERN GERMANY                 |
| HI344  | MODERN DIPLOMACY               |
| HI364  | MODERN WESTERN EUROPE          |
| HI368  | MOD CENTRAL & E. EUR,1896-1989 |
| HI391  | WORLD RELIGIONS                |
| LN482H | SPOKEN HEBREW                  |
| LW410  | COMPARATIVE LEGAL SYSTEMS      |
| LW473  | ENVIRONMENTAL LAW              |
| LW481  | INTERNATIONAL LAW              |
| SS366  | COMPARATIVE POLITICS           |
| SS377  | POLITICS & GOV OF EUROPE       |
| SS465  | TERRORISM: NEW CHALLENGES      |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS484  | INTERNATIONAL ECONOMICS        |
| SS486  | STATE BUILDING                 |
| SS487  | INT'L POLITICAL ECONOMY        |
| WS358  | STRATEGY AND GENERALSHIP       |
| WS385  | WAR AND ITS THEORISTS          |
| WS455  | COMPARATIVE DEFENSE POLICY     |
| XH405  | THE HOLOCAUST AND ITS LEGACY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490G | GERMAN LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

**2029 Foreign Language Major: German & Portuguese w/ Honors Curriculum**

| Code  | Short Description                           | Description                                           | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|-------------------------------------------------------|---------------------------------------------|--------------|--------------|
| FGP2H | Foreign Lang: German & Portuguese w/ Honors | Foreign Language Major: German & Portuguese w/ Honors | Foreign Lang: German & Portuguese w/ Honors | 1            | 0            |

**2029 Foreign Language Major: German & Portuguese w/ Honors Tracks**

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |

**AND**

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

**2029 Foreign Language Major: German & Russian Curriculum**

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FGR2 | Foreign Language: German & Russian | Foreign Language Major: German & Russian | Foreign Language: German & Russian | 2            | 13           |

**2029 Foreign Language Major: German & Russian Tracks**

| Subject Area           | Description   |
|------------------------|---------------|
| <b>Required Course</b> | Choose 1 of 1 |

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| LN380                                             | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                        |                                |
| <b>Electives</b>                                  | Choose 6 of 19                 |
| LG371                                             | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                                             | GERMAN FOR ORAL & WRITTEN COMM |
| LG470                                             | SPECIAL TOPIC IN GERMAN        |
| LG475                                             | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                                             | MILITARY SPKG/RDG - GERMAN     |
| LG483                                             | GERMAN CIVILIZATION I          |
| LG484                                             | GERMAN CIVILIZATION II         |
| LG485                                             | SURVEY OF GERMAN LIT I         |
| LG486                                             | SURVEY OF GERMAN LIT II        |
| LG492                                             | 20TH & 21ST CENTURY GERMANY    |
| LN440G                                            | GERMAN IN CULTURAL CONTEXT     |
| LN451                                             | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                             | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                             | SEM ABROAD: ADV LANG & CULT II |
| LN493                                             | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                             | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                             | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                                        |                                |
| <b>Secondary Language Track</b>                   | Choose 4 of 13                 |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LR203                                             | RUSSIAN I (STANDARD)           |
| LR204                                             | RUSSIAN II (STANDARD)          |
| LR371                                             | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                                             | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475                                             | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                                             | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                                             | RUSSIAN CIV I                  |
| LR484                                             | RUSSIAN CIV II                 |
| LR485                                             | SURVEY OF RUSSIAN LITERATURE I |
| LR486                                             | SURVEY OF RUSSIAN LIT. II      |
| LR492                                             | RUSSIAN LIFE IN FICTION        |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Courses</b>              | Choose 3 of 26                 |
| Choose 3 of 26, each from a different discipline. |                                |
| EN300                                             | LITERARY METHODOLOGIES         |
| EN351                                             | WORLD LITERATURE               |
| EN362                                             | FILM AND FILM THEORY           |
| GE340                                             | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                             | GEOGRAPHY OF EUROPE            |
| GE491                                             | MILITARY GEOGRAPHY             |
| HI343                                             | MODERN GERMANY                 |
| HI344                                             | MODERN DIPLOMACY               |
| HI364                                             | MODERN WESTERN EUROPE          |
| HI368                                             | MOD CENTRAL & E. EUR,1896-1989 |
| HI391                                             | WORLD RELIGIONS                |
| LN482H                                            | SPOKEN HEBREW                  |
| LW410                                             | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                             | ENVIRONMENTAL LAW              |

|       |                              |
|-------|------------------------------|
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS377 | POLITICS & GOV OF EUROPE     |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS385 | WAR AND ITS THEORISTS        |
| WS455 | COMPARATIVE DEFENSE POLICY   |
| XH405 | THE HOLOCAUST AND ITS LEGACY |

**AND**


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**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490G | GERMAN LANGUAGE & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: German & Russian w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FGR2H | Foreign Language: German & Russian w/ Honors | Foreign Language Major: German & Russian w/ Honors | Foreign Language: German & Russian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: German & Russian w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: German & Spanish Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FGS2 | Foreign Language: German & Spanish | Foreign Language Major: German & Spanish | Foreign Language: German & Spanish | 2            | 13           |

### 2029 Foreign Language Major: German & Spanish Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LG371                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG470                           | SPECIAL TOPIC IN GERMAN        |
| LG475                           | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                           | MILITARY SPKG/RDG - GERMAN     |
| LG483                           | GERMAN CIVILIZATION I          |
| LG484                           | GERMAN CIVILIZATION II         |
| LG485                           | SURVEY OF GERMAN LIT I         |
| LG486                           | SURVEY OF GERMAN LIT II        |
| LG492                           | 20TH & 21ST CENTURY GERMANY    |
| LN440G                          | GERMAN IN CULTURAL CONTEXT     |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |

|       |                                |
|-------|--------------------------------|
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LS203 | SPANISH I (STANDARD)           |
| LS204 | SPANISH II (STANDARD)          |
| LS371 | INTENSIVE INTERMEDIATE SPANISH |
| LS372 | SPANISH FOR ORAL & WRIT COMM   |
| LS475 | SPANISH RDG/WRTG THRU MEDIA    |
| LS476 | MILITARY SPKG/RDG - SPANISH    |
| LS483 | SPANISH CIV AND CULTURE        |
| LS484 | SPANISH AMERICAN CIV AND CULT  |
| LS485 | SPANISH-AMERICAN LITERATURE    |
| LS486 | THE LITERATURE OF SPAIN        |
| LS492 | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Complementary Support Courses**

Choose 3 of 26

Choose 3 of 26, each from a different discipline.

|        |                                |
|--------|--------------------------------|
| EN300  | LITERARY METHODOLOGIES         |
| EN351  | WORLD LITERATURE               |
| EN362  | FILM AND FILM THEORY           |
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346  | GEOGRAPHY OF EUROPE            |
| GE491  | MILITARY GEOGRAPHY             |
| HI343  | MODERN GERMANY                 |
| HI344  | MODERN DIPLOMACY               |
| HI364  | MODERN WESTERN EUROPE          |
| HI368  | MOD CENTRAL & E. EUR,1896-1989 |
| HI391  | WORLD RELIGIONS                |
| LN482H | SPOKEN HEBREW                  |
| LW410  | COMPARATIVE LEGAL SYSTEMS      |
| LW473  | ENVIRONMENTAL LAW              |
| LW481  | INTERNATIONAL LAW              |
| SS366  | COMPARATIVE POLITICS           |
| SS377  | POLITICS & GOV OF EUROPE       |
| SS465  | TERRORISM: NEW CHALLENGES      |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS484  | INTERNATIONAL ECONOMICS        |
| SS486  | STATE BUILDING                 |
| SS487  | INT'L POLITICAL ECONOMY        |
| WS358  | STRATEGY AND GENERALSHIP       |
| WS385  | WAR AND ITS THEORISTS          |
| WS455  | COMPARATIVE DEFENSE POLICY     |
| XH405  | THE HOLOCAUST AND ITS LEGACY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490G

GERMAN LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                   |
|-------|-------------------|
| CY305 | CYBER FOUNDATIONS |
|-------|-------------------|

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Curriculum Requirement</b>                                                                                                                                                |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: German & Spanish w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FGS2H | Foreign Language: German & Spanish w/ Honors | Foreign Language Major: German & Spanish w/ Honors | Foreign Language: German & Spanish w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: German & Spanish w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: German w/ Honors Curriculum

| Code  | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|-------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FLG2H | Foreign Language: German w/ Honors | Foreign Language Major: German w/ Honors | Foreign Language: German w/ Honors | 1            | 1            |

## 2029 Foreign Language Major: German w/ Honors Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|



|                                                                                |                                |
|--------------------------------------------------------------------------------|--------------------------------|
| <b>Additional Elective</b>                                                     | Choose 1 of 16                 |
| Take an additional advanced-level elective, not already taken, from this list. |                                |
| LG470                                                                          | SPECIAL TOPIC IN GERMAN        |
| LG475                                                                          | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                                                                          | MILITARY SPKG/RDG - GERMAN     |
| LG483                                                                          | GERMAN CIVILIZATION I          |
| LG484                                                                          | GERMAN CIVILIZATION II         |
| LG485                                                                          | SURVEY OF GERMAN LIT I         |
| LG486                                                                          | SURVEY OF GERMAN LIT II        |
| LG492                                                                          | 20TH & 21ST CENTURY GERMANY    |
| LN440G                                                                         | GERMAN IN CULTURAL CONTEXT     |
| LN451                                                                          | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                                                          | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                                                          | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                                                          | SEM ABROAD: ADV LANG & CULT II |
| LN493                                                                          | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                                                          | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                                                          | SEM ABROAD: ADV LANG & CULT V  |

**AND**

|                      |               |
|----------------------|---------------|
| <b>Honors Thesis</b> | Choose 1 of 1 |
|----------------------|---------------|

Write an honors thesis under the direction of a senior faculty member.

|       |                             |
|-------|-----------------------------|
| LN488 | ADV IND STUDY-FOREIGN LANGS |
|-------|-----------------------------|

**AND**

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.

## 2029 Foreign Language Major: Persian Curriculum

| Code | Short Description         | Description                     | Transcript Description    | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------|---------------------------------|---------------------------|--------------|--------------|
| FLZ1 | Foreign Language: Persian | Foreign Language Major: Persian | Foreign Language: Persian | 2            | 11           |

### 2029 Foreign Language Major: Persian Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 8 of 18                 |
| LN440Z                 | PERSIAN IN CULTURAL CONTEXT    |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |
| LN494                  | SEM ABROAD: ADV LANG & CULT IV |
| LN495                  | SEM ABROAD: ADV LANG & CULT V  |
| LZ371                  | INTENSIVE INTERMEDIATE PERSIAN |

|       |                              |
|-------|------------------------------|
| LZ372 | PERSIAN FOR ORAL & WRIT COMM |
| LZ470 | SPECIAL TOPIC IN PERSIAN     |
| LZ475 | PERSIAN RDG/WRTG THRU MEDIA  |
| LZ476 | MILITARY SPKG/RDG - PERSIAN  |
| LZ483 | PERSIAN CIVILIZATION I       |
| LZ484 | PERSIAN CIVILIZATION II      |
| LZ485 | PERSIAN LITERATURE I         |
| LZ486 | PERSIAN LITERATURE II        |

**AND****Complementary Support Courses**

Choose 3 of 44

Choose 3 of 36, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN311 | ANCIENT TO EARLY MODERN LIT    |
| EN340 | CONTEMPORARY LITERATURE        |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN363 | THE NOVEL                      |
| EN364 | DRAMA                          |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE343 | GEOGRAPHY OF RUSSIA            |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491 | MILITARY GEOGRAPHY             |
| HI339 | THE MODERN MIDDLE EAST         |
| HI344 | MODERN DIPLOMACY               |
| HI367 | IMPERIAL AND SOVIET RUSSIA     |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI383 | MIDDLE EASTERN WARFARE         |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS375 | POL OF THE POST-SOVIET STATES  |
| SS383 | POLITICS OF THE MIDDLE EAST    |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS477 | ECONOMICS OF NATIONAL SECURITY |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND**

**Integrative Experience for the Major** Choose 1 of 1

LN490Z

PERSIAN LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

CH102

GENERAL CHEMISTRY II

CH275

BIOLOGY

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

**2029 Foreign Language Major: Persian & Chinese with Honors Curriculum**

| Code  | Short Description                             | Description                                           | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-------------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FZC1H | Foreign Language: Persian & Chinese w/ Honors | Foreign Language Major: Persian & Chinese with Honors | Foreign Language: Persian & Chinese w/ Honors | 1            | 0            |

**2029 Foreign Language Major: Persian & Chinese with Honors Tracks**

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Persian & French with Honors Curriculum

| Code  | Short Description                            | Description                                          | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FZF1H | Foreign Language: Persian & French w/ Honors | Foreign Language Major: Persian & French with Honors | Foreign Language: Persian & French w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Persian & French with Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Persian & German with Honors Curriculum

| Code  | Short Description                            | Description                                          | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FZG1H | Foreign Language: Persian & German w/ Honors | Foreign Language Major: Persian & German with Honors | Foreign Language: Persian & German w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Persian & German with Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Persian & Portuguese w/ Honors Curriculum

| Code  | Short Description                            | Description                                            | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FZP1H | Foreign Lang: Persian & Portuguese w/ Honors | Foreign Language Major: Persian & Portuguese w/ Honors | Foreign Lang: Persian & Portuguese w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Persian & Portuguese w/ Honors Tracks

| Subject Area         | Description                                                            |
|----------------------|------------------------------------------------------------------------|
| Honors Thesis Course | Choose 1 of 1                                                          |
|                      | Write an honors thesis under the direction of a senior faculty member. |
| LN488                | ADV IND STUDY-FOREIGN LANGS                                            |
| AND                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Persian & Russian w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FZR1H | Foreign Language: Persian & Russian w/ Honors | Foreign Language Major: Persian & Russian w/ Honors | Foreign Language: Persian & Russian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Persian & Russian w/ Honors Tracks

| Subject Area         | Description                                                            |
|----------------------|------------------------------------------------------------------------|
| Honors Thesis Course | Choose 1 of 1                                                          |
|                      | Write an honors thesis under the direction of a senior faculty member. |
| LN488                | ADV IND STUDY-FOREIGN LANGS                                            |
| AND                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Persian & Spanish w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FZS1H | Foreign Language: Persian & Spanish w/ Honors | Foreign Language Major: Persian & Spanish w/ Honors | Foreign Language: Persian & Spanish w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Persian & Spanish w/ Honors Tracks

| Subject Area         | Description   |
|----------------------|---------------|
| Honors Thesis Course | Choose 1 of 1 |

Write an honors thesis under the direction of a senior faculty member.

LN488

ADV IND STUDY-FOREIGN LANGS

AND

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Persian and Chinese Curriculum

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------|--------------|
| FZC1 | Foreign Language: Persian and Chinese | Foreign Language Major: Persian and Chinese | Foreign Language: Persian and Chinese | 2            | 13           |

### 2029 Foreign Language Major: Persian and Chinese Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| AND                             |                                |
| <b>Electives</b>                | Choose 6 of 18                 |
| LN440Z                          | PERSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LZ371                           | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                           | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                           | SPECIAL TOPIC IN PERSIAN       |
| LZ475                           | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                           | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                           | PERSIAN CIVILIZATION I         |
| LZ484                           | PERSIAN CIVILIZATION II        |
| LZ485                           | PERSIAN LITERATURE I           |
| LZ486                           | PERSIAN LITERATURE II          |
| AND                             |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LC203                           | CHINESE I (STANDARD)           |
| LC204                           | CHINESE II (STANDARD)          |
| LC371                           | INTENSIVE INTERMEDIATE CHINESE |
| LC372                           | CHINESE FOR ORAL & WRIT COMM   |
| LC475                           | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                           | MILITARY SPKG/RDG - CHINESE    |
| LC483                           | CHINESE CIVILIZATION I         |
| LC484                           | CHINESE CIVILIZATION II        |

|                                                                                                                                                  |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| LC485                                                                                                                                            | CHINESE LITERATURE I           |
| LC486                                                                                                                                            | CHINESE LITERATURE II          |
| LC492                                                                                                                                            | CHINESE LITERATURE III         |
| LN487                                                                                                                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                                                            | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Complementary Support Courses</b>                                                                                                             | Choose 3 of 22                 |
| Choose 3 of 22, each from a different discipline.                                                                                                |                                |
| EN300                                                                                                                                            | LITERARY METHODOLOGIES         |
| EN351                                                                                                                                            | WORLD LITERATURE               |
| GE491                                                                                                                                            | MILITARY GEOGRAPHY             |
| HI339                                                                                                                                            | THE MODERN MIDDLE EAST         |
| HI344                                                                                                                                            | MODERN DIPLOMACY               |
| HI381                                                                                                                                            | HISTORY OF IRREGULAR WARFARE   |
| HI383                                                                                                                                            | MIDDLE EASTERN WARFARE         |
| HI391                                                                                                                                            | WORLD RELIGIONS                |
| LW410                                                                                                                                            | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                                                                                                                            | ENVIRONMENTAL LAW              |
| LW481                                                                                                                                            | INTERNATIONAL LAW              |
| SS366                                                                                                                                            | COMPARATIVE POLITICS           |
| SS383                                                                                                                                            | POLITICS OF THE MIDDLE EAST    |
| SS465                                                                                                                                            | TERRORISM: NEW CHALLENGES      |
| SS475                                                                                                                                            | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                                                            | INTERNATIONAL ECONOMICS        |
| SS486                                                                                                                                            | STATE BUILDING                 |
| SS487                                                                                                                                            | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                                                            | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                                                            | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Integrative Experience for the Major</b>                                                                                                      | Choose 1 of 1                  |
| LN490Z                                                                                                                                           | PERSIAN LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Science Depth</b>                                                                                                                             |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                          |                                |
| CH102                                                                                                                                            | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                            | BIOLOGY                        |
| PH202                                                                                                                                            | PHYSICS II                     |
| PH252                                                                                                                                            | ADVANCED PHYSICS II            |
| PH275                                                                                                                                            | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                |                                |
| Cadets in the major must take CY305/355 to satisfy the STEM Depth requirement.                                                                   |                                |
| CY305                                                                                                                                            | CYBER FOUNDATIONS              |
| CY355                                                                                                                                            | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Curriculum Requirement</b>                                                                                                                    |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                         |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                      |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355. |                                |
| CY305                                                                                                                                            | CYBER FOUNDATIONS              |
| CY355                                                                                                                                            | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                 |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                  |                                |
| <b>Writing-in-the-Major</b>                                                                                                                      |                                |

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Persian and French Curriculum

| Code | Short Description                    | Description                                | Transcript Description               | Req Crse Cnt | Opt Crse Cnt |
|------|--------------------------------------|--------------------------------------------|--------------------------------------|--------------|--------------|
| FZF1 | Foreign Language: Persian and French | Foreign Language Major: Persian and French | Foreign Language: Persian and French | 2            | 13           |

## 2029 Foreign Language Major: Persian and French Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 18                 |
| LN440Z                          | PERSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LZ371                           | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                           | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                           | SPECIAL TOPIC IN PERSIAN       |
| LZ475                           | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                           | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                           | PERSIAN CIVILIZATION I         |
| LZ484                           | PERSIAN CIVILIZATION II        |
| LZ485                           | PERSIAN LITERATURE I           |
| LZ486                           | PERSIAN LITERATURE II          |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 14                 |
| LF203                           | FRENCH I (STANDARD)            |
| LF204                           | FRENCH II (STANDARD)           |
| LF371                           | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                           | FRENCH FOR ORAL & WRITTEN COMM |
| LF475                           | FRENCH RDG/WRTG THRU MEDIA     |
| LF476                           | MILITARY SPKG/RDG - FRENCH     |
| LF483                           | FRENCH CIVILIZATION I          |
| LF484                           | FRENCH CIVILIZATION II         |
| LF485                           | SURVEY OF FRENCH LIT I         |
| LF486                           | SURVEY OF FRENCH LIT II        |



|                                                                                                                                                  |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| LF492                                                                                                                                            | MASTERWORKS OF FRENCH LIT      |
| LN487                                                                                                                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                                                            | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Complementary Support Courses</b>                                                                                                             | Choose 3 of 22                 |
| Choose 3 of 22, each from a different discipline.                                                                                                |                                |
| EN300                                                                                                                                            | LITERARY METHODOLOGIES         |
| EN351                                                                                                                                            | WORLD LITERATURE               |
| GE340                                                                                                                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE345                                                                                                                                            | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491                                                                                                                                            | MILITARY GEOGRAPHY             |
| HI339                                                                                                                                            | THE MODERN MIDDLE EAST         |
| HI344                                                                                                                                            | MODERN DIPLOMACY               |
| HI381                                                                                                                                            | HISTORY OF IRREGULAR WARFARE   |
| HI383                                                                                                                                            | MIDDLE EASTERN WARFARE         |
| HI391                                                                                                                                            | WORLD RELIGIONS                |
| LW410                                                                                                                                            | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                                                                                                                            | ENVIRONMENTAL LAW              |
| LW481                                                                                                                                            | INTERNATIONAL LAW              |
| SS366                                                                                                                                            | COMPARATIVE POLITICS           |
| SS383                                                                                                                                            | POLITICS OF THE MIDDLE EAST    |
| SS465                                                                                                                                            | TERRORISM: NEW CHALLENGES      |
| SS475                                                                                                                                            | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                                                            | INTERNATIONAL ECONOMICS        |
| SS486                                                                                                                                            | STATE BUILDING                 |
| SS487                                                                                                                                            | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                                                            | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                                                            | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Integrative Experience for the Major</b>                                                                                                      | Choose 1 of 1                  |
| LN490Z                                                                                                                                           | PERSIAN LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Science Depth</b>                                                                                                                             |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                          |                                |
| CH102                                                                                                                                            | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                            | BIOLOGY                        |
| PH202                                                                                                                                            | PHYSICS II                     |
| PH252                                                                                                                                            | ADVANCED PHYSICS II            |
| PH275                                                                                                                                            | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                |                                |
| Cadets in the major must take CY305/355 to satisfy the STEM Depth requirement.                                                                   |                                |
| CY305                                                                                                                                            | CYBER FOUNDATIONS              |
| CY355                                                                                                                                            | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Curriculum Requirement</b>                                                                                                                    |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                         |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                      |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355. |                                |
| CY305                                                                                                                                            | CYBER FOUNDATIONS              |
| CY355                                                                                                                                            | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                 |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                  |                                |
| <b>Writing-in-the-Major</b>                                                                                                                      |                                |

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Persian and German Curriculum

| Code | Short Description                    | Description                                | Transcript Description               | Req Crse Cnt | Opt Crse Cnt |
|------|--------------------------------------|--------------------------------------------|--------------------------------------|--------------|--------------|
| FZG1 | Foreign Language: Persian and German | Foreign Language Major: Persian and German | Foreign Language: Persian and German | 2            | 13           |

## 2029 Foreign Language Major: Persian and German Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 18                 |
| LN440Z                          | PERSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LZ371                           | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                           | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                           | SPECIAL TOPIC IN PERSIAN       |
| LZ475                           | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                           | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                           | PERSIAN CIVILIZATION I         |
| LZ484                           | PERSIAN CIVILIZATION II        |
| LZ485                           | PERSIAN LITERATURE I           |
| LZ486                           | PERSIAN LITERATURE II          |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LG203                           | GERMAN I (STANDARD)            |
| LG204                           | GERMAN II (STANDARD)           |
| LG371                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG475                           | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                           | MILITARY SPKG/RDG - GERMAN     |
| LG483                           | GERMAN CIVILIZATION I          |
| LG484                           | GERMAN CIVILIZATION II         |
| LG485                           | SURVEY OF GERMAN LIT I         |
| LG486                           | SURVEY OF GERMAN LIT II        |

|                                                                                                                                                  |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| LG492                                                                                                                                            | 20TH & 21ST CENTURY GERMANY    |
| LN487                                                                                                                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                                                            | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Complementary Support Courses</b>                                                                                                             | Choose 3 of 26                 |
| Choose 3 of 26, each from a different discipline.                                                                                                |                                |
| EN300                                                                                                                                            | LITERARY METHODOLOGIES         |
| EN351                                                                                                                                            | WORLD LITERATURE               |
| EN362                                                                                                                                            | FILM AND FILM THEORY           |
| GE340                                                                                                                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE345                                                                                                                                            | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491                                                                                                                                            | MILITARY GEOGRAPHY             |
| HI339                                                                                                                                            | THE MODERN MIDDLE EAST         |
| HI344                                                                                                                                            | MODERN DIPLOMACY               |
| HI368                                                                                                                                            | MOD CENTRAL & E. EUR,1896-1989 |
| HI381                                                                                                                                            | HISTORY OF IRREGULAR WARFARE   |
| HI383                                                                                                                                            | MIDDLE EASTERN WARFARE         |
| HI391                                                                                                                                            | WORLD RELIGIONS                |
| LW410                                                                                                                                            | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                                                                                                                            | ENVIRONMENTAL LAW              |
| LW481                                                                                                                                            | INTERNATIONAL LAW              |
| SS366                                                                                                                                            | COMPARATIVE POLITICS           |
| SS383                                                                                                                                            | POLITICS OF THE MIDDLE EAST    |
| SS465                                                                                                                                            | TERRORISM: NEW CHALLENGES      |
| SS475                                                                                                                                            | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                                                            | INTERNATIONAL ECONOMICS        |
| SS486                                                                                                                                            | STATE BUILDING                 |
| SS487                                                                                                                                            | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                                                            | STRATEGY AND GENERALSHIP       |
| WS385                                                                                                                                            | WAR AND ITS THEORISTS          |
| WS455                                                                                                                                            | COMPARATIVE DEFENSE POLICY     |
| XH405                                                                                                                                            | THE HOLOCAUST AND ITS LEGACY   |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Integrative Experience for the Major</b>                                                                                                      | Choose 1 of 1                  |
| LN490Z                                                                                                                                           | PERSIAN LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Science Depth</b>                                                                                                                             |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                          |                                |
| CH102                                                                                                                                            | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                            | BIOLOGY                        |
| PH202                                                                                                                                            | PHYSICS II                     |
| PH252                                                                                                                                            | ADVANCED PHYSICS II            |
| PH275                                                                                                                                            | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                |                                |
| Cadets in the major must take CY305/355 to satisfy the STEM Depth requirement.                                                                   |                                |
| CY305                                                                                                                                            | CYBER FOUNDATIONS              |
| CY355                                                                                                                                            | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Curriculum Requirement</b>                                                                                                                    |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                         |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                      |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355. |                                |
| CY305                                                                                                                                            | CYBER FOUNDATIONS              |

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

## 2029 Foreign Language Major: Persian and Portuguese Curriculum

| Code | Short Description                        | Description                                    | Transcript Description                   | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------------|------------------------------------------------|------------------------------------------|--------------|--------------|
| FZP1 | Foreign Language: Persian and Portuguese | Foreign Language Major: Persian and Portuguese | Foreign Language: Persian and Portuguese | 2            | 13           |

## 2029 Foreign Language Major: Persian and Portuguese Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 18                 |
| LN440Z                          | PERSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LZ371                           | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                           | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                           | SPECIAL TOPIC IN PERSIAN       |
| LZ475                           | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                           | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                           | PERSIAN CIVILIZATION I         |
| LZ484                           | PERSIAN CIVILIZATION II        |
| LZ485                           | PERSIAN LITERATURE I           |
| LZ486                           | PERSIAN LITERATURE II          |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 12                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LP203                           | PORTUGUESE I (STANDARD)        |
| LP204                           | PORTUGUESE II (STANDARD)       |
| LP371                           | INTENSIVE INTERMED. PORTUGUESE |
| LP372                           | PORTUGUESE FOR ORAL/WRIT COMM  |

|       |                                |
|-------|--------------------------------|
| LP475 | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476 | MILITARY SPKG/RDG - PORTUGUESE |
| LP483 | PORTUGUESE CIVILIZATION I      |
| LP484 | PORTUGUESE CIVILIZATION II     |
| LP485 | SURVEY OF PORTUGUESE LIT I     |
| LP492 | LIT OF PORT-SPKG WORLD         |

**AND****Complementary Support Courses**

Choose 3 of 22

Choose 3 of 22, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST |
| GE491 | MILITARY GEOGRAPHY           |
| HI339 | THE MODERN MIDDLE EAST       |
| HI344 | MODERN DIPLOMACY             |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI383 | MIDDLE EASTERN WARFARE       |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS383 | POLITICS OF THE MIDDLE EAST  |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490Z | PERSIAN LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in the major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

**2029 Foreign Language Major: Persian and Russian Curriculum**

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------|--------------|
| FZR1 | Foreign Language: Persian and Russian | Foreign Language Major: Persian and Russian | Foreign Language: Persian and Russian | 2            | 13           |

**2029 Foreign Language Major: Persian and Russian Tracks**

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 18                 |
| LN440Z                          | PERSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LZ371                           | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                           | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                           | SPECIAL TOPIC IN PERSIAN       |
| LZ475                           | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                           | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                           | PERSIAN CIVILIZATION I         |
| LZ484                           | PERSIAN CIVILIZATION II        |
| LZ485                           | PERSIAN LITERATURE I           |
| LZ486                           | PERSIAN LITERATURE II          |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LR203                           | RUSSIAN I (STANDARD)           |
| LR204                           | RUSSIAN II (STANDARD)          |
| LR371                           | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                           | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475                           | RUSSIAN RDG/WRTG THRU MEDIA    |

|       |                                |
|-------|--------------------------------|
| LR476 | MILITARY SPKG/RDG - RUSSIAN    |
| LR483 | RUSSIAN CIV I                  |
| LR484 | RUSSIAN CIV II                 |
| LR485 | SURVEY OF RUSSIAN LITERATURE I |
| LR486 | SURVEY OF RUSSIAN LIT. II      |
| LR492 | RUSSIAN LIFE IN FICTION        |

**AND****Complementary Support Courses**

Choose 3 of 22

Choose 3 of 22, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST |
| GE491 | MILITARY GEOGRAPHY           |
| HI339 | THE MODERN MIDDLE EAST       |
| HI344 | MODERN DIPLOMACY             |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI383 | MIDDLE EASTERN WARFARE       |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS383 | POLITICS OF THE MIDDLE EAST  |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490Z

PERSIAN LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in the major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

**2029 Foreign Language Major: Persian and Spanish Curriculum**

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|-----------------|-----------------|
| FZS1 | Foreign Language: Persian and Spanish | Foreign Language Major: Persian and Spanish | Foreign Language: Persian and Spanish | 2               | 13              |

**2029 Foreign Language Major: Persian and Spanish Tracks**

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 18                 |
| LN440Z                          | PERSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LZ371                           | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                           | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470                           | SPECIAL TOPIC IN PERSIAN       |
| LZ475                           | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                           | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                           | PERSIAN CIVILIZATION I         |
| LZ484                           | PERSIAN CIVILIZATION II        |
| LZ485                           | PERSIAN LITERATURE I           |
| LZ486                           | PERSIAN LITERATURE II          |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LS203                           | SPANISH I (STANDARD)           |
| LS204                           | SPANISH II (STANDARD)          |
| LS371                           | INTENSIVE INTERMEDIATE SPANISH |
| LS372                           | SPANISH FOR ORAL & WRIT COMM   |
| LS475                           | SPANISH RDG/WRTG THRU MEDIA    |



|       |                                |
|-------|--------------------------------|
| LS476 | MILITARY SPKG/RDG - SPANISH    |
| LS483 | SPANISH CIV AND CULTURE        |
| LS484 | SPANISH AMERICAN CIV AND CULT  |
| LS485 | SPANISH-AMERICAN LITERATURE    |
| LS486 | THE LITERATURE OF SPAIN        |
| LS492 | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Complementary Support Courses**

Choose 3 of 22

Choose 3 of 22, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST |
| GE491 | MILITARY GEOGRAPHY           |
| HI339 | THE MODERN MIDDLE EAST       |
| HI344 | MODERN DIPLOMACY             |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI383 | MIDDLE EASTERN WARFARE       |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS383 | POLITICS OF THE MIDDLE EAST  |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490Z

PERSIAN LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in the major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

**2029 Foreign Language Major: Persian w/ Honors Curriculum**

| Code  | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FLZ1H | Foreign Language: Persian w/ Honors | Foreign Language Major: Persian w/ Honors | Foreign Language: Persian w/ Honors | 1            | 1            |

**2029 Foreign Language Major: Persian w/ Honors Tracks**

| Subject Area                                                                   | Description                    |
|--------------------------------------------------------------------------------|--------------------------------|
| <b>Additional Elective</b>                                                     | Choose 1 of 15                 |
| Take an additional advanced-level elective, not already taken, from this list. |                                |
| LN440Z                                                                         | PERSIAN IN CULTURAL CONTEXT    |
| LN451                                                                          | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                                                          | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                                                          | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                                                          | SEM ABROAD: ADV LANG & CULT II |
| LN493                                                                          | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                                                          | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                                                          | SEM ABROAD: ADV LANG & CULT V  |
| LZ470                                                                          | SPECIAL TOPIC IN PERSIAN       |
| LZ475                                                                          | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476                                                                          | MILITARY SPKG/RDG - PERSIAN    |
| LZ483                                                                          | PERSIAN CIVILIZATION I         |
| LZ484                                                                          | PERSIAN CIVILIZATION II        |
| LZ485                                                                          | PERSIAN LITERATURE I           |
| LZ486                                                                          | PERSIAN LITERATURE II          |
| <b>AND</b>                                                                     |                                |
| <b>Honors Thesis</b>                                                           | Choose 1 of 1                  |
| Write an honors thesis under the direction of a senior faculty member.         |                                |
| LN488                                                                          | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                     |                                |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and a minimum APSC of 3.5 in the major.

**2029 Foreign Language Major: Portuguese Curriculum**

| Code | Short Description            | Description                        | Transcript Description       | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------|------------------------------------|------------------------------|--------------|--------------|
| FLP2 | Foreign Language: Portuguese | Foreign Language Major: Portuguese | Foreign Language: Portuguese | 2            | 11           |

### 2029 Foreign Language Major: Portuguese Tracks

| Subject Area                                 | Description                    |
|----------------------------------------------|--------------------------------|
| <b>Required Course</b>                       | Choose 1 of 1                  |
| LN380                                        | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                   |                                |
| <b>Electives</b>                             | Choose 8 of 18                 |
| LN440P                                       | PORTUGUESE IN CULTURAL CONTEXT |
| LN451                                        | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                        | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                        | SEM ABROAD: ADV LANG & CULT II |
| LN493                                        | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                        | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                        | SEM ABROAD: ADV LANG & CULT V  |
| LP371                                        | INTENSIVE INTERMED. PORTUGUESE |
| LP372                                        | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP470                                        | SPECIAL TOPIC IN PORTUGUESE    |
| LP475                                        | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476                                        | MILITARY SPKG/RDG - PORTUGUESE |
| LP483                                        | PORTUGUESE CIVILIZATION I      |
| LP484                                        | PORTUGUESE CIVILIZATION II     |
| LP485                                        | SURVEY OF PORTUGUESE LIT I     |
| LP492                                        | LIT OF PORT-SPKG WORLD         |
| <b>AND</b>                                   |                                |
| <b>Complementary Support Courses</b>         | Choose 3 of 41                 |
| Choose one each from a different discipline. |                                |
| EN300                                        | LITERARY METHODOLOGIES         |
| EN311                                        | ANCIENT TO EARLY MODERN LIT    |
| EN340                                        | CONTEMPORARY LITERATURE        |
| EN351                                        | WORLD LITERATURE               |
| EN353                                        | WAR LITERATURE                 |
| EN361                                        | POETRY                         |
| EN362                                        | FILM AND FILM THEORY           |
| EN363                                        | THE NOVEL                      |
| EN364                                        | DRAMA                          |
| GE340                                        | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                        | GEOGRAPHY OF AFRICA            |
| GE346                                        | GEOGRAPHY OF EUROPE            |
| GE347                                        | GEOGRAPHY OF LATIN AMERICA     |
| GE491                                        | MILITARY GEOGRAPHY             |
| HI344                                        | MODERN DIPLOMACY               |
| HI345                                        | MODERN AFRICA                  |
| HI348                                        | MODERN LATIN AMERICA           |
| HI364                                        | MODERN WESTERN EUROPE          |
| HI381                                        | HISTORY OF IRREGULAR WARFARE   |
| HI391                                        | WORLD RELIGIONS                |
| LW410                                        | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                        | ENVIRONMENTAL LAW              |

|       |                                |
|-------|--------------------------------|
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS384 | POLITICS & GOVT-LATIN AMER     |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

**AND**


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**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490P | PORTUGUESE LANG & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: Portuguese & Arabic Curriculum

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------|--------------|
| FPA1 | Foreign Language: Portuguese & Arabic | Foreign Language Major: Portuguese & Arabic | Foreign Language: Portuguese & Arabic | 2            | 13           |

## 2029 Foreign Language Major: Portuguese & Arabic Tracks

| Subject Area                                      | Description                    |
|---------------------------------------------------|--------------------------------|
| <b>Required Course</b>                            | Choose 1 of 1                  |
| LN380                                             | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                        |                                |
| <b>Electives</b>                                  | Choose 6 of 18                 |
| LN440P                                            | PORTUGUESE IN CULTURAL CONTEXT |
| LN451                                             | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                             | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                             | SEM ABROAD: ADV LANG & CULT II |
| LN493                                             | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                             | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                             | SEM ABROAD: ADV LANG & CULT V  |
| LP371                                             | INTENSIVE INTERMED. PORTUGUESE |
| LP372                                             | PORTUGUESE FOR ORAL/WRT COMM   |
| LP470                                             | SPECIAL TOPIC IN PORTUGUESE    |
| LP475                                             | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476                                             | MILITARY SPKG/RDG - PORTUGUESE |
| LP483                                             | PORTUGUESE CIVILIZATION I      |
| LP484                                             | PORTUGUESE CIVILIZATION II     |
| LP485                                             | SURVEY OF PORTUGUESE LIT I     |
| LP492                                             | LIT OF PORT-SPKG WORLD         |
| <b>AND</b>                                        |                                |
| <b>Secondary Language Track</b>                   | Choose 4 of 14                 |
| LA203                                             | ARABIC I (STANDARD)            |
| LA204                                             | ARABIC II (STANDARD)           |
| LA371                                             | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                             | ARABIC FOR ORAL & WRITTEN COMM |
| LA472                                             | COLLOQUIAL ARABIC              |
| LA475                                             | ARABIC RDG/WRTG THRU MEDIA     |
| LA476                                             | MILITARY SPKG/RDG - ARABIC     |
| LA483                                             | ARAB CIVILIZATION I            |
| LA484                                             | ARAB CIVILIZATION II           |
| LA485                                             | ARABIC LITERATURE I            |
| LA486                                             | ARABIC LITERATURE II           |
| LA492                                             | ARABIC LITERATURE III          |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Courses</b>              | Choose 3 of 42                 |
| Choose 3 of 42, each from a different discipline. |                                |
| EN300                                             | LITERARY METHODOLOGIES         |
| EN351                                             | WORLD LITERATURE               |
| GE340                                             | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                             | GEOGRAPHY OF AFRICA            |
| GE346                                             | GEOGRAPHY OF EUROPE            |
| GE347                                             | GEOGRAPHY OF LATIN AMERICA     |
| GE491                                             | MILITARY GEOGRAPHY             |
| HI339                                             | THE MODERN MIDDLE EAST         |

|       |                                |
|-------|--------------------------------|
| HI344 | MODERN DIPLOMACY               |
| HI345 | MODERN AFRICA                  |
| HI348 | MODERN LATIN AMERICA           |
| HI364 | MODERN WESTERN EUROPE          |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI383 | MIDDLE EASTERN WARFARE         |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS384 | POLITICS & GOVT-LATIN AMER     |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS477 | ECONOMICS OF NATIONAL SECURITY |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490P

PORTUGUESE LANG &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

### Core Engineering Sequence

Cadets in this major may choose any approved three course engineering sequence.

### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Portuguese & Arabic w/ Honors Curriculum

| Code  | Short Description                           | Description                                           | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|-------------------------------------------------------|---------------------------------------------|--------------|--------------|
| FPA1H | Foreign Lang: Portuguese & Arabic w/ Honors | Foreign Language Major: Portuguese & Arabic w/ Honors | Foreign Lang: Portuguese & Arabic w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Portuguese & Arabic w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Portuguese & Chinese Curriculum

| Code | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FPC1 | Foreign Language: Portuguese & Chinese | Foreign Language Major: Portuguese & Chinese | Foreign Language: Portuguese & Chinese | 2            | 13           |

### 2029 Foreign Language Major: Portuguese & Chinese Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 18                 |
| LN440P                 | PORTUGUESE IN CULTURAL CONTEXT |

|       |                                |
|-------|--------------------------------|
| LN451 | ADV LANG & CULTURE IN CONTEXT  |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LN491 | SEM ABROAD: ADV LANG & CULT I  |
| LN492 | SEM ABROAD: ADV LANG & CULT II |
| LN493 | SEM ABROAD: ADV LANG&CULT III  |
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LP371 | INTENSIVE INTERMED. PORTUGUESE |
| LP372 | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP470 | SPECIAL TOPIC IN PORTUGUESE    |
| LP475 | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476 | MILITARY SPKG/RDG - PORTUGUESE |
| LP483 | PORTUGUESE CIVILIZATION I      |
| LP484 | PORTUGUESE CIVILIZATION II     |
| LP485 | SURVEY OF PORTUGUESE LIT I     |
| LP492 | LIT OF PORT-SPKG WORLD         |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LC203 | CHINESE I (STANDARD)           |
| LC204 | CHINESE II (STANDARD)          |
| LC371 | INTENSIVE INTERMEDIATE CHINESE |
| LC372 | CHINESE FOR ORAL & WRIT COMM   |
| LC475 | CHINESE RDG/WRTG THRU MEDIA    |
| LC476 | MILITARY SPKG/RDG - CHINESE    |
| LC483 | CHINESE CIVILIZATION I         |
| LC484 | CHINESE CIVILIZATION II        |
| LC485 | CHINESE LITERATURE I           |
| LC486 | CHINESE LITERATURE II          |
| LC492 | CHINESE LITERATURE III         |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 27

Choose 3 of 27, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE344 | GEOGRAPHY OF AFRICA          |
| GE346 | GEOGRAPHY OF EUROPE          |
| GE347 | GEOGRAPHY OF LATIN AMERICA   |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI345 | MODERN AFRICA                |
| HI348 | MODERN LATIN AMERICA         |
| HI364 | MODERN WESTERN EUROPE        |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS377 | POLITICS & GOV OF EUROPE     |
| SS384 | POLITICS & GOVT-LATIN AMER   |
| SS465 | TERRORISM: NEW CHALLENGES    |



|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| SS475                                                                                                                                                                        | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                                                                                        | INTERNATIONAL ECONOMICS        |
| SS485                                                                                                                                                                        | GOV & POLITICS SUB-SAHARAN AFR |
| SS486                                                                                                                                                                        | STATE BUILDING                 |
| SS487                                                                                                                                                                        | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                                                                                        | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                                                                                        | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                  |
| LN490P                                                                                                                                                                       | PORTUGUESE LANG & CULTURE CAP  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Curriculum Requirement</b>                                                                                                                                                |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES     |

## 2029 Foreign Language Major: Portuguese & Chinese w/ Honors Curriculum

| Code  | Short Description                            | Description                                            | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FPC1H | Foreign Lang: Portuguese & Chinese w/ Honors | Foreign Language Major: Portuguese & Chinese w/ Honors | Foreign Lang: Portuguese & Chinese w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Portuguese & Chinese w/ Honors Tracks

| Subject Area         | Description   |
|----------------------|---------------|
| Honors Thesis Course | Choose 1 of 1 |

Write an honors thesis under the direction of a senior faculty member.

LN488

ADV IND STUDY-FOREIGN LANGS

AND

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Portuguese & French Curriculum

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------|--------------|
| FPF1 | Foreign Language: Portuguese & French | Foreign Language Major: Portuguese & French | Foreign Language: Portuguese & French | 2            | 13           |

## 2029 Foreign Language Major: Portuguese & French Tracks

| Subject Area                    | Description                     |
|---------------------------------|---------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                   |
| LN380                           | NATURE OF MODERN LANGUAGES      |
| AND                             |                                 |
| <b>Electives</b>                | Choose 6 of 18                  |
| LN440P                          | PORTUGUESE IN CULTURAL CONTEXT  |
| LN451                           | ADV LANG & CULTURE IN CONTEXT   |
| LN487                           | ADV IND STUDY-FOREIGN LANGS     |
| LN488                           | ADV IND STUDY-FOREIGN LANGS     |
| LN491                           | SEM ABROAD: ADV LANG & CULT I   |
| LN492                           | SEM ABROAD: ADV LANG & CULT II  |
| LN493                           | SEM ABROAD: ADV LANG&CULT III   |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV  |
| LN495                           | SEM ABROAD: ADV LANG & CULT V   |
| LP371                           | INTENSIVE INTERMED. PORTUGUESE  |
| LP372                           | PORTUGUESE FOR ORAL/WRIT COMM   |
| LP470                           | SPECIAL TOPIC IN PORTUGUESE     |
| LP475                           | PORTUGUESE RDG/WRITG THRU MEDIA |
| LP476                           | MILITARY SPKG/RDG - PORTUGUESE  |
| LP483                           | PORTUGUESE CIVILIZATION I       |
| LP484                           | PORTUGUESE CIVILIZATION II      |
| LP485                           | SURVEY OF PORTUGUESE LIT I      |
| LP492                           | LIT OF PORT-SPKG WORLD          |
| AND                             |                                 |
| <b>Secondary Language Track</b> | Choose 4 of 13                  |
| LF203                           | FRENCH I (STANDARD)             |
| LF204                           | FRENCH II (STANDARD)            |
| LF371                           | INTENSIVE INTERMEDIATE FRENCH   |
| LF372                           | FRENCH FOR ORAL & WRITTEN COMM  |
| LF475                           | FRENCH RDG/WRITG THRU MEDIA     |
| LF476                           | MILITARY SPKG/RDG - FRENCH      |
| LF483                           | FRENCH CIVILIZATION I           |
| LF484                           | FRENCH CIVILIZATION II          |

|                                                                                                         |                                |
|---------------------------------------------------------------------------------------------------------|--------------------------------|
| LF485                                                                                                   | SURVEY OF FRENCH LIT I         |
| LF486                                                                                                   | SURVEY OF FRENCH LIT II        |
| LF492                                                                                                   | MASTERWORKS OF FRENCH LIT      |
| LN487                                                                                                   | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                   | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                                              |                                |
| <hr/>                                                                                                   |                                |
| <b>Complementary Support Courses</b>                                                                    | Choose 3 of 27                 |
| Choose 3 of 27, each from a different discipline.                                                       |                                |
| EN300                                                                                                   | LITERARY METHODOLOGIES         |
| EN351                                                                                                   | WORLD LITERATURE               |
| GE340                                                                                                   | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                                                                                   | GEOGRAPHY OF AFRICA            |
| GE346                                                                                                   | GEOGRAPHY OF EUROPE            |
| GE347                                                                                                   | GEOGRAPHY OF LATIN AMERICA     |
| GE491                                                                                                   | MILITARY GEOGRAPHY             |
| HI344                                                                                                   | MODERN DIPLOMACY               |
| HI345                                                                                                   | MODERN AFRICA                  |
| HI348                                                                                                   | MODERN LATIN AMERICA           |
| HI364                                                                                                   | MODERN WESTERN EUROPE          |
| HI381                                                                                                   | HISTORY OF IRREGULAR WARFARE   |
| HI391                                                                                                   | WORLD RELIGIONS                |
| LW410                                                                                                   | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                                                                                   | ENVIRONMENTAL LAW              |
| LW481                                                                                                   | INTERNATIONAL LAW              |
| SS366                                                                                                   | COMPARATIVE POLITICS           |
| SS377                                                                                                   | POLITICS & GOV OF EUROPE       |
| SS384                                                                                                   | POLITICS & GOVT-LATIN AMER     |
| SS465                                                                                                   | TERRORISM: NEW CHALLENGES      |
| SS475                                                                                                   | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                   | INTERNATIONAL ECONOMICS        |
| SS485                                                                                                   | GOV & POLITICS SUB-SAHARAN AFR |
| SS486                                                                                                   | STATE BUILDING                 |
| SS487                                                                                                   | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                   | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                   | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                              |                                |
| <hr/>                                                                                                   |                                |
| <b>Integrative Experience for the Major</b>                                                             | Choose 1 of 1                  |
| LN490P                                                                                                  | PORTUGUESE LANG & CULTURE CAP  |
| <b>AND</b>                                                                                              |                                |
| <hr/>                                                                                                   |                                |
| <b>Science Depth</b>                                                                                    |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement. |                                |
| CH102                                                                                                   | GENERAL CHEMISTRY II           |
| CH275                                                                                                   | BIOLOGY                        |
| PH202                                                                                                   | PHYSICS II                     |
| PH252                                                                                                   | ADVANCED PHYSICS II            |
| PH275                                                                                                   | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                       |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                         |                                |
| CY305                                                                                                   | CYBER FOUNDATIONS              |
| CY355                                                                                                   | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                              |                                |
| <hr/>                                                                                                   |                                |
| <b>Curriculum Requirement</b>                                                                           |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                |                                |
| <b>IT/CYBER Requirement</b>                                                                             |                                |

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

### Core Engineering Sequence

Cadets in this major may choose any approved three course engineering sequence.

### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Portuguese & French w/ Honors Curriculum

| Code  | Short Description                           | Description                                           | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|-------------------------------------------------------|---------------------------------------------|--------------|--------------|
| FPF1H | Foreign Lang: Portuguese & French w/ Honors | Foreign Language Major: Portuguese & French w/ Honors | Foreign Lang: Portuguese & French w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Portuguese & French w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Portuguese & German Curriculum

| Code | Short Description                     | Description                                 | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------------------|---------------------------------------------|---------------------------------------|--------------|--------------|
| FPG1 | Foreign Language: Portuguese & German | Foreign Language Major: Portuguese & German | Foreign Language: Portuguese & German | 2            | 13           |

### 2029 Foreign Language Major: Portuguese & German Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 18                 |
| LN440P                 | PORTUGUESE IN CULTURAL CONTEXT |

|       |                                |
|-------|--------------------------------|
| LN451 | ADV LANG & CULTURE IN CONTEXT  |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LN491 | SEM ABROAD: ADV LANG & CULT I  |
| LN492 | SEM ABROAD: ADV LANG & CULT II |
| LN493 | SEM ABROAD: ADV LANG&CULT III  |
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LP371 | INTENSIVE INTERMED. PORTUGUESE |
| LP372 | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP470 | SPECIAL TOPIC IN PORTUGUESE    |
| LP475 | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476 | MILITARY SPKG/RDG - PORTUGUESE |
| LP483 | PORTUGUESE CIVILIZATION I      |
| LP484 | PORTUGUESE CIVILIZATION II     |
| LP485 | SURVEY OF PORTUGUESE LIT I     |
| LP492 | LIT OF PORT-SPKG WORLD         |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LG203 | GERMAN I (STANDARD)            |
| LG204 | GERMAN II (STANDARD)           |
| LG371 | INTENSIVE INTERMEDIATE GERMAN  |
| LG372 | GERMAN FOR ORAL & WRITTEN COMM |
| LG475 | GERMAN RDG/WRTG THRU MEDIA     |
| LG476 | MILITARY SPKG/RDG - GERMAN     |
| LG483 | GERMAN CIVILIZATION I          |
| LG484 | GERMAN CIVILIZATION II         |
| LG485 | SURVEY OF GERMAN LIT I         |
| LG486 | SURVEY OF GERMAN LIT II        |
| LG492 | 20TH & 21ST CENTURY GERMANY    |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 31

Choose 3 of 31, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN351 | WORLD LITERATURE               |
| EN362 | FILM AND FILM THEORY           |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE346 | GEOGRAPHY OF EUROPE            |
| GE347 | GEOGRAPHY OF LATIN AMERICA     |
| GE491 | MILITARY GEOGRAPHY             |
| HI344 | MODERN DIPLOMACY               |
| HI345 | MODERN AFRICA                  |
| HI348 | MODERN LATIN AMERICA           |
| HI364 | MODERN WESTERN EUROPE          |
| HI368 | MOD CENTRAL & E. EUR,1896-1989 |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |

|       |                                |
|-------|--------------------------------|
| SS384 | POLITICS & GOVT-LATIN AMER     |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS385 | WAR AND ITS THEORISTS          |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490P | PORTUGUESE LANG & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: Portuguese & German w/ Honors Curriculum

| Code  | Short Description                           | Description                                           | Transcript Description                      | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------------|-------------------------------------------------------|---------------------------------------------|--------------|--------------|
| FPG1H | Foreign Lang: Portuguese & German w/ Honors | Foreign Language Major: Portuguese & German w/ Honors | Foreign Lang: Portuguese & German w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Portuguese & German w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Portuguese & Persian Curriculum

| Code | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FPZ2 | Foreign Language: Portuguese & Persian | Foreign Language Major: Portuguese & Persian | Foreign Language: Portuguese & Persian | 2            | 13           |

### 2029 Foreign Language Major: Portuguese & Persian Tracks

| Subject Area                    | Description                     |
|---------------------------------|---------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                   |
| LN380                           | NATURE OF MODERN LANGUAGES      |
| <b>AND</b>                      |                                 |
| <b>Electives</b>                | Choose 6 of 18                  |
| LN440P                          | PORTUGUESE IN CULTURAL CONTEXT  |
| LN451                           | ADV LANG & CULTURE IN CONTEXT   |
| LN487                           | ADV IND STUDY-FOREIGN LANGS     |
| LN488                           | ADV IND STUDY-FOREIGN LANGS     |
| LN491                           | SEM ABROAD: ADV LANG & CULT I   |
| LN492                           | SEM ABROAD: ADV LANG & CULT II  |
| LN493                           | SEM ABROAD: ADV LANG&CULT III   |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV  |
| LN495                           | SEM ABROAD: ADV LANG & CULT V   |
| LP371                           | INTENSIVE INTERMED. PORTUGUESE  |
| LP372                           | PORTUGUESE FOR ORAL/WRIT COMM   |
| LP470                           | SPECIAL TOPIC IN PORTUGUESE     |
| LP475                           | PORTUGUESE RDG/WRITG THRU MEDIA |
| LP476                           | MILITARY SPKG/RDG - PORTUGUESE  |
| LP483                           | PORTUGUESE CIVILIZATION I       |
| LP484                           | PORTUGUESE CIVILIZATION II      |
| LP485                           | SURVEY OF PORTUGUESE LIT I      |
| LP492                           | LIT OF PORT-SPKG WORLD          |
| <b>AND</b>                      |                                 |
| <b>Secondary Language Track</b> | Choose 4 of 13                  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS     |
| LN488                           | ADV IND STUDY-FOREIGN LANGS     |

|       |                                |
|-------|--------------------------------|
| LZ203 | PERSIAN I (STANDARD)           |
| LZ204 | PERSIAN II (STANDARD)          |
| LZ371 | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372 | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470 | SPECIAL TOPIC IN PERSIAN       |
| LZ475 | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476 | MILITARY SPKG/RDG - PERSIAN    |
| LZ483 | PERSIAN CIVILIZATION I         |
| LZ484 | PERSIAN CIVILIZATION II        |
| LZ485 | PERSIAN LITERATURE I           |
| LZ486 | PERSIAN LITERATURE II          |

**AND****Complementary Support Courses**

Choose 3 of 27

Choose 3 of 27, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN351 | WORLD LITERATURE               |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE346 | GEOGRAPHY OF EUROPE            |
| GE347 | GEOGRAPHY OF LATIN AMERICA     |
| GE491 | MILITARY GEOGRAPHY             |
| HI344 | MODERN DIPLOMACY               |
| HI345 | MODERN AFRICA                  |
| HI348 | MODERN LATIN AMERICA           |
| HI364 | MODERN WESTERN EUROPE          |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS384 | POLITICS & GOVT-LATIN AMER     |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490P

PORTUGUESE LANG &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                   |
|-------|-------------------|
| CY305 | CYBER FOUNDATIONS |
|-------|-------------------|



CY355 CYBER FOUNDATIONS - COMPUTING

**AND**

**Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305 CYBER FOUNDATIONS

CY355 CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Portuguese & Persian w/ Honors Curriculum

| Code  | Short Description                            | Description                                            | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FPZ2H | Foreign Lang: Portuguese & Persian w/ Honors | Foreign Language Major: Portuguese & Persian w/ Honors | Foreign Lang: Portuguese & Persian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Portuguese & Persian w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Portuguese & Russian Curriculum

| Code | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FPR2 | Foreign Language: Portuguese & Russian | Foreign Language Major: Portuguese & Russian | Foreign Language: Portuguese & Russian | 2            | 13           |

### 2029 Foreign Language Major: Portuguese & Russian Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| <b>Required Course</b>                            | Choose 1 of 1                  |
| LN380                                             | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                        |                                |
| <b>Electives</b>                                  | Choose 6 of 18                 |
| LN440P                                            | PORTUGUESE IN CULTURAL CONTEXT |
| LN451                                             | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                             | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                             | SEM ABROAD: ADV LANG & CULT II |
| LN493                                             | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                             | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                             | SEM ABROAD: ADV LANG & CULT V  |
| LP371                                             | INTENSIVE INTERMED. PORTUGUESE |
| LP372                                             | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP470                                             | SPECIAL TOPIC IN PORTUGUESE    |
| LP475                                             | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476                                             | MILITARY SPKG/RDG - PORTUGUESE |
| LP483                                             | PORTUGUESE CIVILIZATION I      |
| LP484                                             | PORTUGUESE CIVILIZATION II     |
| LP485                                             | SURVEY OF PORTUGUESE LIT I     |
| LP492                                             | LIT OF PORT-SPKG WORLD         |
| <b>AND</b>                                        |                                |
| <b>Secondary Language Track</b>                   | Choose 4 of 13                 |
| LN487                                             | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                             | ADV IND STUDY-FOREIGN LANGS    |
| LR203                                             | RUSSIAN I (STANDARD)           |
| LR204                                             | RUSSIAN II (STANDARD)          |
| LR371                                             | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                                             | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475                                             | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                                             | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                                             | RUSSIAN CIV I                  |
| LR484                                             | RUSSIAN CIV II                 |
| LR485                                             | SURVEY OF RUSSIAN LITERATURE I |
| LR486                                             | SURVEY OF RUSSIAN LIT. II      |
| LR492                                             | RUSSIAN LIFE IN FICTION        |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Courses</b>              | Choose 3 of 27                 |
| Choose 3 of 27, each from a different discipline. |                                |
| EN300                                             | LITERARY METHODOLOGIES         |
| EN351                                             | WORLD LITERATURE               |
| GE340                                             | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                             | GEOGRAPHY OF AFRICA            |
| GE346                                             | GEOGRAPHY OF EUROPE            |
| GE347                                             | GEOGRAPHY OF LATIN AMERICA     |
| GE491                                             | MILITARY GEOGRAPHY             |
| HI344                                             | MODERN DIPLOMACY               |
| HI345                                             | MODERN AFRICA                  |
| HI348                                             | MODERN LATIN AMERICA           |
| HI364                                             | MODERN WESTERN EUROPE          |
| HI381                                             | HISTORY OF IRREGULAR WARFARE   |
| HI391                                             | WORLD RELIGIONS                |
| LW410                                             | COMPARATIVE LEGAL SYSTEMS      |

|       |                                |
|-------|--------------------------------|
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS384 | POLITICS & GOVT-LATIN AMER     |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490P | PORTUGUESE LANG & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: Portuguese & Russian w/ Honors Curriculum

| Code  | Short Description                            | Description                                            | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FPR2H | Foreign Lang: Portuguese & Russian w/ Honors | Foreign Language Major: Portuguese & Russian w/ Honors | Foreign Lang: Portuguese & Russian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Portuguese & Russian w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Portuguese & Spanish Curriculum

| Code | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FPS2 | Foreign Language: Portuguese & Spanish | Foreign Language Major: Portuguese & Spanish | Foreign Language: Portuguese & Spanish | 2            | 13           |

### 2029 Foreign Language Major: Portuguese & Spanish Tracks

| Subject Area                    | Description                     |
|---------------------------------|---------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                   |
| LN380                           | NATURE OF MODERN LANGUAGES      |
| <b>AND</b>                      |                                 |
| <b>Electives</b>                | Choose 6 of 18                  |
| LN440P                          | PORTUGUESE IN CULTURAL CONTEXT  |
| LN451                           | ADV LANG & CULTURE IN CONTEXT   |
| LN487                           | ADV IND STUDY-FOREIGN LANGS     |
| LN488                           | ADV IND STUDY-FOREIGN LANGS     |
| LN491                           | SEM ABROAD: ADV LANG & CULT I   |
| LN492                           | SEM ABROAD: ADV LANG & CULT II  |
| LN493                           | SEM ABROAD: ADV LANG&CULT III   |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV  |
| LN495                           | SEM ABROAD: ADV LANG & CULT V   |
| LP371                           | INTENSIVE INTERMED. PORTUGUESE  |
| LP372                           | PORTUGUESE FOR ORAL/WRIT COMM   |
| LP470                           | SPECIAL TOPIC IN PORTUGUESE     |
| LP475                           | PORTUGUESE RDG/WRITG THRU MEDIA |
| LP476                           | MILITARY SPKG/RDG - PORTUGUESE  |
| LP483                           | PORTUGUESE CIVILIZATION I       |
| LP484                           | PORTUGUESE CIVILIZATION II      |
| LP485                           | SURVEY OF PORTUGUESE LIT I      |
| LP492                           | LIT OF PORT-SPKG WORLD          |
| <b>AND</b>                      |                                 |
| <b>Secondary Language Track</b> | Choose 4 of 13                  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS     |
| LN488                           | ADV IND STUDY-FOREIGN LANGS     |

|       |                                |
|-------|--------------------------------|
| LS203 | SPANISH I (STANDARD)           |
| LS204 | SPANISH II (STANDARD)          |
| LS371 | INTENSIVE INTERMEDIATE SPANISH |
| LS372 | SPANISH FOR ORAL & WRIT COMM   |
| LS475 | SPANISH RDG/WRTG THRU MEDIA    |
| LS476 | MILITARY SPKG/RDG - SPANISH    |
| LS483 | SPANISH CIV AND CULTURE        |
| LS484 | SPANISH AMERICAN CIV AND CULT  |
| LS485 | SPANISH-AMERICAN LITERATURE    |
| LS486 | THE LITERATURE OF SPAIN        |
| LS492 | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Complementary Support Courses**

Choose 3 of 27

Choose 3 of 27, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN351 | WORLD LITERATURE               |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE346 | GEOGRAPHY OF EUROPE            |
| GE347 | GEOGRAPHY OF LATIN AMERICA     |
| GE491 | MILITARY GEOGRAPHY             |
| HI344 | MODERN DIPLOMACY               |
| HI345 | MODERN AFRICA                  |
| HI348 | MODERN LATIN AMERICA           |
| HI364 | MODERN WESTERN EUROPE          |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS384 | POLITICS & GOVT-LATIN AMER     |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|        |                               |
|--------|-------------------------------|
| LN490P | PORTUGUESE LANG & CULTURE CAP |
|--------|-------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                   |
|-------|-------------------|
| CY305 | CYBER FOUNDATIONS |
|-------|-------------------|

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>AND</b>                                                                                                                                                                   |                               |
| <b>Curriculum Requirement</b>                                                                                                                                                |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                               |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

### 2029 Foreign Language Major: Portuguese & Spanish w/ Honors Curriculum

| Code  | Short Description                            | Description                                            | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FPS2H | Foreign Lang: Portuguese & Spanish w/ Honors | Foreign Language Major: Portuguese & Spanish w/ Honors | Foreign Lang: Portuguese & Spanish w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Portuguese & Spanish w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Portuguese w/ Honors Curriculum

| Code  | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FLP2H | Foreign Language: Portuguese w/ Honors | Foreign Language Major: Portuguese w/ Honors | Foreign Language: Portuguese w/ Honors | 1            | 1            |

### 2029 Foreign Language Major: Portuguese w/ Honors Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

|                                                                                |                                |
|--------------------------------------------------------------------------------|--------------------------------|
| <b>Additional Elective</b>                                                     | Choose 1 of 15                 |
| Take an additional advanced-level elective, not already taken, from this list. |                                |
| LN440P                                                                         | PORTUGUESE IN CULTURAL CONTEXT |
| LN451                                                                          | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                                                          | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                                                          | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                                                          | SEM ABROAD: ADV LANG & CULT II |
| LN493                                                                          | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                                                          | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                                                          | SEM ABROAD: ADV LANG & CULT V  |
| LP470                                                                          | SPECIAL TOPIC IN PORTUGUESE    |
| LP475                                                                          | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476                                                                          | MILITARY SPKG/RDG - PORTUGUESE |
| LP483                                                                          | PORTUGUESE CIVILIZATION I      |
| LP484                                                                          | PORTUGUESE CIVILIZATION II     |
| LP485                                                                          | SURVEY OF PORTUGUESE LIT I     |
| LP492                                                                          | LIT OF PORT-SPKG WORLD         |

**AND**

|                      |               |
|----------------------|---------------|
| <b>Honors Thesis</b> | Choose 1 of 1 |
|----------------------|---------------|

Write an honors thesis under the direction of a senior faculty member.

|       |                             |
|-------|-----------------------------|
| LN488 | ADV IND STUDY-FOREIGN LANGS |
|-------|-----------------------------|

**AND**

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and a minimum APSC of 3.5 in the major.

## 2029 Foreign Language Major: Russian Curriculum

| Code | Short Description         | Description                     | Transcript Description    | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------|---------------------------------|---------------------------|--------------|--------------|
| FLR2 | Foreign Language: Russian | Foreign Language Major: Russian | Foreign Language: Russian | 2            | 11           |

## 2029 Foreign Language Major: Russian Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 8 of 19                 |
| LN440R                 | RUSSIAN IN CULTURAL CONTEXT    |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |
| LN494                  | SEM ABROAD: ADV LANG & CULT IV |
| LN495                  | SEM ABROAD: ADV LANG & CULT V  |
| LR371                  | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                  | RUSSIAN FOR ORAL & WRIT COMM   |

|       |                                |
|-------|--------------------------------|
| LR470 | SPECIAL TOPIC IN RUSSIAN       |
| LR475 | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476 | MILITARY SPKG/RDG - RUSSIAN    |
| LR483 | RUSSIAN CIV I                  |
| LR484 | RUSSIAN CIV II                 |
| LR485 | SURVEY OF RUSSIAN LITERATURE I |
| LR486 | SURVEY OF RUSSIAN LIT. II      |
| LR492 | RUSSIAN LIFE IN FICTION        |

**AND****Complementary Support Courses**

Choose 3 of 27

Choose one each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE343 | GEOGRAPHY OF RUSSIA          |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI367 | IMPERIAL AND SOVIET RUSSIA   |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS375 | GOV & POL RUSSIA & NEIGHBORS |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490R

RUSSIAN LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**



Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

### Core Engineering Sequence

Cadets in this major may choose any approved three course engineering sequence.

### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Russian & Arabic Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FRA1 | Foreign Language: Russian & Arabic | Foreign Language Major: Russian & Arabic | Foreign Language: Russian & Arabic | 2            | 13           |

### 2029 Foreign Language Major: Russian & Arabic Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LN440R                          | RUSSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LR371                           | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                           | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470                           | SPECIAL TOPIC IN RUSSIAN       |
| LR475                           | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                           | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                           | RUSSIAN CIV I                  |
| LR484                           | RUSSIAN CIV II                 |
| LR485                           | SURVEY OF RUSSIAN LITERATURE I |
| LR486                           | SURVEY OF RUSSIAN LIT. II      |
| LR492                           | RUSSIAN LIFE IN FICTION        |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 14                 |
| LA203                           | ARABIC I (STANDARD)            |
| LA204                           | ARABIC II (STANDARD)           |
| LA371                           | INTENSIVE INTERMEDIATE ARABIC  |

|       |                                |
|-------|--------------------------------|
| LA372 | ARABIC FOR ORAL & WRITTEN COMM |
| LA472 | COLLOQUIAL ARABIC              |
| LA475 | ARABIC RDG/WRTG THRU MEDIA     |
| LA476 | MILITARY SPKG/RDG - ARABIC     |
| LA483 | ARAB CIVILIZATION I            |
| LA484 | ARAB CIVILIZATION II           |
| LA485 | ARABIC LITERATURE I            |
| LA486 | ARABIC LITERATURE II           |
| LA492 | ARABIC LITERATURE III          |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 40

Choose 3 CSCs, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN363 | THE NOVEL                      |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE343 | GEOGRAPHY OF RUSSIA            |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE491 | MILITARY GEOGRAPHY             |
| HI339 | THE MODERN MIDDLE EAST         |
| HI344 | MODERN DIPLOMACY               |
| HI367 | IMPERIAL AND SOVIET RUSSIA     |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI383 | MIDDLE EASTERN WARFARE         |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS375 | GOV & POL RUSSIA & NEIGHBORS   |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS477 | ECONOMICS OF NATIONAL SECURITY |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| LN490R                                                                                                                                                                       | RUSSIAN LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Curriculum Requirement</b>                                                                                                                                                |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES     |

## 2029 Foreign Language Major: Russian & Arabic w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FRA1H | Foreign Language: Russian & Arabic w/ Honors | Foreign Language Major: Russian & Arabic w/ Honors | Foreign Language: Russian & Arabic w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Russian & Arabic w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Russian & Chinese Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FRC1 | Foreign Language: Russian & Chinese | Foreign Language Major: Russian & Chinese | Foreign Language: Russian & Chinese | 2            | 13           |

## 2029 Foreign Language Major: Russian & Chinese Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LN440R                          | RUSSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LR371                           | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                           | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470                           | SPECIAL TOPIC IN RUSSIAN       |
| LR475                           | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                           | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                           | RUSSIAN CIV I                  |
| LR484                           | RUSSIAN CIV II                 |
| LR485                           | SURVEY OF RUSSIAN LITERATURE I |
| LR486                           | SURVEY OF RUSSIAN LIT. II      |
| LR492                           | RUSSIAN LIFE IN FICTION        |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LC203                           | CHINESE I (STANDARD)           |
| LC204                           | CHINESE II (STANDARD)          |
| LC371                           | INTENSIVE INTERMEDIATE CHINESE |
| LC372                           | CHINESE FOR ORAL & WRIT COMM   |
| LC475                           | CHINESE RDG/WRTG THRU MEDIA    |
| LC476                           | MILITARY SPKG/RDG - CHINESE    |
| LC483                           | CHINESE CIVILIZATION I         |
| LC484                           | CHINESE CIVILIZATION II        |
| LC485                           | CHINESE LITERATURE I           |
| LC486                           | CHINESE LITERATURE II          |
| LC492                           | CHINESE LITERATURE III         |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 24

Choose 3 CSCs, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE343 | GEOGRAPHY OF RUSSIA          |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI367 | IMPERIAL AND SOVIET RUSSIA   |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS375 | GOV & POL RUSSIA & NEIGHBORS |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490R | RUSSIAN LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Russian & Chinese w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FRC1H | Foreign Language: Russian & Chinese w/ Honors | Foreign Language Major: Russian & Chinese w/ Honors | Foreign Language: Russian & Chinese w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Russian & Chinese w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Russian & French Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FRF1 | Foreign Language: Russian & French | Foreign Language Major: Russian & French | Foreign Language: Russian & French | 2            | 13           |

## 2029 Foreign Language Major: Russian & French Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LN440R                 | RUSSIAN IN CULTURAL CONTEXT    |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |

|       |                                |
|-------|--------------------------------|
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LR371 | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372 | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470 | SPECIAL TOPIC IN RUSSIAN       |
| LR475 | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476 | MILITARY SPKG/RDG - RUSSIAN    |
| LR483 | RUSSIAN CIV I                  |
| LR484 | RUSSIAN CIV II                 |
| LR485 | SURVEY OF RUSSIAN LITERATURE I |
| LR486 | SURVEY OF RUSSIAN LIT. II      |
| LR492 | RUSSIAN LIFE IN FICTION        |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LF203 | FRENCH I (STANDARD)            |
| LF204 | FRENCH II (STANDARD)           |
| LF371 | INTENSIVE INTERMEDIATE FRENCH  |
| LF372 | FRENCH FOR ORAL & WRITTEN COMM |
| LF475 | FRENCH RDG/WRTG THRU MEDIA     |
| LF476 | MILITARY SPKG/RDG - FRENCH     |
| LF483 | FRENCH CIVILIZATION I          |
| LF484 | FRENCH CIVILIZATION II         |
| LF485 | SURVEY OF FRENCH LIT I         |
| LF486 | SURVEY OF FRENCH LIT II        |
| LF492 | MASTERWORKS OF FRENCH LIT      |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 24

Choose 3 CSCs, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE343 | GEOGRAPHY OF RUSSIA          |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI367 | IMPERIAL AND SOVIET RUSSIA   |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS375 | GOV & POL RUSSIA & NEIGHBORS |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND**

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                  |
| LN490R                                                                                                                                                                       | RUSSIAN LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Curriculum Requirement</b>                                                                                                                                                |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES     |

## 2029 Foreign Language Major: Russian & French w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FRF1H | Foreign Language: Russian & French w/ Honors | Foreign Language Major: Russian & French w/ Honors | Foreign Language: Russian & French w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Russian & French w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.



## 2029 Foreign Language Major: Russian & German Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|-----------------|-----------------|
| FRG1 | Foreign Language: Russian & German | Foreign Language Major: Russian & German | Foreign Language: Russian & German | 2               | 13              |

## 2029 Foreign Language Major: Russian & German Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LN440R                          | RUSSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LR371                           | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                           | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470                           | SPECIAL TOPIC IN RUSSIAN       |
| LR475                           | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                           | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                           | RUSSIAN CIV I                  |
| LR484                           | RUSSIAN CIV II                 |
| LR485                           | SURVEY OF RUSSIAN LITERATURE I |
| LR486                           | SURVEY OF RUSSIAN LIT. II      |
| LR492                           | RUSSIAN LIFE IN FICTION        |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LG203                           | GERMAN I (STANDARD)            |
| LG204                           | GERMAN II (STANDARD)           |
| LG371                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG475                           | GERMAN RDG/WRTG THRU MEDIA     |
| LG476                           | MILITARY SPKG/RDG - GERMAN     |
| LG483                           | GERMAN CIVILIZATION I          |
| LG484                           | GERMAN CIVILIZATION II         |
| LG485                           | SURVEY OF GERMAN LIT I         |
| LG486                           | SURVEY OF GERMAN LIT II        |
| LG492                           | 20TH & 21ST CENTURY GERMANY    |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 28

Choose 3 CSCs, each from a different discipline.

|       |                                 |
|-------|---------------------------------|
| EN300 | LITERARY METHODOLOGIES          |
| EN351 | WORLD LITERATURE                |
| EN353 | WAR LITERATURE                  |
| EN361 | POETRY                          |
| EN362 | FILM AND FILM THEORY            |
| EN363 | THE NOVEL                       |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES    |
| GE343 | GEOGRAPHY OF RUSSIA             |
| GE491 | MILITARY GEOGRAPHY              |
| HI344 | MODERN DIPLOMACY                |
| HI367 | IMPERIAL AND SOVIET RUSSIA      |
| HI368 | MOD CENTRAL & E. EUR, 1896-1989 |
| HI381 | HISTORY OF IRREGULAR WARFARE    |
| HI391 | WORLD RELIGIONS                 |
| LW410 | COMPARATIVE LEGAL SYSTEMS       |
| LW473 | ENVIRONMENTAL LAW               |
| LW481 | INTERNATIONAL LAW               |
| SS366 | COMPARATIVE POLITICS            |
| SS375 | GOV & POL RUSSIA & NEIGHBORS    |
| SS465 | TERRORISM: NEW CHALLENGES       |
| SS475 | COMP POLITICAL INSTITUTIONS     |
| SS484 | INTERNATIONAL ECONOMICS         |
| SS486 | STATE BUILDING                  |
| SS487 | INT'L POLITICAL ECONOMY         |
| WS358 | STRATEGY AND GENERALSHIP        |
| WS385 | WAR AND ITS THEORISTS           |
| WS455 | COMPARATIVE DEFENSE POLICY      |
| XH405 | THE HOLOCAUST AND ITS LEGACY    |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490R | RUSSIAN LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

**2029 Foreign Language Major: Russian & German w/ Honors Curriculum**

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FRG1H | Foreign Language: Russian & German w/ Honors | Foreign Language Major: Russian & German w/ Honors | Foreign Language: Russian & German w/ Honors | 1            | 0            |

**2029 Foreign Language Major: Russian & German w/ Honors Tracks**

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

**2029 Foreign Language Major: Russian & Persian Curriculum**

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FRZ2 | Foreign Language: Russian & Persian | Foreign Language Major: Russian & Persian | Foreign Language: Russian & Persian | 2            | 13           |

**2029 Foreign Language Major: Russian & Persian Tracks**

| Subject Area           | Description                   |
|------------------------|-------------------------------|
| <b>Required Course</b> | Choose 1 of 1                 |
| LN380                  | NATURE OF MODERN LANGUAGES    |
| <b>AND</b>             |                               |
| <b>Electives</b>       | Choose 6 of 19                |
| LN440R                 | RUSSIAN IN CULTURAL CONTEXT   |
| LN451                  | ADV LANG & CULTURE IN CONTEXT |
| LN487                  | ADV IND STUDY-FOREIGN LANGS   |
| LN488                  | ADV IND STUDY-FOREIGN LANGS   |

|       |                                |
|-------|--------------------------------|
| LN491 | SEM ABROAD: ADV LANG & CULT I  |
| LN492 | SEM ABROAD: ADV LANG & CULT II |
| LN493 | SEM ABROAD: ADV LANG&CULT III  |
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LR371 | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372 | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470 | SPECIAL TOPIC IN RUSSIAN       |
| LR475 | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476 | MILITARY SPKG/RDG - RUSSIAN    |
| LR483 | RUSSIAN CIV I                  |
| LR484 | RUSSIAN CIV II                 |
| LR485 | SURVEY OF RUSSIAN LITERATURE I |
| LR486 | SURVEY OF RUSSIAN LIT. II      |
| LR492 | RUSSIAN LIFE IN FICTION        |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LZ203 | PERSIAN I (STANDARD)           |
| LZ204 | PERSIAN II (STANDARD)          |
| LZ371 | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372 | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470 | SPECIAL TOPIC IN PERSIAN       |
| LZ475 | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476 | MILITARY SPKG/RDG - PERSIAN    |
| LZ483 | PERSIAN CIVILIZATION I         |
| LZ484 | PERSIAN CIVILIZATION II        |
| LZ485 | PERSIAN LITERATURE I           |
| LZ486 | PERSIAN LITERATURE II          |

**AND****Complementary Support Courses**

Choose 3 of 24

Choose 3 CSCs, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE343 | GEOGRAPHY OF RUSSIA          |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI367 | IMPERIAL AND SOVIET RUSSIA   |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS375 | GOV & POL RUSSIA & NEIGHBORS |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| WS358                                                                                                                                                                        | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                                                                                        | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                  |
| LN490R                                                                                                                                                                       | RUSSIAN LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Curriculum Requirement</b>                                                                                                                                                |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES     |

## 2029 Foreign Language Major: Russian & Persian w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FRZ2H | Foreign Language: Russian & Persian w/ Honors | Foreign Language Major: Russian & Persian w/ Honors | Foreign Language: Russian & Persian w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Russian & Persian w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Russian & Portuguese Curriculum

| Code | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FRP1 | Foreign Language: Russian & Portuguese | Foreign Language Major: Russian & Portuguese | Foreign Language: Russian & Portuguese | 2            | 13           |

### 2029 Foreign Language Major: Russian & Portuguese Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LN440R                          | RUSSIAN IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LR371                           | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                           | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470                           | SPECIAL TOPIC IN RUSSIAN       |
| LR475                           | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                           | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                           | RUSSIAN CIV I                  |
| LR484                           | RUSSIAN CIV II                 |
| LR485                           | SURVEY OF RUSSIAN LITERATURE I |
| LR486                           | SURVEY OF RUSSIAN LIT. II      |
| LR492                           | RUSSIAN LIFE IN FICTION        |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 12                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LP203                           | PORTUGUESE I (STANDARD)        |
| LP204                           | PORTUGUESE II (STANDARD)       |
| LP371                           | INTENSIVE INTERMED. PORTUGUESE |
| LP372                           | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475                           | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476                           | MILITARY SPKG/RDG - PORTUGUESE |
| LP483                           | PORTUGUESE CIVILIZATION I      |
| LP484                           | PORTUGUESE CIVILIZATION II     |
| LP485                           | SURVEY OF PORTUGUESE LIT I     |
| LP492                           | LIT OF PORT-SPKG WORLD         |
| <b>AND</b>                      |                                |

**Complementary Support Courses**

Choose 3 of 24

Choose 3 CSCs, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE343 | GEOGRAPHY OF RUSSIA          |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI367 | IMPERIAL AND SOVIET RUSSIA   |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS375 | GOV & POL RUSSIA & NEIGHBORS |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490R | RUSSIAN LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

### 2029 Foreign Language Major: Russian & Portuguese w/ Honors Curriculum

| Code  | Short Description                            | Description                                            | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FRP1H | Foreign Lang: Russian & Portuguese w/ Honors | Foreign Language Major: Russian & Portuguese w/ Honors | Foreign Lang: Russian & Portuguese w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Russian & Portuguese w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Russian & Spanish Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FRS2 | Foreign Language: Russian & Spanish | Foreign Language Major: Russian & Spanish | Foreign Language: Russian & Spanish | 2            | 13           |

### 2029 Foreign Language Major: Russian & Spanish Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LN440R                 | RUSSIAN IN CULTURAL CONTEXT    |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |
| LN494                  | SEM ABROAD: ADV LANG & CULT IV |
| LN495                  | SEM ABROAD: ADV LANG & CULT V  |
| LR371                  | INTENSIVE INTERMEDIATE RUSSIAN |



|       |                                |
|-------|--------------------------------|
| LR372 | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470 | SPECIAL TOPIC IN RUSSIAN       |
| LR475 | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476 | MILITARY SPKG/RDG - RUSSIAN    |
| LR483 | RUSSIAN CIV I                  |
| LR484 | RUSSIAN CIV II                 |
| LR485 | SURVEY OF RUSSIAN LITERATURE I |
| LR486 | SURVEY OF RUSSIAN LIT. II      |
| LR492 | RUSSIAN LIFE IN FICTION        |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LS203 | SPANISH I (STANDARD)           |
| LS204 | SPANISH II (STANDARD)          |
| LS371 | INTENSIVE INTERMEDIATE SPANISH |
| LS372 | SPANISH FOR ORAL & WRIT COMM   |
| LS475 | SPANISH RDG/WRTG THRU MEDIA    |
| LS476 | MILITARY SPKG/RDG - SPANISH    |
| LS483 | SPANISH CIV AND CULTURE        |
| LS484 | SPANISH AMERICAN CIV AND CULT  |
| LS485 | SPANISH-AMERICAN LITERATURE    |
| LS486 | THE LITERATURE OF SPAIN        |
| LS492 | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Complementary Support Courses**

Choose 3 of 24

Choose 3 CSCs, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE343 | GEOGRAPHY OF RUSSIA          |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI367 | IMPERIAL AND SOVIET RUSSIA   |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS375 | GOV & POL RUSSIA & NEIGHBORS |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490R | RUSSIAN LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Russian & Spanish w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FRS2H | Foreign Language: Russian & Spanish w/ Honors | Foreign Language Major: Russian & Spanish w/ Honors | Foreign Language: Russian & Spanish w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Russian & Spanish w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Russian w/ Honors Curriculum

| Code  | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FLR2H | Foreign Language: Russian w/ Honors | Foreign Language Major: Russian w/ Honors | Foreign Language: Russian w/ Honors | 1            | 1            |

## 2029 Foreign Language Major: Russian w/ Honors Tracks

| Subject Area               | Description                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------|
| <b>Additional Elective</b> | Choose 1 of 16<br>Take an additional advanced-level elective, not already taken, from this list. |
| LN440R                     | RUSSIAN IN CULTURAL CONTEXT                                                                      |
| LN451                      | ADV LANG & CULTURE IN CONTEXT                                                                    |
| LN487                      | ADV IND STUDY-FOREIGN LANGS                                                                      |
| LN491                      | SEM ABROAD: ADV LANG & CULT I                                                                    |
| LN492                      | SEM ABROAD: ADV LANG & CULT II                                                                   |
| LN493                      | SEM ABROAD: ADV LANG&CULT III                                                                    |
| LN494                      | SEM ABROAD: ADV LANG & CULT IV                                                                   |
| LN495                      | SEM ABROAD: ADV LANG & CULT V                                                                    |
| LR470                      | SPECIAL TOPIC IN RUSSIAN                                                                         |
| LR475                      | RUSSIAN RDG/WRTG THRU MEDIA                                                                      |
| LR476                      | MILITARY SPKG/RDG - RUSSIAN                                                                      |
| LR483                      | RUSSIAN CIV I                                                                                    |
| LR484                      | RUSSIAN CIV II                                                                                   |
| LR485                      | SURVEY OF RUSSIAN LITERATURE I                                                                   |
| LR486                      | SURVEY OF RUSSIAN LIT. II                                                                        |
| LR492                      | RUSSIAN LIFE IN FICTION                                                                          |
| <b>AND</b>                 |                                                                                                  |
| <b>Honors Thesis</b>       | Choose 1 of 1<br>Write an honors thesis under the direction of a senior faculty member.          |
| LN488                      | ADV IND STUDY-FOREIGN LANGS                                                                      |
| <b>AND</b>                 |                                                                                                  |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and a minimum APSC of 3.5 in the major.

## 2029 Foreign Language Major: Spanish Curriculum

| Code | Short Description         | Description                     | Transcript Description    | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------------|---------------------------------|---------------------------|--------------|--------------|
| FLS2 | Foreign Language: Spanish | Foreign Language Major: Spanish | Foreign Language: Spanish | 2            | 11           |

## 2029 Foreign Language Major: Spanish Tracks

| Subject Area                                 | Description                    |
|----------------------------------------------|--------------------------------|
| <b>Required Course</b>                       | Choose 1 of 1                  |
| LN380                                        | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                   |                                |
| <b>Electives</b>                             | Choose 8 of 19                 |
| LN440S                                       | SPANISH IN CULTURAL CONTEXT    |
| LN451                                        | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                        | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                        | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                        | SEM ABROAD: ADV LANG & CULT II |
| LN493                                        | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                        | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                        | SEM ABROAD: ADV LANG & CULT V  |
| LS371                                        | INTENSIVE INTERMEDIATE SPANISH |
| LS372                                        | SPANISH FOR ORAL & WRIT COMM   |
| LS470                                        | SPECIAL TOPIC IN SPANISH       |
| LS475                                        | SPANISH RDG/WRTG THRU MEDIA    |
| LS476                                        | MILITARY SPKG/RDG - SPANISH    |
| LS483                                        | SPANISH CIV AND CULTURE        |
| LS484                                        | SPANISH AMERICAN CIV AND CULT  |
| LS485                                        | SPANISH-AMERICAN LITERATURE    |
| LS486                                        | THE LITERATURE OF SPAIN        |
| LS492                                        | 20TH/21ST CENTURY HISPANIC LIT |
| <b>AND</b>                                   |                                |
| <b>Complementary Support Courses</b>         | Choose 3 of 28                 |
| Choose one each from a different discipline. |                                |
| EN300                                        | LITERARY METHODOLOGIES         |
| EN311                                        | ANCIENT TO EARLY MODERN LIT    |
| EN351                                        | WORLD LITERATURE               |
| EN353                                        | WAR LITERATURE                 |
| EN361                                        | POETRY                         |
| EN363                                        | THE NOVEL                      |
| GE340                                        | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                        | GEOGRAPHY OF EUROPE            |
| GE347                                        | GEOGRAPHY OF LATIN AMERICA     |
| GE491                                        | MILITARY GEOGRAPHY             |
| HI344                                        | MODERN DIPLOMACY               |
| HI348                                        | MODERN LATIN AMERICA           |
| HI364                                        | MODERN WESTERN EUROPE          |
| HI381                                        | HISTORY OF IRREGULAR WARFARE   |
| HI391                                        | WORLD RELIGIONS                |
| LW410                                        | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                        | ENVIRONMENTAL LAW              |
| LW481                                        | INTERNATIONAL LAW              |
| SS366                                        | COMPARATIVE POLITICS           |
| SS377                                        | POLITICS & GOV OF EUROPE       |
| SS384                                        | POLITICS & GOVT-LATIN AMER     |
| SS465                                        | TERRORISM: NEW CHALLENGES      |
| SS475                                        | COMP POLITICAL INSTITUTIONS    |
| SS484                                        | INTERNATIONAL ECONOMICS        |
| SS486                                        | STATE BUILDING                 |

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| SS487                                                                                                                                                                        | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                                                                                        | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                                                                                        | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                  |
| LN490S                                                                                                                                                                       | SPANISH LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Curriculum Requirements</b>                                                                                                                                               |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES     |

## 2029 Foreign Language Major: Spanish & Arabic Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FSB1 | Foreign Language: Spanish & Arabic | Foreign Language Major: Spanish & Arabic | Foreign Language: Spanish & Arabic | 2            | 13           |

## 2029 Foreign Language Major: Spanish & Arabic Tracks

| Subject Area           | Description                |
|------------------------|----------------------------|
| <b>Required Course</b> | Choose 1 of 1              |
| LN380                  | NATURE OF MODERN LANGUAGES |
| <b>AND</b>             |                            |
| <hr/>                  |                            |
| <b>Electives</b>       | Choose 6 of 19             |

|        |                                |
|--------|--------------------------------|
| LN440S | SPANISH IN CULTURAL CONTEXT    |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN487  | ADV IND STUDY-FOREIGN LANGS    |
| LN488  | ADV IND STUDY-FOREIGN LANGS    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |
| LS371  | INTENSIVE INTERMEDIATE SPANISH |
| LS372  | SPANISH FOR ORAL & WRIT COMM   |
| LS470  | SPECIAL TOPIC IN SPANISH       |
| LS475  | SPANISH RDG/WRTG THRU MEDIA    |
| LS476  | MILITARY SPKG/RDG - SPANISH    |
| LS483  | SPANISH CIV AND CULTURE        |
| LS484  | SPANISH AMERICAN CIV AND CULT  |
| LS485  | SPANISH-AMERICAN LITERATURE    |
| LS486  | THE LITERATURE OF SPAIN        |
| LS492  | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Secondary Language Track**

Choose 4 of 14

|       |                                |
|-------|--------------------------------|
| LA203 | ARABIC I (STANDARD)            |
| LA204 | ARABIC II (STANDARD)           |
| LA371 | INTENSIVE INTERMEDIATE ARABIC  |
| LA372 | ARABIC FOR ORAL & WRITTEN COMM |
| LA472 | COLLOQUIAL ARABIC              |
| LA475 | ARABIC RDG/WRTG THRU MEDIA     |
| LA476 | MILITARY SPKG/RDG - ARABIC     |
| LA483 | ARAB CIVILIZATION I            |
| LA484 | ARAB CIVILIZATION II           |
| LA485 | ARABIC LITERATURE I            |
| LA486 | ARABIC LITERATURE II           |
| LA492 | ARABIC LITERATURE III          |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 44

Choose 3 CSCs, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN311 | ANCIENT TO EARLY MODERN LIT  |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE344 | GEOGRAPHY OF AFRICA          |
| GE346 | GEOGRAPHY OF EUROPE          |
| GE347 | GEOGRAPHY OF LATIN AMERICA   |
| GE491 | MILITARY GEOGRAPHY           |
| HI339 | THE MODERN MIDDLE EAST       |
| HI344 | MODERN DIPLOMACY             |
| HI348 | MODERN LATIN AMERICA         |
| HI364 | MODERN WESTERN EUROPE        |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI383 | MIDDLE EASTERN WARFARE       |

|       |                                |
|-------|--------------------------------|
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS384 | POLITICS & GOVT-LATIN AMER     |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS477 | ECONOMICS OF NATIONAL SECURITY |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

**AND****Integrative Experience for the Major** Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490S | SPANISH LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Spanish & Arabic w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FSB1H | Foreign Language: Spanish & Arabic w/ Honors | Foreign Language Major: Spanish & Arabic w/ Honors | Foreign Language: Spanish & Arabic w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Spanish & Arabic w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Spanish & Chinese Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FSC1 | Foreign Language: Spanish & Chinese | Foreign Language Major: Spanish & Chinese | Foreign Language: Spanish & Chinese | 2            | 13           |

## 2029 Foreign Language Major: Spanish & Chinese Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LN440S                 | SPANISH IN CULTURAL CONTEXT    |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |



|       |                                |
|-------|--------------------------------|
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LS371 | INTENSIVE INTERMEDIATE SPANISH |
| LS372 | SPANISH FOR ORAL & WRIT COMM   |
| LS470 | SPECIAL TOPIC IN SPANISH       |
| LS475 | SPANISH RDG/WRTG THRU MEDIA    |
| LS476 | MILITARY SPKG/RDG - SPANISH    |
| LS483 | SPANISH CIV AND CULTURE        |
| LS484 | SPANISH AMERICAN CIV AND CULT  |
| LS485 | SPANISH-AMERICAN LITERATURE    |
| LS486 | THE LITERATURE OF SPAIN        |
| LS492 | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LC203 | CHINESE I (STANDARD)           |
| LC204 | CHINESE II (STANDARD)          |
| LC371 | INTENSIVE INTERMEDIATE CHINESE |
| LC372 | CHINESE FOR ORAL & WRIT COMM   |
| LC475 | CHINESE RDG/WRTG THRU MEDIA    |
| LC476 | MILITARY SPKG/RDG - CHINESE    |
| LC483 | CHINESE CIVILIZATION I         |
| LC484 | CHINESE CIVILIZATION II        |
| LC485 | CHINESE LITERATURE I           |
| LC486 | CHINESE LITERATURE II          |
| LC492 | CHINESE LITERATURE III         |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 28

Choose 3 CSCs, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN311 | ANCIENT TO EARLY MODERN LIT  |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE346 | GEOGRAPHY OF EUROPE          |
| GE347 | GEOGRAPHY OF LATIN AMERICA   |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI348 | MODERN LATIN AMERICA         |
| HI364 | MODERN WESTERN EUROPE        |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS377 | POLITICS & GOV OF EUROPE     |
| SS384 | POLITICS & GOVT-LATIN AMER   |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| SS487                                                                                                                                                                        | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                                                                                        | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                                                                                        | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                  |
| LN490S                                                                                                                                                                       | SPANISH LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take IT305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Curriculum Requirement</b>                                                                                                                                                |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and IT305.                                 |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES     |

## 2029 Foreign Language Major: Spanish & Chinese w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FSC1H | Foreign Language: Spanish & Chinese w/ Honors | Foreign Language Major: Spanish & Chinese w/ Honors | Foreign Language: Spanish & Chinese w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Spanish & Chinese w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Spanish & French Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FSF1 | Foreign Language: Spanish & French | Foreign Language Major: Spanish & French | Foreign Language: Spanish & French | 2            | 13           |

## 2029 Foreign Language Major: Spanish & French Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LN440S                          | SPANISH IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LS371                           | INTENSIVE INTERMEDIATE SPANISH |
| LS372                           | SPANISH FOR ORAL & WRIT COMM   |
| LS470                           | SPECIAL TOPIC IN SPANISH       |
| LS475                           | SPANISH RDG/WRTG THRU MEDIA    |
| LS476                           | MILITARY SPKG/RDG - SPANISH    |
| LS483                           | SPANISH CIV AND CULTURE        |
| LS484                           | SPANISH AMERICAN CIV AND CULT  |
| LS485                           | SPANISH-AMERICAN LITERATURE    |
| LS486                           | THE LITERATURE OF SPAIN        |
| LS492                           | 20TH/21ST CENTURY HISPANIC LIT |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LF203                           | FRENCH I (STANDARD)            |
| LF204                           | FRENCH II (STANDARD)           |
| LF371                           | INTENSIVE INTERMEDIATE FRENCH  |
| LF372                           | FRENCH FOR ORAL & WRITTEN COMM |
| LF475                           | FRENCH RDG/WRTG THRU MEDIA     |
| LF476                           | MILITARY SPKG/RDG - FRENCH     |
| LF483                           | FRENCH CIVILIZATION I          |
| LF484                           | FRENCH CIVILIZATION II         |
| LF485                           | SURVEY OF FRENCH LIT I         |
| LF486                           | SURVEY OF FRENCH LIT II        |
| LF492                           | MASTERWORKS OF FRENCH LIT      |

|                                                                                                                                                  |                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| LN487                                                                                                                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                                                            | ADV IND STUDY-FOREIGN LANGS    |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Complementary Support Courses</b>                                                                                                             | Choose 3 of 28                 |
| Choose 3 CSCs, each from a different discipline.                                                                                                 |                                |
| EN300                                                                                                                                            | LITERARY METHODOLOGIES         |
| EN311                                                                                                                                            | ANCIENT TO EARLY MODERN LIT    |
| EN351                                                                                                                                            | WORLD LITERATURE               |
| EN353                                                                                                                                            | WAR LITERATURE                 |
| EN361                                                                                                                                            | POETRY                         |
| EN363                                                                                                                                            | THE NOVEL                      |
| GE340                                                                                                                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                                                                                                                            | GEOGRAPHY OF EUROPE            |
| GE347                                                                                                                                            | GEOGRAPHY OF LATIN AMERICA     |
| GE491                                                                                                                                            | MILITARY GEOGRAPHY             |
| HI344                                                                                                                                            | MODERN DIPLOMACY               |
| HI348                                                                                                                                            | MODERN LATIN AMERICA           |
| HI364                                                                                                                                            | MODERN WESTERN EUROPE          |
| HI381                                                                                                                                            | HISTORY OF IRREGULAR WARFARE   |
| HI391                                                                                                                                            | WORLD RELIGIONS                |
| LW410                                                                                                                                            | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                                                                                                                            | ENVIRONMENTAL LAW              |
| LW481                                                                                                                                            | INTERNATIONAL LAW              |
| SS366                                                                                                                                            | COMPARATIVE POLITICS           |
| SS377                                                                                                                                            | POLITICS & GOV OF EUROPE       |
| SS384                                                                                                                                            | POLITICS & GOVT-LATIN AMER     |
| SS465                                                                                                                                            | TERRORISM: NEW CHALLENGES      |
| SS475                                                                                                                                            | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                                                            | INTERNATIONAL ECONOMICS        |
| SS486                                                                                                                                            | STATE BUILDING                 |
| SS487                                                                                                                                            | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                                                            | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                                                            | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Integrative Experience for the Major</b>                                                                                                      | Choose 1 of 1                  |
| LN490S                                                                                                                                           | SPANISH LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Science Depth</b>                                                                                                                             |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                          |                                |
| CH102                                                                                                                                            | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                            | BIOLOGY                        |
| PH202                                                                                                                                            | PHYSICS II                     |
| PH252                                                                                                                                            | ADVANCED PHYSICS II            |
| PH275                                                                                                                                            | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                  |                                |
| CY305                                                                                                                                            | CYBER FOUNDATIONS              |
| CY355                                                                                                                                            | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                       |                                |
| <hr/>                                                                                                                                            |                                |
| <b>Curriculum Requirement</b>                                                                                                                    |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                         |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                      |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355. |                                |

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS             |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major may choose any approved three course engineering sequence.                                                                                              |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| LN380                                                                                                                                                                        | NATURE OF MODERN LANGUAGES    |

### 2029 Foreign Language Major: Spanish & French w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FSF1H | Foreign Language: Spanish & French w/ Honors | Foreign Language Major: Spanish & French w/ Honors | Foreign Language: Spanish & French w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Spanish & French w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Spanish & German Curriculum

| Code | Short Description                  | Description                              | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------------|------------------------------------------|------------------------------------|--------------|--------------|
| FSG1 | Foreign Language: Spanish & German | Foreign Language Major: Spanish & German | Foreign Language: Spanish & German | 2            | 13           |

### 2029 Foreign Language Major: Spanish & German Tracks

| Subject Area           | Description                   |
|------------------------|-------------------------------|
| <b>Required Course</b> | Choose 1 of 1                 |
| LN380                  | NATURE OF MODERN LANGUAGES    |
| <b>AND</b>             |                               |
| <b>Electives</b>       | Choose 6 of 19                |
| LN440S                 | SPANISH IN CULTURAL CONTEXT   |
| LN451                  | ADV LANG & CULTURE IN CONTEXT |

|       |                                |
|-------|--------------------------------|
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LN491 | SEM ABROAD: ADV LANG & CULT I  |
| LN492 | SEM ABROAD: ADV LANG & CULT II |
| LN493 | SEM ABROAD: ADV LANG&CULT III  |
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LS371 | INTENSIVE INTERMEDIATE SPANISH |
| LS372 | SPANISH FOR ORAL & WRIT COMM   |
| LS470 | SPECIAL TOPIC IN SPANISH       |
| LS475 | SPANISH RDG/WRTG THRU MEDIA    |
| LS476 | MILITARY SPKG/RDG - SPANISH    |
| LS483 | SPANISH CIV AND CULTURE        |
| LS484 | SPANISH AMERICAN CIV AND CULT  |
| LS485 | SPANISH-AMERICAN LITERATURE    |
| LS486 | THE LITERATURE OF SPAIN        |
| LS492 | 20TH/21ST CENTURY HISPANIC LIT |

**AND****Secondary Language Track**

Choose 4 of 13

|       |                                |
|-------|--------------------------------|
| LG203 | GERMAN I (STANDARD)            |
| LG204 | GERMAN II (STANDARD)           |
| LG371 | INTENSIVE INTERMEDIATE GERMAN  |
| LG372 | GERMAN FOR ORAL & WRITTEN COMM |
| LG475 | GERMAN RDG/WRTG THRU MEDIA     |
| LG476 | MILITARY SPKG/RDG - GERMAN     |
| LG483 | GERMAN CIVILIZATION I          |
| LG484 | GERMAN CIVILIZATION II         |
| LG485 | SURVEY OF GERMAN LIT I         |
| LG486 | SURVEY OF GERMAN LIT II        |
| LG492 | 20TH & 21ST CENTURY GERMANY    |
| LN487 | ADV IND STUDY-FOREIGN LANGS    |
| LN488 | ADV IND STUDY-FOREIGN LANGS    |

**AND****Complementary Support Courses**

Choose 3 of 32

Choose 3 CSCs, each from a different discipline.

|       |                                |
|-------|--------------------------------|
| EN300 | LITERARY METHODOLOGIES         |
| EN311 | ANCIENT TO EARLY MODERN LIT    |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN361 | POETRY                         |
| EN362 | FILM AND FILM THEORY           |
| EN363 | THE NOVEL                      |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346 | GEOGRAPHY OF EUROPE            |
| GE347 | GEOGRAPHY OF LATIN AMERICA     |
| GE491 | MILITARY GEOGRAPHY             |
| HI344 | MODERN DIPLOMACY               |
| HI348 | MODERN LATIN AMERICA           |
| HI364 | MODERN WESTERN EUROPE          |
| HI368 | MOD CENTRAL & E. EUR,1896-1989 |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |

|       |                              |
|-------|------------------------------|
| SS366 | COMPARATIVE POLITICS         |
| SS377 | POLITICS & GOV OF EUROPE     |
| SS384 | POLITICS & GOVT-LATIN AMER   |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS385 | WAR AND ITS THEORISTS        |
| WS455 | COMPARATIVE DEFENSE POLICY   |
| XH405 | THE HOLOCAUST AND ITS LEGACY |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490S | SPANISH LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

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## 2029 Foreign Language Major: Spanish & German w/ Honors Curriculum

| Code  | Short Description                            | Description                                        | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|----------------------------------------------------|----------------------------------------------|--------------|--------------|
| FSG1H | Foreign Language: Spanish & German w/ Honors | Foreign Language Major: Spanish & German w/ Honors | Foreign Language: Spanish & German w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Spanish & German w/ Honors Tracks

| Subject Area                                                           | Description                 |
|------------------------------------------------------------------------|-----------------------------|
| <b>Honors Thesis Course</b>                                            | Choose 1 of 1               |
| Write an honors thesis under the direction of a senior faculty member. |                             |
| LN488                                                                  | ADV IND STUDY-FOREIGN LANGS |
| <b>AND</b>                                                             |                             |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 Foreign Language Major: Spanish & Persian Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FSZ2 | Foreign Language: Spanish & Persian | Foreign Language Major: Spanish & Persian | Foreign Language: Spanish & Persian | 2            | 13           |

### 2029 Foreign Language Major: Spanish & Persian Tracks

| Subject Area                    | Description                    |
|---------------------------------|--------------------------------|
| <b>Required Course</b>          | Choose 1 of 1                  |
| LN380                           | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                      |                                |
| <b>Electives</b>                | Choose 6 of 19                 |
| LN440S                          | SPANISH IN CULTURAL CONTEXT    |
| LN451                           | ADV LANG & CULTURE IN CONTEXT  |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |
| LN488                           | ADV IND STUDY-FOREIGN LANGS    |
| LN491                           | SEM ABROAD: ADV LANG & CULT I  |
| LN492                           | SEM ABROAD: ADV LANG & CULT II |
| LN493                           | SEM ABROAD: ADV LANG&CULT III  |
| LN494                           | SEM ABROAD: ADV LANG & CULT IV |
| LN495                           | SEM ABROAD: ADV LANG & CULT V  |
| LS371                           | INTENSIVE INTERMEDIATE SPANISH |
| LS372                           | SPANISH FOR ORAL & WRIT COMM   |
| LS470                           | SPECIAL TOPIC IN SPANISH       |
| LS475                           | SPANISH RDG/WRTG THRU MEDIA    |
| LS476                           | MILITARY SPKG/RDG - SPANISH    |
| LS483                           | SPANISH CIV AND CULTURE        |
| LS484                           | SPANISH AMERICAN CIV AND CULT  |
| LS485                           | SPANISH-AMERICAN LITERATURE    |
| LS486                           | THE LITERATURE OF SPAIN        |
| LS492                           | 20TH/21ST CENTURY HISPANIC LIT |
| <b>AND</b>                      |                                |
| <b>Secondary Language Track</b> | Choose 4 of 13                 |
| LN487                           | ADV IND STUDY-FOREIGN LANGS    |



|       |                                |
|-------|--------------------------------|
| LN488 | ADV IND STUDY-FOREIGN LANGS    |
| LZ203 | PERSIAN I (STANDARD)           |
| LZ204 | PERSIAN II (STANDARD)          |
| LZ371 | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372 | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470 | SPECIAL TOPIC IN PERSIAN       |
| LZ475 | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476 | MILITARY SPKG/RDG - PERSIAN    |
| LZ483 | PERSIAN CIVILIZATION I         |
| LZ484 | PERSIAN CIVILIZATION II        |
| LZ485 | PERSIAN LITERATURE I           |
| LZ486 | PERSIAN LITERATURE II          |

**AND****Complementary Support Courses**

Choose 3 of 28

Choose 3 CSCs, each from a different discipline.

|       |                              |
|-------|------------------------------|
| EN300 | LITERARY METHODOLOGIES       |
| EN311 | ANCIENT TO EARLY MODERN LIT  |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN361 | POETRY                       |
| EN363 | THE NOVEL                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE346 | GEOGRAPHY OF EUROPE          |
| GE347 | GEOGRAPHY OF LATIN AMERICA   |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI348 | MODERN LATIN AMERICA         |
| HI364 | MODERN WESTERN EUROPE        |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS377 | POLITICS & GOV OF EUROPE     |
| SS384 | POLITICS & GOVT-LATIN AMER   |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND****Integrative Experience for the Major**

Choose 1 of 1

LN490S

SPANISH LANGUAGE &amp; CULTURE CAP

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**AND**

### Curriculum Requirement

This section describes how cadets in this major satisfy various curriculum requirements.

### IT/CYBER Requirement

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

### Core Engineering Sequence

Cadets in this major may choose any approved three course engineering sequence.

### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

LN380

NATURE OF MODERN LANGUAGES

## 2029 Foreign Language Major: Spanish & Persian w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FSZ2H | Foreign Language: Spanish & Persian w/ Honors | Foreign Language Major: Spanish & Persian w/ Honors | Foreign Language: Spanish & Persian w/ Honors | 1            | 0            |

## 2029 Foreign Language Major: Spanish & Persian w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |

**AND**

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Spanish & Portuguese Curriculum

| Code | Short Description                      | Description                                  | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------------------------|----------------------------------------------|----------------------------------------|--------------|--------------|
| FSP1 | Foreign Language: Spanish & Portuguese | Foreign Language Major: Spanish & Portuguese | Foreign Language: Spanish & Portuguese | 2            | 13           |

## 2029 Foreign Language Major: Spanish & Portuguese Tracks

| Subject Area                                     | Description                    |
|--------------------------------------------------|--------------------------------|
| <b>Required Course</b>                           | Choose 1 of 1                  |
| LN380                                            | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>                                       |                                |
| <b>Electives</b>                                 | Choose 6 of 19                 |
| LN440S                                           | SPANISH IN CULTURAL CONTEXT    |
| LN451                                            | ADV LANG & CULTURE IN CONTEXT  |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN491                                            | SEM ABROAD: ADV LANG & CULT I  |
| LN492                                            | SEM ABROAD: ADV LANG & CULT II |
| LN493                                            | SEM ABROAD: ADV LANG&CULT III  |
| LN494                                            | SEM ABROAD: ADV LANG & CULT IV |
| LN495                                            | SEM ABROAD: ADV LANG & CULT V  |
| LS371                                            | INTENSIVE INTERMEDIATE SPANISH |
| LS372                                            | SPANISH FOR ORAL & WRIT COMM   |
| LS470                                            | SPECIAL TOPIC IN SPANISH       |
| LS475                                            | SPANISH RDG/WRTG THRU MEDIA    |
| LS476                                            | MILITARY SPKG/RDG - SPANISH    |
| LS483                                            | SPANISH CIV AND CULTURE        |
| LS484                                            | SPANISH AMERICAN CIV AND CULT  |
| LS485                                            | SPANISH-AMERICAN LITERATURE    |
| LS486                                            | THE LITERATURE OF SPAIN        |
| LS492                                            | 20TH/21ST CENTURY HISPANIC LIT |
| <b>AND</b>                                       |                                |
| <b>Secondary Language Track</b>                  | Choose 4 of 12                 |
| LN487                                            | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                            | ADV IND STUDY-FOREIGN LANGS    |
| LP203                                            | PORTUGUESE I (STANDARD)        |
| LP204                                            | PORTUGUESE II (STANDARD)       |
| LP371                                            | INTENSIVE INTERMED. PORTUGUESE |
| LP372                                            | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475                                            | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476                                            | MILITARY SPKG/RDG - PORTUGUESE |
| LP483                                            | PORTUGUESE CIVILIZATION I      |
| LP484                                            | PORTUGUESE CIVILIZATION II     |
| LP485                                            | SURVEY OF PORTUGUESE LIT I     |
| LP492                                            | LIT OF PORT-SPKG WORLD         |
| <b>AND</b>                                       |                                |
| <b>Complementary Support Courses</b>             | Choose 3 of 28                 |
| Choose 3 CSCs, each from a different discipline. |                                |
| EN300                                            | LITERARY METHODOLOGIES         |
| EN311                                            | ANCIENT TO EARLY MODERN LIT    |
| EN351                                            | WORLD LITERATURE               |
| EN353                                            | WAR LITERATURE                 |
| EN361                                            | POETRY                         |
| EN363                                            | THE NOVEL                      |
| GE340                                            | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                            | GEOGRAPHY OF EUROPE            |
| GE347                                            | GEOGRAPHY OF LATIN AMERICA     |

|       |                              |
|-------|------------------------------|
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI348 | MODERN LATIN AMERICA         |
| HI364 | MODERN WESTERN EUROPE        |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS377 | POLITICS & GOV OF EUROPE     |
| SS384 | POLITICS & GOVT-LATIN AMER   |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**AND**


---

**Integrative Experience for the Major** Choose 1 of 1

|        |                                |
|--------|--------------------------------|
| LN490S | SPANISH LANGUAGE & CULTURE CAP |
|--------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Spanish & Portuguese w/ Honors Curriculum

| Code  | Short Description                            | Description                                            | Transcript Description                       | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------|--------------|--------------|
| FSP1H | Foreign Lang: Spanish & Portuguese w/ Honors | Foreign Language Major: Spanish & Portuguese w/ Honors | Foreign Lang: Spanish & Portuguese w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Spanish & Portuguese w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Spanish & Russian Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FSR1 | Foreign Language: Spanish & Russian | Foreign Language Major: Spanish & Russian | Foreign Language: Spanish & Russian | 2            | 13           |

### 2029 Foreign Language Major: Spanish & Russian Tracks

| Subject Area           | Description                    |
|------------------------|--------------------------------|
| <b>Required Course</b> | Choose 1 of 1                  |
| LN380                  | NATURE OF MODERN LANGUAGES     |
| <b>AND</b>             |                                |
| <b>Electives</b>       | Choose 6 of 19                 |
| LN440S                 | SPANISH IN CULTURAL CONTEXT    |
| LN451                  | ADV LANG & CULTURE IN CONTEXT  |
| LN487                  | ADV IND STUDY-FOREIGN LANGS    |
| LN488                  | ADV IND STUDY-FOREIGN LANGS    |
| LN491                  | SEM ABROAD: ADV LANG & CULT I  |
| LN492                  | SEM ABROAD: ADV LANG & CULT II |
| LN493                  | SEM ABROAD: ADV LANG&CULT III  |
| LN494                  | SEM ABROAD: ADV LANG & CULT IV |
| LN495                  | SEM ABROAD: ADV LANG & CULT V  |
| LS371                  | INTENSIVE INTERMEDIATE SPANISH |
| LS372                  | SPANISH FOR ORAL & WRIT COMM   |
| LS470                  | SPECIAL TOPIC IN SPANISH       |
| LS475                  | SPANISH RDG/WRTG THRU MEDIA    |
| LS476                  | MILITARY SPKG/RDG - SPANISH    |
| LS483                  | SPANISH CIV AND CULTURE        |
| LS484                  | SPANISH AMERICAN CIV AND CULT  |
| LS485                  | SPANISH-AMERICAN LITERATURE    |

|                                                                                                         |                                |
|---------------------------------------------------------------------------------------------------------|--------------------------------|
| LS486                                                                                                   | THE LITERATURE OF SPAIN        |
| LS492                                                                                                   | 20TH/21ST CENTURY HISPANIC LIT |
| <b>AND</b>                                                                                              |                                |
| <b>Secondary Language Track</b>                                                                         | Choose 4 of 13                 |
| LN487                                                                                                   | ADV IND STUDY-FOREIGN LANGS    |
| LN488                                                                                                   | ADV IND STUDY-FOREIGN LANGS    |
| LR203                                                                                                   | RUSSIAN I (STANDARD)           |
| LR204                                                                                                   | RUSSIAN II (STANDARD)          |
| LR371                                                                                                   | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                                                                                                   | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475                                                                                                   | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476                                                                                                   | MILITARY SPKG/RDG - RUSSIAN    |
| LR483                                                                                                   | RUSSIAN CIV I                  |
| LR484                                                                                                   | RUSSIAN CIV II                 |
| LR485                                                                                                   | SURVEY OF RUSSIAN LITERATURE I |
| LR486                                                                                                   | SURVEY OF RUSSIAN LIT. II      |
| LR492                                                                                                   | RUSSIAN LIFE IN FICTION        |
| <b>AND</b>                                                                                              |                                |
| <b>Complementary Support Courses</b>                                                                    | Choose 3 of 28                 |
| Choose 3 CSCs, each from a different discipline.                                                        |                                |
| EN300                                                                                                   | LITERARY METHODOLOGIES         |
| EN311                                                                                                   | ANCIENT TO EARLY MODERN LIT    |
| EN351                                                                                                   | WORLD LITERATURE               |
| EN353                                                                                                   | WAR LITERATURE                 |
| EN361                                                                                                   | POETRY                         |
| EN363                                                                                                   | THE NOVEL                      |
| GE340                                                                                                   | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                                                                                   | GEOGRAPHY OF EUROPE            |
| GE347                                                                                                   | GEOGRAPHY OF LATIN AMERICA     |
| GE491                                                                                                   | MILITARY GEOGRAPHY             |
| HI344                                                                                                   | MODERN DIPLOMACY               |
| HI348                                                                                                   | MODERN LATIN AMERICA           |
| HI364                                                                                                   | MODERN WESTERN EUROPE          |
| HI381                                                                                                   | HISTORY OF IRREGULAR WARFARE   |
| HI391                                                                                                   | WORLD RELIGIONS                |
| LW410                                                                                                   | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                                                                                   | ENVIRONMENTAL LAW              |
| LW481                                                                                                   | INTERNATIONAL LAW              |
| SS366                                                                                                   | COMPARATIVE POLITICS           |
| SS377                                                                                                   | POLITICS & GOV OF EUROPE       |
| SS384                                                                                                   | POLITICS & GOVT-LATIN AMER     |
| SS465                                                                                                   | TERRORISM: NEW CHALLENGES      |
| SS475                                                                                                   | COMP POLITICAL INSTITUTIONS    |
| SS484                                                                                                   | INTERNATIONAL ECONOMICS        |
| SS486                                                                                                   | STATE BUILDING                 |
| SS487                                                                                                   | INT'L POLITICAL ECONOMY        |
| WS358                                                                                                   | STRATEGY AND GENERALSHIP       |
| WS455                                                                                                   | COMPARATIVE DEFENSE POLICY     |
| <b>AND</b>                                                                                              |                                |
| <b>Integrative Experience for the Major</b>                                                             | Choose 1 of 1                  |
| LN490S                                                                                                  | SPANISH LANGUAGE & CULTURE CAP |
| <b>AND</b>                                                                                              |                                |
| <b>Science Depth</b>                                                                                    |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement. |                                |

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirement**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any approved three course engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                            |
|-------|----------------------------|
| LN380 | NATURE OF MODERN LANGUAGES |
|-------|----------------------------|

## 2029 Foreign Language Major: Spanish & Russian w/ Honors Curriculum

| Code  | Short Description                             | Description                                         | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------|--------------|--------------|
| FSR1H | Foreign Language: Spanish & Russian w/ Honors | Foreign Language Major: Spanish & Russian w/ Honors | Foreign Language: Spanish & Russian w/ Honors | 1            | 0            |

### 2029 Foreign Language Major: Spanish & Russian w/ Honors Tracks

| Subject Area                | Description                                                            |
|-----------------------------|------------------------------------------------------------------------|
| <b>Honors Thesis Course</b> | Choose 1 of 1                                                          |
|                             | Write an honors thesis under the direction of a senior faculty member. |
| LN488                       | ADV IND STUDY-FOREIGN LANGS                                            |
| <b>AND</b>                  |                                                                        |

Complete the requirements of the major as shown above, AND attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Foreign Language Major: Spanish w/ Honors Curriculum

| Code  | Short Description                   | Description                               | Transcript Description              | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------------|-------------------------------------------|-------------------------------------|--------------|--------------|
| FLS2H | Foreign Language: Spanish w/ Honors | Foreign Language Major: Spanish w/ Honors | Foreign Language: Spanish w/ Honors | 1            | 1            |

### 2029 Foreign Language Major: Spanish w/ Honors Tracks

| Subject Area               | Description                                                                                      |
|----------------------------|--------------------------------------------------------------------------------------------------|
| <b>Additional Elective</b> | Choose 1 of 16<br>Take an additional advanced-level elective, not already taken, from this list. |
| LN440S                     | SPANISH IN CULTURAL CONTEXT                                                                      |
| LN451                      | ADV LANG & CULTURE IN CONTEXT                                                                    |
| LN487                      | ADV IND STUDY-FOREIGN LANGS                                                                      |
| LN491                      | SEM ABROAD: ADV LANG & CULT I                                                                    |
| LN492                      | SEM ABROAD: ADV LANG & CULT II                                                                   |
| LN493                      | SEM ABROAD: ADV LANG&CULT III                                                                    |
| LN494                      | SEM ABROAD: ADV LANG & CULT IV                                                                   |
| LN495                      | SEM ABROAD: ADV LANG & CULT V                                                                    |
| LS470                      | SPECIAL TOPIC IN SPANISH                                                                         |
| LS475                      | SPANISH RDG/WRTG THRU MEDIA                                                                      |
| LS476                      | MILITARY SPKG/RDG - SPANISH                                                                      |
| LS483                      | SPANISH CIV AND CULTURE                                                                          |
| LS484                      | SPANISH AMERICAN CIV AND CULT                                                                    |
| LS485                      | SPANISH-AMERICAN LITERATURE                                                                      |
| LS486                      | THE LITERATURE OF SPAIN                                                                          |
| LS492                      | 20TH/21ST CENTURY HISPANIC LIT                                                                   |
| <b>AND</b>                 |                                                                                                  |
| <b>Honors Thesis</b>       | Choose 1 of 1<br>Write an honors thesis under the direction of a senior faculty member.          |
| LN488                      | ADV IND STUDY-FOREIGN LANGS                                                                      |
| <b>AND</b>                 |                                                                                                  |

Complete the requirements of the major as shown above, AND attain a minimum APSC of 3.0 in the core curriculum and a minimum APSC of 3.5 in the major.

### 2029 Regional Studies Minor - Africa Curriculum

| Code  | Short Description               | Description                     | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------|---------------------------------|---------------------------------|--------------|--------------|
| RSI1N | Regional Studies Minor - Africa | Regional Studies Minor - Africa | Regional Studies Minor - Africa | 0            | 5            |

### 2029 Regional Studies Minor - Africa Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|



The course of instruction for the Regional Studies Minor will consist of five courses that form a coherent, interdisciplinary study of a geographical region. Three of the courses must be in a foreign language at the 300 or 400 level. The other two courses must be regional or comparative course electives. For any Single or Dual Foreign Language majors choosing this minor, exceptions can be made to the language requirement if all possible language courses for the Cadets' chosen region have been completed as part of the major. Cadets who choose the Foreign Area Studies: Africa (FSI1) major are not allowed to choose this minor.

### Language Courses

Take three courses at the 300-level (or higher) in one of the following languages. Any courses taken to fulfill the requirements for this minor are eligible for double-counting practices, including any that are taken to fulfill core, major or minor curricular components. Three courses must be unique to this minor.

|                   |                                |
|-------------------|--------------------------------|
| <b>French</b>     | Choose 3 of 16                 |
| LF371             | INTENSIVE INTERMEDIATE FRENCH  |
| LF372             | FRENCH FOR ORAL & WRITTEN COMM |
| LF475             | FRENCH RDG/WRTG THRU MEDIA     |
| LF476             | MILITARY SPKG/RDG - FRENCH     |
| LF483             | FRENCH CIVILIZATION I          |
| LF484             | FRENCH CIVILIZATION II         |
| LF485             | SURVEY OF FRENCH LIT I         |
| LF486             | SURVEY OF FRENCH LIT II        |
| LF492             | MASTERWORKS OF FRENCH LIT      |
| LN440F            | FRENCH IN CULTURAL CONTEXT     |
| LN451             | ADV LANG & CULTURE IN CONTEXT  |
| LN491             | SEM ABROAD: ADV LANG & CULT I  |
| LN492             | SEM ABROAD: ADV LANG & CULT II |
| LN493             | SEM ABROAD: ADV LANG&CULT III  |
| LN494             | SEM ABROAD: ADV LANG & CULT IV |
| LN495             | SEM ABROAD: ADV LANG & CULT V  |
| <b>OR</b>         |                                |
| <b>Portuguese</b> | Choose 3 of 15                 |
| LN440P            | PORTUGUESE IN CULTURAL CONTEXT |
| LN451             | ADV LANG & CULTURE IN CONTEXT  |
| LN491             | SEM ABROAD: ADV LANG & CULT I  |
| LN492             | SEM ABROAD: ADV LANG & CULT II |
| LN493             | SEM ABROAD: ADV LANG&CULT III  |
| LN494             | SEM ABROAD: ADV LANG & CULT IV |
| LN495             | SEM ABROAD: ADV LANG & CULT V  |
| LP371             | INTENSIVE INTERMED. PORTUGUESE |
| LP372             | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475             | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476             | MILITARY SPKG/RDG - PORTUGUESE |
| LP483             | PORTUGUESE CIVILIZATION I      |
| LP484             | PORTUGUESE CIVILIZATION II     |
| LP485             | SURVEY OF PORTUGUESE LIT I     |
| LP492             | LIT OF PORT-SPKG WORLD         |
| <b>AND</b>        |                                |

Choose one course from the following regional track and one course from the regional and comparative studies course track. Any course taken from the Regional Track cannot double-count with the course chosen for the Regional and Comparative Studies Track. All pre-requisites for any courses on these lists should be taken prior to taking the chosen course from either track.

|                              |                                |
|------------------------------|--------------------------------|
| <b>Africa Regional Track</b> | Choose 1 of 4                  |
| GE344                        | GEOGRAPHY OF AFRICA            |
| HI341                        | THE AGE OF EXPLORATION         |
| HI345                        | MODERN AFRICA                  |
| SS485                        | GOV & POLITICS SUB-SAHARAN AFR |

**AND****Regional and Comparative Studies Track**

Choose 1 of 22

|       |                                |
|-------|--------------------------------|
| EN351 | WORLD LITERATURE               |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE491 | MILITARY GEOGRAPHY             |
| HI341 | THE AGE OF EXPLORATION         |
| HI344 | MODERN DIPLOMACY               |
| HI345 | MODERN AFRICA                  |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| SS366 | COMPARATIVE POLITICS           |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

**2029 Regional Studies Minor - East Asia Curriculum**

| Code  | Short Description                  | Description                        | Transcript Description             | Req Crse Cnt | Opt Crse Cnt |
|-------|------------------------------------|------------------------------------|------------------------------------|--------------|--------------|
| RSA1N | Regional Studies Minor - East Asia | Regional Studies Minor - East Asia | Regional Studies Minor - East Asia | 0            | 5            |

**2029 Regional Studies Minor - East Asia Tracks**

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

The course of instruction for the Regional Studies Minor will consist of five courses that form a coherent, interdisciplinary study of a geographical region. Three of the courses must be in a foreign language at the 300 or 400 level. The other two courses must be regional or comparative electives. For any Single or Dual Foreign Language majors choosing this minor, exceptions can be made to the language requirement if all possible language courses for the Cadets' chosen region have been completed as part of the major. Cadets who choose the Foreign Area Studies: East Asia (FSA1) major are not allowed to choose this minor.

**Language Courses**

Take three courses at the 300-level (or higher) in the following language. Any courses taken to fulfill the requirements for this minor are eligible for double-counting practices, including any that are taken to fulfill core, major or minor curricular components. Three courses must be unique to this minor.

|                |                                |
|----------------|--------------------------------|
| <b>Chinese</b> | Choose 3 of 16                 |
| LC371          | INTENSIVE INTERMEDIATE CHINESE |
| LC372          | CHINESE FOR ORAL & WRIT COMM   |
| LC475          | CHINESE RDG/WRTG THRU MEDIA    |

|        |                                |
|--------|--------------------------------|
| LC476  | MILITARY SPKG/RDG - CHINESE    |
| LC483  | CHINESE CIVILIZATION I         |
| LC484  | CHINESE CIVILIZATION II        |
| LC485  | CHINESE LITERATURE I           |
| LC486  | CHINESE LITERATURE II          |
| LC492  | CHINESE LITERATURE III         |
| LN440C | CHINESE IN CULTURAL CONTEXT    |
| LN451  | ADV LANG & CULTURE IN CONTEXT  |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LN493  | SEM ABROAD: ADV LANG&CULT III  |
| LN494  | SEM ABROAD: ADV LANG & CULT IV |
| LN495  | SEM ABROAD: ADV LANG & CULT V  |

**AND**

Choose one course from the following regional track and one course from the regional and comparative studies course track. Any course taken from the Regional Track cannot double-count with the course chosen for the Regional and Comparative Studies Track. All pre-requisites for any courses on these lists should be taken prior to taking the chosen course from either track.

**East Asia Regional Track**

Choose 1 of 6

|       |                                |
|-------|--------------------------------|
| GE342 | GEOGRAPHY OF ASIA              |
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI347 | ASIAN WARFARE AND POLITICS     |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS372 | POLITICS AND GOV OF CHINA      |
| SS374 | POL & GOV OF KOREAS & JAPAN    |

**AND****Regional And Comparative Studies Track**

Choose 1 of 24

|       |                                |
|-------|--------------------------------|
| EN351 | WORLD LITERATURE               |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342 | GEOGRAPHY OF ASIA              |
| GE491 | MILITARY GEOGRAPHY             |
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI344 | MODERN DIPLOMACY               |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| SS366 | COMPARATIVE POLITICS           |
| SS372 | POLITICS AND GOV OF CHINA      |
| SS374 | POL & GOV OF KOREAS & JAPAN    |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS455 | COMPARATIVE DEFENSE POLICY     |

## 2029 Regional Studies Minor - Eurasia Curriculum

| Code  | Short Description                | Description                      | Transcript Description           | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------|----------------------------------|----------------------------------|--------------|--------------|
| RSU1N | Regional Studies Minor - Eurasia | Regional Studies Minor - Eurasia | Regional Studies Minor - Eurasia | 0            | 5            |

## 2029 Regional Studies Minor - Eurasia Tracks

### Subject Area Description

The course of instruction for the Regional Studies Minor will consist of five courses that form a coherent, interdisciplinary study of a geographical region. Three of the courses must be in a foreign language at the 300 or 400 level. The other two courses must be regional or comparative electives. For any Single or Dual Foreign Language majors choosing this minor, exceptions can be made to the language requirement if all possible language courses for the Cadets' chosen region have been completed as part of the major. Cadets who choose the Foreign Area Studies: Eurasia (FSU1) major are not allowed to choose this minor.

#### Language Courses

Take three courses at the 300-level (or higher) in the following languages. You must choose one of the following two language tracks. Any courses taken to fulfill the requirements for this minor are eligible for double-counting practices, including any that are taken to fulfill core, major or minor curricular components. Three courses must be unique to this minor.

|                |                                |
|----------------|--------------------------------|
| <b>Russian</b> | Choose 3 of 19                 |
| LN440R         | RUSSIAN IN CULTURAL CONTEXT    |
| LN451          | ADV LANG & CULTURE IN CONTEXT  |
| LN487          | ADV IND STUDY-FOREIGN LANGS    |
| LN488          | ADV IND STUDY-FOREIGN LANGS    |
| LN491          | SEM ABROAD: ADV LANG & CULT I  |
| LN492          | SEM ABROAD: ADV LANG & CULT II |
| LN493          | SEM ABROAD: ADV LANG&CULT III  |
| LN494          | SEM ABROAD: ADV LANG & CULT IV |
| LN495          | SEM ABROAD: ADV LANG & CULT V  |
| LR371          | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372          | RUSSIAN FOR ORAL & WRIT COMM   |
| LR470          | SPECIAL TOPIC IN RUSSIAN       |
| LR475          | RUSSIAN RDG/WRTG THRU MEDIA    |
| LR476          | MILITARY SPKG/RDG - RUSSIAN    |
| LR483          | RUSSIAN CIV I                  |
| LR484          | RUSSIAN CIV II                 |
| LR485          | SURVEY OF RUSSIAN LITERATURE I |
| LR486          | SURVEY OF RUSSIAN LIT. II      |
| LR492          | RUSSIAN LIFE IN FICTION        |

#### OR

|                |                               |
|----------------|-------------------------------|
| <b>Persian</b> | Choose 3 of 18                |
| LN440Z         | PERSIAN IN CULTURAL CONTEXT   |
| LN451          | ADV LANG & CULTURE IN CONTEXT |
| LN487          | ADV IND STUDY-FOREIGN LANGS   |
| LN488          | ADV IND STUDY-FOREIGN LANGS   |
| LN491          | SEM ABROAD: ADV LANG & CULT I |

|       |                                |
|-------|--------------------------------|
| LN492 | SEM ABROAD: ADV LANG & CULT II |
| LN493 | SEM ABROAD: ADV LANG&CULT III  |
| LN494 | SEM ABROAD: ADV LANG & CULT IV |
| LN495 | SEM ABROAD: ADV LANG & CULT V  |
| LZ371 | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372 | PERSIAN FOR ORAL & WRIT COMM   |
| LZ470 | SPECIAL TOPIC IN PERSIAN       |
| LZ475 | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476 | MILITARY SPKG/RDG - PERSIAN    |
| LZ483 | PERSIAN CIVILIZATION I         |
| LZ484 | PERSIAN CIVILIZATION II        |
| LZ485 | PERSIAN LITERATURE I           |
| LZ486 | PERSIAN LITERATURE II          |

**AND**

Choose one course from the following regional track and one course from the regional and comparative studies course track. Any course taken from the Regional Track cannot double-count with the course chosen for the Regional and Comparative Studies Track. All pre-requisites for any courses on these lists should be taken prior to taking the chosen course from either track.

**Eurasia Regional Track**

Choose 1 of 3

|       |                              |
|-------|------------------------------|
| GE343 | GEOGRAPHY OF RUSSIA          |
| HI367 | IMPERIAL AND SOVIET RUSSIA   |
| SS375 | GOV & POL RUSSIA & NEIGHBORS |

**AND****Regional and Comparative Studies Track**

Choose 1 of 21

|       |                              |
|-------|------------------------------|
| EN351 | WORLD LITERATURE             |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE343 | GEOGRAPHY OF RUSSIA          |
| GE491 | MILITARY GEOGRAPHY           |
| HI344 | MODERN DIPLOMACY             |
| HI367 | IMPERIAL AND SOVIET RUSSIA   |
| HI381 | HISTORY OF IRREGULAR WARFARE |
| HI391 | WORLD RELIGIONS              |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |
| LW473 | ENVIRONMENTAL LAW            |
| LW481 | INTERNATIONAL LAW            |
| SS366 | COMPARATIVE POLITICS         |
| SS375 | GOV & POL RUSSIA & NEIGHBORS |
| SS465 | TERRORISM: NEW CHALLENGES    |
| SS475 | COMP POLITICAL INSTITUTIONS  |
| SS479 | INTERNATIONAL ORGANIZATION   |
| SS484 | INTERNATIONAL ECONOMICS      |
| SS486 | STATE BUILDING               |
| SS487 | INT'L POLITICAL ECONOMY      |
| WS358 | STRATEGY AND GENERALSHIP     |
| WS455 | COMPARATIVE DEFENSE POLICY   |

**2029 Regional Studies Minor - Europe Curriculum**

| Code  | Short Description               | Description                     | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------|---------------------------------|---------------------------------|--------------|--------------|
| RSE1N | Regional Studies Minor - Europe | Regional Studies Minor - Europe | Regional Studies Minor - Europe | 0            | 5            |

### 2029 Regional Studies Minor - Europe Tracks

| Subject Area | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>The course of instruction for the Regional Studies Minor will consist of five courses that form a coherent, interdisciplinary study of a geographical region. Three of the courses must be in a foreign language at the 300 or 400 level. The other two courses must be regional or comparative electives. For any Single or Dual Foreign Language majors choosing this minor, exceptions can be made to the language requirement if all possible language courses for the Cadets' chosen region have been completed as part of the major. Cadets who choose the Foreign Area Studies: Europe (FSE1) major are not allowed to choose this minor.</p> |

#### Language Courses

Take three courses at the 300-level (or higher) in one of the following languages. Any courses taken to fulfill the requirements for this minor are eligible for double-counting practices, including any that are taken to fulfill core, major or minor curricular components. Three courses must be unique to this minor.

|                   |                                |
|-------------------|--------------------------------|
| <b>French</b>     | Choose 3 of 16                 |
| LF371             | INTENSIVE INTERMEDIATE FRENCH  |
| LF372             | FRENCH FOR ORAL & WRITTEN COMM |
| LF475             | FRENCH RDG/WRTG THRU MEDIA     |
| LF476             | MILITARY SPKG/RDG - FRENCH     |
| LF483             | FRENCH CIVILIZATION I          |
| LF484             | FRENCH CIVILIZATION II         |
| LF485             | SURVEY OF FRENCH LIT I         |
| LF486             | SURVEY OF FRENCH LIT II        |
| LF492             | MASTERWORKS OF FRENCH LIT      |
| LN440F            | FRENCH IN CULTURAL CONTEXT     |
| LN451             | ADV LANG & CULTURE IN CONTEXT  |
| LN491             | SEM ABROAD: ADV LANG & CULT I  |
| LN492             | SEM ABROAD: ADV LANG & CULT II |
| LN493             | SEM ABROAD: ADV LANG&CULT III  |
| LN494             | SEM ABROAD: ADV LANG & CULT IV |
| LN495             | SEM ABROAD: ADV LANG & CULT V  |
| <b>OR</b>         |                                |
| <b>German</b>     | Choose 3 of 16                 |
| LG371             | INTENSIVE INTERMEDIATE GERMAN  |
| LG372             | GERMAN FOR ORAL & WRITTEN COMM |
| LG475             | GERMAN RDG/WRTG THRU MEDIA     |
| LG476             | MILITARY SPKG/RDG - GERMAN     |
| LG483             | GERMAN CIVILIZATION I          |
| LG484             | GERMAN CIVILIZATION II         |
| LG485             | SURVEY OF GERMAN LIT I         |
| LG486             | SURVEY OF GERMAN LIT II        |
| LG492             | 20TH & 21ST CENTURY GERMANY    |
| LN440G            | GERMAN IN CULTURAL CONTEXT     |
| LN451             | ADV LANG & CULTURE IN CONTEXT  |
| LN491             | SEM ABROAD: ADV LANG & CULT I  |
| LN492             | SEM ABROAD: ADV LANG & CULT II |
| LN493             | SEM ABROAD: ADV LANG&CULT III  |
| LN494             | SEM ABROAD: ADV LANG & CULT IV |
| LN495             | SEM ABROAD: ADV LANG & CULT V  |
| <b>OR</b>         |                                |
| <b>Portuguese</b> | Choose 3 of 15                 |

|                |                                |
|----------------|--------------------------------|
| LN440P         | PORTUGUESE IN CULTURAL CONTEXT |
| LN451          | ADV LANG & CULTURE IN CONTEXT  |
| LN491          | SEM ABROAD: ADV LANG & CULT I  |
| LN492          | SEM ABROAD: ADV LANG & CULT II |
| LN493          | SEM ABROAD: ADV LANG&CULT III  |
| LN494          | SEM ABROAD: ADV LANG & CULT IV |
| LN495          | SEM ABROAD: ADV LANG & CULT V  |
| LP371          | INTENSIVE INTERMED. PORTUGUESE |
| LP372          | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475          | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476          | MILITARY SPKG/RDG - PORTUGUESE |
| LP483          | PORTUGUESE CIVILIZATION I      |
| LP484          | PORTUGUESE CIVILIZATION II     |
| LP485          | SURVEY OF PORTUGUESE LIT I     |
| LP492          | LIT OF PORT-SPKG WORLD         |
| <b>OR</b>      |                                |
| <b>Spanish</b> | Choose 3 of 16                 |
| LN440S         | SPANISH IN CULTURAL CONTEXT    |
| LN451          | ADV LANG & CULTURE IN CONTEXT  |
| LN491          | SEM ABROAD: ADV LANG & CULT I  |
| LN492          | SEM ABROAD: ADV LANG & CULT II |
| LN493          | SEM ABROAD: ADV LANG&CULT III  |
| LN494          | SEM ABROAD: ADV LANG & CULT IV |
| LN495          | SEM ABROAD: ADV LANG & CULT V  |
| LS371          | INTENSIVE INTERMEDIATE SPANISH |
| LS372          | SPANISH FOR ORAL & WRIT COMM   |
| LS475          | SPANISH RDG/WRTG THRU MEDIA    |
| LS476          | MILITARY SPKG/RDG - SPANISH    |
| LS483          | SPANISH CIV AND CULTURE        |
| LS484          | SPANISH AMERICAN CIV AND CULT  |
| LS485          | SPANISH-AMERICAN LITERATURE    |
| LS486          | THE LITERATURE OF SPAIN        |
| LS492          | 20TH/21ST CENTURY HISPANIC LIT |
| <b>AND</b>     |                                |

Choose one course from the following regional track and one course from the regional and comparative studies course track. Any course taken from the Regional Track cannot double-count with the course chosen for the Regional and Comparative Studies Track. All pre-requisites for any courses on these lists should be taken prior to taking the chosen course from either track. Note: Cadets who choose any language other than German to fulfill their language requirement may not choose HI343 from this list.

|                                               |                                |
|-----------------------------------------------|--------------------------------|
| <b>Europe Regional Track</b>                  | Choose 1 of 8                  |
| GE346                                         | GEOGRAPHY OF EUROPE            |
| HI341                                         | THE AGE OF EXPLORATION         |
| HI343                                         | MODERN GERMANY                 |
| HI364                                         | MODERN WESTERN EUROPE          |
| HI368                                         | MOD CENTRAL & E. EUR,1896-1989 |
| PY375                                         | KANT & 19TH CENTURY PHILOSOPHY |
| SS377                                         | POLITICS & GOV OF EUROPE       |
| XH405                                         | THE HOLOCAUST AND ITS LEGACY   |
| <b>AND</b>                                    |                                |
| <b>Regional and Comparative Studies Track</b> | Choose 1 of 28                 |
| EN351                                         | WORLD LITERATURE               |
| EN362                                         | FILM AND FILM THEORY           |
| GE340                                         | GEOGRAPHY OF GLOBAL CULTURES   |
| GE346                                         | GEOGRAPHY OF EUROPE            |

|       |                                |
|-------|--------------------------------|
| GE491 | MILITARY GEOGRAPHY             |
| HI341 | THE AGE OF EXPLORATION         |
| HI343 | MODERN GERMANY                 |
| HI344 | MODERN DIPLOMACY               |
| HI364 | MODERN WESTERN EUROPE          |
| HI368 | MOD CENTRAL & E. EUR,1896-1989 |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI391 | WORLD RELIGIONS                |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW481 | INTERNATIONAL LAW              |
| PY375 | KANT & 19TH CENTURY PHILOSOPHY |
| SS366 | COMPARATIVE POLITICS           |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS479 | INTERNATIONAL ORGANIZATION     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS385 | WAR AND ITS THEORISTS          |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |

### 2029 Regional Studies Minor - Latin America Curriculum

| Code  | Short Description                      | Description                            | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------|----------------------------------------|----------------------------------------|--------------|--------------|
| RSL1N | Regional Studies Minor - Latin America | Regional Studies Minor - Latin America | Regional Studies Minor - Latin America | 0            | 5            |

### 2029 Regional Studies Minor - Latin America Tracks

| Subject Area | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | <p>The course of instruction for the Regional Studies Minor will consist of five courses that form a coherent, interdisciplinary study of a geographical region. Three of the courses must be in a foreign language at the 300 or 400 level. The other two courses must be regional or comparative electives. For any Single or Dual Foreign Language majors choosing this minor, exceptions can be made to the language requirement if all possible language courses for the Cadets' chosen region have been completed as part of the major. Cadets who choose the Foreign Area Studies: Latin America (FSL1) major are not allowed to choose this minor.</p> |

#### Language Courses

Take three courses at the 300-level (or higher) in one of the following languages. Any courses taken to fulfill the requirements for this minor are eligible for double-counting practices, including any that are taken to fulfill core, major or minor curricular components. Three courses must be unique to this minor.

|                   |                                |
|-------------------|--------------------------------|
| <b>Portuguese</b> | Choose 3 of 15                 |
| LN440P            | PORTUGUESE IN CULTURAL CONTEXT |
| LN451             | ADV LANG & CULTURE IN CONTEXT  |
| LN491             | SEM ABROAD: ADV LANG & CULT I  |



|                |                                |
|----------------|--------------------------------|
| LN492          | SEM ABROAD: ADV LANG & CULT II |
| LN493          | SEM ABROAD: ADV LANG&CULT III  |
| LN494          | SEM ABROAD: ADV LANG & CULT IV |
| LN495          | SEM ABROAD: ADV LANG & CULT V  |
| LP371          | INTENSIVE INTERMED. PORTUGUESE |
| LP372          | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475          | PORTUGUESE RDG/WRTG THRU MEDIA |
| LP476          | MILITARY SPKG/RDG - PORTUGUESE |
| LP483          | PORTUGUESE CIVILIZATION I      |
| LP484          | PORTUGUESE CIVILIZATION II     |
| LP485          | SURVEY OF PORTUGUESE LIT I     |
| LP492          | LIT OF PORT-SPKG WORLD         |
| <b>OR</b>      |                                |
| <b>Spanish</b> | Choose 3 of 16                 |
| LN440S         | SPANISH IN CULTURAL CONTEXT    |
| LN451          | ADV LANG & CULTURE IN CONTEXT  |
| LN491          | SEM ABROAD: ADV LANG & CULT I  |
| LN492          | SEM ABROAD: ADV LANG & CULT II |
| LN493          | SEM ABROAD: ADV LANG&CULT III  |
| LN494          | SEM ABROAD: ADV LANG & CULT IV |
| LN495          | SEM ABROAD: ADV LANG & CULT V  |
| LS371          | INTENSIVE INTERMEDIATE SPANISH |
| LS372          | SPANISH FOR ORAL & WRIT COMM   |
| LS475          | SPANISH RDG/WRTG THRU MEDIA    |
| LS476          | MILITARY SPKG/RDG - SPANISH    |
| LS483          | SPANISH CIV AND CULTURE        |
| LS484          | SPANISH AMERICAN CIV AND CULT  |
| LS485          | SPANISH-AMERICAN LITERATURE    |
| LS486          | THE LITERATURE OF SPAIN        |
| LS492          | 20TH/21ST CENTURY HISPANIC LIT |
| <b>AND</b>     |                                |

Choose one course from the following regional track and one course from the regional and comparative studies course track. Any course taken from the Regional Track cannot double-count with the course chosen for the Regional and Comparative Studies Track. All pre-requisites for any courses on these lists should be taken prior to taking the chosen course from either track.

|                                               |                              |
|-----------------------------------------------|------------------------------|
| <b>Latin America Track</b>                    | Choose 1 of 4                |
| GE347                                         | GEOGRAPHY OF LATIN AMERICA   |
| HI341                                         | THE AGE OF EXPLORATION       |
| HI348                                         | MODERN LATIN AMERICA         |
| SS384                                         | POLITICS & GOVT-LATIN AMER   |
| <b>AND</b>                                    |                              |
| <b>Regional and Comparative Studies Track</b> | Choose 1 of 22               |
| EN351                                         | WORLD LITERATURE             |
| GE340                                         | GEOGRAPHY OF GLOBAL CULTURES |
| GE347                                         | GEOGRAPHY OF LATIN AMERICA   |
| GE491                                         | MILITARY GEOGRAPHY           |
| HI341                                         | THE AGE OF EXPLORATION       |
| HI344                                         | MODERN DIPLOMACY             |
| HI348                                         | MODERN LATIN AMERICA         |
| HI381                                         | HISTORY OF IRREGULAR WARFARE |
| HI391                                         | WORLD RELIGIONS              |
| LW410                                         | COMPARATIVE LEGAL SYSTEMS    |
| LW473                                         | ENVIRONMENTAL LAW            |

|       |                             |
|-------|-----------------------------|
| LW481 | INTERNATIONAL LAW           |
| SS366 | COMPARATIVE POLITICS        |
| SS384 | POLITICS & GOVT-LATIN AMER  |
| SS465 | TERRORISM: NEW CHALLENGES   |
| SS475 | COMP POLITICAL INSTITUTIONS |
| SS479 | INTERNATIONAL ORGANIZATION  |
| SS484 | INTERNATIONAL ECONOMICS     |
| SS486 | STATE BUILDING              |
| SS487 | INT'L POLITICAL ECONOMY     |
| WS358 | STRATEGY AND GENERALSHIP    |
| WS455 | COMPARATIVE DEFENSE POLICY  |

### 2029 Regional Studies Minor - Middle East Curriculum

| Code  | Short Description                    | Description                          | Transcript Description               | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|--------------------------------------|--------------------------------------|--------------------------------------|-----------------|-----------------|
| RSM1N | Regional Studies Minor - Middle East | Regional Studies Minor - Middle East | Regional Studies Minor - Middle East | 0               | 5               |

### 2029 Regional Studies Minor - Middle East Tracks

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

The course of instruction for the Regional Studies Minor will consist of five courses that form a coherent, interdisciplinary study of a geographical region. Three of the courses must be in a foreign language at the 300 or 400 level. The other two courses must be regional or comparative electives. For any Single or Dual Foreign Language majors choosing this minor, exceptions can be made to the language requirement if all possible language courses for the Cadets' chosen region have been completed as part of the major. Cadets who choose the Foreign Area Studies: Middle East (FSM1) major are not allowed to choose this minor.

#### Language Courses

Take three courses at the 300-level (or higher) in one of the following languages. Any courses taken to fulfill the requirements for this minor are eligible for double-counting practices, including any that are taken to fulfill core, major or minor curricular components. Three courses must be unique to this minor.

|               |                                |
|---------------|--------------------------------|
| <b>Arabic</b> | Choose 3 of 17                 |
| LA371         | INTENSIVE INTERMEDIATE ARABIC  |
| LA372         | ARABIC FOR ORAL & WRITTEN COMM |
| LA472         | COLLOQUIAL ARABIC              |
| LA475         | ARABIC RDG/WRTG THRU MEDIA     |
| LA476         | MILITARY SPKG/RDG - ARABIC     |
| LA483         | ARAB CIVILIZATION I            |
| LA484         | ARAB CIVILIZATION II           |
| LA485         | ARABIC LITERATURE I            |
| LA486         | ARABIC LITERATURE II           |
| LA492         | ARABIC LITERATURE III          |
| LN440A        | ARABIC IN CULTURAL CONTEXT     |
| LN451         | ADV LANG & CULTURE IN CONTEXT  |
| LN491         | SEM ABROAD: ADV LANG & CULT I  |
| LN492         | SEM ABROAD: ADV LANG & CULT II |
| LN493         | SEM ABROAD: ADV LANG&CULT III  |
| LN494         | SEM ABROAD: ADV LANG & CULT IV |

|                |                                |
|----------------|--------------------------------|
| LN495          | SEM ABROAD: ADV LANG & CULT V  |
| <b>OR</b>      |                                |
| <b>Persian</b> | Choose 3 of 15                 |
| LN440Z         | PERSIAN IN CULTURAL CONTEXT    |
| LN451          | ADV LANG & CULTURE IN CONTEXT  |
| LN491          | SEM ABROAD: ADV LANG & CULT I  |
| LN492          | SEM ABROAD: ADV LANG & CULT II |
| LN493          | SEM ABROAD: ADV LANG&CULT III  |
| LN494          | SEM ABROAD: ADV LANG & CULT IV |
| LN495          | SEM ABROAD: ADV LANG & CULT V  |
| LZ371          | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372          | PERSIAN FOR ORAL & WRIT COMM   |
| LZ475          | PERSIAN RDG/WRTG THRU MEDIA    |
| LZ476          | MILITARY SPKG/RDG - PERSIAN    |
| LZ483          | PERSIAN CIVILIZATION I         |
| LZ484          | PERSIAN CIVILIZATION II        |
| LZ485          | PERSIAN LITERATURE I           |
| LZ486          | PERSIAN LITERATURE II          |
| <b>AND</b>     |                                |

Choose one course from the following regional track and one course from the regional and comparative studies course track. Any course taken from the Regional Track cannot double-count with the course chosen for the Regional and Comparative Studies Track. All pre-requisites for any courses on these lists should be taken prior to taking the chosen course from either track.

|                                   |                              |
|-----------------------------------|------------------------------|
| <b>Middle East Regional Track</b> | Choose 1 of 5                |
| GE344                             | GEOGRAPHY OF AFRICA          |
| GE345                             | GEOGRAPHY OF THE MIDDLE EAST |
| HI339                             | THE MODERN MIDDLE EAST       |
| HI383                             | MIDDLE EASTERN WARFARE       |
| SS383                             | POLITICS & GOVT-MIDDLE EAST  |
| <b>AND</b>                        |                              |

|                                               |                                |
|-----------------------------------------------|--------------------------------|
| <b>Regional and Comparative Studies Track</b> | Choose 1 of 35                 |
| EN351                                         | WORLD LITERATURE               |
| GE340                                         | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                         | GEOGRAPHY OF AFRICA            |
| GE345                                         | GEOGRAPHY OF THE MIDDLE EAST   |
| GE491                                         | MILITARY GEOGRAPHY             |
| HI339                                         | THE MODERN MIDDLE EAST         |
| HI344                                         | MODERN DIPLOMACY               |
| HI381                                         | HISTORY OF IRREGULAR WARFARE   |
| HI383                                         | MIDDLE EASTERN WARFARE         |
| HI391                                         | WORLD RELIGIONS                |
| LW410                                         | COMPARATIVE LEGAL SYSTEMS      |
| LW473                                         | ENVIRONMENTAL LAW              |
| LW481                                         | INTERNATIONAL LAW              |
| SS366                                         | COMPARATIVE POLITICS           |
| SS383                                         | POLITICS & GOVT-MIDDLE EAST    |
| SS465                                         | TERRORISM: NEW CHALLENGES      |
| SS475                                         | COMP POLITICAL INSTITUTIONS    |
| SS477                                         | ECONOMICS OF NATIONAL SECURITY |
| SS479                                         | INTERNATIONAL ORGANIZATION     |
| SS484                                         | INTERNATIONAL ECONOMICS        |
| SS486                                         | STATE BUILDING                 |
| SS487                                         | INT'L POLITICAL ECONOMY        |
| WS358                                         | STRATEGY AND GENERALSHIP       |

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|       |                                |
|-------|--------------------------------|
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| ZH337 | REGIONAL POLITICAL SYSTEMS     |
| ZH347 | INT'L ORGNZTNS & INSTITUTIONS  |
| ZH367 | TOPICS IN MICROECONOMICS       |
| ZH377 | TOPICS IN MACROECONOMICS       |
| ZH407 | TOPICS/AMERICAN FOREIGN POLICY |
| ZH427 | TOPICS IN COMPARATIVE POLITICS |
| ZH447 | TOPICS: INTERNATIONAL POLITICS |
| ZH467 | TOPICS-INTERNATIONAL ECONOMICS |
| ZH477 | TOPICS-INT'L BUSINESS/FINANCE  |

## Geography and Earth Sciences

### 2029 Environmental Science Major Curriculum

| Code | Short Description     | Description                 | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-----------------------|-----------------------------|------------------------|-----------------|-----------------|
| ESC3 | Environmental Science | Environmental Science Major | Environmental Science  | 6               | 7               |

### 2029 Environmental Science Major Tracks

| Subject Area              | Description                    |
|---------------------------|--------------------------------|
| <b>Foundation Courses</b> | Choose 4 of 4                  |
| GE300                     | GLOBAL ENVIRONMENTAL SYSTEMS   |
| GE331                     | AQUATIC SCIENCE                |
| GE333                     | PHYSICAL GEOLOGY               |
| GE431                     | ECOLOGY                        |
| <b>AND</b>                |                                |
| <b>Atmosphere Course</b>  | Choose 1 of 2                  |
| GE321                     | CLIMATOLOGY                    |
| GE324                     | METEOROLOGY                    |
| <b>AND</b>                |                                |
| <b>Tools Elective</b>     | Choose 1 of 3                  |
| CH387                     | HUMAN PHYSIOLOGY               |
| GE310                     | GEOG INFORMATION SYSTEMS       |
| GE373                     | REMOTE SENSING                 |
| <b>AND</b>                |                                |
| <b>Depth Electives</b>    | Choose 2 of 6                  |
| CH383                     | ORGANIC CHEMISTRY I            |
| CH384                     | ORGANIC CHEMISTRY II           |
| GE310                     | GEOG INFORMATION SYSTEMS       |
| GE351                     | LAND USE PLAN & MGT            |
| GE354                     | NATURAL HAZARDS AND RISK       |
| XS391                     | PRIN & APPL OF ENV CHEM        |
| <b>AND</b>                |                                |
| <b>Field Elective</b>     | Choose 1 of 45                 |
| CH383                     | ORGANIC CHEMISTRY I            |
| CH384                     | ORGANIC CHEMISTRY II           |
| CH385                     | INTRODUCTION TO CELL BIOLOGY   |
| CH387                     | HUMAN PHYSIOLOGY               |
| CH388                     | GENETICS                       |
| CH457                     | MICROBIOLOGY                   |
| CH460                     | HUMAN ANATOMY                  |
| CH473                     | BIOCHEMISTRY                   |
| CH481                     | PHYSICAL CHEMISTRY I           |
| EM381                     | ENGINEERING ECONOMY            |
| EV394                     | HYDROGEOLOGY/HYDRAULIC SYSTEMS |
| EV396                     | ENVIRONMENTAL BIOLOGICAL SYS   |
| EV397                     | AIR POLLUTION ENGINEERING      |
| EV489A                    | ADVANCED INDIVIDUAL STUDY I    |
| GE310                     | GEOG INFORMATION SYSTEMS       |

|        |                                |
|--------|--------------------------------|
| GE324  | METEOROLOGY                    |
| GE332  | GEOMORPHOLOGY                  |
| GE341  | GEOGRAPHY OF NORTH AMERICA     |
| GE342  | GEOGRAPHY OF ASIA              |
| GE343  | GEOGRAPHY OF RUSSIA            |
| GE344  | GEOGRAPHY OF AFRICA            |
| GE345  | GEOGRAPHY OF THE MIDDLE EAST   |
| GE346  | GEOGRAPHY OF EUROPE            |
| GE347  | GEOGRAPHY OF LATIN AMERICA     |
| GE351  | LAND USE PLAN & MGT            |
| GE352  | URBAN GEOGRAPHY                |
| GE354  | NATURAL HAZARDS AND RISK       |
| GE373  | REMOTE SENSING                 |
| GE378  | CARTOGRAPHY                    |
| GE380  | SURVEYING                      |
| GE491  | MILITARY GEOGRAPHY             |
| GE499A | ADVANCED INDIVIDUAL STUDY I    |
| LW473  | ENVIRONMENTAL LAW              |
| LW481  | INTERNATIONAL LAW              |
| MA363  | ORDINARY DIFFERENTIAL EQUATION |
| MA366  | APPLIED ENGINEERING MATH       |
| MA376  | APPLIED STATISTICS             |
| MA391  | MATHEMATICAL MODELING          |
| MA396  | NUM METH SOLUTIONS DIFF EQNS   |
| MA476  | MATHEMATICAL STATISTICS        |
| PY329  | TOPICS IN ETHICS               |
| PY350  | PHILOSOPHY OF SCIENCE          |
| SS387  | PUBLIC FINANCE                 |
| SS485  | GOV & POLITICS SUB-SAHARAN AFR |
| WS350  | STRATEGIC COMMUNICATION        |
| XS391  | PRIN & APPL OF ENV CHEM        |

**AND****Complementary Support Course - Required** Choose 1 of 1

The CSCs for this major are divided between one required and two elective courses. The CSCs will provide foundational math and science support for cadets in this major

|       |                              |
|-------|------------------------------|
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
|-------|------------------------------|

**AND****Complementary Support Course - Biology** Choose 1 of 2

|       |                  |
|-------|------------------|
| CH275 | BIOLOGY          |
| CH375 | ADVANCED BIOLOGY |

**AND****Complementary Support Course - Math-Science** Choose 1 of 10

|       |                              |
|-------|------------------------------|
| CH457 | MICROBIOLOGY                 |
| CH473 | BIOCHEMISTRY                 |
| EV396 | ENVIRONMENTAL BIOLOGICAL SYS |
| MA204 | CALCULUS I AND II            |
| MA205 | CALCULUS II                  |
| MA255 | ADV MULTIVARIABLE CALCULUS   |
| MA376 | APPLIED STATISTICS           |
| PH202 | PHYSICS II                   |
| PH252 | ADVANCED PHYSICS II          |
| PH275 | PHYSICS II: SPACE            |

**AND**

**Integrative Experience for the Major** Choose 1 of 1

GE494 ENVIRONMENTAL SECURITY

**AND****Science Depth**

Cadets in this major must take CH102 to satisfy the Science Depth requirement.

CH102 GENERAL CHEMISTRY II

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305 CYBER FOUNDATIONS

CY355 CYBER FOUNDATIONS - COMPUTING

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305 CYBER FOUNDATIONS

CY355 CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major are directed to take the Environmental Engineering sequence to support their major.

EV300 ENVIRONMENTAL SCIENCE

EV350 ENVIRONMNTL ENGR TECHNOLOGIES

EV450 ENV ENG FOR COMMUNITY DEVELOP

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

GE300 GLOBAL ENVIRONMENTAL SYSTEMS

**2029 Environmental Science Major w/ Honors Curriculum**

| Code  | Short Description               | Description                           | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------|---------------------------------------|---------------------------------|--------------|--------------|
| ESC3H | Environmental Science w/ Honors | Environmental Science Major w/ Honors | Environmental Science w/ Honors | 2            | 0            |

**2029 Environmental Science Major w/ Honors Tracks**

| Subject Area                                                                                                                                                                                                                  | Description                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Required Course</b>                                                                                                                                                                                                        | Choose 1 of 1               |
| Take GE499A which requires individual research, a written report, and a formal presentation of research, analysis, and conclusions. You cannot double count GE499A in satisfying both field elective and honors requirements. |                             |
| GE499A                                                                                                                                                                                                                        | ADVANCED INDIVIDUAL STUDY I |
| <b>AND</b>                                                                                                                                                                                                                    |                             |
| <b>Additional Field Elective</b>                                                                                                                                                                                              |                             |
| Take one additional Field Elective from the Environmental Science Field Elective list above.                                                                                                                                  |                             |
| <b>AND</b>                                                                                                                                                                                                                    |                             |
| <b>Grade Requirements</b>                                                                                                                                                                                                     |                             |

Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Geography Major Curriculum

| Code | Short Description | Description     | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------|-----------------|------------------------|--------------|--------------|
| GEO1 | Geography         | Geography Major | Geography              | 5            | 8            |

## 2029 Geography Major Tracks

| Subject Area                              | Description                    |
|-------------------------------------------|--------------------------------|
| <b>Required Foundation Courses</b>        | Choose 4 of 4                  |
| GE205                                     | FOUNDATIONS IN GEOGRAPHY       |
| GE305                                     | GEOGRAPHIC RESEARCH METHODS    |
| GE310                                     | GEOG INFORMATION SYSTEMS       |
| GE340                                     | GEOGRAPHY OF GLOBAL CULTURES   |
| <b>AND</b>                                |                                |
| <b>Required Regional Geography Course</b> | Choose 1 of 6                  |
| GE342                                     | GEOGRAPHY OF ASIA              |
| GE343                                     | GEOGRAPHY OF RUSSIA            |
| GE344                                     | GEOGRAPHY OF AFRICA            |
| GE345                                     | GEOGRAPHY OF THE MIDDLE EAST   |
| GE346                                     | GEOGRAPHY OF EUROPE            |
| GE347                                     | GEOGRAPHY OF LATIN AMERICA     |
| <b>AND</b>                                |                                |
| <b>Geography Electives</b>                | Choose 4 of 27                 |
| GE300                                     | GLOBAL ENVIRONMENTAL SYSTEMS   |
| GE321                                     | CLIMATOLOGY                    |
| GE324                                     | METEOROLOGY                    |
| GE332                                     | GEOMORPHOLOGY                  |
| GE333                                     | PHYSICAL GEOLOGY               |
| GE342                                     | GEOGRAPHY OF ASIA              |
| GE343                                     | GEOGRAPHY OF RUSSIA            |
| GE344                                     | GEOGRAPHY OF AFRICA            |
| GE345                                     | GEOGRAPHY OF THE MIDDLE EAST   |
| GE346                                     | GEOGRAPHY OF EUROPE            |
| GE347                                     | GEOGRAPHY OF LATIN AMERICA     |
| GE351                                     | LAND USE PLAN & MGT            |
| GE352                                     | URBAN GEOGRAPHY                |
| GE354                                     | NATURAL HAZARDS AND RISK       |
| GE373                                     | REMOTE SENSING                 |
| GE378                                     | CARTOGRAPHY                    |
| GE379                                     | PHOTOGRAMMETRY                 |
| GE380                                     | SURVEYING                      |
| GE431                                     | ECOLOGY                        |
| GE462                                     | ENVIRONMENT AND DEVELOPMENT    |
| GE477                                     | ADVANCED REMOTE SENSING        |
| GE478                                     | MILITARY GEOSPATIAL OPERATIONS |



|        |                                |
|--------|--------------------------------|
| GE490  | SPEC TOPICS-GEOG & ENVRNMNT    |
| GE494  | ENVIRONMENTAL SECURITY         |
| GE496  | COLLOQUIUM IN GEOGRAPHY        |
| GE498  | ADV GEOGRAPHIC INFORMATION SYS |
| GE499A | ADVANCED INDIVIDUAL STUDY I    |

**AND****Complementary Support Courses**

Choose 3 of 97

|       |                                |
|-------|--------------------------------|
| CH375 | ADVANCED BIOLOGY               |
| CH387 | HUMAN PHYSIOLOGY               |
| CY300 | PROGRAMMING FUNDAMENTALS       |
| EN321 | AMERICAN LITERATURE I          |
| EN331 | BRITISH LITERATURE I           |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI338 | WARFARE IN AGE OF REVOLUTIONS  |
| HI339 | THE MODERN MIDDLE EAST         |
| HI340 | COLONIAL AMERICA               |
| HI341 | THE AGE OF EXPLORATION         |
| HI343 | MODERN GERMANY                 |
| HI344 | MODERN DIPLOMACY               |
| HI345 | MODERN AFRICA                  |
| HI346 | INDIA, PAKISTAN, & BANGLADESH  |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI348 | MODERN LATIN AMERICA           |
| HI355 | WARFARE, INDUSTRY, & EMPIRE    |
| HI356 | WAR AT SEA, SKY, SPACE         |
| HI357 | ARMED CONFLICT IN THE COLD WAR |
| HI359 | ERA OF THE SECOND WORLD WAR    |
| HI361 | MEDIEVAL EUROPE                |
| HI362 | POLITICS/SOC-EARLY MOD EURO    |
| HI364 | MODERN WESTERN EUROPE          |
| HI367 | IMPERIAL AND SOVIET RUSSIA     |
| HI368 | MOD CENTRAL & E. EUR,1896-1989 |
| HI369 | AMERICAN FRONTIERS             |
| HI370 | ANCIENT & MEDIEVAL WARFARE     |
| HI372 | US FGN RELATIONS SINCE 1898    |
| HI374 | HISTORY OF AFRICA              |
| HI376 | EARLY MODERN WARFARE           |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI391 | WORLD RELIGIONS                |
| HI394 | REVOLUTIONARY AMERICA          |
| HI395 | HIST OF CIVIL WAR AMERICA      |
| HI396 | THEODORE ROOSEVELT'S AMERICA   |
| HI397 | THE US FROM COLD WAR TO TODAY  |
| HI398 | CIVIL RIGHTS IN AMER HIST      |
| HI462 | THE HISTORY OF INNOVATION      |
| HI487 | EARLY AMERICAN REPUB 1787-1826 |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW473 | ENVIRONMENTAL LAW              |
| LW474 | LAW OF ARMED CONFLICT          |
| LW481 | INTERNATIONAL LAW              |
| LX300 | 3RD SEMESTER FOREIGN LANG      |
| LX400 | 4TH SEMESTER FOREIGN LANG      |
| MA376 | APPLIED STATISTICS             |
| PL376 | PSYCHOPATHOLOGY                |

|       |                                |
|-------|--------------------------------|
| PL383 | SOCIAL PSYCHOLOGY              |
| PL392 | COGNITIVE PSYCHOLOGY           |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM    |
| PL482 | ARMED FORCES AND SOCIETY       |
| PY300 | PHILOSOPHICAL METHODS          |
| PY305 | LOGICAL REASONING              |
| PY325 | MILITARY ETHICS                |
| PY330 | POLITICAL PHILOSOPHY           |
| PY345 | PHILOSOPHY OF RELIGION         |
| PY350 | PHILOSOPHY OF SCIENCE          |
| PY355 | PHILOSOPHY OF MIND             |
| PY360 | ANCIENT PHILOSOPHY             |
| PY370 | 17TH & 18TH CENTURY PHILOSOPHY |
| PY375 | KANT & 19TH CENTURY PHILOSOPHY |
| SS364 | GAME THEORY                    |
| SS366 | COMPARATIVE POLITICS           |
| SS368 | ECONOMETRICS I                 |
| SS372 | POLITICS OF CHINA              |
| SS374 | POLITICS OF NORTHEAST ASIA     |
| SS375 | POL OF THE POST-SOVIET STATES  |
| SS377 | POLITICS OF EUROPE             |
| SS378 | GREAT POWER COMPETITION        |
| SS383 | POLITICS OF THE MIDDLE EAST    |
| SS384 | POLITICS OF LATIN AMERICA      |
| SS385 | HISTORY OF ECONOMICS           |
| SS386 | POLITICAL THOUGHT AND IDEAS    |
| SS462 | ECONOMIC GROWTH & DEVELOPMENT  |
| SS464 | HOMELAND SECURITY              |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS466 | COMBATING TERRORISM            |
| SS472 | SOLDIER & STATE: AM CIV-MIL RE |
| SS473 | AMERICAN FOREIGN POLICY        |
| SS475 | COMP POLITICAL INSTITUTIONS    |
| SS476 | CIVIL WARS                     |
| SS481 | POLITICS OF DEFENSE POLICY     |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | POLITICS SUB-SAHARAN AFR       |
| SS486 | STATE BUILDING                 |
| SS487 | INT'L POLITICAL ECONOMY        |
| WS301 | INTRODUCTION TO WAR STUDIES    |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS370 | US STRATEGY AND POLICY         |
| WS385 | WAR AND ITS THEORISTS          |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |
| XH415 | GENOCIDE AND MASS ATROCITY     |

**AND****Integrative Experience for the Major**

Choose 1 of 1

GE491

MILITARY GEOGRAPHY

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

CH102

GENERAL CHEMISTRY II

|       |                          |
|-------|--------------------------|
| CH152 | ADV GENERAL CHEMISTRY II |
| CH275 | BIOLOGY                  |
| PH202 | PHYSICS II               |
| PH252 | ADVANCED PHYSICS II      |
| PH275 | PHYSICS II: SPACE        |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major are required to take the Environmental Engineering Sequence.

|       |                               |
|-------|-------------------------------|
| EV300 | ENVIRONMENTAL SCIENCE         |
| EV350 | ENVIRONMNTL ENGR TECHNOLOGIES |
| EV450 | ENV ENG FOR COMMUNITY DEVELOP |

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                             |
|-------|-----------------------------|
| GE305 | GEOGRAPHIC RESEARCH METHODS |
|-------|-----------------------------|

## 2029 Geography Major w/ Honors Curriculum

| Code  | Short Description   | Description               | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------|---------------------------|------------------------|--------------|--------------|
| GEO1H | Geography w/ Honors | Geography Major w/ Honors | Geography w/ Honors    | 2            | 0            |

## 2029 Geography Major w/ Honors Tracks

| Subject Area                                | Description                  |
|---------------------------------------------|------------------------------|
| <b>Senior Thesis/Project Requirement</b>    |                              |
| Approval by the Geography Program Director. |                              |
| <b>Required Course</b>                      | Choose 2 of 2                |
| GE495                                       | HONORS SEMINAR IN GEOGRAPHY  |
| GE499B                                      | ADVANCED INDIVIDUAL STUDY II |
| <b>AND</b>                                  |                              |

**Grade Requirements**

Cadets must complete the requirements of the major as shown above, and achieve a final APSC of at least 3.0 in the core curriculum and a final APSC of at least 3.5 in the major.

## 2029 Geospatial Science Major Curriculum

| Code | Short Description  | Description              | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|--------------------|--------------------------|------------------------|-----------------|-----------------|
| GSC0 | Geospatial Science | Geospatial Science Major | Geospatial Science     | 7               | 6               |

## 2029 Geospatial Science Major Tracks

| Subject Area                                                                                                                                                                                                         | Description                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                                                              | Choose 6 of 6                  |
| GE310                                                                                                                                                                                                                | GEOG INFORMATION SYSTEMS       |
| GE373                                                                                                                                                                                                                | REMOTE SENSING                 |
| GE378                                                                                                                                                                                                                | CARTOGRAPHY                    |
| GE477                                                                                                                                                                                                                | ADVANCED REMOTE SENSING        |
| GE478                                                                                                                                                                                                                | MILITARY GEOSPATIAL OPERATIONS |
| GE498                                                                                                                                                                                                                | ADV GEOGRAPHIC INFORMATION SYS |
| <b>AND</b>                                                                                                                                                                                                           |                                |
| <b>Spatial Data Acquisition Block</b>                                                                                                                                                                                | Choose 1 of 2                  |
| GE379                                                                                                                                                                                                                | PHOTOGRAMMETRY                 |
| GE380                                                                                                                                                                                                                | SURVEYING                      |
| <b>AND</b>                                                                                                                                                                                                           |                                |
| <b>Foundational Geography Elective</b>                                                                                                                                                                               | Choose 1 of 2                  |
| GE205                                                                                                                                                                                                                | FOUNDATIONS IN GEOGRAPHY       |
| GE340                                                                                                                                                                                                                | GEOGRAPHY OF GLOBAL CULTURES   |
| <b>AND</b>                                                                                                                                                                                                           |                                |
| <b>Restrictive Environmental Analysis Elective</b>                                                                                                                                                                   | Choose 1 of 11                 |
| **Courses selected in the Spatial Data Acquisition Block may not be selected from the Restricted Environmental Analysis Electives list.                                                                              |                                |
| EV300                                                                                                                                                                                                                | ENVIRONMENTAL SCIENCE          |
| EV397                                                                                                                                                                                                                | AIR POLLUTION ENGINEERING      |
| EV481                                                                                                                                                                                                                | WATER RESOURCES PLAN & DESIGN  |
| GE321                                                                                                                                                                                                                | CLIMATOLOGY                    |
| GE332                                                                                                                                                                                                                | GEOMORPHOLOGY                  |
| GE333                                                                                                                                                                                                                | PHYSICAL GEOLOGY               |
| GE351                                                                                                                                                                                                                | LAND USE PLAN & MGT            |
| GE352                                                                                                                                                                                                                | URBAN GEOGRAPHY                |
| GE354                                                                                                                                                                                                                | NATURAL HAZARDS AND RISK       |
| GE379                                                                                                                                                                                                                | PHOTOGRAMMETRY                 |
| GE380                                                                                                                                                                                                                | SURVEYING                      |
| <b>AND</b>                                                                                                                                                                                                           |                                |
| <b>Complementary Support Course - Electives</b>                                                                                                                                                                      | Choose 3 of 40                 |
| **Courses selected in the Spatial Data Acquisition Block, Foundational Geography Elective, and Restrictive Environmental Analysis Elective groupings may not be selected from the Complementary Support Course list. |                                |
| CE350                                                                                                                                                                                                                | INFRASTRUCTURE ENGINEERING     |
| CE371                                                                                                                                                                                                                | SOIL MECHANICS/FNDTN ENGNRG    |
| CE380                                                                                                                                                                                                                | HYDROLOGY/HYDRAULIC DESIGN     |

|       |                               |
|-------|-------------------------------|
| CS393 | DATABASE SYSTEMS              |
| CY300 | PROGRAMMING FUNDAMENTALS      |
| CY383 | SECURE INTERFACE DESIGN       |
| GE205 | FOUNDATIONS IN GEOGRAPHY      |
| GE324 | METEOROLOGY                   |
| GE332 | GEOMORPHOLOGY                 |
| GE333 | PHYSICAL GEOLOGY              |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE341 | GEOGRAPHY OF NORTH AMERICA    |
| GE342 | GEOGRAPHY OF ASIA             |
| GE343 | GEOGRAPHY OF RUSSIA           |
| GE344 | GEOGRAPHY OF AFRICA           |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST  |
| GE346 | GEOGRAPHY OF EUROPE           |
| GE347 | GEOGRAPHY OF LATIN AMERICA    |
| GE354 | NATURAL HAZARDS AND RISK      |
| GE431 | ECOLOGY                       |
| GE494 | ENVIRONMENTAL SECURITY        |
| MA204 | CALCULUS I AND II             |
| MA205 | CALCULUS II                   |
| MA366 | APPLIED ENGINEERING MATH      |
| MA376 | APPLIED STATISTICS            |
| MA383 | FOUNDATIONS OF MATH           |
| ME202 | COMPUTER AIDED DESIGN         |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM   |
| SE301 | FNDTN ENGIN DSGN & SYS MGMT   |
| SE370 | COMPUTER AIDED SYSTEMS ENG    |
| SP471 | ASTRONAUTICS                  |
| SP472 | SPACE PHYSICS                 |
| SS372 | POLITICS OF CHINA             |
| SS374 | POLITICS OF NORTHEAST ASIA    |
| SS375 | POL OF THE POST-SOVIET STATES |
| SS377 | POLITICS OF EUROPE            |
| SS383 | POLITICS OF THE MIDDLE EAST   |
| SS384 | POLITICS OF LATIN AMERICA     |
| SS464 | HOMELAND SECURITY             |
| SS465 | TERRORISM: NEW CHALLENGES     |

**AND****Integrative Experience for the Major**

Choose 1 of 1

GE491

MILITARY GEOGRAPHY

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

CH102

GENERAL CHEMISTRY II

CH275

BIOLOGY

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may choose any authorized engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

GE477

ADVANCED REMOTE SENSING

**2029 Geospatial Science Major w/ Honors Curriculum**

| Code  | Short Description            | Description                        | Transcript Description       | Req Crse Cnt | Opt Crse Cnt |
|-------|------------------------------|------------------------------------|------------------------------|--------------|--------------|
| GSC0H | Geospatial Science w/ Honors | Geospatial Science Major w/ Honors | Geospatial Science w/ Honors | 2            | 0            |

**2029 Geospatial Science Major w/ Honors Tracks**

| Subject Area                                                                                                                                                                      | Description                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Senior Thesis/Project Requirement</b>                                                                                                                                          |                              |
| Completion of a senior thesis or project is required. To graduate w/ Honors the following two options are available.                                                              |                              |
| <b>Required Course</b>                                                                                                                                                            | Choose 2 of 2                |
| GE499A                                                                                                                                                                            | ADVANCED INDIVIDUAL STUDY I  |
| GE499B                                                                                                                                                                            | ADVANCED INDIVIDUAL STUDY II |
| <b>OR</b>                                                                                                                                                                         |                              |
| <b>Alternate Course</b>                                                                                                                                                           |                              |
| GE499A is designed to satisfy the research/design project requirement. GE499A and an additional course from the GIS electives list in lieu of GE499B may be taken.                |                              |
| <b>AND</b>                                                                                                                                                                        |                              |
| <b>Grade Requirements</b>                                                                                                                                                         |                              |
| Cadets must complete the requirements of the major as shown above, and achieve a final APSC of at least 3.0 in the core curriculum and a final APSC of at least 3.5 in the major. |                              |

**Geography and Environmental Engineering**

No Major found

History

No Major found



## History and War Studies

### 2029 General History Minor Curriculum

| Code  | Short Description     | Description           | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|-----------------------|-----------------------|------------------------|-----------------|-----------------|
| GHM1N | General History Minor | General History Minor | General History Minor  | 0               | 5               |

### 2029 General History Minor Tracks

| Subject Area                                                          | Description                     |
|-----------------------------------------------------------------------|---------------------------------|
| Cadets majoring in History may not pursue a minor in History as well. |                                 |
| <b>General History Electives</b>                                      | Choose 5 of 46                  |
| HI301                                                                 | HIST OF MILITARY ART TO 1900    |
| HI337                                                                 | CHINA: EMPIRE, REPUBLIC, & MAO  |
| HI338                                                                 | WARFARE IN AGE OF REVOLUTIONS   |
| HI339                                                                 | THE MODERN MIDDLE EAST          |
| HI340                                                                 | COLONIAL AMERICA                |
| HI341                                                                 | THE AGE OF EXPLORATION          |
| HI343                                                                 | MODERN GERMANY                  |
| HI344                                                                 | MODERN DIPLOMACY                |
| HI345                                                                 | MODERN AFRICA                   |
| HI346                                                                 | INDIA, PAKISTAN, & BANGLADESH   |
| HI347                                                                 | ASIAN WARFARE AND POLITICS      |
| HI348                                                                 | MODERN LATIN AMERICA            |
| HI355                                                                 | WARFARE, INDUSTRY, & EMPIRE     |
| HI357                                                                 | ARMED CONFLICT IN THE COLD WAR  |
| HI359                                                                 | ERA OF THE SECOND WORLD WAR     |
| HI361                                                                 | MEDIEVAL EUROPE                 |
| HI364                                                                 | MODERN WESTERN EUROPE           |
| HI367                                                                 | IMPERIAL AND SOVIET RUSSIA      |
| HI368                                                                 | MOD CENTRAL & E. EUR, 1896-1989 |
| HI369                                                                 | AMERICAN FRONTIERS              |
| HI370                                                                 | ANCIENT & MEDIEVAL WARFARE      |
| HI372                                                                 | US FGN RELATIONS SINCE 1898     |
| HI376                                                                 | EARLY MODERN WARFARE            |
| HI377                                                                 | EMPIRES & REVOLUTIONS           |
| HI378                                                                 | INDUSTRY, IDEOLOGY & THE STATE  |
| HI381                                                                 | HISTORY OF IRREGULAR WARFARE    |
| HI383                                                                 | MIDDLE EASTERN WARFARE          |
| HI387                                                                 | TOPICS IN AMERICAN HISTORY      |
| HI388                                                                 | TOPICS IN INTERNATIONAL HISTORY |
| HI389                                                                 | TOPICS IN MILITARY HISTORY      |
| HI391                                                                 | WORLD RELIGIONS                 |
| HI392                                                                 | AMERICAN HISTORICAL MEMORY      |
| HI394                                                                 | REVOLUTIONARY AMERICA           |
| HI395                                                                 | HIST OF CIVIL WAR AMERICA       |
| HI396                                                                 | THEODORE ROOSEVELT'S AMERICA    |
| HI397                                                                 | THE US FROM COLD WAR TO TODAY   |
| HI398                                                                 | CIVIL RIGHTS IN AMER HIST       |
| HI399                                                                 | HISTORY STAFF RIDE              |

|       |                                |
|-------|--------------------------------|
| HI460 | SENIOR FACULTY COURSE          |
| HI462 | THE HISTORY OF INNOVATION      |
| HI464 | VISITING PROFESSOR ELECTIVE    |
| HI487 | EARLY AMERICAN REPUB 1787-1826 |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS385 | WAR AND ITS THEORISTS          |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |
| XH415 | GENOCIDE AND MASS ATROCITY     |

### 2029 History Major: International Curriculum

| Code | Short Description            | Description                  | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|------------------------------|------------------------------|------------------------|-----------------|-----------------|
| HNT3 | History Major: International | History Major: International | History: International | 13              | 0               |

### 2029 History Major: International Tracks

| Subject Area                                                                                                                                                                   | Description                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Required Courses</b>                                                                                                                                                        | Choose 2 of 2                   |
| HI301                                                                                                                                                                          | HIST OF MILITARY ART TO 1900    |
| HI498                                                                                                                                                                          | COLLOQUIUM IN HISTORY           |
| <b>AND</b>                                                                                                                                                                     |                                 |
| <b>International History</b>                                                                                                                                                   | Choose 5 of 21                  |
| HI337                                                                                                                                                                          | CHINA: EMPIRE, REPUBLIC, & MAO  |
| HI339                                                                                                                                                                          | THE MODERN MIDDLE EAST          |
| HI341                                                                                                                                                                          | THE AGE OF EXPLORATION          |
| HI343                                                                                                                                                                          | MODERN GERMANY                  |
| HI344                                                                                                                                                                          | MODERN DIPLOMACY                |
| HI345                                                                                                                                                                          | MODERN AFRICA                   |
| HI346                                                                                                                                                                          | INDIA, PAKISTAN, & BANGLADESH   |
| HI347                                                                                                                                                                          | ASIAN WARFARE AND POLITICS      |
| HI348                                                                                                                                                                          | MODERN LATIN AMERICA            |
| HI361                                                                                                                                                                          | MEDIEVAL EUROPE                 |
| HI364                                                                                                                                                                          | MODERN WESTERN EUROPE           |
| HI367                                                                                                                                                                          | IMPERIAL AND SOVIET RUSSIA      |
| HI368                                                                                                                                                                          | MOD CENTRAL & E. EUR,1896-1989  |
| HI377                                                                                                                                                                          | EMPIRES & REVOLUTIONS           |
| HI378                                                                                                                                                                          | INDUSTRY, IDEOLOGY & THE STATE  |
| HI388                                                                                                                                                                          | TOPICS IN INTERNATIONAL HISTORY |
| HI391                                                                                                                                                                          | WORLD RELIGIONS                 |
| HI496                                                                                                                                                                          | ADV IND STUDY IN HISTORY (3CR)  |
| XH491                                                                                                                                                                          | INTERDISCIPLINARY CAPSTONE-HIST |
| ZH315                                                                                                                                                                          | MODERN REGIONAL HISTORY         |
| ZH325                                                                                                                                                                          | TOPICS IN REGIONAL HISTORY      |
| <b>AND</b>                                                                                                                                                                     |                                 |
| <b>Out-of-Stem History Electives</b>                                                                                                                                           | Choose 2 of 32                  |
| For cadets who will graduate w/ Honors, one of their courses must be from the HI400 series and they will take an additional elective from all Department of History offerings. |                                 |
| HI338                                                                                                                                                                          | WARFARE IN AGE OF REVOLUTIONS   |

|       |                                |
|-------|--------------------------------|
| HI340 | COLONIAL AMERICA               |
| HI355 | WARFARE, INDUSTRY, & EMPIRE    |
| HI356 | WAR AT SEA, SKY, SPACE         |
| HI357 | ARMED CONFLICT IN THE COLD WAR |
| HI359 | ERA OF THE SECOND WORLD WAR    |
| HI369 | AMERICAN FRONTIERS             |
| HI370 | ANCIENT & MEDIEVAL WARFARE     |
| HI372 | US FGN RELATIONS SINCE 1898    |
| HI376 | EARLY MODERN WARFARE           |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| HI383 | MIDDLE EASTERN WARFARE         |
| HI387 | TOPICS IN AMERICAN HISTORY     |
| HI389 | TOPICS IN MILITARY HISTORY     |
| HI392 | AMERICAN HISTORICAL MEMORY     |
| HI393 | AMERICA IN DEPRESSION AND WAR  |
| HI394 | REVOLUTIONARY AMERICA          |
| HI395 | HIST OF CIVIL WAR AMERICA      |
| HI396 | THEODORE ROOSEVELT'S AMERICA   |
| HI397 | THE US FROM COLD WAR TO TODAY  |
| HI398 | CIVIL RIGHTS IN AMER HIST      |
| HI460 | SENIOR FACULTY COURSE          |
| HI462 | THE HISTORY OF INNOVATION      |
| HI487 | EARLY AMERICAN REPUB 1787-1826 |
| HI496 | ADV IND STUDY IN HISTORY (3CR) |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS385 | WAR AND ITS THEORISTS          |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |
| XH415 | GENOCIDE AND ETHNIC CLEANSING  |
| XH491 | INTERDISCIPLINARY CAPSTNE-HIST |
| ZH335 | MILITARY HISTORY               |
| ZH345 | TOPICS IN MILITARY HISTORY     |

**AND****Complementary Support Courses**

Cadets in this major will fulfill the Complementary Support Course requirement by taking a third semester in a foreign language and two additional choices.

**AND****Complementary Support Course - Foreign Language**

Choose 1 of 1

LX300

3RD SEMESTER FOREIGN LANG

**AND****Complementary Support Courses - Electives**

Choose 2 of 72

Depending on their particular research interests, cadets will consult with their academic counselor to choose their final two CSCs. The courses found on the following list of courses could all aid in an historian's research abilities and methodologies, but a cadet may deviate from this list as long as they can justify why using a course not found on it will aid them in their historical study. In addition, a cadet may choose to fulfill this requirement by taking two additional courses in their chosen foreign language. This includes the DFL courses on regional civilization (example: LA483 Arab Civilization I), and regional literature (example: LF485 Survey of French Literature I).

|       |                        |
|-------|------------------------|
| EN321 | AMERICAN LITERATURE I  |
| EN322 | AMERICAN LITERATURE II |
| EN331 | BRITISH LITERATURE I   |
| EN332 | BRITISH LITERATURE II  |
| EN351 | WORLD LITERATURE       |
| EN353 | WAR LITERATURE         |

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|       |                                |
|-------|--------------------------------|
| EN355 | CRITICISM COLLOQUIUM           |
| EN362 | FILM AND FILM THEORY           |
| GE310 | GEOG INFORMATION SYSTEMS       |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE341 | GEOGRAPHY OF NORTH AMERICA     |
| GE342 | GEOGRAPHY OF ASIA              |
| GE343 | GEOGRAPHY OF RUSSIA            |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST   |
| GE346 | GEOGRAPHY OF EUROPE            |
| GE347 | GEOGRAPHY OF LATIN AMERICA     |
| GE352 | URBAN GEOGRAPHY                |
| GE378 | CARTOGRAPHY                    |
| GE491 | MILITARY GEOGRAPHY             |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW474 | LAW OF ARMED CONFLICT          |
| LW481 | INTERNATIONAL LAW              |
| LW482 | NATIONAL SECURITY LAW          |
| LX400 | 4TH SEMESTER FOREIGN LANG      |
| MA376 | APPLIED STATISTICS             |
| PL360 | PSYCH ELITE PERFORMANCE        |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM    |
| PL471 | LEADERSHIP IN COMBAT           |
| PL482 | ARMED FORCES AND SOCIETY       |
| PY300 | PHILOSOPHICAL METHODS          |
| PY305 | LOGICAL REASONING              |
| PY330 | POLITICAL PHILOSOPHY           |
| PY345 | PHILOSOPHY OF RELIGION         |
| PY360 | ANCIENT PHILOSOPHY             |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| PY370 | 17TH & 18TH CENTURY PHILOSOPHY |
| PY375 | KANT & 19TH CENTURY PHILOSOPHY |
| SS370 | MASS MEDIA & AMER POLITICS     |
| SS372 | POLITICS AND GOV OF CHINA      |
| SS373 | THE AMERICAN PRESIDENCY        |
| SS374 | POL & GOV OF KOREAS & JAPAN    |
| SS375 | GOV & POL RUSSIA & NEIGHBORS   |
| SS377 | POLITICS & GOV OF EUROPE       |
| SS378 | ADV INTL RELATIONS THEORY      |
| SS379 | LEGISLATIVE POLITICS           |
| SS380 | MANPOWER-LABOR ECONOMICS       |
| SS383 | POLITICS & GOVT-MIDDLE EAST    |
| SS384 | POLITICS & GOVT-LATIN AMER     |
| SS385 | HISTORY OF ECONOMICS           |
| SS386 | POLITICAL THOUGHT              |
| SS387 | PUBLIC FINANCE                 |
| SS462 | ECON OF STABILIZATION & GROWTH |
| SS464 | HOMELAND SECURITY              |
| SS465 | TERRORISM: NEW CHALLENGES      |
| SS472 | THE AM STATE & THE SOLDIER     |
| SS473 | AMERICAN FOREIGN POLICY        |
| SS476 | CIVIL WARS                     |
| SS481 | POLITICS OF DEFENSE POLICY     |
| SS483 | NATIONAL SECURITY SEMINAR      |

|       |                                |
|-------|--------------------------------|
| SS484 | INTERNATIONAL ECONOMICS        |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR |
| SS487 | INT'L POLITICAL ECONOMY        |
| SS493 | SENIOR STUDIES - AMER POLITICS |
| WS345 | MILITARY INNOVATION            |
| WS350 | STRATEGIC COMMUNICATION        |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS370 | US STRATEGY AND POLICY         |
| WS399 | INTERNSHIP IN WAR STUDIES      |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| WS475 | FORECAST AND GAME IN DECISION  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

Cadets majoring in History must complete an integrative experience during their second- or first-class year. Cadets will work with their DACs to identify in which course they intend to complete their integrative experience requirement. Once DACs verify with course directors that cadets have completed integrative experience requirements as planned, they will take administrative action to count that elective for integrative experience credit.

HX400

INTEGRATIVE EXPERIENCE IN HIST

**AND****Science Depth**

Cadets in this major may take any approved choice to satisfy the Science Depth requirement.

CH102

GENERAL CHEMISTRY II

CH275

BIOLOGY

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**AND****STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS -  
COMPUTING**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**AND****IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS -  
COMPUTING**AND****Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**AND****Writing-in-the-Major**

Cadets in this major will satisfy the Writing Across the Disciplines requirement by successfully completing the writing requirements in the core curriculum and several history courses including HI498.

HI498

COLLOQUIUM IN HISTORY

### 2029 History Major: International w/ Thesis Curriculum

| Code  | Short Description                | Description                            | Transcript Description           | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------|----------------------------------------|----------------------------------|--------------|--------------|
| HNT3T | History: International w/ Thesis | History Major: International w/ Thesis | History: International w/ Thesis | 1            | 0            |

### 2029 History Major: International w/ Thesis Tracks

| Subject Area           | Description   |
|------------------------|---------------|
| <b>Required Course</b> | Choose 1 of 1 |
| HI499                  | SENIOR THESIS |

### 2029 History Major: International w/ Thesis (Honors) Curriculum

| Code  | Short Description                         | Description                                     | Transcript Description                    | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------------------|-------------------------------------------------|-------------------------------------------|--------------|--------------|
| HNT3H | History: International w/ Thesis (Honors) | History Major: International w/ Thesis (Honors) | History: International w/ Thesis (Honors) | 1            | 1            |

### 2029 History Major: International w/ Thesis (Honors) Tracks

| Subject Area             | Description                                                                                                                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Elective</b>          |                                                                                                                                                                                                                                   |
|                          | Take one additional elective from the History Department catalogue. For cadets in the honors program at least one of their electives must be at the 400 level (HI4XX). This 400 level elective is in addition to HI498 and HI499. |
| <b>AND</b>               |                                                                                                                                                                                                                                   |
| <b>Required Course</b>   | Choose 1 of 1                                                                                                                                                                                                                     |
| HI499                    | SENIOR THESIS                                                                                                                                                                                                                     |
| <b>AND</b>               |                                                                                                                                                                                                                                   |
| <b>Grade Requirement</b> |                                                                                                                                                                                                                                   |
|                          | Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major. Earn an A- in HI499.                                                   |

### 2029 History Major: Military Curriculum

| Code | Short Description       | Description             | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------|-------------------------|------------------------|--------------|--------------|
| HMH3 | History Major: Military | History Major: Military | History: Military      | 13           | 0            |

### 2029 History Major: Military Tracks

| Subject Area                                                                                                                                                                   | Description                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Required Courses</b>                                                                                                                                                        | Choose 2 of 2                   |
| HI301                                                                                                                                                                          | HIST OF MILITARY ART TO 1900    |
| HI498                                                                                                                                                                          | COLLOQUIUM IN HISTORY           |
| <b>AND</b>                                                                                                                                                                     |                                 |
| <b>Military History</b>                                                                                                                                                        | Choose 5 of 16                  |
| HI338                                                                                                                                                                          | WARFARE IN AGE OF REVOLUTIONS   |
| HI355                                                                                                                                                                          | WARFARE, INDUSTRY, & EMPIRE     |
| HI356                                                                                                                                                                          | WAR AT SEA, SKY, SPACE          |
| HI357                                                                                                                                                                          | ARMED CONFLICT IN THE COLD WAR  |
| HI359                                                                                                                                                                          | ERA OF THE SECOND WORLD WAR     |
| HI370                                                                                                                                                                          | ANCIENT & MEDIEVAL WARFARE      |
| HI376                                                                                                                                                                          | EARLY MODERN WARFARE            |
| HI381                                                                                                                                                                          | HISTORY OF IRREGULAR WARFARE    |
| HI383                                                                                                                                                                          | MIDDLE EASTERN WARFARE          |
| HI389                                                                                                                                                                          | TOPICS IN MILITARY HISTORY      |
| HI496                                                                                                                                                                          | ADV IND STUDY IN HISTORY (3CR)  |
| WS358                                                                                                                                                                          | STRATEGY AND GENERALSHIP        |
| WS385                                                                                                                                                                          | WAR AND ITS THEORISTS           |
| XH491                                                                                                                                                                          | INTERDISCIPLINARY CAPSTONE-HIST |
| ZH335                                                                                                                                                                          | MILITARY HISTORY                |
| ZH345                                                                                                                                                                          | TOPICS IN MILITARY HISTORY      |
| <b>AND</b>                                                                                                                                                                     |                                 |
| <b>Out-of-Stem History Electives</b>                                                                                                                                           | Choose 2 of 38                  |
| For cadets who will graduate w/ Honors, one of their courses must be from the HI400 series and they will take an additional elective from all Department of History offerings. |                                 |
| HI337                                                                                                                                                                          | CHINA: EMPIRE, REPUBLIC, & MAO  |
| HI339                                                                                                                                                                          | THE MODERN MIDDLE EAST          |
| HI340                                                                                                                                                                          | COLONIAL AMERICA                |
| HI341                                                                                                                                                                          | THE AGE OF EXPLORATION          |
| HI343                                                                                                                                                                          | MODERN GERMANY                  |
| HI344                                                                                                                                                                          | MODERN DIPLOMACY                |
| HI345                                                                                                                                                                          | MODERN AFRICA                   |
| HI346                                                                                                                                                                          | INDIA, PAKISTAN, & BANGLADESH   |
| HI347                                                                                                                                                                          | ASIAN WARFARE AND POLITICS      |
| HI348                                                                                                                                                                          | MODERN LATIN AMERICA            |
| HI361                                                                                                                                                                          | MEDIEVAL EUROPE                 |
| HI364                                                                                                                                                                          | MODERN WESTERN EUROPE           |
| HI367                                                                                                                                                                          | IMPERIAL AND SOVIET RUSSIA      |
| HI368                                                                                                                                                                          | MOD CENTRAL & E. EUR, 1896-1989 |
| HI369                                                                                                                                                                          | AMERICAN FRONTIERS              |
| HI372                                                                                                                                                                          | US FGN RELATIONS SINCE 1898     |
| HI377                                                                                                                                                                          | EMPIRES & REVOLUTIONS           |
| HI378                                                                                                                                                                          | INDUSTRY, IDEOLOGY & THE STATE  |
| HI383                                                                                                                                                                          | MIDDLE EASTERN WARFARE          |
| HI387                                                                                                                                                                          | TOPICS IN AMERICAN HISTORY      |
| HI388                                                                                                                                                                          | TOPICS IN INTERNATIONAL HISTORY |
| HI391                                                                                                                                                                          | WORLD RELIGIONS                 |

|       |                                |
|-------|--------------------------------|
| HI392 | AMERICAN HISTORICAL MEMORY     |
| HI393 | AMERICA IN DEPRESSION AND WAR  |
| HI394 | REVOLUTIONARY AMERICA          |
| HI395 | HIST OF CIVIL WAR AMERICA      |
| HI396 | THEODORE ROOSEVELT'S AMERICA   |
| HI397 | THE US FROM COLD WAR TO TODAY  |
| HI398 | CIVIL RIGHTS IN AMER HIST      |
| HI460 | SENIOR FACULTY COURSE          |
| HI462 | THE HISTORY OF INNOVATION      |
| HI487 | EARLY AMERICAN REPUB 1787-1826 |
| HI496 | ADV IND STUDY IN HISTORY (3CR) |
| XH405 | THE HOLOCAUST AND ITS LEGACY   |
| XH415 | GENOCIDE AND ETHNIC CLEANSING  |
| XH491 | INTERDISCIPLINARY CAPSTNE-HIST |
| ZH315 | MODERN REGIONAL HISTORY        |
| ZH325 | TOPICS IN REGIONAL HISTORY     |

**AND****Complementary Support Courses**

Cadets in this major will fulfill the Complementary Support Course requirement by taking a third semester in a foreign language and two additional choices.

**AND****Complementary Support Course - Foreign Language**

Choose 1 of 1

LX300

3RD SEMESTER FOREIGN LANG

**AND****Complementary Support Courses - Electives**

Choose 2 of 72

Depending on their particular research interests, cadets will consult with their academic counselor to choose their final two CSCs. The courses found on the following list of courses could all aid in an historian's research abilities and methodologies, but a cadet may deviate from this list as long as they can justify why using a course not found on it will aid them in their historical study. In addition, a cadet may choose to fulfill this requirement by taking two additional courses in their chosen foreign language. This includes the DFL courses on regional civilization (example: LA483 Arab Civilization I), and regional literature (example: LF485 Survey of French Literature I).

|       |                              |
|-------|------------------------------|
| EN321 | AMERICAN LITERATURE I        |
| EN322 | AMERICAN LITERATURE II       |
| EN331 | BRITISH LITERATURE I         |
| EN332 | BRITISH LITERATURE II        |
| EN351 | WORLD LITERATURE             |
| EN353 | WAR LITERATURE               |
| EN355 | CRITICISM COLLOQUIUM         |
| EN362 | FILM AND FILM THEORY         |
| GE310 | GEOG INFORMATION SYSTEMS     |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE341 | GEOGRAPHY OF NORTH AMERICA   |
| GE342 | GEOGRAPHY OF ASIA            |
| GE343 | GEOGRAPHY OF RUSSIA          |
| GE344 | GEOGRAPHY OF AFRICA          |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST |
| GE346 | GEOGRAPHY OF EUROPE          |
| GE347 | GEOGRAPHY OF LATIN AMERICA   |
| GE352 | URBAN GEOGRAPHY              |
| GE378 | CARTOGRAPHY                  |
| GE491 | MILITARY GEOGRAPHY           |
| LW410 | COMPARATIVE LEGAL SYSTEMS    |



|       |                                   |
|-------|-----------------------------------|
| LW474 | LAW OF ARMED CONFLICT             |
| LW481 | INTERNATIONAL LAW                 |
| LW482 | NATIONAL SECURITY LAW             |
| LX400 | 4TH SEMESTER FOREIGN LANG         |
| MA376 | APPLIED STATISTICS                |
| PL360 | PSYCH ELITE PERFORMANCE           |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM       |
| PL471 | LEADERSHIP IN COMBAT              |
| PL482 | ARMED FORCES AND SOCIETY          |
| PY300 | PHILOSOPHICAL METHODS             |
| PY305 | LOGICAL REASONING                 |
| PY330 | POLITICAL PHILOSOPHY              |
| PY345 | PHILOSOPHY OF RELIGION            |
| PY360 | ANCIENT PHILOSOPHY                |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS    |
| PY370 | 17TH & 18TH CENTURY<br>PHILOSOPHY |
| PY375 | KANT & 19TH CENTURY<br>PHILOSOPHY |
| SS370 | MASS MEDIA & AMER POLITICS        |
| SS372 | POLITICS AND GOV OF CHINA         |
| SS373 | THE AMERICAN PRESIDENCY           |
| SS374 | POL & GOV OF KOREAS & JAPAN       |
| SS375 | GOV & POL RUSSIA & NEIGHBORS      |
| SS377 | POLITICS & GOV OF EUROPE          |
| SS378 | ADV INTL RELATIONS THEORY         |
| SS379 | LEGISLATIVE POLITICS              |
| SS380 | MANPOWER-LABOR ECONOMICS          |
| SS383 | POLITICS & GOVT-MIDDLE EAST       |
| SS384 | POLITICS & GOVT-LATIN AMER        |
| SS385 | HISTORY OF ECONOMICS              |
| SS386 | POLITICAL THOUGHT                 |
| SS387 | PUBLIC FINANCE                    |
| SS462 | ECON OF STABILIZATION &<br>GROWTH |
| SS464 | HOMELAND SECURITY                 |
| SS465 | TERRORISM: NEW CHALLENGES         |
| SS472 | THE AM STATE & THE SOLDIER        |
| SS473 | AMERICAN FOREIGN POLICY           |
| SS476 | CIVIL WARS                        |
| SS481 | POLITICS OF DEFENSE POLICY        |
| SS483 | NATIONAL SECURITY SEMINAR         |
| SS484 | INTERNATIONAL ECONOMICS           |
| SS485 | POLIT & DEV SUB-SAHARAN AFR       |
| SS487 | INT'L POLITICAL ECONOMY           |
| SS493 | SENIOR STUDIES - AMER POLITICS    |
| WS345 | MILITARY INNOVATION               |
| WS350 | STRATEGIC COMMUNICATION           |
| WS360 | SPECIAL OPS IN THEORY & PRACTI    |
| WS365 | REBELLN, INSURGNCY, & CIV WAR     |
| WS370 | US STRATEGY AND POLICY            |
| WS399 | INTERNSHIP IN WAR STUDIES         |
| WS455 | COMPARATIVE DEFENSE POLICY        |
| WS475 | FORECAST AND GAME IN<br>DECISION  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

Cadets majoring in History must complete an integrative experience during their second- or first-class year. Cadets will work with their DACs to identify in which course they intend to complete their integrative experience requirement. Once DACs verify with course directors that cadets have completed integrative experience requirements as planned, they will take administrative action to count that elective for integrative experience credit.

HX400 INTEGRATIVE EXPERIENCE IN HIST

**AND**

#### Science Depth

Cadets in this major may take any approved choice to satisfy the Science Depth requirement.

CH102 GENERAL CHEMISTRY II

CH275 BIOLOGY

PH202 PHYSICS II

PH252 ADVANCED PHYSICS II

PH275 PHYSICS II: SPACE

**AND**

#### STEM Depth

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305 CYBER FOUNDATIONS

CY355 CYBER FOUNDATIONS -  
COMPUTING

**AND**

#### Curriculum Requirements

This section describes how cadets in this major satisfy various curriculum requirements.

**AND**

#### IT/CYBER Requirement

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305 CYBER FOUNDATIONS

CY355 CYBER FOUNDATIONS -  
COMPUTING

**AND**

#### Core Engineering Sequence

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**AND**

#### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

HI498 COLLOQUIUM IN HISTORY

## 2029 History Major: Military w/ Thesis Curriculum

| Code  | Short Description           | Description                       | Transcript Description      | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------|-----------------------------------|-----------------------------|--------------|--------------|
| HMH3T | History: Military w/ Thesis | History Major: Military w/ Thesis | History: Military w/ Thesis | 1            | 0            |

## 2029 History Major: Military w/ Thesis Tracks

| Subject Area    | Description   |
|-----------------|---------------|
| Required Course | Choose 1 of 1 |

HI499

SENIOR THESIS

### 2029 History Major: Military w/ Thesis (Honors) Curriculum

| Code  | Short Description                    | Description                                | Transcript Description               | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------------|--------------------------------------------|--------------------------------------|--------------|--------------|
| HMH3H | History: Military w/ Thesis (Honors) | History Major: Military w/ Thesis (Honors) | History: Military w/ Thesis (Honors) | 1            | 1            |

### 2029 History Major: Military w/ Thesis (Honors) Tracks

| Subject Area             | Description                                                                                                                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Elective</b>          | Take one additional elective from the History Department catalogue. For cadets in the honors program at least one of their electives must be at the 400 level (HI4XX). This 400 level elective is in addition to HI498 and HI499. |
| <b>AND</b>               |                                                                                                                                                                                                                                   |
| <b>Required Course</b>   | Choose 1 of 1                                                                                                                                                                                                                     |
| HI499                    | SENIOR THESIS                                                                                                                                                                                                                     |
| <b>AND</b>               |                                                                                                                                                                                                                                   |
| <b>Grade Requirement</b> | Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major. Earn an A- in HI499.                                                   |

### 2029 History Major: United States Curriculum

| Code | Short Description            | Description                  | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|------------------------------|------------------------------|------------------------|--------------|--------------|
| HUS3 | History Major: United States | History Major: United States | History: United States | 13           | 0            |

### 2029 History Major: United States Tracks

| Subject Area            | Description                   |
|-------------------------|-------------------------------|
| <b>Required Courses</b> | Choose 2 of 2                 |
| HI301                   | HIST OF MILITARY ART TO 1900  |
| HI498                   | COLLOQUIUM IN HISTORY         |
| <b>AND</b>              |                               |
| <b>US History</b>       | Choose 5 of 14                |
| HI340                   | COLONIAL AMERICA              |
| HI369                   | AMERICAN FRONTIERS            |
| HI372                   | US FGN RELATIONS SINCE 1898   |
| HI387                   | TOPICS IN AMERICAN HISTORY    |
| HI392                   | AMERICAN HISTORICAL MEMORY    |
| HI393                   | AMERICA IN DEPRESSION AND WAR |

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|       |                                |
|-------|--------------------------------|
| HI394 | REVOLUTIONARY AMERICA          |
| HI395 | HIST OF CIVIL WAR AMERICA      |
| HI396 | THEODORE ROOSEVELT'S AMERICA   |
| HI397 | THE US FROM COLD WAR TO TODAY  |
| HI398 | CIVIL RIGHTS IN AMER HIST      |
| HI487 | EARLY AMERICAN REPUB 1787-1826 |
| HI496 | ADV IND STUDY IN HISTORY (3CR) |
| XH491 | INTERDISCIPLINARY CAPSTNE-HIST |

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**AND****Out-of-Stem History Electives**

Choose 2 of 39

For cadets who will graduate w/ Honors, one of their courses must be from the HI400 series and they will take an additional elective from all Department of History offerings.

|       |                                 |
|-------|---------------------------------|
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO  |
| HI338 | WARFARE IN AGE OF REVOLUTIONS   |
| HI339 | THE MODERN MIDDLE EAST          |
| HI341 | THE AGE OF EXPLORATION          |
| HI343 | MODERN GERMANY                  |
| HI344 | MODERN DIPLOMACY                |
| HI345 | MODERN AFRICA                   |
| HI346 | INDIA, PAKISTAN, & BANGLADESH   |
| HI347 | ASIAN WARFARE AND POLITICS      |
| HI348 | MODERN LATIN AMERICA            |
| HI355 | WARFARE, INDUSTRY, & EMPIRE     |
| HI356 | WAR AT SEA, SKY, SPACE          |
| HI357 | ARMED CONFLICT IN THE COLD WAR  |
| HI359 | ERA OF THE SECOND WORLD WAR     |
| HI361 | MEDIEVAL EUROPE                 |
| HI364 | MODERN WESTERN EUROPE           |
| HI367 | IMPERIAL AND SOVIET RUSSIA      |
| HI368 | MOD CENTRAL & E. EUR, 1896-1989 |
| HI370 | ANCIENT & MEDIEVAL WARFARE      |
| HI376 | EARLY MODERN WARFARE            |
| HI377 | EMPIRES & REVOLUTIONS           |
| HI378 | INDUSTRY, IDEOLOGY & THE STATE  |
| HI381 | HISTORY OF IRREGULAR WARFARE    |
| HI383 | MIDDLE EASTERN WARFARE          |
| HI388 | TOPICS IN INTERNATIONAL HISTORY |
| HI389 | TOPICS IN MILITARY HISTORY      |
| HI391 | WORLD RELIGIONS                 |
| HI460 | SENIOR FACULTY COURSE           |
| HI462 | THE HISTORY OF INNOVATION       |
| HI496 | ADV IND STUDY IN HISTORY (3CR)  |
| WS358 | STRATEGY AND GENERALSHIP        |
| WS385 | WAR AND ITS THEORISTS           |
| XH405 | THE HOLOCAUST AND ITS LEGACY    |
| XH415 | GENOCIDE AND ETHNIC CLEANSING   |
| XH491 | INTERDISCIPLINARY CAPSTNE-HIST  |
| ZH315 | MODERN REGIONAL HISTORY         |
| ZH325 | TOPICS IN REGIONAL HISTORY      |
| ZH335 | MILITARY HISTORY                |
| ZH345 | TOPICS IN MILITARY HISTORY      |

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**AND****Complementary Support Courses**

Cadets in this major will fulfill the Complementary Support Course requirement by taking a third semester in a foreign language and two additional choices.

**AND****Complementary Support Course - Foreign Language**

Choose 1 of 1

LX300

3RD SEMESTER FOREIGN LANG

**AND****Complementary Support Courses - Electives**

Choose 2 of 72

Depending on their particular research interests, cadets will consult with their academic counselor to choose their final two CSCs. The courses found on the following list of courses could all aid in an historian's research abilities and methodologies, but a cadet may deviate from this list as long as they can justify why using a course not found on it will aid them in their historical study. In addition, a cadet may choose to fulfill this requirement by taking two additional courses in their chosen foreign language. This includes the DFL courses on regional civilization (example: LA483 Arab Civilization I), and regional literature (example: LF485 Survey of French Literature I).

|       |                                |
|-------|--------------------------------|
| EN321 | AMERICAN LITERATURE I          |
| EN322 | AMERICAN LITERATURE II         |
| EN331 | BRITISH LITERATURE I           |
| EN332 | BRITISH LITERATURE II          |
| EN351 | WORLD LITERATURE               |
| EN353 | WAR LITERATURE                 |
| EN355 | CRITICISM COLLOQUIUM           |
| EN362 | FILM AND FILM THEORY           |
| GE310 | GEOG INFORMATION SYSTEMS       |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE341 | GEOGRAPHY OF NORTH AMERICA     |
| GE342 | GEOGRAPHY OF ASIA              |
| GE343 | GEOGRAPHY OF RUSSIA            |
| GE344 | GEOGRAPHY OF AFRICA            |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST   |
| GE346 | GEOGRAPHY OF EUROPE            |
| GE347 | GEOGRAPHY OF LATIN AMERICA     |
| GE352 | URBAN GEOGRAPHY                |
| GE378 | CARTOGRAPHY                    |
| GE491 | MILITARY GEOGRAPHY             |
| LW410 | COMPARATIVE LEGAL SYSTEMS      |
| LW474 | LAW OF ARMED CONFLICT          |
| LW481 | INTERNATIONAL LAW              |
| LW482 | NATIONAL SECURITY LAW          |
| LX400 | 4TH SEMESTER FOREIGN LANG      |
| MA376 | APPLIED STATISTICS             |
| PL360 | PSYCH ELITE PERFORMANCE        |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM    |
| PL471 | LEADERSHIP IN COMBAT           |
| PL482 | ARMED FORCES AND SOCIETY       |
| PY300 | PHILOSOPHICAL METHODS          |
| PY305 | LOGICAL REASONING              |
| PY330 | POLITICAL PHILOSOPHY           |
| PY345 | PHILOSOPHY OF RELIGION         |
| PY360 | ANCIENT PHILOSOPHY             |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| PY370 | 17TH & 18TH CENTURY PHILOSOPHY |
| PY375 | KANT & 19TH CENTURY PHILOSOPHY |
| SS370 | MASS MEDIA & AMER POLITICS     |
| SS372 | POLITICS AND GOV OF CHINA      |

|       |                                  |
|-------|----------------------------------|
| SS373 | THE AMERICAN PRESIDENCY          |
| SS374 | POL & GOV OF KOREAS & JAPAN      |
| SS375 | GOV & POL RUSSIA & NEIGHBORS     |
| SS377 | POLITICS & GOV OF EUROPE         |
| SS378 | ADV INTL RELATIONS THEORY        |
| SS379 | LEGISLATIVE POLITICS             |
| SS380 | MANPOWER-LABOR ECONOMICS         |
| SS383 | POLITICS & GOVT-MIDDLE EAST      |
| SS384 | POLITICS & GOVT-LATIN AMER       |
| SS385 | HISTORY OF ECONOMICS             |
| SS386 | POLITICAL THOUGHT                |
| SS387 | PUBLIC FINANCE                   |
| SS462 | ECON OF STABILIZATION & GROWTH   |
| SS464 | HOMELAND SECURITY                |
| SS465 | TERRORISM: NEW CHALLENGES        |
| SS472 | THE AM STATE & THE SOLDIER       |
| SS473 | AMERICAN FOREIGN POLICY          |
| SS476 | CIVIL WARS                       |
| SS481 | POLITICS OF DEFENSE POLICY       |
| SS483 | NATIONAL SECURITY SEMINAR        |
| SS484 | INTERNATIONAL ECONOMICS          |
| SS485 | GOV & POLITICS SUB-SAHARAN AFR   |
| SS487 | INT'L POLITICAL ECONOMY          |
| SS493 | SENIOR STUDIES - AMER POLITICS   |
| WS345 | MILITARY INNOVATION              |
| WS350 | STRATEGIC COMMUNICATION          |
| WS360 | SPECIAL OPS IN THEORY & PRACTICE |
| WS365 | REBELLN, INSURGNCY, & CIV WAR    |
| WS370 | US STRATEGY AND POLICY           |
| WS399 | INTERNSHIP IN WAR STUDIES        |
| WS455 | COMPARATIVE DEFENSE POLICY       |
| WS475 | FORECAST AND GAME IN DECISION    |

**AND****Integrative Experience for the Major**

Choose 1 of 1

Cadets majoring in History must complete an integrative experience during their second- or first-class year. Cadets will work with their DACs to identify in which course they intend to complete their integrative experience requirement. Once DACs verify with course directors that cadets have completed integrative experience requirements as planned, they will take administrative action to count that elective for integrative experience credit.

HX400

INTEGRATIVE EXPERIENCE IN HIST

**AND****Science Depth**

Cadets in this major may take any approved choice to satisfy the Science Depth requirement.

CH102

GENERAL CHEMISTRY II

CH275

BIOLOGY

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**AND****STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS -  
COMPUTING

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**AND****IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS -  
COMPUTING**AND****Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**AND****Writing-in-the-Major**

Cadets in this major will satisfy the Writing Across the Disciplines requirement by successfully completing the writing requirements in the core curriculum and several history courses including HI498.

HI498

COLLOQUIUM IN HISTORY

**2029 History Major: United States w/ Thesis Curriculum**

| Code  | Short Description                | Description                            | Transcript Description           | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|----------------------------------|----------------------------------------|----------------------------------|-----------------|-----------------|
| HUS3T | History: United States w/ Thesis | History Major: United States w/ Thesis | History: United States w/ Thesis | 1               | 0               |

**2029 History Major: United States w/ Thesis Tracks**

| Subject Area           | Description   |
|------------------------|---------------|
| <b>Required Course</b> | Choose 1 of 1 |
| HI499                  | SENIOR THESIS |

**2029 History Major: United States w/ Thesis (Honors) Curriculum**

| Code  | Short Description                         | Description                                     | Transcript Description                    | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|-------------------------------------------|-------------------------------------------------|-------------------------------------------|-----------------|-----------------|
| HUS3H | History: United States w/ Thesis (Honors) | History Major: United States w/ Thesis (Honors) | History: United States w/ Thesis (Honors) | 1               | 1               |

**2029 History Major: United States w/ Thesis (Honors) Tracks**

| Subject Area    | Description |
|-----------------|-------------|
| <b>Elective</b> |             |

Take one additional elective from the History Department catalogue. For cadets in the honors program at least one of their electives must be at the 400 level (HI4XX). This 400 level elective is in addition to HI498 and HI499.

**AND****Required Course**

Choose 1 of 1

HI499

SENIOR THESIS

**AND****Grade Requirement**

Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major. Earn an A- in HI499.

## 2029 War Studies Major Curriculum

| Code | Short Description | Description       | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------|-------------------|------------------------|--------------|--------------|
| WST0 | War Studies Major | War Studies Major | War Studies            | 4            | 9            |

## 2029 War Studies Major Tracks

| Subject Area                                                                                                                                                                                                                                                     | Description                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Cornerstone Course</b>                                                                                                                                                                                                                                        | Choose 1 of 1                  |
| WS301                                                                                                                                                                                                                                                            | INTRODUCTION TO WAR STUDIES    |
| <b>AND</b>                                                                                                                                                                                                                                                       |                                |
| <b>Foundational Courses</b>                                                                                                                                                                                                                                      |                                |
| Cadets in this major will take a total of three foundational courses to develop an understanding of the field in three frames: past, present, and future.                                                                                                        |                                |
| <b>Foundation - Past</b>                                                                                                                                                                                                                                         | Choose 1 of 2                  |
| WS358                                                                                                                                                                                                                                                            | STRATEGY AND GENERALSHIP       |
| WS385                                                                                                                                                                                                                                                            | WAR AND ITS THEORISTS          |
| <b>AND</b>                                                                                                                                                                                                                                                       |                                |
| <b>Foundation - Present</b>                                                                                                                                                                                                                                      | Choose 1 of 1                  |
| WS370                                                                                                                                                                                                                                                            | US STRATEGY AND POLICY         |
| <b>AND</b>                                                                                                                                                                                                                                                       |                                |
| <b>Foundation - Future</b>                                                                                                                                                                                                                                       | Choose 1 of 1                  |
| WS398                                                                                                                                                                                                                                                            | FUTURE WAR                     |
| <b>AND</b>                                                                                                                                                                                                                                                       |                                |
| <b>Concentration</b>                                                                                                                                                                                                                                             |                                |
| Cadets in this major will select one concentration from a choice of three. Within their concentration, cadets will select a total of five electives. Cadets must include courses from at least two different disciplines among their five concentration courses. |                                |
| <b>Warfare and Operations</b>                                                                                                                                                                                                                                    | Choose 5 of 26                 |
| CY460                                                                                                                                                                                                                                                            | CYBER POLICY, STRATEGY, & OPNS |
| GE491                                                                                                                                                                                                                                                            | MILITARY GEOGRAPHY             |
| HI301                                                                                                                                                                                                                                                            | HIST OF MILITARY ART TO 1900   |
| HI338                                                                                                                                                                                                                                                            | WARFARE IN AGE OF REVOLUTIONS  |
| HI355                                                                                                                                                                                                                                                            | WARFARE, INDUSTRY, & EMPIRE    |
| HI356                                                                                                                                                                                                                                                            | WAR AT SEA, SKY, SPACE         |
| HI357                                                                                                                                                                                                                                                            | ARMED CONFLICT IN THE COLD WAR |



|                          |                                          |
|--------------------------|------------------------------------------|
| HI359                    | ERA OF THE SECOND WORLD WAR              |
| HI370                    | ANCIENT & MEDIEVAL WARFARE               |
| HI376                    | EARLY MODERN WARFARE                     |
| HI381                    | HISTORY OF IRREGULAR WARFARE             |
| HI383                    | MIDDLE EASTERN WARFARE                   |
| LW474                    | LAW OF ARMED CONFLICT                    |
| LW476                    | ADVANCED LAW OF ARMED CONFLICT           |
| PL471                    | LEADERSHIP IN COMBAT                     |
| PY325                    | MILITARY ETHICS                          |
| SS466                    | COMBATING TERRORISM                      |
| WS345                    | MILITARY INNOVATION                      |
| WS358                    | STRATEGY AND GENERALSHIP                 |
| WS360                    | SPECIAL OPS IN THEORY & PRACTICE         |
| WS365                    | REBELLION, INSURGENCE, & CIVIL WAR       |
| WS385                    | WAR AND ITS THEORISTS                    |
| WS389                    | WAR IN THE CONTEMPORARY WORLD            |
| WS485                    | JOINT AND MULTINATIONAL OPS              |
| WS489                    | INDIVIDUAL STUDY IN WAR STUDIES          |
| WS490                    | SPECIAL TOPICS IN WAR STUDIES            |
| <b>OR</b>                |                                          |
| <b>Strategic Studies</b> | Choose 5 of 33                           |
| CY460                    | CYBER POLICY, STRATEGY, & OPERATIONS     |
| HI301                    | HISTORY OF MILITARY ART TO 1900          |
| HI344                    | MODERN DIPLOMACY                         |
| HI347                    | ASIAN WARFARE AND POLITICS               |
| HI372                    | US FOREIGN RELATIONS SINCE 1898          |
| HI378                    | INDUSTRY, IDEOLOGY & THE STATE           |
| LW474                    | LAW OF ARMED CONFLICT                    |
| LW476                    | ADVANCED LAW OF ARMED CONFLICT           |
| LW481                    | INTERNATIONAL LAW                        |
| LW482                    | NATIONAL SECURITY LAW                    |
| LX476                    | MILITARY STRATEGIC/DOCTRINE IN WORLD WAR |
| SE385                    | DECISION ANALYSIS                        |
| SS366                    | COMPARATIVE POLITICS                     |
| SS370                    | MASS MEDIA & AMERICAN POLITICS           |
| SS395                    | INTERNATIONAL SECURITY                   |
| SS466                    | COMBATING TERRORISM                      |
| SS473                    | AMERICAN FOREIGN POLICY                  |
| SS476                    | CIVIL WARS                               |
| SS486                    | STATE BUILDING                           |
| WS330                    | INTRODUCTION TO GRAND STRATEGY           |
| WS335                    | TOPICS IN GRAND STRATEGY                 |
| WS345                    | MILITARY INNOVATION                      |
| WS350                    | STRATEGIC COMMUNICATION                  |
| WS360                    | SPECIAL OPS IN THEORY & PRACTICE         |
| WS365                    | REBELLION, INSURGENCE, & CIVIL WAR       |
| WS389                    | WAR IN THE CONTEMPORARY WORLD            |
| WS455                    | COMPARATIVE DEFENSE POLICY               |
| WS475                    | FORECAST AND GAME IN DECISION            |
| WS485                    | JOINT AND MULTINATIONAL OPS              |
| WS489                    | INDIVIDUAL STUDY IN WAR STUDIES          |
| WS490                    | SPECIAL TOPICS IN WAR STUDIES            |

|                                       |                                |
|---------------------------------------|--------------------------------|
| <b>OR</b>                             |                                |
| <b>War, Society, and Institutions</b> | Choose 5 of 33                 |
| EN353                                 | WAR LITERATURE                 |
| GE340                                 | GEOGRAPHY OF GLOBAL CULTURES   |
| HI301                                 | HIST OF MILITARY ART TO 1900   |
| HI338                                 | WARFARE IN AGE OF REVOLUTIONS  |
| HI347                                 | ASIAN WARFARE AND POLITICS     |
| HI355                                 | WARFARE, INDUSTRY, & EMPIRE    |
| HI359                                 | ERA OF THE SECOND WORLD WAR    |
| HI370                                 | ANCIENT & MEDIEVAL WARFARE     |
| HI376                                 | EARLY MODERN WARFARE           |
| HI383                                 | MIDDLE EASTERN WARFARE         |
| HI393                                 | AMERICA IN DEPRESSION AND WAR  |
| HI394                                 | REVOLUTIONARY AMERICA          |
| HI395                                 | HIST OF CIVIL WAR AMERICA      |
| HI396                                 | THEODORE ROOSEVELT'S AMERICA   |
| LW474                                 | LAW OF ARMED CONFLICT          |
| LW482                                 | NATIONAL SECURITY LAW          |
| LX476                                 | MILITARY SPKG/RDG IN WORLD LAN |
| PL482                                 | ARMED FORCES AND SOCIETY       |
| PY325                                 | MILITARY ETHICS                |
| SS464                                 | HOMELAND SECURITY              |
| SS465                                 | TERRORISM: NEW CHALLENGES      |
| SS472                                 | SOLDIER & STATE: AM CIV-MIL RE |
| SS481                                 | POLITICS OF DEFENSE POLICY     |
| SS486                                 | STATE BUILDING                 |
| SS487                                 | INT'L POLITICAL ECONOMY        |
| WS330                                 | INTRODUCTION TO GRAND STRATEGY |
| WS365                                 | REBELLN, INSURGENCY, & CIV WAR |
| WS389                                 | WAR IN THE CONTEMPORARY WORLD  |
| WS455                                 | COMPARATIVE DEFENSE POLICY     |
| WS489                                 | IND STUDY IN WAR STUDIES       |
| WS490                                 | SPECIAL TOPICS IN WAR STUDIES  |
| XH405                                 | THE HOLOCAUST AND ITS LEGACY   |
| XH415                                 | GENOCIDE AND MASS ATROCITY     |
| <b>AND</b>                            |                                |

### Complementary Support Courses

Cadets in this major will select one specialization from a choice of six. Within their specialization, cadets will take three complementary support courses. All pre-requisites must be completed prior to taking a course from this list.

|                                      |                                |
|--------------------------------------|--------------------------------|
| <b>Regional and Cultural Studies</b> | Choose 3 of 35                 |
| GE305                                | GEOGRAPHIC RESEARCH METHODS    |
| GE340                                | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342                                | GEOGRAPHY OF ASIA              |
| GE343                                | GEOGRAPHY OF RUSSIA            |
| GE344                                | GEOGRAPHY OF AFRICA            |
| GE345                                | GEOGRAPHY OF THE MIDDLE EAST   |
| GE346                                | GEOGRAPHY OF EUROPE            |
| GE347                                | GEOGRAPHY OF LATIN AMERICA     |
| HI337                                | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI339                                | THE MODERN MIDDLE EAST         |
| HI343                                | MODERN GERMANY                 |

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|       |                                   |
|-------|-----------------------------------|
| HI345 | MODERN AFRICA                     |
| HI346 | INDIA, PAKISTAN, & BANGLADESH     |
| HI347 | ASIAN WARFARE AND POLITICS        |
| HI348 | MODERN LATIN AMERICA              |
| HI361 | MEDIEVAL EUROPE                   |
| HI364 | MODERN WESTERN EUROPE             |
| HI367 | IMPERIAL AND SOVIET RUSSIA        |
| HI368 | MOD CENTRAL & E. EUR,1896-1989    |
| HI383 | MIDDLE EASTERN WARFARE            |
| HI391 | WORLD RELIGIONS                   |
| LW410 | COMPARATIVE LEGAL SYSTEMS         |
| LX300 | 3RD SEMESTER FOREIGN LANG         |
| LX400 | 4TH SEMESTER FOREIGN LANG         |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS    |
| PY375 | KANT & 19TH CENTURY<br>PHILOSOPHY |
| SS372 | POLITICS OF CHINA                 |
| SS374 | POLITICS OF NORTHEAST ASIA        |
| SS375 | POL OF THE POST-SOVIET STATES     |
| SS377 | POLITICS OF EUROPE                |
| SS383 | POLITICS OF THE MIDDLE EAST       |
| SS384 | POLITICS OF LATIN AMERICA         |
| SS485 | POLITICS SUB-SAHARAN AFR          |

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**OR****American Society and Institutions**

Choose 3 of 16

|       |                                  |
|-------|----------------------------------|
| HI340 | COLONIAL AMERICA                 |
| HI369 | AMERICAN FRONTIERS               |
| HI392 | AMERICAN HISTORICAL MEMORY       |
| HI397 | THE US FROM COLD WAR TO<br>TODAY |
| HI487 | EARLY AMERICAN REPUB 1787-1826   |
| LW475 | ADV CONSTITUTIONAL LAW SEM       |
| LW482 | NATIONAL SECURITY LAW            |
| PL482 | ARMED FORCES AND SOCIETY         |
| SS370 | MASS MEDIA & AMER POLITICS       |
| SS373 | THE AMERICAN PRESIDENCY          |
| SS379 | UNITED STATES CONGRESS           |
| SS464 | HOMELAND SECURITY                |
| SS468 | POLITICAL PARTICIPATION          |
| SS472 | SOLDIER & STATE: AM CIV-MIL RE   |
| SS473 | AMERICAN FOREIGN POLICY          |

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**OR****Adaptation and Innovation**

Choose 3 of 11

|       |                                   |
|-------|-----------------------------------|
| CS486 | ARTIFICIAL INTELLIGENCE           |
| CY460 | CYBER POLICY, STRATEGY, & OPNS    |
| GE478 | MILITARY GEOSPATIAL<br>OPERATIONS |
| MG463 | AGILE INNOVATION IN DEFENSE       |
| PH456 | SCIENCE AND POLICY                |
| PL398 | LEADERSHIP THEORY & PRACTICE      |
| PL479 | LEADING CHANGING<br>ORGANIZATIONS |
| SE302 | FUNDAMENTALS OF SYSTEMS ENG       |
| SE370 | COMPUTER AIDED SYSTEMS ENG        |
| SE385 | DECISION ANALYSIS                 |
| XE492 | DISRUPTIVE INNOVATIONS            |

---

**OR**

|                                                                                                         |                                |
|---------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Conflict Analysis and Resolution</b>                                                                 | Choose 3 of 16                 |
| CE350                                                                                                   | INFRASTRUCTURE ENGINEERING     |
| GE494                                                                                                   | ENVIRONMENTAL SECURITY         |
| LW462                                                                                                   | CYBER LAW                      |
| LW474                                                                                                   | LAW OF ARMED CONFLICT          |
| LW476                                                                                                   | ADVANCED LAW OF ARMED CONFLICT |
| LW481                                                                                                   | INTERNATIONAL LAW              |
| MG390                                                                                                   | NEGOTIATION FOR LEADERS        |
| PL379                                                                                                   | GROUP DYNAMICS                 |
| PL383                                                                                                   | SOCIAL PSYCHOLOGY              |
| PY325                                                                                                   | MILITARY ETHICS                |
| PY326                                                                                                   | ETHICS OF TECHNOLOGY           |
| SS360                                                                                                   | POL SCI RESEARCH METHODS       |
| SS476                                                                                                   | CIVIL WARS                     |
| SS486                                                                                                   | STATE BUILDING                 |
| XH405                                                                                                   | THE HOLOCAUST AND ITS LEGACY   |
| XH415                                                                                                   | GENOCIDE AND MASS ATROCITY     |
| <b>OR</b>                                                                                               |                                |
| <b>Unconventional and Gray-Zone Warfare</b>                                                             | Choose 3 of 14                 |
| CY450                                                                                                   | CYBER SECURITY ENGINEERING     |
| CY460                                                                                                   | CYBER POLICY, STRATEGY, & OPNS |
| HI357                                                                                                   | ARMED CONFLICT IN THE COLD WAR |
| HI377                                                                                                   | EMPIRES & REVOLUTIONS          |
| HI381                                                                                                   | HISTORY OF IRREGULAR WARFARE   |
| LW462                                                                                                   | CYBER LAW                      |
| PY325                                                                                                   | MILITARY ETHICS                |
| PY326                                                                                                   | ETHICS OF TECHNOLOGY           |
| SS464                                                                                                   | HOMELAND SECURITY              |
| SS465                                                                                                   | TERRORISM: NEW CHALLENGES      |
| SS466                                                                                                   | COMBATING TERRORISM            |
| SS476                                                                                                   | CIVIL WARS                     |
| SS486                                                                                                   | STATE BUILDING                 |
| XH415                                                                                                   | GENOCIDE AND MASS ATROCITY     |
| <b>OR</b>                                                                                               |                                |
| <b>Data Analysis and Modeling</b>                                                                       | Choose 3 of 7                  |
| MA367                                                                                                   | MATH FOR THE SOCIAL SCIENCES   |
| MA376                                                                                                   | APPLIED STATISTICS             |
| SE370                                                                                                   | COMPUTER AIDED SYSTEMS ENG     |
| SE385                                                                                                   | DECISION ANALYSIS              |
| SE485                                                                                                   | COMBAT MODELING                |
| SM484                                                                                                   | SYSTEM DYNAMICS SIMULATION     |
| SS364                                                                                                   | GAME THEORY                    |
| <b>AND</b>                                                                                              |                                |
| <b>Capstone Requirement - Integrative Experience</b>                                                    | Choose 1 of 1                  |
| WS495                                                                                                   | METHODS & CAPSTONE WAR STUDIES |
| <b>AND</b>                                                                                              |                                |
| <b>Science Depth</b>                                                                                    |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement. |                                |
| CH102                                                                                                   | GENERAL CHEMISTRY II           |
| CH275                                                                                                   | BIOLOGY                        |
| PH202                                                                                                   | PHYSICS II                     |

|                                                                                                                                                                                                   |                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| PH252                                                                                                                                                                                             | ADVANCED PHYSICS II              |
| PH275                                                                                                                                                                                             | PHYSICS II: SPACE                |
| <b>STEM Depth</b>                                                                                                                                                                                 |                                  |
| Cadets in this major must take CY305/CY355 to satisfy the STEM Depth requirement.                                                                                                                 |                                  |
| CY305                                                                                                                                                                                             | CYBER FOUNDATIONS                |
| CY355                                                                                                                                                                                             | CYBER FOUNDATIONS -<br>COMPUTING |
| <b>AND</b>                                                                                                                                                                                        |                                  |
| <b>Curriculum Requirements</b>                                                                                                                                                                    |                                  |
| This section describes how cadets in this major will satisfy various curriculum requirements.                                                                                                     |                                  |
| <b>IT/CYBER Requirement</b>                                                                                                                                                                       |                                  |
| Cadets in this major will satisfy the IT/CYBER requirement through CY305/355.                                                                                                                     |                                  |
| CY305                                                                                                                                                                                             | CYBER FOUNDATIONS                |
| CY355                                                                                                                                                                                             | CYBER FOUNDATIONS -<br>COMPUTING |
| <b>Core Engineering Sequence</b>                                                                                                                                                                  |                                  |
| Cadets in this major may choose any approved three-course Engineering Sequence.                                                                                                                   |                                  |
| <b>Writing-in-the-Major</b>                                                                                                                                                                       |                                  |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and one of the following two courses in the major. |                                  |
| WS358                                                                                                                                                                                             | STRATEGY AND GENERALSHIP         |
| WS385                                                                                                                                                                                             | WAR AND ITS THEORISTS            |

## 2029 War Studies Major w/ Honors Curriculum

| Code  | Short Description           | Description                 | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|-----------------------------|-----------------------------|------------------------|-----------------|-----------------|
| WST0H | War Studies Major w/ Honors | War Studies Major w/ Honors | War Studies w/ Honors  | 0               | 2               |

## 2029 War Studies Major w/ Honors Tracks

| Subject Area                                                                                                                                                         | Description                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Elective</b>                                                                                                                                                      |                             |
| Take one additional elective, selecting from any of the major's three concentration tables.                                                                          |                             |
| <b>AND</b>                                                                                                                                                           |                             |
| <b>Research Requirement</b>                                                                                                                                          | Choose 1 of 2               |
| In addition to the elective noted above, cadets majoring in War Studies (Honors) must take one of the courses listed below, for a total of 42 courses at graduation. |                             |
| WS496                                                                                                                                                                | THESIS IN WAR STUDIES       |
| WS497                                                                                                                                                                | CLIENT-BASED CAPSTONE IN WS |
| <b>AND</b>                                                                                                                                                           |                             |
| <b>Grade Requirement</b>                                                                                                                                             |                             |
| To qualify for honors distinction, cadets must earn at least a 3.0 APSC and a 3.5 APS within the major, as well as earn an A- or better in WS496 or WS497.           |                             |

2029 War Studies w/ Thesis Major Curriculum

| Code  | Short Description           | Description                 | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------|-----------------------------|------------------------|--------------|--------------|
| WST0T | War Studies w/ Thesis Major | War Studies w/ Thesis Major | War Studies w/ Thesis  | 0            | 1            |

2029 War Studies w/ Thesis Major Tracks

| Subject Area                                                                                                                                                                            | Description                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Research Requirement                                                                                                                                                                    | Choose 1 of 2               |
| Cadets majoring in War Studies with Thesis or Client-Based Capstone must take one additional course above the base major, from the list below, for a total of 41 courses at graduation. |                             |
| WS496                                                                                                                                                                                   | THESIS IN WAR STUDIES       |
| WS497                                                                                                                                                                                   | CLIENT-BASED CAPSTONE IN WS |

## Kinesiology

### 2029 Kinesiology Major Curriculum

| Code | Short Description | Description       | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------|-------------------|------------------------|-----------------|-----------------|
| KIN1 | Kinesiology       | Kinesiology Major | Kinesiology            | 10              | 3               |

### 2029 Kinesiology Major Tracks

| Subject Area                                                                    | Description                    |
|---------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                         | Choose 9 of 9                  |
| CH275                                                                           | BIOLOGY                        |
| CH387                                                                           | HUMAN PHYSIOLOGY               |
| KN355                                                                           | FUNCTIONAL ANATOMY             |
| KN360                                                                           | BIOMECHANICS OF HUMAN MOVEMENT |
| KN365                                                                           | NUTRITION FOR PERFORMANCE      |
| KN455                                                                           | PSYCHOLOGY OF EXERCISE         |
| KN460                                                                           | EXERCISE PHYSIOLOGY            |
| KN470                                                                           | FITNESS ASSESSMENT AND RX      |
| KN475                                                                           | MUSCULAR FUNCTION & ADAPTATION |
| <b>AND</b>                                                                      |                                |
| <b>Complementary Support Courses</b>                                            | Choose 3 of 11                 |
| CH383                                                                           | ORGANIC CHEMISTRY I            |
| CH384                                                                           | ORGANIC CHEMISTRY II           |
| CH385                                                                           | INTRODUCTION TO CELL BIOLOGY   |
| CH460                                                                           | HUMAN ANATOMY                  |
| CH473                                                                           | BIOCHEMISTRY                   |
| PL250                                                                           | NEUROCOG FNDTNS OF BEHAVIOR    |
| PL361                                                                           | RESEARCH METHODS I             |
| PL376                                                                           | PSYCHOPATHOLOGY                |
| PL383                                                                           | SOCIAL PSYCHOLOGY              |
| PL387                                                                           | FOUNDATIONS OF COUNSELING      |
| PL390                                                                           | BIOLOGICAL PSYCHOLOGY          |
| PL392                                                                           | COGNITIVE PSYCHOLOGY           |
| PL394                                                                           | ANTHROPOMETRICS & BIOMECHANICS |
| PY350                                                                           | PHILOSOPHY OF SCIENCE          |
| <b>AND</b>                                                                      |                                |
| <b>Integrative Experience for the Major</b>                                     | Choose 1 of 1                  |
| KN480                                                                           | T/P OF ADVANCED PERFORMANCE    |
| <b>AND</b>                                                                      |                                |
| <b>Science Depth</b>                                                            |                                |
| CH102                                                                           | GENERAL CHEMISTRY II           |
| <b>AND</b>                                                                      |                                |
| <b>STEM Depth</b>                                                               |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement. |                                |
| CY305                                                                           | CYBER FOUNDATIONS              |
| CY355                                                                           | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                      |                                |
| <b>Curriculum Requirements</b>                                                  |                                |

This section describes how cadets in this major satisfy various curriculum requirements.

#### IT/CYBER Requirement

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

#### Core Engineering Sequence

Cadets in this major can choose any approved three-course Core Engineering Sequence.

#### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

KN470

FITNESS ASSESSMENT AND RX

### 2029 Kinesiology Major w/ Honors Curriculum

| Code  | Short Description     | Description                 | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|-----------------------|-----------------------------|------------------------|-----------------|-----------------|
| KIN1H | Kinesiology w/ Honors | Kinesiology Major w/ Honors | Kinesiology w/ Honors  | 2               | 0               |

### 2029 Kinesiology Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                                            | Description        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Required Courses</b>                                                                                                                                                                                                 | Choose 2 of 2      |
| *If PL361 is chosen to represent 1 of the 3 required Complementary Support Courses, an additional elective approved by the DPE must be taken along with KN495 to fulfill the Major with Honors curriculum requirements. |                    |
| KN495                                                                                                                                                                                                                   | HONORS THESIS      |
| PL361                                                                                                                                                                                                                   | RESEARCH METHODS I |
| <b>AND</b>                                                                                                                                                                                                              |                    |

Complete the requirements of the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.



## Law & Philosophy

### 2029 Law and Legal Studies Major Curriculum

| Code | Short Description     | Description                 | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-----------------------|-----------------------------|------------------------|-----------------|-----------------|
| LLS3 | Law and Legal Studies | Law and Legal Studies Major | Law and Legal Studies  | 3               | 10              |

### 2029 Law and Legal Studies Major Tracks

| Subject Area                                                   | Description                       |
|----------------------------------------------------------------|-----------------------------------|
| <b>Required Courses</b>                                        | Choose 2 of 2                     |
| LW310                                                          | INTRO TO LEGAL METHOD             |
| LW474                                                          | LAW OF ARMED CONFLICT             |
| <b>AND</b>                                                     |                                   |
| <b>Law Electives</b>                                           | Choose 7 of 14                    |
| LW410                                                          | COMPARATIVE LEGAL SYSTEMS         |
| LW461                                                          | CIVIL RIGHTS AND THE LAW          |
| LW462                                                          | CYBER LAW                         |
| LW472                                                          | CRIMINAL LAW                      |
| LW473                                                          | ENVIRONMENTAL LAW                 |
| LW475                                                          | ADV CONSTITUTIONAL LAW SEM        |
| LW476                                                          | ADVANCED LAW OF ARMED<br>CONFLICT |
| LW477                                                          | MILITARY JUSTICE: FOUNDATIONS     |
| LW481                                                          | INTERNATIONAL LAW                 |
| LW482                                                          | NATIONAL SECURITY LAW             |
| LW488                                                          | BUSINESS LAW                      |
| LW490                                                          | SPECIAL TOPICS IN THE LAW         |
| LW490A                                                         | SPECIAL TOPICS IN THE LAW - A     |
| LW490B                                                         | SPECIAL TOPICS IN THE LAW - B     |
| <b>AND</b>                                                     |                                   |
| <b>Complementary Support Courses</b>                           |                                   |
| You must select one of the following two specialty law tracks. |                                   |
| <b>CSC - International Law and Legal<br/>Systems</b>           | Choose 3 of 18                    |
| GE340                                                          | GEOGRAPHY OF GLOBAL CULTURES      |
| GE342                                                          | GEOGRAPHY OF ASIA                 |
| GE343                                                          | GEOGRAPHY OF RUSSIA               |
| GE344                                                          | GEOGRAPHY OF AFRICA               |
| GE345                                                          | GEOGRAPHY OF THE MIDDLE EAST      |
| GE346                                                          | GEOGRAPHY OF EUROPE               |
| GE347                                                          | GEOGRAPHY OF LATIN AMERICA        |
| HI344                                                          | MODERN DIPLOMACY                  |
| HI372                                                          | US FGN RELATIONS SINCE 1898       |
| HI391                                                          | WORLD RELIGIONS                   |
| MG390                                                          | NEGOTIATION FOR LEADERS           |
| SS366                                                          | COMPARATIVE POLITICS              |
| SS385                                                          | HISTORY OF ECONOMICS              |
| SS465                                                          | TERRORISM: NEW CHALLENGES         |
| SS466                                                          | COMBATING TERRORISM               |

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| SS473                                                                                                                                                                        | AMERICAN FOREIGN POLICY        |
| SS483                                                                                                                                                                        | NATIONAL SECURITY SEMINAR      |
| SS486                                                                                                                                                                        | STATE BUILDING                 |
| <b>OR</b>                                                                                                                                                                    |                                |
| <b>CSC - American Law and Society</b>                                                                                                                                        | Choose 3 of 15                 |
| HI394                                                                                                                                                                        | REVOLUTIONARY AMERICA          |
| HI395                                                                                                                                                                        | HIST OF CIVIL WAR AMERICA      |
| HI396                                                                                                                                                                        | THEODORE ROOSEVELT'S AMERICA   |
| HI398                                                                                                                                                                        | CIVIL RIGHTS IN AMER HIST      |
| HI487                                                                                                                                                                        | EARLY AMERICAN REPUB 1787-1826 |
| PL372                                                                                                                                                                        | MILITARY FAMILIES              |
| PL393                                                                                                                                                                        | CRIMINOLOGY-CRIM JUST SYSTM    |
| PY305                                                                                                                                                                        | LOGICAL REASONING              |
| PY330                                                                                                                                                                        | POLITICAL PHILOSOPHY           |
| SS373                                                                                                                                                                        | THE AMERICAN PRESIDENCY        |
| SS379                                                                                                                                                                        | LEGISLATIVE POLITICS           |
| SS386                                                                                                                                                                        | POLITICAL THOUGHT              |
| SS464                                                                                                                                                                        | HOMELAND SECURITY              |
| SS472                                                                                                                                                                        | THE AM STATE & THE SOLDIER     |
| SS483                                                                                                                                                                        | NATIONAL SECURITY SEMINAR      |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                  |
| LW495                                                                                                                                                                        | JURISPRUDENCE AND LEGAL THEORY |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may take any approved choice to satisfy the Science Depth requirement.                                                                                  |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Curriculum Requirements</b>                                                                                                                                               |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major can choose any approved three-course Core Engineering Sequence.                                                                                         |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| LW495                                                                                                                                                                        | JURISPRUDENCE AND LEGAL THEORY |

### 2029 Law and Legal Studies Major w/ Honors Curriculum

| Code  | Short Description               | Description                           | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------|---------------------------------------|---------------------------------|--------------|--------------|
| LLS3H | Law and Legal Studies w/ Honors | Law and Legal Studies Major w/ Honors | Law and Legal Studies w/ Honors | 2            | 0            |

### 2029 Law and Legal Studies Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                               | Description                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Required Courses</b>                                                                                                                                                                                    | Choose 2 of 2                 |
| LW498                                                                                                                                                                                                      | THESIS I: PROPOSAL & RESEARCH |
| LW499                                                                                                                                                                                                      | THESIS II: PAPER & DEFENSE    |
| <b>AND</b>                                                                                                                                                                                                 |                               |
| <b>Grade Requirements</b>                                                                                                                                                                                  |                               |
| Complete the requirements of the major as shown above, attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major, and an average of at least 3.33 in LW498 and LW499. |                               |

### 2029 Philosophy Major Curriculum

| Code | Short Description | Description      | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------|------------------|------------------------|--------------|--------------|
| PYL2 | Philosophy Major  | Philosophy Major | Philosophy             | 4            | 9            |

### 2029 Philosophy Major Tracks

| Subject Area                       | Description            |
|------------------------------------|------------------------|
| <b>Required Courses</b>            | Choose 2 of 2          |
| PY300                              | PHILOSOPHICAL METHODS  |
| PY305                              | LOGICAL REASONING      |
| <b>AND</b>                         |                        |
| <b>Philosophy Core Area Course</b> | Choose 1 of 2          |
| PY310                              | REALITY AND KNOWLEDGE  |
| PY320                              | ETHICS                 |
| <b>AND</b>                         |                        |
| <b>Philosophy Electives</b>        | Choose 6 of 16         |
| PY325                              | MILITARY ETHICS        |
| PY326                              | ETHICS OF TECHNOLOGY   |
| PY329                              | TOPICS IN ETHICS       |
| PY330                              | POLITICAL PHILOSOPHY   |
| PY345                              | PHILOSOPHY OF RELIGION |

|       |                                |
|-------|--------------------------------|
| PY350 | PHILOSOPHY OF SCIENCE          |
| PY355 | PHILOSOPHY OF MIND             |
| PY360 | ANCIENT PHILOSOPHY             |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |
| PY370 | 17TH & 18TH CENTURY PHILOSOPHY |
| PY375 | KANT & 19TH CENTURY PHILOSOPHY |
| PY380 | 20TH CENTURY PHILOSOPHY        |
| PY390 | INTER-DEPARTMENT SEMINAR       |
| PY395 | SPECIAL TOPICS IN PHILOSOPHY   |
| PY495 | INDEPENDENT STUDY: PHILOSOPHY  |
| XH400 | INTERCOLLEGIATE SEMINAR        |

**AND****Complementary Support Courses**

Cadets chose a track and then pick three courses from the track. Cadets may pick outside of their chosen track with Program Director approval.

**History Track**

Choose 3 of 8

|       |                                |
|-------|--------------------------------|
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI339 | THE MODERN MIDDLE EAST         |
| HI345 | MODERN AFRICA                  |
| HI346 | INDIA, PAKISTAN, & BANGLADESH  |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI364 | MODERN WESTERN EUROPE          |
| HI367 | IMPERIAL AND SOVIET RUSSIA     |
| HI392 | AMERICAN HISTORICAL MEMORY     |

**OR****Law Track**

Choose 3 of 6

|       |                           |
|-------|---------------------------|
| HI344 | MODERN DIPLOMACY          |
| LW410 | COMPARATIVE LEGAL SYSTEMS |
| LW461 | CIVIL RIGHTS AND THE LAW  |
| LW462 | CYBER LAW                 |
| LW472 | CRIMINAL LAW              |
| WS385 | WAR AND ITS THEORISTS     |

**OR****Psychology Track**

Choose 3 of 3

|       |                             |
|-------|-----------------------------|
| PL376 | PSYCHOPATHOLOGY             |
| PL392 | COGNITIVE PSYCHOLOGY        |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM |

**OR****Literature Track**

Choose 3 of 5

Cadets who select this track may also choose other courses from the English Major's catalogue.

|       |                             |
|-------|-----------------------------|
| EN300 | LITERARY METHODOLOGIES      |
| LG485 | SURVEY OF GERMAN LIT I      |
| LG486 | SURVEY OF GERMAN LIT II     |
| LG492 | 20TH & 21ST CENTURY GERMANY |
| LN380 | NATURE OF MODERN LANGUAGES  |

**OR****Strategic Studies Track**

Choose 3 of 5

|       |                               |
|-------|-------------------------------|
| GE494 | ENVIRONMENTAL SECURITY        |
| HI351 | ADV HISTORY OF MILITARY ART   |
| SS465 | TERRORISM: NEW CHALLENGES     |
| WS365 | REBELLN, INSURGNCY, & CIV WAR |
| WS455 | COMPARATIVE DEFENSE POLICY    |

**OR****Politics Track**

Choose 3 of 5

|       |                               |
|-------|-------------------------------|
| HI347 | ASIAN WARFARE AND POLITICS    |
| SS366 | COMPARATIVE POLITICS          |
| SS386 | POLITICAL THOUGHT             |
| SS465 | TERRORISM: NEW CHALLENGES     |
| XH415 | GENOCIDE AND ETHNIC CLEANSING |

**OR****Cultural Studies Track**

Choose 3 of 5

Cadets who select this track may also choose courses from D/GENE's list of regional geography courses.

|       |                              |
|-------|------------------------------|
| GE343 | GEOGRAPHY OF RUSSIA          |
| HI391 | WORLD RELIGIONS              |
| LG483 | GERMAN CIVILIZATION I        |
| LG484 | GERMAN CIVILIZATION II       |
| XH405 | THE HOLOCAUST AND ITS LEGACY |

**AND****Integrative Experience for the Major**

Choose 1 of 1

|       |                              |
|-------|------------------------------|
| PY400 | SENIOR SEMINAR IN PHILOSOPHY |
|-------|------------------------------|

**AND****Science Depth**

Cadets in this major may take any approved choice to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major can choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                       |
|-------|-----------------------|
| PY300 | PHILOSOPHICAL METHODS |
|-------|-----------------------|

## 2029 Philosophy Major w/ Honors Curriculum

| Code | Short Description | Description | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------|-------------|------------------------|-----------------|-----------------|
|------|-------------------|-------------|------------------------|-----------------|-----------------|

|                            |                            |                      |   |   |
|----------------------------|----------------------------|----------------------|---|---|
| PYL2H Philosophy w/ Honors | Philosophy Major w/ Honors | Philosophy w/ Honors | 2 | 0 |
|----------------------------|----------------------------|----------------------|---|---|

2029 Philosophy Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                | Description     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Required Courses</b>                                                                                                                                                                     | Choose 2 of 2   |
| Cadets must achieve at least a B in PY490 in order to proceed into PY491. The thesis adviser will normally recommend the cadet for Honors consideration if the PY491 grade is A- or better. |                 |
| PY490                                                                                                                                                                                       | THESIS RESEARCH |
| PY491                                                                                                                                                                                       | SENIOR THESIS   |
| <b>AND</b>                                                                                                                                                                                  |                 |
| Complete the requirements of the major as shown above; attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.                                                            |                 |
| <b>AND</b>                                                                                                                                                                                  |                 |
| <b>Approval of the Department Head</b>                                                                                                                                                      |                 |
| On the basis of recommendations by the cadet's DAC and senior-thesis adviser.                                                                                                               |                 |

## Mathematical Sciences

### 2029 Applied Statistics and Data Science Major Curriculum

| Code | Short Description                   | Description                               | Transcript Description              | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------------------------|-------------------------------------------|-------------------------------------|-----------------|-----------------|
| ASD0 | Applied Statistics and Data Science | Applied Statistics and Data Science Major | Applied Statistics and Data Science | 9               | 4               |

### 2029 Applied Statistics and Data Science Major Tracks

| Subject Area                                                                                                                      | Description                   |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Required Courses</b>                                                                                                           | Choose 6 of 6                 |
| MA371                                                                                                                             | LINEAR ALGEBRA                |
| MA376                                                                                                                             | APPLIED STATISTICS            |
| MA476                                                                                                                             | MATHEMATICAL STATISTICS       |
| MA477                                                                                                                             | THEORY & APPL OF DATA SCIENCE |
| MA478                                                                                                                             | GENERALIZED LINEAR MODELS     |
| MA486                                                                                                                             | MATHEMATICAL COMPUTATION      |
| <b>AND</b>                                                                                                                        |                               |
| <b>Discipline Electives</b>                                                                                                       | Choose 2 of 17                |
| CS473                                                                                                                             | COMPUTER GRAPHICS             |
| CS486                                                                                                                             | ARTIFICIAL INTELLIGENCE       |
| CY383                                                                                                                             | SECURE INTERFACE DESIGN       |
| MA372                                                                                                                             | INTRODUCTION TO DISCRETE MATH |
| MA383                                                                                                                             | FOUNDATIONS OF MATH           |
| MA386                                                                                                                             | INTRO TO NUMERICAL ANALYSIS   |
| MA387                                                                                                                             | MATHEMATICAL ANALYSIS I       |
| MA388                                                                                                                             | SABERMETRICS                  |
| MA391                                                                                                                             | MATHEMATICAL MODELING         |
| MA394                                                                                                                             | FUNDAMENTALS/NETWORK SCIENCE  |
| MA461                                                                                                                             | GRAPH THEORY AND NETWORKS     |
| MA462                                                                                                                             | COMBINATORICS                 |
| MA488                                                                                                                             | SPECIAL TOPICS IN MATHEMATICS |
| MA489                                                                                                                             | ADV INDIV STUDY IN MATH       |
| PY326                                                                                                                             | ETHICS OF TECHNOLOGY          |
| XE402                                                                                                                             | INTEGRATIVE SYSTEM DESIGN II  |
| XE492                                                                                                                             | DISRUPTIVE INNOVATIONS        |
| <b>AND</b>                                                                                                                        |                               |
| <b>Research Course</b>                                                                                                            |                               |
| Choose a 2-course research sequence in another department as approved by the ASDS Program Director or choose one of these tracks. |                               |
| <b>Math Research</b>                                                                                                              | Choose 1 of 1                 |
| MA491                                                                                                                             | RESEARCH SEMNR-APPLD MATH     |
| <b>OR</b>                                                                                                                         |                               |
| <b>Integrated Research Course</b>                                                                                                 | Choose 1 of 1                 |
| XE401                                                                                                                             | INTEGRATIVE SYSTEM DESIGN I   |
| <b>AND</b>                                                                                                                        |                               |
| <b>Complementary Support Course - Required</b>                                                                                    | Choose 1 of 2                 |

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Complementary Support Courses - Computer Science</b>                                                                                                                      | Choose 2 of 2                  |
| CS393                                                                                                                                                                        | DATABASE SYSTEMS               |
| SE370                                                                                                                                                                        | COMPUTER AIDED SYSTEMS ENG     |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                  |
| MA490                                                                                                                                                                        | ETHICS IN MATH, DATA SCI & ENG |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take MA204, MA205 or MA255 to satisfy the STEM Depth requirement.                                                                                  |                                |
| MA204                                                                                                                                                                        | CALCULUS I AND II              |
| MA205                                                                                                                                                                        | CALCULUS II                    |
| MA255                                                                                                                                                                        | ADV MULTIVARIABLE CALCULUS     |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Curriculum Requirements</b>                                                                                                                                               |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major are directed to take the Cyber Engineering sequence to support their major.                                                                             |                                |
| CY300                                                                                                                                                                        | PROGRAMMING FUNDAMENTALS       |
| CY350                                                                                                                                                                        | COMPUTER NETWORKS              |
| CY450                                                                                                                                                                        | CYBER SECURITY ENGINEERING     |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| MA376                                                                                                                                                                        | APPLIED STATISTICS             |

## 2029 Applied Statistics and Data Science Major with Honors Curriculum

| Code  | Short Description                             | Description                                           | Transcript Description                        | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------------|-------------------------------------------------------|-----------------------------------------------|--------------|--------------|
| ASD0H | Applied Statistics and Data Science w/ Honors | Applied Statistics and Data Science Major with Honors | Applied Statistics and Data Science w/ Honors | 1            | 1            |



## 2029 Applied Statistics and Data Science Major with Honors Tracks

| Subject Area                                                                                                                                                                                                                                                          | Description                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Required Courses</b>                                                                                                                                                                                                                                               |                               |
| Applied Statistics and Data Science with Honors majors will replace MA491 with the two-course sequence of MA498 and MA499 or will take the two-course sequence of XE401 and XE402. For honors majors, neither of these courses may count as one of the two electives. |                               |
| <b>Math Research</b>                                                                                                                                                                                                                                                  | Choose 1 of 1                 |
| MA499                                                                                                                                                                                                                                                                 | SR THESIS II: PAPER & DEFENSE |
| <b>OR</b>                                                                                                                                                                                                                                                             |                               |
| <b>Integrated Research</b>                                                                                                                                                                                                                                            | Choose 1 of 1                 |
| XE402                                                                                                                                                                                                                                                                 | INTEGRATIVE SYSTEM DESIGN II  |
| <b>AND</b>                                                                                                                                                                                                                                                            |                               |
| <b>Foundations of Math</b>                                                                                                                                                                                                                                            | Choose 1 of 1                 |
| Applied Statistics and Data Science with Honors majors will take MA383. For honors majors, this may not count as one of the two electives.                                                                                                                            |                               |
| MA383                                                                                                                                                                                                                                                                 | FOUNDATIONS OF MATH           |
| <b>AND</b>                                                                                                                                                                                                                                                            |                               |
| <b>Grade Requirements</b>                                                                                                                                                                                                                                             |                               |
| Complete the 42 required courses of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major to be awarded the Applied Statistics and Data Science with Honors Degree.                            |                               |

## 2029 Applied Statistics Minor Curriculum

| Code  | Short Description        | Description              | Transcript Description   | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------|--------------------------|--------------------------|--------------|--------------|
| AST1N | Applied Statistics Minor | Applied Statistics Minor | Applied Statistics Minor | 1            | 4            |

## 2029 Applied Statistics Minor Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                                                                               | Description              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Required Course</b>                                                                                                                                                                                                                                                                                                                                                                     | Choose 1 of 1            |
| Cadets majoring in Applied Statistics and Data Science may not pursue a minor in Applied Statistics as well.                                                                                                                                                                                                                                                                               |                          |
| MA476                                                                                                                                                                                                                                                                                                                                                                                      | MATHEMATICAL STATISTICS  |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                 |                          |
| <b>Foundation Course</b>                                                                                                                                                                                                                                                                                                                                                                   | Choose 1 of 2            |
| MA376                                                                                                                                                                                                                                                                                                                                                                                      | APPLIED STATISTICS       |
| SE375                                                                                                                                                                                                                                                                                                                                                                                      | STATISTICS FOR ENGINEERS |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                 |                          |
| <b>Electives</b>                                                                                                                                                                                                                                                                                                                                                                           | Choose 3 of 14           |
| Cadets may also choose from a variety of seminars, colloquia, summer AIADs for credit, and independent studies in any department, when topics are offered that are relevant to the Applied Statistics Minor. Approval authority for inclusion of these courses is the Applied Statistics DAC in the Department of Mathematical Sciences, who will coordinate with the offering department. |                          |
| EM481                                                                                                                                                                                                                                                                                                                                                                                      | SYSTEMS SIMULATION       |

|       |                               |
|-------|-------------------------------|
| MA204 | CALCULUS I AND II             |
| MA205 | CALCULUS II                   |
| MA255 | ADV MULTIVARIABLE CALCULUS    |
| MA371 | LINEAR ALGEBRA                |
| MA388 | SABERMETRICS                  |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE  |
| MA488 | SPECIAL TOPICS IN MATHEMATICS |
| MA489 | ADV INDIV STUDY IN MATH       |
| PH361 | EXPERIMENTAL PHYSICS          |
| PH481 | STATISTICAL PHYSICS           |
| PL386 | EXPERIMENTAL PSYCHOLOGY       |
| PL497 | SEMINAR IN BEHAVIORAL SCI     |
| SE388 | STOCHASTIC MODELS             |
| SS368 | ECONOMETRICS I                |
| SS469 | ECONOMETRICS II               |

### 2029 Mathematical Sciences Major Curriculum

| Code | Short Description     | Description                 | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-----------------------|-----------------------------|------------------------|-----------------|-----------------|
| MSC1 | Mathematical Sciences | Mathematical Sciences Major | Mathematical Sciences  | 8               | 6               |

### 2029 Mathematical Sciences Major Tracks

| Subject Area                 | Description                    |
|------------------------------|--------------------------------|
| <b>Required Courses</b>      | Choose 7 of 7                  |
| MA363                        | ORDINARY DIFFERENTIAL EQUATION |
| MA371                        | LINEAR ALGEBRA                 |
| MA383                        | FOUNDATIONS OF MATH            |
| MA386                        | INTRO TO NUMERICAL ANALYSIS    |
| MA387                        | MATHEMATICAL ANALYSIS I        |
| MA391                        | MATHEMATICAL MODELING          |
| MA491                        | RESEARCH SEMNR-APPLD MATH      |
| <b>AND</b>                   |                                |
| <b>Statistics Elective</b>   | Choose 1 of 2                  |
| MA376                        | APPLIED STATISTICS             |
| MA476                        | MATHEMATICAL STATISTICS        |
| <b>AND</b>                   |                                |
| <b>Mathematics Electives</b> | Choose 2 of 26                 |
| MA372                        | INTRODUCTION TO DISCRETE MATH  |
| MA376                        | APPLIED STATISTICS             |
| MA381                        | NONLINEAR OPTIMIZATION         |
| MA385                        | CHAOS AND FRACTALS             |
| MA388                        | SABERMETRICS                   |
| MA394                        | FUNDAMENTALS/NETWORK SCIENCE   |
| MA396                        | NUM METH SOLUTIONS DIFF EQNS   |
| MA461                        | GRAPH THEORY AND NETWORKS      |
| MA462                        | COMBINATORICS                  |
| MA464                        | APPLIED ALGEBRA W/ CRYPTOLOGY  |
| MA466                        | ABSTRACT ALGEBRA               |

|        |                               |
|--------|-------------------------------|
| MA476  | MATHEMATICAL STATISTICS       |
| MA477  | THEORY & APPL OF DATA SCIENCE |
| MA478  | GENERALIZED LINEAR MODELS     |
| MA481  | LINEAR OPTIMIZATION           |
| MA484  | PARTIAL DIFF EQUATIONS        |
| MA485  | APPLIED COMPLEX VARIABLES     |
| MA486  | MATHEMATICAL COMPUTATION      |
| MA487  | MATHEMATICAL ANALYSIS II      |
| MA488  | SPECIAL TOPICS IN MATHEMATICS |
| MA489  | ADV INDIV STUDY IN MATH       |
| MA493A | OPNL CALC AND TRANSFORMS      |
| MA493B | REAL VARIABLE THEORY          |
| MA493C | TOPICS IN NUMERICAL ANALYSIS  |
| MA493D | INTRODUCTION TO TOPOLOGY      |
| MA493E | TOPICS IN ANALYSIS            |

**AND**

**Complementary Support Course - Required** Choose 1 of 2

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Complementary Support Courses - Elective Tracks**

Cadets may choose from these following courses to satisfy the remaining two CSC requirements. The list is organized into groups/themes to guide the selection process. DACs, in most cases, will guide cadets to choose two courses within a particular theme. If cadets are taking any of the courses on the following lists as part of their Core Engineering Sequence, these courses cannot be used to fulfill the two CSC requirements for the major.

**Computational Mathematics Tools** Choose 2 of 7

|       |                                |
|-------|--------------------------------|
| AI350 | DATA ENGINEERING               |
| CS384 | DATA STRUCTURES                |
| CS385 | DESIGN & ANALYSIS-ALGORITHMS   |
| CS393 | DATABASE SYSTEMS               |
| CS403 | SOFTWARE TESTING & DEVELOPMENT |
| CS473 | COMPUTER GRAPHICS              |
| CY300 | PROGRAMMING FUNDAMENTALS       |

**OR**

**Finance Applications** Choose 2 of 5

|       |                     |
|-------|---------------------|
| EM381 | ENGINEERING ECONOMY |
| MG410 | MANAGERIAL FINANCE  |
| SS368 | ECONOMETRICS I      |
| SS382 | MICROECONOMICS      |
| SS388 | MACROECONOMICS      |

**OR**

**Cyber Applications** Choose 2 of 10

|       |                                |
|-------|--------------------------------|
| CS483 | DIGITAL FORENSICS              |
| CY350 | COMPUTER NETWORKS              |
| CY384 | NETWORK SYSTEMS PROGRAMMING    |
| CY450 | CYBER SECURITY ENGINEERING     |
| CY460 | CYBER POLICY, STRATEGY, & OPNS |
| LW482 | NATIONAL SECURITY LAW          |
| PY395 | SPECIAL TOPICS IN PHILOSOPHY   |
| XE401 | INTEGRATIVE SYSTEM DESIGN I    |
| XE402 | INTEGRATIVE SYSTEM DESIGN II   |
| XE492 | DISRUPTIVE INNOVATIONS         |

**OR****Engineering Applications**

Choose 2 of 16

|       |                                |
|-------|--------------------------------|
| CH481 | PHYSICAL CHEMISTRY I           |
| EE383 | ELECTROMAGN FIELDS & WAVES     |
| EM384 | ANYL METH FOR ENGR MANAGEMENT  |
| EM481 | SYSTEMS SIMULATION             |
| EV350 | ENVIRONMNTL ENGR TECHNOLOGIES  |
| EV394 | HYDROGEOLOGY/HYDRAULIC SYSTEMS |
| EV397 | AIR POLLUTION ENGINEERING      |
| GE373 | REMOTE SENSING                 |
| GE379 | PHOTOGRAMMETRY                 |
| GE477 | ADVANCED REMOTE SENSING        |
| ME202 | COMPUTER AIDED DESIGN          |
| ME480 | HEAT TRANSFER                  |
| NE355 | NUCLEAR REACTOR ENGINEERING    |
| PH381 | INTRMED CLASSICAL MECHANICS    |
| SA473 | INTRODUCTION TO ASTRONAUTICS   |
| SE370 | COMPUTER AIDED SYSTEMS ENG     |

**OR****Network Science**

Choose 2 of 4

|       |                              |
|-------|------------------------------|
| CY350 | COMPUTER NETWORKS            |
| EM482 | SUPPLY CHAIN ENG & INFO MGMT |
| GE310 | GEOG INFORMATION SYSTEMS     |
| SE387 | DETERMINISTIC MODELS         |

**OR****Applied Statistics**

Choose 2 of 3

|       |                          |
|-------|--------------------------|
| PH481 | STATISTICAL PHYSICS      |
| SE375 | STATISTICS FOR ENGINEERS |
| SS469 | ECONOMETRICS II          |

**AND****Integrative Experience for the Major**

Choose 1 of 1

Cadets may take MA490 or any other major's integrative experience.

|       |                                |
|-------|--------------------------------|
| MA490 | ETHICS IN MATH, DATA SCI & ENG |
|-------|--------------------------------|

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take MA204 or MA205 to satisfy the STEM Depth requirement.

|       |                   |
|-------|-------------------|
| MA204 | CALCULUS I AND II |
| MA205 | CALCULUS II       |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any authorized engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

MA383

FOUNDATIONS OF MATH

**2029 Mathematical Sciences Major w/ Honors Curriculum**

| Code  | Short Description               | Description                           | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------|---------------------------------------|---------------------------------|--------------|--------------|
| MSC1H | Mathematical Sciences w/ Honors | Mathematical Sciences Major w/ Honors | Mathematical Sciences w/ Honors | 1            | 0            |

**2029 Mathematical Sciences Major w/ Honors Tracks**

| Subject Area                                                                                                                                                                 | Description                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                      | Choose 1 of 2                  |
| Mathematical Sciences with Honors majors will replace MA491 with the two-course sequence of MA498 and MA499. Neither of these courses may count as one of the two electives. |                                |
| MA498                                                                                                                                                                        | SR THESIS I: RSCRCH & PROPOSAL |
| MA499                                                                                                                                                                        | SR THESIS II: PAPER & DEFENSE  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Grade Requirements</b>                                                                                                                                                    |                                |
| Complete the requirements of the major (excepting MA491) as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major. |                                |

**2029 Mathematical Studies Major Curriculum**

| Code | Short Description    | Description                | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------|----------------------------|------------------------|--------------|--------------|
| MST1 | Mathematical Studies | Mathematical Studies Major | Mathematical Studies   | 7            | 6            |

**2029 Mathematical Studies Major Tracks**

| Subject Area            | Description                    |
|-------------------------|--------------------------------|
| <b>Required Courses</b> | Choose 6 of 6                  |
| MA363                   | ORDINARY DIFFERENTIAL EQUATION |
| MA371                   | LINEAR ALGEBRA                 |
| MA383                   | FOUNDATIONS OF MATH            |
| MA386                   | INTRO TO NUMERICAL ANALYSIS    |
| MA391                   | MATHEMATICAL MODELING          |
| MA491                   | RESEARCH SEMNR-APPLD MATH      |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |
| <b>Statistics Elective</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | Choose 1 of 2                  |
| MA376                                                                                                                                                                                                                                                                                                                                                                                                                                                     | APPLIED STATISTICS             |
| MA476                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MATHEMATICAL STATISTICS        |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |
| <b>Mathematics Electives</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | Choose 2 of 27                 |
| MA372                                                                                                                                                                                                                                                                                                                                                                                                                                                     | INTRODUCTION TO DISCRETE MATH  |
| MA376                                                                                                                                                                                                                                                                                                                                                                                                                                                     | APPLIED STATISTICS             |
| MA381                                                                                                                                                                                                                                                                                                                                                                                                                                                     | NONLINEAR OPTIMIZATION         |
| MA385                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CHAOS AND FRACTALS             |
| MA387                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MATHEMATICAL ANALYSIS I        |
| MA388                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SABERMETRICS                   |
| MA394                                                                                                                                                                                                                                                                                                                                                                                                                                                     | FUNDAMENTALS/NETWORK SCIENCE   |
| MA396                                                                                                                                                                                                                                                                                                                                                                                                                                                     | NUM METH SOLUTIONS DIFF EQNS   |
| MA461                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GRAPH THEORY AND NETWORKS      |
| MA462                                                                                                                                                                                                                                                                                                                                                                                                                                                     | COMBINATORICS                  |
| MA464                                                                                                                                                                                                                                                                                                                                                                                                                                                     | APPLIED ALGEBRA W/ CRYPTOLOGY  |
| MA466                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ABSTRACT ALGEBRA               |
| MA476                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MATHEMATICAL STATISTICS        |
| MA477                                                                                                                                                                                                                                                                                                                                                                                                                                                     | THEORY & APPL OF DATA SCIENCE  |
| MA478                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GENERALIZED LINEAR MODELS      |
| MA481                                                                                                                                                                                                                                                                                                                                                                                                                                                     | LINEAR OPTIMIZATION            |
| MA484                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PARTIAL DIFF EQUATIONS         |
| MA485                                                                                                                                                                                                                                                                                                                                                                                                                                                     | APPLIED COMPLEX VARIABLES      |
| MA486                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MATHEMATICAL COMPUTATION       |
| MA487                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MATHEMATICAL ANALYSIS II       |
| MA488                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SPECIAL TOPICS IN MATHEMATICS  |
| MA489                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ADV INDIV STUDY IN MATH        |
| MA493A                                                                                                                                                                                                                                                                                                                                                                                                                                                    | OPNL CALC AND TRANSFORMS       |
| MA493B                                                                                                                                                                                                                                                                                                                                                                                                                                                    | REAL VARIABLE THEORY           |
| MA493C                                                                                                                                                                                                                                                                                                                                                                                                                                                    | TOPICS IN NUMERICAL ANALYSIS   |
| MA493D                                                                                                                                                                                                                                                                                                                                                                                                                                                    | INTRODUCTION TO TOPOLOGY       |
| MA493E                                                                                                                                                                                                                                                                                                                                                                                                                                                    | TOPICS IN ANALYSIS             |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |
| <b>Complementary Support Course - Required</b>                                                                                                                                                                                                                                                                                                                                                                                                            | Choose 1 of 2                  |
| CY305                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |
| <b>Complementary Support Courses - Elective Tracks</b>                                                                                                                                                                                                                                                                                                                                                                                                    |                                |
| Cadets may choose from these following courses to satisfy the remaining two CSC requirements. The list is organized into groups/themes to guide the selection process. DACs, in most cases, will guide cadets to choose two courses within a particular theme. If cadets are taking any of the courses on the following lists as part of their Core Engineering Sequence, these courses cannot be used to fulfill the two CSC requirements for the major. |                                |
| <b>Computational Mathematics Tools</b>                                                                                                                                                                                                                                                                                                                                                                                                                    | Choose 2 of 7                  |
| AI350                                                                                                                                                                                                                                                                                                                                                                                                                                                     | DATA ENGINEERING               |
| CS384                                                                                                                                                                                                                                                                                                                                                                                                                                                     | DATA STRUCTURES                |
| CS385                                                                                                                                                                                                                                                                                                                                                                                                                                                     | DESIGN & ANALYS-ALGORITHMS     |
| CS393                                                                                                                                                                                                                                                                                                                                                                                                                                                     | DATABASE SYSTEMS               |
| CS403                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SOFTWARE TESTING & DEVELOPMENT |
| CS473                                                                                                                                                                                                                                                                                                                                                                                                                                                     | COMPUTER GRAPHICS              |
| CY300                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PROGRAMMING FUNDAMENTALS       |
| <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |

|                                                                    |                                |
|--------------------------------------------------------------------|--------------------------------|
| <b>Finance Applications</b>                                        | Choose 2 of 5                  |
| EM381                                                              | ENGINEERING ECONOMY            |
| MG410                                                              | MANAGERIAL FINANCE             |
| SS368                                                              | ECONOMETRICS I                 |
| SS382                                                              | MICROECONOMICS                 |
| SS388                                                              | MACROECONOMICS                 |
| <b>OR</b>                                                          |                                |
| <b>Cyber Applications</b>                                          | Choose 2 of 10                 |
| CS483                                                              | DIGITAL FORENSICS              |
| CY350                                                              | COMPUTER NETWORKS              |
| CY384                                                              | NETWORK SYSTEMS PROGRAMMING    |
| CY450                                                              | CYBER SECURITY ENGINEERING     |
| CY460                                                              | CYBER POLICY, STRATEGY, & OPNS |
| LW482                                                              | NATIONAL SECURITY LAW          |
| PY395                                                              | SPECIAL TOPICS IN PHILOSOPHY   |
| XE401                                                              | INTEGRATIVE SYSTEM DESIGN I    |
| XE402                                                              | INTEGRATIVE SYSTEM DESIGN II   |
| XE492                                                              | DISRUPTIVE INNOVATIONS         |
| <b>OR</b>                                                          |                                |
| <b>Engineering Applications</b>                                    | Choose 2 of 16                 |
| CH481                                                              | PHYSICAL CHEMISTRY I           |
| EE383                                                              | ELECTROMAGN FIELDS & WAVES     |
| EM384                                                              | ANYL METH FOR ENGR MANAGEMENT  |
| EM481                                                              | SYSTEMS SIMULATION             |
| EV350                                                              | ENVIRONMNTL ENGR TECHNOLOGIES  |
| EV394                                                              | HYDROGEOLOGY/HYDRAULIC SYSTEMS |
| EV397                                                              | AIR POLLUTION ENGINEERING      |
| GE373                                                              | REMOTE SENSING                 |
| GE379                                                              | PHOTOGRAMMETRY                 |
| GE477                                                              | ADVANCED REMOTE SENSING        |
| ME370                                                              | COMPUTER AIDED DESIGN          |
| ME480                                                              | HEAT TRANSFER                  |
| NE355                                                              | NUCLEAR REACTOR ENGINEERING    |
| PH381                                                              | INTRMED CLASSICAL MECHANICS    |
| SA473                                                              | INTRODUCTION TO ASTRONAUTICS   |
| SE370                                                              | COMPUTER AIDED SYSTEMS ENG     |
| <b>OR</b>                                                          |                                |
| <b>Network Science</b>                                             | Choose 2 of 4                  |
| CY350                                                              | COMPUTER NETWORKS              |
| EM482                                                              | SUPPLY CHAIN ENG & INFO MGMT   |
| GE310                                                              | GEOG INFORMATION SYSTEMS       |
| SE387                                                              | DETERMINISTIC MODELS           |
| <b>OR</b>                                                          |                                |
| <b>Applied Statistics</b>                                          | Choose 2 of 3                  |
| PH481                                                              | STATISTICAL PHYSICS            |
| SE375                                                              | STATISTICS FOR ENGINEERS       |
| SS469                                                              | ECONOMETRICS II                |
| <b>AND</b>                                                         |                                |
| <b>Integrative Experience for the Major</b>                        | Choose 1 of 1                  |
| Cadets may take MA490 or any other major's integrative experience. |                                |
| MA490                                                              | ETHICS IN MATH, DATA SCI & ENG |
| <b>AND</b>                                                         |                                |
| <b>Science Depth</b>                                               |                                |

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

#### STEM Depth

Cadets in this major must take MA204 or MA205 to satisfy the STEM Depth requirement.

|       |                   |
|-------|-------------------|
| MA204 | CALCULUS I AND II |
| MA205 | CALCULUS II       |

#### AND

#### Curriculum Requirements

This section describes how cadets in this major satisfy various curriculum requirements.

#### IT/CYBER Requirement

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

#### Core Engineering Sequence

Cadets in this major may choose any authorized engineering sequence.

#### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                     |
|-------|---------------------|
| MA383 | FOUNDATIONS OF MATH |
|-------|---------------------|

## 2029 Mathematical Studies Major Remarks Alternate major for Mathematical Sciences.

### 2029 Mathematics Minor Curriculum

| Code  | Short Description | Description       | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------|-------------------|------------------------|--------------|--------------|
| MSMON | Mathematics Minor | Mathematics Minor | Mathematics Minor      | 1            | 5            |

### 2029 Mathematics Minor Tracks

| Subject Area                                                                            | Description                    |
|-----------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Course</b>                                                                  | Choose 1 of 3                  |
| Cadets majoring in Mathematical Sciences may not pursue a minor in Mathematics as well. |                                |
| MA204                                                                                   | CALCULUS I AND II              |
| MA205                                                                                   | CALCULUS II                    |
| MA255                                                                                   | ADV MULTIVARIABLE CALCULUS     |
| <b>AND</b>                                                                              |                                |
| <b>Mathematics Minor Electives</b>                                                      | Choose 5 of 34                 |
| MA363                                                                                   | ORDINARY DIFFERENTIAL EQUATION |
| MA364                                                                                   | ENGINEERING MATHEMATICS        |
| MA365                                                                                   | ADV MATH FOR ENGRS/SCIENTISTS  |
| MA367                                                                                   | MATH FOR THE SOCIAL SCIENCES   |
| MA371                                                                                   | LINEAR ALGEBRA                 |



|        |                               |
|--------|-------------------------------|
| MA372  | INTRODUCTION TO DISCRETE MATH |
| MA376  | APPLIED STATISTICS            |
| MA381  | NONLINEAR OPTIMIZATION        |
| MA383  | FOUNDATIONS OF MATH           |
| MA385  | CHAOS AND FRACTALS            |
| MA386  | INTRO TO NUMERICAL ANALYSIS   |
| MA387  | MATHEMATICAL ANALYSIS I       |
| MA388  | SABERMETRICS                  |
| MA389  | INDIV STUDY IN MATHEMATICS    |
| MA391  | MATHEMATICAL MODELING         |
| MA394  | FUNDAMENTALS/NETWORK SCIENCE  |
| MA396  | NUM METH SOLUTIONS DIFF EQNS  |
| MA461  | GRAPH THEORY AND NETWORKS     |
| MA462  | COMBINATORICS                 |
| MA464  | APPLIED ALGEBRA W/ CRYPTOLOGY |
| MA466  | ABSTRACT ALGEBRA              |
| MA476  | MATHEMATICAL STATISTICS       |
| MA477  | THEORY & APPL OF DATA SCIENCE |
| MA478  | GENERALIZED LINEAR MODELS     |
| MA481  | LINEAR OPTIMIZATION           |
| MA484  | PARTIAL DIFF EQUATIONS        |
| MA485  | APPLIED COMPLEX VARIABLES     |
| MA486  | MATHEMATICAL COMPUTATION      |
| MA487  | MATHEMATICAL ANALYSIS II      |
| MA488  | SPECIAL TOPICS IN MATHEMATICS |
| MA489  | ADV INDIV STUDY IN MATH       |
| MA491  | RESEARCH SEMNR-APPLD MATH     |
| MA493D | INTRODUCTION TO TOPOLOGY      |
| SE375  | STATISTICS FOR ENGINEERS      |

## 2029 Operations Research Major Curriculum

| Code | Short Description   | Description               | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|---------------------|---------------------------|------------------------|-----------------|-----------------|
| ORE2 | Operations Research | Operations Research Major | Operations Research    | 13              | 4               |

## 2029 Operations Research Major Tracks

| Subject Area            | Description                   |
|-------------------------|-------------------------------|
| <b>Required Courses</b> | Choose 8 of 8                 |
| MA371                   | LINEAR ALGEBRA                |
| MA376                   | APPLIED STATISTICS            |
| MA381                   | NONLINEAR OPTIMIZATION        |
| MA476                   | MATHEMATICAL STATISTICS       |
| MA477                   | THEORY & APPL OF DATA SCIENCE |
| MA481                   | LINEAR OPTIMIZATION           |
| MA486                   | MATHEMATICAL COMPUTATION      |
| SE385                   | DECISION ANALYSIS             |

**AND**

**Discipline Threads**

|                                   |                                |
|-----------------------------------|--------------------------------|
| <b>Operations Management</b>      | Choose 3 of 6                  |
| EM381                             | ENGINEERING ECONOMY            |
| EM411                             | PROJECT MANAGEMENT             |
| EM420                             | PRODUCTION OPERATIONS MGMT     |
| EM482                             | SUPPLY CHAIN ENG & INFO MGMT   |
| SE370                             | COMPUTER AIDED SYSTEMS ENG     |
| SE489                             | AD IND STY IN SYS ENG/ENG MGMT |
| <b>OR</b>                         |                                |
| <b>Data Science</b>               | Choose 3 of 6                  |
| CS393                             | DATABASE SYSTEMS               |
| CS486                             | ARTIFICIAL INTELLIGENCE        |
| CY300                             | PROGRAMMING FUNDAMENTALS       |
| MA388                             | SABERMETRICS                   |
| MA478                             | GENERALIZED LINEAR MODELS      |
| SE370                             | COMPUTER AIDED SYSTEMS ENG     |
| <b>OR</b>                         |                                |
| <b>Networks, Computing and IT</b> | Choose 3 of 8                  |
| CS384                             | DATA STRUCTURES                |
| CS393                             | DATABASE SYSTEMS               |
| CS486                             | ARTIFICIAL INTELLIGENCE        |
| CY300                             | PROGRAMMING FUNDAMENTALS       |
| CY460                             | CYBER POLICY, STRATEGY, & OPNS |
| MA372                             | INTRODUCTION TO DISCRETE MATH  |
| MA394                             | FUNDAMENTALS/NETWORK SCIENCE   |
| MA461                             | GRAPH THEORY AND NETWORKS      |
| <b>OR</b>                         |                                |
| <b>Simulation</b>                 | Choose 3 of 3                  |
| CY300                             | PROGRAMMING FUNDAMENTALS       |
| SE485                             | COMBAT MODELING                |
| SM484                             | SYSTEM DYNAMICS SIMULATION     |
| <b>OR</b>                         |                                |
| <b>Finance</b>                    | Choose 3 of 8                  |
| EM381                             | ENGINEERING ECONOMY            |
| MG410                             | MANAGERIAL FINANCE             |
| SS368                             | ECONOMETRICS I                 |
| SS382                             | MICROECONOMICS                 |
| SS388                             | MACROECONOMICS                 |
| SS394                             | FUND. FINANCIAL DATA ANALYSIS  |
| SS463                             | INVESTMENT THEORY & APPL       |
| SS469                             | ECONOMETRICS II                |
| <b>AND</b>                        |                                |
| <b>Discipline Elective</b>        | Choose 1 of 36                 |
| CS384                             | DATA STRUCTURES                |
| CS393                             | DATABASE SYSTEMS               |
| CS486                             | ARTIFICIAL INTELLIGENCE        |
| CY300                             | PROGRAMMING FUNDAMENTALS       |
| CY460                             | CYBER POLICY, STRATEGY, & OPNS |
| EM381                             | ENGINEERING ECONOMY            |
| EM411                             | PROJECT MANAGEMENT             |
| EM420                             | PRODUCTION OPERATIONS MGMT     |
| EM482                             | SUPPLY CHAIN ENG & INFO MGMT   |
| MA372                             | INTRODUCTION TO DISCRETE MATH  |
| MA383                             | FOUNDATIONS OF MATH            |
| MA386                             | INTRO TO NUMERICAL ANALYSIS    |

|       |                                |
|-------|--------------------------------|
| MA387 | MATHEMATICAL ANALYSIS I        |
| MA388 | SABERMETRICS                   |
| MA391 | MATHEMATICAL MODELING          |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE   |
| MA461 | GRAPH THEORY AND NETWORKS      |
| MA462 | COMBINATORICS                  |
| MA478 | GENERALIZED LINEAR MODELS      |
| MA488 | SPECIAL TOPICS IN MATHEMATICS  |
| MA489 | ADV INDIV STUDY IN MATH        |
| MA490 | ETHICS IN MATH, DATA SCI & ENG |
| MG410 | MANAGERIAL FINANCE             |
| SE301 | FNDTN ENGIN DSGN & SYS MGMT    |
| SE370 | COMPUTER AIDED SYSTEMS ENG     |
| SE485 | COMBAT MODELING                |
| SE489 | AD IND STY IN SYS ENG/ENG MGMT |
| SE490 | AD TOPICS IN SYS ENG/ENG MGMT  |
| SE491 | RSRCH PROJ IN SYS ENG/ENG MGMT |
| SM484 | SYSTEM DYNAMICS SIMULATION     |
| SS368 | ECONOMETRICS I                 |
| SS382 | MICROECONOMICS                 |
| SS388 | MACROECONOMICS                 |
| SS394 | FUND. FINANCIAL DATA ANALYSIS  |
| SS463 | INVESTMENT THEORY & APPL       |
| SS469 | ECONOMETRICS II                |

**AND****Complementary Support Course - Cyber**

Choose 1 of 2

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**AND****Complementary Support Course - Engineering**

Choose 2 of 2

EM481

SYSTEMS SIMULATION

SE388

STOCHASTIC MODELS

**AND****Integrative Experience for the Major**

Choose 2 of 2

SE402

SYSTEMS DESIGN &amp; MANAGEMENT I

SE403

SYSTEMS DESIGN &amp; MANAGEMENT II

**OR****Integrative Experience - Research**

Choose 2 of 2

Or choose a 2-course research sequence in another department as approved by the OR Program Director.

MA498

SR THESIS I: RSCRCH &amp; PROPOSAL

MA499

SR THESIS II: PAPER &amp; DEFENSE

**AND****Science Depth**

Cadets in this major may choose any one of the following authorized courses to satisfy the Science Depth requirement.

CH102

GENERAL CHEMISTRY II

CH275

BIOLOGY

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take MA204/MA205 or MA255 to satisfy the STEM Depth requirement.

|            |                            |
|------------|----------------------------|
| MA204      | CALCULUS I AND II          |
| MA205      | CALCULUS II                |
| MA255      | ADV MULTIVARIABLE CALCULUS |
| <b>AND</b> |                            |

#### Curriculum Requirements

This section describes how cadets in this major satisfy various curriculum requirements.

#### IT/CYBER Requirement

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

#### Core Engineering Sequence

Cadets in this major have embedded engineering courses that enable it to satisfy the core engineering sequence requirement.

#### Writing-in-the-Major

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                    |
|-------|--------------------|
| MA376 | APPLIED STATISTICS |
|-------|--------------------|

## 2029 Operations Research Studies Major Curriculum

| Code | Short Description           | Description                       | Transcript Description      | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------|-----------------------------------|-----------------------------|--------------|--------------|
| ORS1 | Operations Research Studies | Operations Research Studies Major | Operations Research Studies | 12           | 4            |

### 2029 Operations Research Studies Major Tracks

| Subject Area                 | Description                    |
|------------------------------|--------------------------------|
| <b>Required Courses</b>      | Choose 7 of 7                  |
| MA371                        | LINEAR ALGEBRA                 |
| MA376                        | APPLIED STATISTICS             |
| MA381                        | NONLINEAR OPTIMIZATION         |
| MA477                        | THEORY & APPL OF DATA SCIENCE  |
| MA481                        | LINEAR OPTIMIZATION            |
| MA486                        | MATHEMATICAL COMPUTATION       |
| SE385                        | DECISION ANALYSIS              |
| <b>AND</b>                   |                                |
| <b>Discipline Threads</b>    |                                |
| <b>Operations Management</b> | Choose 3 of 6                  |
| EM381                        | ENGINEERING ECONOMY            |
| EM411                        | PROJECT MANAGEMENT             |
| EM420                        | PRODUCTION OPERATIONS MGMT     |
| EM482                        | SUPPLY CHAIN ENG & INFO MGMT   |
| SE370                        | COMPUTER AIDED SYSTEMS ENG     |
| SE489                        | AD IND STY IN SYS ENG/ENG MGMT |
| <b>OR</b>                    |                                |

|                                   |                                |
|-----------------------------------|--------------------------------|
| <b>Data Science</b>               | Choose 3 of 6                  |
| CS393                             | DATABASE SYSTEMS               |
| CS486                             | ARTIFICIAL INTELLIGENCE        |
| CY300                             | PROGRAMMING FUNDAMENTALS       |
| MA388                             | SABERMETRICS                   |
| MA478                             | GENERALIZED LINEAR MODELS      |
| SE370                             | COMPUTER AIDED SYSTEMS ENG     |
| <b>OR</b>                         |                                |
| <b>Networks, Computing and IT</b> | Choose 3 of 8                  |
| CS384                             | DATA STRUCTURES                |
| CS393                             | DATABASE SYSTEMS               |
| CS486                             | ARTIFICIAL INTELLIGENCE        |
| CY300                             | PROGRAMMING FUNDAMENTALS       |
| CY460                             | CYBER POLICY, STRATEGY, & OPNS |
| MA372                             | INTRODUCTION TO DISCRETE MATH  |
| MA394                             | FUNDAMENTALS/NETWORK SCIENCE   |
| MA461                             | GRAPH THEORY AND NETWORKS      |
| <b>OR</b>                         |                                |
| <b>Simulation</b>                 | Choose 3 of 3                  |
| CY300                             | PROGRAMMING FUNDAMENTALS       |
| SE485                             | COMBAT MODELING                |
| SM484                             | SYSTEM DYNAMICS SIMULATION     |
| <b>OR</b>                         |                                |
| <b>Finance</b>                    | Choose 3 of 8                  |
| EM381                             | ENGINEERING ECONOMY            |
| MG410                             | MANAGERIAL FINANCE             |
| SS368                             | ECONOMETRICS I                 |
| SS382                             | MICROECONOMICS                 |
| SS388                             | MACROECONOMICS                 |
| SS394                             | FUND. FINANCIAL DATA ANALYSIS  |
| SS463                             | INVESTMENT THEORY & APPL       |
| SS469                             | ECONOMETRICS II                |
| <b>AND</b>                        |                                |
| <b>Discipline Elective</b>        | Choose 1 of 37                 |
| CS384                             | DATA STRUCTURES                |
| CS393                             | DATABASE SYSTEMS               |
| CS486                             | ARTIFICIAL INTELLIGENCE        |
| CY300                             | PROGRAMMING FUNDAMENTALS       |
| CY460                             | CYBER POLICY, STRATEGY, & OPNS |
| EM381                             | ENGINEERING ECONOMY            |
| EM411                             | PROJECT MANAGEMENT             |
| EM420                             | PRODUCTION OPERATIONS MGMT     |
| EM482                             | SUPPLY CHAIN ENG & INFO MGMT   |
| MA372                             | INTRODUCTION TO DISCRETE MATH  |
| MA383                             | FOUNDATIONS OF MATH            |
| MA386                             | INTRO TO NUMERICAL ANALYSIS    |
| MA387                             | MATHEMATICAL ANALYSIS I        |
| MA388                             | SABERMETRICS                   |
| MA391                             | MATHEMATICAL MODELING          |
| MA394                             | FUNDAMENTALS/NETWORK SCIENCE   |
| MA461                             | GRAPH THEORY AND NETWORKS      |
| MA462                             | COMBINATORICS                  |
| MA476                             | MATHEMATICAL STATISTICS        |
| MA478                             | GENERALIZED LINEAR MODELS      |
| MA488                             | SPECIAL TOPICS IN MATHEMATICS  |

|       |                                |
|-------|--------------------------------|
| MA489 | ADV INDIV STUDY IN MATH        |
| MA490 | ETHICS IN MATH, DATA SCI & ENG |
| MG410 | MANAGERIAL FINANCE             |
| SE301 | FNDTN ENGIN DSGN & SYS MGMT    |
| SE370 | COMPUTER AIDED SYSTEMS ENG     |
| SE485 | COMBAT MODELING                |
| SE489 | AD IND STY IN SYS ENG/ENG MGMT |
| SE490 | AD TOPICS IN SYS ENG/ENG MGMT  |
| SE491 | RSRCH PROJ IN SYS ENG/ENG MGMT |
| SM484 | SYSTEM DYNAMICS SIMULATION     |
| SS368 | ECONOMETRICS I                 |
| SS382 | MICROECONOMICS                 |
| SS388 | MACROECONOMICS                 |
| SS394 | FUND. FINANCIAL DATA ANALYSIS  |
| SS463 | INVESTMENT THEORY & APPL       |
| SS469 | ECONOMETRICS II                |

**AND****Complementary Support Course - Cyber**

Choose 1 of 2

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Complementary Support Course - Engineering**

Choose 2 of 2

|       |                    |
|-------|--------------------|
| EM481 | SYSTEMS SIMULATION |
| SE388 | STOCHASTIC MODELS  |

**AND****Integrative Experience for the Major**

Choose 2 of 2

|       |                                |
|-------|--------------------------------|
| SE402 | SYSTEMS DESIGN & MANAGEMENT I  |
| SE403 | SYSTEMS DESIGN & MANAGEMENT II |

**OR****Integrative Experience - Research**

Choose 2 of 2

Or choose a 2-course research sequence in another department as approved by the OR Program Director.

|       |                                |
|-------|--------------------------------|
| MA498 | SR THESIS I: RSCRCH & PROPOSAL |
| MA499 | SR THESIS II: PAPER & DEFENSE  |

**AND****Science Depth**

Cadets in this major may choose any one of the following authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take MA204/MA205 or MA255 to satisfy the STEM Depth requirement.

|       |                            |
|-------|----------------------------|
| MA204 | CALCULUS I AND II          |
| MA205 | CALCULUS II                |
| MA255 | ADV MULTIVARIABLE CALCULUS |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major have embedded engineering courses that enable it to satisfy the core engineering sequence requirement.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

MA376

APPLIED STATISTICS

## 2029 Operations Research Studies Major Remarks Alternate major for Operations Research since Class of 2019.

### 2029 Operations Research with Honors Curriculum

| Code  | Short Description         | Description                     | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------|---------------------------------|---------------------------------|--------------|--------------|
| ORE2H | Operations Rsch w/ Honors | Operations Research with Honors | Operations Research with Honors | 0            | 1            |

### 2029 Operations Research with Honors Tracks

| Subject Area                                                                                                                                          | Description                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                               | Choose 1 of 23                 |
| Operations Research with Honors majors will add an additional elective in the cadets' chosen thread, bringing the total thread electives to four (4). |                                |
| CS384                                                                                                                                                 | DATA STRUCTURES                |
| CS393                                                                                                                                                 | DATABASE SYSTEMS               |
| CS486                                                                                                                                                 | ARTIFICIAL INTELLIGENCE        |
| CY300                                                                                                                                                 | PROGRAMMING FUNDAMENTALS       |
| CY460                                                                                                                                                 | CYBER POLICY, STRATEGY, & OPNS |
| EM381                                                                                                                                                 | ENGINEERING ECONOMY            |
| EM411                                                                                                                                                 | PROJECT MANAGEMENT             |
| EM420                                                                                                                                                 | PRODUCTION OPERATIONS MGMT     |
| EM482                                                                                                                                                 | SUPPLY CHAIN ENG & INFO MGMT   |
| MA372                                                                                                                                                 | INTRODUCTION TO DISCRETE MATH  |
| MA388                                                                                                                                                 | SABERMETRICS                   |
| MA394                                                                                                                                                 | FUNDAMENTALS/NETWORK SCIENCE   |
| MA461                                                                                                                                                 | GRAPH THEORY AND NETWORKS      |
| MA478                                                                                                                                                 | GENERALIZED LINEAR MODELS      |
| MG410                                                                                                                                                 | MANAGERIAL FINANCE             |
| SE370                                                                                                                                                 | COMPUTER AIDED SYSTEMS ENG     |
| SE485                                                                                                                                                 | COMBAT MODELING                |
| SE489                                                                                                                                                 | AD IND STY IN SYS ENG/ENG MGMT |
| SM484                                                                                                                                                 | SYSTEM DYNAMICS SIMULATION     |
| SS368                                                                                                                                                 | ECONOMETRICS I                 |
| SS382                                                                                                                                                 | INTERMEDIATE MICROECONOMICS    |
| SS388                                                                                                                                                 | INTERMEDIATE MACROECONOMICS    |
| SS469                                                                                                                                                 | ECONOMETRICS II                |
| <b>AND</b>                                                                                                                                            |                                |
| <b>Grade Requirements</b>                                                                                                                             |                                |

Complete the 42 required courses of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major to be awarded the Operations Research with Honors Degree.



## Mechanical and Aerospace Engineering

### 2029 Aerospace Engineering Major Curriculum

| Code | Short Description           | Description                 | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-----------------------------|-----------------------------|------------------------|-----------------|-----------------|
| AEN0 | Aerospace Engineering Major | Aerospace Engineering Major | Aerospace Engineering  | 12              | 7               |

### 2029 Aerospace Engineering Major Tracks

| Subject Area                                 | Description                    |
|----------------------------------------------|--------------------------------|
| <b>Required Courses</b>                      | Choose 12 of 12                |
| AE201                                        | INTRO TO AEROSPACE ENGINEERING |
| AE354                                        | PROPULSION                     |
| AE364                                        | AEROSPACE STRUCTURES           |
| AE400                                        | AEROSPACE SEMINAR              |
| AE473                                        | STABILITY AND CONTROL          |
| AE481                                        | AERODYNAMICS                   |
| AE483                                        | AEROSPACE SYSTEM DESIGN        |
| ME301                                        | THERMODYNAMICS                 |
| ME306                                        | DYNAMICS                       |
| ME362                                        | FLUID MECHANICS                |
| ME380                                        | ENGINEERING MATERIALS          |
| XE472                                        | DYNAMIC MODELING AND CONTROL   |
| <b>AND</b>                                   |                                |
| <b>Introductory Elective</b>                 | Choose 1 of 2                  |
| AE287                                        | INTRODUCTION TO AERONAUTICS    |
| SP471                                        | ASTRONAUTICS                   |
| <b>AND</b>                                   |                                |
| <b>Technical Electives</b>                   | Choose 1 of 8                  |
| AE287                                        | INTRODUCTION TO AERONAUTICS    |
| AE388                                        | VTOL AERONAUTICS               |
| AE389                                        | IND STUDY IN AEROSPACE ENG     |
| AE489                                        | ADV IND STUDY IN AEROSPACE ENG |
| EE381                                        | SIGNALS AND SYSTEMS            |
| ME480                                        | HEAT TRANSFER                  |
| SP471                                        | ASTRONAUTICS                   |
| SP472                                        | SPACE PHYSICS                  |
| <b>AND</b>                                   |                                |
| <b>Complementary Support Courses</b>         | Choose 3 of 4                  |
| EE301                                        | FUNDAMENTALS OF ELEC ENGIN     |
| MA364                                        | ENGINEERING MATHEMATICS        |
| MA365                                        | ADV MATH FOR ENGRS/SCIENTISTS  |
| MC300                                        | FUND OF ENGR MECH AND DESIGN   |
| <b>AND</b>                                   |                                |
| <b>Integrative Experience - First Course</b> | Choose 1 of 2                  |
| AE498                                        | AEROSPACE ENGINEERING DESIGN I |
| XE485                                        | ENGINEERING CAPSTONE DESIGN I  |
| <b>AND</b>                                   |                                |

**Integrative Experience - Second Course**

Choose 1 of 2

AE499

AEROSPACE ENGINEERING DESIGN II

XE495

ENGINEERING CAPSTONE DESIGN II

**AND****Science Depth**

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take MA204 or MA205/255 to satisfy the STEM Depth requirement.

MA204

CALCULUS I AND II

MA205

CALCULUS II

MA255

ADV MULTIVARIABLE CALCULUS

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing EE301 and ME202.

EE301

FUNDAMENTALS OF ELEC ENGIN

ME202

INTRO COMPUTATIONAL ANALYSIS

**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

AE400

AEROSPACE SEMINAR

**2029 Aerospace Engineering Major Remarks GY2028 First Official Class****2029 Aerospace Engineering Major w/ Honors Curriculum**

| Code  | Short Description                     | Description                           | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|---------------------------------------|---------------------------------------|---------------------------------|--------------|--------------|
| AEN0H | Aerospace Engineering Major w/ Honors | Aerospace Engineering Major w/ Honors | Aerospace Engineering w/ Honors | 0            | 0            |

**2029 Aerospace Engineering Major w/ Honors Tracks**

| Subject Area                   | Description                                                                                                                                                                           |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scholarship Requirement</b> | Submit evidence of an engineering-based scholarly activity outside of the normal curricular requirements for the aerospace engineering major, as approved by the AE Program Director. |
| <b>Grade Requirements</b>      | Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.                            |

## 2029 Mechanical Engineering Major Curriculum

| Code | Short Description      | Description                  | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|------------------------|------------------------------|------------------------|-----------------|-----------------|
| MEN3 | Mechanical Engineering | Mechanical Engineering Major | Mechanical Engineering | 14              | 5               |

### 2029 Mechanical Engineering Major Tracks

| Subject Area                                                                                                                                                                                                                                                                                                           | Description                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                                                                                                                                                                | Choose 8 of 8                  |
| CE364                                                                                                                                                                                                                                                                                                                  | MECHANICS OF MATERIALS         |
| ME201                                                                                                                                                                                                                                                                                                                  | INTRO TO MECH ENGINEERING      |
| ME202                                                                                                                                                                                                                                                                                                                  | COMPUTER AIDED DESIGN          |
| ME306                                                                                                                                                                                                                                                                                                                  | DYNAMICS                       |
| ME380                                                                                                                                                                                                                                                                                                                  | ENGINEERING MATERIALS          |
| ME400                                                                                                                                                                                                                                                                                                                  | MECHANICAL ENGINEERING SEMINAR |
| ME403                                                                                                                                                                                                                                                                                                                  | MANUFACTURING/MACHINE COMP DSN |
| ME480                                                                                                                                                                                                                                                                                                                  | HEAT TRANSFER                  |
| <b>AND</b>                                                                                                                                                                                                                                                                                                             |                                |
| <b>Thermal-Fluid Track</b>                                                                                                                                                                                                                                                                                             | Choose 2 of 2                  |
| ME301                                                                                                                                                                                                                                                                                                                  | THERMODYNAMICS                 |
| ME362                                                                                                                                                                                                                                                                                                                  | FLUID MECHANICS                |
| <b>AND</b>                                                                                                                                                                                                                                                                                                             |                                |
| <b>Dynamic Systems Elective</b>                                                                                                                                                                                                                                                                                        | Choose 1 of 2                  |
| ME486                                                                                                                                                                                                                                                                                                                  | VIBRATION ENGINEERING          |
| XE472                                                                                                                                                                                                                                                                                                                  | DYNAMIC MODELING AND CONTROL   |
| <b>AND</b>                                                                                                                                                                                                                                                                                                             |                                |
| <b>General Electives</b>                                                                                                                                                                                                                                                                                               | Choose 3 of 23                 |
| The electives selected from this list must together contain at least 3.0 credit hours of Engineering Topics (ET). One elective may be chosen from outside this list with approval of the ME program director and if it contains at least 3.0 credit hours of Math (MA), Basic Science (BS) or Engineering Topics (ET). |                                |
| AE287                                                                                                                                                                                                                                                                                                                  | INTRODUCTION TO AERONAUTICS    |
| CE350                                                                                                                                                                                                                                                                                                                  | INFRASTRUCTURE ENGINEERING     |
| CE403                                                                                                                                                                                                                                                                                                                  | STRUCTURAL ANALYSIS            |
| EE360                                                                                                                                                                                                                                                                                                                  | FUNDAMENTALS OF DIGITAL LOGIC  |
| EM381                                                                                                                                                                                                                                                                                                                  | ENGINEERING ECONOMY            |
| ME388                                                                                                                                                                                                                                                                                                                  | HELICOPTER AERONAUTICS         |
| ME389                                                                                                                                                                                                                                                                                                                  | INTRO TO ADV STUDY IN MECH ENG |
| ME389A                                                                                                                                                                                                                                                                                                                 | INTRO TO ADV STUDY IN MECH ENG |
| ME472                                                                                                                                                                                                                                                                                                                  | ENERGY CONVERSION SYSTEMS      |
| ME481                                                                                                                                                                                                                                                                                                                  | AIRCRAFT AERODYNAMICS AND DSGN |
| ME486                                                                                                                                                                                                                                                                                                                  | VIBRATION ENGINEERING          |
| ME489                                                                                                                                                                                                                                                                                                                  | ADV STUDY IN MECH ENGRNG       |
| ME489A                                                                                                                                                                                                                                                                                                                 | ADV STUDY IN MECH ENGRNG       |
| ME490                                                                                                                                                                                                                                                                                                                  | TOPICS IN MECHANICAL ENGRNG    |
| ME491                                                                                                                                                                                                                                                                                                                  | AUTOMOTIVE POWERPLANTS         |
| ME492                                                                                                                                                                                                                                                                                                                  | PWR TRAINS & VEH DYNAMICS      |

|       |                              |
|-------|------------------------------|
| ME493 | WEAPONS ENGINEERING          |
| NE300 | FUNDAMENTALS OF NUCLEAR ENGR |
| XE310 | INTRO TO BIOMECH ENGINEERING |
| XE365 | ADV EXP METHODS & DATA PROC  |
| XE465 | TOPICS: ADVANCED TECHNOLOGY  |
| XE472 | DYNAMIC MODELING AND CONTROL |
| XE475 | MECHATRONICS                 |

**AND****Complementary Support Courses** Choose 3 of 4

The following courses provide foundational math and applied science for cadets in the mechanical engineering major. For the math requirement, a cadet should take MA364 but may substitute MA365 if qualified.

|       |                               |
|-------|-------------------------------|
| EE301 | FUNDAMENTALS OF ELEC ENGIN    |
| MA364 | ENGINEERING MATHEMATICS       |
| MA365 | ADV MATH FOR ENGRS/SCIENTISTS |
| MC300 | FUND OF ENGR MECH AND DESIGN  |

**AND****Integrative Experience - First Course** Choose 1 of 2

|       |                               |
|-------|-------------------------------|
| ME404 | MECHANICAL ENGINEERING DESIGN |
| XE485 | ENGINEERING CAPSTONE DESIGN I |

**AND****Integrative Experience - Second Course** Choose 1 of 2

|       |                                |
|-------|--------------------------------|
| ME496 | MECHANICAL SYSTEM DESIGN       |
| XE495 | ENGINEERING CAPSTONE DESIGN II |

**AND****Science Depth**

Cadets in this major must take PH202, PH252, or PH275 to satisfy the Science Depth requirement.

|       |                     |
|-------|---------------------|
| PH202 | PHYSICS II          |
| PH252 | ADVANCED PHYSICS II |
| PH275 | PHYSICS II: SPACE   |

**STEM Depth**

Cadets in this major must take MA204/MA205 or MA255 to satisfy the STEM Depth requirement.

|       |                            |
|-------|----------------------------|
| MA204 | CALCULUS I AND II          |
| MA205 | CALCULUS II                |
| MA255 | ADV MULTIVARIABLE CALCULUS |

**AND****Curriculum Requirements**

This section describes how cadets in this major will satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the core curriculum, EE301 and ME202.

|       |                            |
|-------|----------------------------|
| EE301 | FUNDAMENTALS OF ELEC ENGIN |
| ME202 | COMPUTER AIDED DESIGN      |

**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and ME400.

|       |                                |
|-------|--------------------------------|
| ME400 | MECHANICAL ENGINEERING SEMINAR |
|-------|--------------------------------|

## 2029 Mechanical Engineering Major w/ Honors Curriculum

| Code  | Short Description                      | Description                            | Transcript Description                 | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------------|----------------------------------------|----------------------------------------|--------------|--------------|
| MEN3H | Mechanical Engineering Major w/ Honors | Mechanical Engineering Major w/ Honors | Mechanical Engineering Major w/ Honors | 0            | 0            |

## 2029 Mechanical Engineering Major w/ Honors Tracks

| Subject Area                   | Description                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Examination Requirement</b> | Pass the Fundamentals of Engineering Examination in the final semester of study, unless waived by the ME Program Director.                                                             |
| <b>Scholarship Requirement</b> | Submit evidence of an engineering-based scholarly activity outside of the normal curricular requirements for the mechanical engineering major, as approved by the ME Program Director. |
| <b>Grade Requirements</b>      | Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.                             |

## 2029 Mechanical Engineering Studies Major Curriculum

| Code | Short Description              | Description                          | Transcript Description         | Req Crse Cnt | Opt Crse Cnt |
|------|--------------------------------|--------------------------------------|--------------------------------|--------------|--------------|
| MES3 | Mechanical Engineering Studies | Mechanical Engineering Studies Major | Mechanical Engineering Studies | 11           | 6            |

## 2029 Mechanical Engineering Studies Major Tracks

| Subject Area               | Description                    |
|----------------------------|--------------------------------|
| <b>Required Courses</b>    | Choose 7 of 7                  |
| CE364                      | MECHANICS OF MATERIALS         |
| ME201                      | INTRO TO MECH ENGINEERING      |
| ME202                      | COMPUTER AIDED DESIGN          |
| ME306                      | DYNAMICS                       |
| ME380                      | ENGINEERING MATERIALS          |
| ME400                      | MECHANICAL ENGINEERING SEMINAR |
| ME403                      | MANUFACTURING/MACHINE COMP DSN |
| <b>AND</b>                 |                                |
| <b>Thermal-Fluid Track</b> | Choose 2 of 2                  |
| ME301                      | THERMODYNAMICS                 |
| ME362                      | FLUID MECHANICS                |

|                                                                                                                                                                                                                                                                                        |                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>AND</b>                                                                                                                                                                                                                                                                             |                                |
| <b>General Electives</b>                                                                                                                                                                                                                                                               |                                |
|                                                                                                                                                                                                                                                                                        | Choose 3 of 24                 |
| AE287                                                                                                                                                                                                                                                                                  | INTRODUCTION TO AERONAUTICS    |
| CE350                                                                                                                                                                                                                                                                                  | INFRASTRUCTURE ENGINEERING     |
| CE403                                                                                                                                                                                                                                                                                  | STRUCTURAL ANALYSIS            |
| EE360                                                                                                                                                                                                                                                                                  | FUNDAMENTALS OF DIGITAL LOGIC  |
| EM381                                                                                                                                                                                                                                                                                  | ENGINEERING ECONOMY            |
| ME388                                                                                                                                                                                                                                                                                  | HELICOPTER AERONAUTICS         |
| ME389                                                                                                                                                                                                                                                                                  | INTRO TO ADV STUDY IN MECH ENG |
| ME389A                                                                                                                                                                                                                                                                                 | INTRO TO ADV STUDY IN MECH ENG |
| ME472                                                                                                                                                                                                                                                                                  | ENERGY CONVERSION SYSTEMS      |
| ME480                                                                                                                                                                                                                                                                                  | HEAT TRANSFER                  |
| ME481                                                                                                                                                                                                                                                                                  | AIRCRAFT AERODYNAMICS AND DSGN |
| ME486                                                                                                                                                                                                                                                                                  | VIBRATION ENGINEERING          |
| ME489                                                                                                                                                                                                                                                                                  | ADV STUDY IN MECH ENGRNG       |
| ME489A                                                                                                                                                                                                                                                                                 | ADV STUDY IN MECH ENGRNG       |
| ME490                                                                                                                                                                                                                                                                                  | TOPICS IN MECHANICAL ENGRNG    |
| ME491                                                                                                                                                                                                                                                                                  | AUTOMOTIVE POWERPLANTS         |
| ME492                                                                                                                                                                                                                                                                                  | PWR TRAINS & VEH DYNAMICS      |
| ME493                                                                                                                                                                                                                                                                                  | WEAPONS ENGINEERING            |
| NE300                                                                                                                                                                                                                                                                                  | FUNDAMENTALS OF NUCLEAR ENGR   |
| XE310                                                                                                                                                                                                                                                                                  | INTRO TO BIOMECH ENGINEERING   |
| XE365                                                                                                                                                                                                                                                                                  | ADV EXP METHODS & DATA PROC    |
| XE465                                                                                                                                                                                                                                                                                  | TOPICS: ADVANCED TECHNOLOGY    |
| XE472                                                                                                                                                                                                                                                                                  | DYNAMIC MODELING AND CONTROL   |
| XE475                                                                                                                                                                                                                                                                                  | MECHATRONICS                   |
| <b>AND</b>                                                                                                                                                                                                                                                                             |                                |
| <b>Complementary Support Courses</b>                                                                                                                                                                                                                                                   |                                |
|                                                                                                                                                                                                                                                                                        | Choose 3 of 6                  |
| The following courses provide foundational math and applied science for cadets in the mechanical engineering major. For the math requirement, a cadet should take MA364 but may substitute MA365 if qualified. For the IT/CYBER requirement, a cadet may take EE301 or CY305 or CY355. |                                |
| CY305                                                                                                                                                                                                                                                                                  | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                                                                                                                                  | CYBER FOUNDATIONS - COMPUTING  |
| EE301                                                                                                                                                                                                                                                                                  | FUNDAMENTALS OF ELEC ENGIN     |
| MA364                                                                                                                                                                                                                                                                                  | ENGINEERING MATHEMATICS        |
| MA365                                                                                                                                                                                                                                                                                  | ADV MATH FOR ENGRS/SCIENTISTS  |
| MC300                                                                                                                                                                                                                                                                                  | FUND OF ENGR MECH AND DESIGN   |
| <b>AND</b>                                                                                                                                                                                                                                                                             |                                |
| <b>Integrative Experience - ME Sequence</b>                                                                                                                                                                                                                                            |                                |
|                                                                                                                                                                                                                                                                                        | Choose 2 of 2                  |
| ME404                                                                                                                                                                                                                                                                                  | MECHANICAL ENGINEERING DESIGN  |
| ME496                                                                                                                                                                                                                                                                                  | MECHANICAL SYSTEM DESIGN       |
| <b>OR</b>                                                                                                                                                                                                                                                                              |                                |
| <b>Integrative Experience - XE Sequence</b>                                                                                                                                                                                                                                            |                                |
|                                                                                                                                                                                                                                                                                        | Choose 2 of 2                  |
| XE485                                                                                                                                                                                                                                                                                  | ENGINEERING CAPSTONE DESIGN I  |
| XE495                                                                                                                                                                                                                                                                                  | ENGINEERING CAPSTONE DESIGN II |
| <b>AND</b>                                                                                                                                                                                                                                                                             |                                |
| <b>Science Depth</b>                                                                                                                                                                                                                                                                   |                                |
| Cadets in this major must take PH206, PH256, or PH275 to satisfy the Science Depth requirement.                                                                                                                                                                                        |                                |
| PH202                                                                                                                                                                                                                                                                                  | PHYSICS II                     |
| PH252                                                                                                                                                                                                                                                                                  | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                                                                                                                                  | PHYSICS II: SPACE              |

**STEM Depth**

Cadets in this major must take MA204/MA205 or MA255 to satisfy the STEM Depth requirement.

|       |                            |
|-------|----------------------------|
| MA204 | CALCULUS I AND II          |
| MA205 | CALCULUS II                |
| MA255 | ADV MULTIVARIABLE CALCULUS |

**AND****Curriculum Requirements**

This section describes how cadets in this major will satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the core curriculum, EE301 (or CY305/355) and ME202.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |
| EE301 | FUNDAMENTALS OF ELEC ENGIN    |
| ME202 | COMPUTER AIDED DESIGN         |

**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and ME400.

|       |                                |
|-------|--------------------------------|
| ME400 | MECHANICAL ENGINEERING SEMINAR |
|-------|--------------------------------|

**2029 Mechanical Engineering Studies Major Remarks Alternate major for Mechanical Engineering.**

**Military Instruction**

No Major found



## Physics and Nuclear Engineering

### 2029 Nuclear Engineering Major Curriculum

| Code | Short Description   | Description               | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|---------------------|---------------------------|------------------------|-----------------|-----------------|
| NEN1 | Nuclear Engineering | Nuclear Engineering Major | Nuclear Engineering    | 17              | 1               |

### 2029 Nuclear Engineering Major Tracks

| Subject Area                                                                               | Description                    |
|--------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                    | Choose 13 of 13                |
| CE364                                                                                      | MECHANICS OF MATERIALS         |
| MC300                                                                                      | FUND OF ENGR MECH AND DESIGN   |
| ME301                                                                                      | THERMODYNAMICS                 |
| ME362                                                                                      | FLUID MECHANICS                |
| NE300                                                                                      | FUNDAMENTALS OF NUCLEAR ENGR   |
| NE355                                                                                      | NUCLEAR REACTOR ENGINEERING    |
| NE361                                                                                      | COMPUTATIONAL DESIGN IN NE     |
| NE400                                                                                      | NUCLEAR ENGINEERING SEMINAR    |
| NE450                                                                                      | NUCLEAR WEAPONS EFFECTS        |
| NE452                                                                                      | INSTRUMENTATION AND SHIELDING  |
| NE461                                                                                      | ADV COMPUTATIONAL DESIGN IN NE |
| NE474                                                                                      | RADIOLOGICAL SAFETY            |
| PH365                                                                                      | MODERN PHYSICS                 |
| <b>AND</b>                                                                                 |                                |
| <b>Complementary Support Courses</b>                                                       | Choose 3 of 4                  |
| For the math requirement, a cadet should take MA364 but may substitute MA365 if qualified. |                                |
| EE301                                                                                      | FUNDAMENTALS OF ELEC ENGIN     |
| MA364                                                                                      | ENGINEERING MATHEMATICS        |
| MA365                                                                                      | ADV MATH FOR ENGRS/SCIENTISTS  |
| ME480                                                                                      | HEAT TRANSFER                  |
| <b>AND</b>                                                                                 |                                |
| <b>Integrative Experience for the Major</b>                                                | Choose 2 of 2                  |
| NE495                                                                                      | ADV NUC SYSTEM DESIGN PROJ I   |
| NE496                                                                                      | ADV NUC SYSTEM DESIGN PROJ II  |
| <b>AND</b>                                                                                 |                                |
| <b>Science Depth</b>                                                                       |                                |
| Cadets in this major must take PH202 or PH275 to satisfy the Science Depth requirement.    |                                |
| PH202                                                                                      | PHYSICS II                     |
| PH275                                                                                      | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                          |                                |
| Cadets in this major must take MA204 or MA205 to satisfy the STEM Depth requirement.       |                                |
| MA204                                                                                      | CALCULUS I AND II              |
| MA205                                                                                      | CALCULUS II                    |
| <b>AND</b>                                                                                 |                                |
| <b>Curriculum Requirements</b>                                                             |                                |

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum, EE301 and NE361.

EE301

FUNDAMENTALS OF ELEC ENGIN

NE361

COMPUTATIONAL DESIGN IN NE

**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

NE400

NUCLEAR ENGINEERING SEMINAR

**2029 Nuclear Engineering Major w/ Honors Curriculum**

| Code  | Short Description             | Description                         | Transcript Description        | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------|-------------------------------------|-------------------------------|--------------|--------------|
| NEN1H | Nuclear Engineering w/ Honors | Nuclear Engineering Major w/ Honors | Nuclear Engineering w/ Honors | 0            | 1            |

**2029 Nuclear Engineering Major w/ Honors Tracks**

| Subject Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Description                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Required Course</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Choose 1 of 2               |
| NE389                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | INDIVIDUAL STUDY IN NE      |
| NE489                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ADV IND STDY NUCLEAR ENGNRG |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             |
| <b>Research/Writing Requirement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                             |
| Cadets must demonstrate excellence by writing a paper based upon the results of the NE389 or NE489, an AIAD, or their integrative experience that is suitable for submission to an undergraduate level journal. Present this paper at, for example, USMA Project's Day, a PaNE colloquium, or an ANS Student Conference. The Head of the Department of Physics and Nuclear Engineering will determine whether the quality of the work completed is sufficient to merit successful completion of the honors program. |                             |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                             |
| <b>Grade Requirement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                             |
| Attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.                                                                                                                                                                                                                                                                                                                                                                                                                                           |                             |

**2029 Nuclear Engineering Science Major Curriculum**

| Code | Short Description           | Description                       | Transcript Description      | Req Crse Cnt | Opt Crse Cnt |
|------|-----------------------------|-----------------------------------|-----------------------------|--------------|--------------|
| NES1 | Nuclear Engineering Science | Nuclear Engineering Science Major | Nuclear Engineering Science | 16           | 1            |

## 2029 Nuclear Engineering Science Major Tracks

| Subject Area                                                                                                                                                                 | Description                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                      | Choose 12 of 12                |
| CE364                                                                                                                                                                        | MECHANICS OF MATERIALS         |
| MC300                                                                                                                                                                        | FUND OF ENGR MECH AND DESIGN   |
| ME362                                                                                                                                                                        | FLUID MECHANICS                |
| NE300                                                                                                                                                                        | FUNDAMENTALS OF NUCLEAR ENGR   |
| NE355                                                                                                                                                                        | NUCLEAR REACTOR ENGINEERING    |
| NE361                                                                                                                                                                        | COMPUTATIONAL DESIGN IN NE     |
| NE400                                                                                                                                                                        | NUCLEAR ENGINEERING SEMINAR    |
| NE450                                                                                                                                                                        | NUCLEAR WEAPONS EFFECTS        |
| NE452                                                                                                                                                                        | INSTRUMENTATION AND SHIELDING  |
| NE461                                                                                                                                                                        | ADV COMPUTATIONAL DESIGN IN NE |
| NE474                                                                                                                                                                        | RADIOLOGICAL SAFETY            |
| PH365                                                                                                                                                                        | MODERN PHYSICS                 |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Complementary Support Courses</b>                                                                                                                                         | Choose 3 of 4                  |
| For the math requirement, a cadet should take MA364 but may substitute MA365 if qualified.                                                                                   |                                |
| EE301                                                                                                                                                                        | FUNDAMENTALS OF ELEC ENGIN     |
| MA364                                                                                                                                                                        | ENGINEERING MATHEMATICS        |
| MA365                                                                                                                                                                        | ADV MATH FOR ENGRS/SCIENTISTS  |
| ME301                                                                                                                                                                        | THERMODYNAMICS                 |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 2 of 2                  |
| NE495                                                                                                                                                                        | ADV NUC SYSTEM DESIGN PROJ I   |
| NE496                                                                                                                                                                        | ADV NUC SYSTEM DESIGN PROJ II  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major must take PH202 or PH275 to satisfy the Science Depth requirement.                                                                                      |                                |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take MA204 or MA205 to satisfy the STEM Depth requirement.                                                                                         |                                |
| MA204                                                                                                                                                                        | CALCULUS I AND II              |
| MA205                                                                                                                                                                        | CALCULUS II                    |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Curriculum Requirements</b>                                                                                                                                               |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum, EE301 and NE361.                          |                                |
| EE301                                                                                                                                                                        | FUNDAMENTALS OF ELEC ENGIN     |
| NE361                                                                                                                                                                        | COMPUTATIONAL DESIGN IN NE     |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major will satisfy the core engineering requirement as part of their major courses.                                                                           |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| NE400                                                                                                                                                                        | NUCLEAR ENGINEERING SEMINAR    |

## 2029 Nuclear Engineering Science Major Remarks Alternate major for Nuclear Engineering.

### 2029 Nuclear Science Minor Curriculum

| Code  | Short Description     | Description           | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|-----------------------|-----------------------|------------------------|-----------------|-----------------|
| NSM0N | Nuclear Science Minor | Nuclear Science Minor | Nuclear Science Minor  | 3               | 3               |

### 2029 Nuclear Science Minor Tracks

#### Subject Area

#### Description

Cadets enrolled in the minor must take four courses: NE300, NE355, NE452, and either NE361 or NE474 (no course substitution authorized). Two courses may be substituted for NE350, NE450, and either NE361 or NE474 at the discretion of the Head of the Department of Physics and Nuclear Engineering. Course substitutes must include significant content of basic science (physical, biological, and chemical), mathematics, or engineering science and engineering topics.

Choose 3 of 3

(no substitution)

|       |                               |
|-------|-------------------------------|
| NE300 | FUNDAMENTALS OF NUCLEAR ENGR  |
| NE355 | NUCLEAR REACTOR ENGINEERING   |
| NE452 | INSTRUMENTATION AND SHIELDING |

**AND**

Choose 1 of 2

(no substitution)

|       |                            |
|-------|----------------------------|
| NE361 | COMPUTATIONAL DESIGN IN NE |
| NE474 | RADIOLOGICAL SAFETY        |

**AND**

Choose 2 of 3

For Cadets who chose NE474 above: (course substitution at the discretion of the Head, D/PaNE)

|       |                            |
|-------|----------------------------|
| NE350 | RADIOLOGICAL ENGR DESIGN   |
| NE361 | COMPUTATIONAL DESIGN IN NE |
| NE450 | NUCLEAR WEAPONS EFFECTS    |

**OR**

Choose 2 of 3

For Cadets who chose NE361 rather than NE474: (course substitution at the discretion of the Head, D/PaNE)

|       |                          |
|-------|--------------------------|
| NE350 | RADIOLOGICAL ENGR DESIGN |
| NE450 | NUCLEAR WEAPONS EFFECTS  |
| NE474 | RADIOLOGICAL SAFETY      |

### 2029 Photonics Minor Curriculum

| Code  | Short Description | Description     | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|-------------------|-----------------|------------------------|-----------------|-----------------|
| PHT0N | Photonics Minor   | Photonics Minor | Photonics Minor        | 4               | 2               |

### 2029 Photonics Minor Tracks

| Subject Area                                          | Description                  |
|-------------------------------------------------------|------------------------------|
| <b>Required Courses</b>                               | Choose 4 of 4                |
| EE362                                                 | INTRODUCTION TO ELECTRONICS  |
| EE483                                                 | PHOTONICS ENGINEERING        |
| PH384                                                 | APPLIED OPTICS               |
| PH485                                                 | LASER PHYSICS                |
| <b>AND</b>                                            |                              |
| <b>Required Course</b>                                | Choose 1 of 2                |
| Research project must be optics or photonics related. |                              |
| PH365                                                 | MODERN PHYSICS               |
| PH489                                                 | ADV INDIV STUDY IN PHYSICS   |
| <b>AND</b>                                            |                              |
| <b>Required Course</b>                                | Choose 1 of 2                |
| Research project must be optics or photonics related. |                              |
| EE480                                                 | OPTICAL FIBER COMMUNICATIONS |
| EE489                                                 | ADV IND STUDY IN ELECT ENGR  |

### 2029 Physics Major Curriculum

| Code | Short Description | Description   | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------|---------------|------------------------|-----------------|-----------------|
| PHY1 | Physics           | Physics Major | Physics                | 14              | 0               |

### 2029 Physics Major Tracks

| Subject Area                                                                                                                                                                                                                                | Description                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                                                                                                                                                     | Choose 10 of 10                |
| PH365                                                                                                                                                                                                                                       | MODERN PHYSICS                 |
| PH381                                                                                                                                                                                                                                       | INTRMED CLASSICAL MECHANICS    |
| PH382                                                                                                                                                                                                                                       | INTERMEDIATE ELECTRODYNAMICS   |
| PH384                                                                                                                                                                                                                                       | APPLIED OPTICS                 |
| PH481                                                                                                                                                                                                                                       | STATISTICAL PHYSICS            |
| PH482                                                                                                                                                                                                                                       | ADVANCED CLASSICAL MECHAN      |
| PH484                                                                                                                                                                                                                                       | INTERMEDIATE QUANTUM MECHANICS |
| PH485                                                                                                                                                                                                                                       | LASER PHYSICS                  |
| PH486                                                                                                                                                                                                                                       | EXPERIMENTAL PHYSICS           |
| PH487                                                                                                                                                                                                                                       | ADVANCED QUANTUM MECHANICS     |
| <b>AND</b>                                                                                                                                                                                                                                  |                                |
| <b>Complementary Support Courses</b>                                                                                                                                                                                                        | Choose 3 of 5                  |
| The following courses provide foundational math and applied science support for physics majors. Cadets placement in mathematics dictates the choice of MA364 or MA365; cadet performance in PH382 will guide placement in EE486 over EE383. |                                |
| EE301                                                                                                                                                                                                                                       | FUNDAMENTALS OF ELEC ENGR      |

|                                                                                                                                                                              |                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| EE383                                                                                                                                                                        | ELECTROMAGN FIELDS & WAVES    |
| EE486                                                                                                                                                                        | SOLID STATE ELECTRONICS       |
| MA364                                                                                                                                                                        | ENGINEERING MATHEMATICS       |
| MA365                                                                                                                                                                        | ADV MATH FOR ENGRS/SCIENTISTS |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 1 of 1                 |
| PH456                                                                                                                                                                        | SCIENCE AND POLICY            |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Science Depth</b>                                                                                                                                                         |                               |
| Cadets in this major must take PH202, PH252, or PH275 to satisfy the Science Depth requirement.                                                                              |                               |
| PH202                                                                                                                                                                        | PHYSICS II                    |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II           |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE             |
| <b>STEM Depth</b>                                                                                                                                                            |                               |
| Cadets in this major must take MA204/MA205 or MA255 to satisfy the STEM Depth requirement.                                                                                   |                               |
| MA204                                                                                                                                                                        | CALCULUS I AND II             |
| MA205                                                                                                                                                                        | CALCULUS II                   |
| MA255                                                                                                                                                                        | ADV MULTIVARIABLE CALCULUS    |
| <b>AND</b>                                                                                                                                                                   |                               |
| <hr/>                                                                                                                                                                        |                               |
| <b>Curriculum Requirements</b>                                                                                                                                               |                               |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                               |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                               |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum, EE301 and PH456.                          |                               |
| EE301                                                                                                                                                                        | FUNDAMENTALS OF ELEC ENGIN    |
| PH456                                                                                                                                                                        | SCIENCE AND POLICY            |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                               |
| Cadets in this major can choose any approved three-course Core Engineering Sequence.                                                                                         |                               |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                               |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                               |
| PH456                                                                                                                                                                        | SCIENCE AND POLICY            |

## 2029 Physics Major w/ Honors Curriculum

| Code  | Short Description | Description             | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------|-------------------------|------------------------|--------------|--------------|
| PHY1H | Physics w/ Honors | Physics Major w/ Honors | Physics w/ Honors      | 1            | 1            |

## 2029 Physics Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                                             | Description   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Honors Program</b>                                                                                                                                                                                                    |               |
| The honors program in physics entails the completion of two courses beyond the 14-course major. An essential component of this program is cadet participation in physics, space science or nuclear engineering research. |               |
| <b>Research Requirement</b>                                                                                                                                                                                              | Choose 1 of 6 |

|       |                                |
|-------|--------------------------------|
| NE389 | INDIVIDUAL STUDY IN NE         |
| NE489 | ADV IND STDY NUCLEAR ENGNRG    |
| PH389 | INDIVIDUAL STUDY IN PHYSICS    |
| PH489 | ADV INDIV STUDY IN PHYSICS     |
| SP389 | IND RESEARCH SPACE SCIENCE     |
| SP489 | ADV IND RESEARCH SPACE SCIENCE |

**AND****Courses**

Choose 1 of 27

Complete one course from the following list.

|        |                                |
|--------|--------------------------------|
| MA371  | LINEAR ALGEBRA                 |
| MA376  | APPLIED STATISTICS             |
| MA385  | CHAOS AND FRACTALS             |
| MA386  | INTRO TO NUMERICAL ANALYSIS    |
| MA396  | NUM METH SOLUTIONS DIFF EQNS   |
| MA476  | MATHEMATICAL STATISTICS        |
| MA484  | PARTIAL DIFF EQUATIONS         |
| MA485  | APPLIED COMPLEX VARIABLES      |
| NE389  | INDIVIDUAL STUDY IN NE         |
| NE389A | INDIVIDUAL STUDY IN NE         |
| NE474  | RADIOLOGICAL SAFETY            |
| NE489  | ADV IND STDY NUCLEAR ENGNRG    |
| NE489A | ADV IND STUDY NUCLEAR ENGNRG   |
| PH389  | INDIVIDUAL STUDY IN PHYSICS    |
| PH389A | INDIVIDUAL STUDY IN PHYSICS-A  |
| PH489  | ADV INDIV STUDY IN PHYSICS     |
| PH489A | ADV INDIV STUDY IN PHYSICS     |
| PH489B | ADV INDIV STUDY IN PHYSICS     |
| PH495  | SPECIAL TOPICS IN PHYSICS      |
| SP389  | IND RESEARCH SPACE SCIENCE     |
| SP389A | IND RESEARCH SPACE SCIENCE - A |
| SP389B | IND RESEARCH SPACE SCIENCE - B |
| SP471  | ASTRONAUTICS                   |
| SP472  | SPACE PHYSICS                  |
| SP489  | ADV IND RESEARCH SPACE SCIENCE |
| SP489A | ADV IND RESEARCH SPACE SCIENCE |
| SP489B | ADV IND RESEARCH SPACE SCIENCE |

**AND****Grade Requirements**

Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Physics Studies Major Curriculum

| Code | Short Description     | Description           | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-----------------------|-----------------------|------------------------|-----------------|-----------------|
| PHS0 | Physics Studies Major | Physics Studies Major | Physics Studies        | 9               | 4               |

## 2029 Physics Studies Major Tracks

| Subject Area     | Description   |
|------------------|---------------|
| Required Courses | Choose 5 of 5 |

|                                                                                                                                                                                                                                             |                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| PH365                                                                                                                                                                                                                                       | MODERN PHYSICS                 |
| PH381                                                                                                                                                                                                                                       | INTRMED CLASSICAL MECHANICS    |
| PH382                                                                                                                                                                                                                                       | INTERMEDIATE ELECTRODYNAMICS   |
| PH384                                                                                                                                                                                                                                       | APPLIED OPTICS                 |
| PH484                                                                                                                                                                                                                                       | INTERMEDIATE QUANTUM MECHANICS |
| <b>AND</b>                                                                                                                                                                                                                                  |                                |
| <b>Required Physics Electives</b>                                                                                                                                                                                                           | Choose 4 of 5                  |
| PH481                                                                                                                                                                                                                                       | STATISTICAL PHYSICS            |
| PH482                                                                                                                                                                                                                                       | ADVANCED CLASSICAL MECHAN      |
| PH485                                                                                                                                                                                                                                       | LASER PHYSICS                  |
| PH486                                                                                                                                                                                                                                       | EXPERIMENTAL PHYSICS           |
| PH487                                                                                                                                                                                                                                       | ADVANCED QUANTUM MECHANICS     |
| <b>AND</b>                                                                                                                                                                                                                                  |                                |
| <b>Complementary Support Courses</b>                                                                                                                                                                                                        | Choose 3 of 5                  |
| The following courses provide foundational math and applied science support for physics majors. Cadets placement in mathematics dictates the choice of MA364 or MA365; cadet performance in PH382 will guide placement in EE486 over EE383. |                                |
| EE301                                                                                                                                                                                                                                       | FUNDAMENTALS OF ELEC ENGIN     |
| EE383                                                                                                                                                                                                                                       | ELECTROMAGN FIELDS & WAVES     |
| EE486                                                                                                                                                                                                                                       | SOLID STATE ELECTRONICS        |
| MA364                                                                                                                                                                                                                                       | ENGINEERING MATHEMATICS        |
| MA365                                                                                                                                                                                                                                       | ADV MATH FOR ENGRS/SCIENTISTS  |
| <b>AND</b>                                                                                                                                                                                                                                  |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                                                                                 | Choose 1 of 1                  |
| PH456                                                                                                                                                                                                                                       | SCIENCE AND POLICY             |
| <b>AND</b>                                                                                                                                                                                                                                  |                                |
| <b>Science Depth</b>                                                                                                                                                                                                                        |                                |
| Cadets in this major must take PH202, PH252, or PH275 to satisfy the Science Depth requirement.                                                                                                                                             |                                |
| PH202                                                                                                                                                                                                                                       | PHYSICS II                     |
| PH252                                                                                                                                                                                                                                       | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                                                                                       | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                                                                                           |                                |
| Cadets in this major must take MA204/MA205 or MA255 to satisfy the STEM Depth requirement.                                                                                                                                                  |                                |
| MA204                                                                                                                                                                                                                                       | CALCULUS I AND II              |
| MA205                                                                                                                                                                                                                                       | CALCULUS II                    |
| MA255                                                                                                                                                                                                                                       | ADV MULTIVARIABLE CALCULUS     |
| <b>AND</b>                                                                                                                                                                                                                                  |                                |
| <b>Curriculum Requirements</b>                                                                                                                                                                                                              |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                                                                                    |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                                                                                 |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum, EE301 and PH456.                                                                                         |                                |
| EE301                                                                                                                                                                                                                                       | FUNDAMENTALS OF ELEC ENGIN     |
| PH456                                                                                                                                                                                                                                       | SCIENCE AND POLICY             |
| <b>Core Engineering Sequence</b>                                                                                                                                                                                                            |                                |
| Cadets in this major can choose any approved three-course Core Engineering Sequence.                                                                                                                                                        |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                                                                                 |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.                                                                |                                |
| PH456                                                                                                                                                                                                                                       | SCIENCE AND POLICY             |



## 2029 Physics Studies Major Remarks Physics Studies Major (PHS0) is the required alternate major to the Physics Major (PHY1).

### 2029 Space Science Major Curriculum

| Code | Short Description | Description         | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------|---------------------|------------------------|--------------|--------------|
| SSC0 | Space Science     | Space Science Major | Space Science          | 12           | 1            |

### 2029 Space Science Major Tracks

| Subject Area                                                                                                                             | Description                    |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Space Science Courses</b>                                                                                                    | Choose 8 of 8                  |
| EE301                                                                                                                                    | FUNDAMENTALS OF ELEC ENGIN     |
| PH382                                                                                                                                    | INTERMEDIATE ELECTRODYNAMICS   |
| PH384                                                                                                                                    | APPLIED OPTICS                 |
| PH485                                                                                                                                    | LASER PHYSICS                  |
| SP471                                                                                                                                    | ASTRONAUTICS                   |
| SP472                                                                                                                                    | SPACE PHYSICS                  |
| SP473                                                                                                                                    | OBSERVATIONAL ASTRONOMY        |
| SP474                                                                                                                                    | ASTROPHYSICS                   |
| <b>AND</b>                                                                                                                               |                                |
| <b>Required Space Science Related Electives</b>                                                                                          | Choose 1 of 2                  |
| PH365                                                                                                                                    | MODERN PHYSICS                 |
| PH381                                                                                                                                    | INTRMED CLASSICAL MECHANICS    |
| <b>AND</b>                                                                                                                               |                                |
| <b>Complementary Support Courses</b>                                                                                                     | Choose 3 of 5                  |
| Some cadets may qualify for MA365 to fulfill their Math CSC requirement. Cadets can take either GE373 or GE478 for the third CSC course. |                                |
| GE373                                                                                                                                    | REMOTE SENSING                 |
| GE477                                                                                                                                    | ADVANCED REMOTE SENSING        |
| GE478                                                                                                                                    | MILITARY GEOSPATIAL OPERATIONS |
| MA364                                                                                                                                    | ENGINEERING MATHEMATICS        |
| MA365                                                                                                                                    | ADV MATH FOR ENGRS/SCIENTISTS  |
| <b>AND</b>                                                                                                                               |                                |
| <b>Integrative Experience</b>                                                                                                            | Choose 1 of 1                  |
| PH456                                                                                                                                    | SCIENCE AND POLICY             |
| <b>AND</b>                                                                                                                               |                                |
| <b>Science Depth</b>                                                                                                                     |                                |
| Cadets in this major must take PH202/252 or PH275 to satisfy the Science Depth requirement.                                              |                                |
| PH202                                                                                                                                    | PHYSICS II                     |
| PH252                                                                                                                                    | ADVANCED PHYSICS II            |
| PH275                                                                                                                                    | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                        |                                |
| Cadets in this major must take MA204/MA205 or MA255 to satisfy the STEM Depth requirement.                                               |                                |
| MA204                                                                                                                                    | CALCULUS I AND II              |

|       |                            |
|-------|----------------------------|
| MA205 | CALCULUS II                |
| MA255 | ADV MULTIVARIABLE CALCULUS |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum, EE301 and PH456.

|       |                            |
|-------|----------------------------|
| EE301 | FUNDAMENTALS OF ELEC ENGIN |
| PH456 | SCIENCE AND POLICY         |

**Core Engineering Sequence**

Cadets in this major may choose any approved three-course Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                    |
|-------|--------------------|
| PH456 | SCIENCE AND POLICY |
|-------|--------------------|

## 2029 Space Science Major w/ Honors Curriculum

| Code  | Short Description       | Description                   | Transcript Description  | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------|-------------------------------|-------------------------|--------------|--------------|
| SSC0H | Space Science w/ Honors | Space Science Major w/ Honors | Space Science w/ Honors | 1            | 1            |

## 2029 Space Science Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                            | Description                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Honors Program</b>                                                                                                                                                                                                                                                                                                                   |                                |
| The honors program in space science entails the completion of two courses beyond the 13-course major. An essential component of this program is cadet participation in space science, physics, or nuclear engineering research. The Required Space Science Related Elective cannot double count towards fulfilling Honors requirements. |                                |
| <b>Required Courses</b>                                                                                                                                                                                                                                                                                                                 | Choose 1 of 6                  |
| NE389                                                                                                                                                                                                                                                                                                                                   | INDIVIDUAL STUDY IN NE         |
| NE489                                                                                                                                                                                                                                                                                                                                   | ADV IND STDY NUCLEAR ENGNRG    |
| PH389                                                                                                                                                                                                                                                                                                                                   | INDIVIDUAL STUDY IN PHYSICS    |
| PH489                                                                                                                                                                                                                                                                                                                                   | ADV INDIV STUDY IN PHYSICS     |
| SP389                                                                                                                                                                                                                                                                                                                                   | IND RESEARCH SPACE SCIENCE     |
| SP489                                                                                                                                                                                                                                                                                                                                   | ADV IND RESEARCH SPACE SCIENCE |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                              |                                |
| <b>Additional Physics Elective</b>                                                                                                                                                                                                                                                                                                      | Choose 1 of 22                 |
| NE389                                                                                                                                                                                                                                                                                                                                   | INDIVIDUAL STUDY IN NE         |
| NE389A                                                                                                                                                                                                                                                                                                                                  | INDIVIDUAL STUDY IN NE         |
| NE489                                                                                                                                                                                                                                                                                                                                   | ADV IND STDY NUCLEAR ENGNRG    |
| NE489A                                                                                                                                                                                                                                                                                                                                  | ADV IND STUDY NUCLEAR ENGNRG   |
| PH365                                                                                                                                                                                                                                                                                                                                   | MODERN PHYSICS                 |
| PH381                                                                                                                                                                                                                                                                                                                                   | INTRMED CLASSICAL MECHANICS    |
| PH389                                                                                                                                                                                                                                                                                                                                   | INDIVIDUAL STUDY IN PHYSICS    |
| PH389A                                                                                                                                                                                                                                                                                                                                  | INDIVIDUAL STUDY IN PHYSICS-A  |
| PH481                                                                                                                                                                                                                                                                                                                                   | STATISTICAL PHYSICS            |

|        |                                |
|--------|--------------------------------|
| PH482  | ADVANCED CLASSICAL MECHAN      |
| PH484  | INTERMEDIATE QUANTUM MECHANICS |
| PH487  | ADVANCED QUANTUM MECHANICS     |
| PH489  | ADV INDIV STUDY IN PHYSICS     |
| PH489A | ADV INDIV STUDY IN PHYSICS     |
| PH489B | ADV INDIV STUDY IN PHYSICS     |
| PH495  | SPECIAL TOPICS IN PHYSICS      |
| SP389  | IND RESEARCH SPACE SCIENCE     |
| SP389A | IND RESEARCH SPACE SCIENCE - A |
| SP389B | IND RESEARCH SPACE SCIENCE - B |
| SP489  | ADV IND RESEARCH SPACE SCIENCE |
| SP489A | ADV IND RESEARCH SPACE SCIENCE |
| SP489B | ADV IND RESEARCH SPACE SCIENCE |

**AND****Grade Requirements**

Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

## 2029 Space Science Minor Curriculum

| Code  | Short Description   | Description         | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|---------------------|---------------------|------------------------|-----------------|-----------------|
| SPA0N | Space Science Minor | Space Science Minor | Space Science Minor    | 3               | 2               |

## 2029 Space Science Minor Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Description                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Space Science Courses</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Choose 3 of 3                  |
| SP471                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ASTRONAUTICS                   |
| SP472                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | SPACE PHYSICS                  |
| SP473                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | OBSERVATIONAL ASTRONOMY        |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |
| <b>Required Space Science Related Discipline Electives</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |
| Choose one of the following Space and Missile Defense (SMD) critical areas of interest; Space Science, Missile Defense, Directed Energy, Cyber Operations, and Policy Development from which to select your two electives for this minor. These related discipline electives are further grouped by root electives that support areas of interest that are inherent in particular majors. These groupings allow the student to synthesize and ponder how to apply these respective electives to SMD critical areas of interest as well as space exploration. Cadets choosing this minor must meet all prerequisites for the courses chosen. Cadets majoring in Space Science are not eligible for this minor. |                                |
| <b>Space Science: Biology, Chemistry, Kinesiology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Choose 2 of 10                 |
| CH275                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | BIOLOGY                        |
| CH375                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ADVANCED BIOLOGY               |
| CH383                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ORGANIC CHEMISTRY I            |
| CH384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ORGANIC CHEMISTRY II           |
| CH387                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | HUMAN PHYSIOLOGY               |
| CH457                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MICROBIOLOGY                   |
| KN355                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | FUNCTIONAL ANATOMY             |
| KN360                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | BIOMECHANICS OF HUMAN MOVEMENT |

|                                                                                   |                                |
|-----------------------------------------------------------------------------------|--------------------------------|
| KN460                                                                             | EXERCISE PHYSIOLOGY            |
| KN475                                                                             | MUSCULAR FUNCTION & ADAPTATION |
| <b>OR</b>                                                                         |                                |
| <b>Space Science: Electronics, Signals, Fields, Communications, and Photonics</b> | Choose 2 of 6                  |
| EE362                                                                             | INTRODUCTION TO ELECTRONICS    |
| EE381                                                                             | SIGNALS AND SYSTEMS            |
| EE383                                                                             | ELECTROMAGN FIELDS & WAVES     |
| EE482                                                                             | WIRELESS COMM SYS ENGINEERING  |
| EE483                                                                             | PHOTONICS ENGINEERING          |
| XE383                                                                             | ELECTROMAGNETIC WAVES          |
| <b>OR</b>                                                                         |                                |
| <b>Space Science: Remote sensing, Geology, Hydrology, and Meteorology</b>         | Choose 2 of 7                  |
| EV394                                                                             | HYDROGEOLOGY/HYDRAULIC SYSTEMS |
| GE324                                                                             | METEOROLOGY                    |
| GE332                                                                             | GEOMORPHOLOGY                  |
| GE333                                                                             | PHYSICAL GEOLOGY               |
| GE373                                                                             | REMOTE SENSING                 |
| GE477                                                                             | ADVANCED REMOTE SENSING        |
| GE478                                                                             | MILITARY GEOSPATIAL OPERATIONS |
| <b>OR</b>                                                                         |                                |
| <b>Space Science: Systems Engineering, Design, and Management</b>                 | Choose 2 of 4                  |
| EM411                                                                             | PROJECT MANAGEMENT             |
| SE301                                                                             | FNDTN ENGIN DSGN & SYS MGMT    |
| SE302                                                                             | FUNDAMENTALS OF SYSTEMS ENG    |
| SE375                                                                             | STATISTICS FOR ENGINEERS       |
| <b>OR</b>                                                                         |                                |
| <b>Missile Defense: Dynamics, Controls, and Thermal/Fluid Systems</b>             | Choose 2 of 7                  |
| CE364                                                                             | MECHANICS OF MATERIALS         |
| ME301                                                                             | THERMODYNAMICS                 |
| ME306                                                                             | DYNAMICS                       |
| ME362                                                                             | FLUID MECHANICS                |
| ME472                                                                             | ENERGY CONVERSION SYSTEMS      |
| ME480                                                                             | HEAT TRANSFER                  |
| ME486                                                                             | VIBRATION ENGINEERING          |
| <b>OR</b>                                                                         |                                |
| <b>Missile Defense: Nuclear Science and Radiation</b>                             | Choose 2 of 4                  |
| NE300                                                                             | FUNDAMENTALS OF NUCLEAR ENGR   |
| NE350                                                                             | RADIOLOGICAL ENGR DESIGN       |
| NE450                                                                             | NUCLEAR WEAPONS EFFECTS        |
| NE452                                                                             | INSTRUMENTATION AND SHIELDING  |
| <b>OR</b>                                                                         |                                |
| <b>Directed Energy: Applied Physics and Astrophysics</b>                          | Choose 2 of 6                  |
| PH365                                                                             | MODERN PHYSICS                 |
| PH381                                                                             | INTRMED CLASSICAL MECHANICS    |
| PH382                                                                             | INTERMEDIATE ELECTRODYNAMICS   |
| PH384                                                                             | APPLIED OPTICS                 |
| PH485                                                                             | LASER PHYSICS                  |

SP474

ASTROPHYSICS

**OR****Cyber Operations: Cyber and  
Network Engineering**

Choose 2 of 3

CY300

PROGRAMMING FUNDAMENTALS

CY350

COMPUTER NETWORKS

CY450

CYBER SECURITY ENGINEERING

**OR****Policy Development: National  
Strategy, Policy, and Law**

Choose 2 of 4

LW482

NATIONAL SECURITY LAW

SS477

ECONOMICS OF NATIONAL SECURITY

SS483

NATIONAL SECURITY SEMINAR

WS370

US STRATEGY AND POLICY

## Social Sciences

### 2029 American Foundations Minor Curriculum

| Code  | Short Description          | Description                | Transcript Description     | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------|----------------------------|----------------------------|--------------|--------------|
| AMF0N | American Foundations Minor | American Foundations Minor | American Foundations Minor | 2            | 4            |

### 2029 American Foundations Minor Tracks

| Subject Area                                                                                                                        | Description                    |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Cornerstone Course</b>                                                                                                  | Choose 1 of 1                  |
| SS386                                                                                                                               | POLITICAL THOUGHT AND IDEAS    |
| <b>AND</b>                                                                                                                          |                                |
| <b>History Foundation Elective</b>                                                                                                  | Choose 1 of 2                  |
| HI394                                                                                                                               | REVOLUTIONARY AMERICA          |
| HI487                                                                                                                               | EARLY AMERICAN REPUB 1787-1826 |
| <b>AND</b>                                                                                                                          |                                |
| <b>Required Capstone Tutorial</b>                                                                                                   | Choose 1 of 1                  |
| XH476                                                                                                                               | TUTORIAL IN AM FNDTNS          |
| <b>AND</b>                                                                                                                          |                                |
| <b>Required Core Electives</b>                                                                                                      | Choose 2 of 8                  |
| The course selected as the history foundation course cannot be selected from this list.                                             |                                |
| EN321                                                                                                                               | AMERICAN LITERATURE I          |
| HI340                                                                                                                               | COLONIAL AMERICA               |
| HI394                                                                                                                               | REVOLUTIONARY AMERICA          |
| HI487                                                                                                                               | EARLY AMERICAN REPUB 1787-1826 |
| LW475                                                                                                                               | ADV CONSTITUTIONAL LAW SEM     |
| PY370                                                                                                                               | 17TH & 18TH CENTURY PHILOSOPHY |
| SS376                                                                                                                               | SEMINAR IN AMERICAN GOVERNMENT |
| SS385                                                                                                                               | HISTORY OF ECONOMICS           |
| <b>AND</b>                                                                                                                          |                                |
| <b>Required Core or Legacies Elective</b>                                                                                           | Choose 1 of 21                 |
| Any courses taken to fulfill the history foundation or core elective requirements for this minor cannot be selected from this list. |                                |
| EN321                                                                                                                               | AMERICAN LITERATURE I          |
| EN322                                                                                                                               | AMERICAN LITERATURE II         |
| EN362                                                                                                                               | FILM AND FILM THEORY           |
| GE341                                                                                                                               | GEOGRAPHY OF NORTH AMERICA     |
| HI340                                                                                                                               | COLONIAL AMERICA               |
| HI369                                                                                                                               | AMERICAN FRONTIERS             |
| HI392                                                                                                                               | AMERICAN HISTORICAL MEMORY     |
| HI394                                                                                                                               | REVOLUTIONARY AMERICA          |
| HI395                                                                                                                               | HIST OF CIVIL WAR AMERICA      |
| HI398                                                                                                                               | CIVIL RIGHTS IN AMER HIST      |
| HI487                                                                                                                               | EARLY AMERICAN REPUB 1787-1826 |
| LW461                                                                                                                               | CIVIL RIGHTS AND THE LAW       |
| LW475                                                                                                                               | ADV CONSTITUTIONAL LAW SEM     |
| PL482                                                                                                                               | ARMED FORCES AND SOCIETY       |
| PY330                                                                                                                               | POLITICAL PHILOSOPHY           |
| PY370                                                                                                                               | 17TH & 18TH CENTURY PHILOSOPHY |

|        |                                |
|--------|--------------------------------|
| SS376  | SEMINAR IN AMERICAN GOVERNMENT |
| SS385  | HISTORY OF ECONOMICS           |
| SS472  | SOLDIER & STATE: AM CIV-MIL RE |
| SS490A | COLLOQUIUM (AMER POLITICS)     |
| SS493  | SENIOR STUDIES - AMER POLITICS |

## 2029 American Politics Major Curriculum

| Code | Short Description | Description             | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|-------------------|-------------------------|------------------------|-----------------|-----------------|
| PAP3 | American Politics | American Politics Major | American Politics      | 5               | 8               |

## 2029 American Politics Major Tracks

| Subject Area                                                         | Description                    |
|----------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                              | Choose 4 of 4                  |
| SS360                                                                | POL SCI RESEARCH METHODS       |
| SS366                                                                | COMPARATIVE POLITICS           |
| SS376                                                                | SEMINAR IN AMERICAN GOVERNMENT |
| SS386                                                                | POLITICAL THOUGHT              |
| <b>AND</b>                                                           |                                |
| <b>American Politics Electives</b>                                   | Choose 4 of 13                 |
| SS370                                                                | MASS MEDIA & AMER POLITICS     |
| SS373                                                                | THE AMERICAN PRESIDENCY        |
| SS379                                                                | LEGISLATIVE POLITICS           |
| SS392                                                                | RIGHTS, LIBERTIES & US CITIZEN |
| SS464                                                                | HOMELAND SECURITY              |
| SS465                                                                | TERRORISM: NEW CHALLENGES      |
| SS466                                                                | COMBATING TERRORISM            |
| SS468                                                                | POLITICAL PARTICIPATION        |
| SS472                                                                | THE AM STATE & THE SOLDIER     |
| SS473                                                                | AMERICAN FOREIGN POLICY        |
| SS481                                                                | POLITICS OF DEFENSE POLICY     |
| SS490A                                                               | COLLOQUIUM (AMER POLITICS)     |
| SS493                                                                | SENIOR STUDIES - AMER POLITICS |
| <b>AND</b>                                                           |                                |
| <b>International Relations and<br/>Comparative Politics Elective</b> | Choose 1 of 17                 |
| SS372                                                                | POLITICS AND GOV OF CHINA      |
| SS374                                                                | POL & GOV OF KOREAS & JAPAN    |
| SS375                                                                | GOV & POL RUSSIA & NEIGHBORS   |
| SS377                                                                | POLITICS & GOV OF EUROPE       |
| SS378                                                                | ADV INTL RELATIONS THEORY      |
| SS383                                                                | POLITICS & GOVT-MIDDLE EAST    |
| SS384                                                                | POLITICS & GOVT-LATIN AMER     |
| SS475                                                                | COMP POLITICAL INSTITUTIONS    |
| SS476                                                                | CIVIL WARS                     |
| SS478                                                                | DIST PROF OF SECURITY STUD SEM |
| SS483                                                                | NATIONAL SECURITY SEMINAR      |
| SS485                                                                | GOV & POLITICS SUB-SAHARAN AFR |

|        |                                |
|--------|--------------------------------|
| SS486  | STATE BUILDING                 |
| SS487  | INT'L POLITICAL ECONOMY        |
| SS490B | COLLOQUIUM (INTERNTL AFFAIRS)  |
| WS330  | INTRODUCTION TO GRAND STRATEGY |
| XH397  | GRAND STRATEGY FIELD STUDY     |

**AND****Complementary Support Courses**

Choose 3 of 68

Complementary Support Courses provide cadets exposure to concepts relevant to the study of American Politics from a different disciplinary perspective. For a study-in-depth, choose three complementary support courses from within any of the focus areas (see the Department Academic Counselor for a list of the current focus areas). For a broadening study, choose any three complementary support courses from this list.

|       |                                |
|-------|--------------------------------|
| EN321 | AMERICAN LITERATURE I          |
| EN322 | AMERICAN LITERATURE II         |
| EN353 | WAR LITERATURE                 |
| EN370 | SHAKESPEARE                    |
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE351 | LAND USE PLAN & MGT            |
| GE352 | URBAN GEOGRAPHY                |
| GE462 | ENVIRONMENT AND DEVELOPMENT    |
| GE491 | MILITARY GEOGRAPHY             |
| GE494 | ENVIRONMENTAL SECURITY         |
| HI344 | MODERN DIPLOMACY               |
| HI372 | US FGN RELATIONS SINCE 1898    |
| HI378 | INDUSTRY, IDEOLOGY & THE STATE |
| HI391 | WORLD RELIGIONS                |
| HI392 | AMERICAN HISTORICAL MEMORY     |
| HI396 | THEODORE ROOSEVELT'S AMERICA   |
| HI397 | THE US FROM COLD WAR TO TODAY  |
| HI398 | CIVIL RIGHTS IN AMER HIST      |
| LW472 | CRIMINAL LAW                   |
| LW473 | ENVIRONMENTAL LAW              |
| LW475 | ADV CONSTITUTIONAL LAW SEM     |
| LW481 | INTERNATIONAL LAW              |
| LW482 | NATIONAL SECURITY LAW          |
| MA367 | MATH FOR THE SOCIAL SCIENCES   |
| MA376 | APPLIED STATISTICS             |
| MA394 | FUNDAMENTALS/NETWORK SCIENCE   |
| MG381 | INTRODUCTION TO MANAGEMENT     |
| MG382 | HUMAN RESOURCE MANAGEMENT      |
| MG390 | NEGOTIATION FOR LEADERS        |
| MG421 | STRATEGIC MANAGEMENT           |
| PL361 | RESEARCH METHODS I             |
| PL372 | MILITARY FAMILIES              |
| PL383 | SOCIAL PSYCHOLOGY              |
| PL393 | CRIMINOLOGY-CRIM JUST SYSTM    |
| PL398 | LEADERSHIP THEORY & PRACTICE   |
| PL479 | LEADING ORGNZS THRU CHANGE     |
| PY305 | LOGICAL REASONING              |
| PY310 | REALITY AND KNOWLEDGE          |
| PY325 | MILITARY ETHICS                |
| PY330 | POLITICAL PHILOSOPHY           |
| PY345 | PHILOSOPHY OF RELIGION         |
| PY350 | PHILOSOPHY OF SCIENCE          |
| PY355 | PHILOSOPHY OF MIND             |
| PY360 | ANCIENT PHILOSOPHY             |
| PY369 | ASIAN PHILOSOPHICAL TRADITIONS |



|       |                                |
|-------|--------------------------------|
| PY370 | 17TH & 18TH CENTURY PHILOSOPHY |
| PY375 | KANT & 19TH CENTURY PHILOSOPHY |
| PY380 | 20TH CENTURY PHILOSOPHY        |
| SS364 | GAME THEORY                    |
| SS368 | ECONOMETRICS I                 |
| SS380 | MANPOWER-LABOR ECONOMICS       |
| SS382 | MICROECONOMICS                 |
| SS385 | HISTORY OF ECONOMICS           |
| SS387 | PUBLIC FINANCE                 |
| SS388 | MACROECONOMICS                 |
| SS394 | FUND. FINANCIAL DATA ANALYSIS  |
| SS462 | ECON OF STABILIZATION & GROWTH |
| SS469 | ECONOMETRICS II                |
| SS470 | MONEY AND BANKING              |
| SS477 | ECONOMICS OF NATIONAL SECURITY |
| SS494 | PRINCIPLES OF FINANCE          |
| WS345 | MILITARY INNOVATION            |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS370 | US STRATEGY AND POLICY         |
| WS385 | WAR AND ITS THEORISTS          |
| WS455 | COMPARATIVE DEFENSE POLICY     |
| WS490 | SPECIAL TOPICS IN WAR STUDIES  |

**AND****Integrative Experience for the Major**

Choose 1 of 1

SS480

PUBLIC POLICYMAKING PROCESS

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**Core Engineering Sequence**

Cadets in this major may choose any authorized engineering sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

|       |                                |
|-------|--------------------------------|
| SS376 | SEMINAR IN AMERICAN GOVERNMENT |
|-------|--------------------------------|

## 2029 American Politics w/ Thesis (Honors) Major Curriculum

| Code  | Short Description                    | Description                                | Transcript Description               | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------------|--------------------------------------------|--------------------------------------|--------------|--------------|
| PAP3H | American Politics w/ Thesis (Honors) | American Politics w/ Thesis (Honors) Major | American Politics w/ Thesis (Honors) | 1            | 1            |

## 2029 American Politics w/ Thesis (Honors) Major Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Description                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Program Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                |
| The American Politics with Thesis (Honors) Program is open to cadets who demonstrate exceptional academic aptitude in the American politics major. Cadets aspiring to graduate with an American Politics major with Honors must complete all requirements of the American Politics with Thesis track and satisfy additional course and grade requirements. Honors cadets must complete all the course requirements for the American Politics major, complete a thesis by taking SS489A and SS498A their Firstie year, and take one additional American Politics Elective to increase the depth of study in their major. The additional elective will normally be taken in the cadet's Cow year.                              |                                |
| <b>Additional American Politics Elective</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Choose 1 of 13                 |
| SS370                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MASS MEDIA & AMER POLITICS     |
| SS373                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | THE AMERICAN PRESIDENCY        |
| SS379                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | UNITED STATES CONGRESS         |
| SS392                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | RIGHTS, LIBERTIES & US CITIZEN |
| SS464                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | HOMELAND SECURITY              |
| SS465                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | TERRORISM: NEW CHALLENGES      |
| SS466                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | COMBATING TERRORISM            |
| SS468                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | POLITICAL PARTICIPATION        |
| SS472                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | SOLDIER & STATE: AM CIV-MIL RE |
| SS473                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AMERICAN FOREIGN POLICY        |
| SS481                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | POLITICS OF DEFENSE POLICY     |
| SS490A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | COLLOQUIUM (AMER POLITICS)     |
| SS493                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | SENIOR STUDIES - AMER POLITICS |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Choose 1 of 1                  |
| Honors students will continue work on their theses in SS498A Senior Thesis in American Politics, taken in the spring of the first class year when they will finish writing and defend their theses. SS498A Senior Thesis in American Politics consists of independent study and weekly meetings between individual cadets and their thesis advisors. Cadets will be responsible for coordinating meetings with their advisor. Course requirements will include a 30-50 page thesis submitted NLT lesson 35, and a defense of the thesis before their entire committee during the final two weeks of classes. Upon completion of the thesis and defense, the thesis committee recommends a final grade to the thesis advisor. |                                |
| SS498A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SENIOR THESIS IN AMER POLITICS |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
| <b>Grade Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                |
| Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                |

### 2029 American Politics w/ Thesis Major Curriculum

| Code  | Short Description           | Description                       | Transcript Description      | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------|-----------------------------------|-----------------------------|--------------|--------------|
| PAP3T | American Politics w/ Thesis | American Politics w/ Thesis Major | American Politics w/ Thesis | 1            | 0            |

### 2029 American Politics w/ Thesis Major Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Description                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Program Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                |
| The American Politics major with Thesis track will consist of a two-course sequence in addition to the requirements of the major, culminating in the cadet writing a thesis and defending it in front of a thesis committee.                                                                                                                                                                                                                                                              |                                |
| Cadets aspiring to graduate with a thesis will take SS489A in the fall and SS498A in the spring of their Firstie year as part of a two-course thesis sequence. Throughout the year, cadets will prepare and write an in-depth research paper that brings together theoretical perspectives acquired during their earlier studies. In SS489A, Thesis students will complete the literature review, an annotated bibliography, and a prospectus for their theses.                           |                                |
| Choose 1 of 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| SS498A: Senior Thesis in American Politics consists of independent study and weekly meetings between individual cadets and their thesis advisors. Cadets will be responsible for coordinating meetings with their advisor. Course requirements will include an approximately 30-50 page thesis and a defense of the thesis before their entire committee on Projects Day. Upon completion of the thesis and defense, the thesis committee recommends a final grade to the thesis advisor. |                                |
| SS498A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SENIOR THESIS IN AMER POLITICS |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |
| <b>Grade Requirements</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                |
| Complete the requirements of the major as shown above, and attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.                                                                                                                                                                                                                                                                                                                                |                                |

### 2029 Economics Major Curriculum

| Code | Short Description | Description     | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------|-----------------|------------------------|--------------|--------------|
| ECN2 | Economics         | Economics Major | Economics              | 6            | 7            |

### 2029 Economics Major Tracks

| Subject Area            | Description    |
|-------------------------|----------------|
| <b>Required Courses</b> | Choose 3 of 3  |
| SS368                   | ECONOMETRICS I |
| SS382                   | MICROECONOMICS |
| SS388                   | MACROECONOMICS |

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| <b>AND</b>                                        |                                |
| <b>Integrative Course</b>                         | Choose 1 of 3                  |
| SS380                                             | MANPOWER-LABOR ECONOMICS       |
| SS387                                             | PUBLIC FINANCE                 |
| SS484                                             | INTERNATIONAL ECONOMICS        |
| <b>AND</b>                                        |                                |
| <b>Economics Electives</b>                        | Choose 5 of 16                 |
| SS364                                             | GAME THEORY                    |
| SS380                                             | MANPOWER-LABOR ECONOMICS       |
| SS385                                             | HISTORY OF ECONOMICS           |
| SS387                                             | PUBLIC FINANCE                 |
| SS390                                             | BEHAVIORAL ECONOMICS           |
| SS394                                             | FUND. FINANCIAL DATA ANALYSIS  |
| SS461                                             | INDUSTRIAL ORGANIZATION        |
| SS462                                             | ECON OF STABILIZATION & GROWTH |
| SS463                                             | INVESTMENTS THEORY & APPL      |
| SS469                                             | ECONOMETRICS II                |
| SS470                                             | MONEY AND BANKING              |
| SS484                                             | INTERNATIONAL ECONOMICS        |
| SS487                                             | INT'L POLITICAL ECONOMY        |
| SS490C                                            | COLLOQUIUM (ECONOMICS)         |
| SS494                                             | PRINCIPLES OF FINANCE          |
| SS497                                             | ISSUES IN MICROECONOMIC THEORY |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Courses - Foundation</b> | Choose 2 of 4                  |
| MA204                                             | CALCULUS I AND II              |
| MA205                                             | CALCULUS II                    |
| MA255                                             | ADV MULTIVARIABLE CALCULUS     |
| MA367                                             | MATH FOR THE SOCIAL SCIENCES   |
| <b>AND</b>                                        |                                |
| <b>Complementary Support Course - Elective</b>    | Choose 1 of 49                 |
| EM381                                             | ENGINEERING ECONOMY            |
| EM420                                             | PRODUCTION OPERATIONS MGMT     |
| GE205                                             | FOUNDATIONS IN GEOGRAPHY       |
| GE340                                             | GEOGRAPHY OF GLOBAL CULTURES   |
| GE341                                             | GEOGRAPHY OF NORTH AMERICA     |
| GE342                                             | GEOGRAPHY OF ASIA              |
| GE343                                             | GEOGRAPHY OF RUSSIA            |
| GE344                                             | GEOGRAPHY OF AFRICA            |
| GE345                                             | GEOGRAPHY OF THE MIDDLE EAST   |
| GE346                                             | GEOGRAPHY OF EUROPE            |
| GE347                                             | GEOGRAPHY OF LATIN AMERICA     |
| GE351                                             | LAND USE PLAN & MGT            |
| GE352                                             | URBAN GEOGRAPHY                |
| HI344                                             | MODERN DIPLOMACY               |
| HI364                                             | MODERN WESTERN EUROPE          |
| HI367                                             | IMPERIAL AND SOVIET RUSSIA     |
| HI372                                             | US FGN RELATIONS SINCE 1898    |
| HI392                                             | AMERICAN HISTORICAL MEMORY     |
| HI396                                             | THEODORE ROOSEVELT'S AMERICA   |
| LW488                                             | BUSINESS LAW                   |

|        |                                   |
|--------|-----------------------------------|
| LX300  | 3RD SEMESTER FOREIGN LANG         |
| MA371  | LINEAR ALGEBRA                    |
| MA376  | APPLIED STATISTICS                |
| MA381  | NONLINEAR OPTIMIZATION            |
| MA387  | MATHEMATICAL ANALYSIS I           |
| MA476  | MATHEMATICAL STATISTICS           |
| MA481  | LINEAR OPTIMIZATION               |
| MA490  | ETHICS IN MATH, DATA SCI & ENG    |
| MA493E | TOPICS IN ANALYSIS                |
| MG380  | MARKETING                         |
| MG390  | NEGOTIATION FOR LEADERS           |
| MG420  | OPERATIONS MANAGEMENT             |
| PL373  | LIFE CYCLE & HUMAN DEVEL          |
| PL383  | EXPERIMENTAL SOCIAL<br>PSYCHOLOGY |
| PL386  | EXPERIMENTAL PSYCHOLOGY           |
| PY350  | PHILOSOPHY OF SCIENCE             |
| SE385  | DECISION ANALYSIS                 |
| SE387  | DETERMINISTIC MODELS              |
| SE388  | STOCHASTIC MODELS                 |
| SS376  | SEMINAR IN AMERICAN<br>GOVERNMENT |
| SS386  | POLITICAL THOUGHT                 |
| SS464  | HOMELAND SECURITY                 |
| SS465  | TERRORISM: NEW CHALLENGES         |
| SS466  | COMBATING TERRORISM               |
| SS473  | AMERICAN FOREIGN POLICY           |
| SS475  | COMP POLITICAL INSTITUTIONS       |
| SS480  | PUBLIC POLICYMAKING PROCESS       |
| SS481  | POLITICS OF DEFENSE POLICY        |
| SS483  | NATIONAL SECURITY SEMINAR         |

**AND****Integrative Experience for the Major**

Choose 1 of 1

SS477

ECONOMICS OF NATIONAL  
SECURITY**AND****Science Depth**

Cadets in this major may take any approved Science Depth course.

CH102

GENERAL CHEMISTRY II

CH275

BIOLOGY

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS -  
COMPUTING**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and IT305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS -  
COMPUTING

**Core Engineering Sequence**

Cadets in this major can choose any approved Core Engineering Sequence.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

SS477

ECONOMICS OF NATIONAL  
SECURITY**2029 Economics Major w/ Thesis Curriculum**

| Code  | Short Description   | Description               | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|---------------------|---------------------------|------------------------|-----------------|-----------------|
| ECN2T | Economics w/ Thesis | Economics Major w/ Thesis | Economics w/ Thesis    | 1               | 0               |

**2029 Economics Major w/ Thesis Tracks**

| Subject Area           | Description                |
|------------------------|----------------------------|
| <b>Required Course</b> | Choose 1 of 1              |
| SS498C                 | SENIOR THESIS IN ECONOMICS |

**2029 Economics Major w/ Thesis (Honors) Curriculum**

| Code  | Short Description            | Description                        | Transcript Description       | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|-------|------------------------------|------------------------------------|------------------------------|-----------------|-----------------|
| ECN2H | Economics w/ Thesis (Honors) | Economics Major w/ Thesis (Honors) | Economics w/ Thesis (Honors) | 1               | 1               |

**2029 Economics Major w/ Thesis (Honors) Tracks**

| Subject Area             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Required Course</b>   | Choose 1 of 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                          | The Economics Honors Program will consist of a two-course sequence culminating in the cadet writing a thesis and defending it in front of a thesis committee. The first course will be an additional 300- or 400-level economics elective or Complementary Support Course relevant to the cadet's desired thesis topic. The second course will be an existing three-credit course, SS498C Senior Thesis in Economics, taken in the spring of the first class year when the honors cadets will finish writing and defend their theses. |
| SS498C                   | SENIOR THESIS IN ECONOMICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>AND</b>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Grade Requirement</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                          | Complete the requirements of the major as shown above, attain a minimum APSC of 3.0 in the core curriculum and a 3.5 in the major.                                                                                                                                                                                                                                                                                                                                                                                                    |

## 2029 International Affairs Major w/ Honors - Foreign Policy and Security Studies Curriculum

| Code  | Short Description                                | Description                                                                 | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------|--------------|--------------|
| PIF0H | International Affairs Major w/ Honors - FP and S | International Affairs Major w/ Honors - Foreign Policy and Security Studies | International Affairs w/ Honors | 2            | 0            |

### 2029 International Affairs Major w/ Honors - Foreign Policy and Security Studies Tracks

| Subject Area                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program Requirements</b> | The International Affairs with Honors designation is available to cadets majoring in International Affairs Foreign Policy and Security Studies (PIF0) and International Affairs - Institutions, Governance, and Development (PII0). Awarding of the Honors designation requires the successful completion of a research project in one of three capstone courses, two additional IA courses beyond the 10 course base major (not including the required three CSC courses), and the achievement of a specified grade requirement in both the core curriculum and the IA major. SS489B and SS498B can fulfill the two additional IA course requirement for thesis cadets. |
| <b>AND</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Grade Requirements</b>   | Complete the requirements of the major as shown above, attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## 2029 International Affairs Major w/ Honors - Institutions, Governance, and Development Curriculum

| Code  | Short Description                                | Description                                                                       | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------|--------------|--------------|
| PII0H | International Affairs Major w/ Honors - I, G & D | International Affairs Major w/ Honors - Institutions, Governance, and Development | International Affairs Major w/ Honors | 2            | 0            |

### 2029 International Affairs Major w/ Honors - Institutions, Governance, and Development Tracks

| Subject Area                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program Requirements</b> | The International Affairs with Honors designation is available to cadets majoring in International Affairs Foreign Policy and Security Studies (PIF0) and International Affairs - Institutions, Governance, and Development (PII0). Awarding of the Honors designation requires the successful completion of a research project in one of three capstone courses, two additional IA courses beyond the 10 course base major (not including the required three CSC courses), and the achievement of a specified grade requirement in both the core curriculum and the IA major. SS489B and SS498B can fulfill the two additional IA course requirement for thesis cadets. |
| <b>AND</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Grade Requirements</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Complete the requirements of the major as shown above, attain an APSC of at least 3.0 in the core curriculum and an APSC of at least 3.5 in the major.

### 2029 International Affairs Major w/ Thesis - Foreign Policy and Security Studies Curriculum

| Code  | Short Description                                | Description                                                                 | Transcript Description          | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------|--------------|--------------|
| PIF0T | International Affairs Major w/ Thesis - FP and S | International Affairs Major w/ Thesis - Foreign Policy and Security Studies | International Affairs w/ Thesis | 2            | 0            |

### 2029 International Affairs Major w/ Thesis - Foreign Policy and Security Studies Tracks

| Subject Area            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | The International Affairs Thesis Program is available to cadets majoring in International Affairs Foreign Policy and Security Studies (PIF0) and International Affairs - Institutions, Governance, and Development (PII0). In addition to the standard course of study, Cadets admitted to the IA Thesis Program will complete a two-course thesis sequence (SS489B and SS498B), defend their work in front of a three-member committee, and present findings during the Academy's Projects Day. |
| <b>AND</b>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Required Courses</b> | Choose 2 of 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| SS489B                  | SENIOR RESEARCH INTERN AFFAIRS                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| SS498B                  | SENIOR THESIS IN INTER AFFAIRS                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### 2029 International Affairs Major w/ Thesis - Institutions, Governance, and Development Curriculum

| Code  | Short Description                                | Description                                                                       | Transcript Description                | Req Crse Cnt | Opt Crse Cnt |
|-------|--------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------|--------------|--------------|
| PII0T | International Affairs Major w/ Thesis - I, G & D | International Affairs Major w/ Thesis - Institutions, Governance, and Development | International Affairs Major w/ Thesis | 2            | 0            |

### 2029 International Affairs Major w/ Thesis - Institutions, Governance, and Development Tracks

| Subject Area                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program Requirements</b> | The International Affairs Thesis Program is available to cadets majoring in International Affairs Foreign Policy and Security Studies (PIF0) and International Affairs - Institutions, Governance, and Development (PII0). In addition to the standard course of study, Cadets admitted to the IA Thesis Program will complete a two-course thesis sequence (SS489B and SS498B), defend their work in front of a three-member committee, and present findings during the Academy's Projects Day. |
| <b>AND</b>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Required Courses</b>     | Choose 2 of 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



SS489B Senior Thesis Preparation and SS498B Senior Thesis in International Affairs.  
1.0 credit research course offered in the spring of each year is optional but encouraged.

|        |                                |
|--------|--------------------------------|
| SS489B | SENIOR RESEARCH INTERN AFFAIRS |
| SS498B | SENIOR THESIS IN INTER AFFAIRS |

## 2029 International Affairs Major: Foreign Policy and Security Studies Curriculum

| Code | Short Description                            | Description                                                      | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|----------------------------------------------|------------------------------------------------------------------|------------------------|--------------|--------------|
| PIF0 | International Affairs: FP & Security Studies | International Affairs Major: Foreign Policy and Security Studies | International Affairs  | 5            | 8            |

## 2029 International Affairs Major: Foreign Policy and Security Studies Tracks

| Subject Area                                                                                                    | Description                    |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                                                                         | Choose 3 of 3                  |
| SS360                                                                                                           | POL SCI RESEARCH METHODS       |
| SS366                                                                                                           | COMPARATIVE POLITICS           |
| SS386                                                                                                           | POLITICAL THOUGHT              |
| <b>AND</b>                                                                                                      |                                |
| <b>Thematic Track Gateway Courses</b>                                                                           | Choose 2 of 2                  |
| Cadets in the Foreign Policy and Security Studies Track must take these two Thematic Gateway courses.           |                                |
| SS395                                                                                                           | INTERNATIONAL SECURITY         |
| SS483                                                                                                           | NATIONAL SECURITY SEMINAR      |
| <b>AND</b>                                                                                                      |                                |
| <b>Thematic Track Elective</b>                                                                                  | Choose 1 of 27                 |
| Cadets in the Foreign Policy and Security Studies Track must choose one thematic track elective from this list. |                                |
| CY460                                                                                                           | CYBER POLICY, STRATEGY, & OPNS |
| SS372                                                                                                           | POLITICS OF CHINA              |
| SS374                                                                                                           | POLITICS OF NORTHEAST ASIA     |
| SS375                                                                                                           | POL OF THE POST-SOVIET STATES  |
| SS377                                                                                                           | POLITICS OF EUROPE             |
| SS383                                                                                                           | POLITICS OF THE MIDDLE EAST    |
| SS384                                                                                                           | POLITICS OF LATIN AMERICA      |
| SS385                                                                                                           | HISTORY OF ECONOMICS           |
| SS399                                                                                                           | SOCSCI INTERNSHIP/PRACTCAL EXP |
| SS462                                                                                                           | ECONOMIC GROWTH & DEVELOPMENT  |
| SS464                                                                                                           | HOMELAND SECURITY              |
| SS465                                                                                                           | TERRORISM: NEW CHALLENGES      |
| SS466                                                                                                           | COMBATING TERRORISM            |
| SS472                                                                                                           | THE AM STATE & THE SOLDIER     |
| SS473                                                                                                           | AMERICAN FOREIGN POLICY        |
| SS476                                                                                                           | CIVIL WARS                     |
| SS477                                                                                                           | ECONOMICS OF NATIONAL SECURITY |
| SS478                                                                                                           | DIST PROF OF SECURITY STUD SEM |
| SS479                                                                                                           | INTERNATIONAL ORGANIZATION     |
| SS483                                                                                                           | NATIONAL SECURITY SEMINAR      |
| SS485                                                                                                           | GOV & POLITICS SUB-SAHARAN AFR |

|                                                                                                                                                                                                                                                                                                                                                                                                   |                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| SS486                                                                                                                                                                                                                                                                                                                                                                                             | STATE BUILDING                 |
| SS487                                                                                                                                                                                                                                                                                                                                                                                             | INT'L POLITICAL ECONOMY        |
| SS490B                                                                                                                                                                                                                                                                                                                                                                                            | COLLOQUIUM (INTERNTL AFFAIRS)  |
| SS490C                                                                                                                                                                                                                                                                                                                                                                                            | COLLOQUIUM (ECONOMICS)         |
| WS330                                                                                                                                                                                                                                                                                                                                                                                             | INTRODUCTION TO GRAND STRATEGY |
| WS335                                                                                                                                                                                                                                                                                                                                                                                             | TOPICS IN GRAND STRATEGY       |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                        |                                |
| <b>Thematic Breadth Requirement</b>                                                                                                                                                                                                                                                                                                                                                               | Choose 1 of 2                  |
| Cadets in the Foreign Policy and Security Studies Track must select one course from the list of Thematic Track Gateway Courses from outside of their Thematic Track.                                                                                                                                                                                                                              |                                |
| SS475                                                                                                                                                                                                                                                                                                                                                                                             | COMP POLITICAL INSTITUTIONS    |
| SS487                                                                                                                                                                                                                                                                                                                                                                                             | INT'L POLITICAL ECONOMY        |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                        |                                |
| <b>Regional Breadth Requirement</b>                                                                                                                                                                                                                                                                                                                                                               | Choose 1 of 7                  |
| SS372                                                                                                                                                                                                                                                                                                                                                                                             | POLITICS OF CHINA              |
| SS374                                                                                                                                                                                                                                                                                                                                                                                             | POLITICS OF NORTHEAST ASIA     |
| SS375                                                                                                                                                                                                                                                                                                                                                                                             | POL OF THE POST-SOVIET STATES  |
| SS377                                                                                                                                                                                                                                                                                                                                                                                             | POLITICS OF EUROPE             |
| SS383                                                                                                                                                                                                                                                                                                                                                                                             | POLITICS OF THE MIDDLE EAST    |
| SS384                                                                                                                                                                                                                                                                                                                                                                                             | POLITICS OF LATIN AMERICA      |
| SS485                                                                                                                                                                                                                                                                                                                                                                                             | GOV & POLITICS SUB-SAHARAN AFR |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                        |                                |
| <b>American Politics Elective</b>                                                                                                                                                                                                                                                                                                                                                                 | Choose 1 of 11                 |
| SS370                                                                                                                                                                                                                                                                                                                                                                                             | MASS MEDIA & AMER POLITICS     |
| SS373                                                                                                                                                                                                                                                                                                                                                                                             | THE AMERICAN PRESIDENCY        |
| SS376                                                                                                                                                                                                                                                                                                                                                                                             | SEMINAR IN AMERICAN GOVERNMENT |
| SS379                                                                                                                                                                                                                                                                                                                                                                                             | LEGISLATIVE POLITICS           |
| SS392                                                                                                                                                                                                                                                                                                                                                                                             | RIGHTS, LIBERTIES & US CITIZEN |
| SS464                                                                                                                                                                                                                                                                                                                                                                                             | HOMELAND SECURITY              |
| SS468                                                                                                                                                                                                                                                                                                                                                                                             | POLITICAL PARTICIPATION        |
| SS472                                                                                                                                                                                                                                                                                                                                                                                             | THE AM STATE & THE SOLDIER     |
| SS480                                                                                                                                                                                                                                                                                                                                                                                             | PUBLIC POLICYMAKING PROCESS    |
| SS481                                                                                                                                                                                                                                                                                                                                                                                             | POLITICS OF DEFENSE POLICY     |
| SS490A                                                                                                                                                                                                                                                                                                                                                                                            | COLLOQUIUM (AMER POLITICS)     |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                                                                        |                                |
| <b>Complementary Support Courses</b>                                                                                                                                                                                                                                                                                                                                                              |                                |
| All majors will choose three classes from beyond the discipline of political science as their complementary support courses. Cadets can request courses not in these groupings based on approval of the DAC or Program Director.                                                                                                                                                                  |                                |
| Cadets should be aware that some of these courses have prerequisites that may be difficult to attain, some courses are only offered at certain times, and course prerequisites may change. If cadets do not meet the prerequisite requirements but still would like to take the course, they are responsible to meet with the course director to see if taking the course will still be feasible. |                                |
| <b>CSC-Foreign Language</b>                                                                                                                                                                                                                                                                                                                                                                       | Choose 3 of 24                 |
| Take three upper division courses in the same language (typically LX371, LX372, and LX475)                                                                                                                                                                                                                                                                                                        |                                |
| LA371                                                                                                                                                                                                                                                                                                                                                                                             | INTENSIVE INTERMEDIATE ARABIC  |
| LA372                                                                                                                                                                                                                                                                                                                                                                                             | ARABIC FOR ORAL & WRITTEN COMM |
| LA475                                                                                                                                                                                                                                                                                                                                                                                             | ARABIC RDG/WRTG THRU MEDIA     |
| LC371                                                                                                                                                                                                                                                                                                                                                                                             | INTENSIVE INTERMEDIATE CHINESE |
| LC372                                                                                                                                                                                                                                                                                                                                                                                             | CHINESE FOR ORAL & WRIT COMM   |
| LC475                                                                                                                                                                                                                                                                                                                                                                                             | CHINESE RDG/WRTG THRU MEDIA    |
| LF371                                                                                                                                                                                                                                                                                                                                                                                             | INTENSIVE INTERMEDIATE FRENCH  |

|                                                                                                 |                                |
|-------------------------------------------------------------------------------------------------|--------------------------------|
| LF372                                                                                           | FRENCH FOR ORAL & WRITTEN COMM |
| LF475                                                                                           | FRENCH RDG/WRTG THRU MEDIA     |
| LG371                                                                                           | INTENSIVE INTERMEDIATE GERMAN  |
| LG372                                                                                           | GERMAN FOR ORAL & WRITTEN COMM |
| LG475                                                                                           | GERMAN RDG/WRTG THRU MEDIA     |
| LP371                                                                                           | INTENSIVE INTERMED. PORTUGUESE |
| LP372                                                                                           | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475                                                                                           | PORTUGUESE RDG/WRTG THRU MEDIA |
| LR371                                                                                           | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372                                                                                           | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475                                                                                           | RUSSIAN RDG/WRTG THRU MEDIA    |
| LS371                                                                                           | INTENSIVE INTERMEDIATE SPANISH |
| LS372                                                                                           | SPANISH FOR ORAL & WRIT COMM   |
| LS475                                                                                           | SPANISH RDG/WRTG THRU MEDIA    |
| LZ371                                                                                           | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372                                                                                           | PERSIAN FOR ORAL & WRIT COMM   |
| LZ475                                                                                           | PERSIAN RDG/WRTG THRU MEDIA    |
| <b>OR</b>                                                                                       |                                |
| <b>CSC-Regional Studies: Europe</b>                                                             |                                |
| Choose 3 of 10                                                                                  |                                |
| LX300/LX400 = an additional language course in French, German, Portuguese, Spanish, or Russian. |                                |
| GE340                                                                                           | GEOGRAPHY OF GLOBAL CULTURES   |
| GE343                                                                                           | GEOGRAPHY OF RUSSIA            |
| GE346                                                                                           | GEOGRAPHY OF EUROPE            |
| HI343                                                                                           | MODERN GERMANY                 |
| HI361                                                                                           | MEDIEVAL EUROPE                |
| HI364                                                                                           | MODERN WESTERN EUROPE          |
| HI367                                                                                           | IMPERIAL AND SOVIET RUSSIA     |
| HI368                                                                                           | MOD CENTRAL & E. EUR,1896-1989 |
| LX300                                                                                           | 3RD SEMESTER FOREIGN LANG      |
| LX400                                                                                           | 4TH SEMESTER FOREIGN LANG      |
| <b>OR</b>                                                                                       |                                |
| <b>CSC-Regional Studies: Asia</b>                                                               |                                |
| Choose 3 of 6                                                                                   |                                |
| LX300 = an additional language course in Chinese.                                               |                                |
| GE340                                                                                           | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342                                                                                           | GEOGRAPHY OF ASIA              |
| HI337                                                                                           | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI346                                                                                           | INDIA, PAKISTAN, & BANGLADESH  |
| HI347                                                                                           | ASIAN WARFARE AND POLITICS     |
| LX300                                                                                           | 3RD SEMESTER FOREIGN LANG      |
| <b>OR</b>                                                                                       |                                |
| <b>CSC-Regional Studies: Middle East and Africa</b>                                             |                                |
| Choose 3 of 7                                                                                   |                                |
| LX300 = an additional language course in Arabic, Hebrew or Persian.                             |                                |
| GE340                                                                                           | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344                                                                                           | GEOGRAPHY OF AFRICA            |
| GE345                                                                                           | GEOGRAPHY OF THE MIDDLE EAST   |
| HI339                                                                                           | THE MODERN MIDDLE EAST         |
| HI345                                                                                           | MODERN AFRICA                  |
| HI383                                                                                           | MIDDLE EASTERN WARFARE         |
| LX300                                                                                           | 3RD SEMESTER FOREIGN LANG      |
| <b>OR</b>                                                                                       |                                |
| <b>CSC-Regional Studies: The Americas</b>                                                       |                                |
| Choose 3 of 9                                                                                   |                                |
| LX300 = an additional language course in Spanish or Portuguese.                                 |                                |

|       |                               |
|-------|-------------------------------|
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE341 | GEOGRAPHY OF NORTH AMERICA    |
| GE347 | GEOGRAPHY OF LATIN AMERICA    |
| HI348 | MODERN LATIN AMERICA          |
| HI372 | US FGN RELATIONS SINCE 1898   |
| HI392 | AMERICAN HISTORICAL MEMORY    |
| HI396 | THEODORE ROOSEVELT'S AMERICA  |
| HI397 | THE US FROM COLD WAR TO TODAY |
| LX300 | 3RD SEMESTER FOREIGN LANG     |

**OR****CSC-Law**

Choose 3 of 7

|       |                            |
|-------|----------------------------|
| LW410 | COMPARATIVE LEGAL SYSTEMS  |
| LW462 | CYBER LAW                  |
| LW473 | ENVIRONMENTAL LAW          |
| LW474 | LAW OF ARMED CONFLICT      |
| LW475 | ADV CONSTITUTIONAL LAW SEM |
| LW481 | INTERNATIONAL LAW          |
| LW482 | NATIONAL SECURITY LAW      |

**OR****CSC-Strategic Studies**

Choose 3 of 21

|       |                                |
|-------|--------------------------------|
| GE491 | MILITARY GEOGRAPHY             |
| HI338 | WARFARE IN AGE OF REVOLUTIONS  |
| HI347 | ASIAN WARFARE AND POLITICS     |
| HI355 | WARFARE, INDUSTRY, & EMPIRE    |
| HI356 | WAR AT SEA, SKY, SPACE         |
| HI357 | ARMED CONFLICT IN THE COLD WAR |
| HI359 | ERA OF THE SECOND WORLD WAR    |
| HI370 | ANCIENT & MEDIEVAL WARFARE     |
| HI376 | EARLY MODERN WARFARE           |
| HI381 | HISTORY OF IRREGULAR WARFARE   |
| PL471 | LEADERSHIP IN COMBAT           |
| WS301 | INTRODUCTION TO WAR STUDIES    |
| WS345 | MILITARY INNOVATION            |
| WS358 | STRATEGY AND GENERALSHIP       |
| WS360 | SPECIAL OPS IN THEORY & PRACTI |
| WS365 | REBELLN, INSURGNCY, & CIV WAR  |
| WS370 | US STRATEGY AND POLICY         |
| WS385 | WAR AND ITS THEORISTS          |
| WS475 | FORECAST AND GAME IN DECISION  |
| WS485 | JOINT AND MULTINATIONAL OPS    |
| XH397 | GRAND STRATEGY FIELD STUDY     |

**OR****CSC-United States Foreign Policy**

Choose 3 of 9

|       |                                |
|-------|--------------------------------|
| HI340 | COLONIAL AMERICA               |
| HI369 | AMERICAN FRONTIERS             |
| HI372 | US FGN RELATIONS SINCE 1898    |
| HI394 | REVOLUTIONARY AMERICA          |
| HI395 | HIST OF CIVIL WAR AMERICA      |
| HI396 | THEODORE ROOSEVELT'S AMERICA   |
| HI397 | THE US FROM COLD WAR TO TODAY  |
| HI398 | CIVIL RIGHTS IN AMER HIST      |
| HI487 | EARLY AMERICAN REPUB 1787-1826 |

**OR****CSC-Culture and Philosophy**

Choose 3 of 9

|                                                                                                                                                                                                                                                                                                  |                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| EN344                                                                                                                                                                                                                                                                                            | CRITICISM                         |
| HI391                                                                                                                                                                                                                                                                                            | WORLD RELIGIONS                   |
| HI392                                                                                                                                                                                                                                                                                            | AMERICAN HISTORICAL MEMORY        |
| LW461                                                                                                                                                                                                                                                                                            | CIVIL RIGHTS AND THE LAW          |
| PL372                                                                                                                                                                                                                                                                                            | MILITARY FAMILIES                 |
| PL383                                                                                                                                                                                                                                                                                            | EXPERIMENTAL SOCIAL<br>PSYCHOLOGY |
| PY305                                                                                                                                                                                                                                                                                            | LOGICAL REASONING                 |
| PY326                                                                                                                                                                                                                                                                                            | ETHICS OF TECHNOLOGY              |
| PY350                                                                                                                                                                                                                                                                                            | PHILOSOPHY OF SCIENCE             |
| <b>OR</b>                                                                                                                                                                                                                                                                                        |                                   |
| <b>CSC-Research Methods: General Track</b> Choose 3 of 7                                                                                                                                                                                                                                         |                                   |
| For the Design Requirement, you must select 2 courses from WS495, GE305, PL361, PL363. For the Formal Theory Requirement, you must select 1 course from MA383, PY305, SS364. Note that the prerequisite requirements for some of these courses may make this a difficult option for some cadets. |                                   |
| GE305                                                                                                                                                                                                                                                                                            | GEOGRAPHIC RESEARCH METHODS       |
| MA383                                                                                                                                                                                                                                                                                            | FOUNDATIONS OF MATH               |
| PL361                                                                                                                                                                                                                                                                                            | RESEARCH METHODS I                |
| PL363                                                                                                                                                                                                                                                                                            | QUALITATIVE RESEARCH METHODS      |
| PY305                                                                                                                                                                                                                                                                                            | LOGICAL REASONING                 |
| SS364                                                                                                                                                                                                                                                                                            | GAME THEORY                       |
| WS495                                                                                                                                                                                                                                                                                            | METHODS & CAPSTONE WAR<br>STUDIES |
| <b>OR</b>                                                                                                                                                                                                                                                                                        |                                   |
| <b>CSC-Research Methods: Quantitative Focus Track</b> Choose 3 of 8                                                                                                                                                                                                                              |                                   |
| For the Quantitative Focus Track, you must select MA204/MA205 and MA367 and choose 1 of GE305, MA376, MA391, PL462, SS468. Note that the prerequisite requirements for some of these courses may make this a difficult option for some cadets.                                                   |                                   |
| GE305                                                                                                                                                                                                                                                                                            | GEOGRAPHIC RESEARCH METHODS       |
| MA204                                                                                                                                                                                                                                                                                            | CALCULUS I AND II                 |
| MA205                                                                                                                                                                                                                                                                                            | CALCULUS II                       |
| MA367                                                                                                                                                                                                                                                                                            | MATH FOR THE SOCIAL SCIENCES      |
| MA376                                                                                                                                                                                                                                                                                            | APPLIED STATISTICS                |
| MA391                                                                                                                                                                                                                                                                                            | MATHEMATICAL MODELING             |
| PL462                                                                                                                                                                                                                                                                                            | EXPERIMENTAL APP IN<br>PSYCHOLOGY |
| SS468                                                                                                                                                                                                                                                                                            | POLITICAL PARTICIPATION           |
| <b>OR</b>                                                                                                                                                                                                                                                                                        |                                   |
| <b>CSC-Economics and Finance</b> Choose 3 of 17                                                                                                                                                                                                                                                  |                                   |
| Cadets in this CSC track must select SS388 and any two of the remaining courses listed.                                                                                                                                                                                                          |                                   |
| SS364                                                                                                                                                                                                                                                                                            | GAME THEORY                       |
| SS368                                                                                                                                                                                                                                                                                            | ECONOMETRICS I                    |
| SS380                                                                                                                                                                                                                                                                                            | MANPOWER-LABOR ECONOMICS          |
| SS382                                                                                                                                                                                                                                                                                            | MICROECONOMICS                    |
| SS385                                                                                                                                                                                                                                                                                            | HISTORY OF ECONOMICS              |
| SS387                                                                                                                                                                                                                                                                                            | PUBLIC FINANCE                    |
| SS388                                                                                                                                                                                                                                                                                            | MACROECONOMICS                    |
| SS390                                                                                                                                                                                                                                                                                            | BEHAVIORAL ECONOMICS              |
| SS394                                                                                                                                                                                                                                                                                            | FUND. FINANCIAL DATA ANALYSIS     |
| SS461                                                                                                                                                                                                                                                                                            | INDUSTRIAL ORGANIZATION           |
| SS462                                                                                                                                                                                                                                                                                            | ECON OF STABILIZATION & GROWTH    |
| SS463                                                                                                                                                                                                                                                                                            | INVESTMENTS THEORY & APPL         |
| SS469                                                                                                                                                                                                                                                                                            | ECONOMETRICS II                   |
| SS470                                                                                                                                                                                                                                                                                            | MONEY AND BANKING                 |
| SS484                                                                                                                                                                                                                                                                                            | INTERNATIONAL ECONOMICS           |
| SS494                                                                                                                                                                                                                                                                                            | PRINCIPLES OF FINANCE             |

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| SS497                                                                                                                                                                        | ISSUES IN MICROECONOMIC THEORY |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Capstone Requirement - Integrative Experience for the Major</b>                                                                                                           | Choose 1 of 3                  |
| Only cadets in the Thesis major can select SS498B to fulfill this requirement.                                                                                               |                                |
| SS491                                                                                                                                                                        | SENIOR PROJECT                 |
| SS495                                                                                                                                                                        | SENIOR FACULTY COLLOQUIUM      |
| SS498B                                                                                                                                                                       | SENIOR THESIS IN INTER AFFAIRS |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.                                                                      |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/CY355 to satisfy the STEM Depth Requirement.                                                                                            |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <b>Curriculum Requirements</b>                                                                                                                                               |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirement</b>                                                                                                                                                  |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| Cadets in this major may select any Core Engineering Sequence. This sequence is selected in the fall semester of their third class year.                                     |                                |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| SS395                                                                                                                                                                        | INTERNATIONAL SECURITY         |

## 2029 International Affairs Major: Institutions, Governance, and Development Curriculum

| Code | Short Description                               | Description                                                            | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------------------------|------------------------------------------------------------------------|------------------------|--------------|--------------|
| PII0 | International Affairs: Inst, Gov, & Development | International Affairs Major: Institutions, Governance, and Development | International Affairs  | 5            | 8            |

## 2029 International Affairs Major: Institutions, Governance, and Development Tracks

| Subject Area     | Description   |
|------------------|---------------|
| Required Courses | Choose 3 of 3 |

|                                                                                                                                                                      |                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| SS360                                                                                                                                                                | POL SCI RESEARCH METHODS       |
| SS366                                                                                                                                                                | COMPARATIVE POLITICS           |
| SS386                                                                                                                                                                | POLITICAL THOUGHT              |
| <b>AND</b>                                                                                                                                                           |                                |
| <b>Thematic Track Gateway Courses</b>                                                                                                                                | Choose 2 of 2                  |
| Cadets in the Institutions, Governance, and Development Track must take these two Thematic Gateway courses.                                                          |                                |
| SS475                                                                                                                                                                | COMP POLITICAL INSTITUTIONS    |
| SS487                                                                                                                                                                | INT'L POLITICAL ECONOMY        |
| <b>AND</b>                                                                                                                                                           |                                |
| <b>Thematic Track Elective</b>                                                                                                                                       | Choose 1 of 27                 |
| Cadets in the Institutions, Governance, and Development must choose one thematic track elective from this list.                                                      |                                |
| CY460                                                                                                                                                                | CYBER POLICY, STRATEGY, & OPNS |
| SS372                                                                                                                                                                | POLITICS OF CHINA              |
| SS374                                                                                                                                                                | POLITICS OF NORTHEAST ASIA     |
| SS375                                                                                                                                                                | POL OF THE POST-SOVIET STATES  |
| SS377                                                                                                                                                                | POLITICS OF EUROPE             |
| SS383                                                                                                                                                                | POLITICS OF THE MIDDLE EAST    |
| SS384                                                                                                                                                                | POLITICS OF LATIN AMERICA      |
| SS385                                                                                                                                                                | HISTORY OF ECONOMICS           |
| SS399                                                                                                                                                                | SOCSCI INTERNSHIP/PRACTCAL EXP |
| SS462                                                                                                                                                                | ECON OF STABILIZATION & GROWTH |
| SS464                                                                                                                                                                | HOMELAND SECURITY              |
| SS465                                                                                                                                                                | TERRORISM: NEW CHALLENGES      |
| SS466                                                                                                                                                                | COMBATING TERRORISM            |
| SS472                                                                                                                                                                | SOLDIER & STATE: AM CIV-MIL RE |
| SS473                                                                                                                                                                | AMERICAN FOREIGN POLICY        |
| SS476                                                                                                                                                                | CIVIL WARS                     |
| SS477                                                                                                                                                                | ECONOMICS OF NATIONAL SECURITY |
| SS478                                                                                                                                                                | DIST PROF OF SECURITY STUD SEM |
| SS479                                                                                                                                                                | INTERNATIONAL ORGANIZATION     |
| SS483                                                                                                                                                                | NATIONAL SECURITY SEMINAR      |
| SS485                                                                                                                                                                | POLITICS SUB-SAHARAN AFR       |
| SS486                                                                                                                                                                | STATE BUILDING                 |
| SS487                                                                                                                                                                | INT'L POLITICAL ECONOMY        |
| SS490B                                                                                                                                                               | COLLOQUIUM (INTERNTL AFFAIRS)  |
| SS490C                                                                                                                                                               | COLLOQUIUM (ECONOMICS)         |
| WS330                                                                                                                                                                | INTRODUCTION TO GRAND STRATEGY |
| WS335                                                                                                                                                                | TOPICS IN GRAND STRATEGY       |
| <b>AND</b>                                                                                                                                                           |                                |
| <b>Thematic Breadth Requirement</b>                                                                                                                                  | Choose 1 of 2                  |
| Cadets in the Institutions, Governance, and Development must select one course from the list of Thematic Track Gateway Courses from outside of their Thematic Track. |                                |
| SS395                                                                                                                                                                | INTERNATIONAL SECURITY         |
| SS483                                                                                                                                                                | NATIONAL SECURITY SEMINAR      |
| <b>AND</b>                                                                                                                                                           |                                |
| <b>Regional Breadth Requirement</b>                                                                                                                                  | Choose 1 of 7                  |
| SS372                                                                                                                                                                | POLITICS OF CHINA              |
| SS374                                                                                                                                                                | POLITICS OF NORTHEAST ASIA     |
| SS375                                                                                                                                                                | POL OF THE POST-SOVIET STATES  |
| SS377                                                                                                                                                                | POLITICS OF EUROPE             |
| SS383                                                                                                                                                                | POLITICS OF THE MIDDLE EAST    |
| SS384                                                                                                                                                                | POLITICS OF LATIN AMERICA      |
| SS485                                                                                                                                                                | GOV & POLITICS SUB-SAHARAN AFR |
| <b>AND</b>                                                                                                                                                           |                                |

**American Politics Elective**

Choose 1 of 11

|        |                                |
|--------|--------------------------------|
| SS370  | MASS MEDIA & AMER POLITICS     |
| SS373  | THE AMERICAN PRESIDENCY        |
| SS376  | SEMINAR IN AMERICAN GOVERNMENT |
| SS379  | LEGISLATIVE POLITICS           |
| SS392  | RIGHTS, LIBERTIES & US CITIZEN |
| SS464  | HOMELAND SECURITY              |
| SS468  | POLITICAL PARTICIPATION        |
| SS472  | THE AM STATE & THE SOLDIER     |
| SS480  | PUBLIC POLICYMAKING PROCESS    |
| SS481  | POLITICS OF DEFENSE POLICY     |
| SS490A | COLLOQUIUM (AMER POLITICS)     |

**AND****Complementary Support Courses**

All majors will choose three classes from beyond the discipline of political science as their complementary support courses. Cadets can request courses not in these groupings based on approval of the DAC or Program Director.

Cadets should be aware that some of these courses have prerequisites that may be difficult to attain, some courses are only offered at certain times, and course prerequisites may change. If cadets do not meet the prerequisite requirements but still would like to take the course, they are responsible to meet with the course director to see if taking the course will still be feasible.

**CSC-Foreign Language**

Choose 3 of 24

Take three upper division courses in the same language (typically LX371, LX372, and LX475)

|       |                                |
|-------|--------------------------------|
| LA371 | INTENSIVE INTERMEDIATE ARABIC  |
| LA372 | ARABIC FOR ORAL & WRITTEN COMM |
| LA475 | ARABIC RDG/WRTG THRU MEDIA     |
| LC371 | INTENSIVE INTERMEDIATE CHINESE |
| LC372 | CHINESE FOR ORAL & WRIT COMM   |
| LC475 | CHINESE RDG/WRTG THRU MEDIA    |
| LF371 | INTENSIVE INTERMEDIATE FRENCH  |
| LF372 | FRENCH FOR ORAL & WRITTEN COMM |
| LF475 | FRENCH RDG/WRTG THRU MEDIA     |
| LG371 | INTENSIVE INTERMEDIATE GERMAN  |
| LG372 | GERMAN FOR ORAL & WRITTEN COMM |
| LG475 | GERMAN RDG/WRTG THRU MEDIA     |
| LP371 | INTENSIVE INTERMED. PORTUGUESE |
| LP372 | PORTUGUESE FOR ORAL/WRIT COMM  |
| LP475 | PORTUGUESE RDG/WRTG THRU MEDIA |
| LR371 | INTENSIVE INTERMEDIATE RUSSIAN |
| LR372 | RUSSIAN FOR ORAL & WRIT COMM   |
| LR475 | RUSSIAN RDG/WRTG THRU MEDIA    |
| LS371 | INTENSIVE INTERMEDIATE SPANISH |
| LS372 | SPANISH FOR ORAL & WRIT COMM   |
| LS475 | SPANISH RDG/WRTG THRU MEDIA    |
| LZ371 | INTENSIVE INTERMEDIATE PERSIAN |
| LZ372 | PERSIAN FOR ORAL & WRIT COMM   |
| LZ475 | PERSIAN RDG/WRTG THRU MEDIA    |

**OR****CSC-Regional Studies: Europe**

Choose 3 of 10



LX300/LX400 = an additional language course in French, German, Portuguese, Spanish, or Russian.

|       |                                 |
|-------|---------------------------------|
| GE340 | GEOGRAPHY OF GLOBAL CULTURES    |
| GE343 | GEOGRAPHY OF RUSSIA             |
| GE346 | GEOGRAPHY OF EUROPE             |
| HI343 | MODERN GERMANY                  |
| HI361 | MEDIEVAL EUROPE                 |
| HI364 | MODERN WESTERN EUROPE           |
| HI367 | IMPERIAL AND SOVIET RUSSIA      |
| HI368 | MOD CENTRAL & E. EUR, 1896-1989 |
| LX300 | 3RD SEMESTER FOREIGN LANG       |
| LX400 | 4TH SEMESTER FOREIGN LANG       |

**OR**

**CSC-Regional Studies: Asia**

Choose 3 of 6

LX300 = an additional language course in Chinese.

|       |                                |
|-------|--------------------------------|
| GE340 | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342 | GEOGRAPHY OF ASIA              |
| HI337 | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI346 | INDIA, PAKISTAN, & BANGLADESH  |
| HI347 | ASIAN WARFARE AND POLITICS     |
| LX300 | 3RD SEMESTER FOREIGN LANG      |

**OR**

**CSC-Regional Studies: Middle East and Africa**

Choose 3 of 7

LX300 = an additional language course in Arabic, Hebrew or Persian.

|       |                              |
|-------|------------------------------|
| GE340 | GEOGRAPHY OF GLOBAL CULTURES |
| GE344 | GEOGRAPHY OF AFRICA          |
| GE345 | GEOGRAPHY OF THE MIDDLE EAST |
| HI339 | THE MODERN MIDDLE EAST       |
| HI345 | MODERN AFRICA                |
| HI383 | MIDDLE EASTERN WARFARE       |
| LX300 | 3RD SEMESTER FOREIGN LANG    |

**OR**

**CSC-Regional Studies: The Americas**

Choose 3 of 9

LX300 = an additional language course in Spanish or Portuguese.

|       |                               |
|-------|-------------------------------|
| GE340 | GEOGRAPHY OF GLOBAL CULTURES  |
| GE341 | GEOGRAPHY OF NORTH AMERICA    |
| GE347 | GEOGRAPHY OF LATIN AMERICA    |
| HI348 | MODERN LATIN AMERICA          |
| HI372 | US FGN RELATIONS SINCE 1898   |
| HI392 | AMERICAN HISTORICAL MEMORY    |
| HI396 | THEODORE ROOSEVELT'S AMERICA  |
| HI397 | THE US FROM COLD WAR TO TODAY |
| LX300 | 3RD SEMESTER FOREIGN LANG     |

**OR**

**CSC-Law**

Choose 3 of 7

|       |                            |
|-------|----------------------------|
| LW410 | COMPARATIVE LEGAL SYSTEMS  |
| LW462 | CYBER LAW                  |
| LW473 | ENVIRONMENTAL LAW          |
| LW474 | LAW OF ARMED CONFLICT      |
| LW475 | ADV CONSTITUTIONAL LAW SEM |
| LW481 | INTERNATIONAL LAW          |
| LW482 | NATIONAL SECURITY LAW      |

**OR**

**CSC-Strategic Studies**

Choose 3 of 21

|       |                               |
|-------|-------------------------------|
| GE491 | MILITARY GEOGRAPHY            |
| HI338 | WARFARE IN AGE OF REVOLUTIONS |

|       |                                    |
|-------|------------------------------------|
| HI347 | ASIAN WARFARE AND POLITICS         |
| HI355 | WARFARE, INDUSTRY, & EMPIRE        |
| HI356 | WAR AT SEA, SKY, SPACE             |
| HI357 | ARMED CONFLICT IN THE COLD WAR     |
| HI359 | ERA OF THE SECOND WORLD WAR        |
| HI370 | ANCIENT & MEDIEVAL WARFARE         |
| HI376 | EARLY MODERN WARFARE               |
| HI381 | HISTORY OF IRREGULAR WARFARE       |
| PL471 | LEADERSHIP IN COMBAT               |
| WS301 | INTRODUCTION TO WAR STUDIES        |
| WS345 | MILITARY INNOVATION                |
| WS358 | STRATEGY AND GENERALSHIP           |
| WS360 | SPECIAL OPS IN THEORY & PRACTICE   |
| WS365 | REBELLION, INSURGENCE, & CIVIL WAR |
| WS370 | US STRATEGY AND POLICY             |
| WS385 | WAR AND ITS THEORISTS              |
| WS475 | FORECAST AND GAME IN DECISION      |
| WS485 | JOINT AND MULTINATIONAL OPS        |
| XH397 | GRAND STRATEGY FIELD STUDY         |

**OR****CSC-United States Foreign Policy**

Choose 3 of 9

|       |                                   |
|-------|-----------------------------------|
| HI340 | COLONIAL AMERICA                  |
| HI369 | AMERICAN FRONTIERS                |
| HI372 | US FOREIGN RELATIONS SINCE 1898   |
| HI394 | REVOLUTIONARY AMERICA             |
| HI395 | HISTORY OF CIVIL WAR AMERICA      |
| HI396 | THEODORE ROOSEVELT'S AMERICA      |
| HI397 | THE US FROM COLD WAR TO TODAY     |
| HI398 | CIVIL RIGHTS IN AMERICAN HISTORY  |
| HI487 | EARLY AMERICAN REPUBLIC 1787-1826 |

**OR****CSC-Culture and Philosophy**

Choose 3 of 9

|       |                                |
|-------|--------------------------------|
| EN344 | CRITICISM                      |
| HI391 | WORLD RELIGIONS                |
| HI392 | AMERICAN HISTORICAL MEMORY     |
| LW461 | CIVIL RIGHTS AND THE LAW       |
| PL372 | MILITARY FAMILIES              |
| PL383 | EXPERIMENTAL SOCIAL PSYCHOLOGY |
| PY305 | LOGICAL REASONING              |
| PY326 | ETHICS OF TECHNOLOGY           |
| PY350 | PHILOSOPHY OF SCIENCE          |

**OR****CSC-Research Methods: General Track**

Choose 3 of 7

For the Design Requirement, you must select 2 courses from WS495, GE305, PL361, PL363. For the Formal Theory Requirement, you must select 1 course from MA383, PY305, SS364. Note that the prerequisite requirements for some of these courses may make this a difficult option for some cadets.

|       |                                |
|-------|--------------------------------|
| GE305 | GEOGRAPHIC RESEARCH METHODS    |
| MA383 | FOUNDATIONS OF MATH            |
| PL361 | RESEARCH METHODS I             |
| PL363 | QUALITATIVE RESEARCH METHODS   |
| PY305 | LOGICAL REASONING              |
| SS364 | GAME THEORY                    |
| WS495 | METHODS & CAPSTONE WAR STUDIES |

**OR****CSC-Research Methods: Quantitative Focus Track**

Choose 3 of 7

For the Quantitative Focus Track, you must select MA204/MA205 and MA367 and choose 1 of GE305, MA376, MA391, PL462, SS468. Note that the prerequisite requirements for some of these courses may make this a difficult option for some cadets.

|       |                                |
|-------|--------------------------------|
| GE305 | GEOGRAPHIC RESEARCH METHODS    |
| MA204 | CALCULUS I AND II              |
| MA205 | CALCULUS II                    |
| MA367 | MATH FOR THE SOCIAL SCIENCES   |
| MA376 | APPLIED STATISTICS             |
| MA391 | MATHEMATICAL MODELING          |
| PL462 | EXPERIMENTAL APP IN PSYCHOLOGY |
| SS468 | POLITICAL PARTICIPATION        |

**OR****CSC-Economics and Finance**

Choose 3 of 17

Cadets in this CSC track must select SS388 and any two of the remaining courses listed.

|       |                                |
|-------|--------------------------------|
| SS364 | GAME THEORY                    |
| SS368 | ECONOMETRICS I                 |
| SS380 | MANPOWER-LABOR ECONOMICS       |
| SS382 | MICROECONOMICS                 |
| SS385 | HISTORY OF ECONOMICS           |
| SS387 | PUBLIC FINANCE                 |
| SS388 | MACROECONOMICS                 |
| SS390 | BEHAVIORAL ECONOMICS           |
| SS394 | FUND. FINANCIAL DATA ANALYSIS  |
| SS461 | INDUSTRIAL ORGANIZATION        |
| SS462 | ECON OF STABILIZATION & GROWTH |
| SS463 | INVESTMENTS THEORY & APPL      |
| SS469 | ECONOMETRICS II                |
| SS470 | MONEY AND BANKING              |
| SS484 | INTERNATIONAL ECONOMICS        |
| SS494 | PRINCIPLES OF FINANCE          |
| SS497 | ISSUES IN MICROECONOMIC THEORY |

**AND****Capstone Requirement - Integrative Experience for the Major**

Choose 1 of 3

Only cadets in the Thesis major can select SS498B to fulfill this requirement.

|        |                                |
|--------|--------------------------------|
| SS491  | SENIOR PROJECT                 |
| SS495  | SENIOR FACULTY COLLOQUIUM      |
| SS498B | SENIOR THESIS IN INTER AFFAIRS |

**AND****Science Depth**

Cadets in this major may choose any of the authorized courses to satisfy the Science Depth requirement.

|       |                      |
|-------|----------------------|
| CH102 | GENERAL CHEMISTRY II |
| CH275 | BIOLOGY              |
| PH202 | PHYSICS II           |
| PH252 | ADVANCED PHYSICS II  |
| PH275 | PHYSICS II: SPACE    |

**STEM Depth**

Cadets in this major must take CY305/CY355 to satisfy the STEM Depth Requirement.

|       |                               |
|-------|-------------------------------|
| CY305 | CYBER FOUNDATIONS             |
| CY355 | CYBER FOUNDATIONS - COMPUTING |

**AND**

**Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirement**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major may select any Core Engineering Sequence. This sequence is selected in the fall semester of their third class year.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

SS475

COMP POLITICAL INSTITUTIONS

**2029 Terrorism Studies Minor Curriculum**

| Code  | Short Description       | Description             | Transcript Description  | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------|-------------------------|-------------------------|--------------|--------------|
| TST1N | Terrorism Studies Minor | Terrorism Studies Minor | Terrorism Studies Minor | 2            | 3            |

**2029 Terrorism Studies Minor Tracks**

| Subject Area | Description |
|--------------|-------------|
|--------------|-------------|

|                         |               |
|-------------------------|---------------|
| <b>Required Courses</b> | Choose 2 of 2 |
|-------------------------|---------------|

SS465

TERRORISM: NEW CHALLENGES

SS466

COMBATING TERRORISM

**AND****Subdisciplines**

Select one of the following five tracks. Cadets desiring to concentrate on counter terrorism studies should choose SS464 and 2 CT electives; those desiring to concentrate on terrorism in the Middle East/Africa should choose 1 mandatory course and 2 electives from the Middle East/Africa Track; those desiring to concentrate on terrorism in Asia should choose 1 mandatory course and 2 electives from the Asia Track; those desiring to concentrate on terrorism in Latin America should choose 1 mandatory course and 2 electives from the Latin America Track; and those desiring to concentrate on terrorism in Eurasia should choose 1 mandatory course and 2 electives from the Eurasia Track.

**AND****CT Track**

Choose 3 of 25

SS464 must be one of the 3 courses.

CY450

CYBER SECURITY ENGINEERING

CY460

CYBER POLICY, STRATEGY, &amp; OPNS

GE494

ENVIRONMENTAL SECURITY

HI381

HISTORY OF IRREGULAR WARFARE

LW474

LAW OF ARMED CONFLICT

LW482

NATIONAL SECURITY LAW

MA394

FUNDAMENTALS/NETWORK SCIENCE

NE450

NUCLEAR WEAPONS EFFECTS

PL482

ARMED FORCES AND SOCIETY

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|        |                                |
|--------|--------------------------------|
| SS399  | SOCSCI INTERNSHIP/PRACTCAL EXP |
| SS464  | HOMELAND SECURITY              |
| SS473  | AMERICAN FOREIGN POLICY        |
| SS476  | CIVIL WARS                     |
| SS477  | ECONOMICS OF NATIONAL SECURITY |
| SS478  | DIST PROF OF SECURITY STUD SEM |
| SS481  | AM GRAND STRAT/DEFENSE POLICY  |
| SS483  | NATIONAL SECURITY SEMINAR      |
| SS486  | STATE BUILDING                 |
| SS490A | COLLOQUIUM (AMER POLITICS)     |
| SS490B | COLLOQUIUM (INTERNTL AFFAIRS)  |
| SS490C | COLLOQUIUM (ECONOMICS)         |
| WS360  | SPECIAL OPS IN THEORY & PRACTI |
| WS365  | REBELLN, INSURGNCY, & CIV WAR  |
| WS370  | US STRATEGY AND POLICY         |
| WS399  | INTERNSHIP IN WAR STUDIES      |

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**OR****Middle East/Africa Track**

Choose 3 of 18

One of the three courses must be either HI339, HI345, or SS383.

|        |                                |
|--------|--------------------------------|
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE344  | GEOGRAPHY OF AFRICA            |
| GE345  | GEOGRAPHY OF THE MIDDLE EAST   |
| HI339  | THE MODERN MIDDLE EAST         |
| HI345  | MODERN AFRICA                  |
| HI383  | MIDDLE EASTERN WARFARE         |
| LA483  | ARAB CIVILIZATION I            |
| LA484  | ARAB CIVILIZATION II           |
| LN440A | ARABIC IN CULTURAL CONTEXT     |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| SS383  | POLITICS & GOVT-MIDDLE EAST    |
| SS399  | SOCSCI INTERNSHIP/PRACTCAL EXP |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS485  | POLIT & DEV SUB-SAHARAN AFR    |
| SS490A | COLLOQUIUM (AMER POLITICS)     |
| SS490B | COLLOQUIUM (INTERNTL AFFAIRS)  |
| SS490C | COLLOQUIUM (ECONOMICS)         |

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**OR****Asia Track**

Choose 3 of 13

HI346 must be one of the 3 courses.

|        |                                |
|--------|--------------------------------|
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342  | GEOGRAPHY OF ASIA              |
| HI337  | CHINA: EMPIRE, REPUBLIC, & MAO |
| HI346  | INDIA, PAKISTAN, & BANGLADESH  |
| HI347  | ASIAN WARFARE AND POLITICS     |
| LC483  | CHINESE CIVILIZATION I         |
| LC484  | CHINESE CIVILIZATION II        |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| PY380  | 20TH CENTURY PHILOSOPHY        |
| SS372  | POLITICS AND GOV OF CHINA      |
| SS374  | POL & GOV OF KOREAS & JAPAN    |
| SS399  | SOCSCI INTERNSHIP/PRACTCAL EXP |
| SS490A | COLLOQUIUM (AMER POLITICS)     |
| SS490B | COLLOQUIUM (INTERNTL AFFAIRS)  |
| SS490C | COLLOQUIUM (ECONOMICS)         |

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**OR**

**Latin America Track**

Choose 3 of 10

One of the 3 courses must be either HI348 or SS384.

|        |                                |
|--------|--------------------------------|
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE347  | GEOGRAPHY OF LATIN AMERICA     |
| HI348  | MODERN LATIN AMERICA           |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LP484  | PORTUGUESE CIVILIZATION II     |
| LS484  | SPANISH AMERICAN CIV AND CULT  |
| SS384  | POLITICS & GOVT-LATIN AMER     |
| SS399  | SOCSCI INTERNSHIP/PRACTCAL EXP |
| SS475  | COMP POLITICAL INSTITUTIONS    |
| SS490A | COLLOQUIUM (AMER POLITICS)     |
| SS490B | COLLOQUIUM (INTERNTL AFFAIRS)  |
| SS490C | COLLOQUIUM (ECONOMICS)         |

**OR****Eurasia Track**

Choose 3 of 23

One of the 3 courses must be either HI364, HI368, or SS377.

|        |                                |
|--------|--------------------------------|
| GE340  | GEOGRAPHY OF GLOBAL CULTURES   |
| GE343  | GEOGRAPHY OF RUSSIA            |
| GE346  | GEOGRAPHY OF EUROPE            |
| HI343  | MODERN GERMANY                 |
| HI361  | MEDIEVAL EUROPE                |
| HI364  | MODERN WESTERN EUROPE          |
| HI367  | IMPERIAL AND SOVIET RUSSIA     |
| HI368  | MOD CENTRAL & E. EUR,1896-1989 |
| LF483  | FRENCH CIVILIZATION I          |
| LF484  | FRENCH CIVILIZATION II         |
| LG483  | GERMAN CIVILIZATION I          |
| LG484  | GERMAN CIVILIZATION II         |
| LG492  | 20TH & 21ST CENTURY GERMANY    |
| LN491  | SEM ABROAD: ADV LANG & CULT I  |
| LN492  | SEM ABROAD: ADV LANG & CULT II |
| LR483  | RUSSIAN CIV I                  |
| LR484  | RUSSIAN CIV II                 |
| SS375  | GOV & POL RUSSIA & NEIGHBORS   |
| SS377  | POLITICS & GOV OF EUROPE       |
| SS399  | SOCSCI INTERNSHIP/PRACTCAL EXP |
| SS490A | COLLOQUIUM (AMER POLITICS)     |
| SS490B | COLLOQUIUM (INTERNTL AFFAIRS)  |
| SS490C | COLLOQUIUM (ECONOMICS)         |

## Systems Engineering

### 2029 Engineering Management Major Curriculum

| Code | Short Description      | Description                  | Transcript Description | Req Crse<br>Cnt | Opt Crse<br>Cnt |
|------|------------------------|------------------------------|------------------------|-----------------|-----------------|
| ENM1 | Engineering Management | Engineering Management Major | Engineering Management | 11              | 6               |

### 2029 Engineering Management Major Tracks

| Subject Area                                                                                                                                         | Description                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Required Courses</b>                                                                                                                              | Choose 10 of 12                  |
| For the math requirement, cadets should take MA204, MA205 or MA255 if eligible.                                                                      |                                  |
| EM381                                                                                                                                                | ENGINEERING ECONOMY              |
| EM384                                                                                                                                                | ANYL METH FOR ENGR<br>MANAGEMENT |
| EM411                                                                                                                                                | PROJECT MANAGEMENT               |
| EM420                                                                                                                                                | PRODUCTION OPERATIONS MGMT       |
| EM482                                                                                                                                                | SUPPLY CHAIN ENG & INFO MGMT     |
| MA204                                                                                                                                                | CALCULUS I AND II                |
| MA205                                                                                                                                                | CALCULUS II                      |
| MA255                                                                                                                                                | ADV MULTIVARIABLE CALCULUS       |
| SE301                                                                                                                                                | FNDTN ENGIN DSGN & SYS MGMT      |
| SE375                                                                                                                                                | STATISTICS FOR ENGINEERS         |
| SE385                                                                                                                                                | DECISION ANALYSIS                |
| SE400                                                                                                                                                | PROFESSIONAL ENGINEERING SEMIN   |
| <b>AND</b>                                                                                                                                           |                                  |
| <b>Simulation Elective</b>                                                                                                                           | Choose 1 of 3                    |
| EM481                                                                                                                                                | SYSTEMS SIMULATION               |
| SE485                                                                                                                                                | COMBAT MODELING                  |
| SM484                                                                                                                                                | SYSTEM DYNAMICS SIMULATION       |
| <b>AND</b>                                                                                                                                           |                                  |
| <b>Math or Basic Science Elective</b>                                                                                                                | Choose 1 of 9                    |
| To satisfy this requirement, Cadets can choose other courses containing at least 1 Math or Basic Science credit hour upon Program Director approval. |                                  |
| CH102                                                                                                                                                | GENERAL CHEMISTRY II             |
| CH275                                                                                                                                                | BIOLOGY                          |
| MA364                                                                                                                                                | ENGINEERING MATHEMATICS          |
| MA365                                                                                                                                                | ADV MATH FOR ENGRS/SCIENTISTS    |
| MA367                                                                                                                                                | MATH FOR THE SOCIAL SCIENCES     |
| MA371                                                                                                                                                | LINEAR ALGEBRA                   |
| MA386                                                                                                                                                | INTRO TO NUMERICAL ANALYSIS      |
| MA391                                                                                                                                                | MATHEMATICAL MODELING            |
| PH365                                                                                                                                                | MODERN PHYSICS                   |
| <b>AND</b>                                                                                                                                           |                                  |
| <b>Complementary Support Courses</b>                                                                                                                 |                                  |
| Choose one of eight CSC tracks below.                                                                                                                |                                  |
| <b>Project Management in Civil Engineering</b>                                                                                                       | Choose 3 of 3                    |
| CE350                                                                                                                                                | INFRASTRUCTURE ENGINEERING       |
| CE450                                                                                                                                                | CONSTRUCTION MANAGEMENT          |
| MC300                                                                                                                                                | FUND OF ENGR MECH AND DESIGN     |

|                                                                                             |                                |
|---------------------------------------------------------------------------------------------|--------------------------------|
| <b>OR</b>                                                                                   |                                |
| <b>Electrical Engineering</b>                                                               | Choose 3 of 3                  |
| EE302                                                                                       | INTRO ELECTRICAL ENGIN         |
| EE360                                                                                       | FUNDAMENTALS OF DIGITAL LOGIC  |
| EE362                                                                                       | INTRODUCTION TO ELECTRONICS    |
| <b>OR</b>                                                                                   |                                |
| <b>Engineering Fundamentals</b>                                                             | Choose 3 of 3                  |
| EE301                                                                                       | FUNDAMENTALS OF ELEC ENGIN     |
| MC300                                                                                       | FUND OF ENGR MECH AND DESIGN   |
| ME301                                                                                       | THERMODYNAMICS                 |
| <b>OR</b>                                                                                   |                                |
| <b>Environmental Engineering</b>                                                            | Choose 3 of 3                  |
| EV350                                                                                       | ENVIRONMNTL ENGR TECHNOLOGIES  |
| EV481                                                                                       | WATER RESOURCES PLAN & DESIGN  |
| GE310                                                                                       | GEOG INFORMATION SYSTEMS       |
| <b>OR</b>                                                                                   |                                |
| <b>Infrastructure Engineering</b>                                                           | Choose 3 of 3                  |
| CE350                                                                                       | INFRASTRUCTURE ENGINEERING     |
| EV350                                                                                       | ENVIRONMNTL ENGR TECHNOLOGIES  |
| GE310                                                                                       | GEOG INFORMATION SYSTEMS       |
| <b>OR</b>                                                                                   |                                |
| <b>Nuclear Engineering</b>                                                                  | Choose 3 of 3                  |
| NE300                                                                                       | FUNDAMENTALS OF NUCLEAR ENGR   |
| NE350                                                                                       | RADIOLOGICAL ENGR DESIGN       |
| NE450                                                                                       | NUCLEAR WEAPONS EFFECTS        |
| <b>OR</b>                                                                                   |                                |
| <b>Software Fundamentals</b>                                                                | Choose 3 of 3                  |
| CS384                                                                                       | DATA STRUCTURES                |
| CS393                                                                                       | DATABASE SYSTEMS               |
| CY300                                                                                       | PROGRAMMING FUNDAMENTALS       |
| <b>OR</b>                                                                                   |                                |
| <b>Chemical Engineering</b>                                                                 | Choose 3 of 3                  |
| CH362                                                                                       | MASS & ENERGY BALANCES         |
| CH363                                                                                       | SEPARATION PROCESSES           |
| CH364                                                                                       | CHEMICAL REACTION ENGINEERING  |
| <b>AND</b>                                                                                  |                                |
| <b>Integrative Experience for the Major</b>                                                 | Choose 2 of 2                  |
| SE402                                                                                       | SYSTEMS DESIGN & MANAGEMENT I  |
| SE403                                                                                       | SYSTEMS DESIGN & MANAGEMENT II |
| <b>AND</b>                                                                                  |                                |
| <b>Science Depth</b>                                                                        |                                |
| Cadets in this major must take PH202/252 or PH275 to satisfy the Science Depth requirement. |                                |
| PH202                                                                                       | PHYSICS II                     |
| PH252                                                                                       | ADVANCED PHYSICS II            |
| PH275                                                                                       | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                           |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.             |                                |
| CY305                                                                                       | CYBER FOUNDATIONS              |
| CY355                                                                                       | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                  |                                |
| <b>Curriculum Requirements</b>                                                              |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.    |                                |



**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.

CY305

CYBER FOUNDATIONS

CY355

CYBER FOUNDATIONS - COMPUTING

**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

SE402

SYSTEMS DESIGN &amp; MANAGEMENT I

**2029 Engineering Management Major w/ Honors Curriculum**

| Code  | Short Description                | Description                            | Transcript Description           | Req Crse Cnt | Opt Crse Cnt |
|-------|----------------------------------|----------------------------------------|----------------------------------|--------------|--------------|
| ENM1H | Engineering Management w/ Honors | Engineering Management Major w/ Honors | Engineering Management w/ Honors | 1            | 1            |

**2029 Engineering Management Major w/ Honors Tracks**

| Subject Area                                                                                                                                                                                                                                                                                                                                            | Description                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Overarching Requirement</b>                                                                                                                                                                                                                                                                                                                          |                                |
| A minimum of 42 total academic courses (3.0 credit hours or higher, taken or validated) are required to achieve the Honors distinction. All Honors Cadets will take SE491 in addition to one extra elective. This 42nd course defaults to SE490 but can be substituted for a different elective track course or other approved by the Program Director. |                                |
| <b>Research Course</b>                                                                                                                                                                                                                                                                                                                                  | Choose 2 of 2                  |
| SE490                                                                                                                                                                                                                                                                                                                                                   | AD TOPICS IN SYS ENG/ENG MGMT  |
| SE491                                                                                                                                                                                                                                                                                                                                                   | RSRCH PROJ IN SYS ENG/ENG MGMT |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                              |                                |
| <b>Grade Requirement</b>                                                                                                                                                                                                                                                                                                                                |                                |
| Complete the requirements of the major as shown above and attain a minimum APSC of 3.0 in the core curriculum and 3.5 in the major.                                                                                                                                                                                                                     |                                |

**2029 Engineering Management Minor Curriculum**

| Code  | Short Description            | Description                  | Transcript Description       | Req Crse Cnt | Opt Crse Cnt |
|-------|------------------------------|------------------------------|------------------------------|--------------|--------------|
| EMN0N | Engineering Management Minor | Engineering Management Minor | Engineering Management Minor | 2            | 3            |

**2029 Engineering Management Minor Tracks**

| Subject Area                                                                                        | Description                   |
|-----------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Required Courses</b>                                                                             | Choose 2 of 2                 |
| Cadets majoring in Engineering Management may not pursue a minor in Engineering Management as well. |                               |
| EM381                                                                                               | ENGINEERING ECONOMY           |
| EM411                                                                                               | PROJECT MANAGEMENT            |
| <b>AND</b>                                                                                          |                               |
| <b>Engineering Design Introductory Course</b>                                                       | Choose 1 of 8                 |
| CY300                                                                                               | PROGRAMMING FUNDAMENTALS      |
| EE300                                                                                               | FUNDAMENTALS OF DIGITAL LOGIC |
| EE362                                                                                               | INTRODUCTION TO ELECTRONICS   |
| EV300                                                                                               | ENVIRONMENTAL SCIENCE         |
| EV401                                                                                               | PHYS & CHEM TREATMENT         |
| MC300                                                                                               | FUND OF ENGR MECH AND DESIGN  |
| NE361                                                                                               | COMPUTATIONAL DESIGN IN NE    |
| SE301                                                                                               | FNDTN ENGIN DSGN & SYS MGMT   |
| <b>AND</b>                                                                                          |                               |
| <b>Quantitative Management Methods Course</b>                                                       | Choose 1 of 7                 |
| CE350                                                                                               | INFRASTRUCTURE ENGINEERING    |
| CE495                                                                                               | TRANSPORTATION ENGINEERING    |
| EM384                                                                                               | ANYL METH FOR ENGR MANAGEMENT |
| MA364                                                                                               | ENGINEERING MATHEMATICS       |
| MA365                                                                                               | ADV MATH FOR ENGRS/SCIENTISTS |
| MA367                                                                                               | MATH FOR THE SOCIAL SCIENCES  |
| SE387                                                                                               | DETERMINISTIC MODELS          |
| <b>AND</b>                                                                                          |                               |
| <b>Engineering Management Elective</b>                                                              | Choose 1 of 12                |
| EM420                                                                                               | PRODUCTION OPERATIONS MGMT    |
| EM481                                                                                               | SYSTEMS SIMULATION            |
| EM482                                                                                               | SUPPLY CHAIN ENG & INFO MGMT  |
| MA376                                                                                               | APPLIED STATISTICS            |
| MA391                                                                                               | MATHEMATICAL MODELING         |
| MA394                                                                                               | FUNDAMENTALS/NETWORK SCIENCE  |
| ME202                                                                                               | COMPUTER AIDED DESIGN         |
| SE370                                                                                               | COMPUTER AIDED SYSTEMS ENG    |
| SE375                                                                                               | STATISTICS FOR ENGINEERS      |
| SE385                                                                                               | DECISION ANALYSIS             |
| SE485                                                                                               | COMBAT MODELING               |
| SM484                                                                                               | SYSTEM DYNAMICS SIMULATION    |

## 2029 Systems and Decision Sciences Major Curriculum

| Code | Short Description             | Description                         | Transcript Description        | Req Crse Cnt | Opt Crse Cnt |
|------|-------------------------------|-------------------------------------|-------------------------------|--------------|--------------|
| SDS1 | Systems and Decision Sciences | Systems and Decision Sciences Major | Systems and Decision Sciences | 9            | 7            |

## 2029 Systems and Decision Sciences Major Tracks

| Subject Area                                                                                                                                | Description                       |
|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>Required Courses</b>                                                                                                                     | Choose 7 of 7                     |
| EM381                                                                                                                                       | ENGINEERING ECONOMY               |
| EM384                                                                                                                                       | ANYL METH FOR ENGR<br>MANAGEMENT  |
| EM411                                                                                                                                       | PROJECT MANAGEMENT                |
| SE301                                                                                                                                       | FNDTN ENGIN DSGN & SYS MGMT       |
| SE375                                                                                                                                       | STATISTICS FOR ENGINEERS          |
| SE385                                                                                                                                       | DECISION ANALYSIS                 |
| SM484                                                                                                                                       | SYSTEM DYNAMICS SIMULATION        |
| <b>AND</b>                                                                                                                                  |                                   |
| <b>Technical Elective</b>                                                                                                                   | Choose 1 of 10                    |
| CY300                                                                                                                                       | PROGRAMMING FUNDAMENTALS          |
| MA204                                                                                                                                       | CALCULUS I AND II                 |
| MA205                                                                                                                                       | CALCULUS II                       |
| MA255                                                                                                                                       | ADV MULTIVARIABLE CALCULUS        |
| MA388                                                                                                                                       | SABERMETRICS                      |
| MA391                                                                                                                                       | MATHEMATICAL MODELING             |
| MA394                                                                                                                                       | FUNDAMENTALS/NETWORK<br>SCIENCE   |
| PY305                                                                                                                                       | LOGICAL REASONING                 |
| SE370                                                                                                                                       | COMPUTER AIDED SYSTEMS ENG        |
| SS364                                                                                                                                       | GAME THEORY                       |
| <b>AND</b>                                                                                                                                  |                                   |
| <b>Areas of Concentration</b>                                                                                                               |                                   |
| Choose one of the following areas of concentration/tracks and courses not already taken above or below. Note that each track has two parts. |                                   |
| <b>Track 1: Management Systems<br/>(Study-in-Depth)</b>                                                                                     | Choose 3 of 8                     |
| EM420                                                                                                                                       | PRODUCTION OPERATIONS MGMT        |
| EM481                                                                                                                                       | SYSTEMS SIMULATION                |
| EM482                                                                                                                                       | SUPPLY CHAIN ENG & INFO MGMT      |
| SE302                                                                                                                                       | FUNDAMENTALS OF SYSTEMS ENG       |
| SE370                                                                                                                                       | COMPUTER AIDED SYSTEMS ENG        |
| SE485                                                                                                                                       | COMBAT MODELING                   |
| SE489                                                                                                                                       | AD IND STY IN SYS ENG/ENG MGMT    |
| SE490                                                                                                                                       | AD TOPICS IN SYS ENG/ENG MGMT     |
| <b>AND</b>                                                                                                                                  |                                   |
| <b>Track 1: Management Systems<br/>(Electives - including CSCs)</b>                                                                         | Choose 3 of 9                     |
| MG379                                                                                                                                       | LEADING TEAMS                     |
| MG380                                                                                                                                       | MARKETING                         |
| MG382                                                                                                                                       | HUMAN RESOURCE MANAGEMENT         |
| MG390                                                                                                                                       | NEGOTIATION FOR LEADERS           |
| MG395                                                                                                                                       | FUNDAMENTALS OF ACCOUNTING        |
| MG410                                                                                                                                       | MANAGERIAL FINANCE                |
| PL383                                                                                                                                       | SOCIAL PSYCHOLOGY                 |
| PL398                                                                                                                                       | LEADERSHIP THEORY & PRACTICE      |
| PL479                                                                                                                                       | LEADING CHANGING<br>ORGANIZATIONS |
| <b>OR</b>                                                                                                                                   |                                   |
| <b>Track 2: Defense Systems<br/>(Study-in-Depth)</b>                                                                                        | Choose 2 of 4                     |

|                                                                     |                                |
|---------------------------------------------------------------------|--------------------------------|
| SE302                                                               | FUNDAMENTALS OF SYSTEMS ENG    |
| SE370                                                               | COMPUTER AIDED SYSTEMS ENG     |
| SE485                                                               | COMBAT MODELING                |
| SE490                                                               | AD TOPICS IN SYS ENG/ENG MGMT  |
| <b>AND</b>                                                          |                                |
| <b>Track 2: Defense Systems (Electives, including CSCs)</b>         | Choose 4 of 12                 |
| HI381                                                               | HISTORY OF IRREGULAR WARFARE   |
| SE489                                                               | AD IND STY IN SYS ENG/ENG MGMT |
| SS464                                                               | HOMELAND SECURITY              |
| SS465                                                               | TERRORISM: NEW CHALLENGES      |
| SS481                                                               | POLITICS OF DEFENSE POLICY     |
| SS483                                                               | NATIONAL SECURITY SEMINAR      |
| WS345                                                               | MILITARY INNOVATION            |
| WS360                                                               | SPECIAL OPS IN THEORY & PRACTI |
| WS365                                                               | REBELLN, INSURGNCY, & CIV WAR  |
| WS370                                                               | US STRATEGY AND POLICY         |
| WS455                                                               | COMPARATIVE DEFENSE POLICY     |
| WS475                                                               | FORECAST AND GAME IN DECISION  |
| <b>OR</b>                                                           |                                |
| <b>Track 3: Financial Systems (Study-in-Depth)</b>                  | Choose 2 of 4                  |
| EM420                                                               | PRODUCTION OPERATIONS MGMT     |
| EM482                                                               | SUPPLY CHAIN ENG & INFO MGMT   |
| SE370                                                               | COMPUTER AIDED SYSTEMS ENG     |
| SE490                                                               | AD TOPICS IN SYS ENG/ENG MGMT  |
| <b>AND</b>                                                          |                                |
| <b>Track 3: Financial Systems (Electives, including CSCs)</b>       | Choose 4 of 8                  |
| MG395                                                               | FUNDAMENTALS OF ACCOUNTING     |
| SE489                                                               | AD IND STY IN SYS ENG/ENG MGMT |
| SS364                                                               | GAME THEORY                    |
| SS368                                                               | ECONOMETRICS I                 |
| SS385                                                               | HISTORY OF ECONOMICS           |
| SS394                                                               | FUND. FINANCIAL DATA ANALYSIS  |
| SS469                                                               | ECONOMETRICS II                |
| SS494                                                               | PRINCIPLES OF FINANCE          |
| <b>OR</b>                                                           |                                |
| <b>Track 4: Socio-Technical Systems (Study-in-Depth)</b>            | Choose 2 of 4                  |
| EM420                                                               | PRODUCTION OPERATIONS MGMT     |
| EM482                                                               | SUPPLY CHAIN ENG & INFO MGMT   |
| SE370                                                               | COMPUTER AIDED SYSTEMS ENG     |
| SE490                                                               | AD TOPICS IN SYS ENG/ENG MGMT  |
| <b>AND</b>                                                          |                                |
| <b>Track 4: Socio-Technical Systems (Electives, including CSCs)</b> | Choose 4 of 20                 |
| GE340                                                               | GEOGRAPHY OF GLOBAL CULTURES   |
| GE342                                                               | GEOGRAPHY OF ASIA              |
| GE343                                                               | GEOGRAPHY OF RUSSIA            |
| GE344                                                               | GEOGRAPHY OF AFRICA            |
| GE345                                                               | GEOGRAPHY OF THE MIDDLE EAST   |
| GE346                                                               | GEOGRAPHY OF EUROPE            |
| GE347                                                               | GEOGRAPHY OF LATIN AMERICA     |
| HI339                                                               | THE MODERN MIDDLE EAST         |
| HI345                                                               | MODERN AFRICA                  |

|                                                                                                                                                                              |                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| HI346                                                                                                                                                                        | INDIA, PAKISTAN, & BANGLADESH  |
| HI347                                                                                                                                                                        | ASIAN WARFARE AND POLITICS     |
| HI348                                                                                                                                                                        | MODERN LATIN AMERICA           |
| SE489                                                                                                                                                                        | AD IND STY IN SYS ENG/ENG MGMT |
| SS372                                                                                                                                                                        | POLITICS OF CHINA              |
| SS374                                                                                                                                                                        | POLITICS OF NORTHEAST ASIA     |
| SS375                                                                                                                                                                        | POL OF THE POST-SOVIET STATES  |
| SS377                                                                                                                                                                        | POLITICS OF EUROPE             |
| SS383                                                                                                                                                                        | POLITICS OF THE MIDDLE EAST    |
| SS384                                                                                                                                                                        | POLITICS OF LATIN AMERICA      |
| SS485                                                                                                                                                                        | POLITICS SUB-SAHARAN AFR       |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Integrative Experience for the Major</b>                                                                                                                                  | Choose 2 of 2                  |
| SE402                                                                                                                                                                        | SYSTEMS DESIGN & MANAGEMENT I  |
| SE403                                                                                                                                                                        | SYSTEMS DESIGN & MANAGEMENT II |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Science Depth</b>                                                                                                                                                         |                                |
| Cadets in this major may choose any approved Science Depth course.                                                                                                           |                                |
| CH102                                                                                                                                                                        | GENERAL CHEMISTRY II           |
| CH275                                                                                                                                                                        | BIOLOGY                        |
| PH202                                                                                                                                                                        | PHYSICS II                     |
| PH252                                                                                                                                                                        | ADVANCED PHYSICS II            |
| PH275                                                                                                                                                                        | PHYSICS II: SPACE              |
| <b>STEM Depth</b>                                                                                                                                                            |                                |
| Cadets in this major must take CY305/355 to satisfy the STEM Depth requirement.                                                                                              |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>AND</b>                                                                                                                                                                   |                                |
| <hr/>                                                                                                                                                                        |                                |
| <b>Curriculum Requirements</b>                                                                                                                                               |                                |
| This section describes how cadets in this major satisfy various curriculum requirements.                                                                                     |                                |
| <b>IT/CYBER Requirements</b>                                                                                                                                                 |                                |
| Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and CY305/355.                             |                                |
| CY305                                                                                                                                                                        | CYBER FOUNDATIONS              |
| CY355                                                                                                                                                                        | CYBER FOUNDATIONS - COMPUTING  |
| <b>Core Engineering Sequence</b>                                                                                                                                             |                                |
| The SDS major achieves the CES guidelines with existing courses of the major. The courses are SE301, EM384, and SE402.                                                       |                                |
| EM384                                                                                                                                                                        | ANYL METH FOR ENGR MANAGEMENT  |
| SE301                                                                                                                                                                        | FNDTN ENGIN DSGN & SYS MGMT    |
| SE402                                                                                                                                                                        | SYSTEMS DESIGN & MANAGEMENT I  |
| <b>Writing-in-the-Major</b>                                                                                                                                                  |                                |
| Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major. |                                |
| SE402                                                                                                                                                                        | SYSTEMS DESIGN & MANAGEMENT I  |

## 2029 Systems and Decision Sciences Major w/ Honors Curriculum

| Code  | Short Description                       | Description                                   | Transcript Description                  | Req Crse Cnt | Opt Crse Cnt |
|-------|-----------------------------------------|-----------------------------------------------|-----------------------------------------|--------------|--------------|
| SDS1H | Systems and Decision Sciences w/ Honors | Systems and Decision Sciences Major w/ Honors | Systems and Decision Sciences w/ Honors | 1            | 1            |

### 2029 Systems and Decision Sciences Major w/ Honors Tracks

| Subject Area                                                                                                                                                                                                                                                                                                                                            | Description                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Overarching Requirements</b>                                                                                                                                                                                                                                                                                                                         |                                |
| A minimum of 42 total academic courses (3.0 credit hours or higher, taken or validated) are required to achieve the Honors distinction. All Honors Cadets will take SE491 in addition to one extra elective. This 42nd course defaults to SE490 but can be substituted for a different elective track course or other approved by the Program Director. |                                |
| <b>Research Course</b>                                                                                                                                                                                                                                                                                                                                  | Choose 2 of 2                  |
| SE490                                                                                                                                                                                                                                                                                                                                                   | AD TOPICS IN SYS ENG/ENG MGMT  |
| SE491                                                                                                                                                                                                                                                                                                                                                   | RSRCH PROJ IN SYS ENG/ENG MGMT |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                              |                                |
| <b>Grade Requirements</b>                                                                                                                                                                                                                                                                                                                               |                                |
| Complete the requirements of the major shown above, attain an APSC of at least 3.0 in the core curriculum, and at least a 3.5 in the major.                                                                                                                                                                                                             |                                |

### 2029 Systems Engineering Major Curriculum

| Code | Short Description   | Description               | Transcript Description | Req Crse Cnt | Opt Crse Cnt |
|------|---------------------|---------------------------|------------------------|--------------|--------------|
| SEN2 | Systems Engineering | Systems Engineering Major | Systems Engineering    | 13           | 4            |

### 2029 Systems Engineering Major Tracks

| Subject Area                                           | Description                    |
|--------------------------------------------------------|--------------------------------|
| <b>Required Courses</b>                                | Choose 9 of 9                  |
| EM411                                                  | PROJECT MANAGEMENT             |
| SE301                                                  | FNDTN ENGIN DSGN & SYS MGMT    |
| SE302                                                  | FUNDAMENTALS OF SYSTEMS ENG    |
| SE370                                                  | COMPUTER AIDED SYSTEMS ENG     |
| SE375                                                  | STATISTICS FOR ENGINEERS       |
| SE385                                                  | DECISION ANALYSIS              |
| SE387                                                  | DETERMINISTIC MODELS           |
| SE388                                                  | STOCHASTIC MODELS              |
| SE400                                                  | PROFESSIONAL ENGINEERING SEMIN |
| <b>AND</b>                                             |                                |
| <b>Simulation Elective</b>                             | Choose 1 of 3                  |
| EM481                                                  | SYSTEMS SIMULATION             |
| SE485                                                  | COMBAT MODELING                |
| SM484                                                  | SYSTEM DYNAMICS SIMULATION     |
| <b>AND</b>                                             |                                |
| <b>Complementary Support Courses - Elective Tracks</b> |                                |

Choose one of the following elective tracks: Aerospace Systems, Artificial Intelligence, Cyber Security Systems, General Engineering, Geographic Information Systems, Human Systems Integration, Nuclear Systems, Robotic Systems, Software Systems, Sustainable Systems, and Cadet Engineered Systems Design. The elective tracks consist of EE301, MC300, and three courses that provide depth in a chosen engineering domain discipline outside the major.

|                                       |                                   |
|---------------------------------------|-----------------------------------|
| <b>All Tracks</b>                     | Choose 2 of 2                     |
| EE301                                 | FUNDAMENTALS OF ELEC ENGIN        |
| MC300                                 | FUND OF ENGR MECH AND DESIGN      |
| <b>AND</b>                            |                                   |
| <b>Aerospace Systems</b>              | Choose 3 of 5                     |
| AE287                                 | INTRODUCTION TO AERONAUTICS       |
| ME301                                 | THERMODYNAMICS                    |
| ME362                                 | FLUID MECHANICS                   |
| SP471                                 | ASTRONAUTICS                      |
| XE472                                 | DYNAMIC MODELING AND CONTROL      |
| <b>OR</b>                             |                                   |
| <b>Artificial Intelligence</b>        | Choose 3 of 3                     |
| CS384                                 | DATA STRUCTURES                   |
| CS486                                 | ARTIFICIAL INTELLIGENCE           |
| CY300                                 | PROGRAMMING FUNDAMENTALS          |
| <b>OR</b>                             |                                   |
| <b>Cyber Security Systems</b>         | Choose 3 of 5                     |
| CY300                                 | PROGRAMMING FUNDAMENTALS          |
| CY350                                 | COMPUTER NETWORKS                 |
| CY392                                 | DEFENSIVE CYBER OPERATIONS        |
| CY450                                 | CYBER SECURITY ENGINEERING        |
| CY460                                 | CYBER POLICY, STRATEGY, & OPNS    |
| <b>OR</b>                             |                                   |
| <b>General Engineering</b>            | Choose 3 of 7                     |
| CY300                                 | PROGRAMMING FUNDAMENTALS          |
| CY305                                 | CYBER FOUNDATIONS                 |
| EE381                                 | SIGNALS AND SYSTEMS               |
| EM381                                 | ENGINEERING ECONOMY               |
| MA364                                 | ENGINEERING MATHEMATICS           |
| ME301                                 | THERMODYNAMICS                    |
| ME362                                 | FLUID MECHANICS                   |
| <b>OR</b>                             |                                   |
| <b>Geographic Information Systems</b> | Choose 3 of 3                     |
| GE310                                 | GEOG INFORMATION SYSTEMS          |
| GE378                                 | CARTOGRAPHY                       |
| GE498                                 | ADV GEOGRAPHIC INFORMATION<br>SYS |
| <b>OR</b>                             |                                   |
| <b>Human Systems Integration</b>      | Choose 3 of 4                     |
| PL392                                 | COGNITIVE PSYCHOLOGY              |
| PL394                                 | ANTHROPOMETRICS &<br>BIOMECHANICS |
| PL475                                 | HUMAN-COMPUTER INTERACTION        |
| XE310                                 | INTRO TO BIOMECH ENGINEERING      |
| <b>OR</b>                             |                                   |
| <b>Nuclear Systems</b>                | Choose 3 of 3                     |
| ME301                                 | THERMODYNAMICS                    |
| NE300                                 | FUNDAMENTALS OF NUCLEAR ENGR      |
| NE350                                 | RADIOLOGICAL ENGR DESIGN          |
| <b>OR</b>                             |                                   |

**Robotic Systems**

Choose 3 of 5

EE360

FUNDAMENTALS OF DIGITAL LOGIC

EE450

MILITARY ROBOTIC SYSTEMS

ME306

DYNAMICS

XE472

DYNAMIC MODELING AND CONTROL

XE475

MECHATRONICS

**OR****Software Systems**

Choose 3 of 4

CS384

DATA STRUCTURES

CS393

DATABASE SYSTEMS

CS403

SOFTWARE TESTING &  
DEVELOPMENT

CY300

PROGRAMMING FUNDAMENTALS

**OR****Sustainable Systems**

Choose 3 of 5

CE350

INFRASTRUCTURE ENGINEERING

EV301

ENVIRONMENTAL SUSTAINABILITY

EV350

ENVIRONMNTL ENGR  
TECHNOLOGIES

SM484

SYSTEM DYNAMICS SIMULATION

XE442

ALTERNATIVE ENERGY  
ENGINEERING**OR****Cadet Engineered Systems Design Track**

Choose 3 of 3

Choose three courses, in addition to the required EE301 and MC300. At least two of the remaining three courses must be outside the department. The three courses must have an engineering domain theme and be approved by the Program Director. The designed track must contribute at least 4.0 Engineering Technology credit-hours.

**AND****Integrative Experience for the Major**

Choose 2 of 2

SE402

SYSTEMS DESIGN &amp; MANAGEMENT I

SE403

SYSTEMS DESIGN & MANAGEMENT  
II**AND****Science Depth**

Cadets in this major must take PH202, PH252, or PH275 to satisfy the Science Depth requirement.

PH202

PHYSICS II

PH252

ADVANCED PHYSICS II

PH275

PHYSICS II: SPACE

**STEM Depth**

Cadets in this major must take MA204/MA205 or MA255 to satisfy the STEM Depth requirement.

MA204

CALCULUS I AND II

MA205

CALCULUS II

MA255

ADV MULTIVARIABLE CALCULUS

**AND****Curriculum Requirements**

This section describes how cadets in this major satisfy various curriculum requirements.

**IT/CYBER Requirements**

Cadets in this major will satisfy the IT/CYBER requirement by successfully completing the IT/CYBER content in the core curriculum and the combination of EE301, SE370 and SE400.

EE301

FUNDAMENTALS OF ELEC ENGIN

SE370

COMPUTER AIDED SYSTEMS ENG

SE400

PROFESSIONAL ENGINEERING  
SEMIN



**Core Engineering Sequence**

Cadets in this major will satisfy the core engineering requirement as part of their major courses.

**Writing-in-the-Major**

Cadets in this major satisfy the West Point Writing Program requirement by successfully completing writing requirements in the core curriculum and this course in the major.

SE402

SYSTEMS DESIGN &amp; MANAGEMENT I

**2029 Systems Engineering Major w/ Honors Curriculum**

| Code  | Short Description             | Description                         | Transcript Description        | Req Crse Cnt | Opt Crse Cnt |
|-------|-------------------------------|-------------------------------------|-------------------------------|--------------|--------------|
| SEN2H | Systems Engineering w/ Honors | Systems Engineering Major w/ Honors | Systems Engineering w/ Honors | 2            | 0            |

**2029 Systems Engineering Major w/ Honors Tracks**

| Subject Area                                                                                                                                                                                                                                                                                                                                            | Description                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Overarching Requirements</b>                                                                                                                                                                                                                                                                                                                         |                                |
| A minimum of 42 total academic courses (3.0 credit hours or higher, taken or validated) are required to achieve the Honors distinction. All Honors Cadets will take SE491 in addition to one extra elective. This 42nd course defaults to SE490 but can be substituted for a different elective track course or other approved by the Program Director. |                                |
| <b>Research Course</b>                                                                                                                                                                                                                                                                                                                                  | Choose 2 of 2                  |
| SE490                                                                                                                                                                                                                                                                                                                                                   | AD TOPICS IN SYS ENG/ENG MGMT  |
| SE491                                                                                                                                                                                                                                                                                                                                                   | RSRCH PROJ IN SYS ENG/ENG MGMT |
| <b>AND</b>                                                                                                                                                                                                                                                                                                                                              |                                |
| <b>Grade Requirement</b>                                                                                                                                                                                                                                                                                                                                |                                |
| Complete the requirements of the major as shown above, attain an APSC of at least 3.0 in the core curriculum, and at least a 3.5 in the major.                                                                                                                                                                                                          |                                |

# APPENDIX

**Substantive and Select Administrative Changes**  
(since the AB last approved the Redbook, 27 June 2024)

**Department of Behavioral Sciences and Leadership:**

New major, Behavioral Sciences (3 tracks), effective for the Class of 2026

Sunset the Psychology Major (2 tracks), with GY2025 as the last official class year

Sunset the Sociology Major, with GY2025 as the last official class year

Archive PL371 Introduction to Sociology (3.0CR), PL377 Social Inequality (3.0CR), PL384 Sociological Theory (3.0CR), and PL488D Colloquium-BSL-Sociology (3.0CR)

**Department of Chemistry and Life Science:**

New minor, Bioengineering, effective for the Class of 2026

Change the name of the major from “Life Science” to “Biology,” effective for the GY2026 and beyond

**Department of Civil and Mechanical Engineering:**

Provisionally approved course, AE201 Introduction to Aerospace Engineering (3.5CR), effective AY26-1

Provisionally approved courses, AE189(A) Ind Study in Aerospace Eng (1.0CR), AE289(A) Ind Study in Aerospace Eng (2.0CR), AE389(A) Ind Study in Aerospace Eng (3.0CR), and AE489(A) Adv Ind Study in Aerospace Eng (3.0CR), effective AY26-1

New course, XE200 Scientific Argument (1.0CR), effective AY26-2

Sunset the Aeronautical Engineering minor, with GY2027 as the last official class year

Course conversion, archive MC306 Dynamics (3.0CR) and replace with ME306 Dynamics (3.0CR), effective AY2026-1

Course conversion, archive MC311 Thermal-Fluid Systems I (3.5CR) and replace with CV311 Thermal-Fluid Systems I (3.5CR), effective AY2027-1

Course conversion, archive MC364 Mechanics of Materials (3.5CR) and replace with CE364 Mechanics of Materials (3.5CR), effective AY2026-2

Course conversion, archive MC380 Engineering Materials (3.5 CR) and replace with ME380 Engineering Materials (3.5CR), effective AY2026-1

Course conversion, archive MC486 Vibration Engineering (3.0CR) and replace with ME486 Vibration Engineering (3.0CR), effective AY2026-2

### **Department of Electrical Engineering and Computer Science:**

Modified major, Cyber Science Major (CYS1) with Cyber Science: Cyber Operations Track (CYO0), effective for the Class of 2028

New core engineering sequence, Artificial Intelligence and Data Engineering (CES-AI), effective for the Class of 2028

New course, EE484 Fundamentals of Robotics (3.0CR), effective AY26-2

Course conversion, archive EE487 Embedded Systems Development (3.0CR) and replace with EE380 Embedded Systems Development, effective AY26-1

### **Department of English and World Languages:**

Sunset the Diversity and Inclusion Studies Minor, with GY2024 as the last official class year

Archive EN352 Power and Difference (3.0CR)

### **Department of Geography and Environmental Engineering:**

Change the name of the major from “Geospatial Information Science” to “Geospatial Science,” effective for the GY2026 and beyond

Sunset the Geography Minor, with GY2024 as the last official class year

New course, GE300 Global Environmental Systems (3.0CR), effective AY26-1

Modified major, Geography (1 track), effective for the Class of 2028

EV to GE course conversions, effective AY26-1, for the following courses:

|        |                              |  |        |                              |
|--------|------------------------------|--|--------|------------------------------|
| EV203  | PHYSICAL GEOGRAPHY           |  | GE203  | GEOGRAPHY                    |
| EV289A | INTRO IND STUDY & RESEARCH   |  | GE299A | INTRO IND STUDY & RESEARCH   |
| EV303  | FOUNDATIONS IN GEOGRAPHY     |  | GE205  | FOUNDATIONS IN GEOGRAPHY     |
| EV310  | AQUATIC SCIENCE              |  | GE331  | AQUATIC SCIENCE              |
| EV365  | GEOGRAPHY OF GLOBAL CULTURES |  | GE340  | GEOGRAPHY OF GLOBAL CULTURES |
| EV367  | GEOGRAPHIC RESEARCH METHODS  |  | GE305  | GEOGRAPHIC RESEARCH METHODS  |

|        |                                |  |        |                                |
|--------|--------------------------------|--|--------|--------------------------------|
| EV371  | GEOGRAPHY OF RUSSIA            |  | GE343  | GEOGRAPHY OF RUSSIA            |
| EV372  | GEOGRAPHY OF ASIA              |  | GE342  | GEOGRAPHY OF ASIA              |
| EV373  | GEOGRAPHY OF LATIN AMERICA     |  | GE347  | GEOGRAPHY OF LATIN AMERICA     |
| EV375  | GEOGRAPHY OF AFRICA            |  | GE344  | GEOGRAPHY OF AFRICA            |
| EV376  | GEOGRAPHY OF THE MIDDLE EAST   |  | GE345  | GEOGRAPHY OF THE MIDDLE EAST   |
| EV377  | REMOTE SENSING                 |  | GE373  | REMOTE SENSING                 |
| EV378  | CARTOGRAPHY                    |  | GE378  | CARTOGRAPHY                    |
| EV379  | PHOTOGRAMMETRY                 |  | GE379  | PHOTOGRAMMETRY                 |
| EV380  | SURVEYING                      |  | GE380  | SURVEYING                      |
| EV384  | GEOGRAPHY OF NORTH AMERICA     |  | GE341  | GEOGRAPHY OF NORTH AMERICA     |
| EV386  | GEOGRAPHY OF EUROPE            |  | GE346  | GEOGRAPHY OF EUROPE            |
| EV387  | METEOROLOGY                    |  | GE324  | METEOROLOGY                    |
| EV388A | PHYSICAL GEOLOGY               |  | GE333  | PHYSICAL GEOLOGY               |
| EV388B | GEOMORPHOLOGY                  |  | GE332  | GEOMORPHOLOGY                  |
| EV389A | INTER IND STUDY & RESEARCH     |  | GE399A | INTER IND STUDY & RESEARCH     |
| EV389B | CLIMATOLOGY                    |  | GE321  | CLIMATOLOGY                    |
| EV389C | INTER IND STUDY & RESEARCH II  |  | GE399B | INTER IND STUDY & RESEARCH II  |
| EV390B | URBAN GEOGRAPHY                |  | GE352  | URBAN GEOGRAPHY                |
| EV391A | LAND USE PLAN & MGT            |  | GE351  | LAND USE PLAN & MGT            |
| EV391B | NATURAL HAZARDS AND RISK       |  | GE354  | NATURAL HAZARDS AND RISK       |
| EV398  | GEOG INFORMATION SYSTEMS       |  | GE310  | GEOG INFORMATION SYSTEMS       |
| EV399A | GEOLOGY FIELD COURSE           |  | GE333A | GEOLOGY FIELD COURSE           |
| EV471  | ECOLOGY                        |  | GE431  | ECOLOGY                        |
| EV477  | ADVANCED REMOTE SENSING        |  | GE477  | ADVANCED REMOTE SENSING        |
| EV478  | MILITARY GEOSPATIAL OPERATIONS |  | GE478  | MILITARY GEOSPATIAL OPERATIONS |
| EV480  | HONORS SEMINAR IN GEOGRAPHY    |  | GE495  | HONORS SEMINAR IN GEOGRAPHY    |
| EV482  | MILITARY GEOGRAPHY             |  | GE491  | MILITARY GEOGRAPHY             |
| EV483  | COLLOQUIUM IN GEOGRAPHY        |  | GE496  | COLLOQUIUM IN GEOGRAPHY        |
| EV485  | SPEC TOPICS-GEOG & ENVRNMNT    |  | GE490  | SPEC TOPICS-GEOG & ENVRNMNT    |
| EV486  | ENVIRONMENT AND DEVELOPMENT    |  | GE462  | ENVIRONMENT AND DEVELOPMENT    |
| EV487  | ENVIRONMENTAL SECURITY         |  | GE494  | ENVIRONMENTAL SECURITY         |
| EV489A | ADVANCED INDIVIDUAL STUDY I    |  | GE499A | ADVANCED INDIVIDUAL STUDY I    |
| EV489B | ADVANCED INDIVIDUAL            |  | GE499B | ADVANCED INDIVIDUAL            |

|       |                                   |  |       |                                   |
|-------|-----------------------------------|--|-------|-----------------------------------|
|       | STUDY II                          |  |       | STUDY II                          |
| EV498 | ADV GEOGRAPHIC<br>INFORMATION SYS |  | GE498 | ADV GEOGRAPHIC<br>INFORMATION SYS |

### Department of History:

New major, War Studies, effective for the Class of 2028

Course conversions tied to the establishment of the War Studies Major:

|       |                                   |  |       |                                              |
|-------|-----------------------------------|--|-------|----------------------------------------------|
| DS320 | INTRO TO STRATEGIC<br>STUDIES     |  | WS301 | INTRODUCTION TO WAR<br>STUDIES               |
| DS345 | MILITARY INNOVATION               |  | WS345 | MILITARY INNOVATION                          |
| DS350 | STRAT & PERSUASIVE<br>COMMUNICATN |  | WS350 | STRATEGIC COMMUNICATION                      |
| DS360 | SPECIAL OPS: THEORY &<br>PRACTICE |  | WS360 | SPECIAL OPERATIONS IN<br>THEORY AND PRACTICE |
| DS370 | US STRATEGY AND POLICY            |  | WS370 | US STRATEGY AND POLICY                       |
| DS399 | STRATEGIC STUDIES<br>INTERNSHIP   |  | WS399 | INTERNSHIP IN WAR STUDIES                    |
| DS455 | COMPARATIVE DEFENSE<br>POLICY     |  | WS455 | COMPARATIVE DEFENSE<br>POLICY                |
| DS460 | IRREGULAR WARFARE<br>THEORY & PRA |  | WS365 | REBELLION, INSURGENCY, AND<br>CIVIL WAR      |
| DS475 | FORECAST & GAM IN<br>DECISION-MAK |  | WS475 | FORECASTING AND GAMING IN<br>DECISION-MAKING |
| DS485 | DOMAINS OF WAR                    |  | WS485 | JOINT AND MULTINATIONAL<br>OPERATIONS        |
| DS489 | ADV IND STUD-DEF/STRAT<br>STUDIES |  | WS489 | INDEPENDENT STUDY IN WAR<br>STUDIES          |
| DS490 | SPECIAL TOPICS: STRAT<br>STUDIES  |  | WS490 | SPECIAL TOPICS IN WAR<br>STUDIES             |
| DS495 | RESEARCH METHODS STRAT<br>STUDIES |  | WS495 | METHODS AND CAPSTONE IN<br>WAR STUDIES       |
| DS496 | STRATEGIC STUDIES THESIS          |  | WS496 | THESIS IN WAR STUDIES                        |
| DS497 | STRATEGIC STUDIES<br>CAPSTONE     |  | WS497 | CLIENT-BASED CAPSTONE IN<br>WAR STUDIES      |
| DS498 | LEADERSHIP IN FUTURE WAR          |  | WS398 | FUTURE WAR                                   |
| HI358 | POLICY, STRATEGY &<br>GENERALSHIP |  | WS358 | STRATEGY AND GENERALSHIP                     |
| HI385 | WAR & ITS THEORISTS               |  | WS385 | WAR AND ITS THEORISTS                        |
| SS457 | INTRODUCTION TO GRAND<br>STRATEGY |  | WS330 | INTRODUCTION TO GRAND<br>STRATEGY            |
| SS458 | TOPICS IN GRAND STRATEGY          |  | WS335 | TOPICS IN GRAND STRATEGY                     |

Sunset the America in the World Minor, with GY2025 as the last official class year

Sunset the War, Technology, and Society Minor, with GY2025 as the last official class year

New core course, HI108W Foundations of the Modern World, effective AY26-1

New course, HI377 Empires & Revolutions (3.0CR), effective AY27-2

New course, HI378 Industry, Ideology & the State (3.0CR), effective AY28-1

New course, HI387 Topics in American History (3.0CR), effective AY26-1

New course, HI388 Topics in International History (3.0CR), effective AY26-1

New course, HI389 Topics in Military History (3.0CR), effective AY26-1

New course, WS301 Introduction to War Studies (3.0CR), effective AY26-1

New course, WS389 War in the Contemporary World (3.0CR), effective AY27-2

Course conversion, archive HI390 Early National America (3.0CR) and replace with HI487 The Early American Republic, 1787-1826 (3.0CR), effective AY27-1

Archive, HI101 The Army of the Republic: Leading Citizen Soldiers (3.0CR), HI108A-U Regional Studies in World History (6 tracks: Africa, East Asia, Latin America, Middle East, Russia, Europe) (3.0CR), HI354 World War I in Global Perspective (3.0CR), HI365 The Ancient World (3.0 CR), HI461 Topics in Gender History (3.0CR), HI463 Race, Ethnicity, Nation (3.0CR), XH341 Intel Cyber History (3.0CR)

### **Department of Law and Philosophy: N/A**

### **Department of Mathematical Sciences:**

New course, MA204 Calculus I and II (4.5CR), effective AY25-2

### **Department of Military Instruction:**

Sunset the Defense and Strategic Studies Major, with GY2027 as the last official class year

New advanced course, MS250 Advanced Fundamentals of Small Unit Operations (1.5CR), effective AY26-1

New advanced course, MS350 Advanced Platoon Operations (1.5CR), effective AY26-1

Archive all DS courses and convert to WS (War Studies) prefix, effective AY28

**Department of Physical Education: N/A**

**Department of Physics and Nuclear Engineering: N/A**

**Department of Social Sciences:**

Change the name of the major from “Political Science: American Politics” to “American Politics,” effective for the GY2025 and beyond

Sunset the Grand Strategy Minor, with GY2025 as the last official class year

Archive SS381 Cultural and Political Anthropology (3.0CR)

**Department of Systems Engineering:**

Sunset the Systems Engineering Minor, with GY2024 as the last official class year

Archive XH102, Introduction to Excel Scholars Program (0.5CR), XH201 Foundations of Excel Program (0.5CR), and XH301 Foundations of Excel Program (0.5CR), effective AY25-2